

ZA-GAS Covariate & Long-Short Model Report

Station: BELO HORIZONTE

Models evaluated: 78

Generated: 2026-06-17 05:24

Model inventory

[1] xi_dewpoint_t [2] xi_dewpoint_lag1 [3] xi_dewpoint_short_lags [4] xi_dewpoint_seasonal_lags [5] xi_temperature_t
[6] xi_temperature_lag1 [7] xi_temperature_short_lags [8] xi_temperature_seasonal_lags [9] xi_dewpoint_temperature_t
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xi_mjo_t [14] xi_mjo_lag1 [15] xi_mjo_short_lags [16] xi_nino34_t [17] xi_nino34_lags [18] xi_nino34_rolling [19]
xi_interaction_dewpoint_temperature_t [20] xi_interaction_dewpoint_mjo [21] xi_interaction_dewpoint_nino34 [22]
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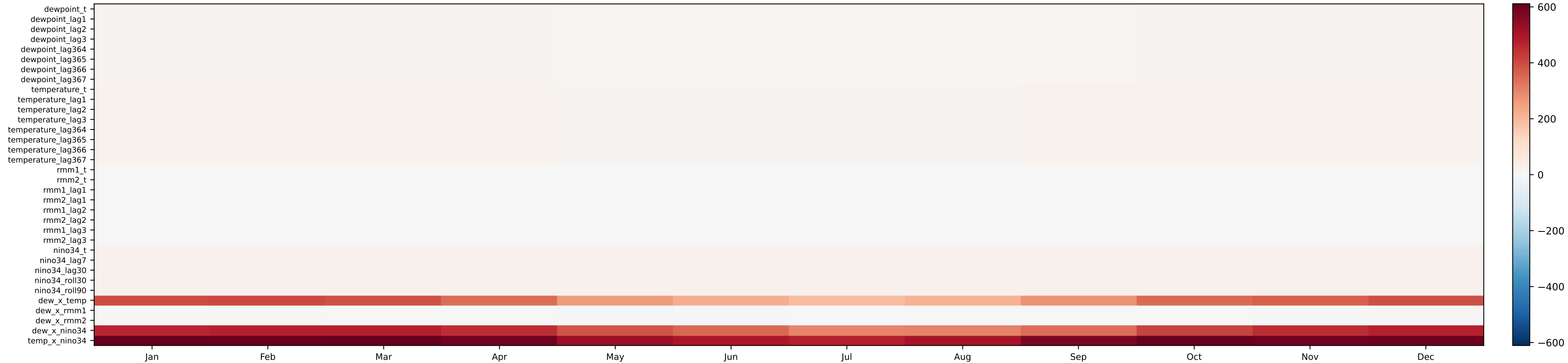
Covariate descriptive statistics — Raw

variable	mean	std	min	q05	q25	q50	q75	q95	max	skew	kurt	acf 1
dewpoint_t	14.932	3.216	2.020	9.269	12.528	15.588	17.631	19.163	20.532	-0.534	-0.491	0.926
dewpoint_lag1	14.931	3.216	2.020	9.269	12.528	15.585	17.630	19.163	20.532	-0.534	-0.491	0.926
dewpoint_lag2	14.930	3.215	2.020	9.269	12.528	15.582	17.629	19.161	20.532	-0.535	-0.491	0.926
dewpoint_lag3	14.928	3.215	2.020	9.269	12.528	15.580	17.626	19.159	20.532	-0.535	-0.491	0.926
dewpoint_lag364	14.900	3.224	2.020	9.270	12.481	15.554	17.582	19.180	20.532	-0.527	-0.478	0.926
dewpoint_lag365	14.899	3.224	2.020	9.270	12.481	15.553	17.580	19.180	20.532	-0.527	-0.478	0.926
dewpoint_lag366	14.898	3.224	2.020	9.270	12.481	15.549	17.579	19.180	20.532	-0.526	-0.478	0.926
dewpoint_lag367	14.897	3.224	2.020	9.270	12.481	15.545	17.579	19.180	20.532	-0.526	-0.479	0.926
temperature_t	20.798	2.657	10.493	16.187	18.934	21.122	22.770	24.613	28.673	-0.329	-0.314	0.922
temperature_lag1	20.797	2.657	10.493	16.186	18.934	21.122	22.771	24.613	28.673	-0.329	-0.315	0.922
temperature_lag2	20.796	2.657	10.493	16.185	18.934	21.121	22.770	24.613	28.673	-0.328	-0.314	0.922
temperature_lag3	20.795	2.656	10.493	16.185	18.934	21.120	22.769	24.609	28.673	-0.328	-0.314	0.922
temperature_lag364	20.759	2.627	10.493	16.208	18.899	21.064	22.732	24.530	27.919	-0.346	-0.357	0.921
temperature_lag365	20.759	2.627	10.493	16.207	18.899	21.065	22.733	24.530	27.919	-0.346	-0.357	0.921
temperature_lag366	20.759	2.628	10.493	16.206	18.899	21.065	22.734	24.531	27.919	-0.346	-0.358	0.921
temperature_lag367	20.759	2.628	10.493	16.205	18.899	21.065	22.735	24.531	27.919	-0.345	-0.359	0.921
rmm1_t	-0.024	1.052	-4.180	-1.721	-0.717	-0.002	0.677	1.658	3.558	-0.149	0.252	0.971
rmm2_t	0.002	1.082	-2.836	-1.656	-0.760	-0.054	0.699	1.908	4.026	0.325	0.017	0.976
rmm1_lag1	-0.024	1.052	-4.180	-1.721	-0.717	-0.001	0.677	1.658	3.558	-0.150	0.252	0.971
rmm2_lag1	0.002	1.081	-2.836	-1.655	-0.759	-0.054	0.700	1.908	4.026	0.326	0.018	0.976
rmm1_lag2	-0.024	1.052	-4.180	-1.721	-0.717	0.000	0.677	1.658	3.558	-0.151	0.254	0.971
rmm2_lag2	0.003	1.081	-2.836	-1.655	-0.758	-0.053	0.701	1.908	4.026	0.326	0.019	0.976
rmm1_lag3	-0.023	1.052	-4.180	-1.721	-0.716	0.001	0.677	1.659	3.558	-0.151	0.256	0.971
rmm2_lag3	0.003	1.081	-2.836	-1.655	-0.756	-0.053	0.701	1.908	4.026	0.325	0.019	0.977
nino34_t	27.187	1.045	24.848	25.635	26.253	27.180	28.069	28.831	29.815	0.095	-0.996	0.999
nino34_lag7	27.184	1.044	24.848	25.634	26.251	27.178	28.059	28.831	29.815	0.098	-0.992	0.999
nino34_lag30	27.175	1.042	24.848	25.633	26.242	27.170	28.044	28.833	29.815	0.110	-0.976	0.999
nino34_roll30	27.175	1.029	25.365	25.659	26.231	27.164	28.043	28.781	29.572	0.121	-1.024	1.000
nino34_roll90	27.153	0.977	25.470	25.719	26.269	27.188	27.849	28.783	29.219	0.159	-0.937	1.000
dew_x_temp	316.111	93.196	30.366	157.003	239.277	334.913	392.633	442.684	506.329	-0.383	-0.851	0.942
dew_x_rmm1	-0.182	16.469	-74.263	-27.722	-9.627	-0.015	9.790	26.685	66.057	-0.155	1.003	0.969
dew_x_rmm2	0.370	17.171	-49.582	-27.142	-10.383	-0.771	9.999	31.248	72.943	0.398	0.638	0.974
dew_x_nino34	405.848	88.344	54.225	251.038	343.490	420.015	474.709	528.576	596.319	-0.466	-0.368	0.928
temp_x_nino34	565.158	74.087	281.445	440.389	515.853	567.491	615.695	684.300	820.974	-0.053	-0.028	0.926

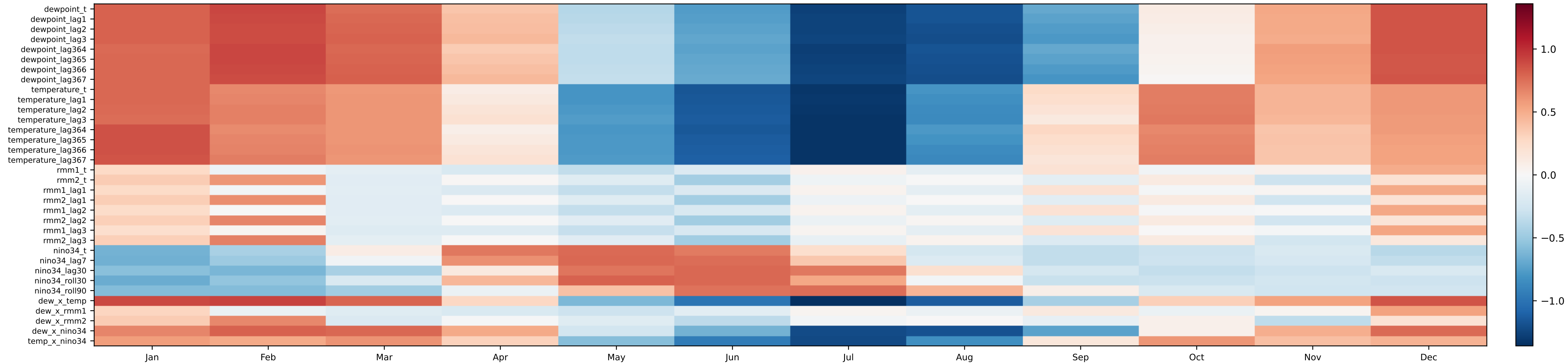
Covariate descriptive statistics — Normalised

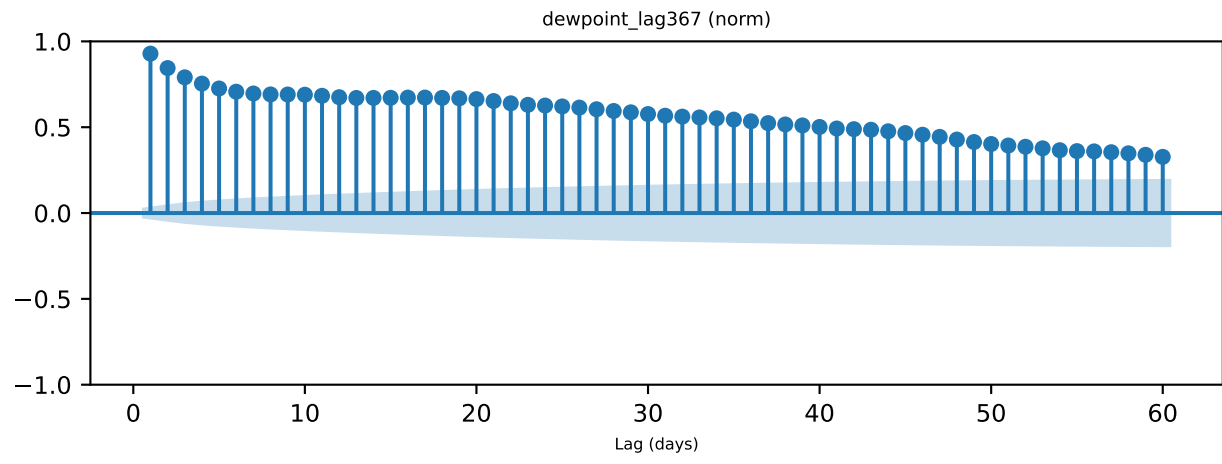
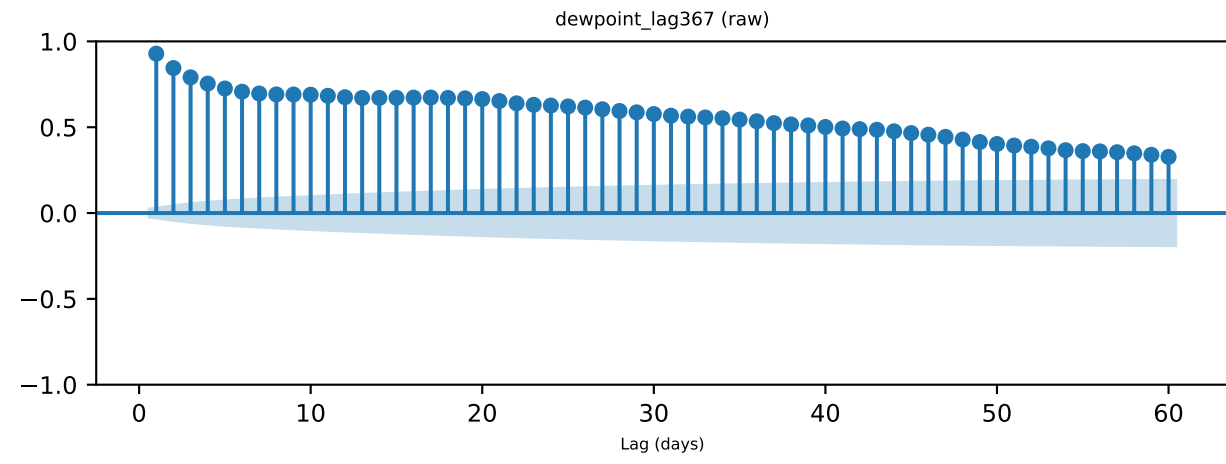
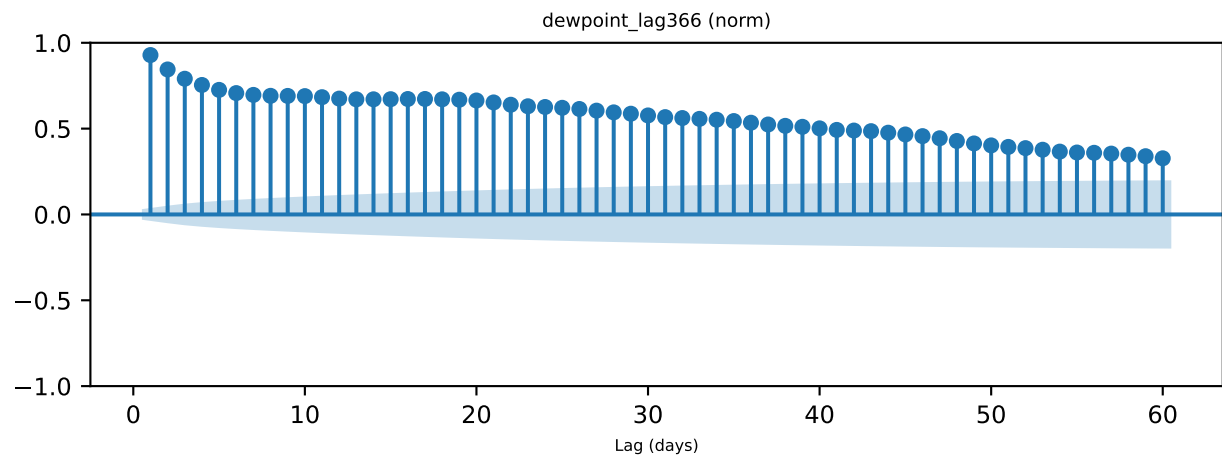
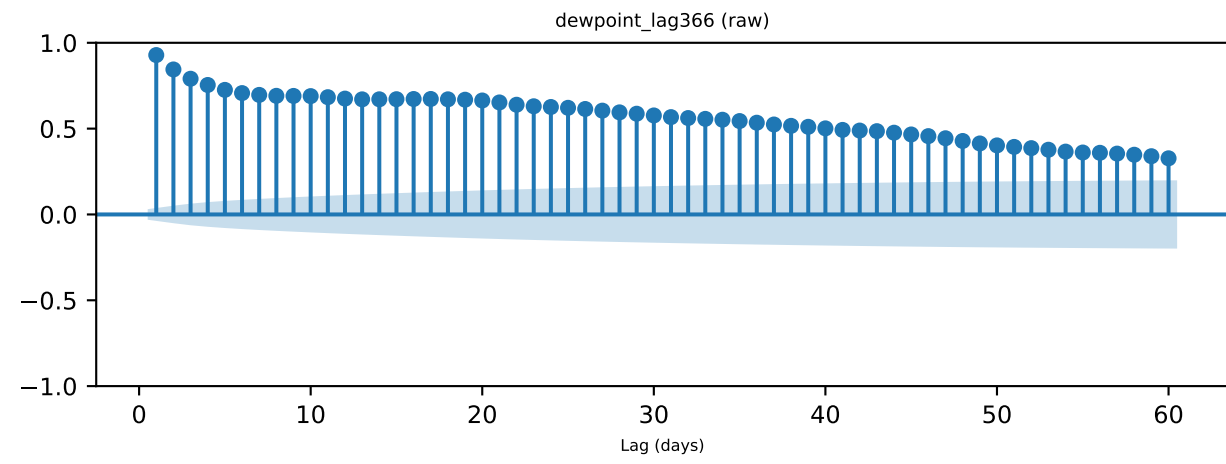
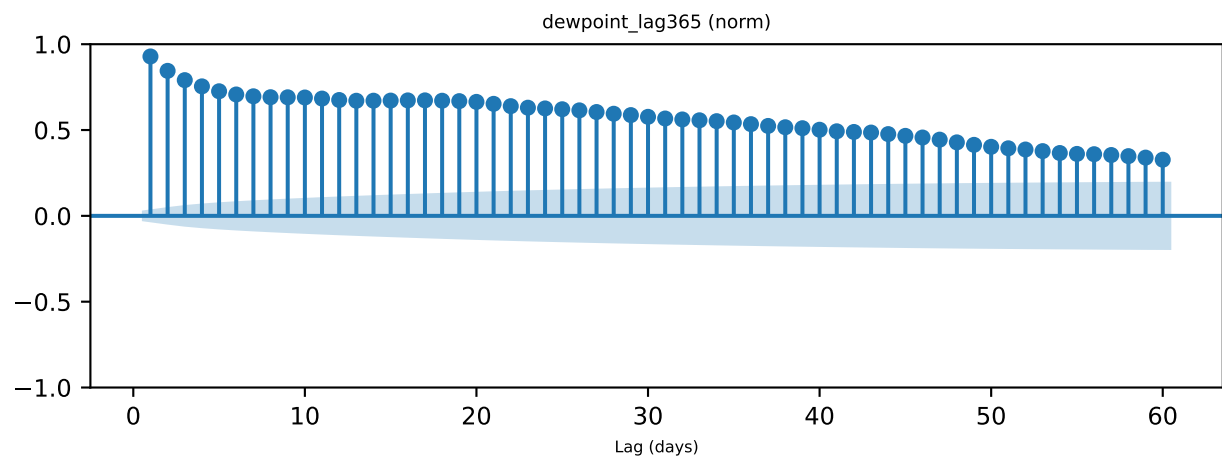
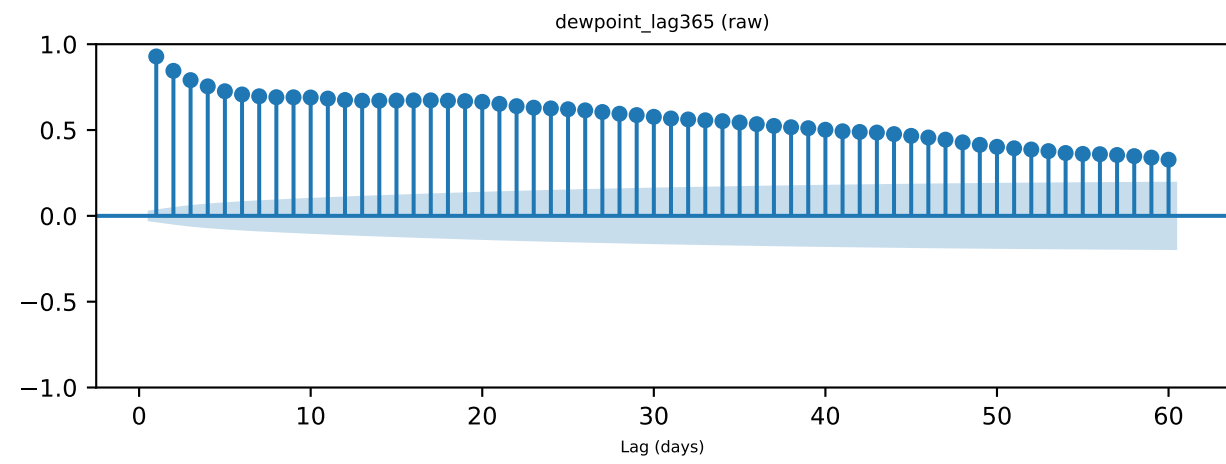
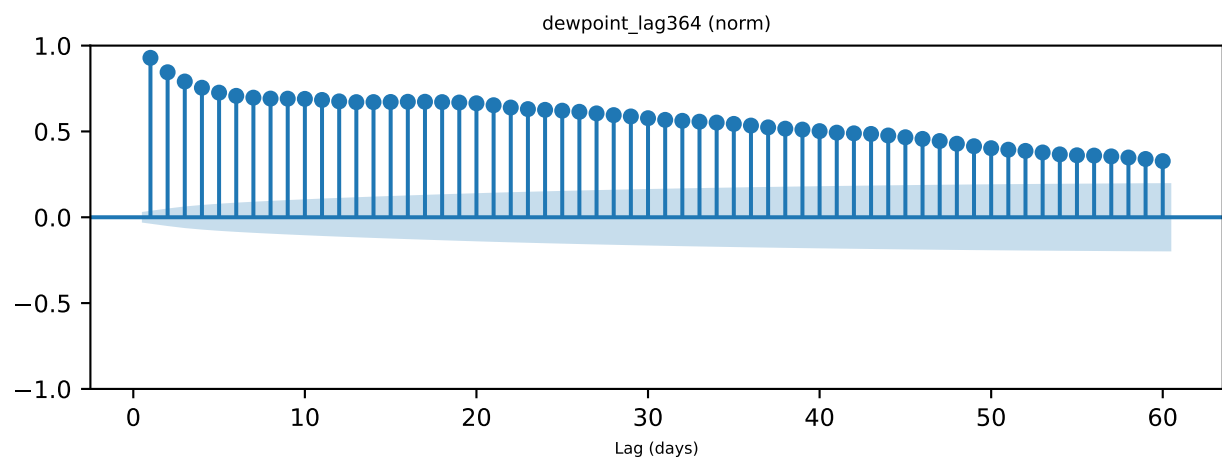
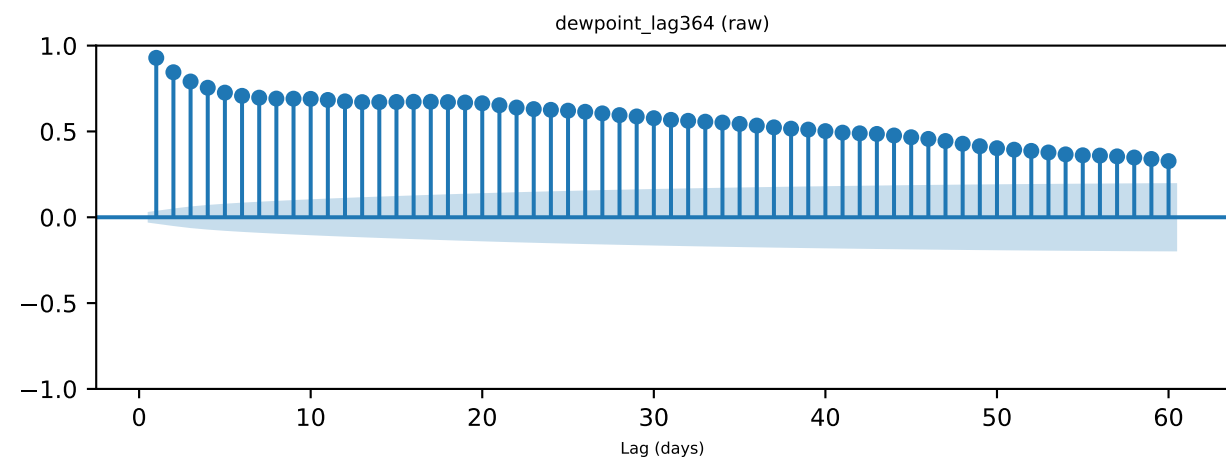
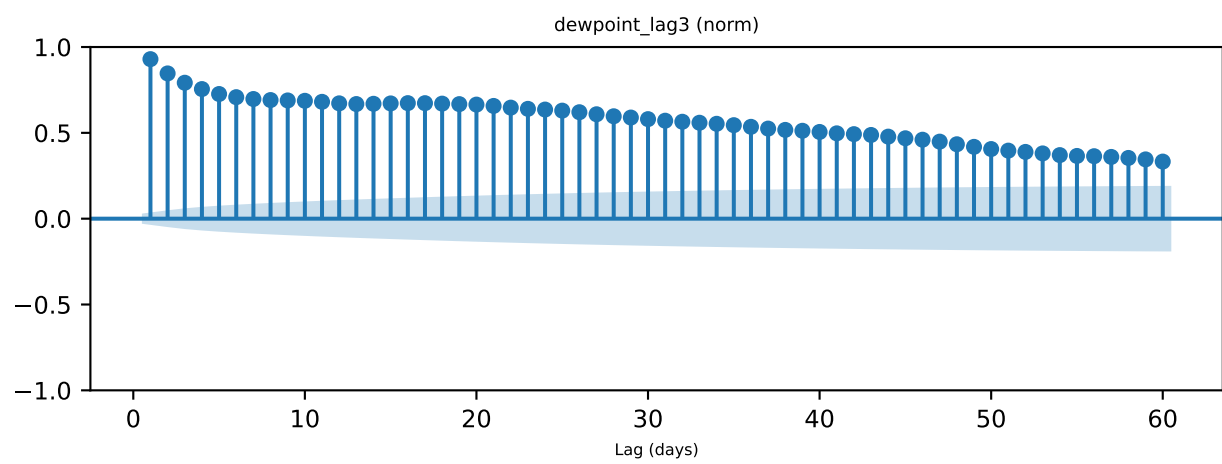
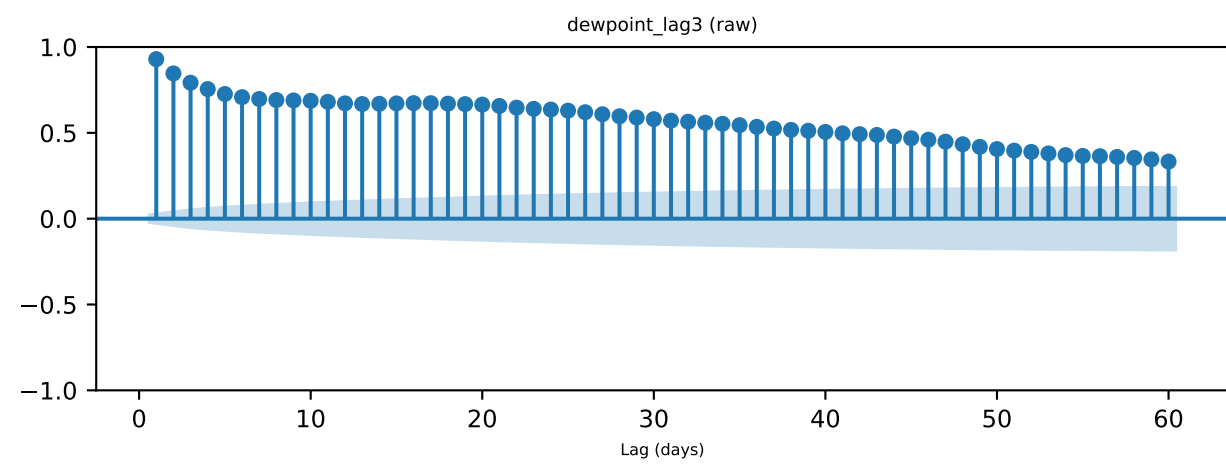
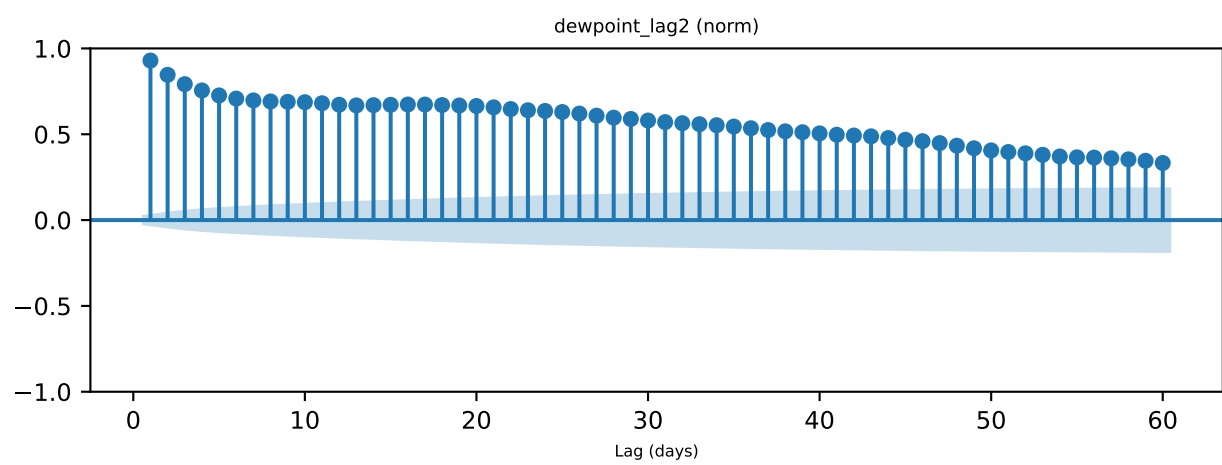
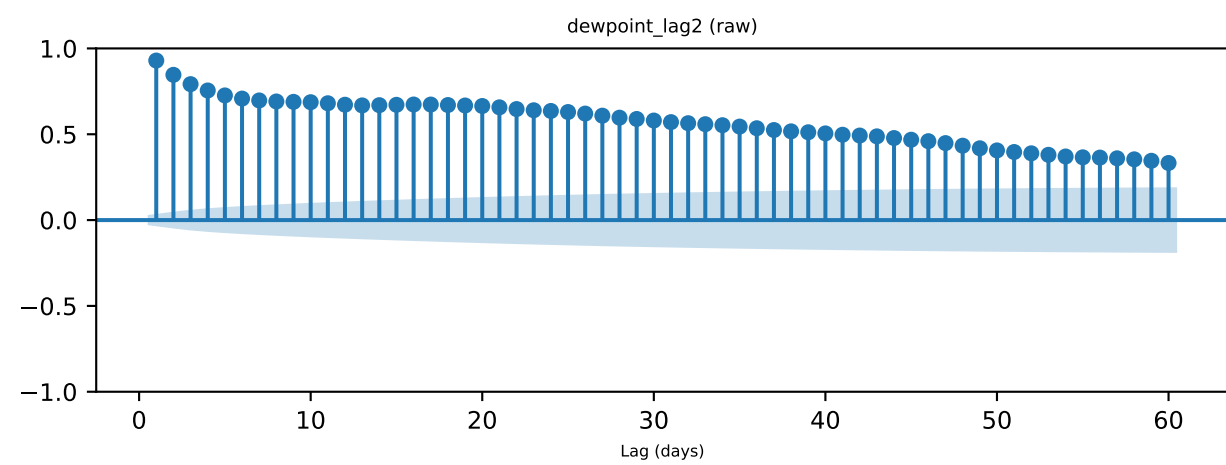
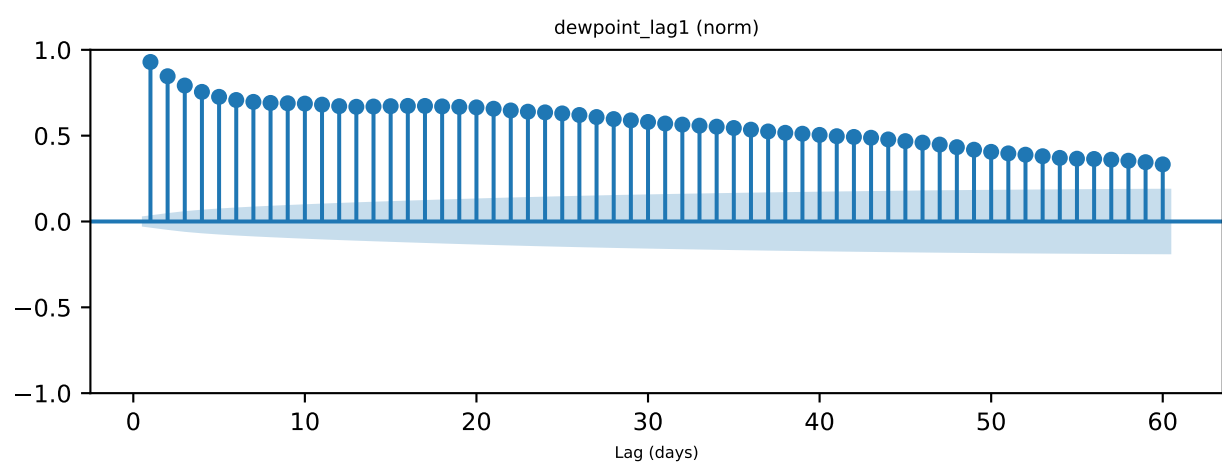
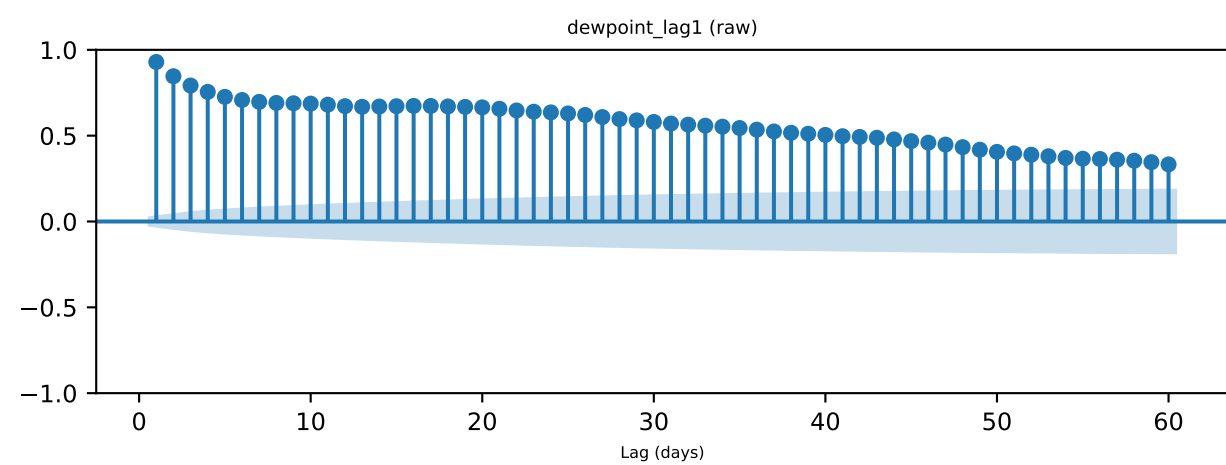
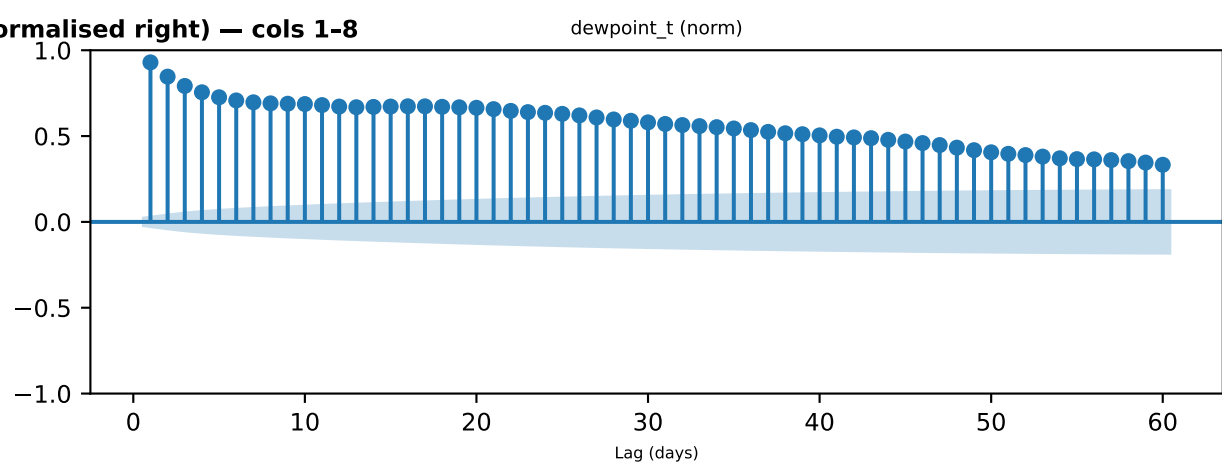
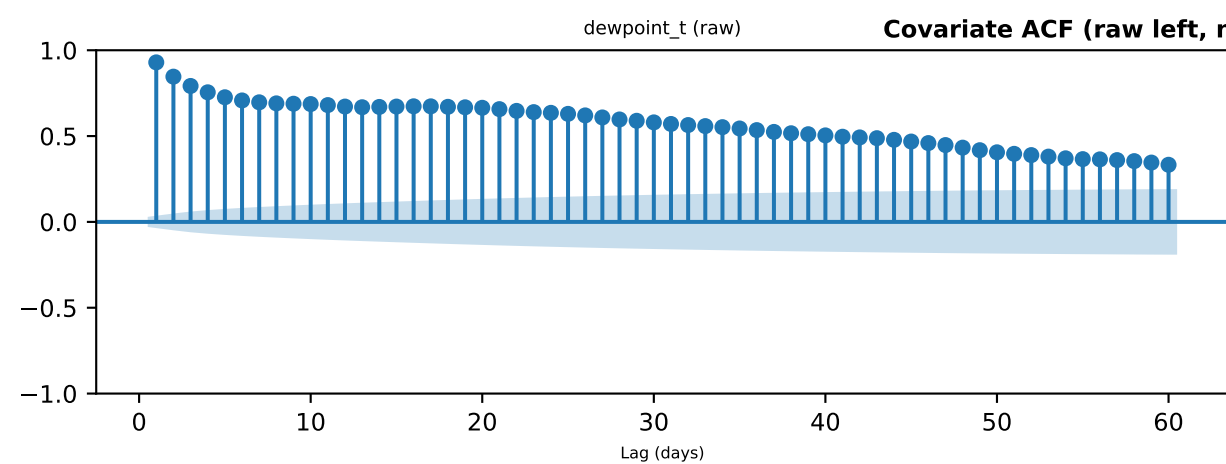
variable	mean	std	min	q05	q25	q50	q75	q95	max	skew	kurt	acf 1
dewpoint_t	-0.000	1.000	-4.015	-1.761	-0.748	0.204	0.839	1.315	1.741	-0.534	-0.491	0.926
dewpoint_lag1	-0.000	1.000	-4.015	-1.761	-0.747	0.203	0.839	1.316	1.741	-0.534	-0.491	0.926
dewpoint_lag2	-0.000	1.000	-4.015	-1.761	-0.747	0.203	0.840	1.316	1.742	-0.535	-0.491	0.926
dewpoint_lag3	-0.000	1.000	-4.016	-1.760	-0.747	0.203	0.839	1.316	1.743	-0.535	-0.491	0.926
dewpoint_lag364	0.000	1.000	-3.995	-1.746	-0.750	0.203	0.832	1.327	1.747	-0.527	-0.478	0.926
dewpoint_lag365	0.000	1.000	-3.995	-1.746	-0.750	0.203	0.832	1.328	1.747	-0.527	-0.478	0.926
dewpoint_lag366	0.000	1.000	-3.994	-1.746	-0.750	0.202	0.831	1.328	1.747	-0.526	-0.478	0.926
dewpoint_lag367	0.000	1.000	-3.994	-1.746	-0.750	0.201	0.832	1.328	1.748	-0.526	-0.479	0.926
temperature_t	0.000	1.000	-3.879	-1.736	-0.701	0.122	0.742	1.436	2.964	-0.329	-0.314	0.922
temperature_lag1	-0.000	1.000	-3.878	-1.736	-0.701	0.122	0.743	1.436	2.964	-0.329	-0.315	0.922
temperature_lag2	-0.000	1.000	-3.878	-1.736	-0.701	0.122	0.743	1.437	2.965	-0.328	-0.314	0.922
temperature_lag3	-0.000	1.000	-3.879	-1.736	-0.701	0.122	0.743	1.436	2.966	-0.328	-0.314	0.922
temperature_lag364	0.000	1.000	-3.908	-1.733	-0.708	0.116	0.751	1.436	2.726	-0.346	-0.357	0.921
temperature_lag365	0.000	1.000	-3.908	-1.733	-0.708	0.116	0.752	1.436	2.725	-0.346	-0.357	0.921
temperature_lag366	0.000	1.000	-3.907	-1.733	-0.708	0.116	0.752	1.435	2.725	-0.346	-0.358	0.921
temperature_lag367	0.000	1.000	-3.906	-1.733	-0.708	0.116	0.752	1.435	2.725	-0.345	-0.359	0.921
rmm1_t	-0.000	1.000	-3.950	-1.613	-0.658	0.021	0.667	1.599	3.405	-0.149	0.252	0.971
rmm2_t	0.000	1.000	-2.623	-1.532	-0.704	-0.052	0.645	1.763	3.721	0.325	0.017	0.976
rmm1_lag1	-0.000	1.000	-3.950	-1.613	-0.659	0.022	0.667	1.599	3.405	-0.150	0.252	0.971
rmm2_lag1	0.000	1.000	-2.624	-1.532	-0.704	-0.052	0.645	1.763	3.721	0.326	0.018	0.976
rmm1_lag2	0.000	1.000	-3.951	-1.614	-0.659	0.023	0.666	1.599	3.405	-0.151	0.254	0.971
rmm2_lag2	0.000	1.000	-2.625	-1.533	-0.704	-0.052	0.646	1.763	3.721	0.326	0.019	0.976
rmm1_lag3	0.000	1.000	-3.952	-1.614	-0.659	0.023	0.666	1.599	3.405	-0.151	0.256	0.971
rmm2_lag3	-0.000	1.000	-2.625	-1.533	-0.702	-0.051	0.646	1.762	3.721	0.325	0.019	0.977
nino34_t	0.000	1.000	-2.238	-1.485	-0.893	-0.006	0.844	1.573	2.515	0.095	-0.996	0.999
nino34_lag7	0.000	1.000	-2.237	-1.484	-0.893	-0.006	0.838	1.577	2.519	0.098	-0.992	0.999
nino34_lag30	0.000	1.000	-2.234	-1.480	-0.896	-0.005	0.834	1.592	2.534	0.110	-0.976	0.999
nino34_rol130	0.000	1.000	-1.760	-1.474	-0.918	-0.011	0.843	1.561	2.329	0.121	-1.024	1.000
nino34_rol190	0.000	1.000	-1.722	-1.467	-0.904	0.036	0.713	1.669	2.115	0.159	-0.937	1.000
dew_x_temp	0.000	1.000	-3.066	-1.707	-0.824	0.202	0.821	1.358	2.041	-0.383	-0.851	0.942
dew_x_rmm1	0.000	1.000	-4.498	-1.672	-0.574	0.010	0.606	1.631	4.022	-0.155	1.003	0.969
dew_x_rmm2	0.000	1.000	-2.909	-1.602	-0.626	-0.066	0.561	1.798	4.226	0.398	0.638	0.974
dew_x_nino34	0.000	1.000	-3.980	-1.752	-0.706	0.160	0.779	1.389	2.156	-0.466	-0.368	0.928
temp_x_nino34	-0.000	1.000	-3.829	-1.684	-0.666	0.031	0.682	1.608	3.453	-0.053	-0.028	0.926

Covariate seasonality (monthly mean, Raw)

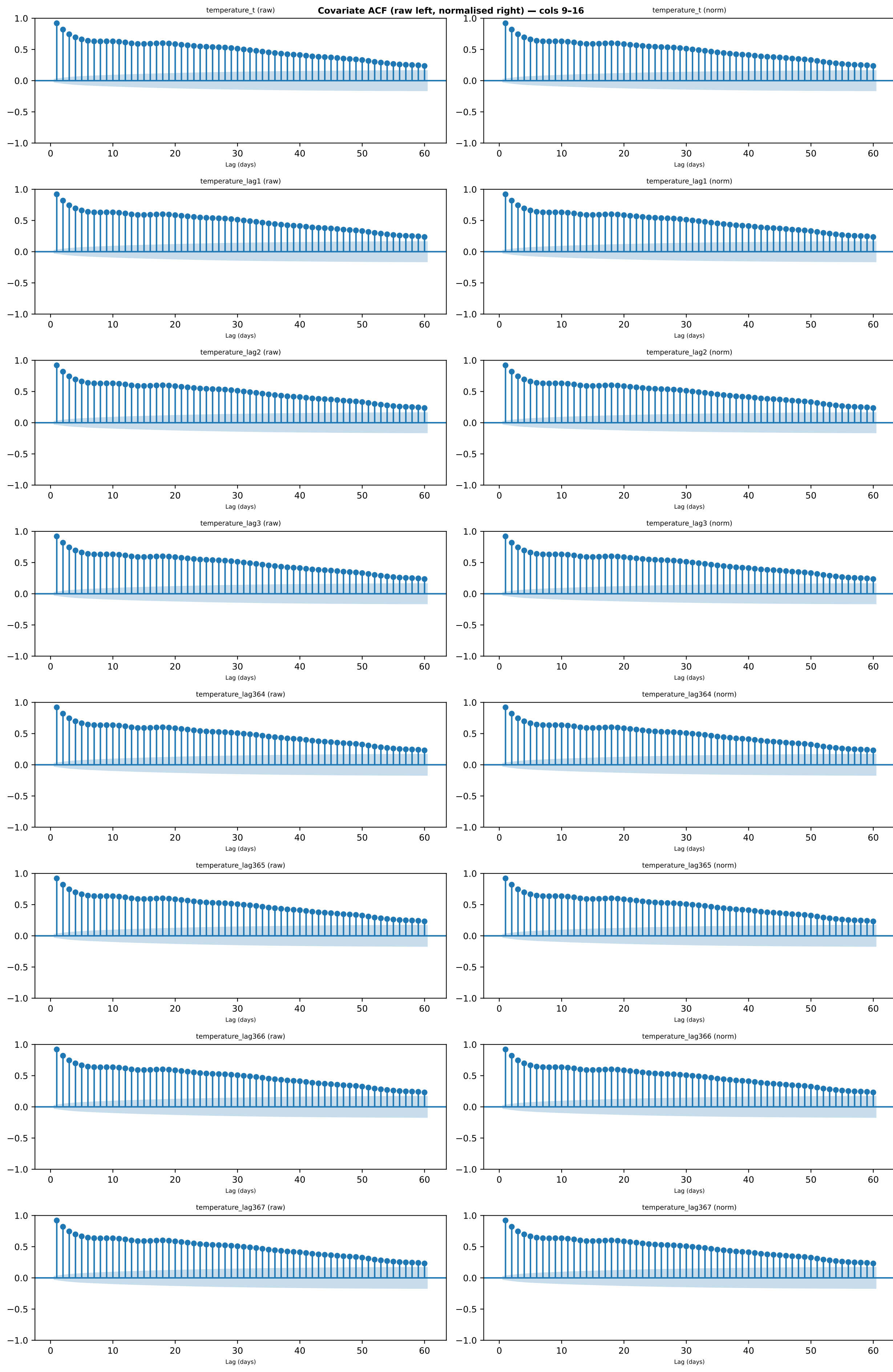


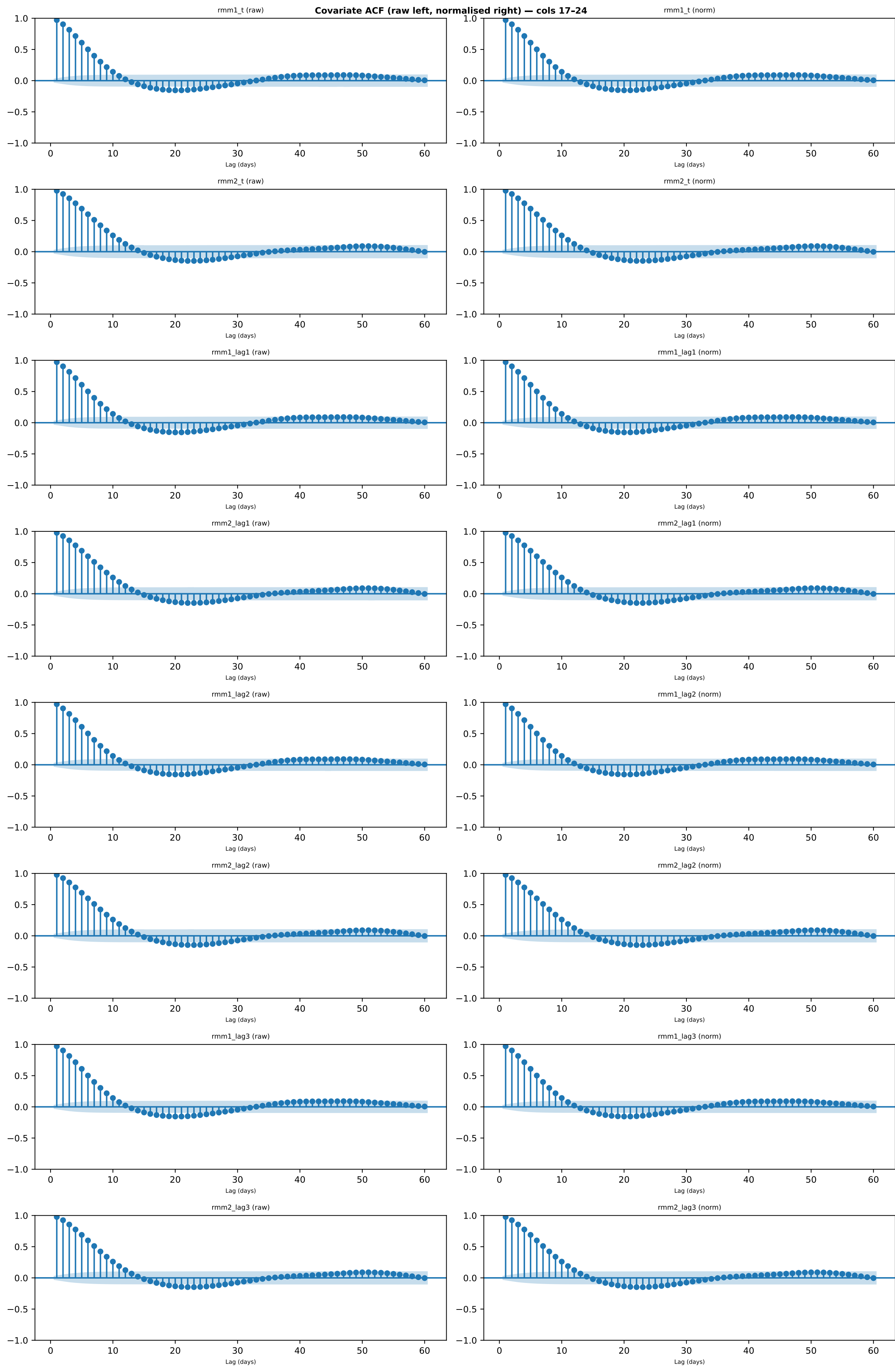
Covariate seasonality (monthly mean, Normalised)



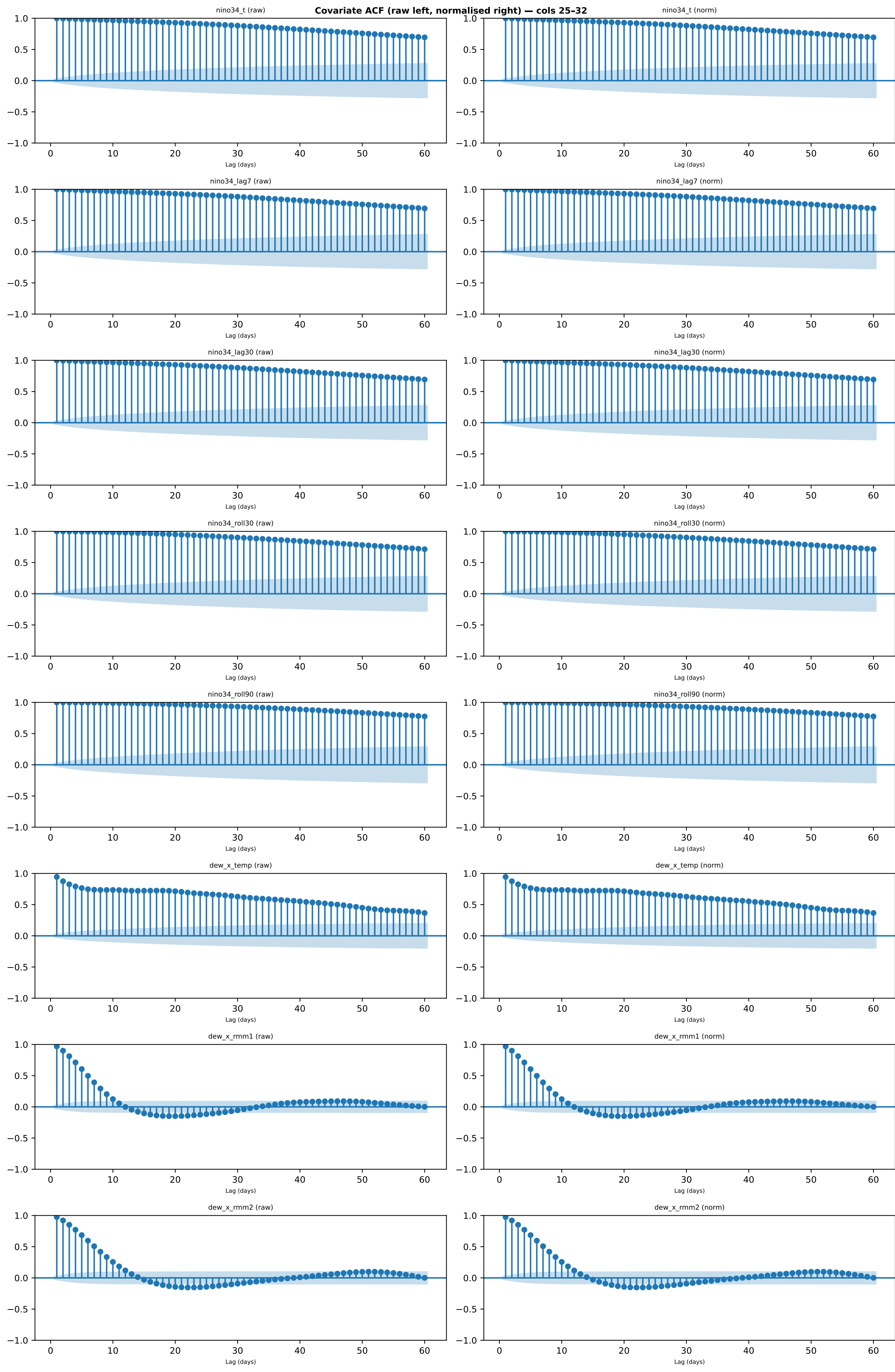


Covariate ACF (raw left, normalised right) — cols 9-16



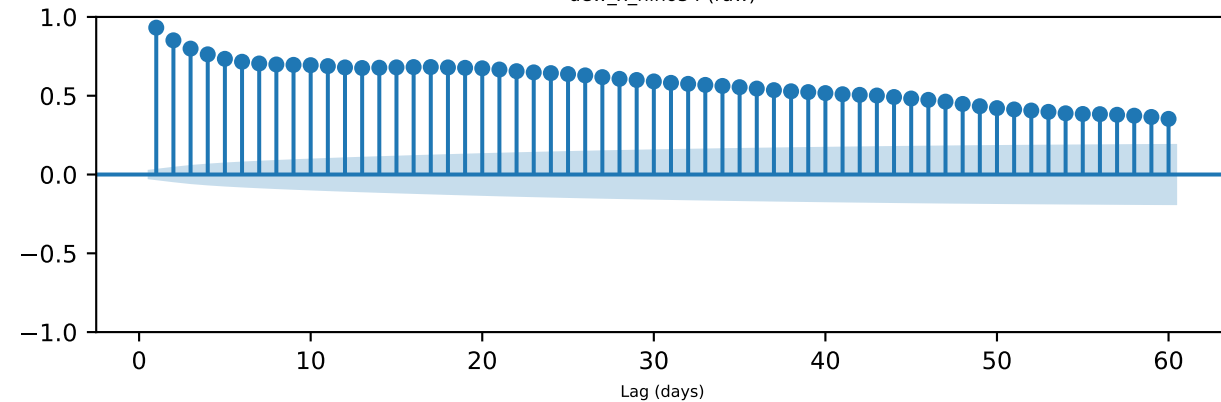


Covariate ACF (raw left, normalised right) — cols 25-32

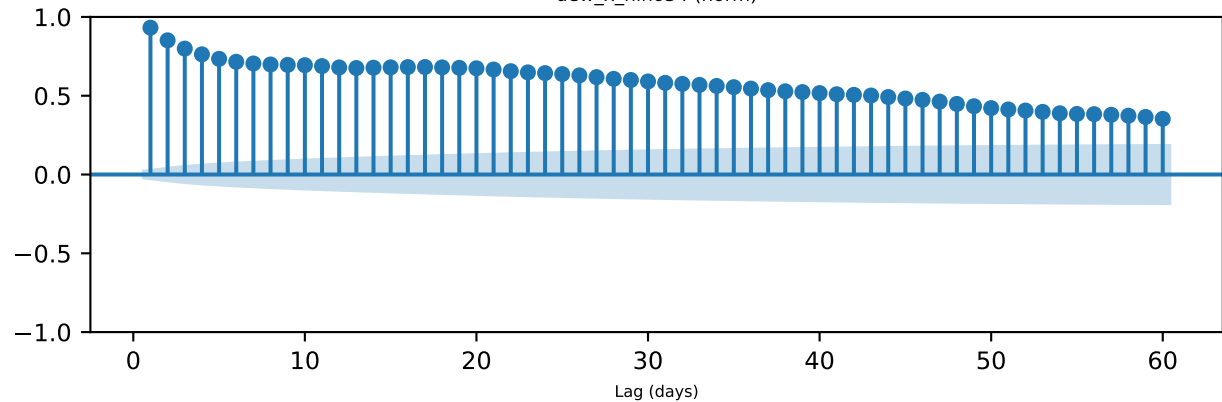


Covariate ACF (raw left, normalised right) — cols 33-34

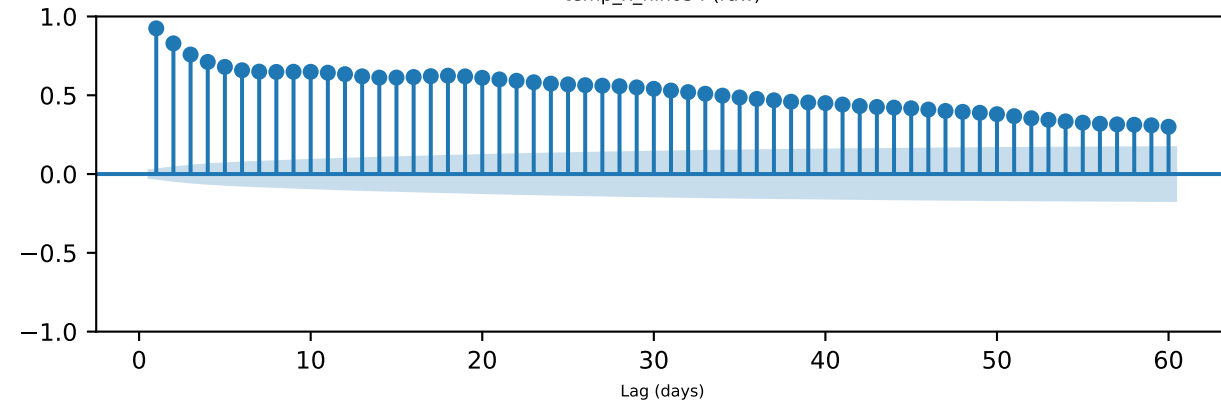
dew_x_nino34 (raw)



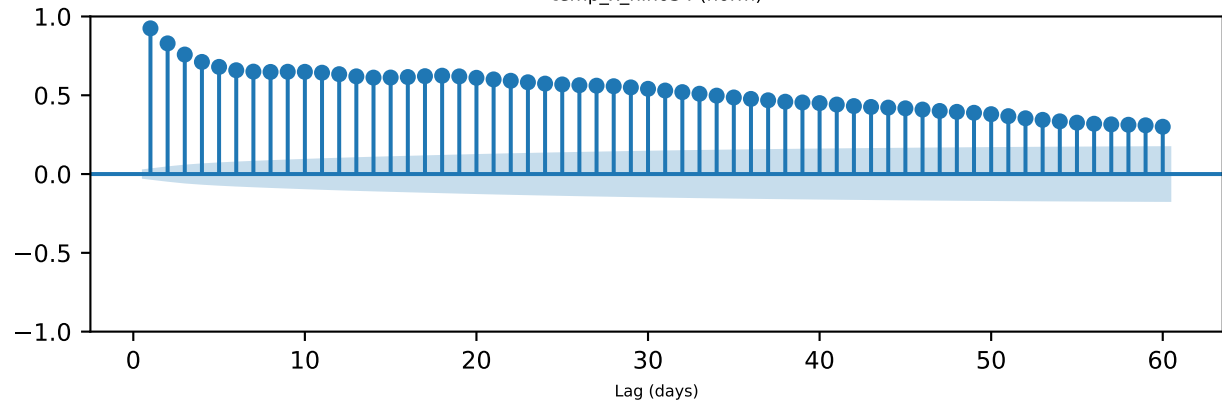
dew_x_nino34 (norm)



temp_x_nino34 (raw)

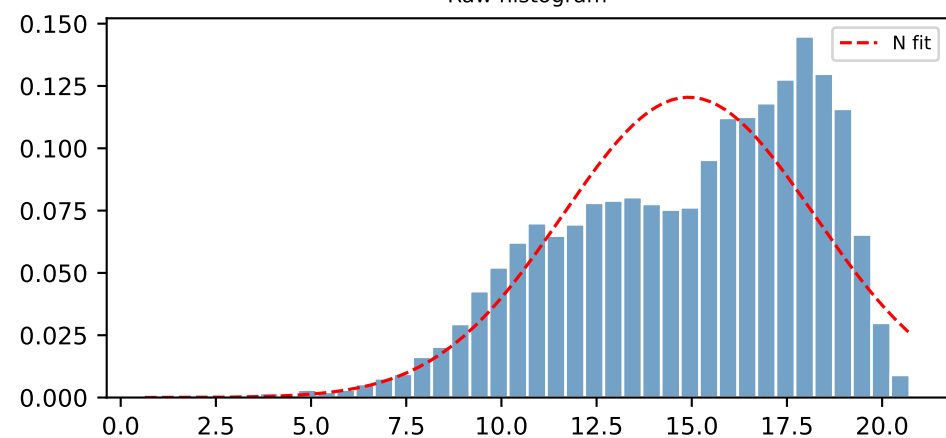


temp_x_nino34 (norm)

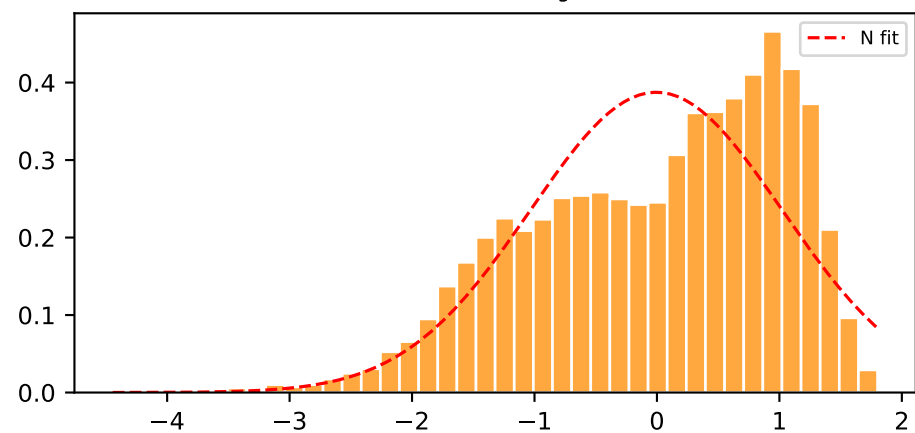


Covariate diagnostics — dewpoint_t

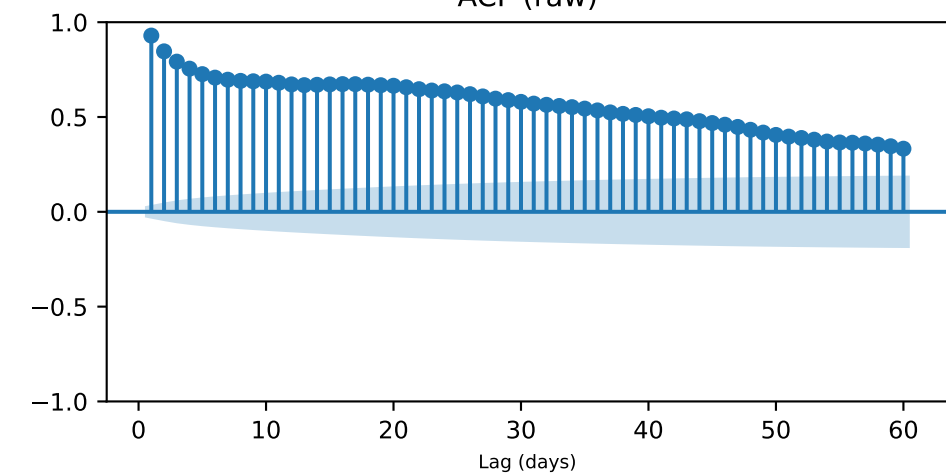
Raw histogram



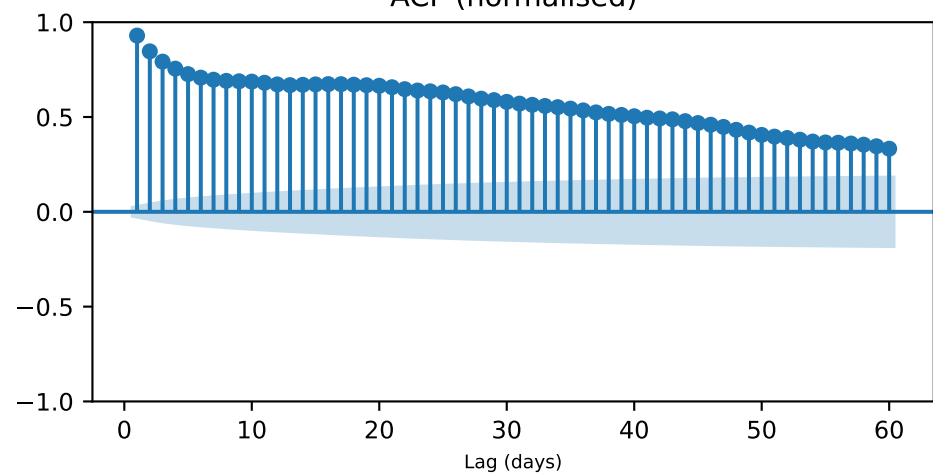
Normalised histogram



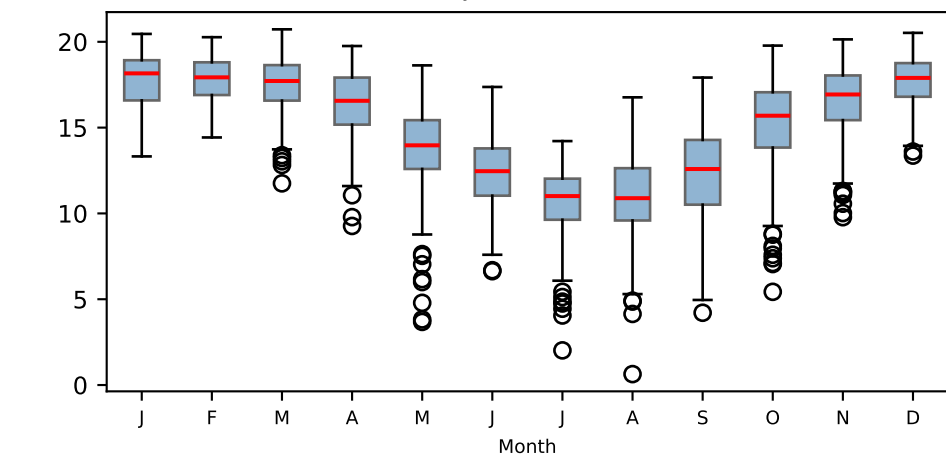
ACF (raw)



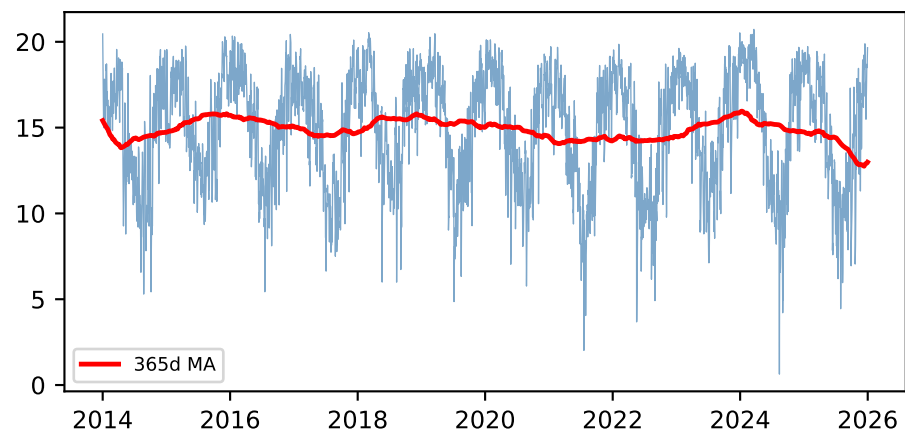
ACF (normalised)



Monthly distribution (raw)

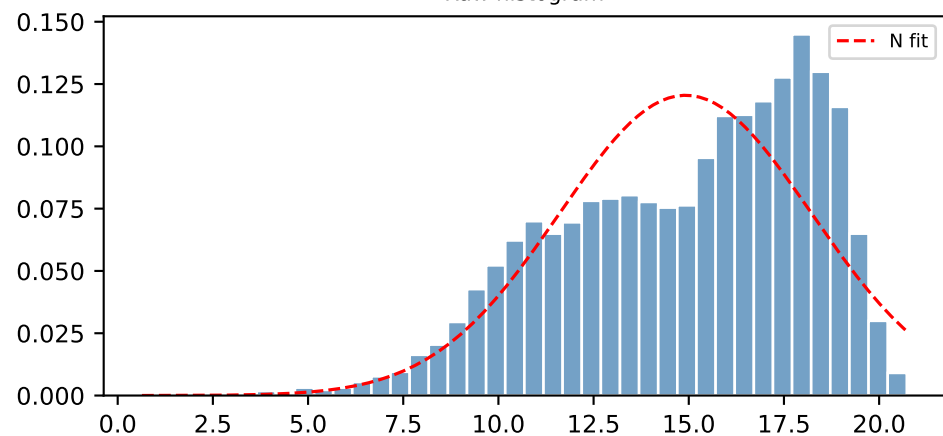


Time series + 365d mean

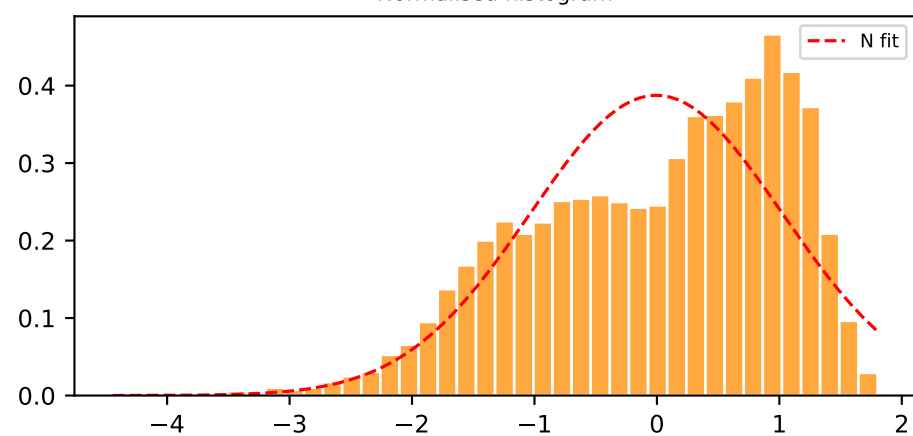


Covariate diagnostics — dewpoint_lag1

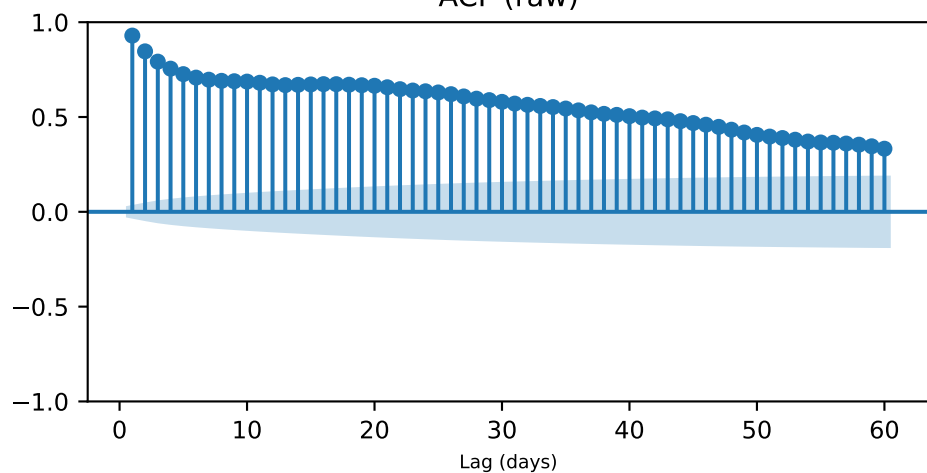
Raw histogram



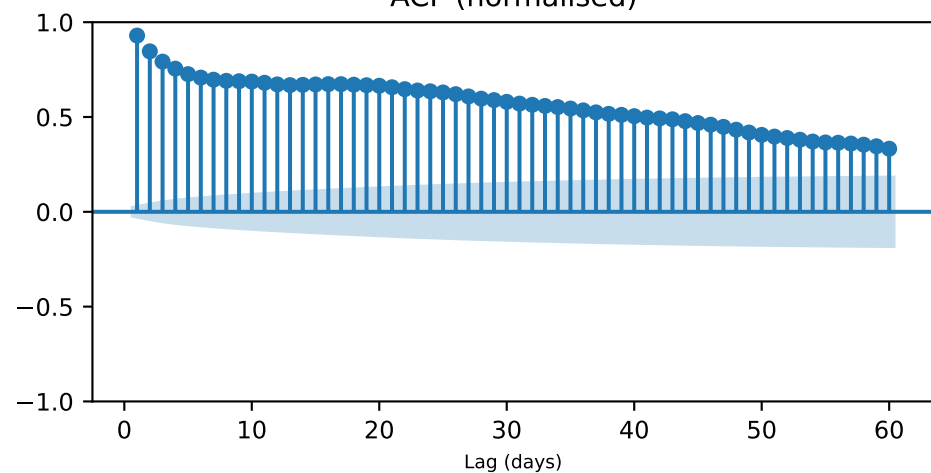
Normalised histogram



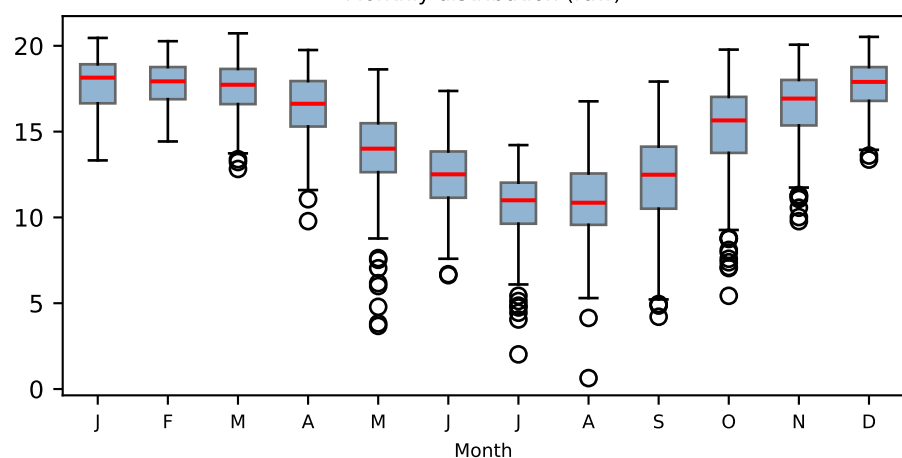
ACF (raw)



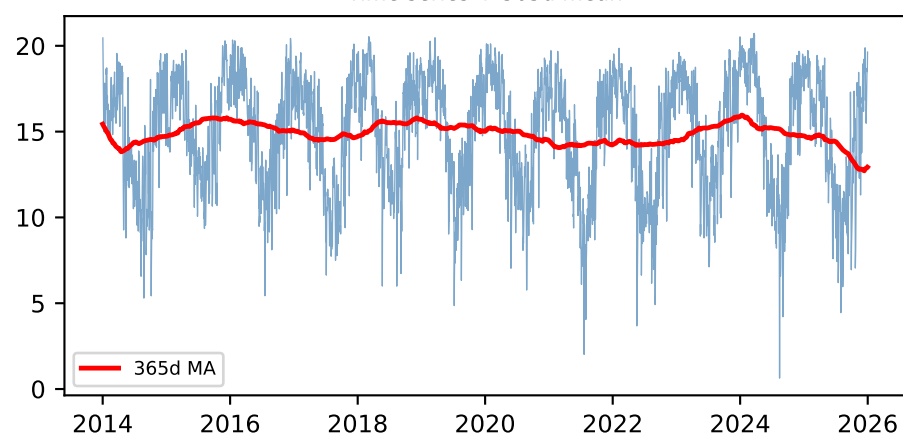
ACF (normalised)



Monthly distribution (raw)

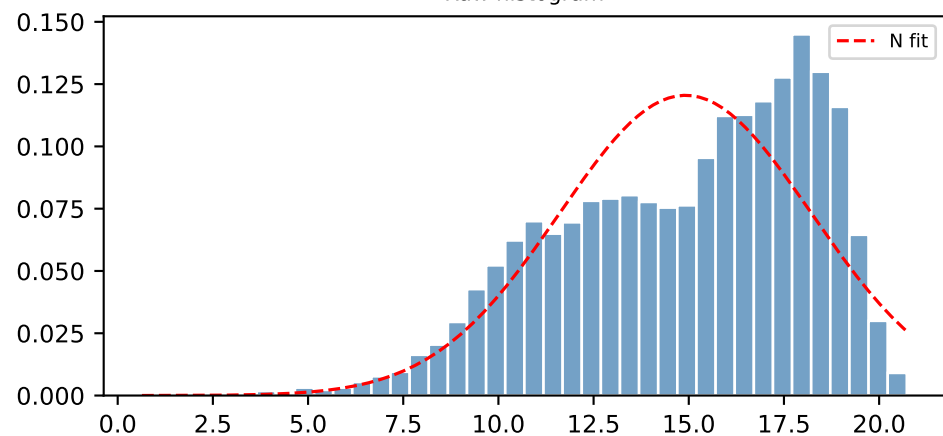


Time series + 365d mean

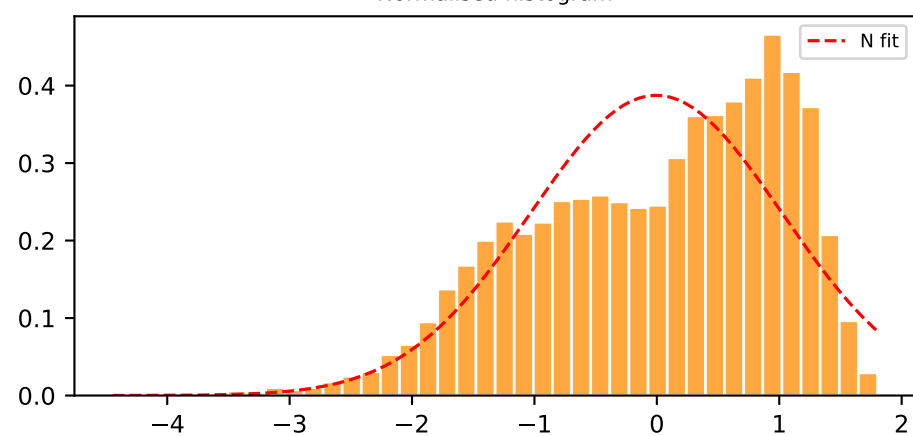


Covariate diagnostics — dewpoint_lag2

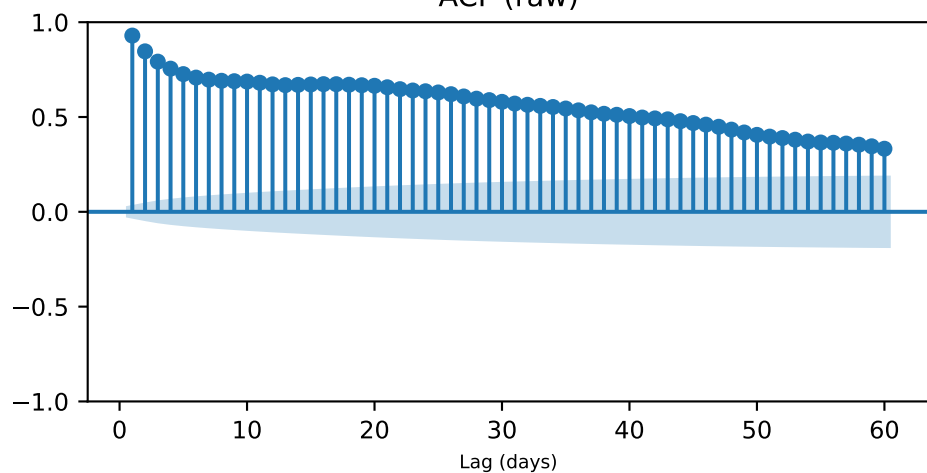
Raw histogram



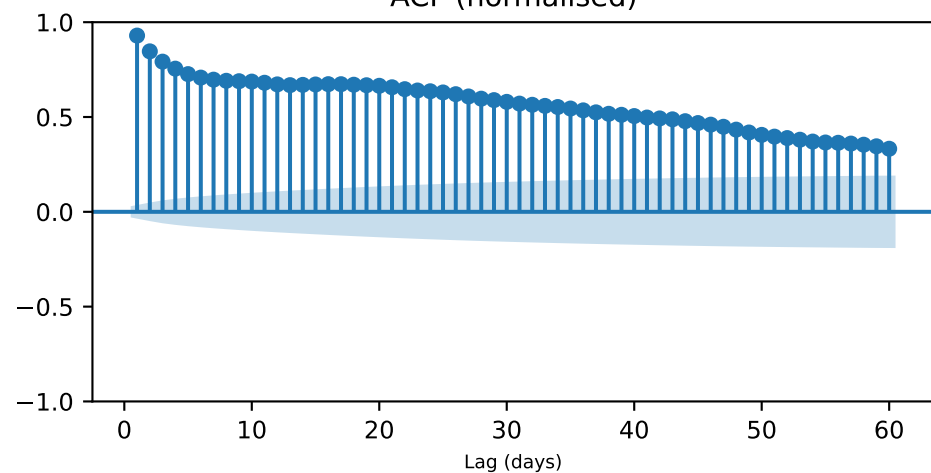
Normalised histogram



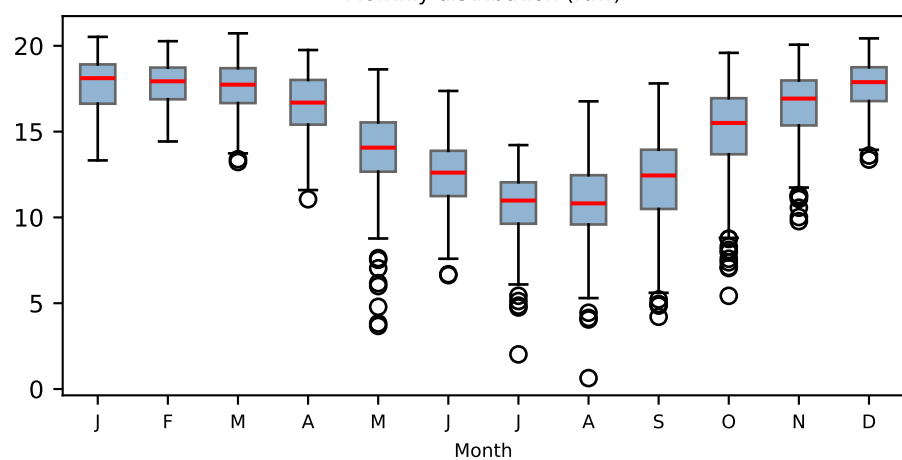
ACF (raw)



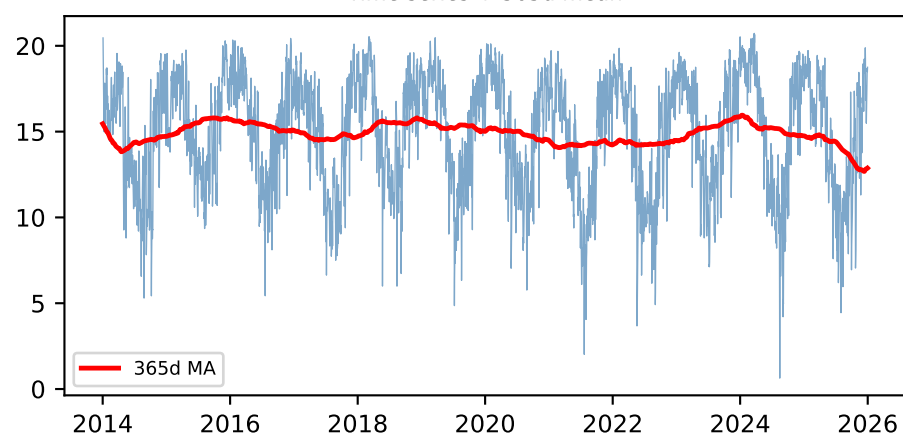
ACF (normalised)



Monthly distribution (raw)

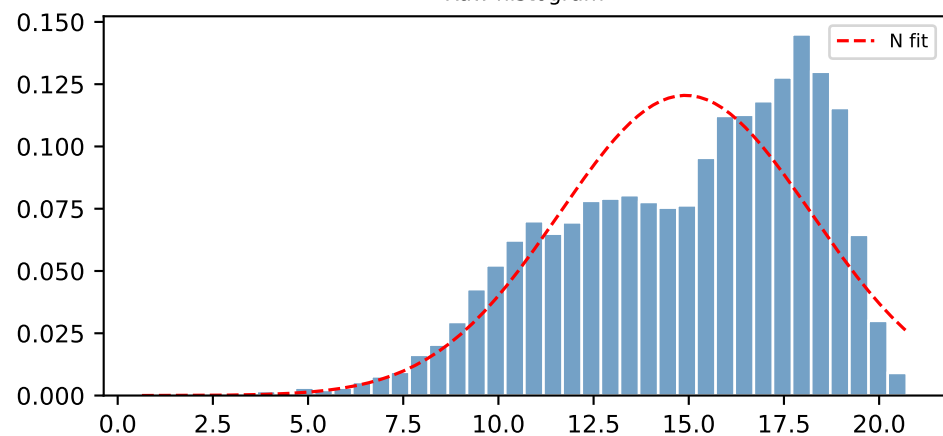


Time series + 365d mean

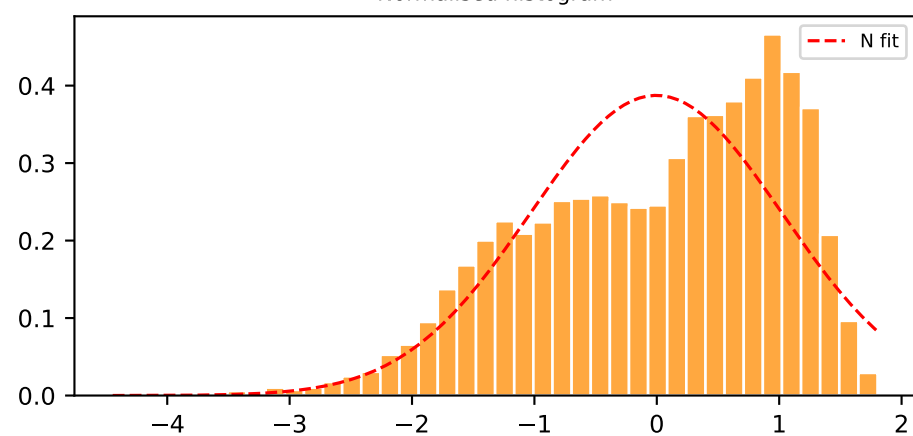


Covariate diagnostics — dewpoint_lag3

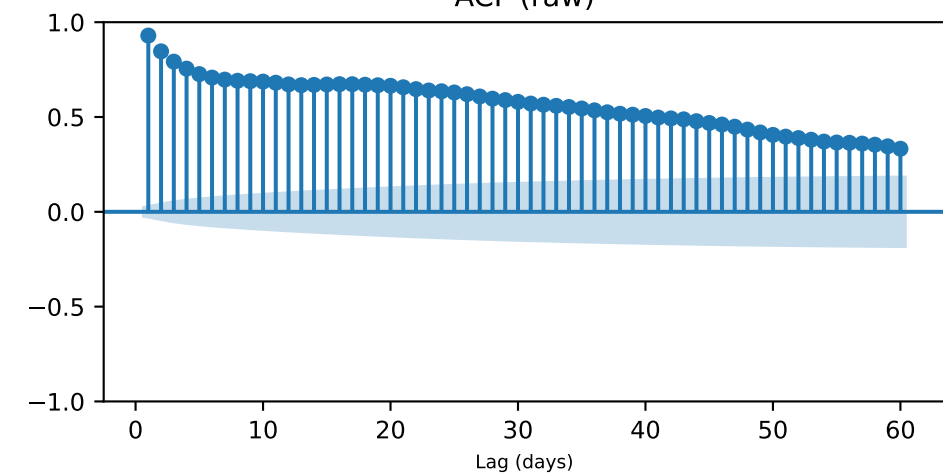
Raw histogram



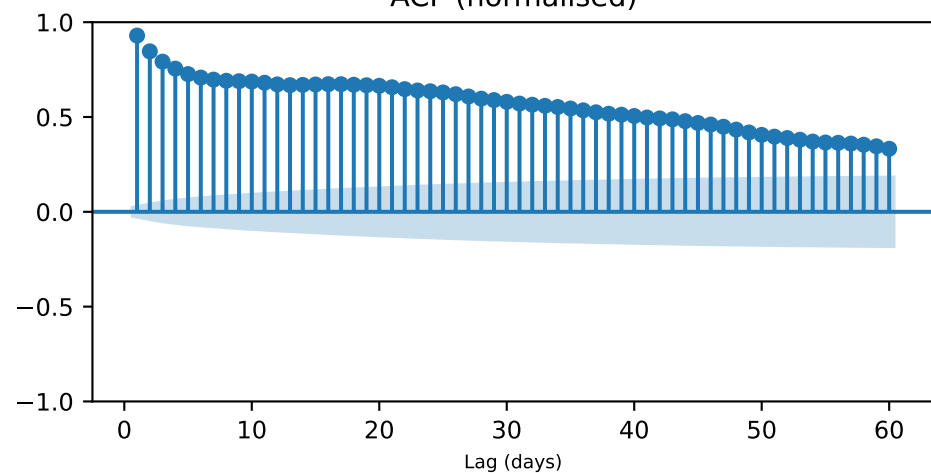
Normalised histogram



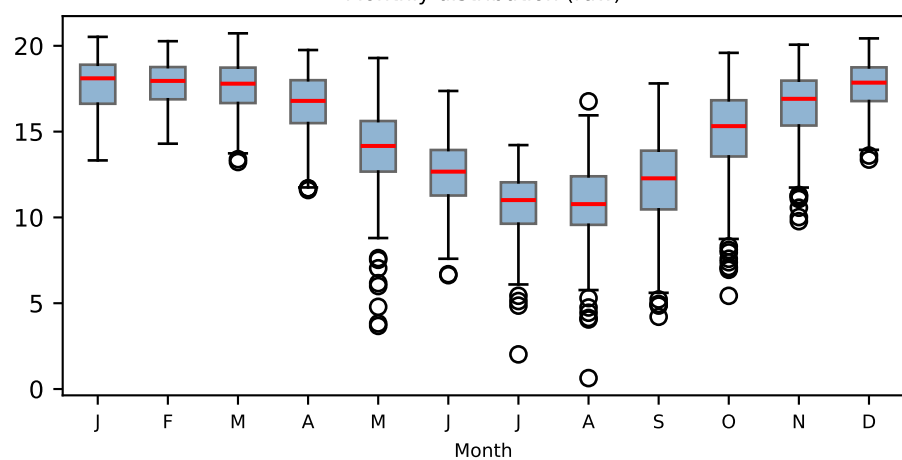
ACF (raw)



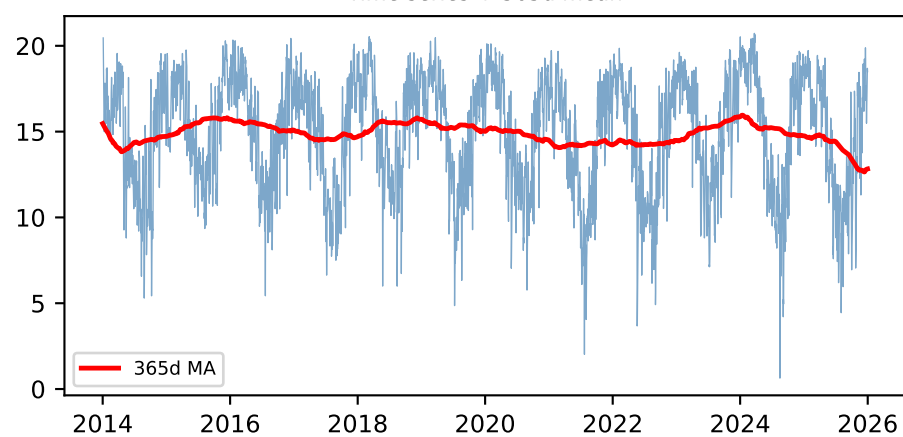
ACF (normalised)



Monthly distribution (raw)

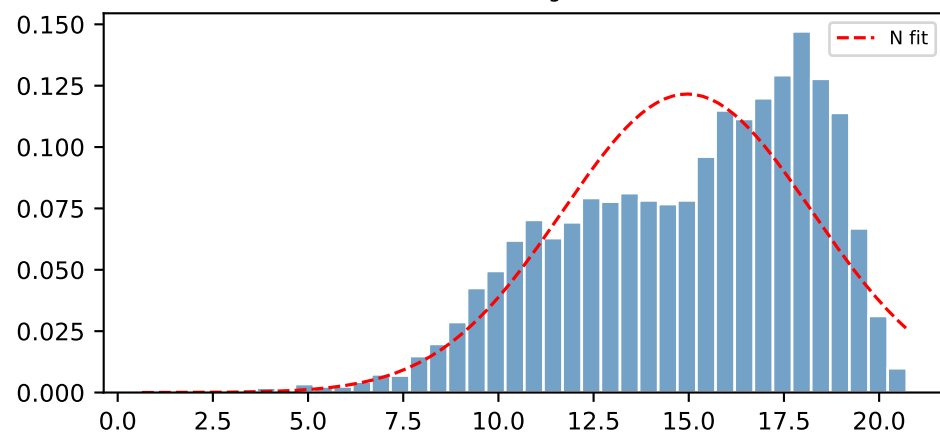


Time series + 365d mean

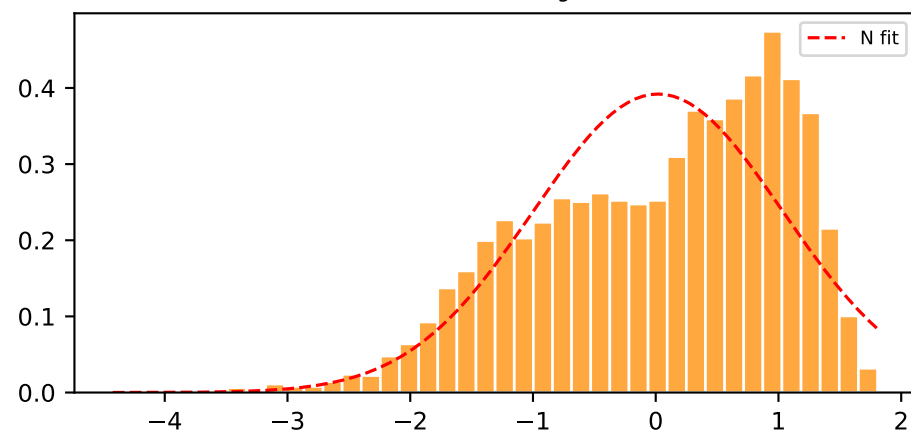


Covariate diagnostics — dewpoint_lag364

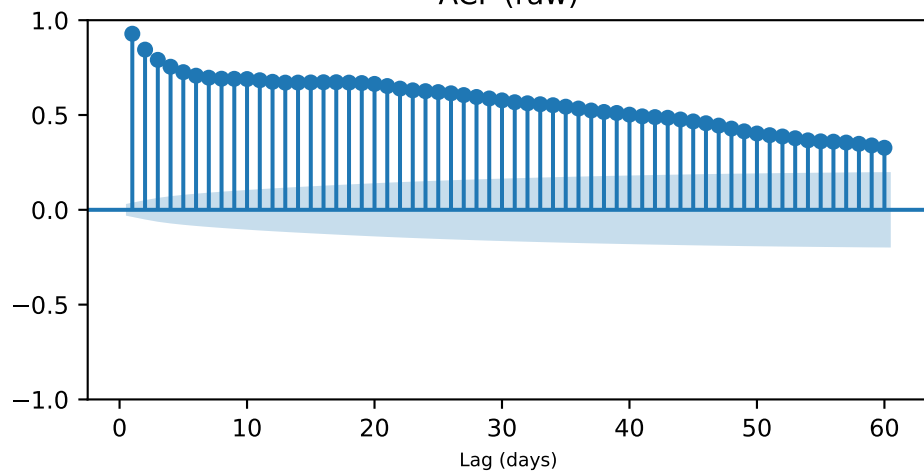
Raw histogram



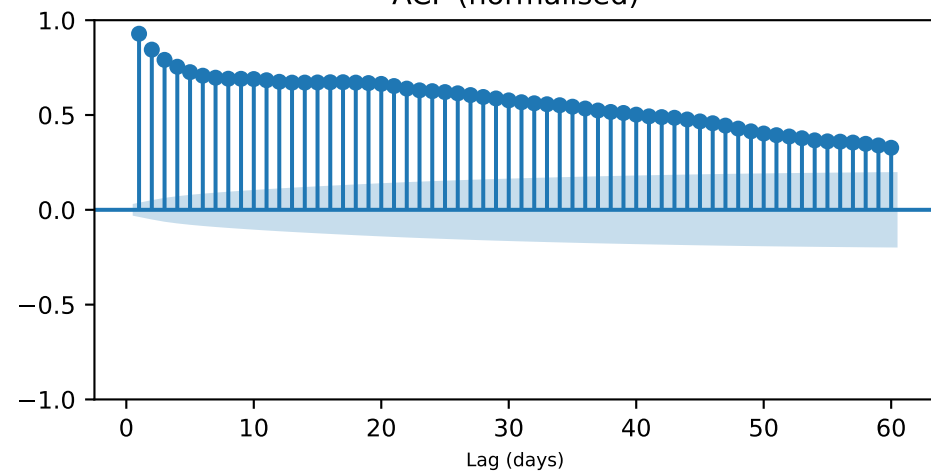
Normalised histogram



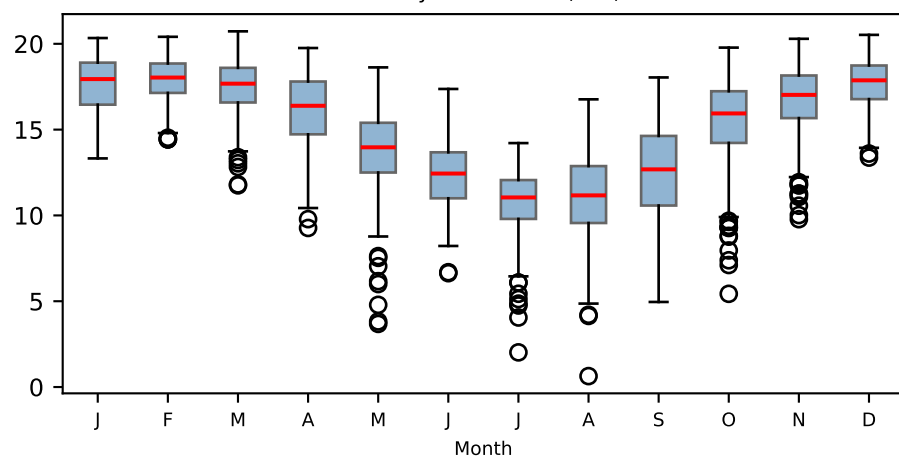
ACF (raw)



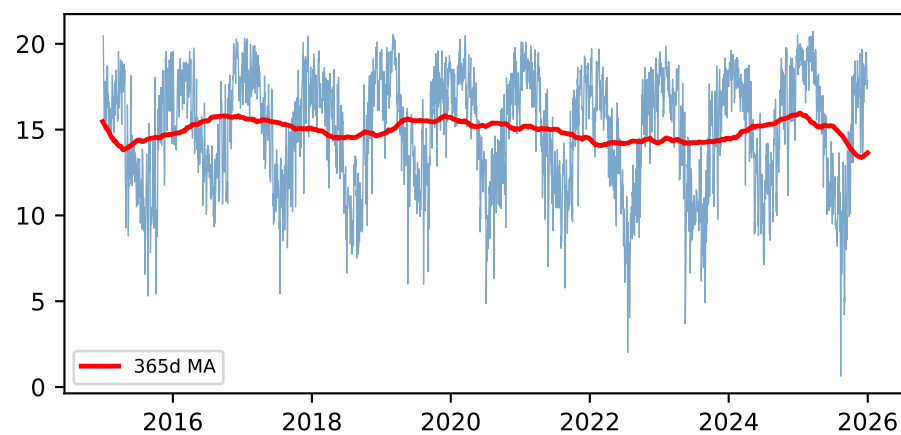
ACF (normalised)



Monthly distribution (raw)

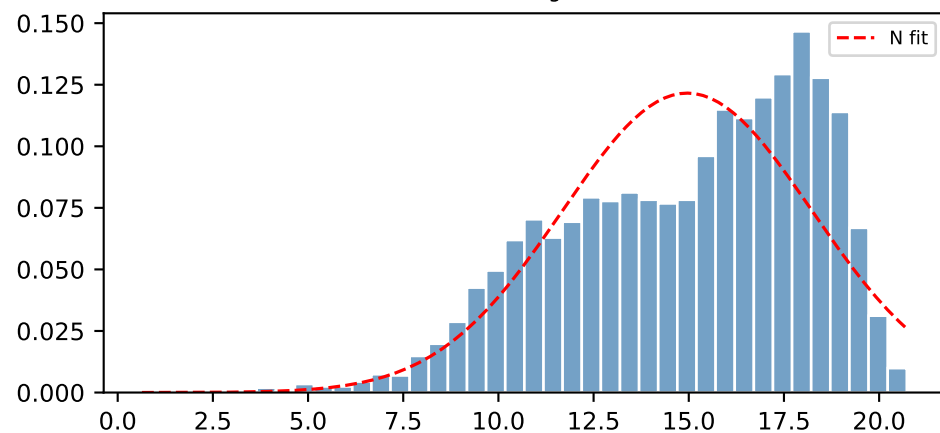


Time series + 365d mean

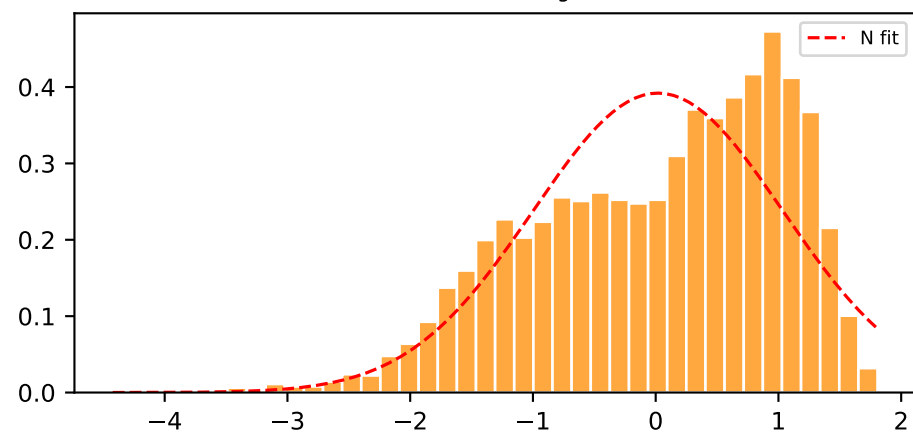


Covariate diagnostics — dewpoint_lag365

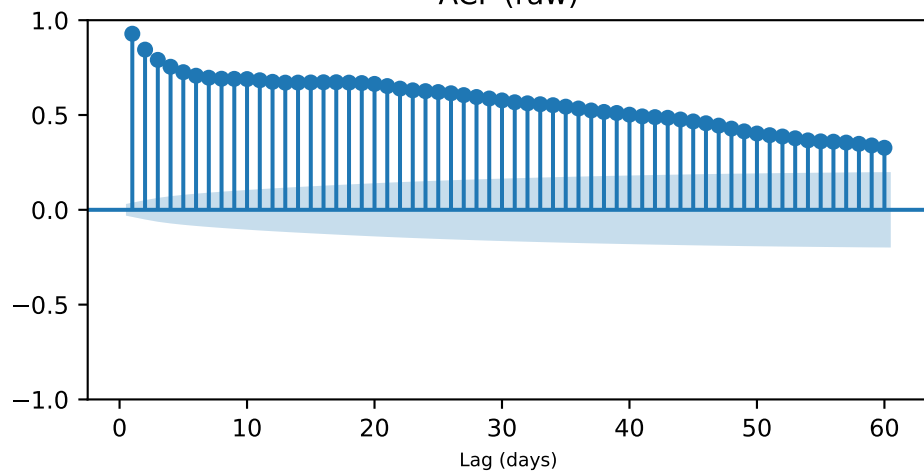
Raw histogram



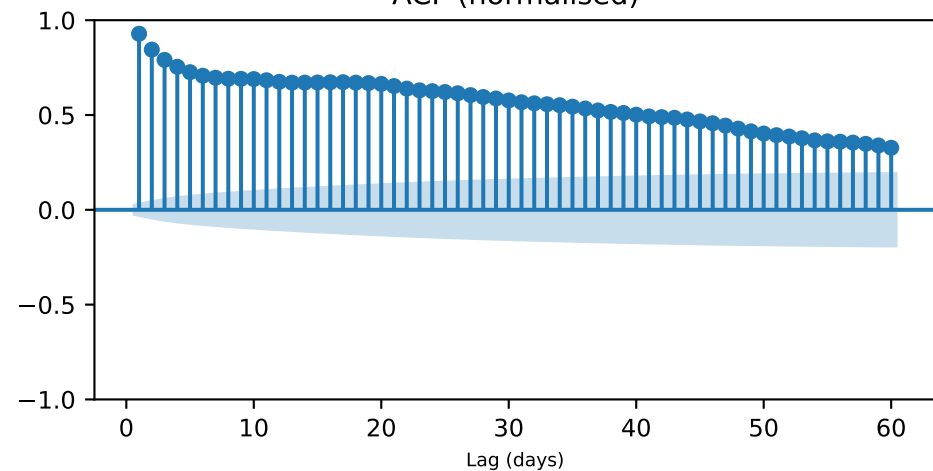
Normalised histogram



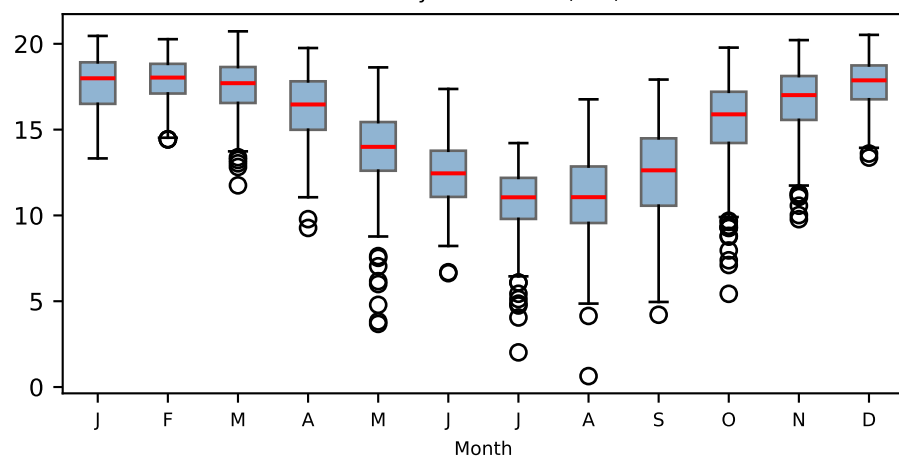
ACF (raw)



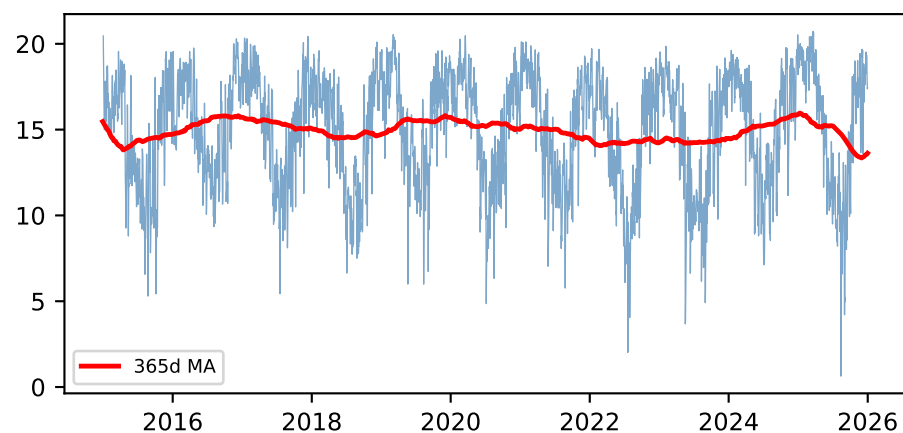
ACF (normalised)



Monthly distribution (raw)

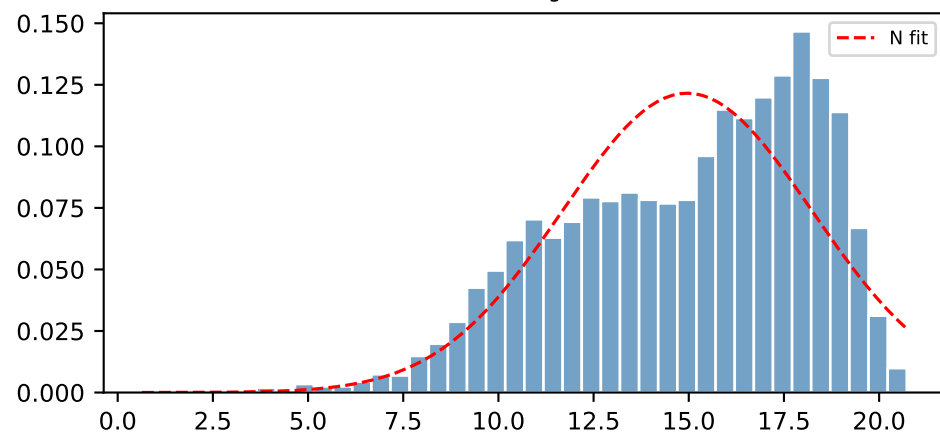


Time series + 365d mean

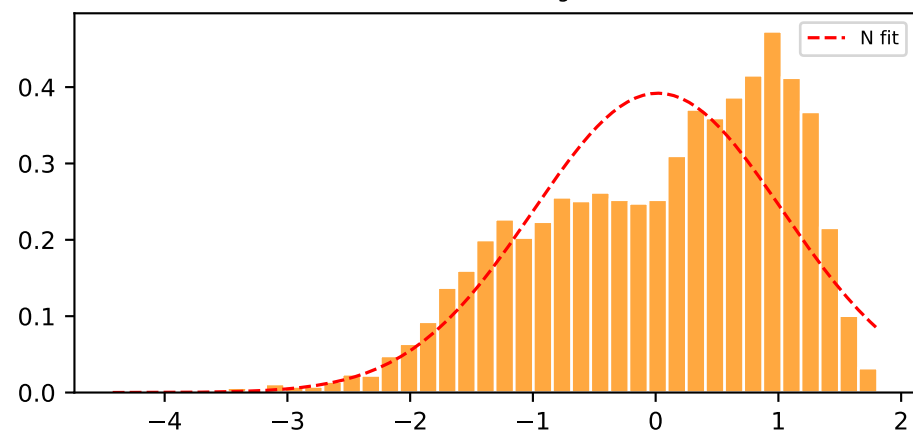


Covariate diagnostics — dewpoint_lag366

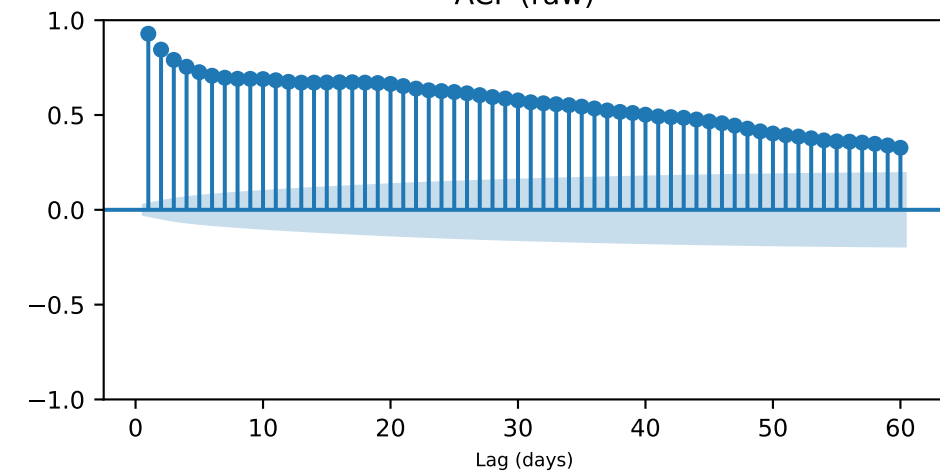
Raw histogram



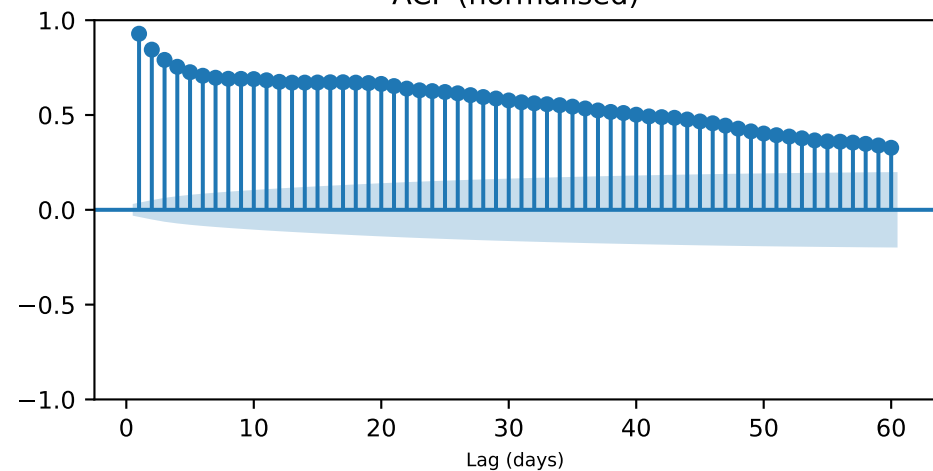
Normalised histogram



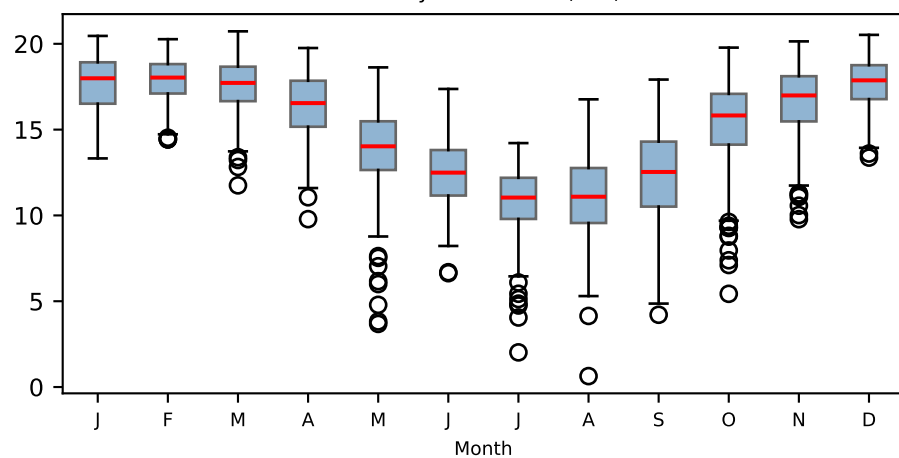
ACF (raw)



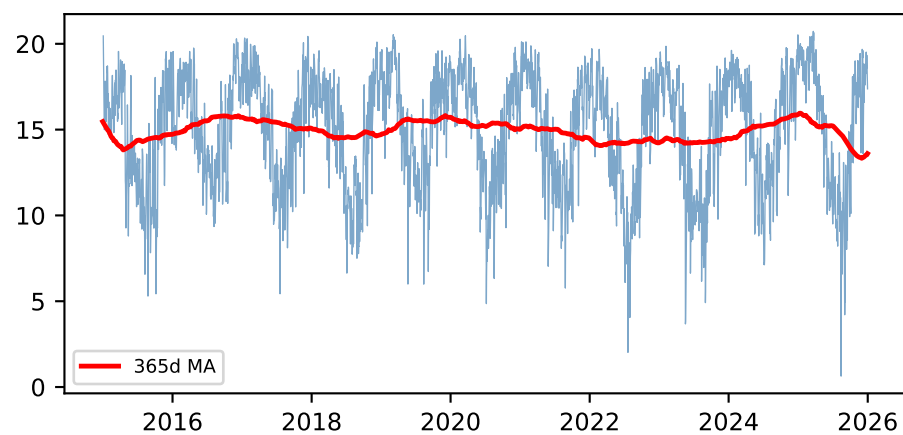
ACF (normalised)



Monthly distribution (raw)

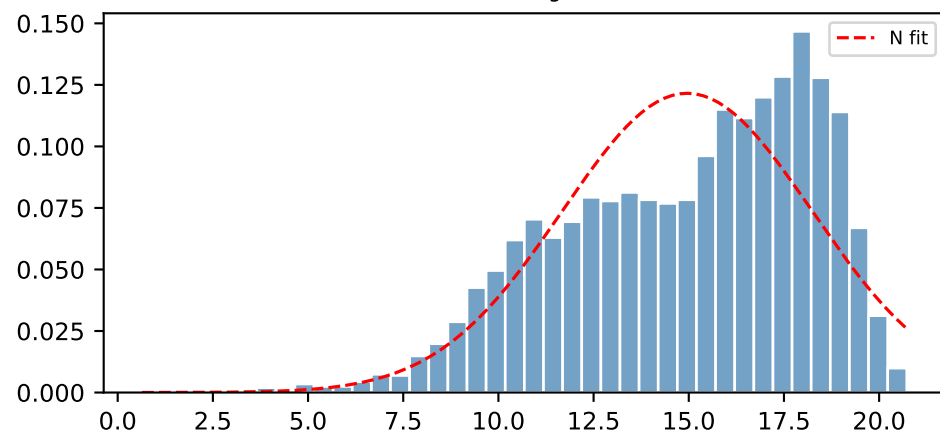


Time series + 365d mean

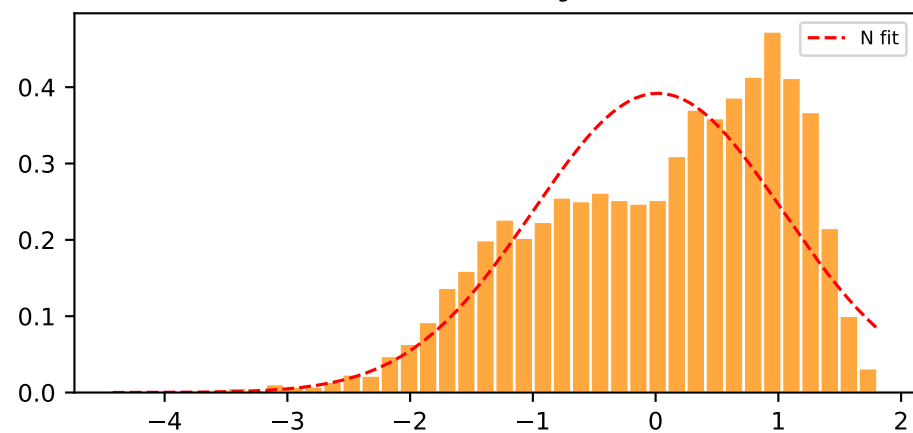


Covariate diagnostics — dewpoint_lag367

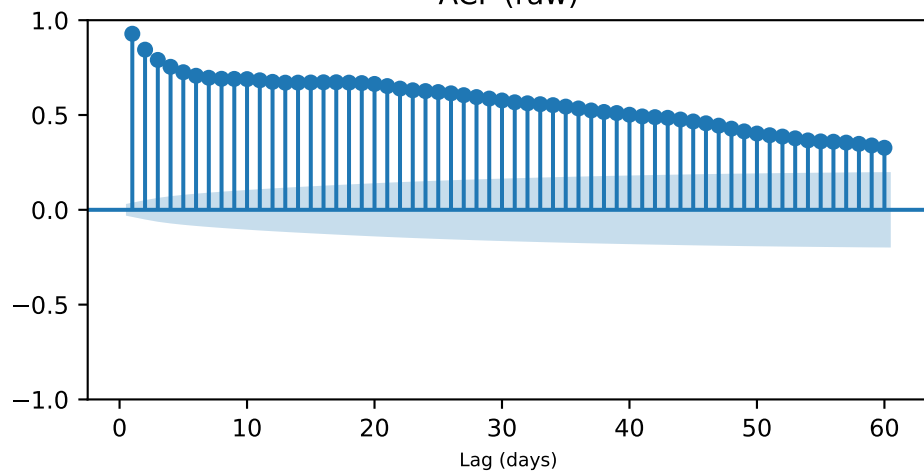
Raw histogram



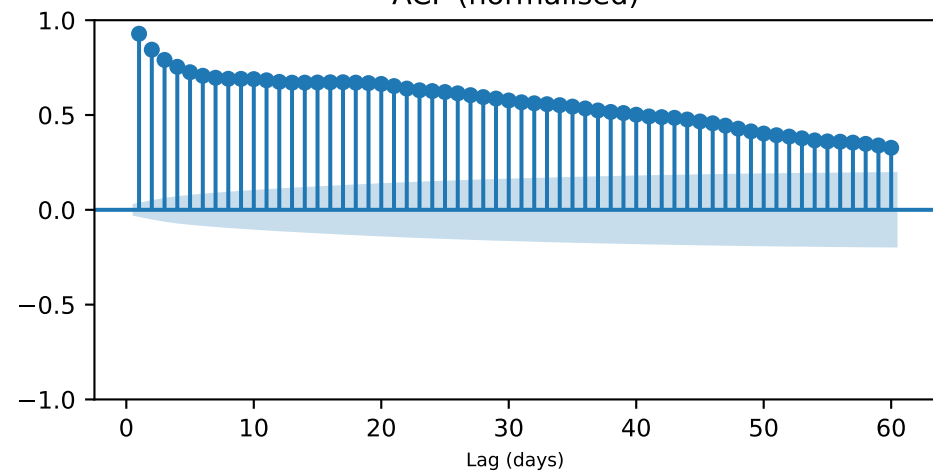
Normalised histogram



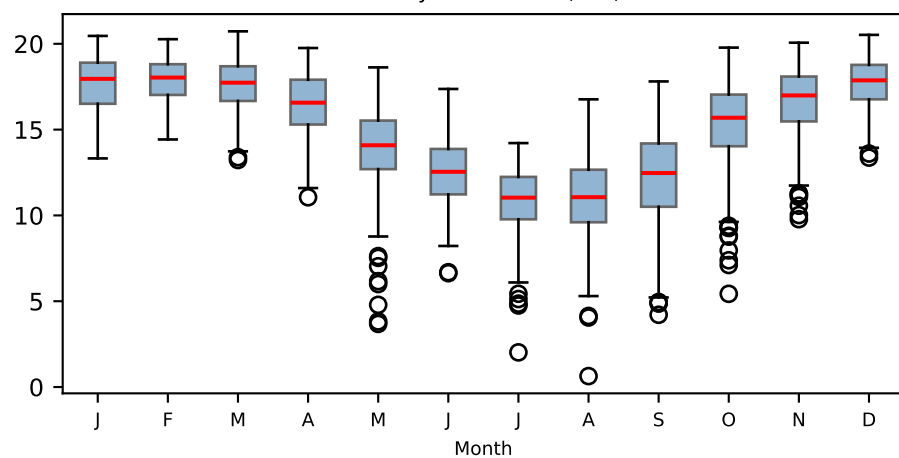
ACF (raw)



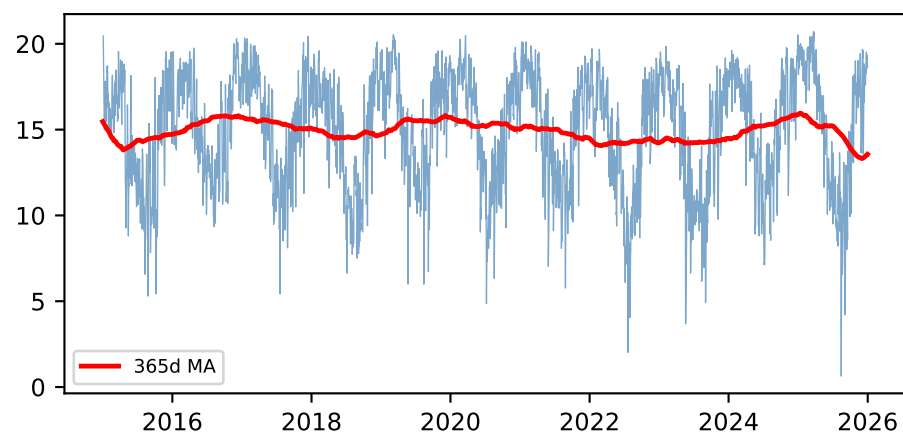
ACF (normalised)



Monthly distribution (raw)

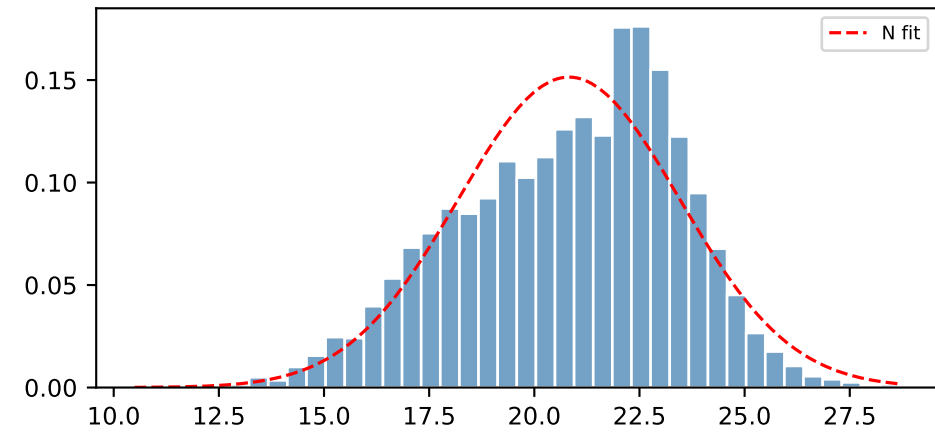


Time series + 365d mean

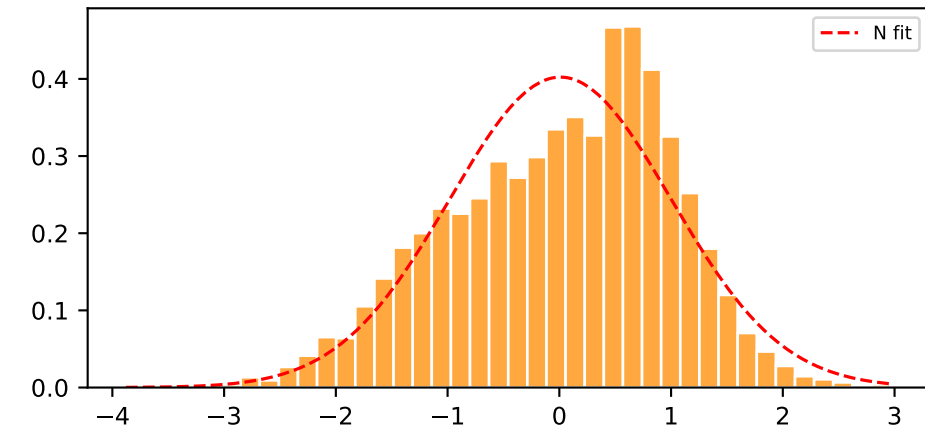


Covariate diagnostics — temperature_t

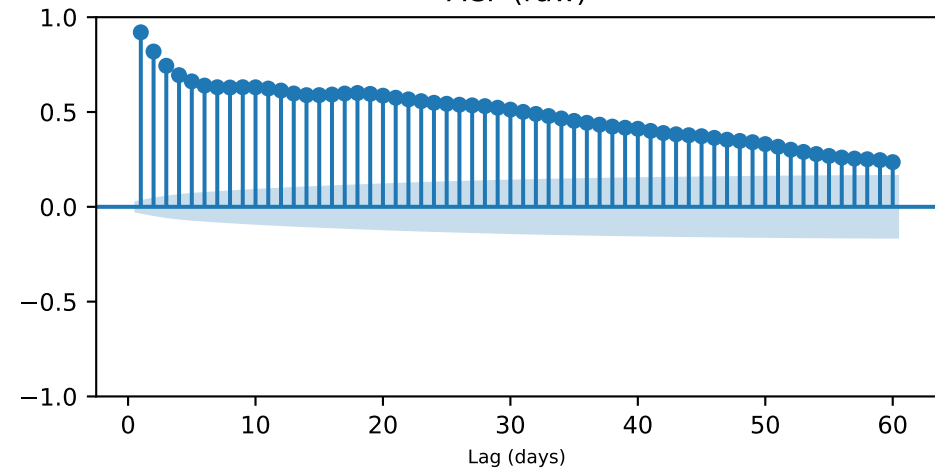
Raw histogram



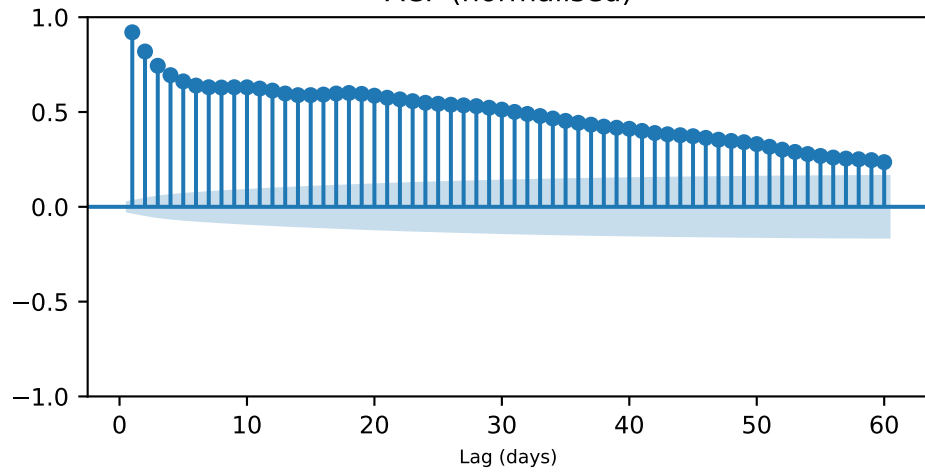
Normalised histogram



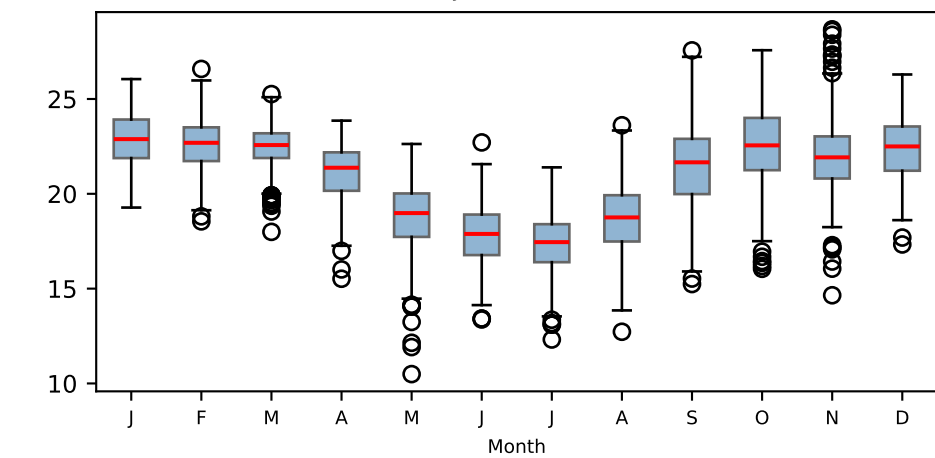
ACF (raw)



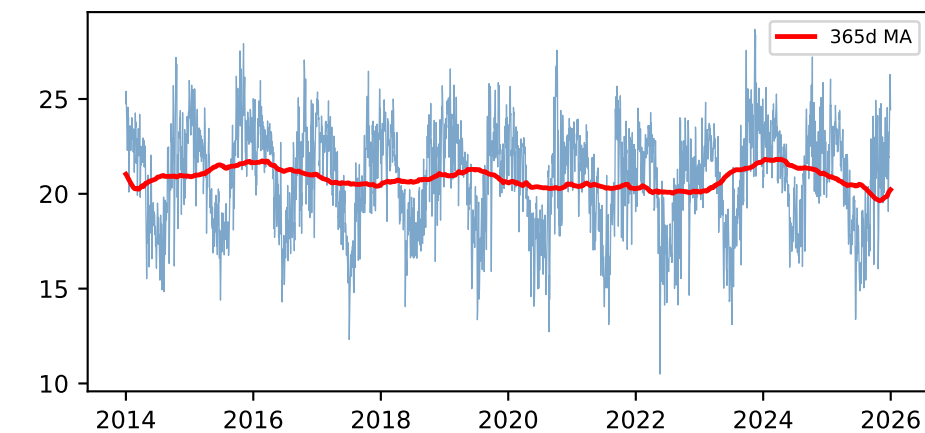
ACF (normalised)



Monthly distribution (raw)

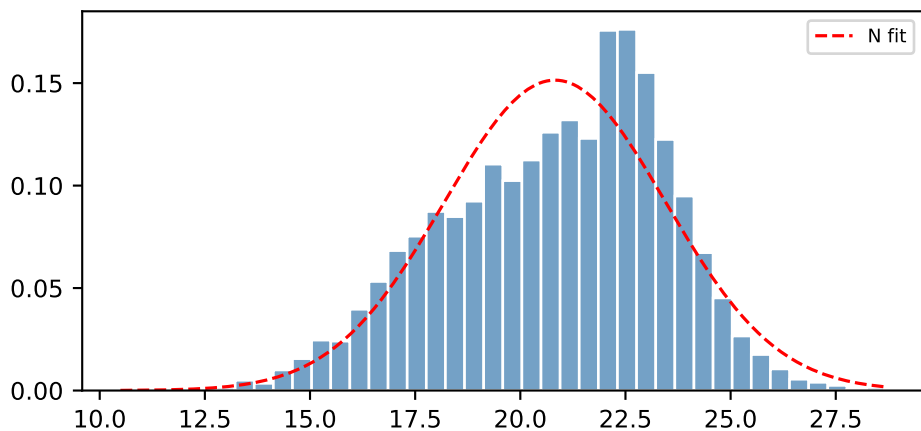


Time series + 365d mean

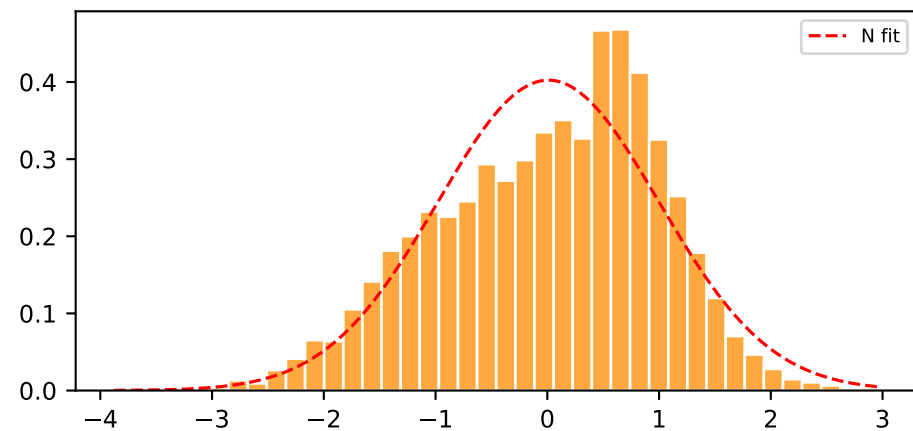


Covariate diagnostics — temperature_lag1

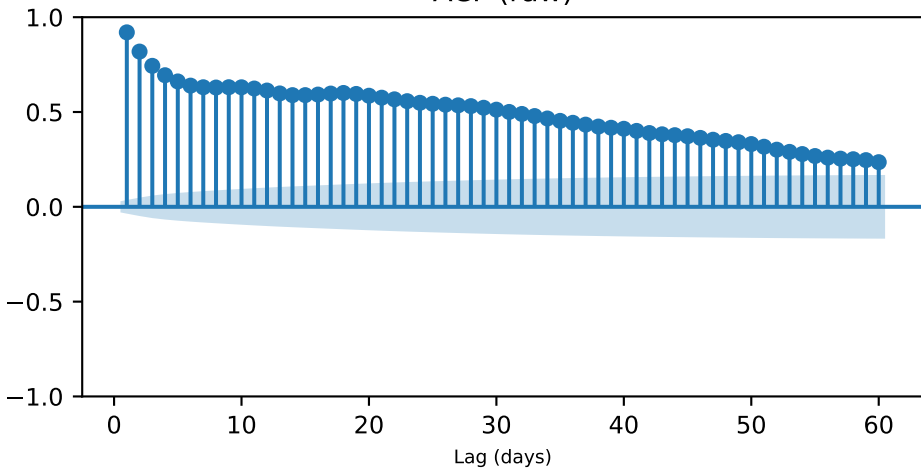
Raw histogram



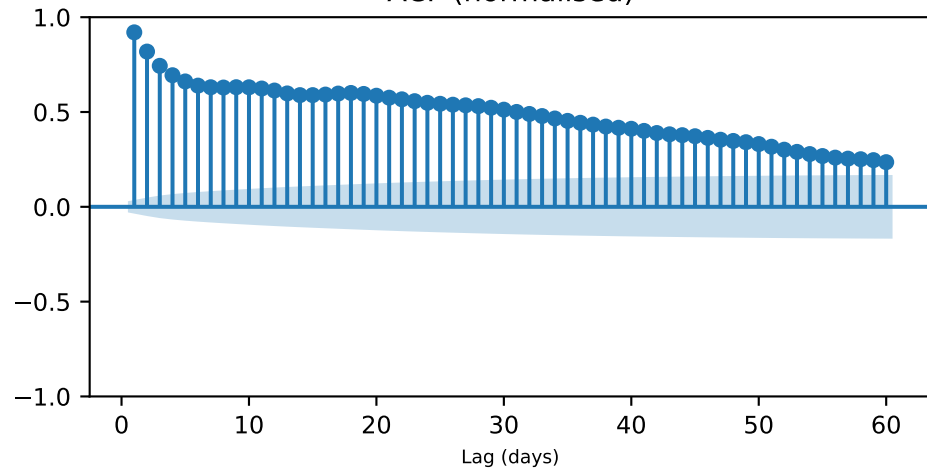
Normalised histogram



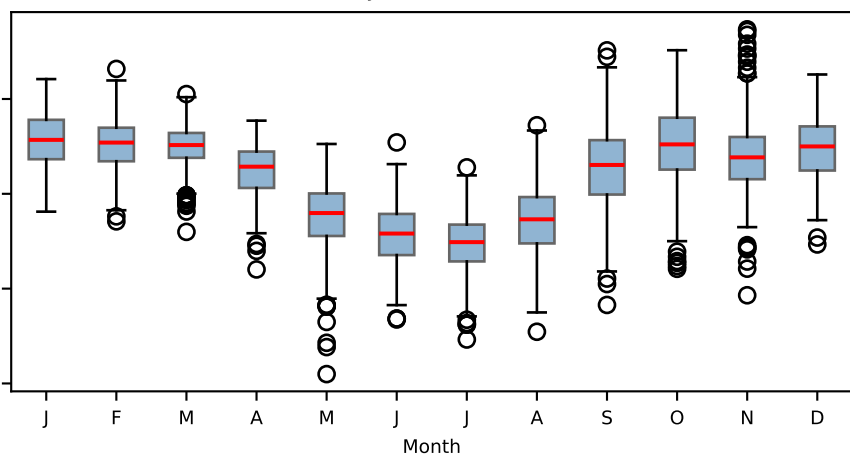
ACF (raw)



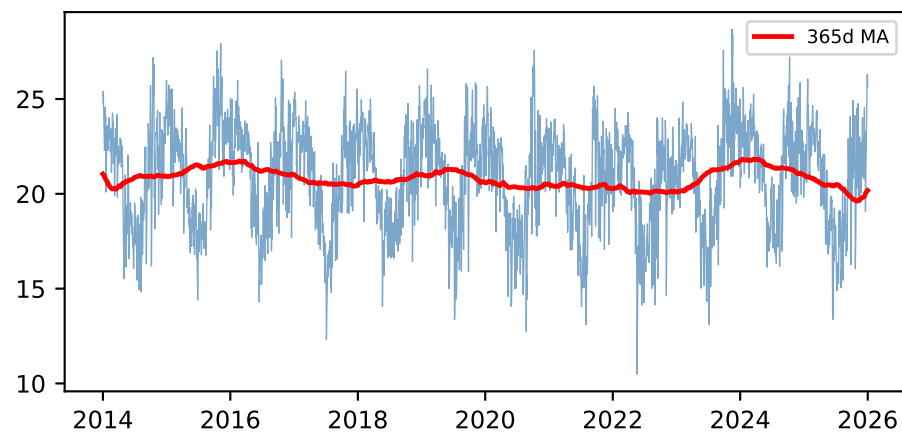
ACF (normalised)



Monthly distribution (raw)

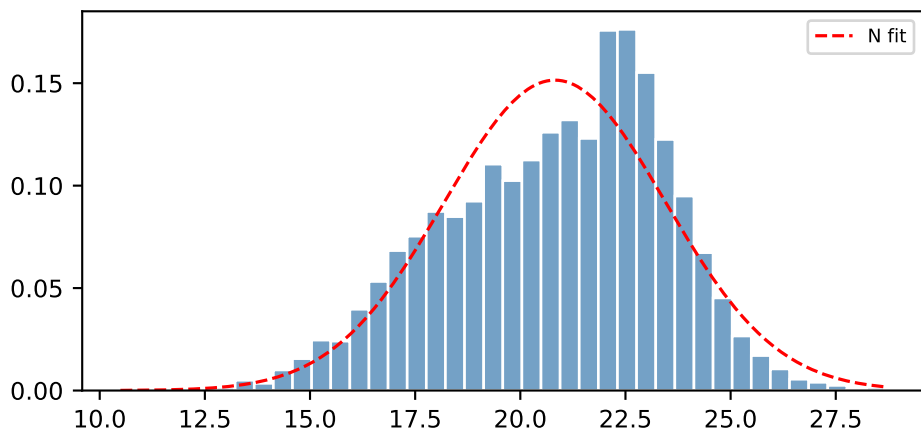


Time series + 365d mean

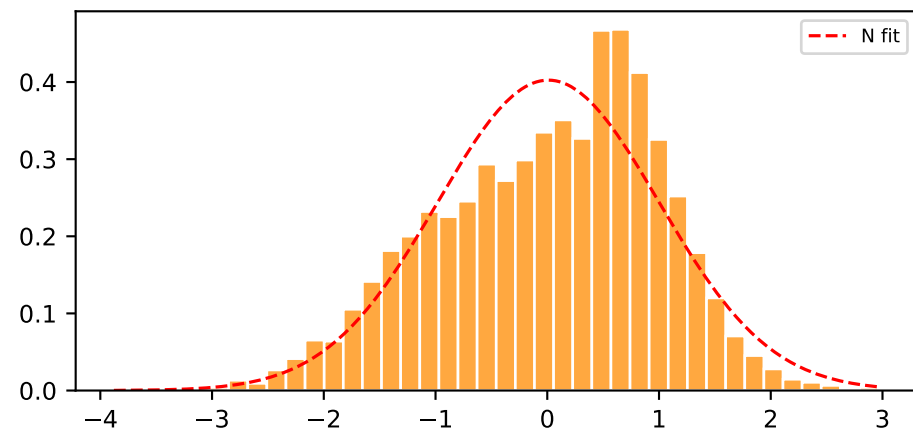


Covariate diagnostics — temperature_lag2

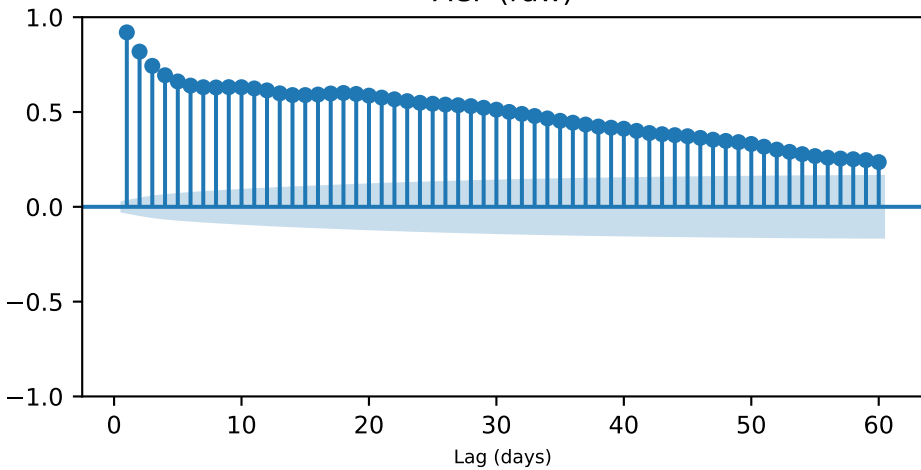
Raw histogram



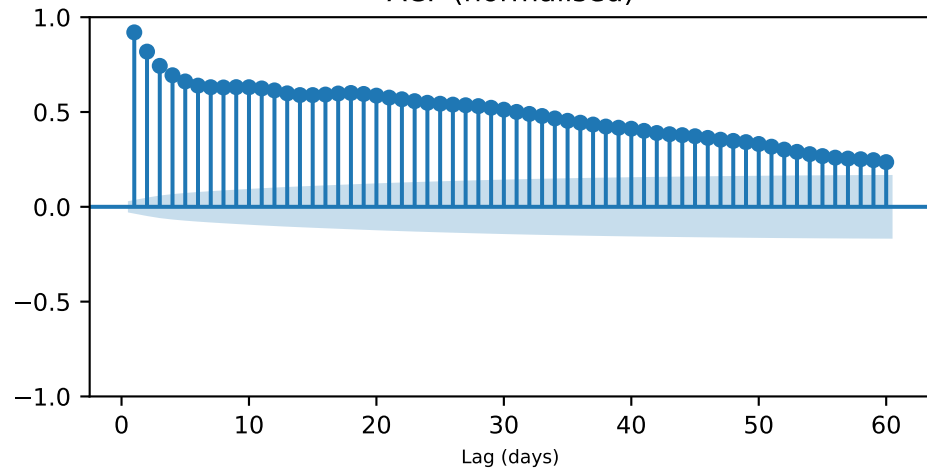
Normalised histogram



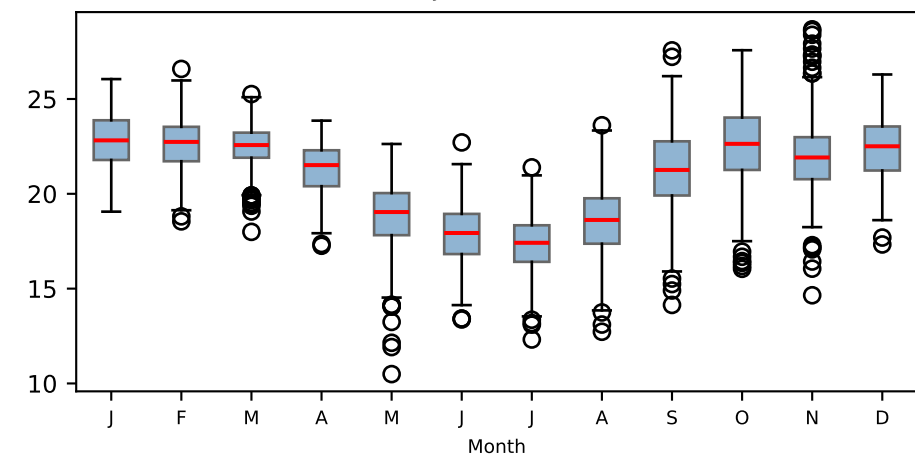
ACF (raw)



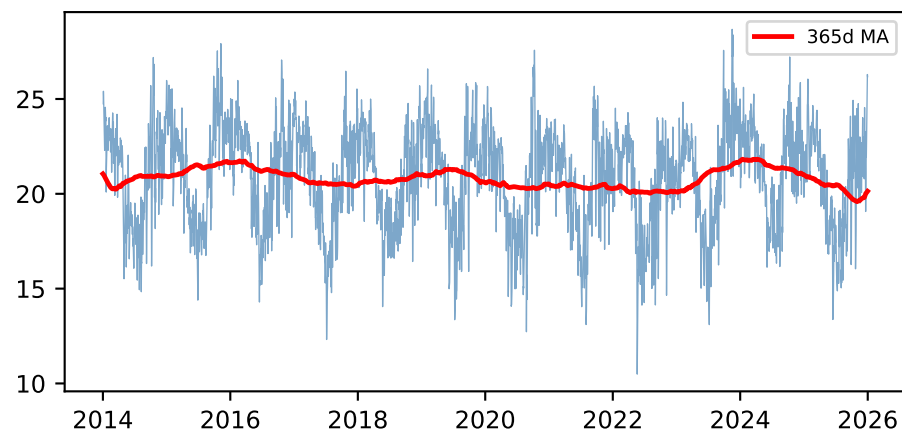
ACF (normalised)



Monthly distribution (raw)

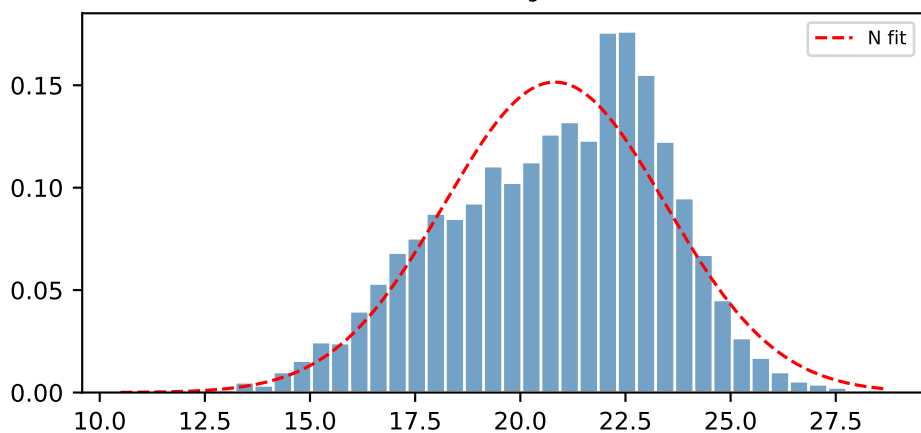


Time series + 365d mean

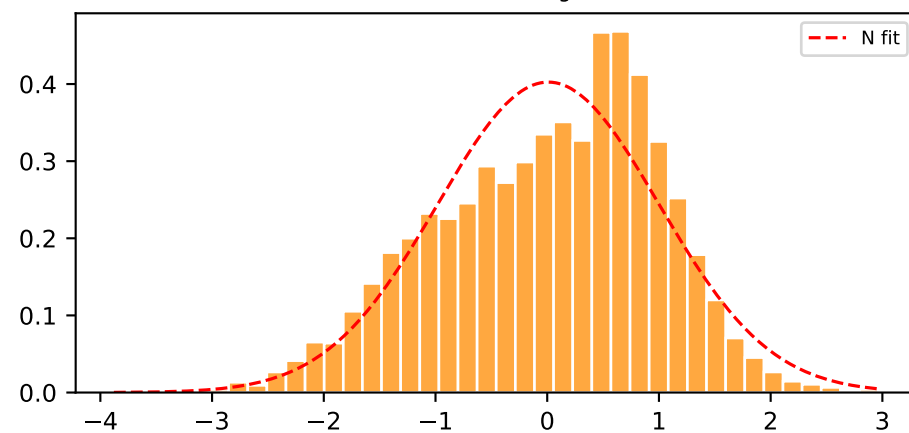


Covariate diagnostics — temperature_lag3

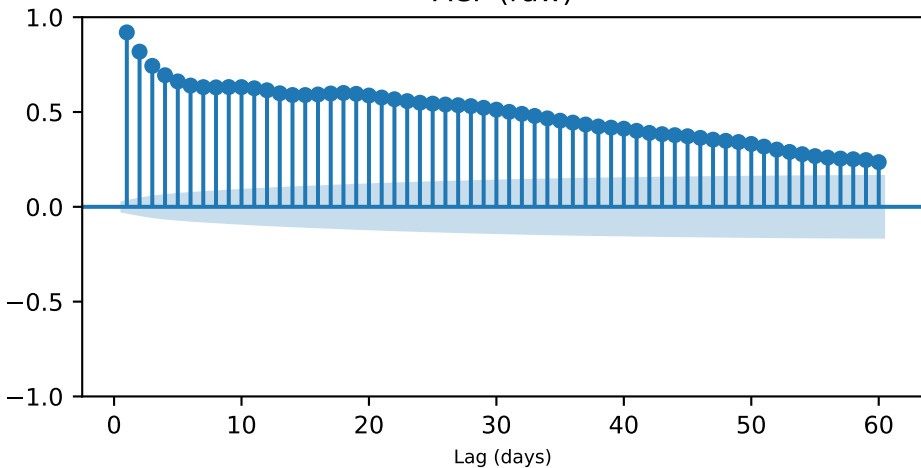
Raw histogram



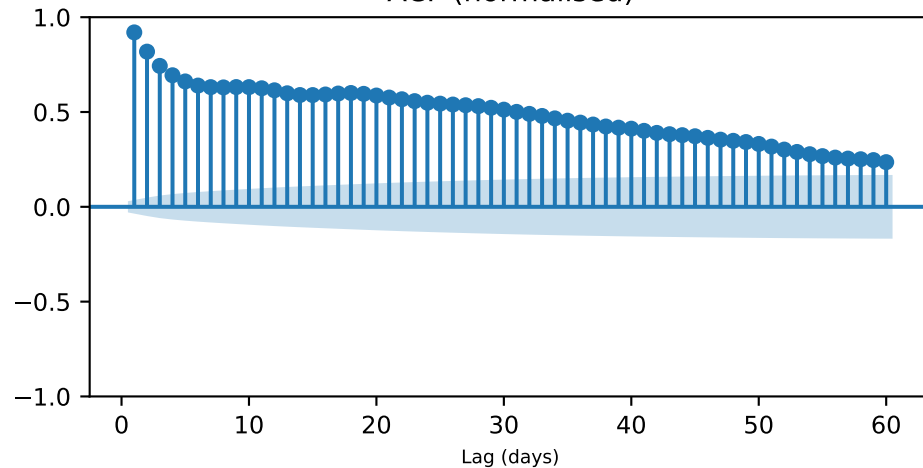
Normalised histogram



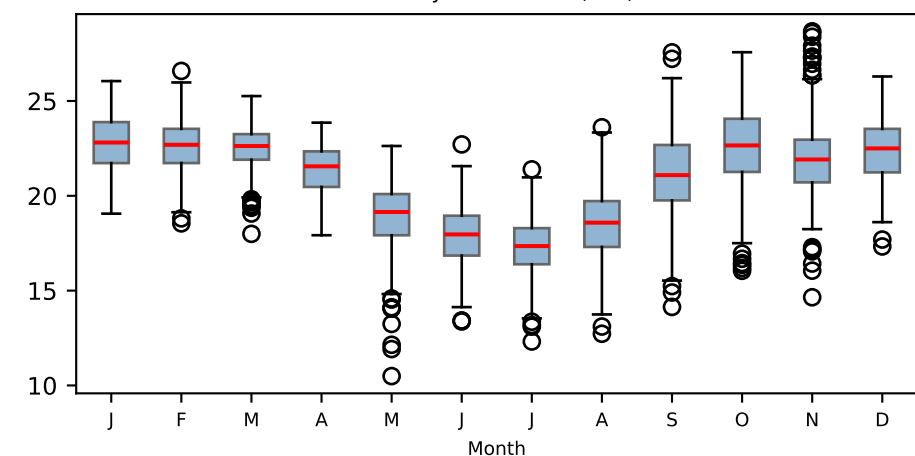
ACF (raw)



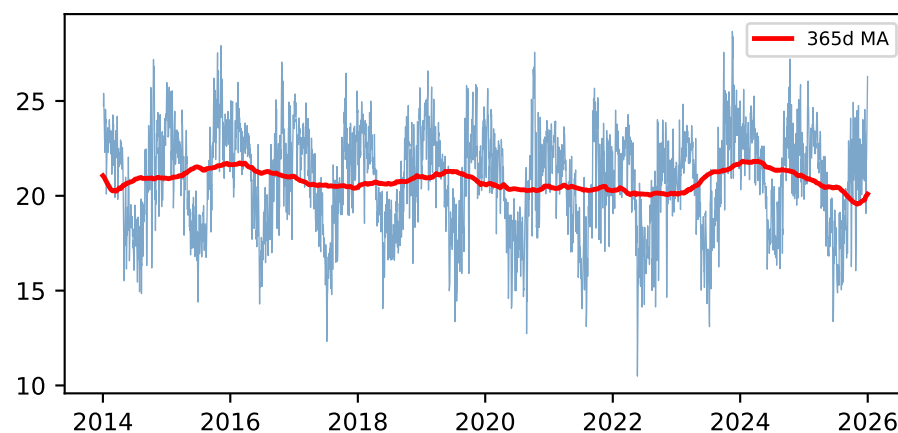
ACF (normalised)



Monthly distribution (raw)

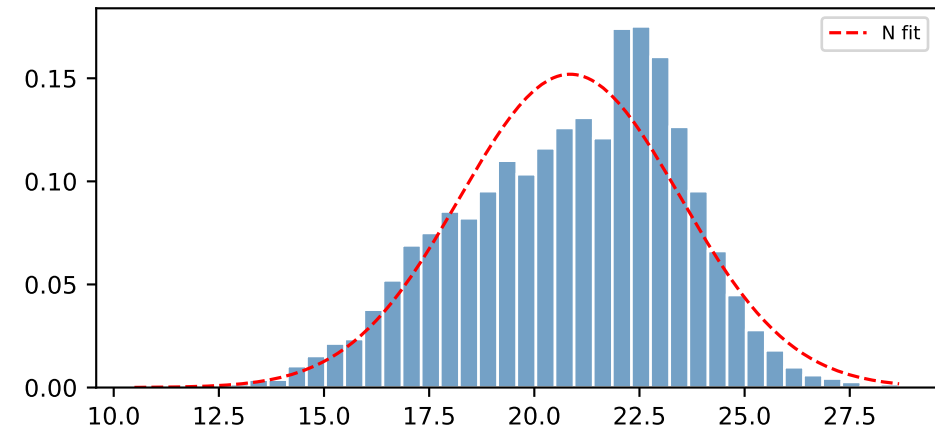


Time series + 365d mean

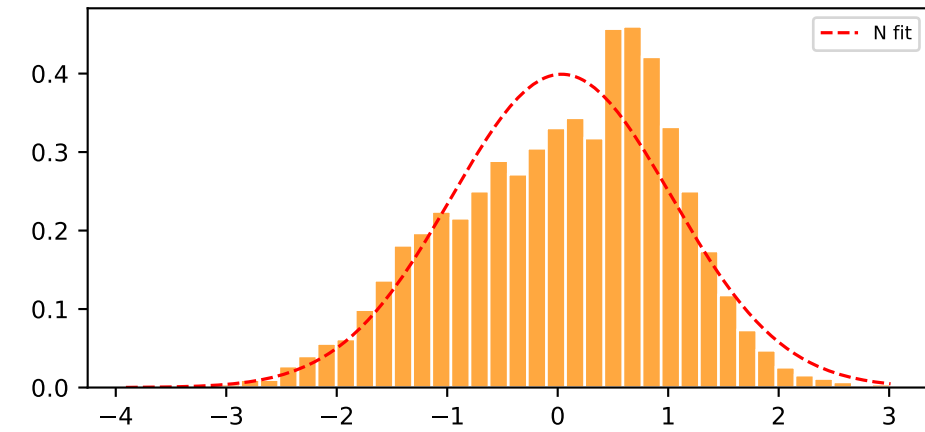


Covariate diagnostics — temperature_lag364

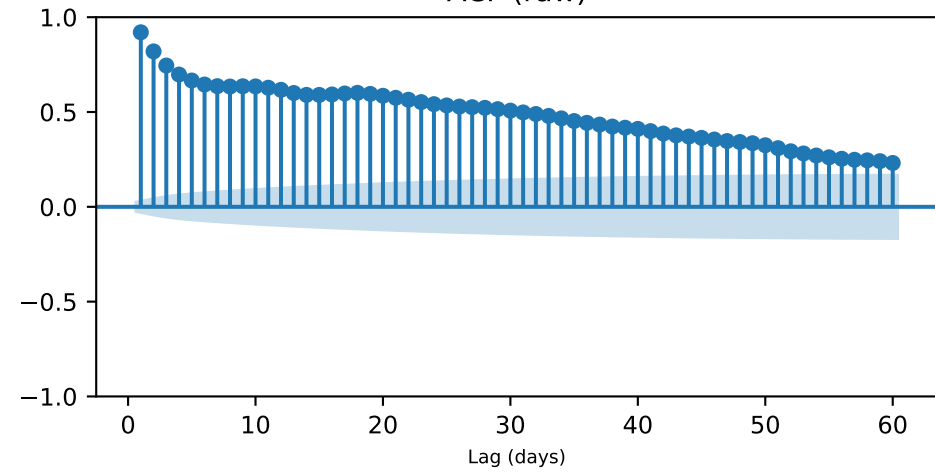
Raw histogram



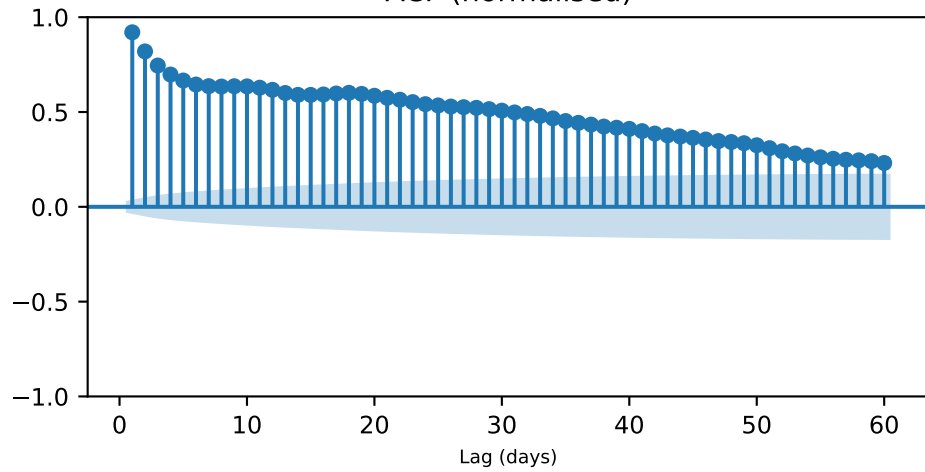
Normalised histogram



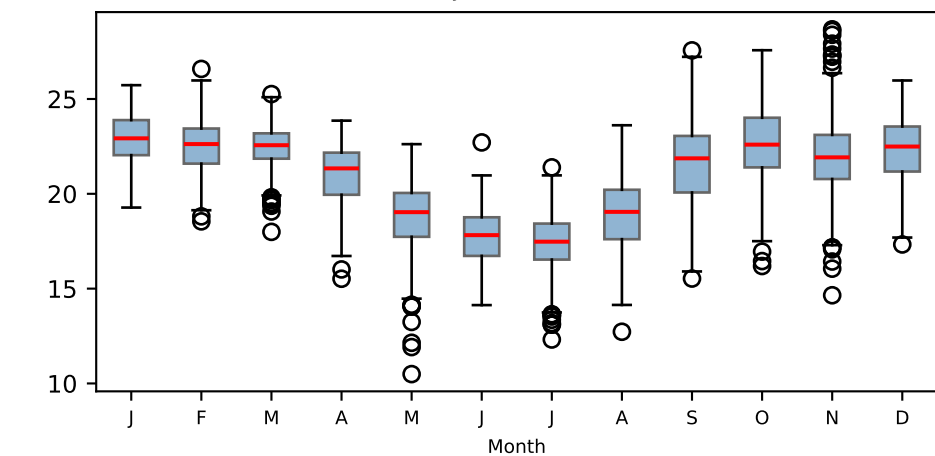
ACF (raw)



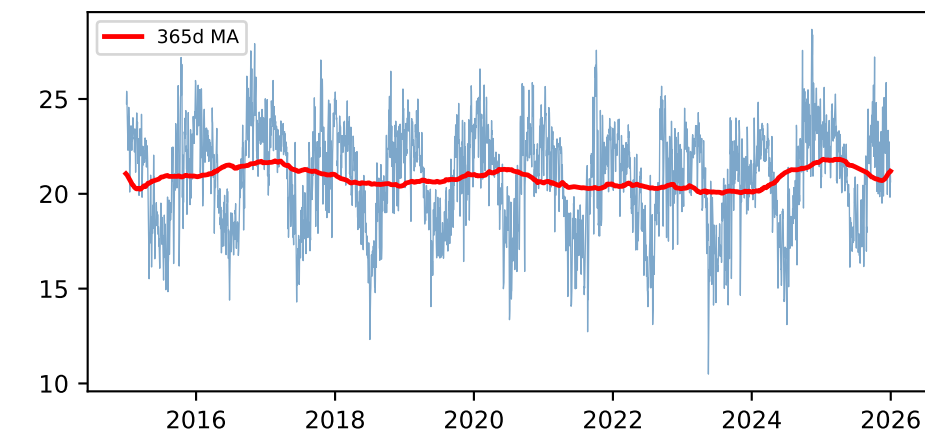
ACF (normalised)



Monthly distribution (raw)

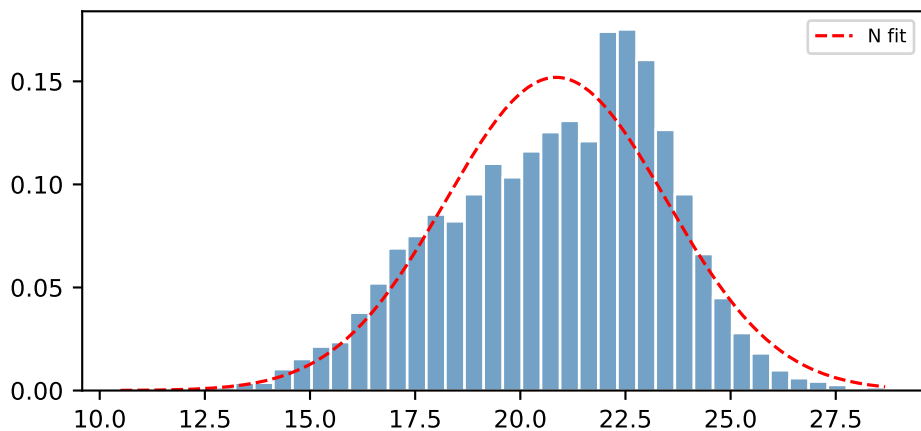


Time series + 365d mean

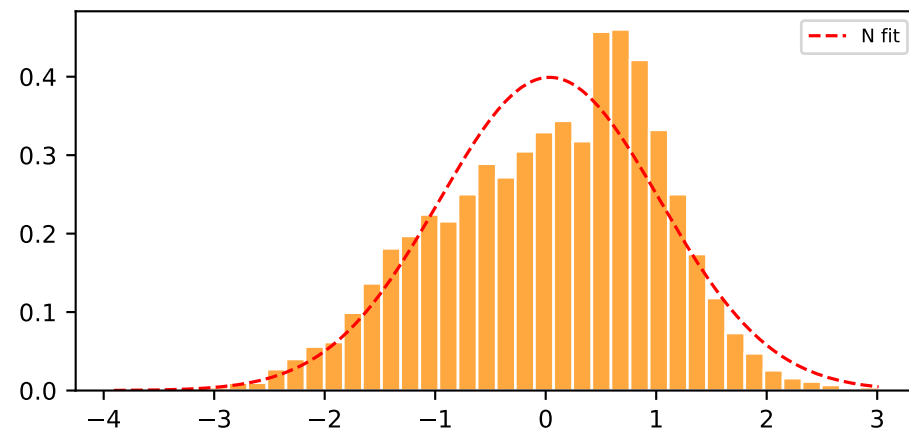


Covariate diagnostics — temperature_lag365

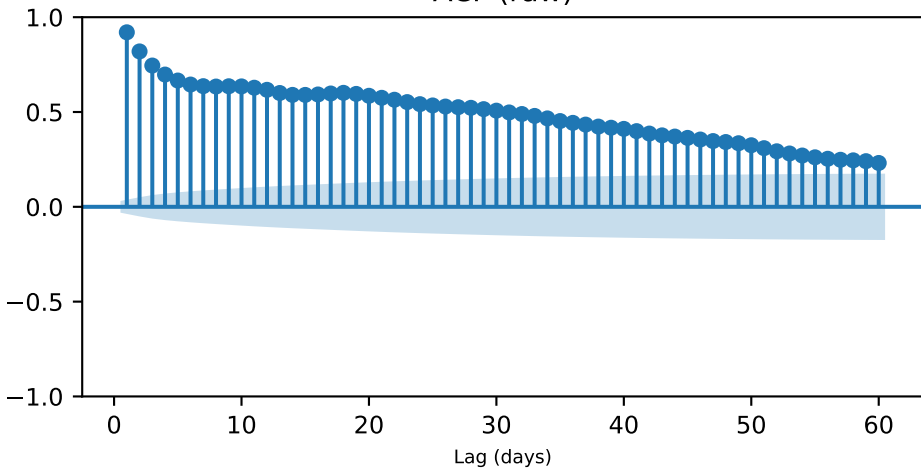
Raw histogram



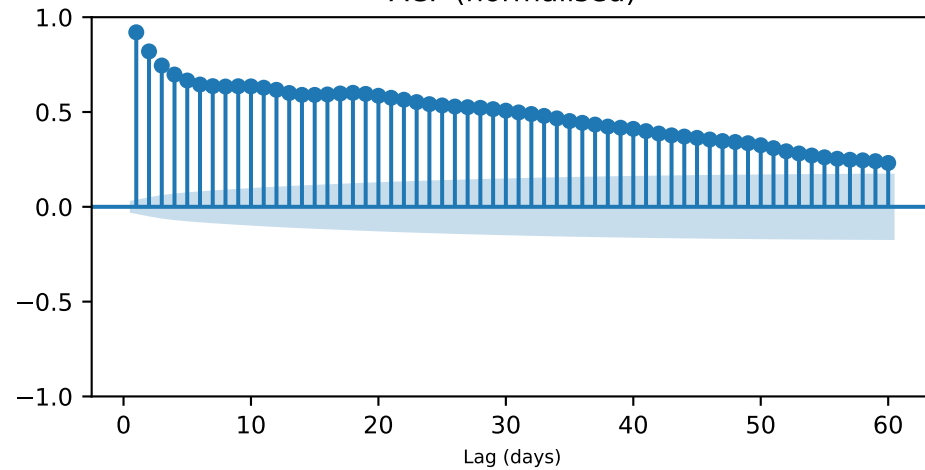
Normalised histogram



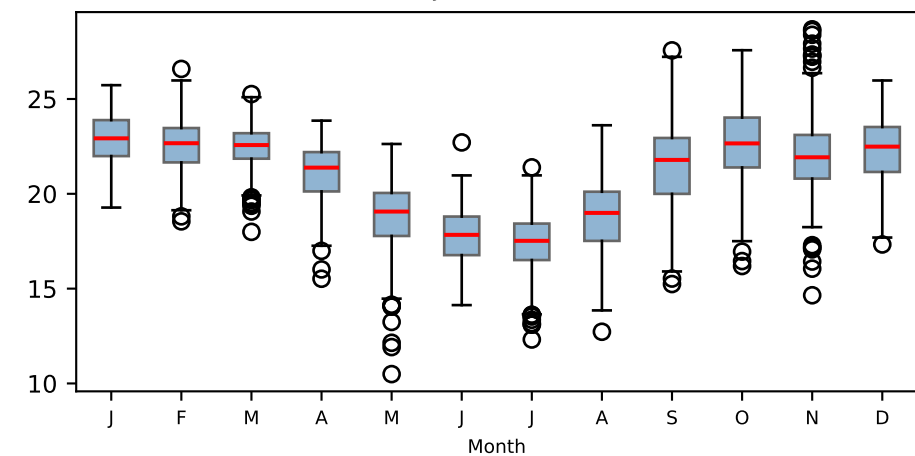
ACF (raw)



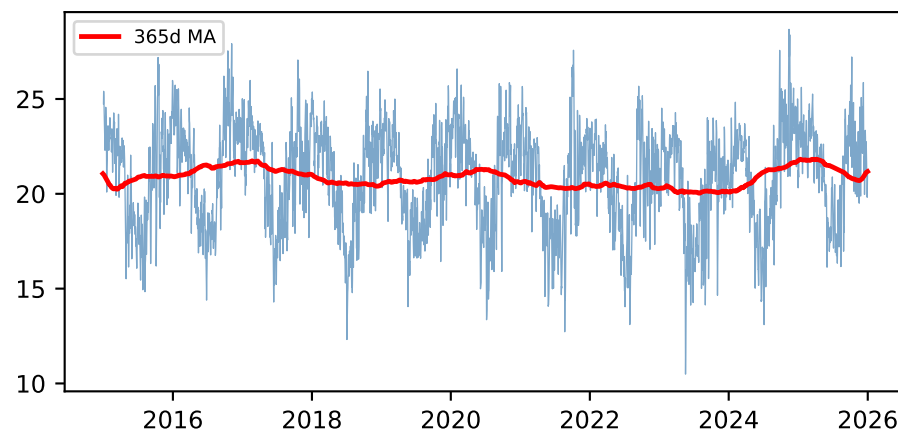
ACF (normalised)



Monthly distribution (raw)

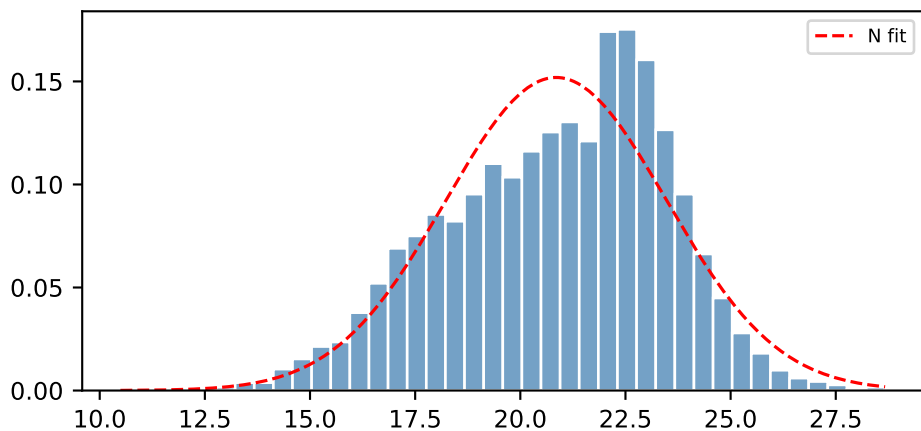


Time series + 365d mean

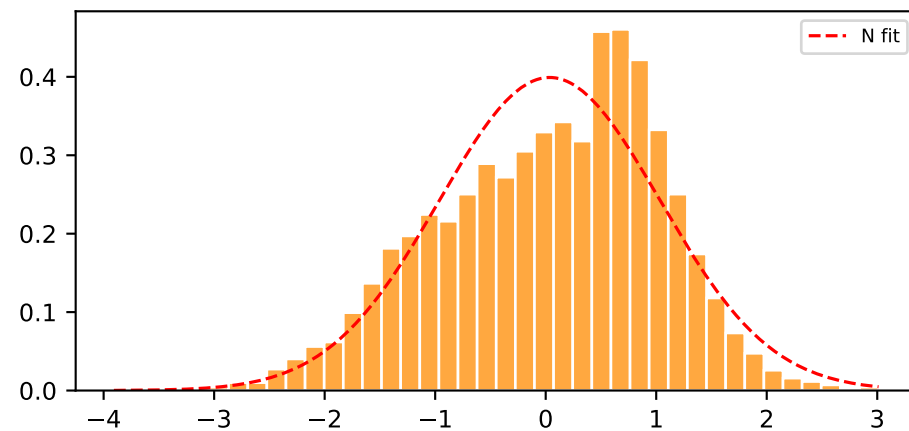


Covariate diagnostics — temperature_lag366

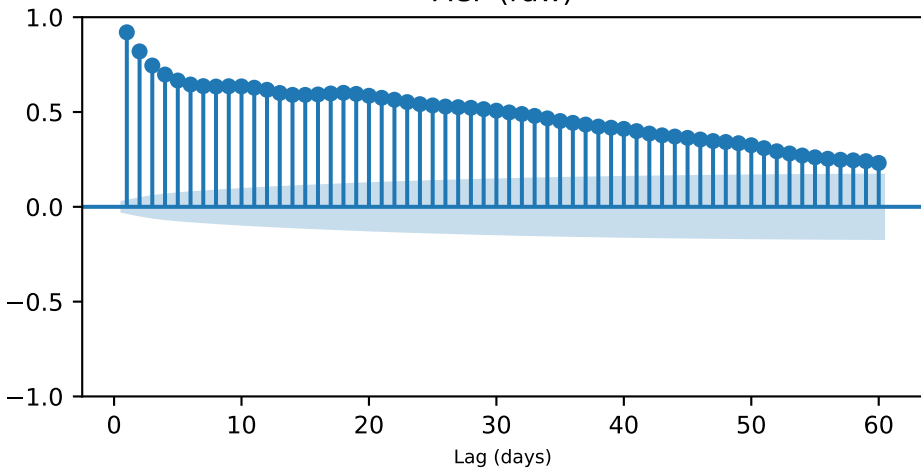
Raw histogram



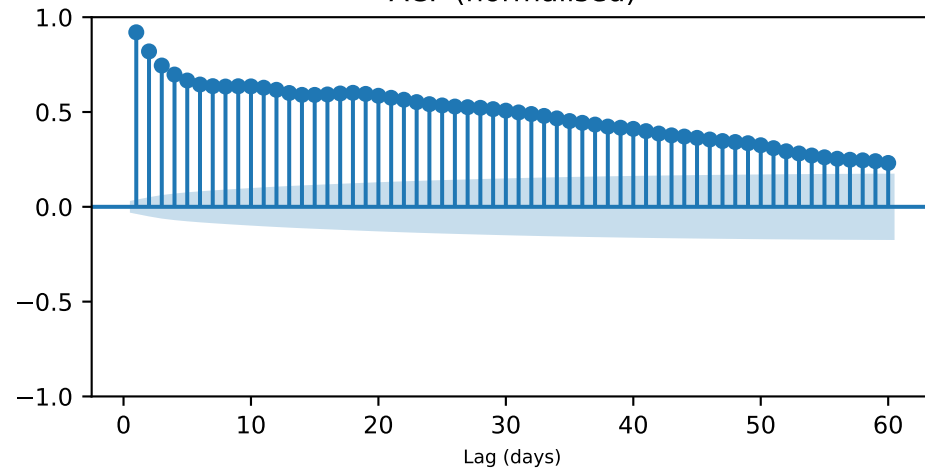
Normalised histogram



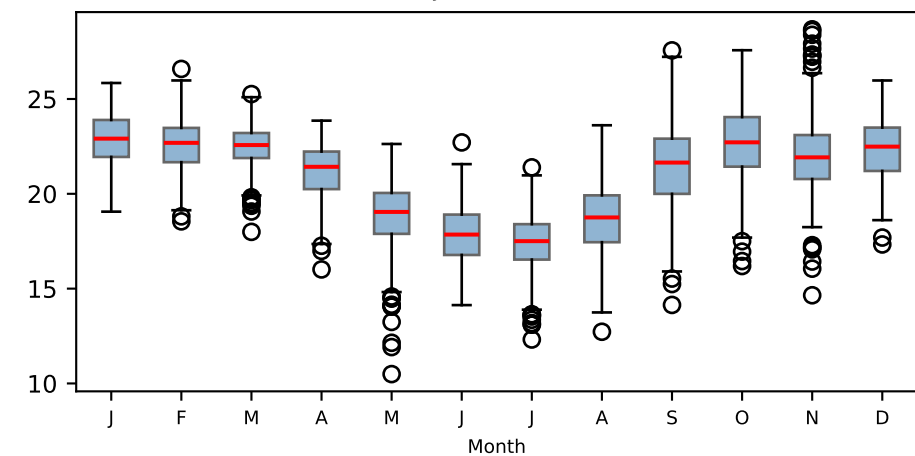
ACF (raw)



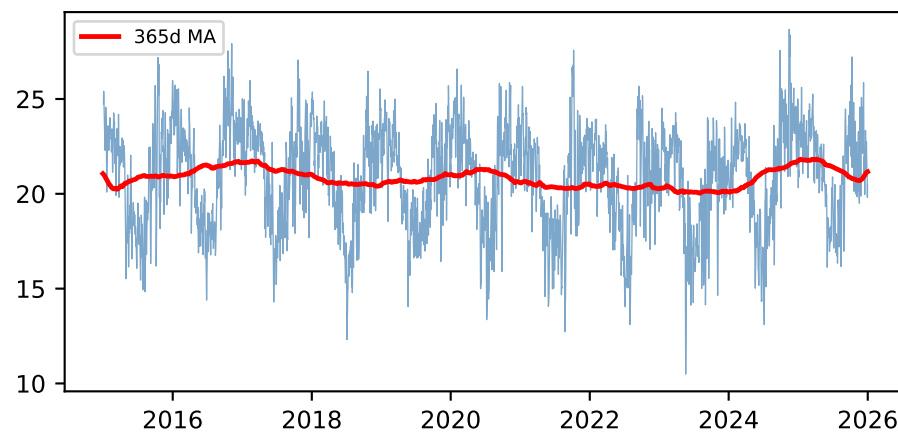
ACF (normalised)



Monthly distribution (raw)

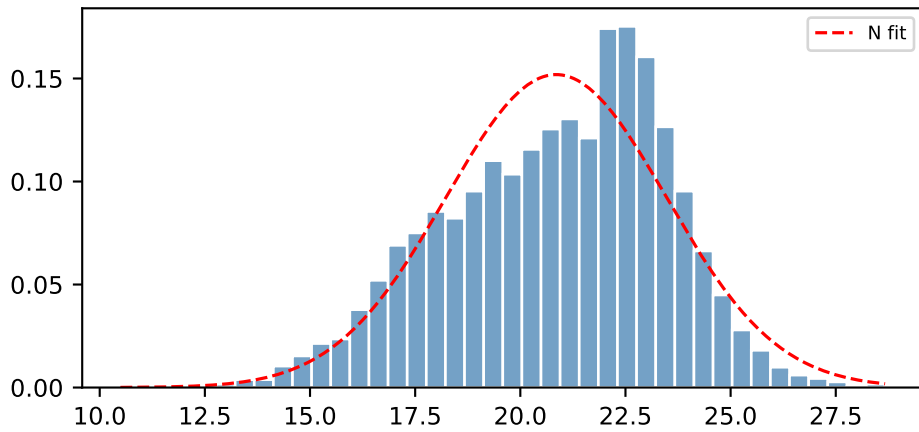


Time series + 365d mean

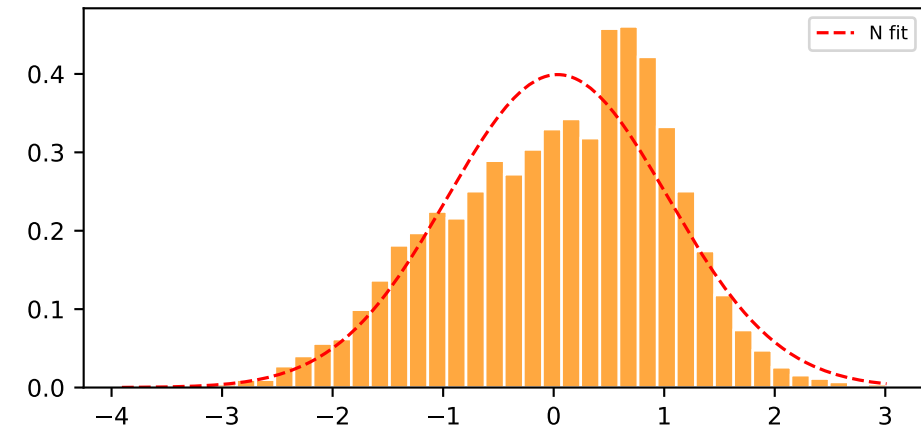


Covariate diagnostics — temperature_lag367

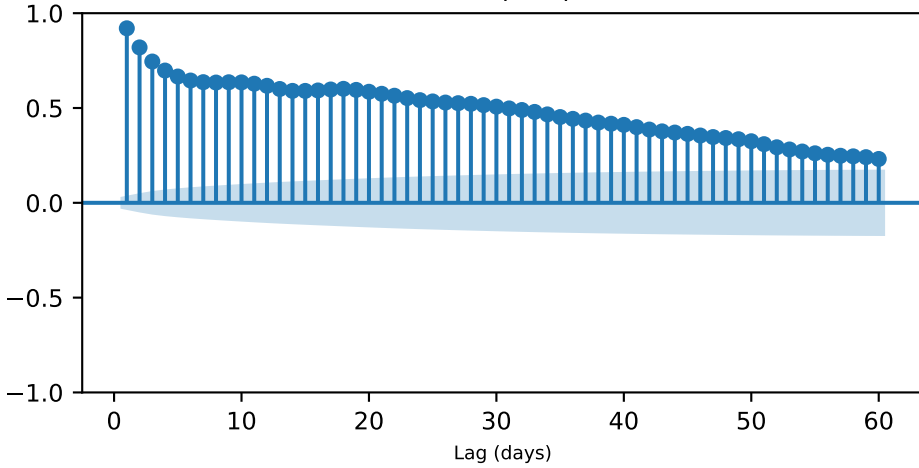
Raw histogram



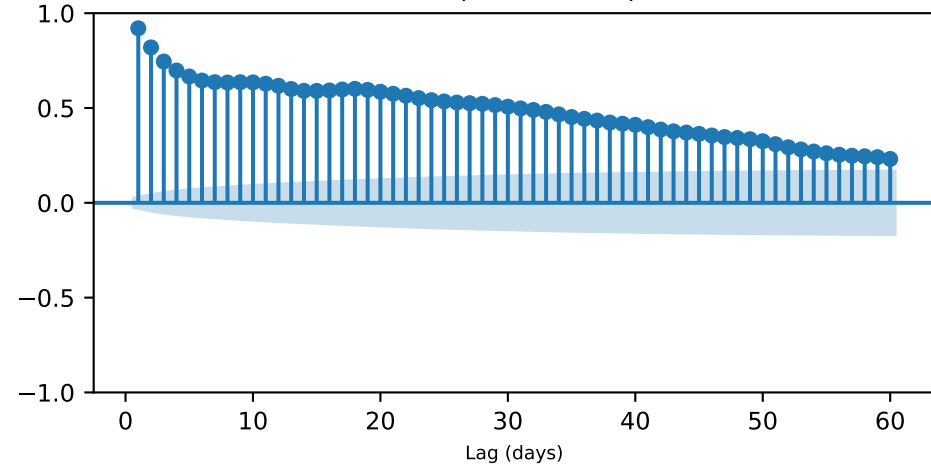
Normalised histogram



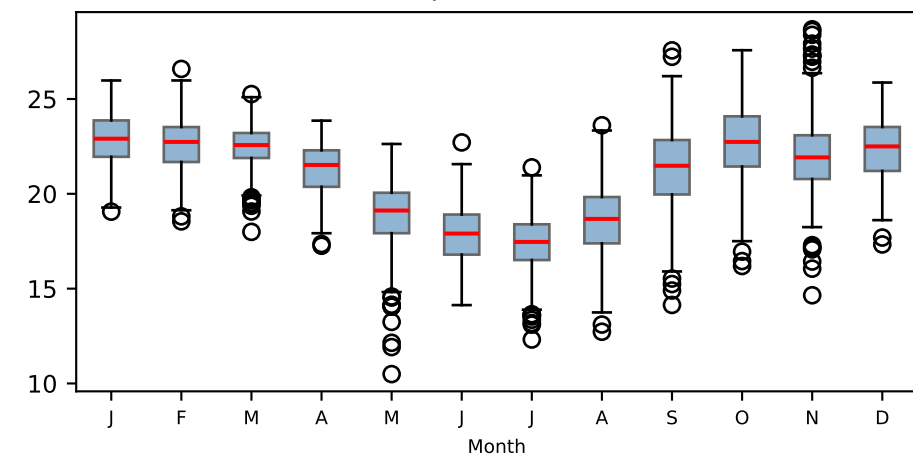
ACF (raw)



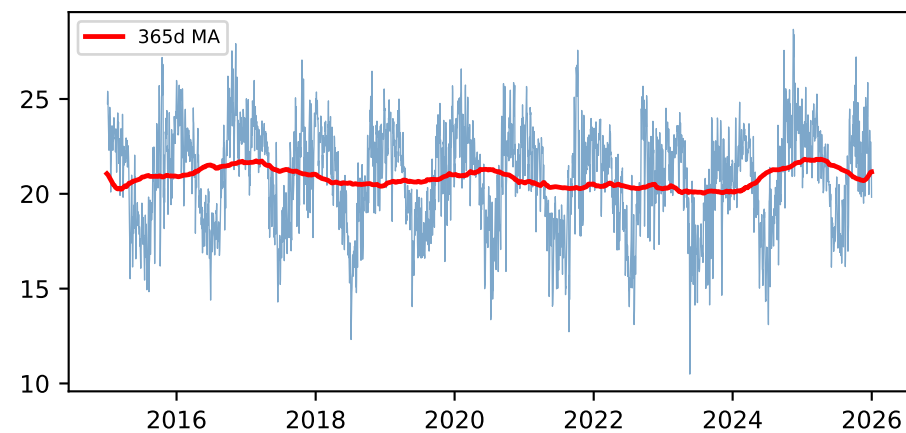
ACF (normalised)



Monthly distribution (raw)

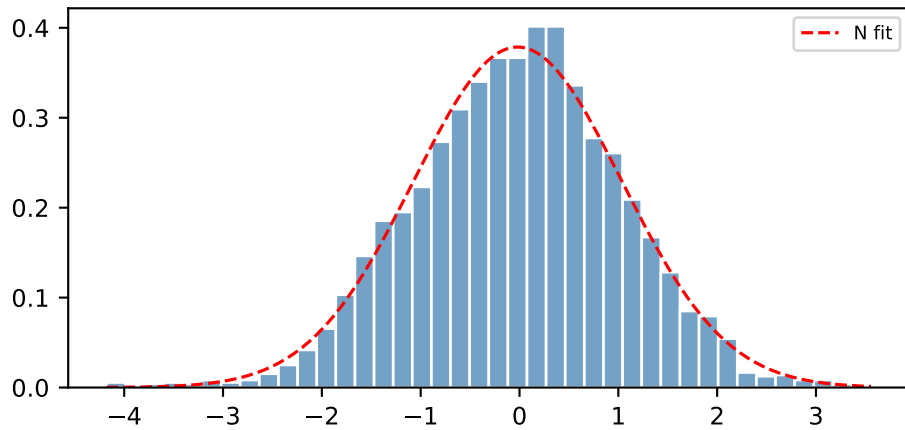


Time series + 365d mean

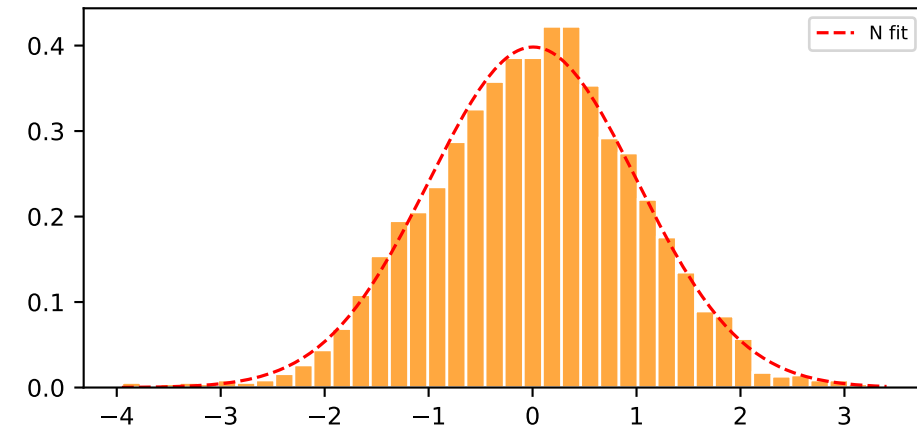


Covariate diagnostics — rmm1_t

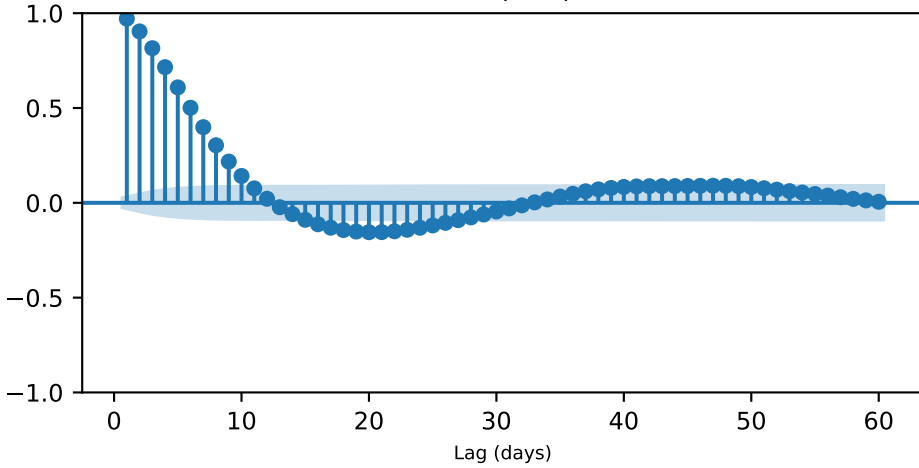
Raw histogram



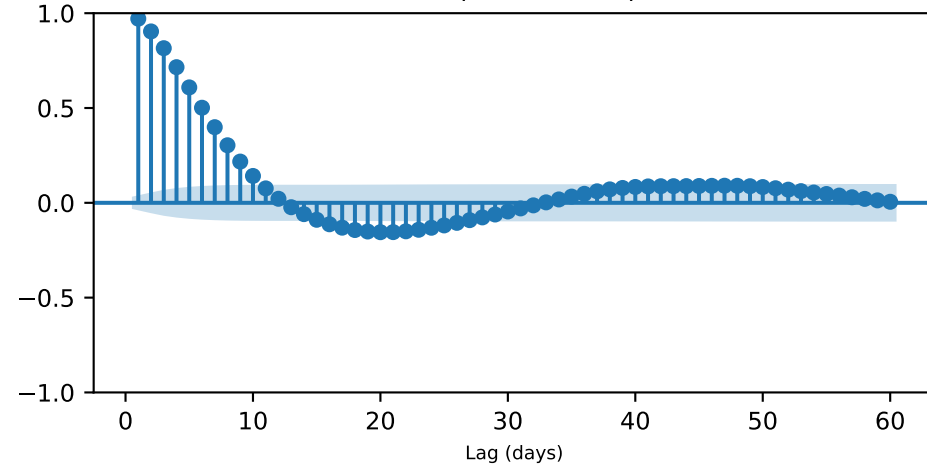
Normalised histogram



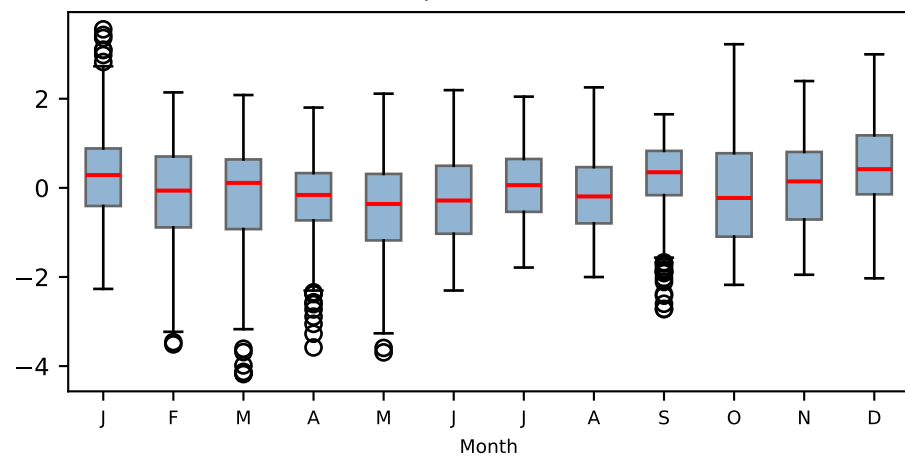
ACF (raw)



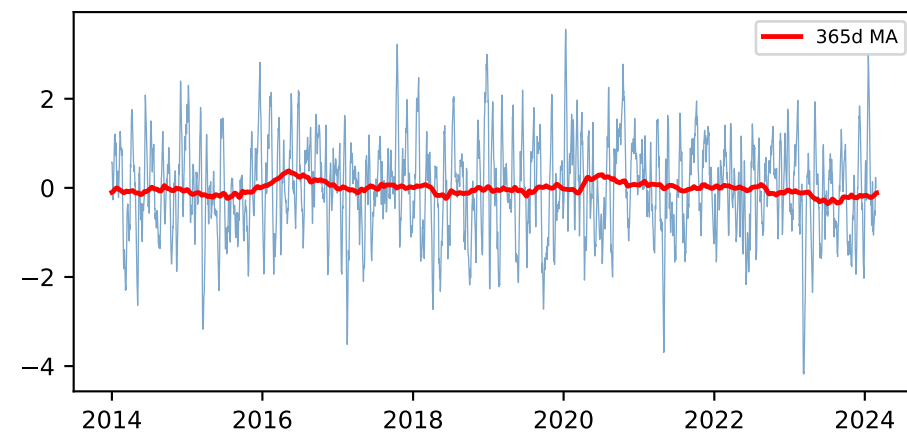
ACF (normalised)



Monthly distribution (raw)

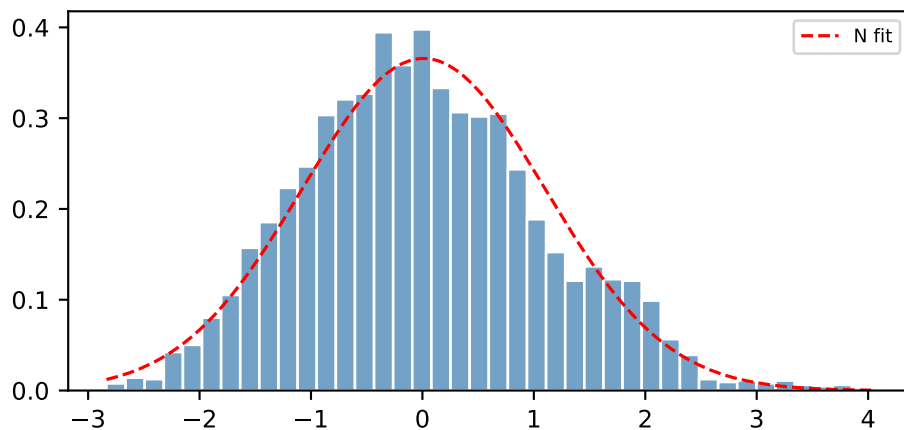


Time series + 365d mean

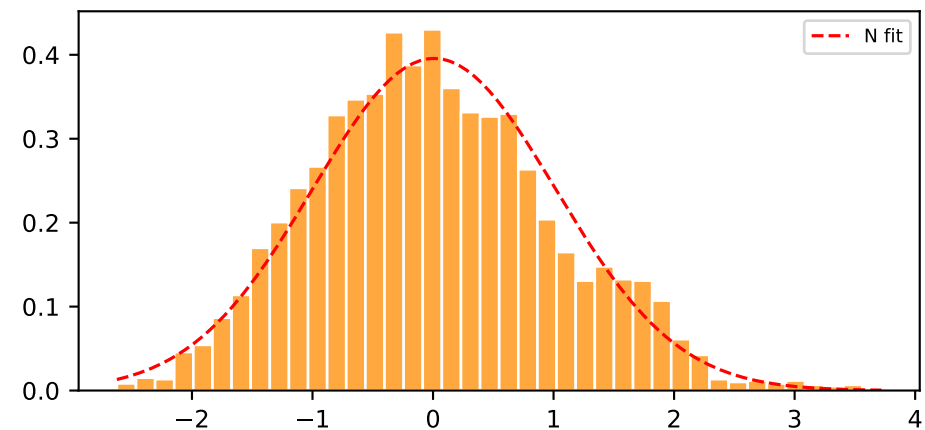


Covariate diagnostics — rmm2_t

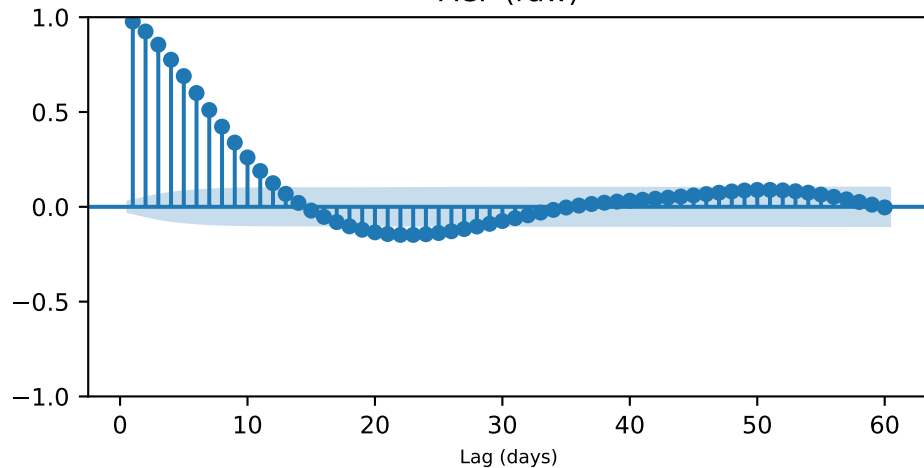
Raw histogram



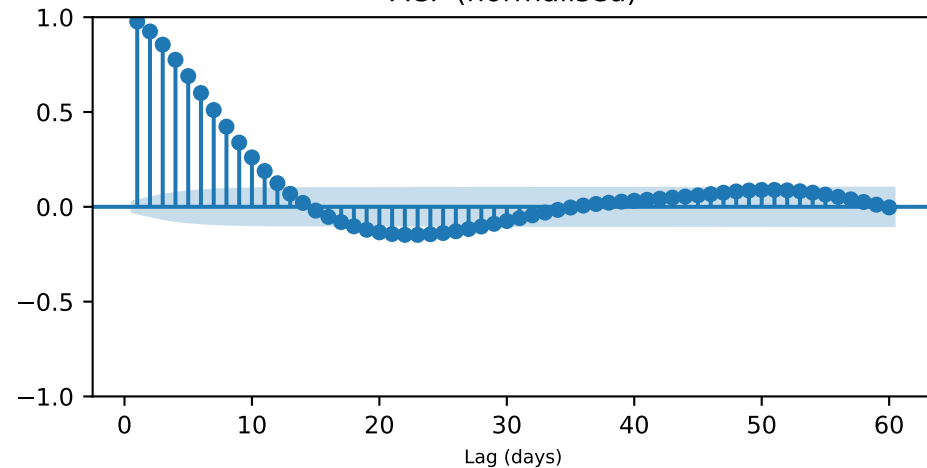
Normalised histogram



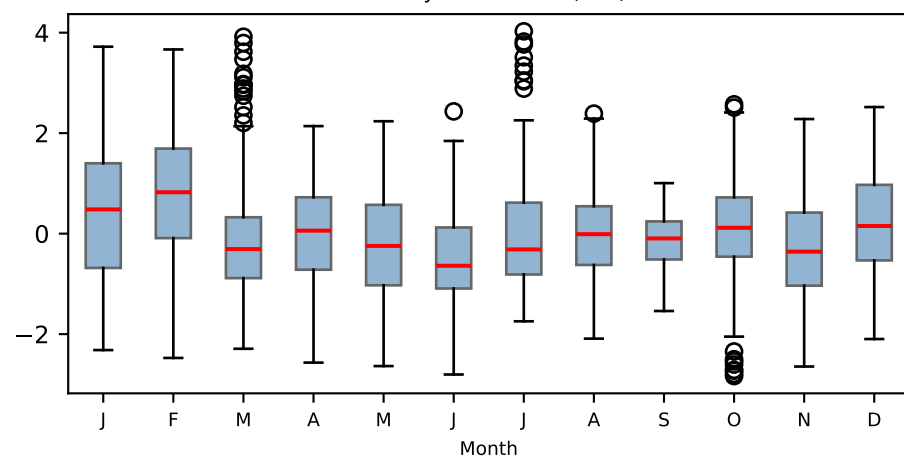
ACF (raw)



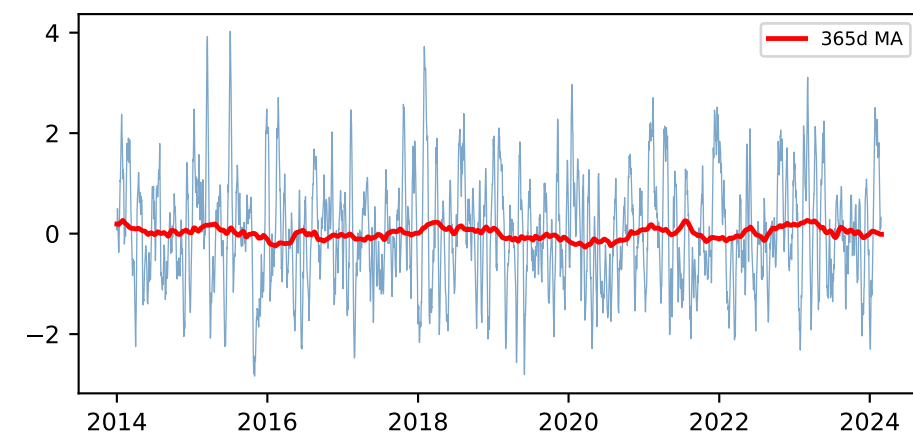
ACF (normalised)



Monthly distribution (raw)

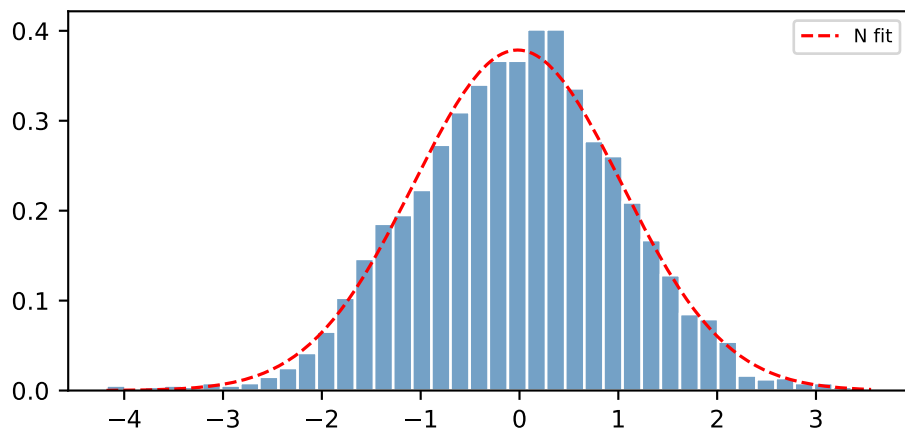


Time series + 365d mean

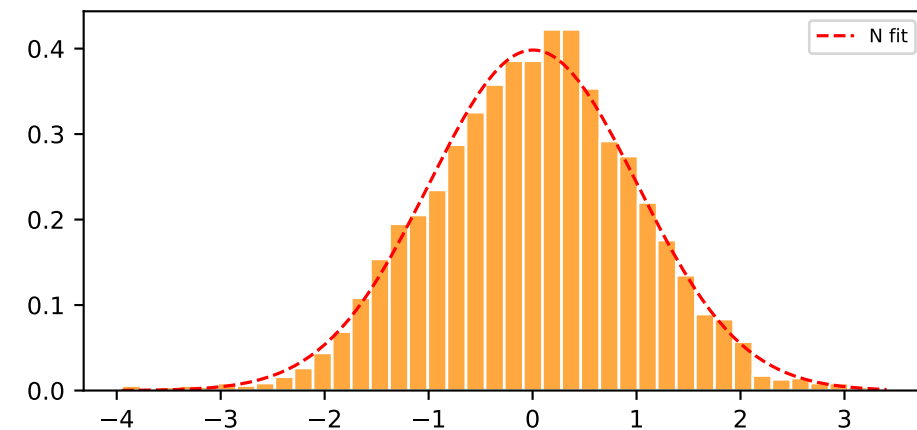


Covariate diagnostics — rmm1_lag1

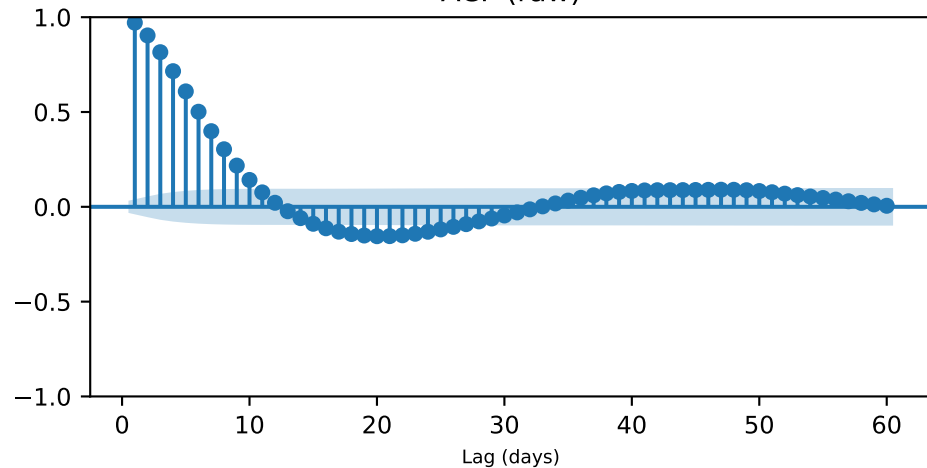
Raw histogram



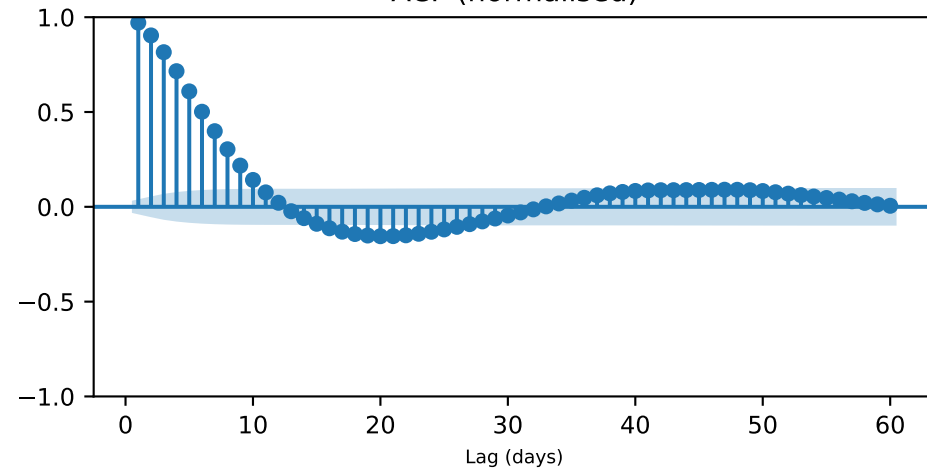
Normalised histogram



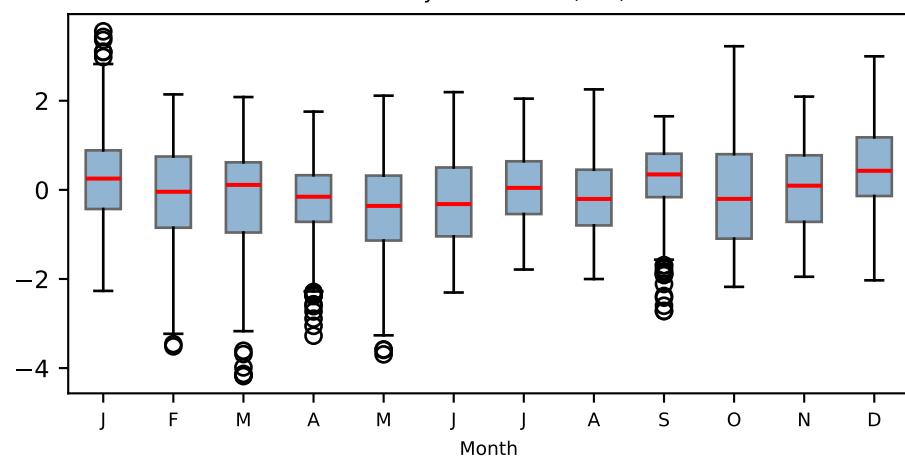
ACF (raw)



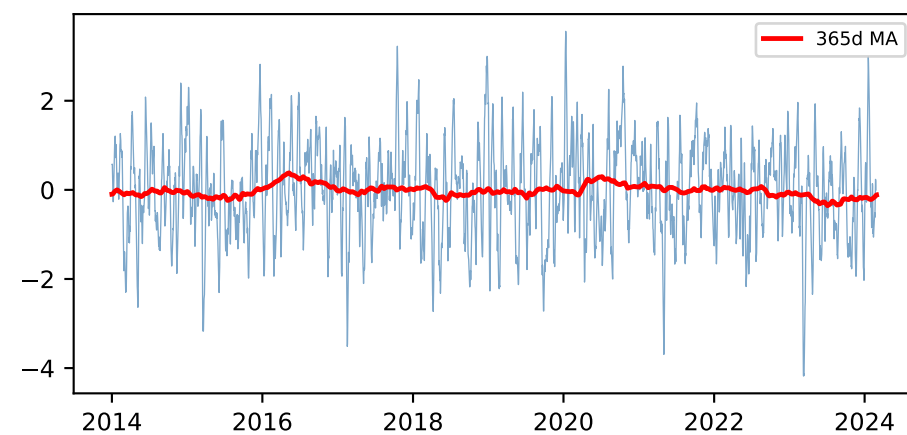
ACF (normalised)



Monthly distribution (raw)

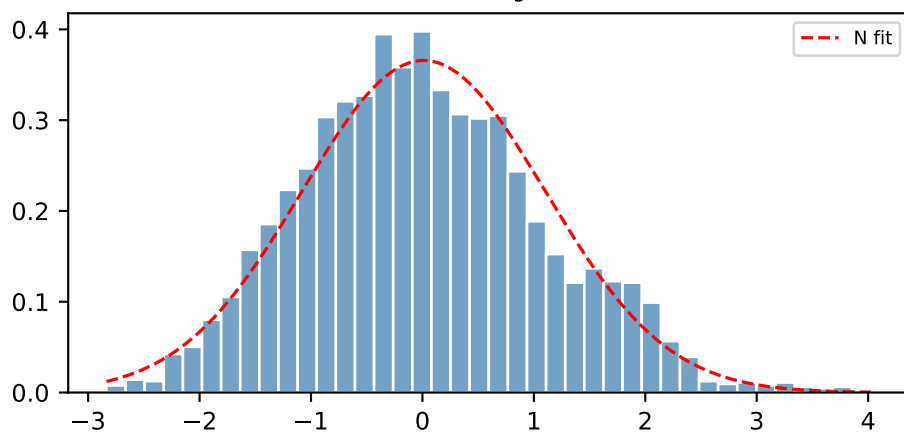


Time series + 365d mean

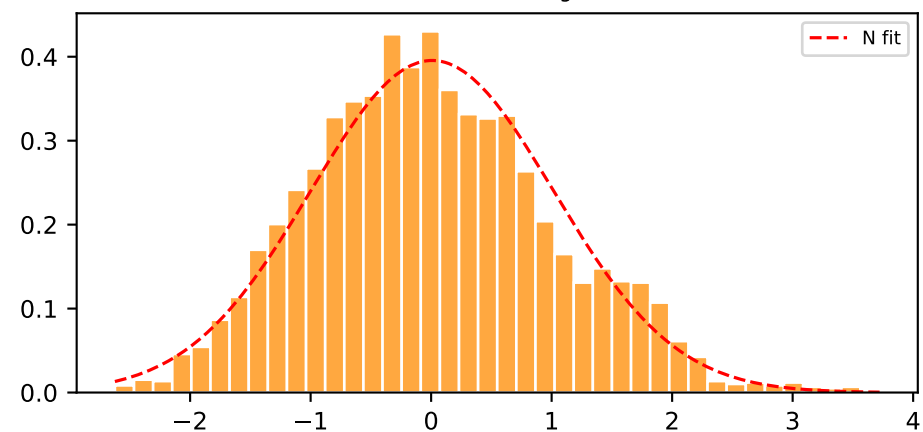


Covariate diagnostics — rmm2_lag1

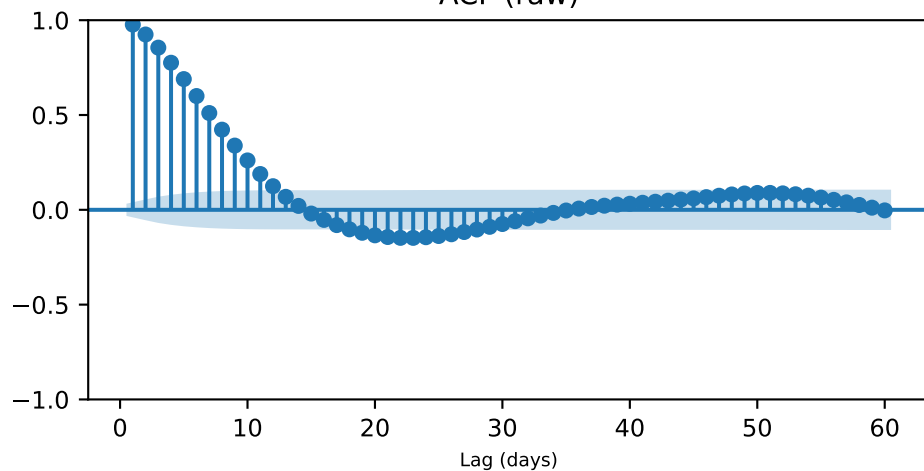
Raw histogram



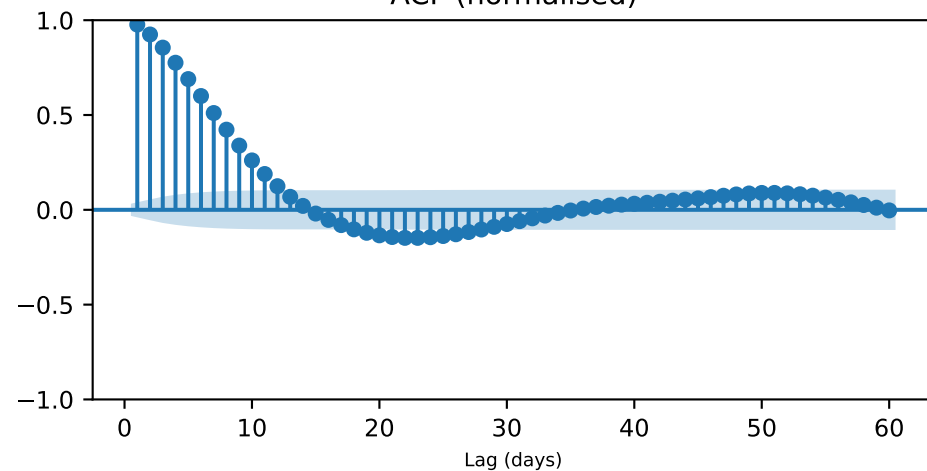
Normalised histogram



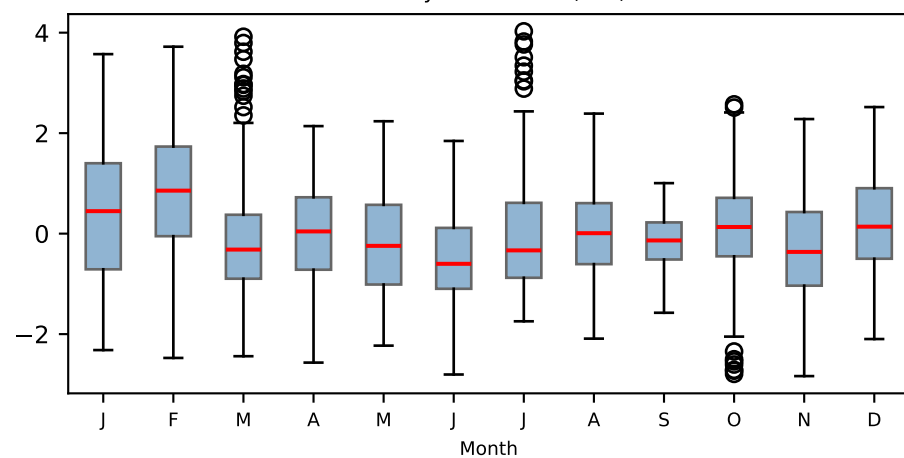
ACF (raw)



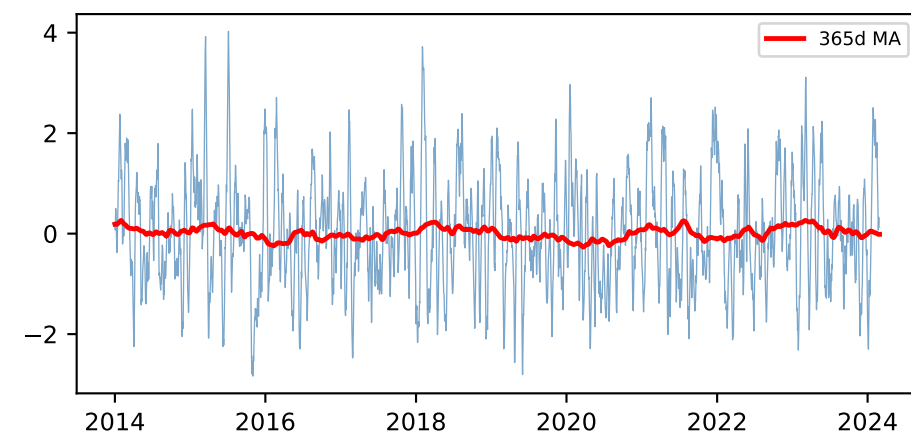
ACF (normalised)



Monthly distribution (raw)

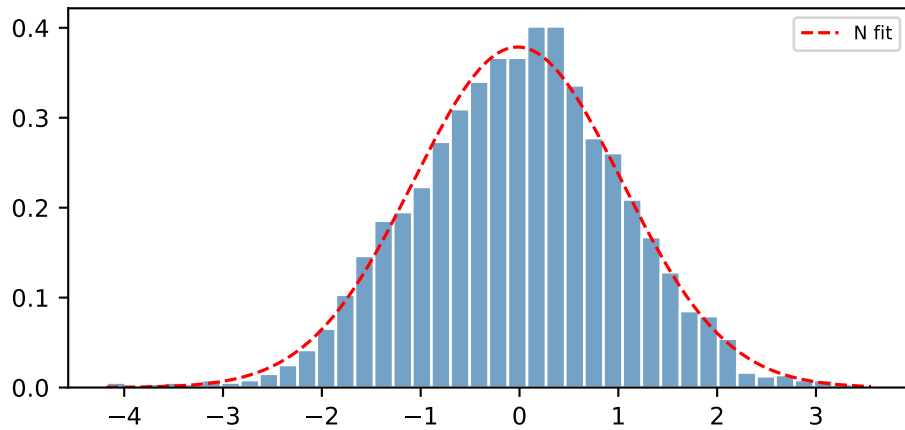


Time series + 365d mean

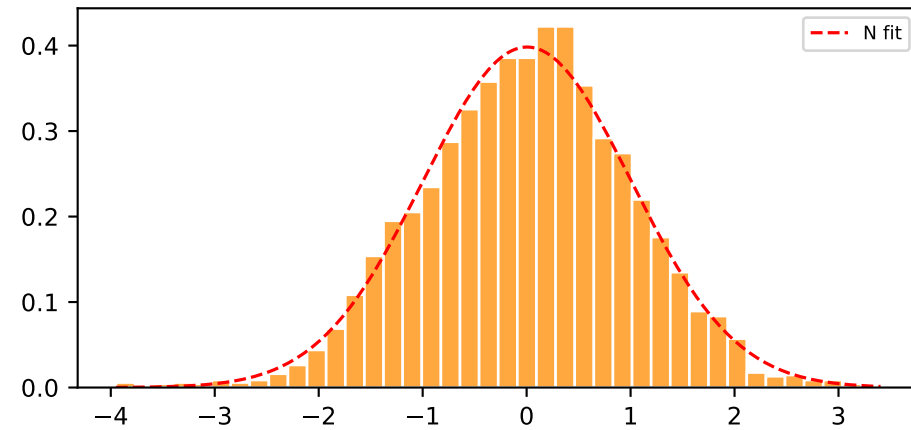


Covariate diagnostics — rmm1_lag2

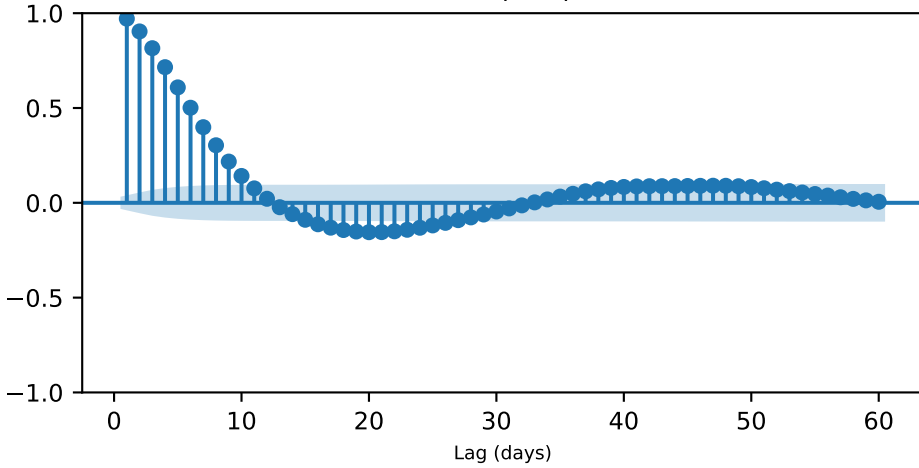
Raw histogram



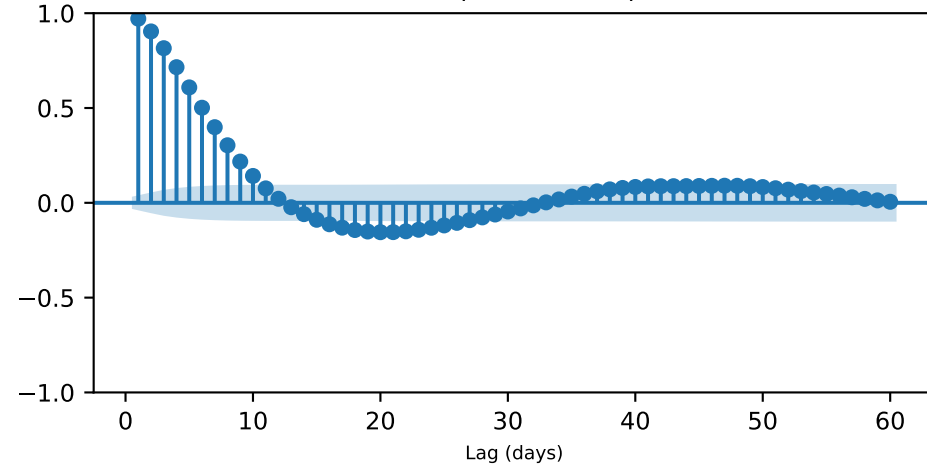
Normalised histogram



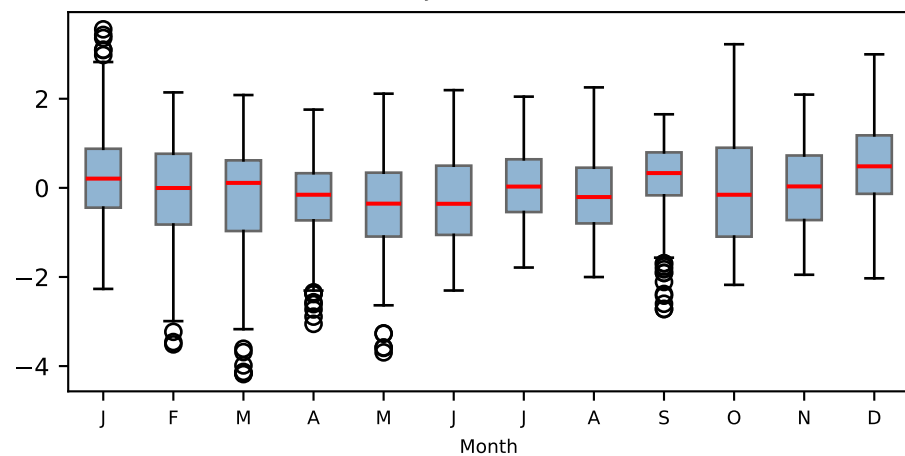
ACF (raw)



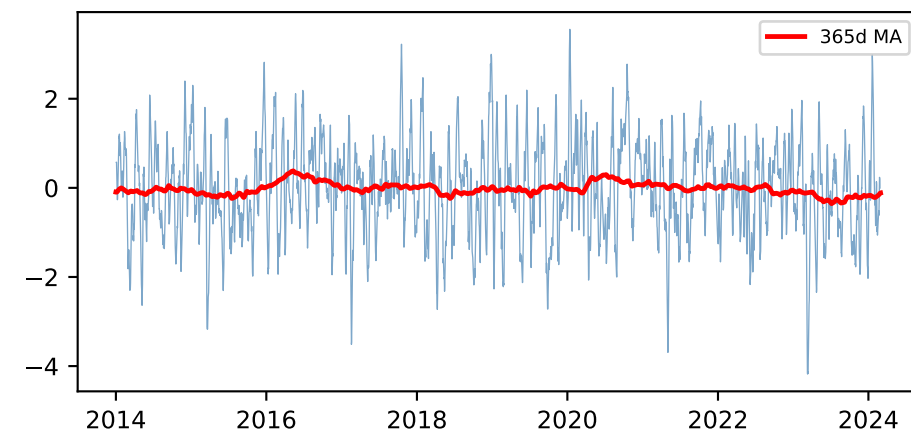
ACF (normalised)



Monthly distribution (raw)

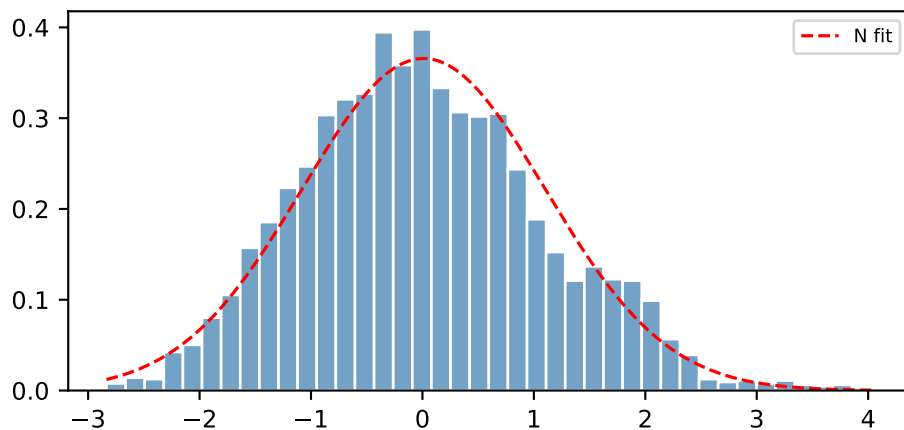


Time series + 365d mean

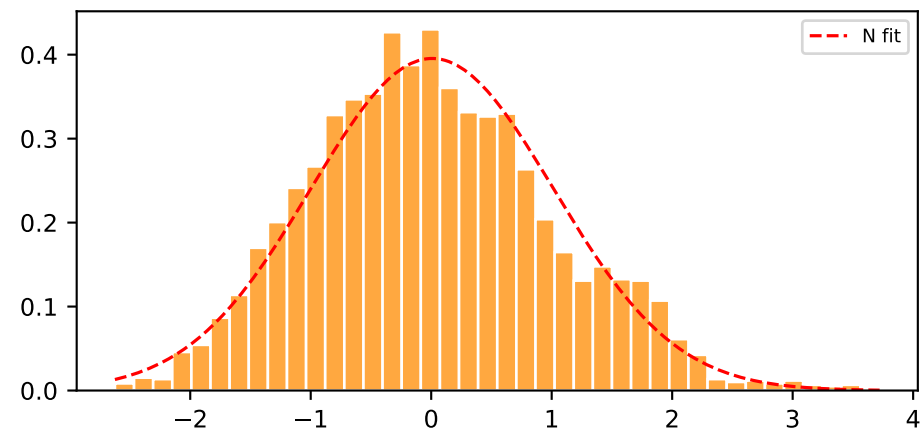


Covariate diagnostics — rmm2_lag2

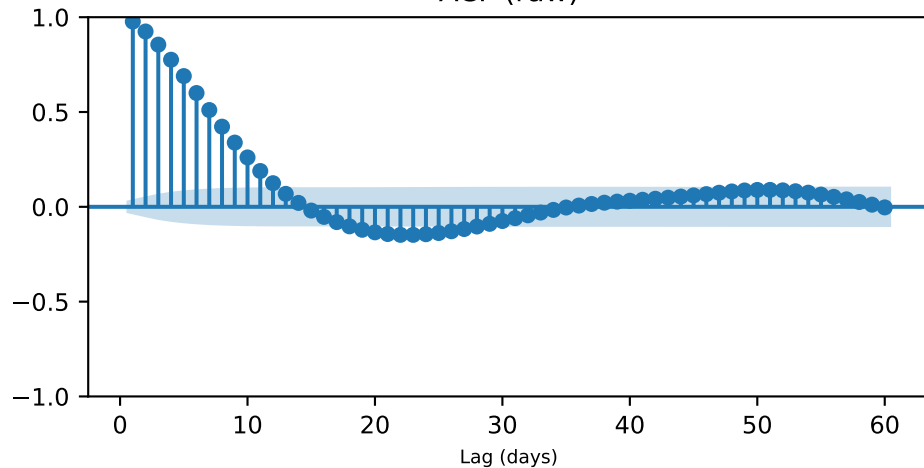
Raw histogram



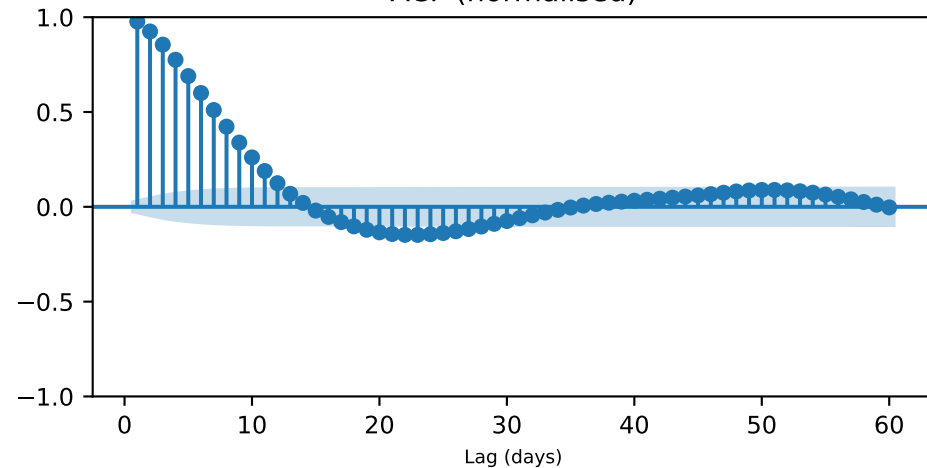
Normalised histogram



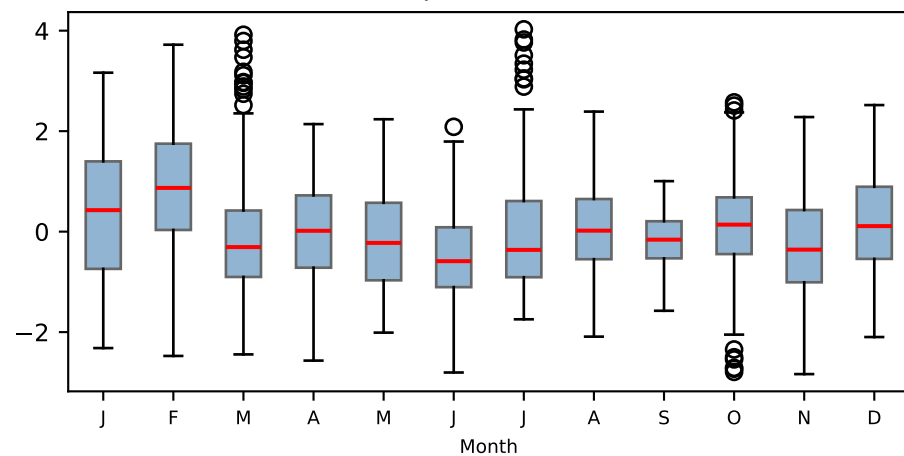
ACF (raw)



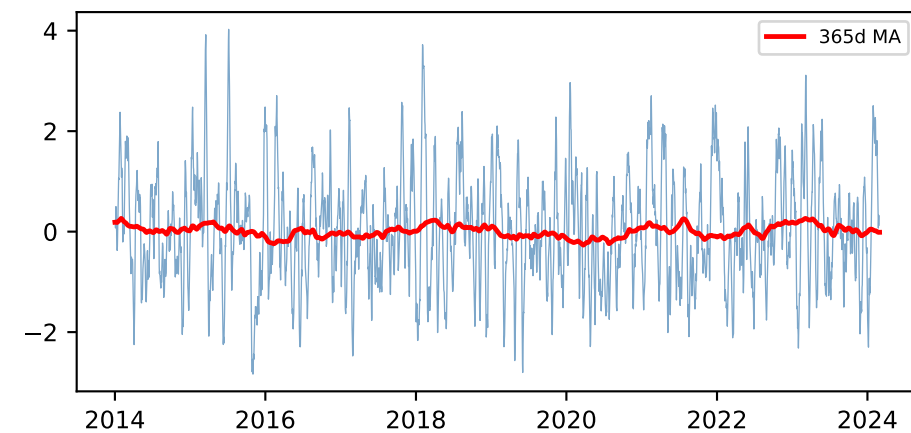
ACF (normalised)



Monthly distribution (raw)

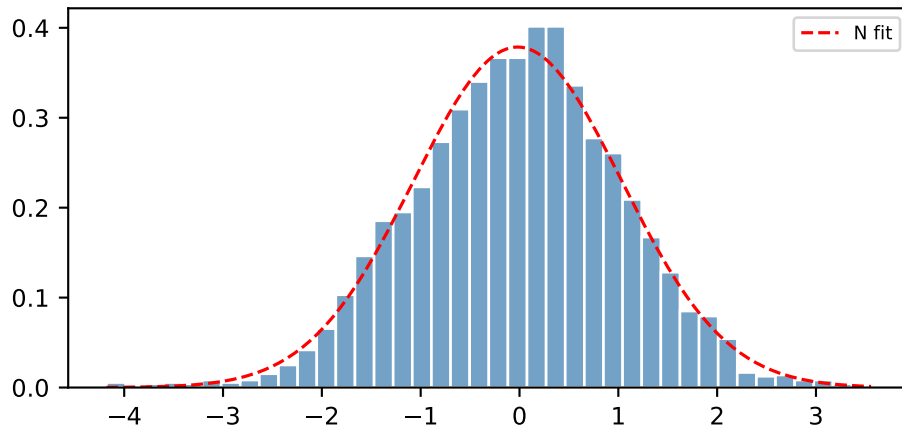


Time series + 365d mean

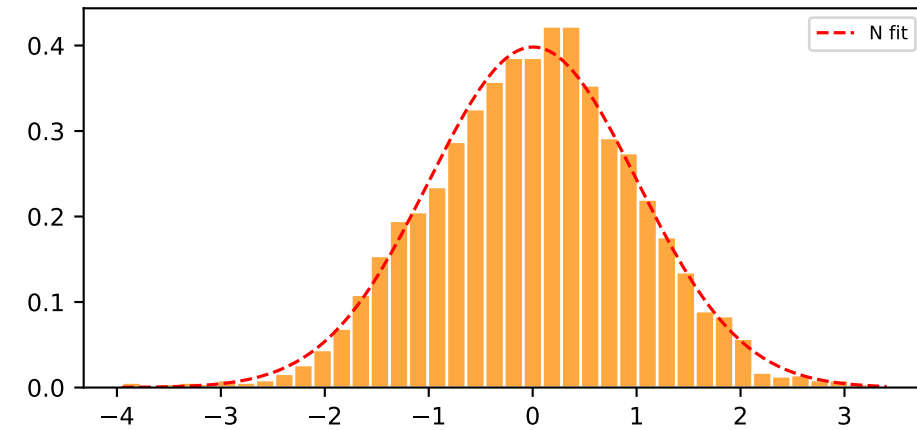


Covariate diagnostics — rmm1_lag3

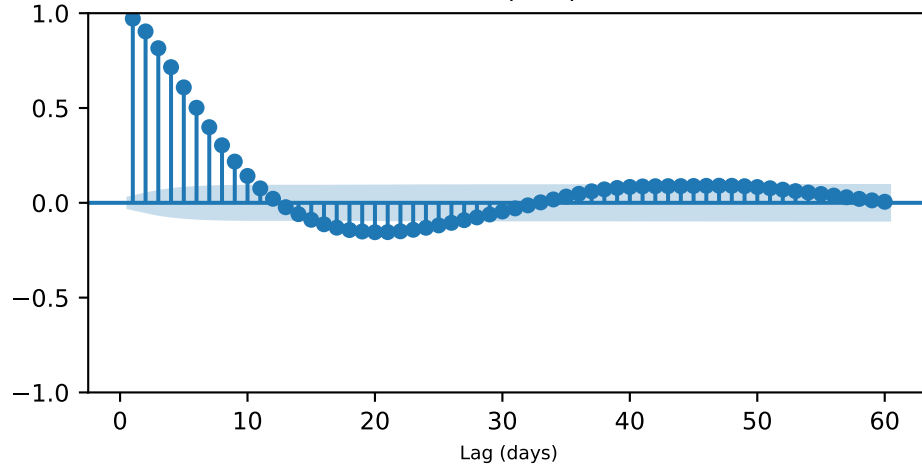
Raw histogram



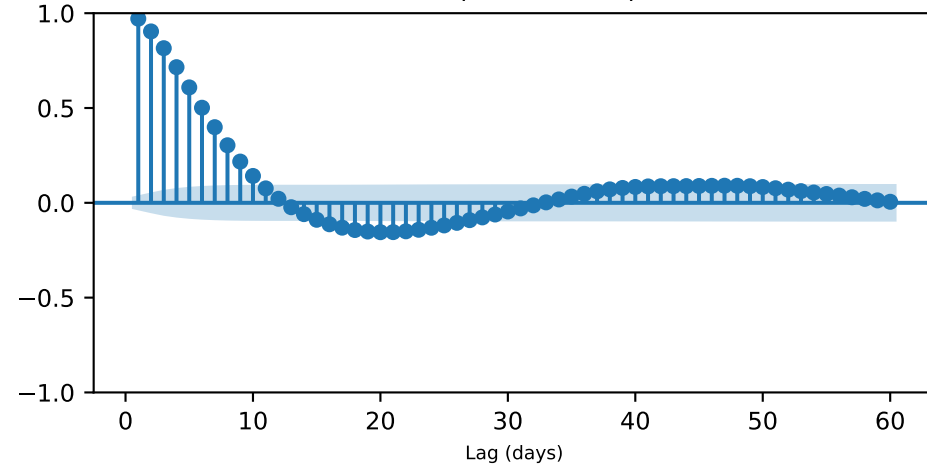
Normalised histogram



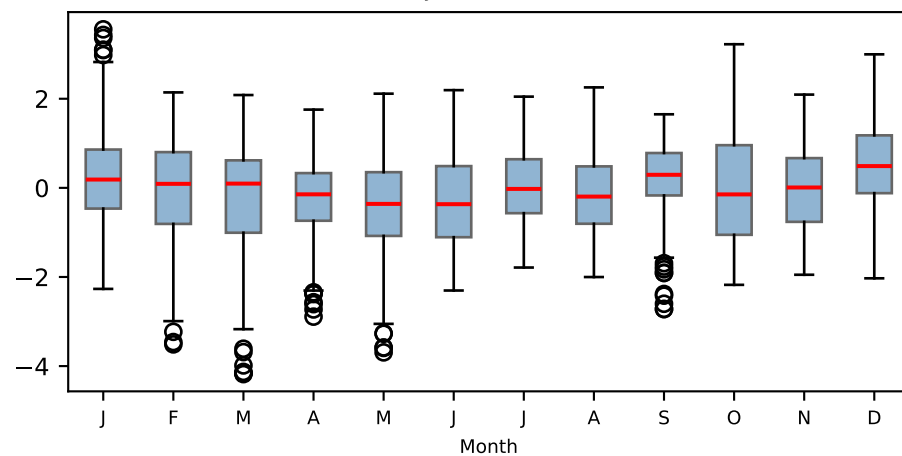
ACF (raw)



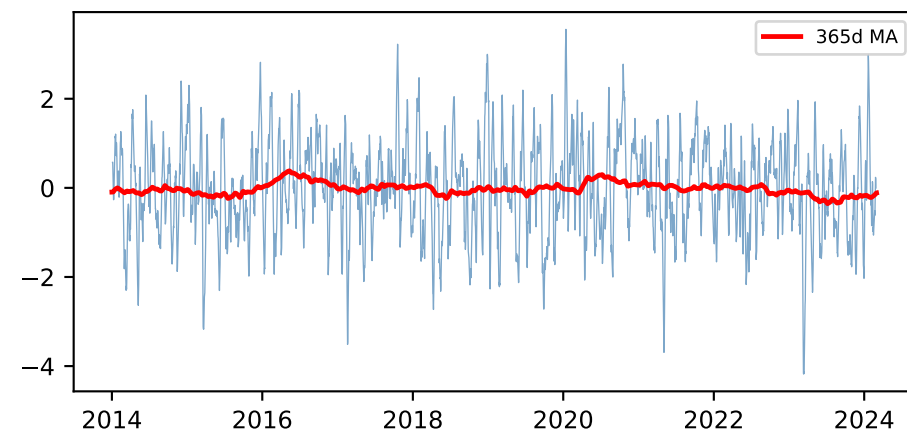
ACF (normalised)



Monthly distribution (raw)

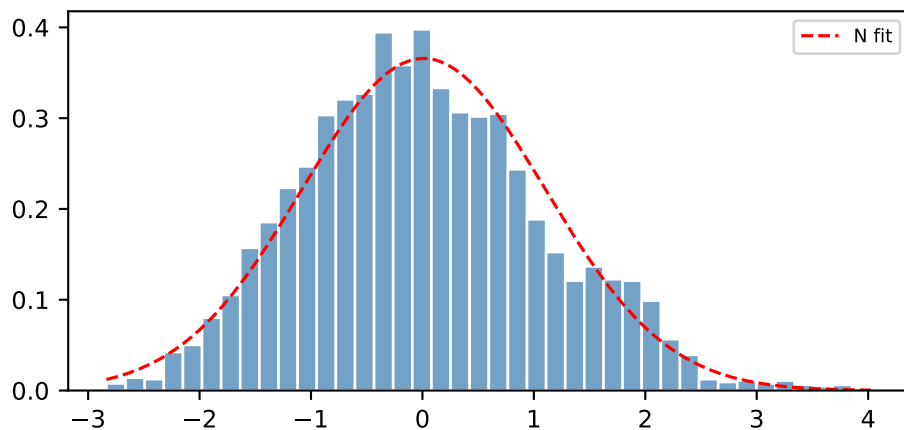


Time series + 365d mean

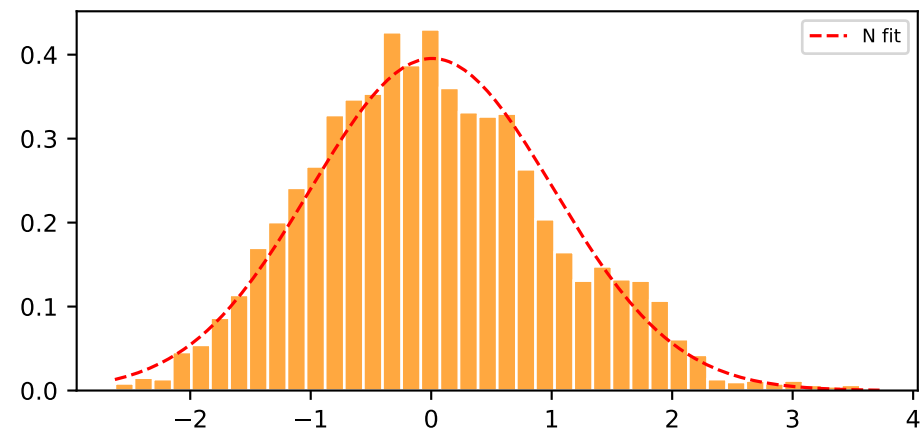


Covariate diagnostics — rmm2_lag3

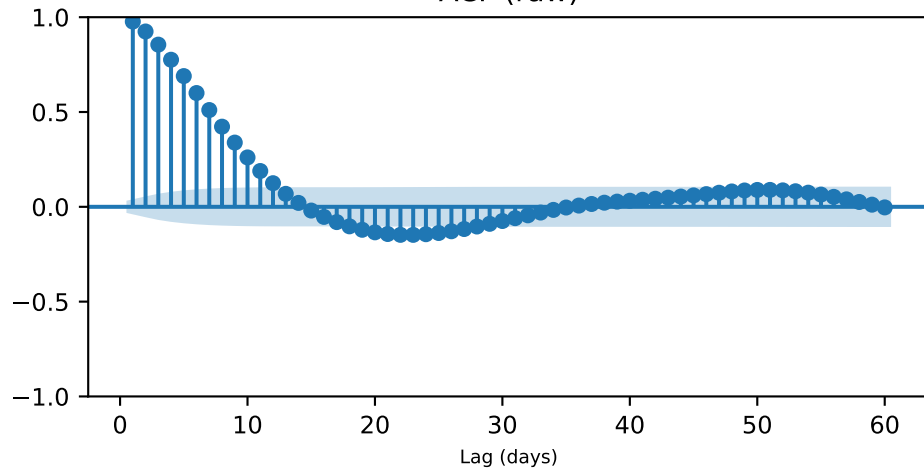
Raw histogram



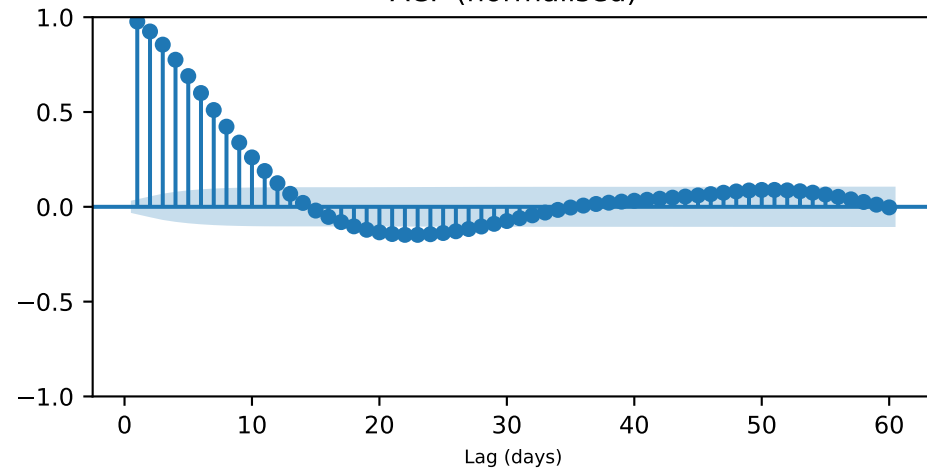
Normalised histogram



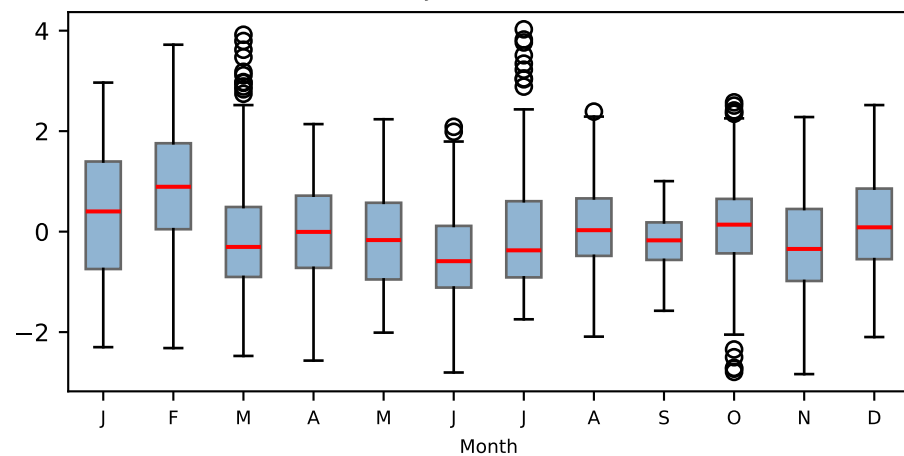
ACF (raw)



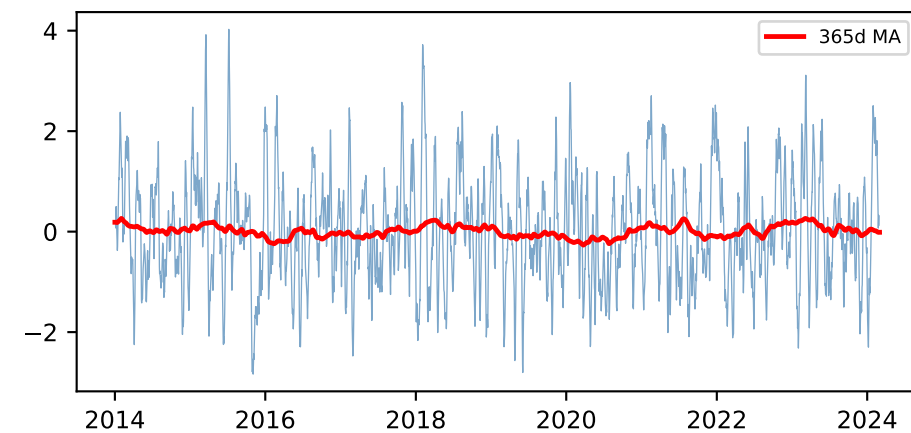
ACF (normalised)



Monthly distribution (raw)

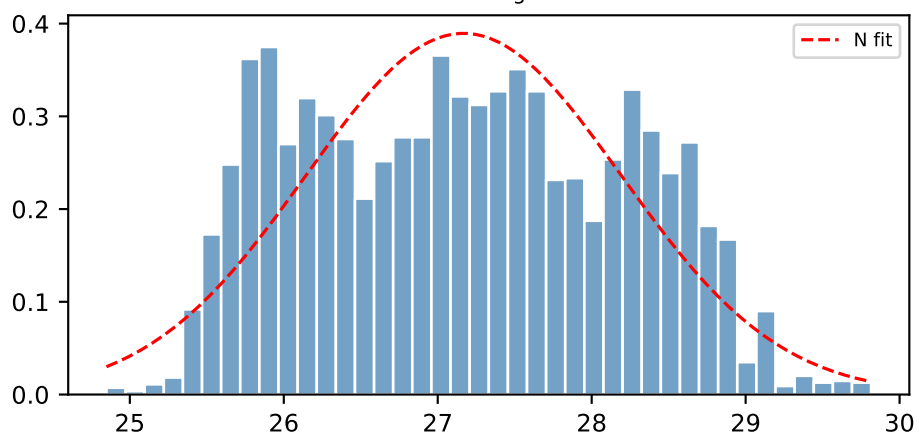


Time series + 365d mean

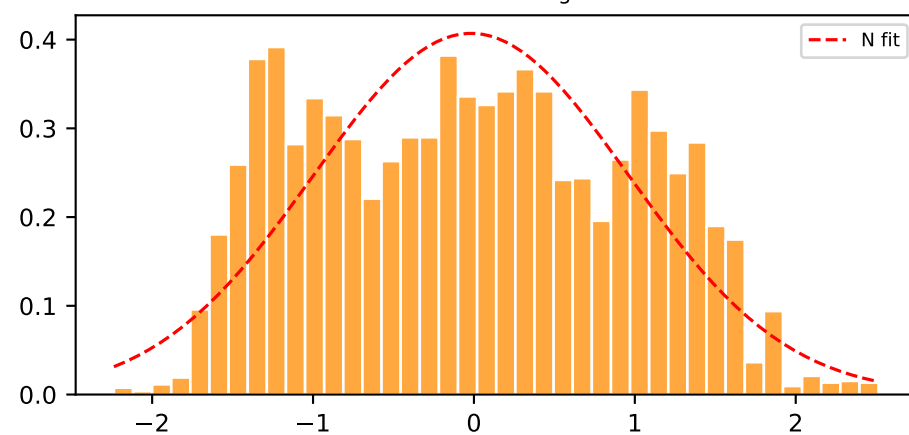


Covariate diagnostics — nino34_t

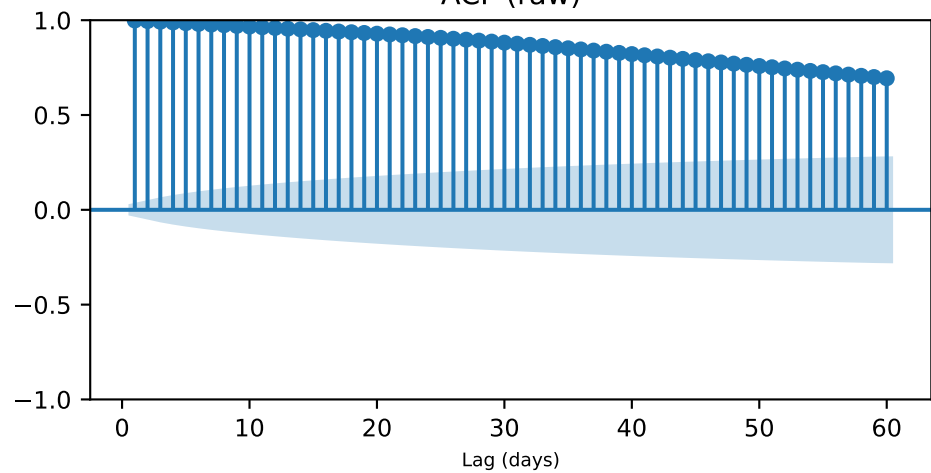
Raw histogram



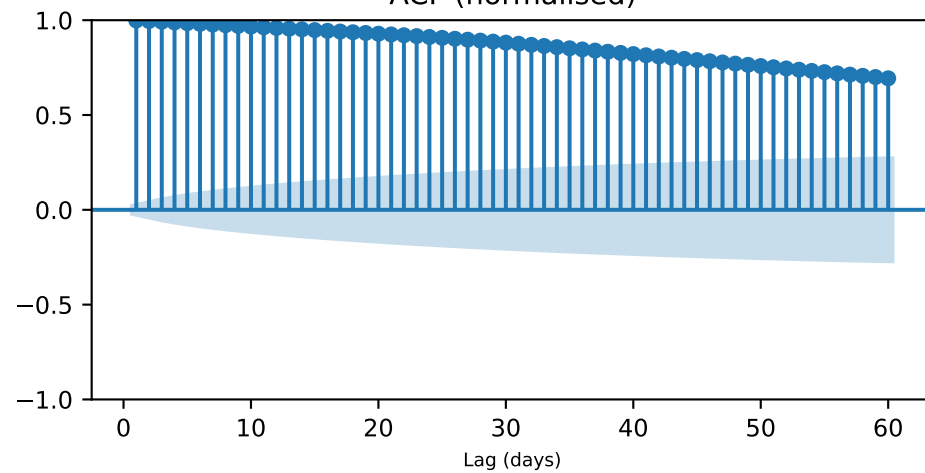
Normalised histogram



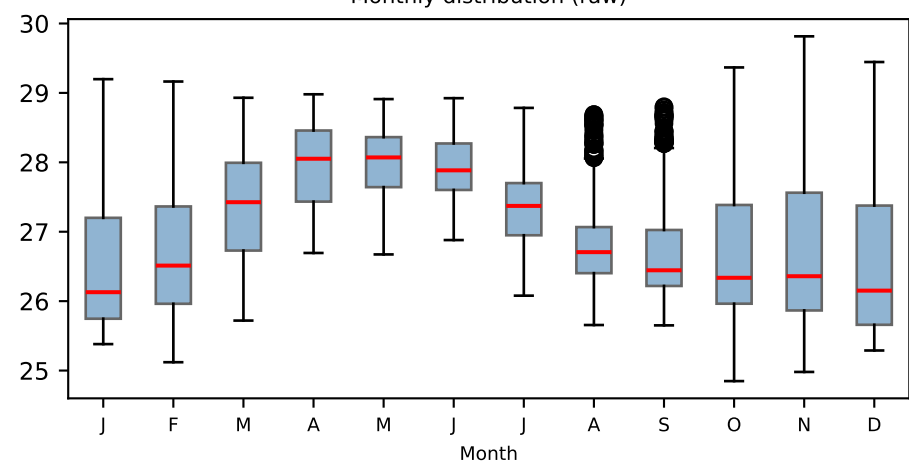
ACF (raw)



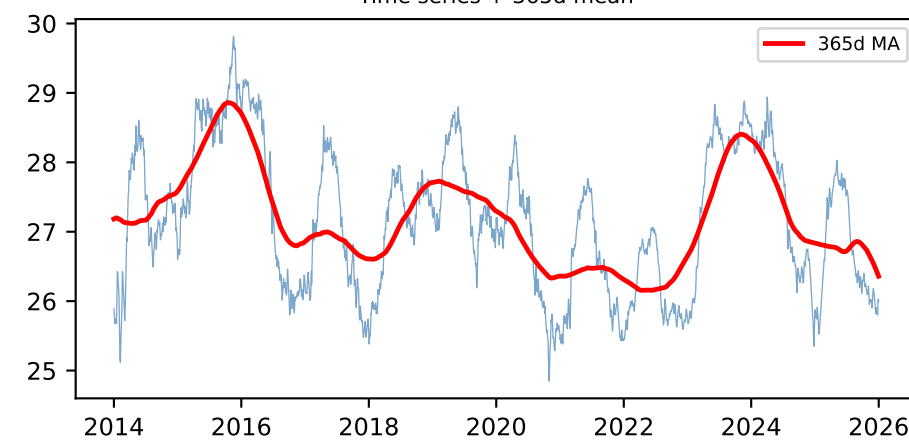
ACF (normalised)



Monthly distribution (raw)

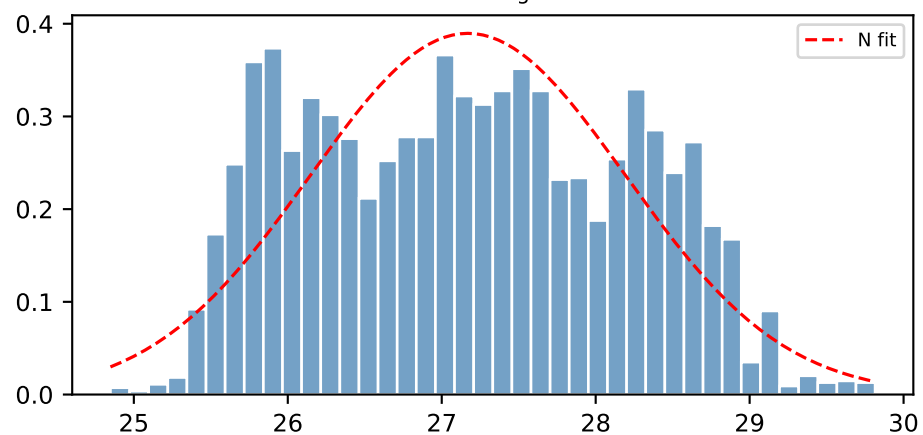


Time series + 365d mean

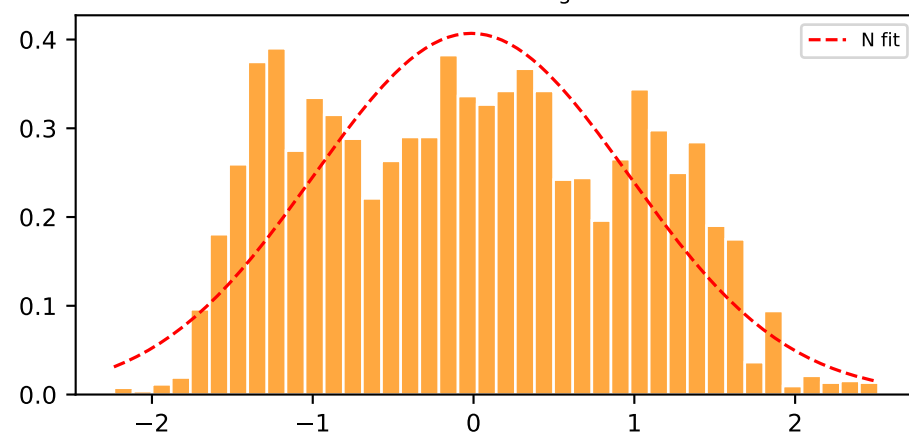


Covariate diagnostics — nino34_lag7

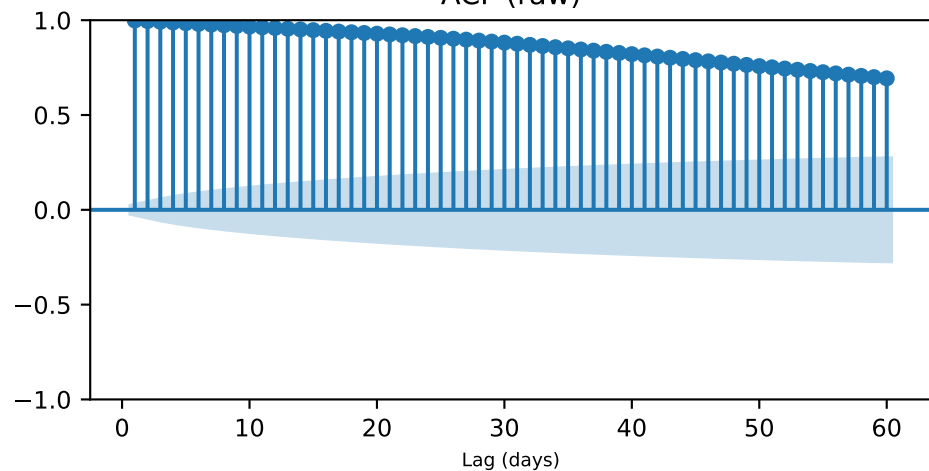
Raw histogram



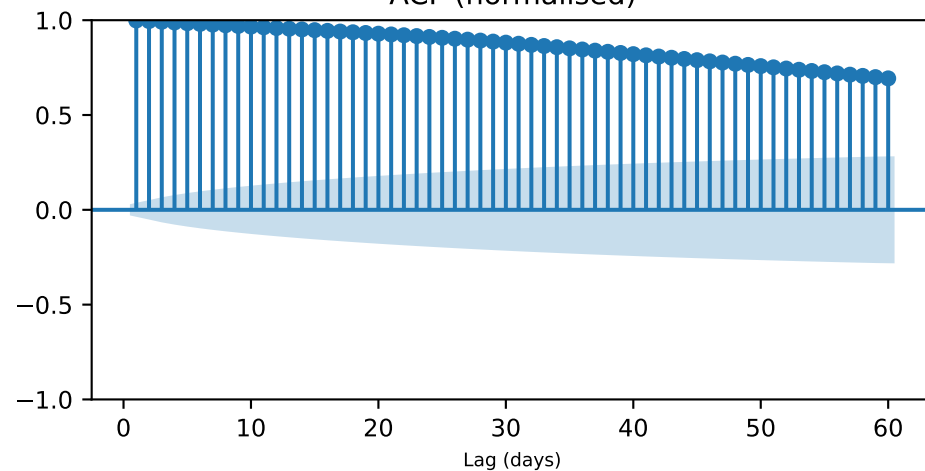
Normalised histogram



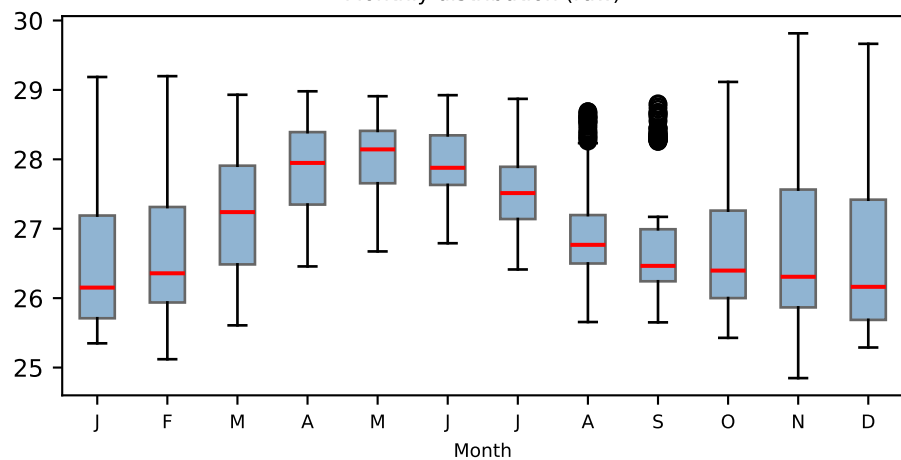
ACF (raw)



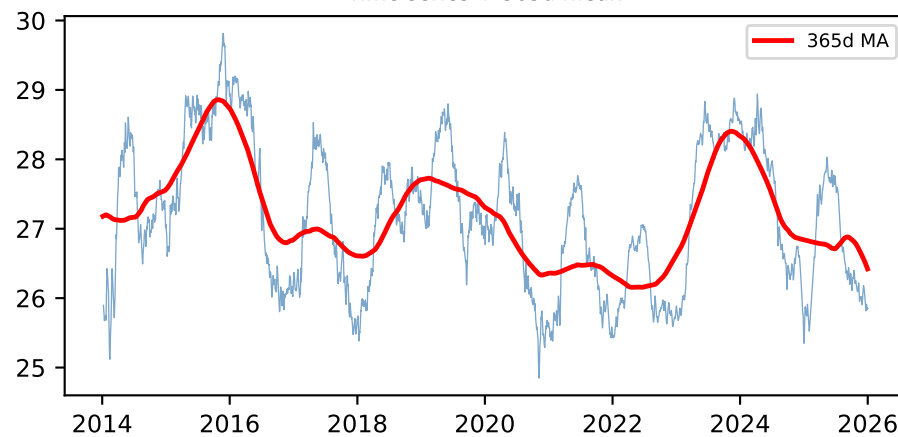
ACF (normalised)



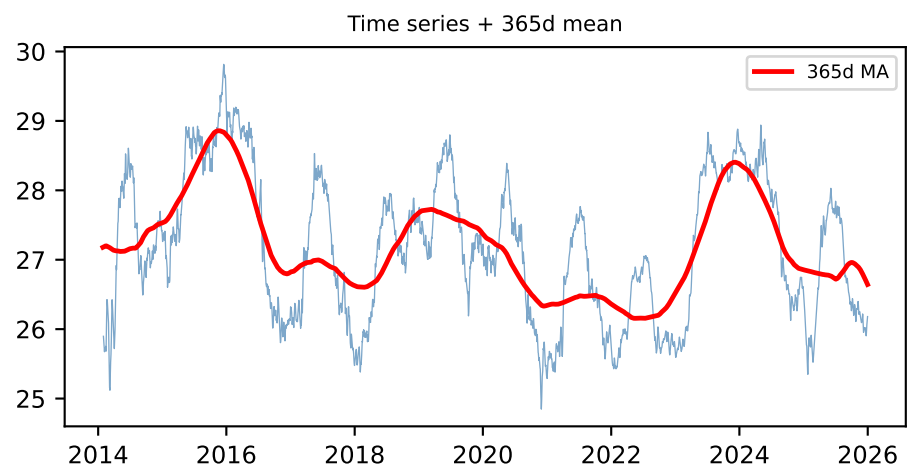
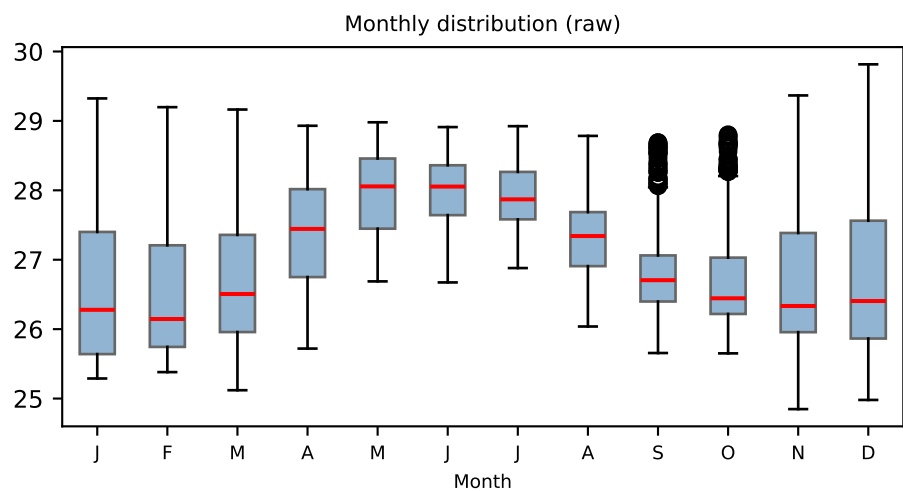
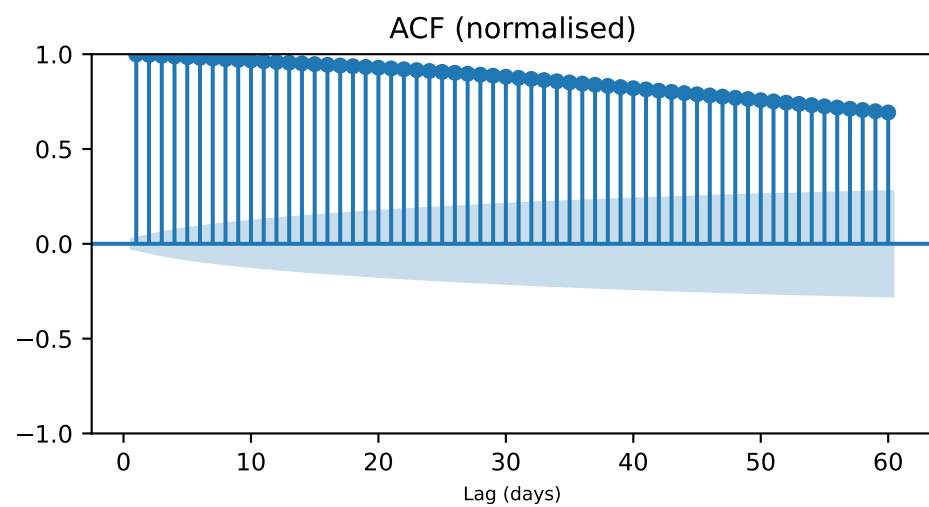
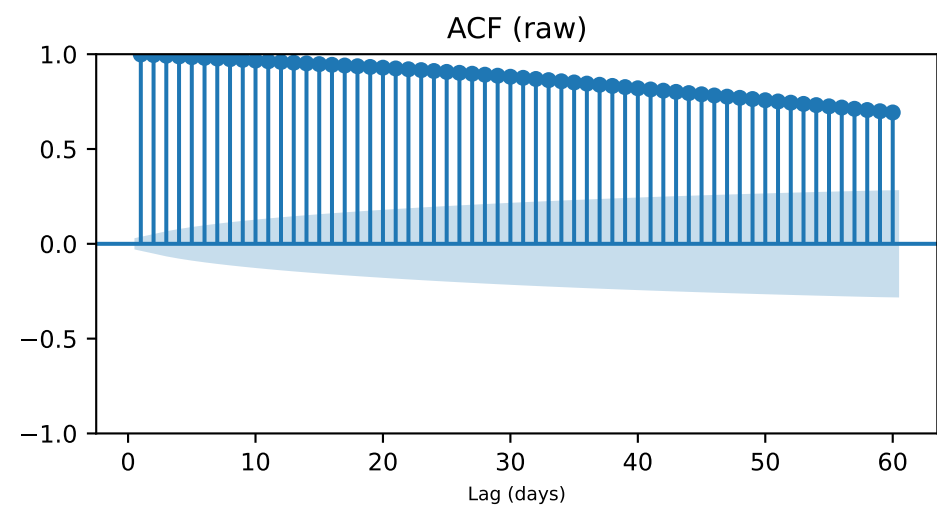
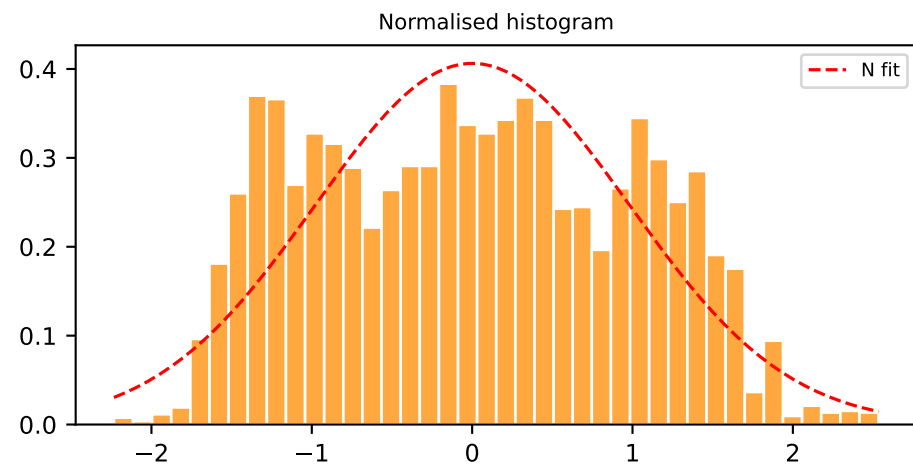
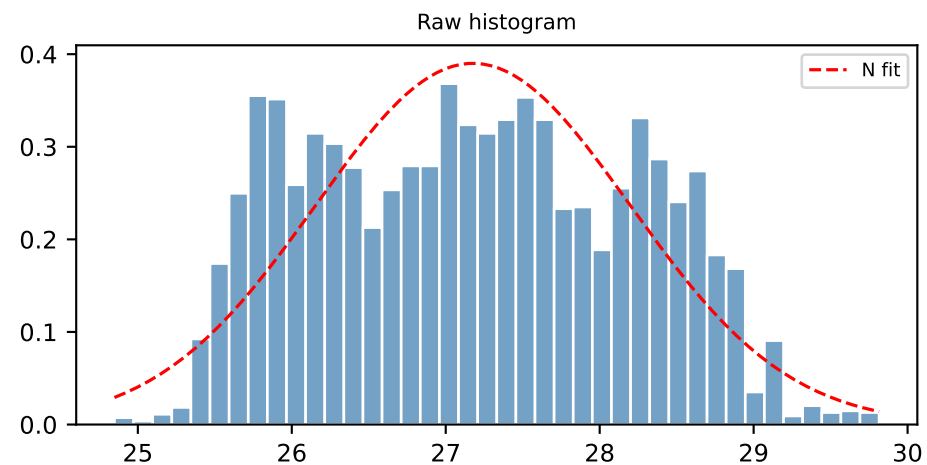
Monthly distribution (raw)



Time series + 365d mean

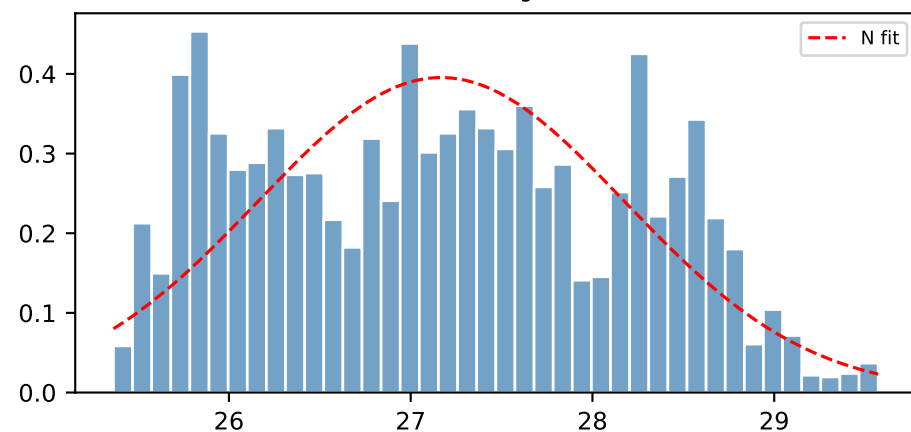


Covariate diagnostics — nino34_lag30

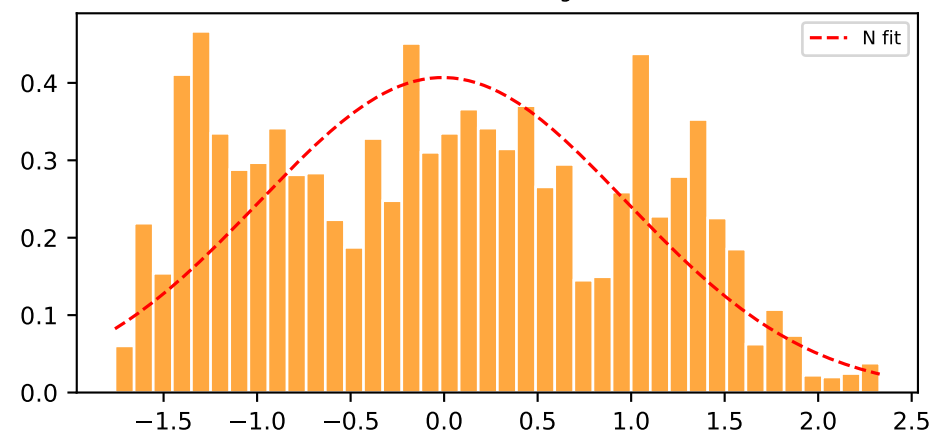


Covariate diagnostics — nino34_roll30

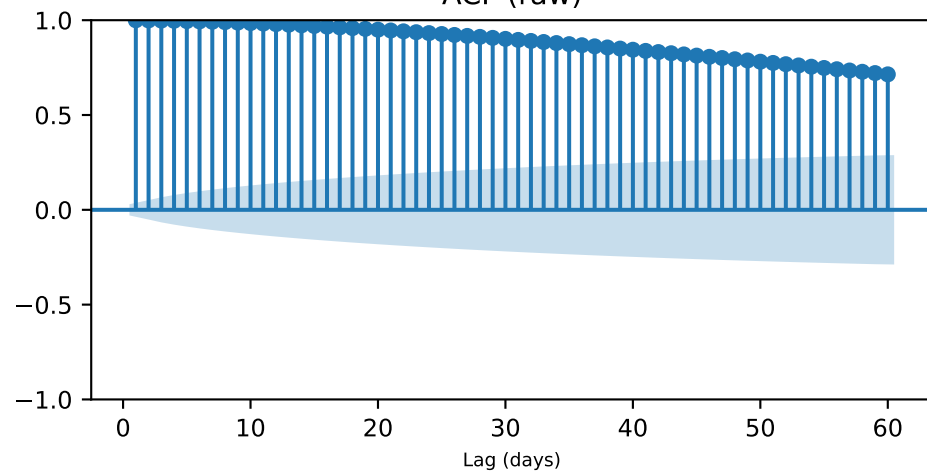
Raw histogram



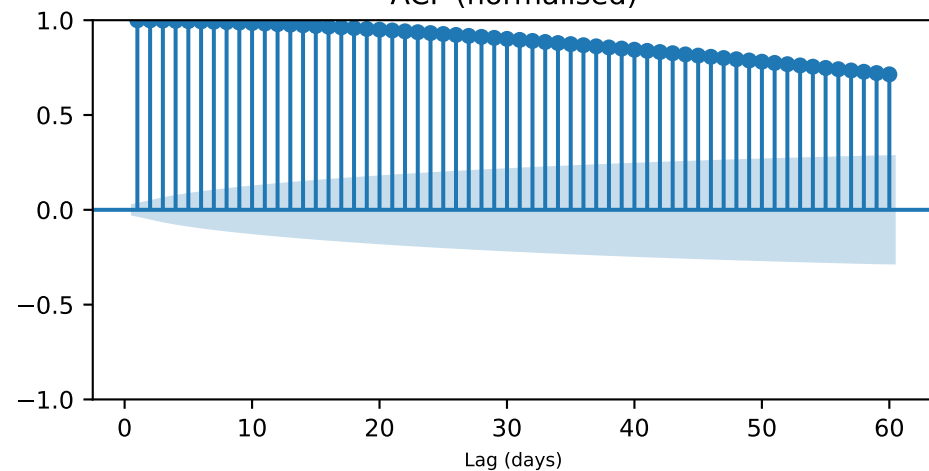
Normalised histogram



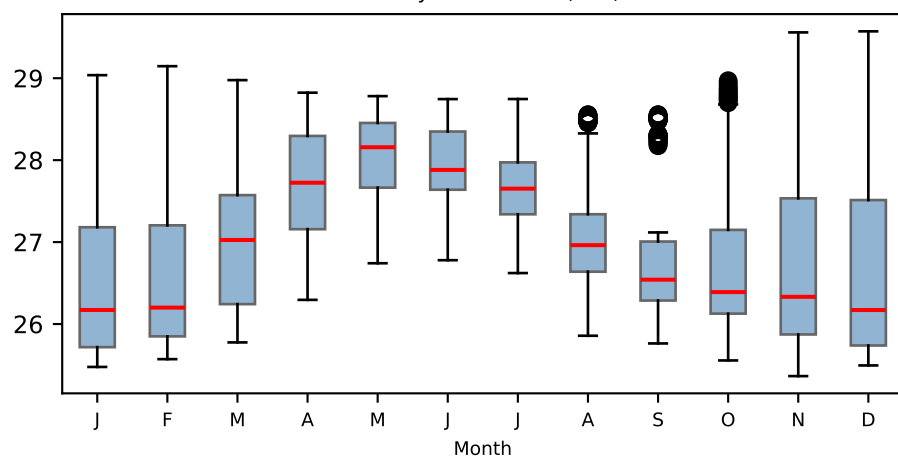
ACF (raw)



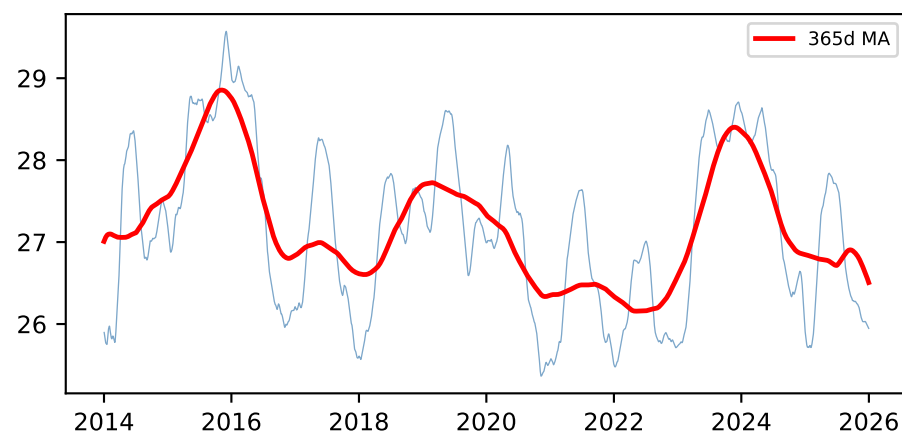
ACF (normalised)



Monthly distribution (raw)

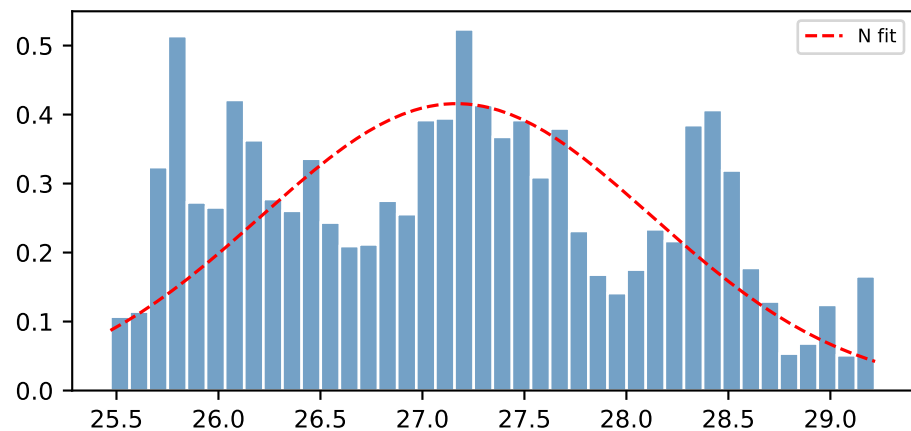


Time series + 365d mean

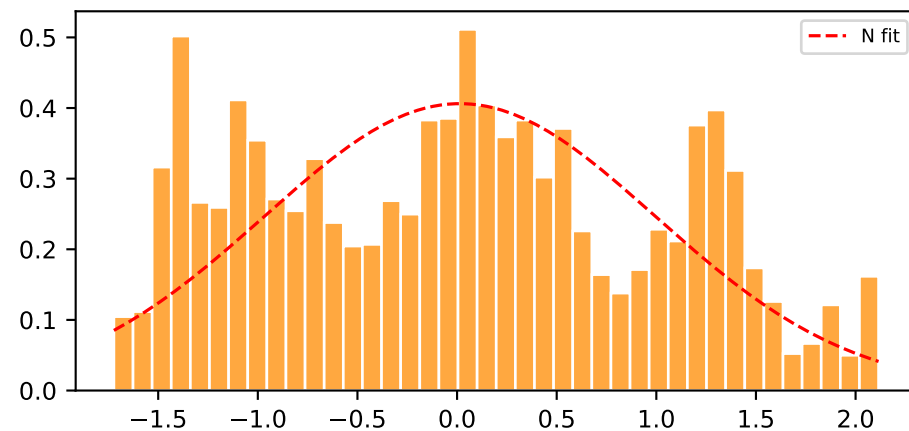


Covariate diagnostics — nino34_roll90

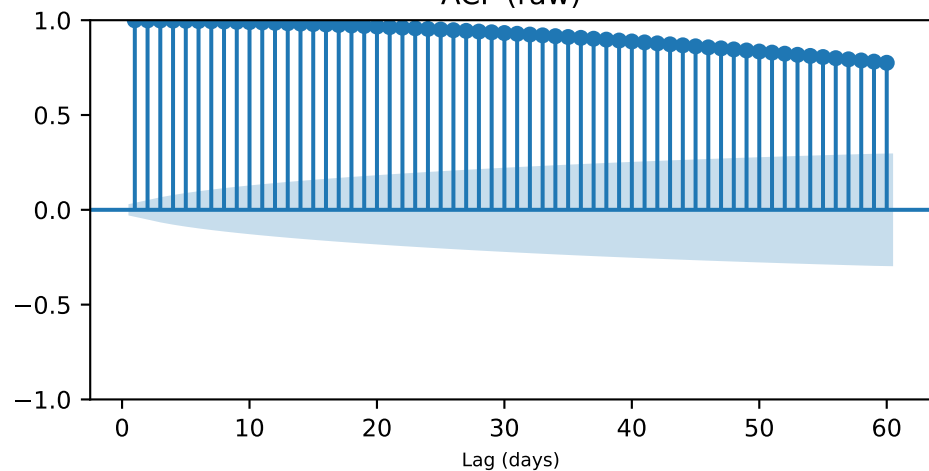
Raw histogram



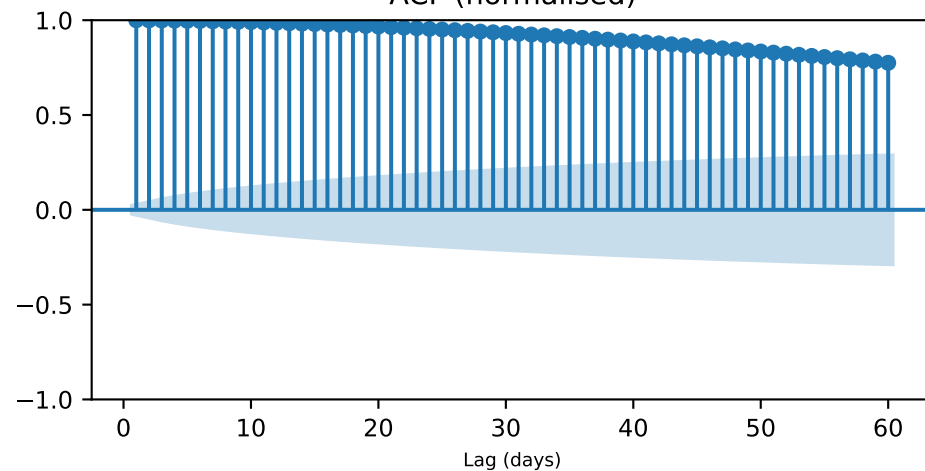
Normalised histogram



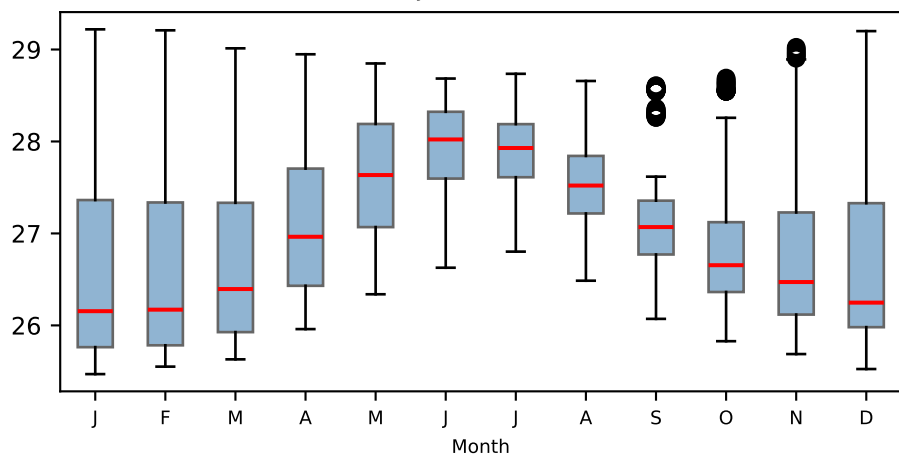
ACF (raw)



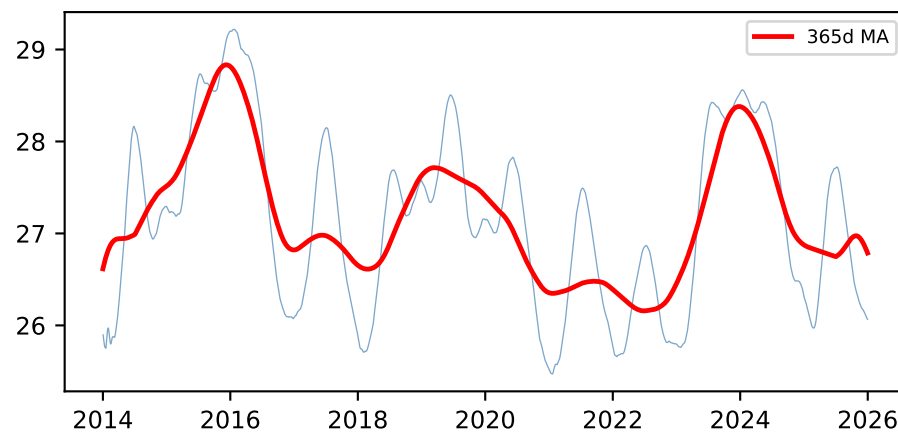
ACF (normalised)



Monthly distribution (raw)

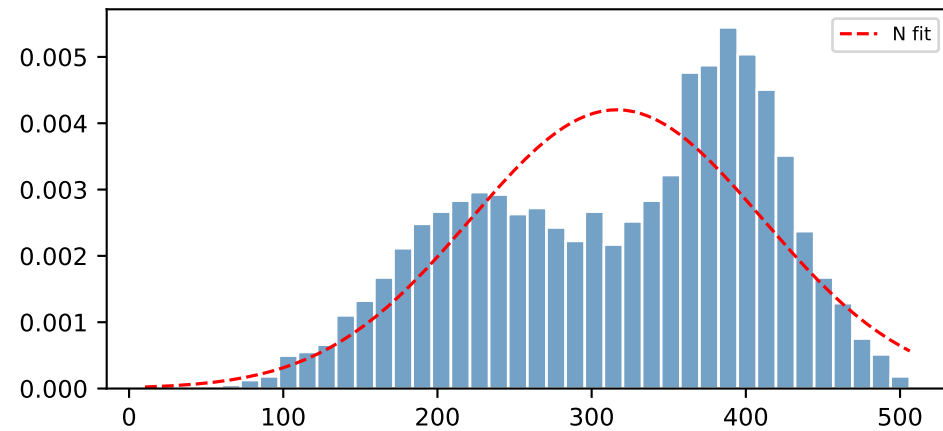


Time series + 365d mean

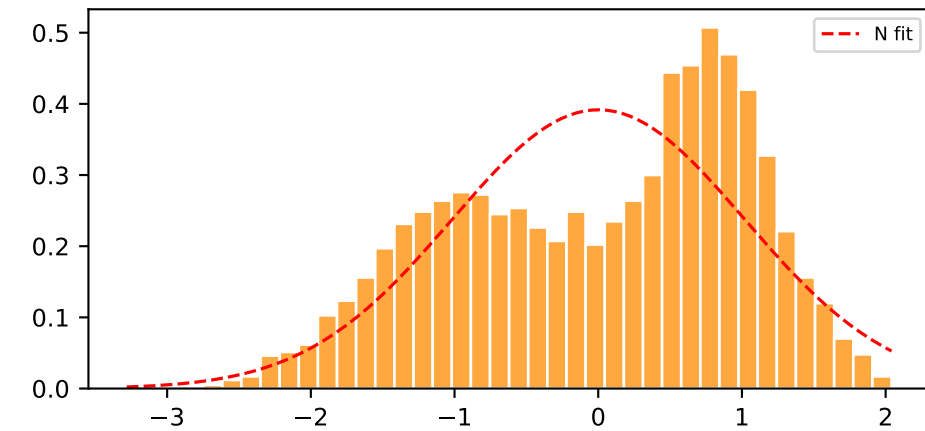


Covariate diagnostics — dew_x_temp

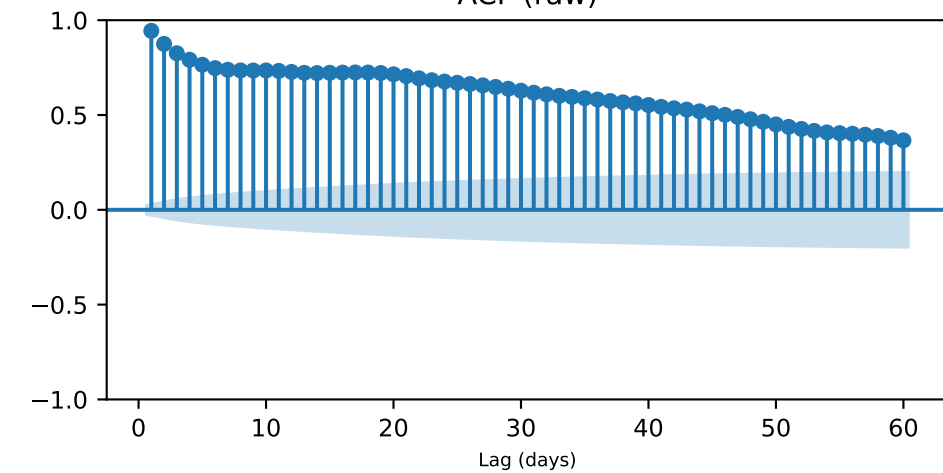
Raw histogram



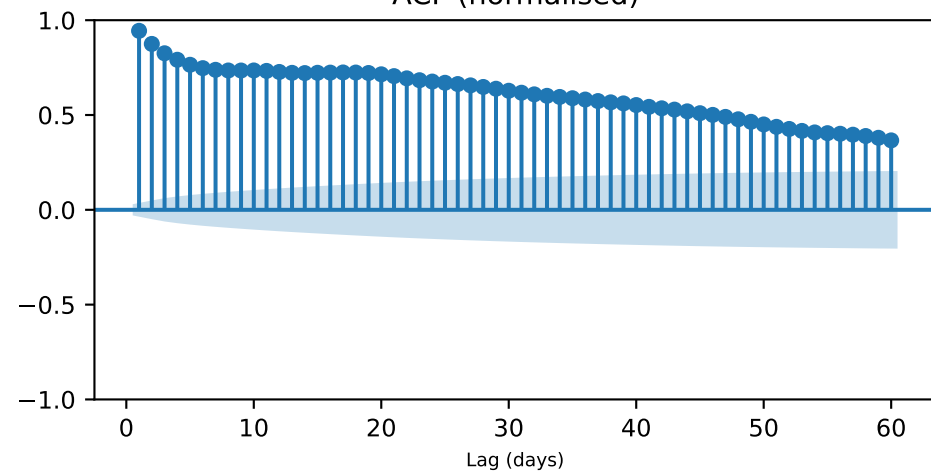
Normalised histogram



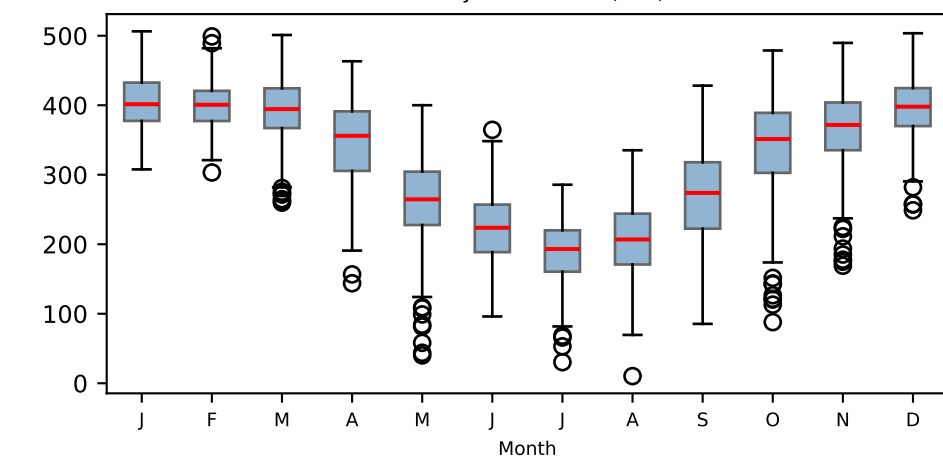
ACF (raw)



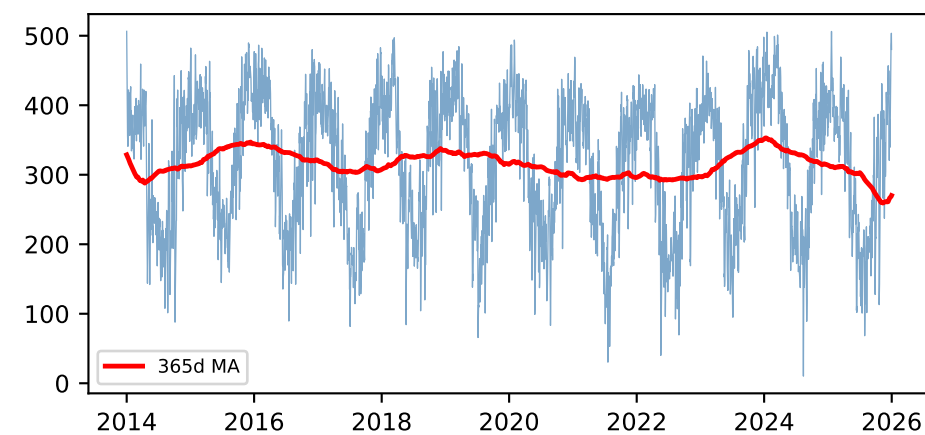
ACF (normalised)



Monthly distribution (raw)

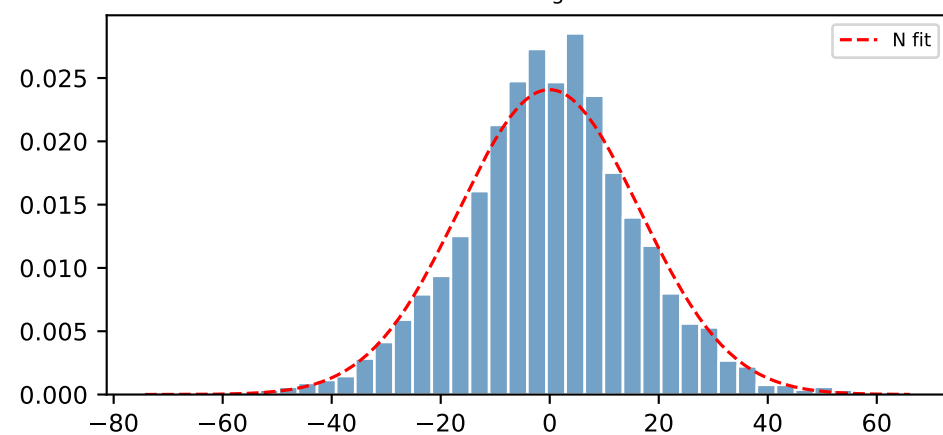


Time series + 365d mean

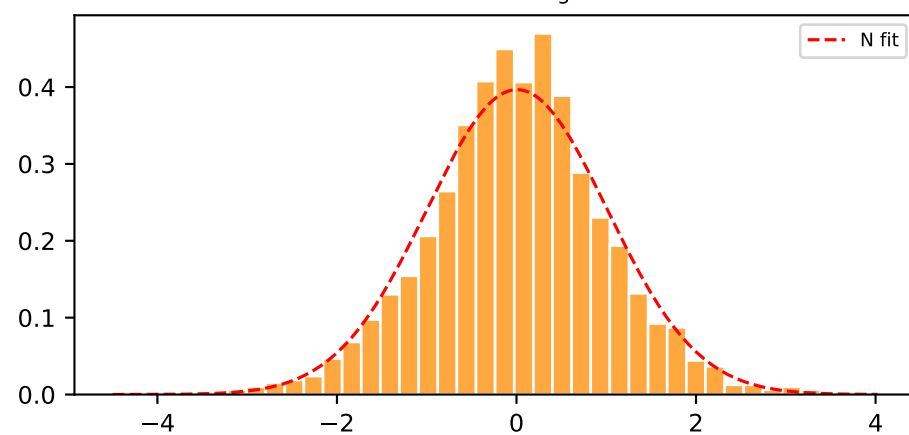


Covariate diagnostics — dew_x_rmm1

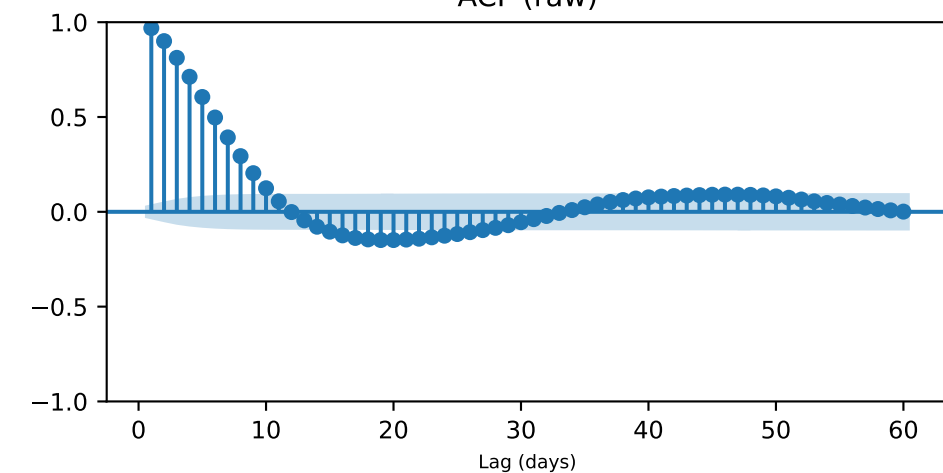
Raw histogram



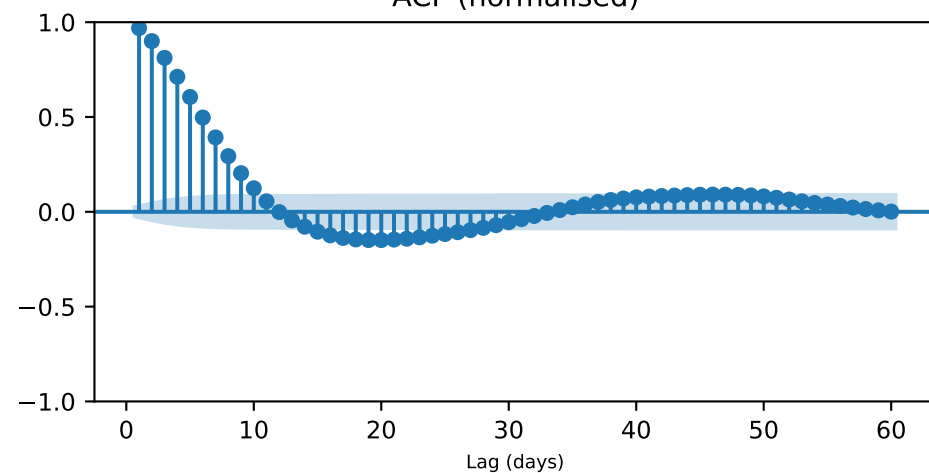
Normalised histogram



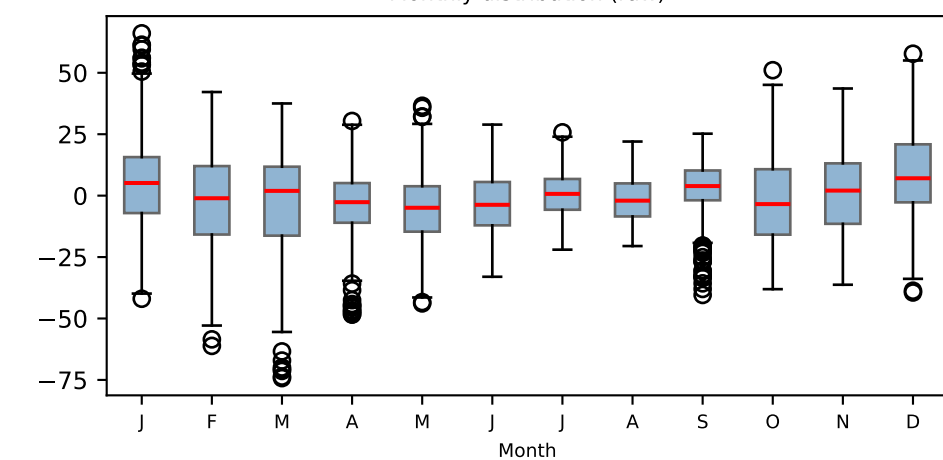
ACF (raw)



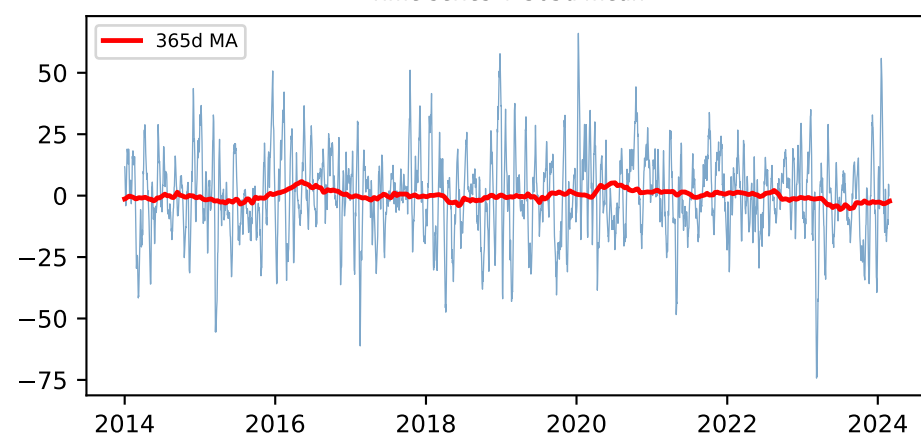
ACF (normalised)



Monthly distribution (raw)

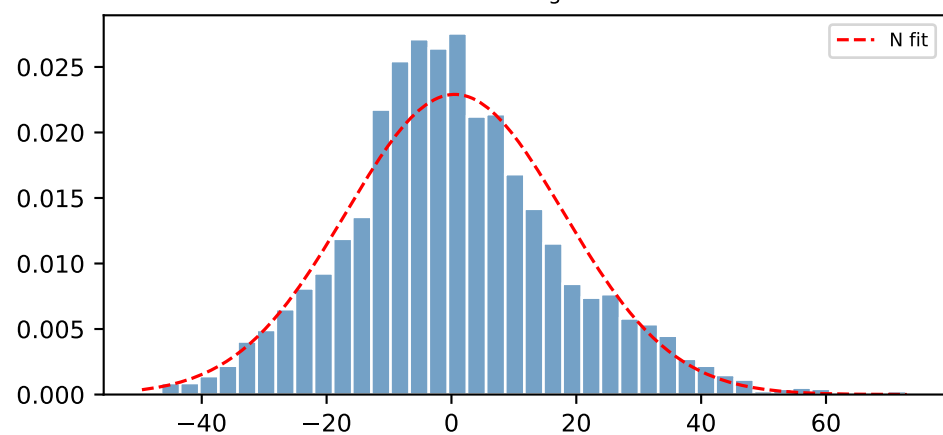


Time series + 365d mean

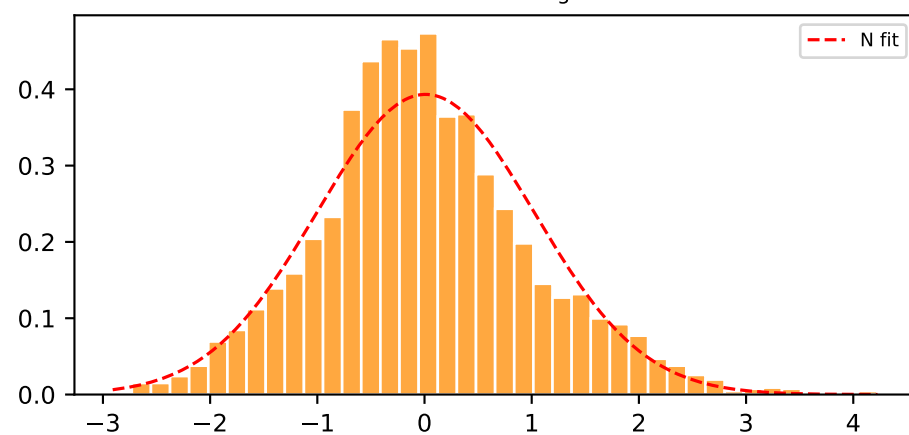


Covariate diagnostics — dew_x_rmm2

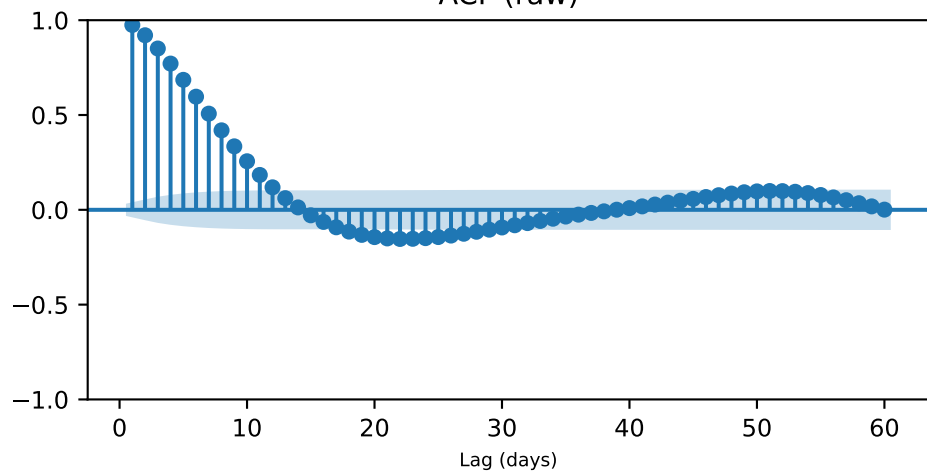
Raw histogram



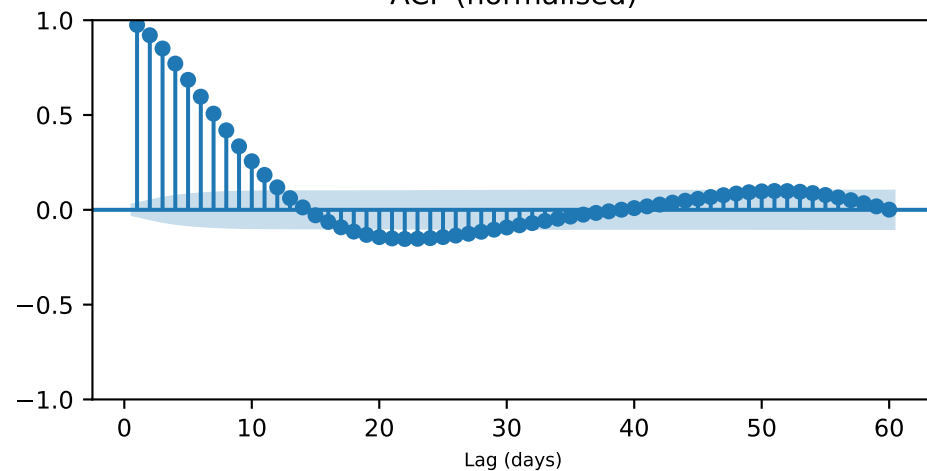
Normalised histogram



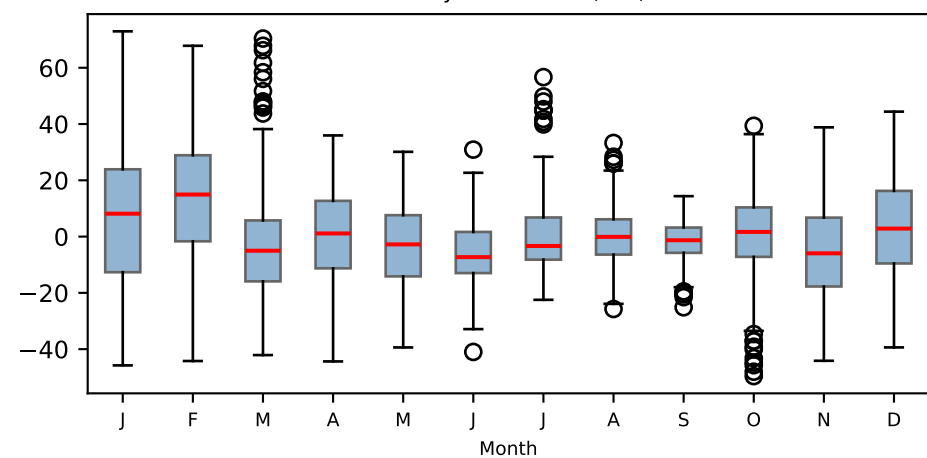
ACF (raw)



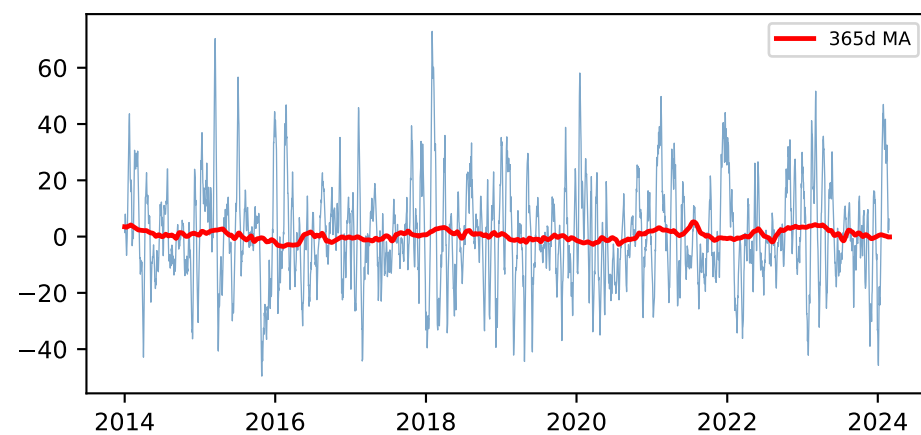
ACF (normalised)



Monthly distribution (raw)

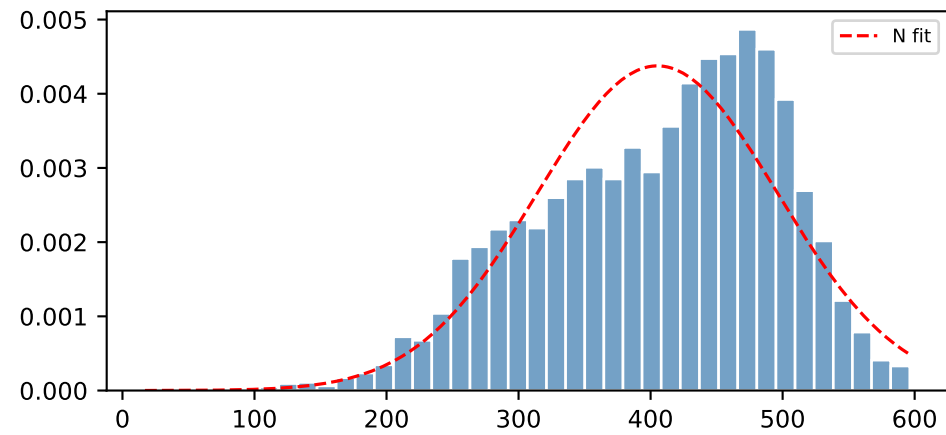


Time series + 365d mean

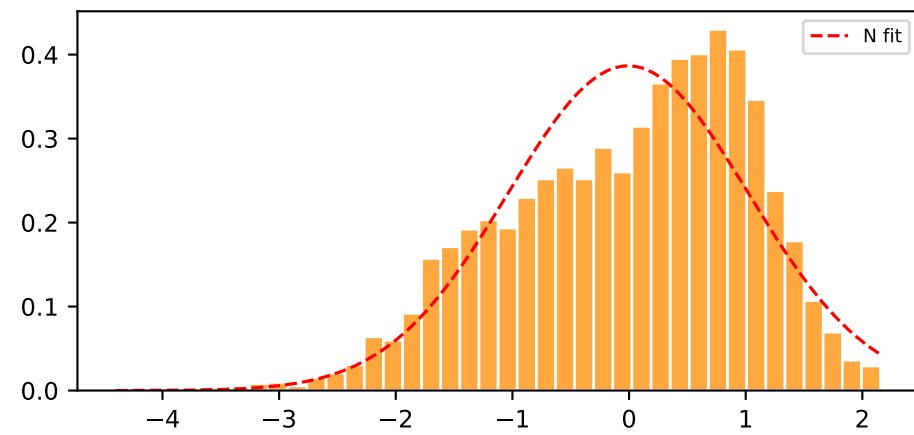


Covariate diagnostics — dew_x_nino34

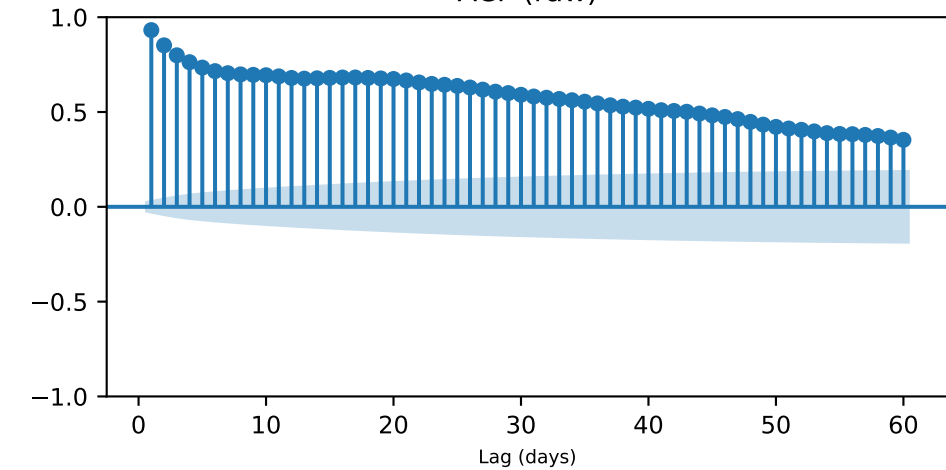
Raw histogram



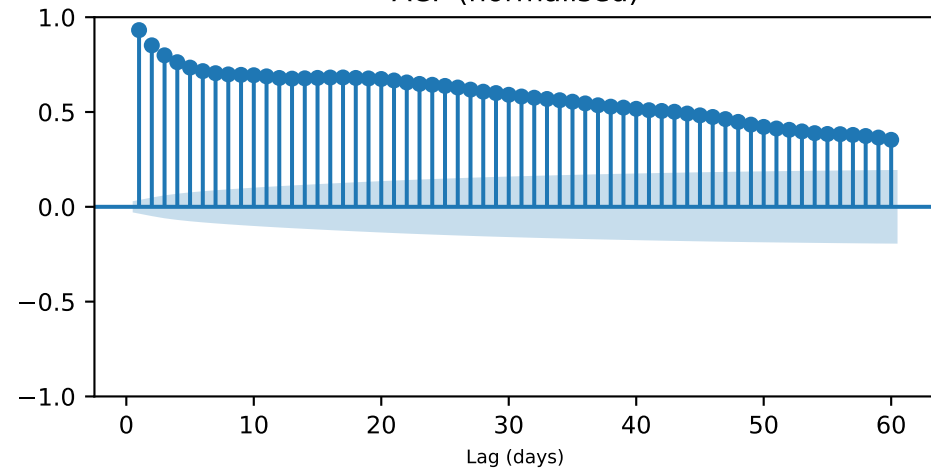
Normalised histogram



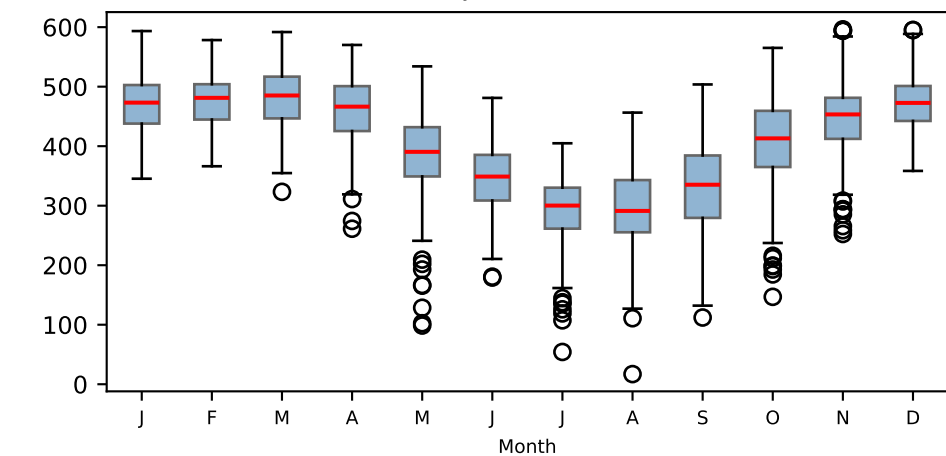
ACF (raw)



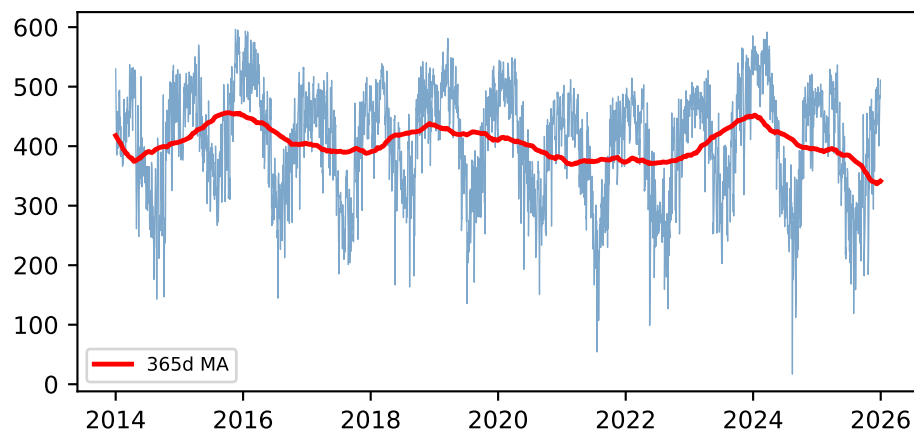
ACF (normalised)



Monthly distribution (raw)

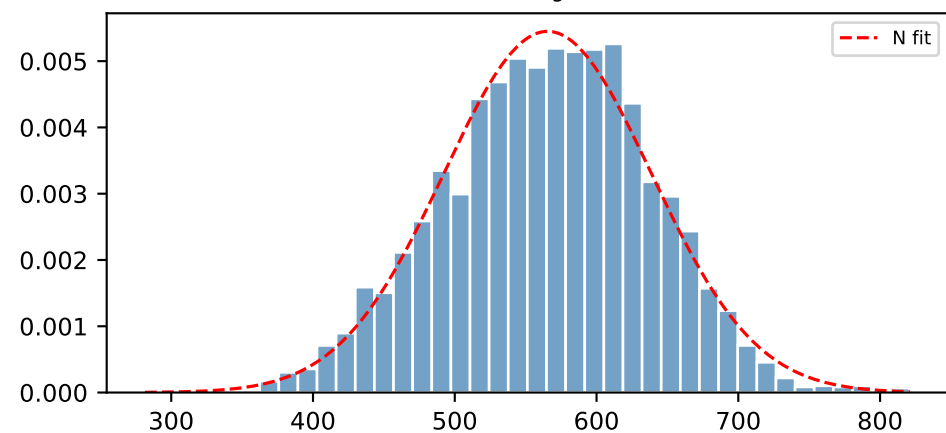


Time series + 365d mean

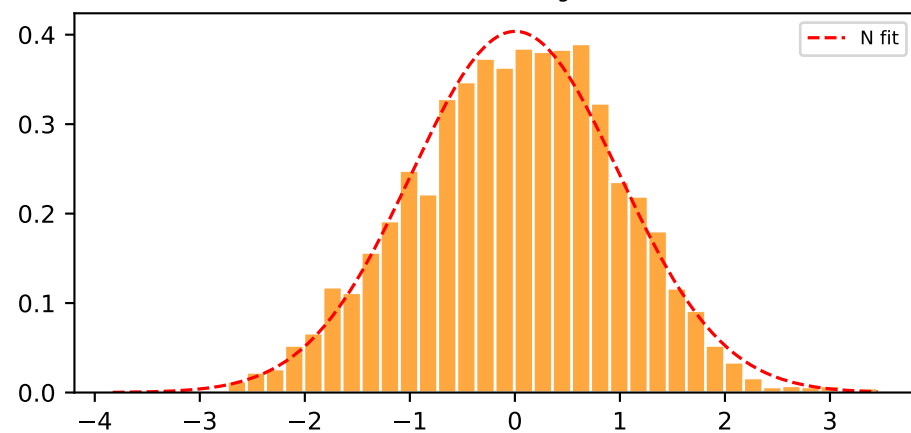


Covariate diagnostics — temp_x_nino34

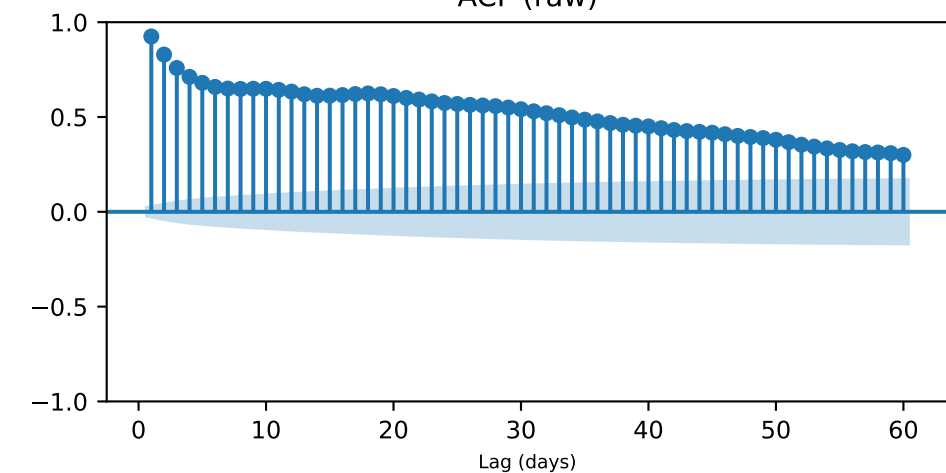
Raw histogram



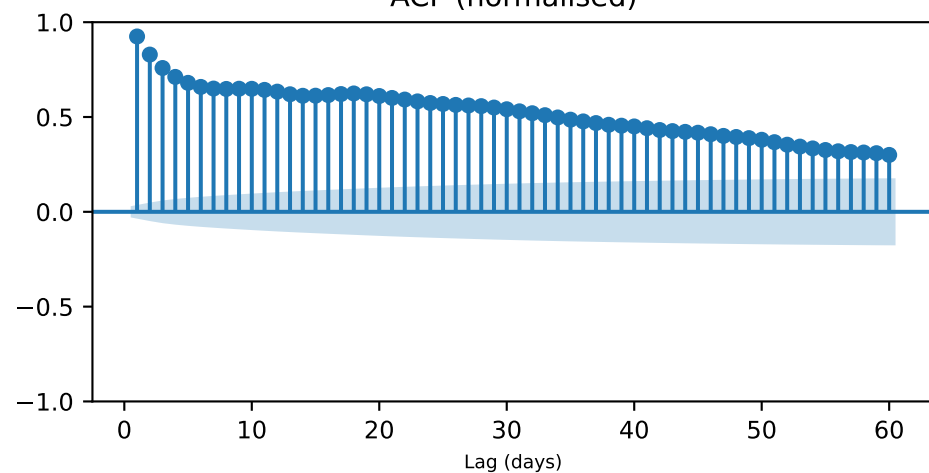
Normalised histogram



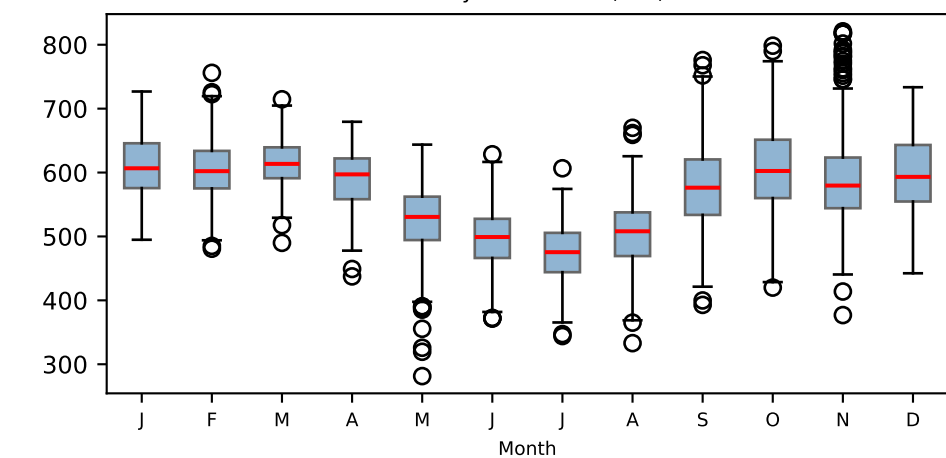
ACF (raw)



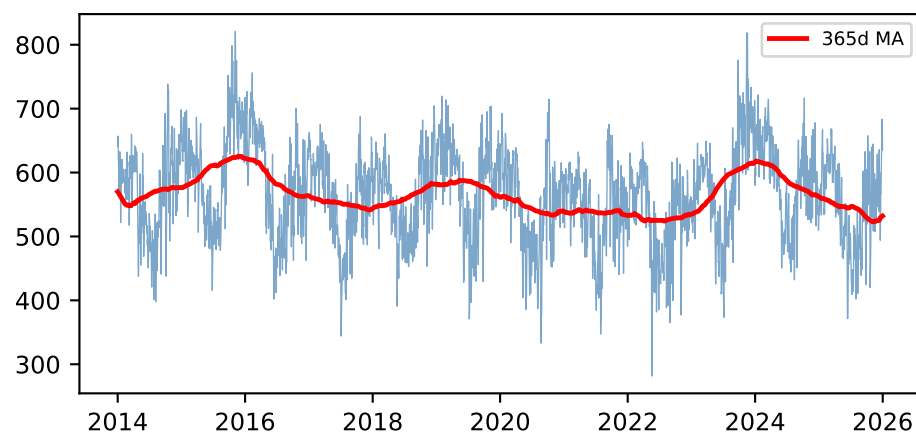
ACF (normalised)



Monthly distribution (raw)



Time series + 365d mean



In-sample metrics

model id	loglik	aic	bic	crps	rmse	mad	twcrps 90	twcrps 95	kup95 pvalue	chrf95 pvalue
xi_dewpoint_t	-5075.3824	10184.7648	10288.4210	2.7382	738672.5019	585842.9316	0.5705	0.4492	0.5941	0.6703
xi_dewpoint_lag1	-5066.6116	10169.2232	10278.9769	2.7278	1124099.4034	881421.4944	0.5038	0.3933	0.5941	0.6703
xi_dewpoint_short_lags	-5062.8379	10165.6759	10287.6244	2.7174	1285.0782	974.8977	0.4526	0.3377	0.5941	0.6703
xi_dewpoint_seasonal_lags	-5060.8898	10169.7795	10316.1177	2.7113	1578869.6795	1256625.0565	0.5141	0.4037	0.5941	0.6703
xi_temperature_t	-5081.8118	10197.6236	10301.2798	2.7586	4048.8473	3199.5003	0.5760	0.4323	0.5941	0.6703
xi_temperature_lag1	-5079.8847	10195.7695	10305.5232	2.7486	641.3098	504.5952	0.5709	0.4283	0.5941	0.6703
xi_temperature_short_lags	-5074.1501	10188.3001	10310.2486	2.7501	686.5928	540.3439	0.5621	0.4216	0.5941	0.6703
xi_temperature_seasonal_lags	-5067.3765	10182.7529	10329.0911	2.6866	19.7866	15.1315	0.5908	0.4479	0.5941	0.6703
xi_dewpoint_temperature_t	-5078.4290	10192.8581	10302.6118	2.7421	3420.3209	2654.1318	0.5452	0.4085	0.5941	0.6703
xi_dewpoint_temperature_lag1	-5057.7261	10155.4521	10277.4006	2.6740	98.0908	72.9433	0.4626	0.3476	0.5941	0.6703
xi_dewpoint_temperature_short_lags	-5051.8708	10151.7417	10298.0799	2.6858	128687.9911	97532.2459	0.4314	0.3262	0.5941	0.6703
xi_dewpoint_temperature_seasonal_lags	-5041.8538	10147.7077	10342.8253	2.6303	1051.6433	768.5659	0.4474	0.3369	0.5406	0.6581
xi_mjo_t	-5088.1755	10212.3511	10322.1047	2.7815	632052.7586	517825.5985	0.6760	0.5306	0.5941	0.6703
xi_mjo_lag1	-5082.6942	10205.3883	10327.3368	2.7474	4890.5263	3931.7988	0.6028	0.4508	0.5941	0.6703
xi_mjo_short_lags	-5078.0583	10204.1165	10350.4548	2.7370	14045.2911	11341.2297	0.6185	0.4664	0.5941	0.6703
xi_nino34_t	-5086.9880	10207.9759	10311.6322	2.7608	182430.6319	148322.3269	0.6548	0.5043	0.5941	0.6703
xi_nino34_lags	-5089.4652	10216.9304	10332.7815	2.7682	78223.5441	63835.2036	0.6525	0.4969	0.5941	0.6703
xi_nino34_rolling	-5087.7217	10213.4433	10329.2944	2.7697	555977.7690	454532.4900	0.6910	0.5430	0.5941	0.6703
xi_interaction_dewpoint_temperature_t	-5082.4101	10198.8203	10302.4765	2.7457	318789.0798	254598.1296	0.6082	0.4712	0.5941	0.6703
xi_interaction_dewpoint_mjo	-5086.9467	10209.8933	10319.6470	2.7655	276625.4427	225096.6009	0.6522	0.5055	0.5941	0.6703
xi_interaction_dewpoint_nino34	-5076.1467	10186.2935	10289.9497	2.7411	5020.9724	3898.3568	0.5097	0.3782	0.5941	0.6703
xi_interaction_temperature_nino34	-5084.6212	10203.2424	10306.8987	2.7651	8033.0972	6386.7668	0.5932	0.4474	0.5941	0.6703
phi_dewpoint_t	-5057.7698	10149.5395	10253.1958	2.6795	42.1698	26.0407	0.3959	0.2976	0.5941	0.6703
phi_dewpoint_lag1	-5060.5222	10157.0445	10266.7982	2.6812	14.7876	8.9314	0.3872	0.2874	0.5941	0.6703
phi_dewpoint_short_lags	-5056.9073	10153.8145	10275.7630	2.6740	21.1146	12.7721	0.3908	0.2949	0.5941	0.6703
phi_dewpoint_seasonal_lags	-5053.1694	10154.3388	10300.6770	2.6509	17.8537	10.8406	0.3928	0.2946	0.5941	0.6703
phi_temperature_t	-5074.4236	10182.8473	10286.5035	2.7166	27.4422	18.6625	0.5081	0.3696	0.5941	0.6703
phi_temperature_lag1	-5069.5245	10175.0491	10284.8027	2.7014	15.8874	10.5287	0.4911	0.3598	0.5941	0.6703
phi_temperature_short_lags	-5066.8525	10173.7050	10295.6535	2.7085	36.8695	24.1780	0.5068	0.3765	0.5941	0.6703
phi_temperature_seasonal_lags	-5062.8558	10173.7116	10320.0498	2.6953	16.7480	11.0044	0.4836	0.3525	0.5941	0.6703
phi_dewpoint_temperature_t	-5055.5677	10147.1354	10256.8891	2.6735	45.6163	28.3108	0.3680	0.2752	0.5941	0.6703
phi_dewpoint_temperature_lag1	-5052.7444	10145.4888	10267.4373	2.6596	18.1012	10.9106	0.3634	0.2702	0.5941	0.6703
phi_dewpoint_temperature_short_lags	-5047.8508	10143.7015	10290.0398	2.6611	116.6189	69.1219	0.3619	0.2727	0.5941	0.6703
phi_dewpoint_temperature_seasonal_lags	-5040.3269	10144.6538	10339.7715	2.6446	12.6269	7.3077	0.3552	0.2619	0.5941	0.6703
phi_mjo_t	-5082.8033	10201.6067	10311.3604	2.7283	55.1296	38.8532	0.5750	0.4221	0.5941	0.6703
phi_mjo_lag1	-5079.2218	10198.4436	10320.3922	2.7277	22.9490	16.1999	0.5167	0.3758	0.5941	0.6703
phi_mjo_short_lags	-5073.9330	10195.8661	10342.2043	2.7029	29.3457	21.0538	0.5487	0.4031	0.5941	0.6703
phi_nino34_t	-5080.4765	10194.9530	10298.6092	2.7324	144.0020	99.4718	0.5308	0.3976	0.5406	0.4994
phi_nino34_lags	-5078.2023	10194.4047	10310.2558	2.7119	63.1931	43.2422	0.5554	0.4139	0.5941	0.6703
phi_nino34_rolling	-5077.2658	10192.5316	10308.3827	2.7058	44.1890	29.8616	0.5525	0.4129	0.5941	0.6703
phi_interaction_dewpoint_temperature_t	-5070.2524	10174.5049	10278.1611	2.6936	14.4968	9.3688	0.4825	0.3543	0.5941	0.6703
phi_interaction_dewpoint_mjo	-5082.8791	10201.7582	10311.5119	2.7316	53.5433	37.7455	0.5710	0.4183	0.5941	0.6703
phi_interaction_dewpoint_nino34	-5074.2768	10182.5535	10286.2098	2.6941	20.9623	13.8829	0.4764	0.3470	0.5941	0.6703
phi_interaction_temperature_nino34	-5078.2722	10190.5444	10294.2006	2.7258	27.0112	18.2781	0.5200	0.3830	0.5941	0.6703
phi_xi_dewpoint_t	-5052.5060	10155.0120	10307.4477	2.6744	58.8666	36.5461	0.3933	0.2938	0.5941	0.6703
phi_xi_dewpoint_lag1	-5049.9257	10153.8515	10318.4820	2.6495	13.3184	8.1663	0.4077	0.2980	0.5941	0.6703
phi_xi_dewpoint_short_lags	-5047.1312	10156.2623	10345.2825	2.6501	45.9574	29.5750	0.4261	0.3117	0.5941	0.6703
phi_xi_dewpoint_seasonal_lags	-5045.9906	10169.9813	10407.7809	2.6463	10.8752	4.8323	0.3829	0.2803	0.5941	0.6703
phi_xi_temperature_t	-5073.7517	10197.5035	10349.9391	2.6925	686.7990	453.7888	0.5916	0.4556	0.5406	0.6581
phi_xi_temperature_lag1	-5066.8456	10187.6912	10352.3217	2.7042	23.2660	15.6432	0.4826	0.3511	0.5941	0.6703
phi_xi_temperature_short_lags	-5063.6646	10189.3291	10378.3493	2.7205	211.7075	148.0449	0.5119	0.3813	0.5941	0.6703
phi_xi_temperature_seasonal_lags	-5052.7593	10183.5186	10421.3183	2.6815	21.7089	14.2913	0.4828	0.3529	0.5941	0.6703
phi_xi_dewpoint_temperature_t	-5047.3588	10148.7175	10313.3480	2.6678	117.8684	74.2303	0.3595	0.2669	0.5941	0.6703
phi_xi_dewpoint_temperature_lag1	-5040.9036	10143.8073	10332.8275	2.6283	46.8693	28.8112	0.3666	0.2692	0.5941	0.6703
phi_xi_dewpoint_temperature_short_lags	-5036.0644	10150.1288	10387.9284	2.6346	38.4537	23.2559	0.3559	0.2622	0.5941	0.6703
phi_xi_dewpoint_temperature_seasonal_lags	-5019.1536	10148.3072	10483.6657	2.6101	28.0894	16.5463	0.3509	0.2576	0.5941	0.6703
phi_xi_mjo_t	-5081.1774	10216.3548	10380.9853	2.7380	136.2628	103.3166	0.6139	0.4514	0.5941	0.6703
phi_xi_mjo_lag1	-5075.9915	10213.9831	10403.0033	2.7174	138.7616	102.6304	0.5858	0.4294	0.5941	0.6703
phi_xi_mjo_short_lags	-5068.9739	10215.9477	10453.7474	2.7030	114.0988	83.7982	0.5583	0.4068	0.5941	0.6703
phi_xi_nino34_t	-5076.3383	10202.6766	10355.1123	2.7255	89.4815	63.3955	0.5206	0.3839	0.5941	0.6703
phi_xi_nino34_lags	-5073.6062	10205.2124	10382.0378	2.7156	201.1050	139.6041	0.5484	0.4051	0.5941	0.6703
phi_xi_nino34_rolling	-5073.9730	10205.9460	10382.7713	2.7143	151.5840	105.5303	0.5309	0.3920	0.5941	0.6703
phi_xi_interaction_dewpoint_temperature_t	-5068.3890	10186.7781	10339.2137	2.7063	32.0314	21.0538	0.4751	0.3461	0.5941	0.6703
phi_xi_interaction_dewpoint_mjo	-5082.4309	10218.8619	10383.4924	2.7383	17.3729	13.1593	0.5946	0.4326	0.5941	0.6703
phi_xi_interaction_dewpoint_nino34	-5064.4376	10178.8753	10331.3109	2.7042	111.3653	73.2424	0.4375	0.3255	0.5941	0.6703
phi_xi_interaction_temperature_nino34	-5076.0777	10202.1555	10354.5911	2.7362	310.4872	239.0725	0.5994	0.4501	0.5941	0.6703
long_short_xi_basic	-5073.9774	10193.9549	10334.1957	2.7135	51814.9377	40275.1039	0.5560	0.4181	0.5941	0.6703
long_short_xi_short_lags	-5037.4432	10144.8864	10358.2963	2.6520	5395.2483	4030.5460	0.3985	0.2992	0.5941	0.6703
long_short_xi_weather_climate	-5035.5422	10145.0844	10370.6892	2.6508	38254.5198	28686.1393	0.3967	0.2975	0.5941	0.6703
long_short_xi_interactions	-5035.0983	10146.1965	10377.8987	2.6433	30910.9520	22786.3147	0.3920	0.2949	0.5941	0.6703
long_short_phi_basic	-5047.5310	10141.0620	10281.3028	2.6523	26.1490	15.8680	0.3656	0.2757	0.5941	0.6703
long_short_phi_short_lags	-5036.5672	10143.1344	10356.5444	2.6356	23.6381	13.8202	0.3349	0.2495	0.5941	0.6703
long_short_phi_weather_climate	-5035.3555	10144.7111	10370.3159	2.6328	18.1172	10.7280	0.3295	0.2442	0.5941	0.6703
long_short_phi_interactions	-5035.8995	10147.7990	10379.5012	2.6319	11.3075	6.1935	0.3259	0.2394	0.5941	0.6703
long_short_phi_xi_basic	-5038.3930	10150.7861	10376.3908	2.6311	11.0299	6.0584	0.3539	0.2615	0.5941	0.6703
long_short_phi_xi_short_lags	-5017.6544	10157.3087	10529.2517	2.5931	426.9170	273.3007	0.3657	0.2691	0.5941	0.6703
long_short_phi_xi_weather_climate	-5010.4776	10150.9553	10547.2880	2.5781	52.9068	29.2499	0.3568	0.2676	0.5941	0.6703
long_short_phi_xi_interactions	-5015.0617	10164.1233	10572.6509	2.5951	90.3693	18.1464	0.5782	0.4892	0.5941	0.6703

Out-of-sample metrics

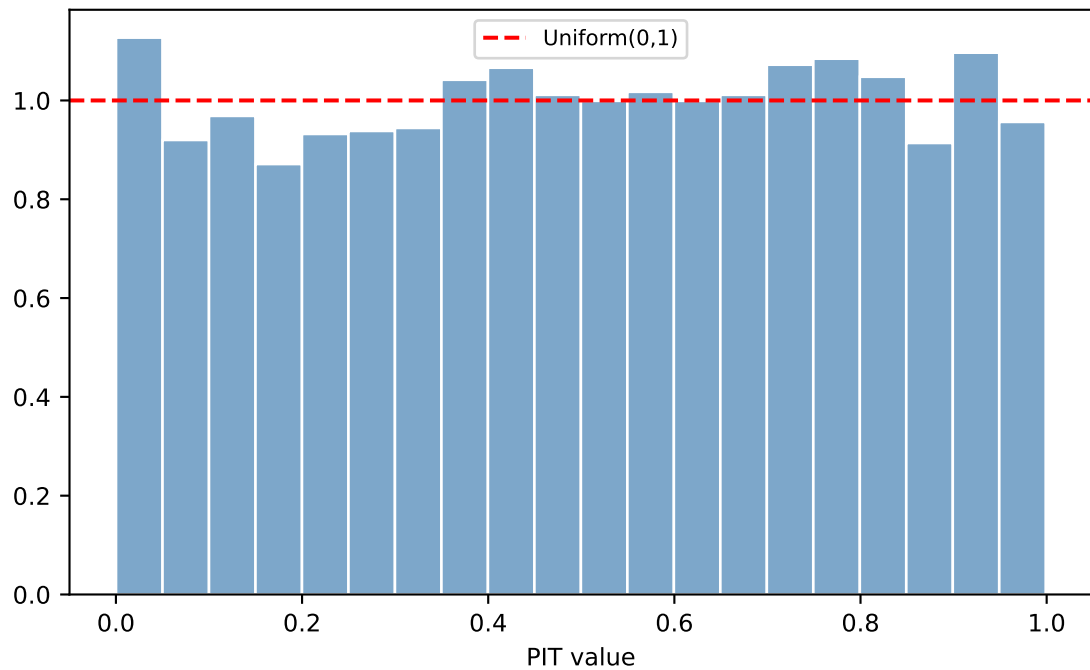
model id	loglik	aic	bic	crps	rmse	mad	twcrps 90	twcrps 95	kup95 pvalue	chrf95 pvalue
xi_dewpoint_t	-1128.9893	2291.9786	2370.0836	2.6651	756744.1147	590047.1578	0.6207	0.4921	0.0898	0.2373
xi_dewpoint_lag1	-1129.7607	2295.5214	2378.2209	2.6617	1151691.9115	887202.2497	0.5618	0.4448	0.1225	0.2444
xi_dewpoint_short_lags	-1128.0920	2296.1841	2388.0723	2.6532	1322.9623	984.3397	0.4914	0.3715	0.3658	0.4035
xi_dewpoint_seasonal_lags	-1138.0485	2324.0971	2434.3630	2.5713	1674058.1734	1297642.0659	0.7104	0.5743	0.2863	0.3736
xi_temperature_t	-1134.9248	2303.8496	2381.9547	2.6558	4259.6186	3262.9351	0.6611	0.4939	0.3658	0.4035
xi_temperature_lag1	-1132.8914	2301.7829	2384.4823	2.6377	676.2017	515.0807	0.6612	0.4983	0.3658	0.4035
xi_temperature_short_lags	-1131.6618	2303.3236	2395.2118	2.6263	722.5110	552.3665	0.6466	0.4896	0.3658	0.4035
xi_temperature_seasonal_lags	-1132.2962	2312.5923	2422.8583	2.7134	19.2384	15.2808	0.6052	0.4530	0.3658	0.4035
xi_dewpoint_temperature_t	-1132.5508	2301.1016	2383.8011	2.6345	3625.7655	2723.6722	0.6415	0.4927	0.5405	0.7505
xi_dewpoint_temperature_lag1	-1128.7287	2297.4574	2389.3457	2.7438	94.9542	70.8846	0.4476	0.3329	0.4306	0.6474
xi_dewpoint_temperature_short_lags	-1124.5728	2297.1456	2407.4115	2.6224	132375.9190	98827.0256	0.4816	0.3689	0.3658	0.4035
xi_dewpoint_temperature_seasonal_lags	-1143.0003	2350.0007	2497.0219	2.6533	1043.5966	774.7679	0.4947	0.3771	0.3658	0.4035
xi_mjo_t	-1134.2302	2304.4604	2387.1599	2.7796	630795.1226	514860.8024	0.6733	0.5219	0.2863	0.3736
xi_mjo_lag1	-1133.8158	2307.6317	2399.5199	2.7159	4955.5720	3903.7719	0.6380	0.4698	0.3658	0.4035
xi_mjo_short_lags	-1134.7968	2317.5936	2427.8595	2.7234	14207.7946	11231.4281	0.6492	0.4837	0.3658	0.4035
xi_nino34_t	-1134.1887	2302.3775	2380.4825	2.7089	186510.5375	148769.9916	0.7067	0.5409	0.3658	0.4035
xi_nino34_lags	-1135.7197	2309.4394	2396.7332	2.7605	77667.6189	63464.2737	0.6831	0.5131	0.3658	0.4035
xi_nino34_rolling	-1132.8329	2303.6659	2390.9597	2.7251	559043.6830	451387.5974	0.7013	0.5489	0.5405	0.7604
xi_interaction_dewpoint_temperature_t	-1133.3634	2300.7268	2378.8318	2.5941	340505.7429	258911.4908	0.7386	0.5809	0.6807	0.7517
xi_interaction_dewpoint_mjo	-1133.4604	2302.9207	2385.6201	2.7628	275888.8844	223177.5624	0.6594	0.5052	0.3658	0.4035
xi_interaction_dewpoint_nino34	-1131.4999	2296.9998	2375.1048	2.6262	5377.3789	4064.1146	0.6080	0.4490	0.3658	0.4035
xi_interaction_temperature_nino34	-1136.2957	2306.5915	2384.6965	2.6584	8496.6807	6547.8820	0.6883	0.5207	0.3658	0.4035
phi_dewpoint_t	-1123.5186	2281.0372	2359.1423	2.5827	47.9066	28.5810	0.5351	0.4163	0.3658	0.4035
phi_dewpoint_lag1	-1123.0520	2282.1041	2364.8035	2.6162	15.2280	9.4153	0.4635	0.3513	0.3658	0.4035
phi_dewpoint_short_lags	-1122.9174	2285.8348	2377.7231	2.5889	23.4233	13.9710	0.5123	0.3946	0.3658	0.4035
phi_dewpoint_seasonal_lags	-1123.3280	2294.6559	2404.9218	2.5487	19.8014	11.8982	0.5495	0.4261	0.3658	0.4035
phi_temperature_t	-1130.7972	2295.5943	2373.6994	2.5841	30.7739	20.1824	0.6849	0.5273	0.3658	0.4035
phi_temperature_lag1	-1127.8214	2291.6428	2374.3423	2.5780	17.1122	11.2621	0.6608	0.5007	0.3658	0.4035
phi_temperature_short_lags	-1127.2340	2294.4680	2386.3563	2.5888	39.6341	25.8787	0.6380	0.4863	0.3658	0.4035
phi_temperature_seasonal_lags	-1127.3166	2302.6332	2412.8991	2.6061	16.9377	11.3784	0.5700	0.4247	0.3658	0.4035
phi_dewpoint_temperature_t	-1121.4102	2278.8205	2361.5199	2.5848	50.8855	30.5464	0.4794	0.3697	0.3658	0.4035
phi_dewpoint_temperature_lag1	-1121.1389	2282.2777	2374.1660	2.6027	19.0345	11.5307	0.4294	0.3253	0.3658	0.4035
phi_dewpoint_temperature_short_lags	-1120.3709	2288.7418	2399.0077	2.6001	119.8170	71.9871	0.4129	0.3194	0.3658	0.4035
phi_dewpoint_temperature_seasonal_lags	-1120.7355	2305.4710	2452.4922	2.5608	12.5807	7.5573	0.4262	0.3175	0.3658	0.4035
phi_mjo_t	-1132.2991	2300.5982	2383.2977	2.6596	57.6627	40.1633	0.6663	0.4988	0.3658	0.4035
phi_mjo_lag1	-1129.3196	2298.6392	2390.5275	2.7259	21.3993	15.2328	0.4658	0.3347	0.3658	0.1942
phi_mjo_short_lags	-1132.2936	2312.5872	2422.8531	2.6911	29.2631	20.4342	0.5498	0.3997	0.3658	0.4035
phi_nino34_t	-1130.5849	2295.1697	2373.2747	2.7150	135.2938	94.8432	0.5073	0.3720	0.3658	0.4035
phi_nino34_lags	-1131.3886	2300.7772	2388.0711	2.6653	63.0091	41.6446	0.5823	0.4378	0.3658	0.4035
phi_nino34_rolling	-1129.8477	2297.6955	2384.9893	2.6513	44.3084	28.6206	0.5776	0.4347	0.3658	0.4035
phi_interaction_dewpoint_temperature_t	-1128.1448	2290.2897	2368.3947	2.5877	15.3607	9.9977	0.6404	0.4876	0.3658	0.4035
phi_interaction_dewpoint_mjo	-1132.5411	2301.0821	2383.7816	2.6580	56.0736	39.3459	0.6755	0.5046	0.6807	0.9171
phi_interaction_dewpoint_nino34	-1127.4811	2288.9621	2367.0672	2.5814	23.5301	15.2620	0.6485	0.4981	0.3658	0.4035
phi_interaction_temperature_nino34	-1131.6182	2297.2364	2375.3414	2.6027	30.0357	19.8796	0.6980	0.5380	0.3658	0.4035
phi_xi_dewpoint_t	-1128.5667	2307.1335	2421.9938	2.5595	69.8220	41.6824	0.5859	0.4558	0.3658	0.4035
phi_xi_dewpoint_lag1	-1135.2686	2324.5372	2448.5864	2.5510	14.3275	8.9539	0.5636	0.4216	0.3658	0.4035
phi_xi_dewpoint_short_lags	-1131.7657	2325.5313	2467.9581	2.5536	53.9200	33.5646	0.6379	0.4941	0.3658	0.4035
phi_xi_dewpoint_seasonal_lags	-1127.1572	2332.3145	2511.4966	2.5453	10.2272	4.9370	0.5286	0.4007	0.6807	0.7517
phi_xi_temperature_t	-1167.9803	2385.9607	2500.8210	2.5107	705.2149	493.1728	0.8744	0.6701	0.3658	0.4035
phi_xi_temperature_lag1	-1129.8384	2313.6768	2437.7260	2.5894	26.2806	17.1590	0.6617	0.4995	0.3658	0.4035
phi_xi_temperature_short_lags	-1132.6566	2327.3133	2469.7401	2.5911	236.0563	161.8757	0.6927	0.5338	0.3658	0.4035
phi_xi_temperature_seasonal_lags	-1131.1922	2340.3845	2519.5666	2.5812	23.2159	15.1934	0.5967	0.4437	0.3658	0.4035
phi_xi_dewpoint_temperature_t	-1127.3899	2308.7798	2432.8289	2.5641	132.0309	81.3013	0.4945	0.3774	0.3658	0.4035
phi_xi_dewpoint_temperature_lag1	-1140.8547	2343.7095	2486.1363	2.4970	52.4456	32.0273	0.5279	0.3929	0.3658	0.4035
phi_xi_dewpoint_temperature_short_lags	-1124.1203	2326.2407	2505.4228	2.5757	39.6005	23.8172	0.4100	0.3088	0.3658	0.4035
phi_xi_dewpoint_temperature_seasonal_lags	-1131.9328	2373.8656	2626.5583	2.4476	34.3201	19.6263	0.5852	0.4544	0.3658	0.4035
phi_xi_mjo_t	-1135.4921	2324.9842	2449.0333	2.7067	139.2757	104.8625	0.6873	0.5043	0.3658	0.4035
phi_xi_mjo_lag1	-1135.7443	2333.4886	2475.9154	2.7333	136.3426	101.6327	0.6253	0.4536	0.3658	0.4035
phi_xi_mjo_short_lags	-1136.9822	2351.9644	2531.1466	2.7019	115.8343	82.0286	0.5890	0.4305	0.3658	0.4035
phi_xi_nino34_t	-1131.1981	2312.3961	2427.2565	2.7397	82.3914	58.1403	0.4609	0.3247	0.3658	0.4035
phi_xi_nino34_lags	-1132.9298	2323.8597	2457.0977	2.6819	200.5420	134.3173	0.5728	0.4291	0.3658	0.4035
phi_xi_nino34_rolling	-1129.9450	2317.8901	2451.1281	2.6916	147.6945	98.3895	0.5049	0.3717	0.3658	0.4035
phi_xi_interaction_dewpoint_temperature_t	-1132.3644	2314.7289	2429.5892	2.6253	35.9819	22.9214	0.6252	0.4665	0.3658	0.4035
phi_xi_interaction_dewpoint_mjo	-1136.3899	2326.7798	2450.8289	2.6976	18.2352	13.7865	0.6883	0.5006	0.2199	0.3348
phi_xi_interaction_dewpoint_nino34	-1132.8073	2315.6146	2430.4750	2.5724	134.3959	84.5028	0.6618	0.5198	0.3658	0.4035
phi_xi_interaction_temperature_nino34	-1137.6192	2325.2384	2440.0987	2.6583	313.3698	237.7073	0.6492	0.4844	0.3658	0.4035
long_short_xi_basic	-1128.9170	2303.8341	2409.5056	2.6393	53484.8411	41074.9072	0.6337	0.4811	0.4585	0.6811
long_short_xi_short_lags	-1121.0184	2312.0369	2472.8414	2.6242	5432.5709	4043.4737	0.4173	0.3156	0.3658	0.4035
long_short_xi_weather_climate	-1121.8980	2317.7960	2487.7893	2.6387	38447.1300	28807.4187	0.4226	0.3188	0.3658	0.4035
long_short_xi_interactions	-1122.3909	2320.7819	2495.3696	2.6135	31304.3886	22894.7555	0.4226	0.3208	0.3658	0.4035
long_short_phi_basic	-1120.2834	2286.5668	2392.2383	2.5994	27.1937	16.4682	0.4243	0.3211	0.3658	0.4035
long_short_phi_short_lags	-1117.2024	2304.4048	2465.2093	2.5872	23.6776	14.2576	0.3744	0.2792	0.8068	0.3893
long_short_phi_weather_climate	-1118.0026	2310.0053	2479.9985	2.5909	18.2371	11.1239	0.3666	0.2726	0.3658	0.4035
long_short_phi_interactions	-1116.6904	2309.3808	2483.9685	2.5791	10.8988	6.2944	0.3562	0.2619	0.3658	0.4035
long_short_phi_xi_basic	-1124.6553	2323.3105	2493.3038	2.5578	11.0538	6.3411	0.4581	0.3433	0.3658	0.4035
long_short_phi_xi_short_lags	-1124.3124	2370.6248	2650.8840	2.4622	480.0910	297.4064	0.5150	0.3969	0.3658	0.4035
long_short_phi_xi_weather_climate	-1133.5077	2397.0154	2695.6523	2.5695	48.4589	29.0635	0.3997	0.2963	0.3658	0.4035
long_short_phi_xi_interactions	-1123.3630	2380.7260	2688.5517	2.5486	27.0150	15.9841	0.3888	0.2874	0.3658	0.4035

OOS tail metrics

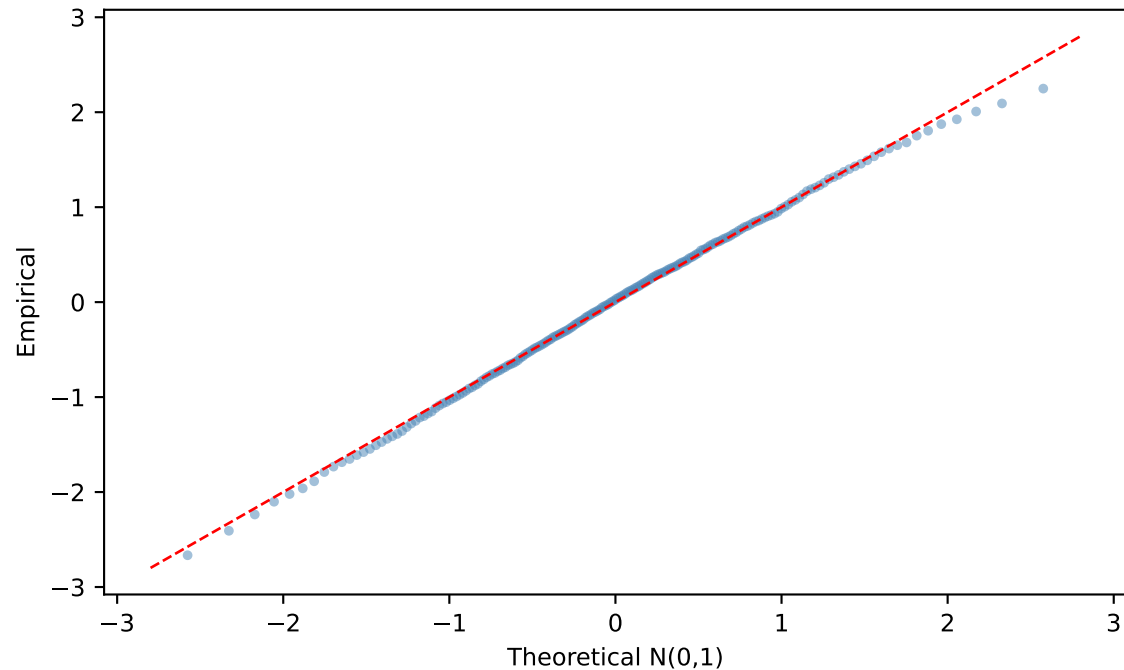
model id	crps	twcrps 90	twcrps 95	qs 900	qs 950	qs 975	qs 990	brier 90	brier 95	brier 99	kup95 pvalue	chrf95 pvalue
xi_dewpoint_t	2.6651	0.6207	0.4921	2.1058	1.4104	0.8814	0.4833	0.0286	0.0147	0.0042	0.0898	0.2373
xi_dewpoint_lag1	2.6617	0.5618	0.4448	2.0733	1.3886	0.8573	0.4733	0.0284	0.0146	0.0041	0.1225	0.2444
xi_dewpoint_short_lags	2.6532	0.4914	0.3715	2.0312	1.3583	0.8518	0.4619	0.0284	0.0146	0.0041	0.3658	0.4035
xi_dewpoint_seasonal_lags	2.5713	0.7104	0.5743	2.0366	1.3701	0.8543	0.4832	0.0286	0.0148	0.0042	0.2863	0.3736
xi_temperature_t	2.6558	0.6611	0.4939	2.2024	1.5376	0.9649	0.5292	0.0289	0.0148	0.0041	0.3658	0.4035
xi_temperature_lag1	2.6377	0.6612	0.4983	2.1940	1.5168	0.9500	0.5183	0.0289	0.0147	0.0041	0.3658	0.4035
xi_temperature_short_lags	2.6263	0.6466	0.4896	2.1624	1.4860	0.9569	0.5224	0.0286	0.0146	0.0041	0.3658	0.4035
xi_temperature_seasonal_lags	2.7134	0.6052	0.4530	2.2579	1.6346	1.0688	0.5688	0.0285	0.0145	0.0041	0.3658	0.4035
xi_dewpoint_temperature_t	2.6345	0.6415	0.4927	2.1645	1.4923	0.9365	0.5031	0.0290	0.0148	0.0041	0.5405	0.7505
xi_dewpoint_temperature_lag1	2.7438	0.4476	0.3329	2.1444	1.5004	0.8944	0.4838	0.0286	0.0147	0.0041	0.4306	0.6474
xi_dewpoint_temperature_short_lags	2.6224	0.4816	0.3689	1.9942	1.2768	0.8015	0.4468	0.0282	0.0145	0.0041	0.3658	0.4035
xi_dewpoint_temperature_seasonal_lags	2.6533	0.4947	0.3771	2.1113	1.5690	1.0403	0.5373	0.0283	0.0146	0.0042	0.3658	0.4035
xi_mjo_t	2.7796	0.6733	0.5219	2.3456	1.6999	1.0782	0.5666	0.0288	0.0147	0.0041	0.2863	0.3736
xi_mjo_lag1	2.7159	0.6380	0.4698	2.3020	1.6684	1.0654	0.5587	0.0289	0.0148	0.0041	0.3658	0.4035
xi_mjo_short_lags	2.7234	0.6492	0.4837	2.3227	1.6741	1.0858	0.5713	0.0290	0.0148	0.0041	0.3658	0.4035
xi_nino34_t	2.7089	0.7067	0.5409	2.3139	1.6852	1.0914	0.5822	0.0290	0.0148	0.0042	0.3658	0.4035
xi_nino34_lags	2.7605	0.6831	0.5131	2.3801	1.7463	1.1479	0.6076	0.0288	0.0147	0.0041	0.3658	0.4035
xi_nino34_rolling	2.7251	0.7013	0.5489	2.2931	1.6379	1.0315	0.5557	0.0289	0.0148	0.0041	0.5405	0.7604
xi_interaction_dewpoint_temperature_t	2.5941	0.7386	0.5809	2.1384	1.4761	0.9178	0.4991	0.0291	0.0149	0.0042	0.6807	0.7517
xi_interaction_dewpoint_mjo	2.7628	0.6594	0.5052	2.3252	1.2768	0.8015	0.5618	0.0288	0.0147	0.0041	0.3658	0.4035
xi_interaction_dewpoint_nino34	2.6262	0.6080	0.4490	2.1058	1.4297	0.8753	0.4831	0.0286	0.0146	0.0041	0.3658	0.4035
xi_interaction_temperature_nino34	2.6584	0.6883	0.5207	2.2252	1.5635	0.9843	0.5420	0.0290	0.0148	0.0042	0.3658	0.4035
phi_dewpoint_t	2.5827	0.5351	0.4163	2.0322	1.3207	0.7997	0.4100	0.0284	0.0146	0.0042	0.3658	0.4035
phi_dewpoint_lag1	2.6162	0.4635	0.3513	2.0603	1.3385	0.8037	0.4177	0.0284	0.0147	0.0042	0.3658	0.4035
phi_dewpoint_short_lags	2.5889	0.5123	0.3946	2.0291	1.3200	0.7880	0.4099	0.0285	0.0147	0.0043	0.3658	0.4035
phi_dewpoint_seasonal_lags	2.5487	0.5495	0.4261	2.0243	1.2985	0.7920	0.3889	0.0286	0.0148	0.0042	0.3658	0.4035
phi_temperature_t	2.5841	0.6849	0.5273	2.1579	1.4986	0.9234	0.4853	0.0290	0.0149	0.0042	0.3658	0.4035
phi_temperature_lag1	2.5780	0.6608	0.5007	2.1553	1.4695	0.8960	0.4697	0.0289	0.0148	0.0042	0.3658	0.4035
phi_temperature_short_lags	2.5888	0.6380	0.4863	2.1668	1.4766	0.9148	0.5014	0.0286	0.0146	0.0042	0.3658	0.4035
phi_temperature_seasonal_lags	2.6061	0.5700	0.4247	2.1734	1.4730	0.9299	0.4992	0.0285	0.0145	0.0042	0.3658	0.4035
phi_dewpoint_temperature_t	2.5848	0.4794	0.3697	2.0025	1.2888	0.7693	0.3946	0.0282	0.0145	0.0042	0.3658	0.4035
phi_dewpoint_temperature_lag1	2.6027	0.4294	0.3253	2.0316	1.3082	0.7896	0.3974	0.0282	0.0147	0.0042	0.3658	0.4035
phi_dewpoint_temperature_short_lags	2.6001	0.4129	0.3194	1.9914	1.2743	0.7731	0.4057	0.0279	0.0145	0.0042	0.3658	0.4035
phi_dewpoint_temperature_seasonal_lags	2.5608	0.4262	0.3175	1.9702	1.2407	0.7649	0.4019	0.0279	0.0143	0.0042	0.3658	0.4035
phi_mjo_t	2.6596	0.6663	0.4988	2.3261	1.7174	1.1153	0.5748	0.0292	0.0150	0.0042	0.3658	0.4035
phi_mjo_lag1	2.7259	0.4658	0.3347	2.2891	1.6263	1.0440	0.5287	0.0288	0.0148	0.0041	0.3658	0.1942
phi_mjo_short_lags	2.6911	0.5498	0.3997	2.3032	1.6619	1.0946	0.5714	0.0290	0.0149	0.0042	0.3658	0.4035
phi_nino34_t	2.7150	0.5073	0.3720	2.2945	1.6251	1.0583	0.5547	0.0289	0.0148	0.0042	0.3658	0.4035
phi_nino34_lags	2.6653	0.5823	0.4378	2.2542	1.6088	1.0151	0.5434	0.0291	0.0149	0.0042	0.3658	0.4035
phi_nino34_rolling	2.6513	0.5776	0.4347	2.2246	1.5655	0.9816	0.5110	0.0291	0.0149	0.0042	0.3658	0.4035
phi_interaction_dewpoint_temperature_t	2.5877	0.6404	0.4876	2.1612	1.4860	0.9176	0.4776	0.0291	0.0149	0.0042	0.3658	0.4035
phi_interaction_dewpoint_mjo	2.6580	0.6755	0.5046	2.3311	1.7244	1.1201	0.5800	0.0292	0.0150	0.0042	0.6807	0.9171
phi_interaction_dewpoint_nino34	2.5814	0.6485	0.4981	2.1709	1.4659	0.9150	0.4677	0.0292	0.0149	0.0042	0.3658	0.4035
phi_interaction_temperature_nino34	2.6027	0.6980	0.5380	2.2063	1.5471	0.9557	0.5028	0.0292	0.0150	0.0042	0.3658	0.4035
phi_xi_dewpoint_t	2.5595	0.5859	0.4558	2.0547	1.3344	0.8110	0.4343	0.0288	0.0148	0.0043	0.3658	0.4035
phi_xi_dewpoint_lag1	2.5510	0.5636	0.4216	2.0254	1.3303	0.8482	0.4610	0.0289	0.0149	0.0042	0.3658	0.4035
phi_xi_dewpoint_short_lags	2.5536	0.6379	0.4941	2.0766	1.3774	0.8575	0.4689	0.0290	0.0147	0.0042	0.3658	0.4035
phi_xi_dewpoint_seasonal_lags	2.5453	0.5286	0.4007	2.0076	1.3151	0.8113	0.4346	0.0288	0.0148	0.0043	0.6807	0.7517
phi_xi_temperature_t	2.5107	0.8744	0.6701	2.2418	1.5749	1.0419	0.5642	0.0298	0.0153	0.0043	0.3658	0.4035
phi_xi_temperature_lag1	2.5894	0.6617	0.4995	2.1750	1.4737	0.9082	0.4812	0.0291	0.0149	0.0042	0.3658	0.4035
phi_xi_temperature_short_lags	2.5911	0.6927	0.5338	2.1756	1.5258	0.9444	0.4765	0.0289	0.0148	0.0042	0.3658	0.4035
phi_xi_temperature_seasonal_lags	2.5812	0.5967	0.4437	2.1844	1.5374	0.9659	0.5132	0.0289	0.0147	0.0042	0.3658	0.4035
phi_xi_dewpoint_temperature_t	2.5641	0.4945	0.3774	1.9778	1.2645	0.7626	0.4000	0.0283	0.0146	0.0042	0.3658	0.4035
phi_xi_dewpoint_temperature_lag1	2.4970	0.5279	0.3929	1.9559	1.2721	0.7802	0.4129	0.0286	0.0149	0.0042	0.3658	0.4035
phi_xi_dewpoint_temperature_short_lags	2.5757	0.4100	0.3088	1.9467	1.2500	0.7817	0.4297	0.0281	0.0146	0.0042	0.3658	0.4035
phi_xi_dewpoint_temperature_seasonal_lags	2.4476	0.5852	0.4544	1.9139	1.2328	0.7807	0.3981	0.0286	0.0145	0.0043	0.3658	0.4035
phi_xi_mjo_t	2.7067	0.6873	0.5043	2.3731	1.7430	1.1257	0.5973	0.0293	0.0149	0.0042	0.3658	0.4035
phi_xi_mjo_lag1	2.7333	0.6253	0.4536	2.3618	1.7412	1.1310	0.5799	0.0293	0.0149	0.0042	0.3658	0.4035
phi_xi_mjo_short_lags	2.7019	0.5890	0.4305	2.3158	1.6776	1.1206	0.6086	0.0291	0.0149	0.0042	0.3658	0.4035
phi_xi_nino34_t	2.7397	0.4609	0.3247	2.2895	1.6168	1.0464	0.5533	0.0288	0.0147	0.0041	0.3658	0.4035
phi_xi_nino34_lags	2.6819	0.5728	0.4291	2.2674	1.6348	1.0531	0.5779	0.0294	0.0150	0.0042	0.3658	0.4035
phi_xi_nino34_rolling	2.6916	0.5049	0.3717	2.2368	1.5475	0.9874	0.5161	0.0290	0.0148	0.0042	0.3658	0.4035
phi_xi_interaction_dewpoint_temperature_t	2.6253	0.6252	0.4665	2.1941	1.4813	0.9501	0.5099	0.0295	0.0151	0.0043	0.3658	0.4035
phi_xi_interaction_dewpoint_mjo	2.6976	0.6883	0.5006	2.3781	1.7649	1.1401	0.5996	0.0292	0.0149	0.0042	0.2199	0.3348
phi_xi_interaction_dewpoint_nino34	2.5724	0.6618	0.5198	2.1209	1.4079	0.8634	0.4835	0.0293	0.0151	0.0043	0.3658	0.4035
phi_xi_interaction_temperature_nino34	2.6583	0.6492	0.4844	2.2189	1.5683	1.0289	0.5666	0.0289	0.0147	0.0042	0.3658	0.4035
long_short_xi_basic	2.6393	0.6337	0.4811	2.1324	1.4789	0.9513	0.4985	0.0286	0.0147	0.0041	0.4585	0.6811
long_short_xi_short_lags	2.6242	0.4173	0.3156	1.9362	1.2297	0.7647	0.4176	0.0279	0.0144	0.0041	0.3658	0.4035
long_short_xi_weather_climate	2.6387	0.4226	0.3188	1.9502	1.2460	0.7875	0.4395	0.0280	0.0145	0.0041	0.3658	0.4035
long_short_xi_interactions	2.6135	0.4226	0.3208	1.9294	1.2576	0.7750	0.4115	0.0281	0.0145	0.0041	0.3658	0.4035
long_short_phi_basic	2.5994	0.4243	0.3211	1.9731	1.2885	0.7734	0.3794	0.0279	0.0144	0.0042	0.3658	0.4035
long_short_phi_short_lags	2.5872	0.3744	0.2792	1.9212	1.2276	0.7624	0.3882	0.0276	0.0143	0.0042	0.8068	0.3893
long_short_phi_weather_climate	2.5909	0.3666	0.2726	1.9251	1.2431	0.7693	0.4205	0.0276	0.0144	0.0042	0.3658	0.4035
long_short_phi_interactions	2.5791	0.3562	0.2619	1.9078	1.2254	0.7591	0.3854	0.0275	0.0143	0.0042	0.3658	0.4035
long_short_phi_xi_basic	2.5578	0.4581	0.3433	1.9139	1.2313	0.7321	0.3684	0.0281	0.0144	0.0042	0.3658	0.4035
long_short_phi_xi_short_lags	2.4622	0.5150	0.3969	1.8138	1.2184	0.7961	0.4282	0.0276	0.0145	0.0041	0.3658	0.4035
long_short_phi_xi_weather_climate	2.5695	0.3997	0.2963	1.9629	1.2515	0.7840	0.4495	0.0282	0.0146	0.0042	0.3658	0.4035
long_short_phi_xi_interactions	2.5486	0.3888	0.2874	1.9182	1.2192	0.7664	0.4094	0.0277	0.0145	0.0042	0.3658	0.4035

xi_dewpoint_t — IS diagnostics

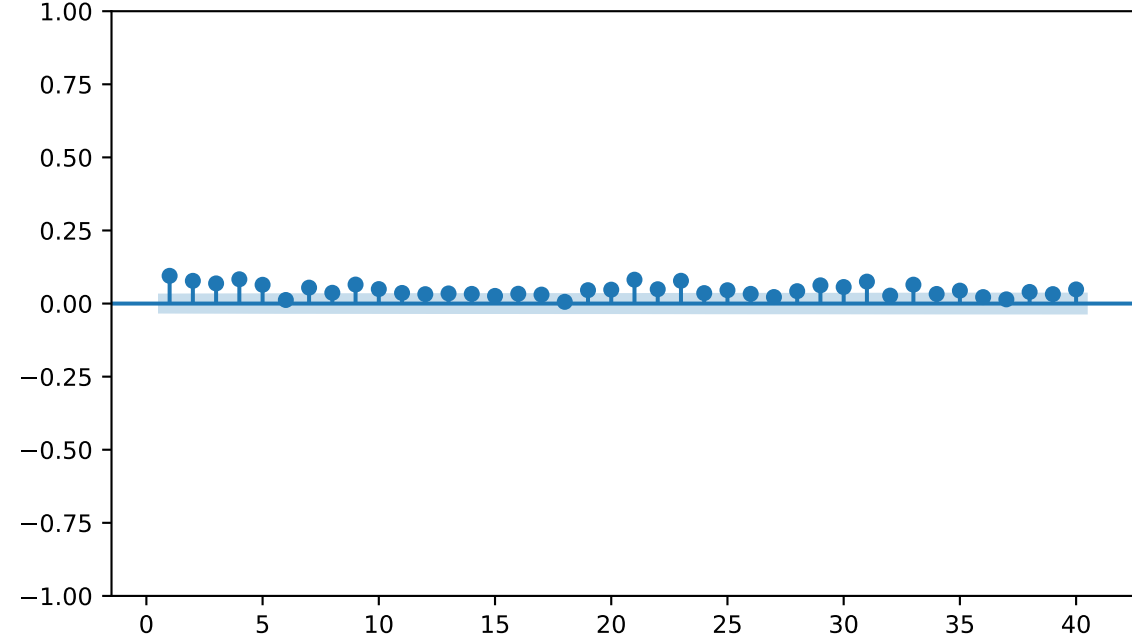
PIT histogram



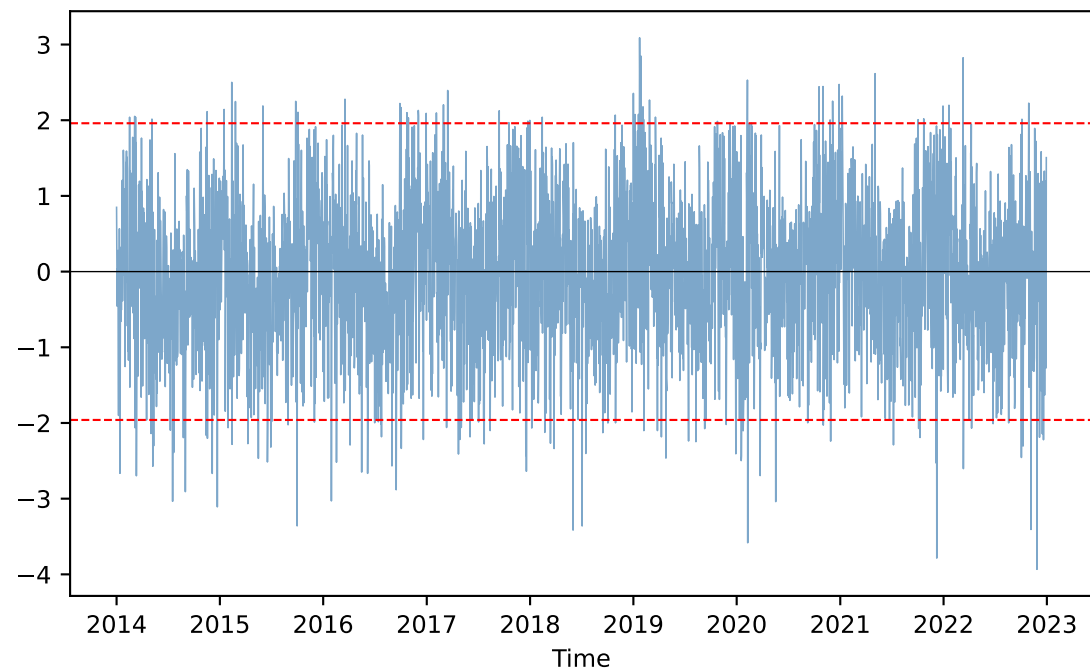
Normal QQ plot



ACF of QR

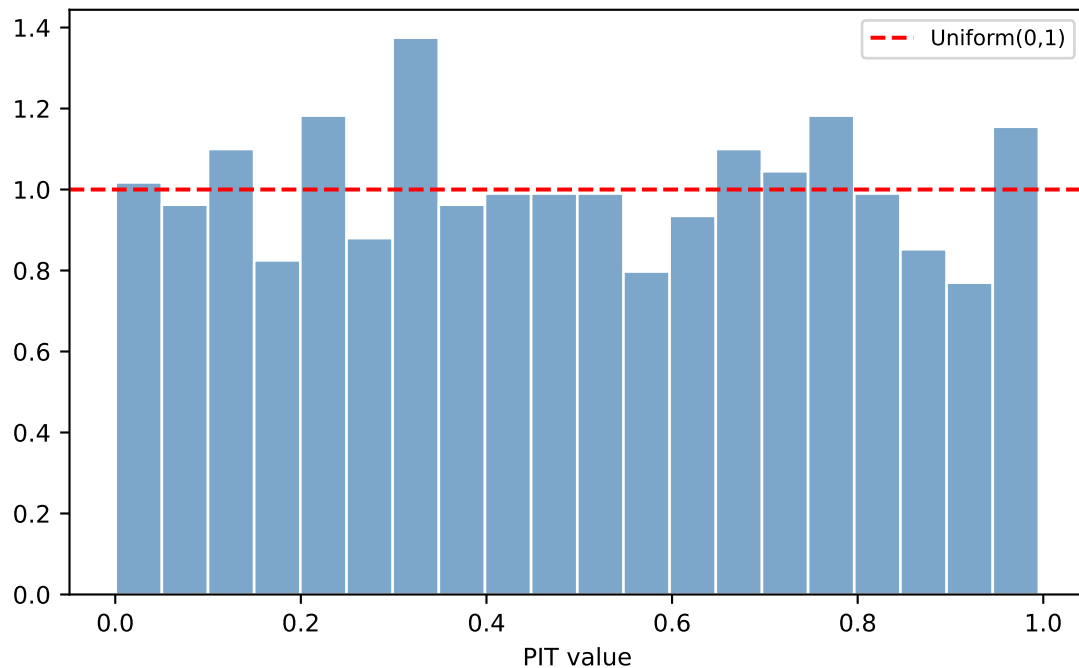


QR time series

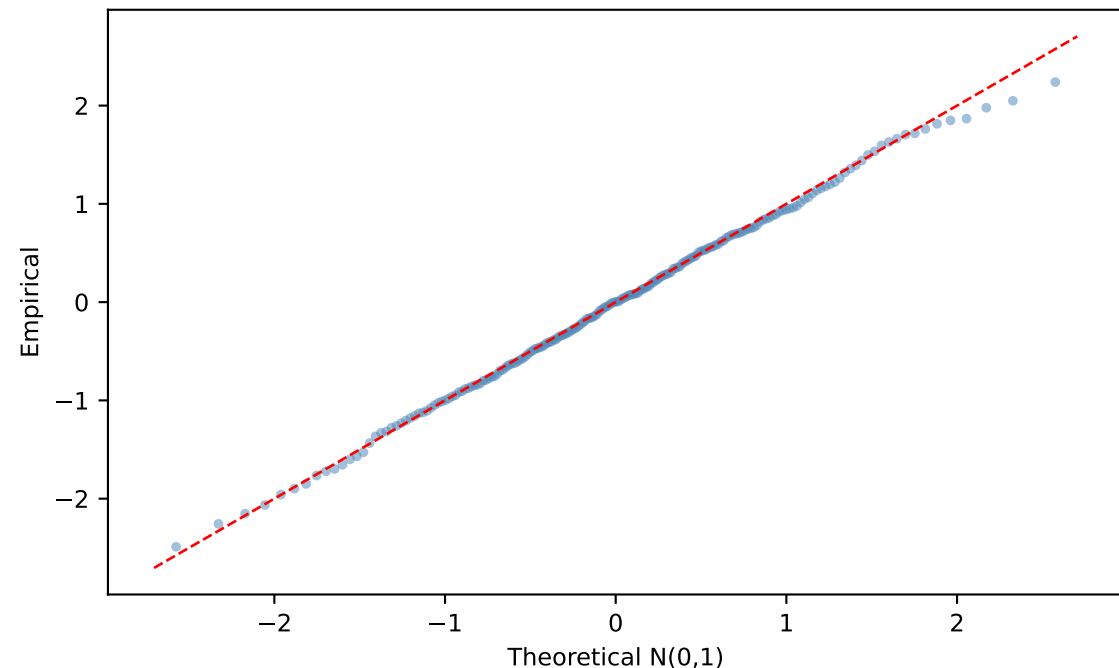


xi_dewpoint_t — OOS diagnostics

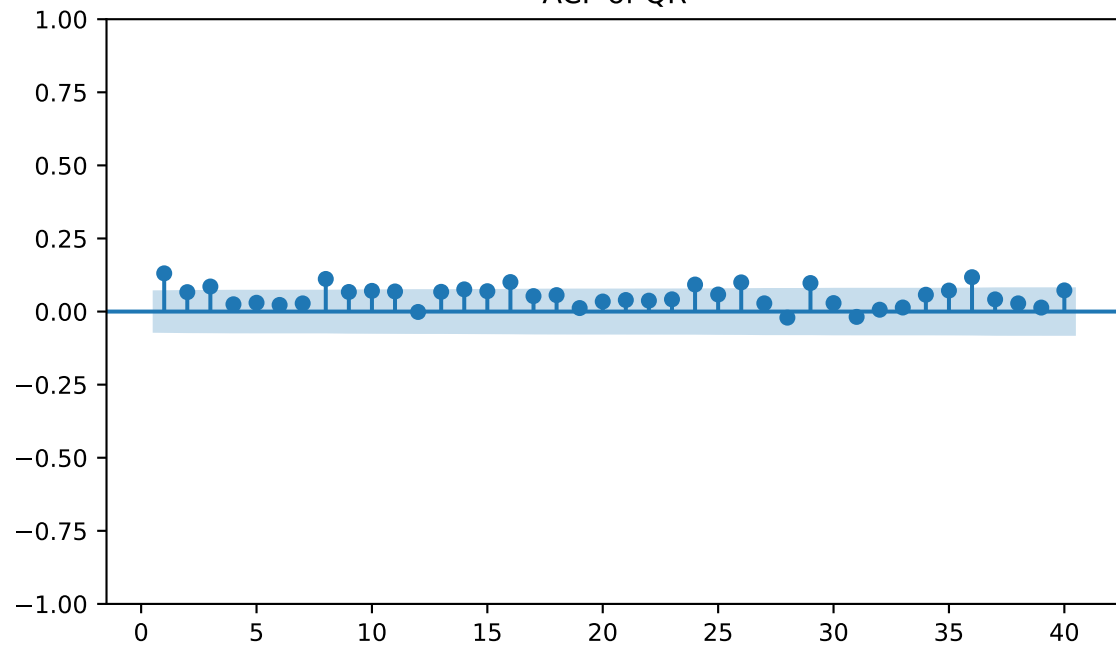
PIT histogram



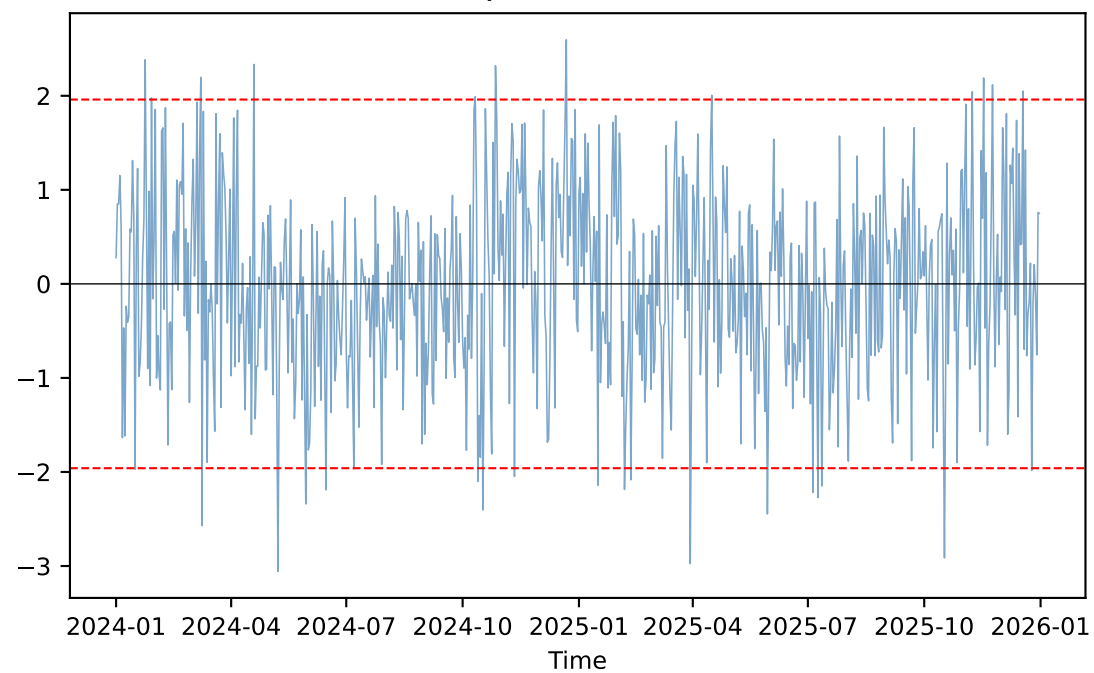
Normal QQ plot



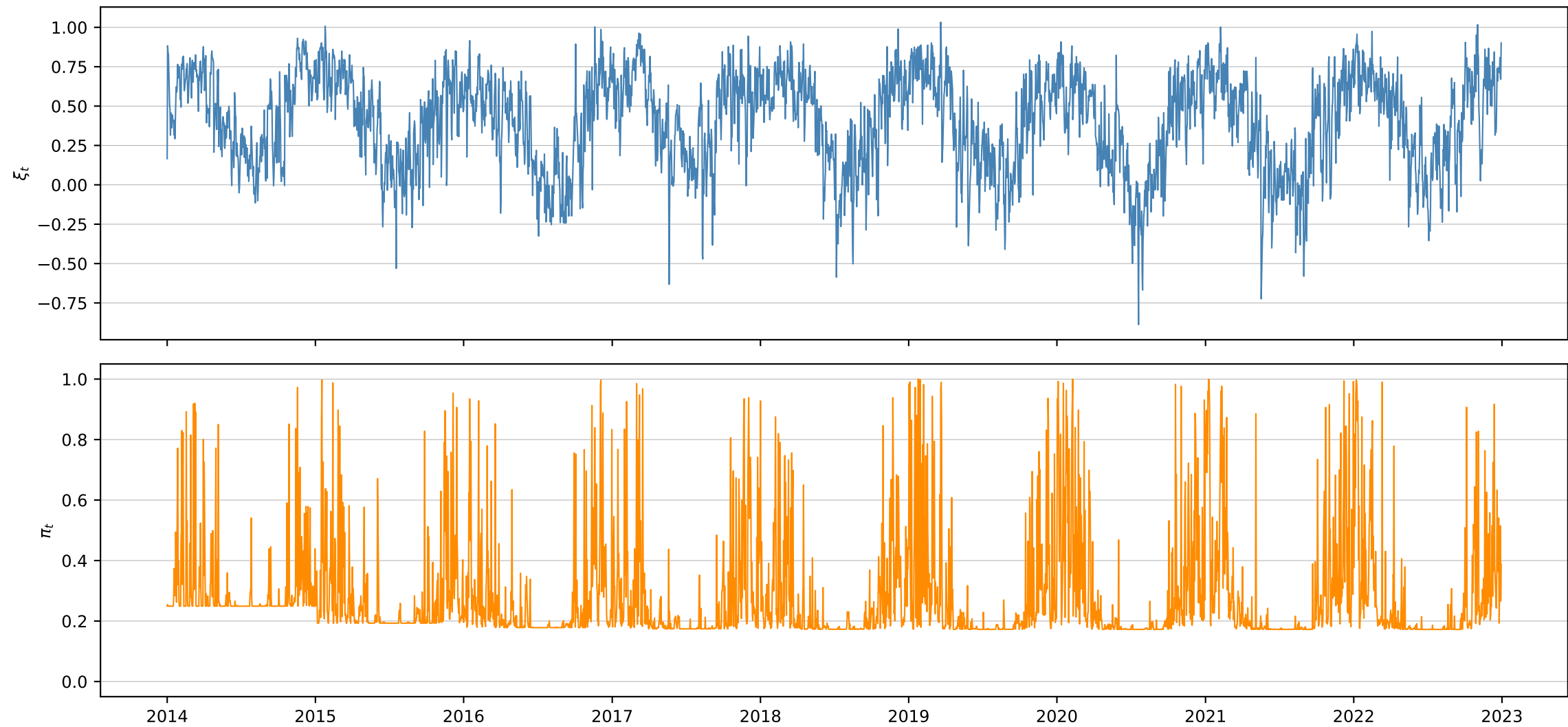
ACF of QR



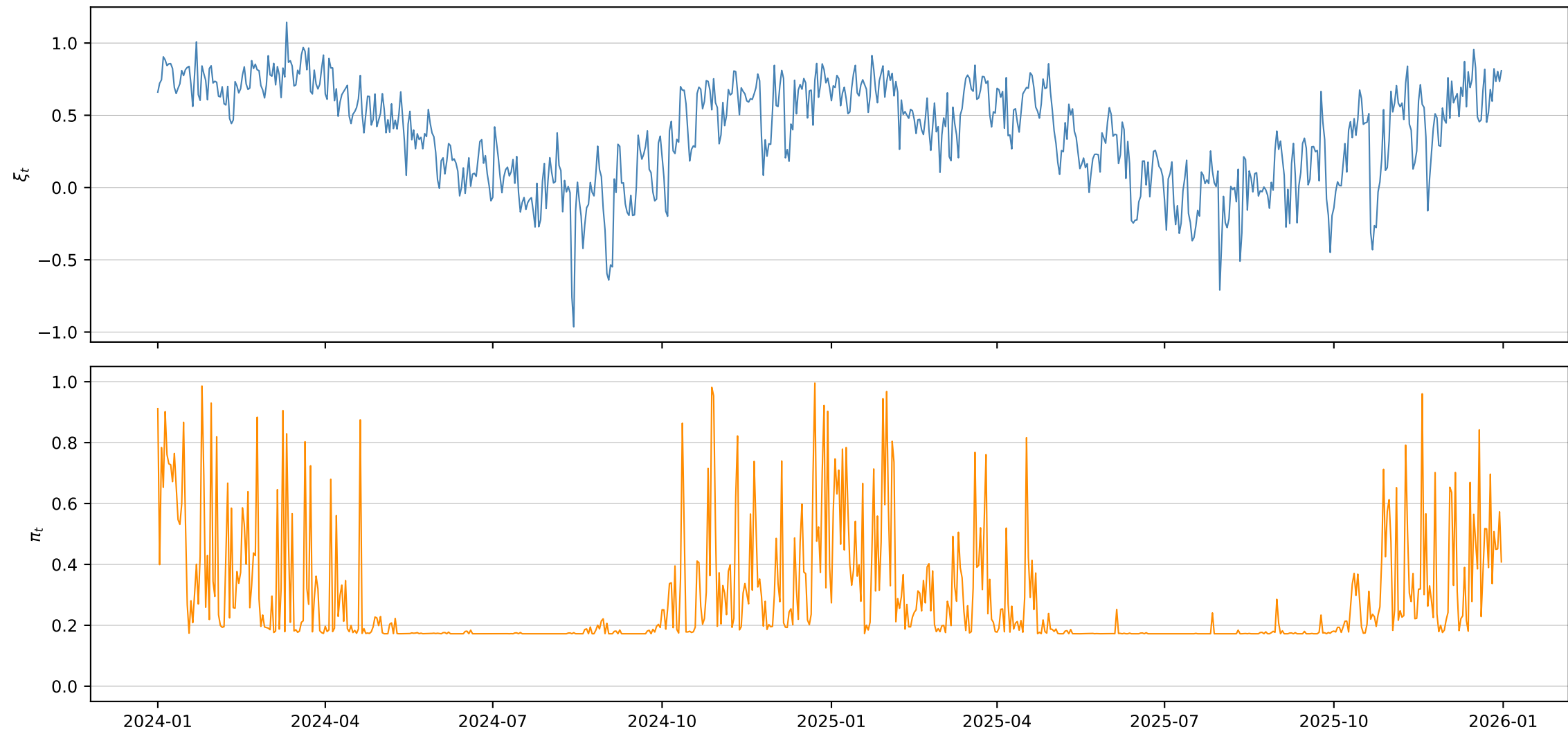
QR time series



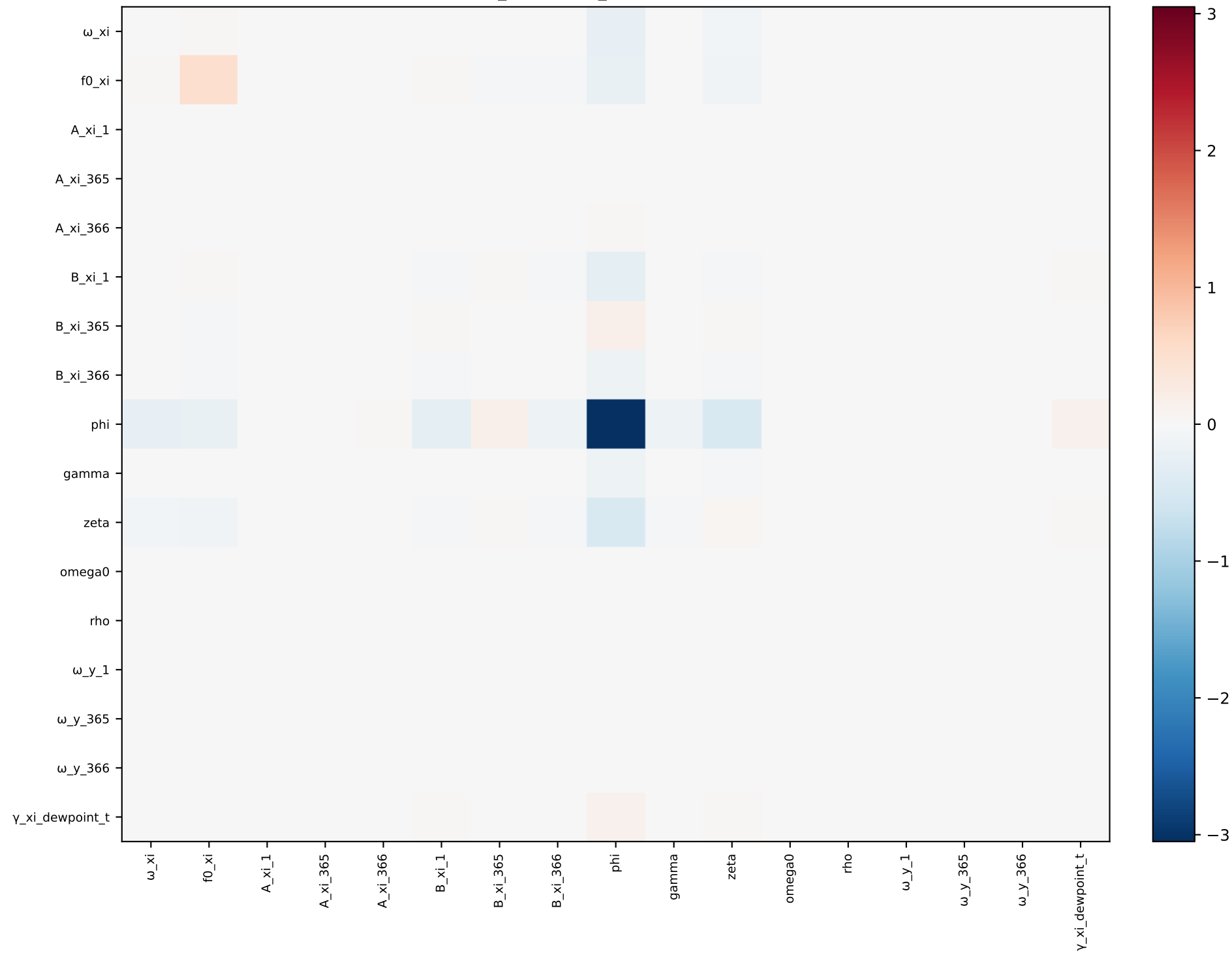
xi_dewpoint_t — filtered states (IS)



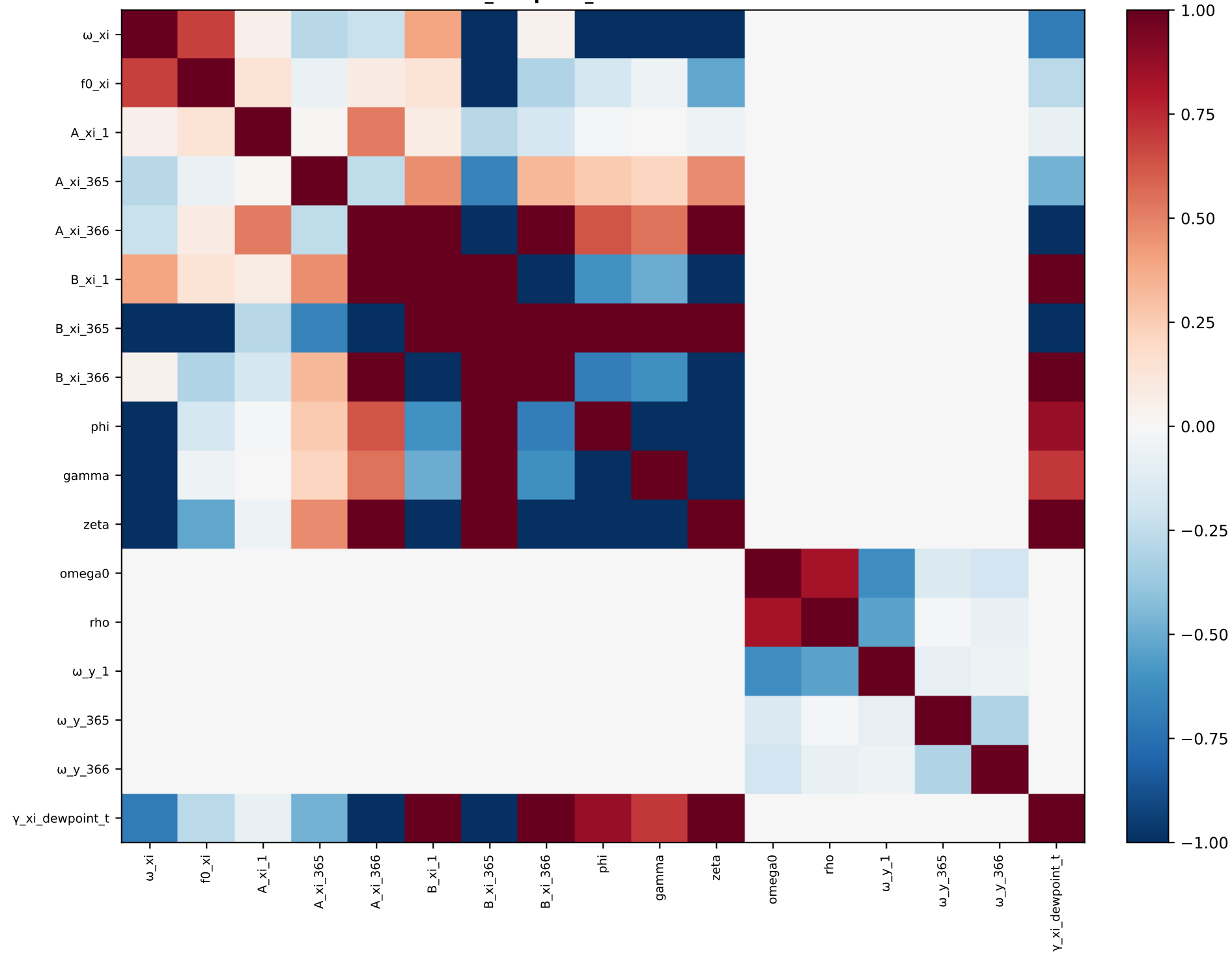
xi_dewpoint_t — filtered states (OOS)



xi_dewpoint_t — covariance

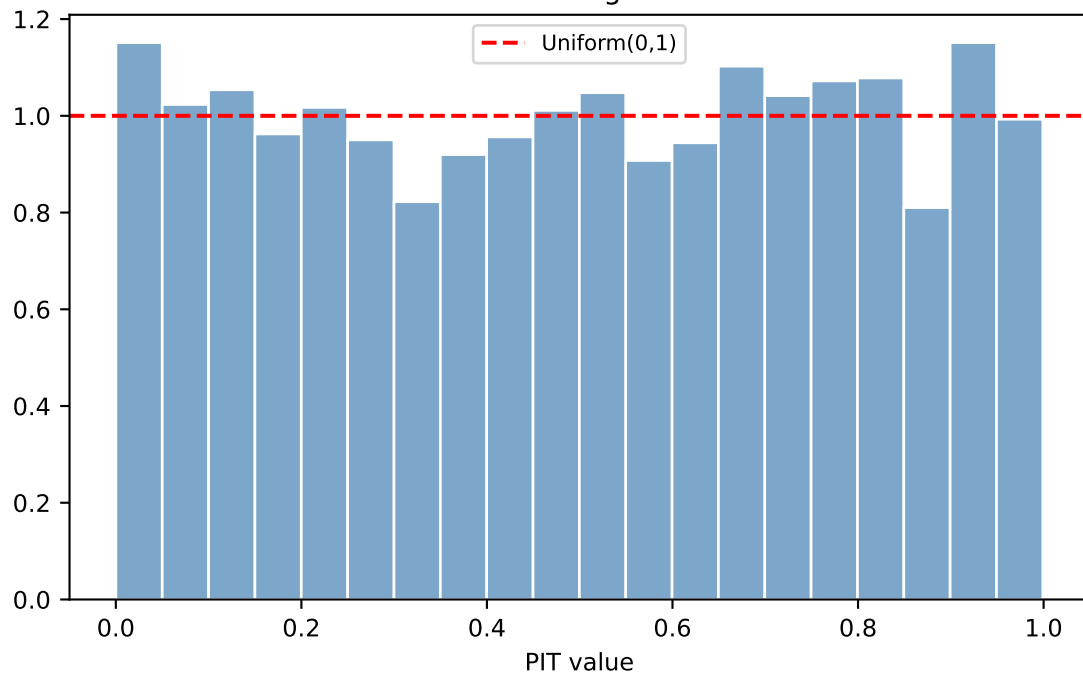


xi_dewpoint_t — correlation

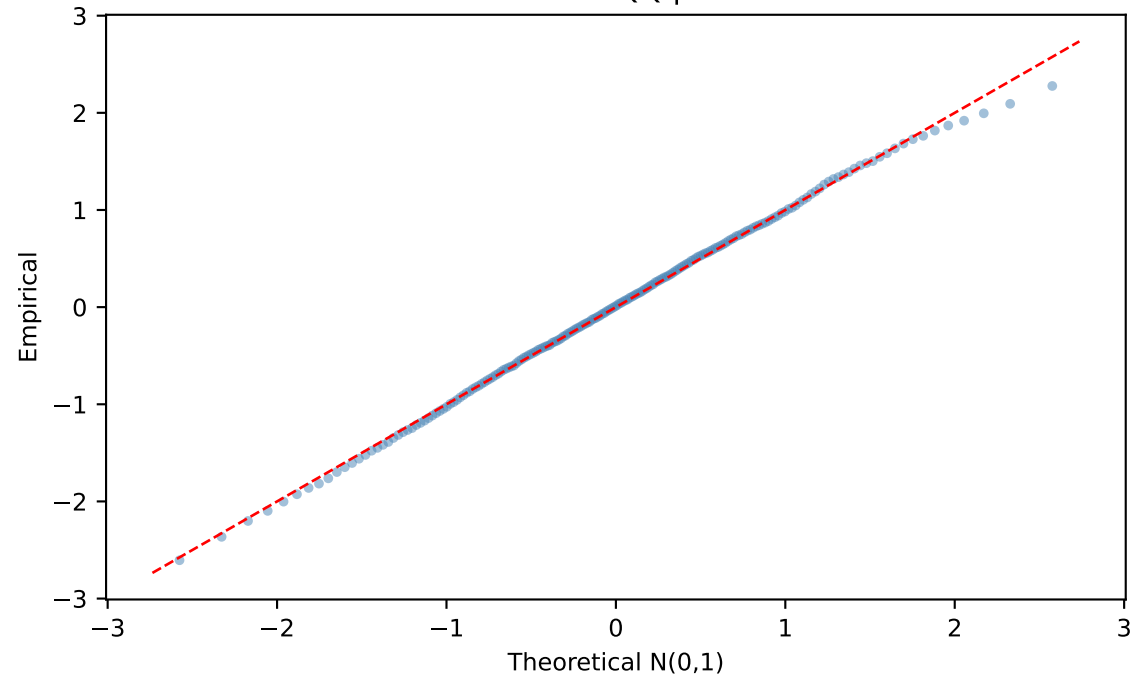


xi_dewpoint_lag1 — IS diagnostics

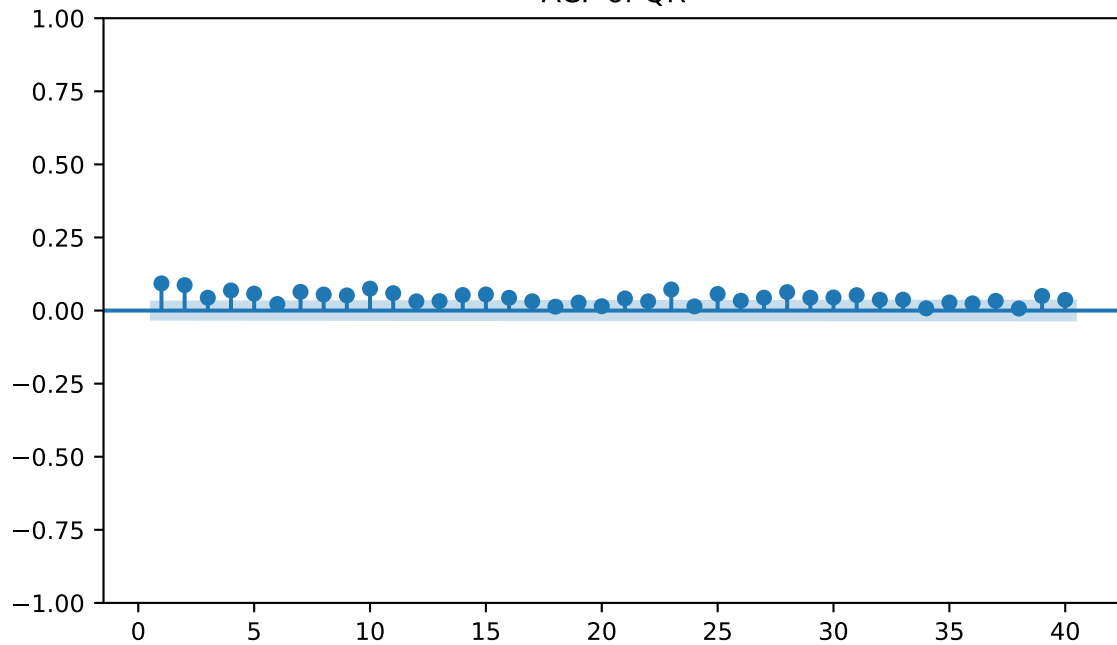
PIT histogram



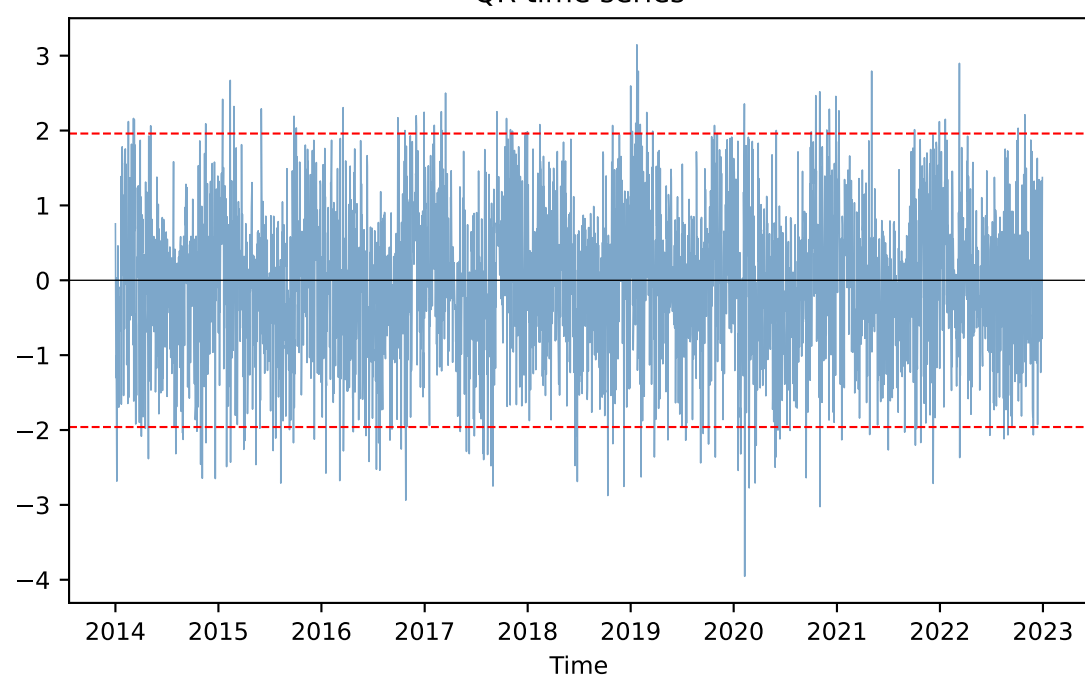
Normal QQ plot



ACF of QR

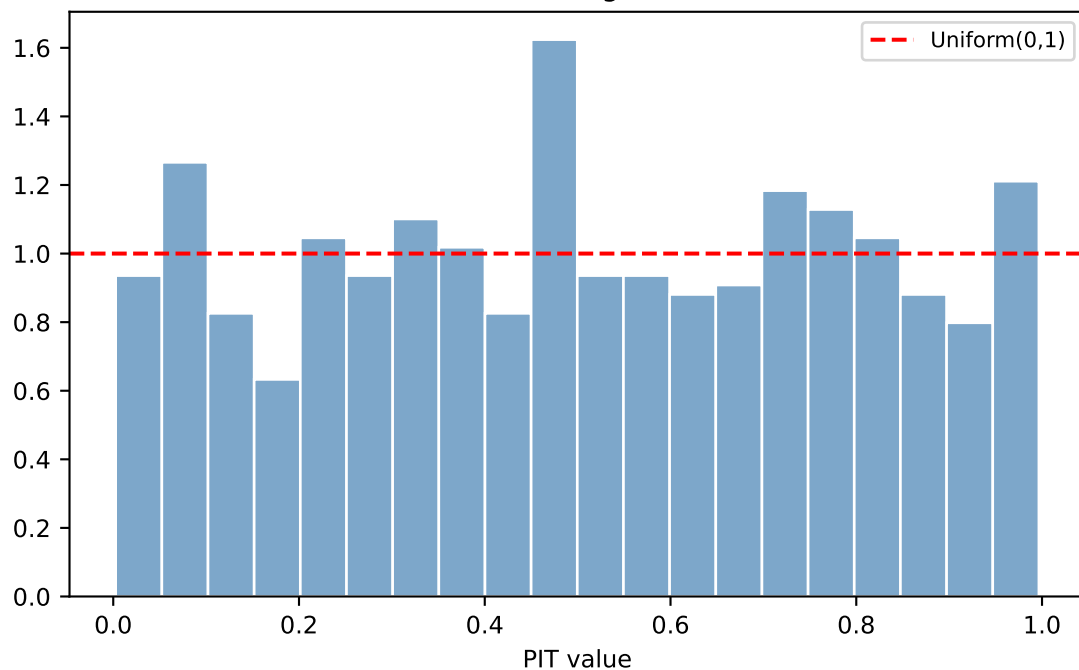


QR time series

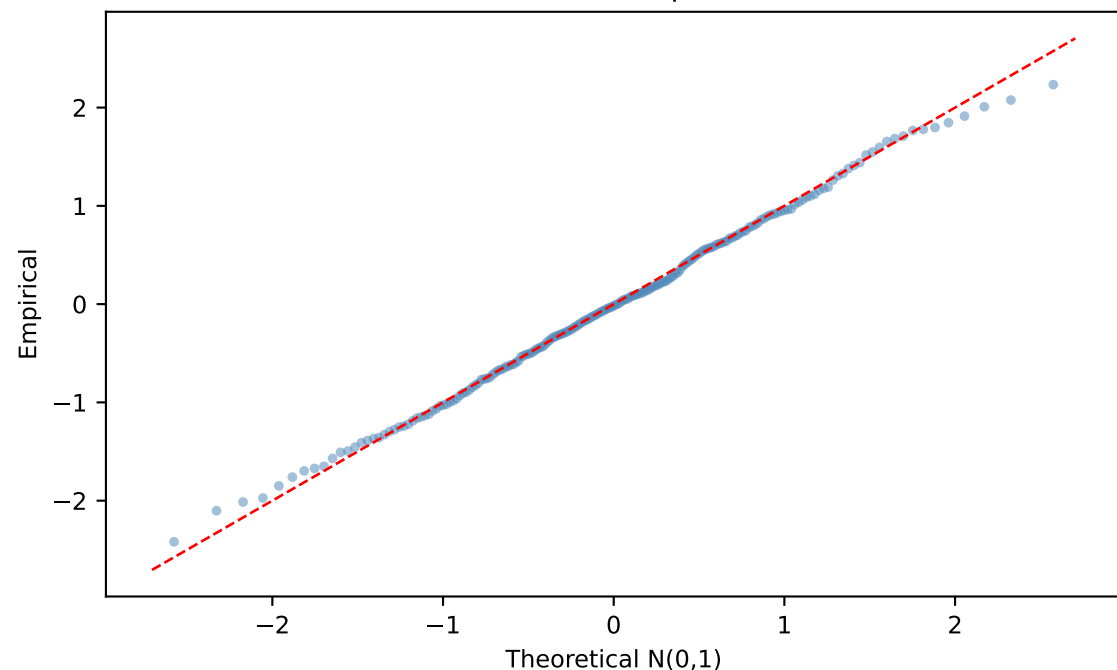


xi_dewpoint_lag1 — OOS diagnostics

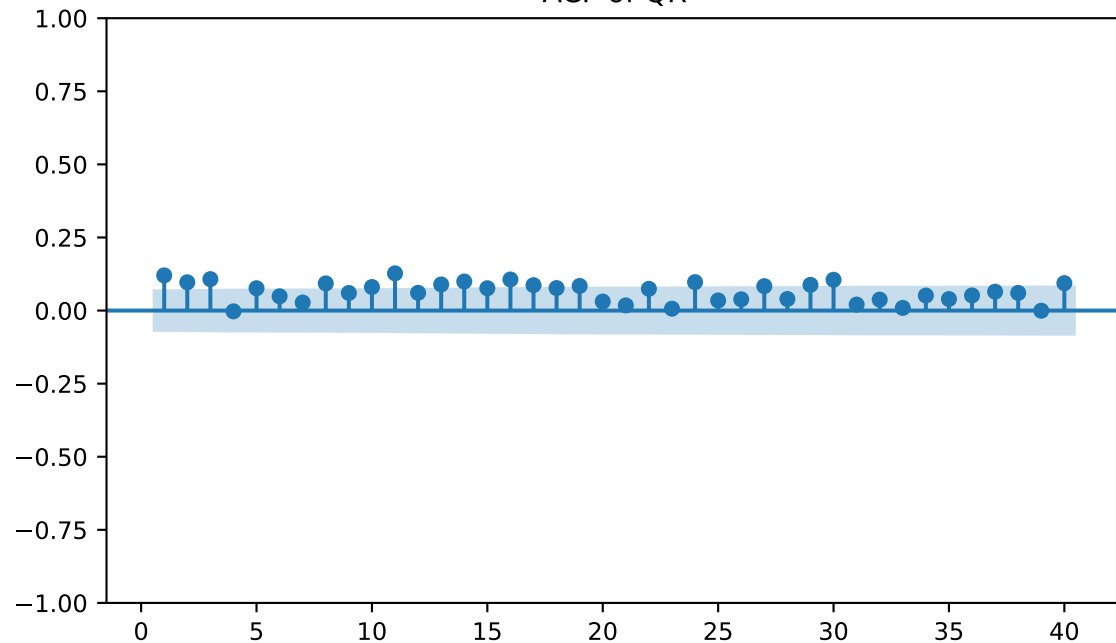
PIT histogram



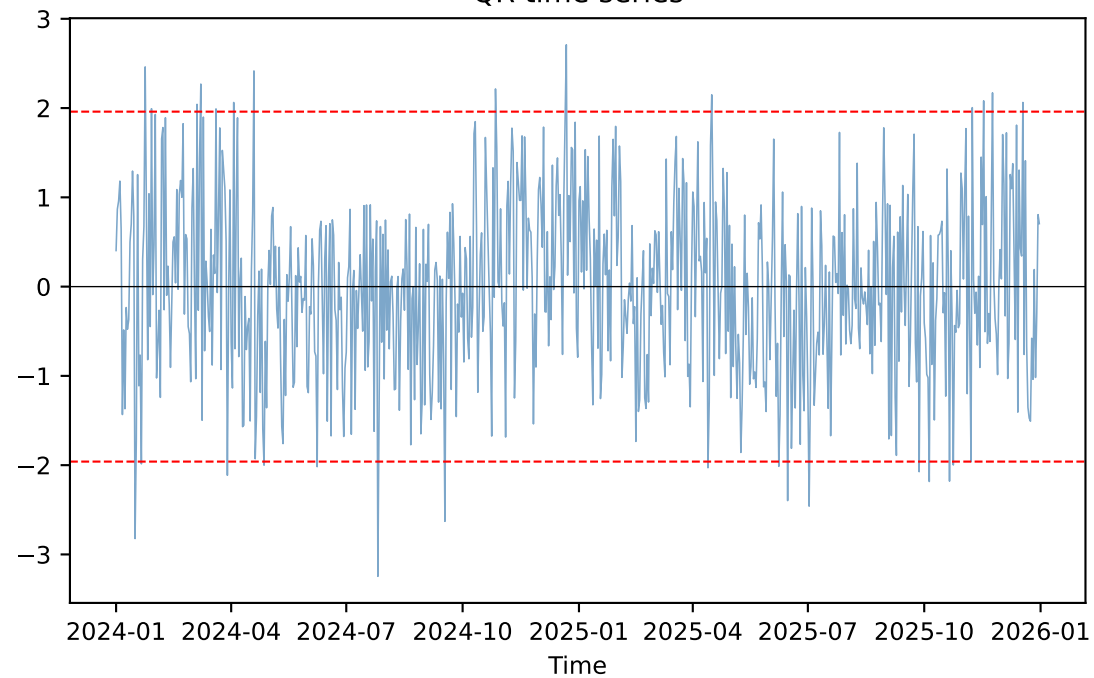
Normal QQ plot



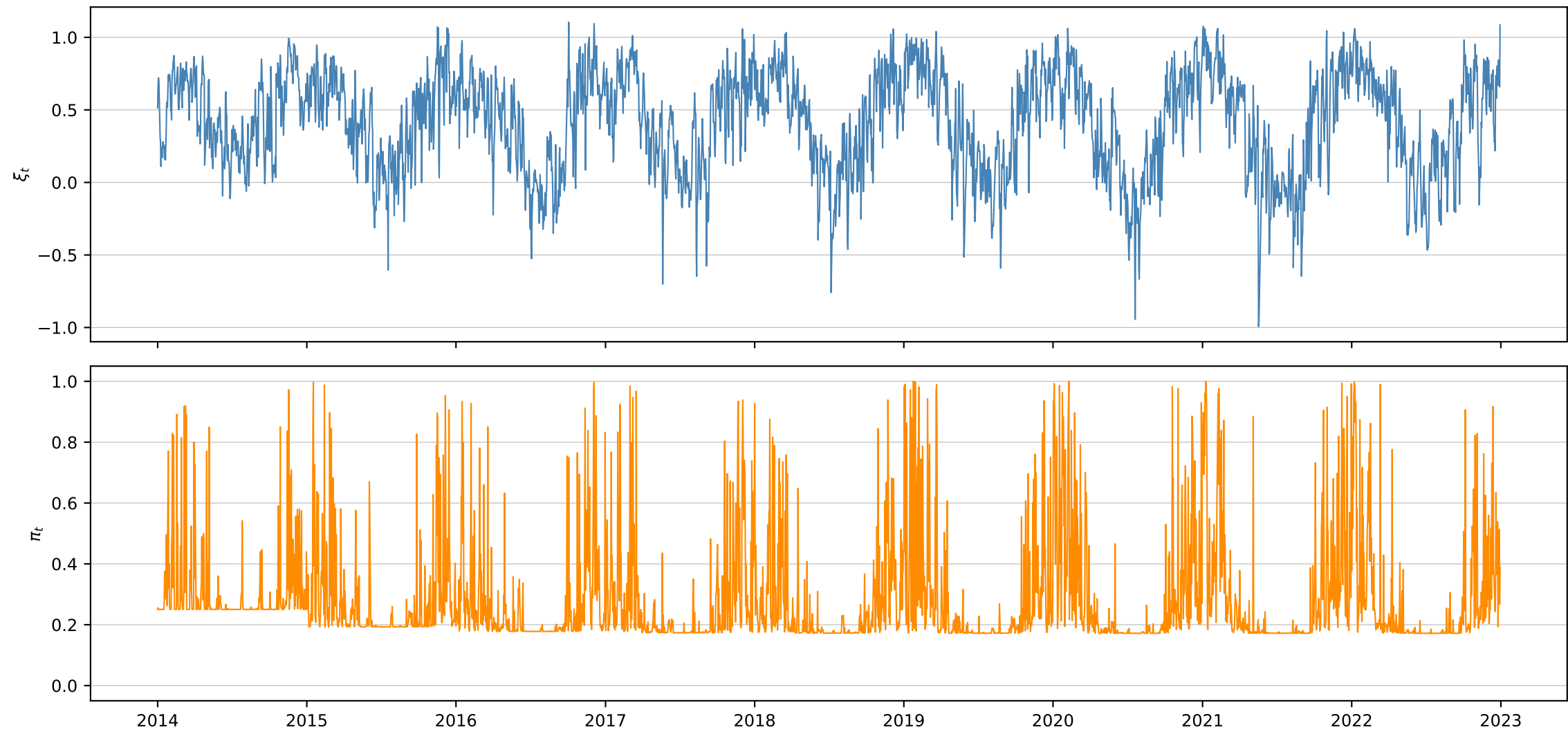
ACF of QR



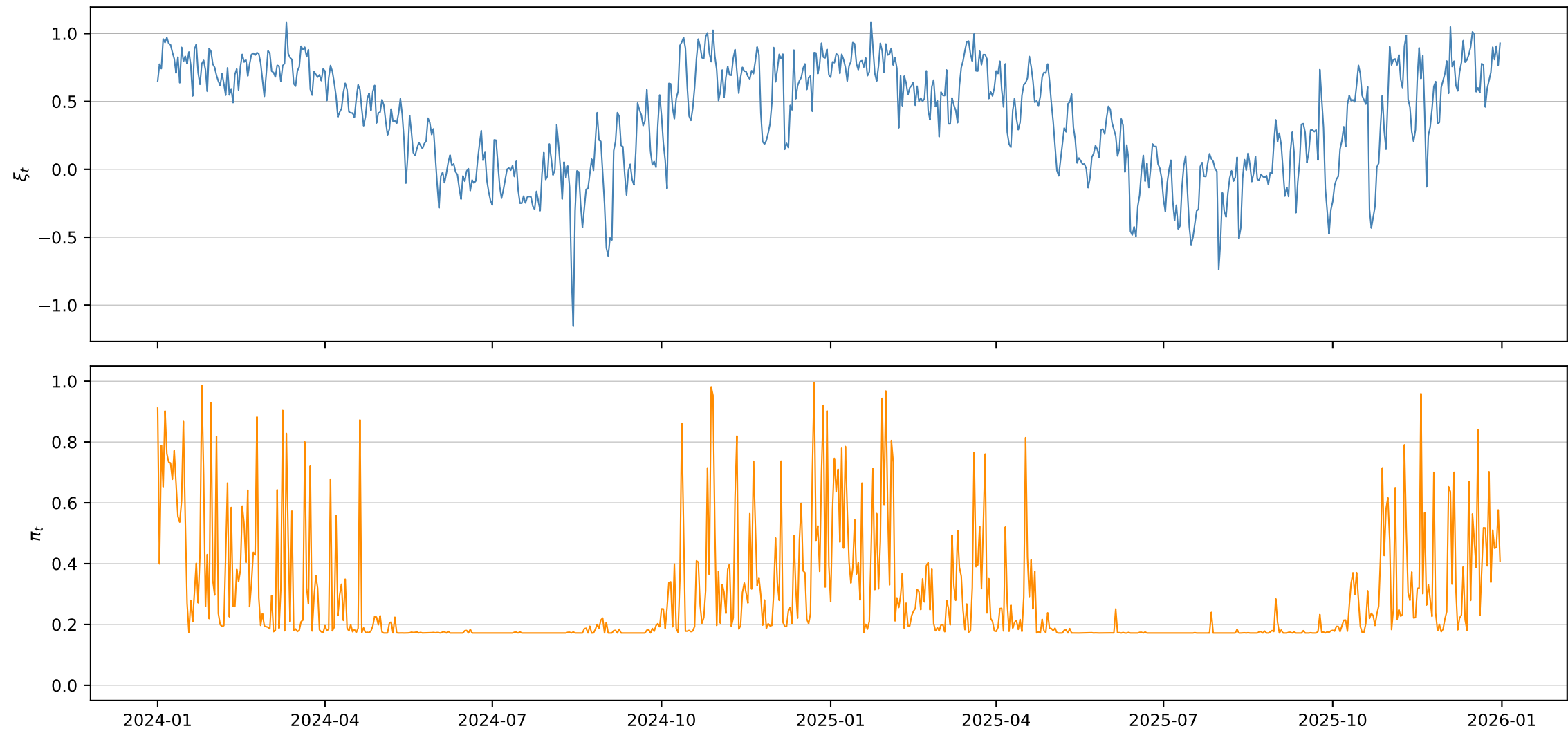
QR time series



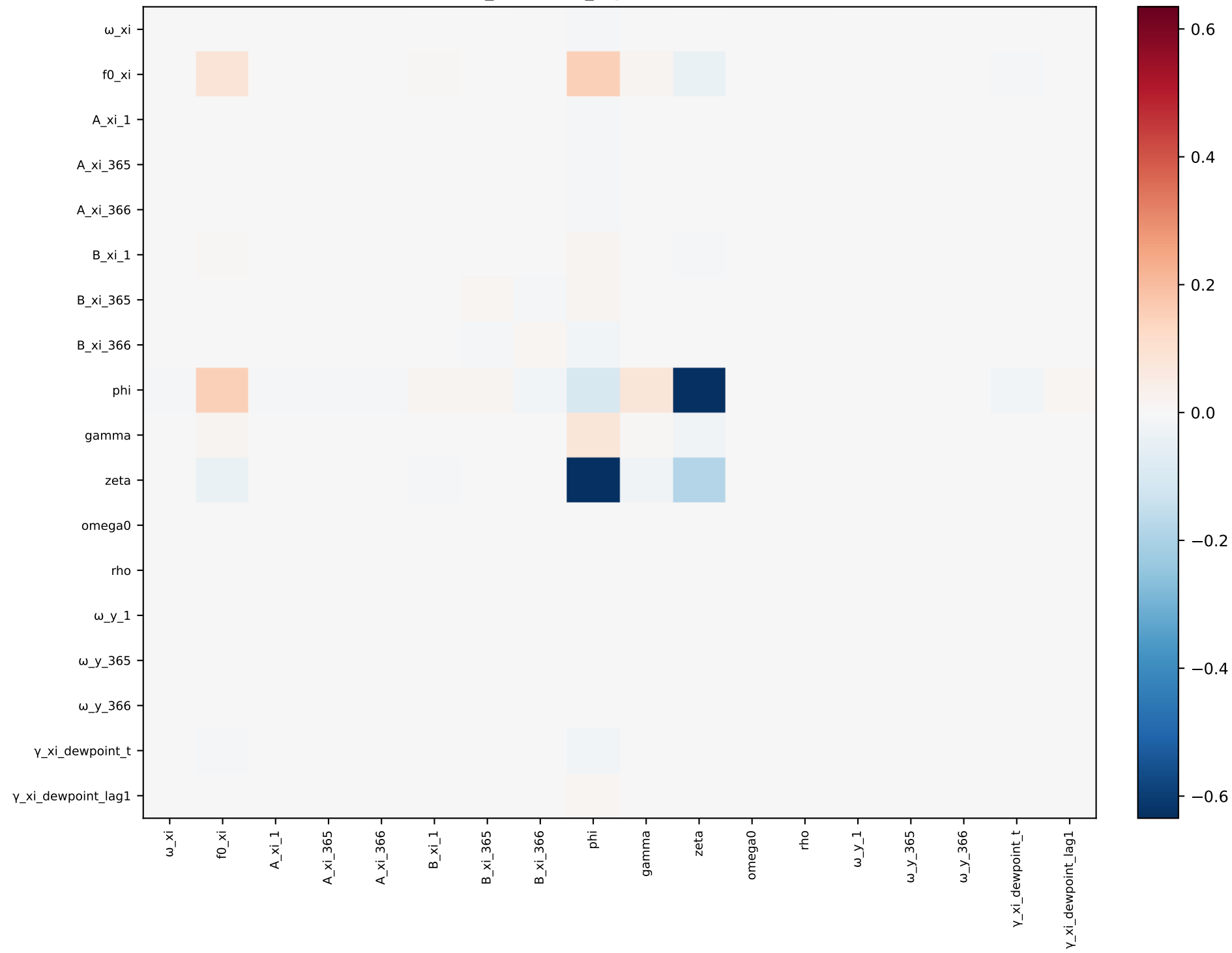
xi_dewpoint_lag1 — filtered states (IS)



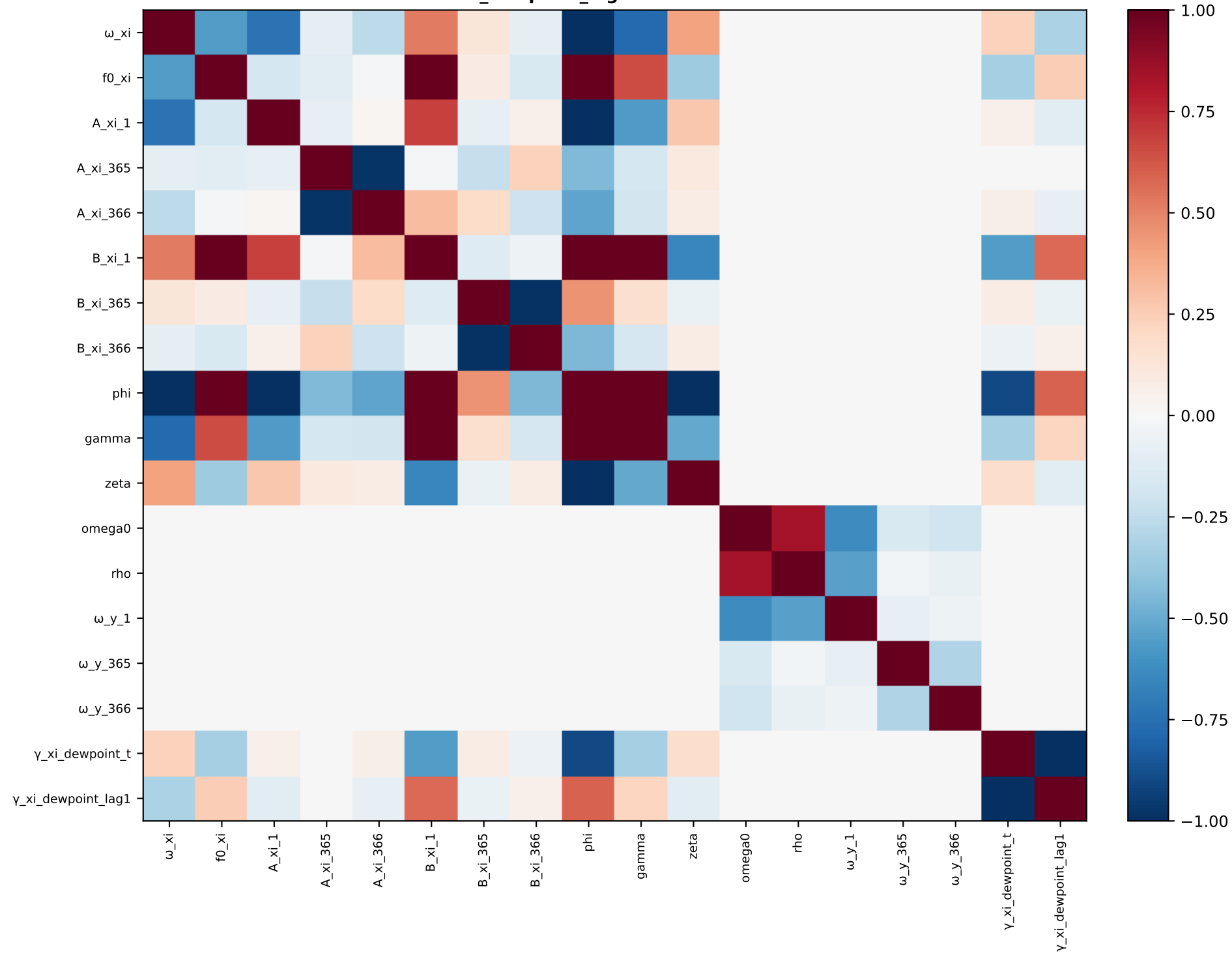
xi_dewpoint_lag1 — filtered states (OOS)



xi_dewpoint_lag1 — covariance

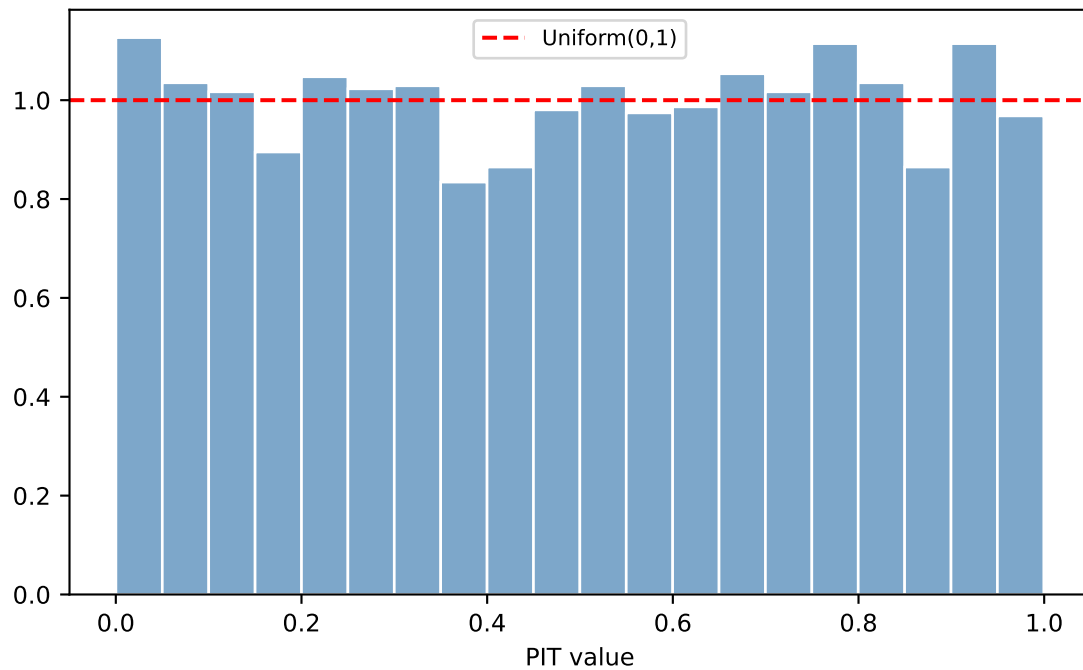


xi_dewpoint_lag1 — correlation

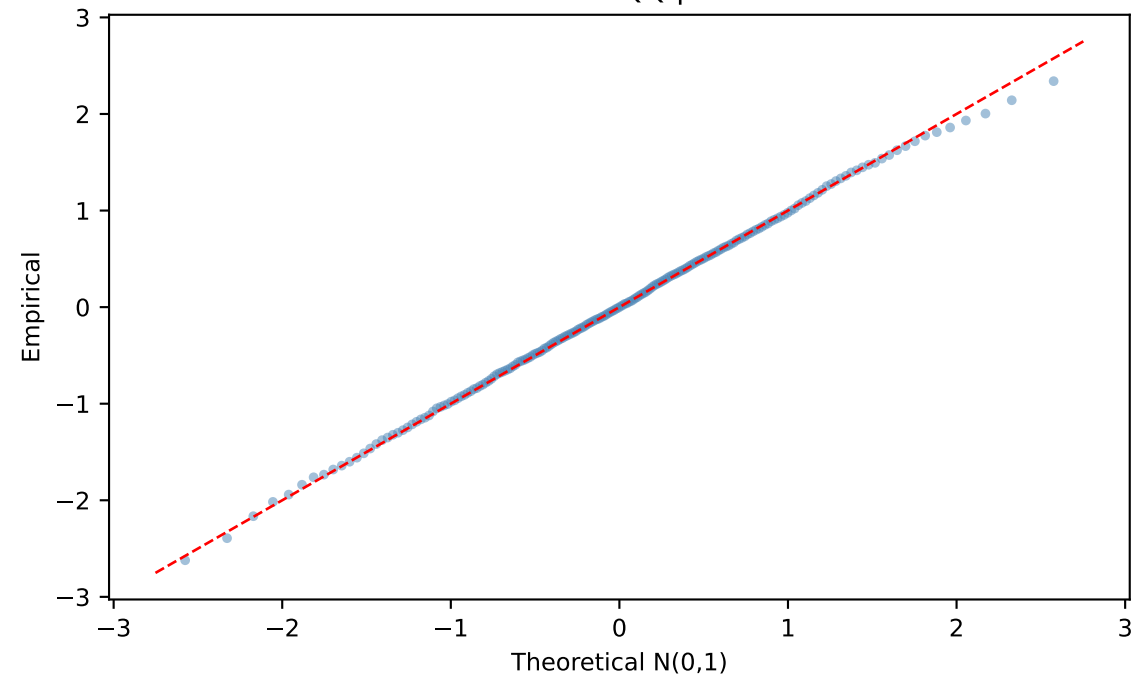


xi_dewpoint_short_lags — IS diagnostics

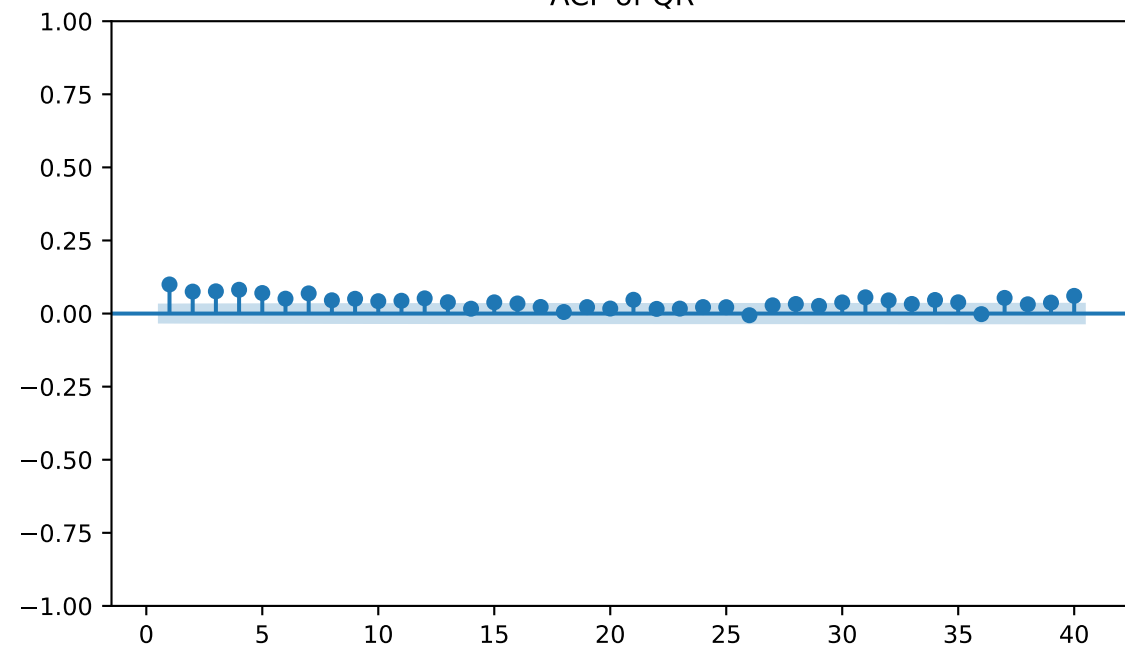
PIT histogram



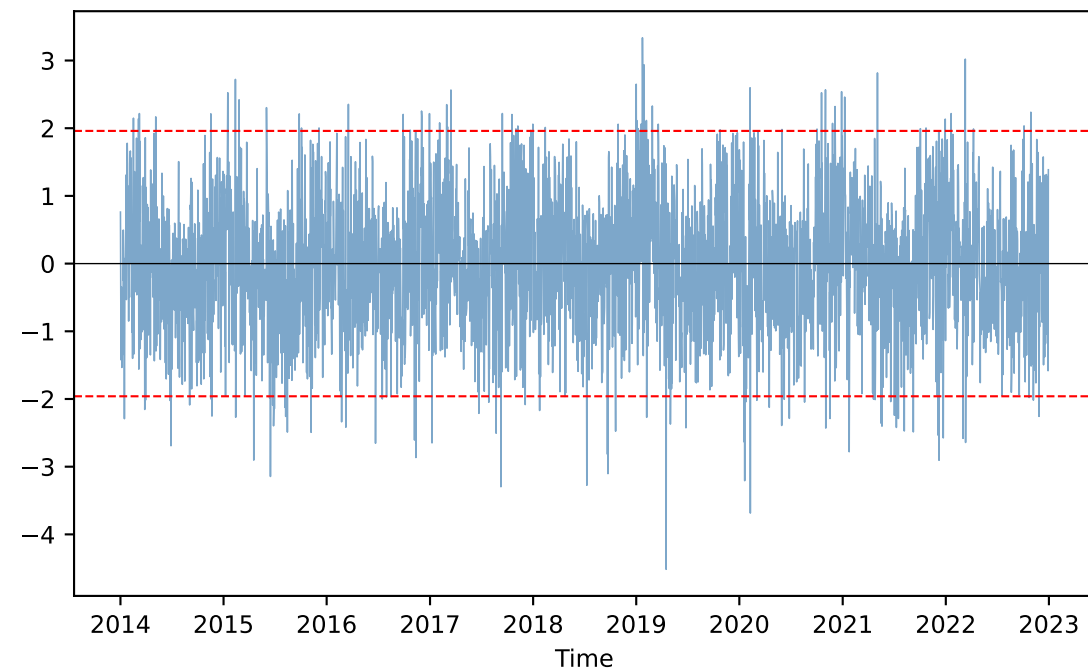
Normal QQ plot



ACF of QR

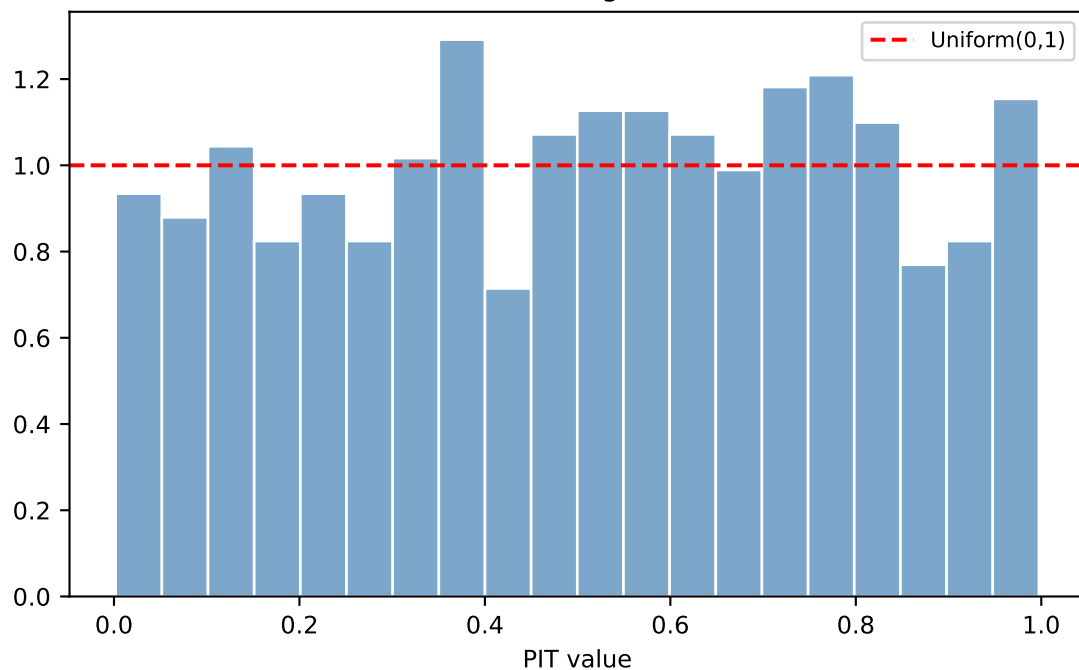


QR time series

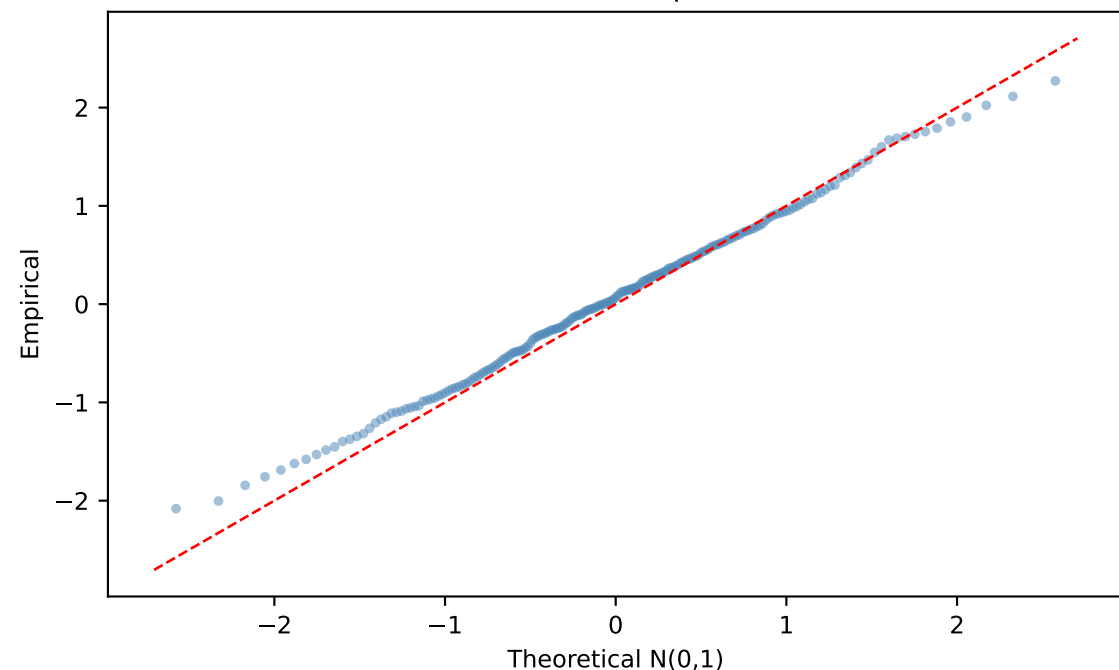


xi_dewpoint_short_lags — OOS diagnostics

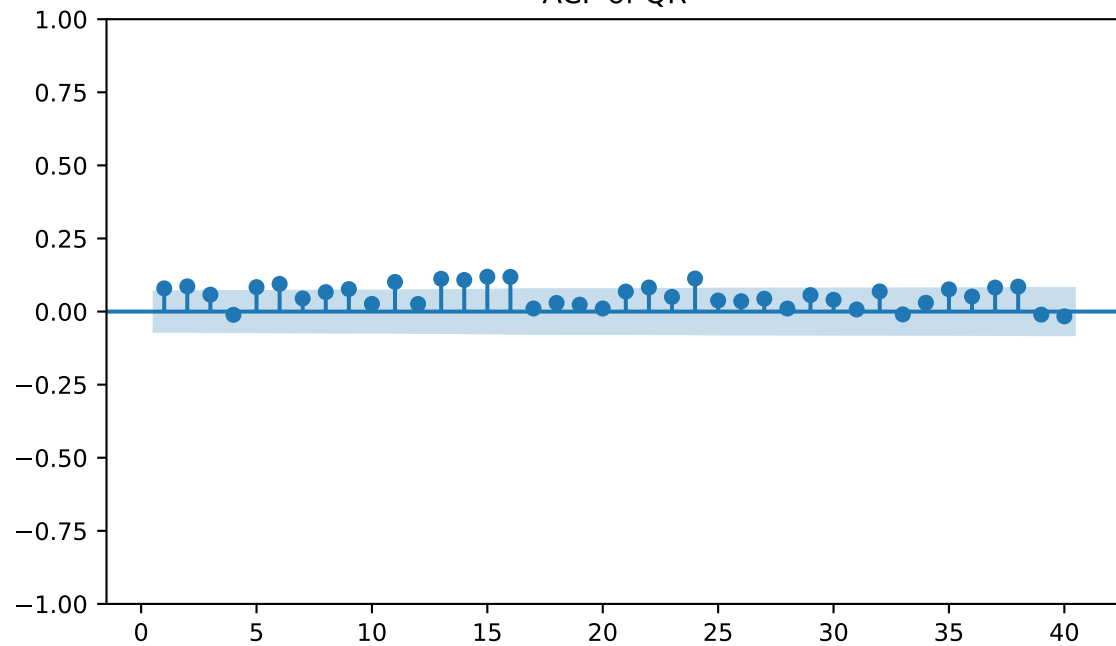
PIT histogram



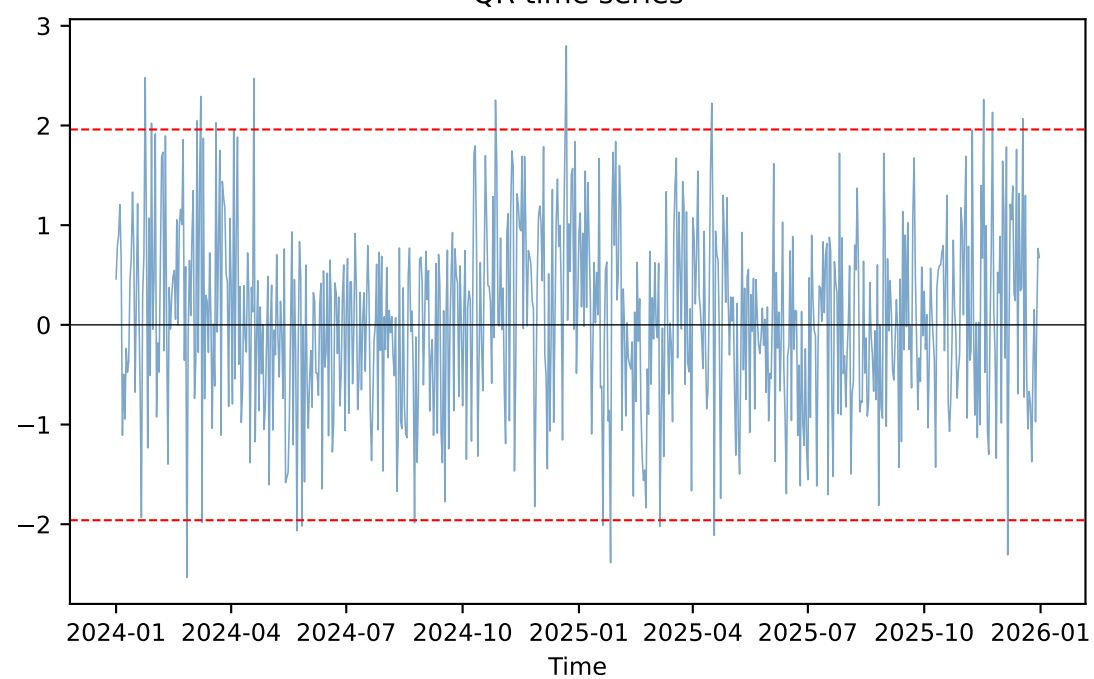
Normal QQ plot



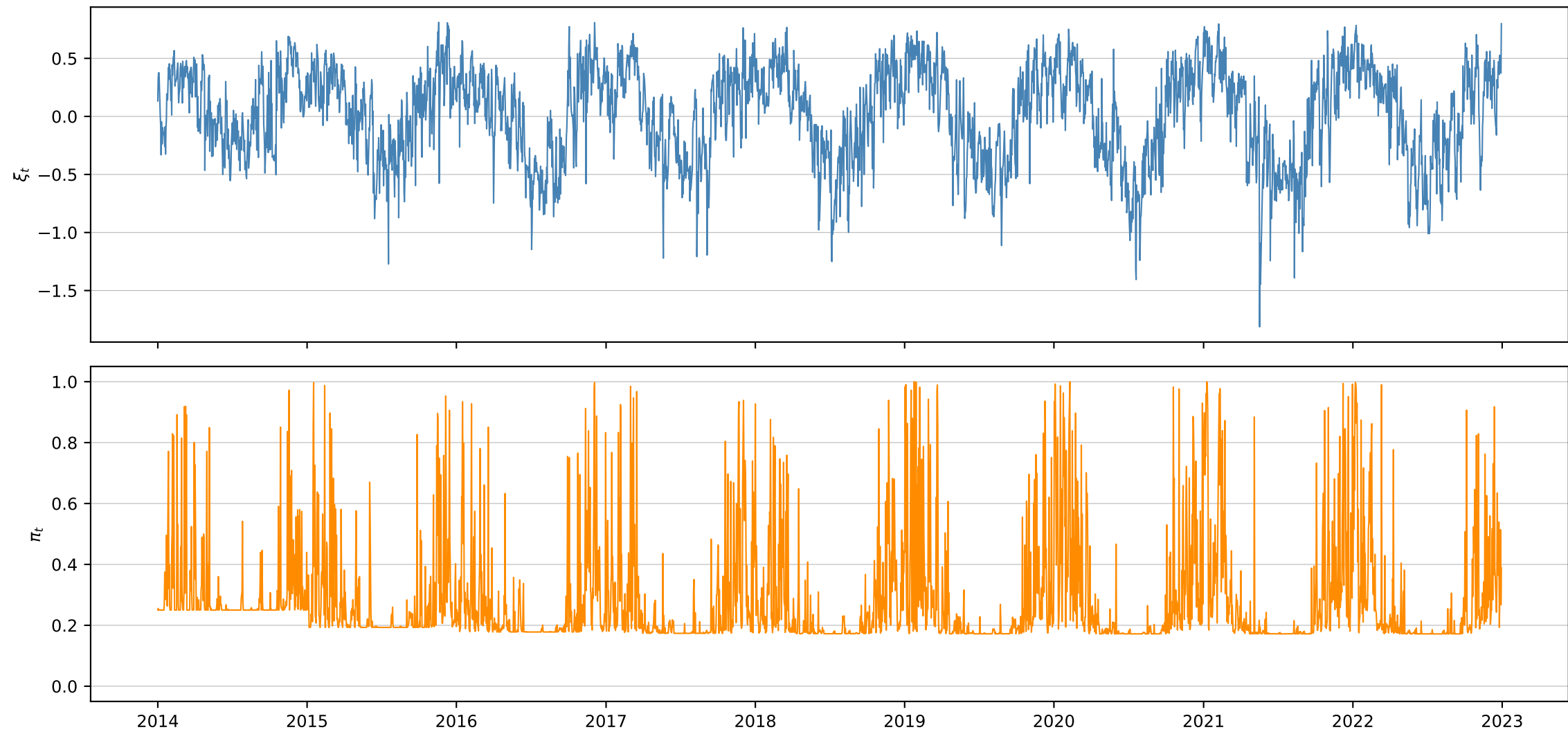
ACF of QR



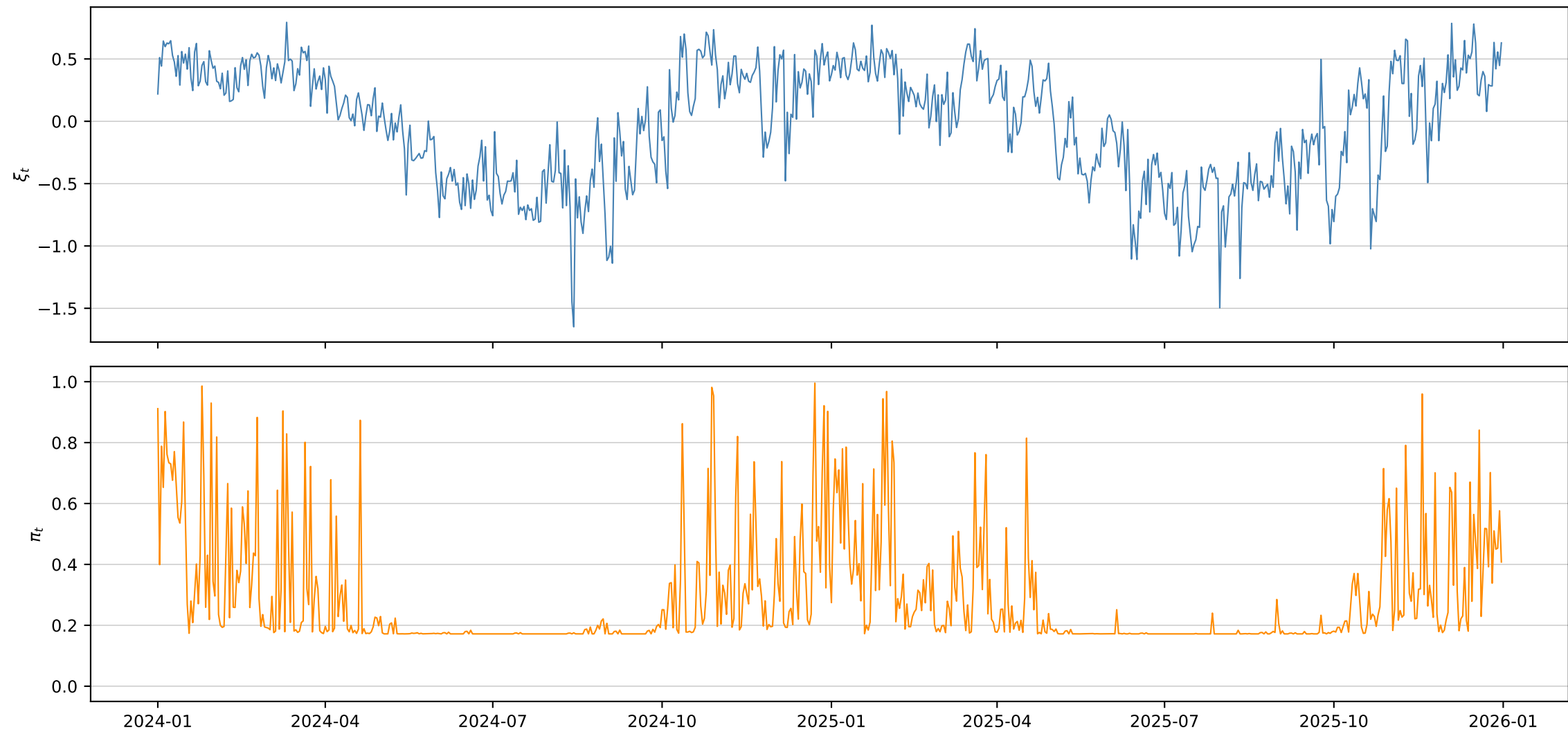
QR time series



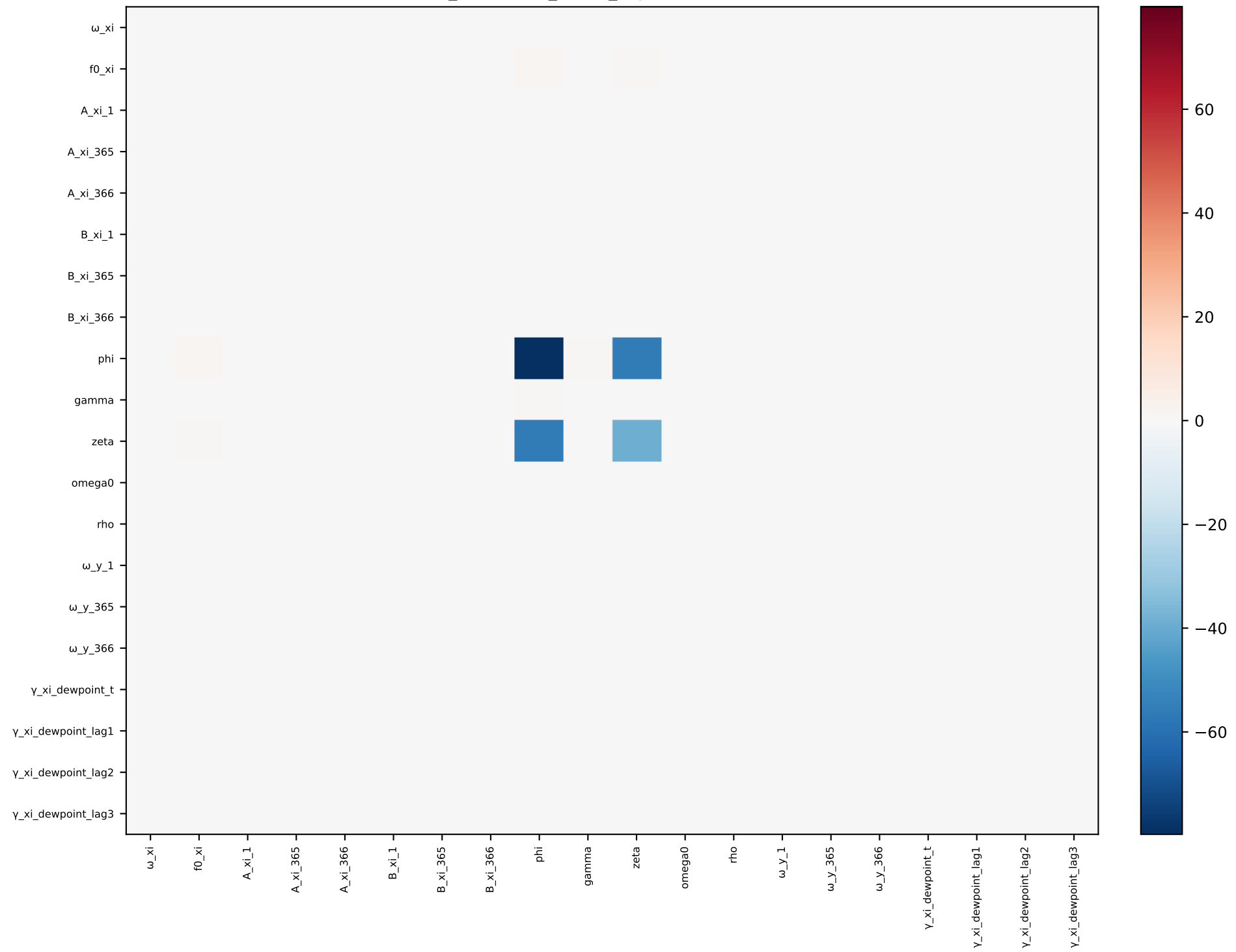
xi_dewpoint_short_lags — filtered states (IS)



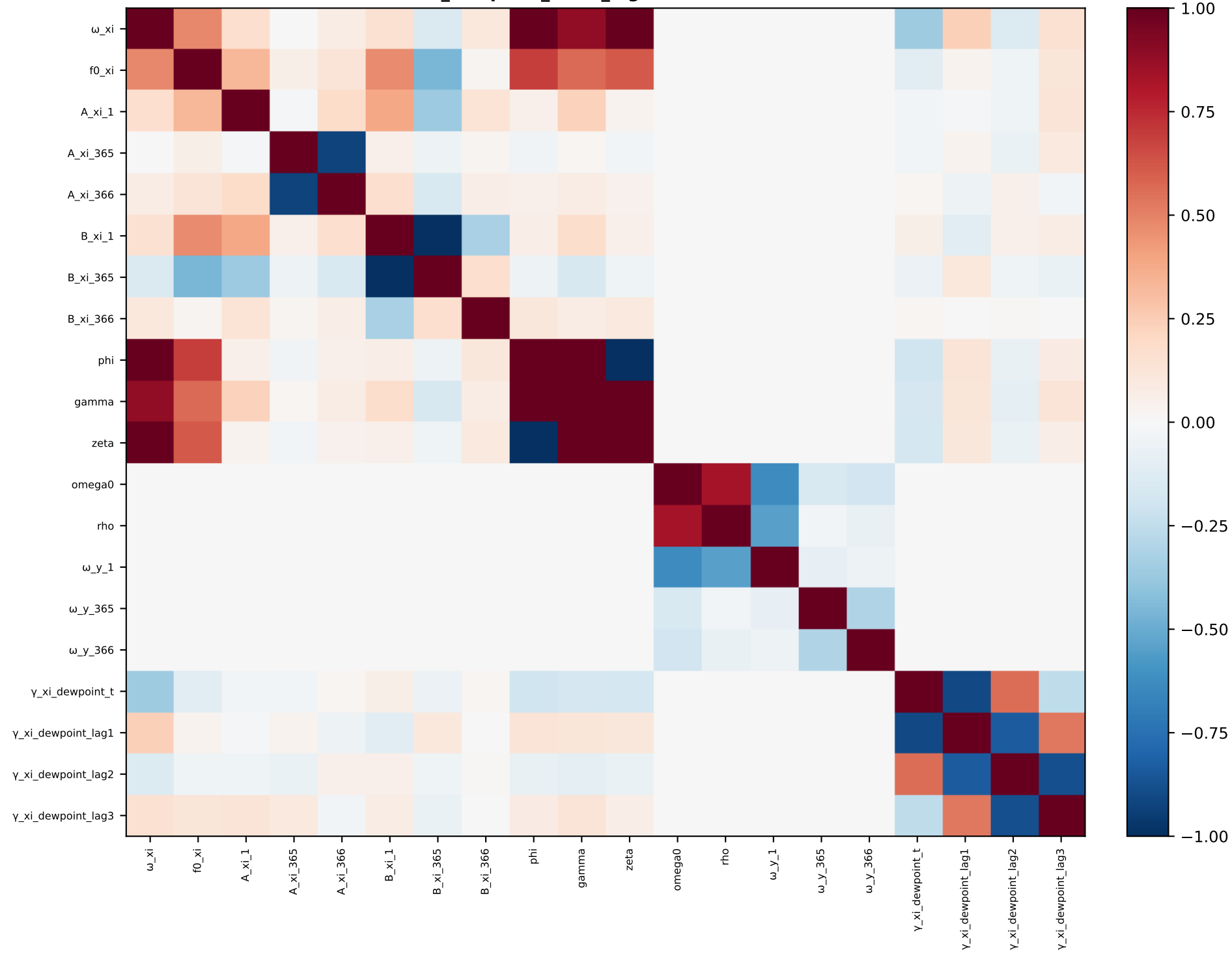
xi_dewpoint_short_lags — filtered states (OOS)



xi_dewpoint_short_lags — covariance

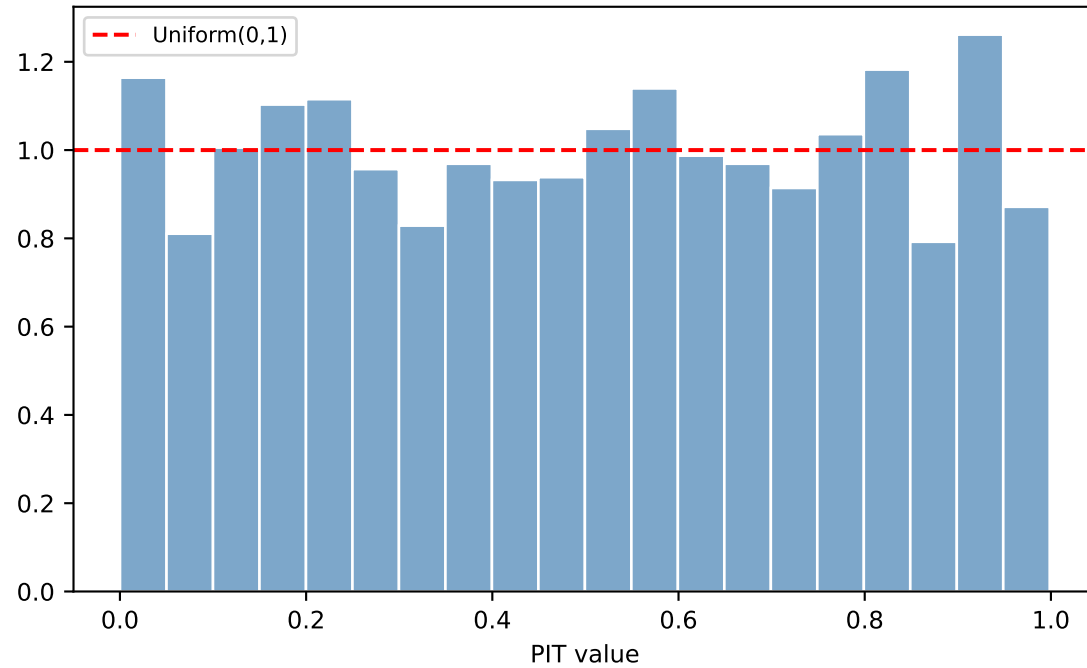


xi_dewpoint_short_lags — correlation

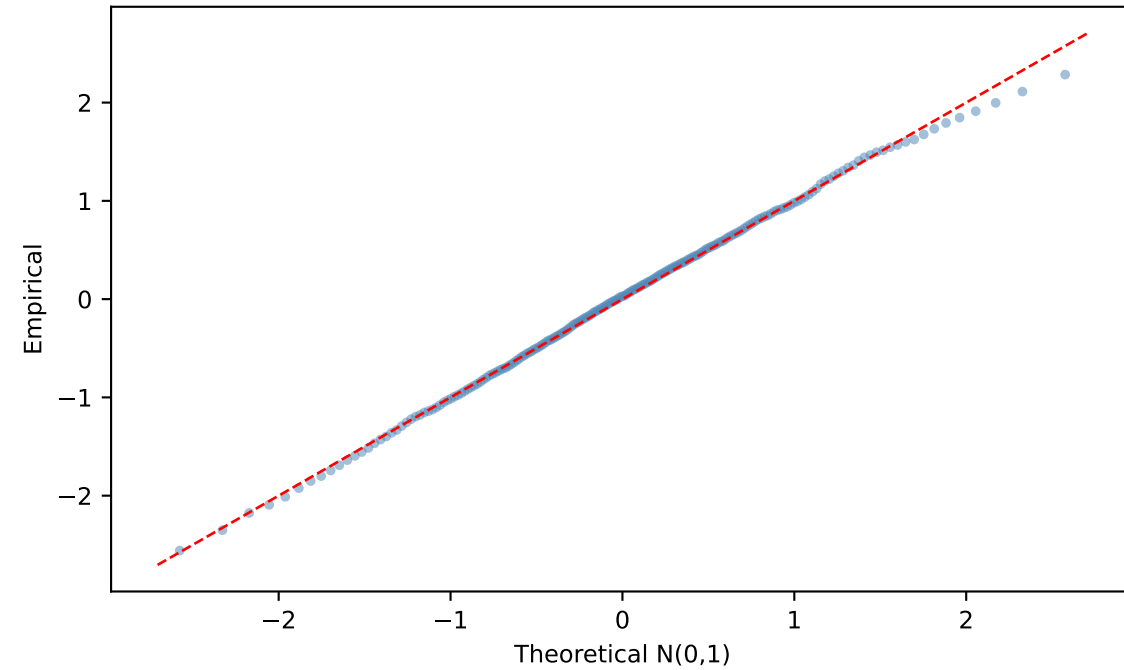


xi_dewpoint_seasonal_lags — IS diagnostics

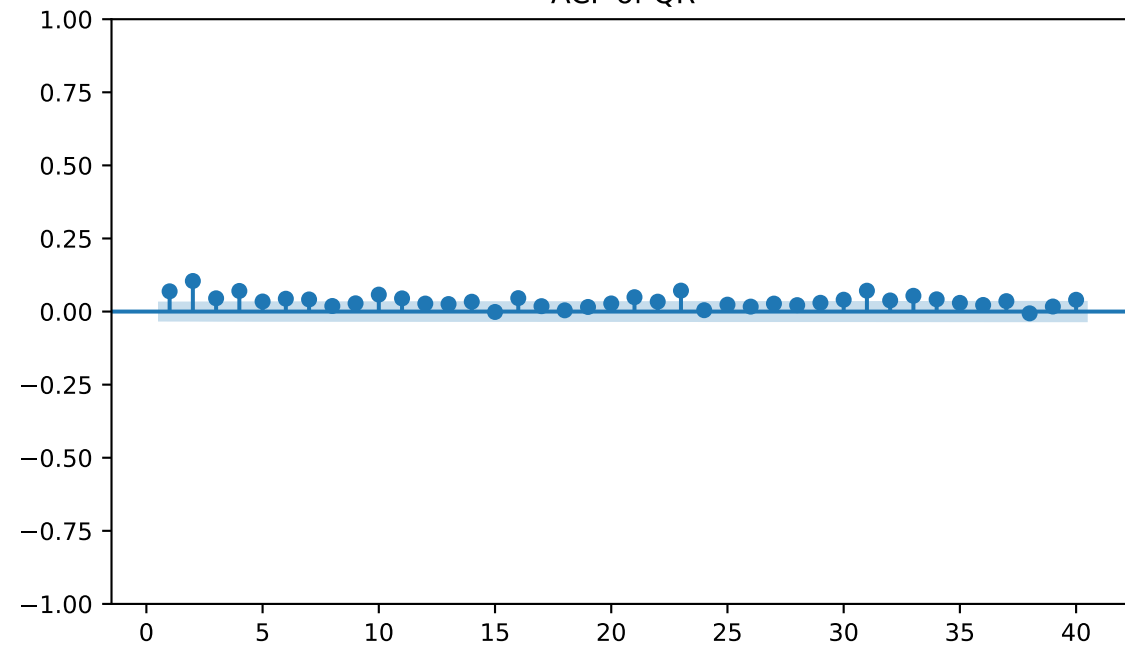
PIT histogram



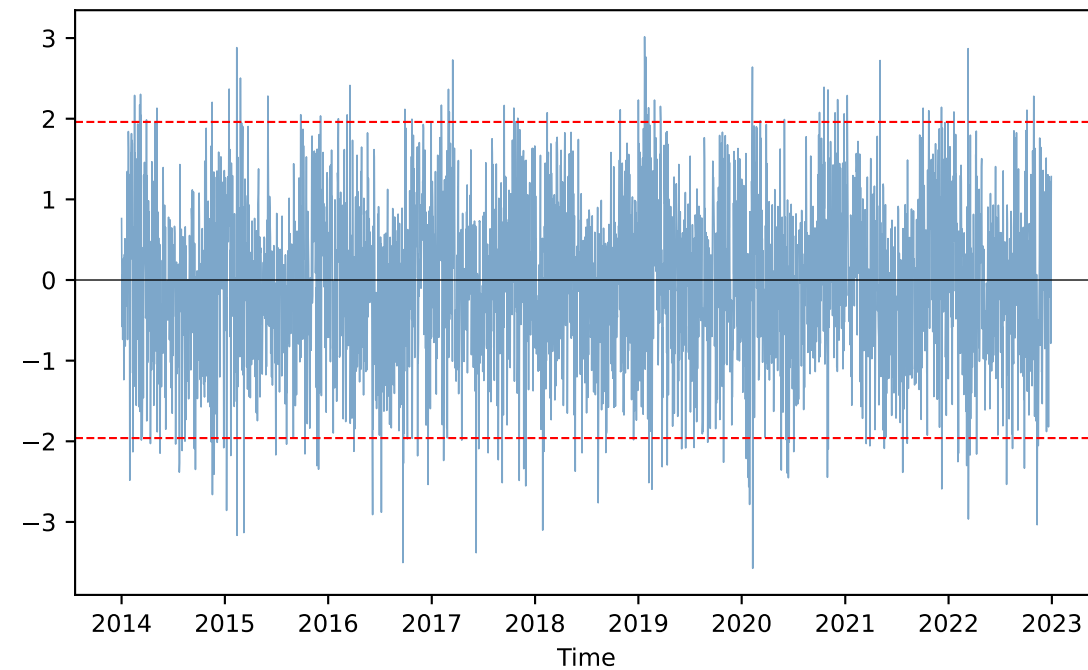
Normal QQ plot



ACF of QR

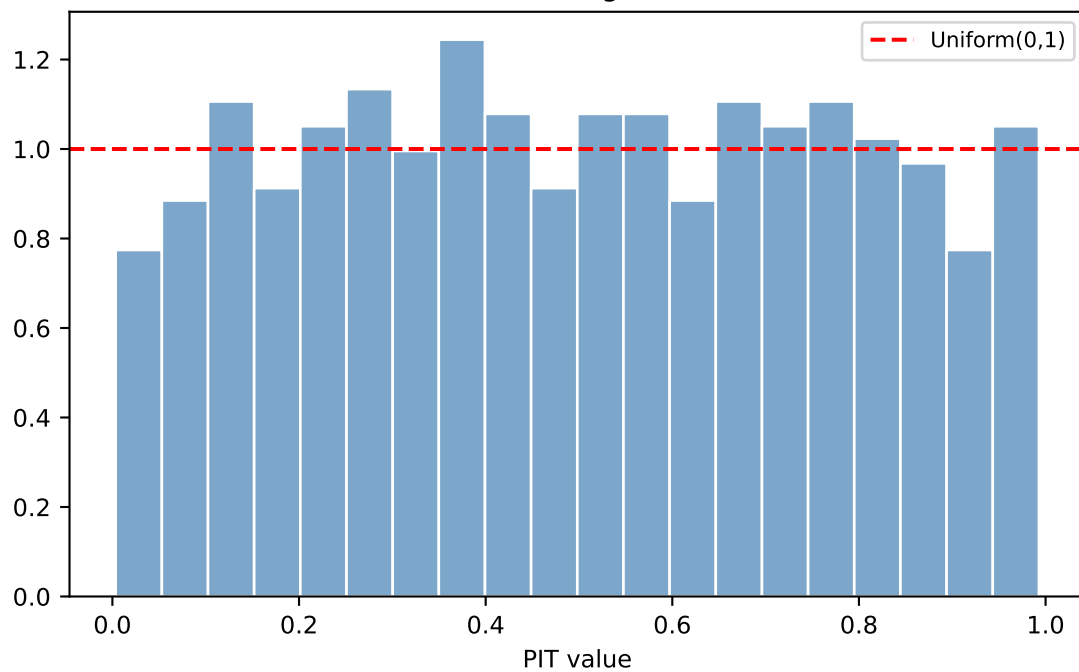


QR time series

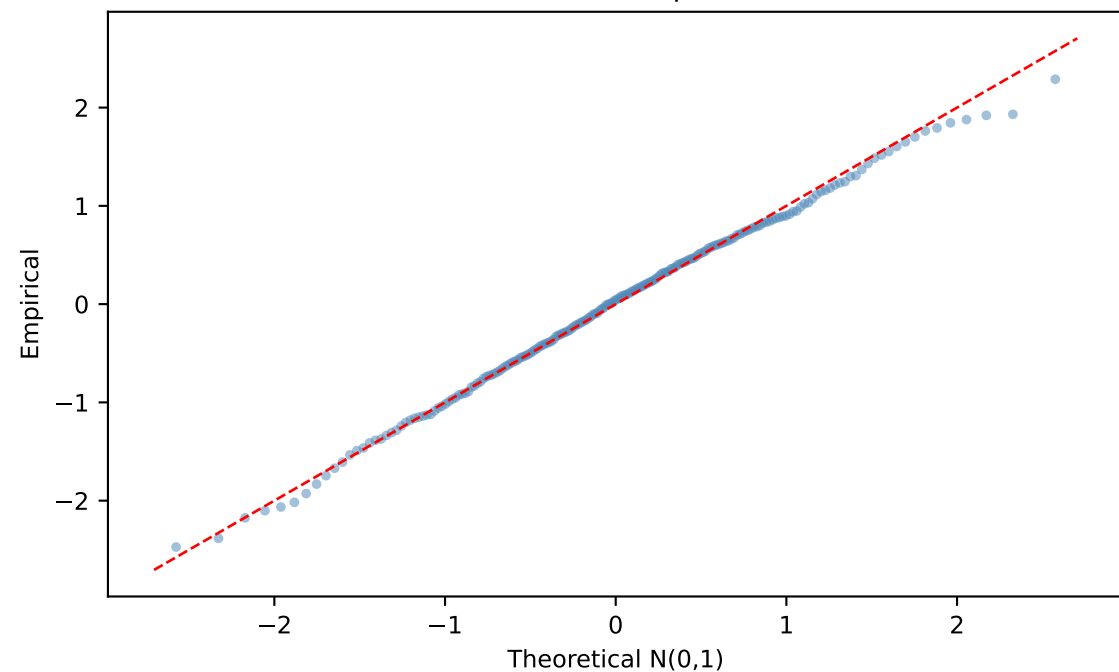


xi_dewpoint_seasonal_lags — OOS diagnostics

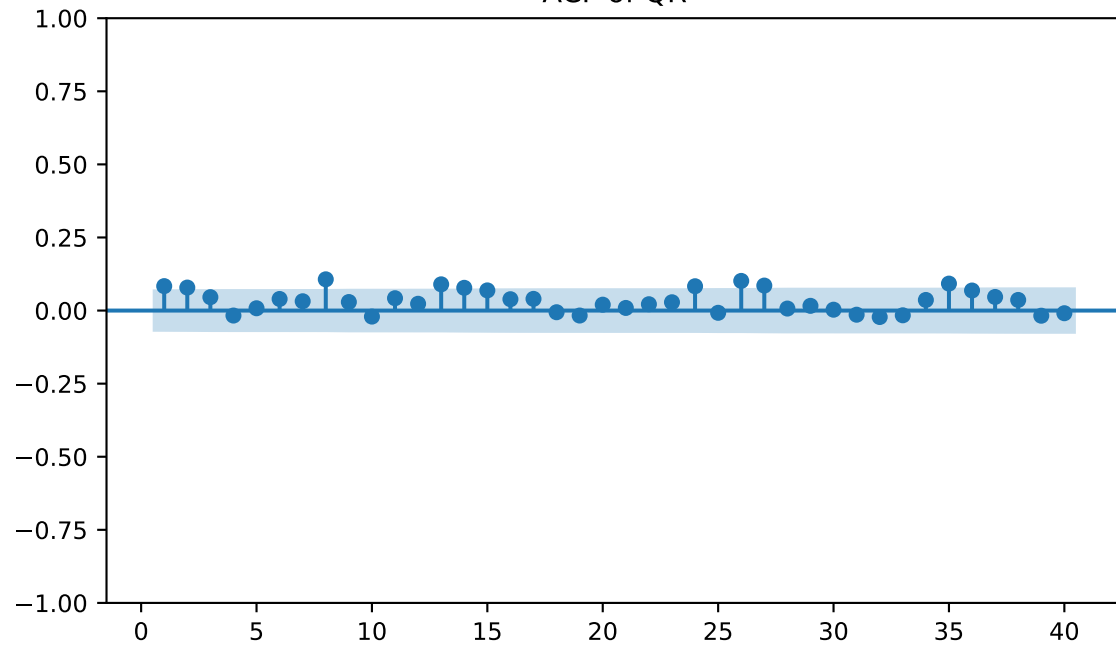
PIT histogram



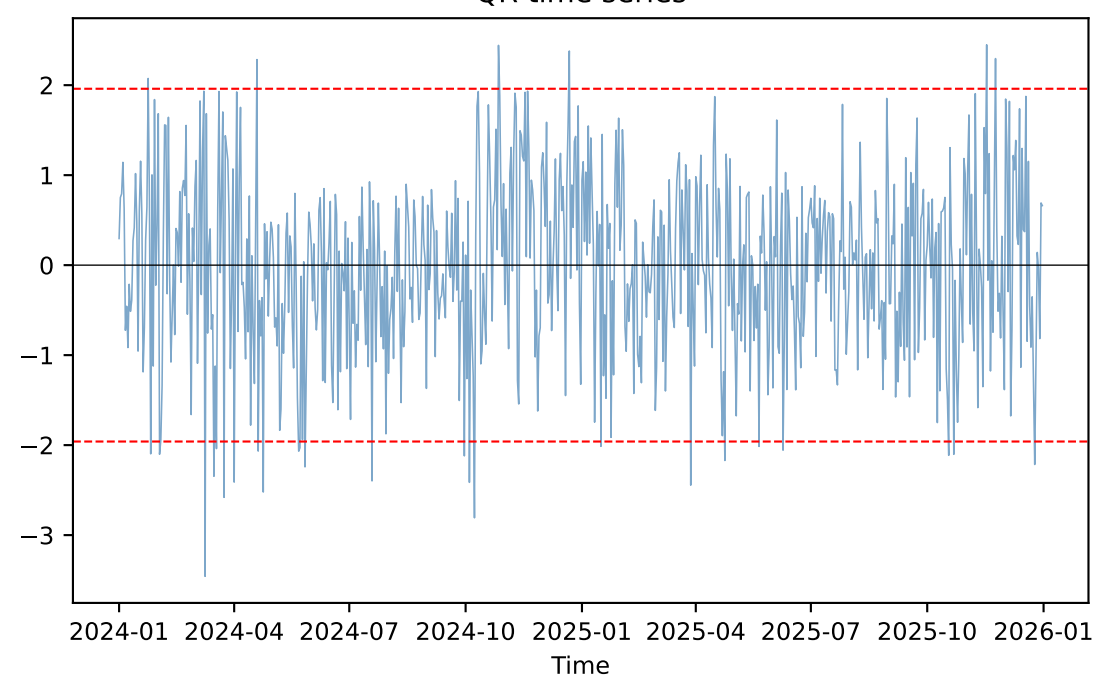
Normal QQ plot



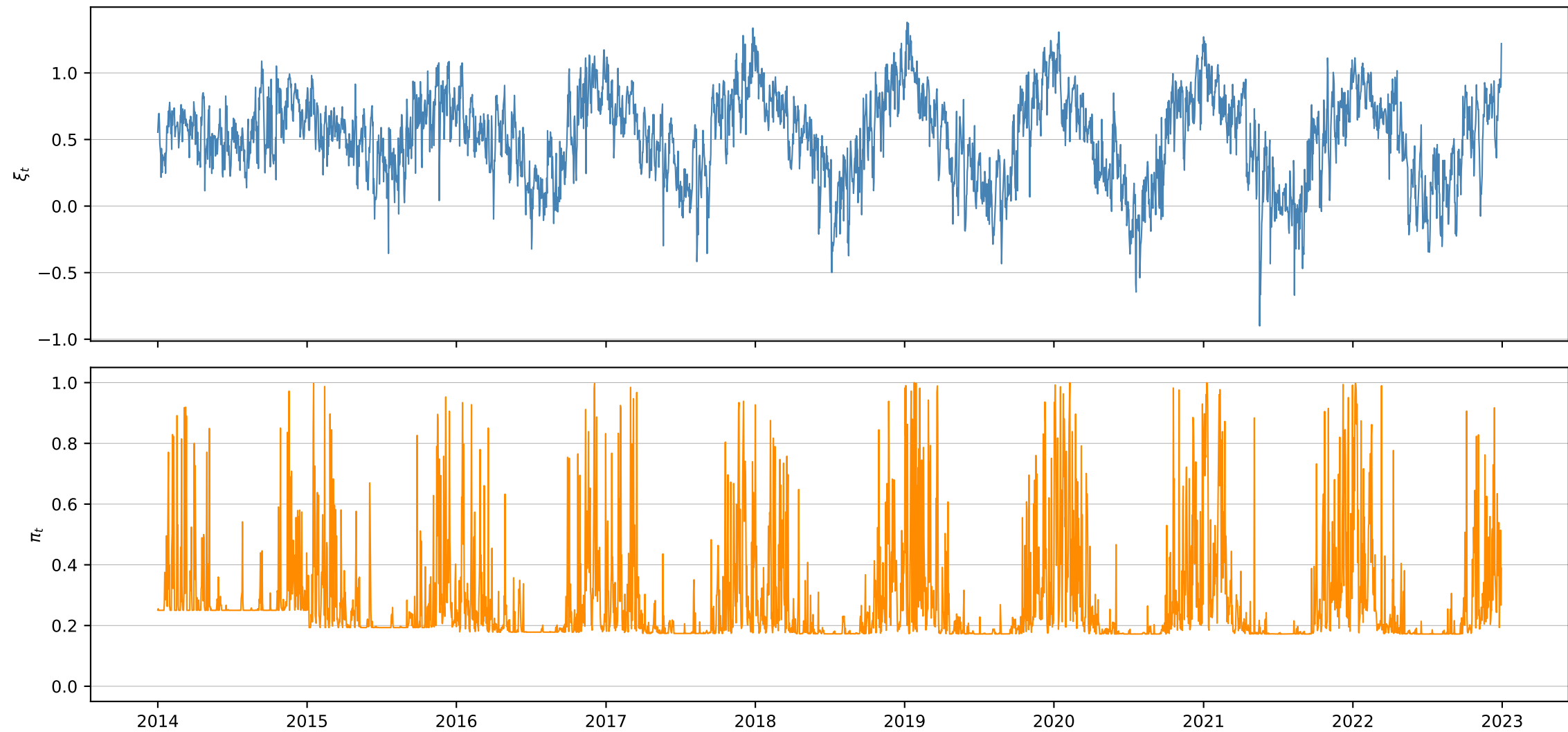
ACF of QR



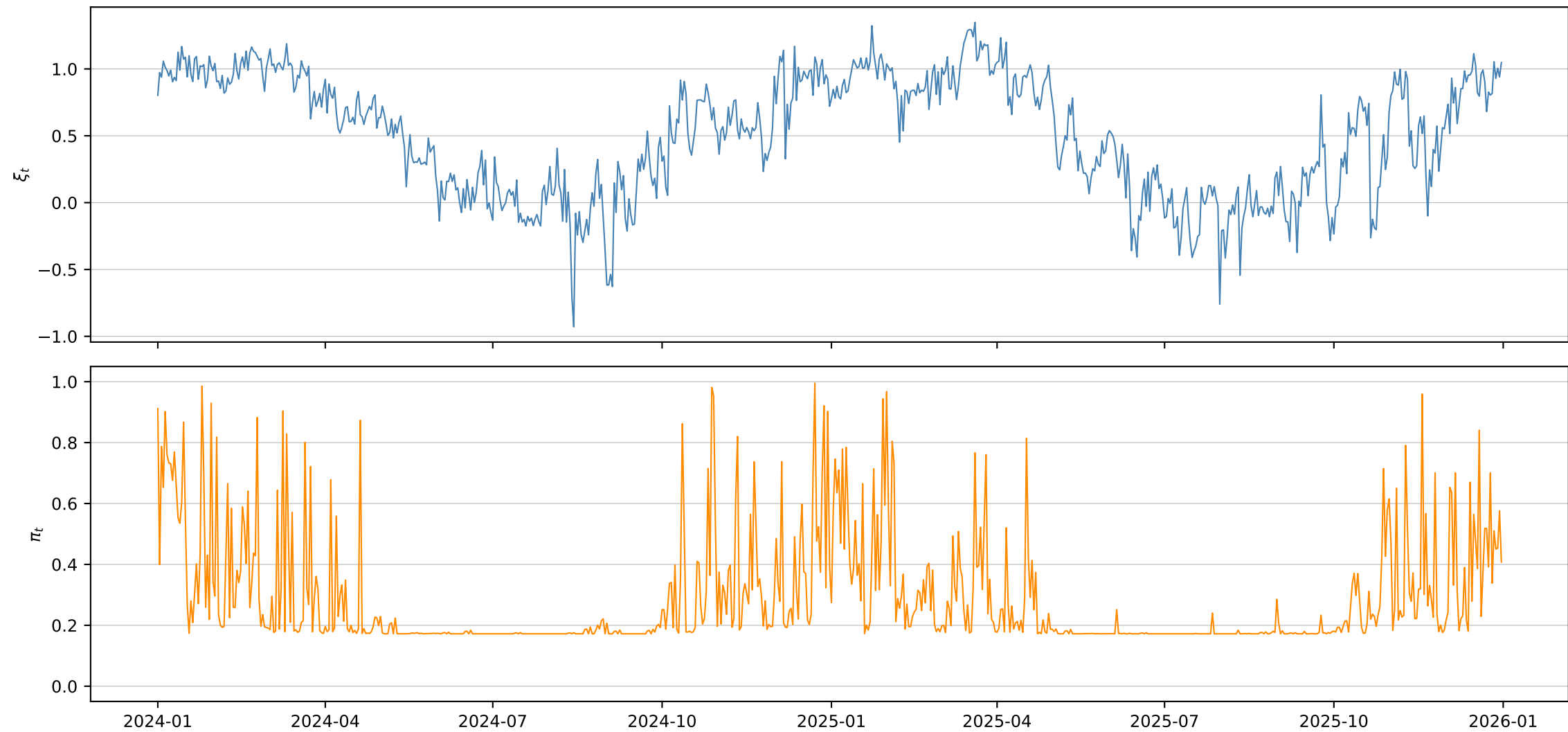
QR time series



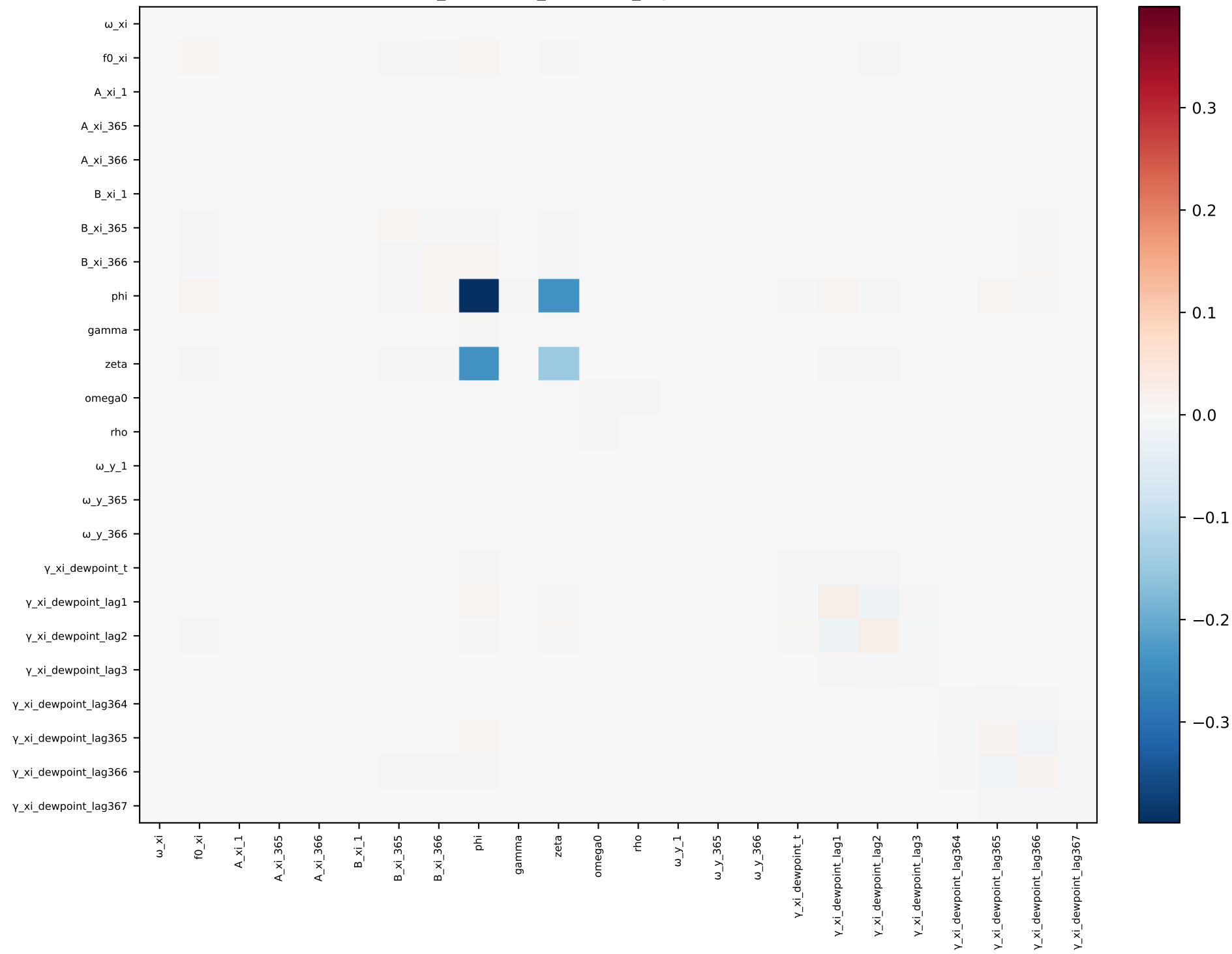
xi_dewpoint_seasonal_lags — filtered states (IS)



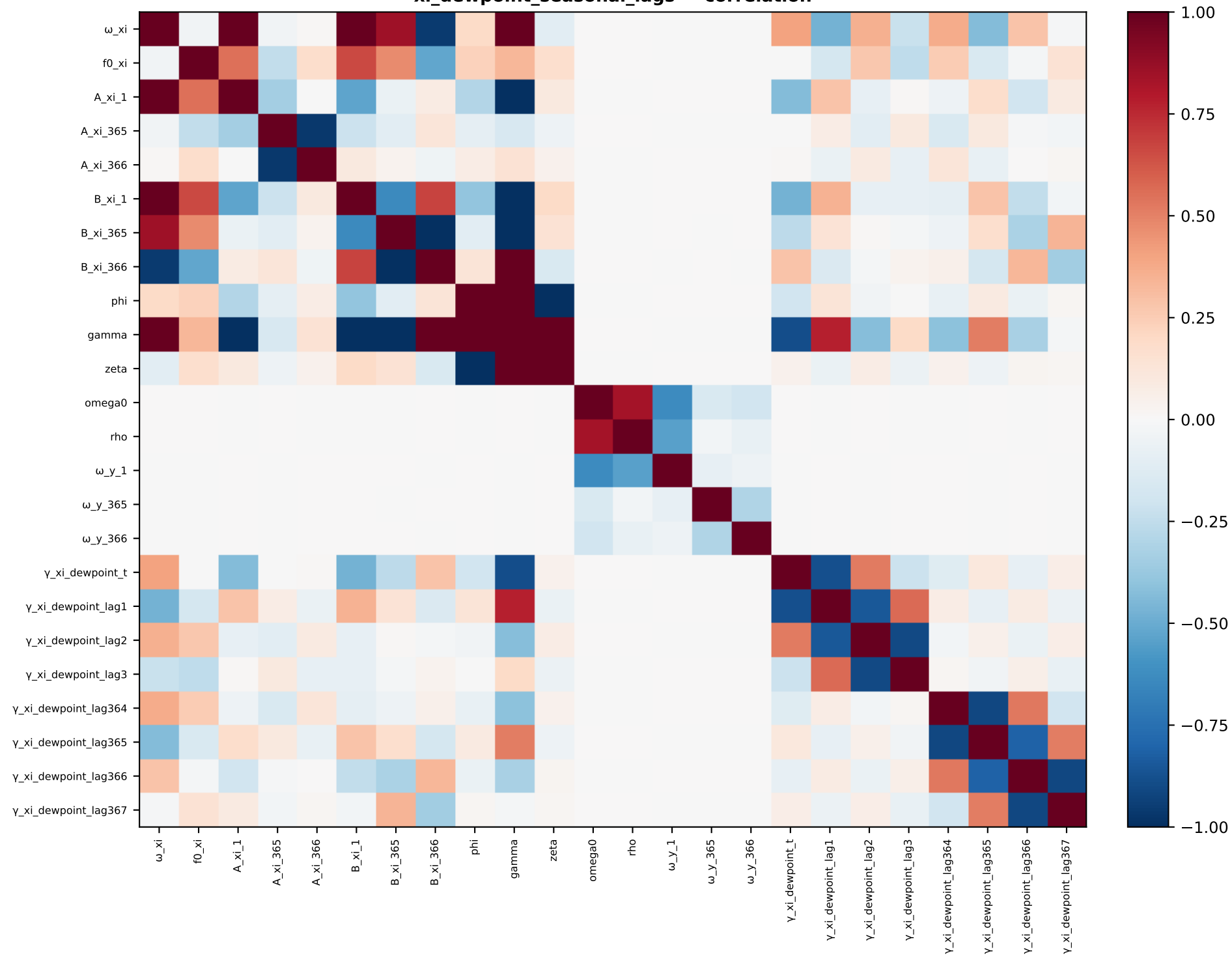
xi_dewpoint_seasonal_lags — filtered states (OOS)



xi_dewpoint_seasonal_lags — covariance

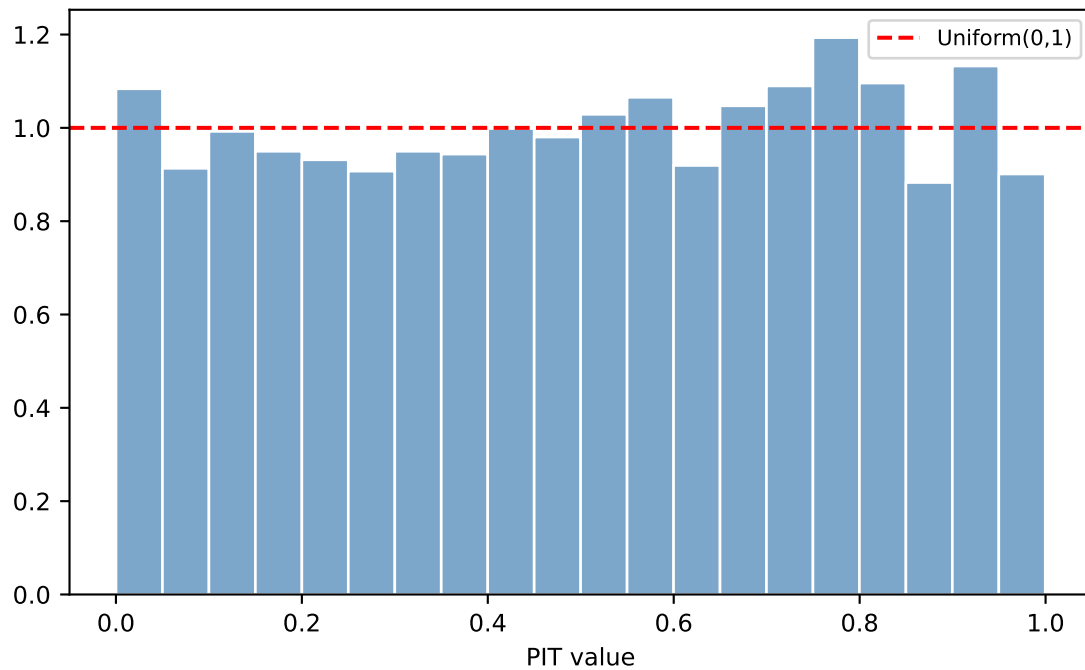


xi_dewpoint_seasonal_lags — correlation

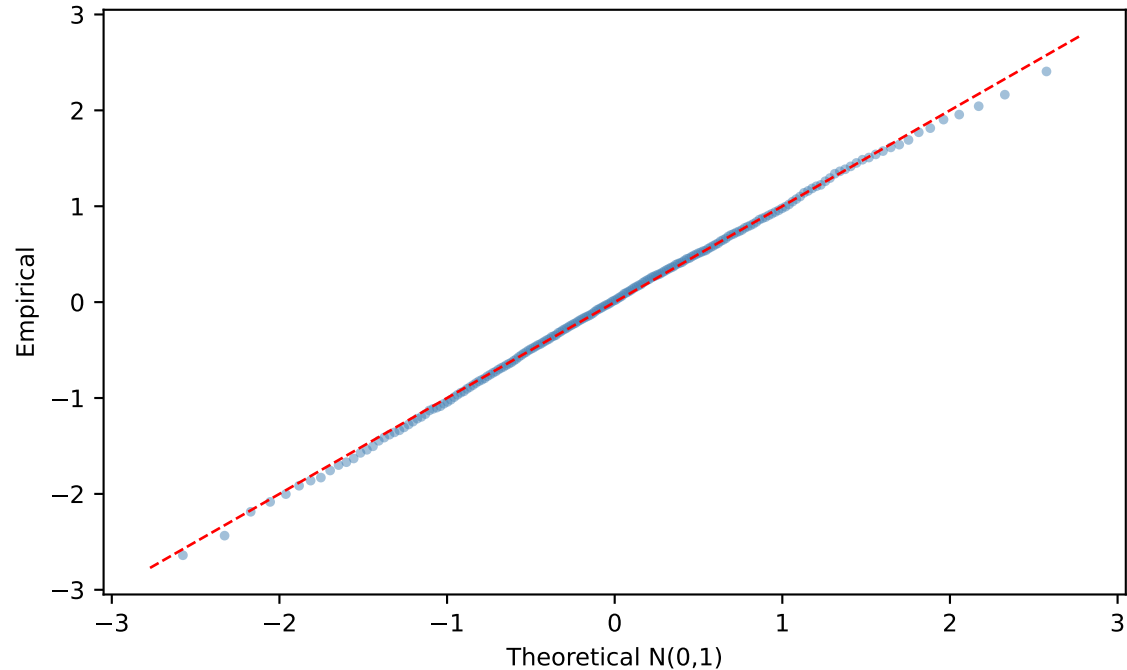


xi_temperature_t — IS diagnostics

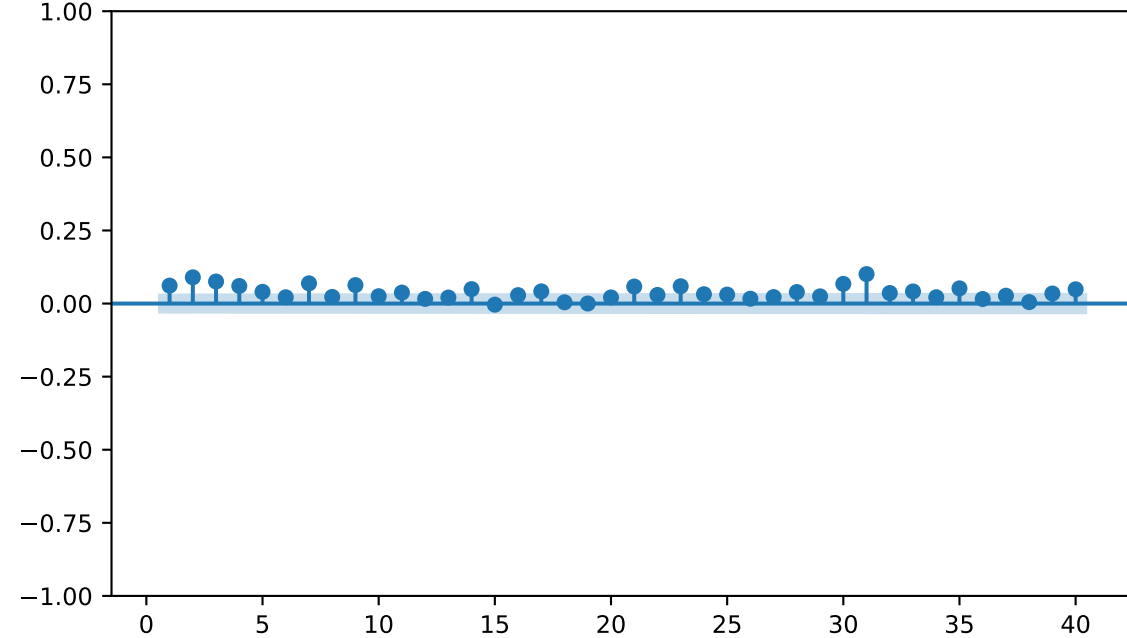
PIT histogram



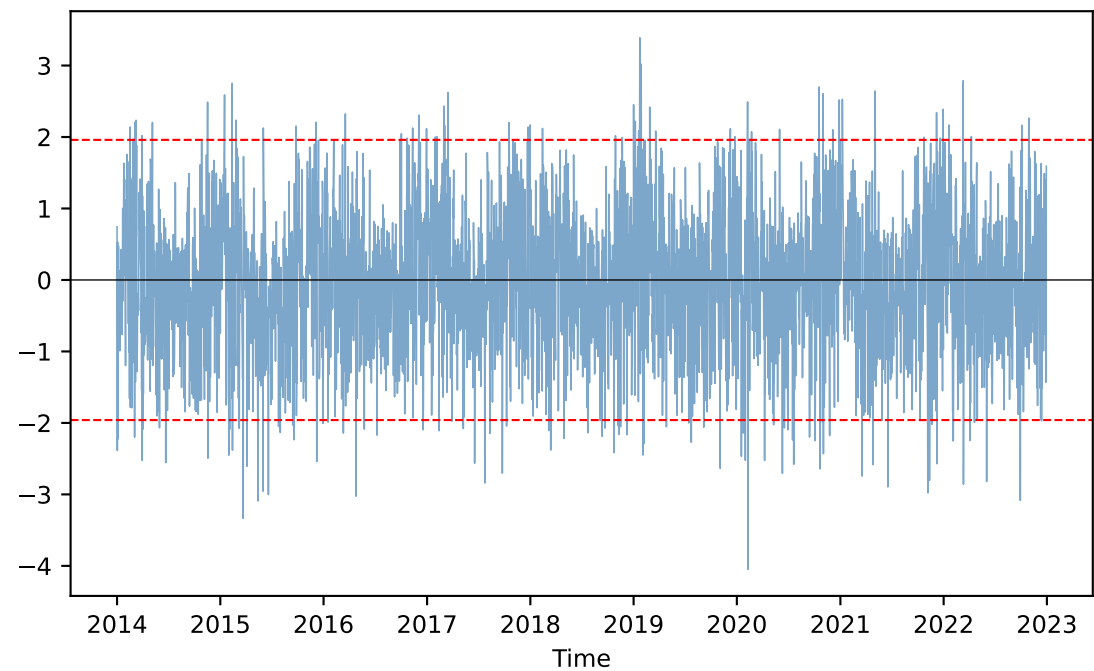
Normal QQ plot



ACF of QR

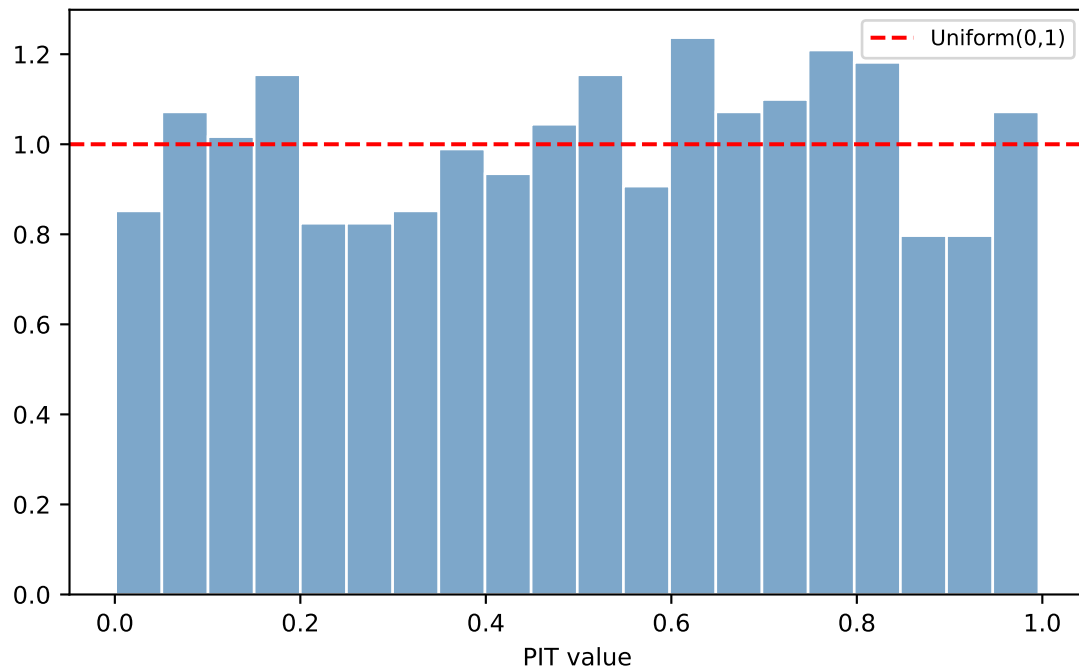


QR time series

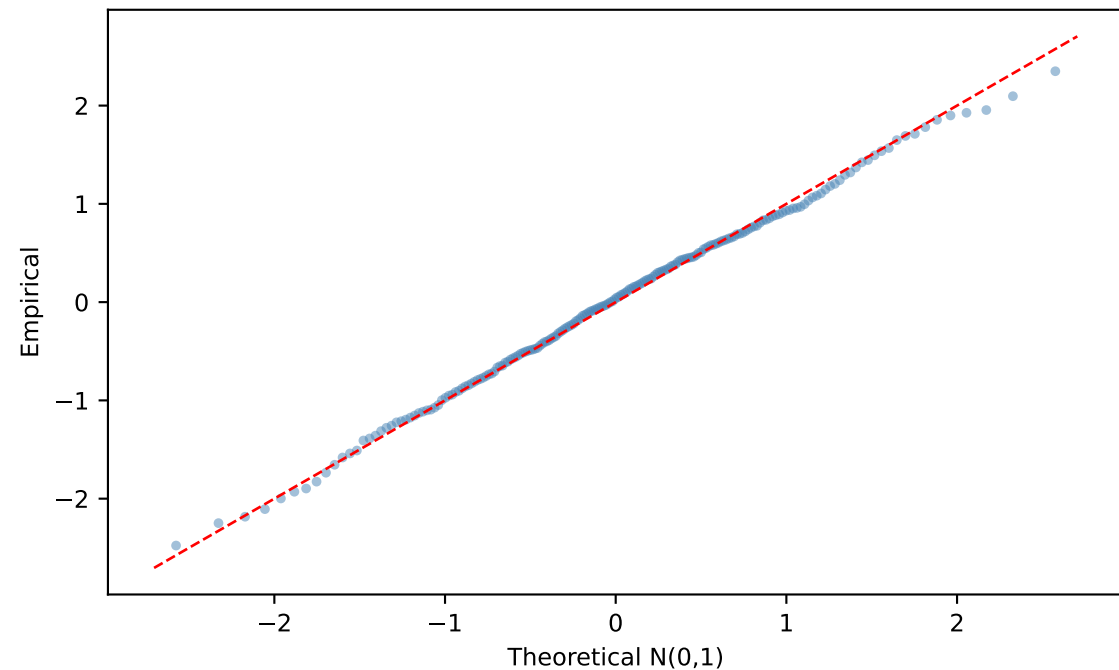


xi_temperature_t — OOS diagnostics

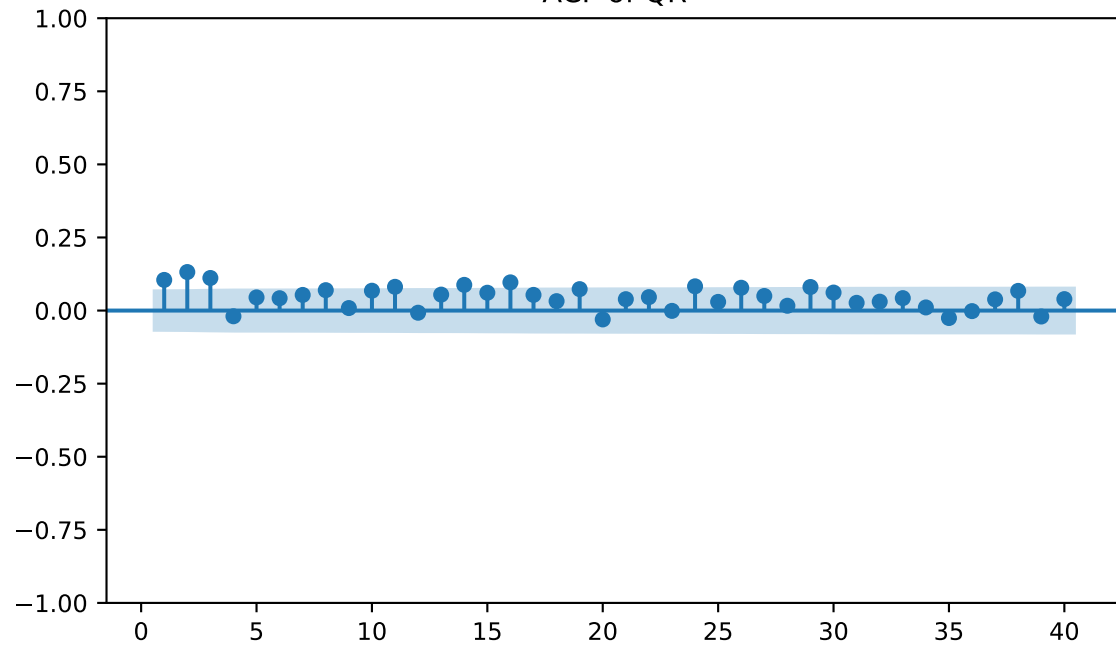
PIT histogram



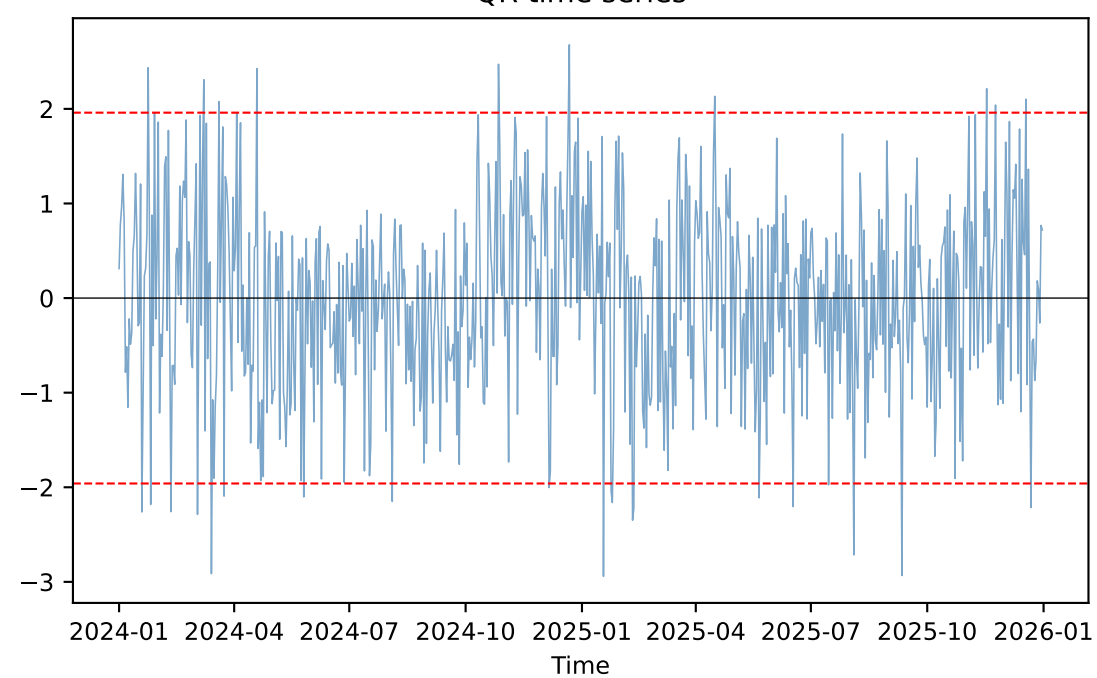
Normal QQ plot



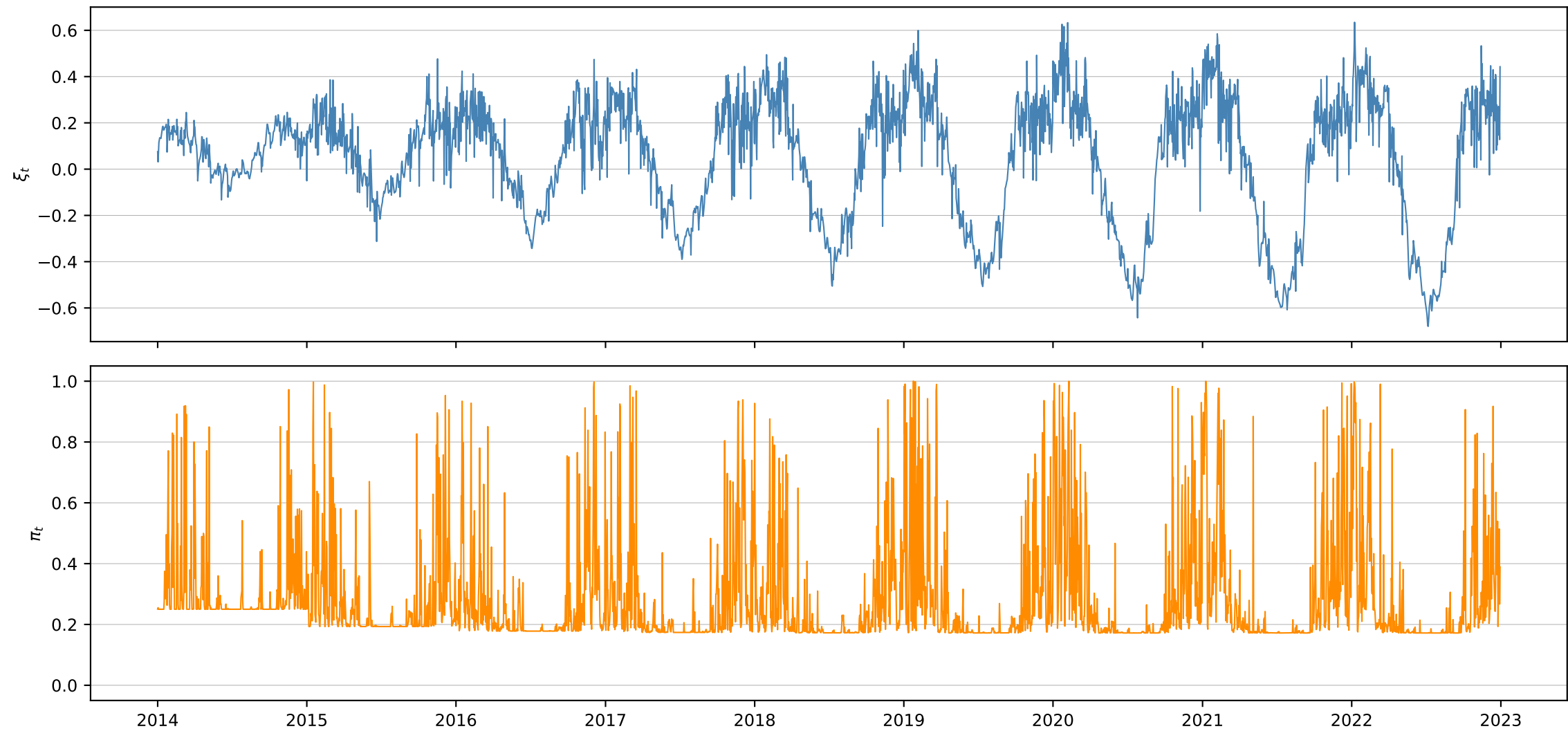
ACF of QR



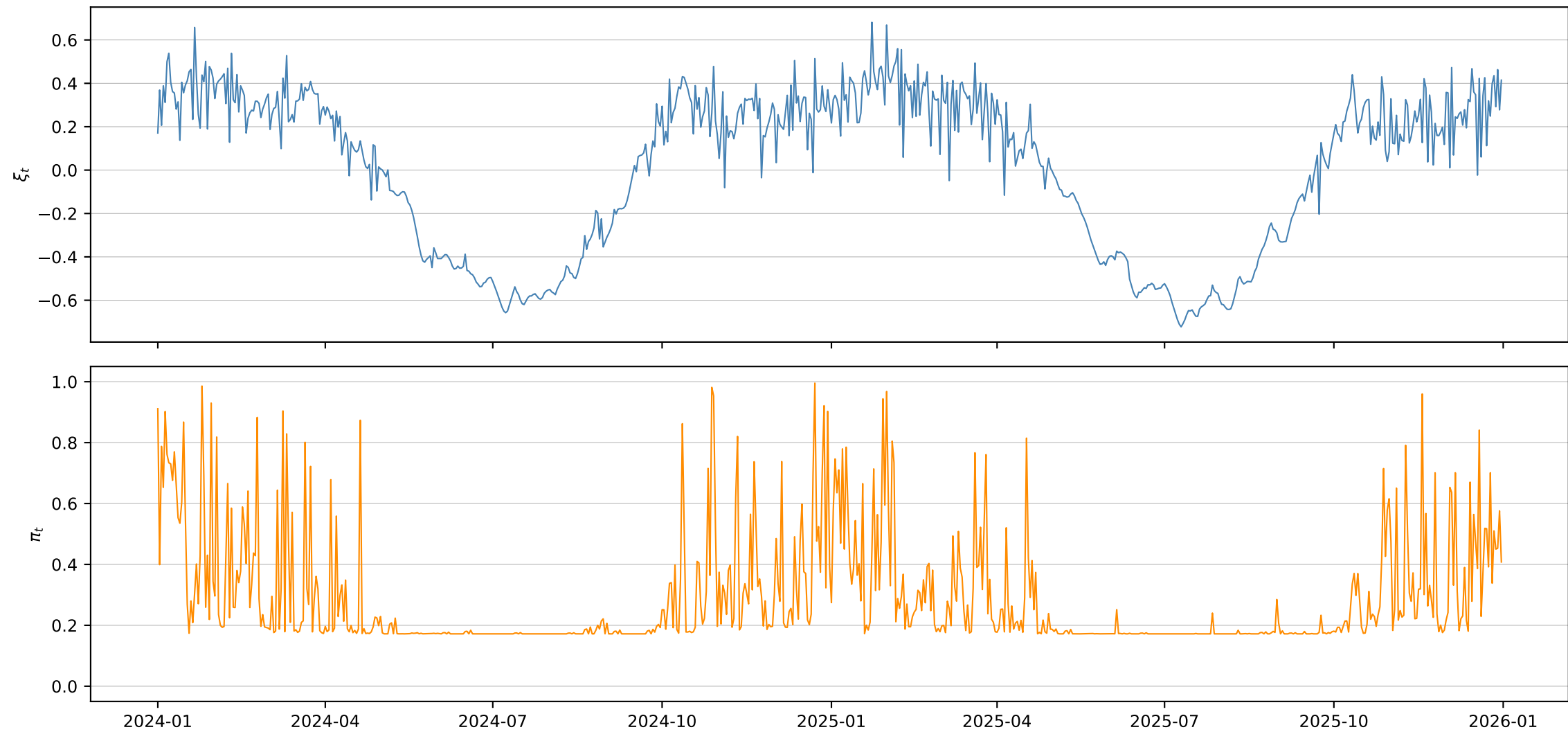
QR time series



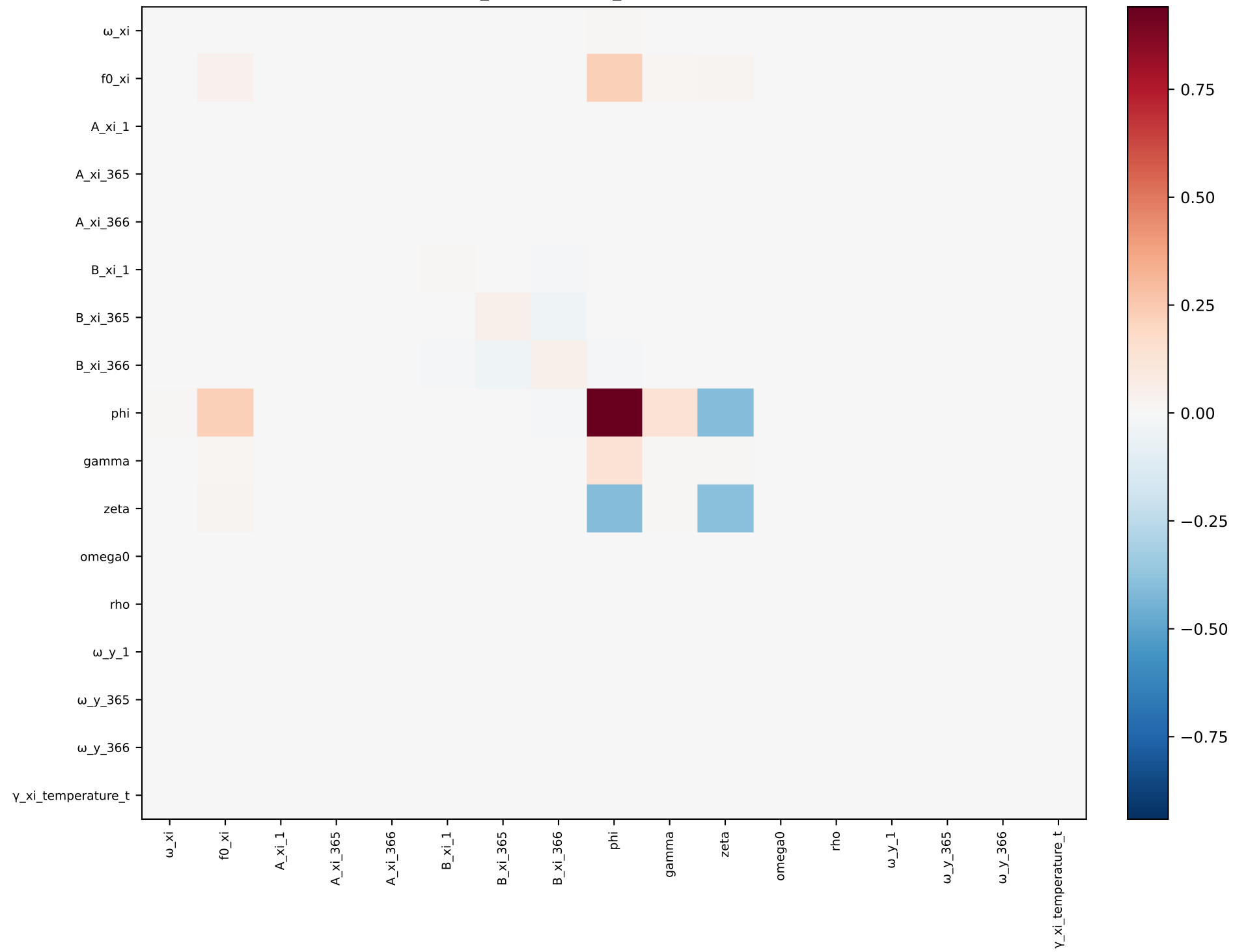
xi_temperature_t — filtered states (IS)



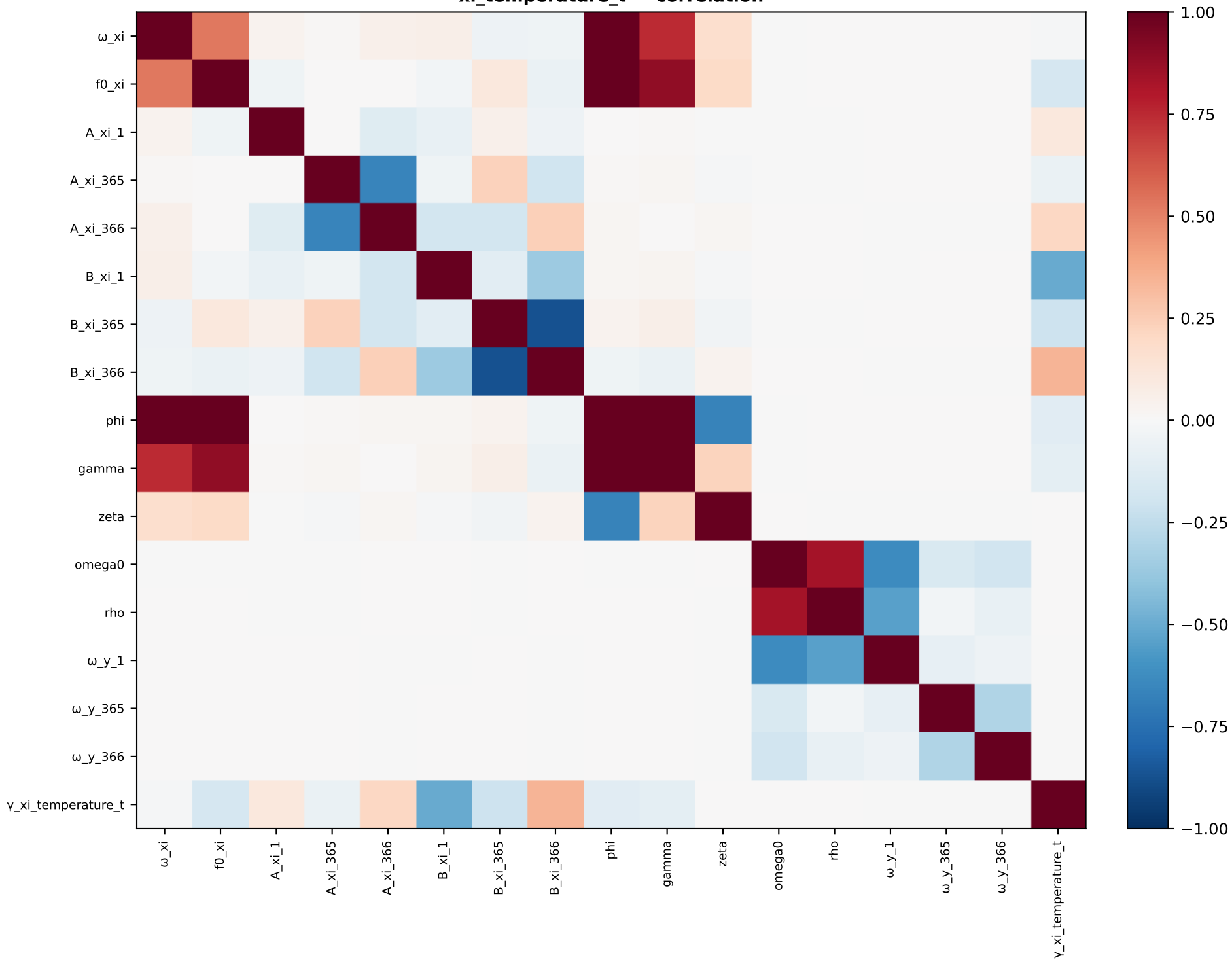
xi_temperature_t — filtered states (OOS)



xi_temperature_t — covariance

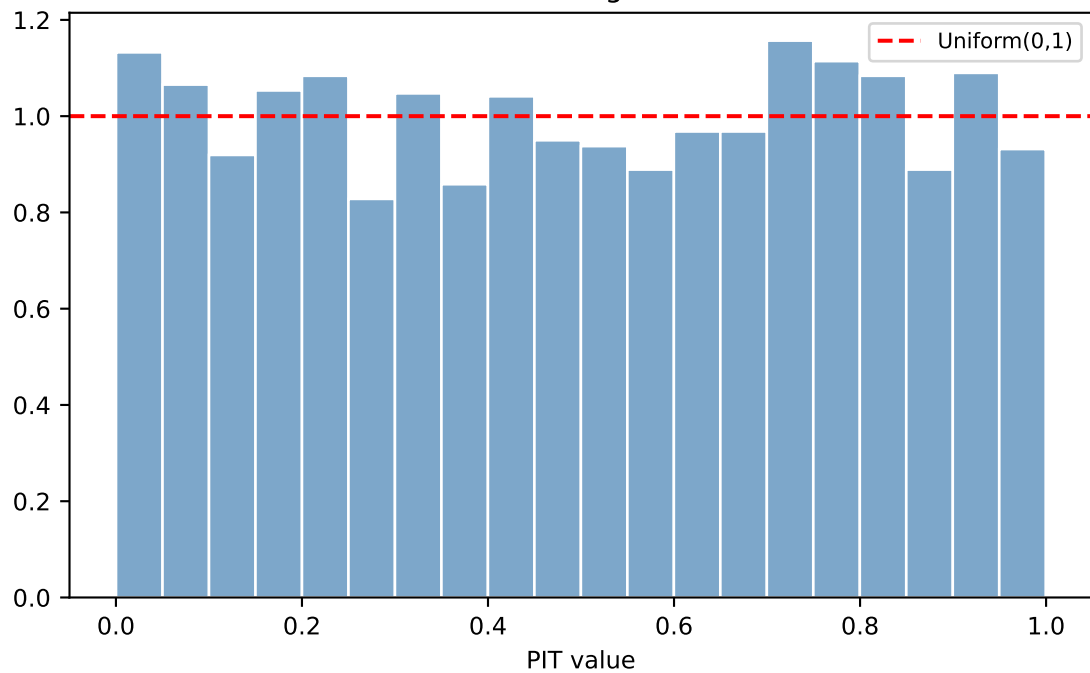


xi_temperature_t — correlation

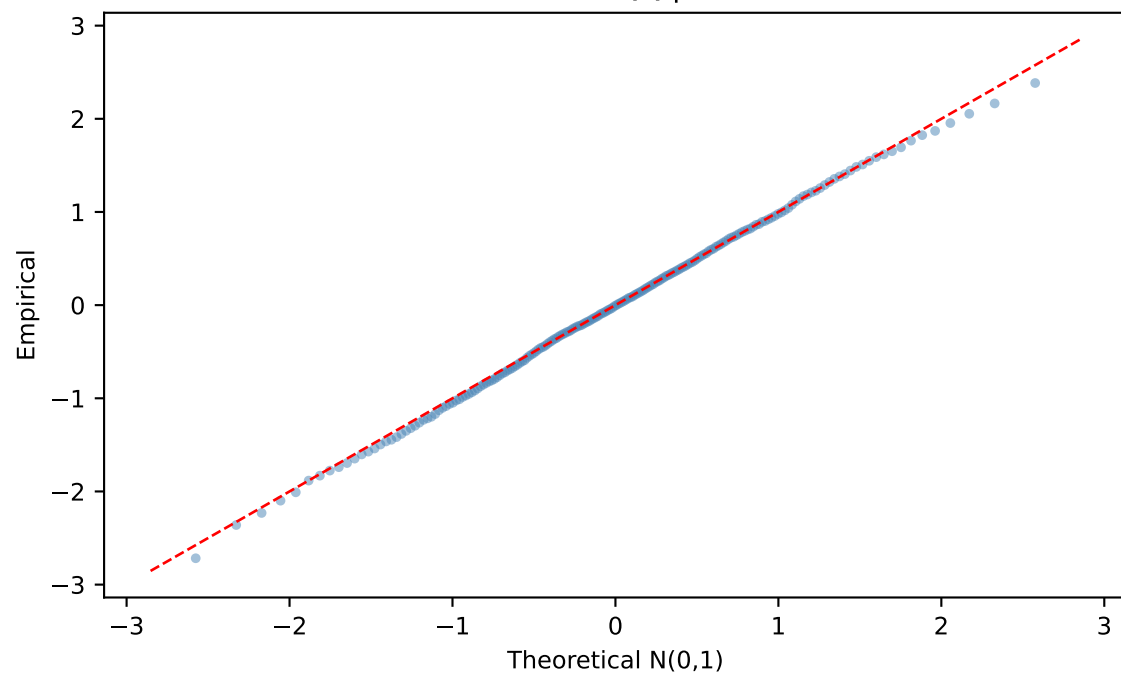


xi_temperature_lag1 — IS diagnostics

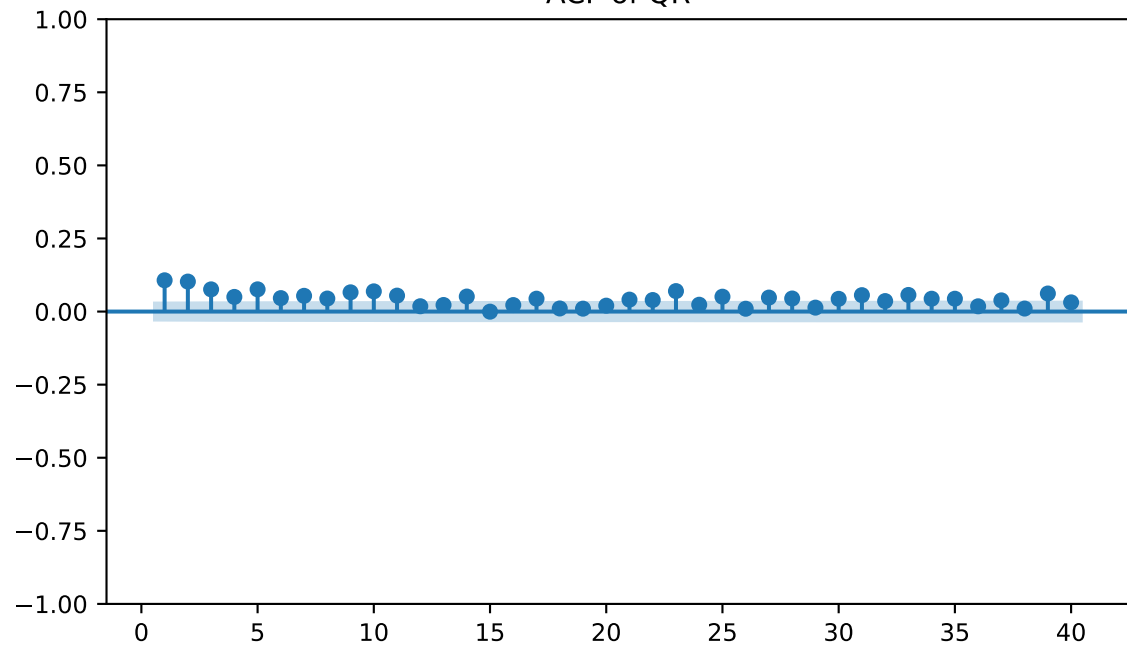
PIT histogram



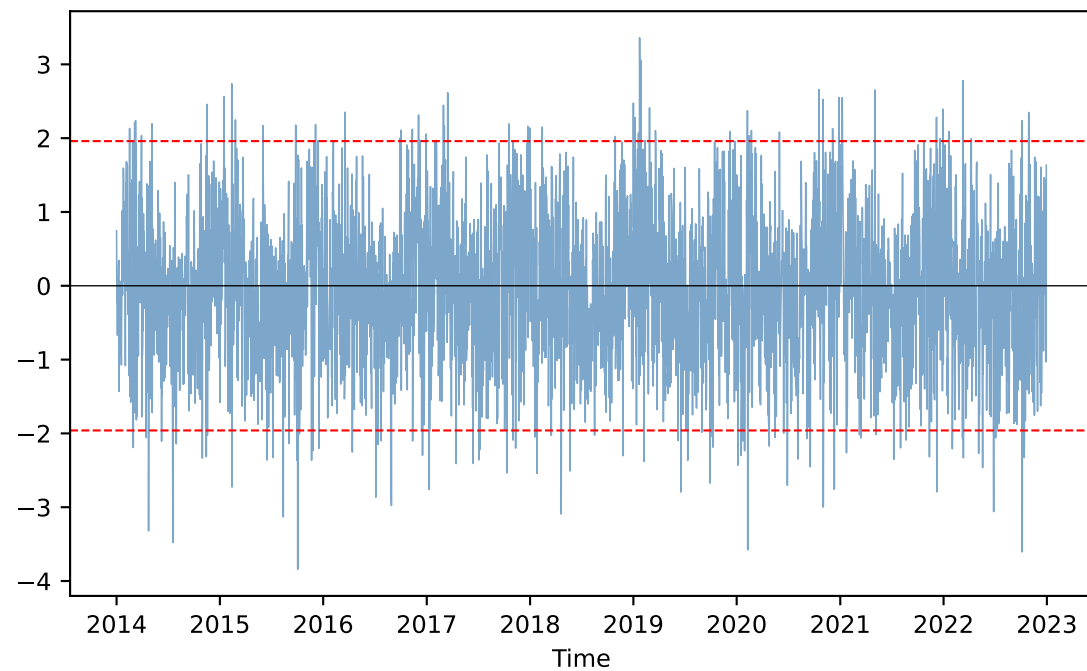
Normal QQ plot



ACF of QR

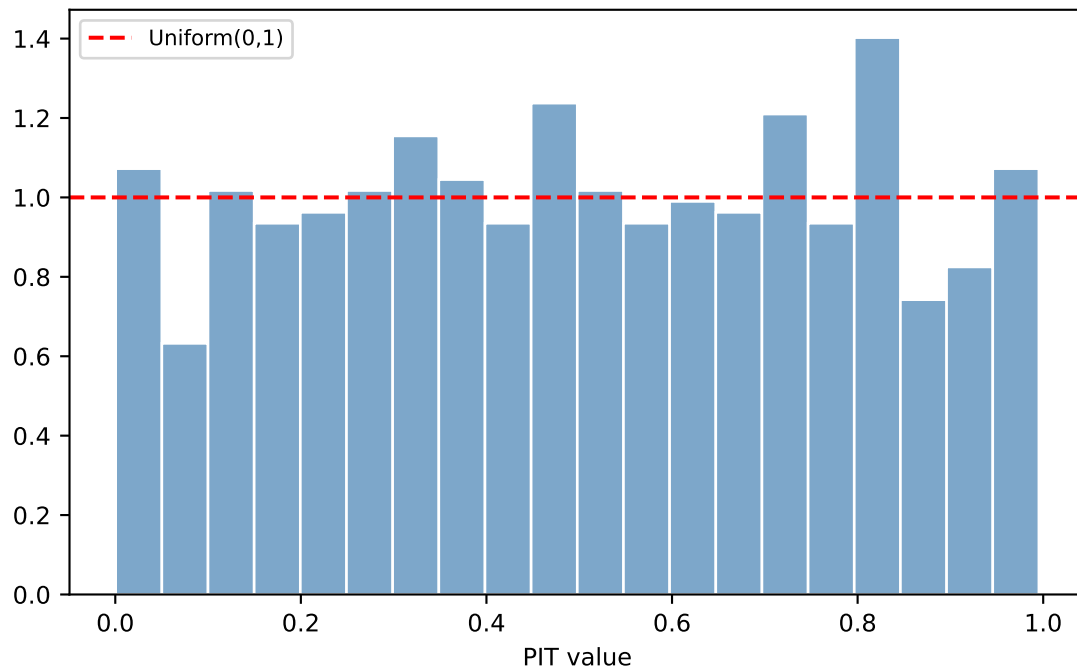


QR time series

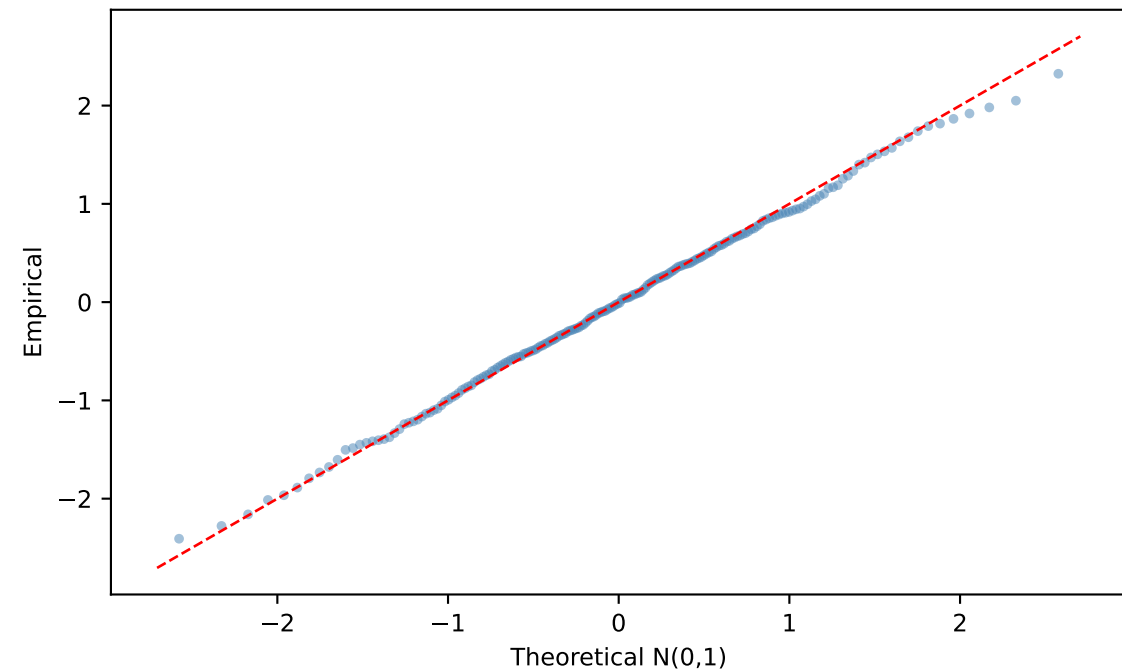


xi_temperature_lag1 — OOS diagnostics

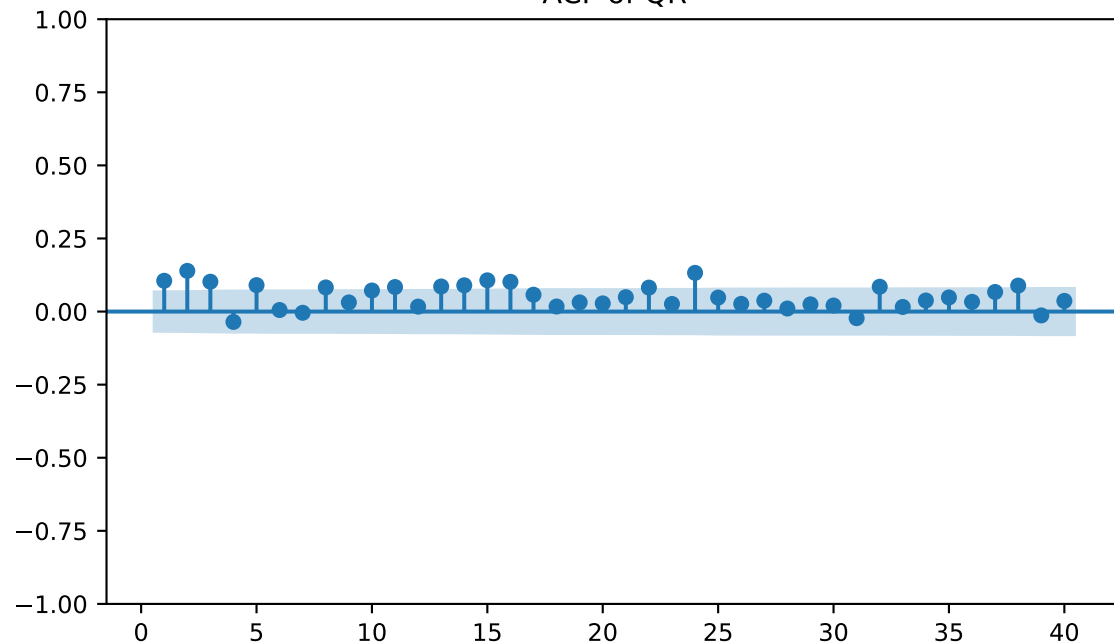
PIT histogram



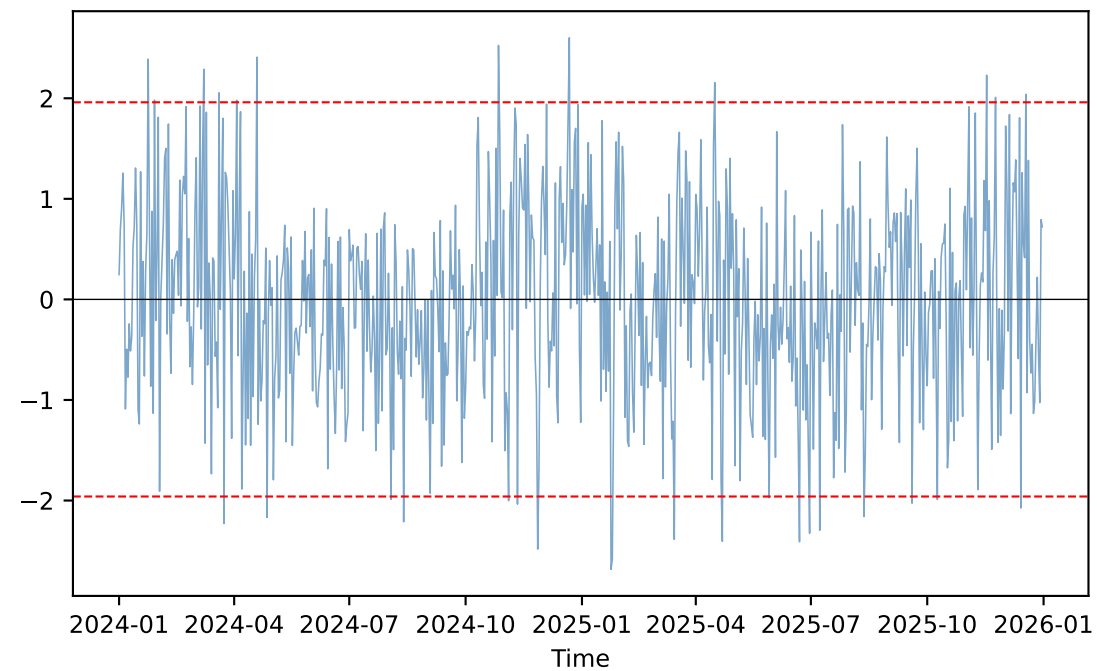
Normal QQ plot



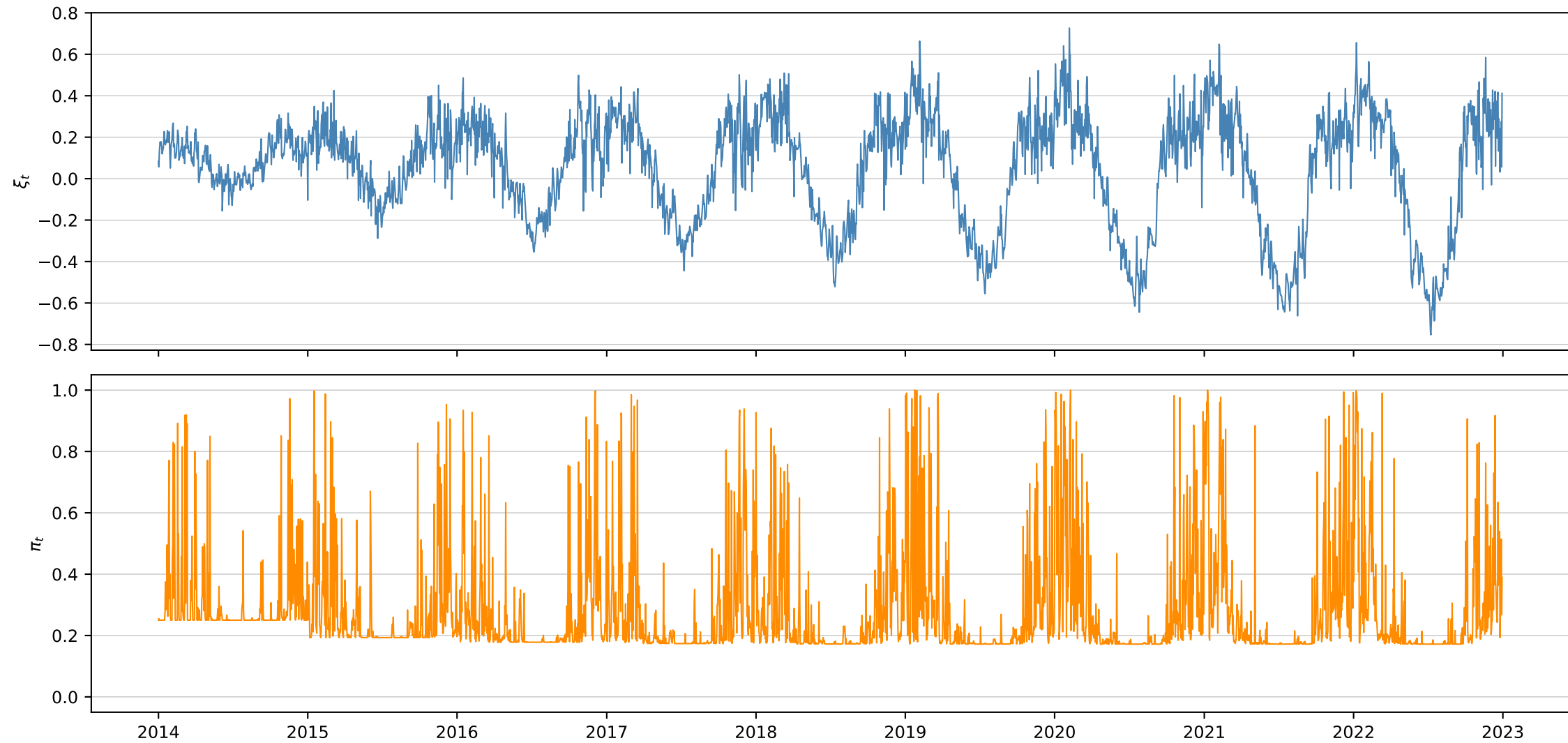
ACF of QR



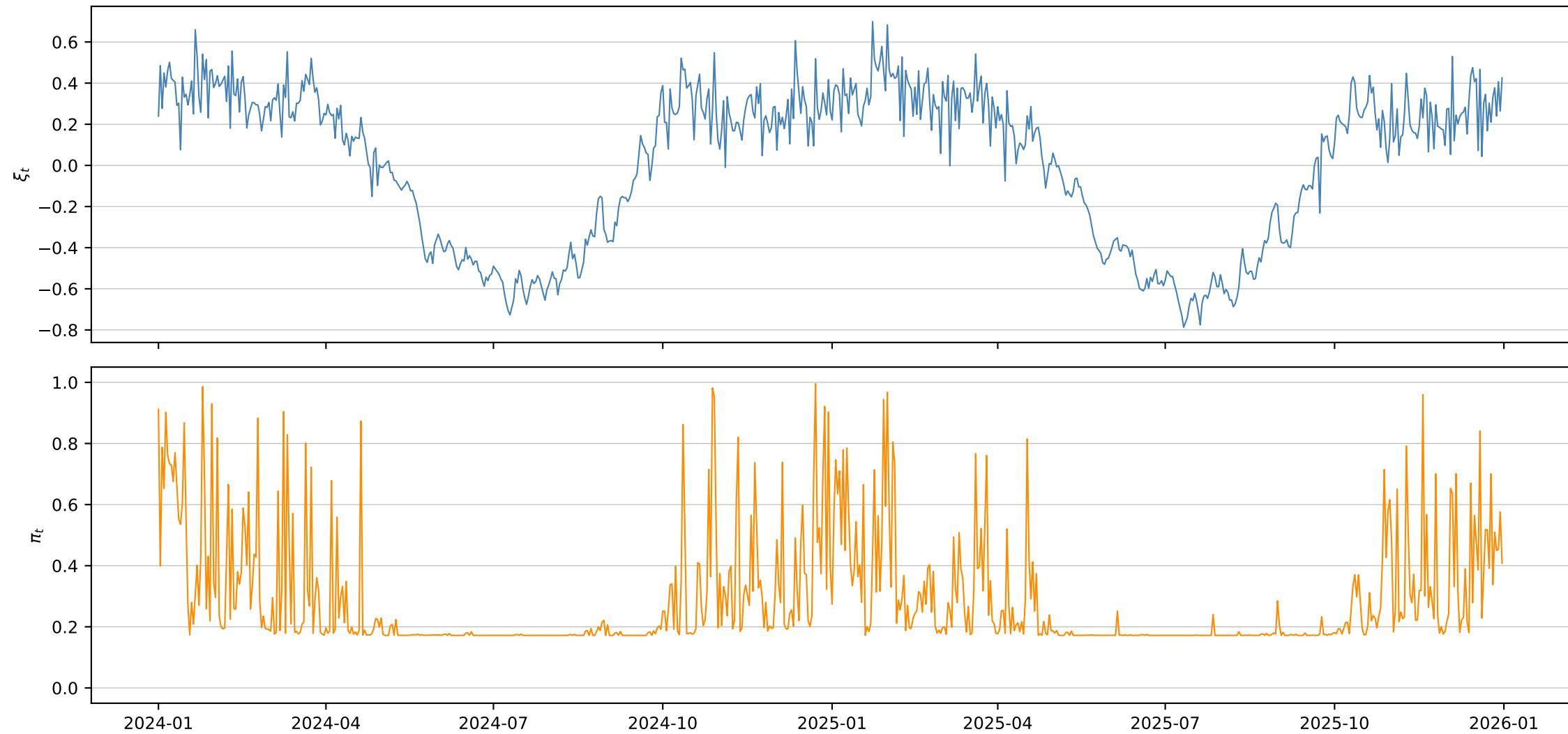
QR time series



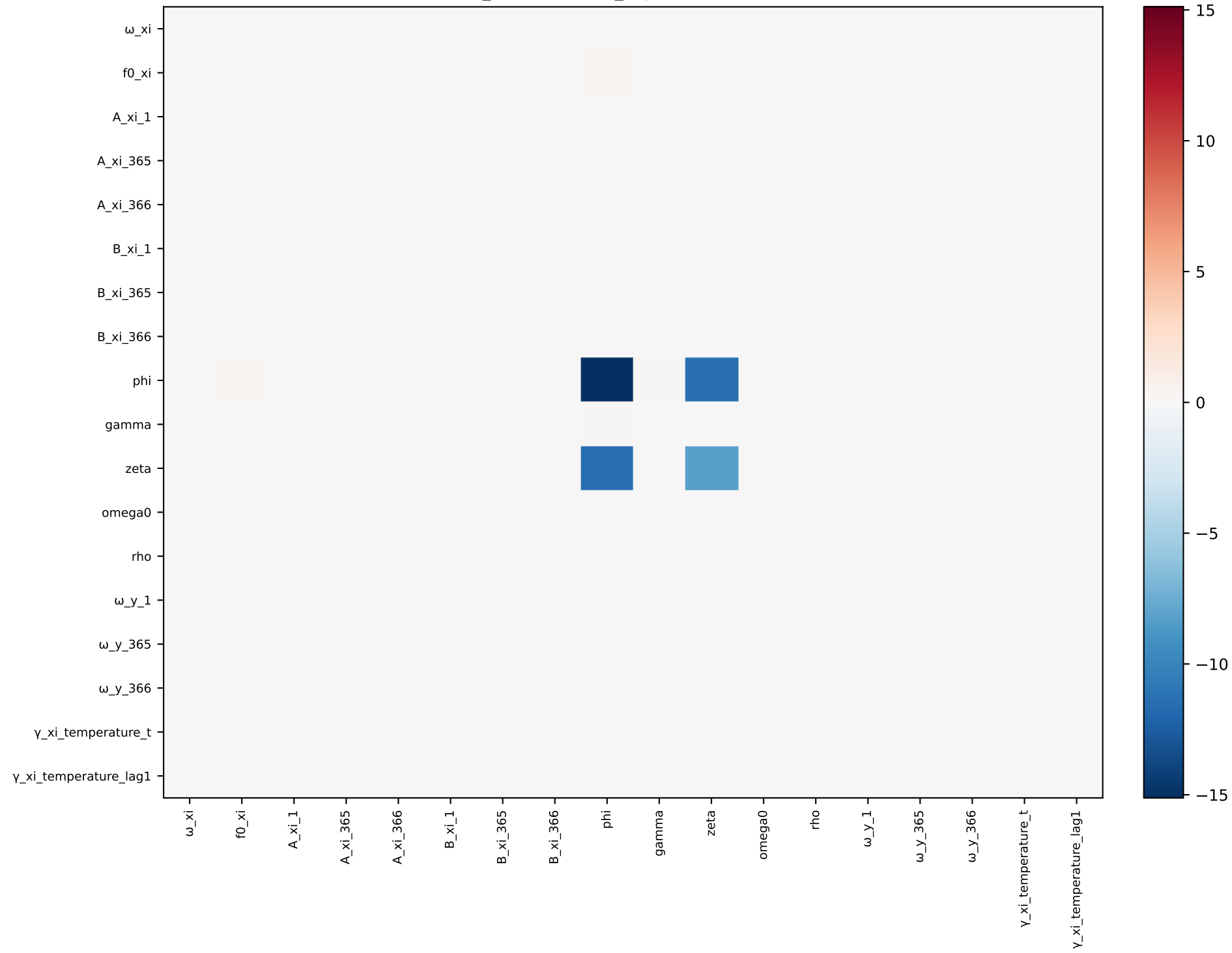
xi_temperature_lag1 — filtered states (IS)

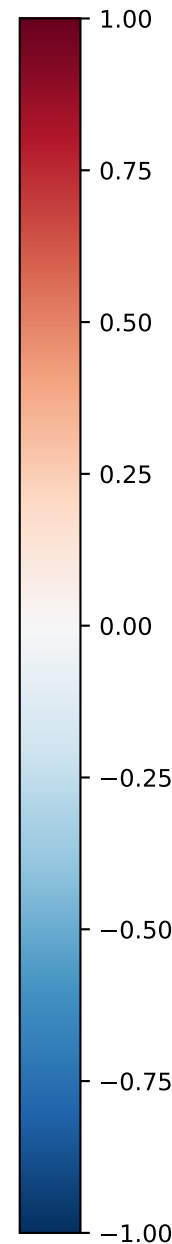


xi_temperature_lag1 — filtered states (OOS)



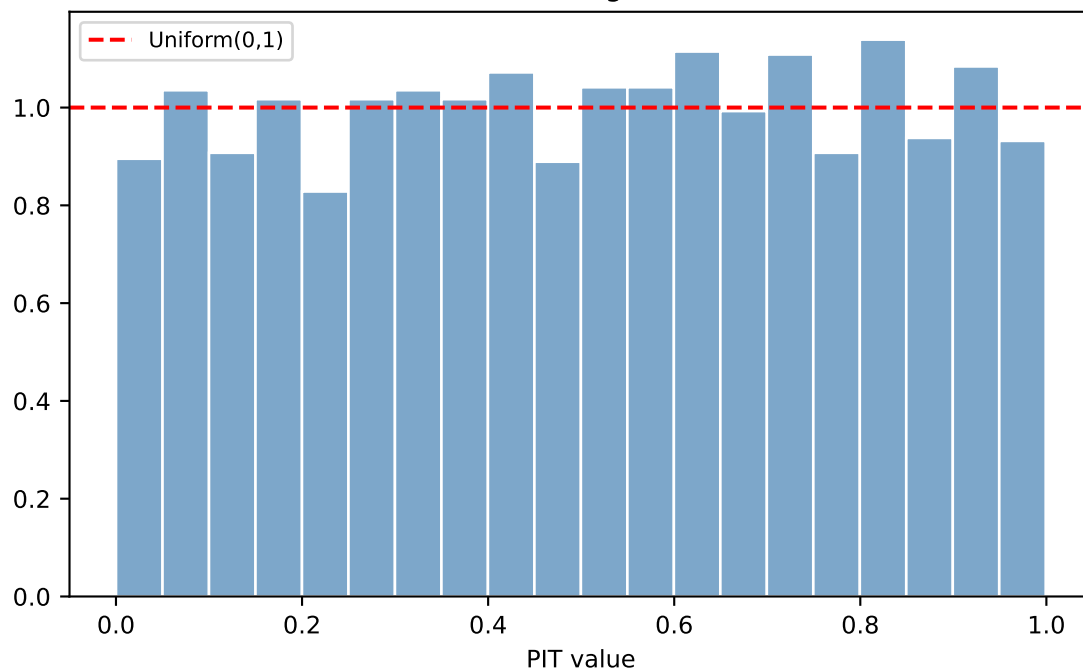
xi_temperature_lag1 — covariance



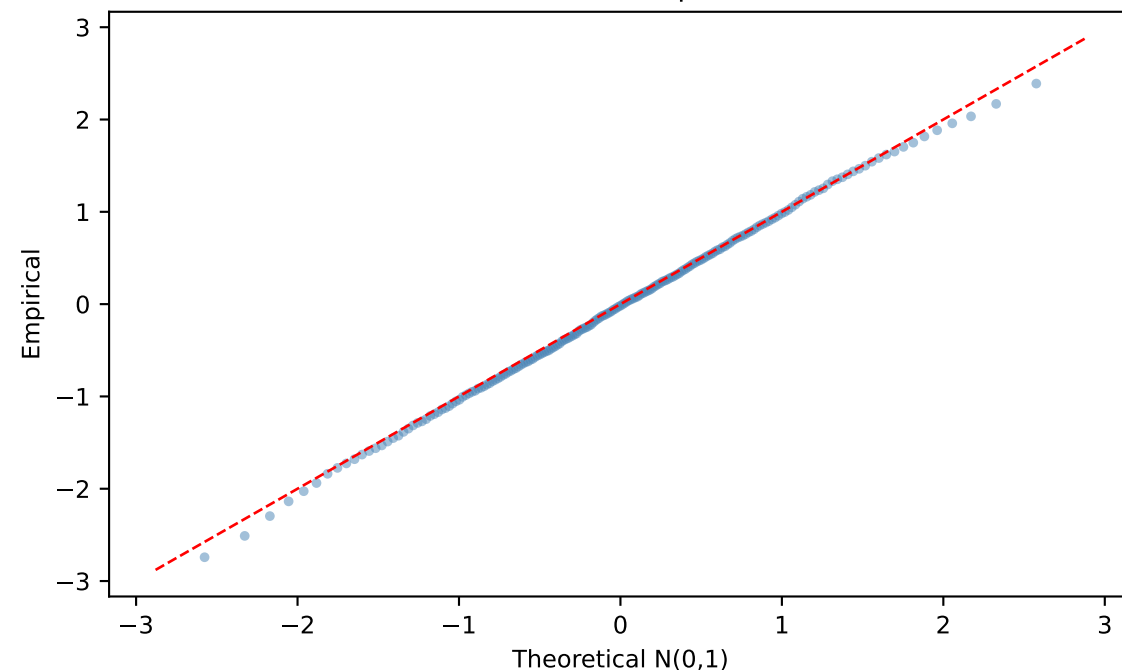


xi_temperature_short_lags — IS diagnostics

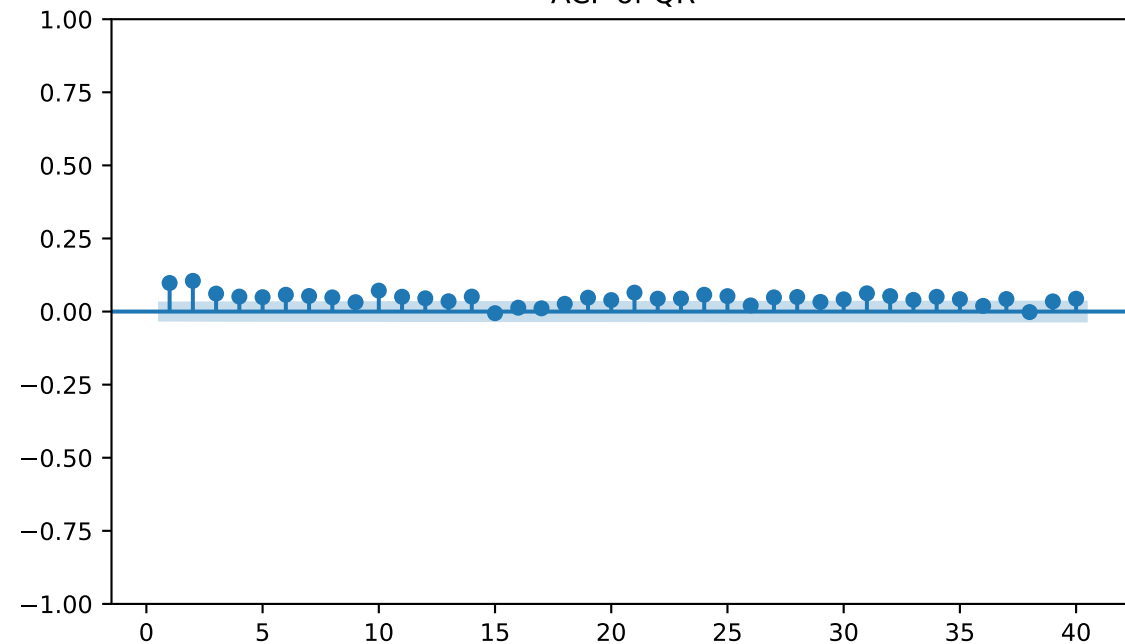
PIT histogram



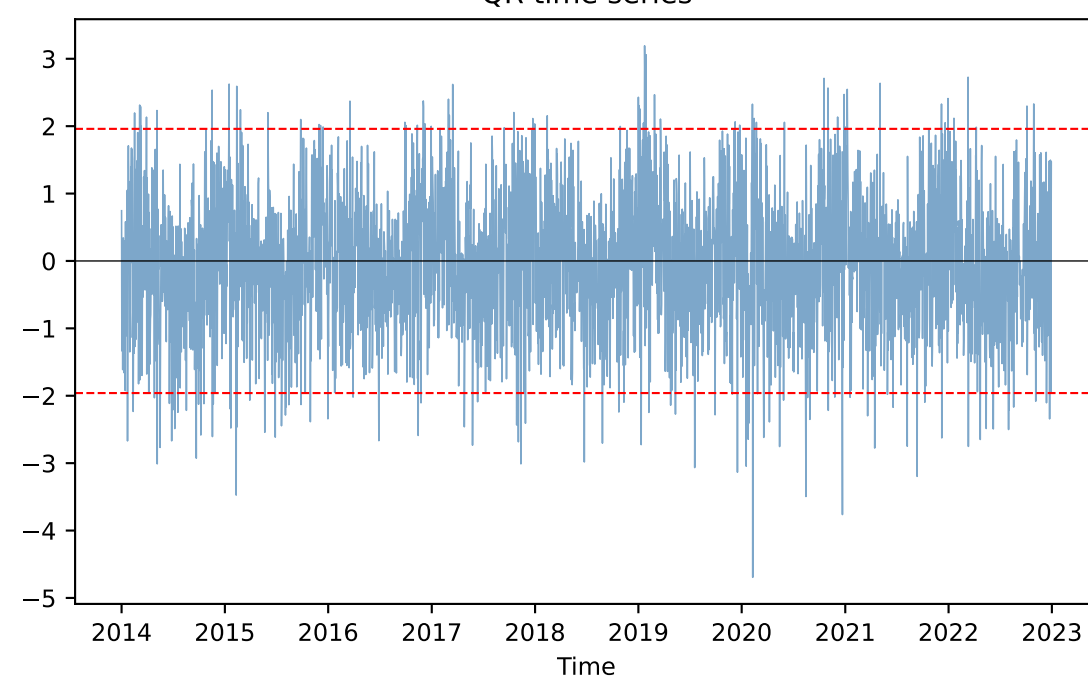
Normal QQ plot



ACF of QR

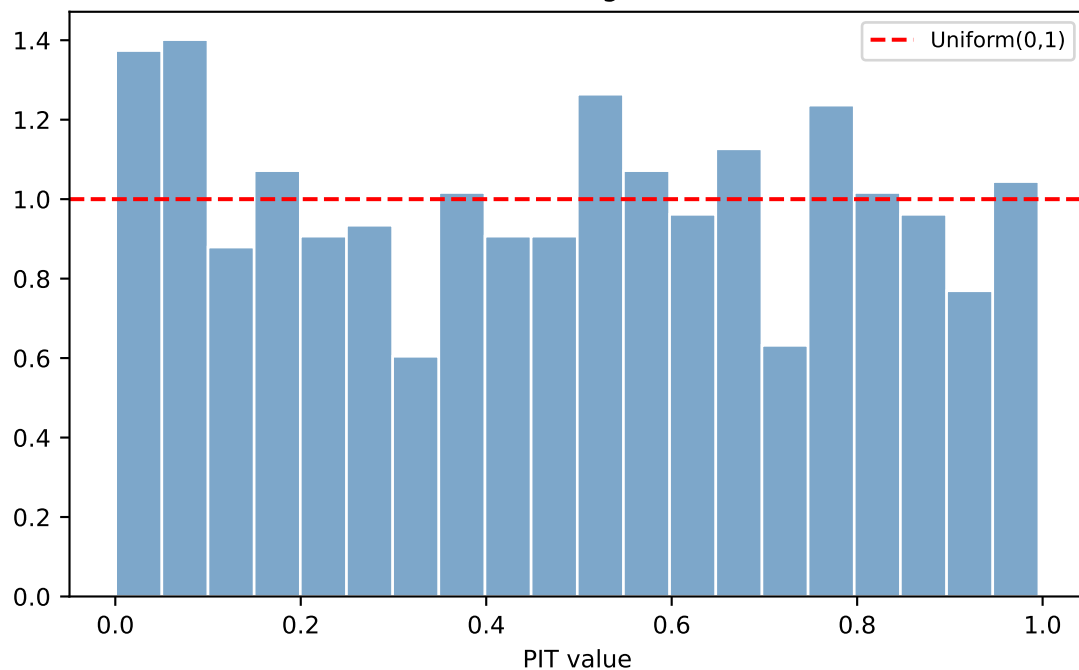


QR time series

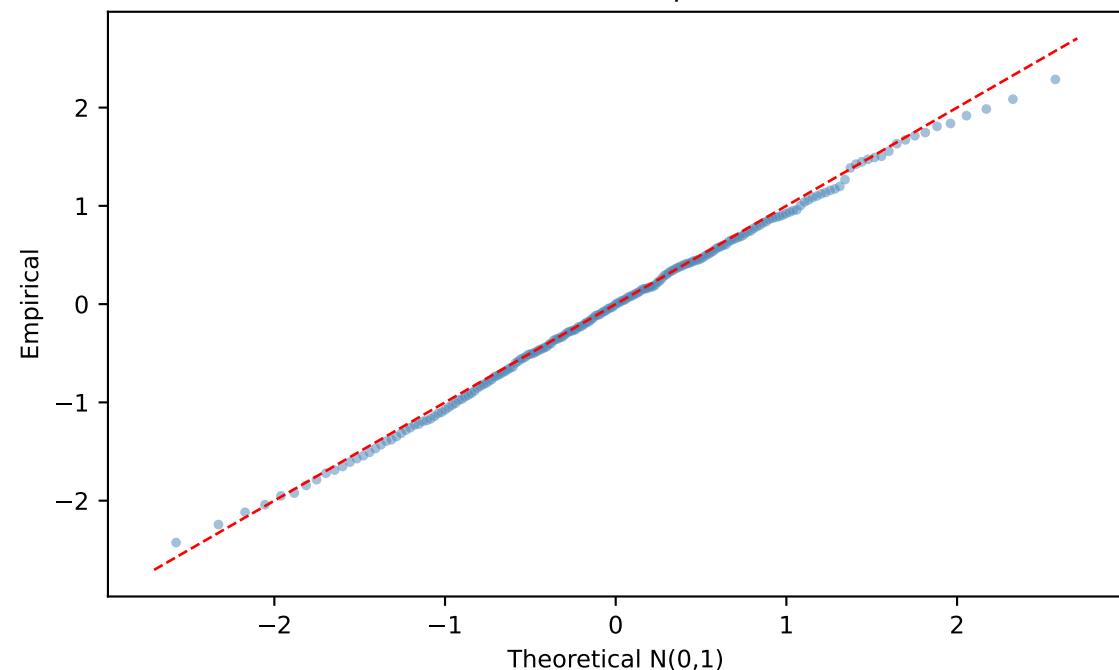


xi_temperature_short_lags — OOS diagnostics

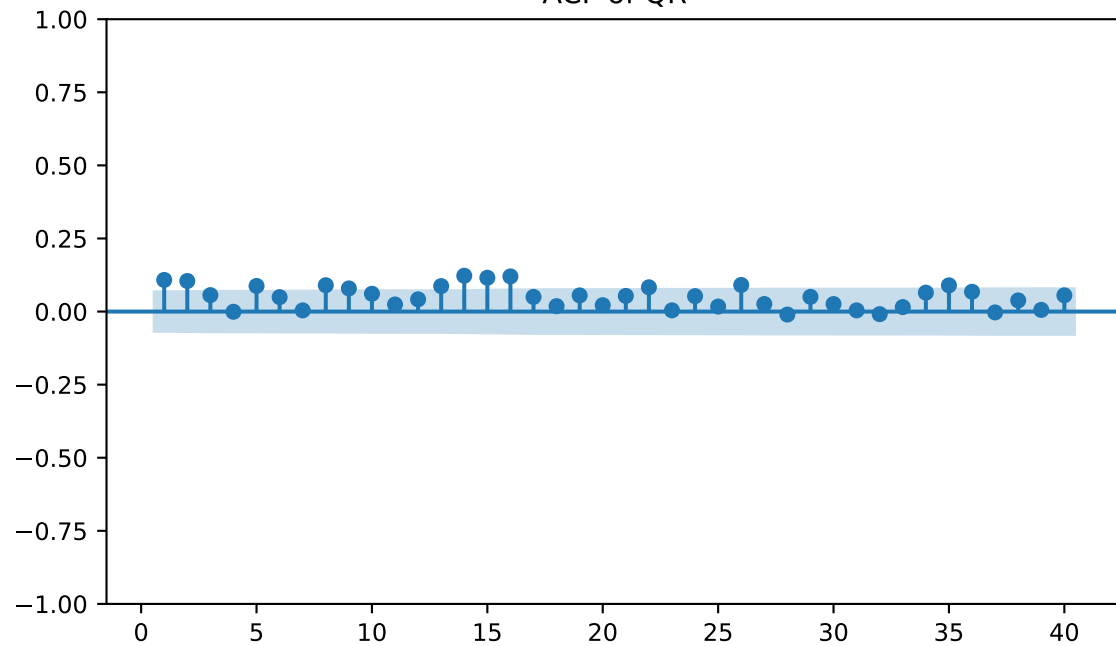
PIT histogram



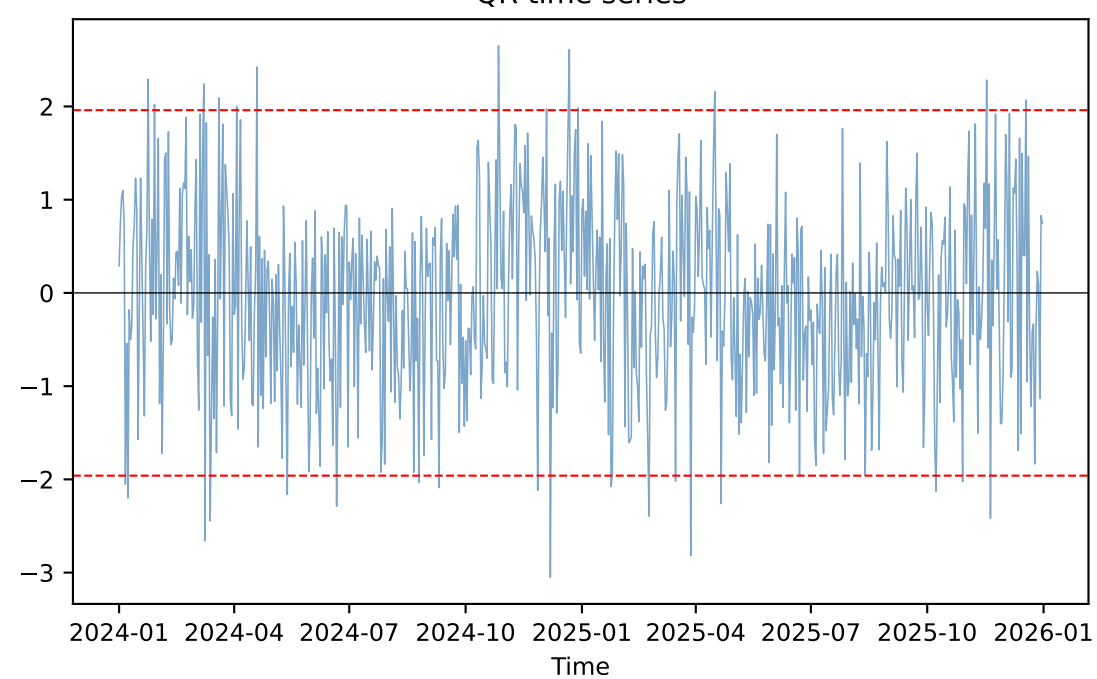
Normal QQ plot



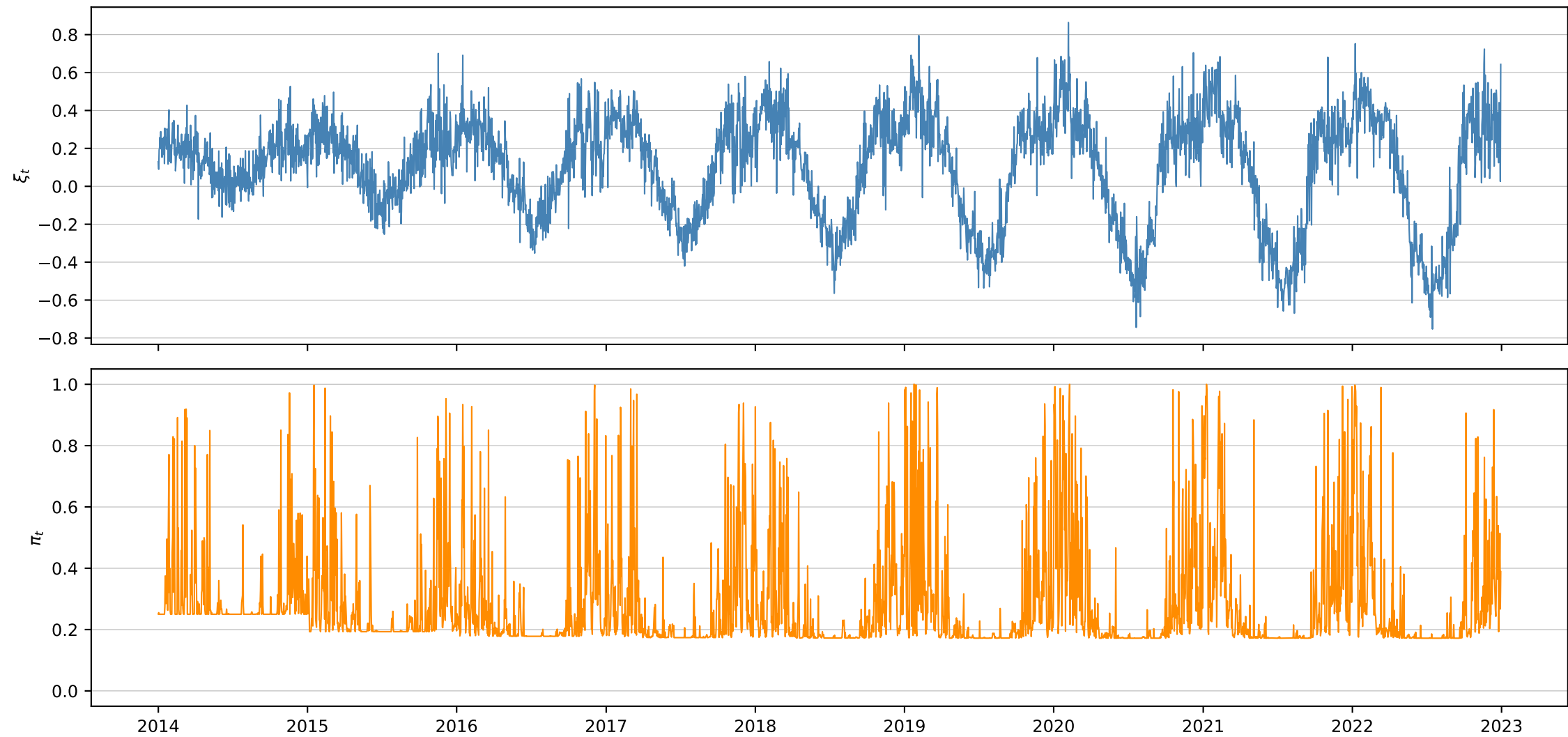
ACF of QR



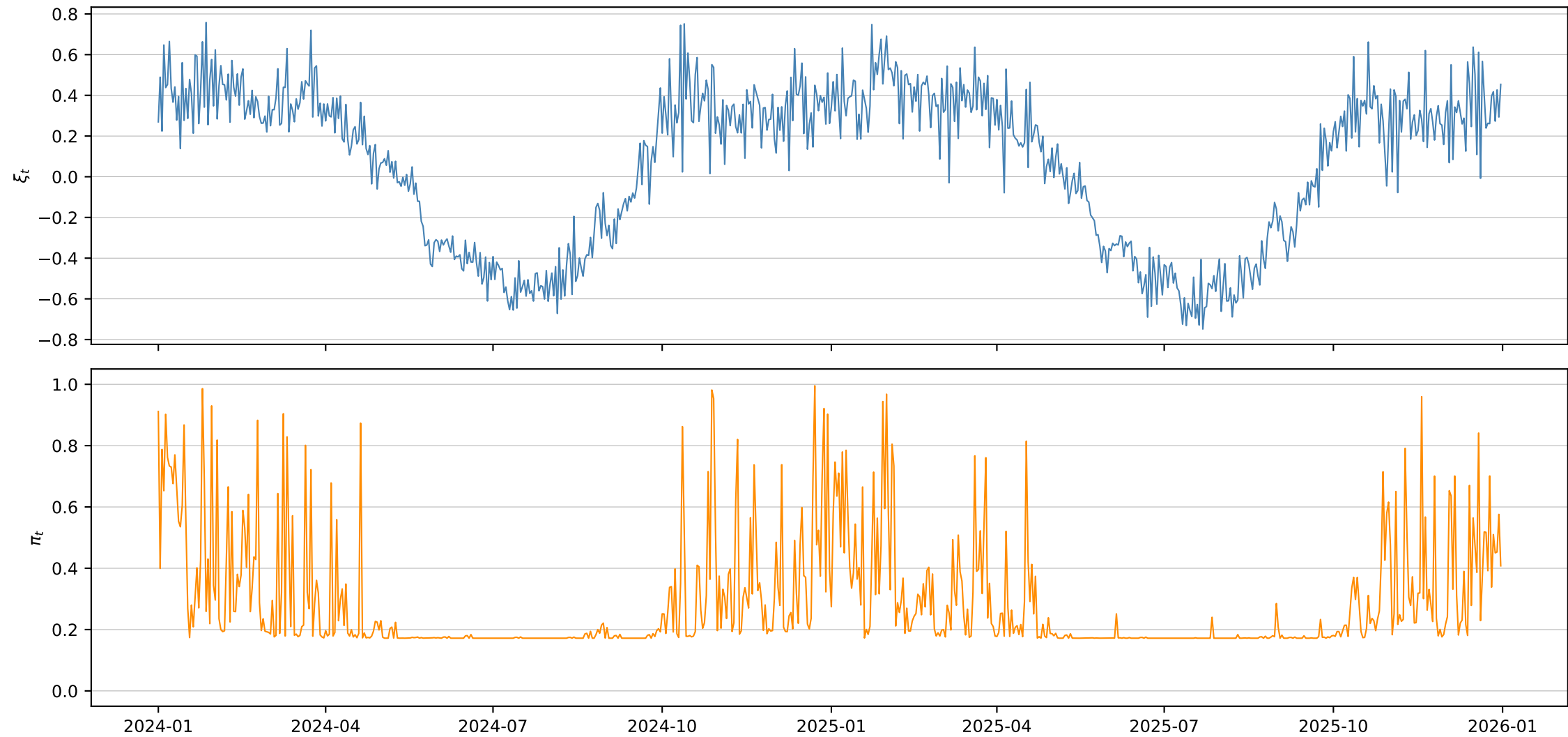
QR time series

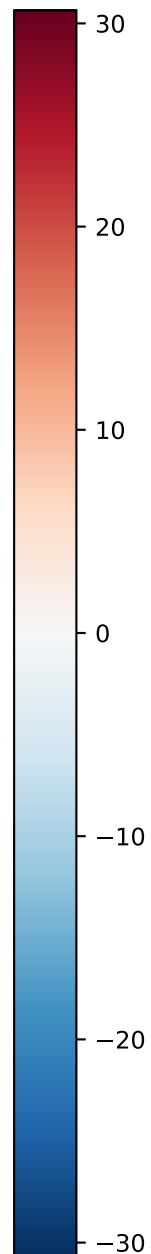


xi_temperature_short_lags — filtered states (IS)

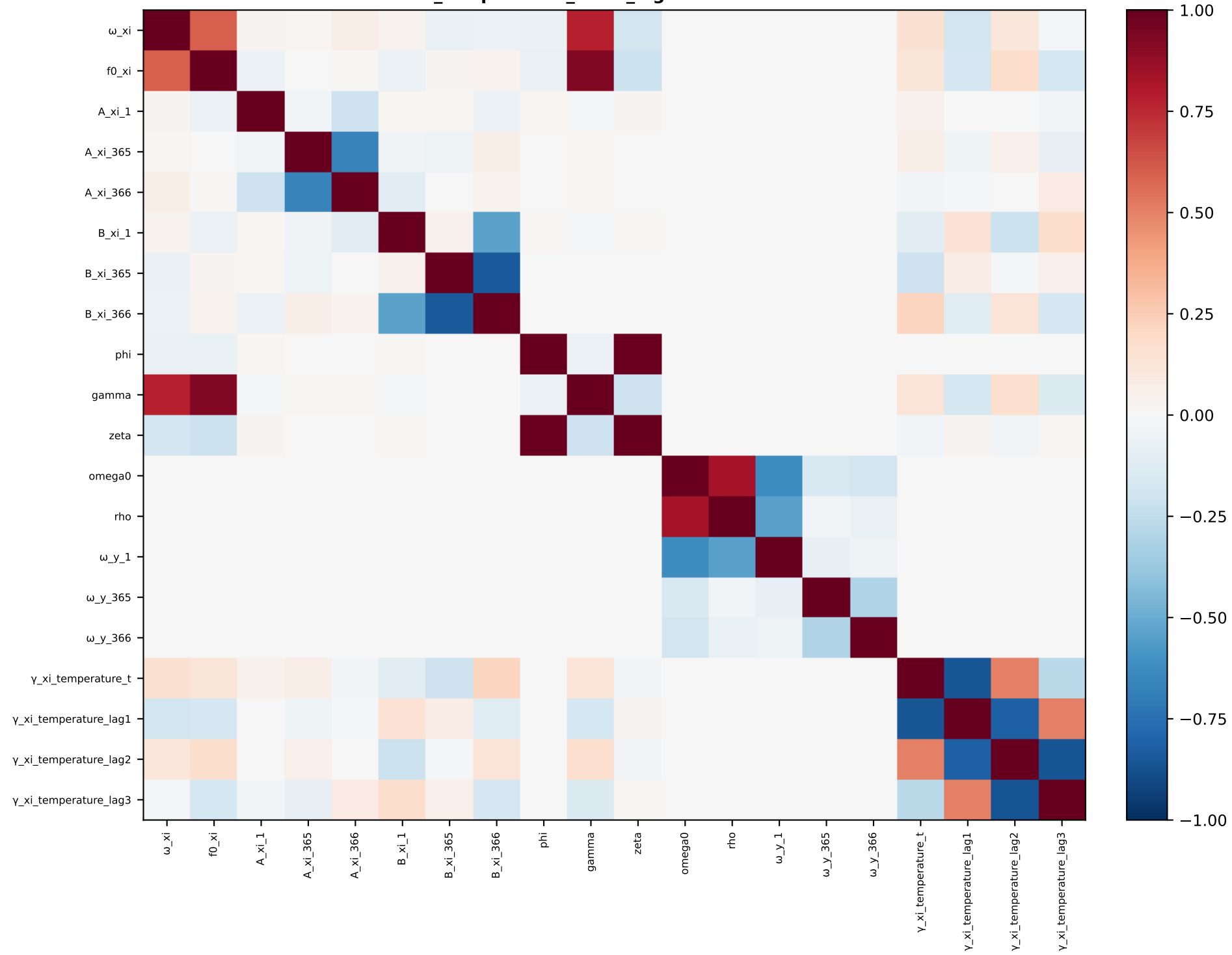


xi_temperature_short_lags — filtered states (OOS)



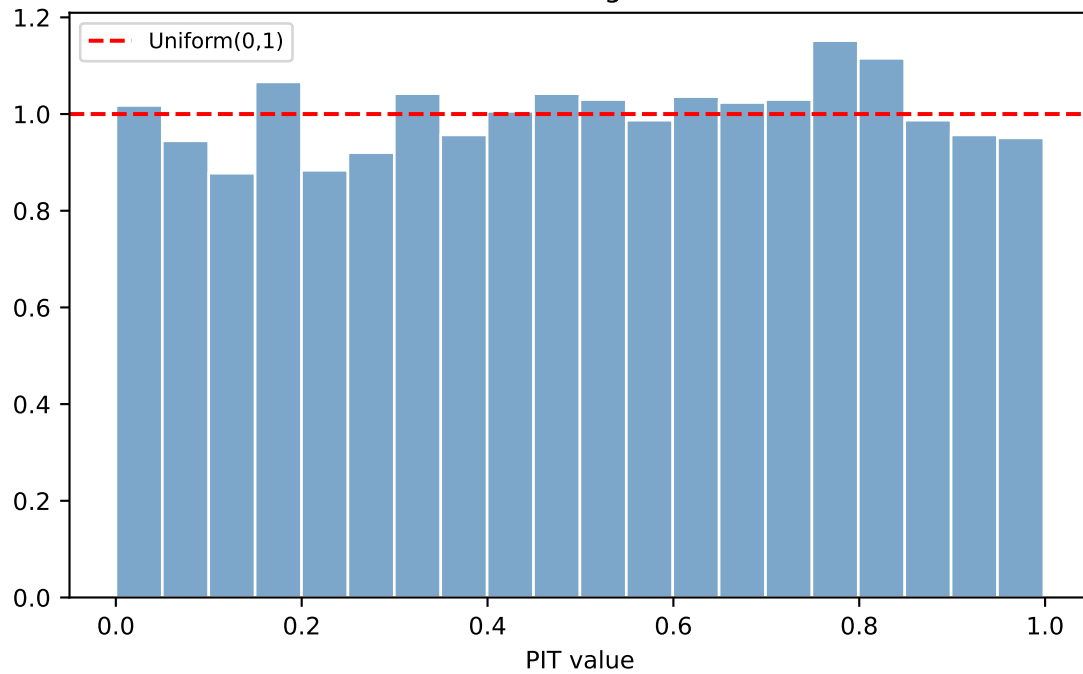


xi_temperature_short_lags — correlation

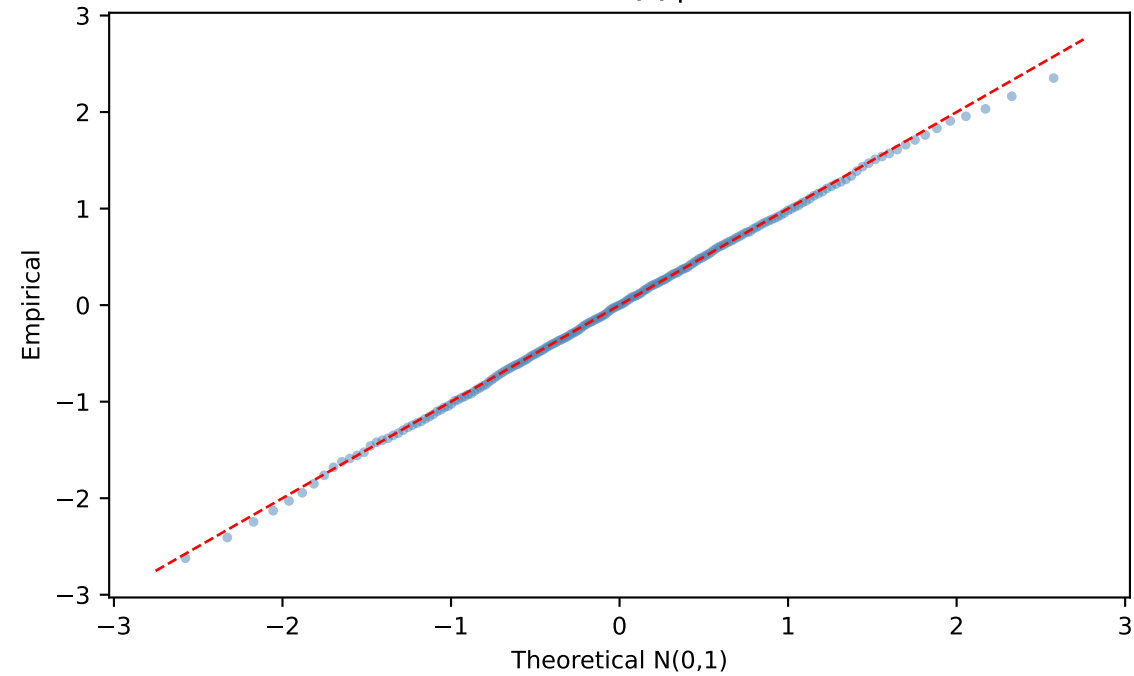


xi_temperature_seasonal_lags — IS diagnostics

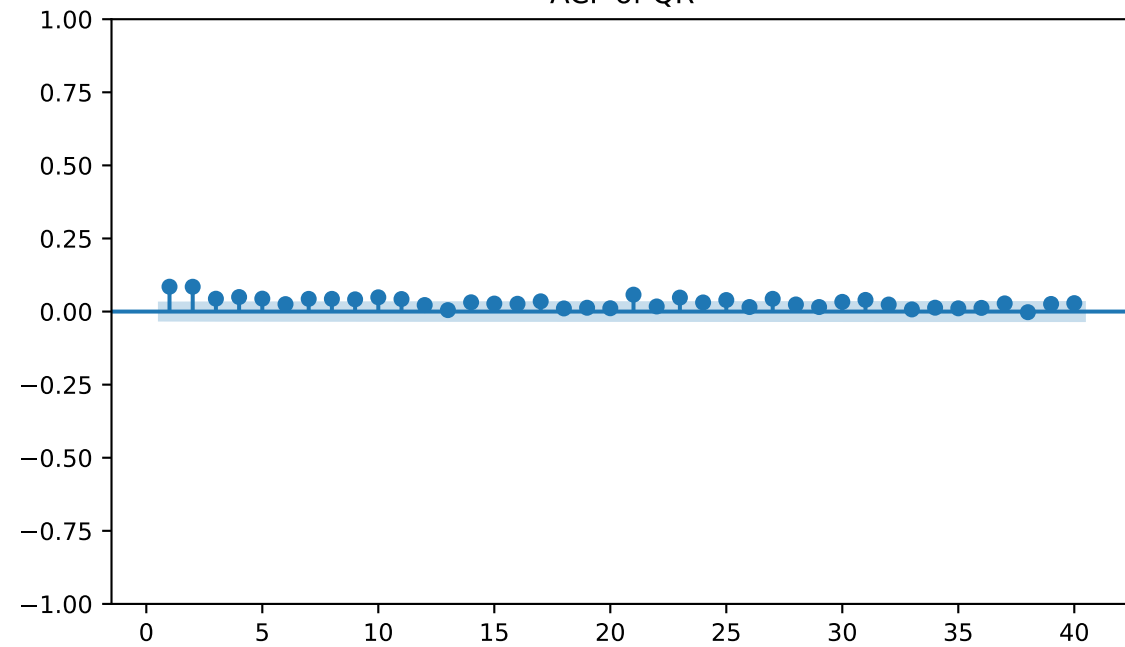
PIT histogram



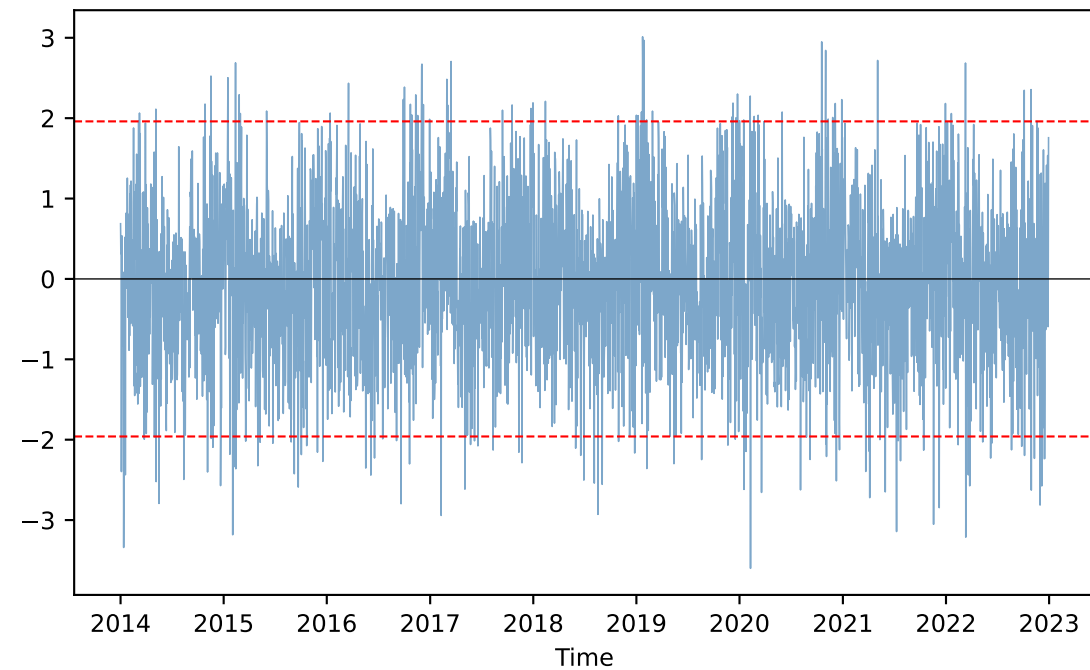
Normal QQ plot



ACF of QR

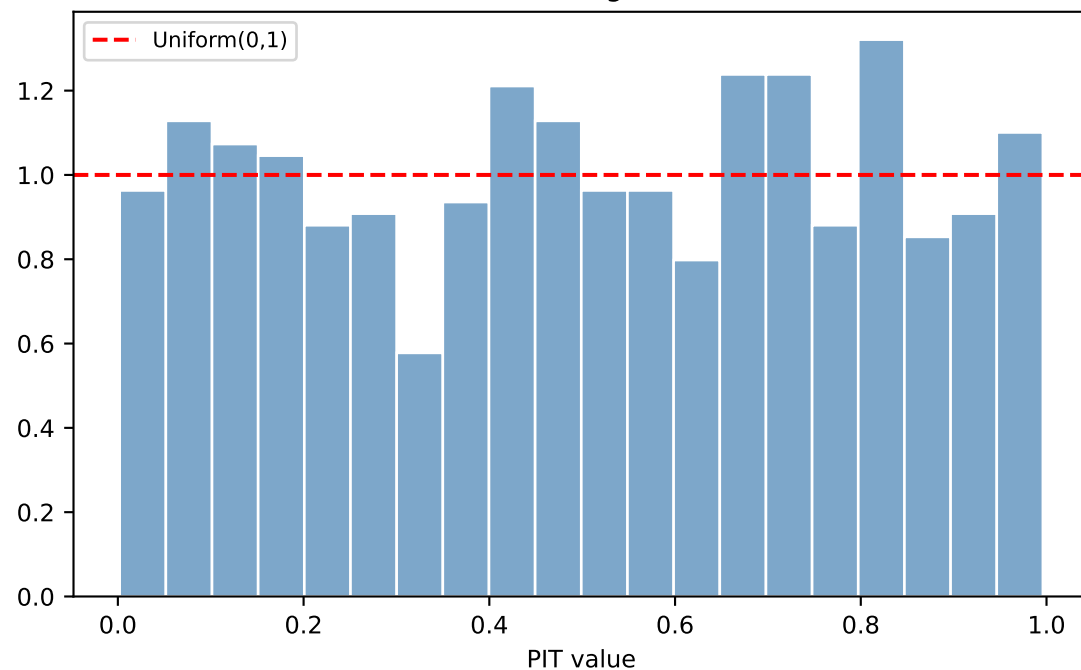


QR time series

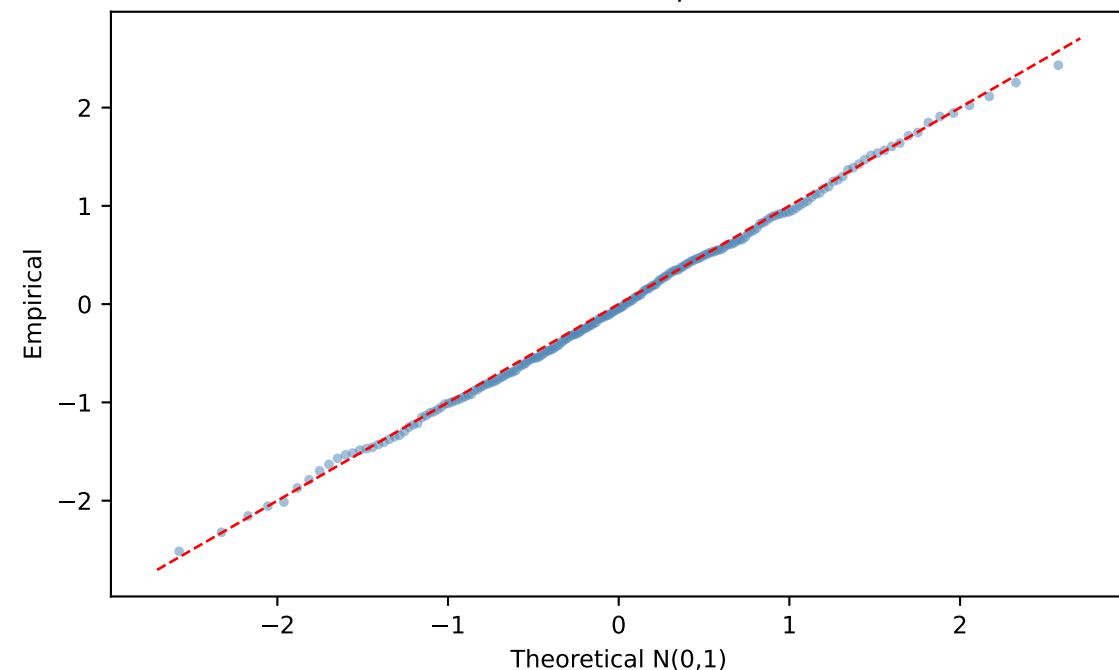


xi_temperature_seasonal_lags — OOS diagnostics

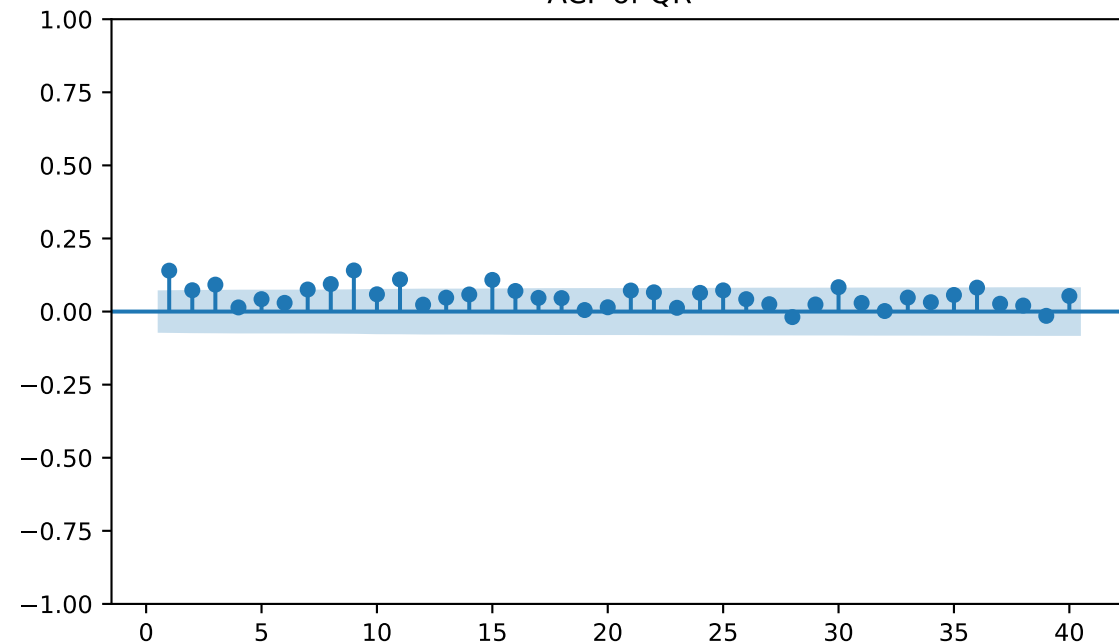
PIT histogram



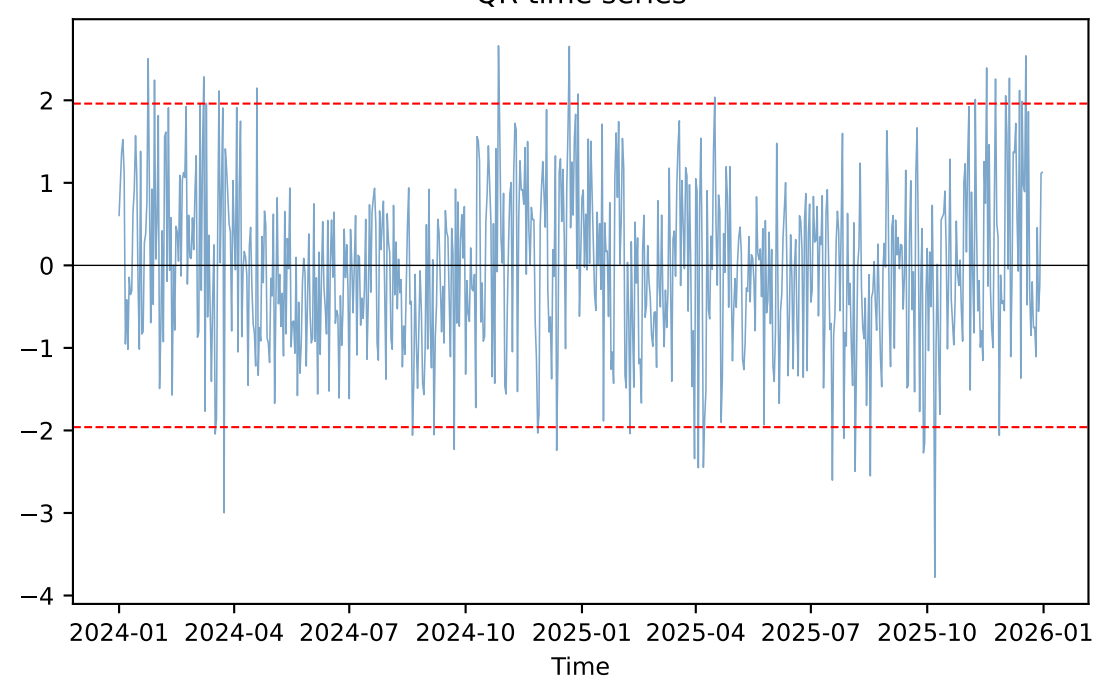
Normal QQ plot



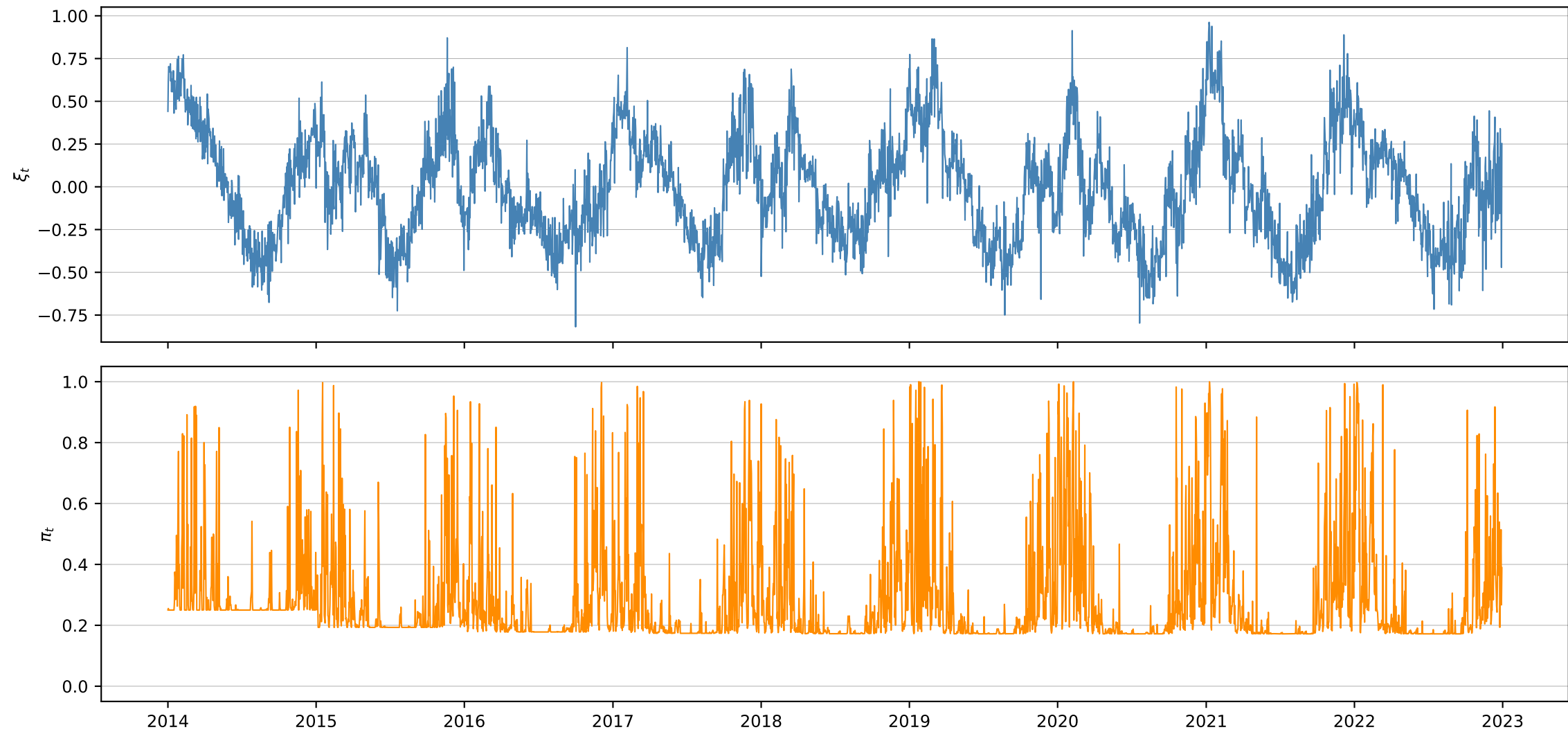
ACF of QR



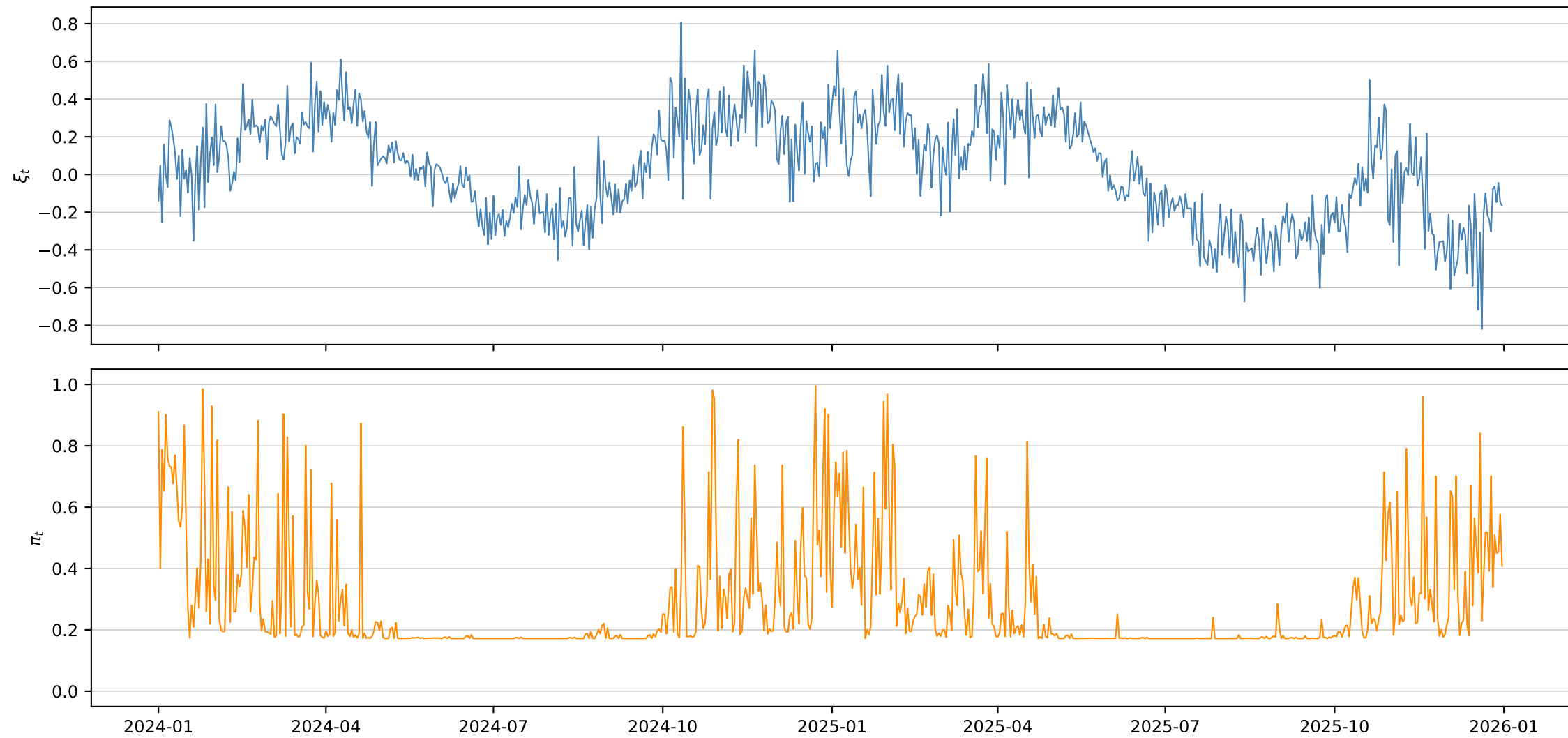
QR time series



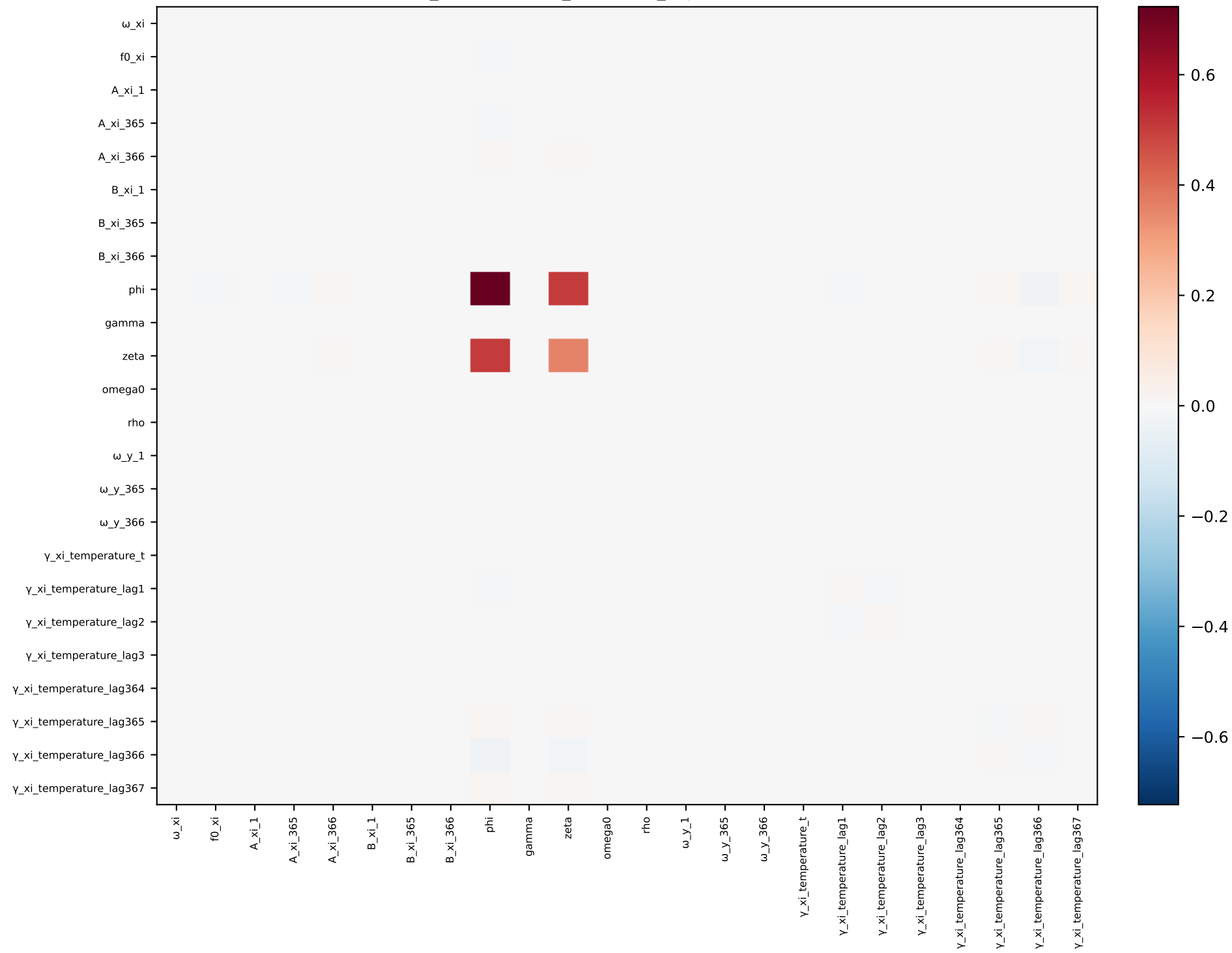
xi_temperature_seasonal_lags — filtered states (IS)



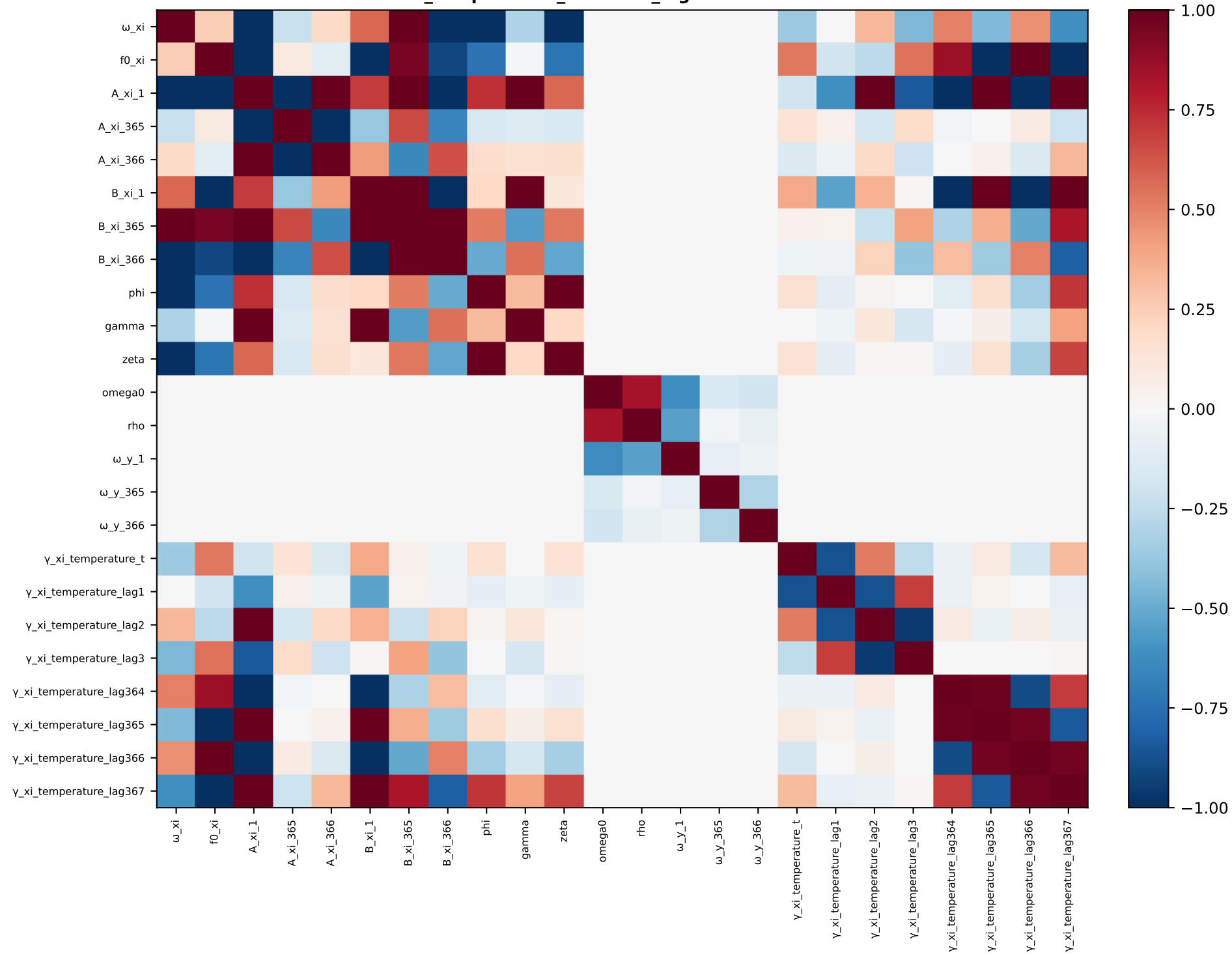
xi_temperature_seasonal_lags — filtered states (OOS)



xi_temperature_seasonal_lags — covariance

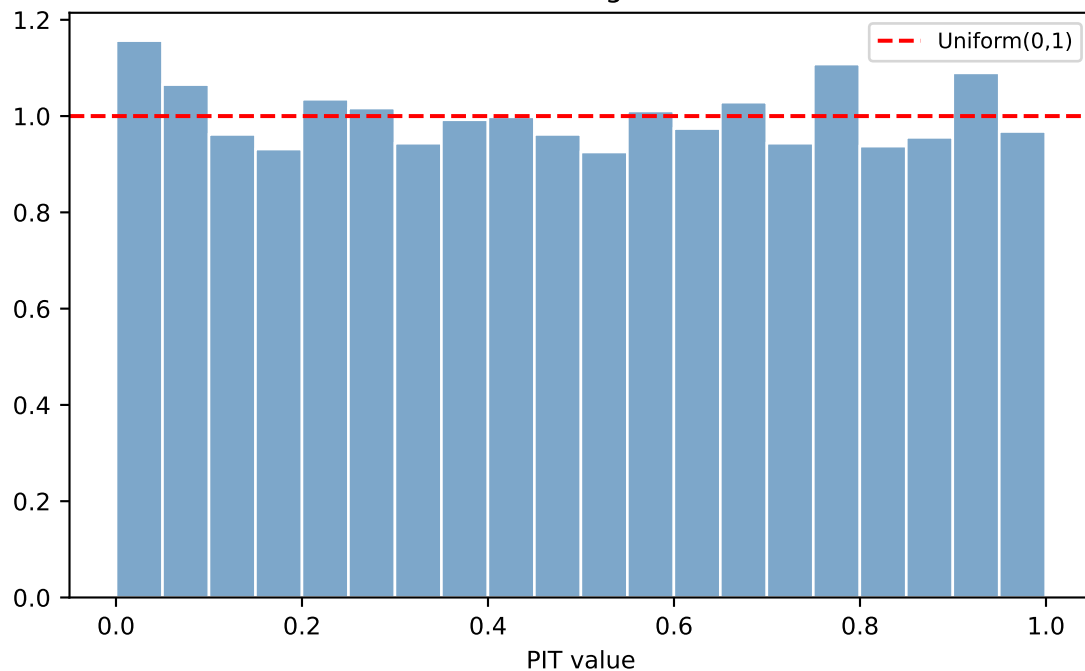


xi_temperature_seasonal_lags — correlation

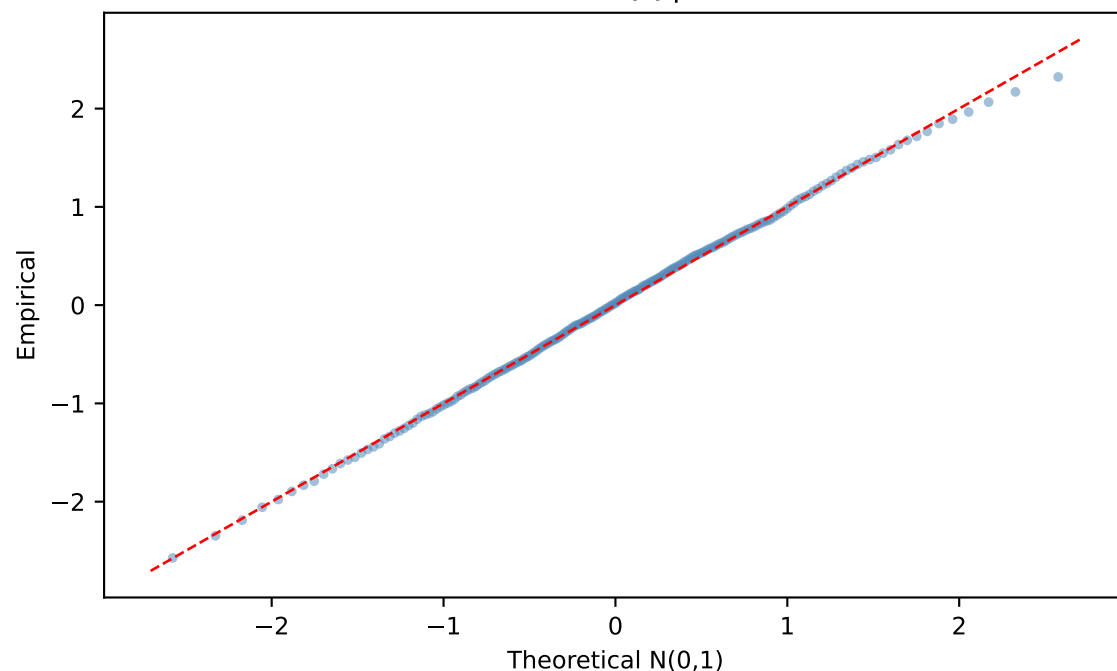


xi_dewpoint_temperature_t — IS diagnostics

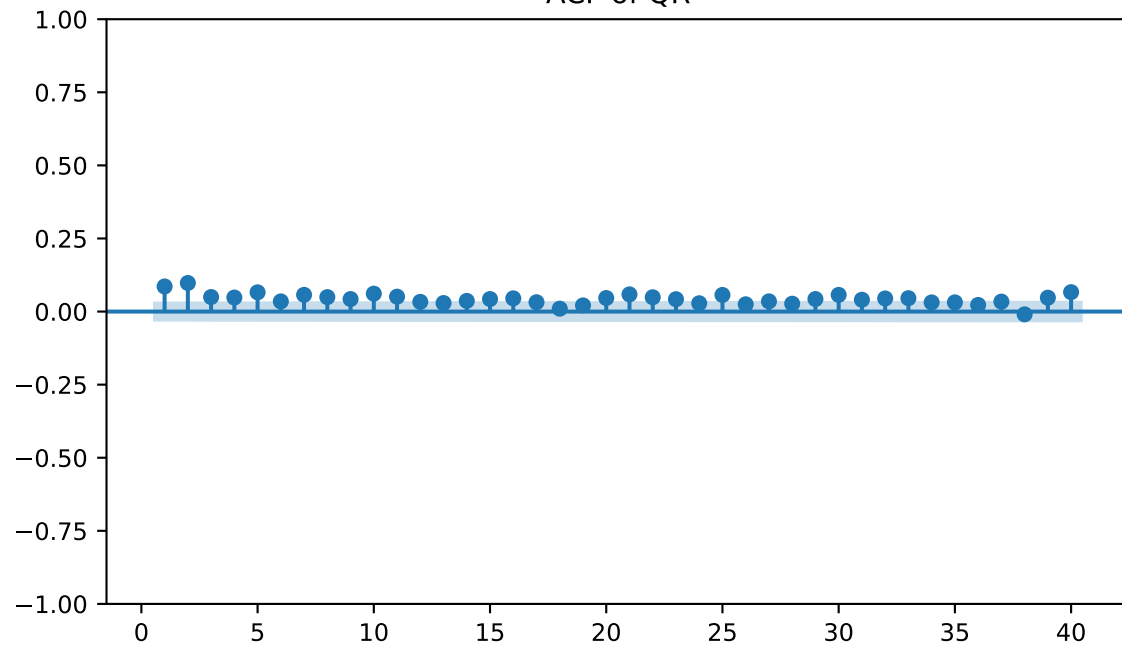
PIT histogram



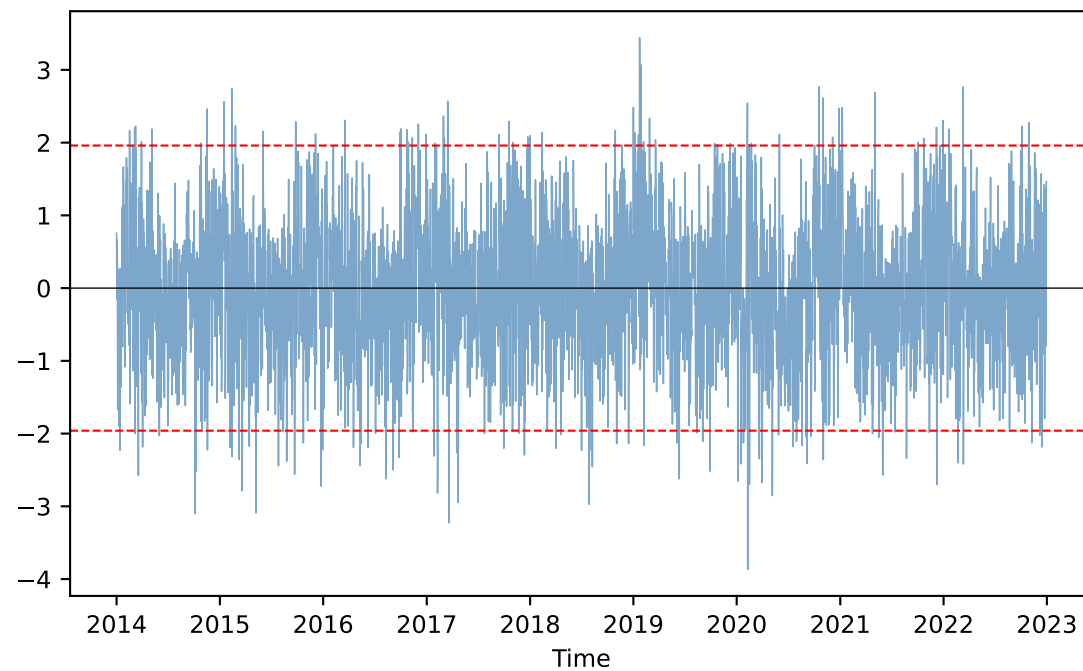
Normal QQ plot



ACF of QR

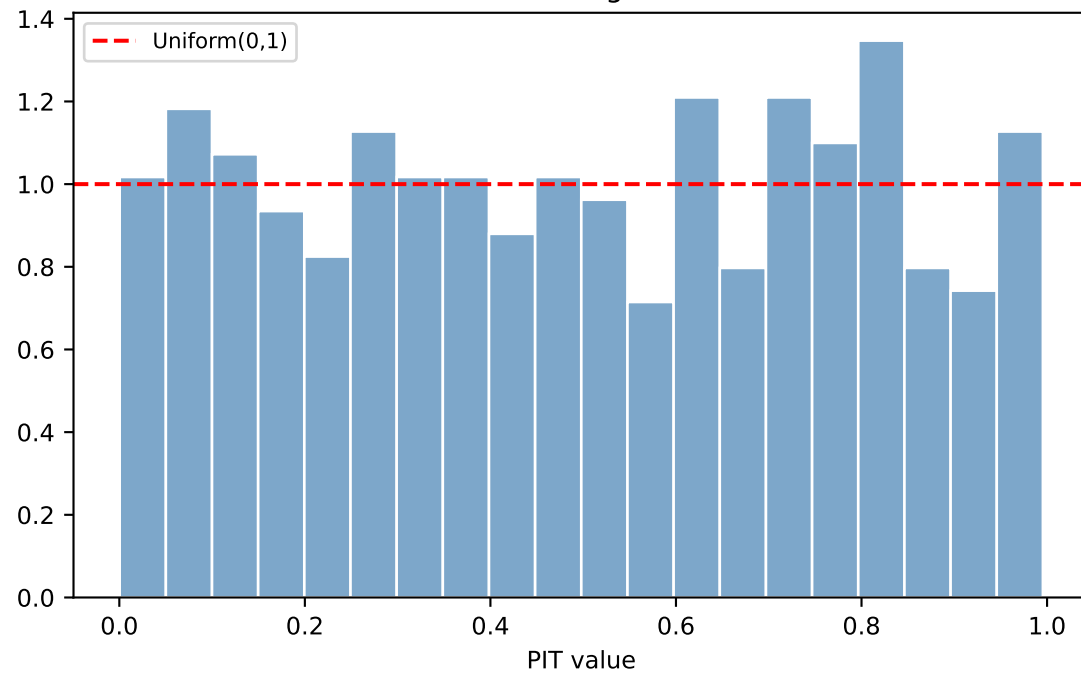


QR time series

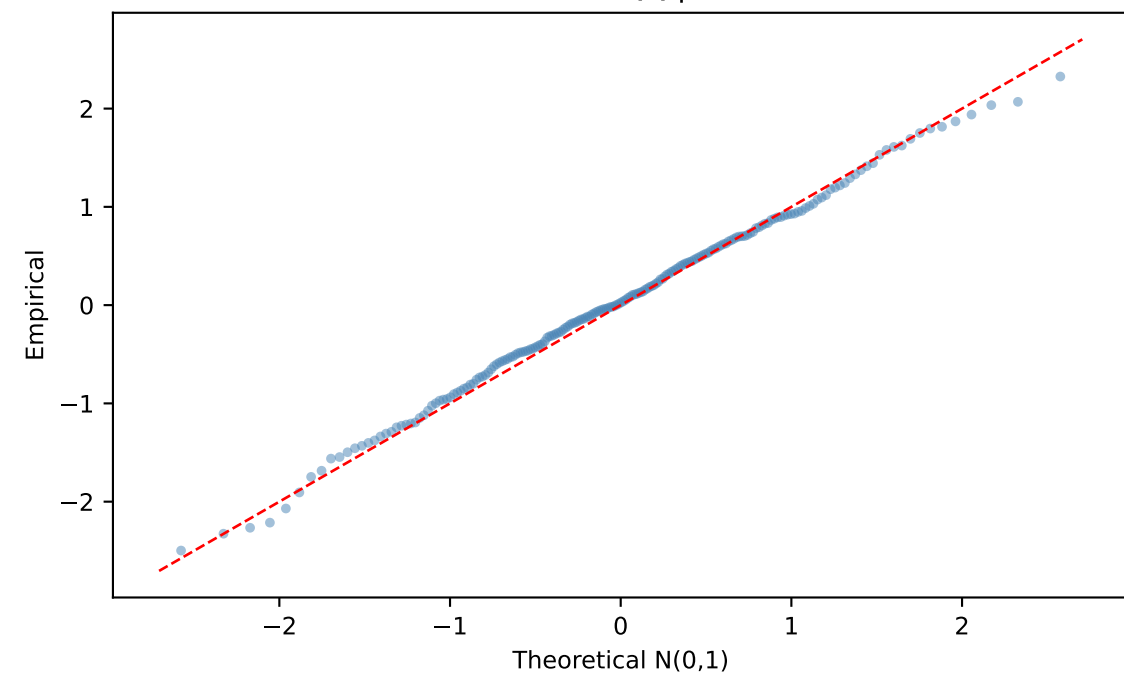


xi_dewpoint_temperature_t — OOS diagnostics

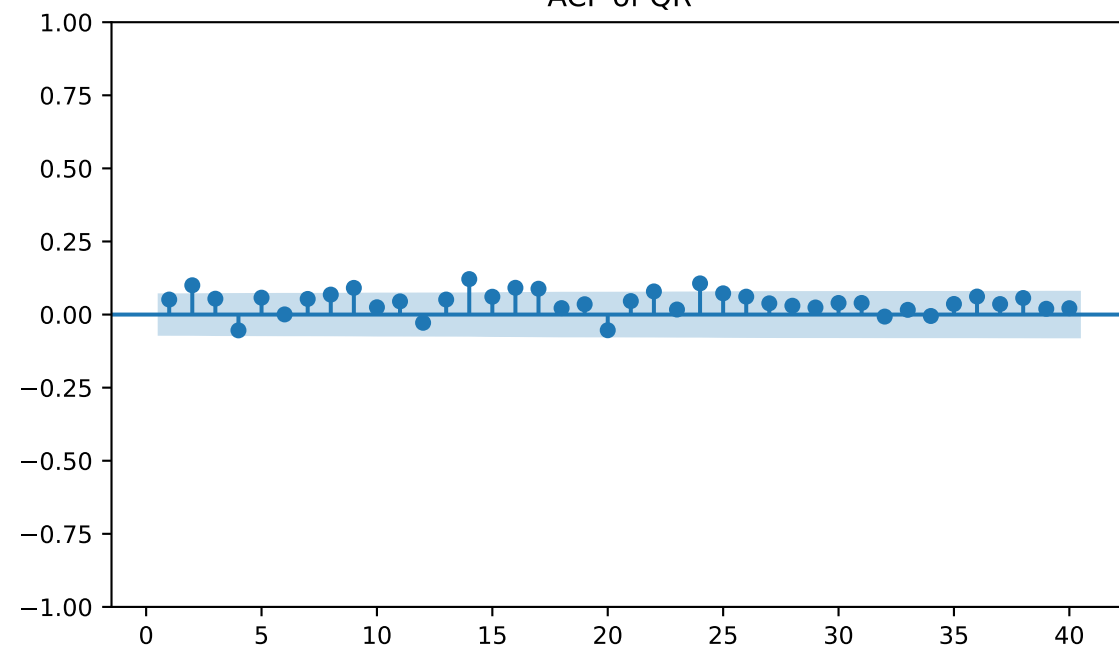
PIT histogram



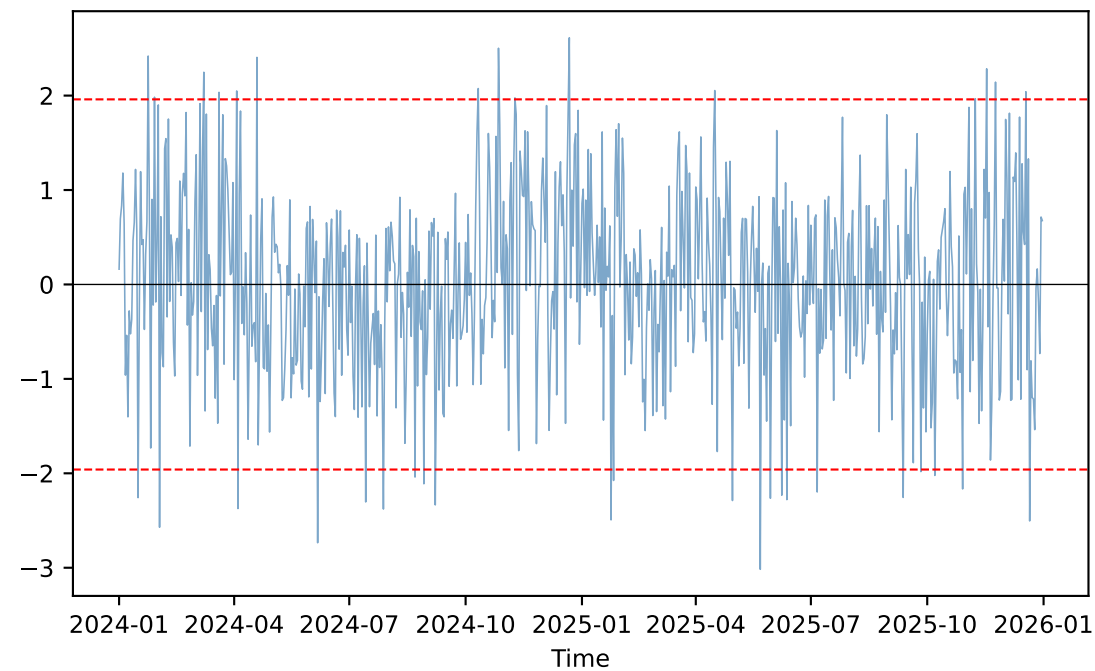
Normal QQ plot



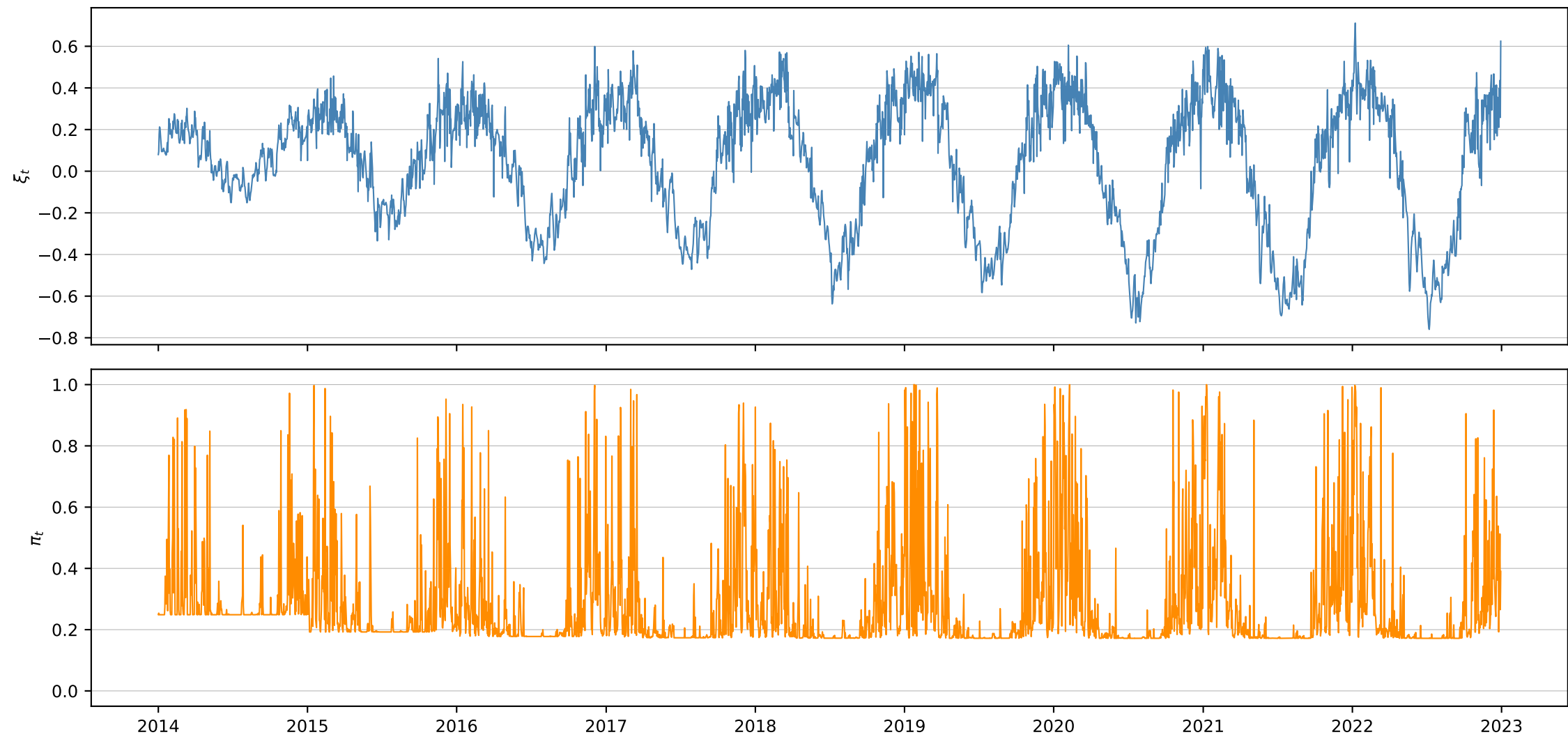
ACF of QR



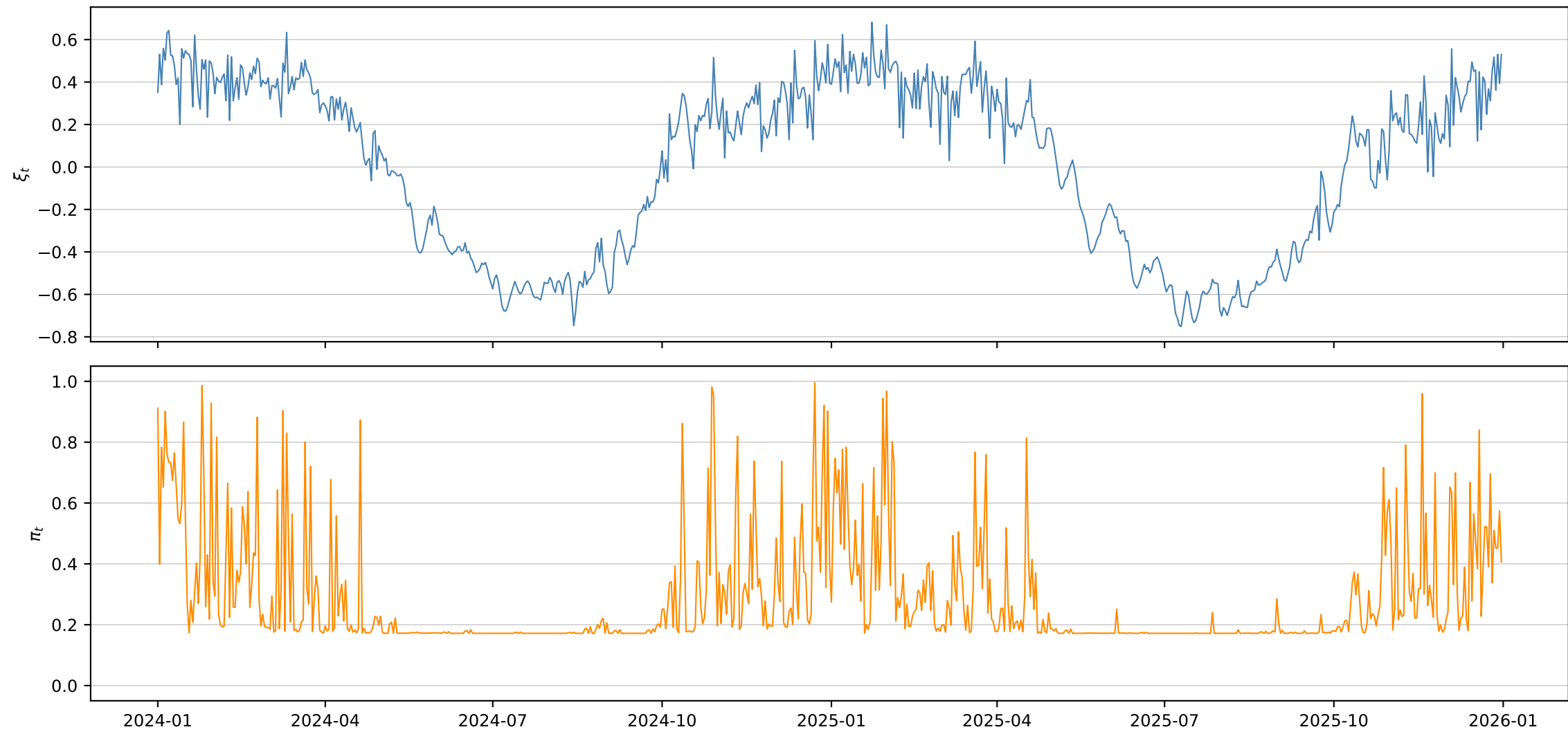
QR time series



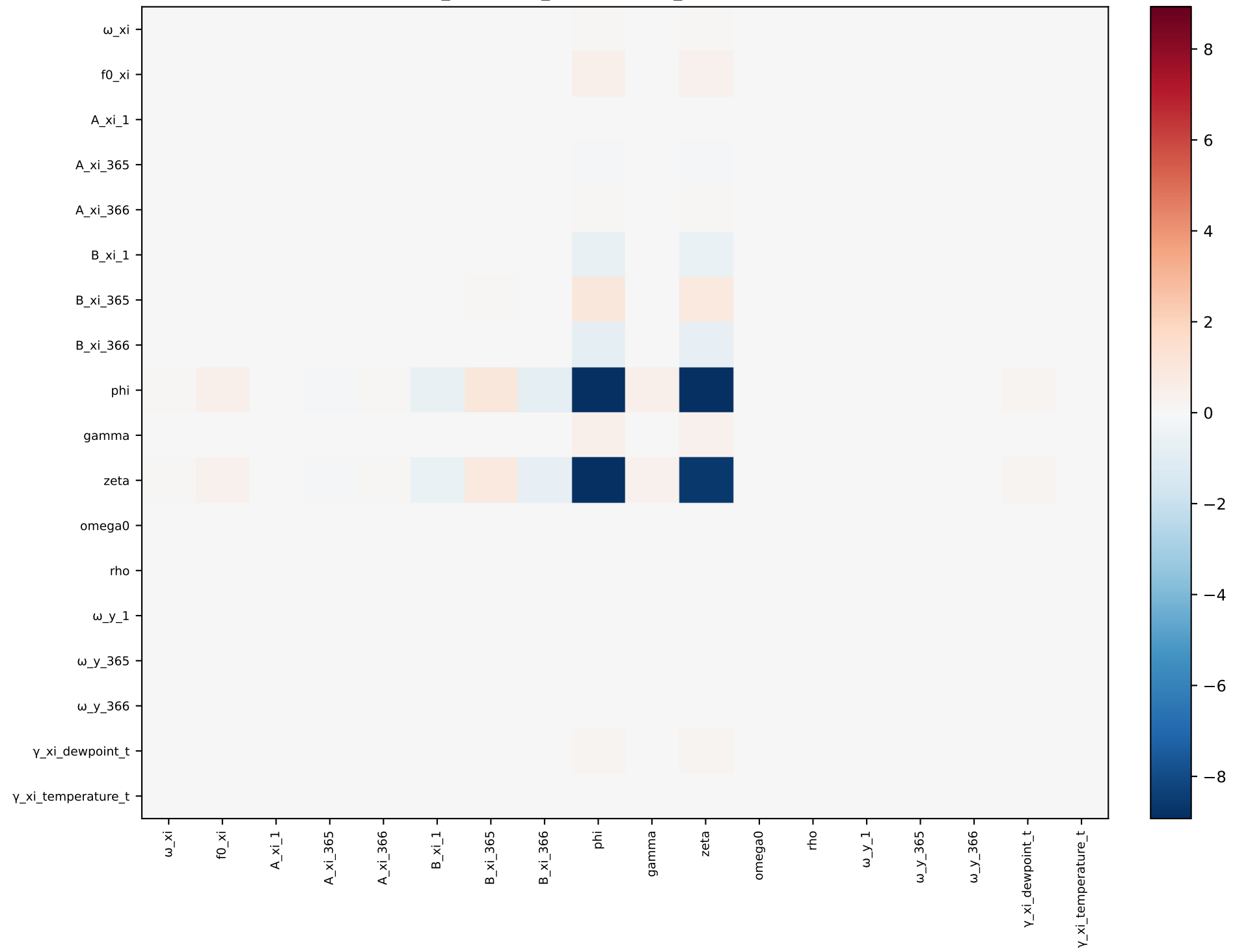
xi_dewpoint_temperature_t — filtered states (IS)



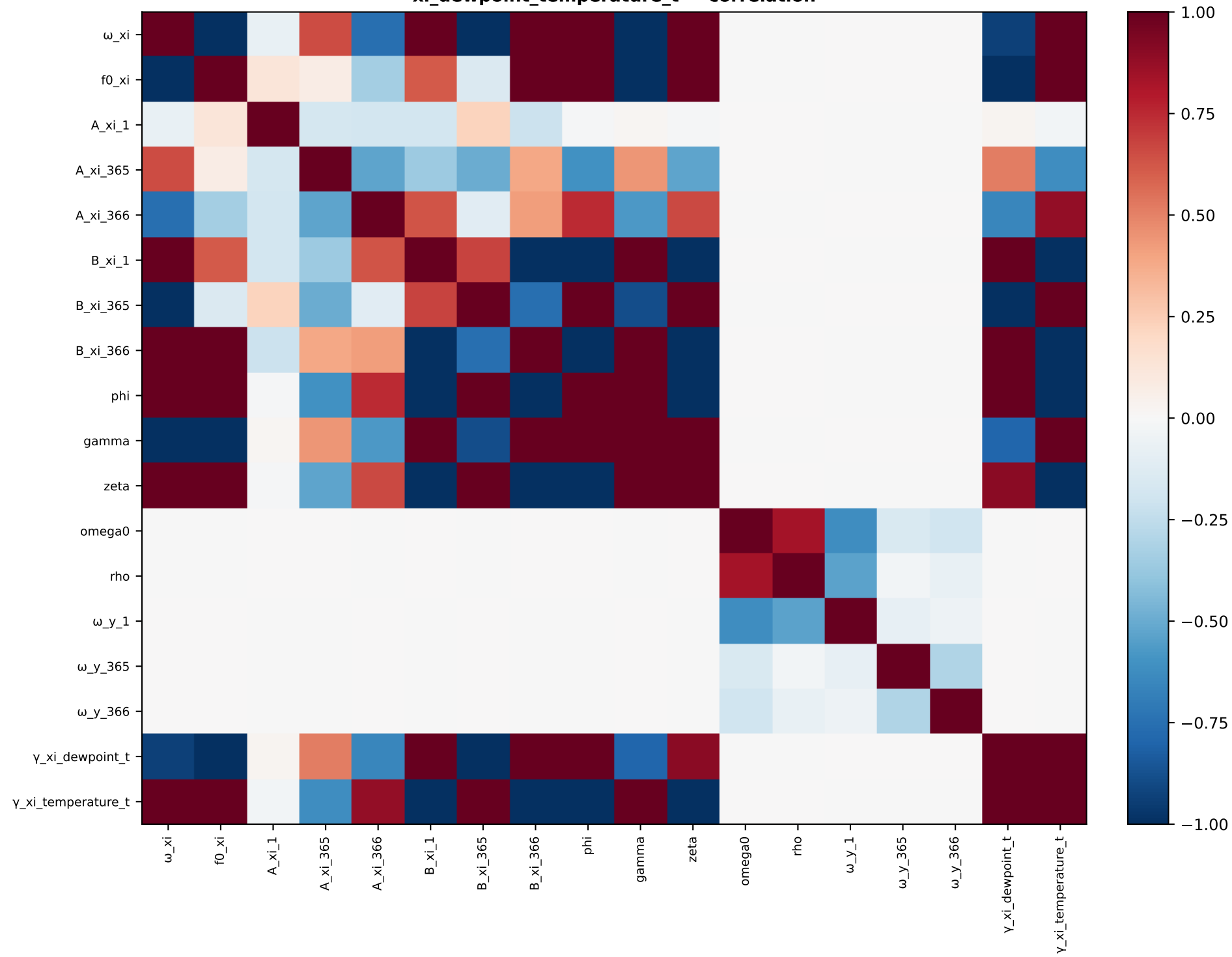
xi_dewpoint_temperature_t — filtered states (OOS)



xi_dewpoint_temperature_t — covariance

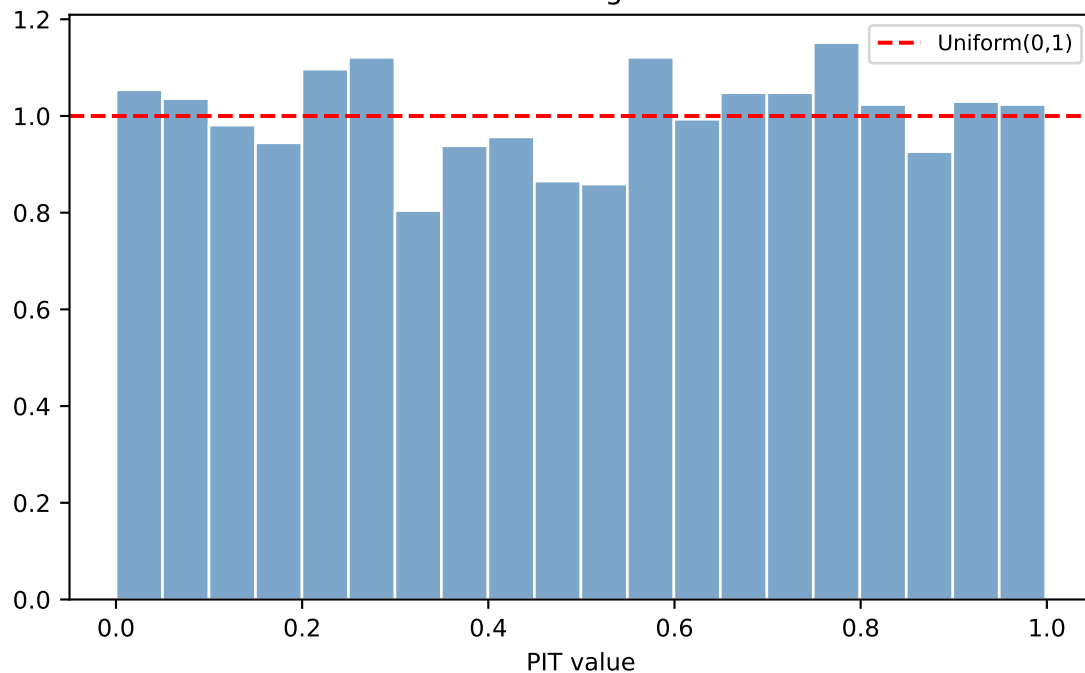


xi_dewpoint_temperature_t — correlation

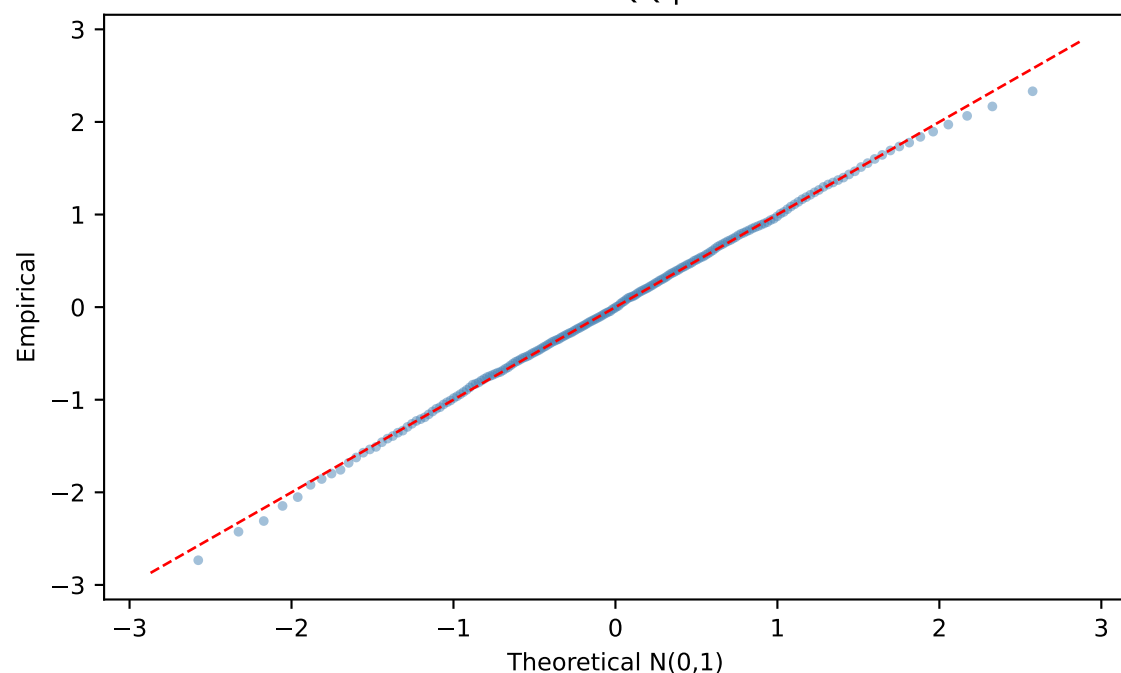


xi_dewpoint_temperature_lag1 — IS diagnostics

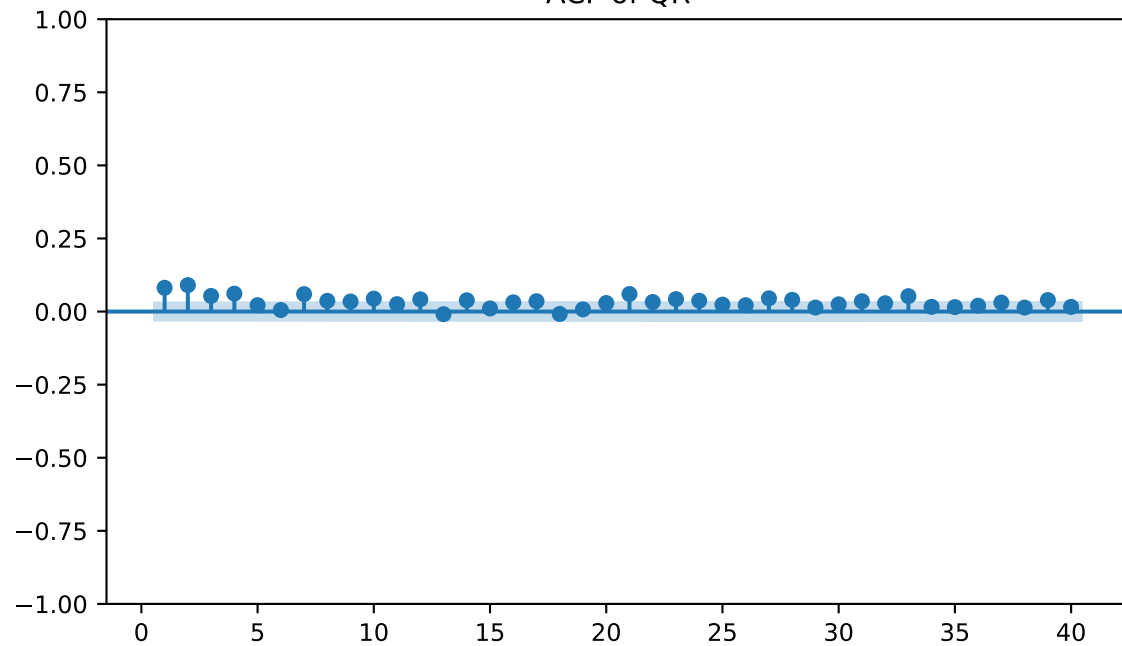
PIT histogram



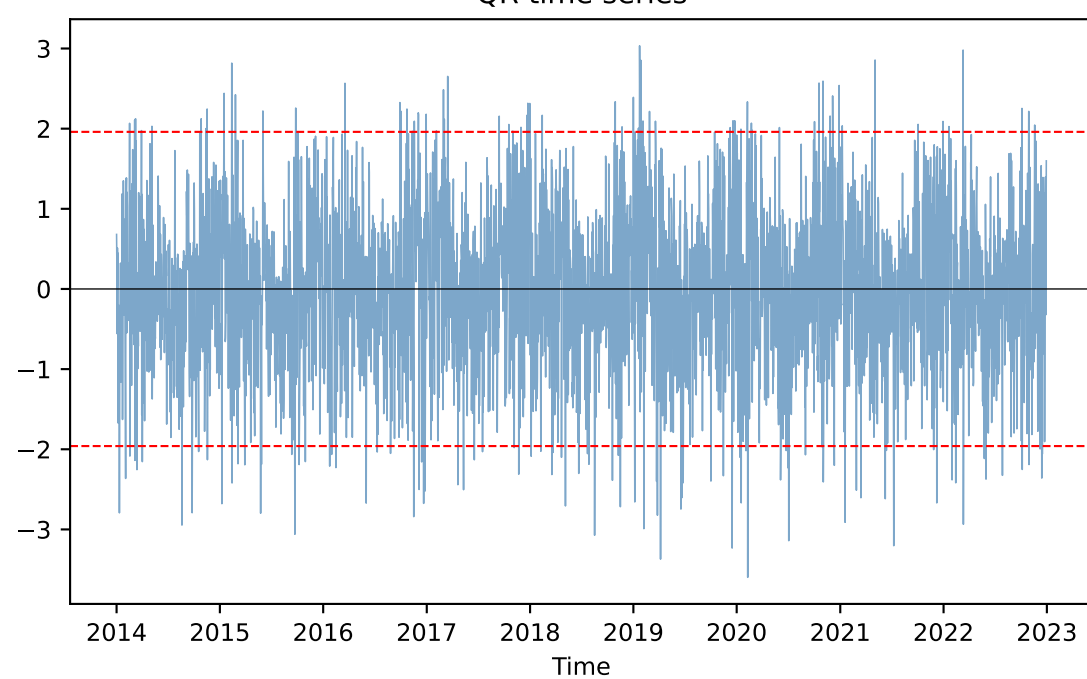
Normal QQ plot



ACF of QR

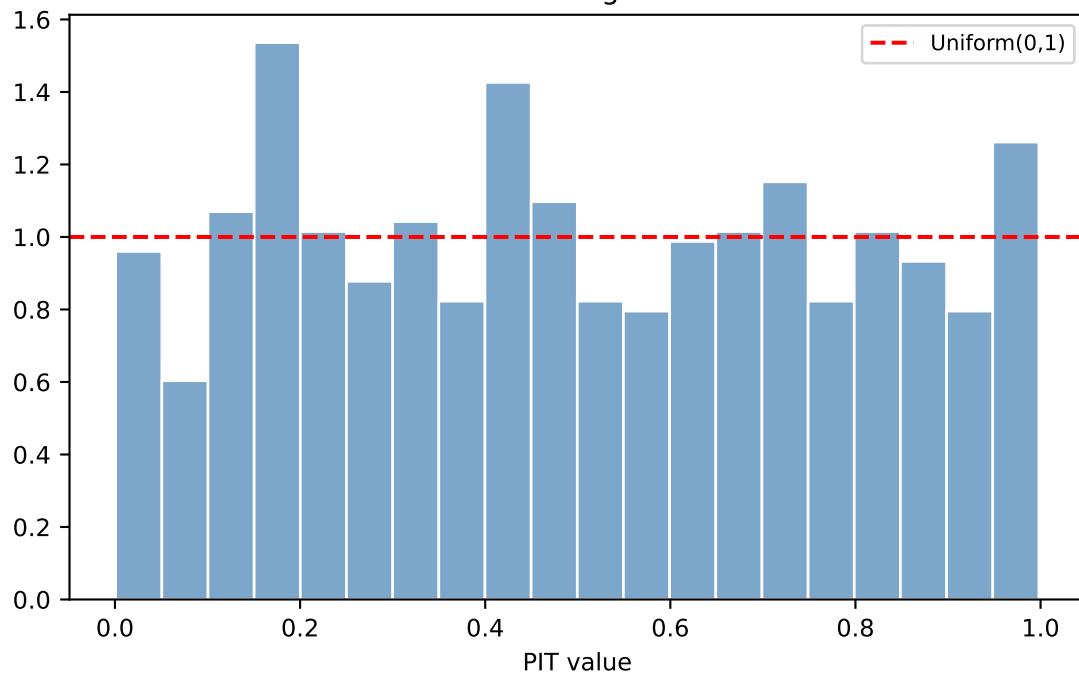


QR time series

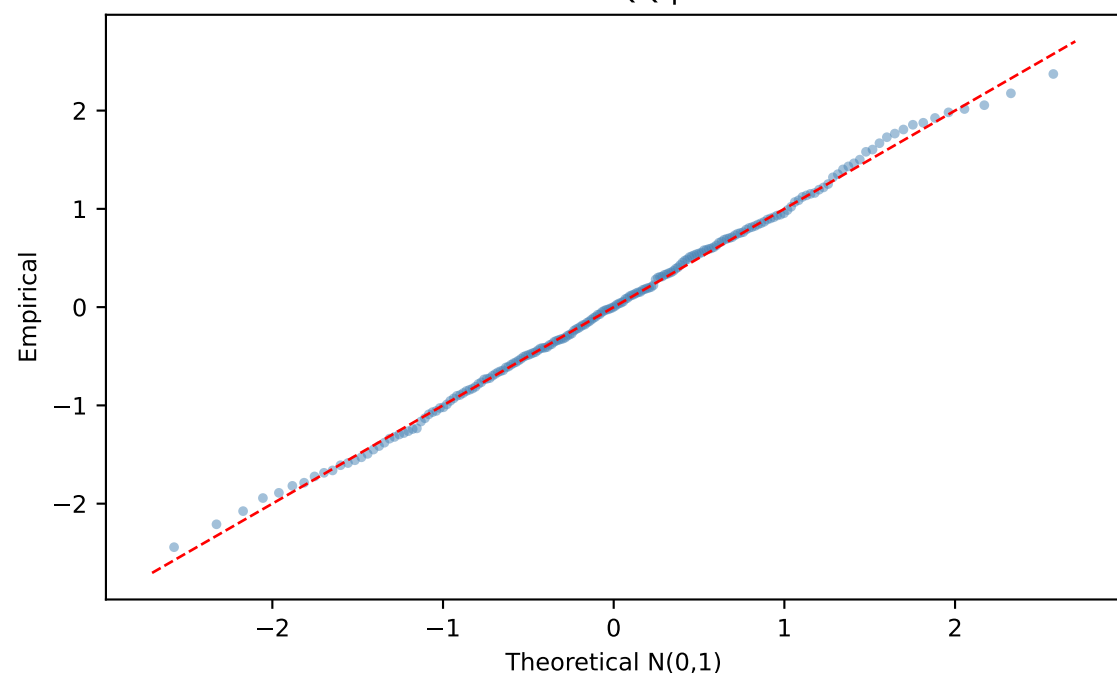


xi_dewpoint_temperature_lag1 — OOS diagnostics

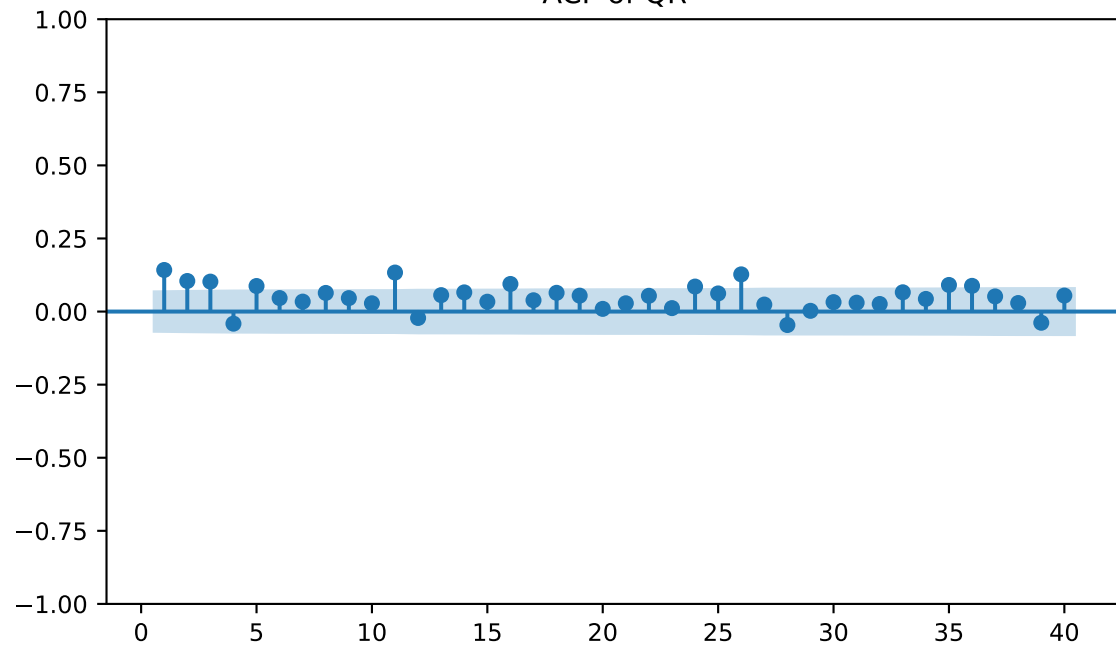
PIT histogram



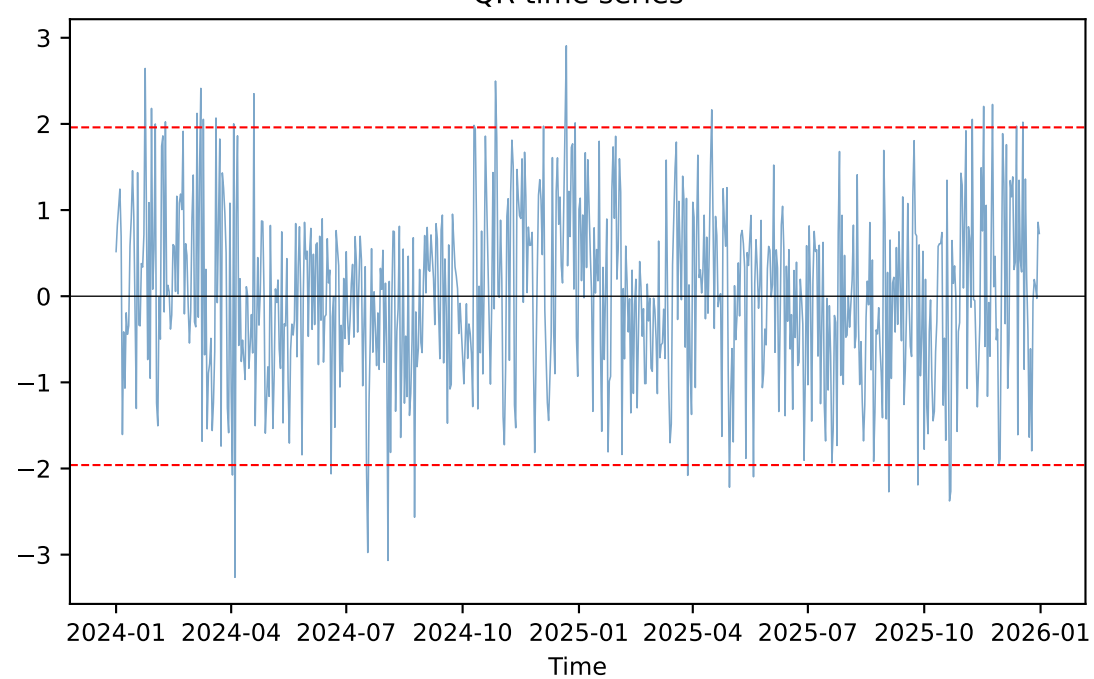
Normal QQ plot



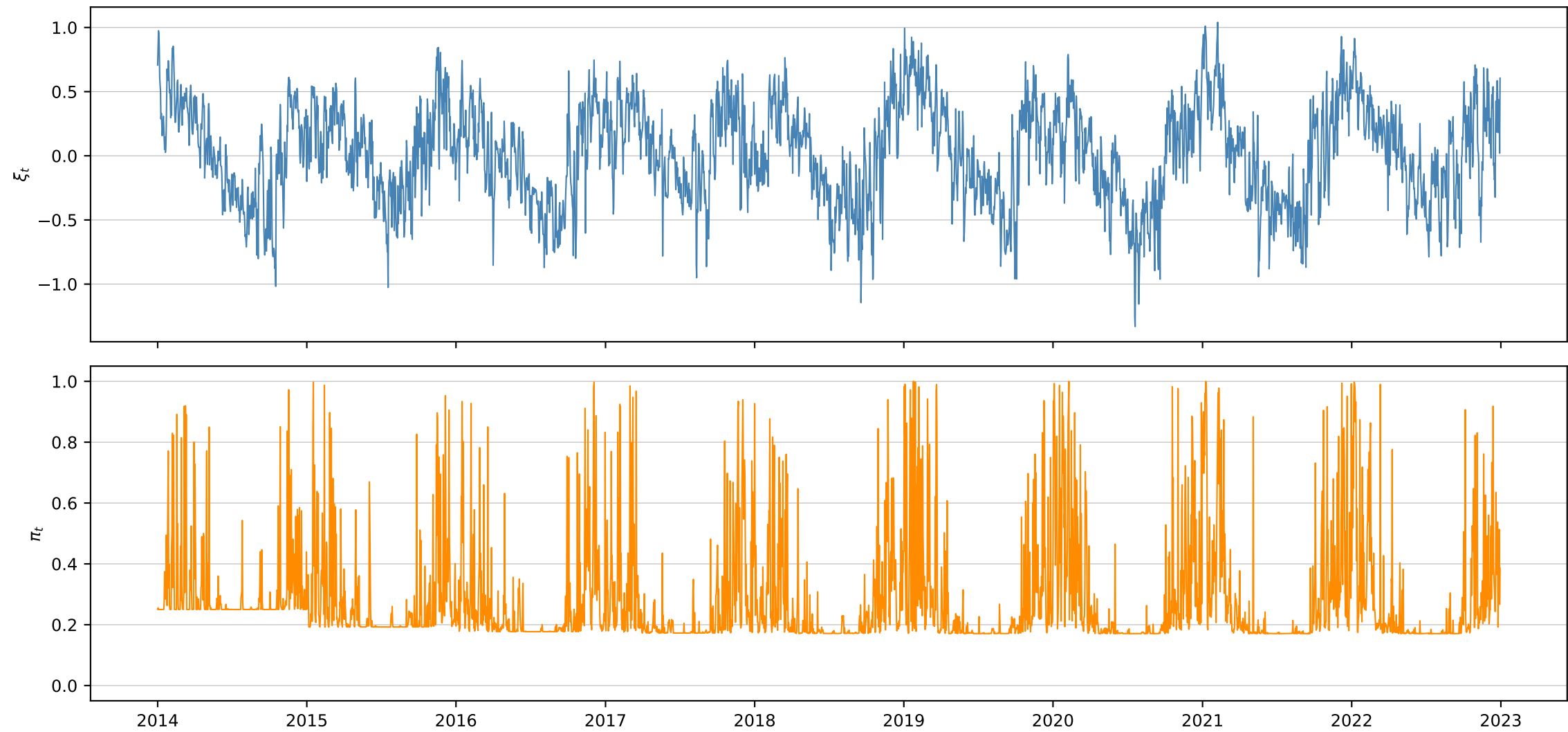
ACF of QR



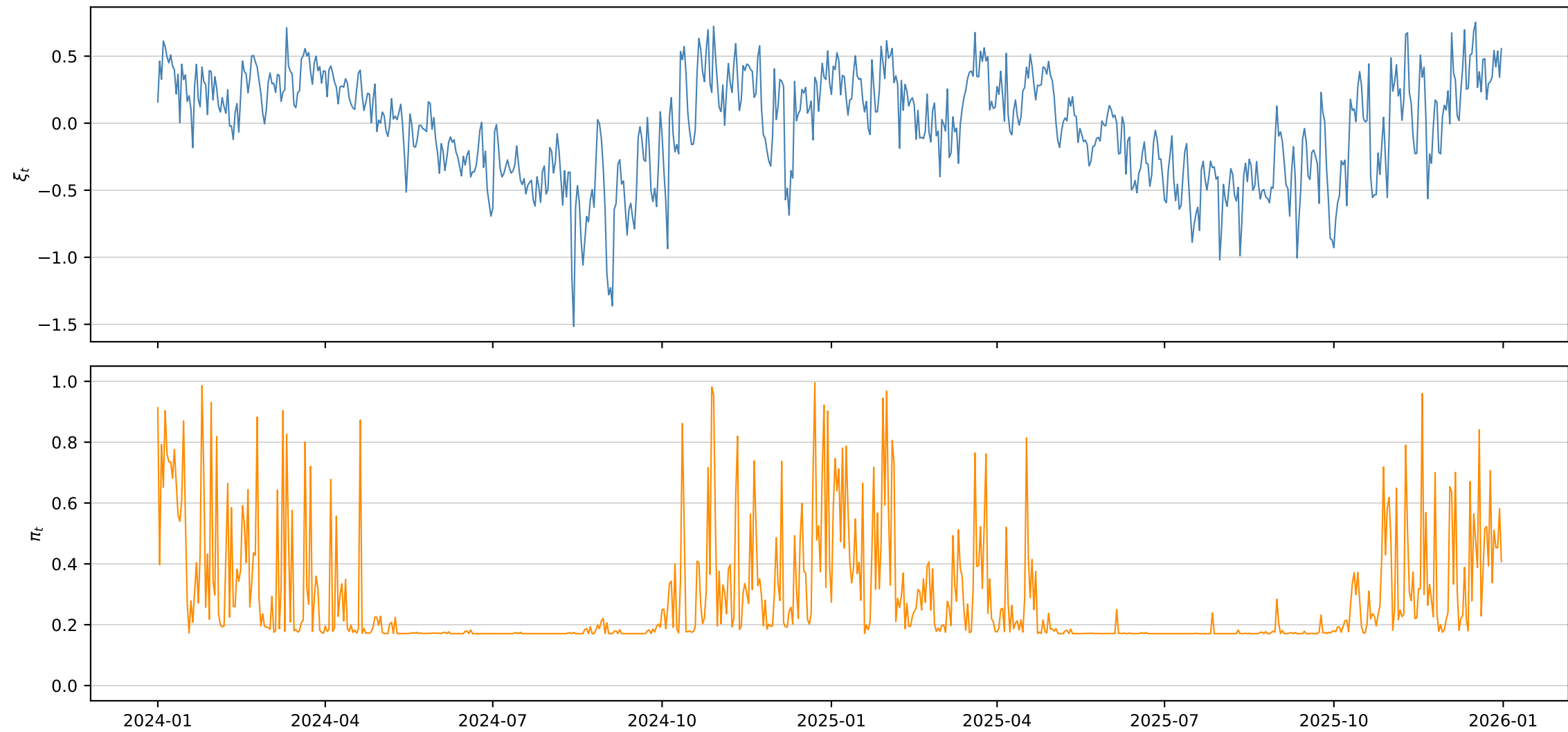
QR time series



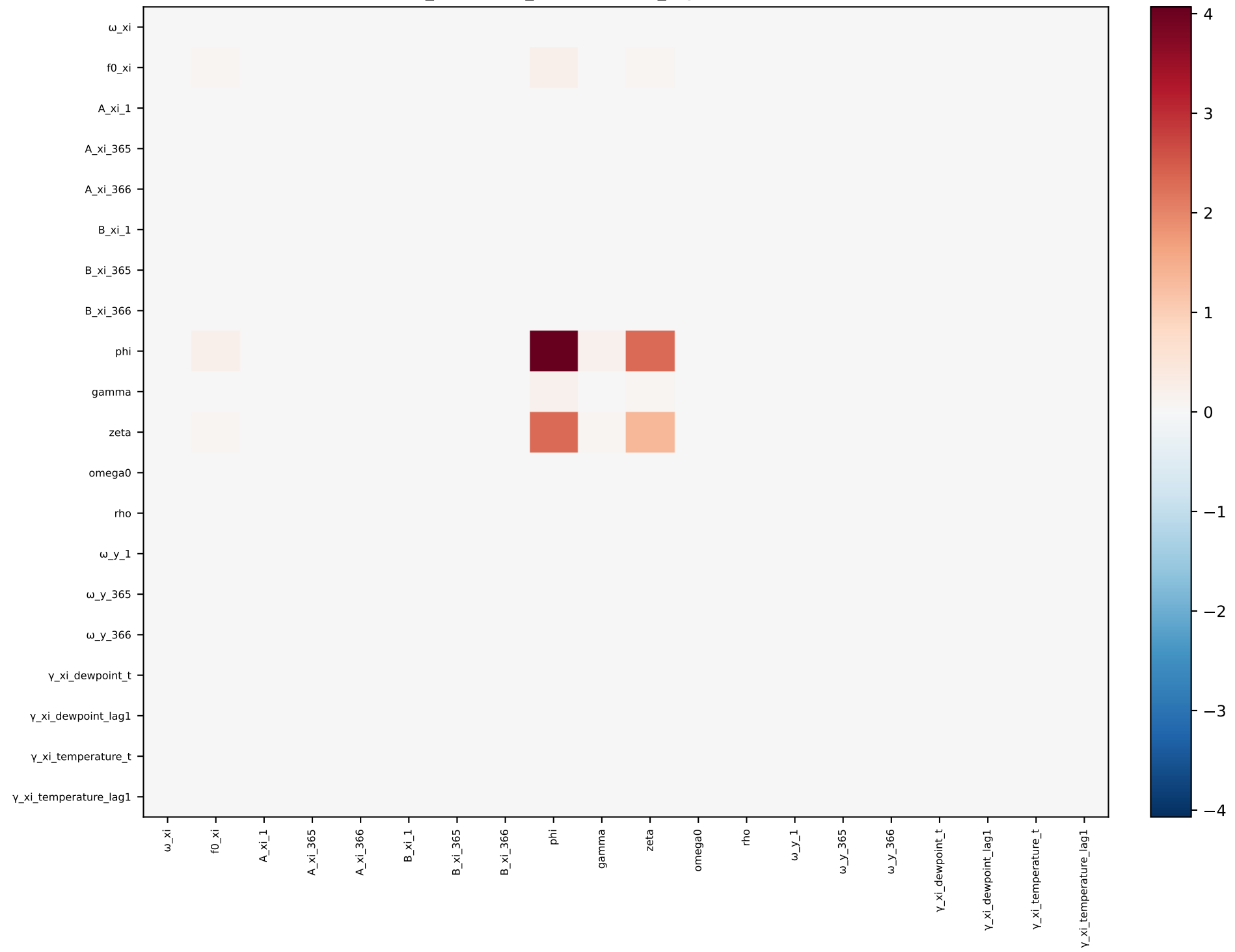
xi_dewpoint_temperature_lag1 — filtered states (IS)



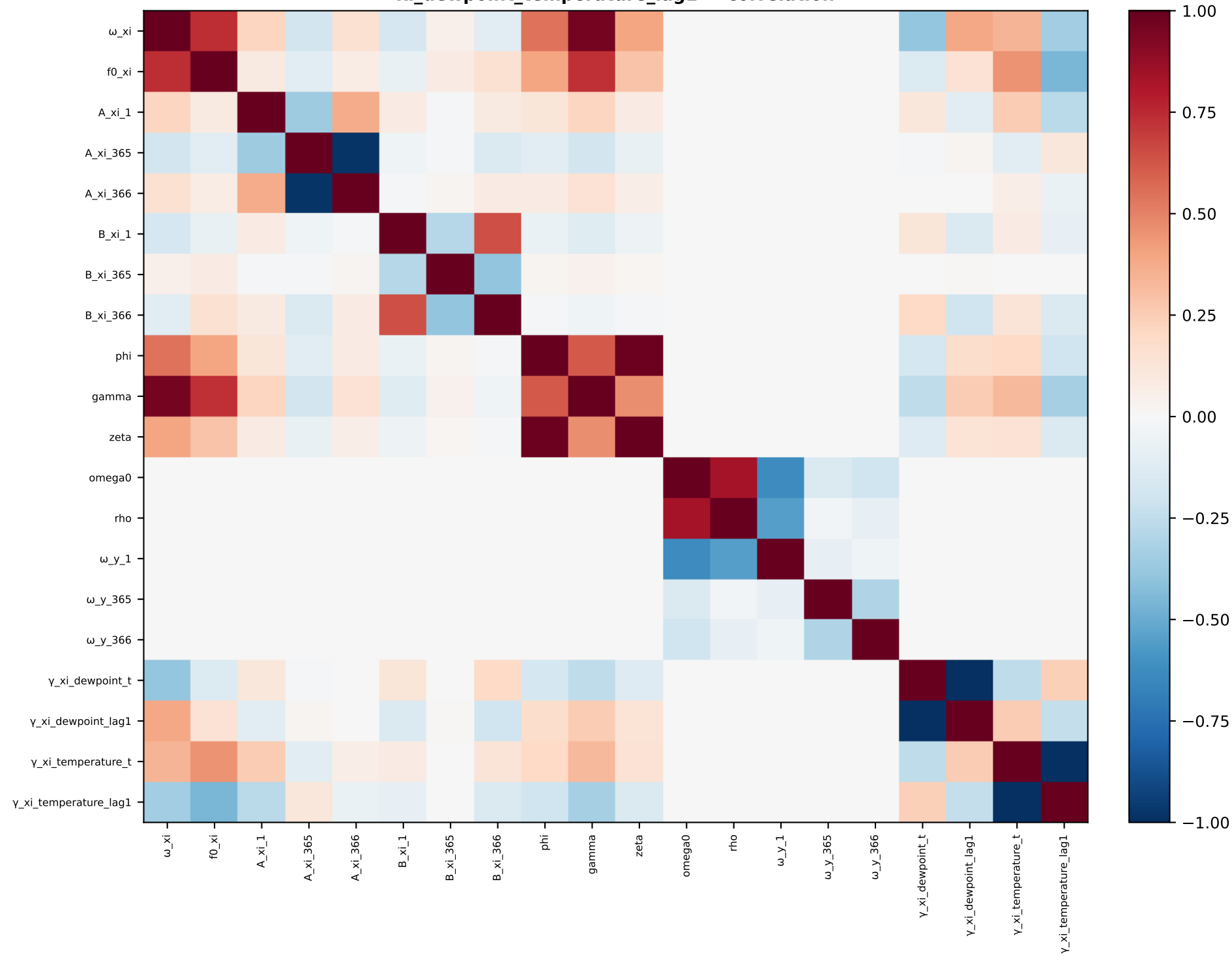
xi_dewpoint_temperature_lag1 — filtered states (OOS)



xi_dewpoint_temperature_lag1 — covariance

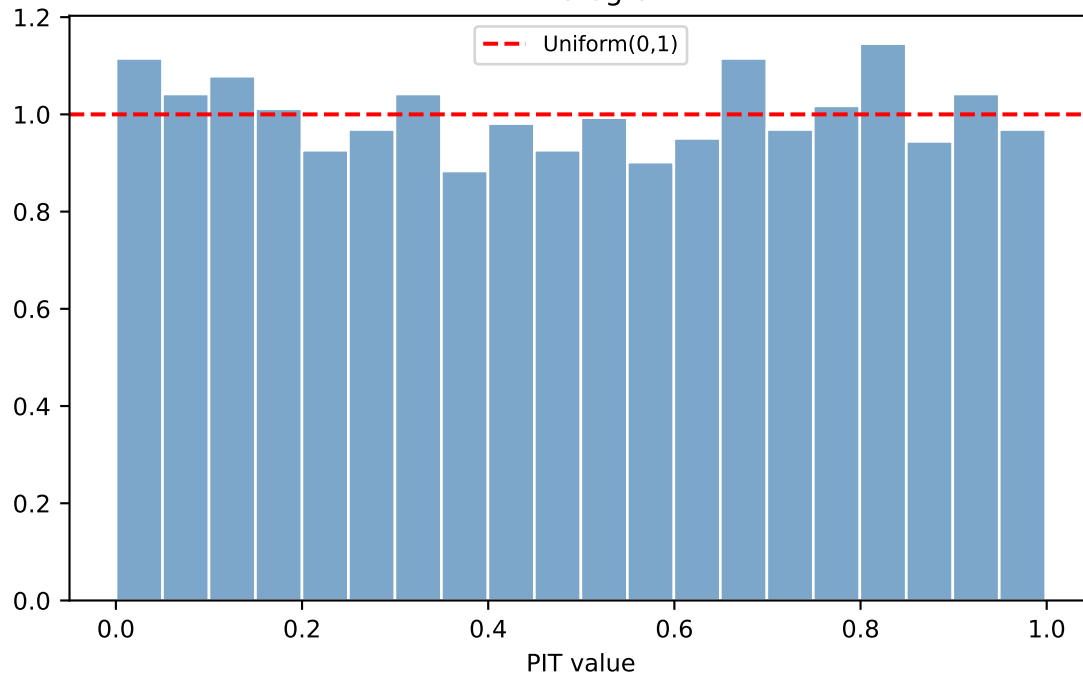


xi_dewpoint_temperature_lag1 — correlation

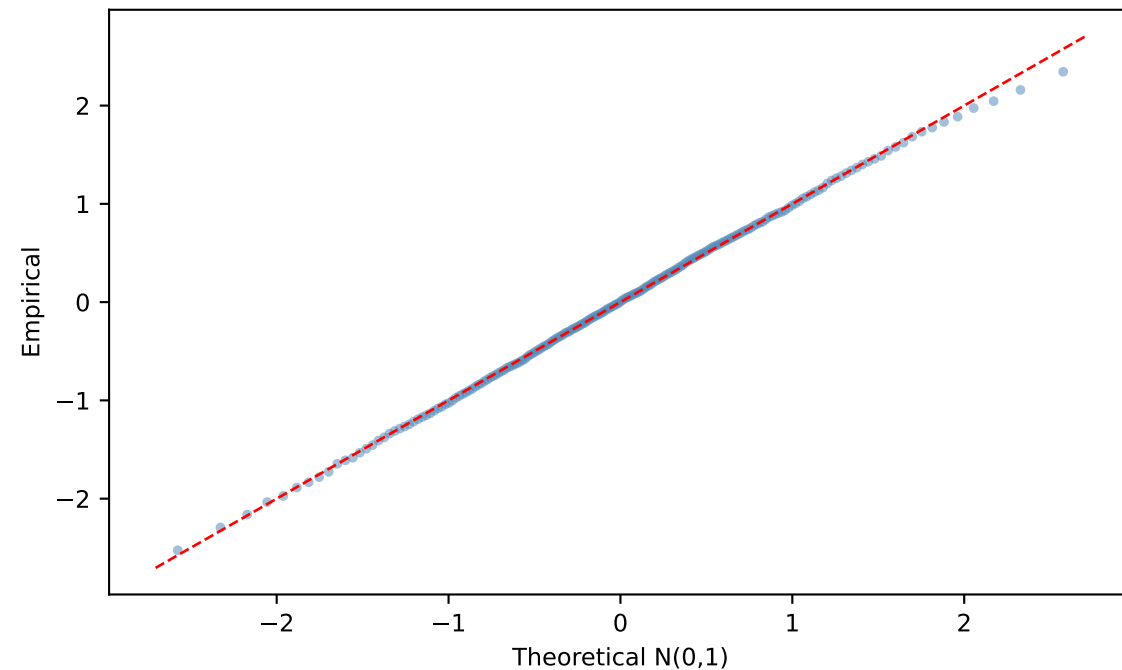


xi_dewpoint_temperature_short_lags — IS diagnostics

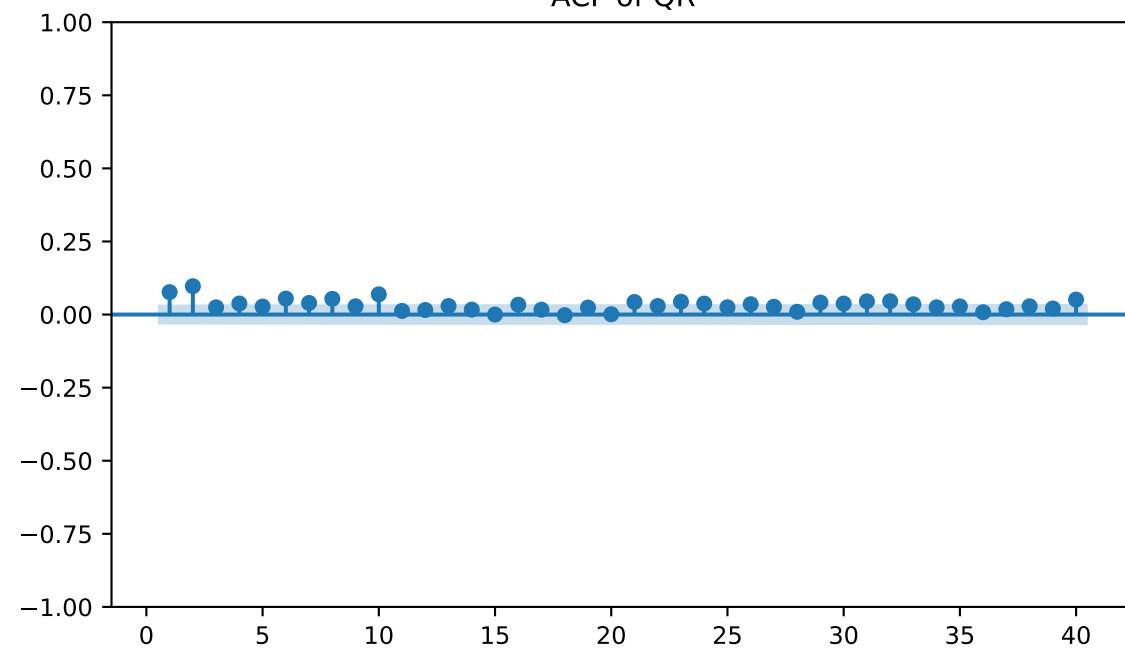
PIT histogram



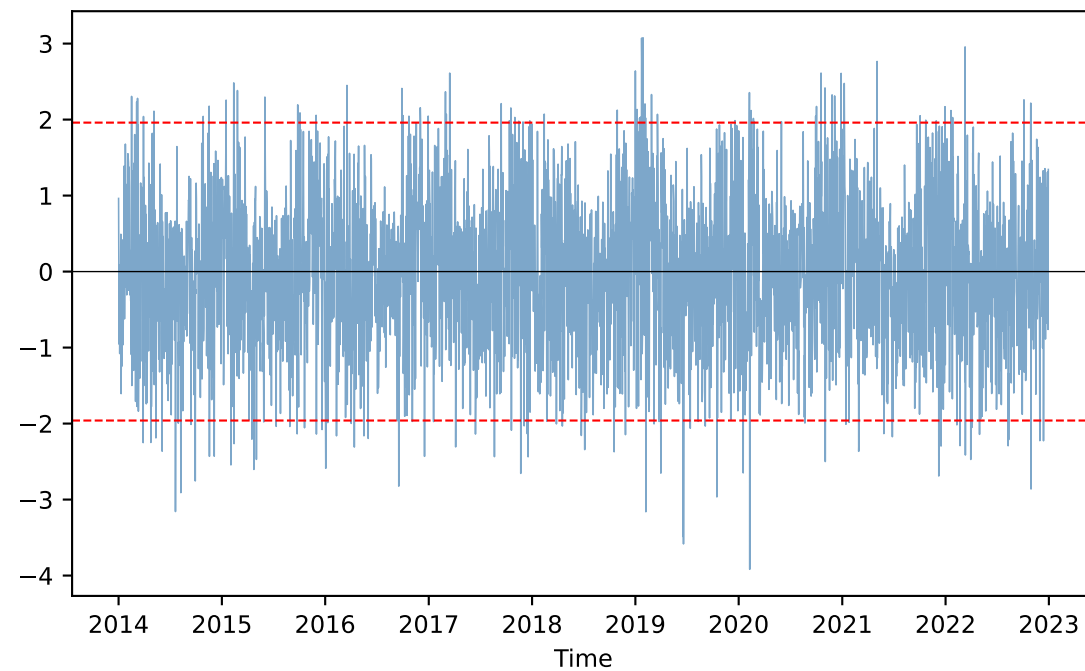
Normal QQ plot



ACF of QR

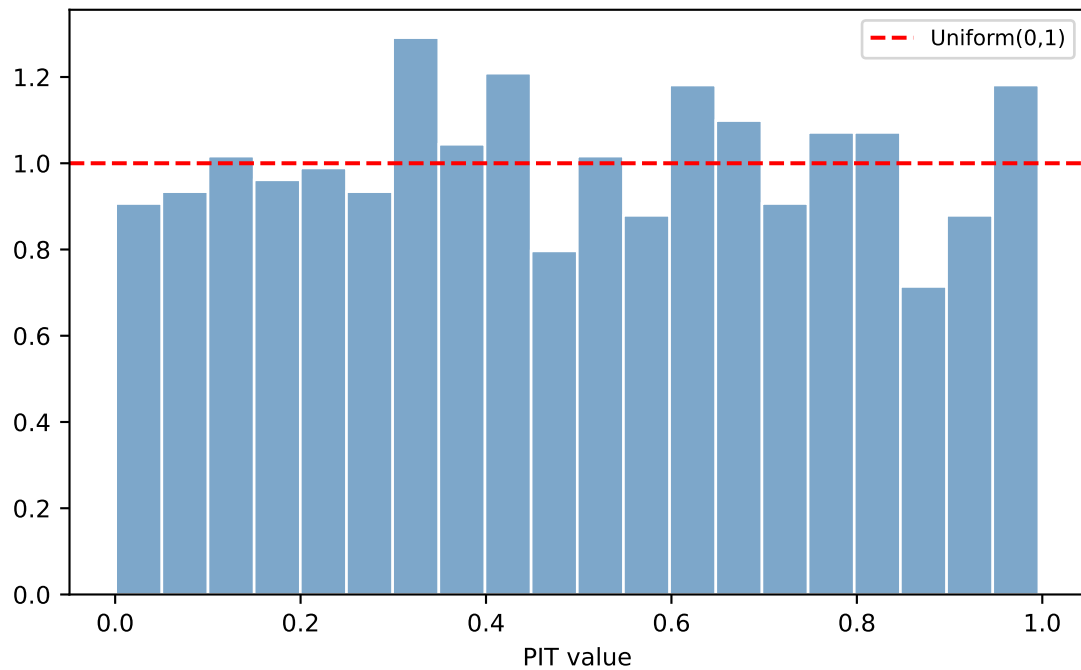


QR time series

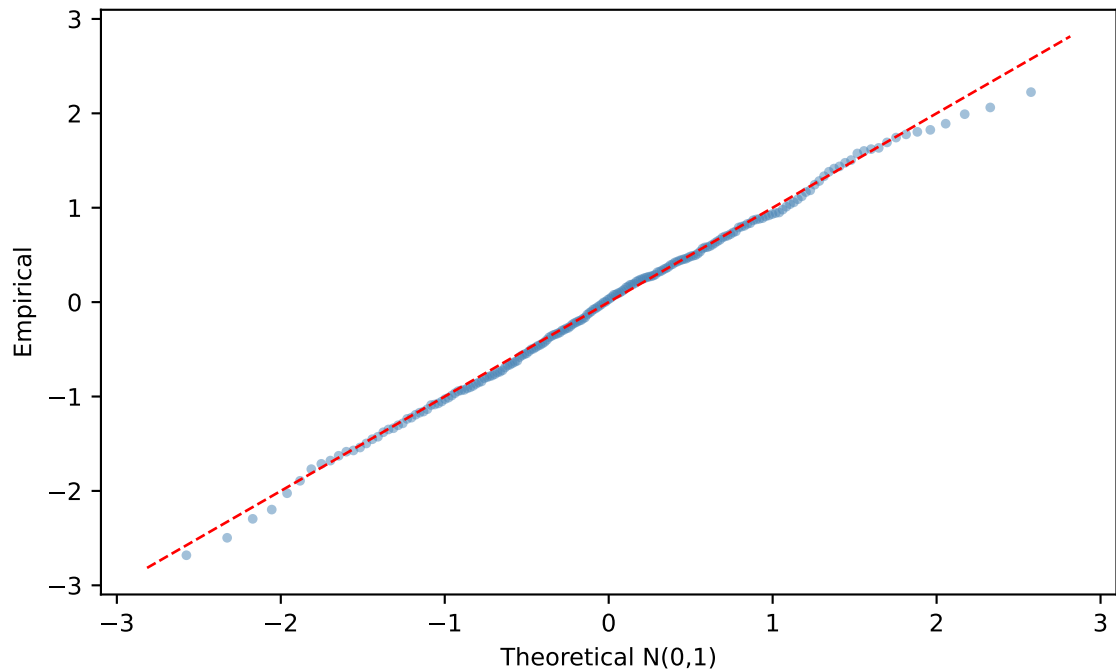


xi_dewpoint_temperature_short_lags — OOS diagnostics

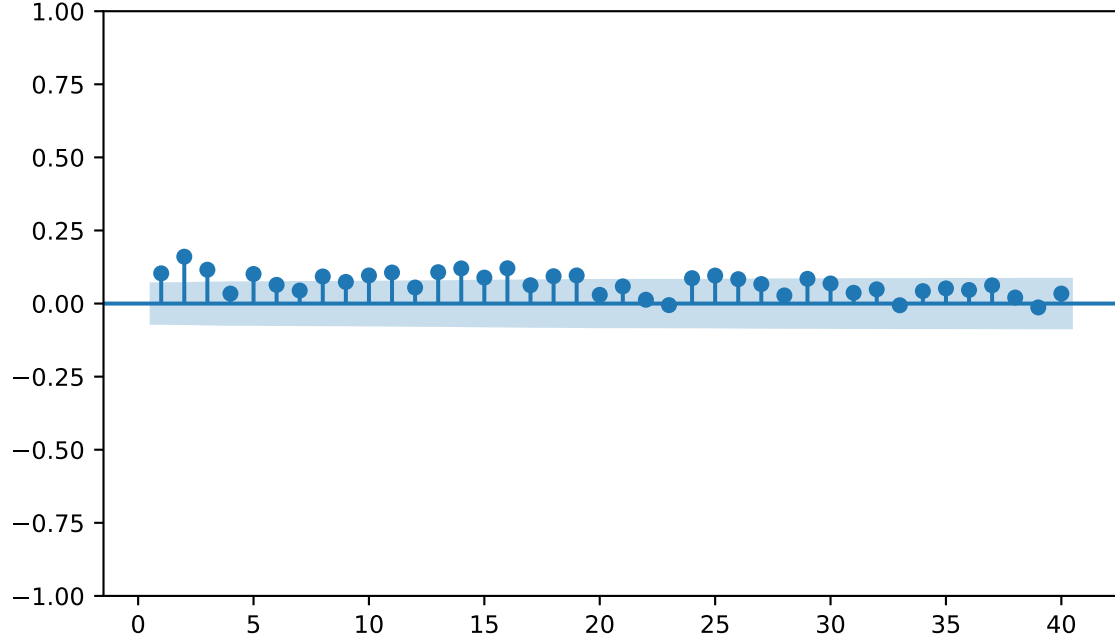
PIT histogram



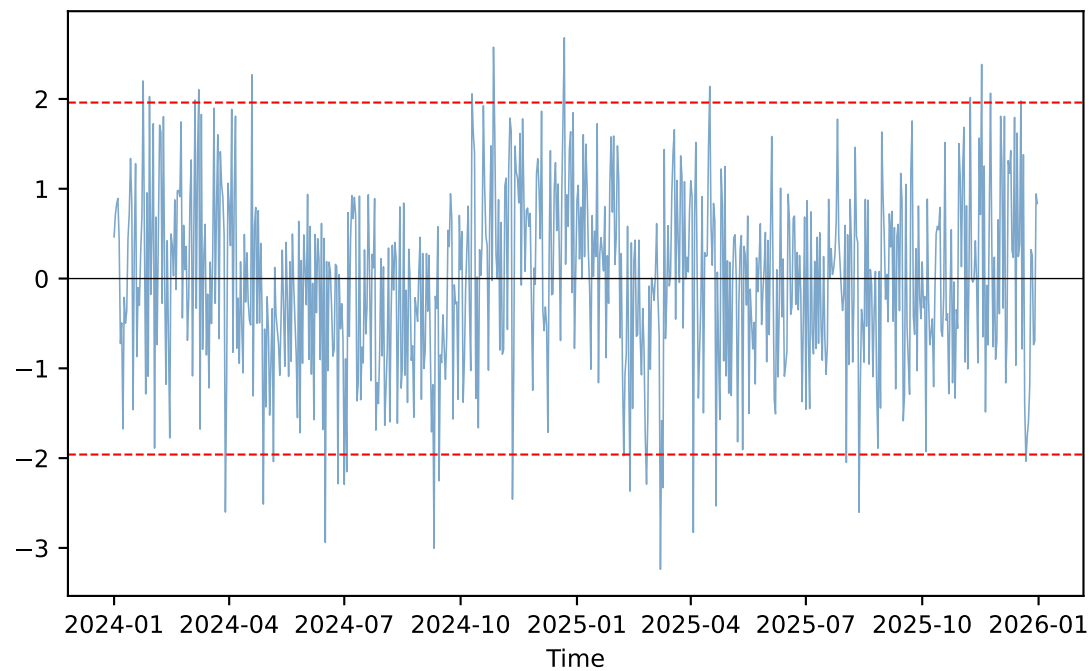
Normal QQ plot



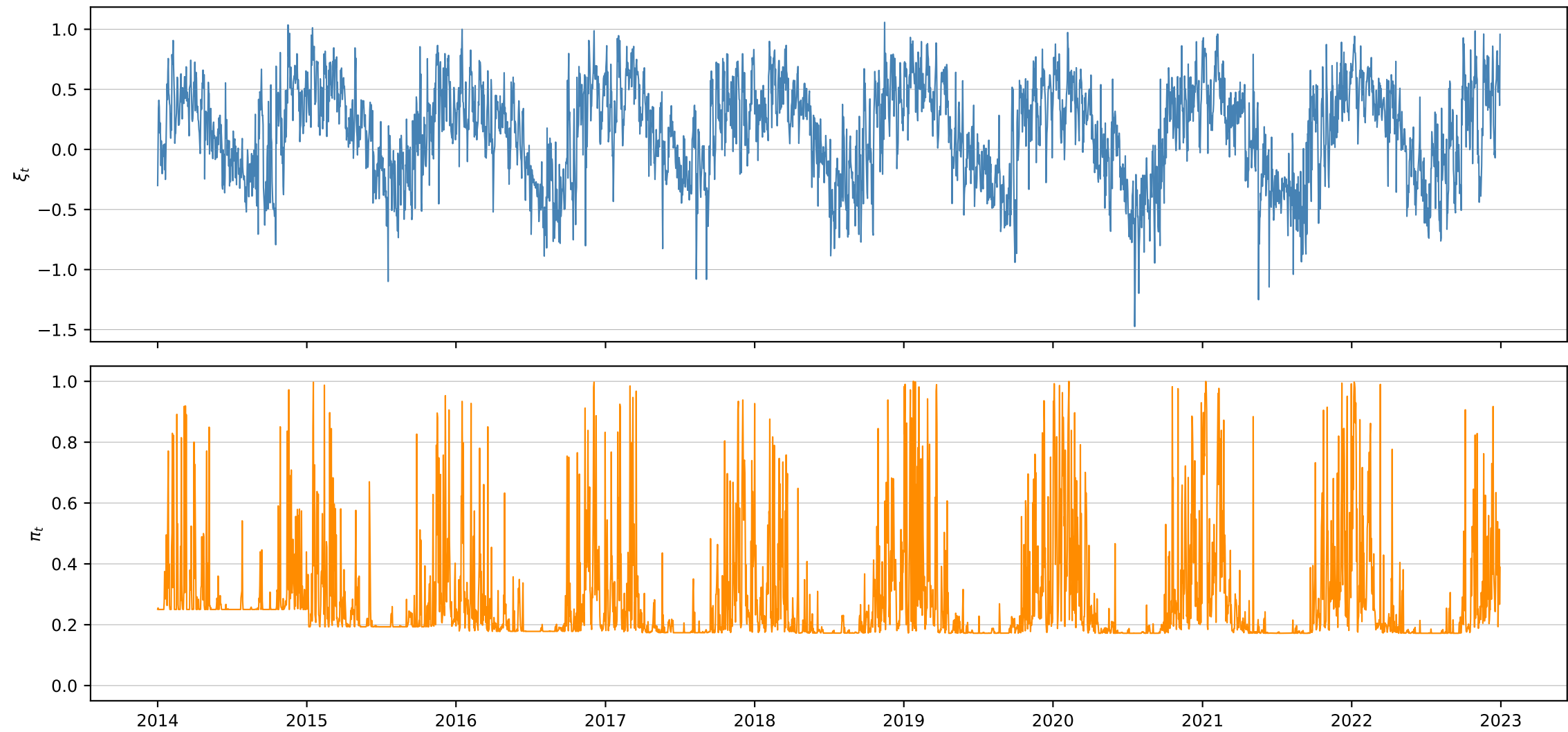
ACF of QR



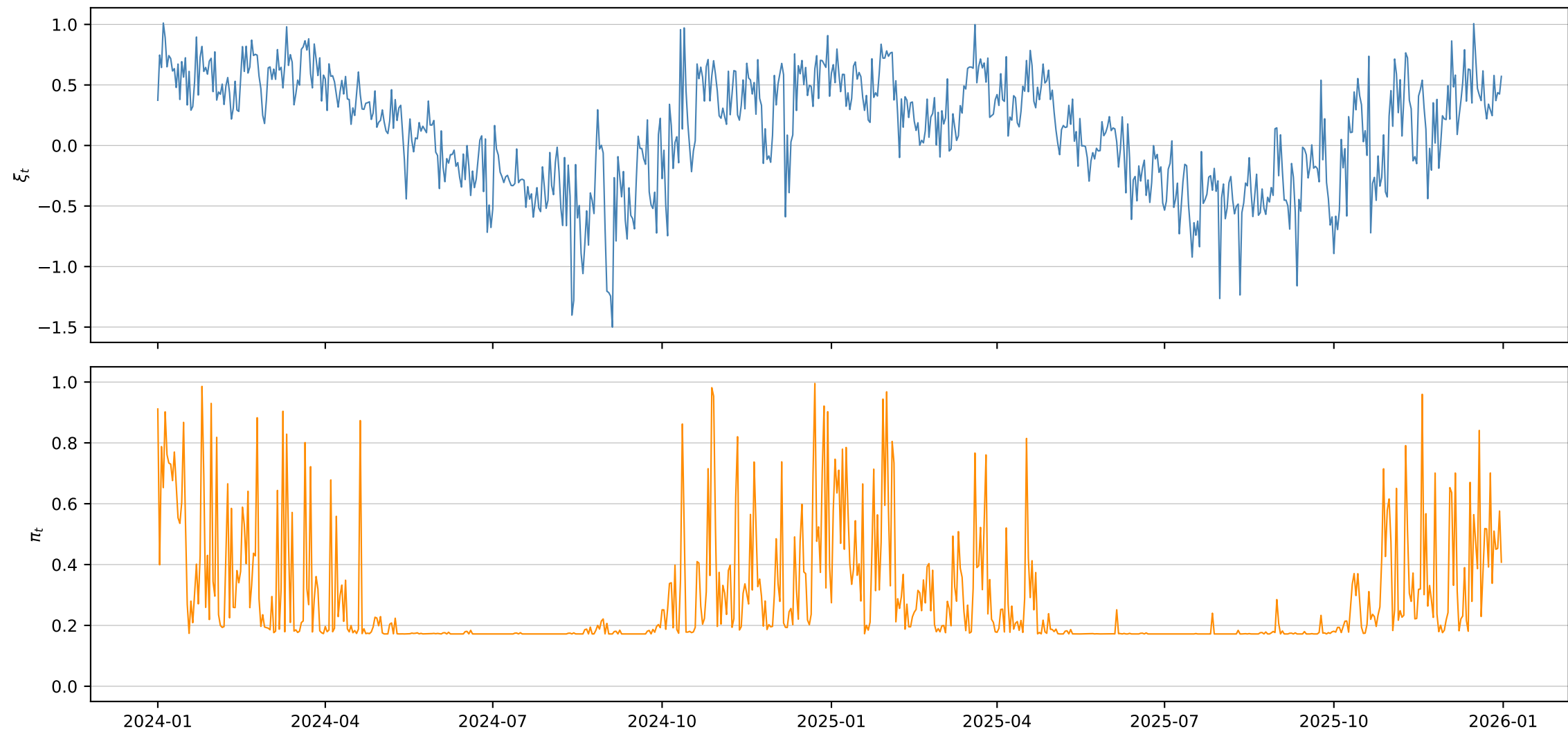
QR time series



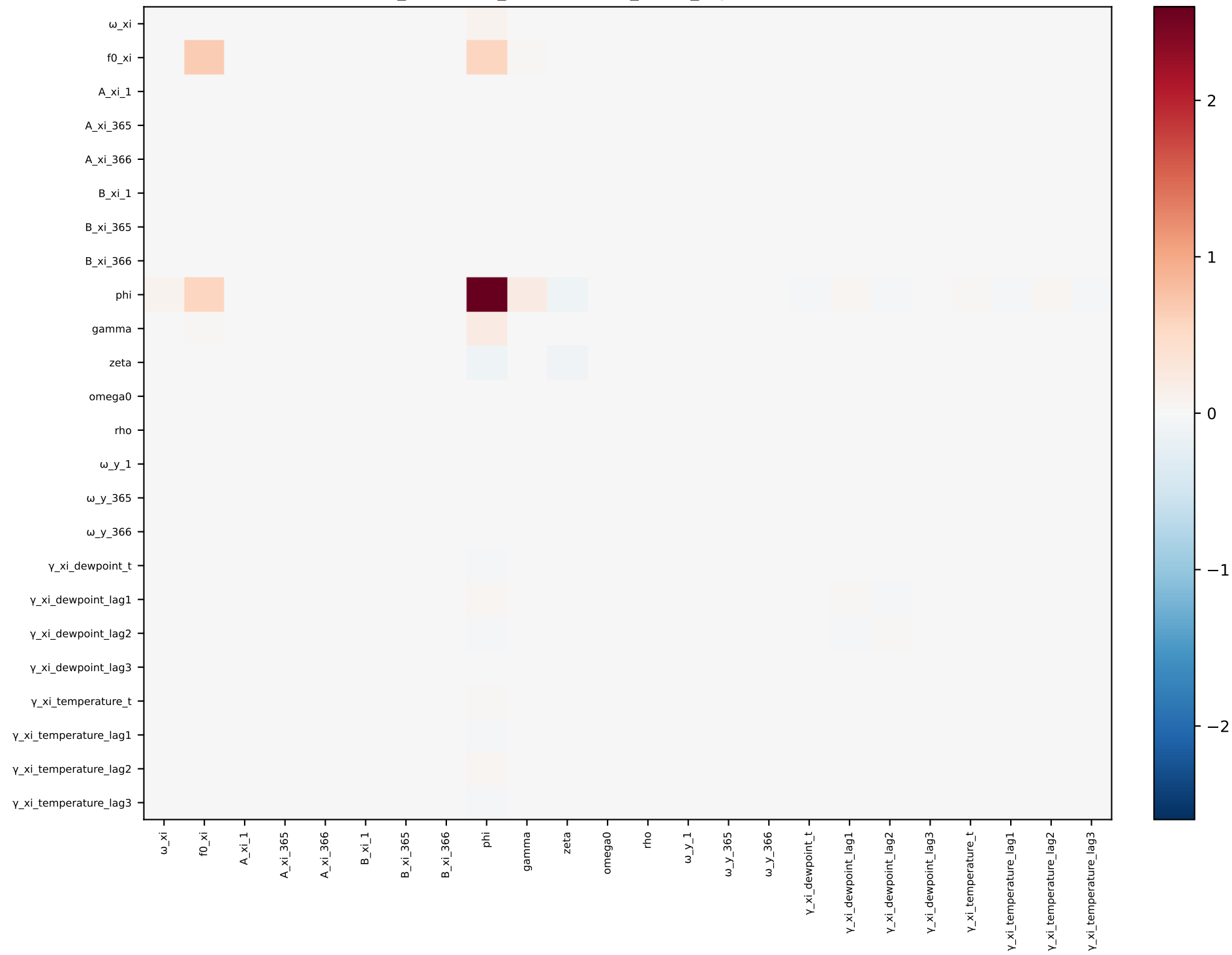
xi_dewpoint_temperature_short_lags — filtered states (IS)



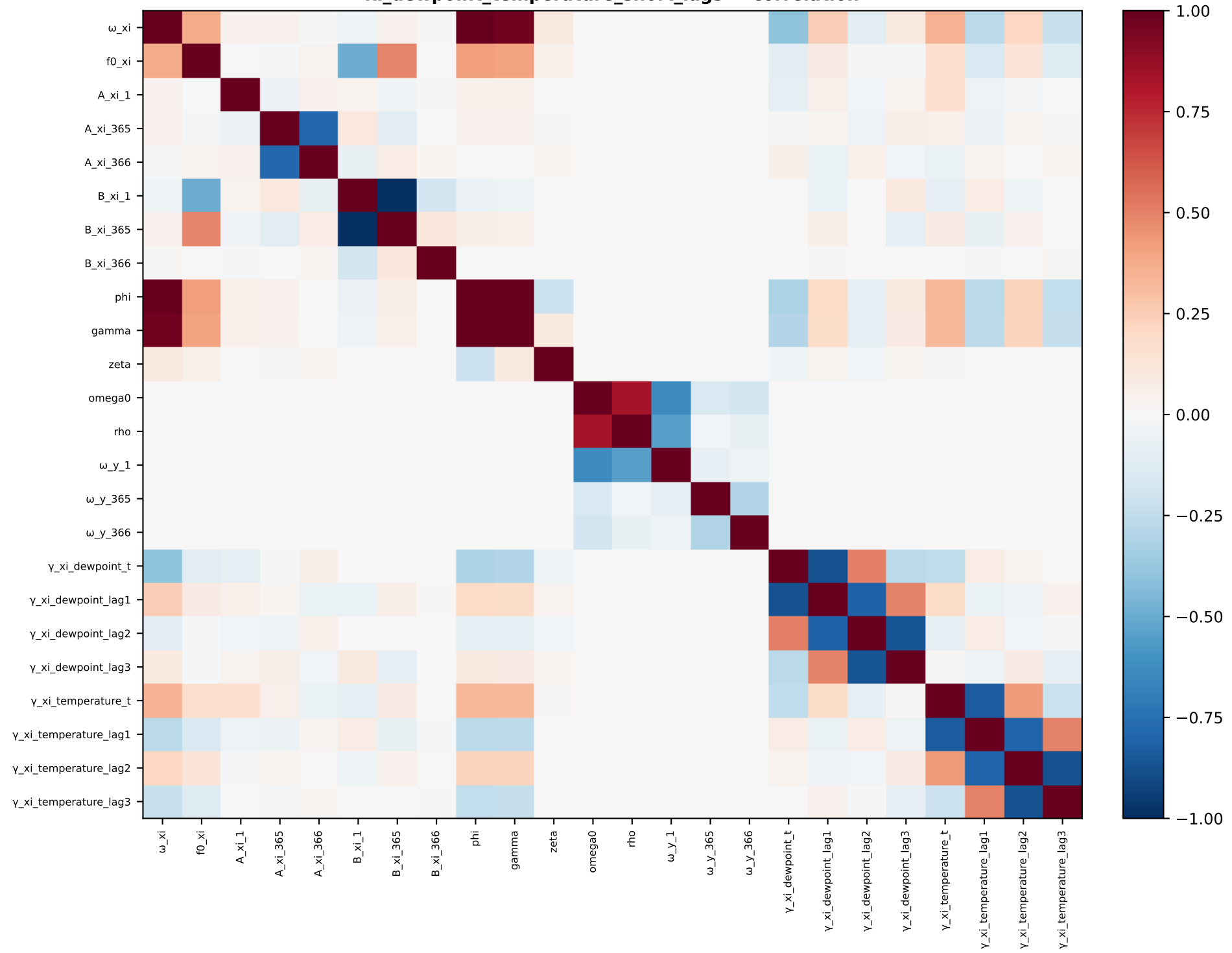
xi_dewpoint_temperature_short_lags — filtered states (OOS)



xi_dewpoint_temperature_short_lags — covariance

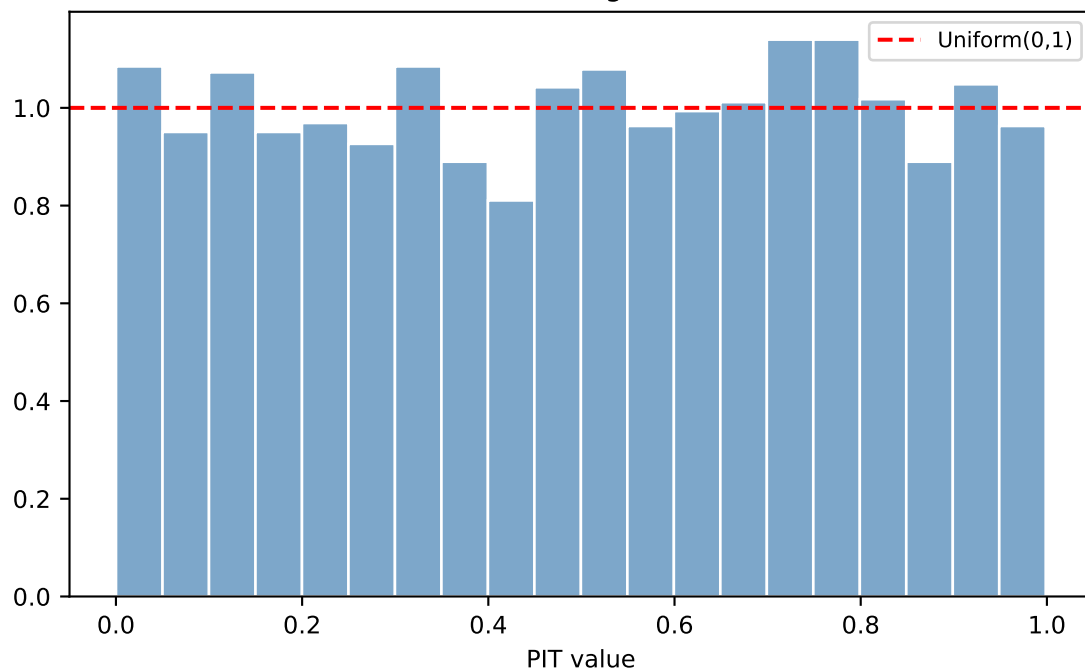


xi_dewpoint_temperature_short_lags — correlation

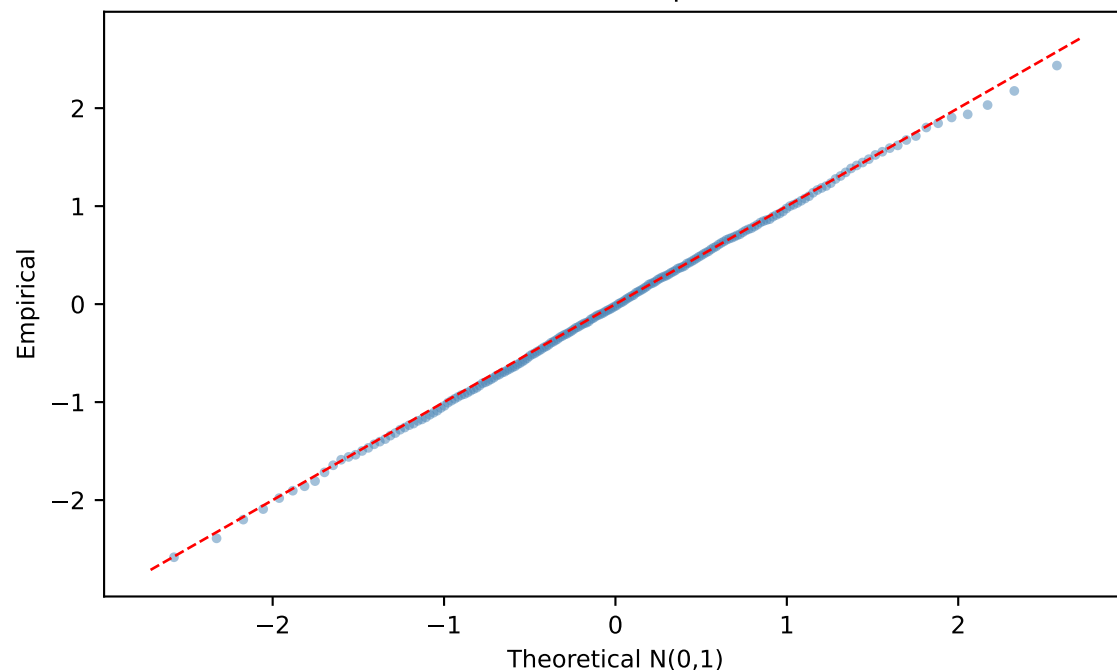


xi_dewpoint_temperature_seasonal_lags — IS diagnostics

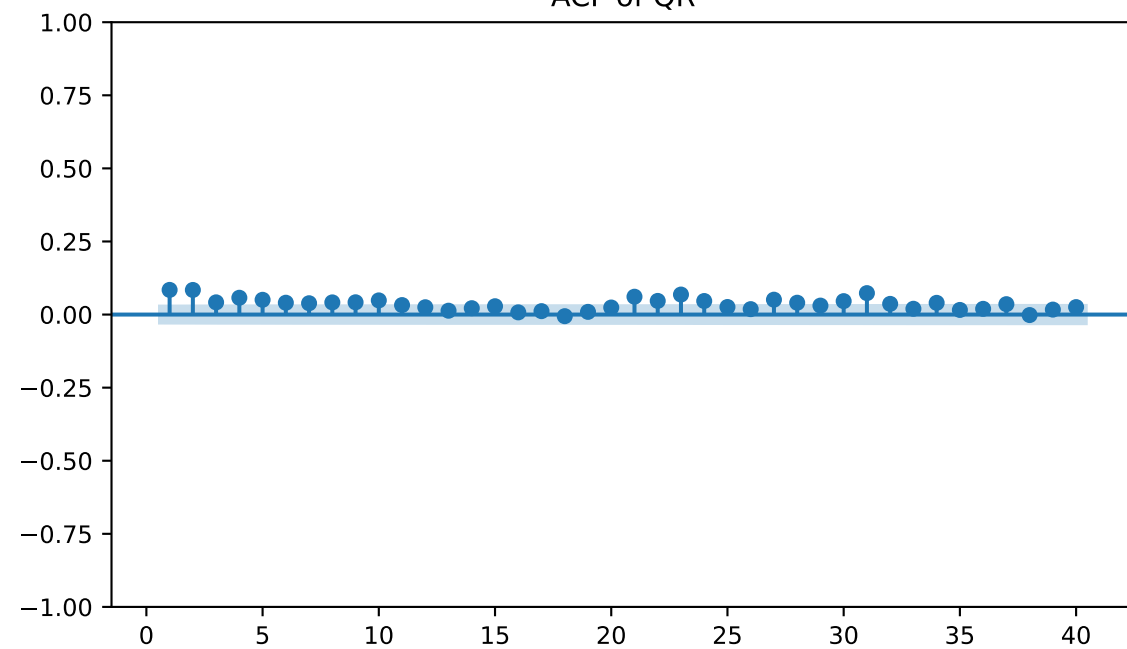
PIT histogram



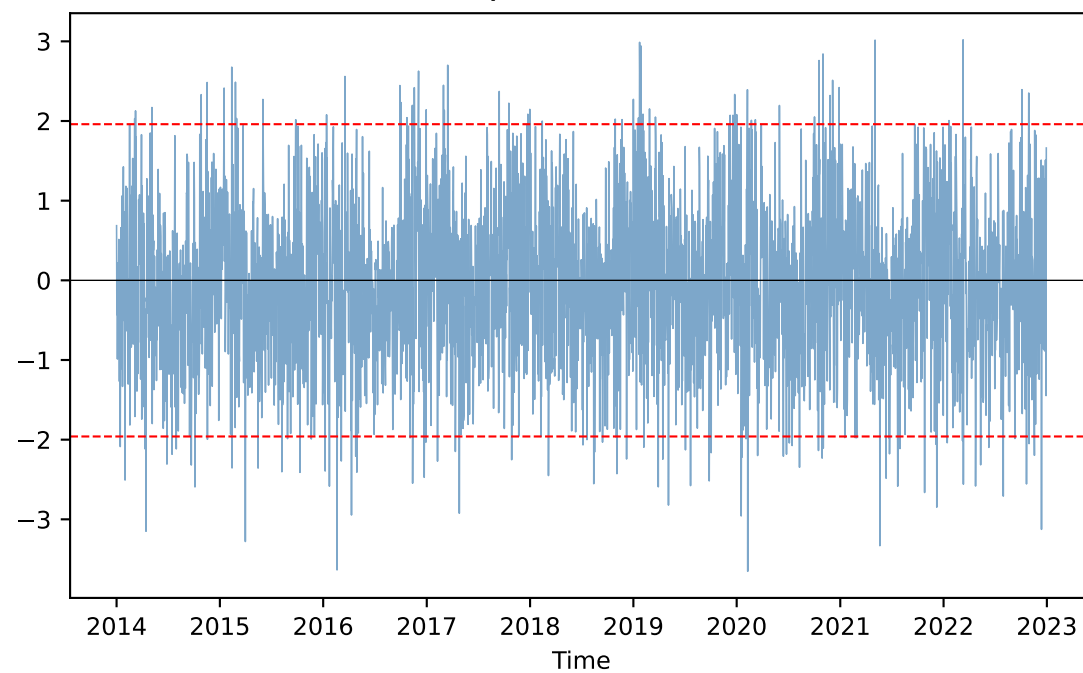
Normal QQ plot



ACF of QR

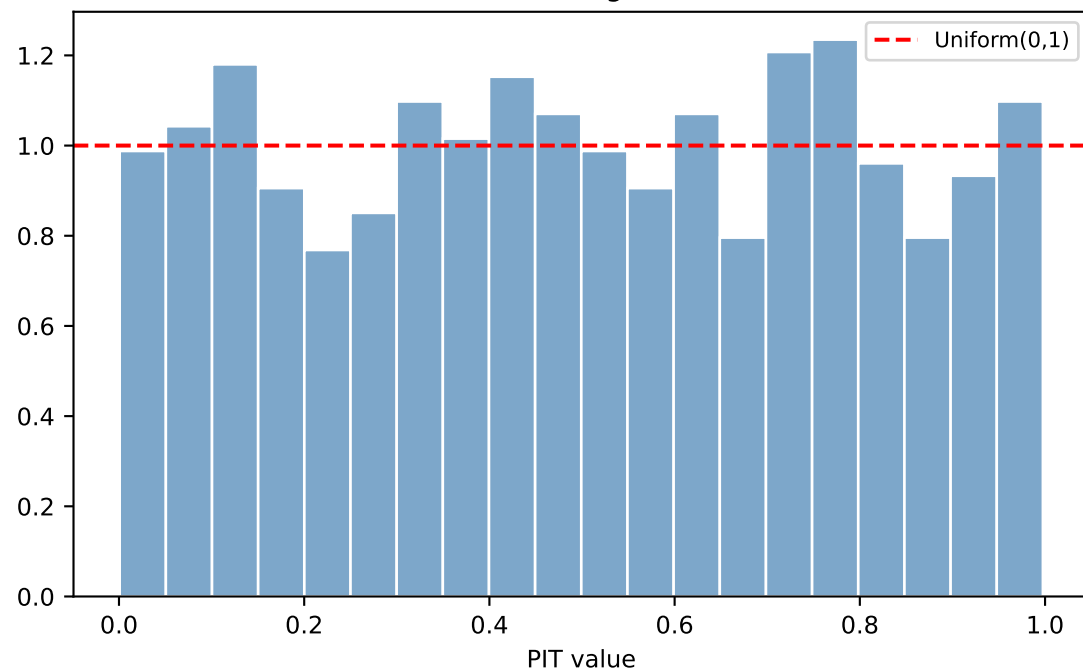


QR time series

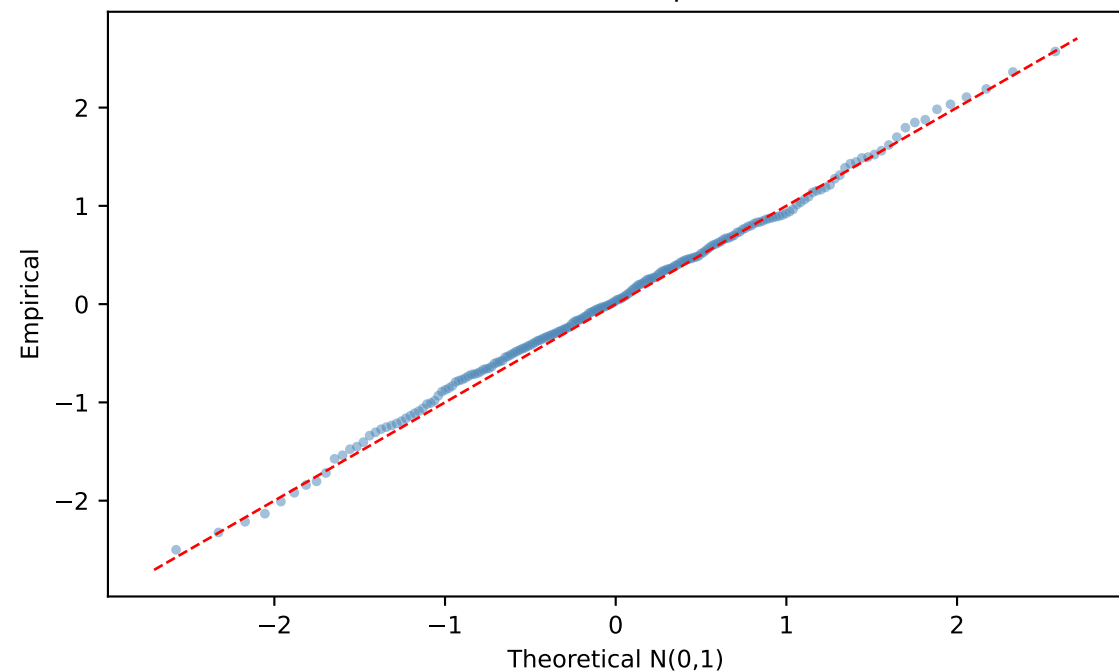


xi_dewpoint_temperature_seasonal_lags — OOS diagnostics

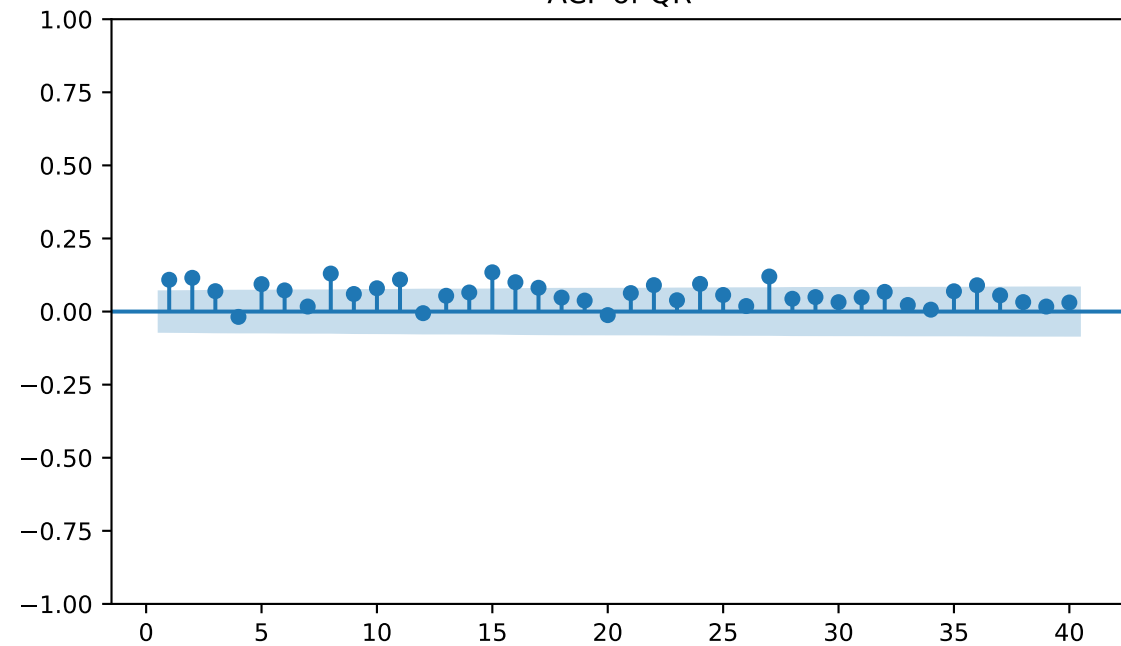
PIT histogram



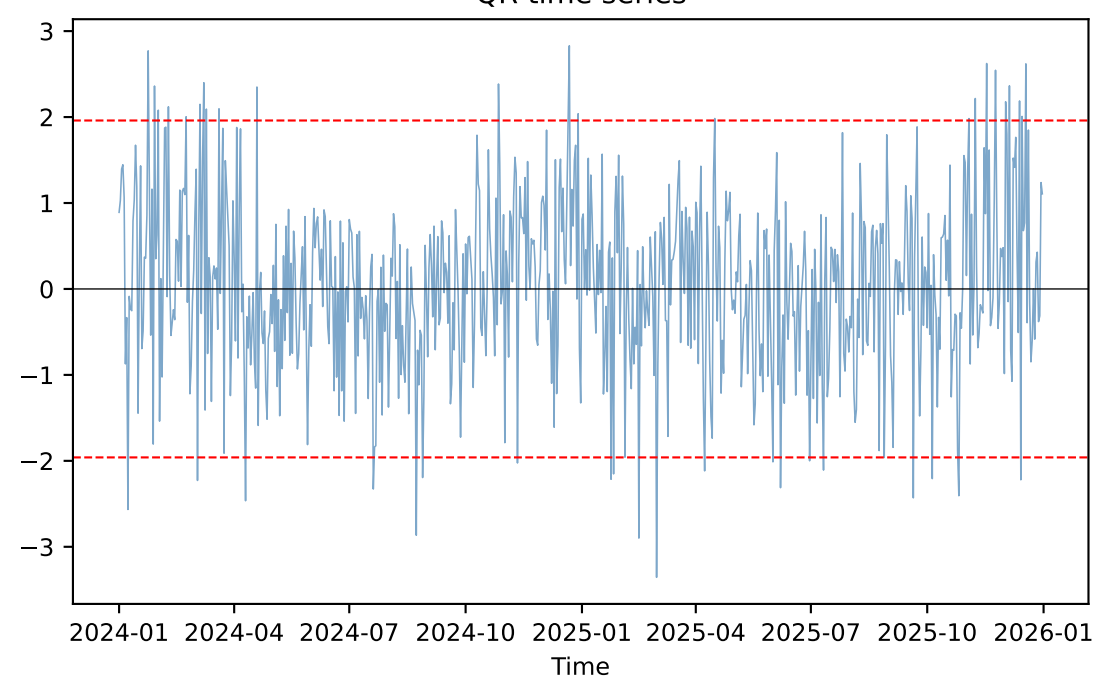
Normal QQ plot



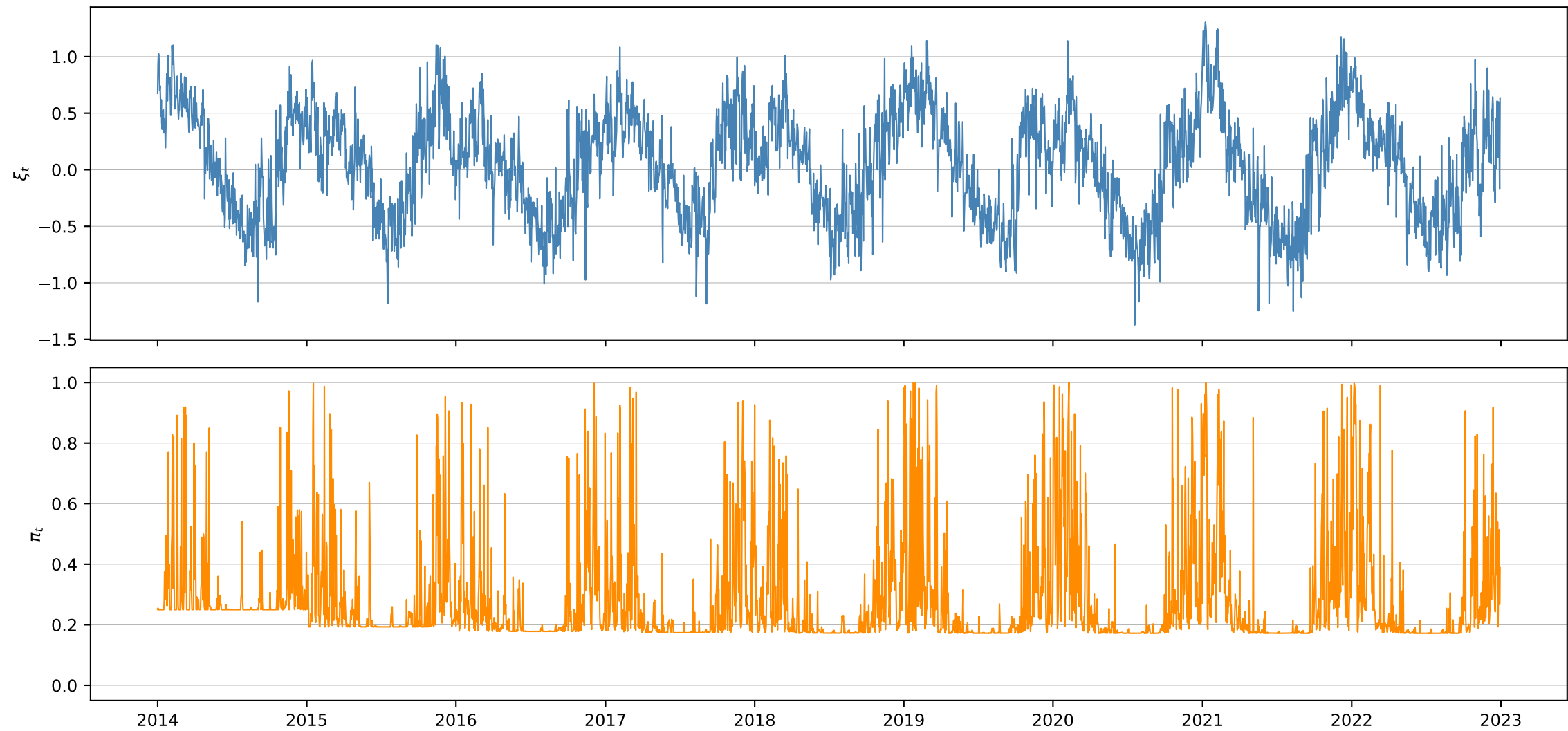
ACF of QR



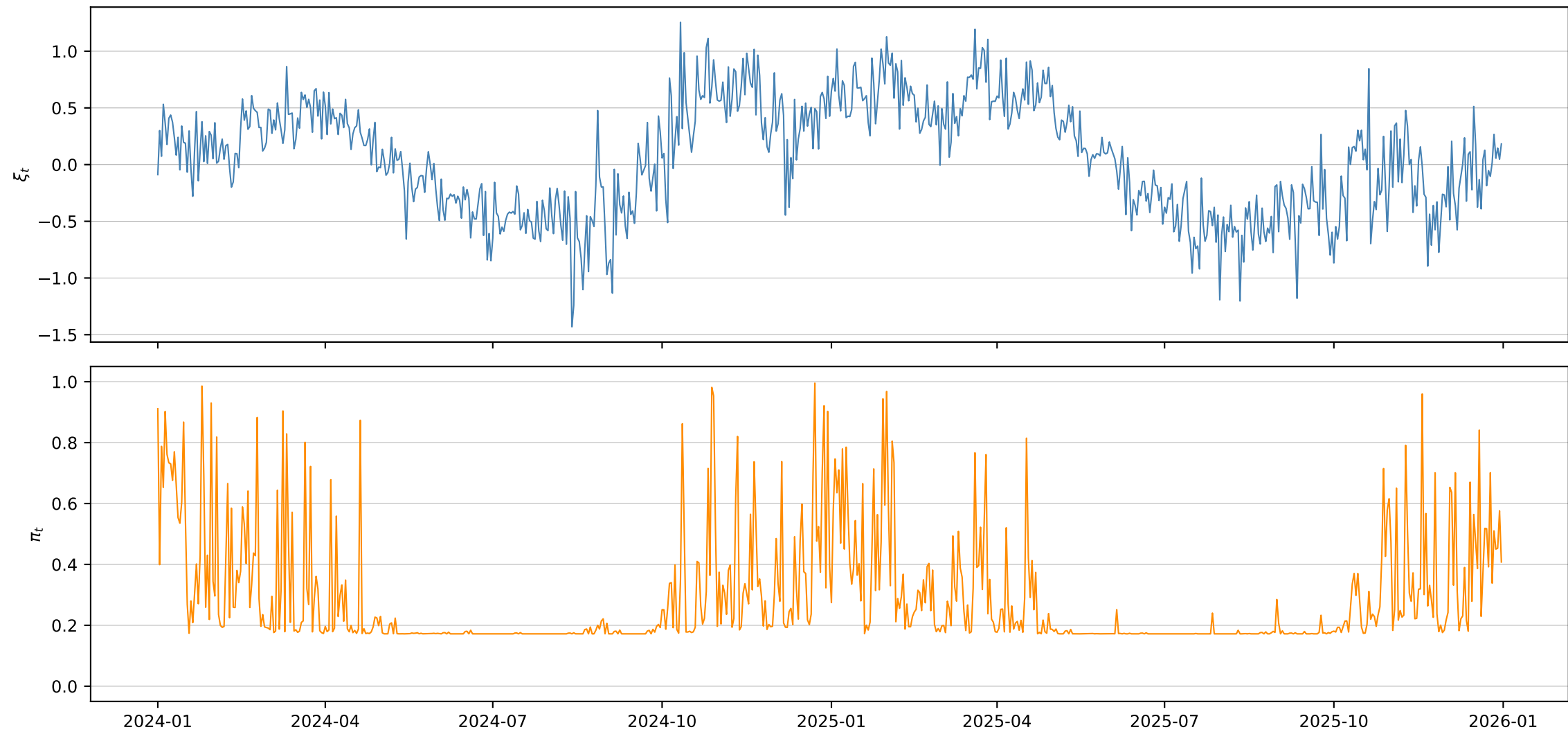
QR time series



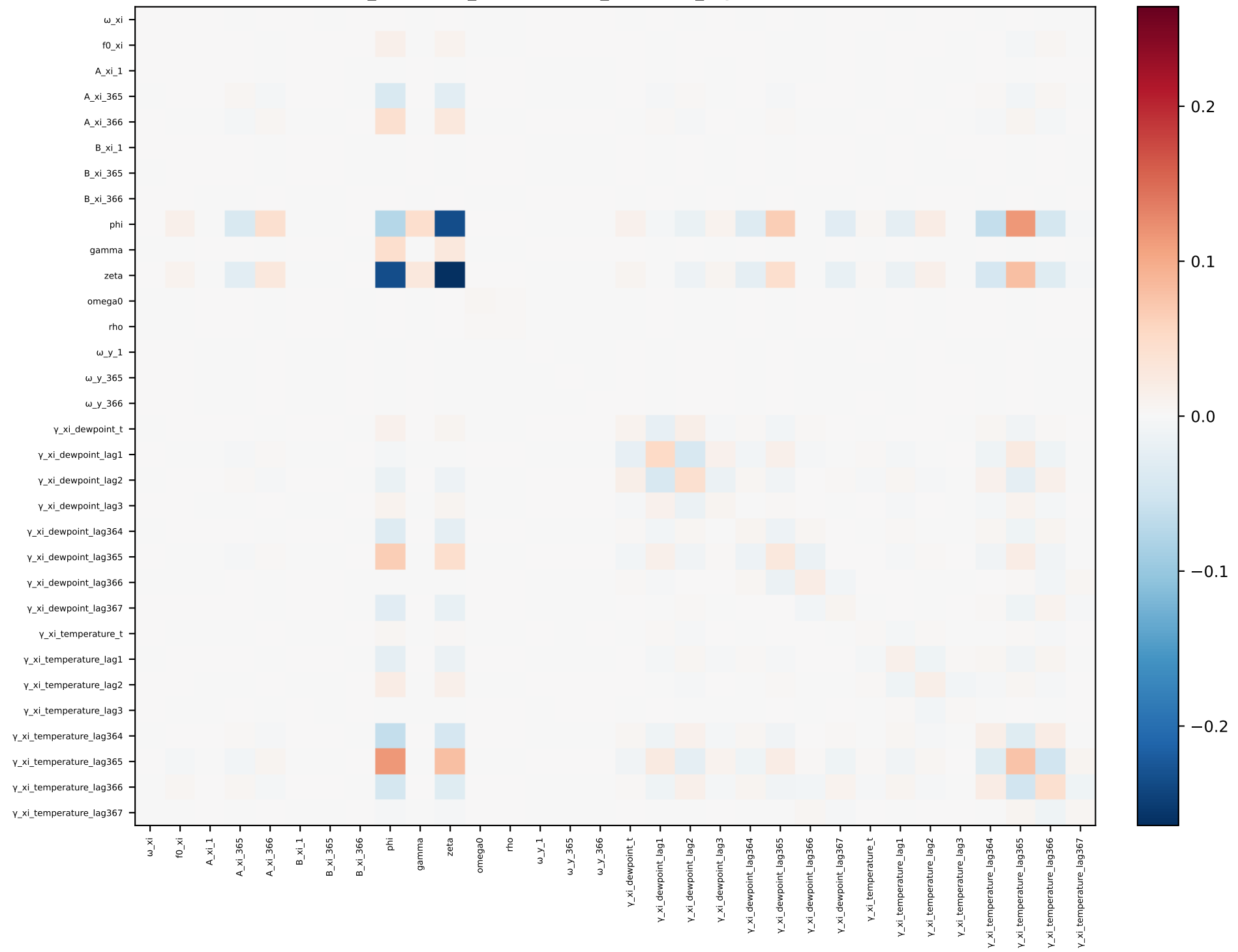
xi_dewpoint_temperature_seasonal_lags — filtered states (IS)



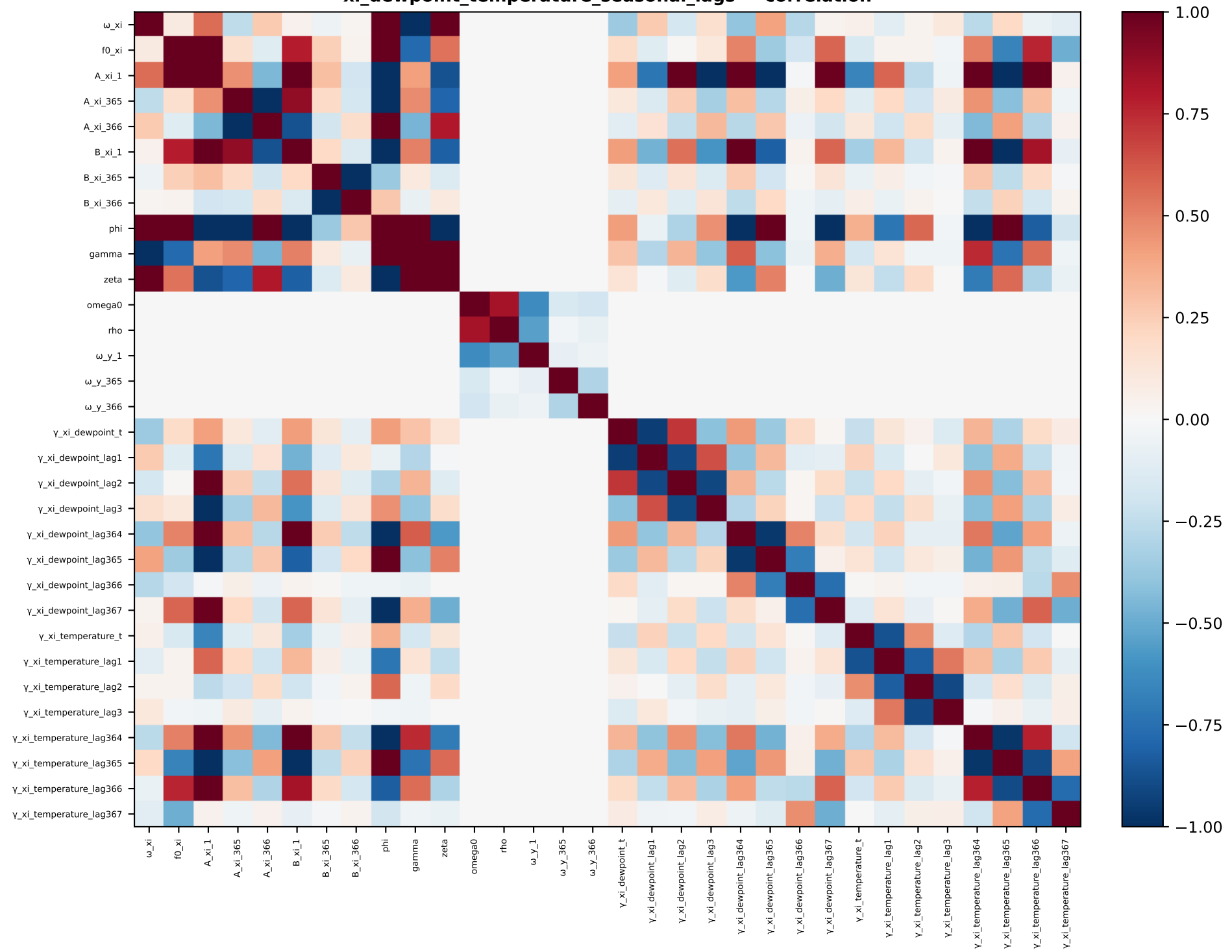
xi_dewpoint_temperature_seasonal_lags — filtered states (OOS)



xi_dewpoint_temperature_seasonal_lags — covariance

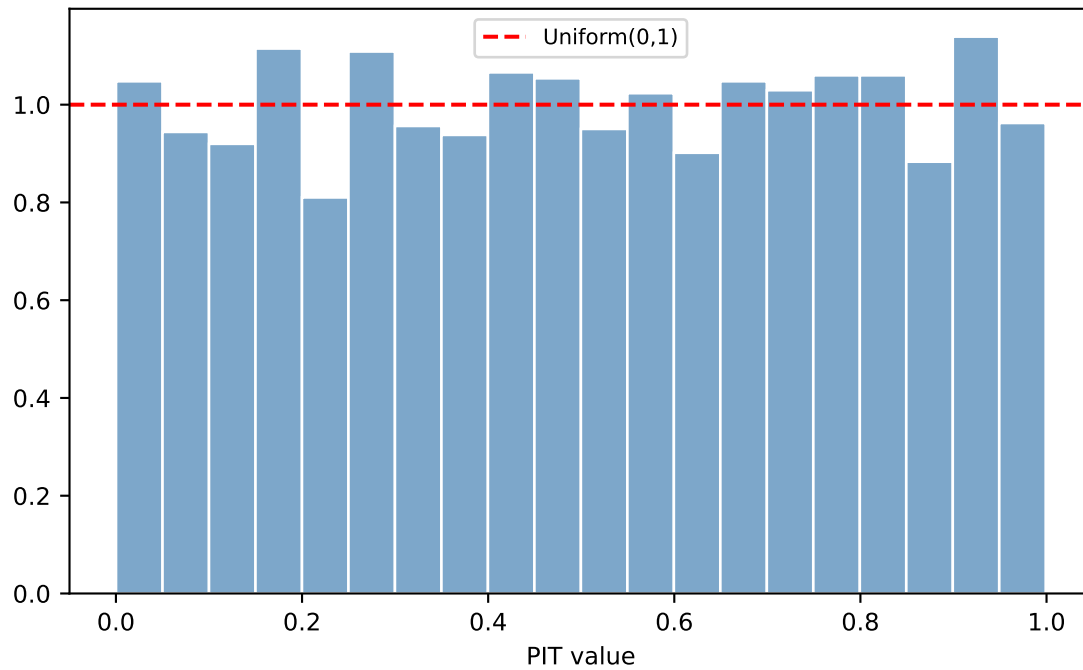


xi_dewpoint_temperature_seasonal_lags — correlation

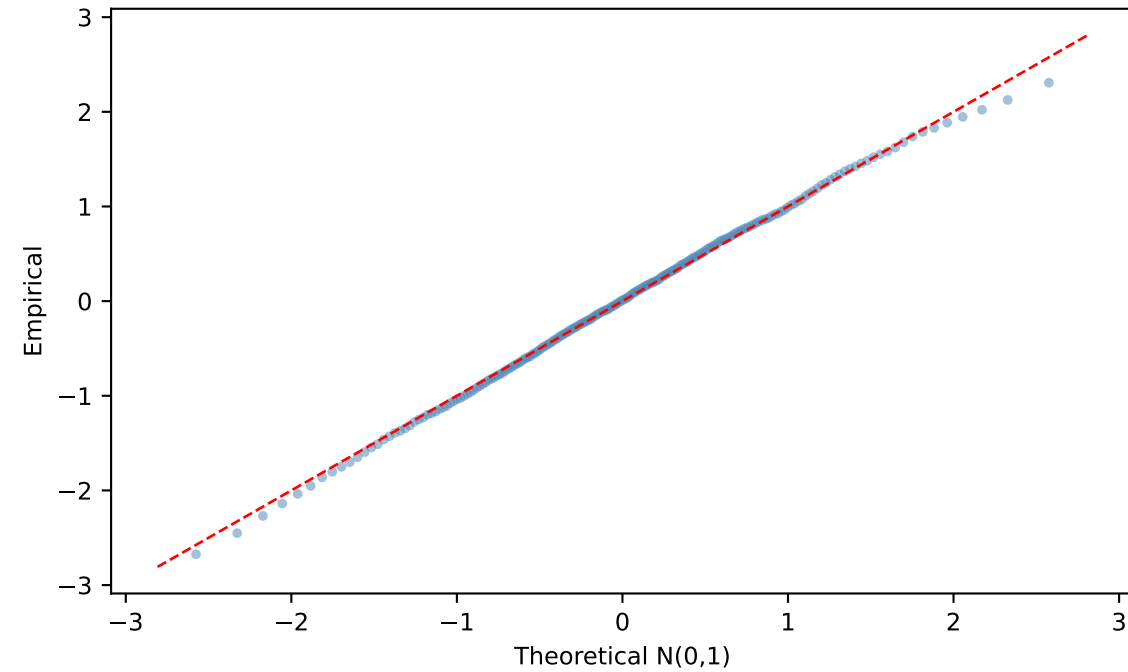


xi_mjo_t — IS diagnostics

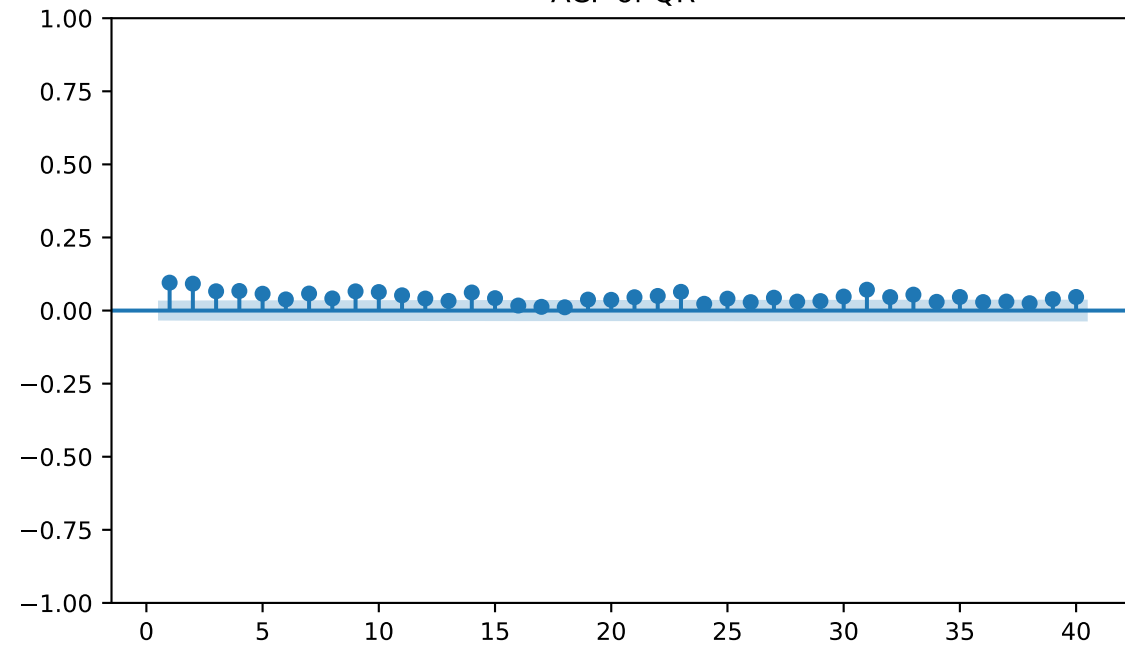
PIT histogram



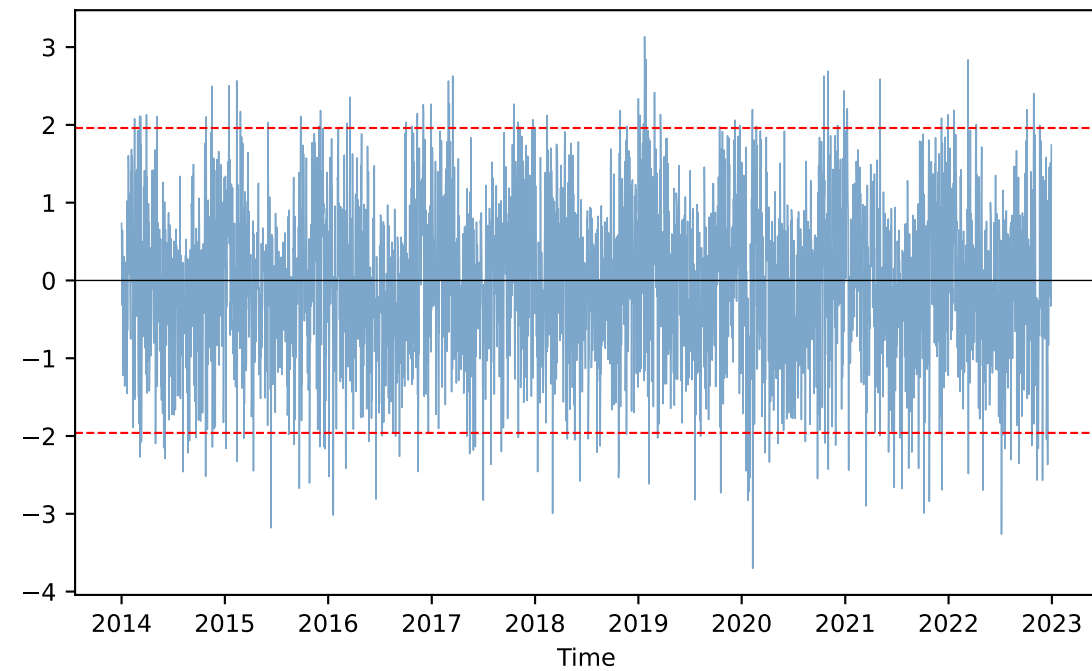
Normal QQ plot



ACF of QR

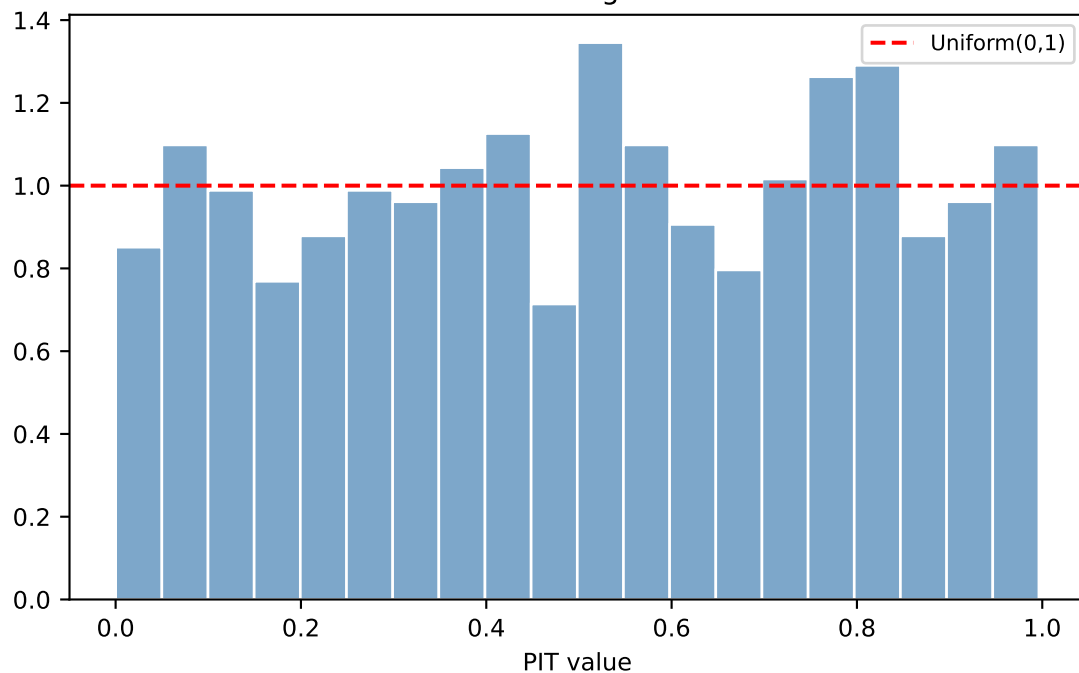


QR time series

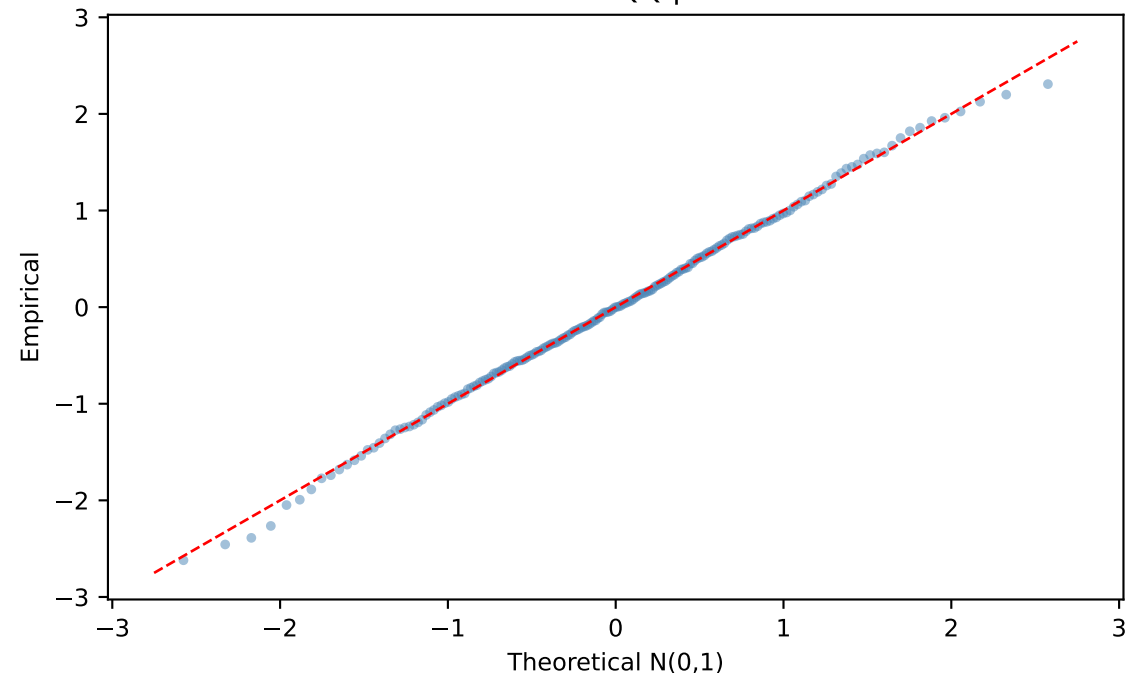


xi_mjo_t — OOS diagnostics

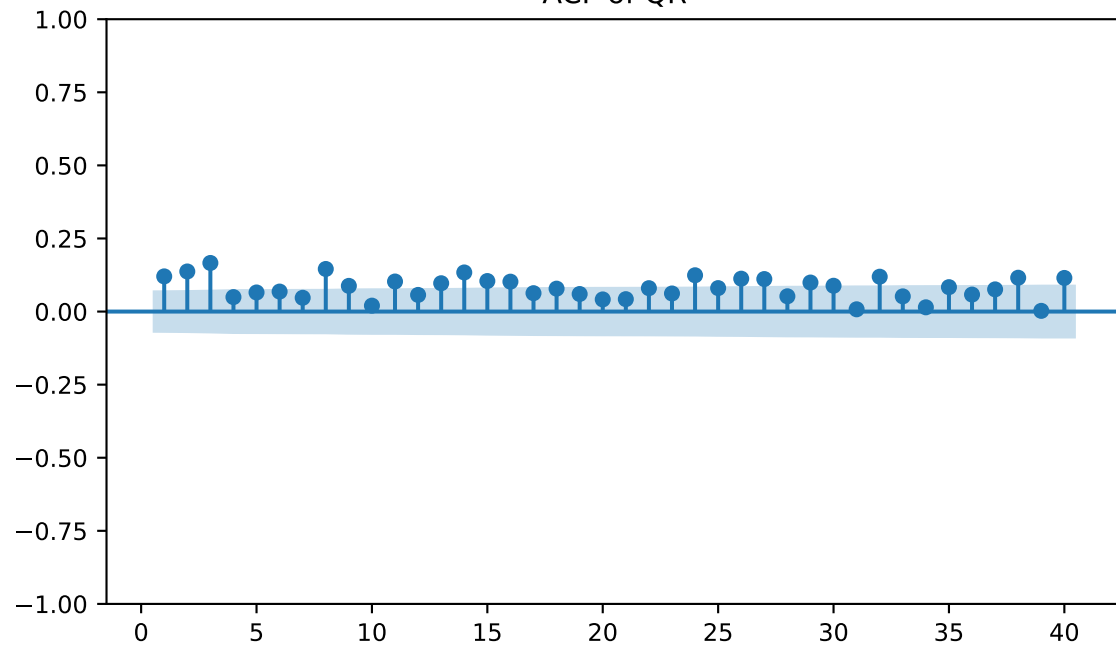
PIT histogram



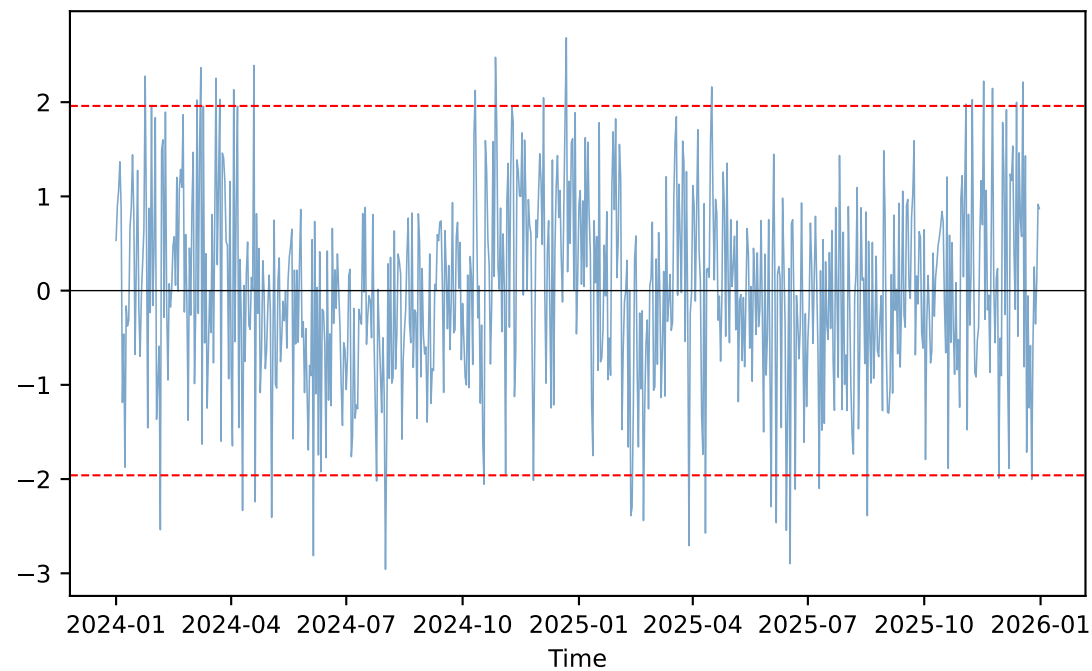
Normal QQ plot



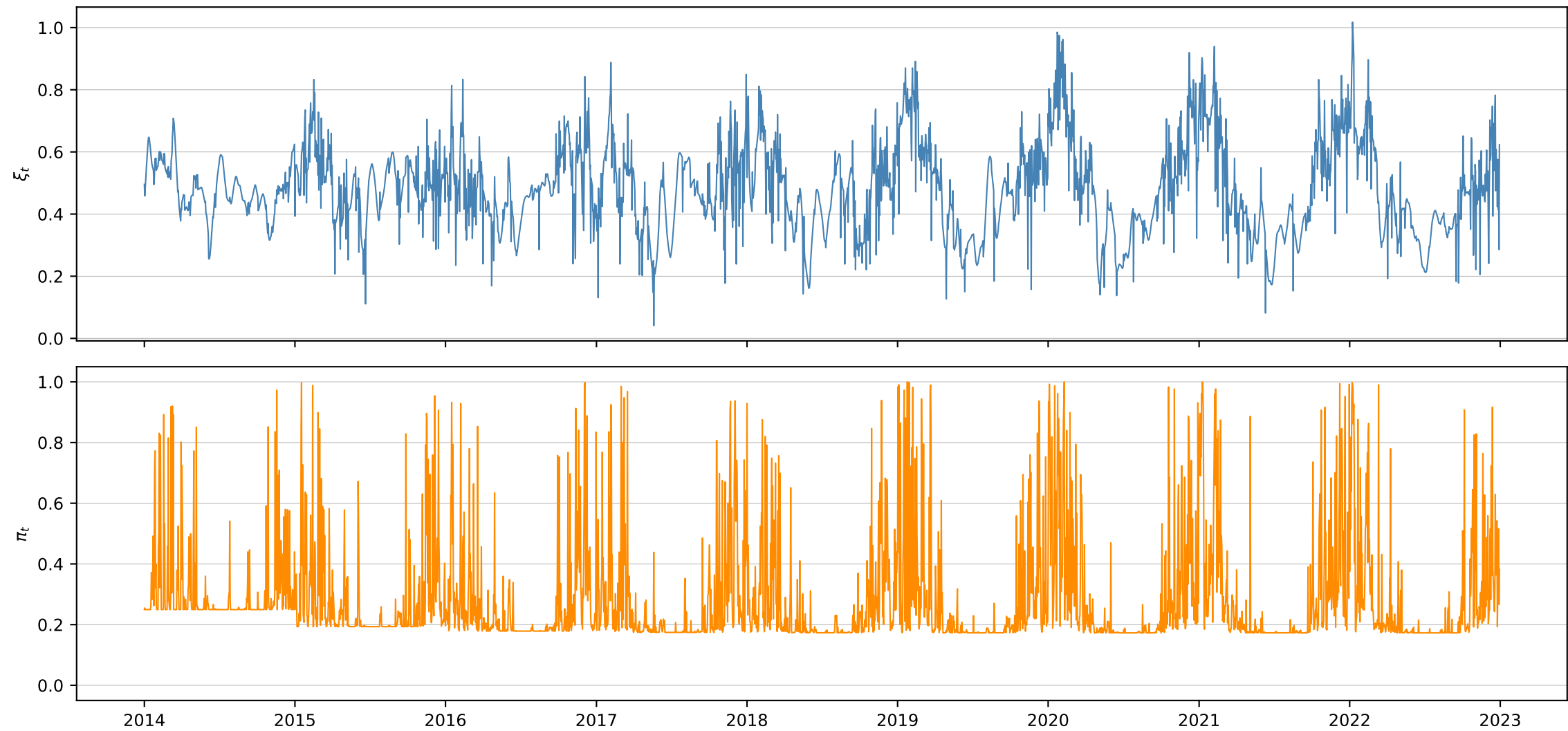
ACF of QR



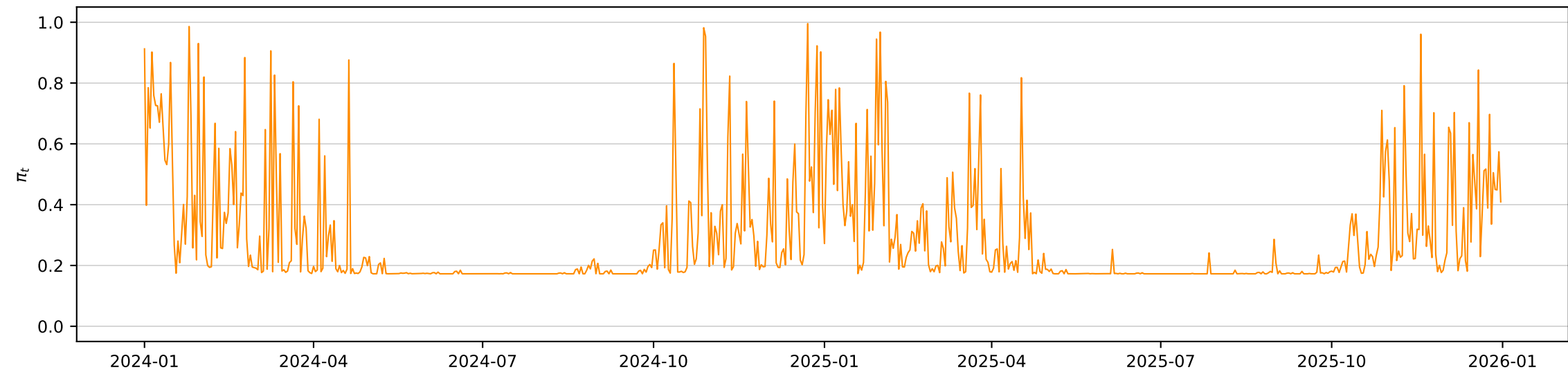
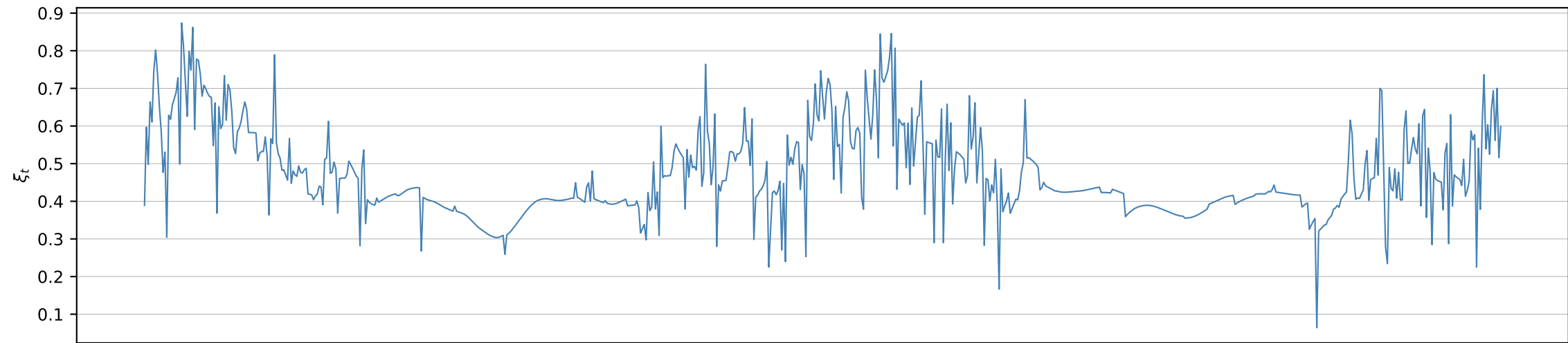
QR time series



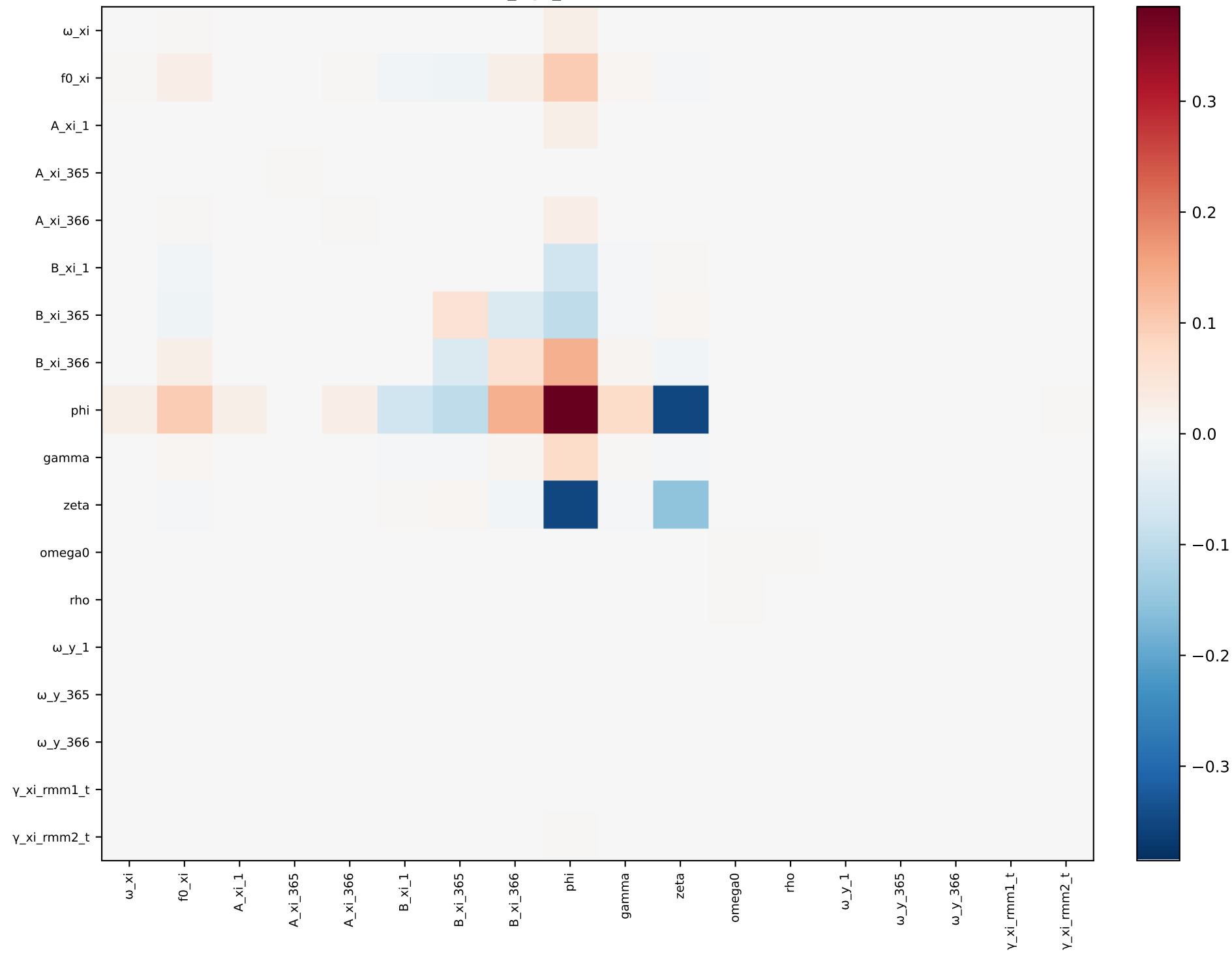
xi_mjo_t — filtered states (IS)



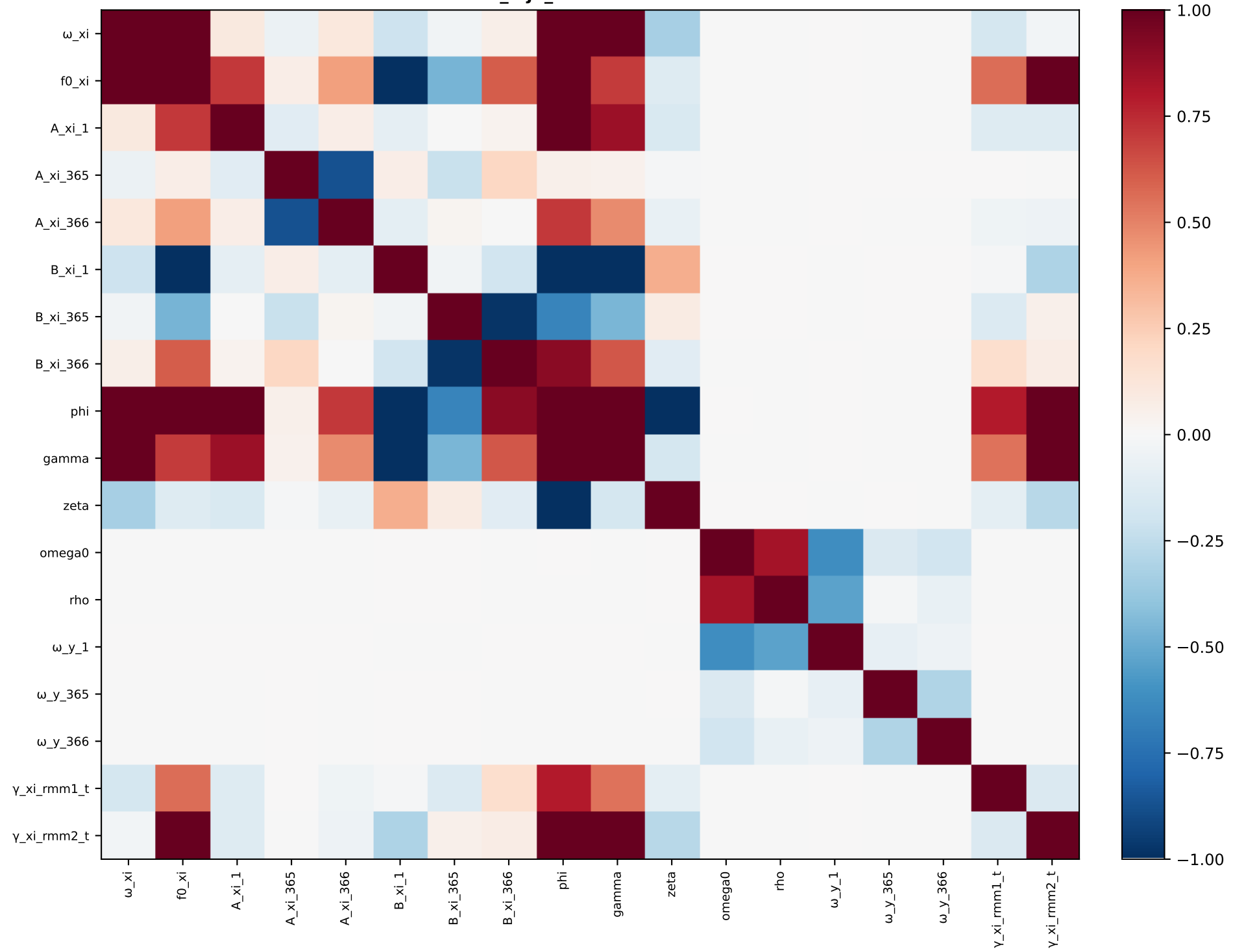
xi_mjo_t — filtered states (OOS)



xi_mjo_t — covariance

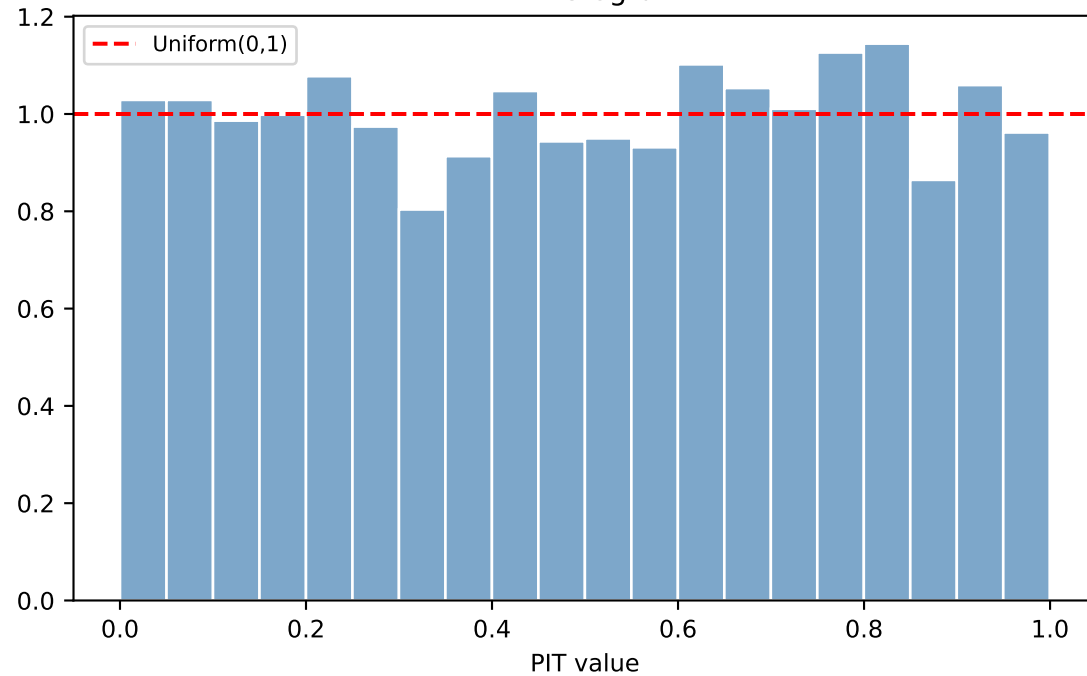


xi_mjo_t — correlation

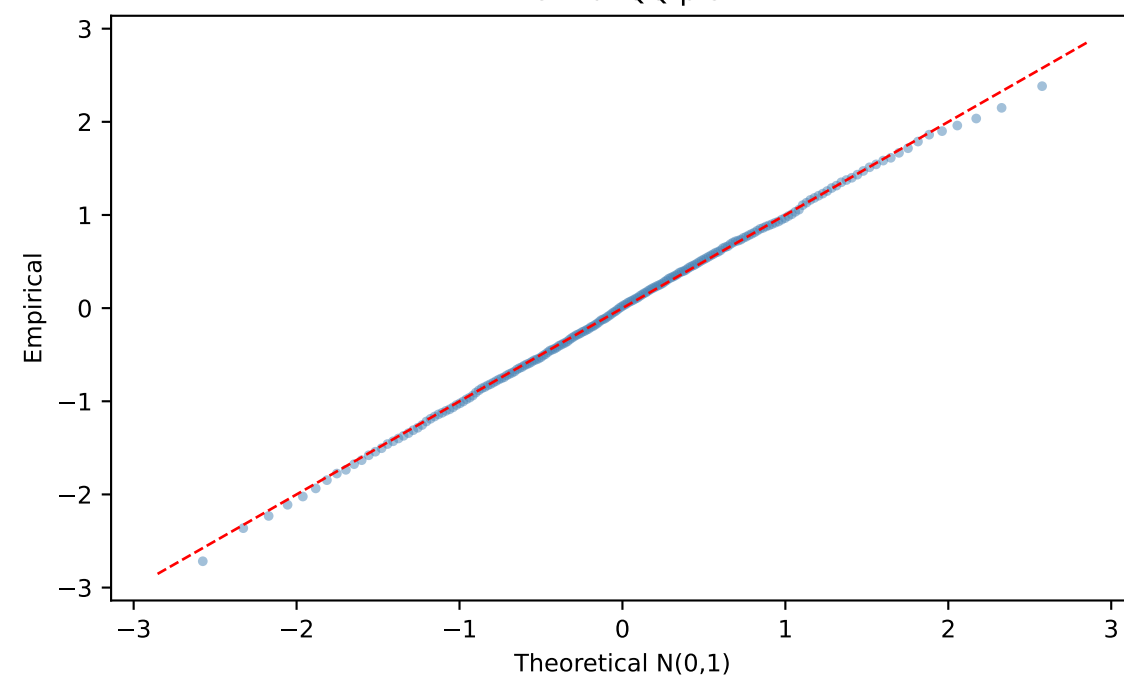


xi_mjo_lag1 — IS diagnostics

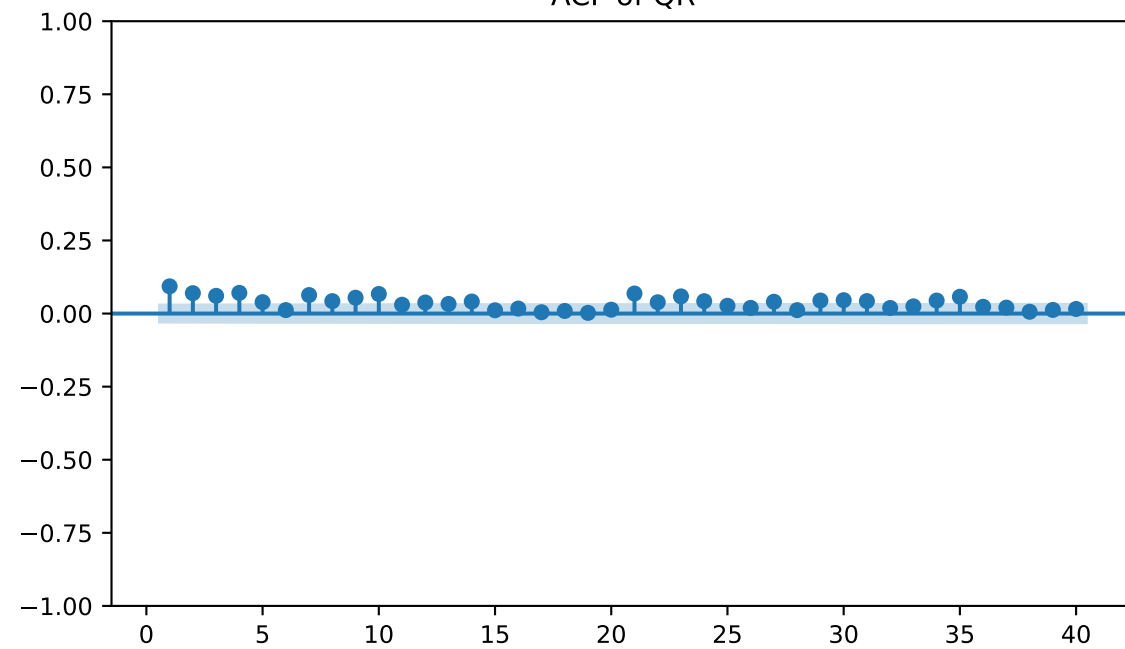
PIT histogram



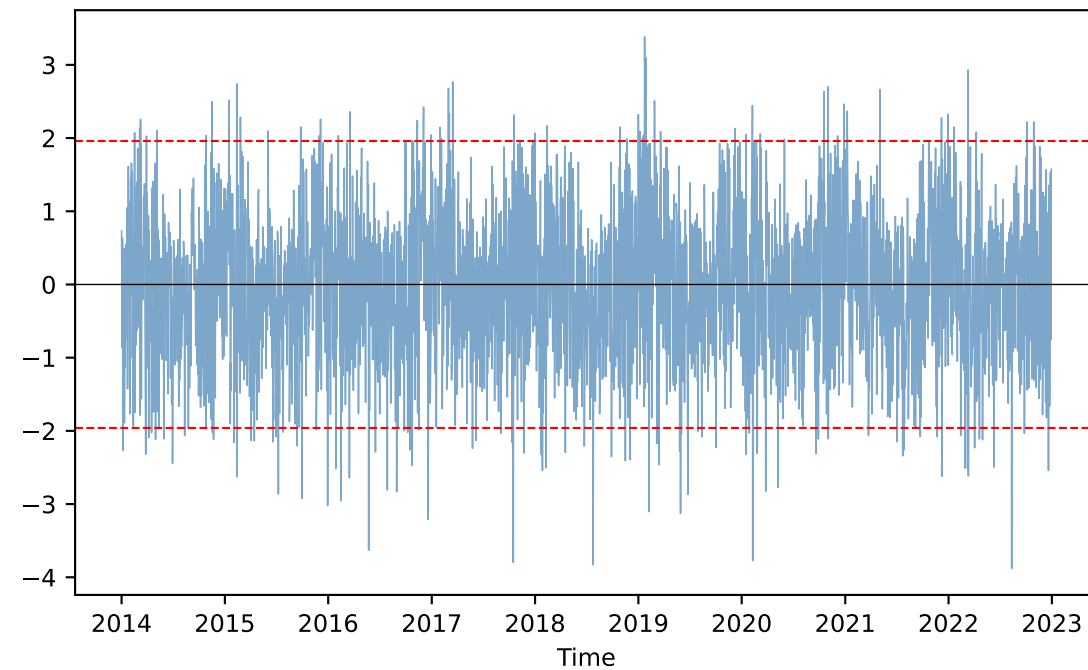
Normal QQ plot



ACF of QR

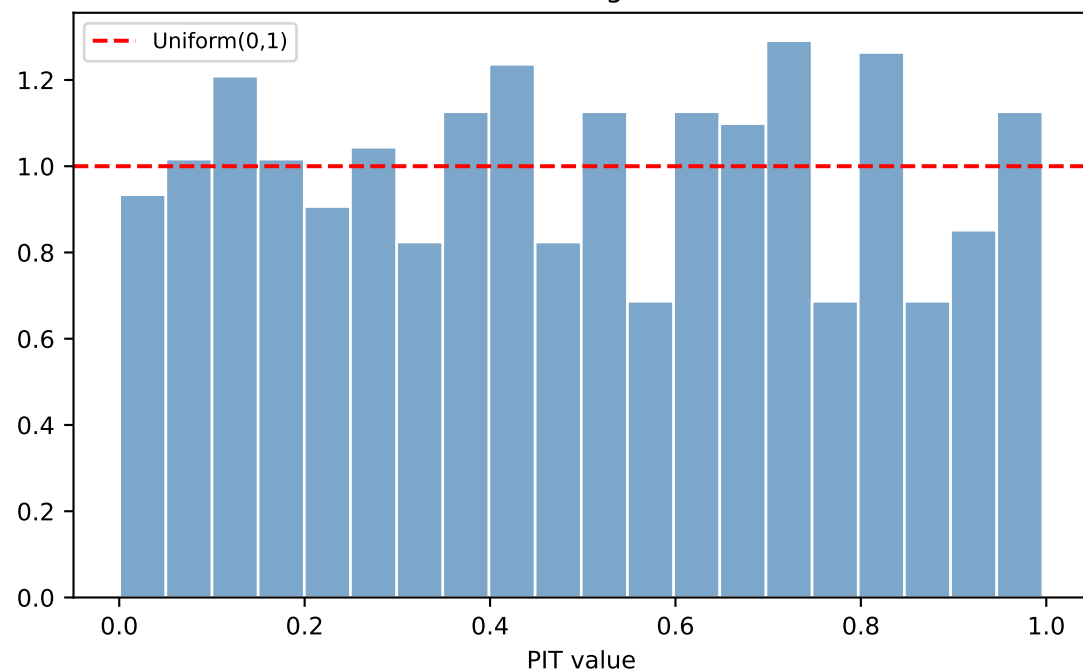


QR time series

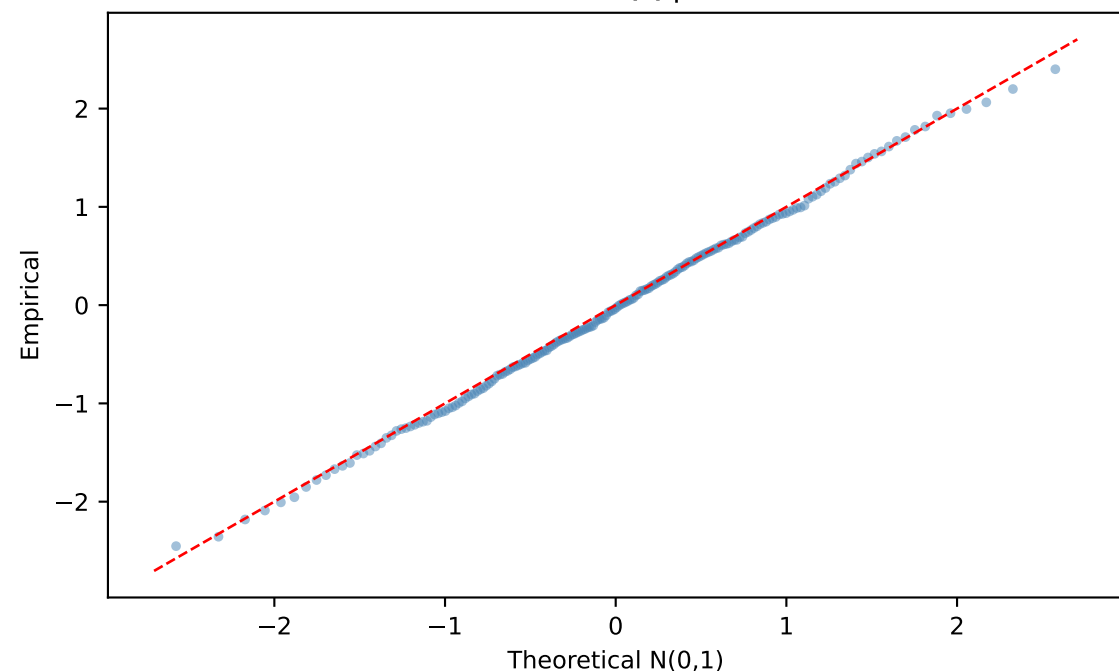


xi_mjo_lag1 — OOS diagnostics

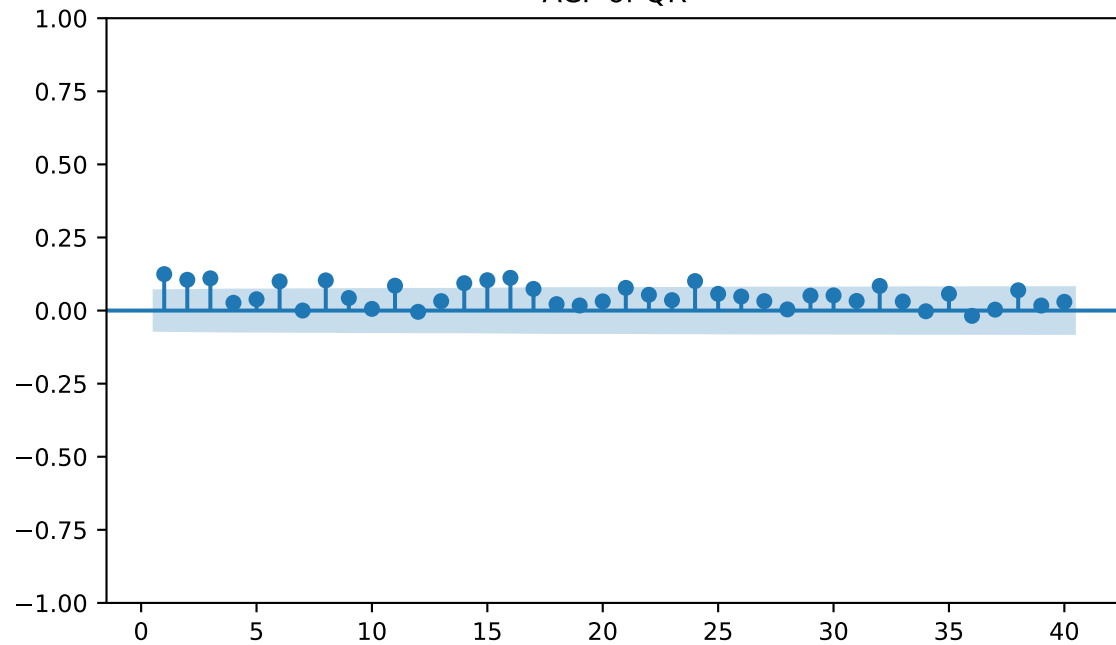
PIT histogram



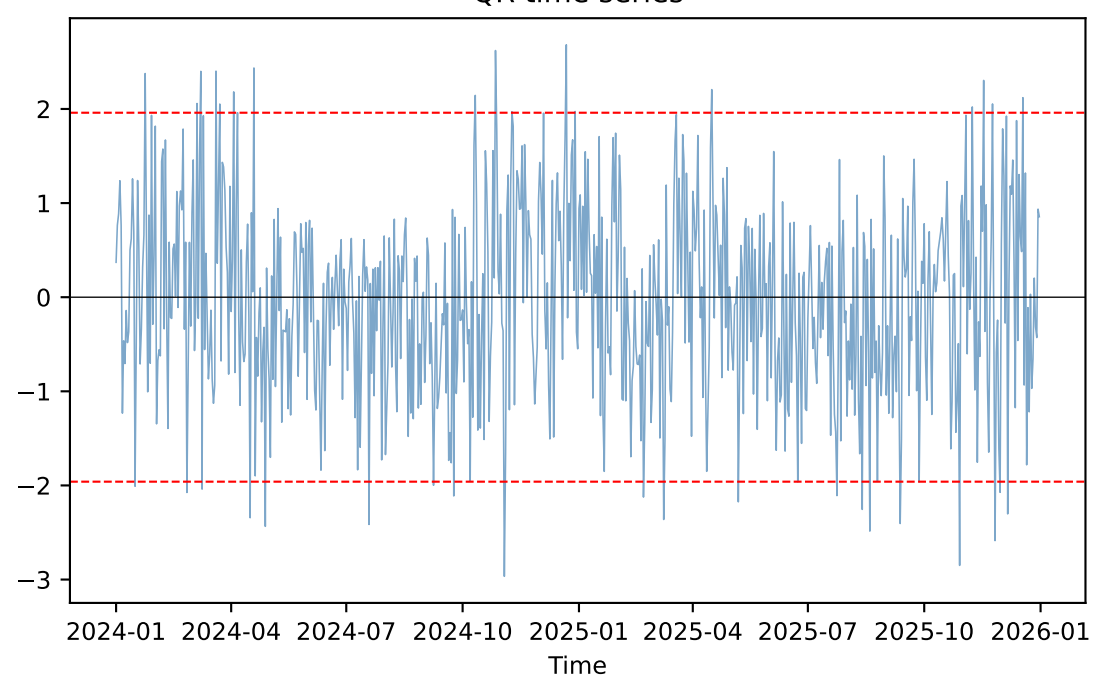
Normal QQ plot



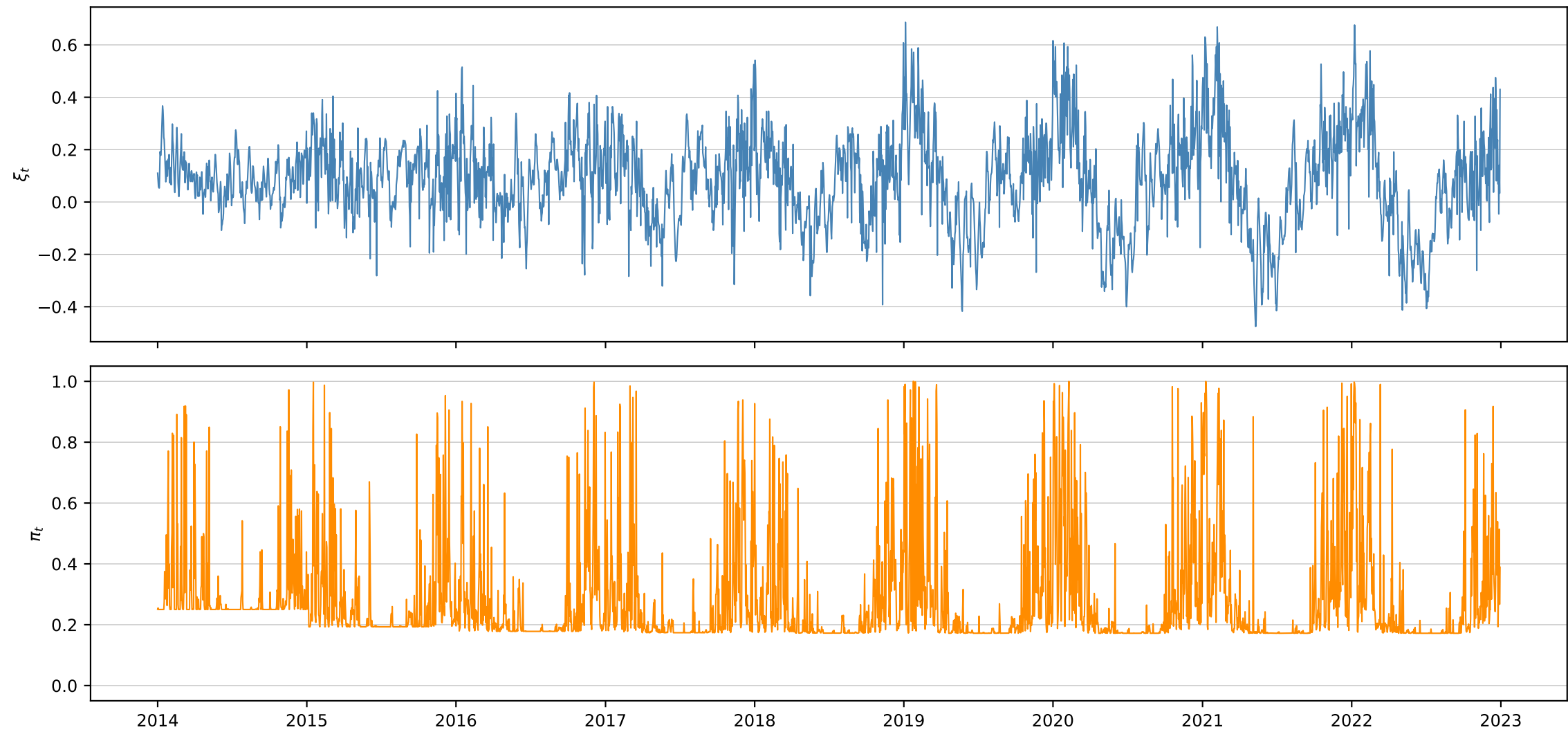
ACF of QR



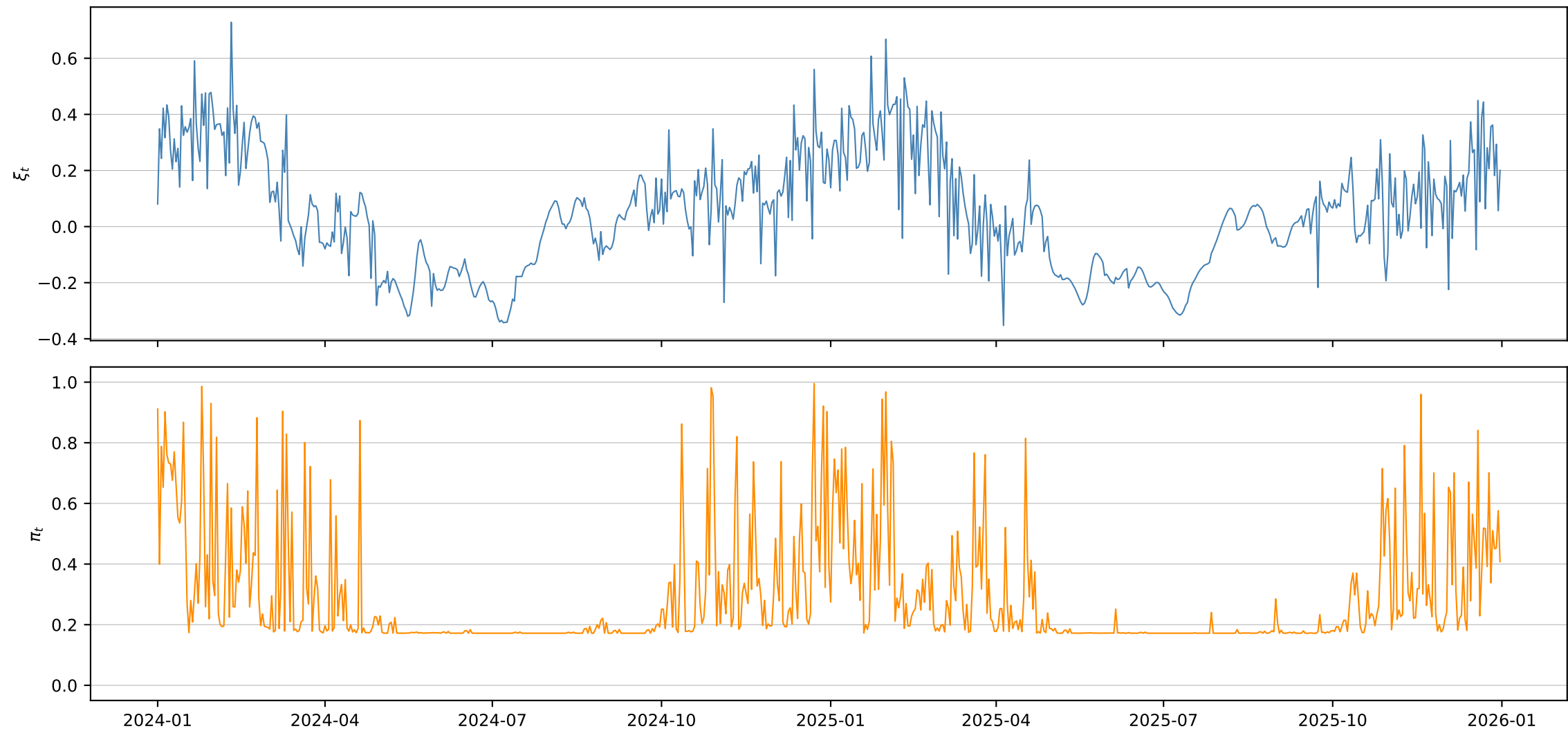
QR time series



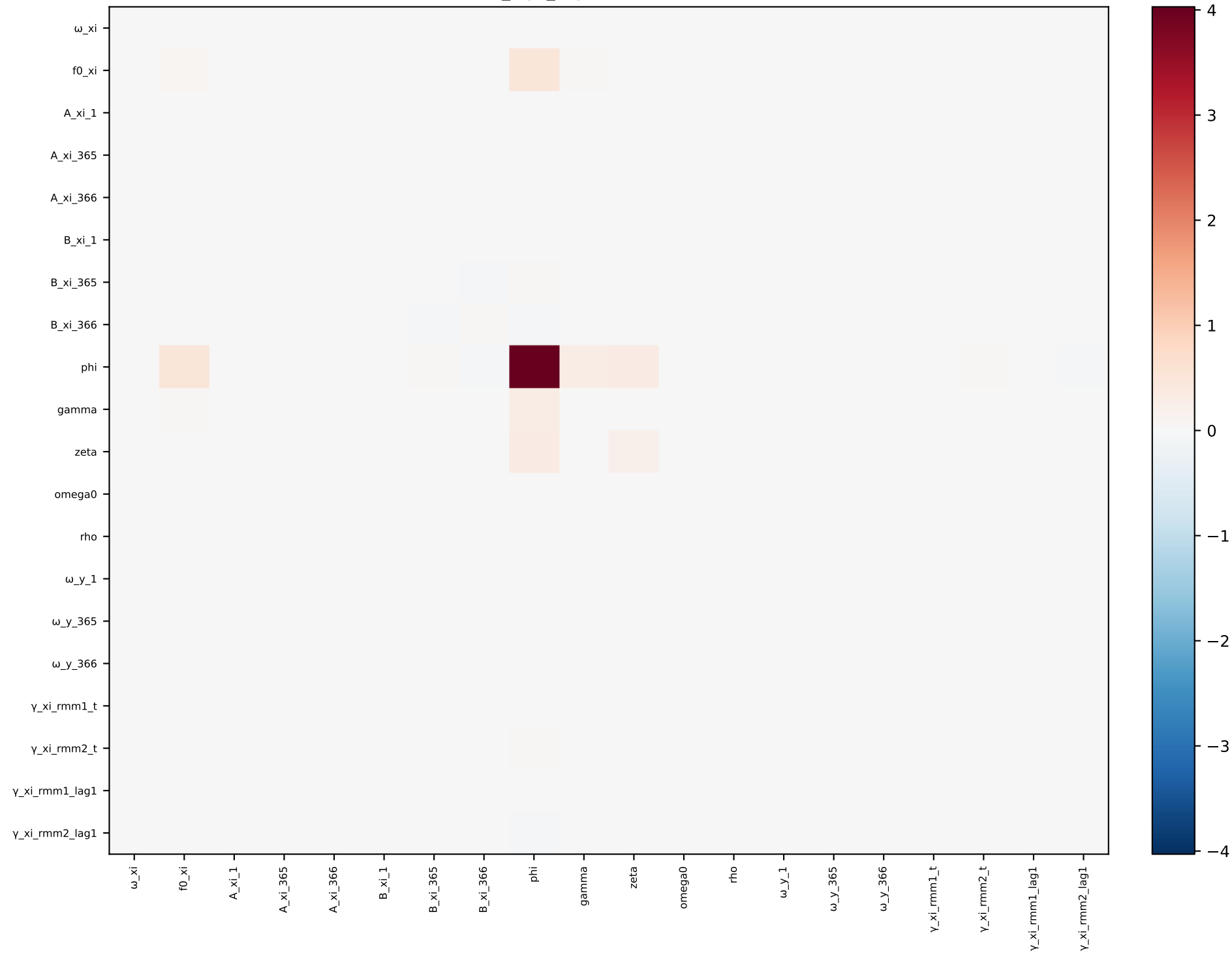
xi_mjo_lag1 — filtered states (IS)



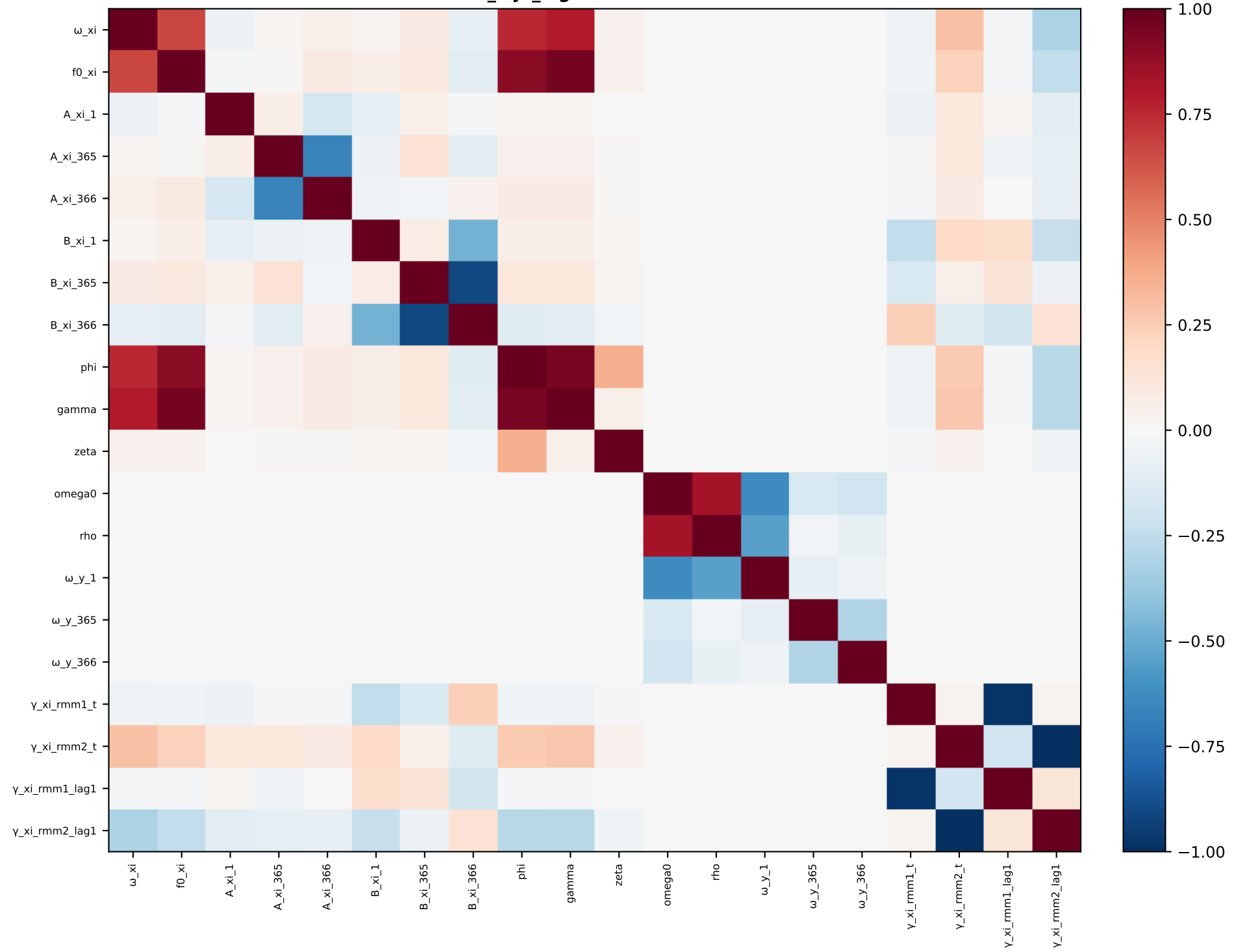
xi_mjo_lag1 — filtered states (OOS)



xi_mjo_lag1 — covariance

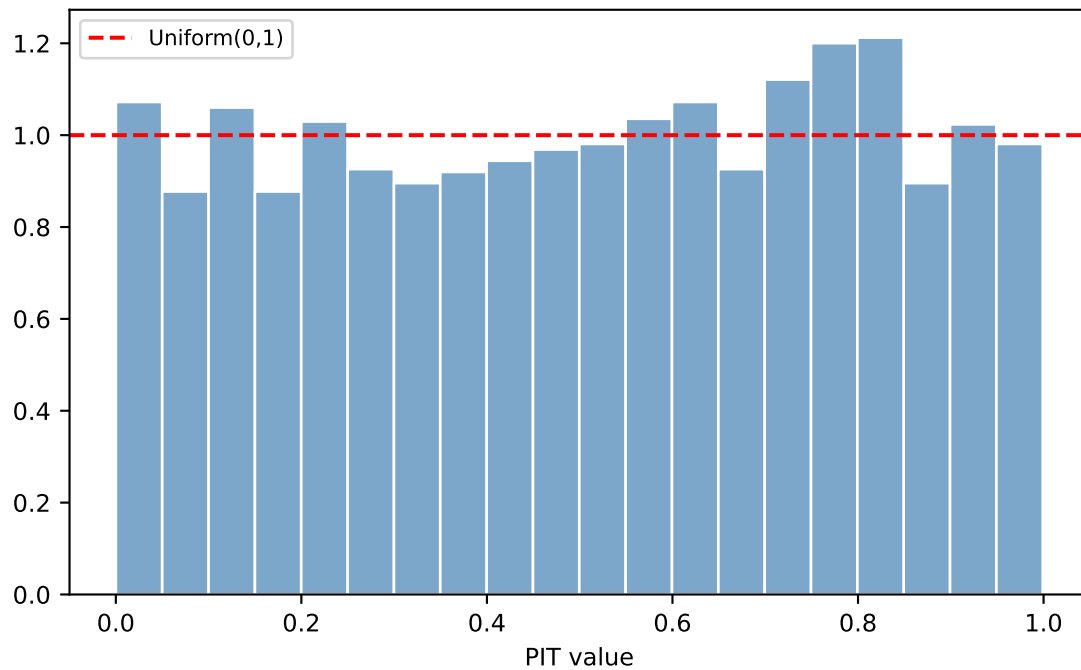


xi_mjo_lag1 — correlation

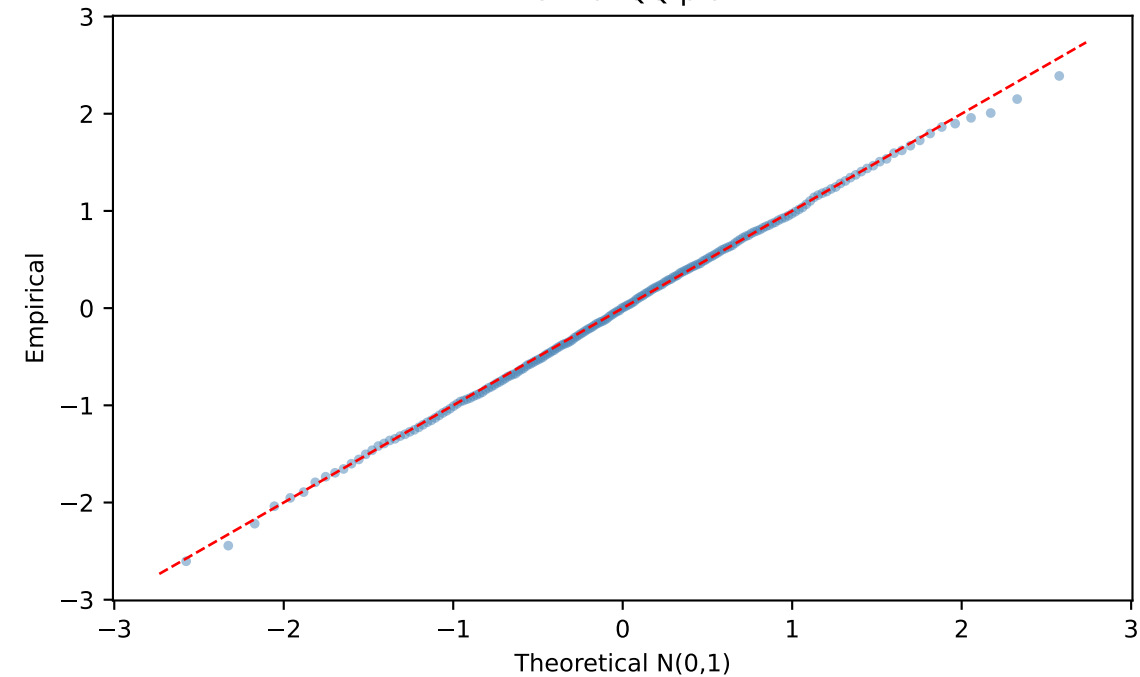


xi_mjo_short_lags — IS diagnostics

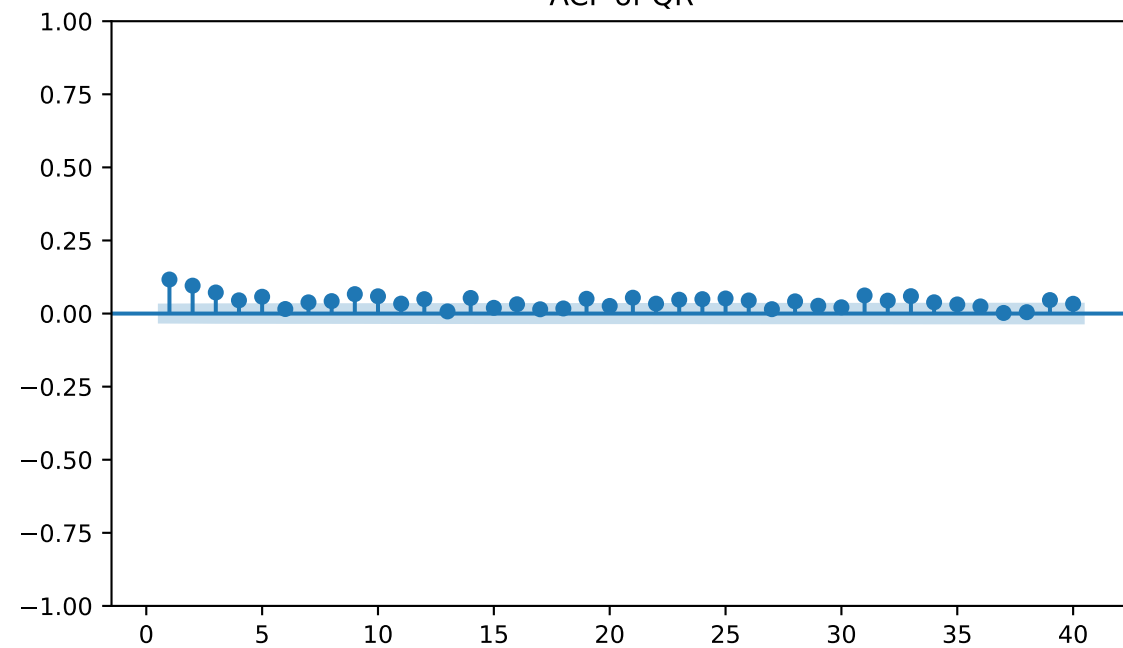
PIT histogram



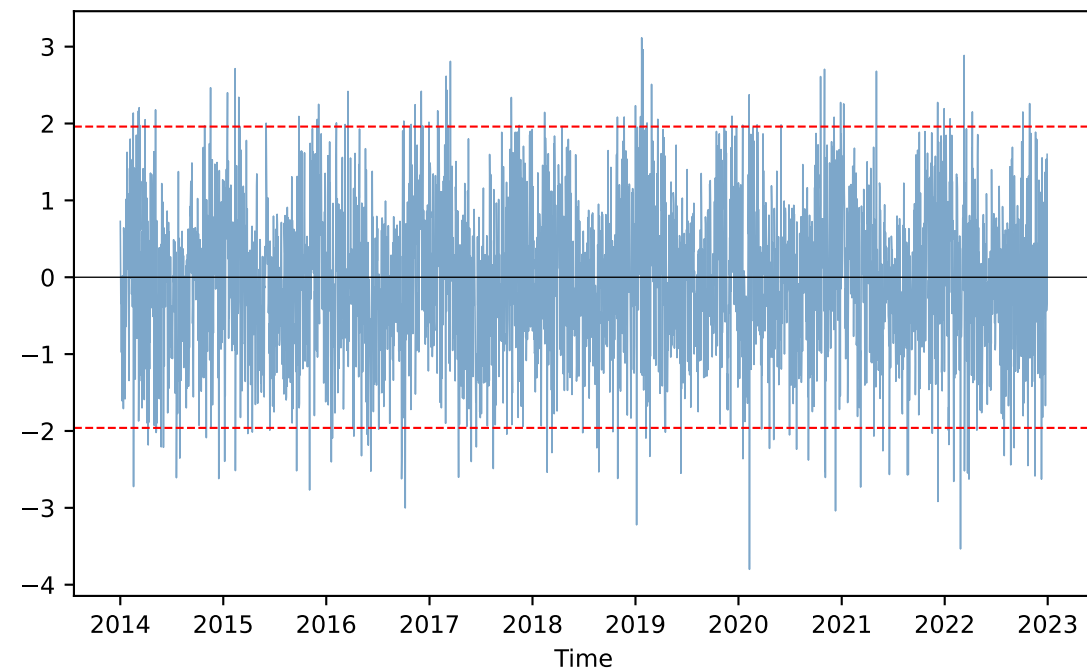
Normal QQ plot



ACF of QR

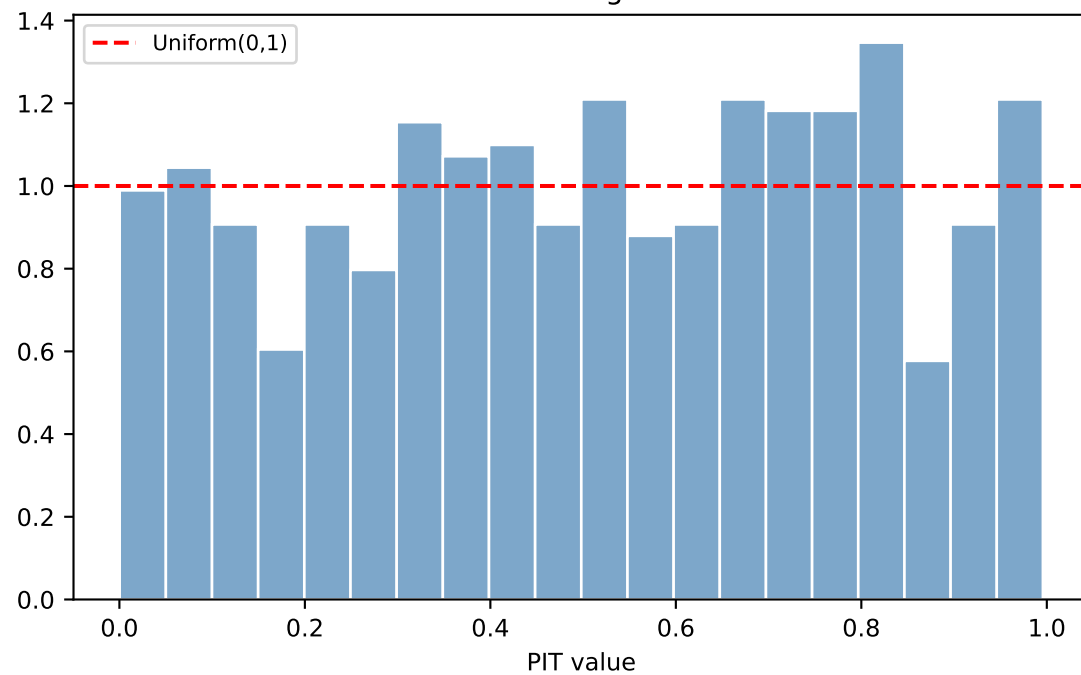


QR time series

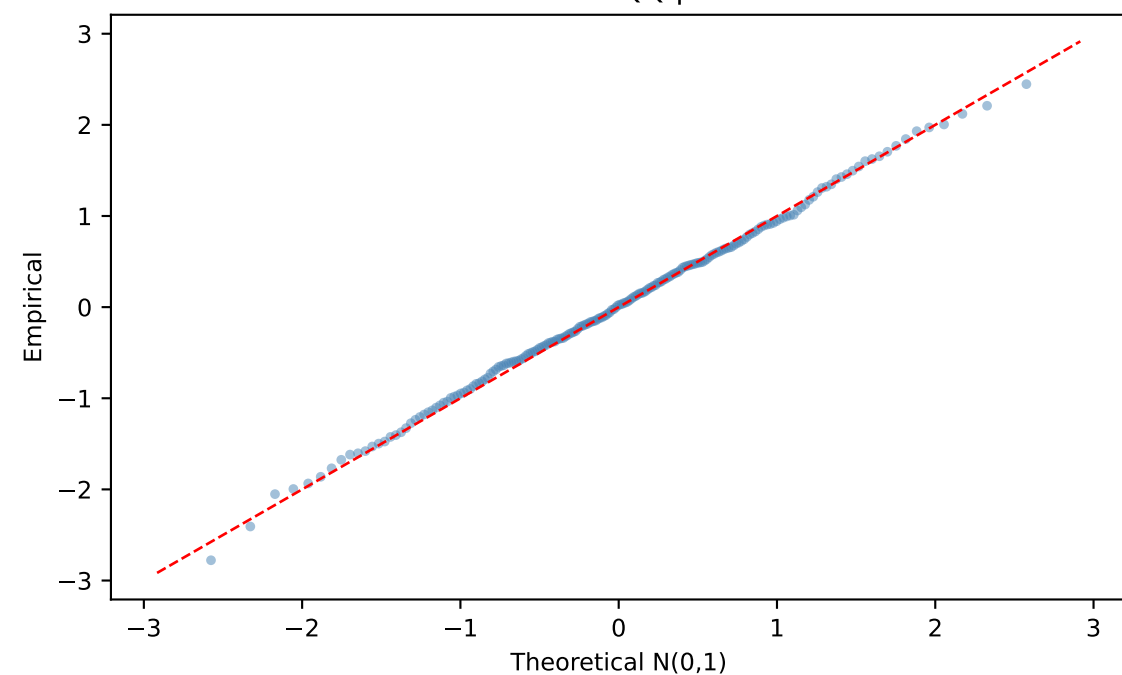


xi_mjo_short_lags — OOS diagnostics

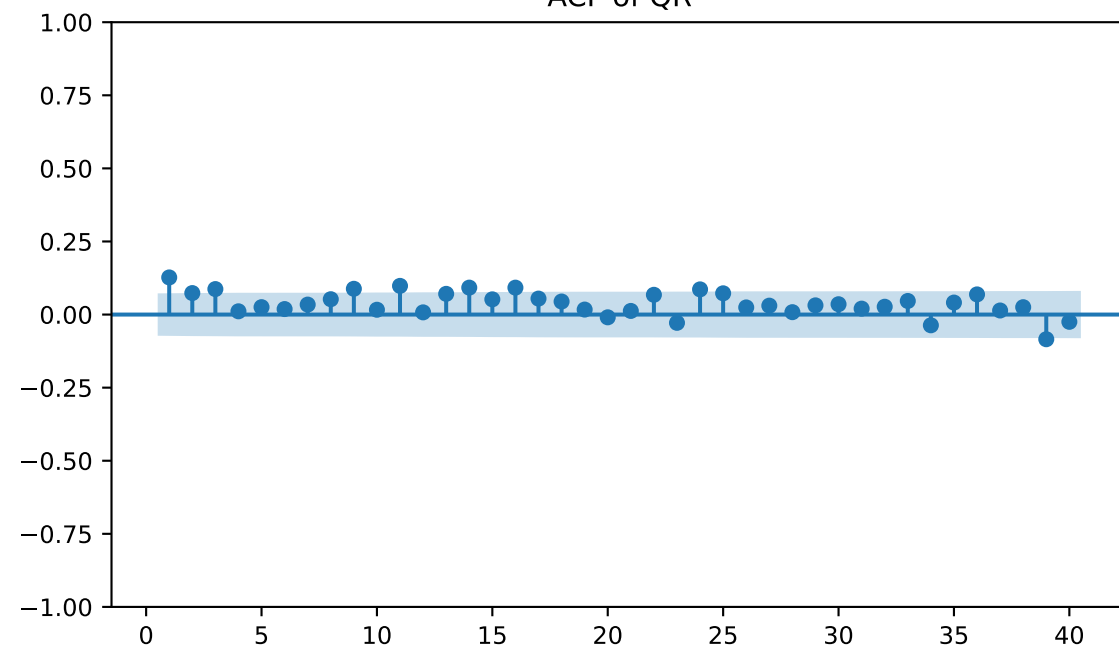
PIT histogram



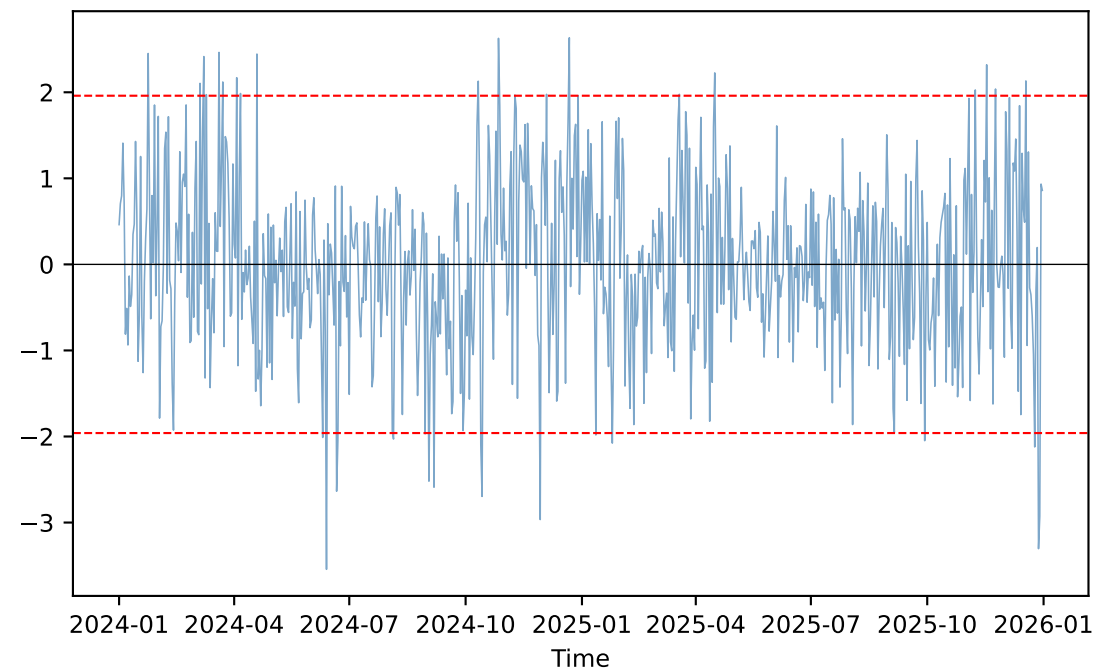
Normal QQ plot



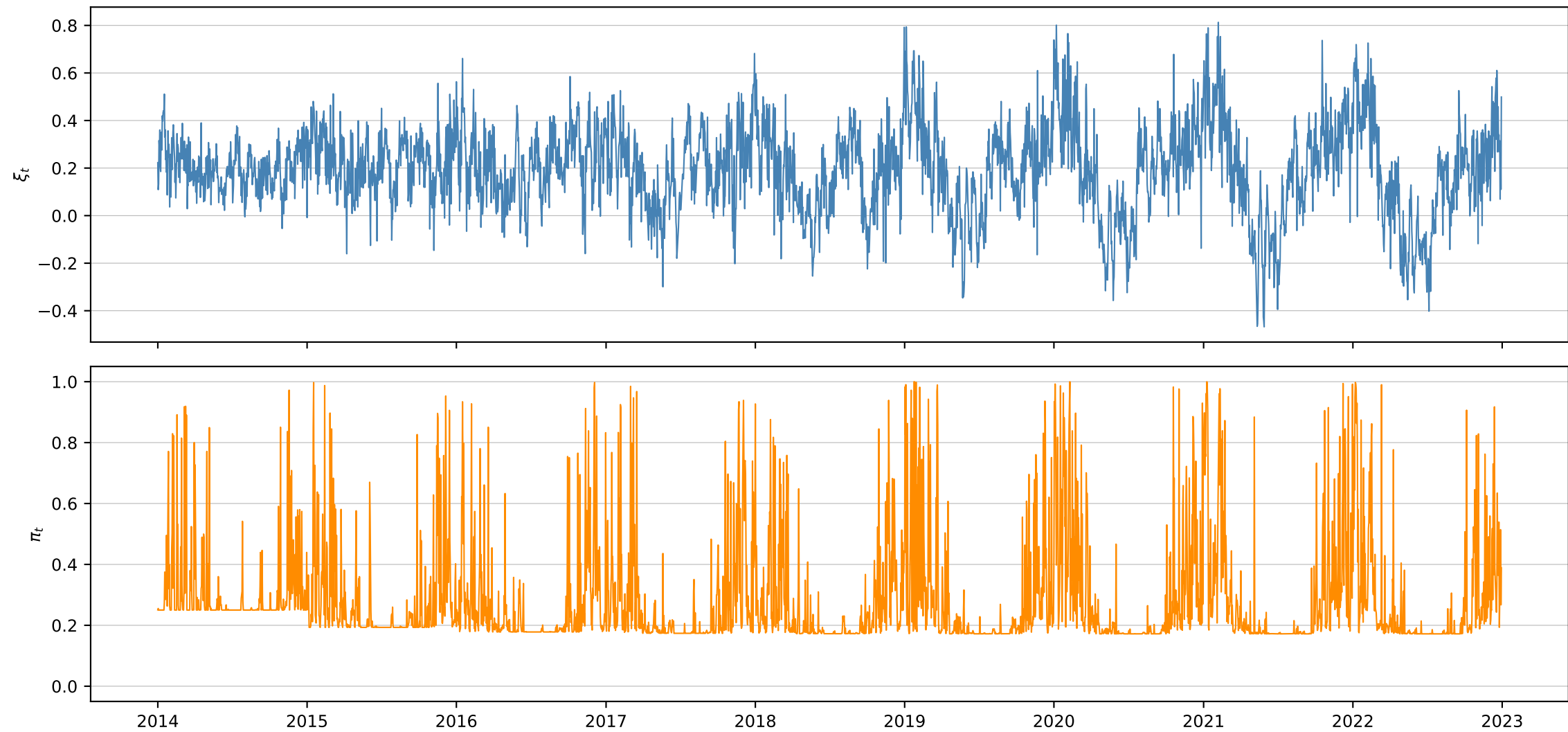
ACF of QR



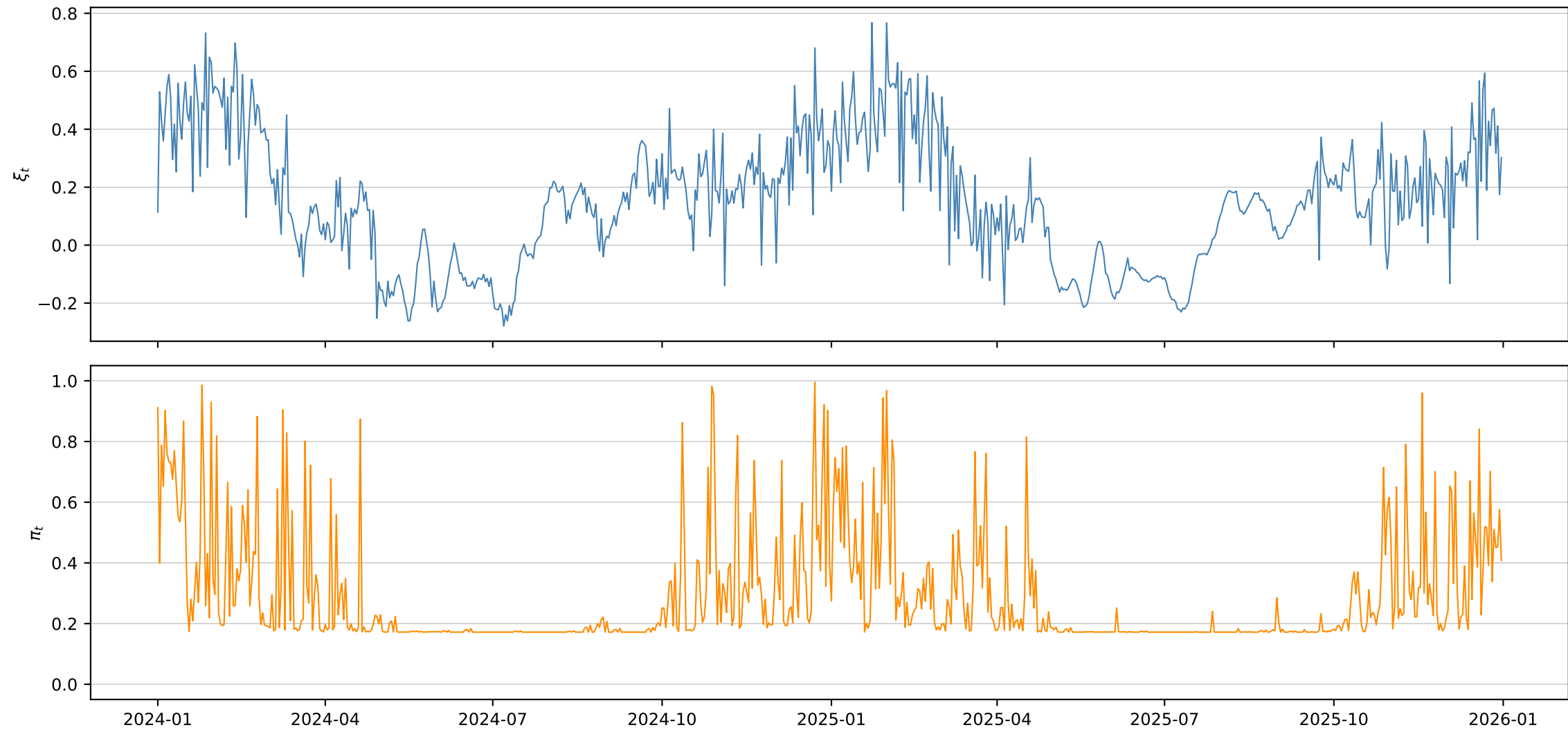
QR time series



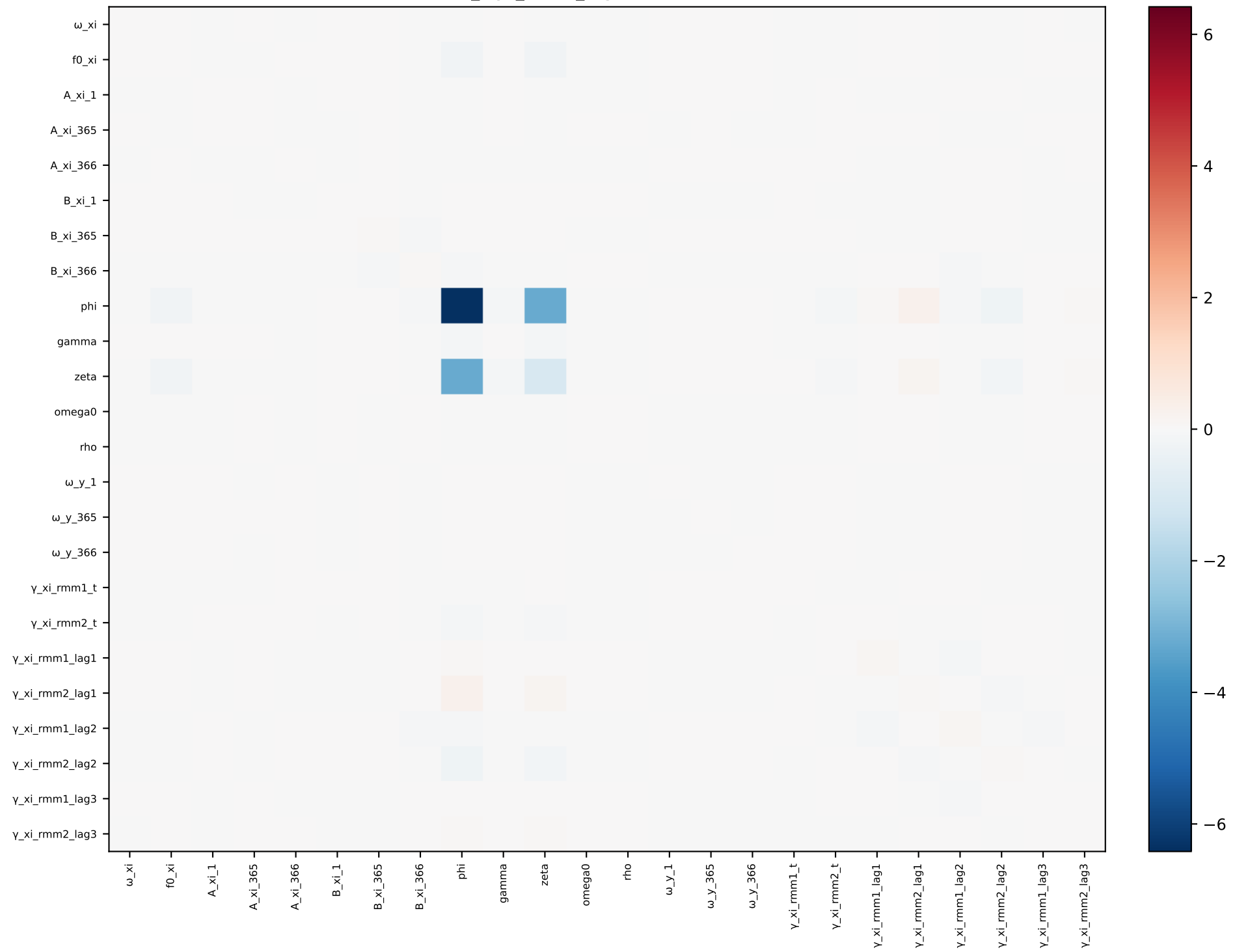
xi_mjo_short_lags — filtered states (IS)



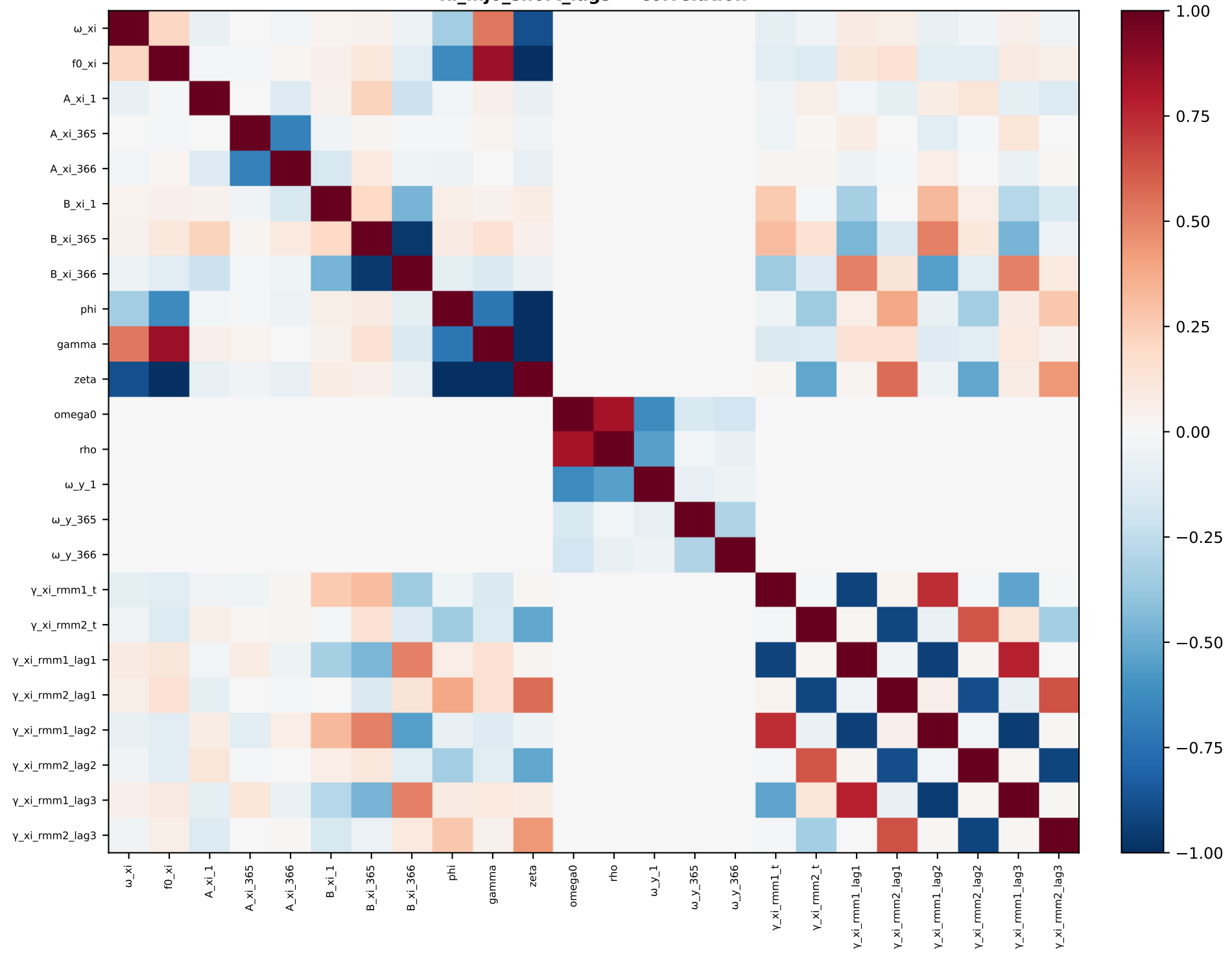
xi_mjo_short_lags — filtered states (OOS)



xi_mjo_short_lags — covariance

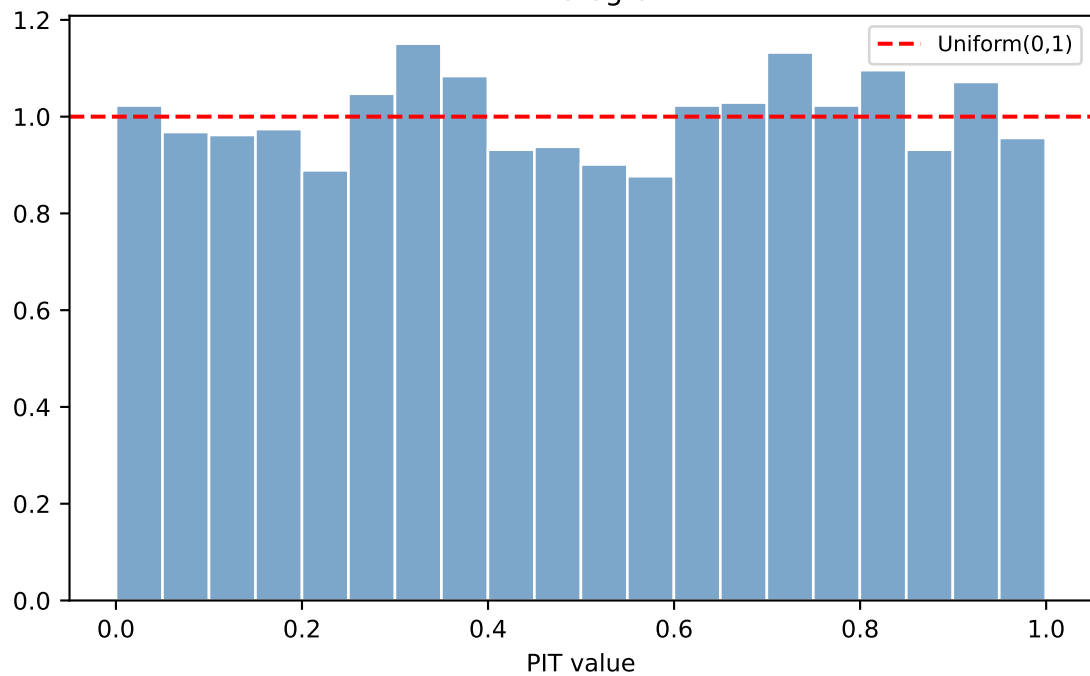


xi_mjo_short_lags — correlation

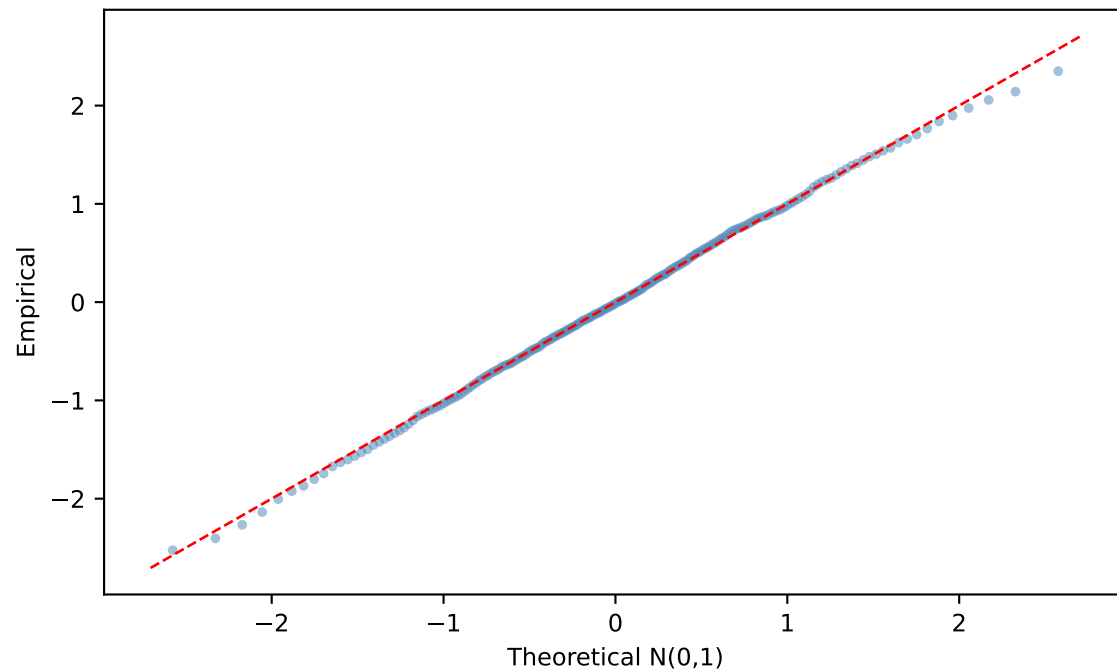


xi_nino34_t — IS diagnostics

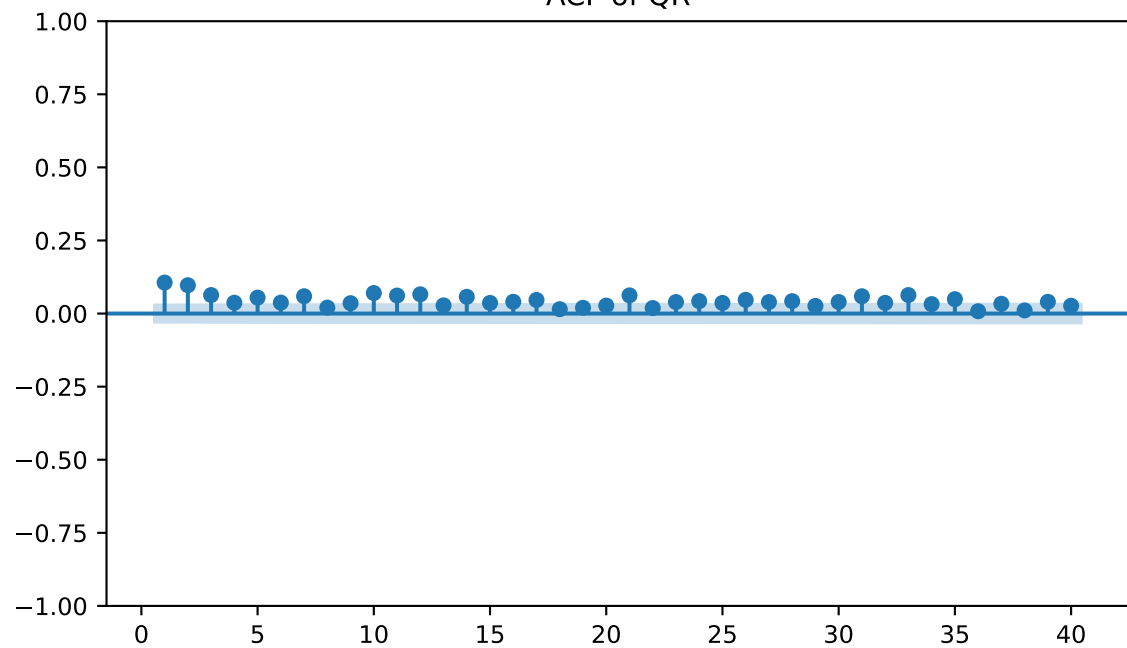
PIT histogram



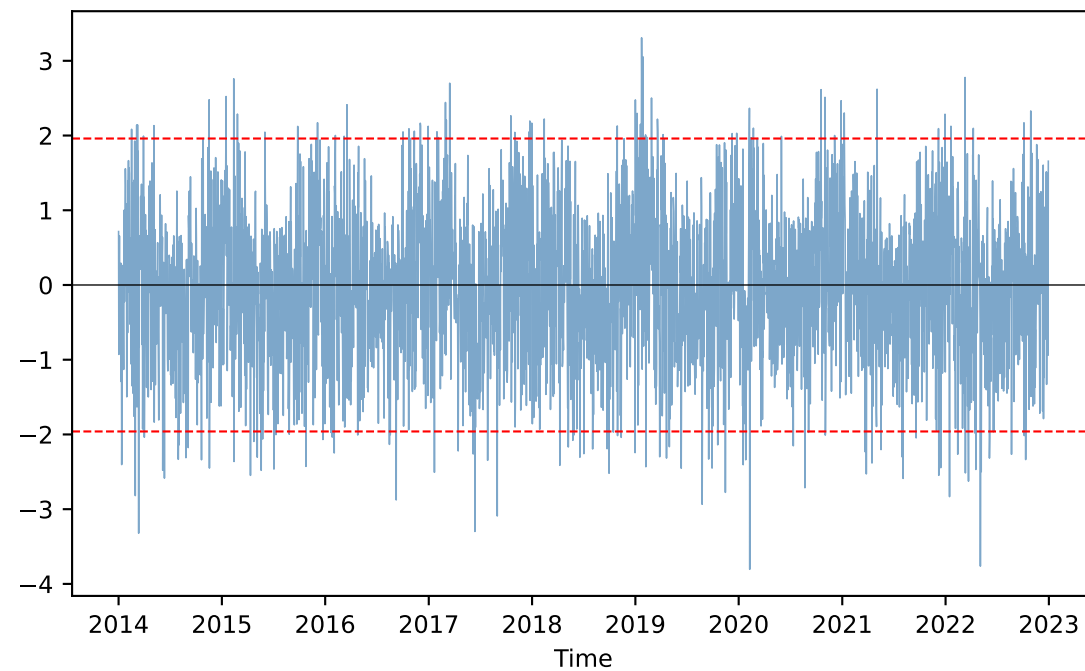
Normal QQ plot



ACF of QR

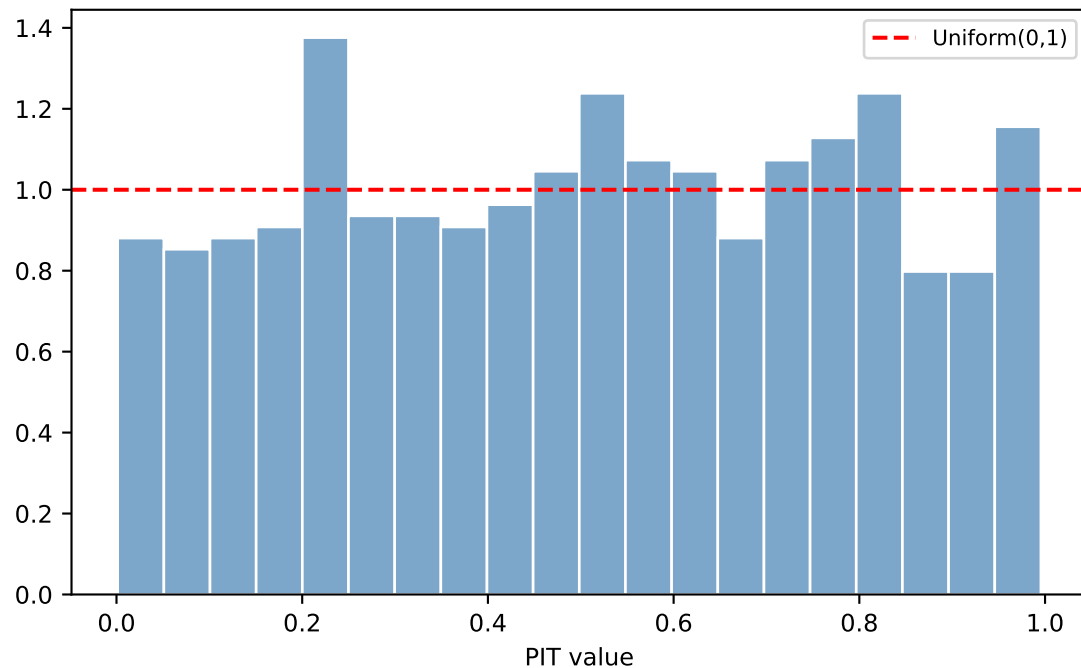


QR time series

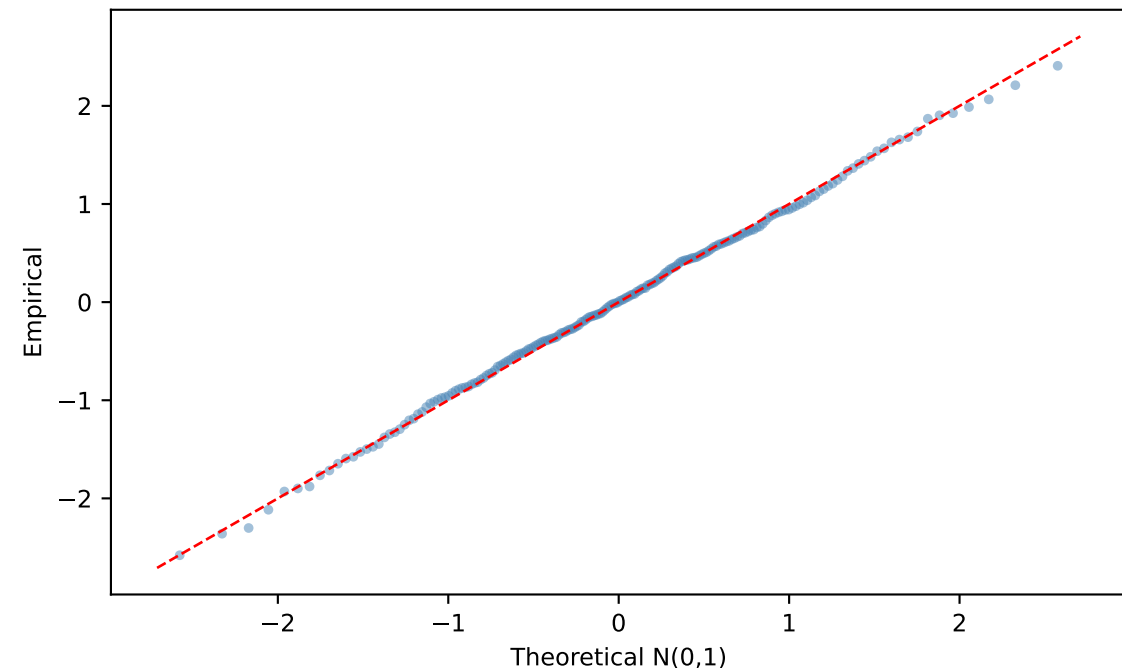


xi_nino34_t — OOS diagnostics

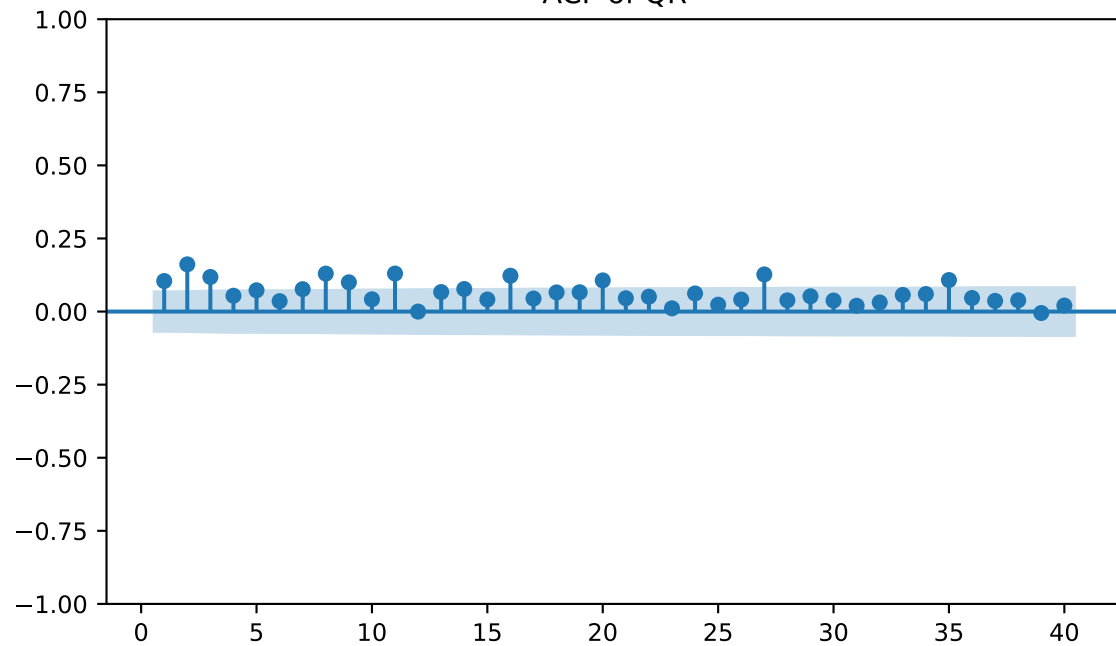
PIT histogram



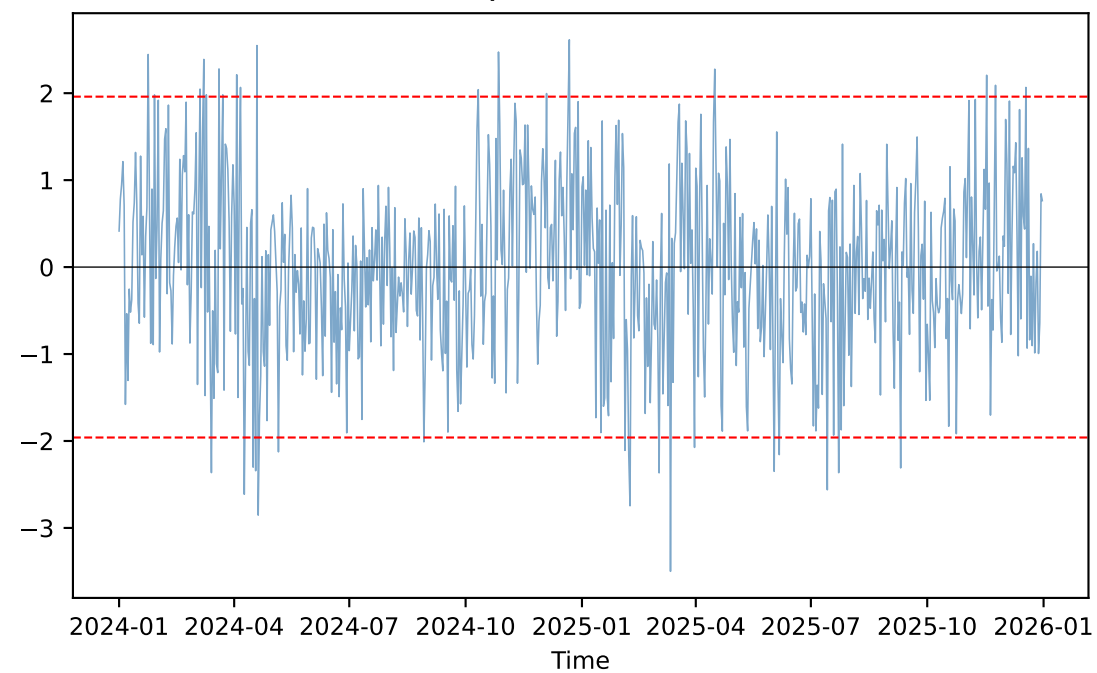
Normal QQ plot



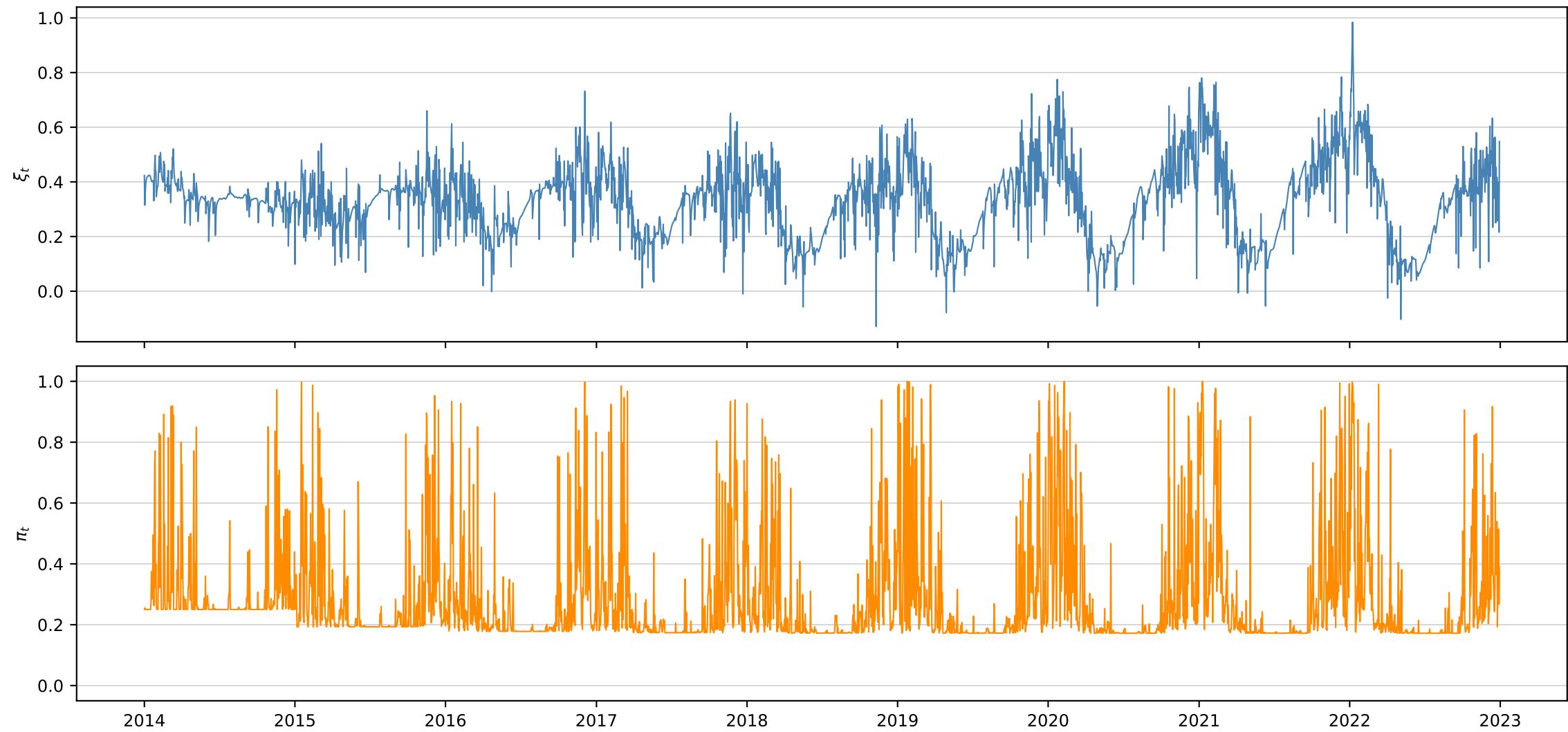
ACF of QR



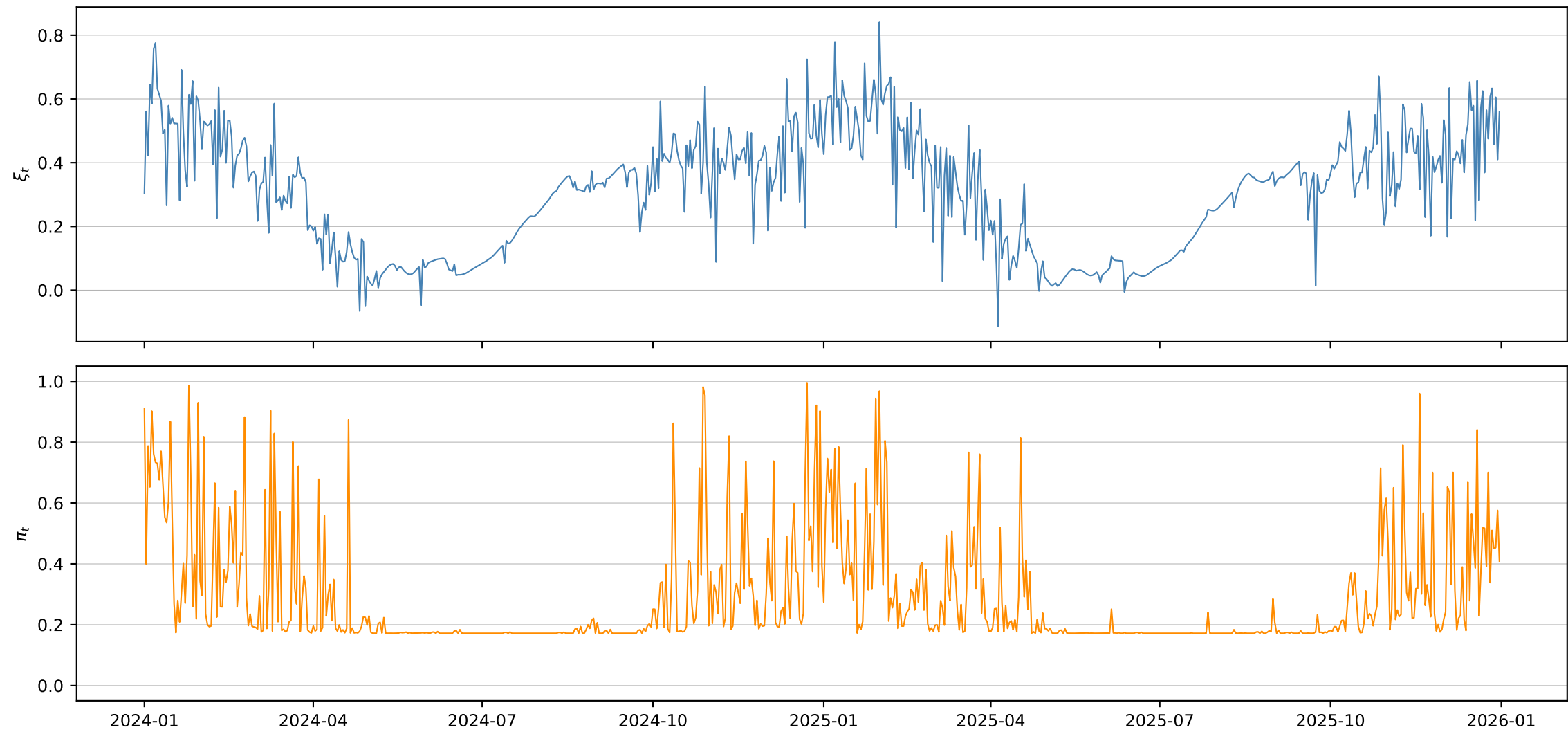
QR time series



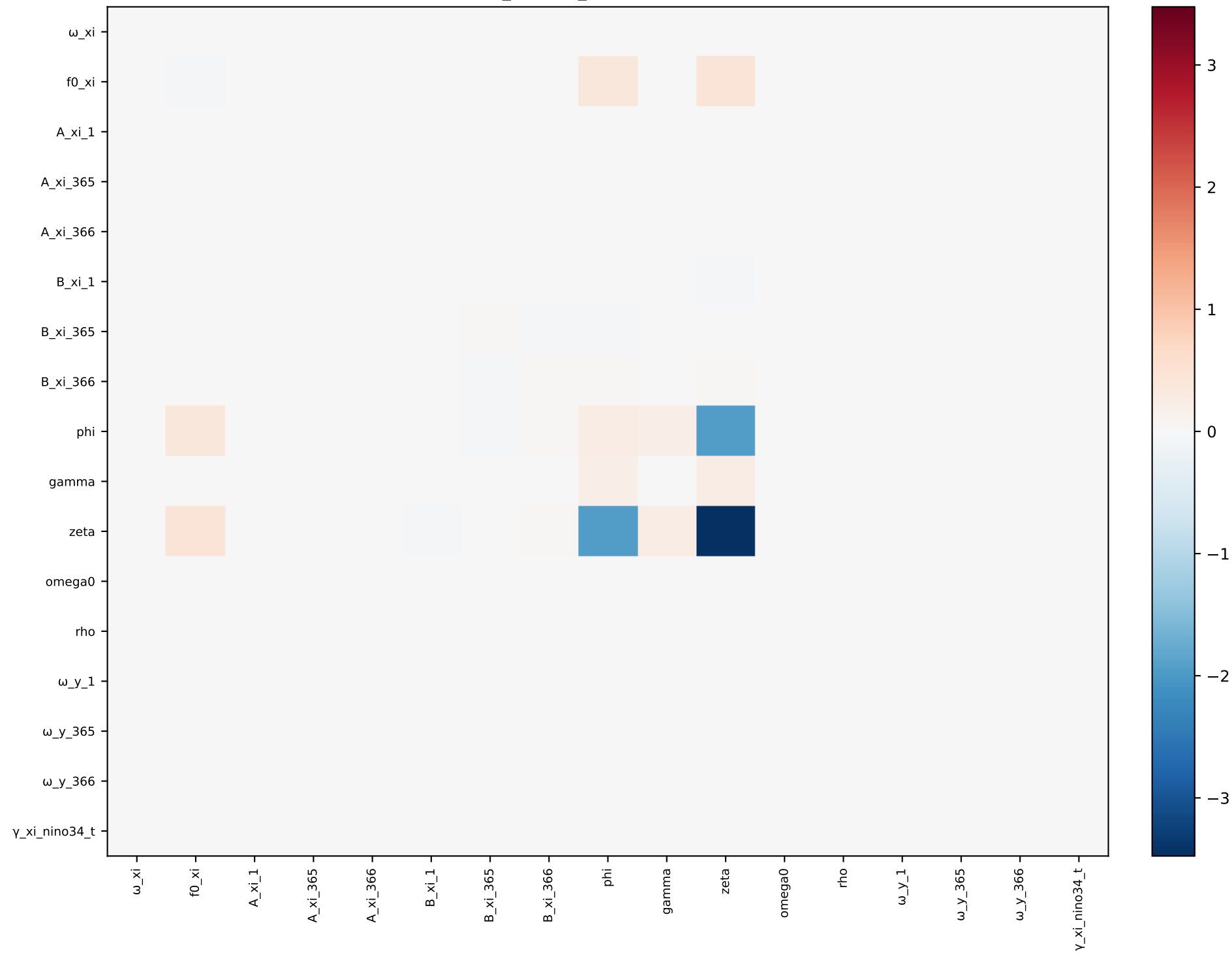
xi_nino34_t — filtered states (IS)



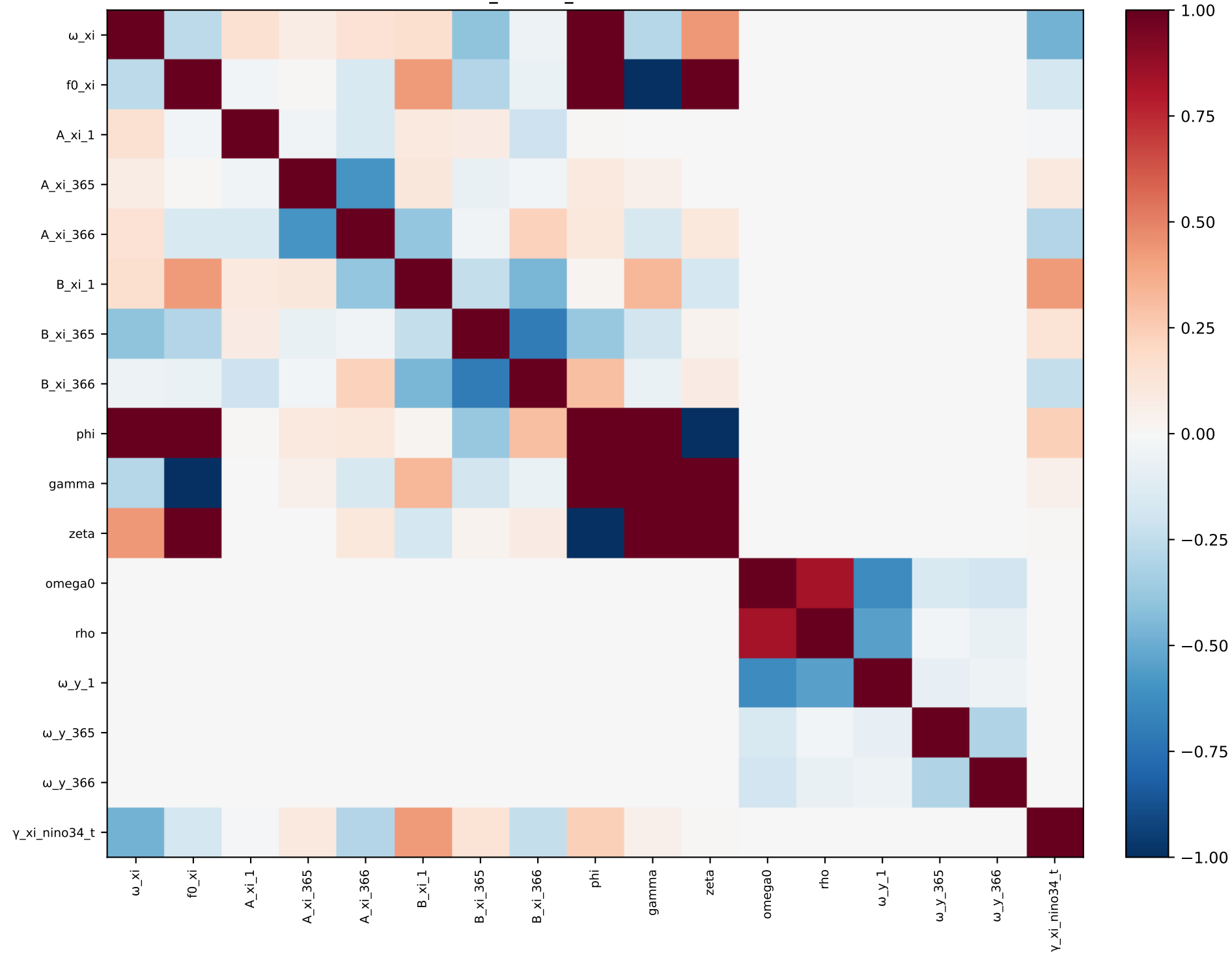
xi_nino34_t — filtered states (OOS)



xi_nino34_t — covariance

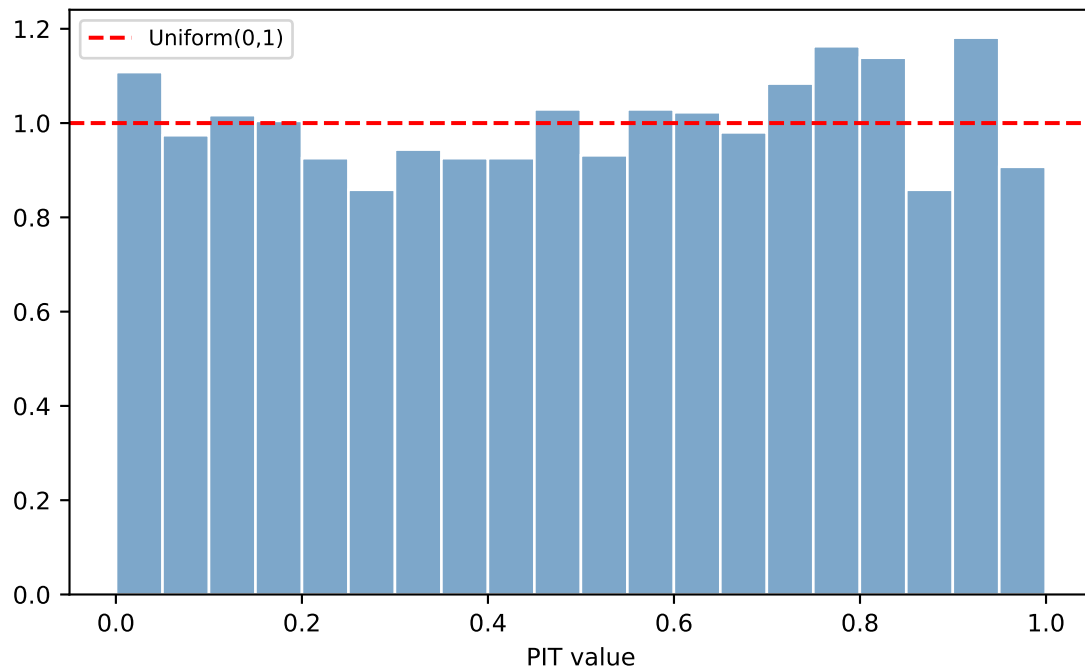


xi_nino34_t — correlation

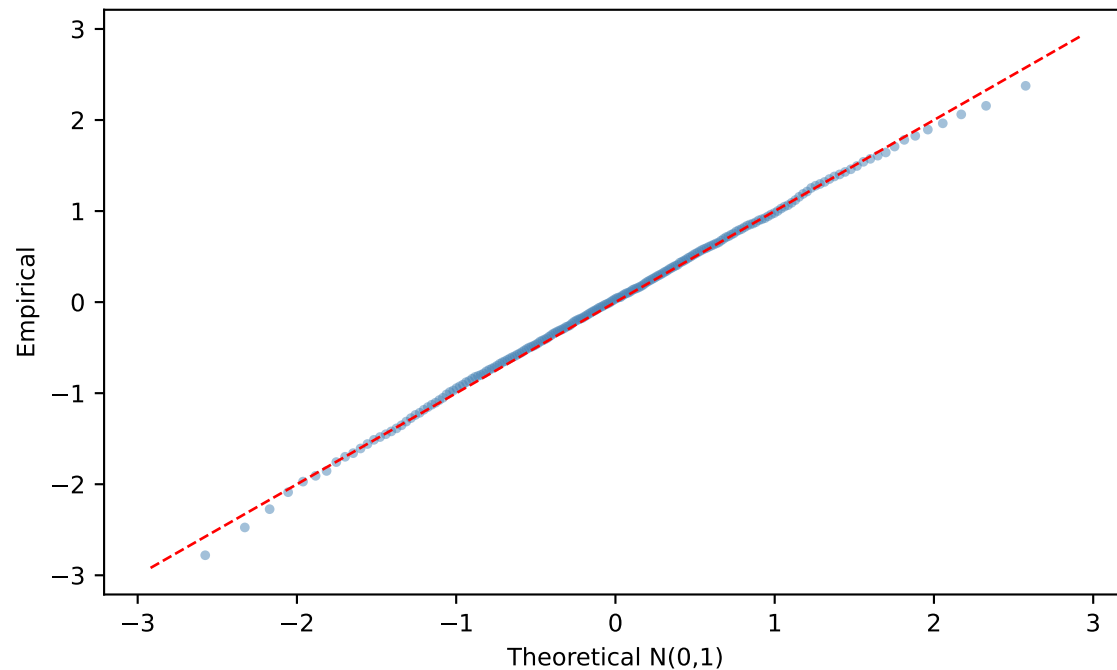


xi_nino34_lags — IS diagnostics

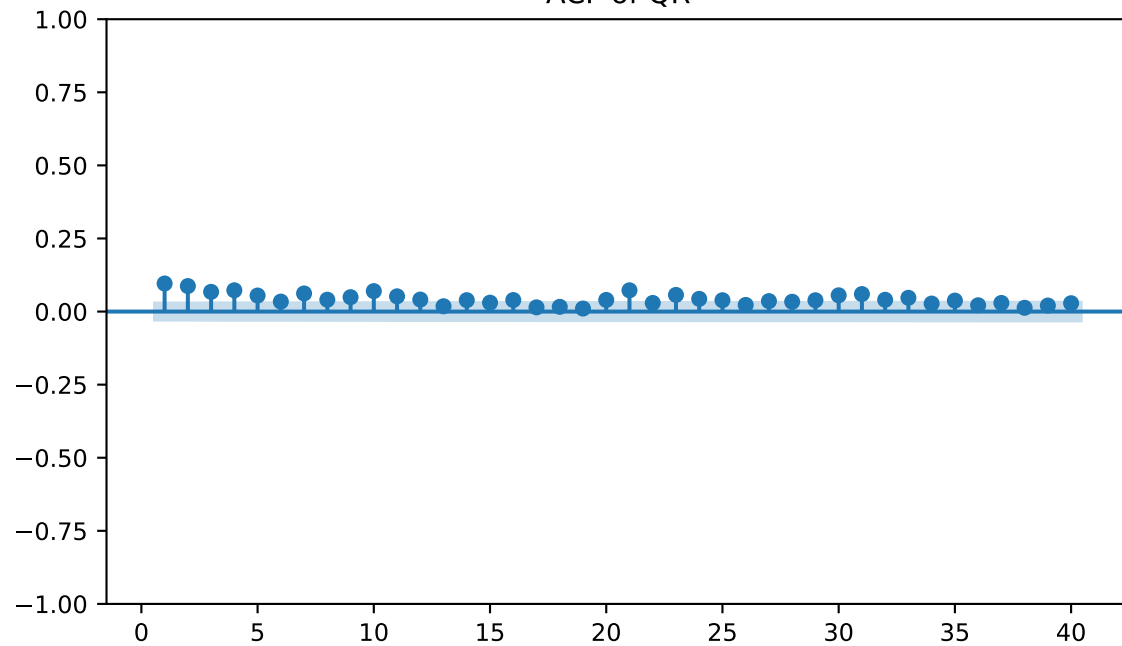
PIT histogram



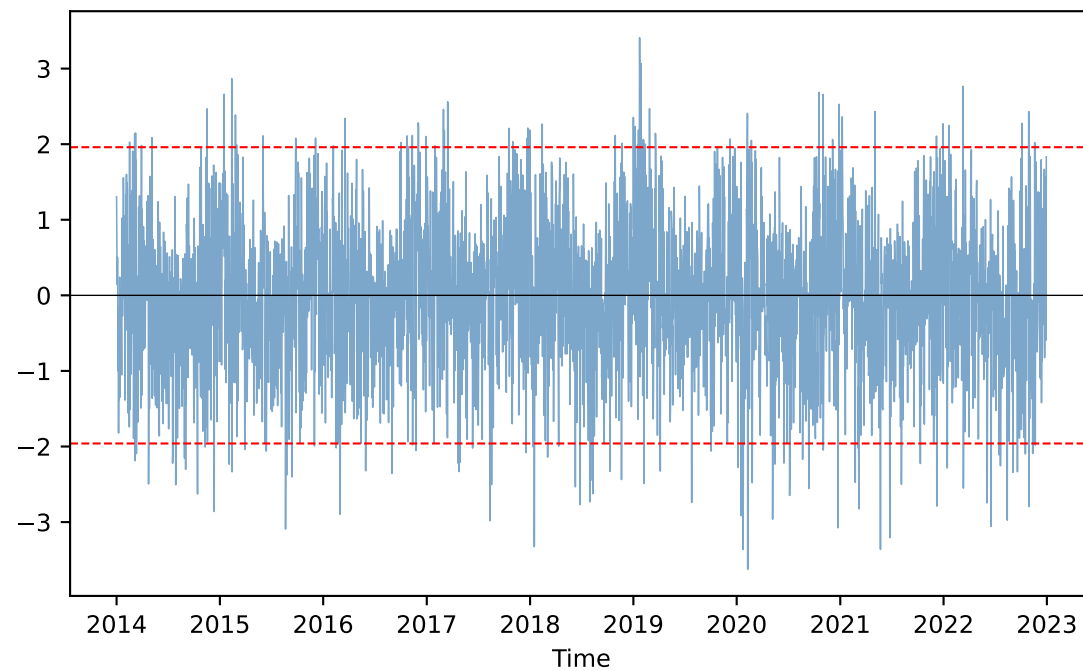
Normal QQ plot



ACF of QR

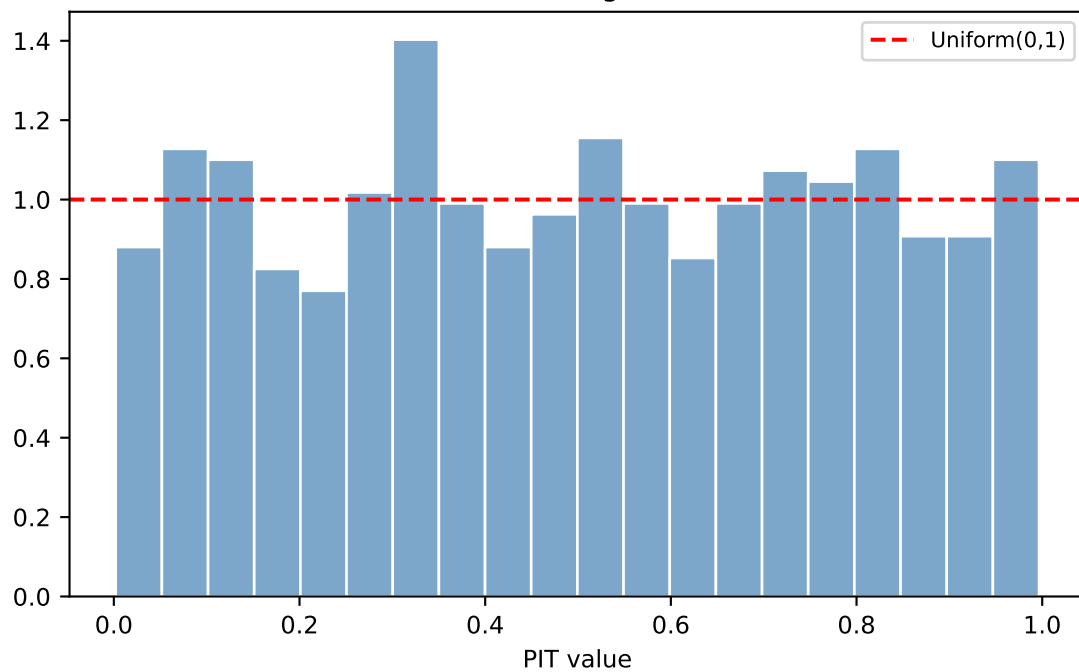


QR time series

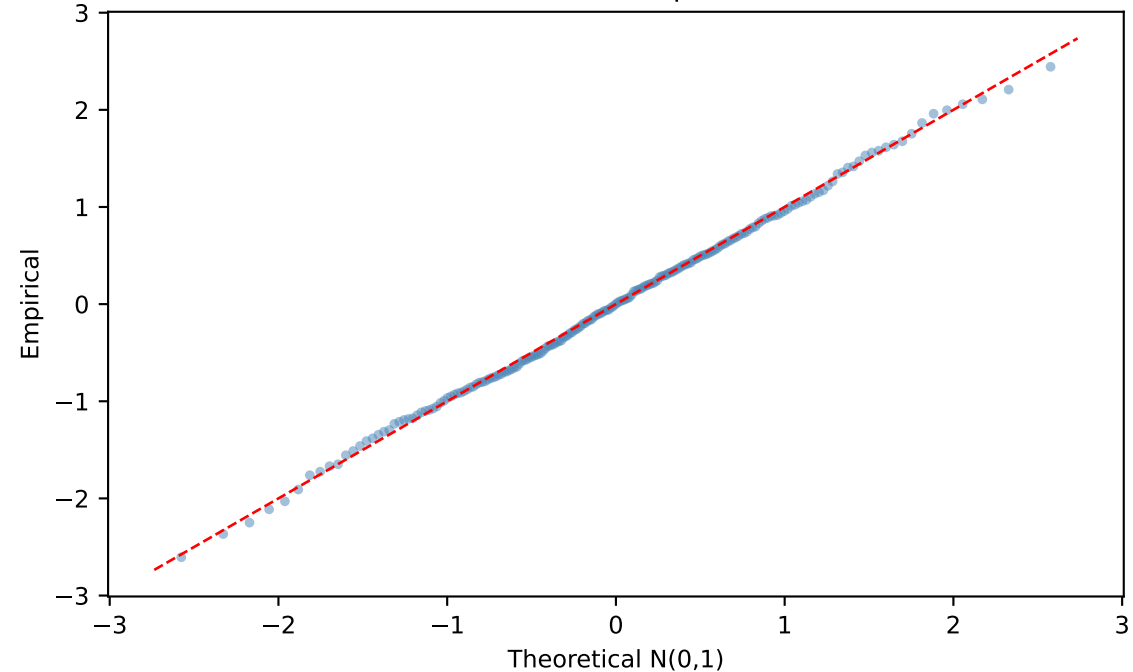


xi_nino34_lags — OOS diagnostics

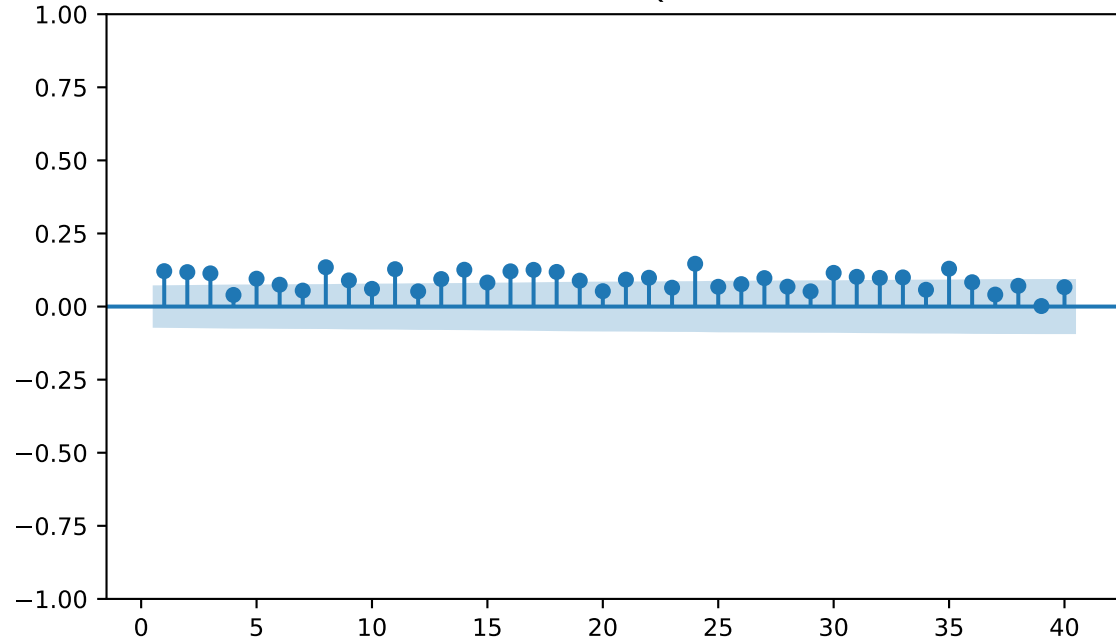
PIT histogram



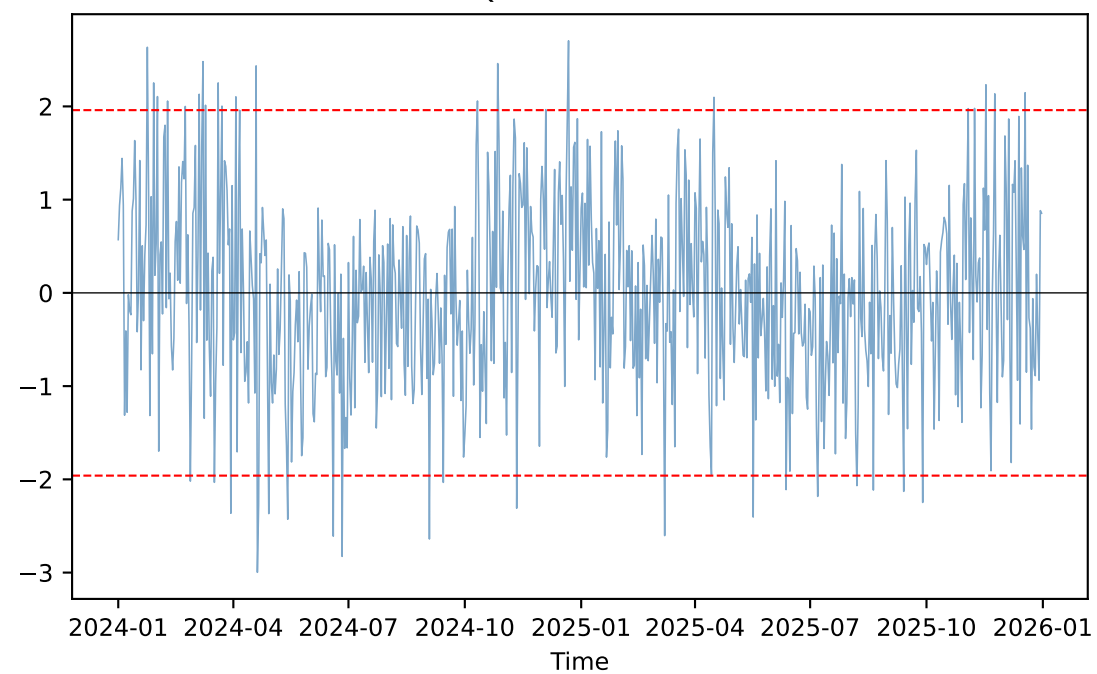
Normal QQ plot



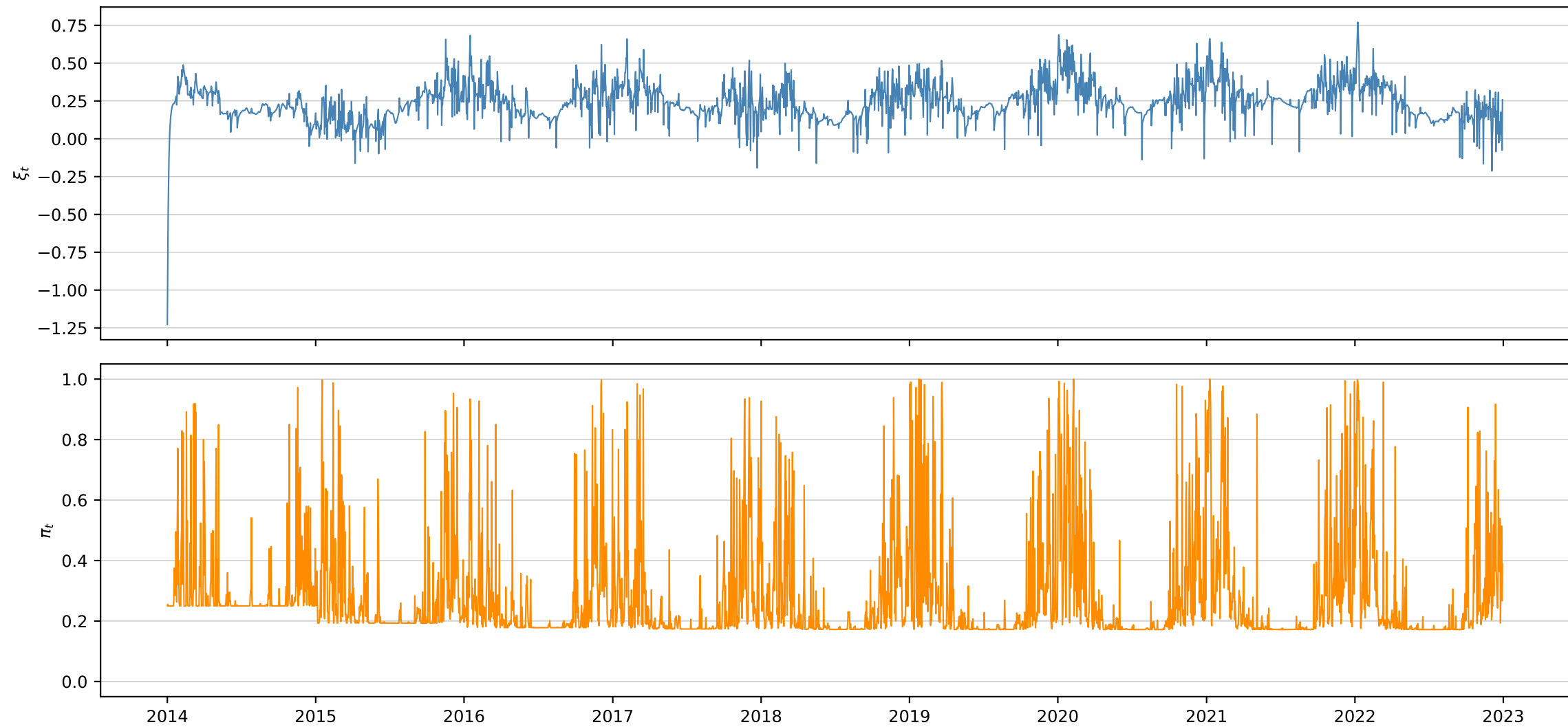
ACF of QR



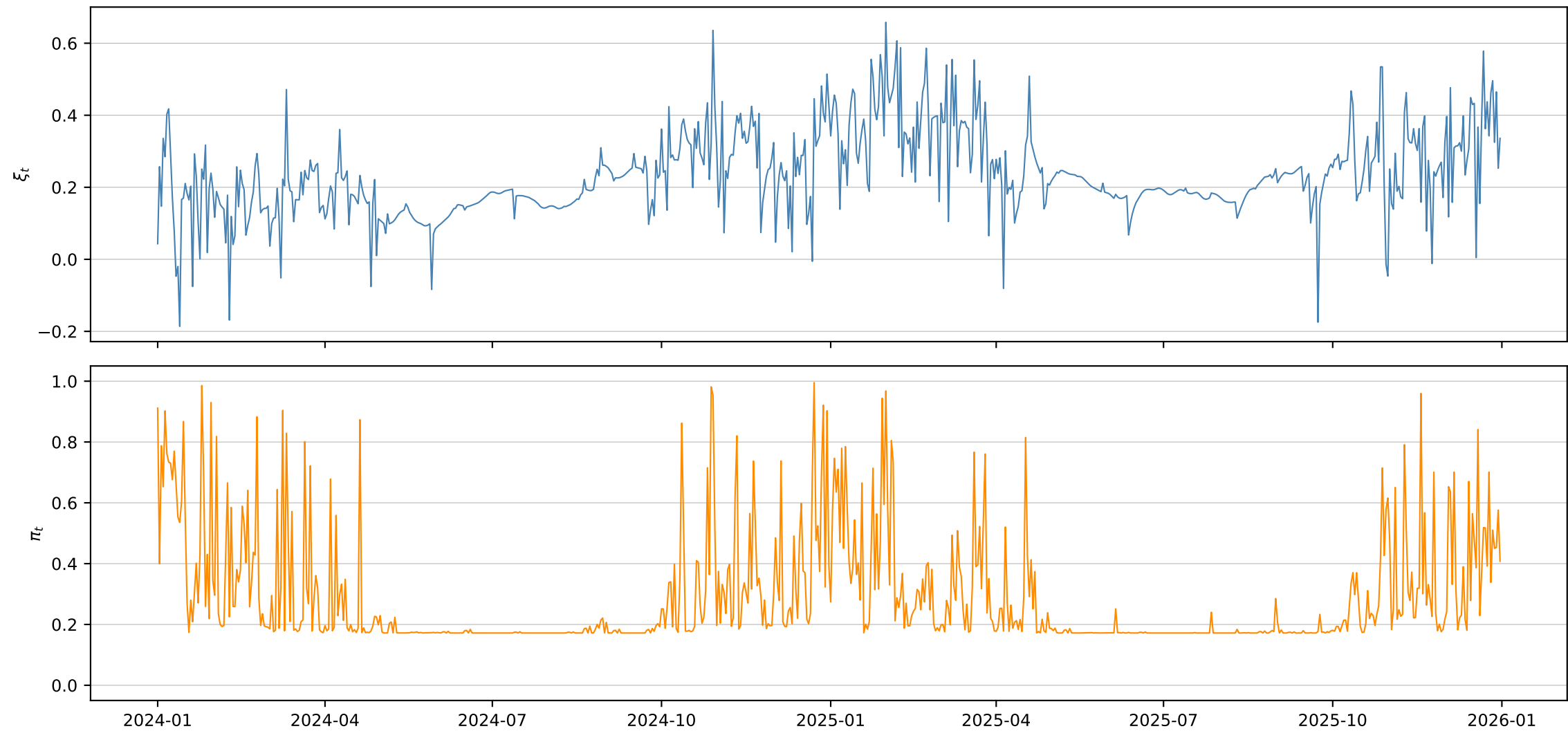
QR time series



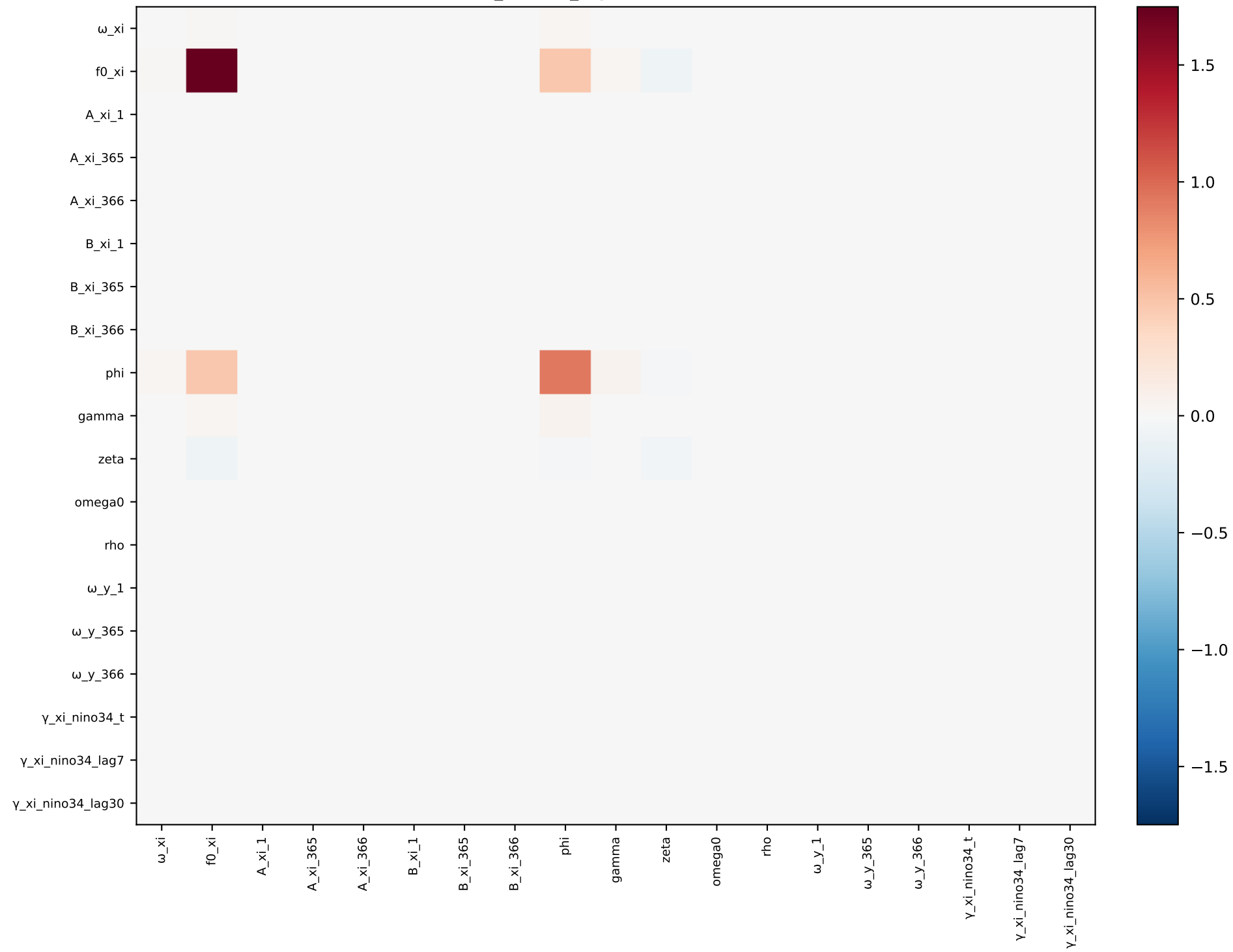
xi_nino34_lags — filtered states (IS)



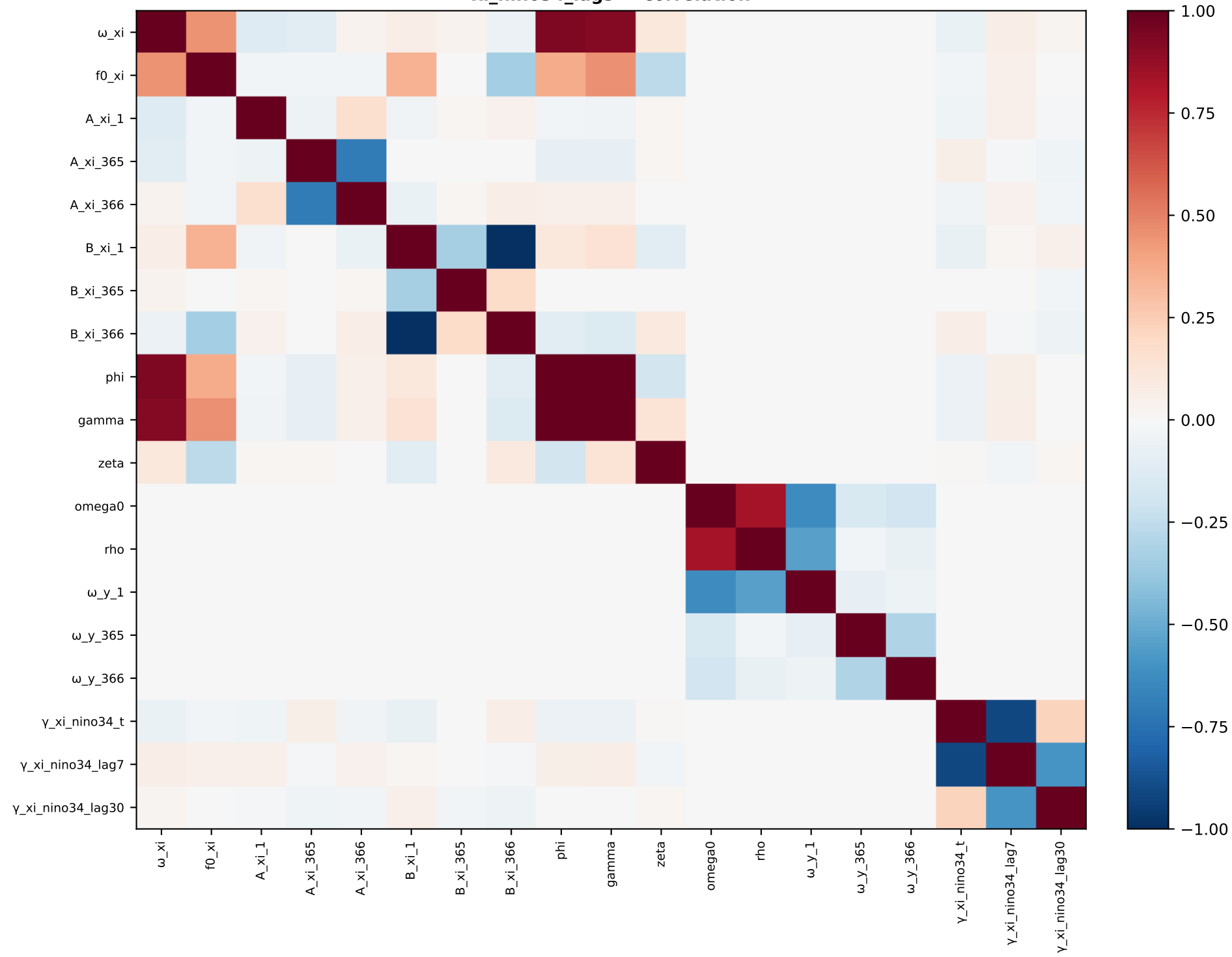
xi_nino34_lags — filtered states (OOS)



xi_nino34_lags — covariance

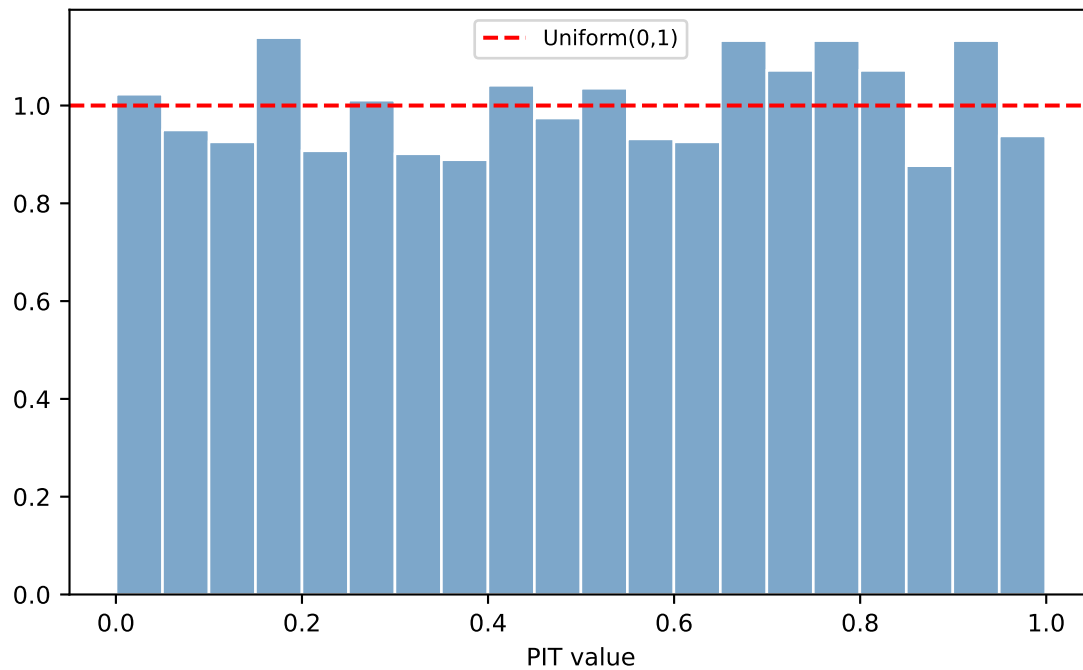


xi_nino34_lags — correlation

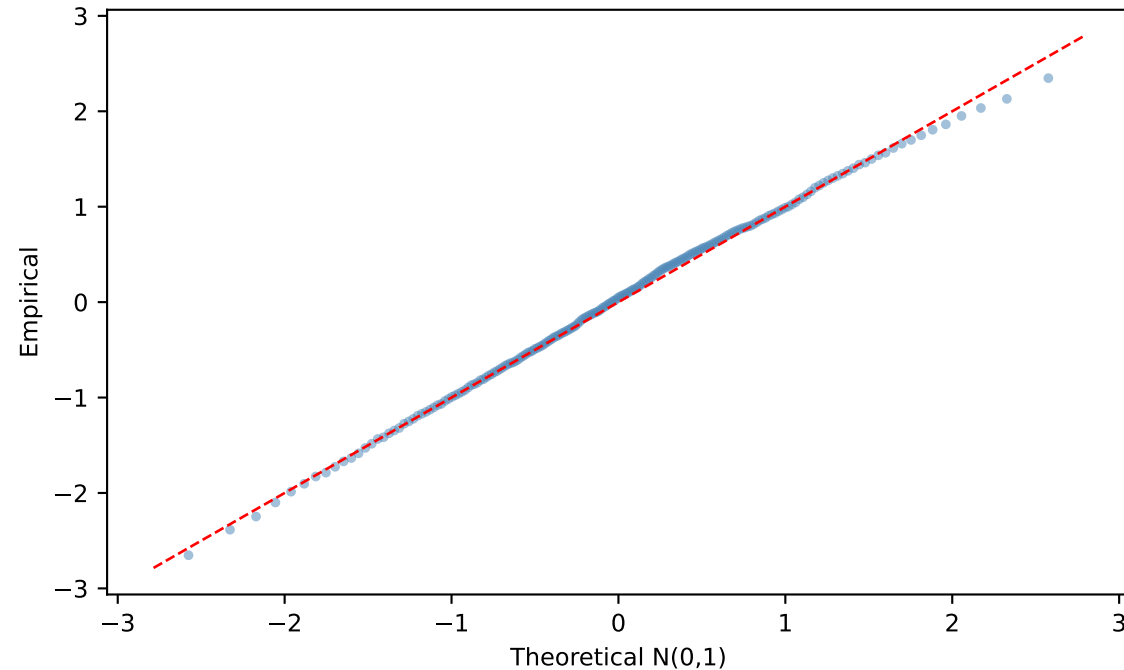


xi_nino34_rolling — IS diagnostics

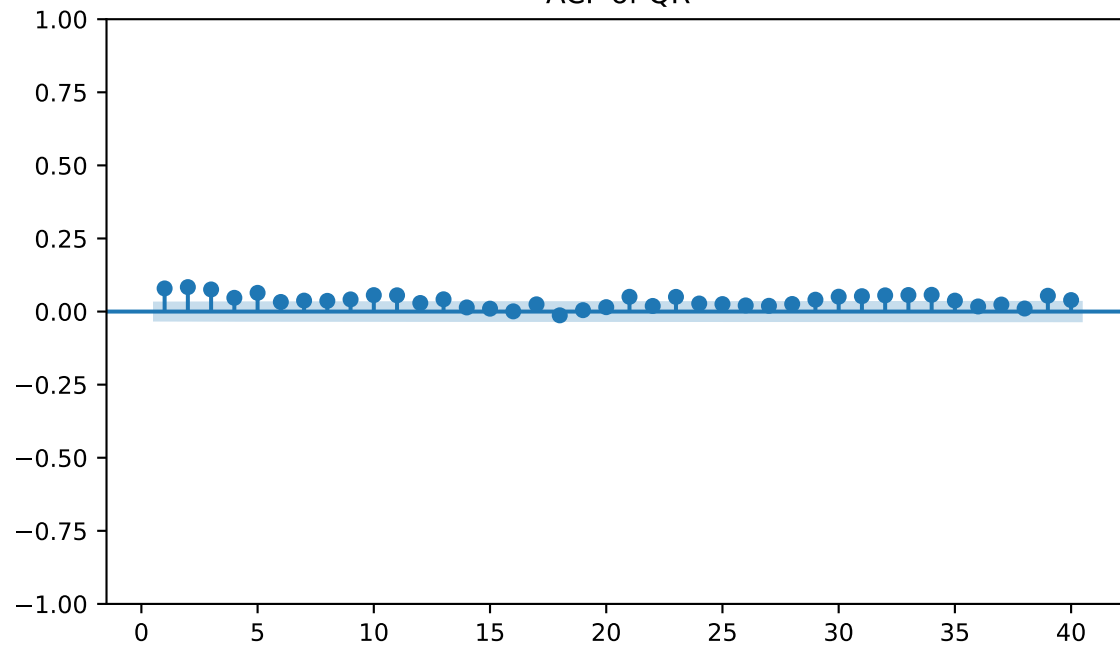
PIT histogram



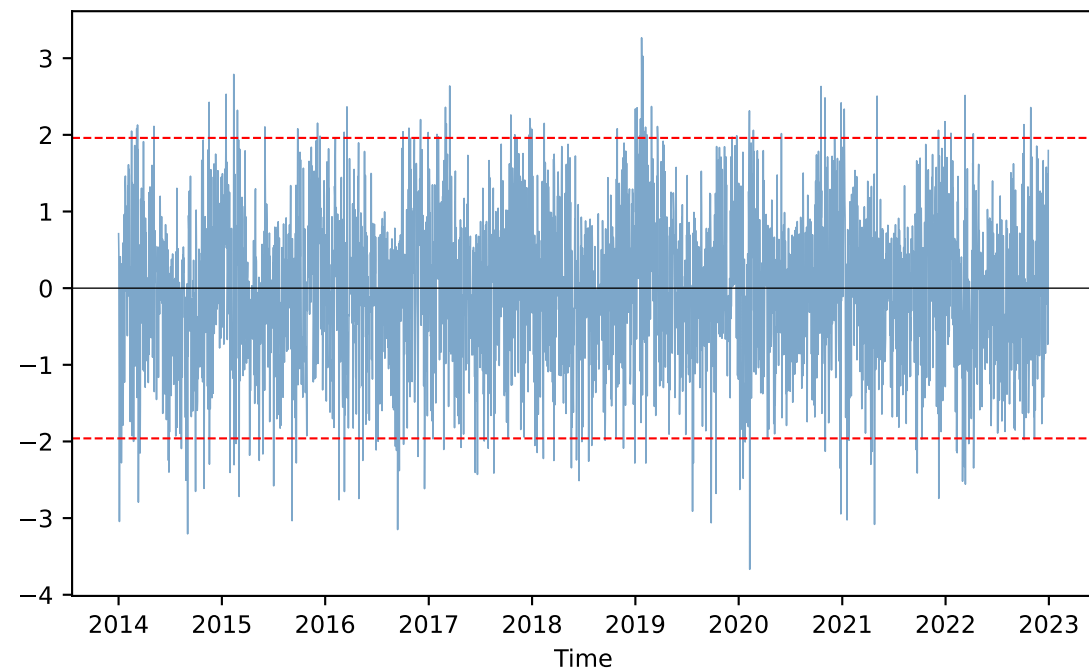
Normal QQ plot



ACF of QR

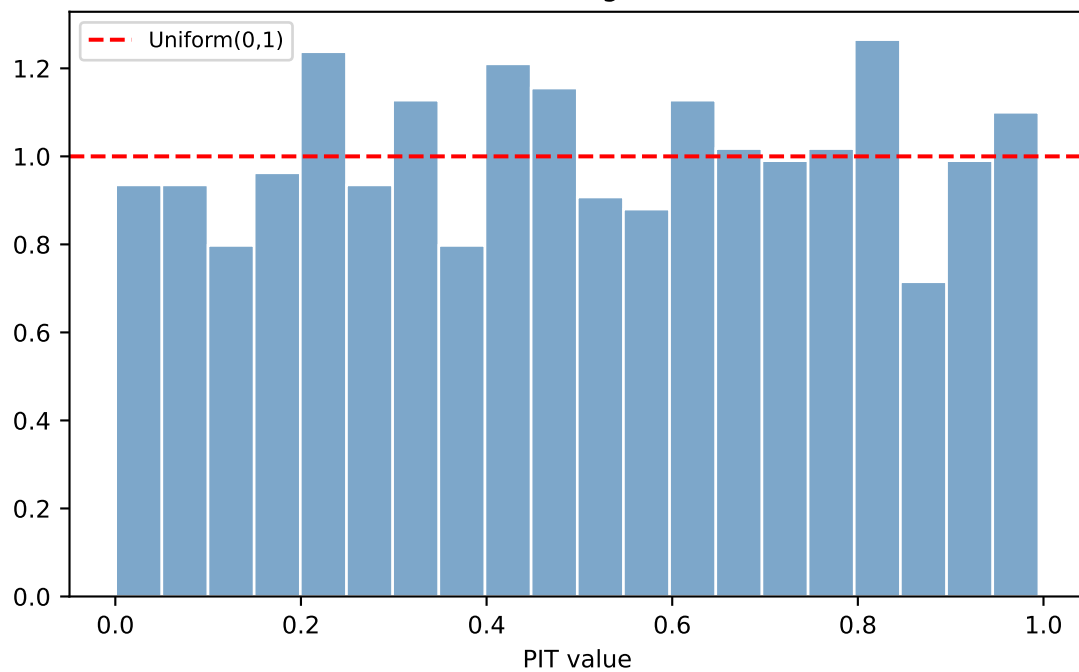


QR time series

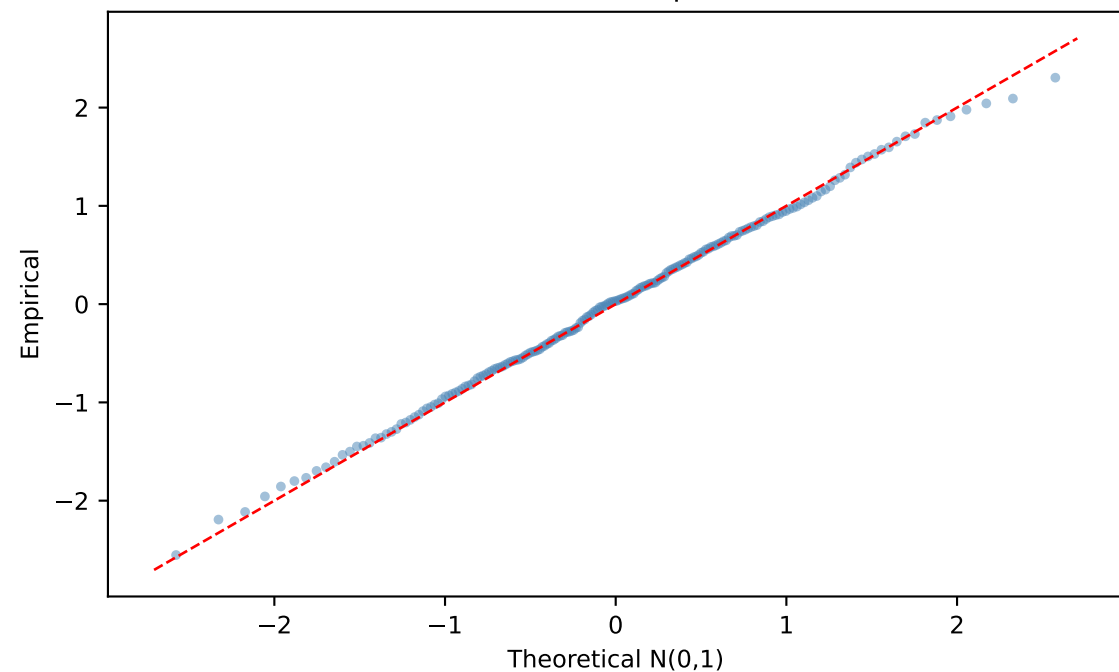


xi_nino34_rolling — OOS diagnostics

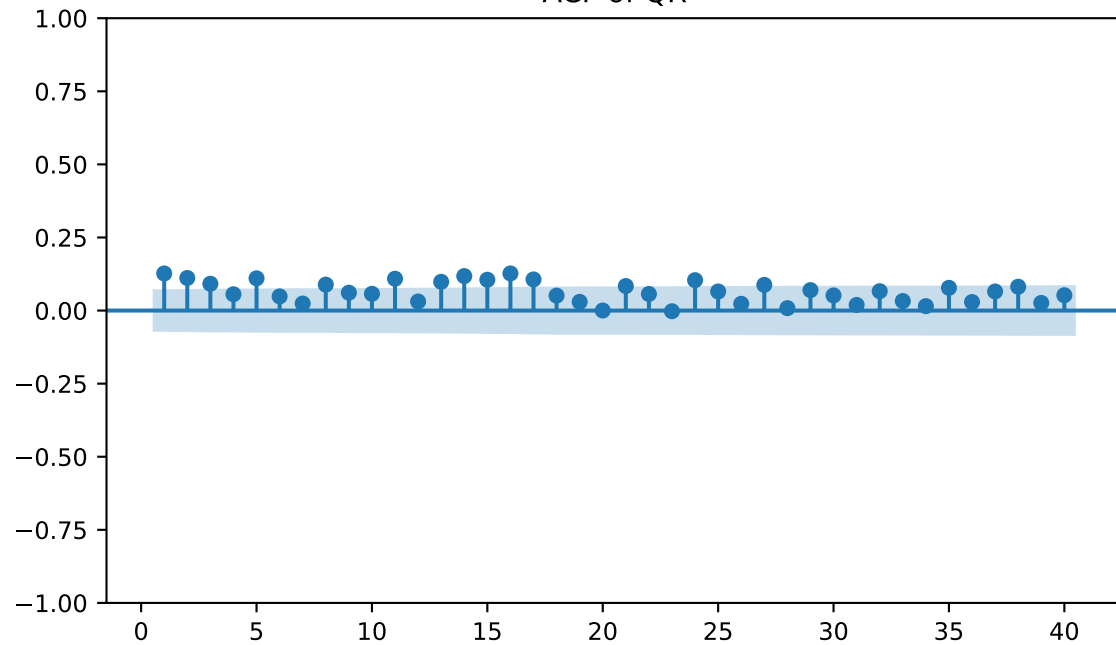
PIT histogram



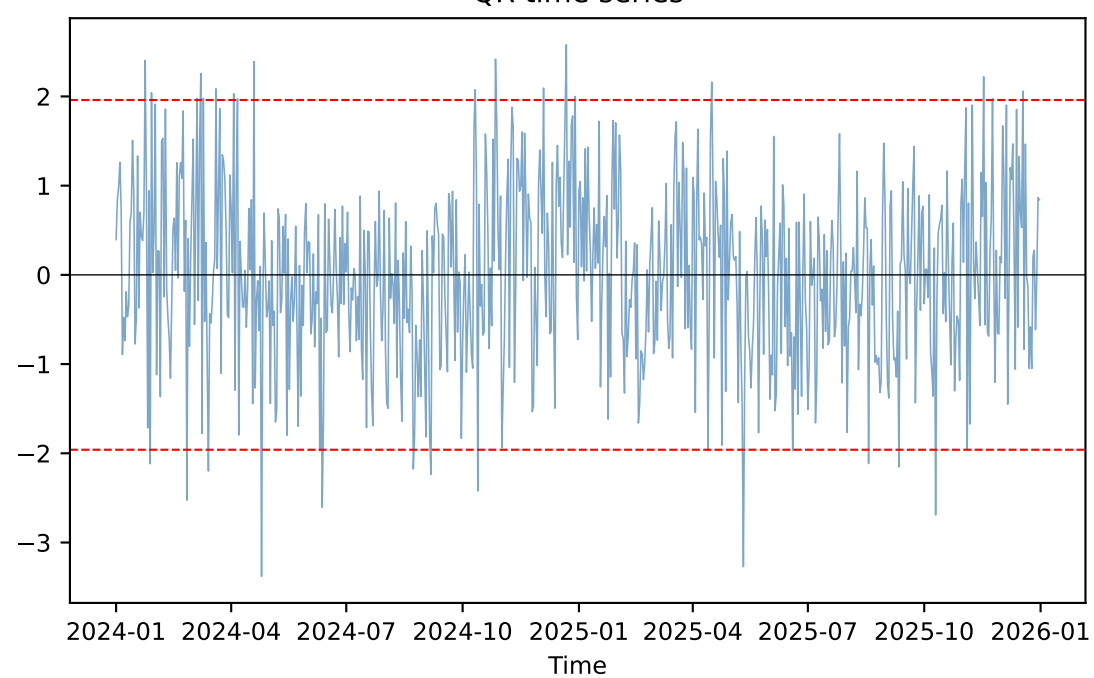
Normal QQ plot



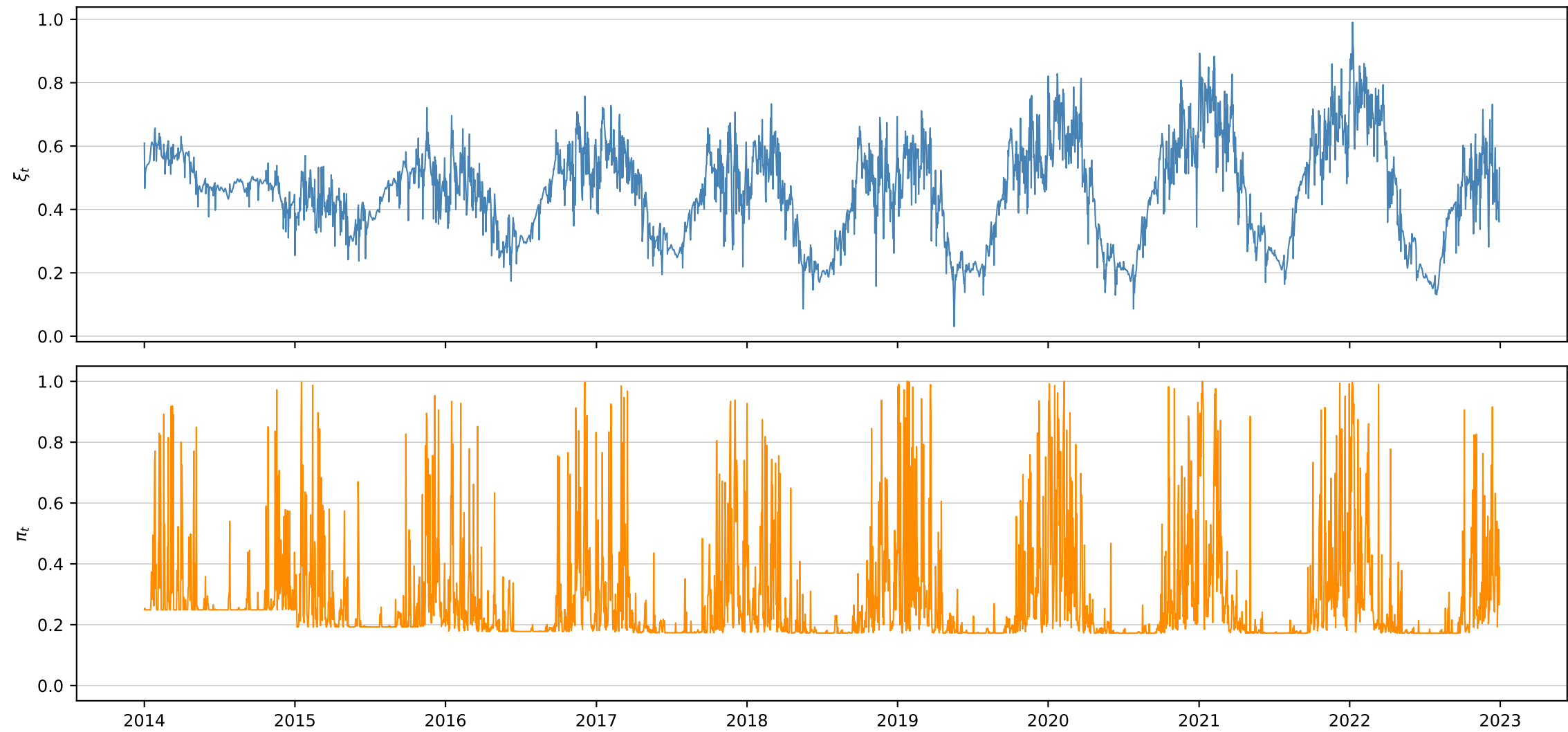
ACF of QR



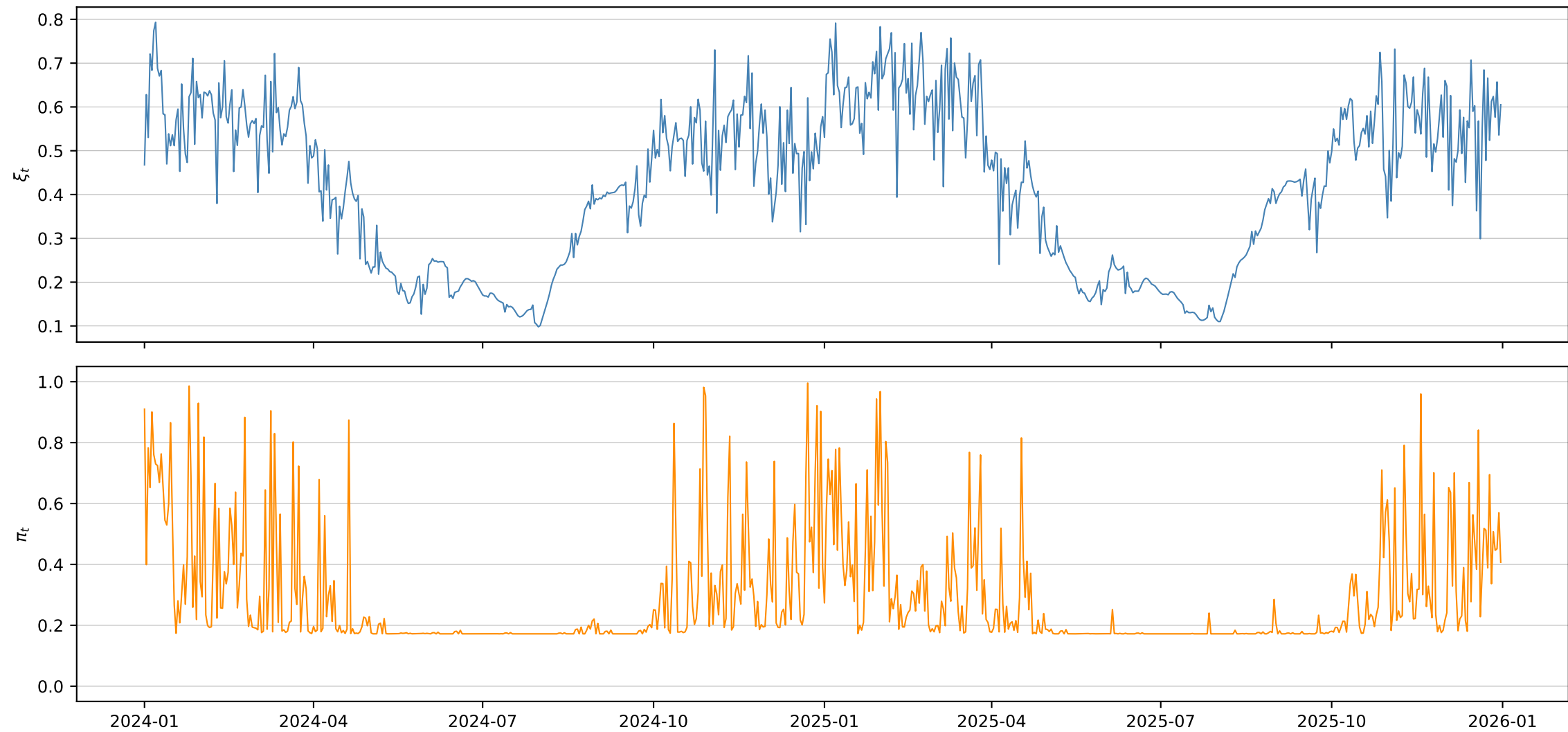
QR time series



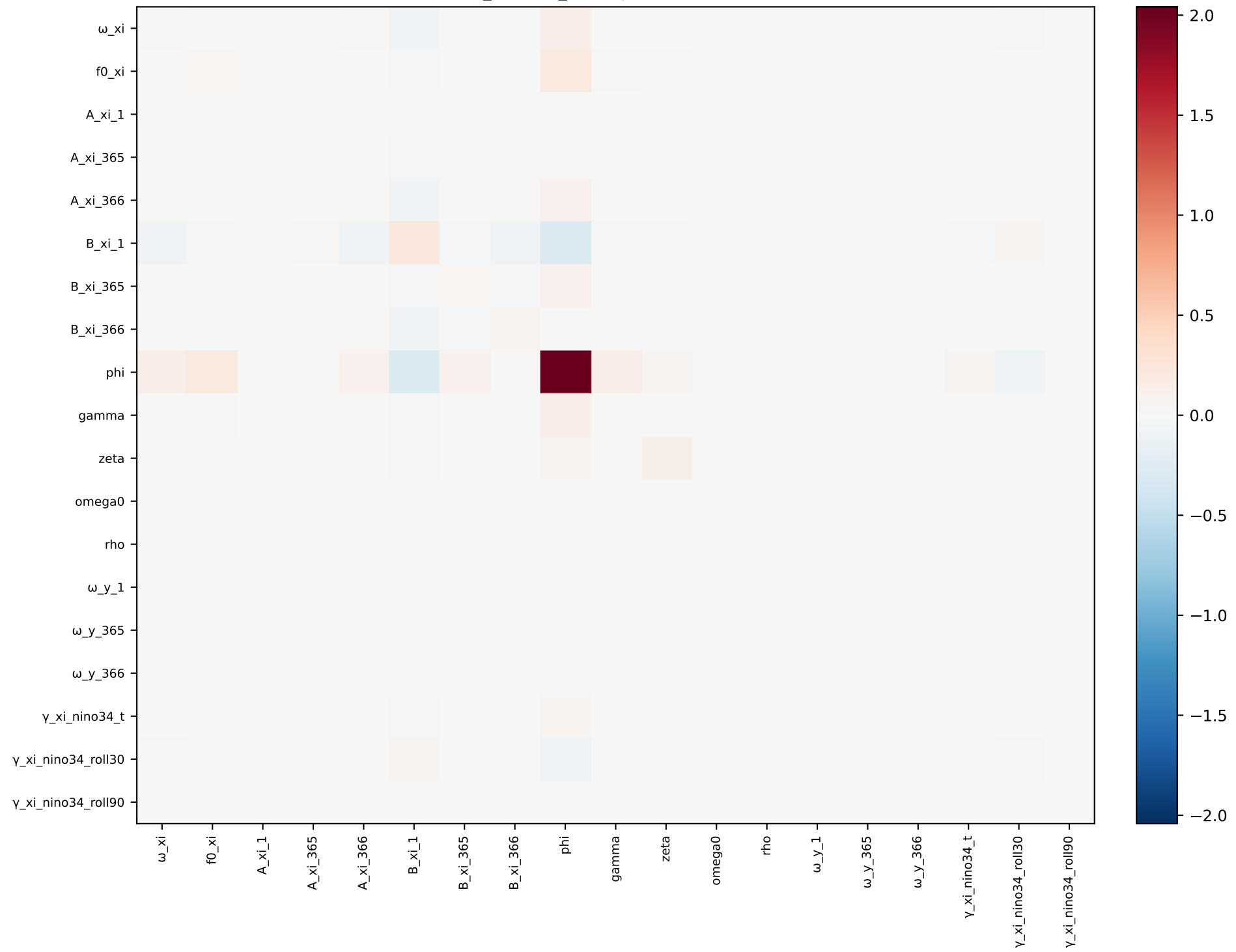
xi_nino34_rolling — filtered states (IS)



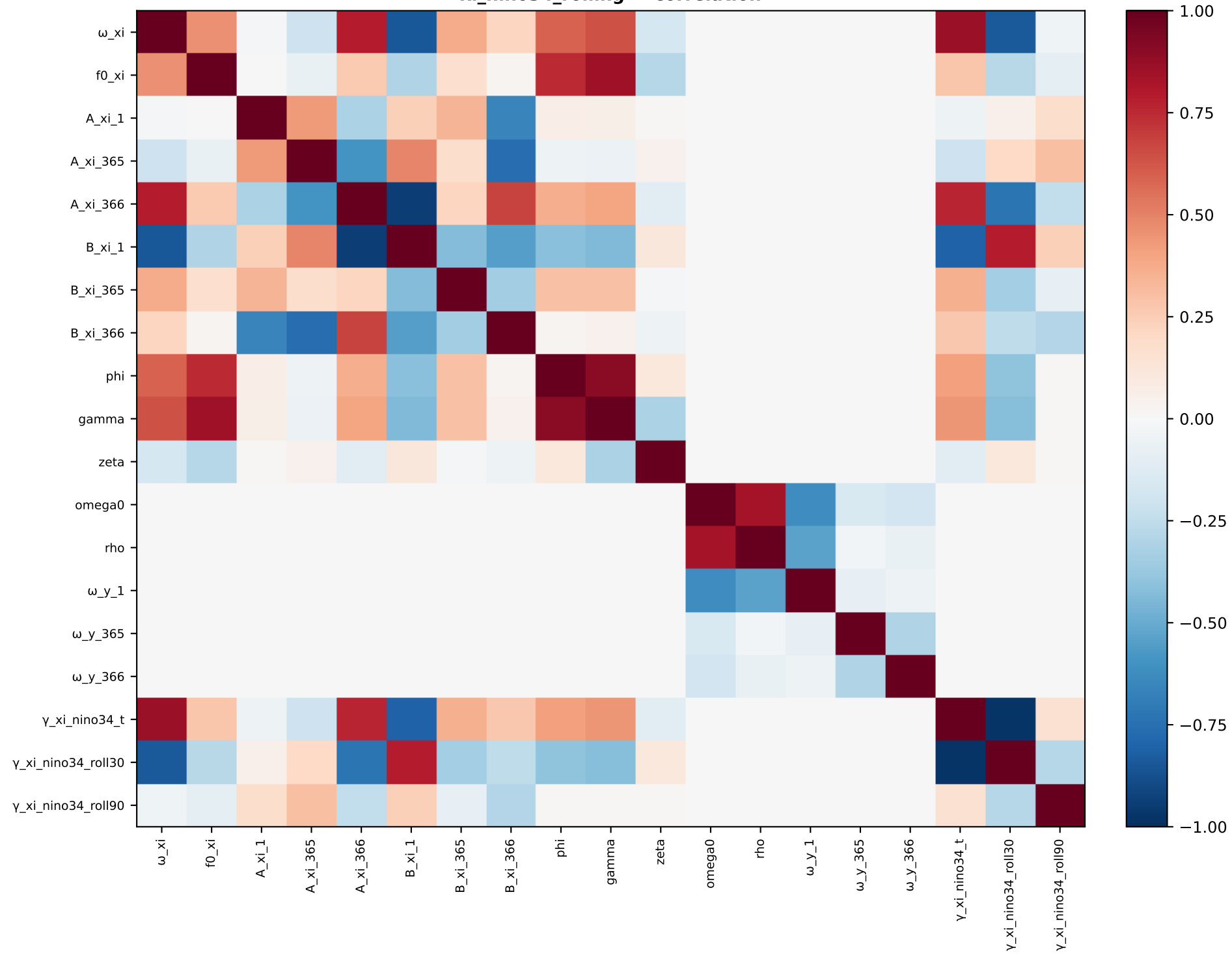
xi_nino34_rolling — filtered states (OOS)



xi nino34 rolling — covariance

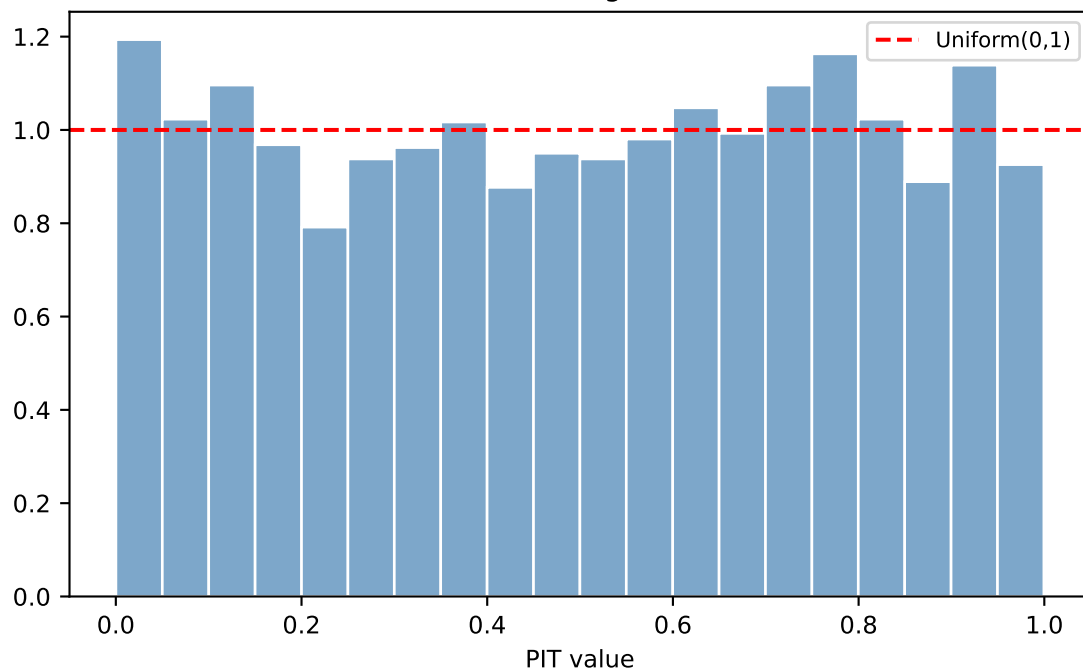


xi_nino34_rolling — correlation

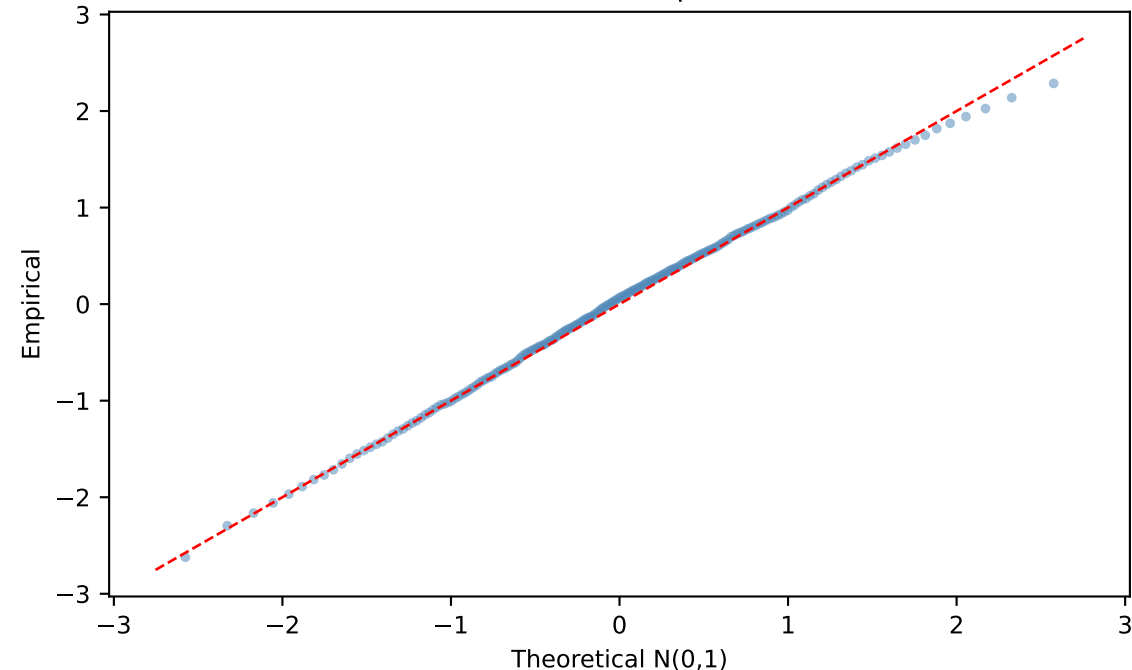


xi_interaction_dewpoint_temperature_t — IS diagnostics

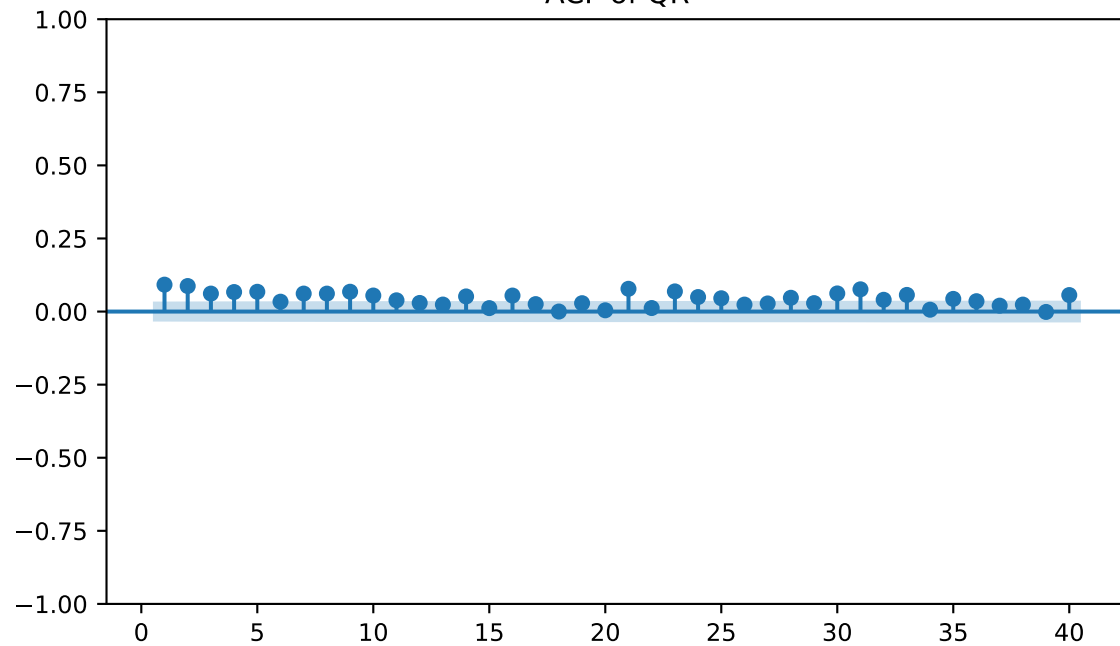
PIT histogram



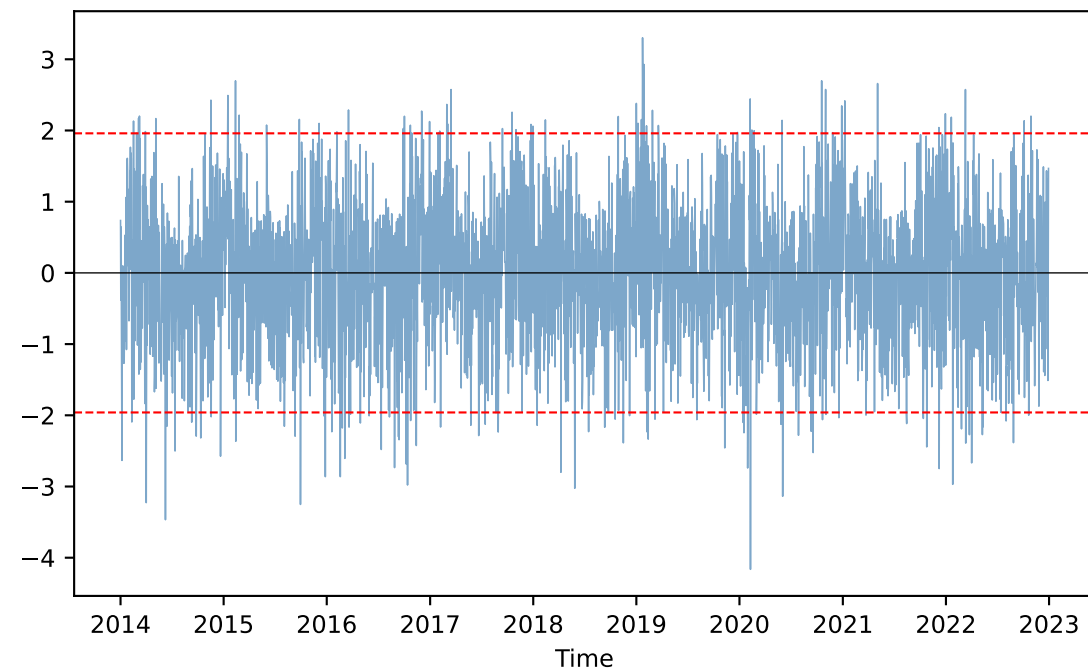
Normal QQ plot



ACF of QR

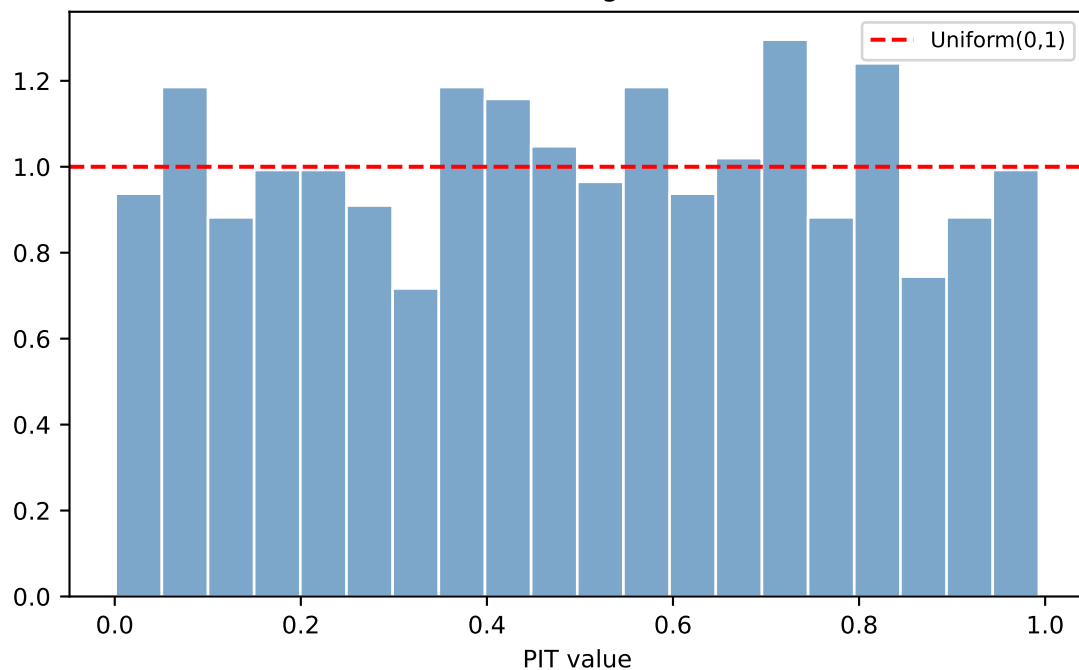


QR time series

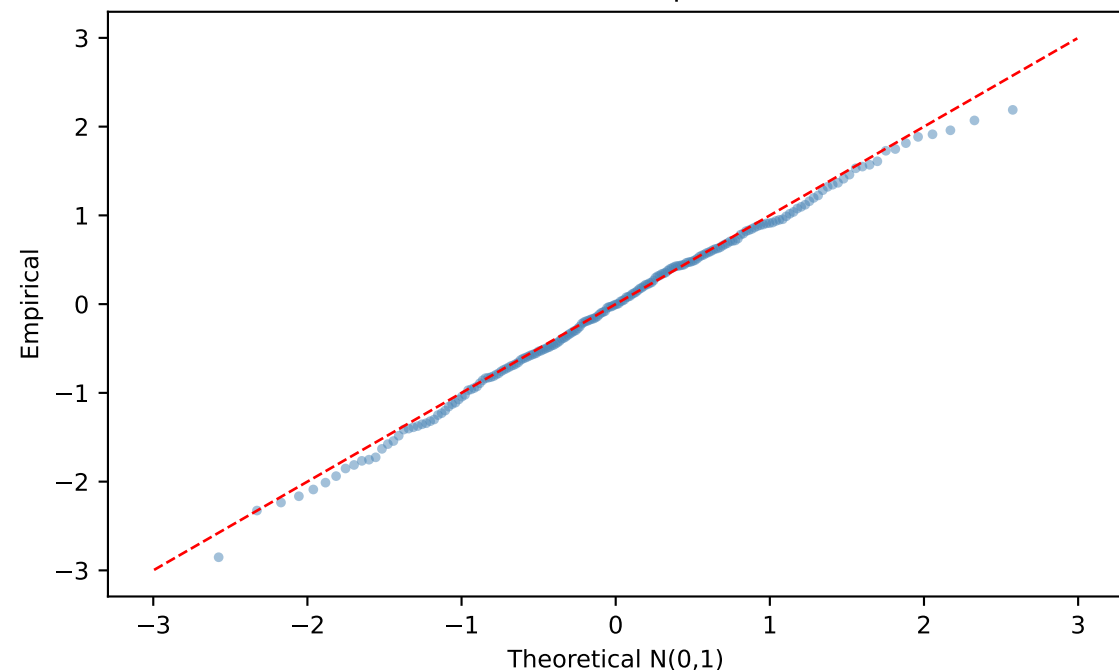


xi_interaction_dewpoint_temperature_t — OOS diagnostics

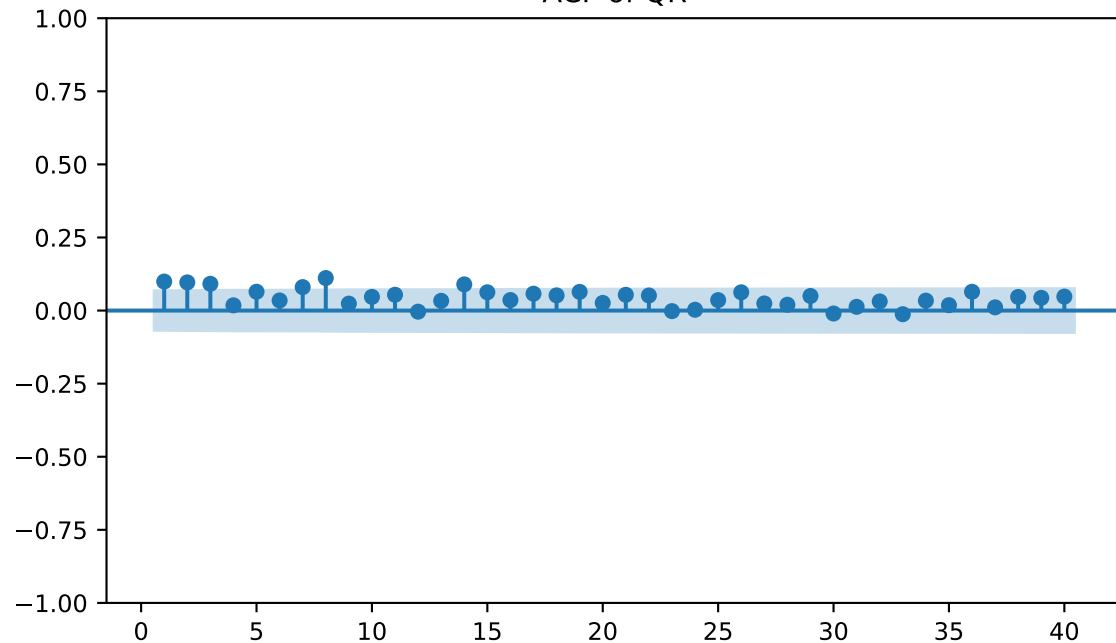
PIT histogram



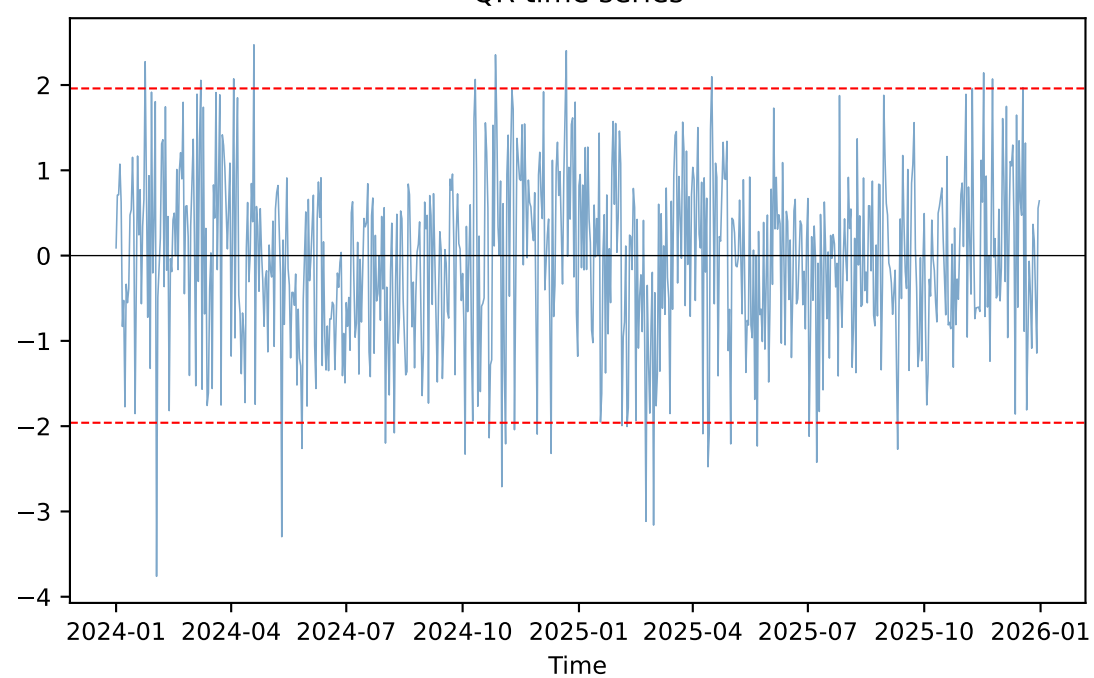
Normal QQ plot



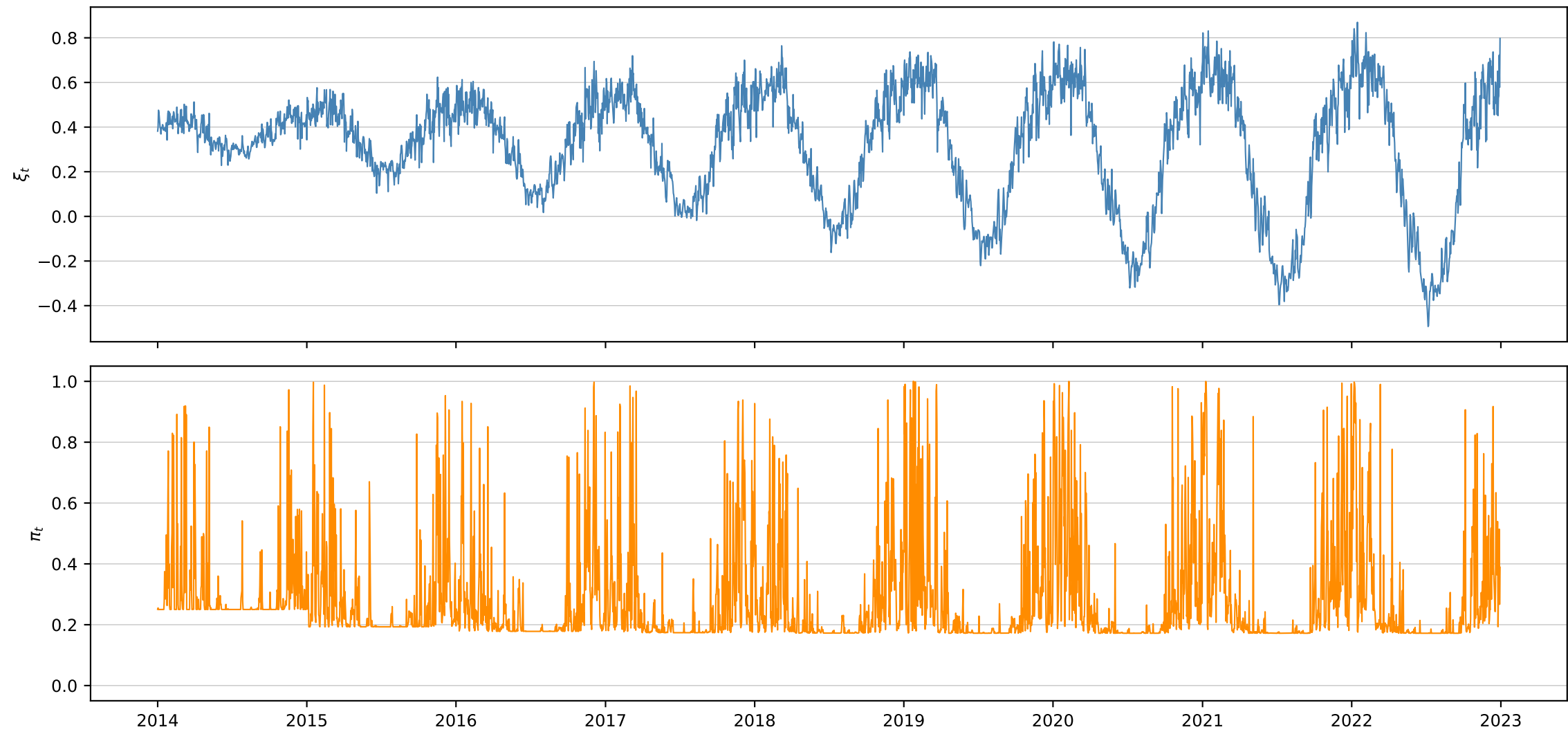
ACF of QR



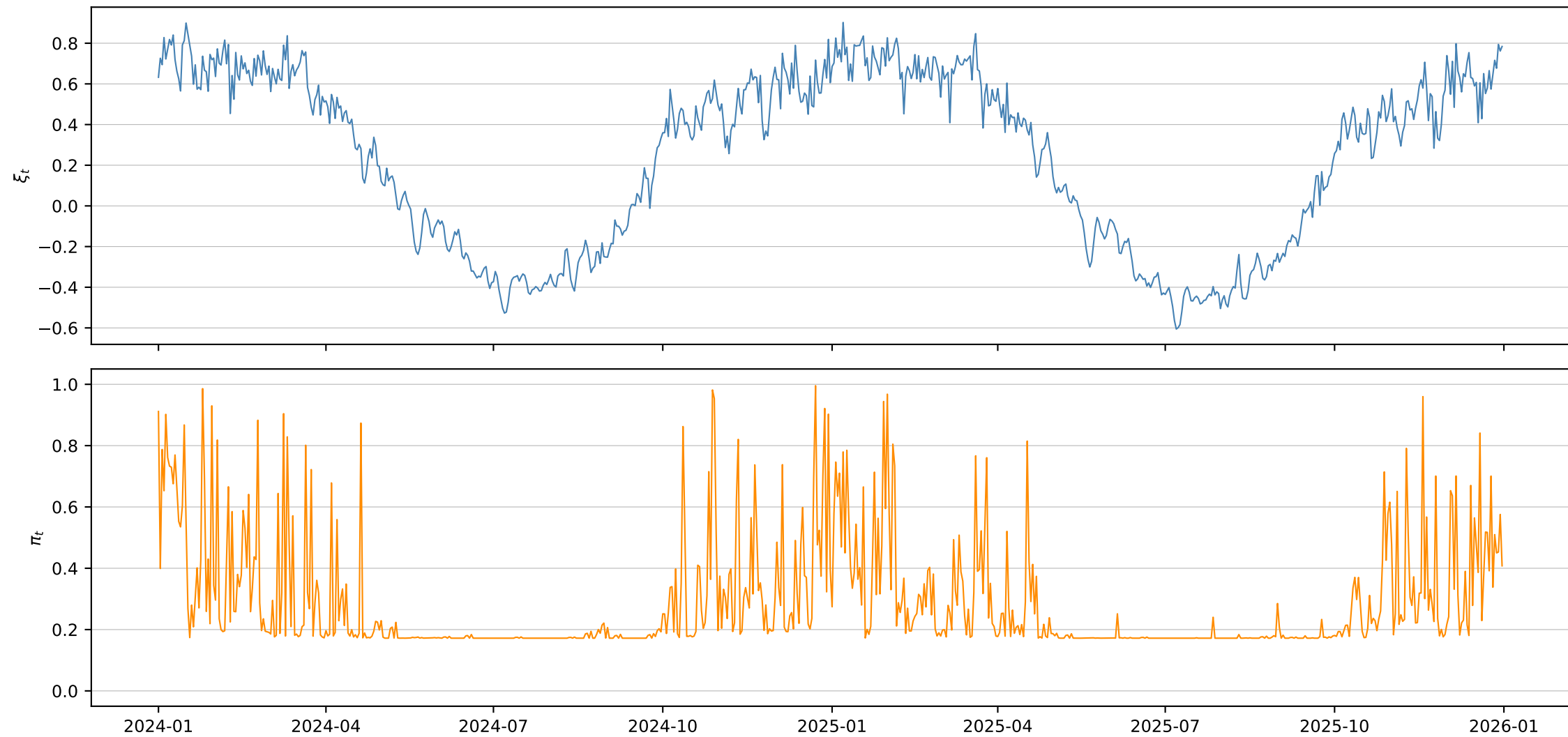
QR time series



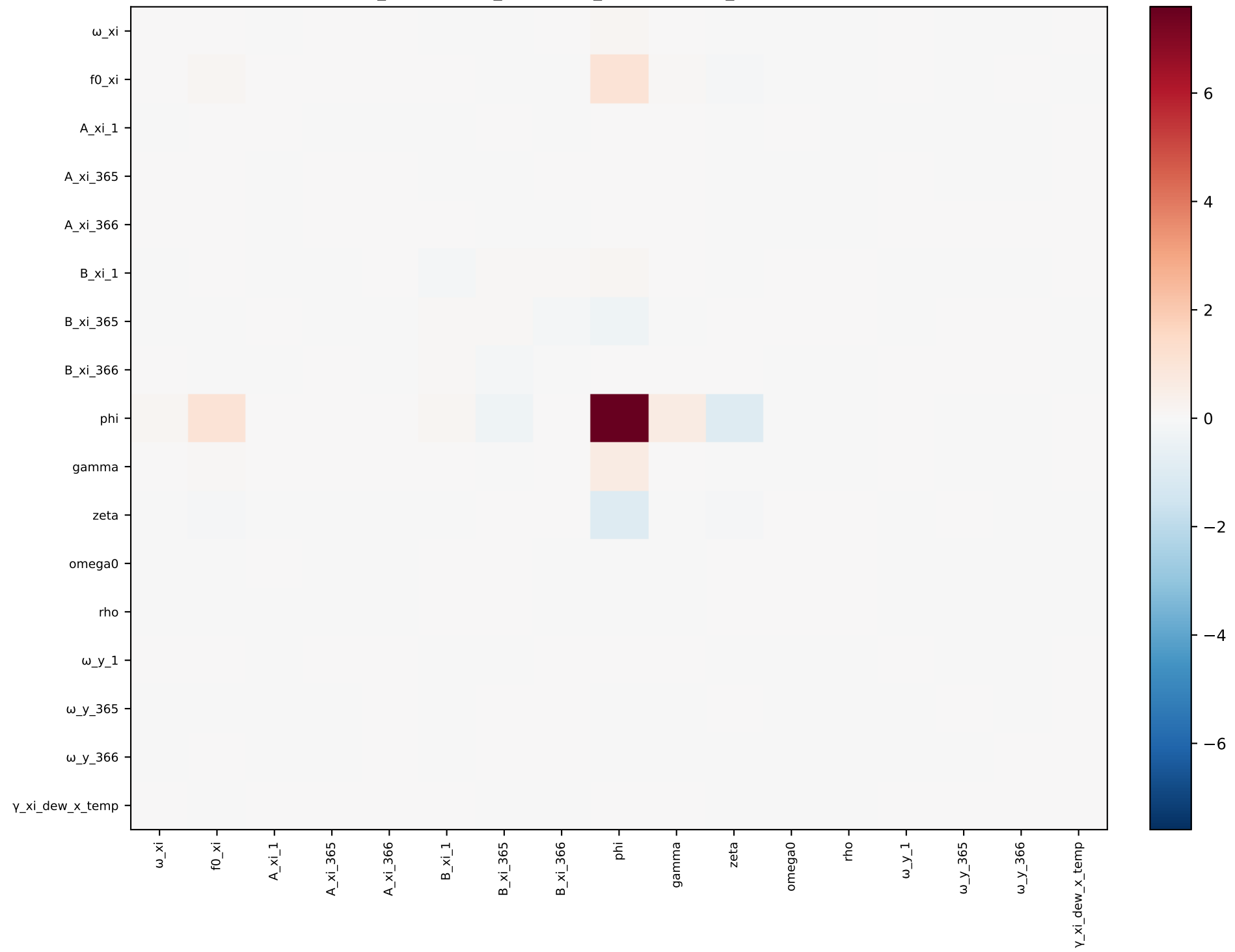
xi_interaction_dewpoint_temperature_t — filtered states (IS)



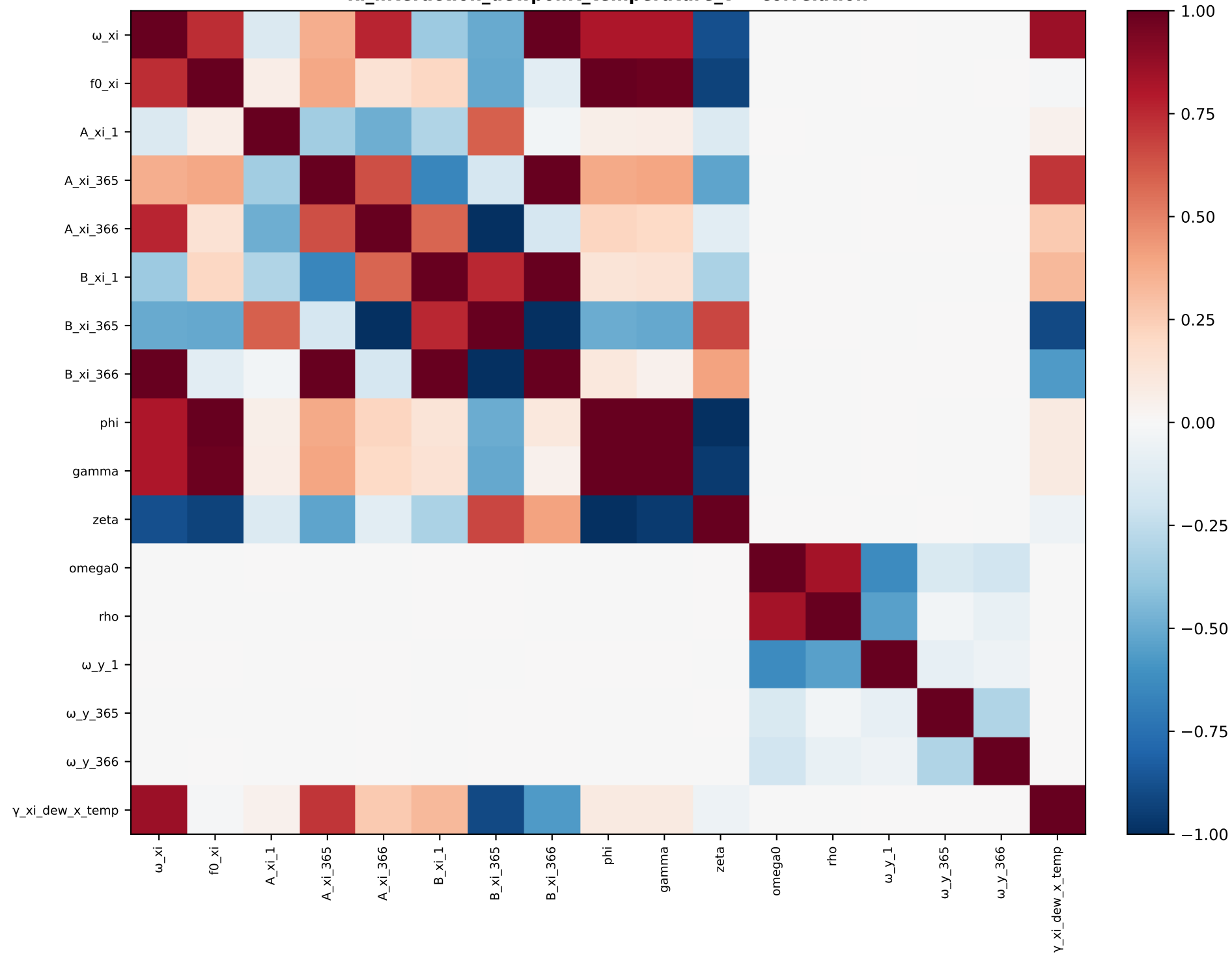
xi_interaction_dewpoint_temperature_t — filtered states (OOS)



xi_interaction_dewpoint_temperature_t — covariance

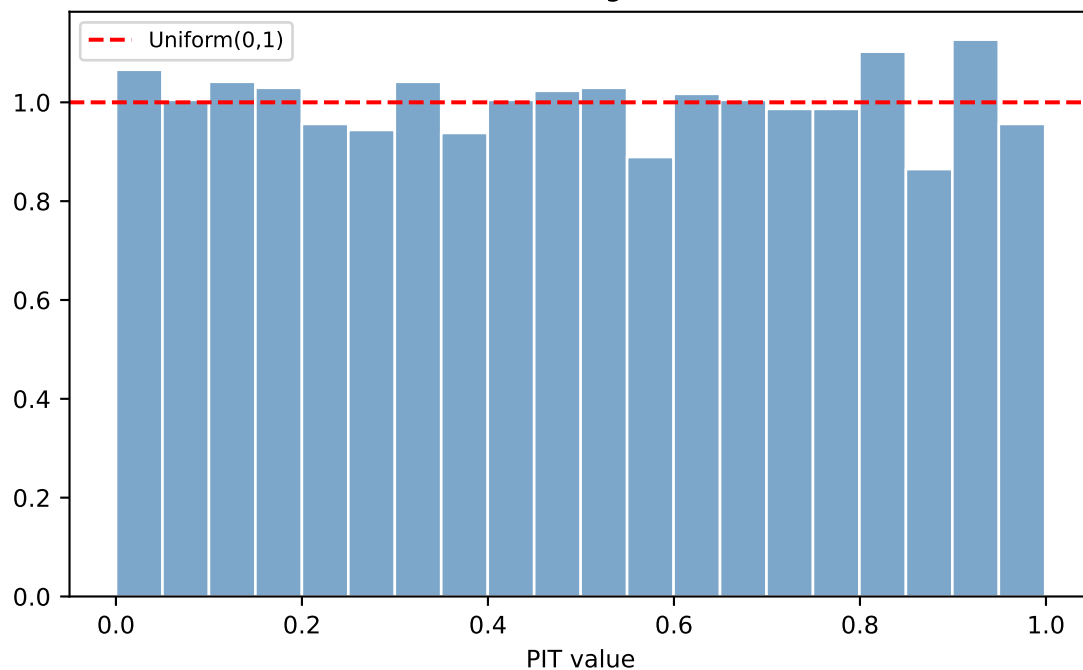


xi_interaction_dewpoint_temperature_t — correlation

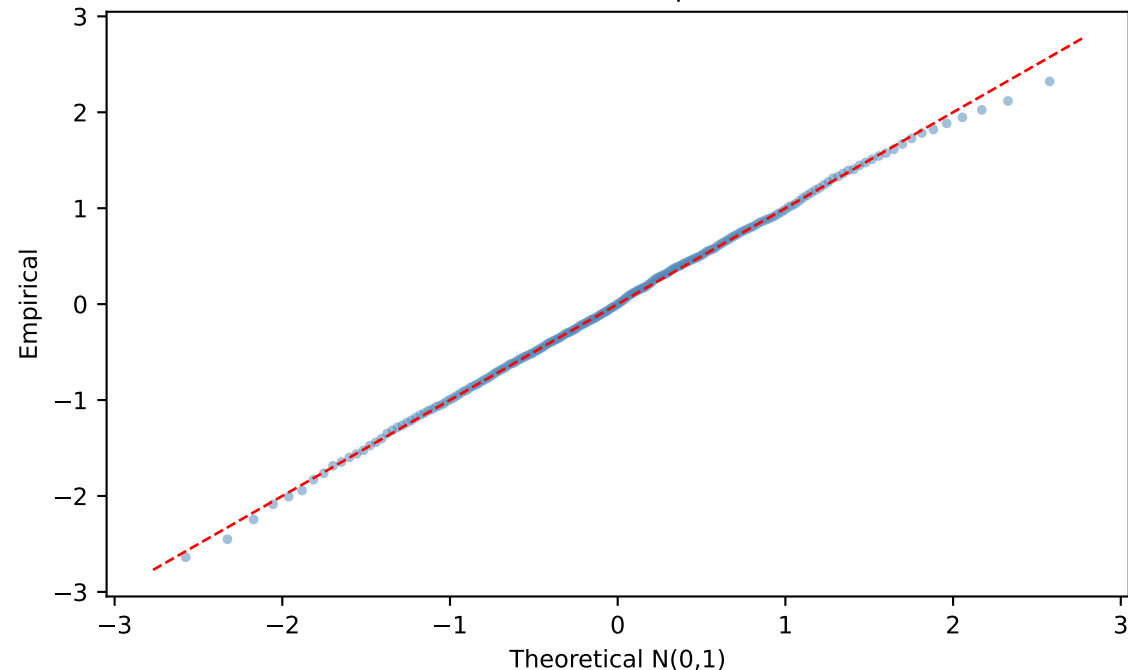


xi_interaction_dewpoint_mjo — IS diagnostics

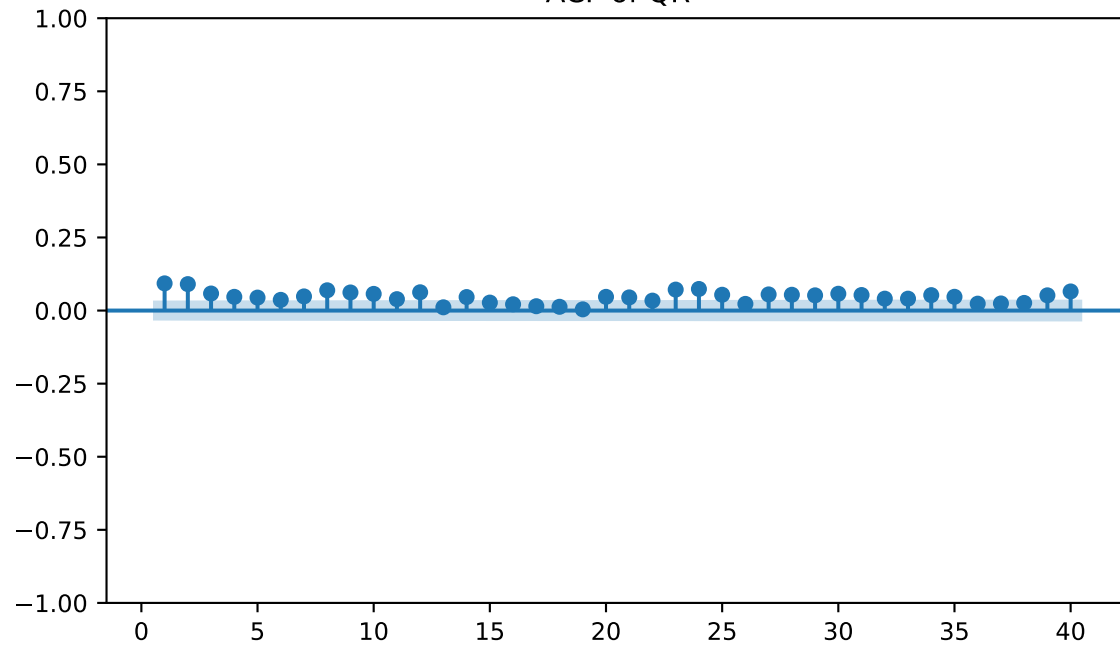
PIT histogram



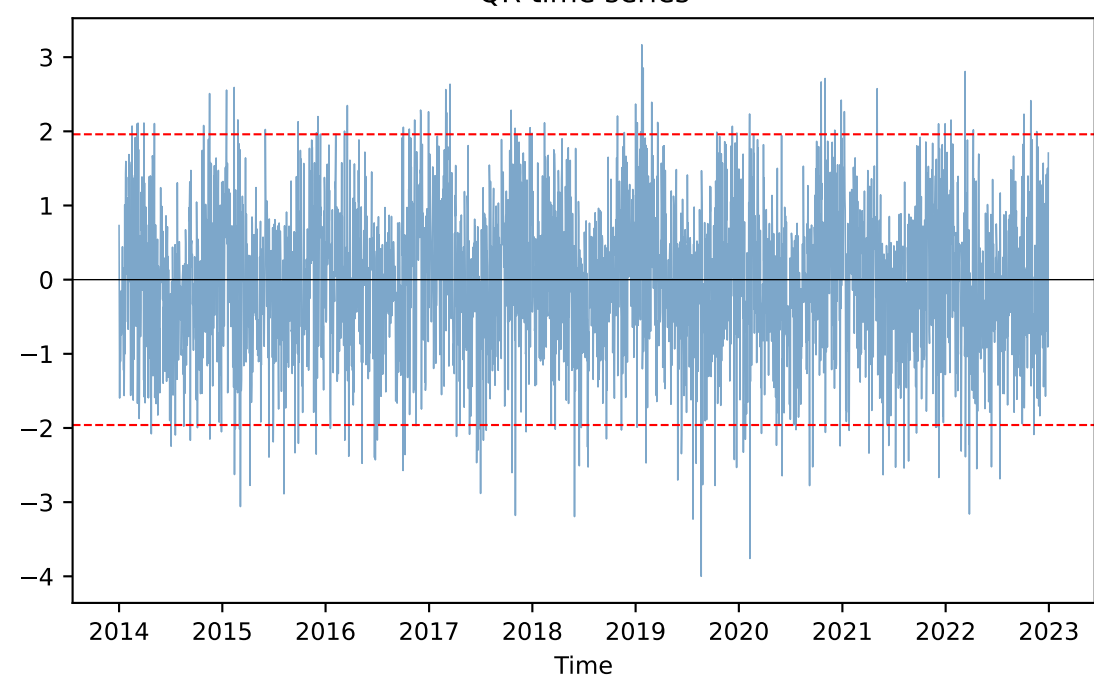
Normal QQ plot



ACF of QR

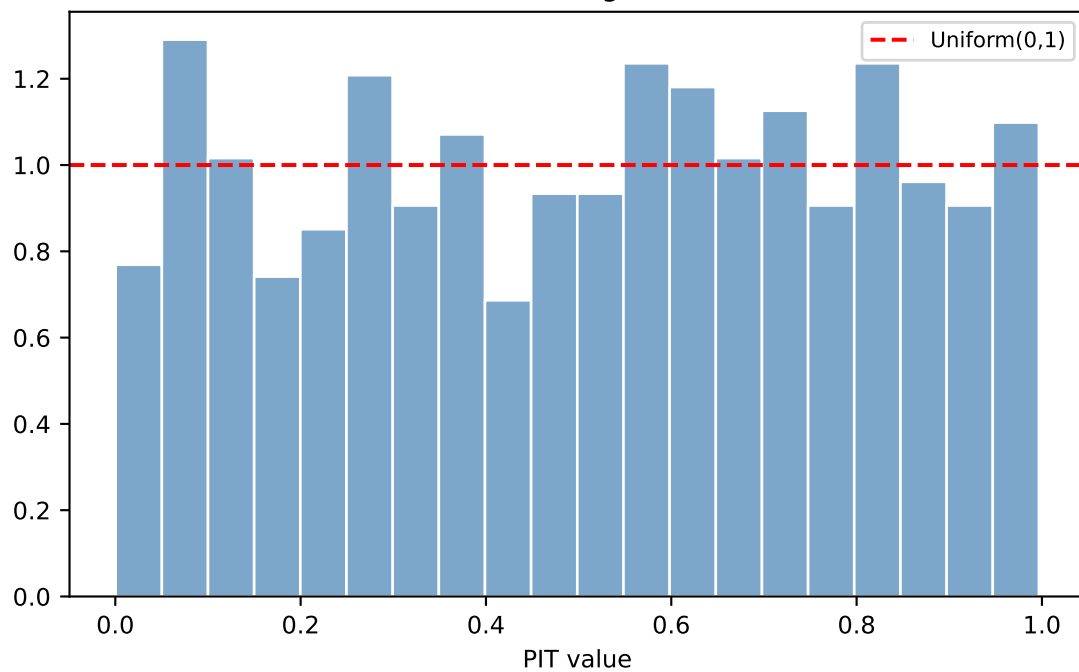


QR time series

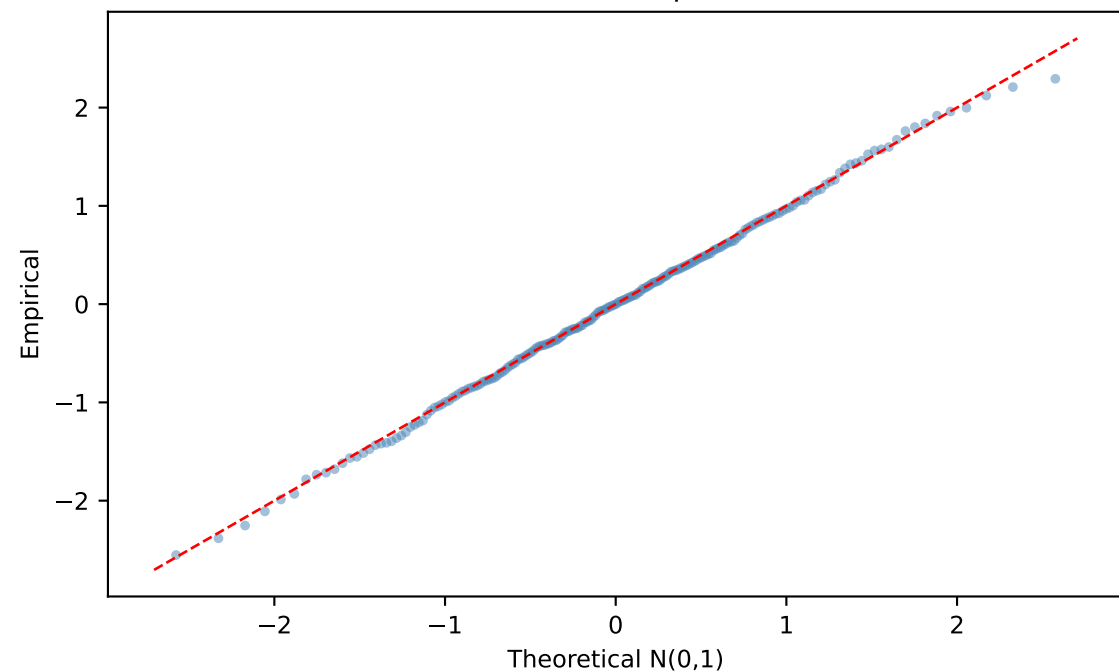


xi_interaction_dewpoint_mjo — OOS diagnostics

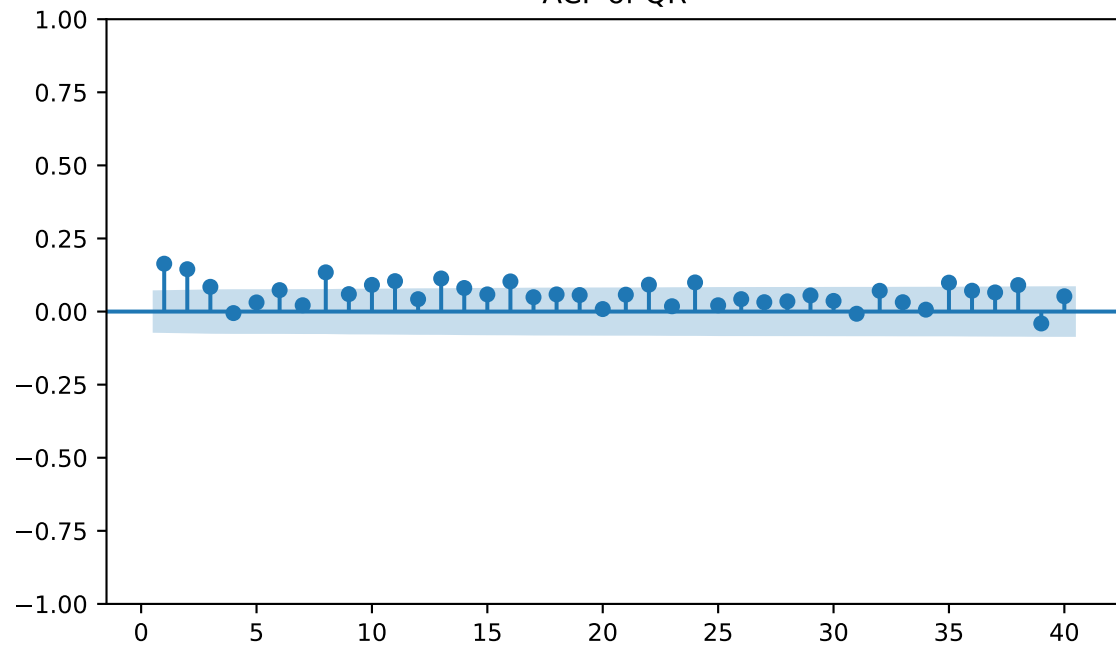
PIT histogram



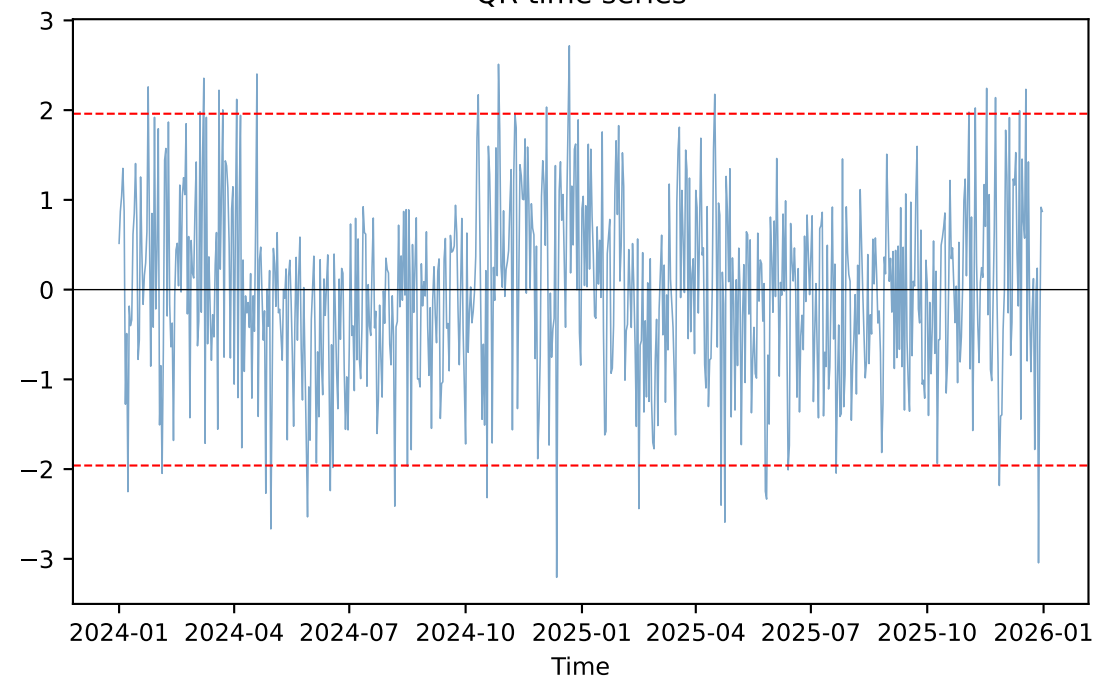
Normal QQ plot



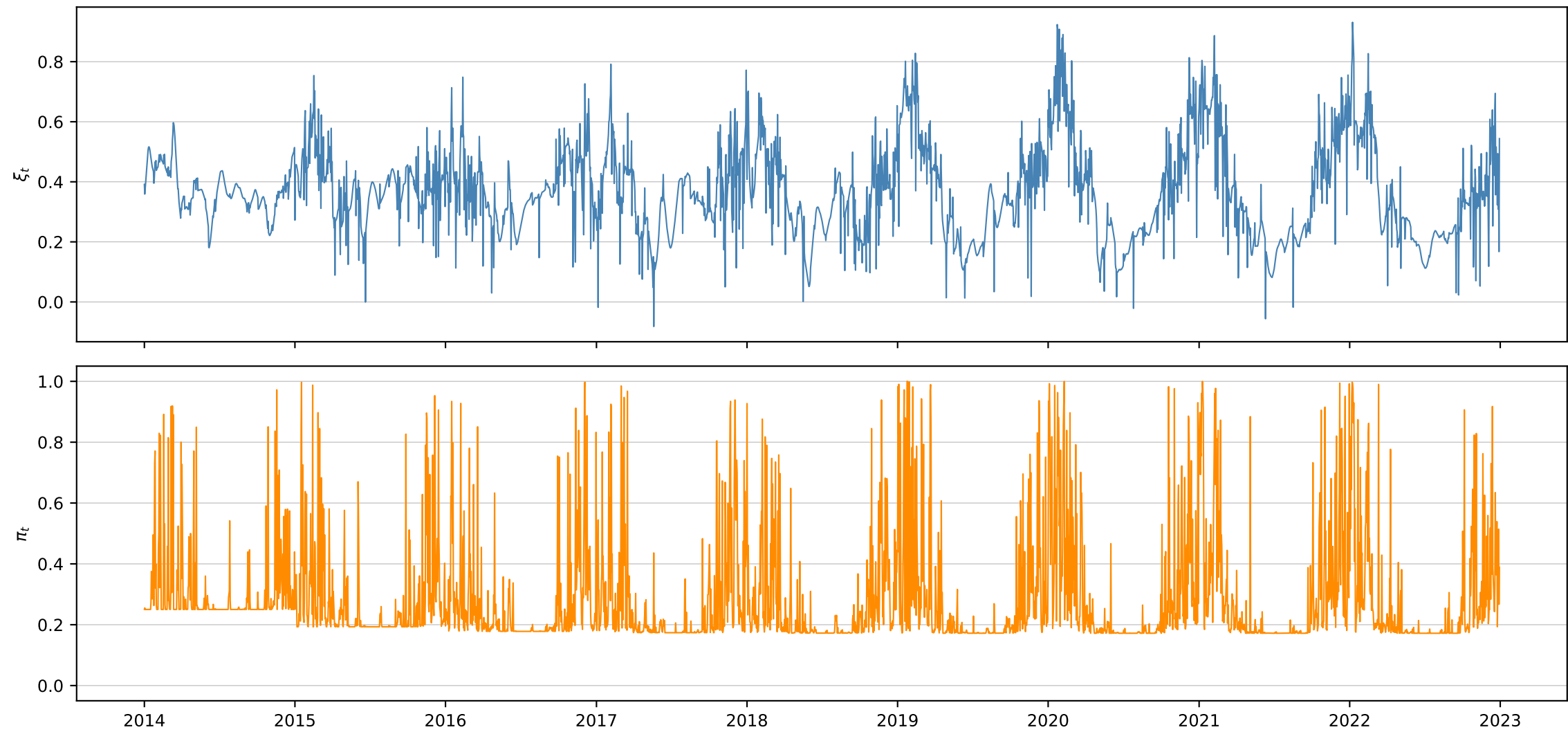
ACF of QR



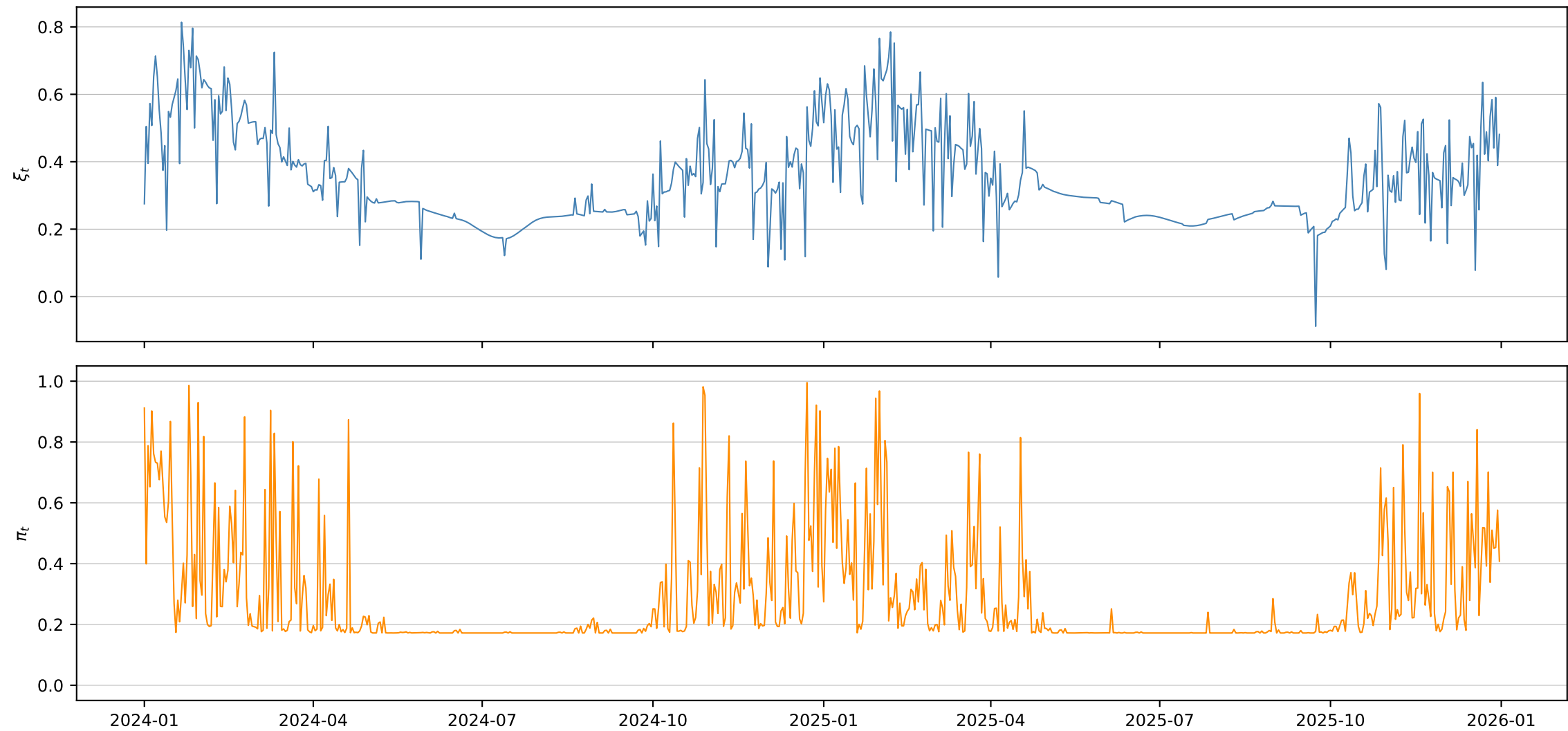
QR time series



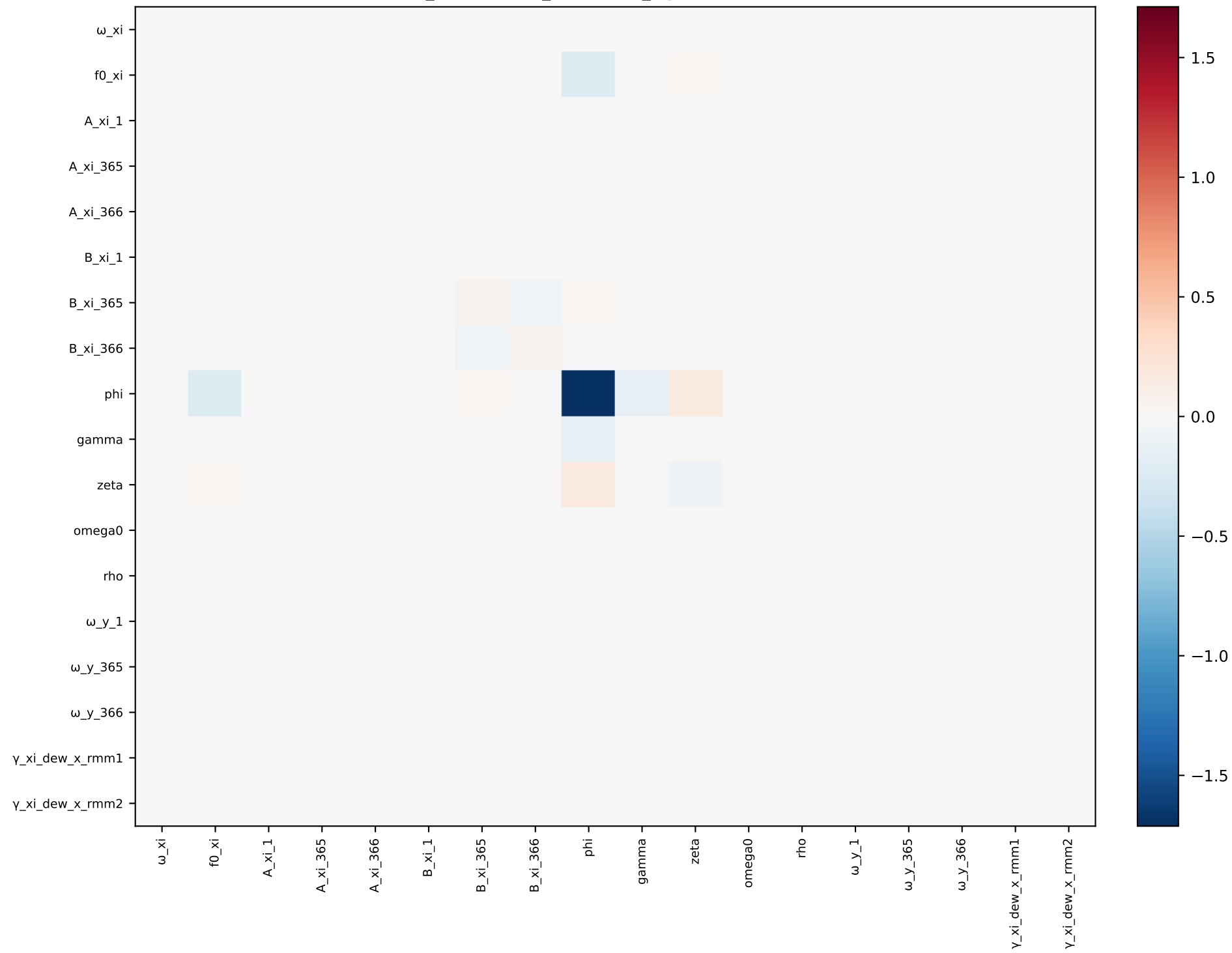
xi_interaction_dewpoint_mjo — filtered states (IS)



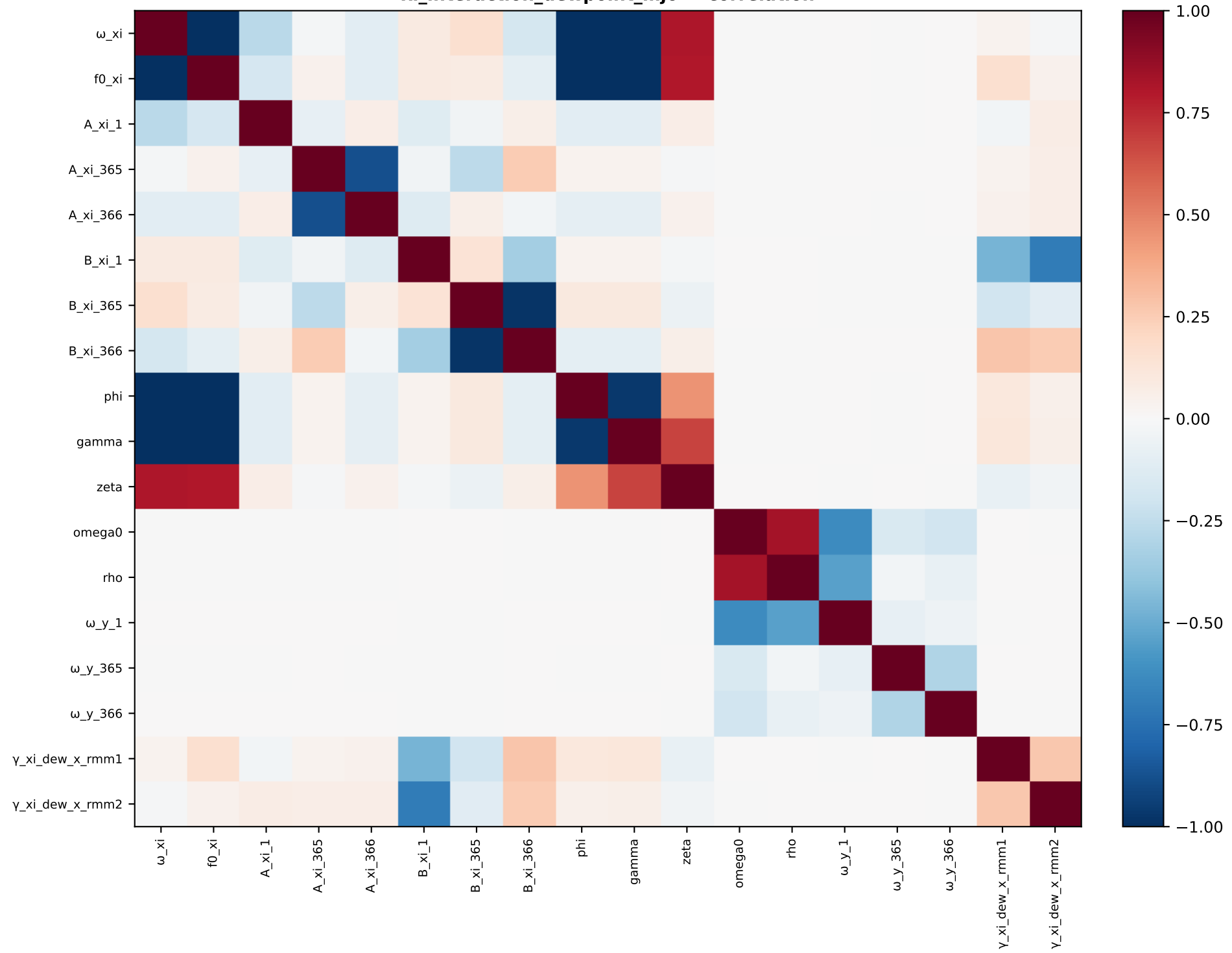
xi_interaction_dewpoint_mjo — filtered states (OOS)



xi_interaction_dewpoint_mjo — covariance

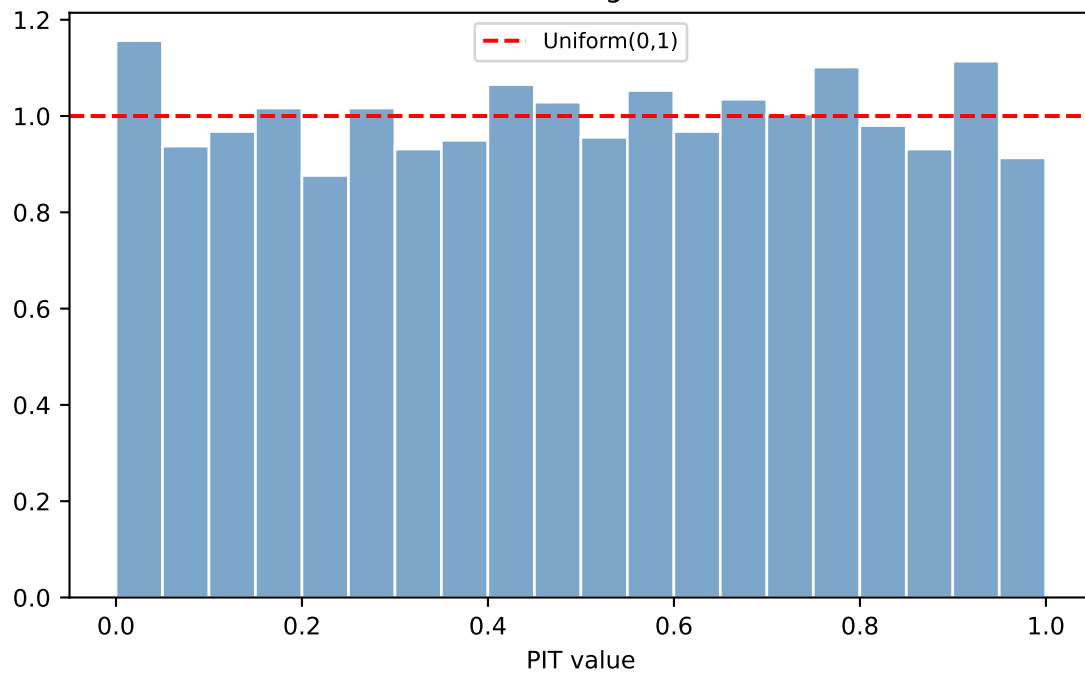


xi_interaction_dewpoint_mjo — correlation

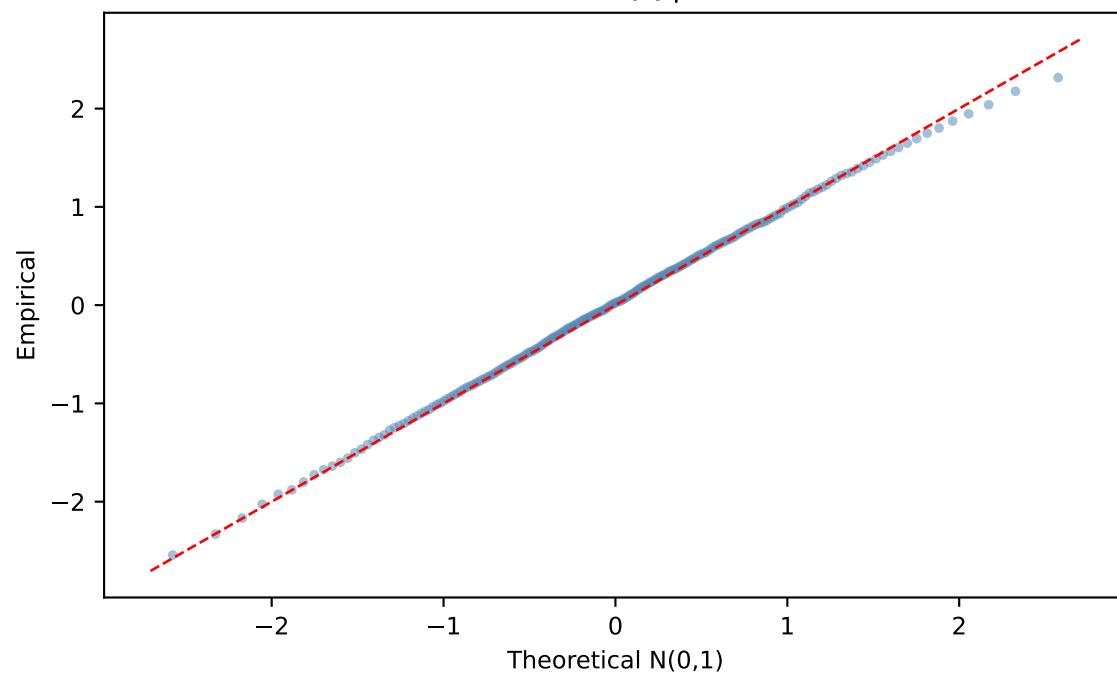


xi_interaction_dewpoint_nino34 — IS diagnostics

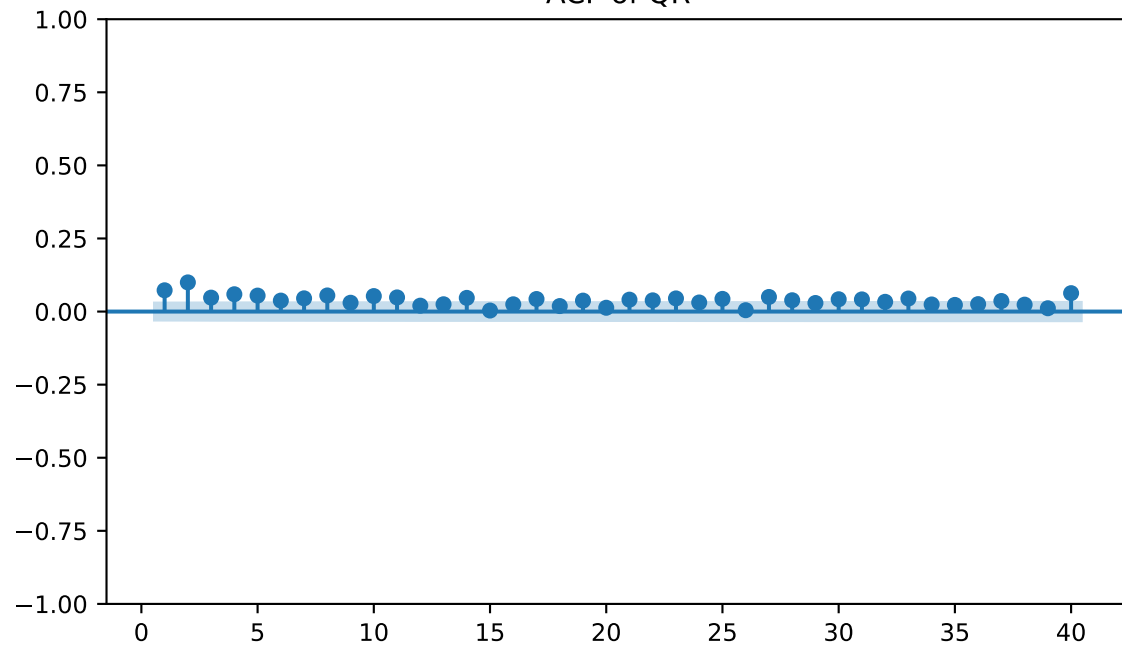
PIT histogram



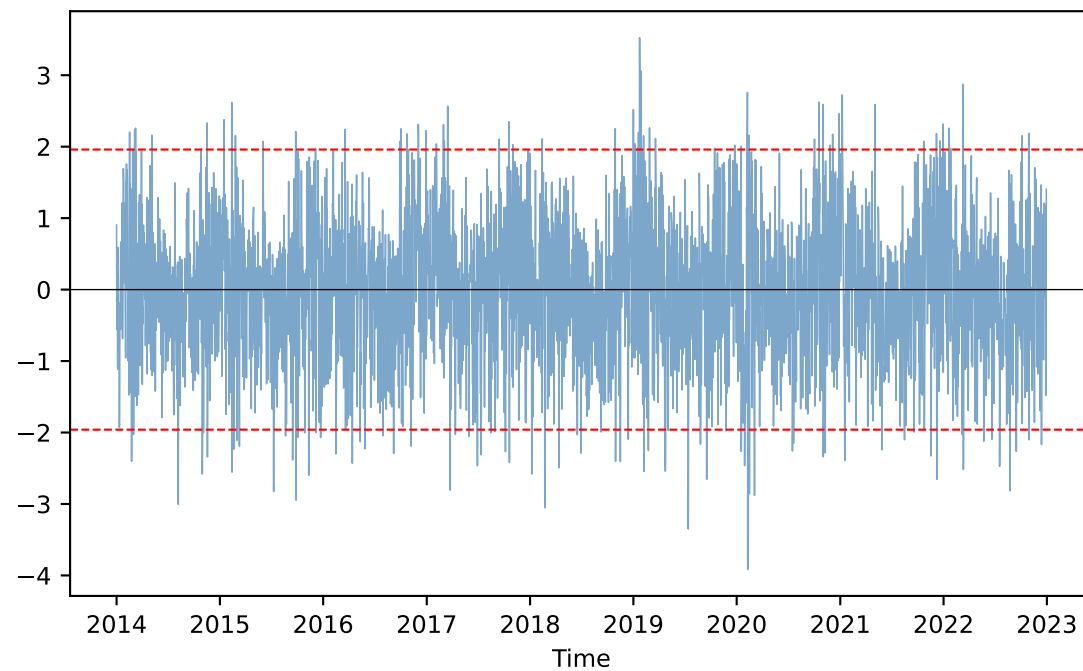
Normal QQ plot



ACF of QR

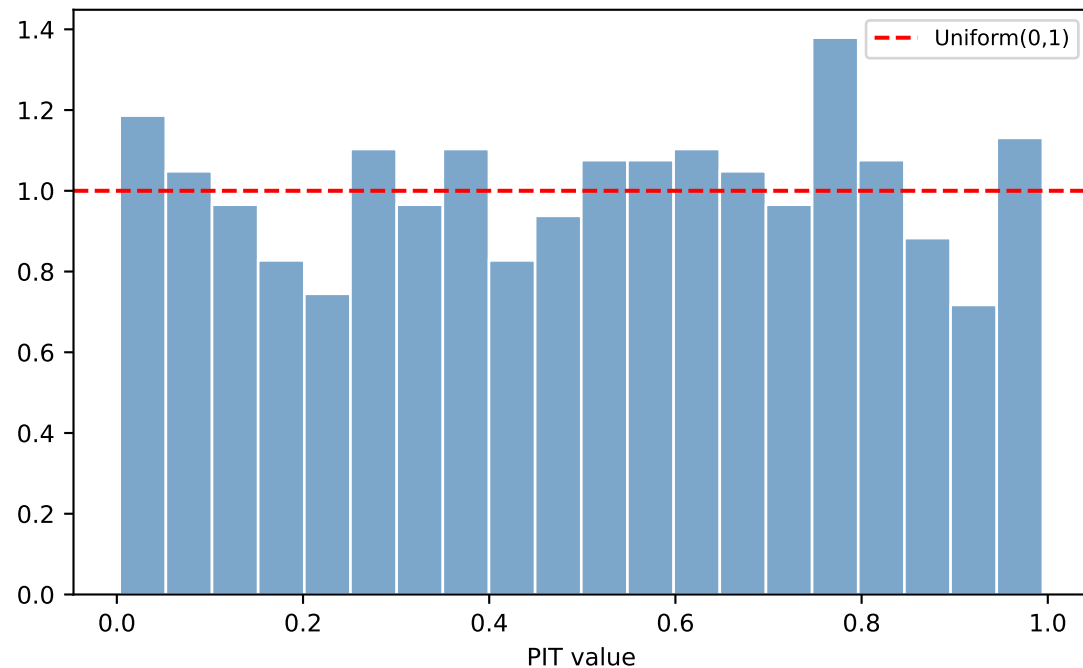


QR time series

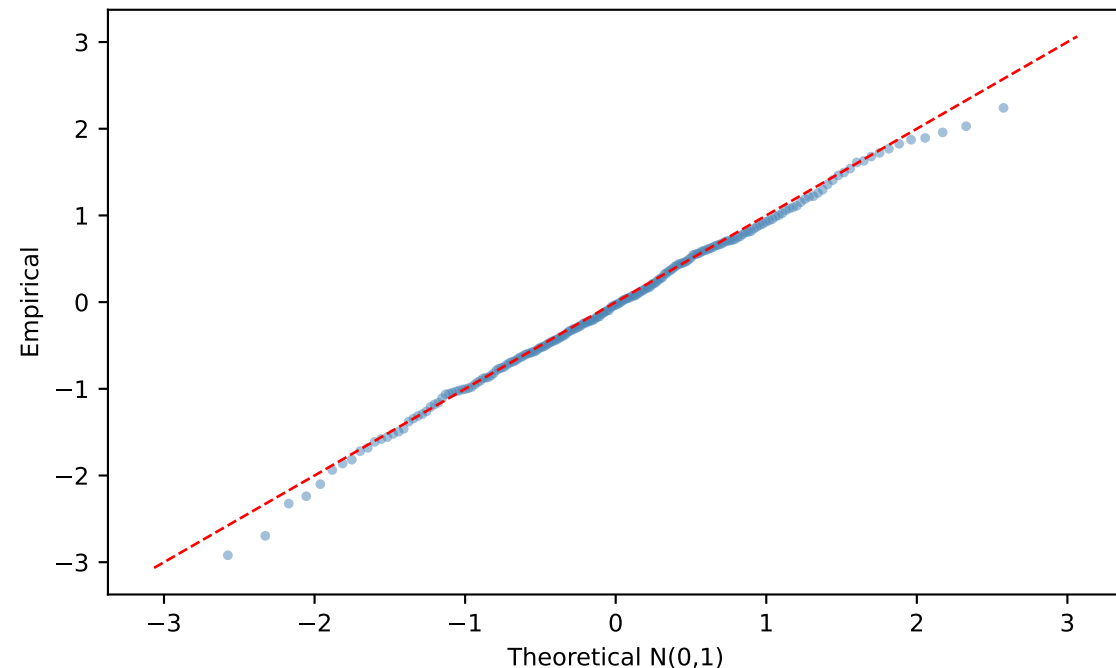


xi_interaction_dewpoint_nino34 — OOS diagnostics

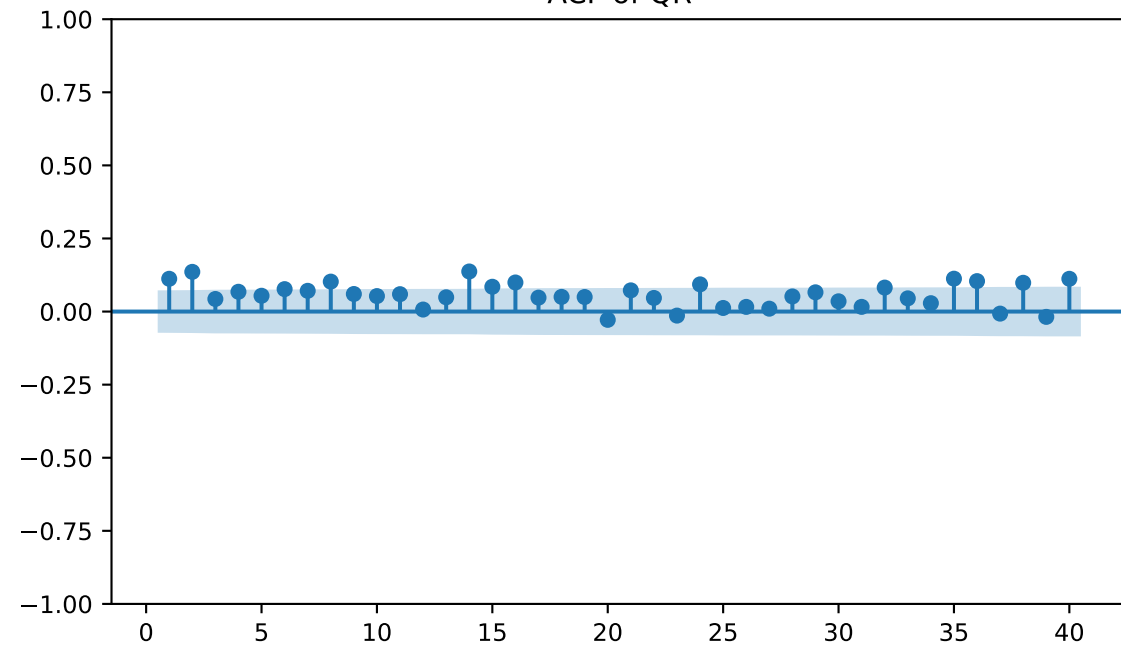
PIT histogram



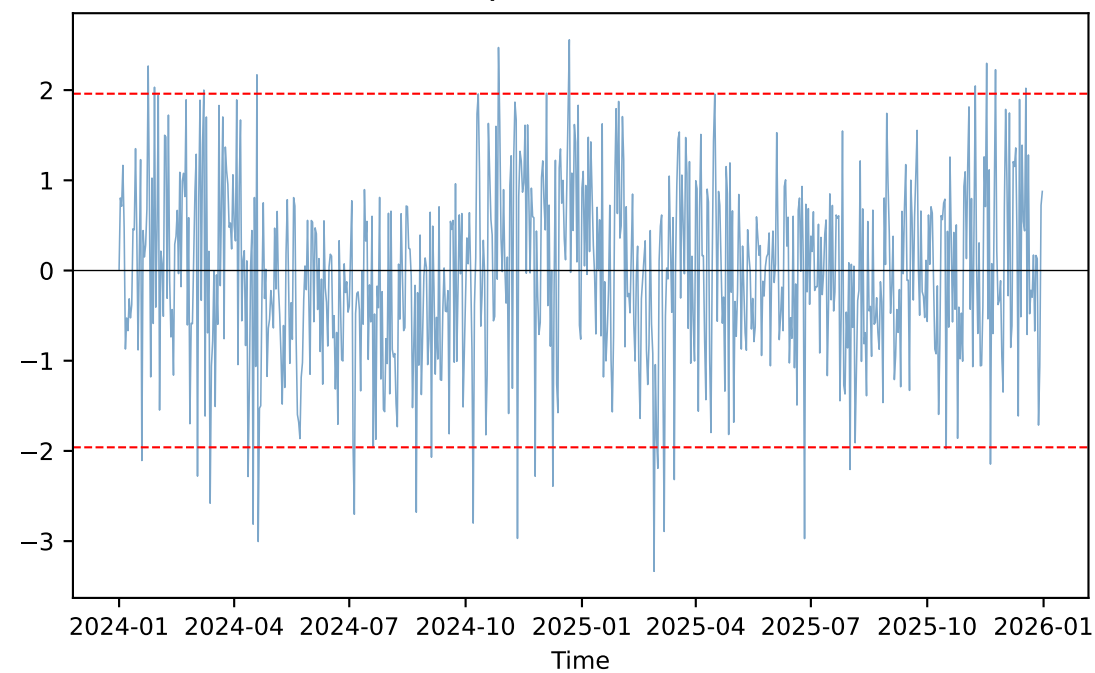
Normal QQ plot



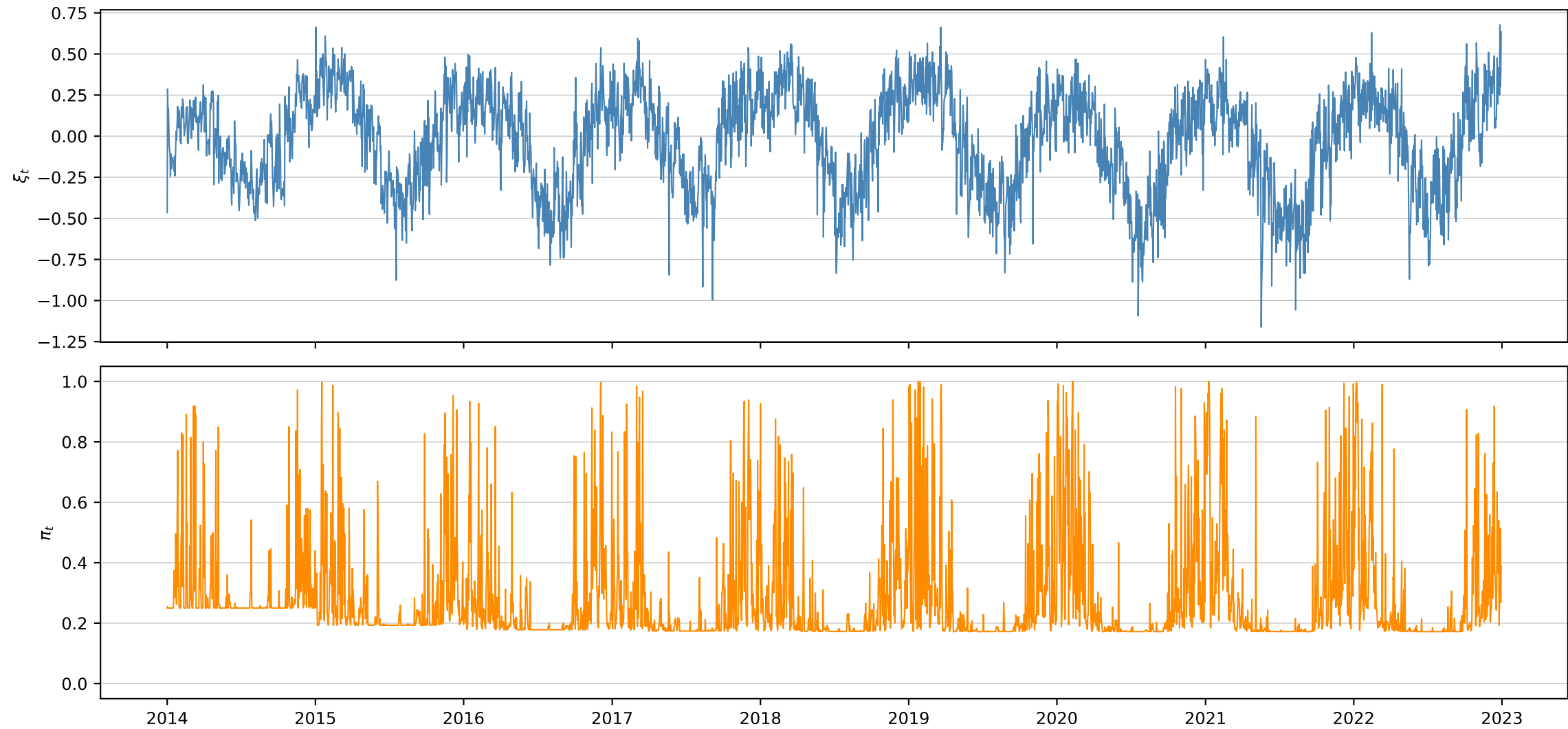
ACF of QR



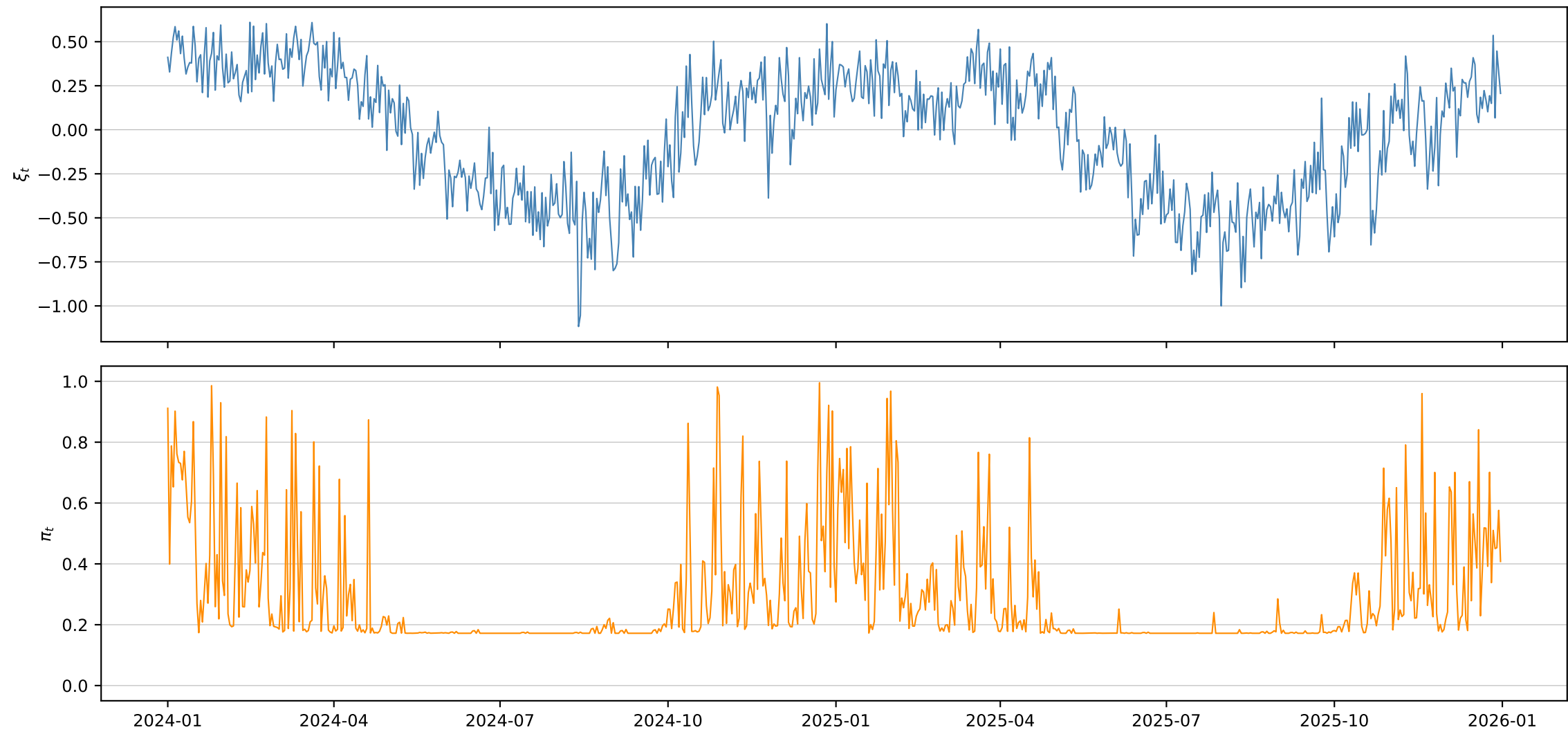
QR time series



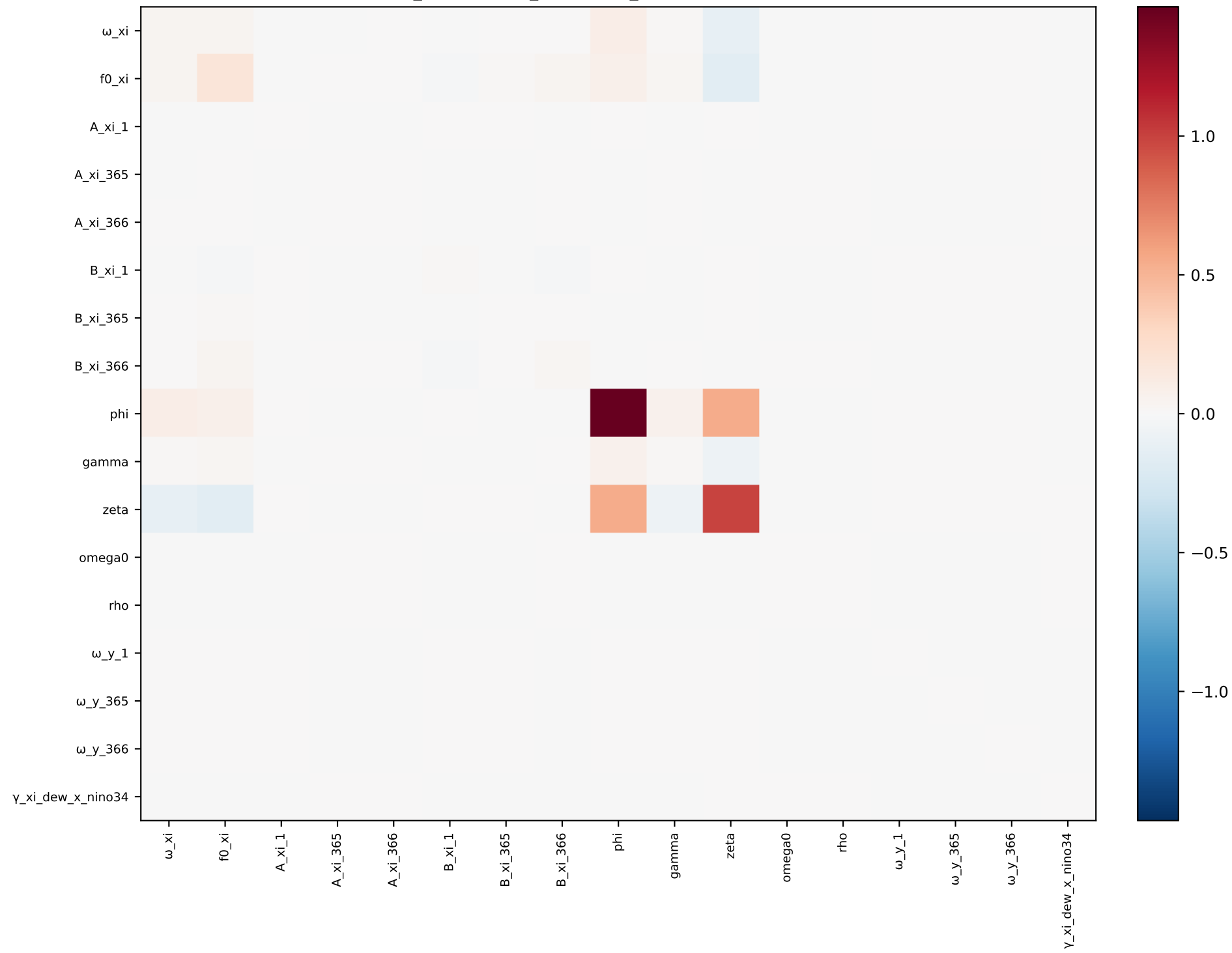
xi_interaction_dewpoint_nino34 — filtered states (IS)



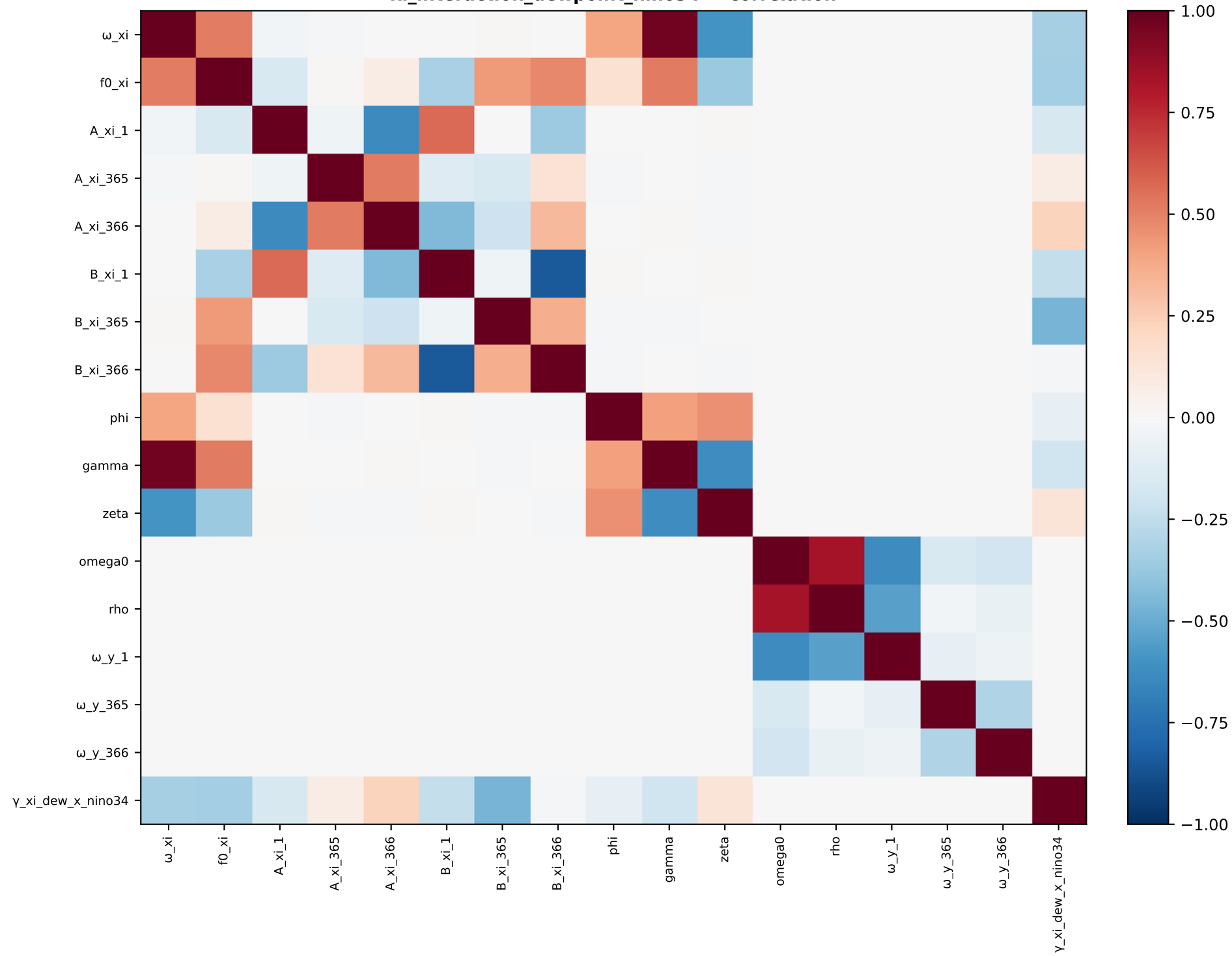
xi_interaction_dewpoint_nino34 — filtered states (OOS)



xi_interaction_dewpoint_nino34 — covariance

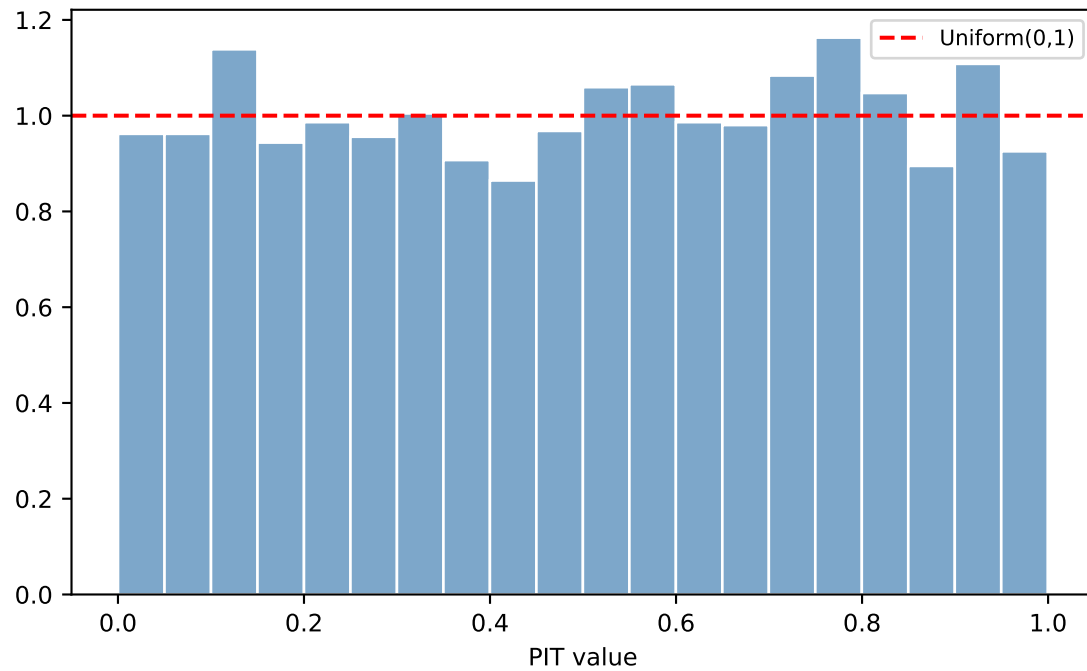


xi_interaction_dewpoint_nino34 — correlation

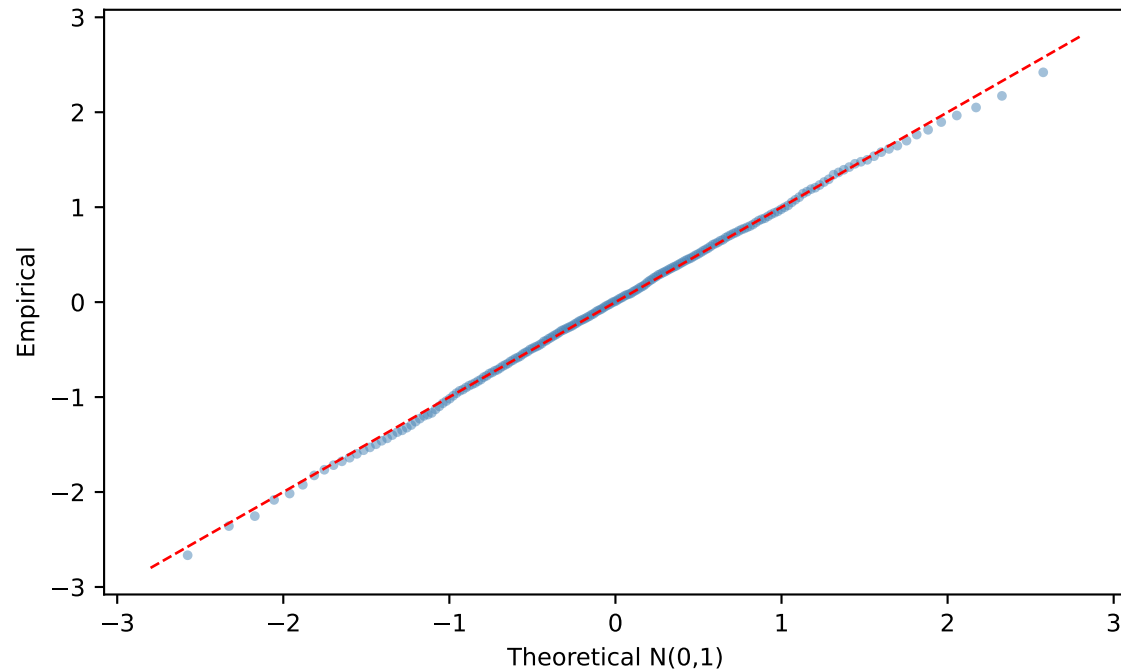


xi_interaction_temperature_nino34 — IS diagnostics

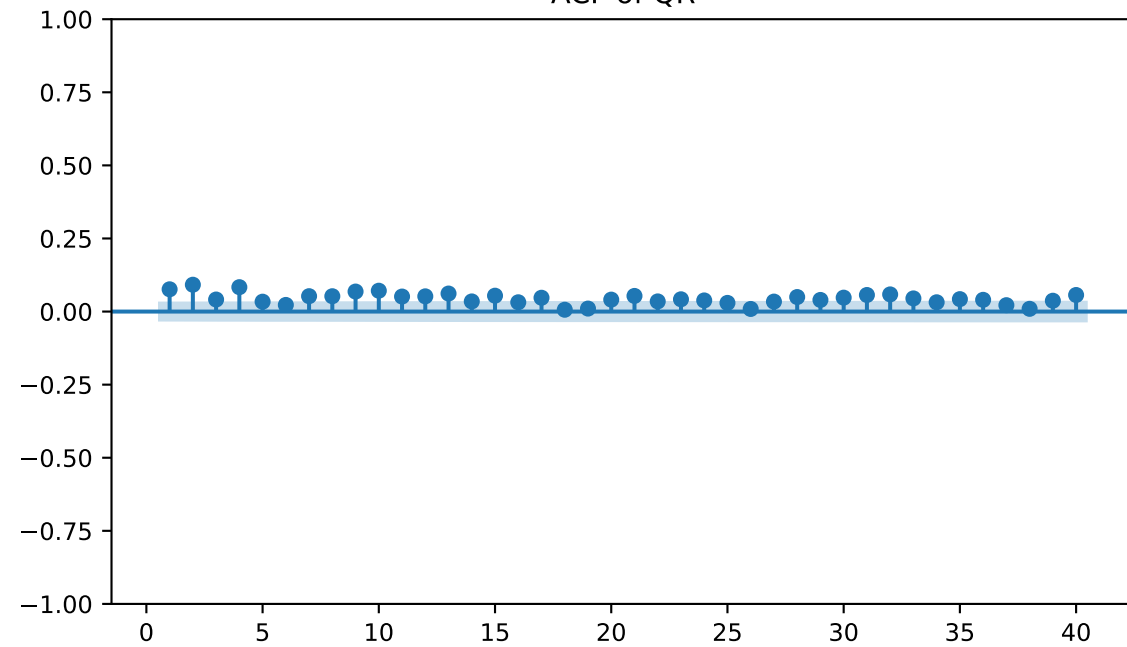
PIT histogram



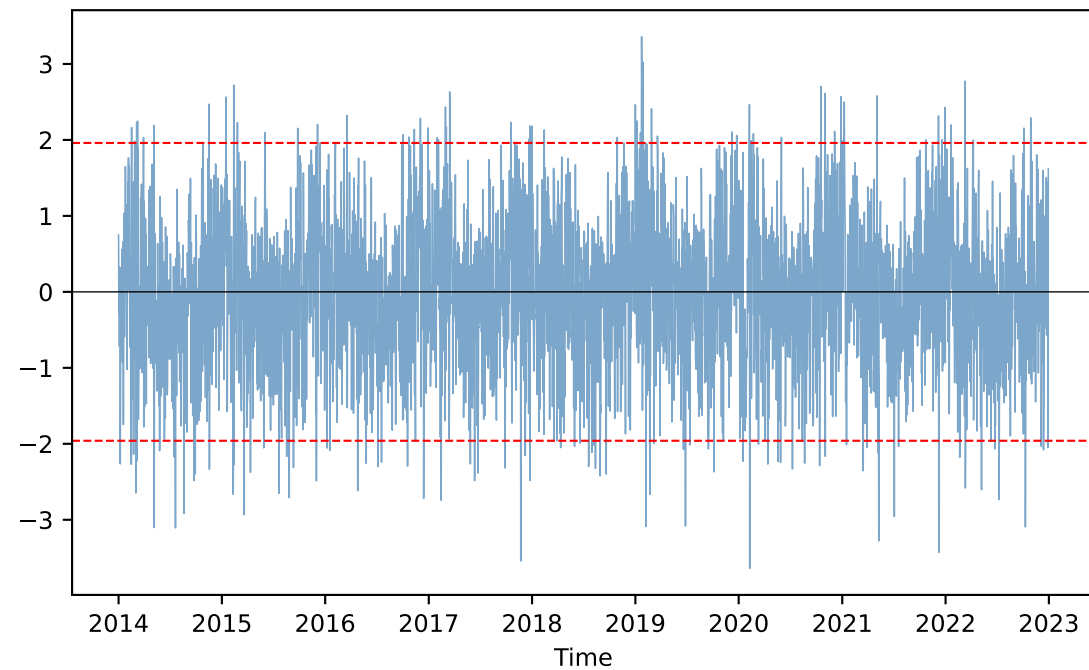
Normal QQ plot



ACF of QR

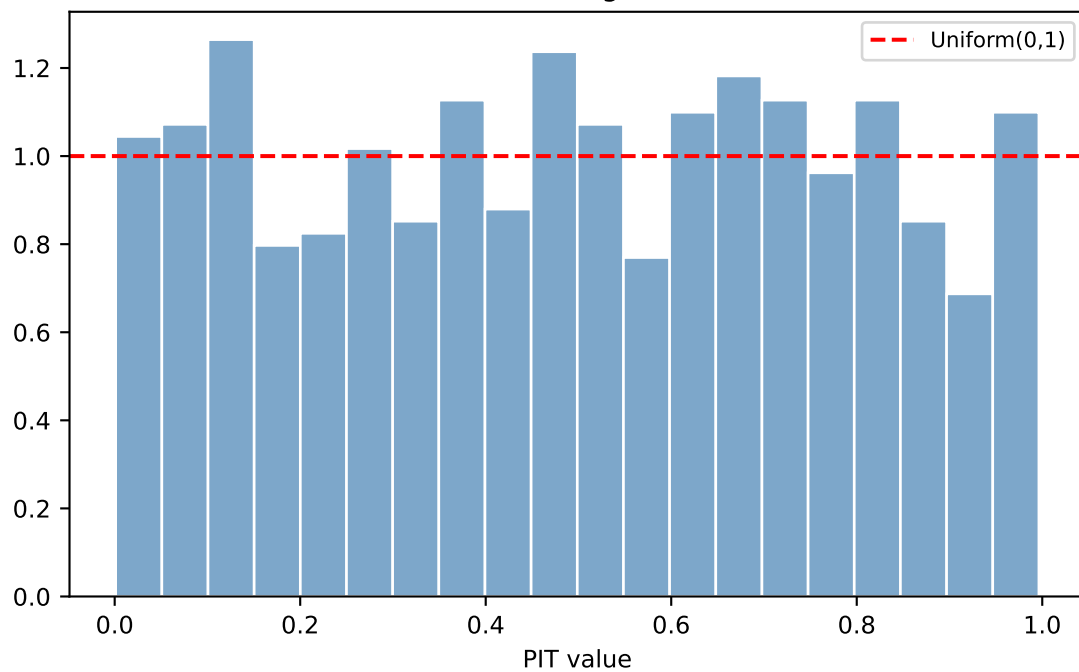


QR time series

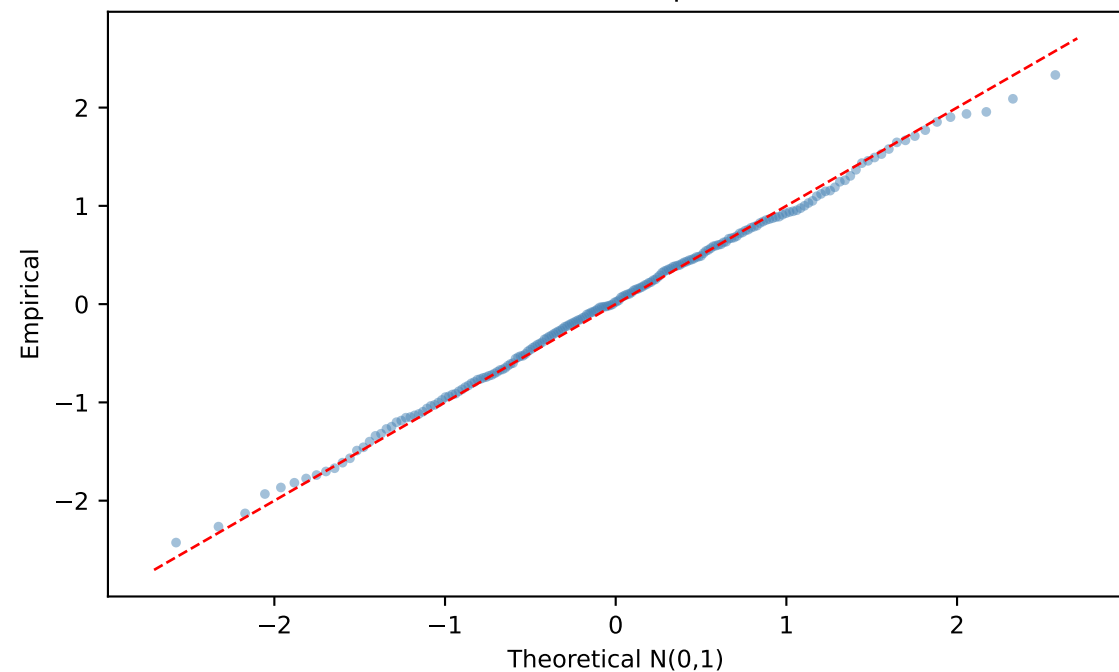


xi_interaction_temperature_nino34 — OOS diagnostics

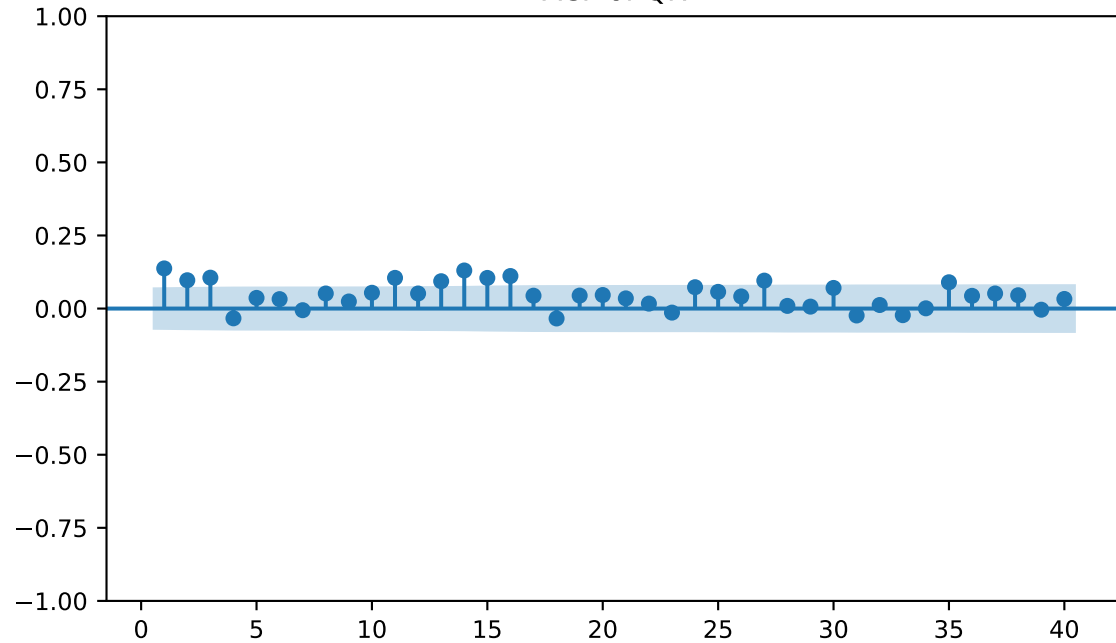
PIT histogram



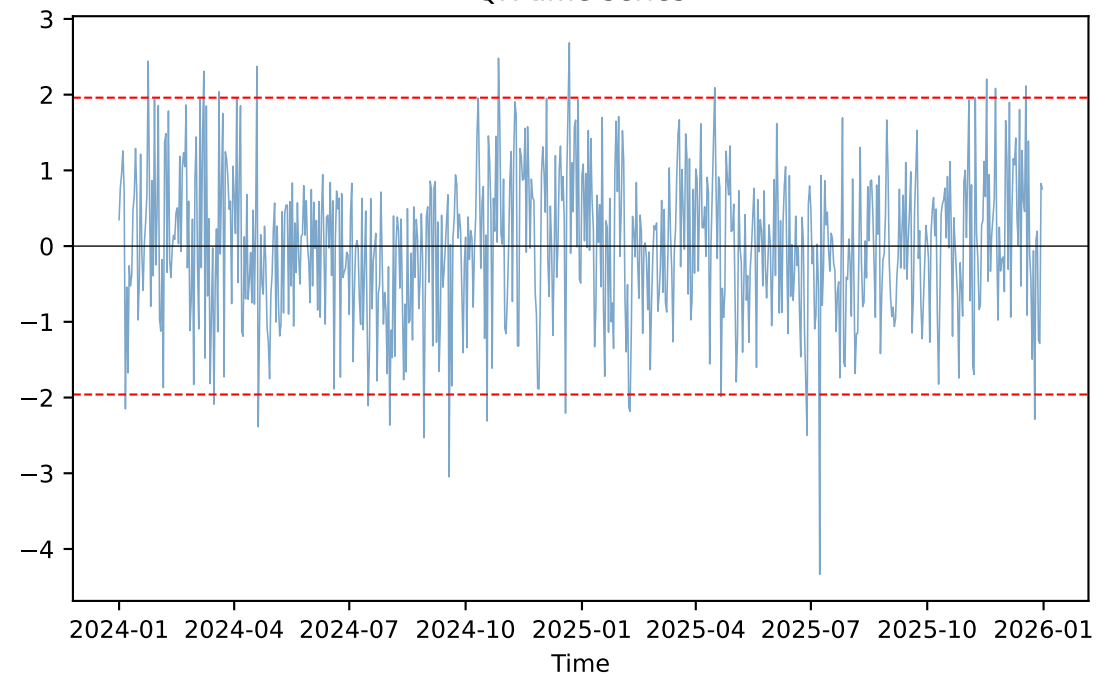
Normal QQ plot



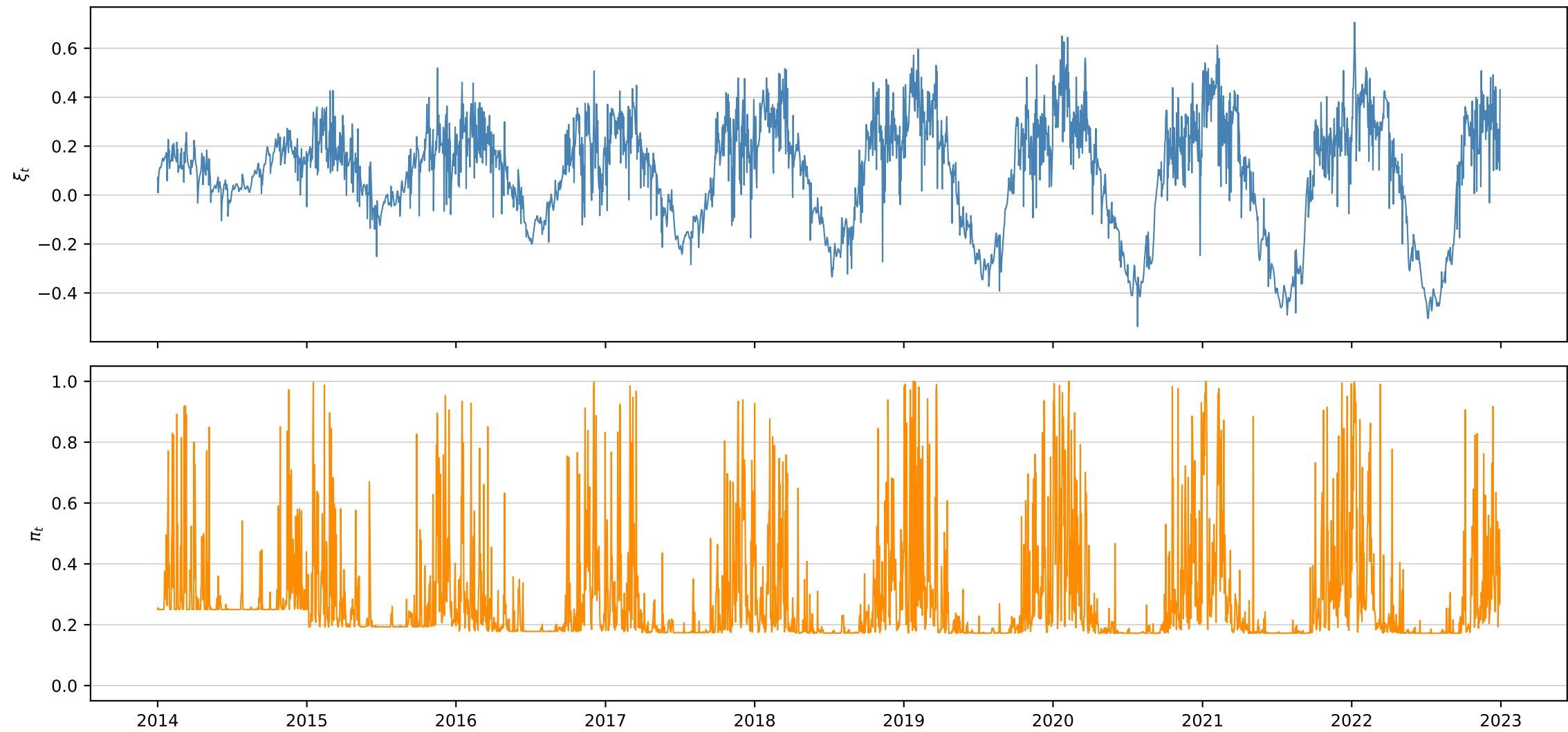
ACF of QR



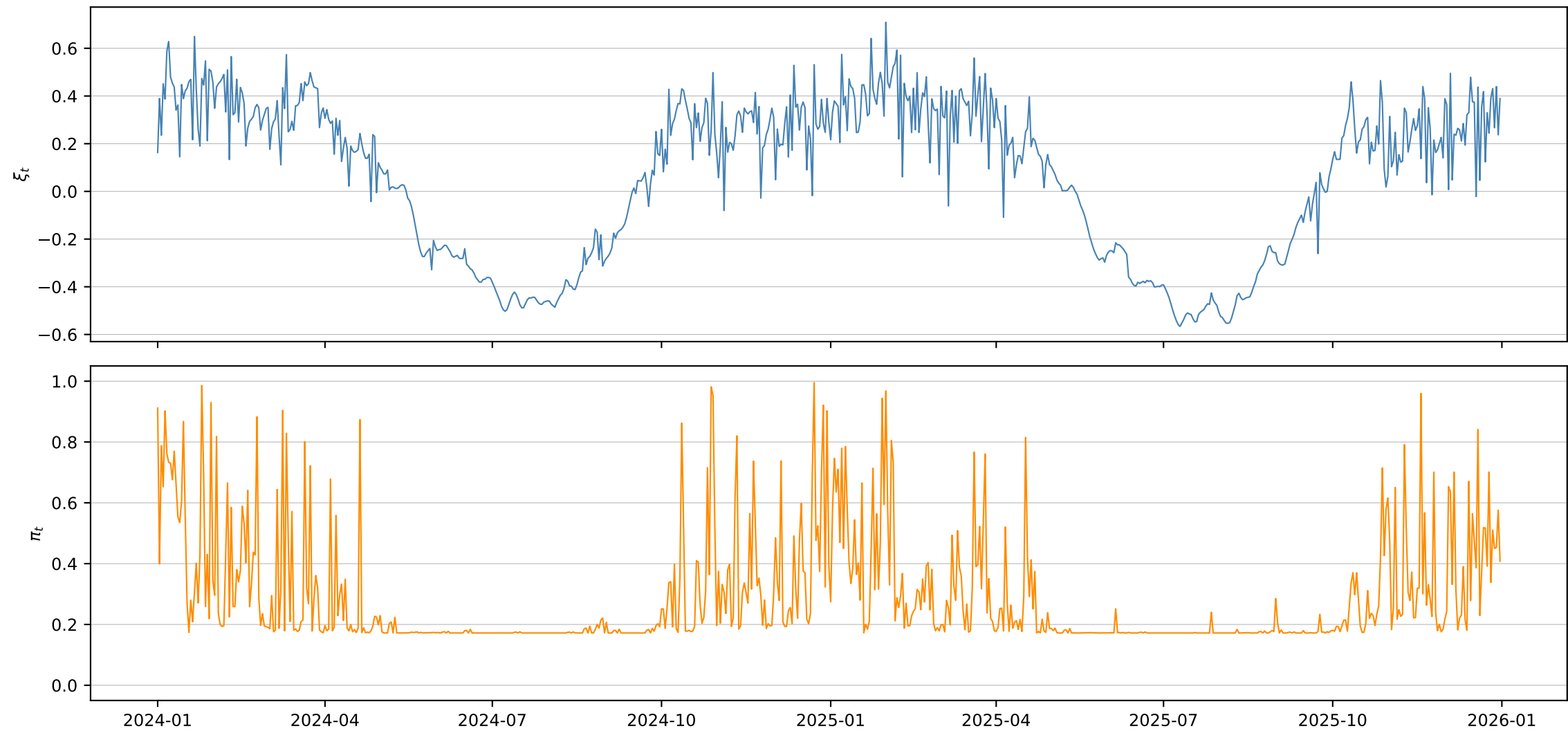
QR time series



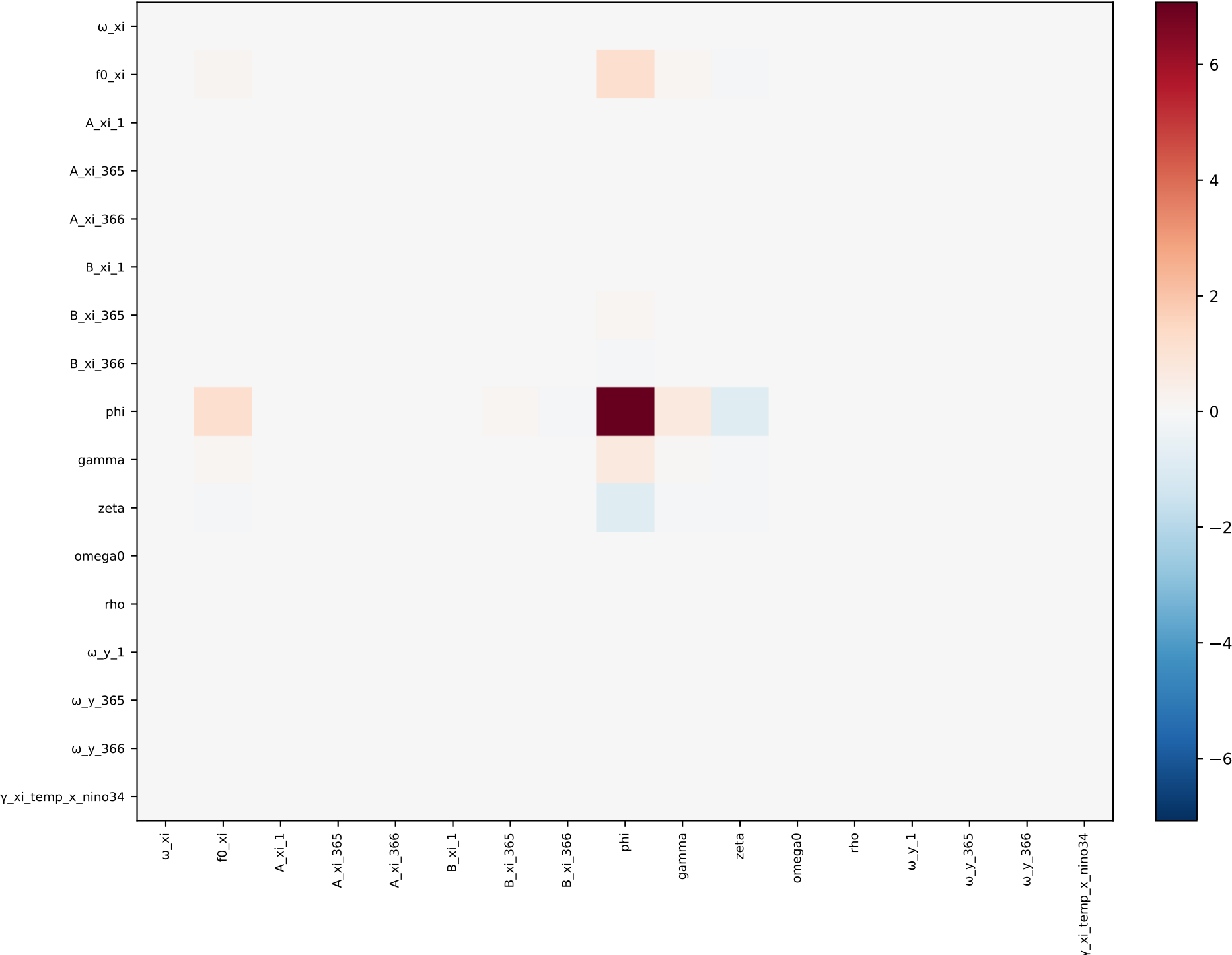
xi_interaction_temperature_nino34 — filtered states (IS)



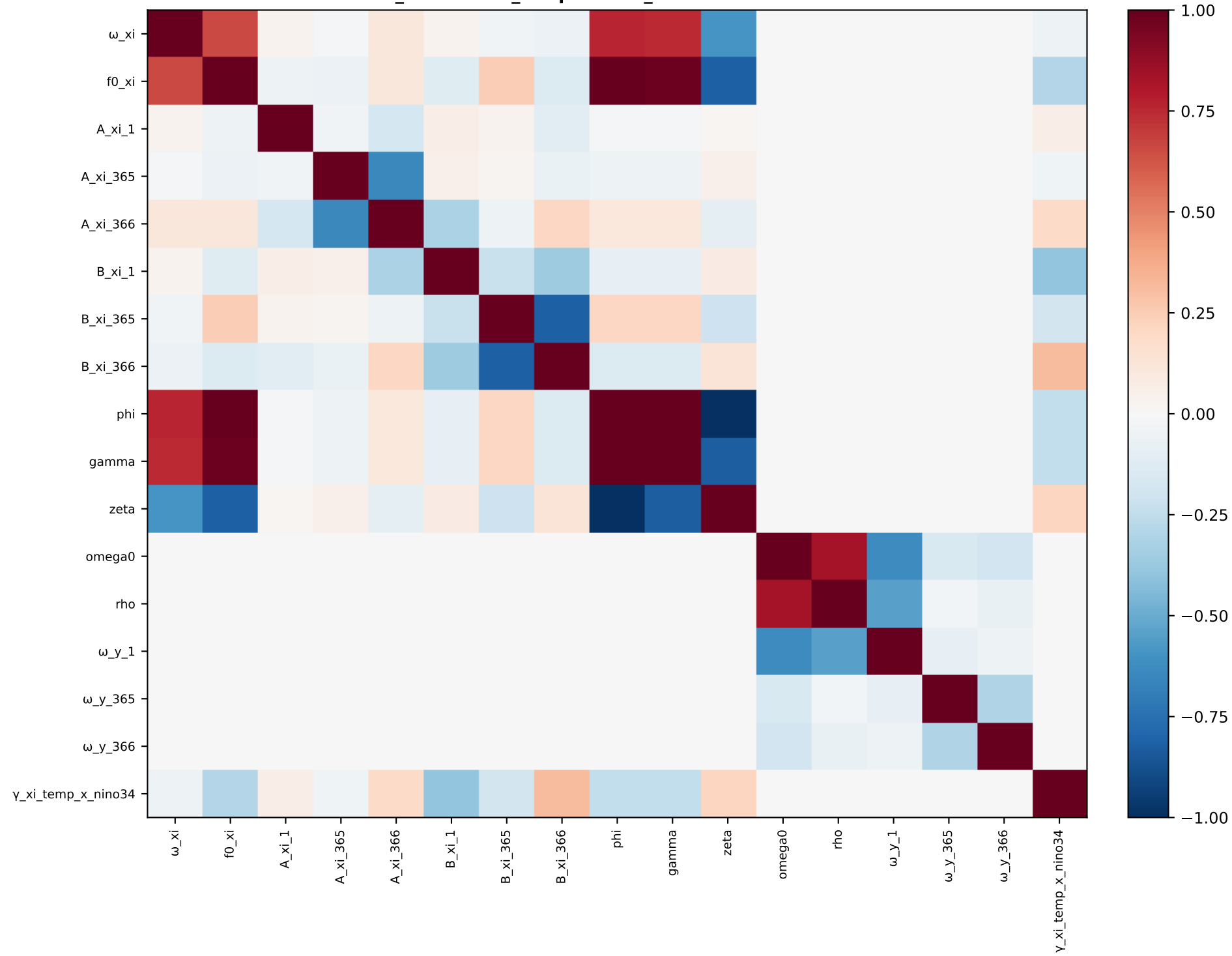
xi_interaction_temperature_nino34 — filtered states (OOS)



xi interaction temperature nino34 — covariance

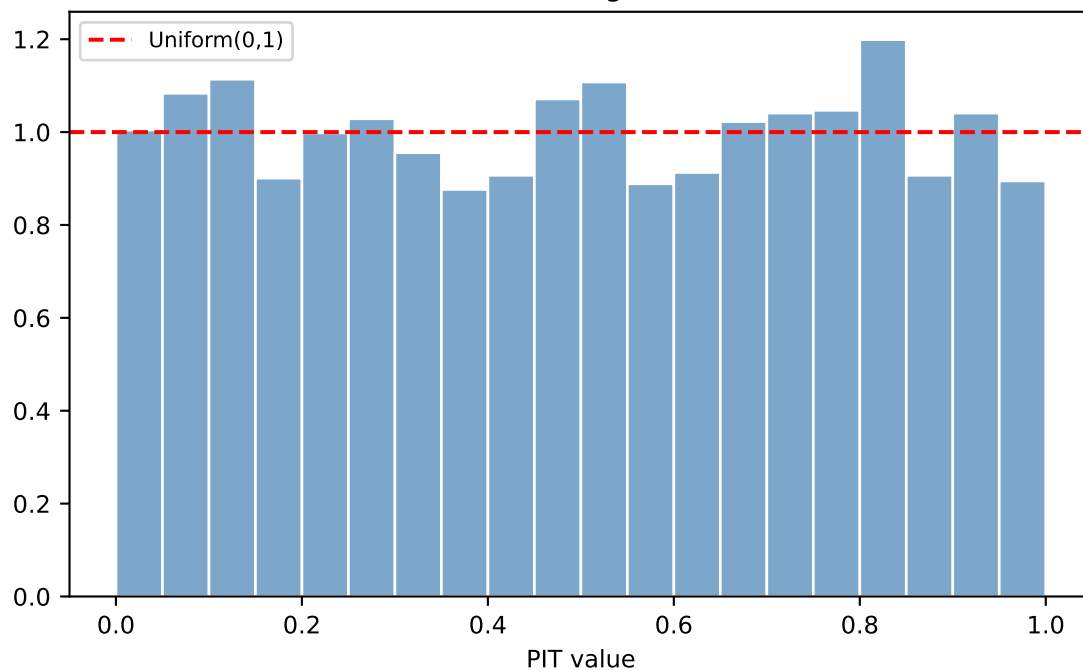


xi_interaction_temperature_nino34 — correlation

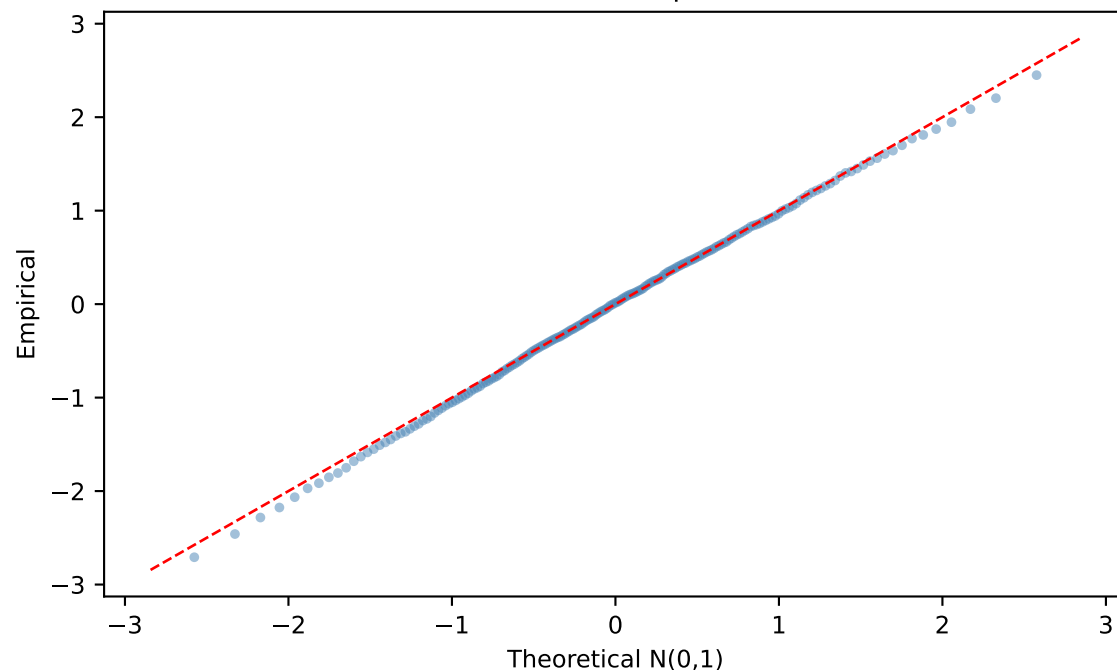


phi_dewpoint_t — IS diagnostics

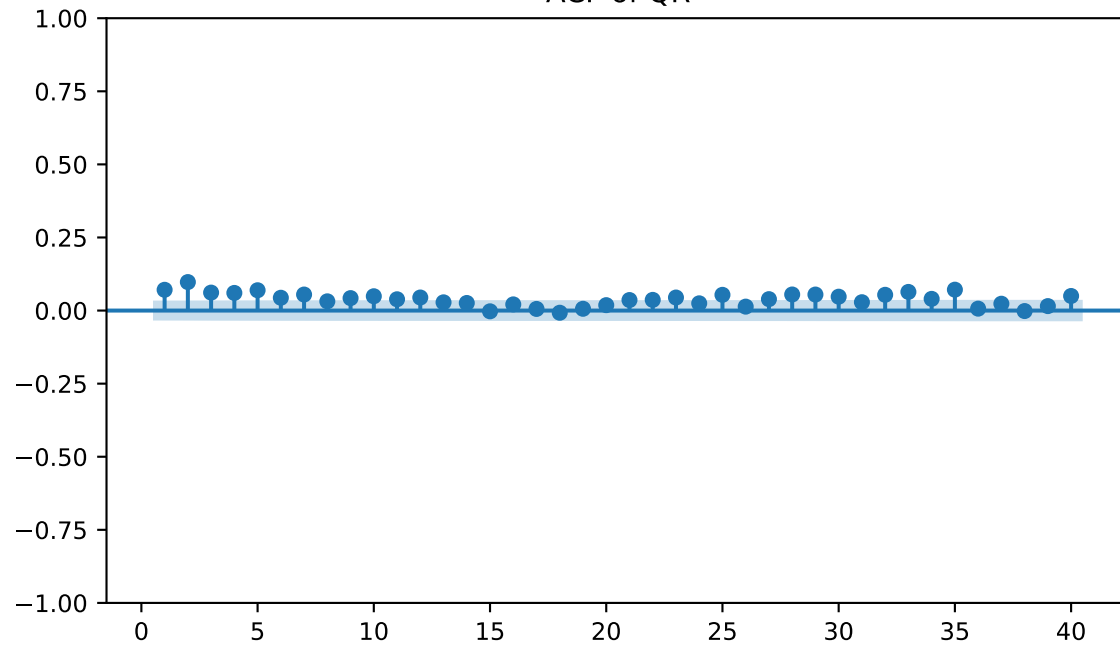
PIT histogram



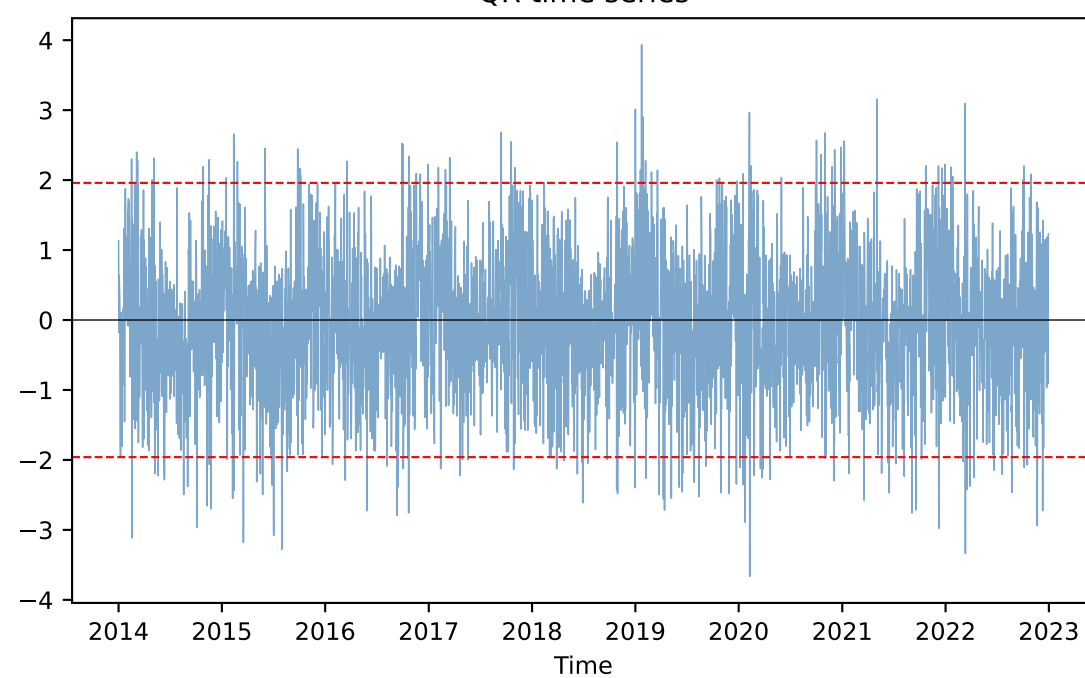
Normal QQ plot



ACF of QR

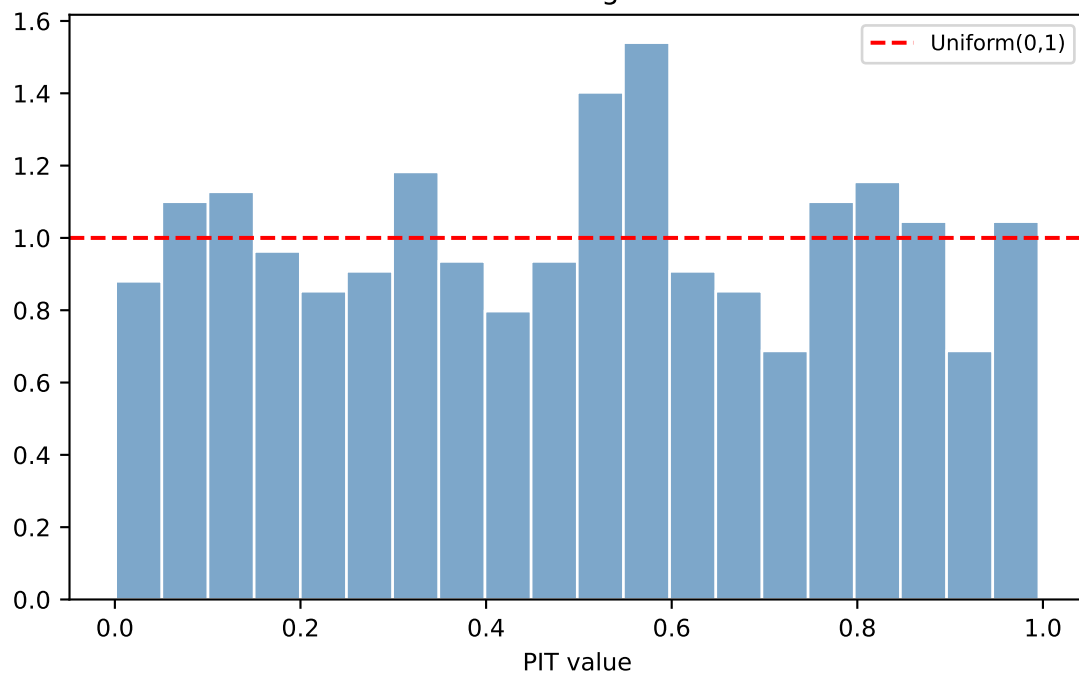


QR time series

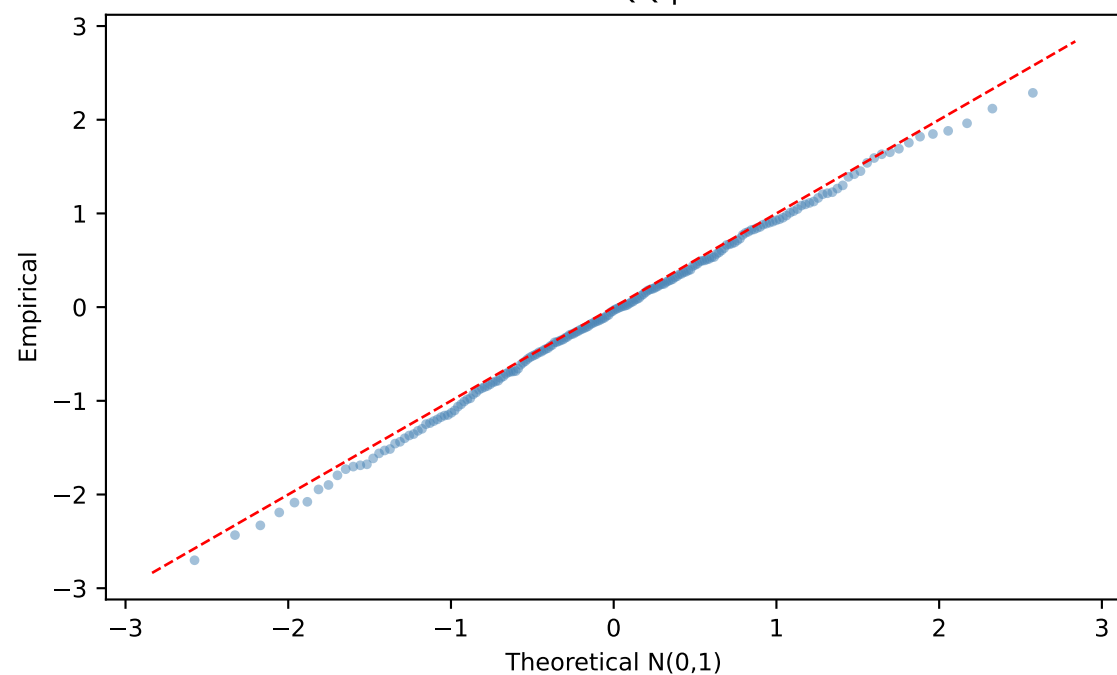


phi_dewpoint_t — OOS diagnostics

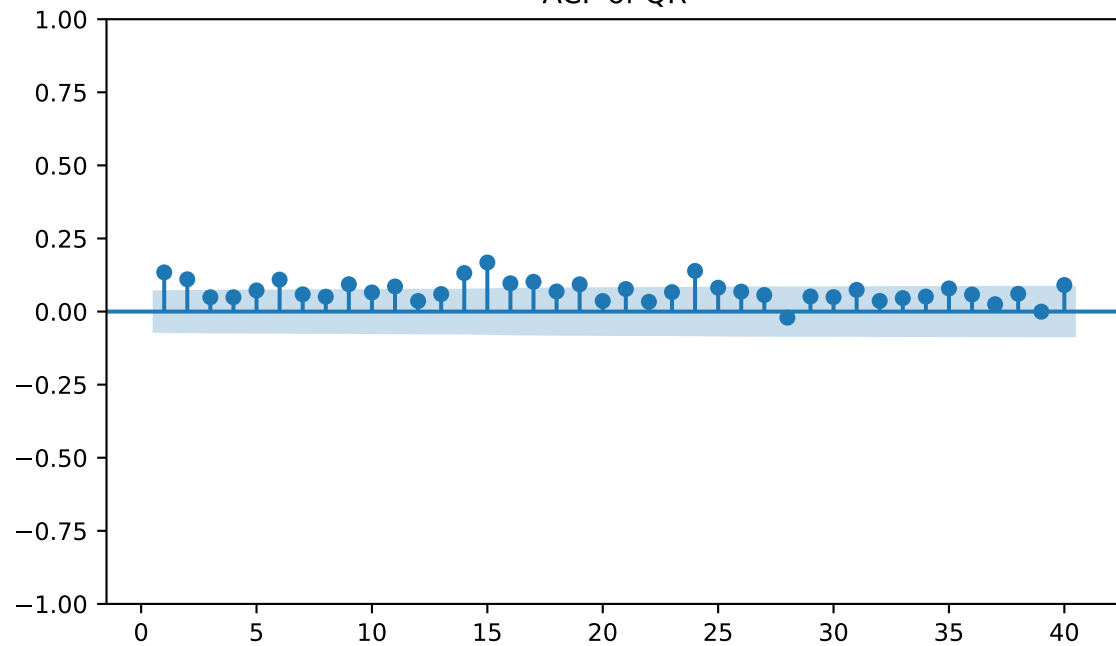
PIT histogram



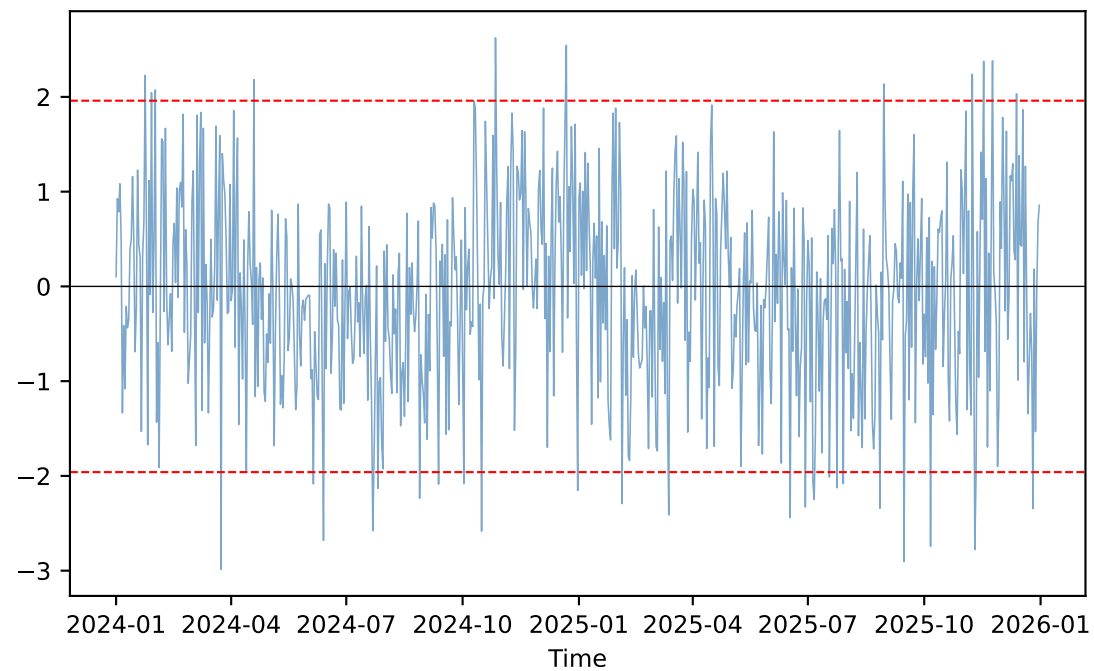
Normal QQ plot



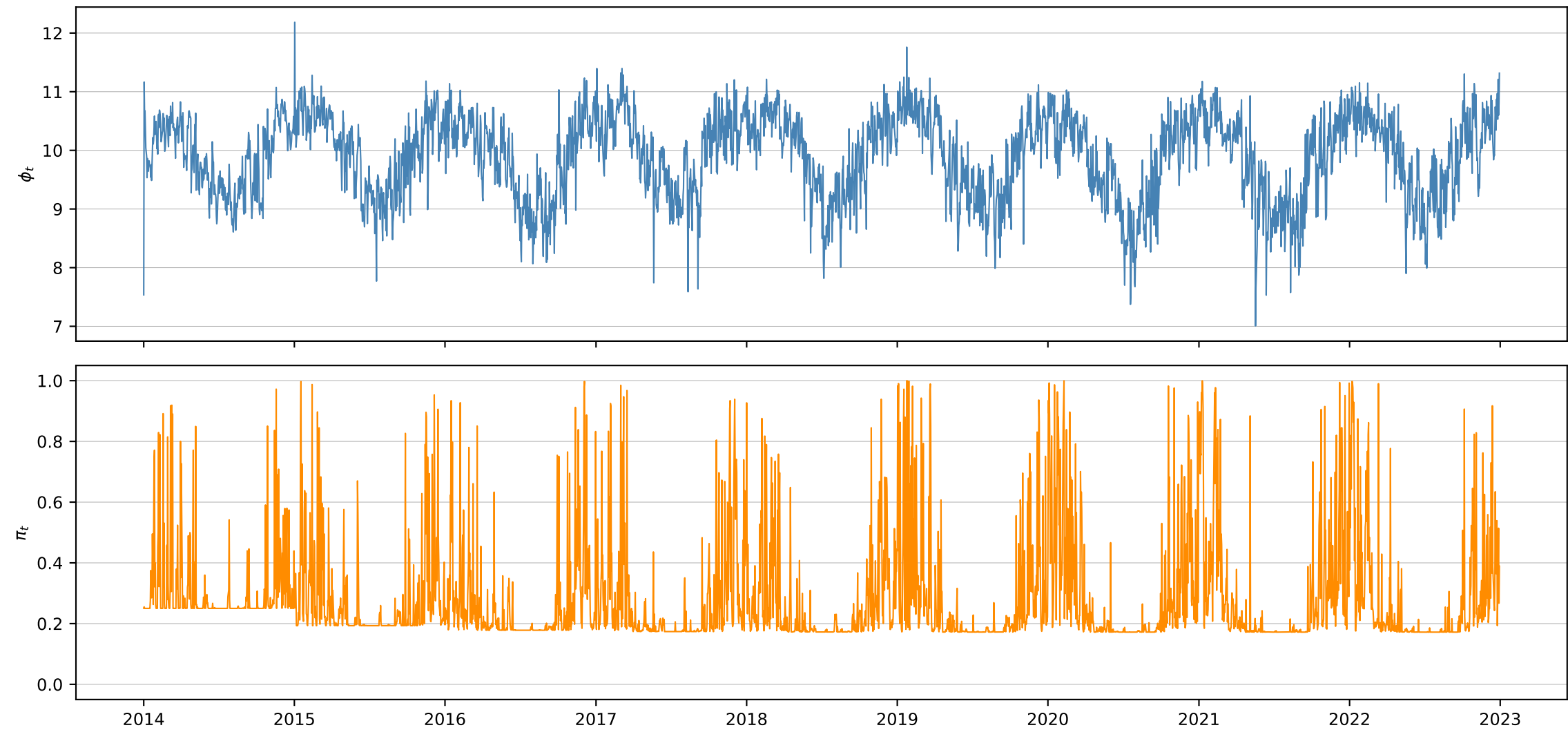
ACF of QR



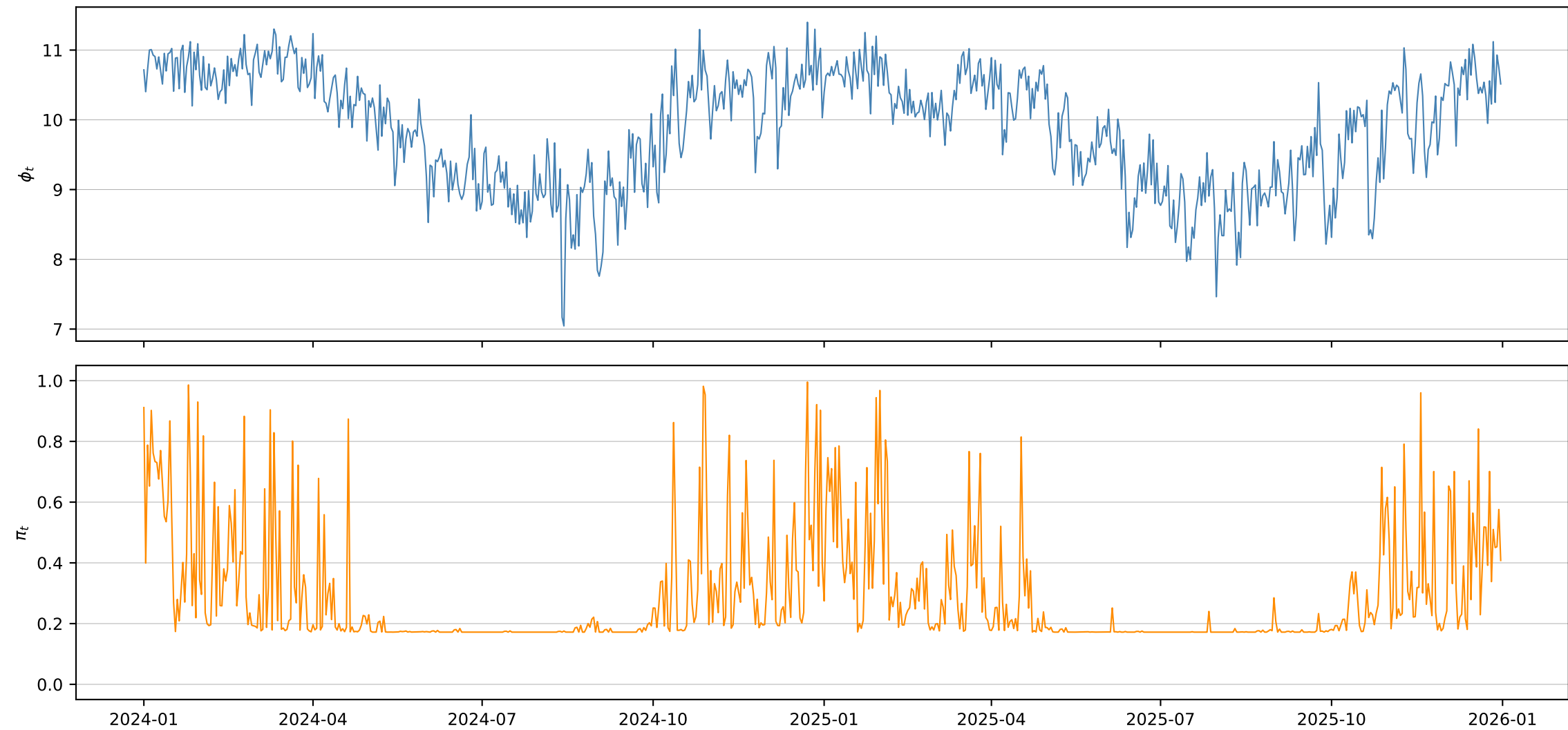
QR time series



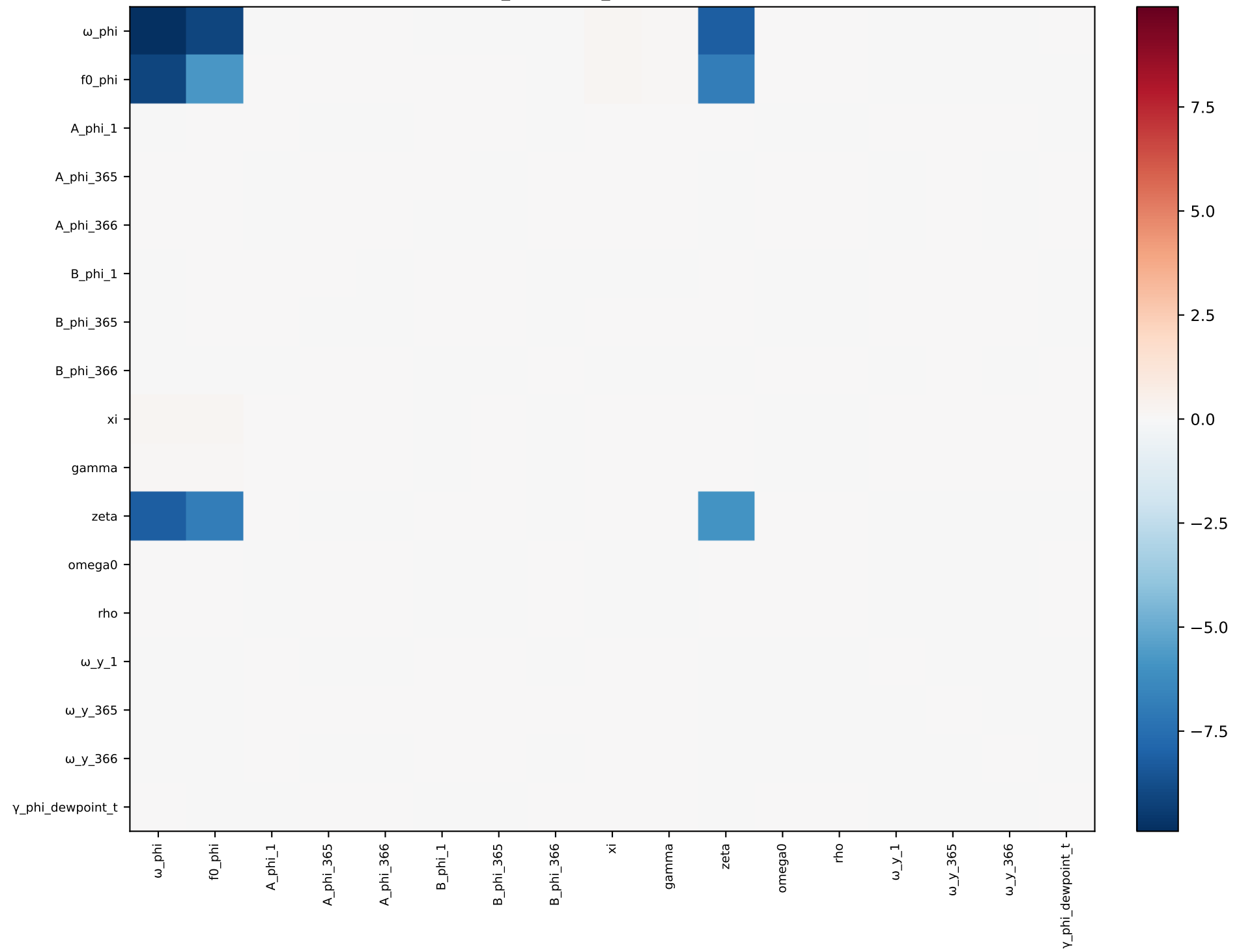
phi_dewpoint_t — filtered states (IS)



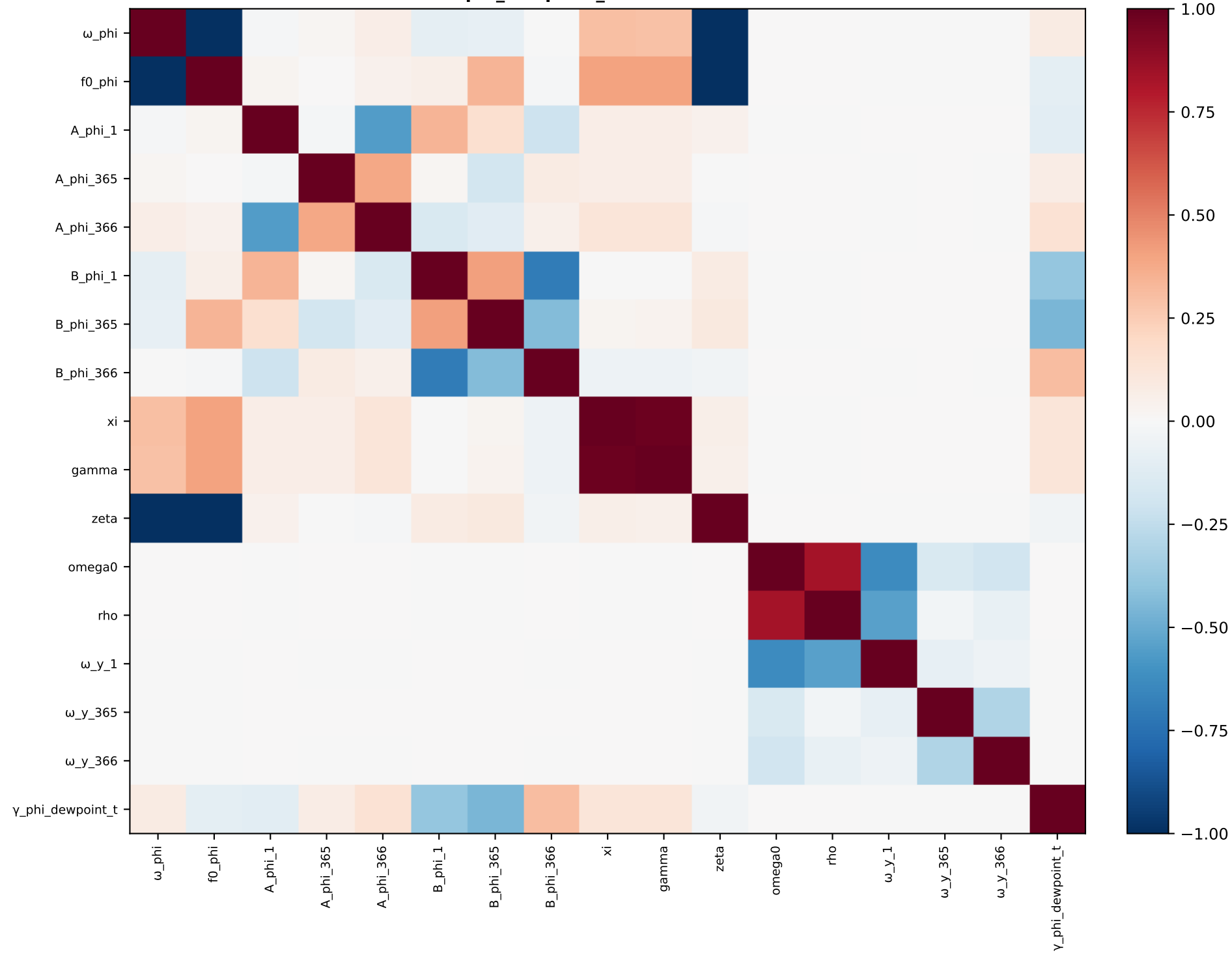
phi_dewpoint_t — filtered states (OOS)



phi_dewpoint_t — covariance

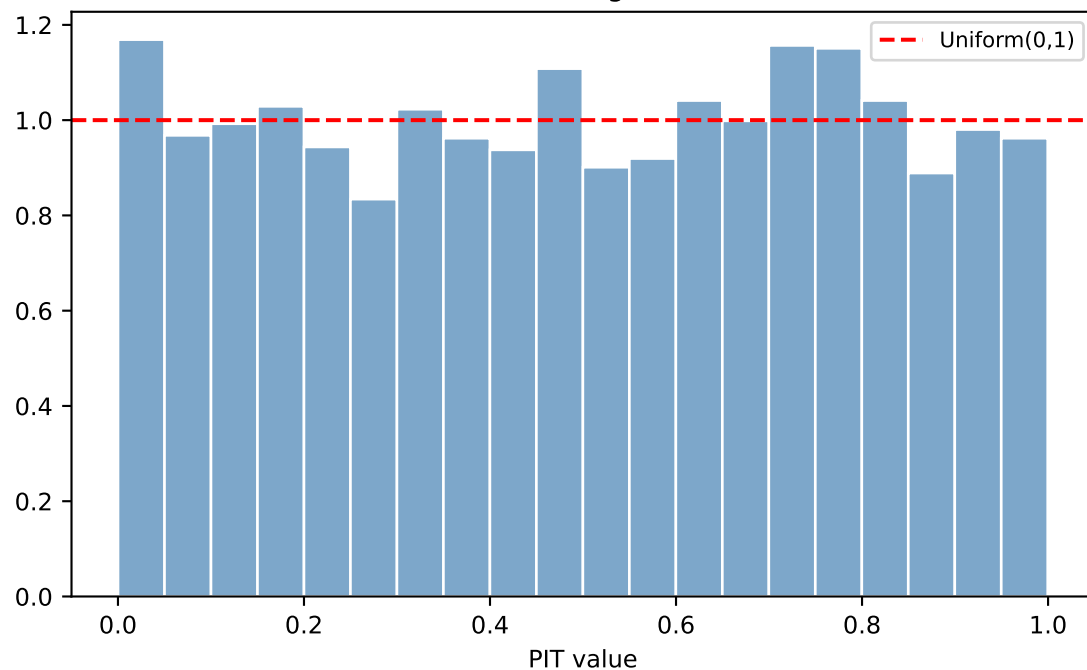


phi_dewpoint_t — correlation

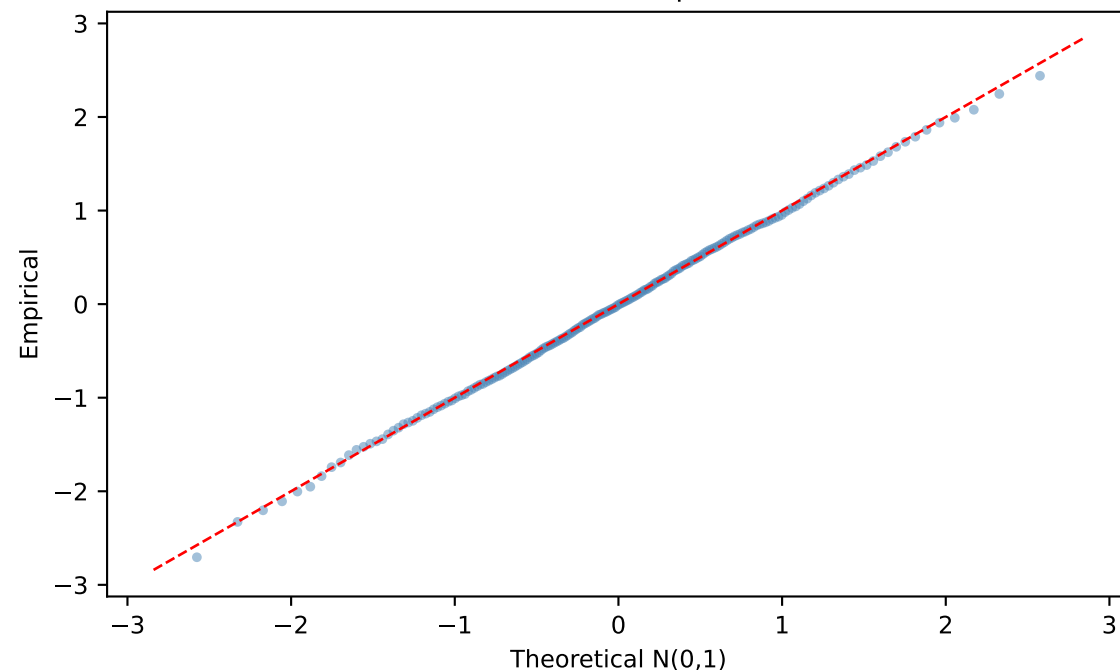


phi_dewpoint_lag1 — IS diagnostics

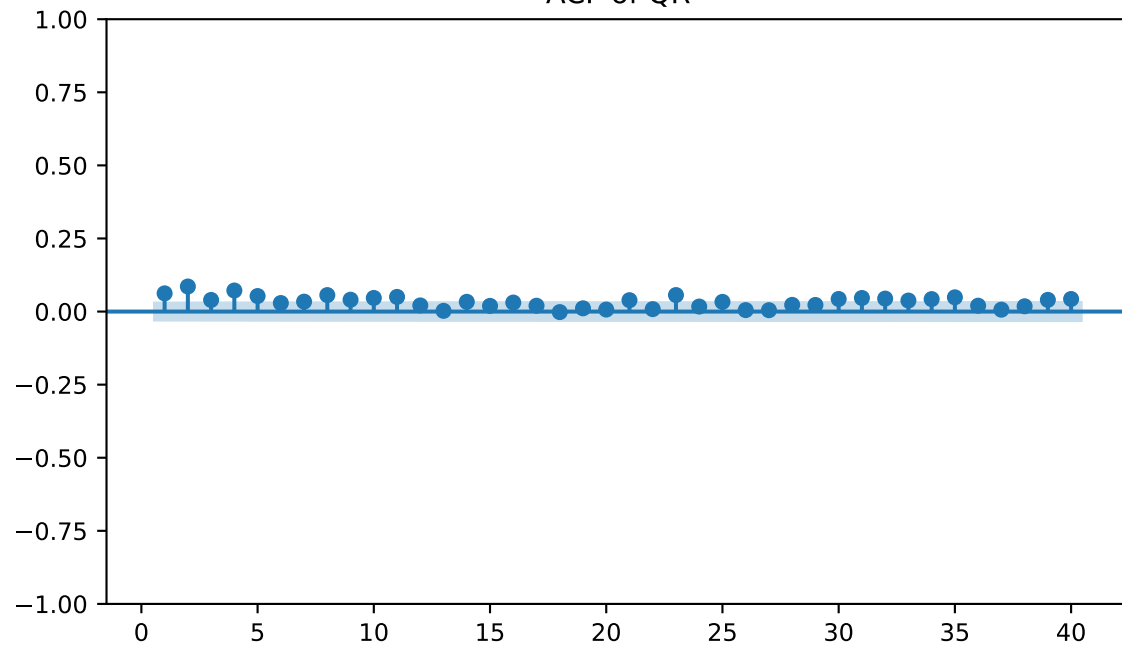
PIT histogram



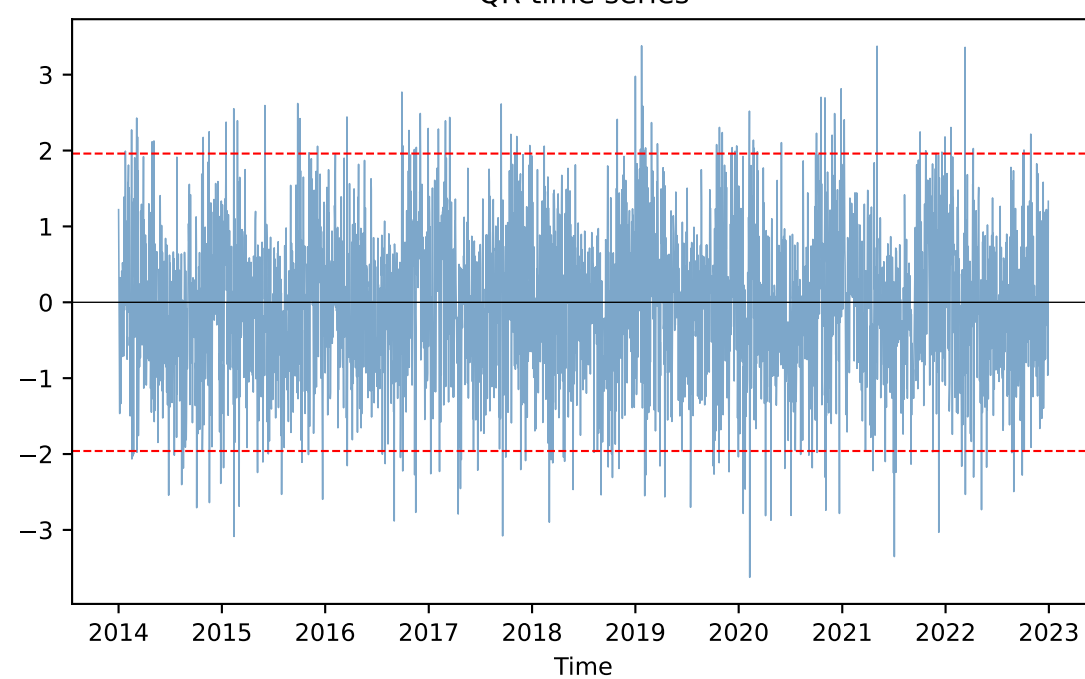
Normal QQ plot



ACF of QR

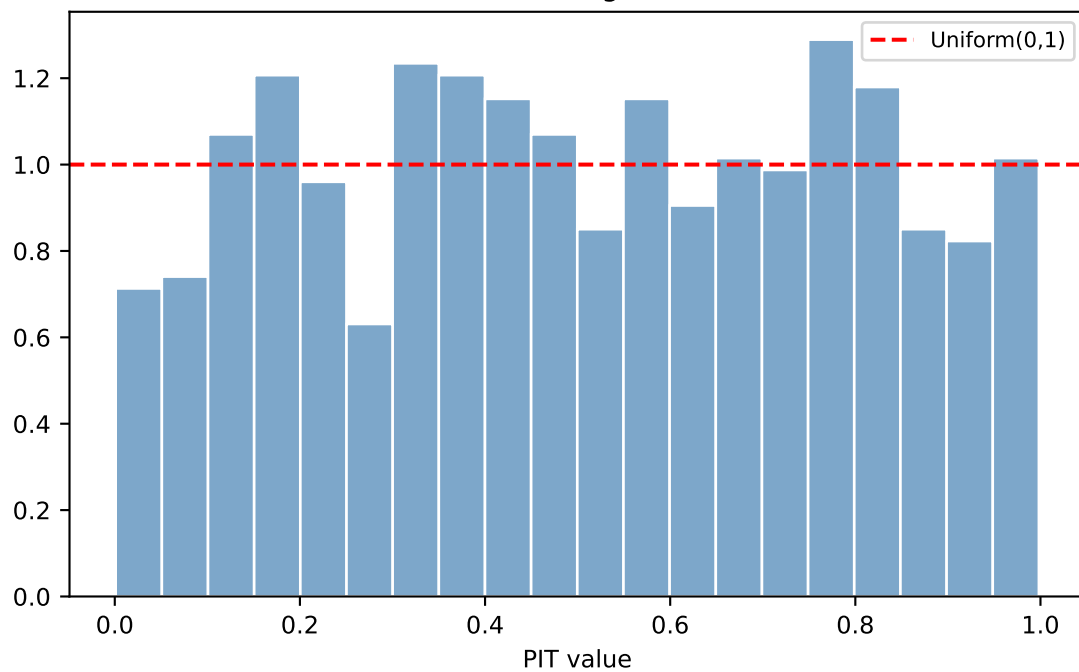


QR time series

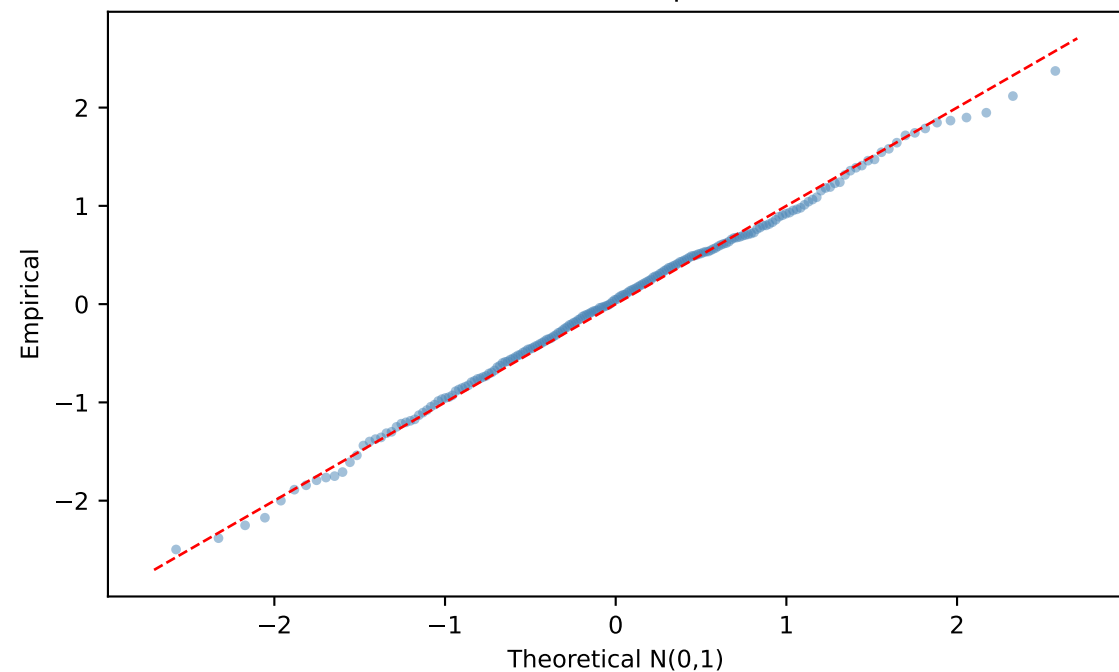


phi_dewpoint_lag1 — OOS diagnostics

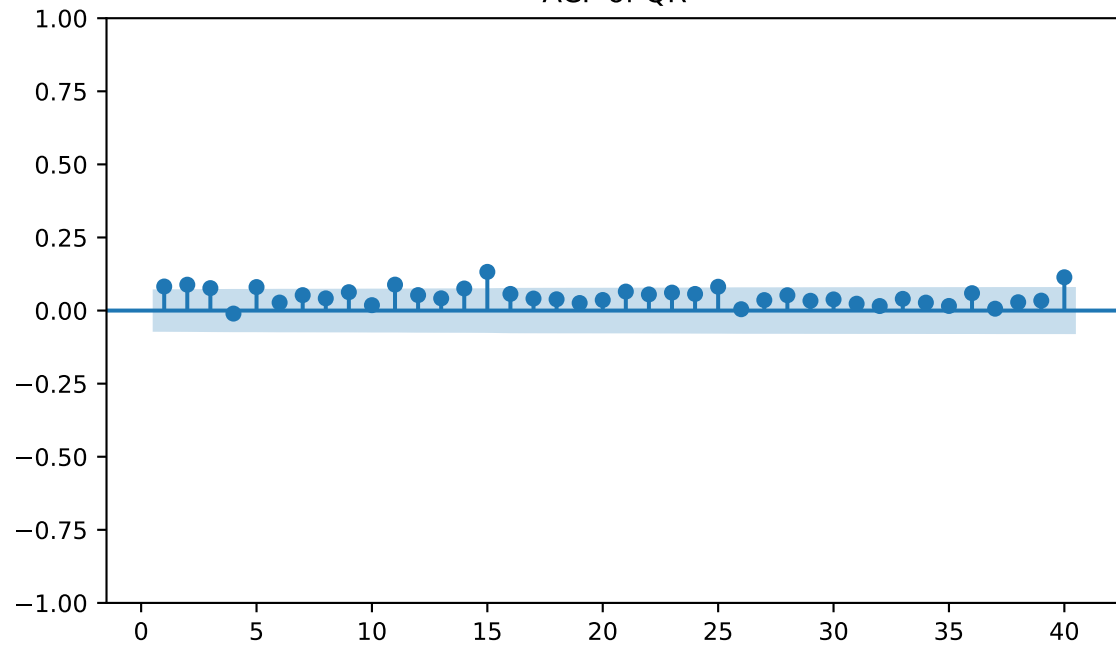
PIT histogram



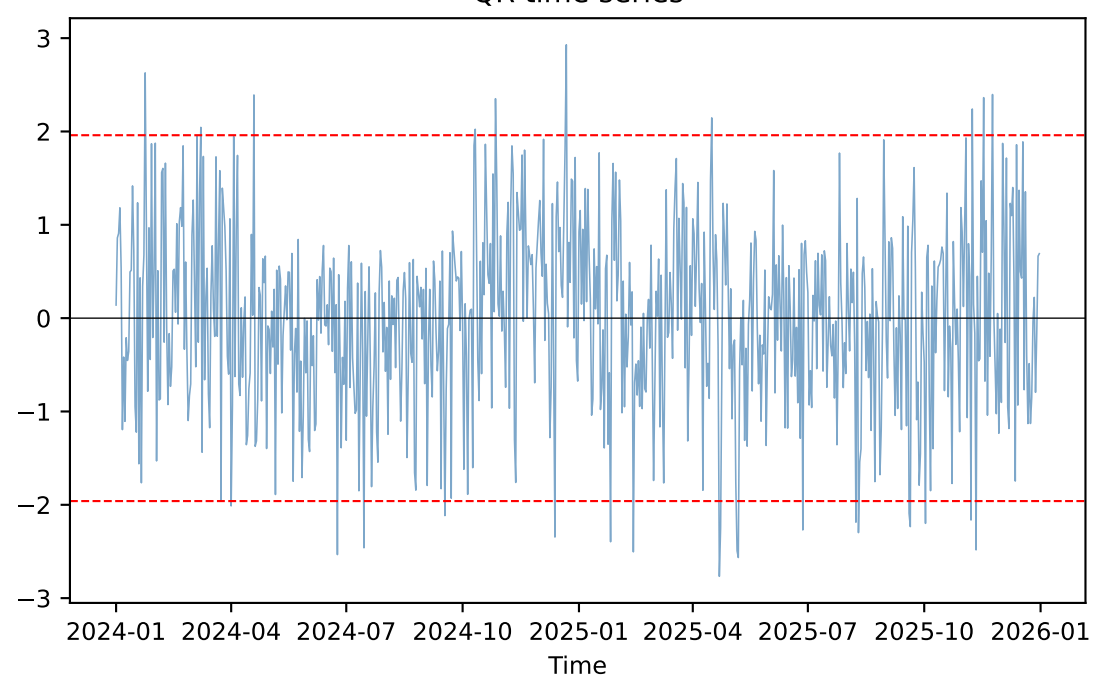
Normal QQ plot



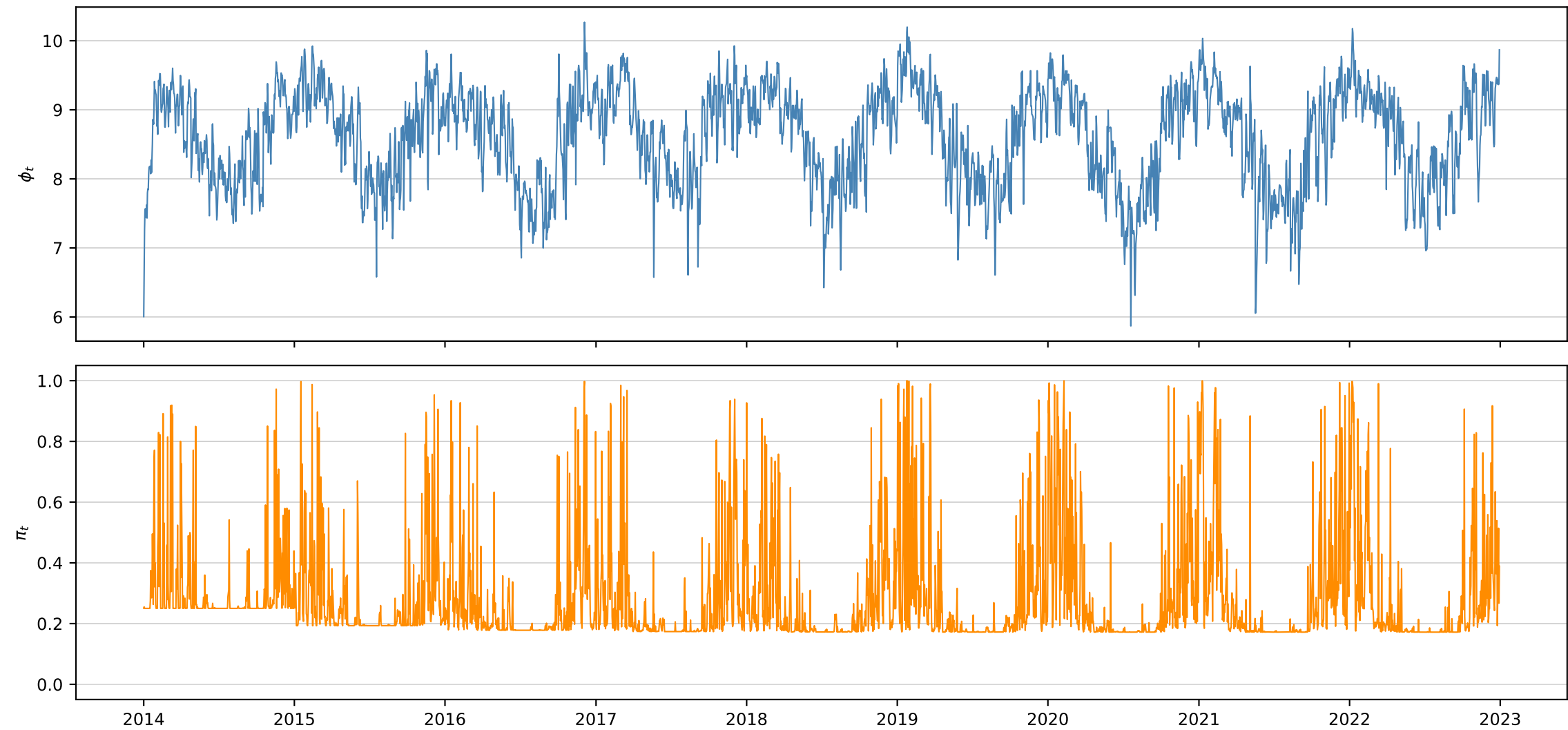
ACF of QR



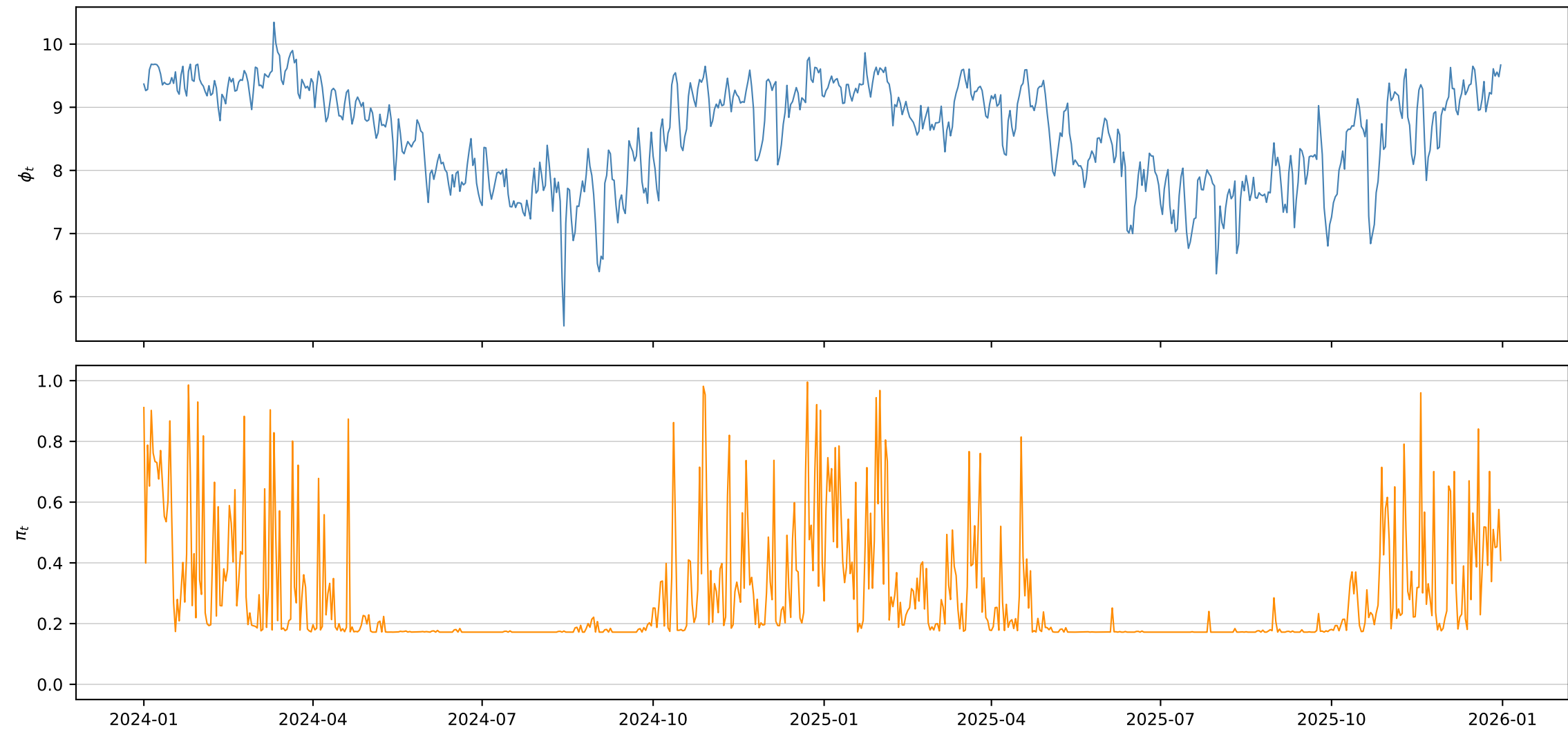
QR time series



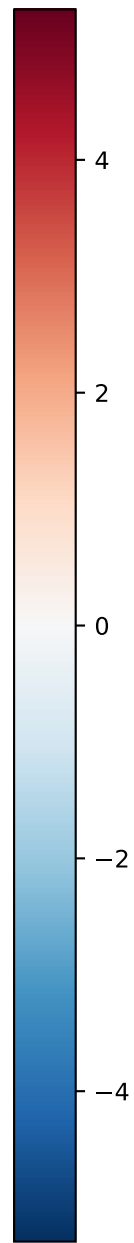
phi_dewpoint_lag1 — filtered states (IS)



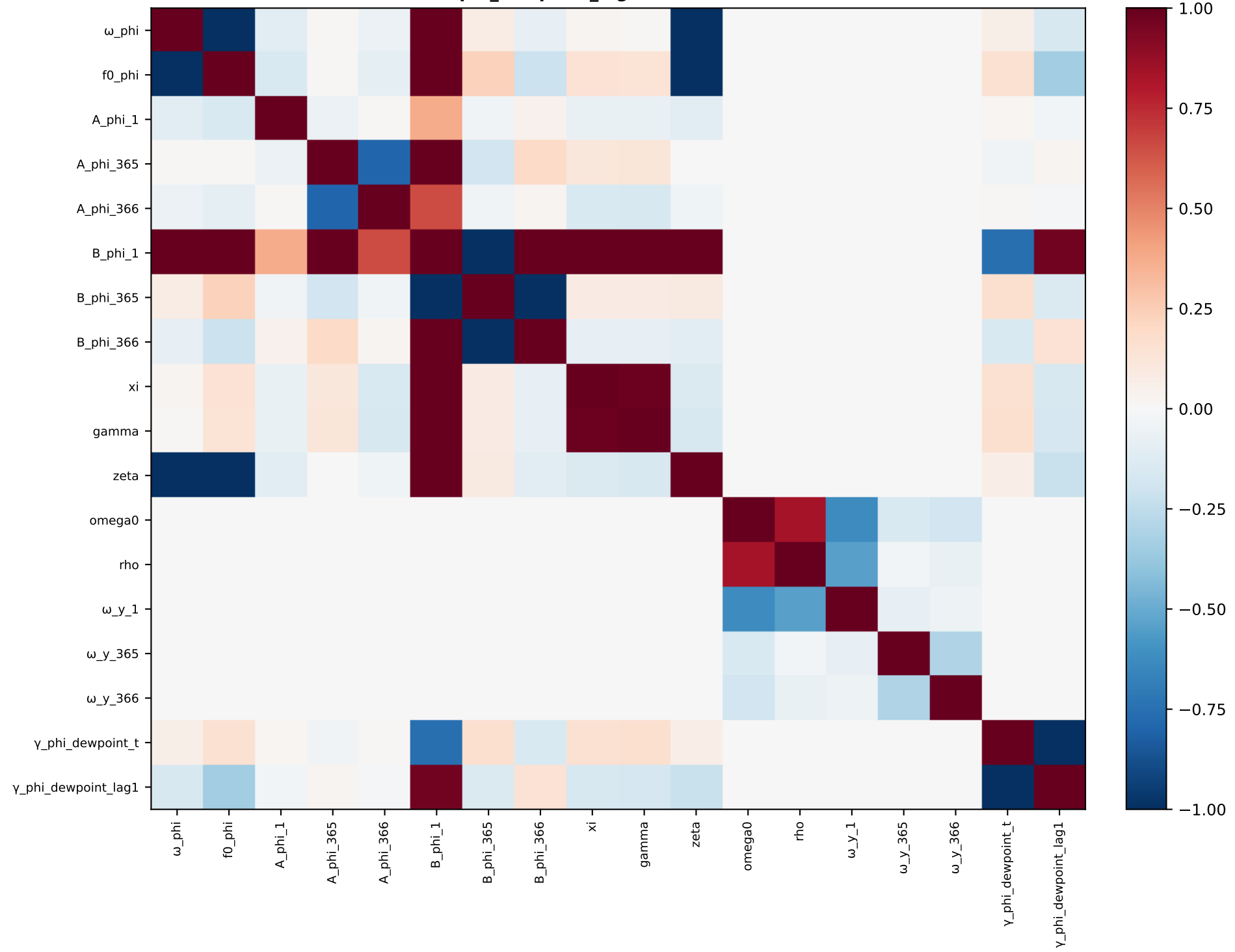
phi_dewpoint_lag1 — filtered states (OOS)



Age	Female	Male
18-24	Low	High
25-34	Low	High
35-44	Low	High
45-54	Low	High
55-64	Low	High
65-74	Low	High
75-84	Low	High
85-94	Low	High
95-104	Low	High
105-114	Low	High

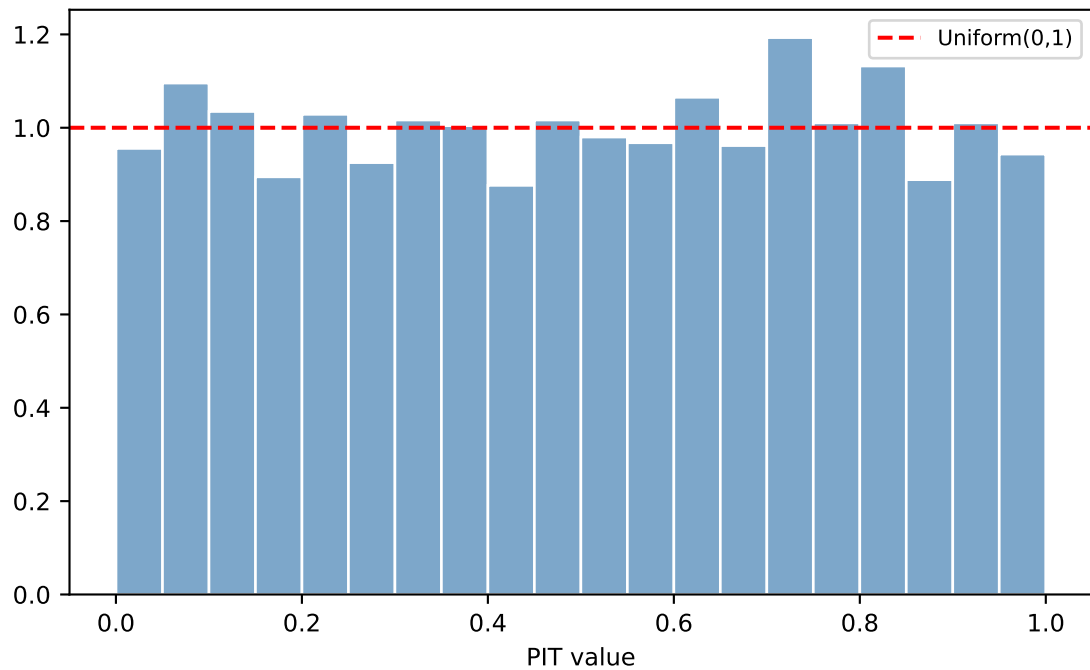


phi_dewpoint_lag1 — correlation

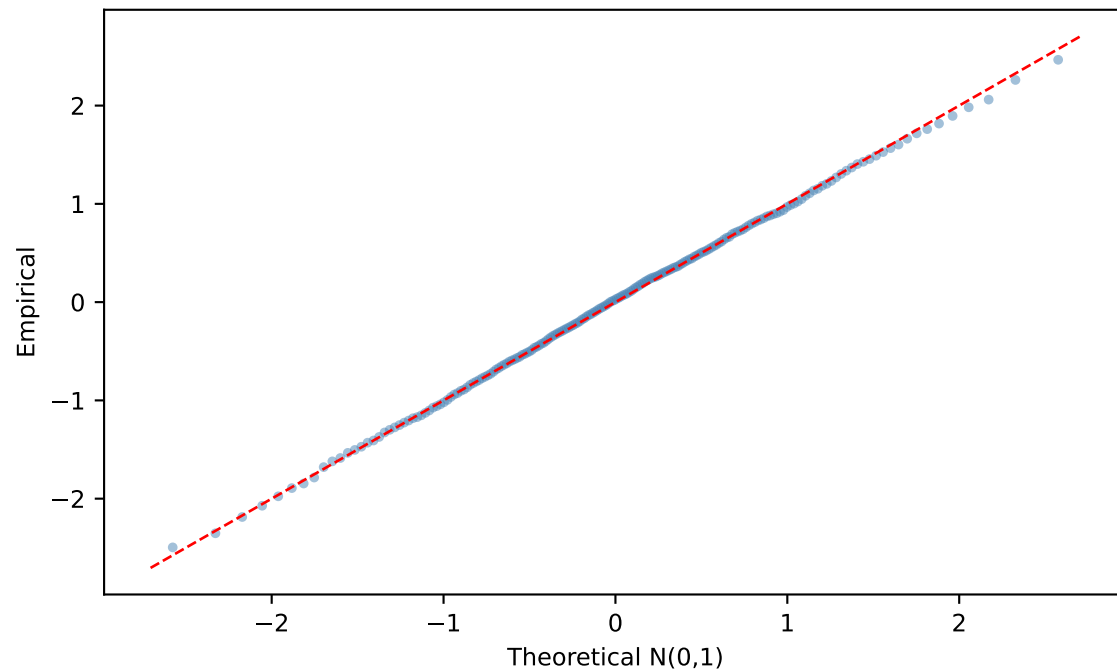


phi_dewpoint_short_lags — IS diagnostics

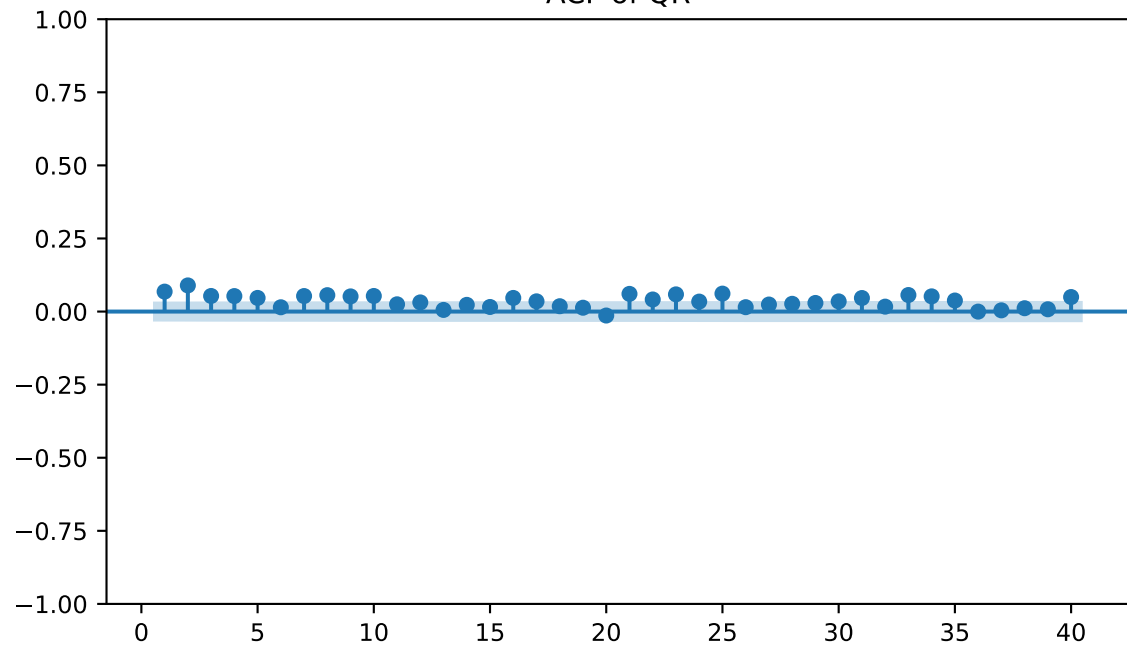
PIT histogram



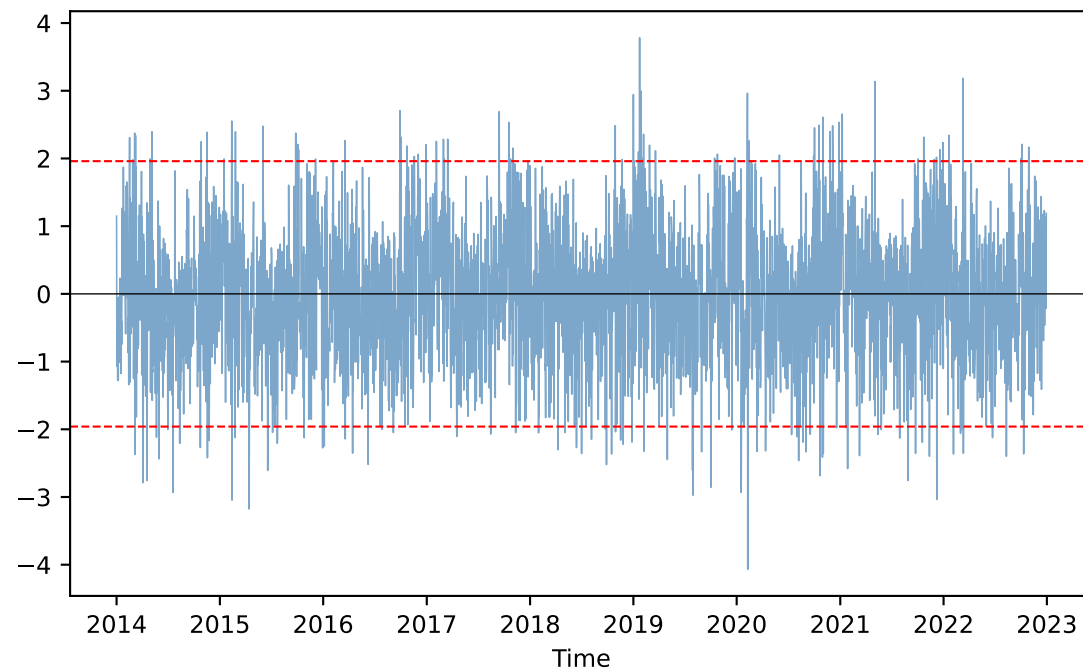
Normal QQ plot



ACF of QR

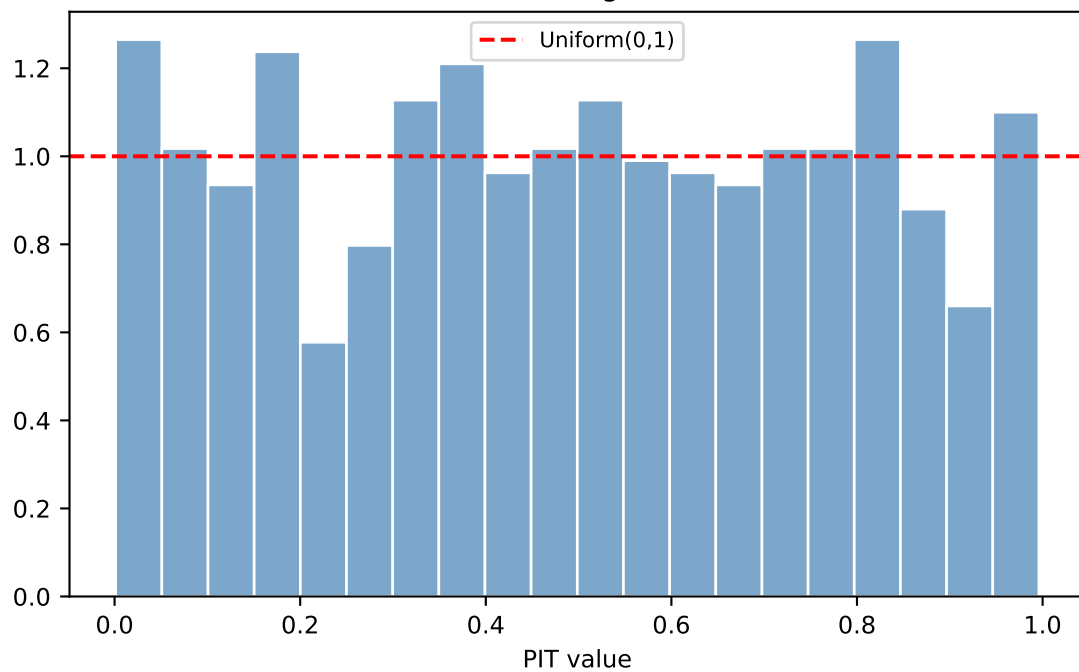


QR time series

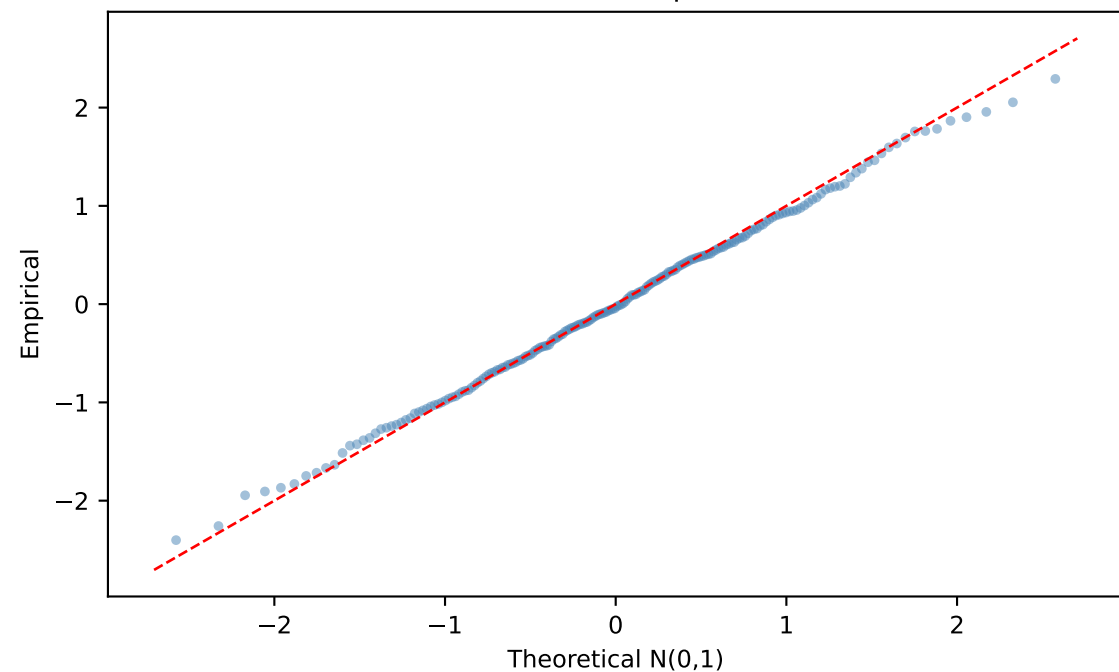


phi_dewpoint_short_lags — OOS diagnostics

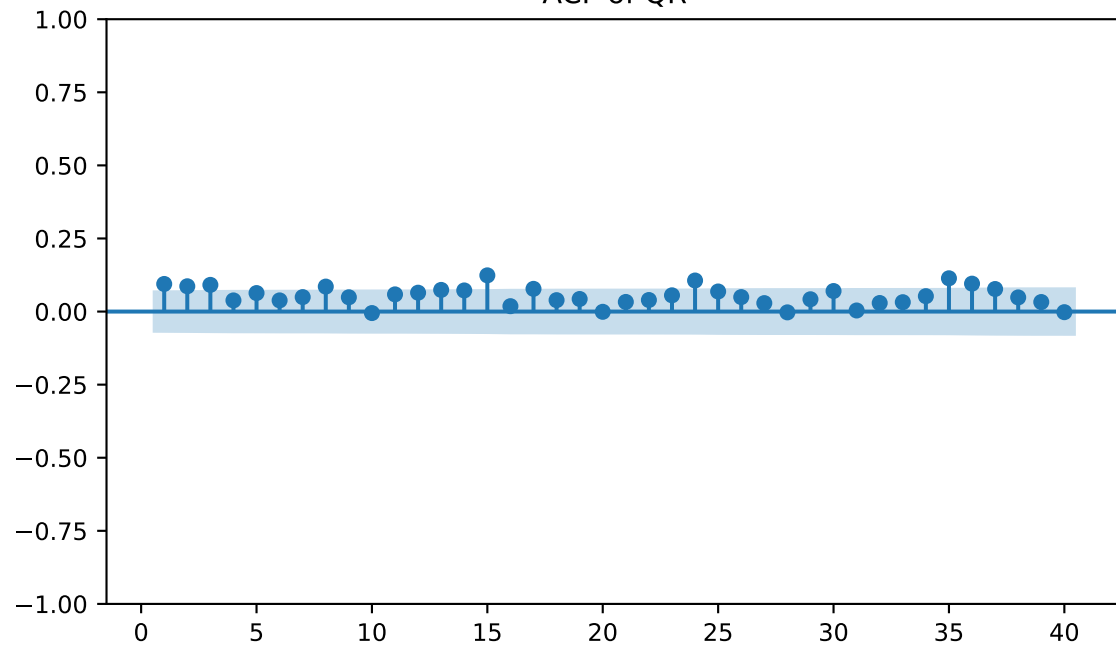
PIT histogram



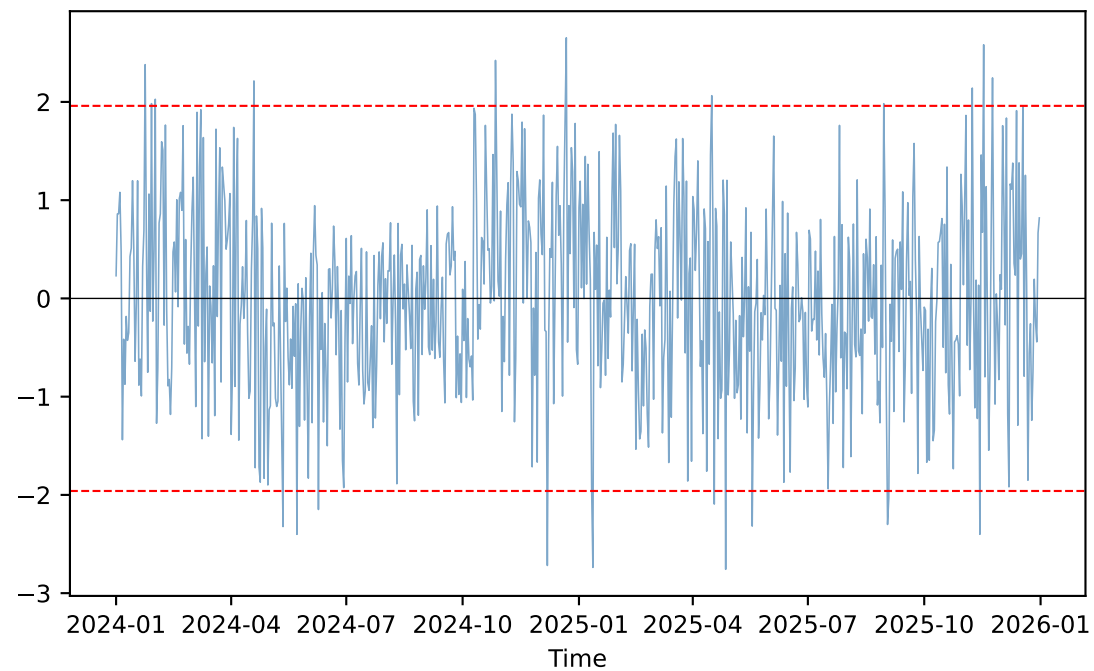
Normal QQ plot



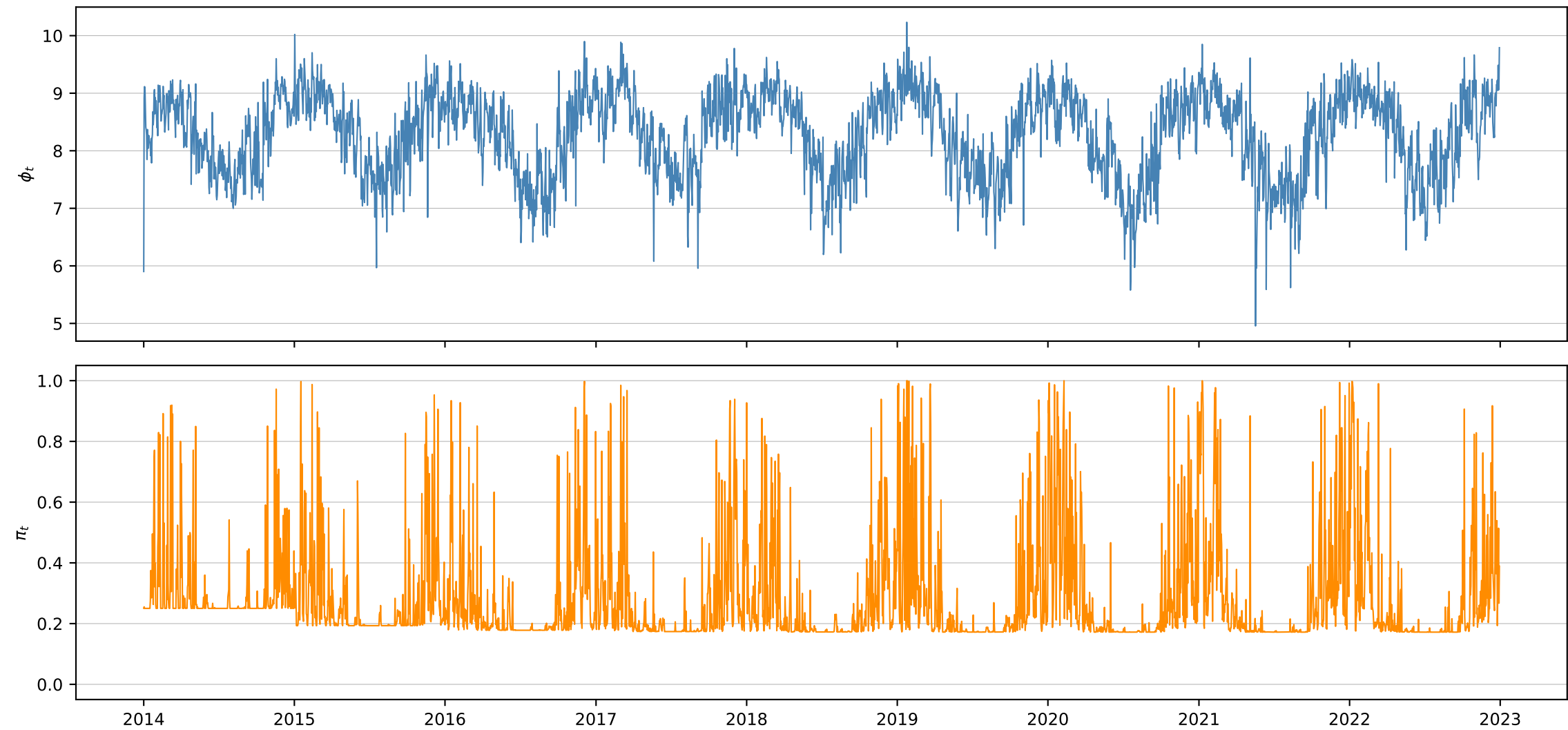
ACF of QR



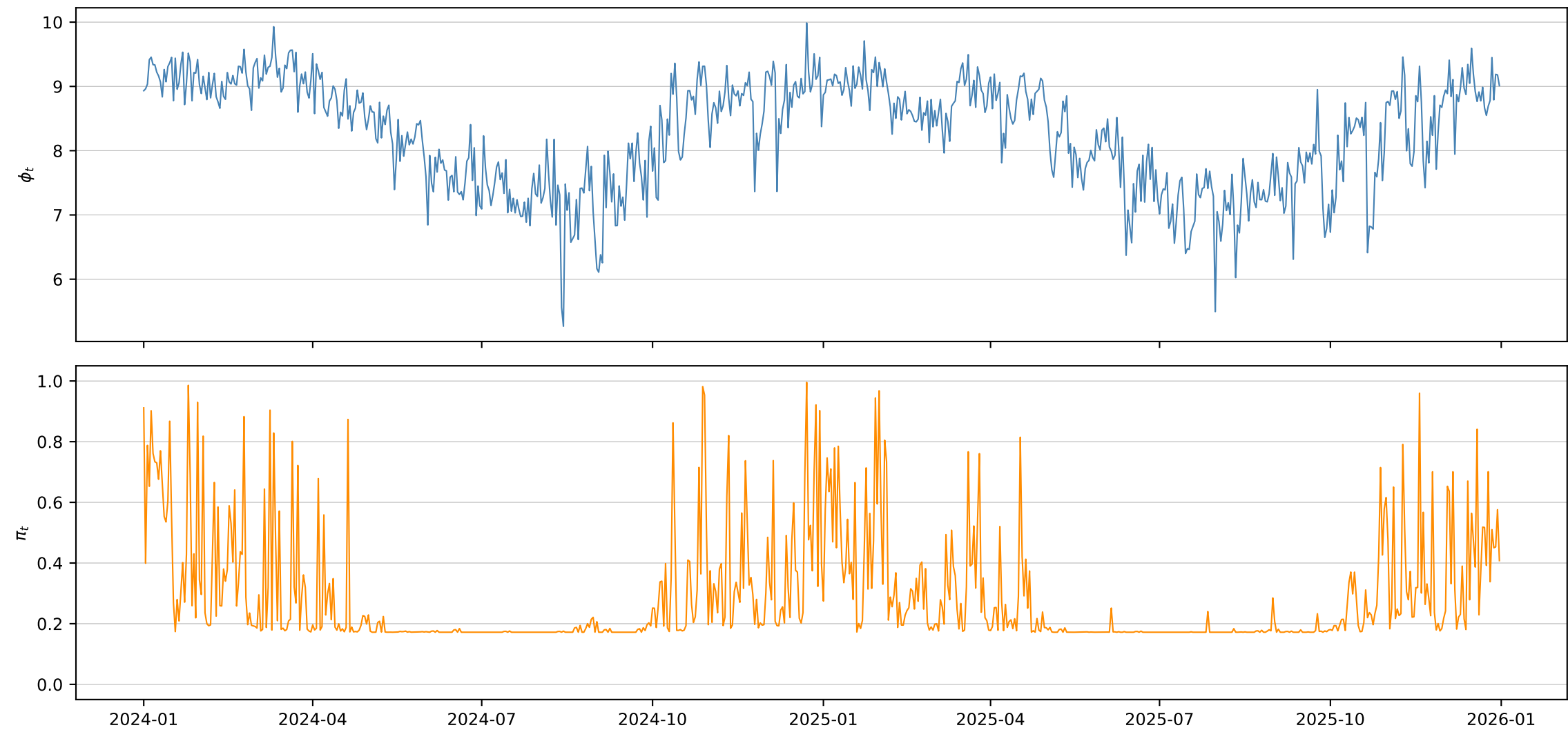
QR time series



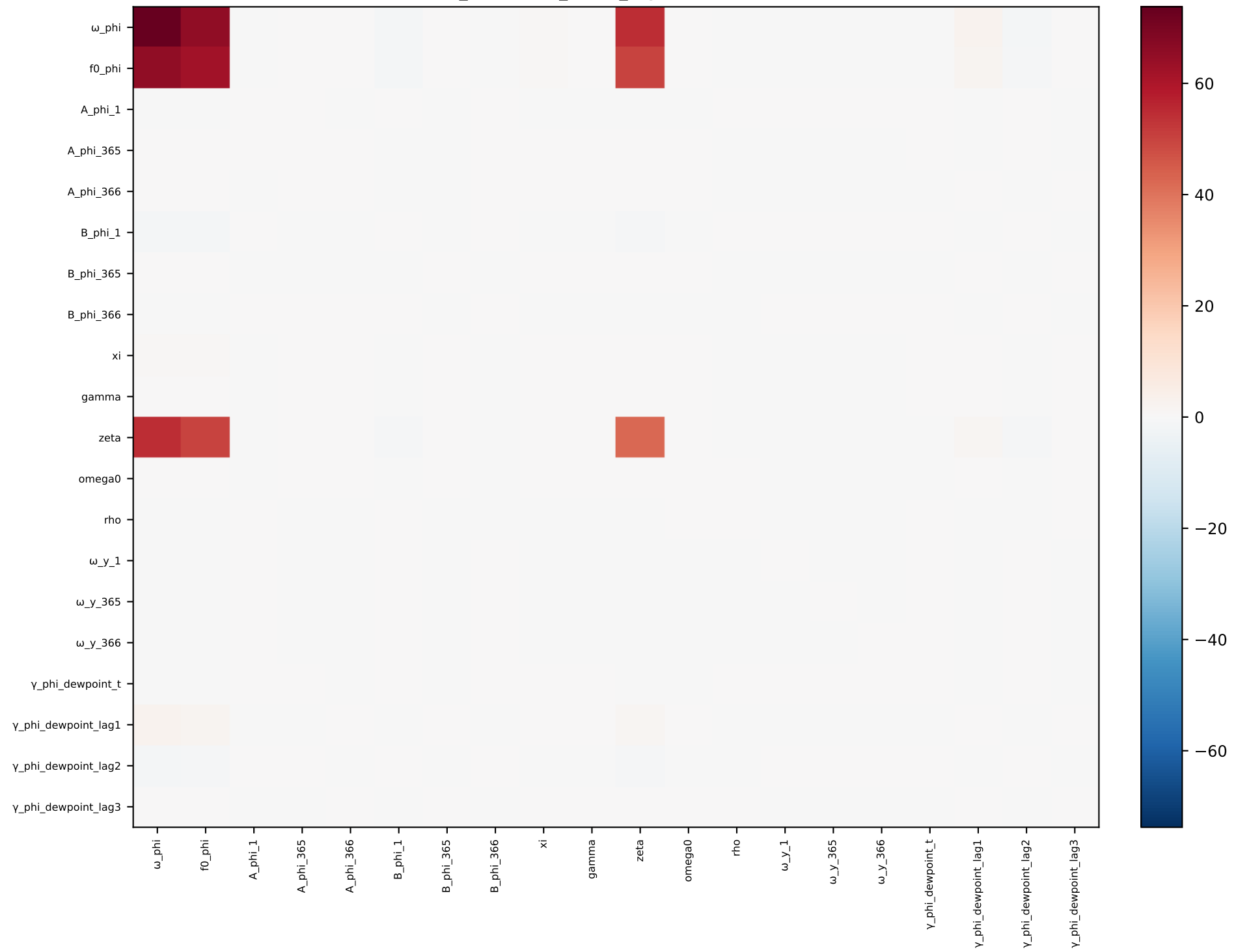
phi_dewpoint_short_lags — filtered states (IS)



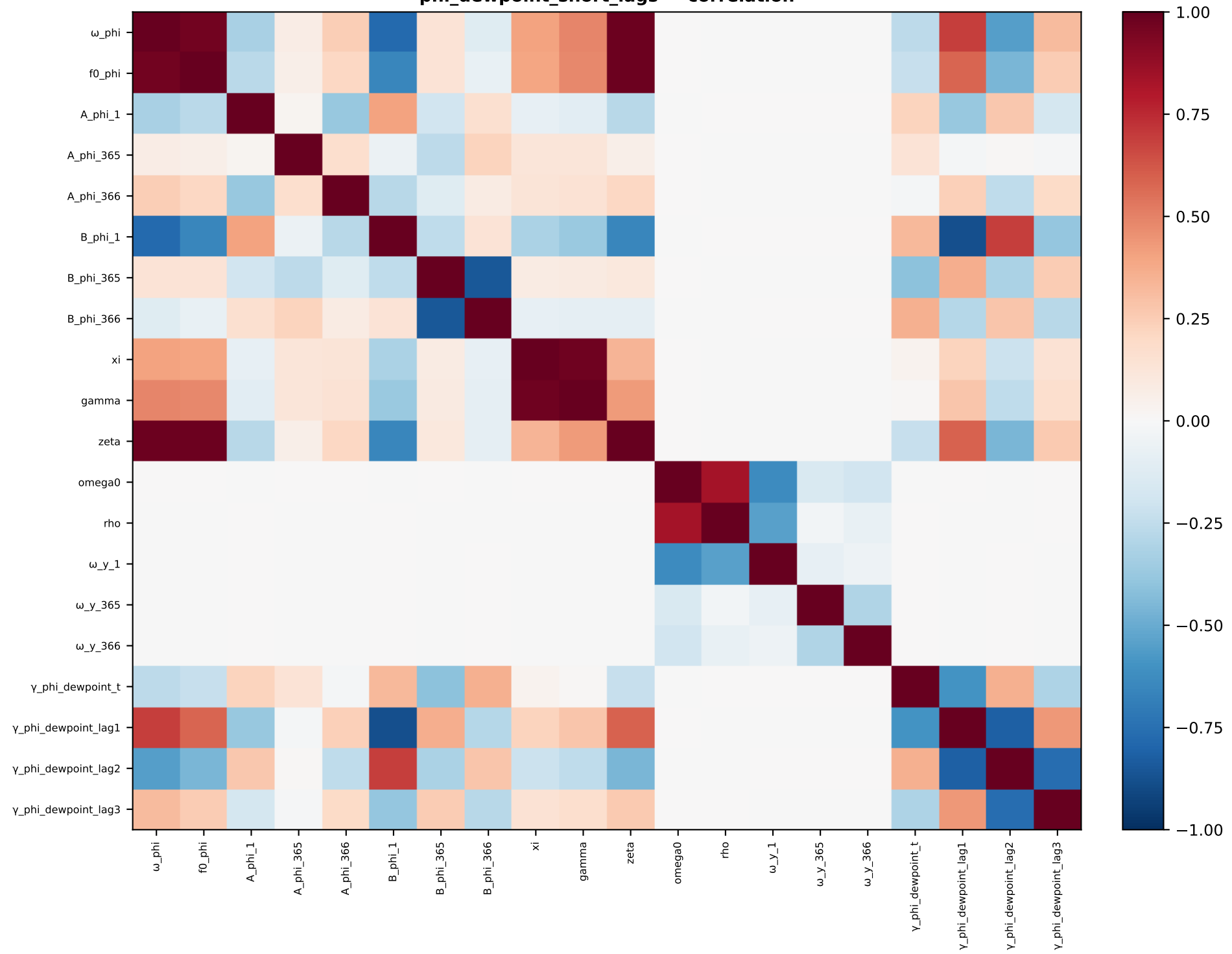
phi_dewpoint_short_lags — filtered states (OOS)



phi_dewpoint_short_lags — covariance

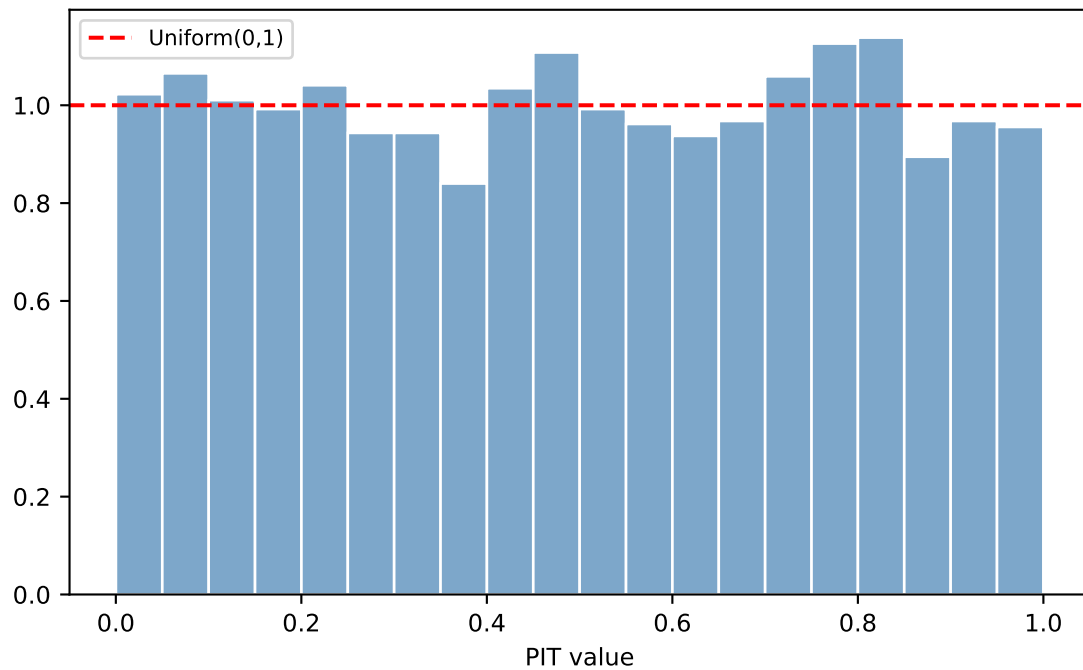


phi_dewpoint_short_lags — correlation

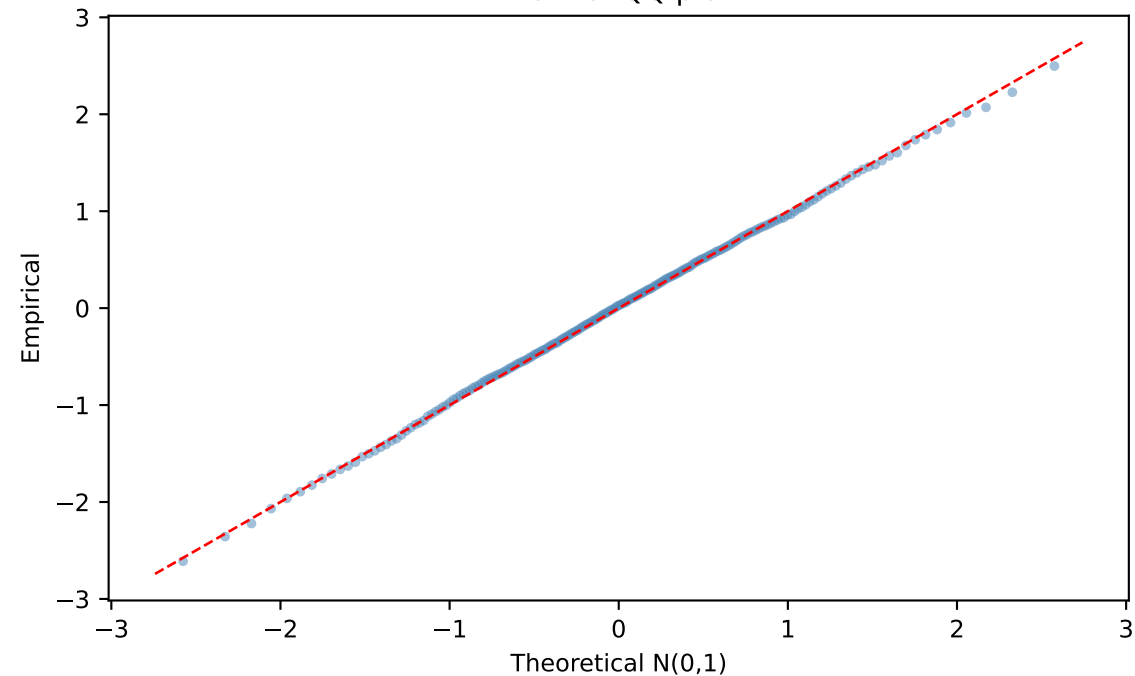


phi_dewpoint_seasonal_lags — IS diagnostics

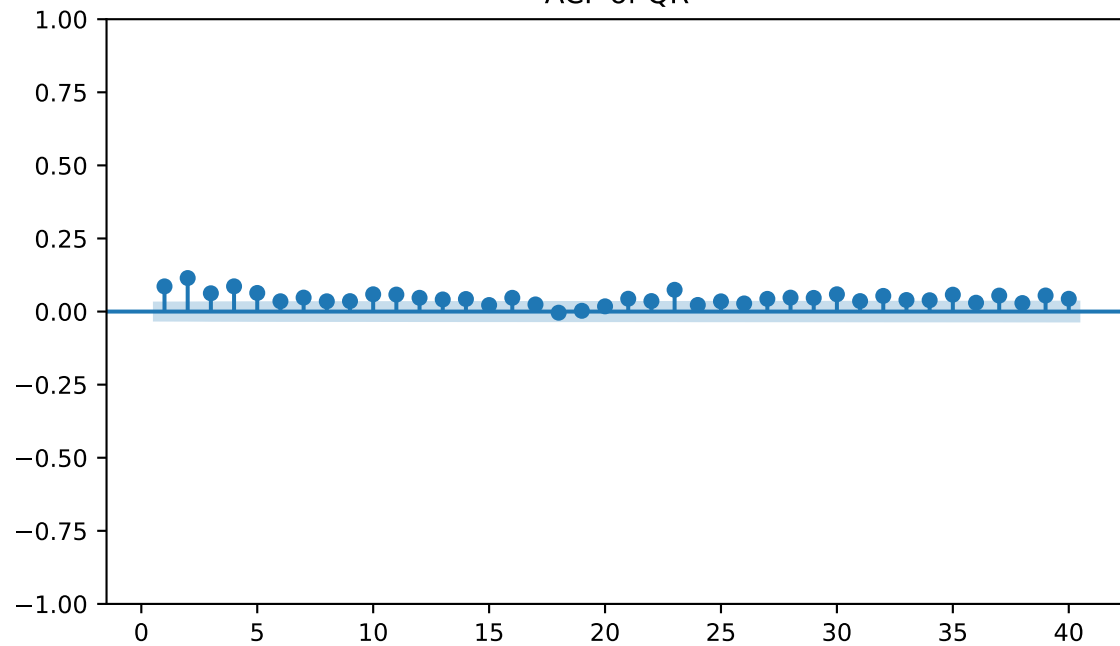
PIT histogram



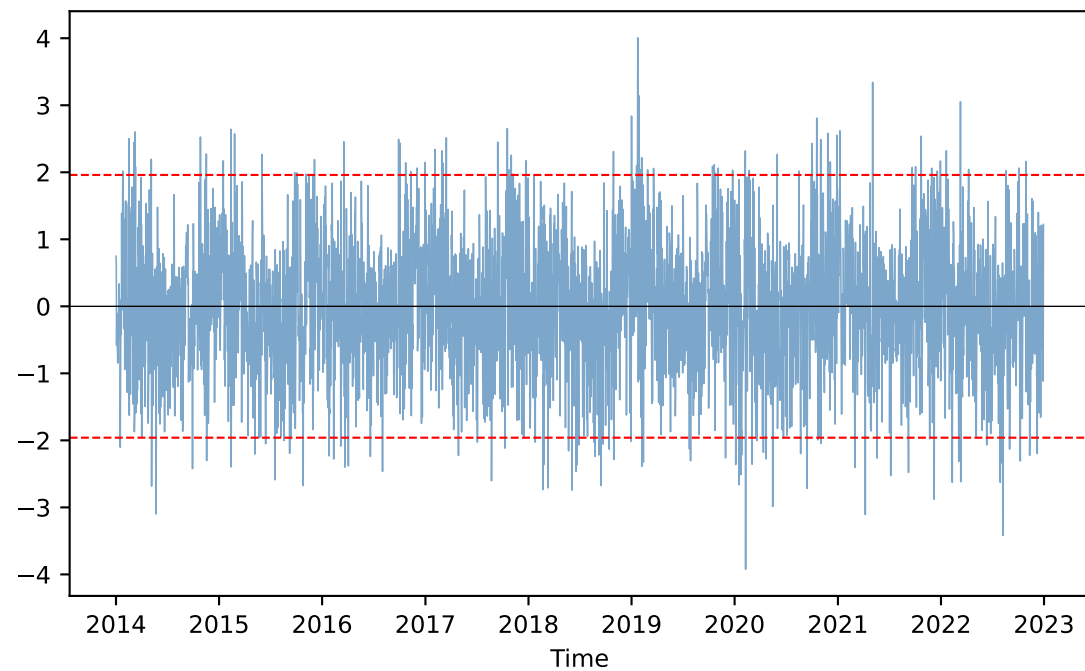
Normal QQ plot



ACF of QR

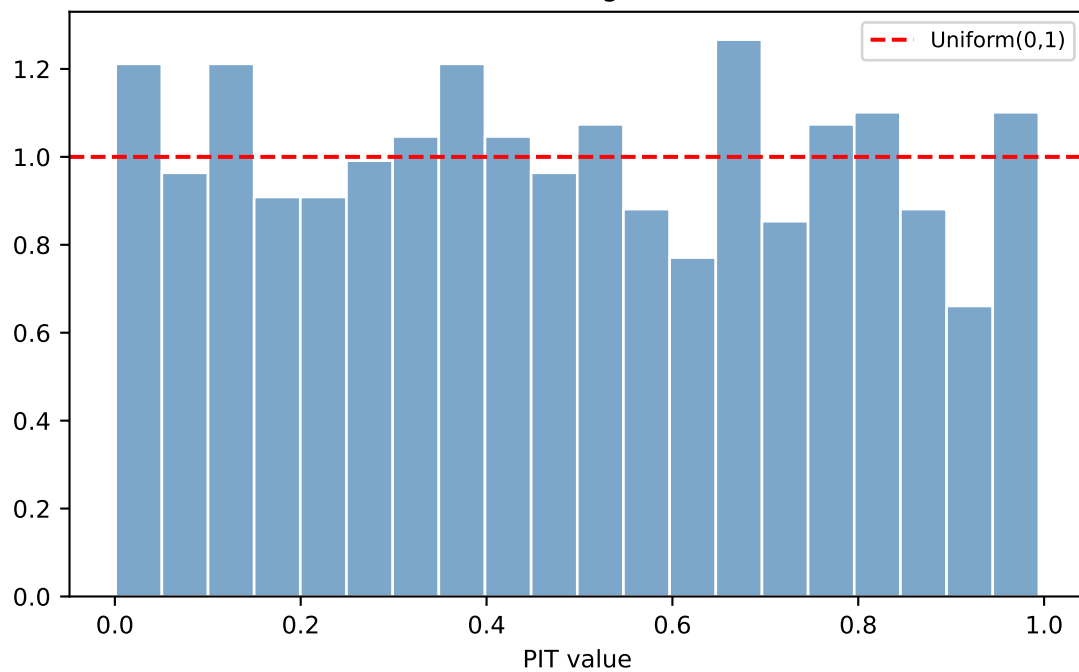


QR time series

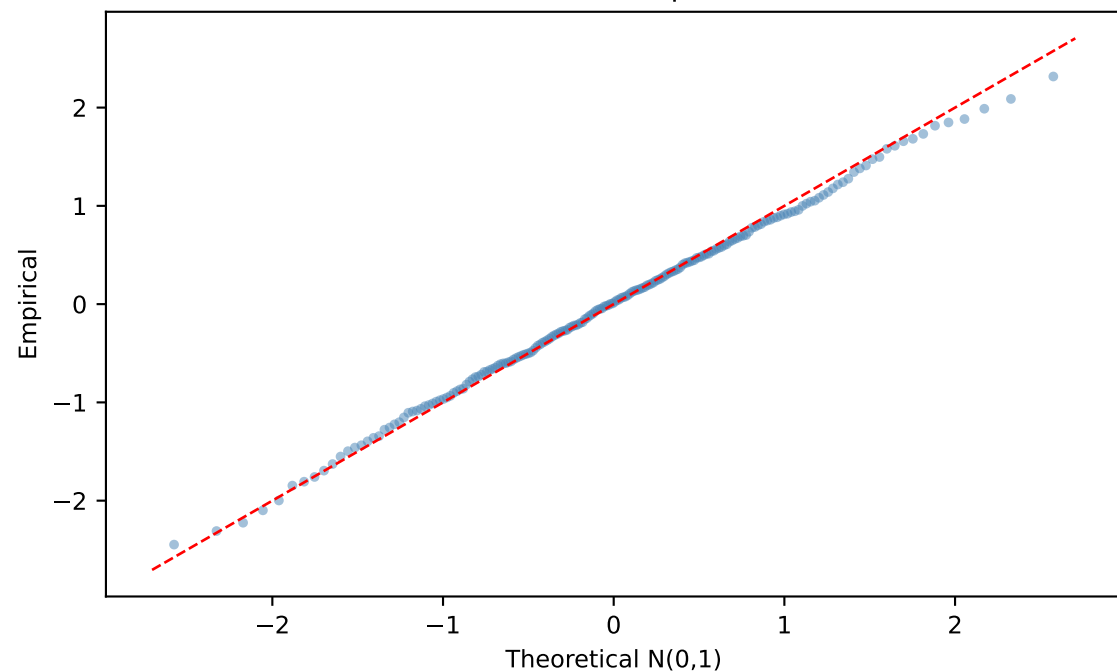


phi_dewpoint_seasonal_lags — OOS diagnostics

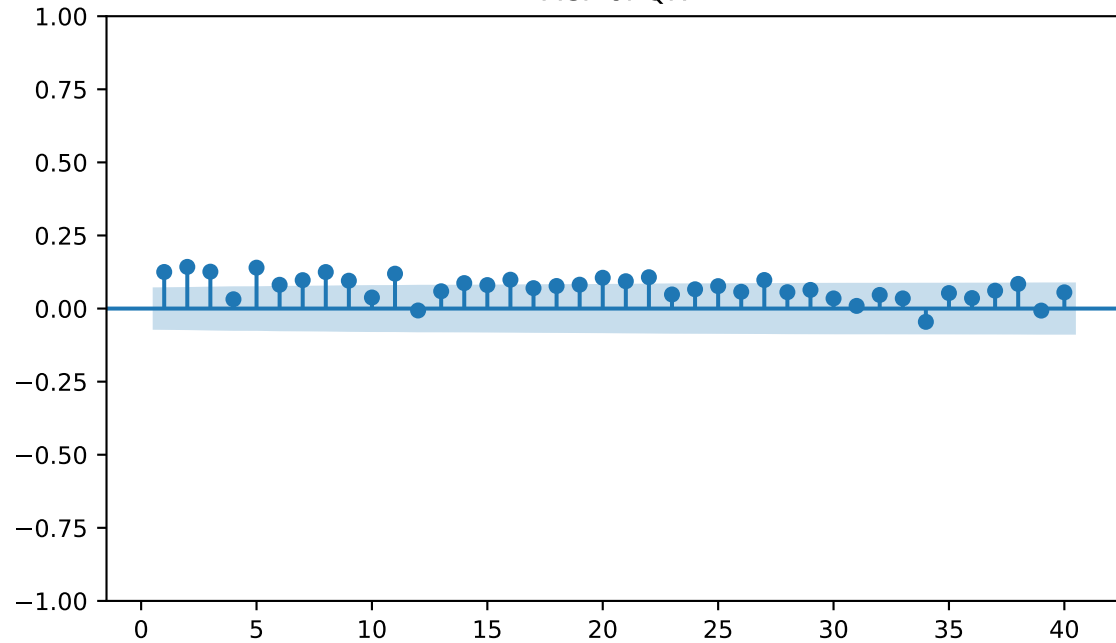
PIT histogram



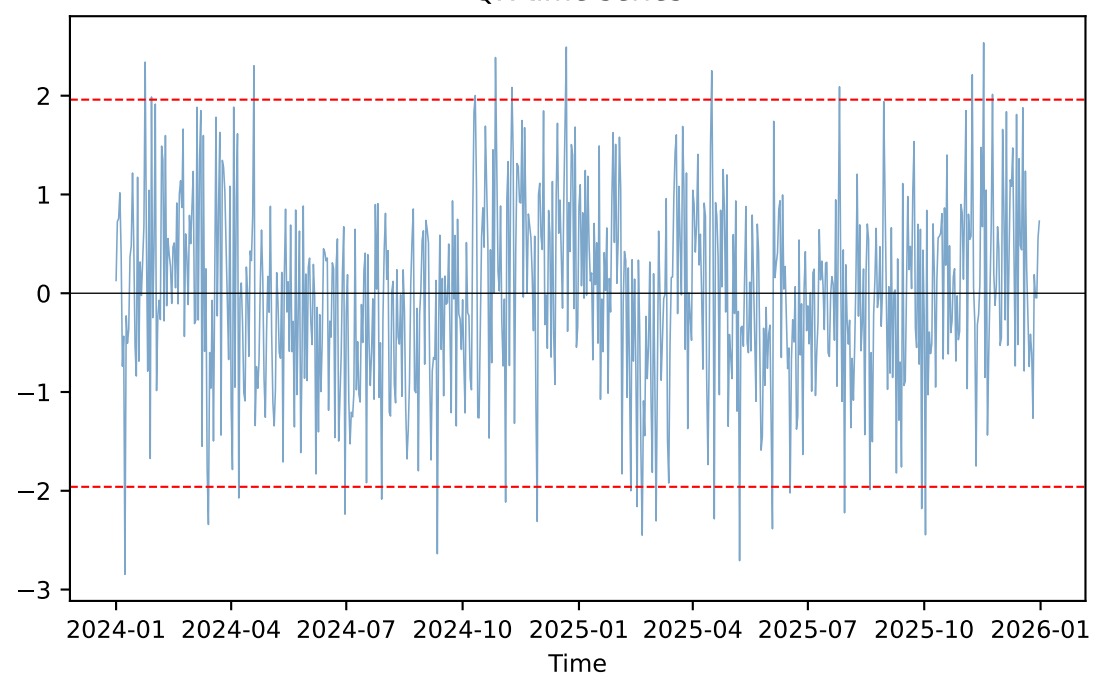
Normal QQ plot



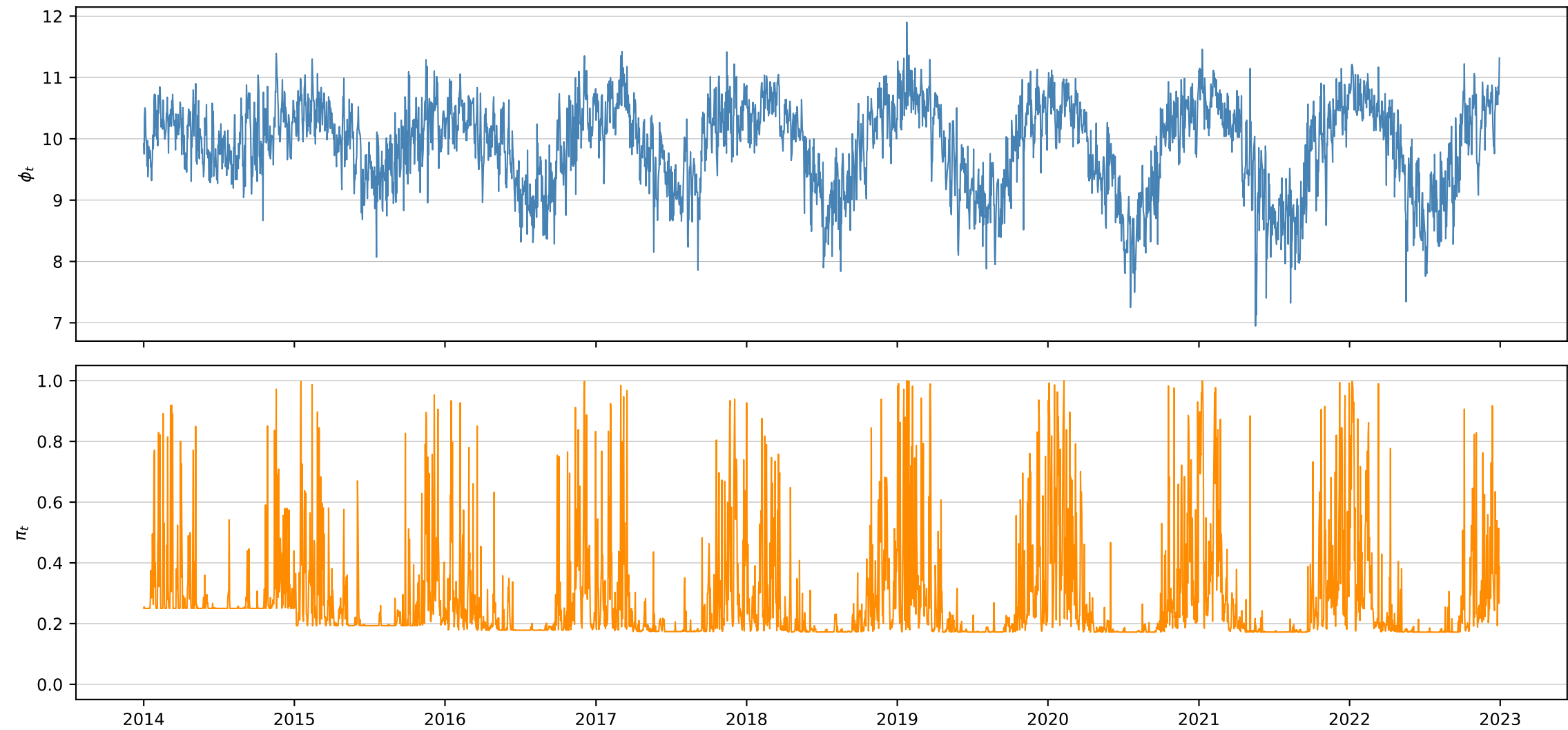
ACF of QR



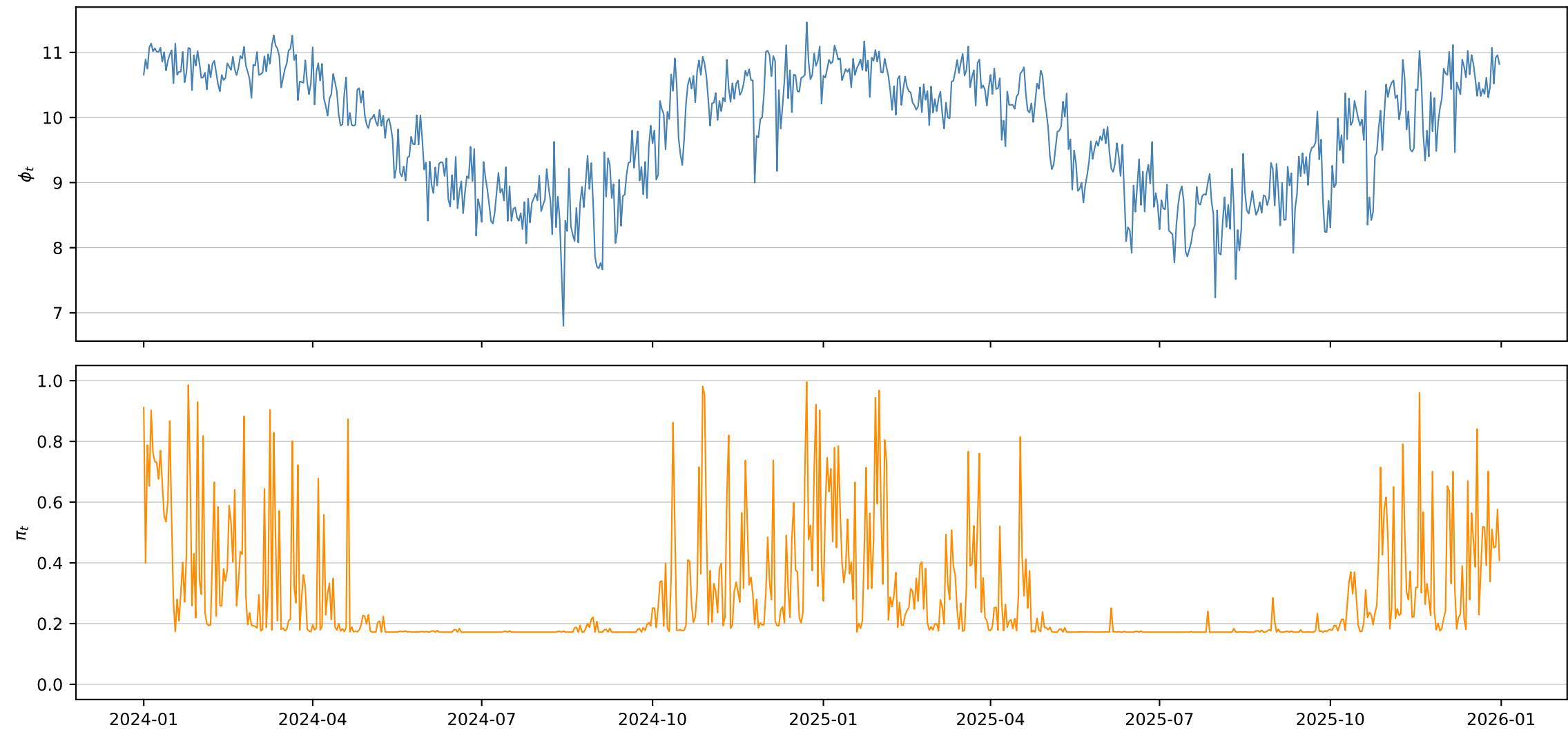
QR time series



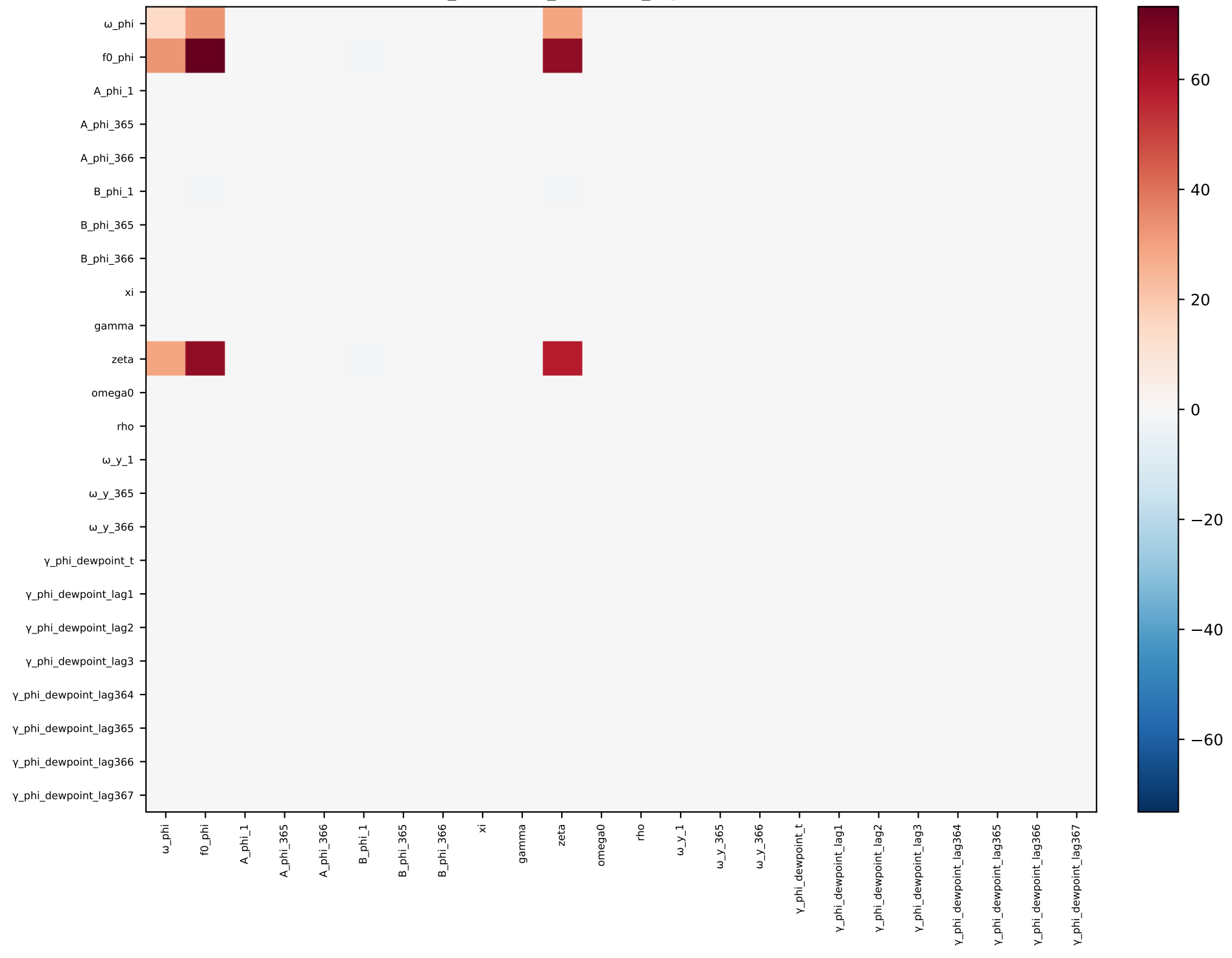
phi_dewpoint_seasonal_lags — filtered states (IS)



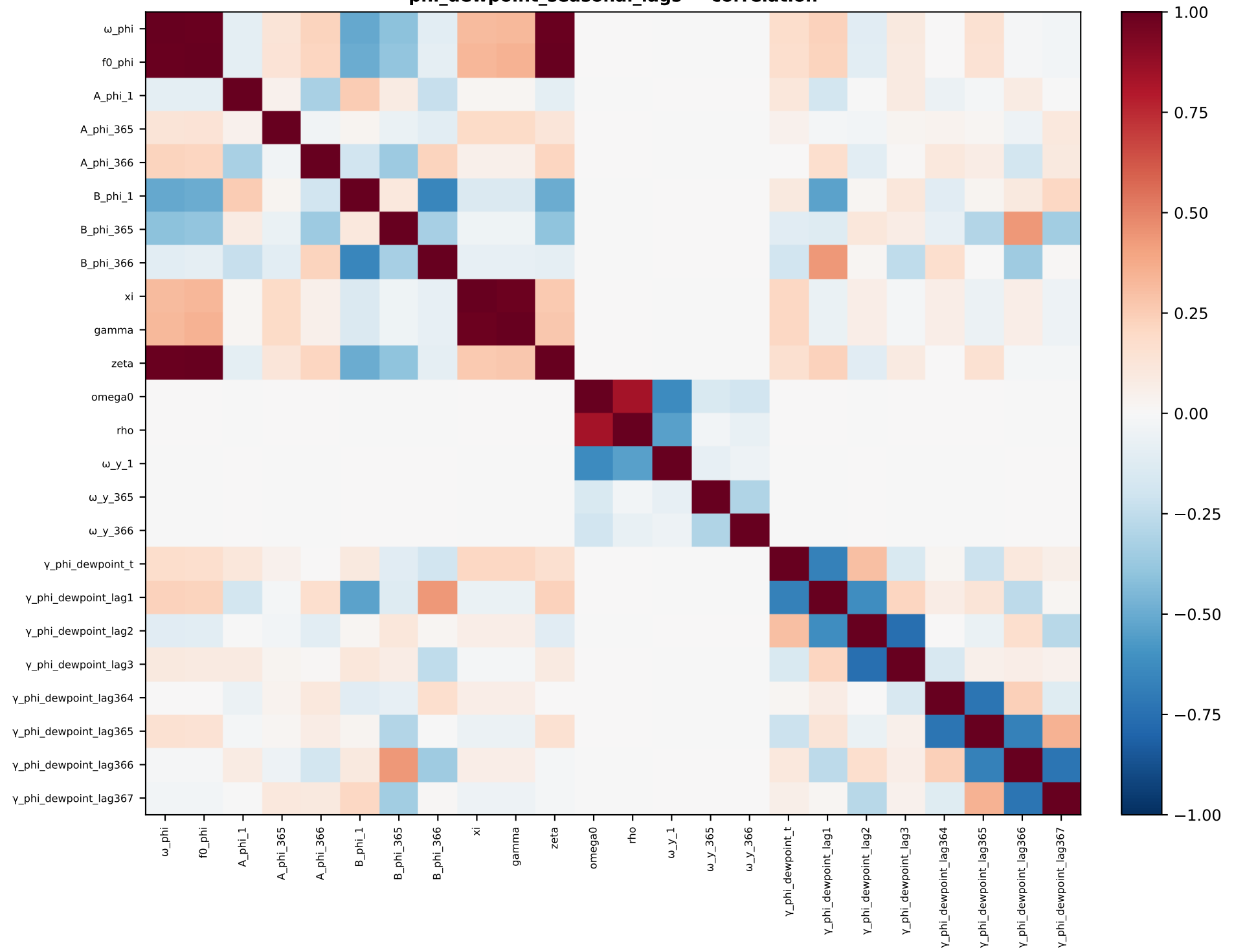
phi_dewpoint_seasonal_lags — filtered states (OOS)



phi_dewpoint_seasonal_lags — covariance

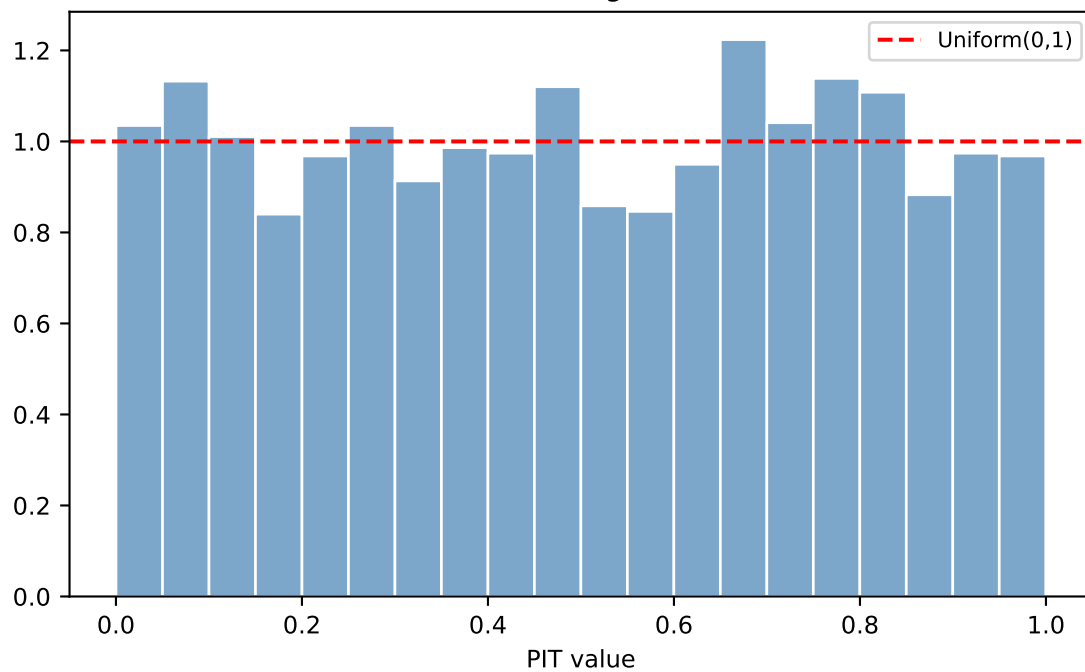


phi_dewpoint_seasonal_lags — correlation

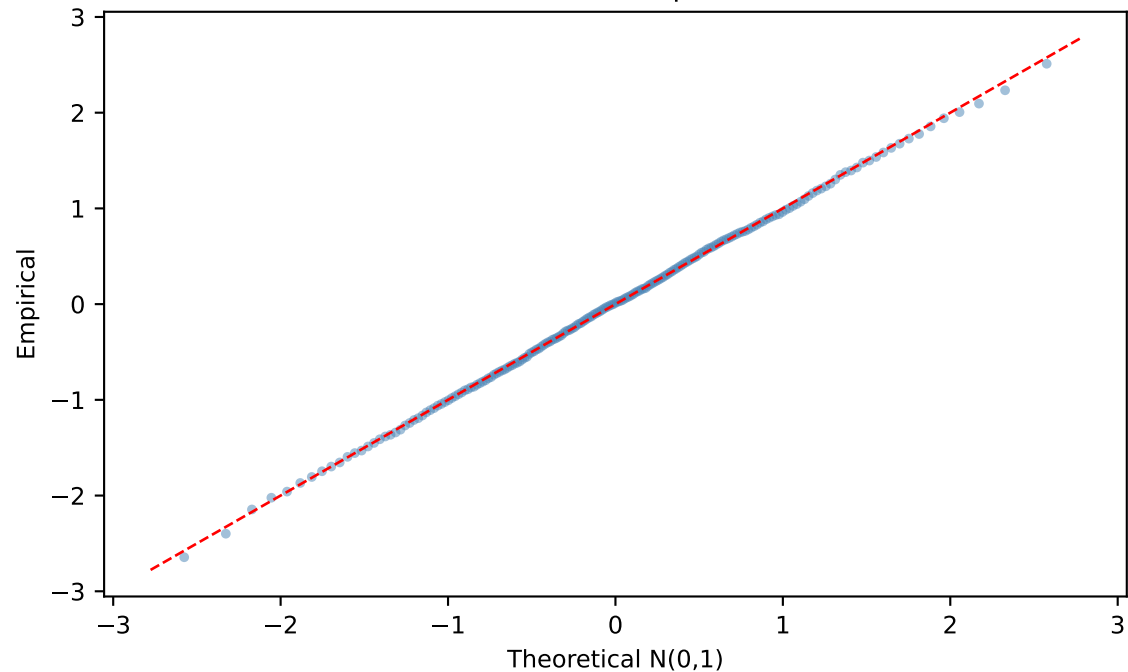


phi_temperature_t — IS diagnostics

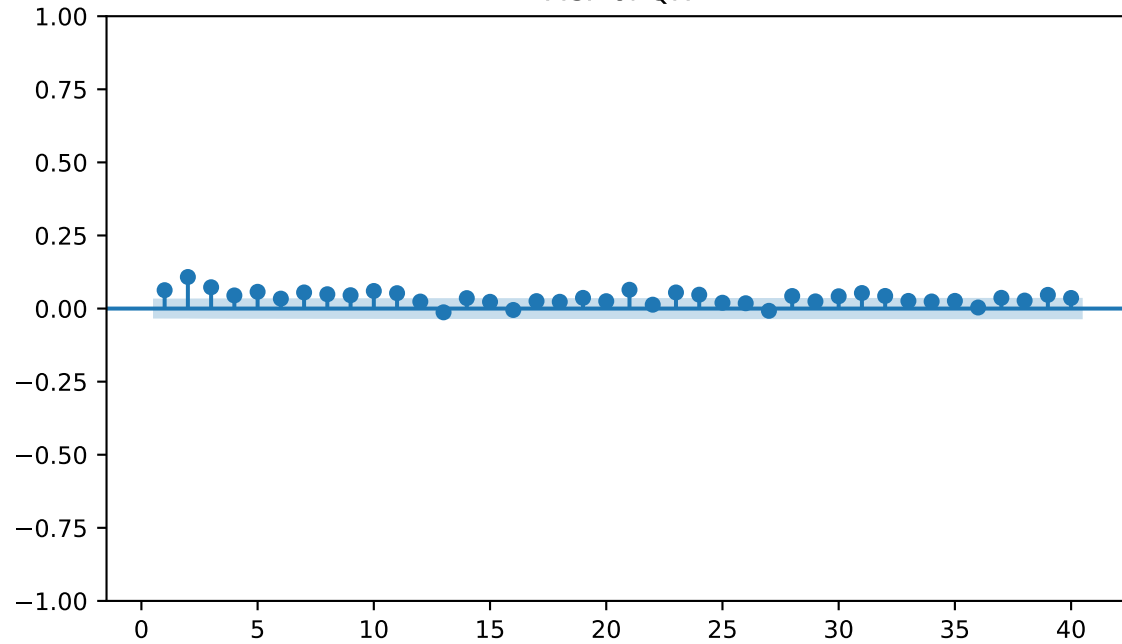
PIT histogram



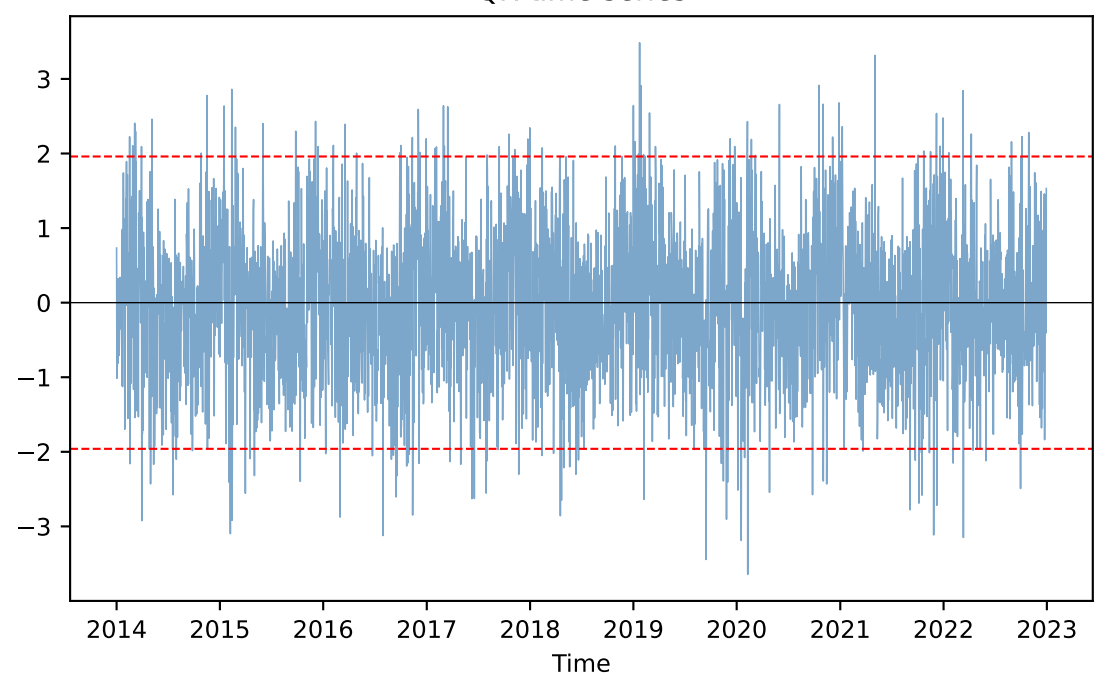
Normal QQ plot



ACF of QR

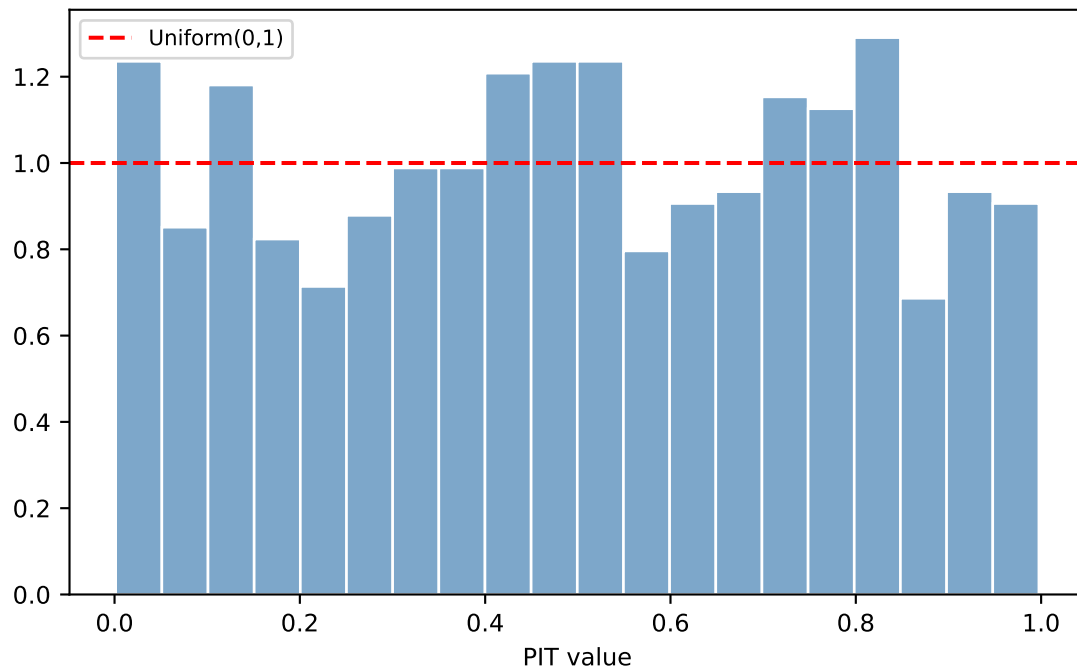


QR time series

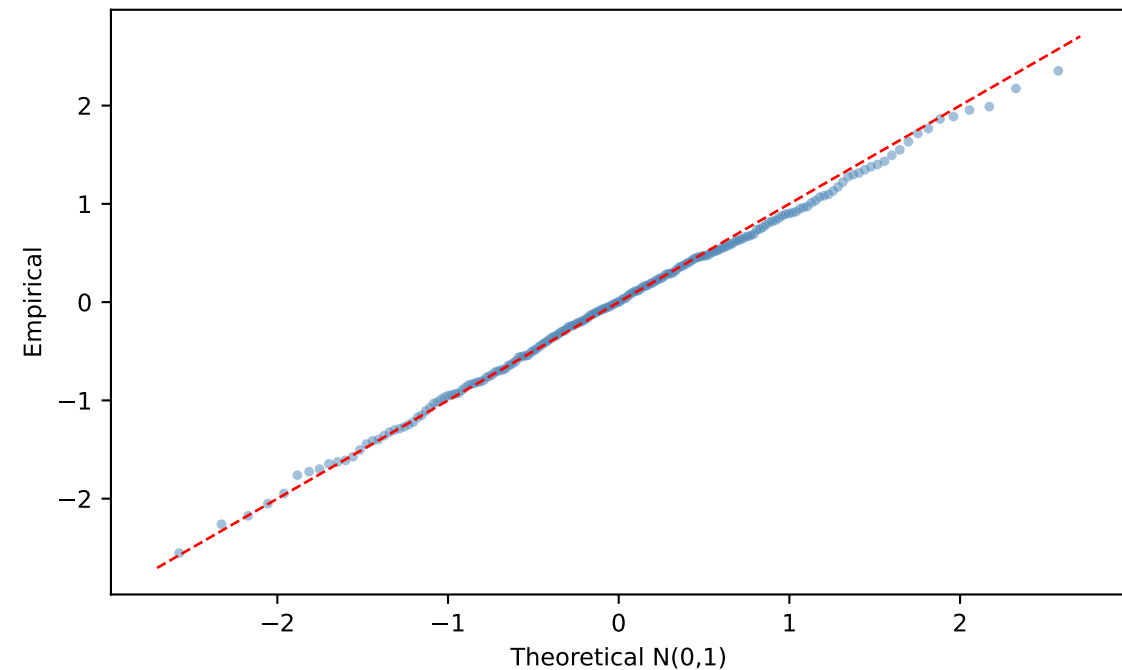


phi_temperature_t — OOS diagnostics

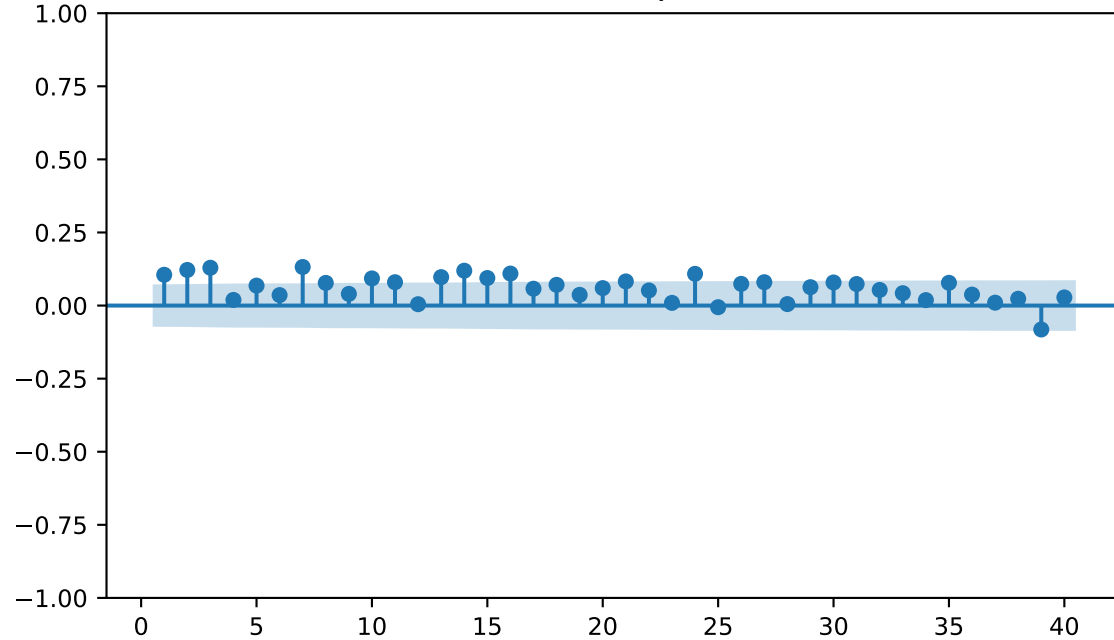
PIT histogram



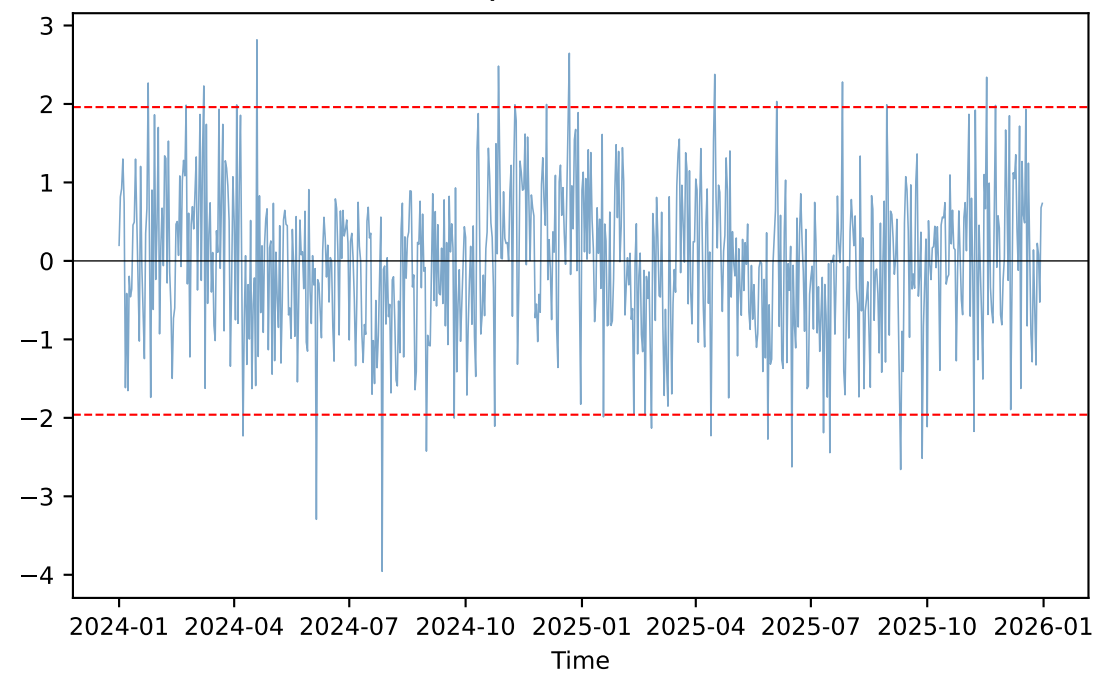
Normal QQ plot



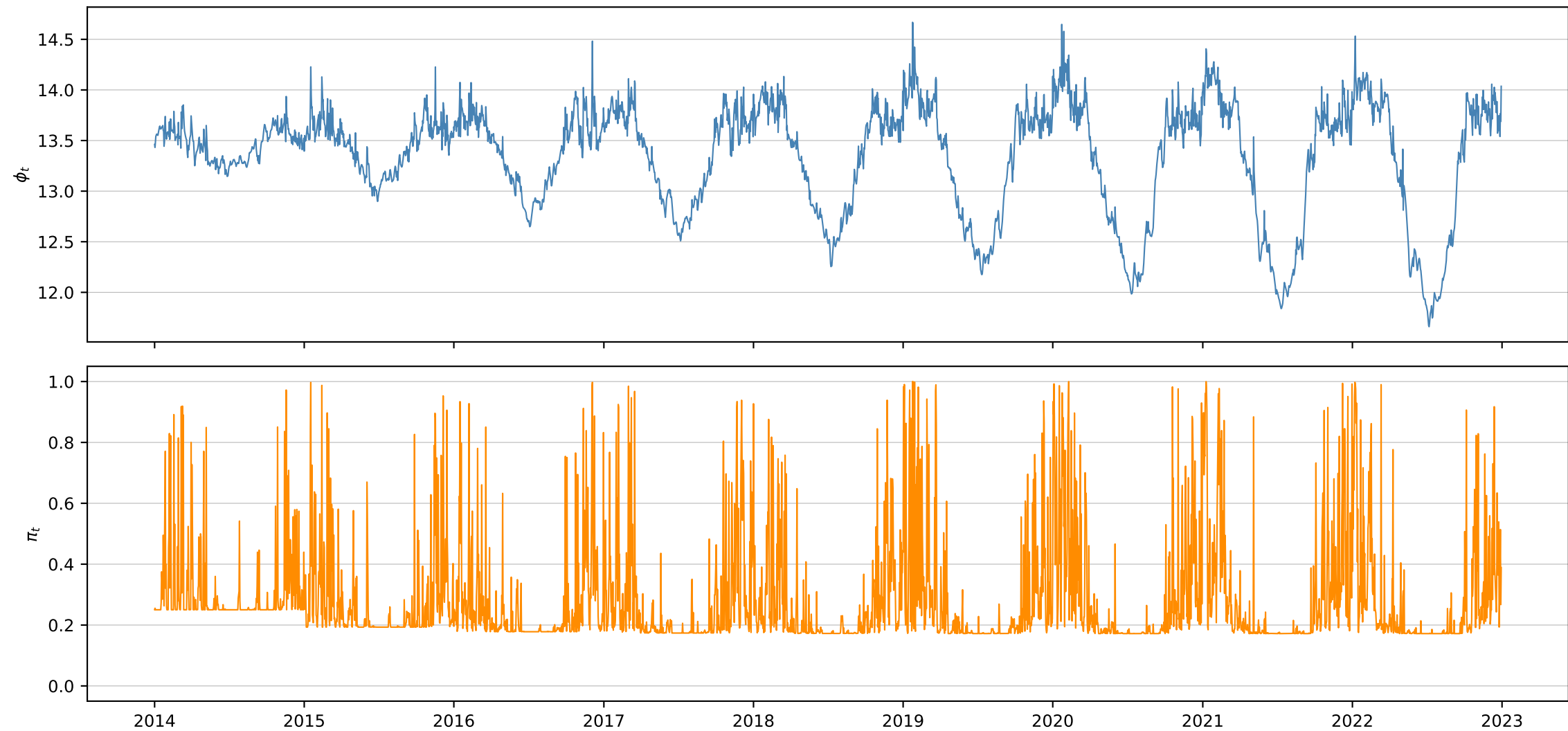
ACF of QR



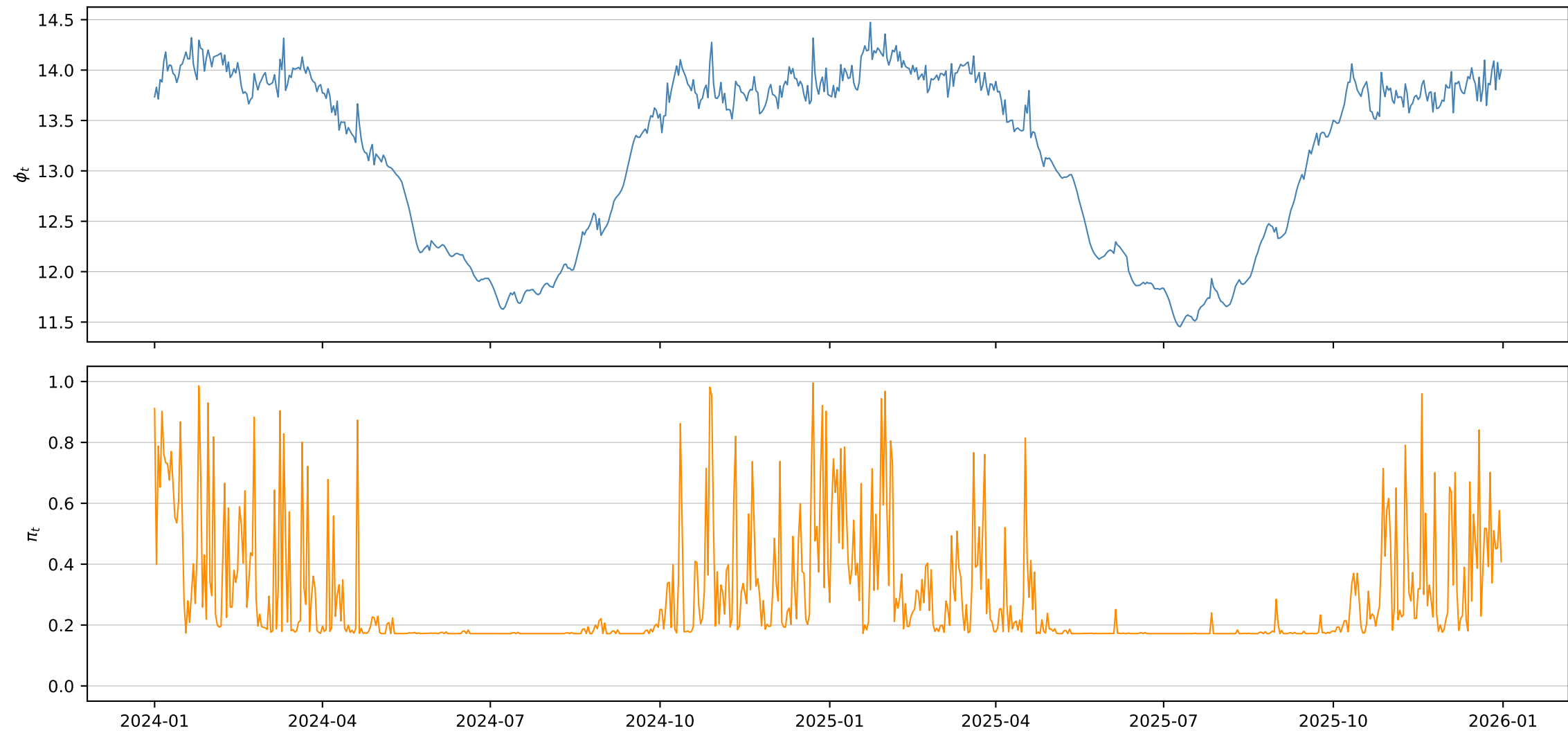
QR time series



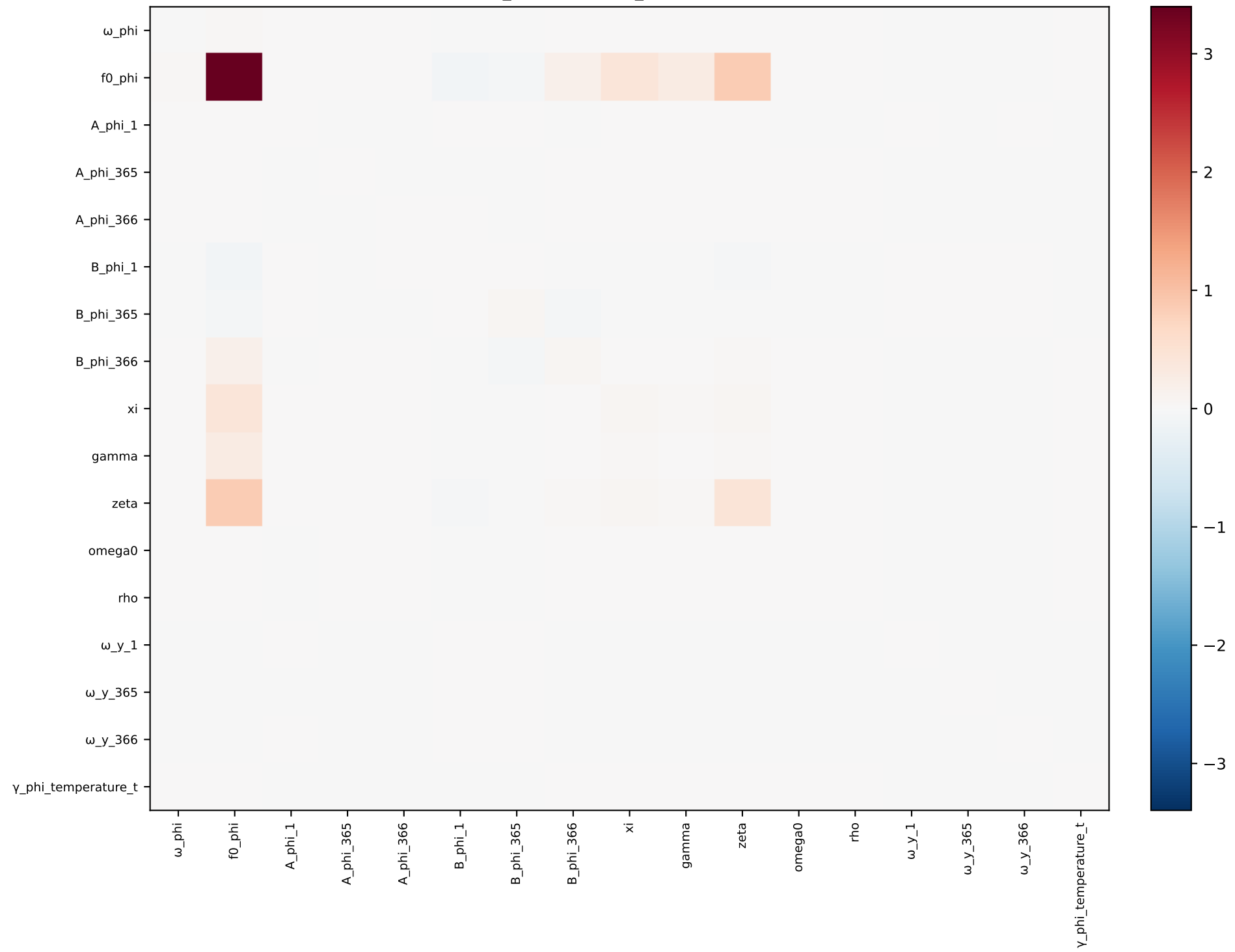
phi_temperature_t — filtered states (IS)



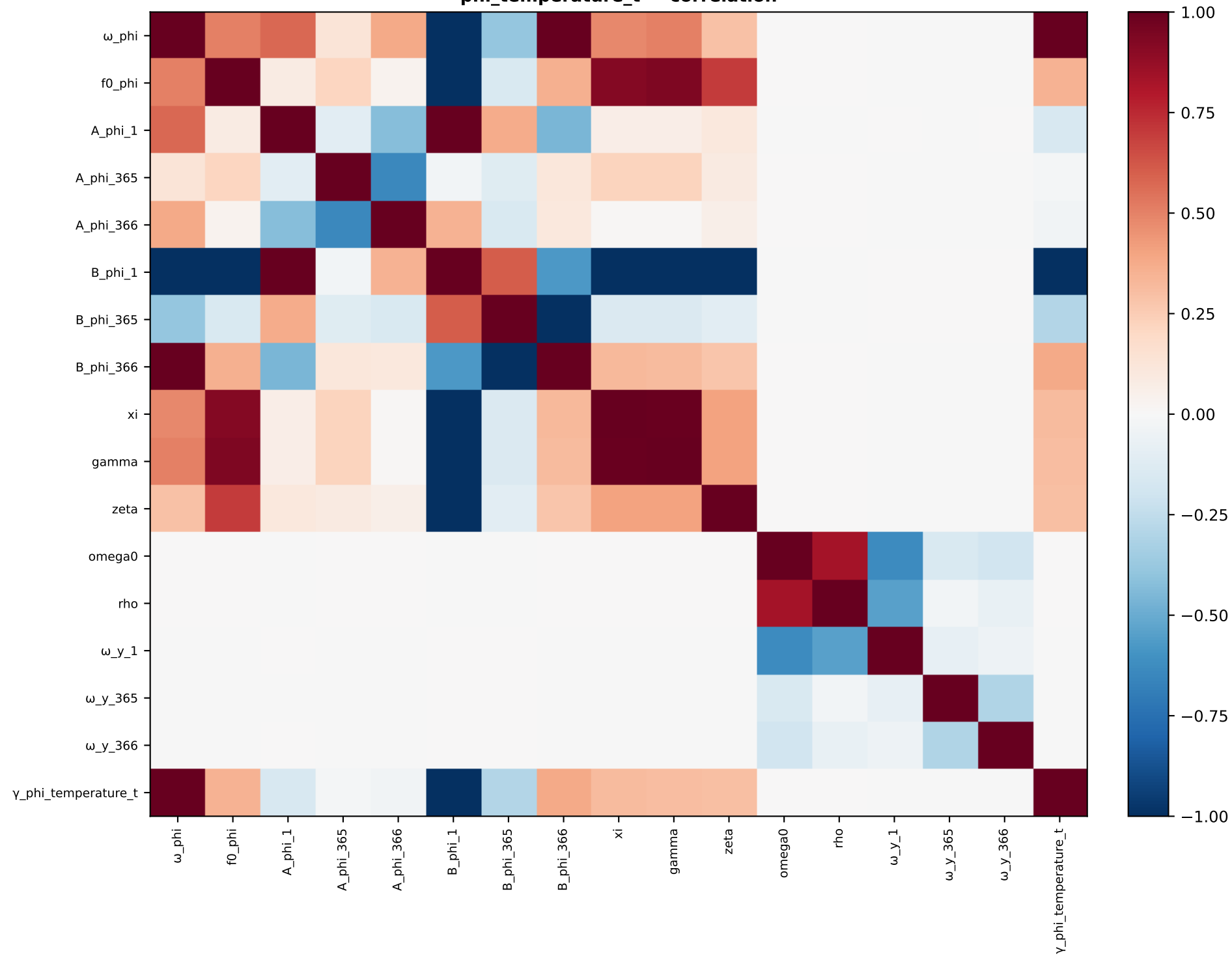
phi_temperature_t — filtered states (OOS)



phi_temperature_t — covariance

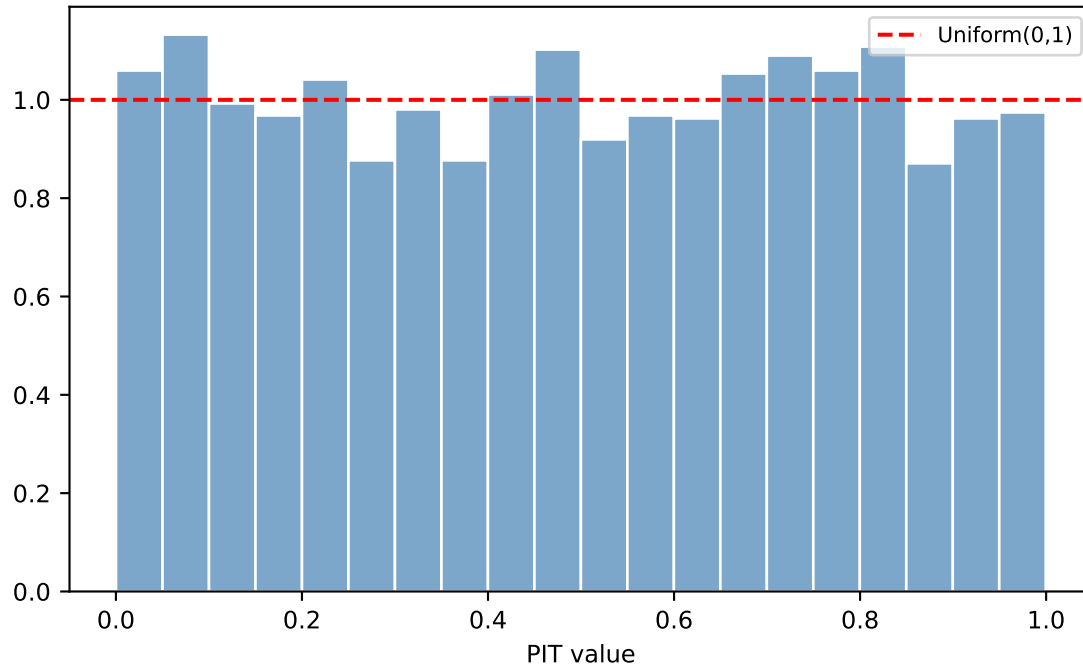


phi_temperature_t — correlation

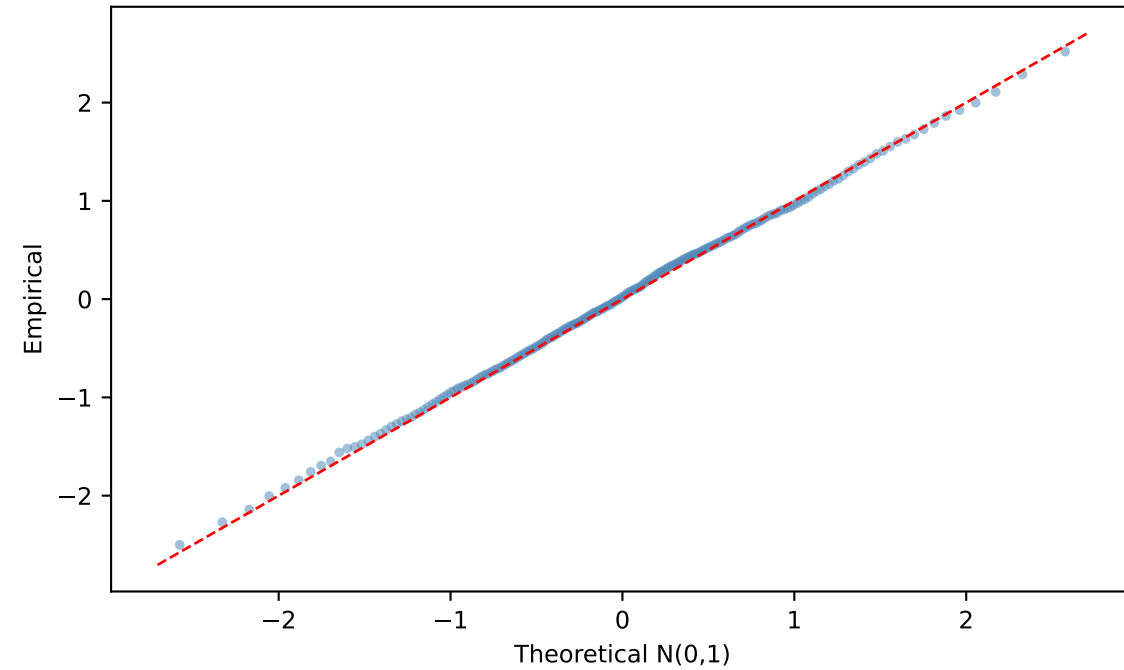


phi_temperature_lag1 — IS diagnostics

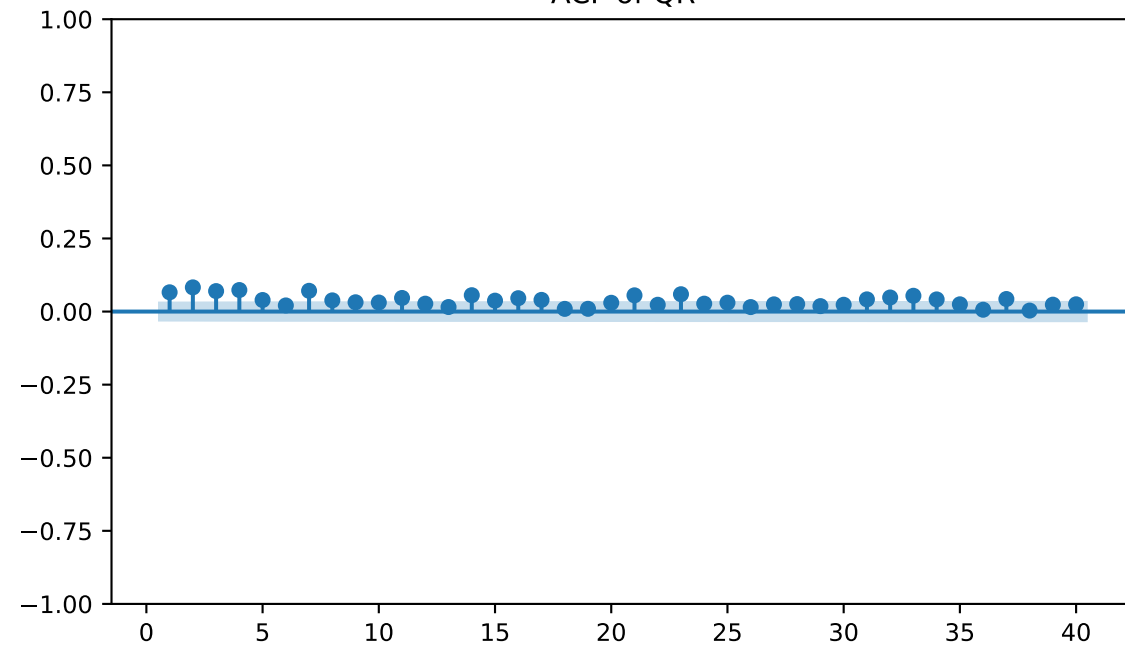
PIT histogram



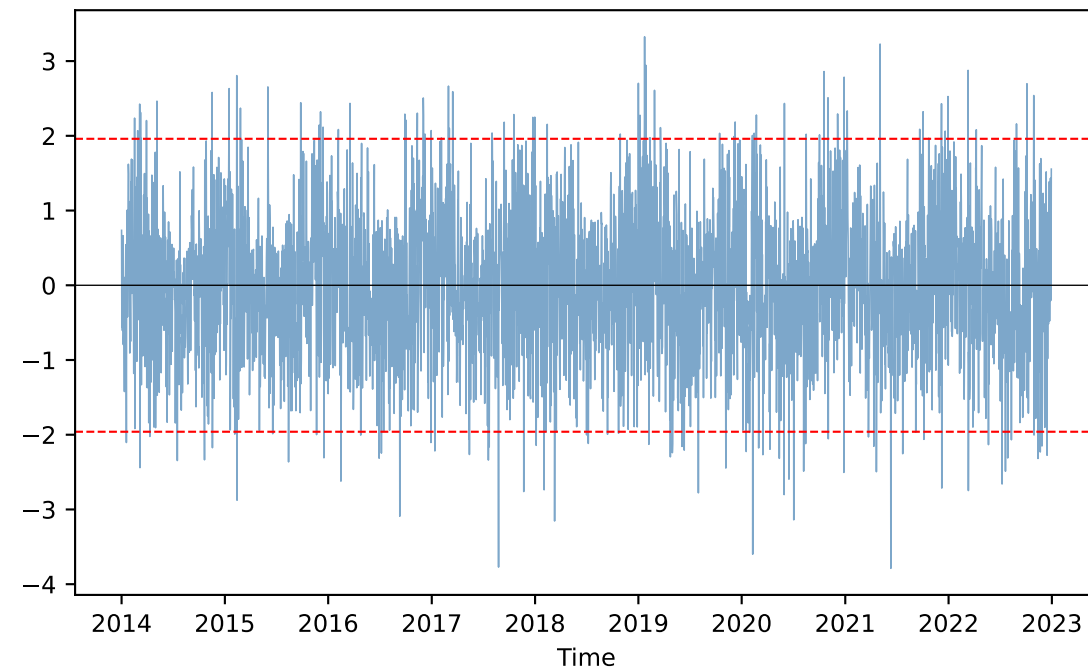
Normal QQ plot



ACF of QR

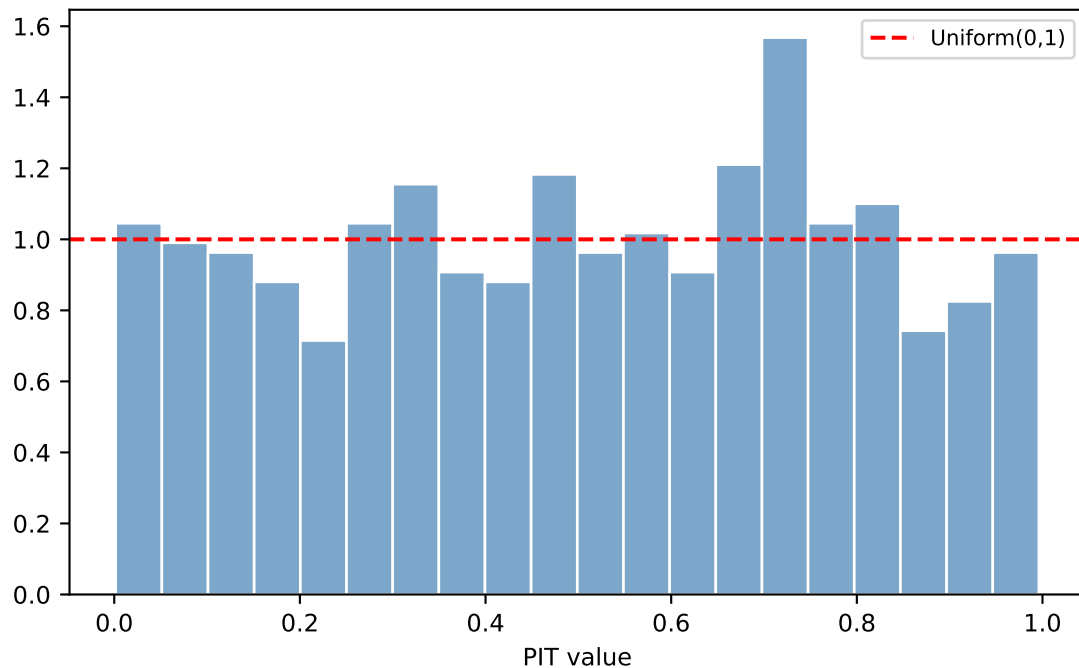


QR time series

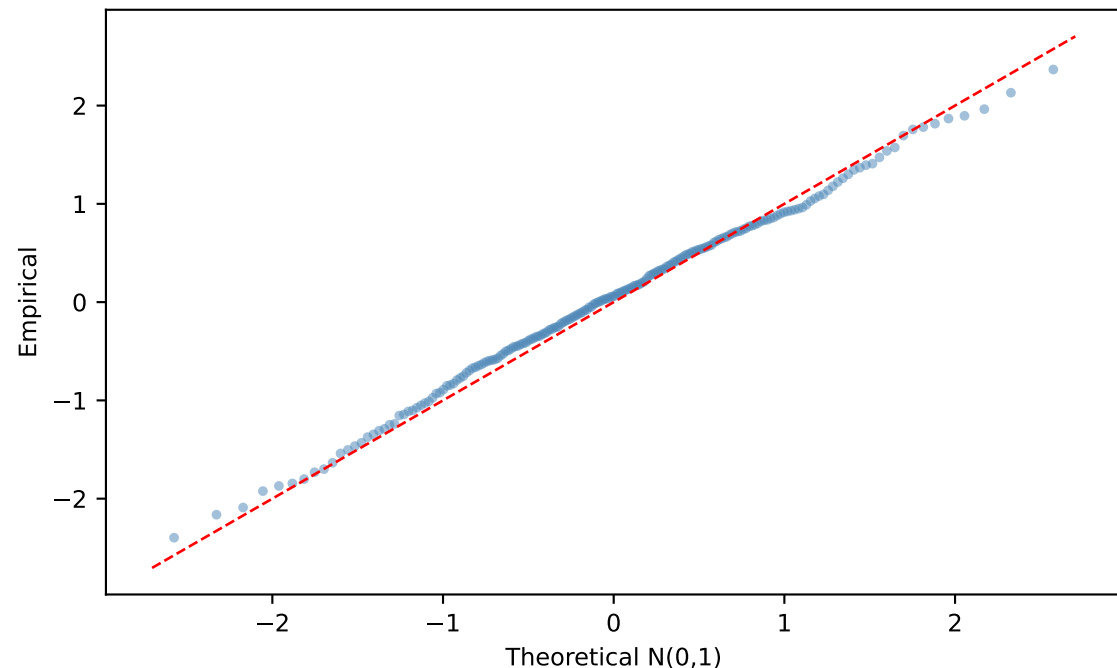


phi_temperature_lag1 — OOS diagnostics

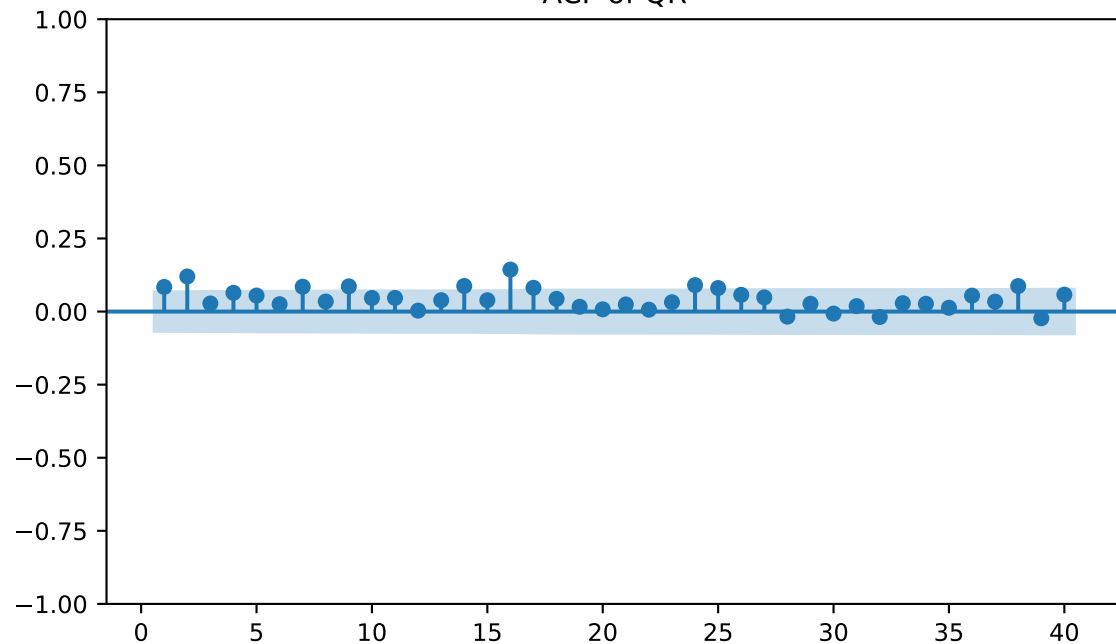
PIT histogram



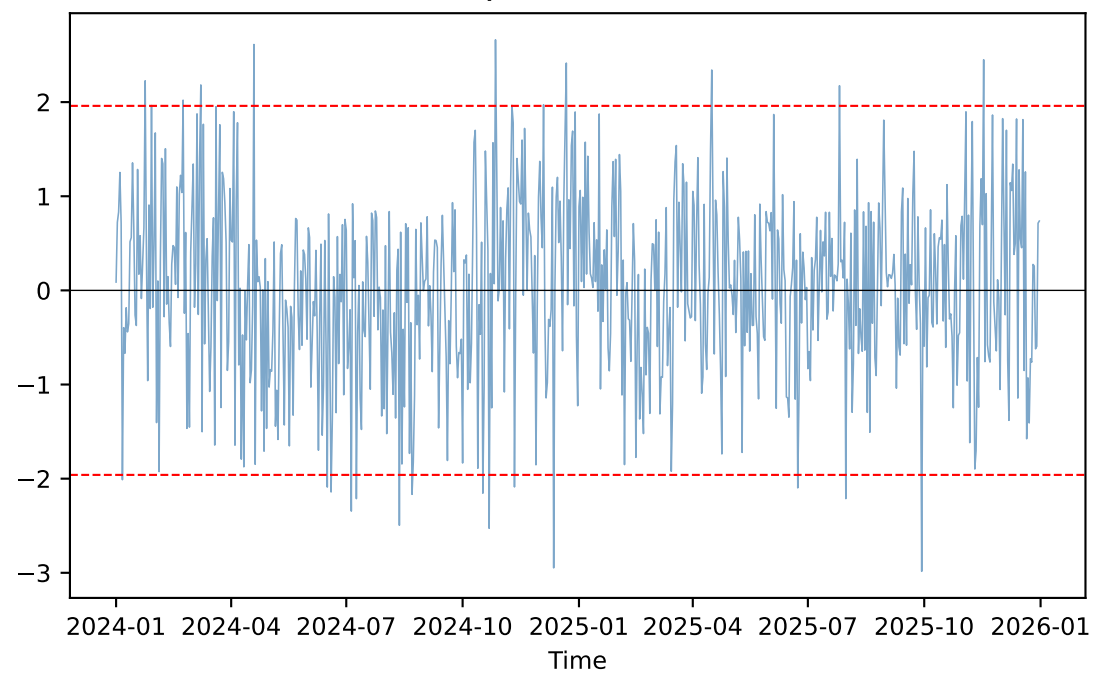
Normal QQ plot



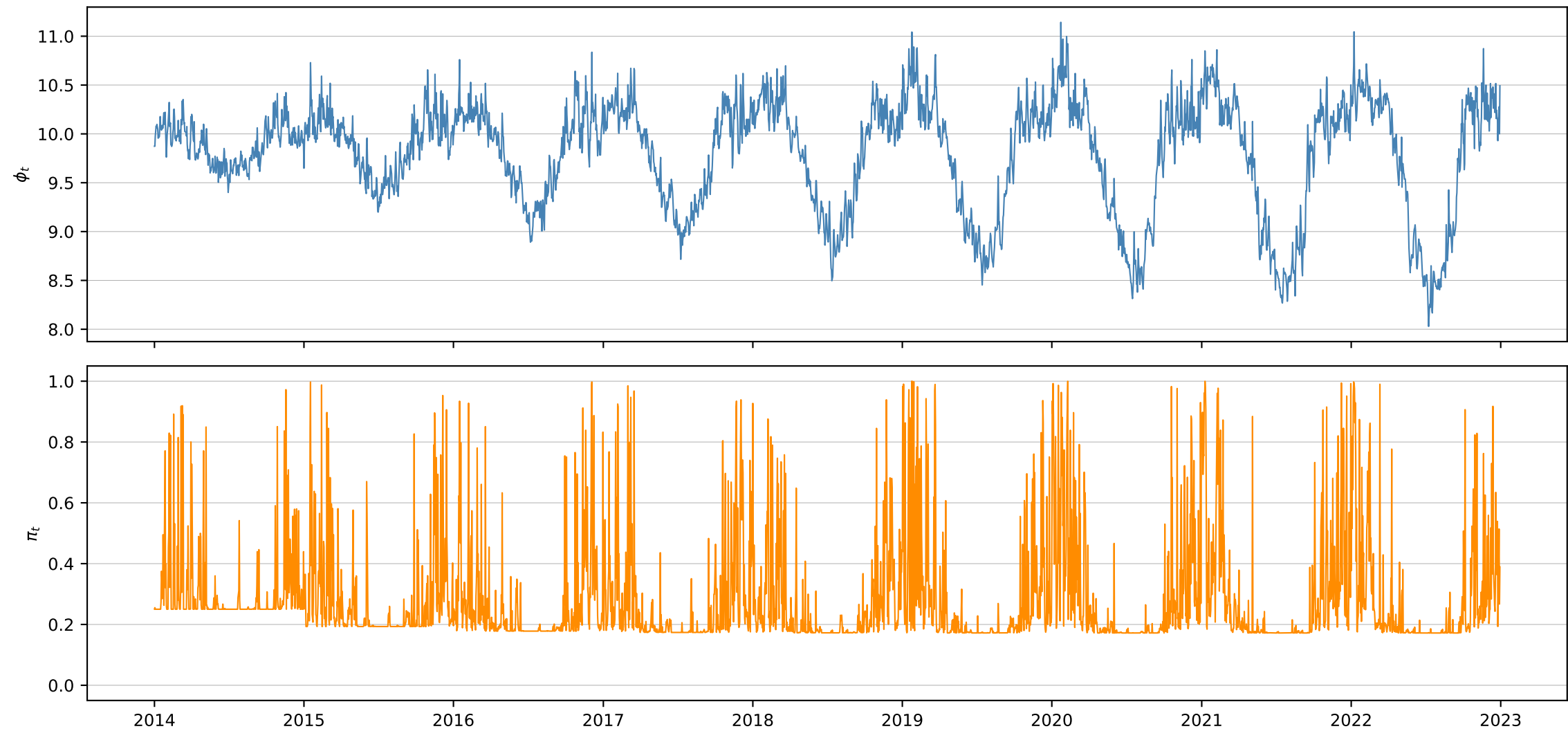
ACF of QR



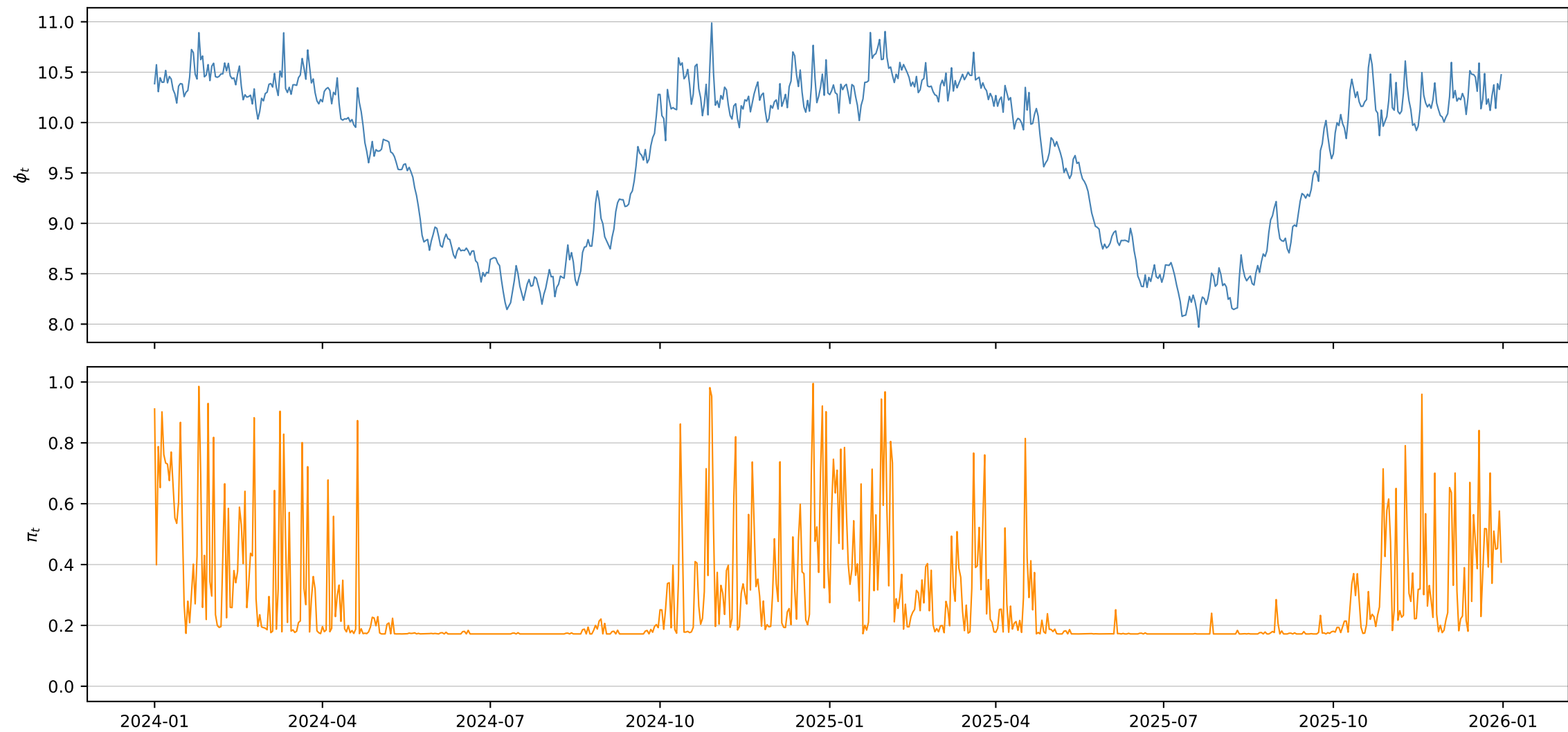
QR time series



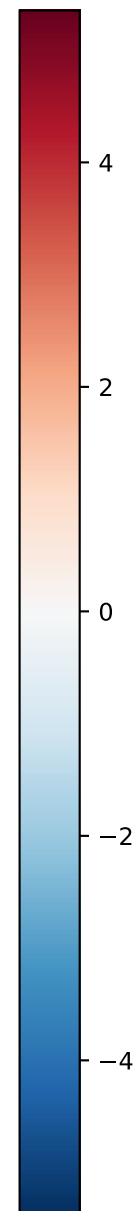
phi_temperature_lag1 — filtered states (IS)



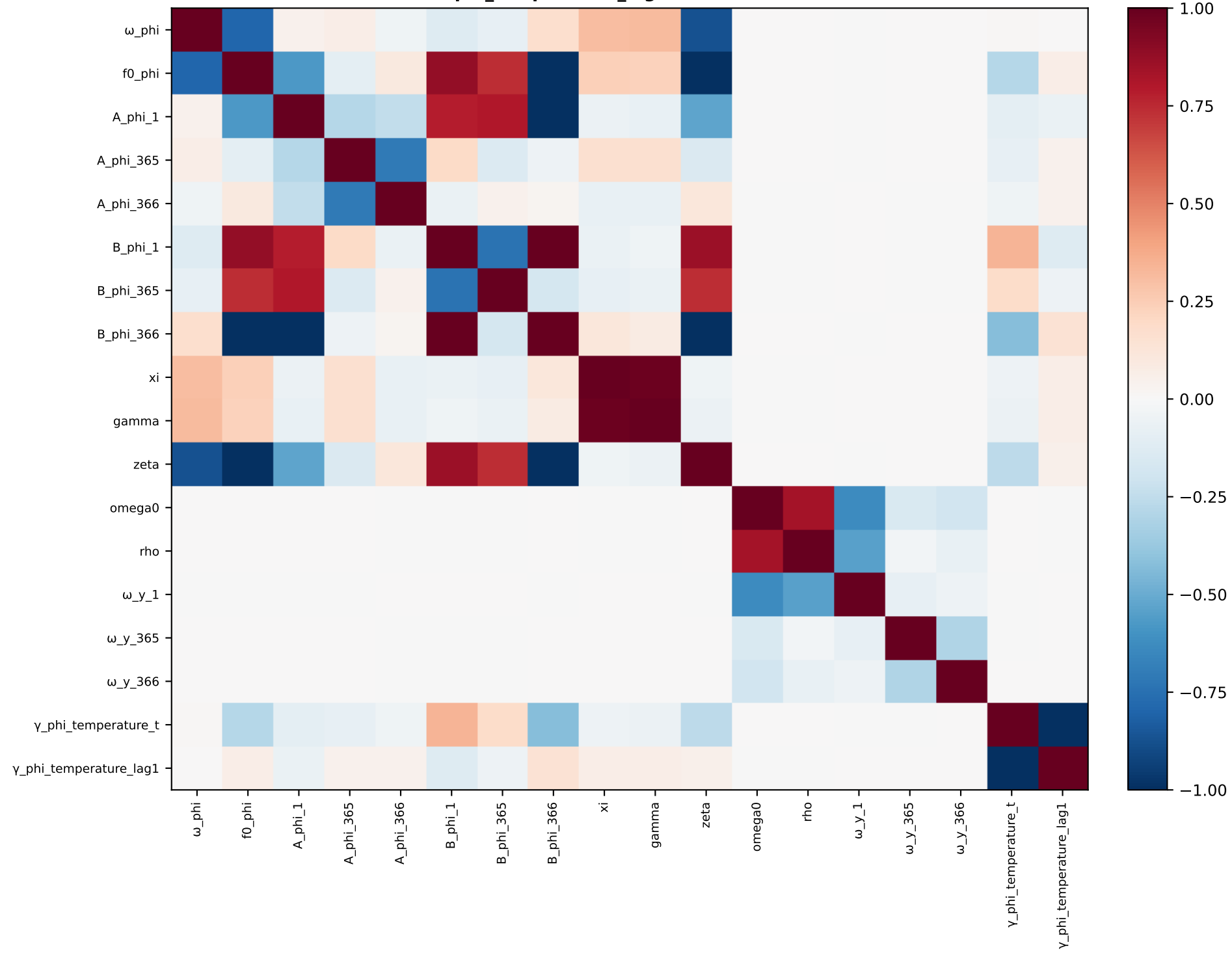
phi_temperature_lag1 — filtered states (OOS)



Heatmap visualization of the estimated partial correlation matrix for 20 variables. The variables are listed on both the x and y axes. The color scale ranges from -4 (dark blue) to 4 (dark red), with 0 being white. The diagonal elements are all 1.0. The matrix shows strong negative correlations between $f0_phi$ and $zeta$, and between B_phi_1 and B_phi_366 . There are also positive correlations between B_phi_1 and B_phi_365 , and between B_phi_365 and B_phi_366 .

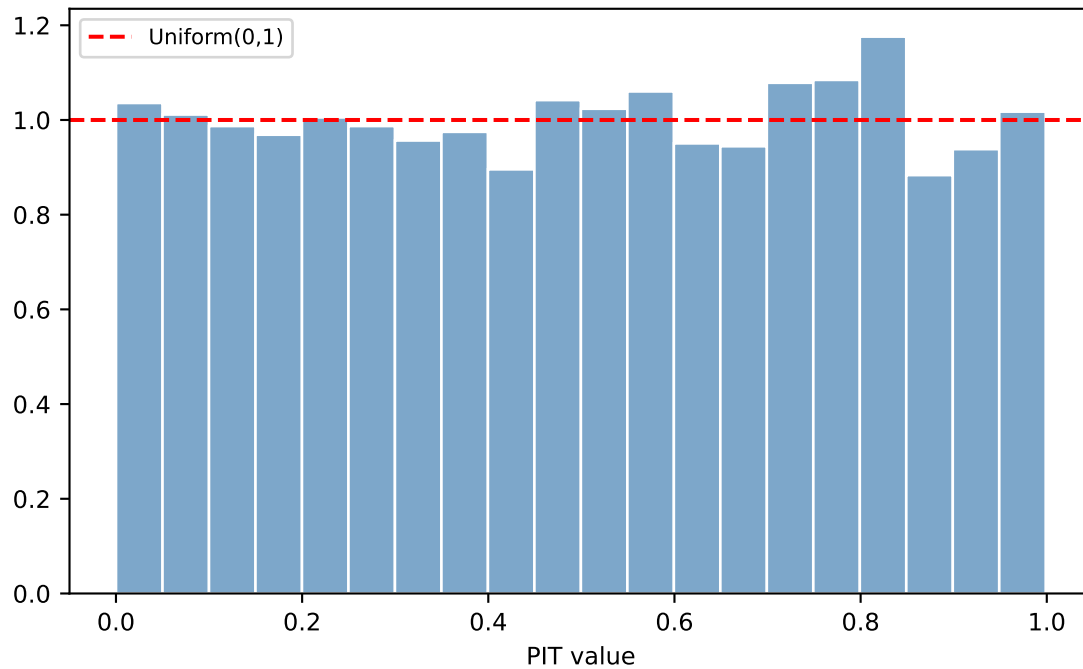


phi_temperature_lag1 — correlation

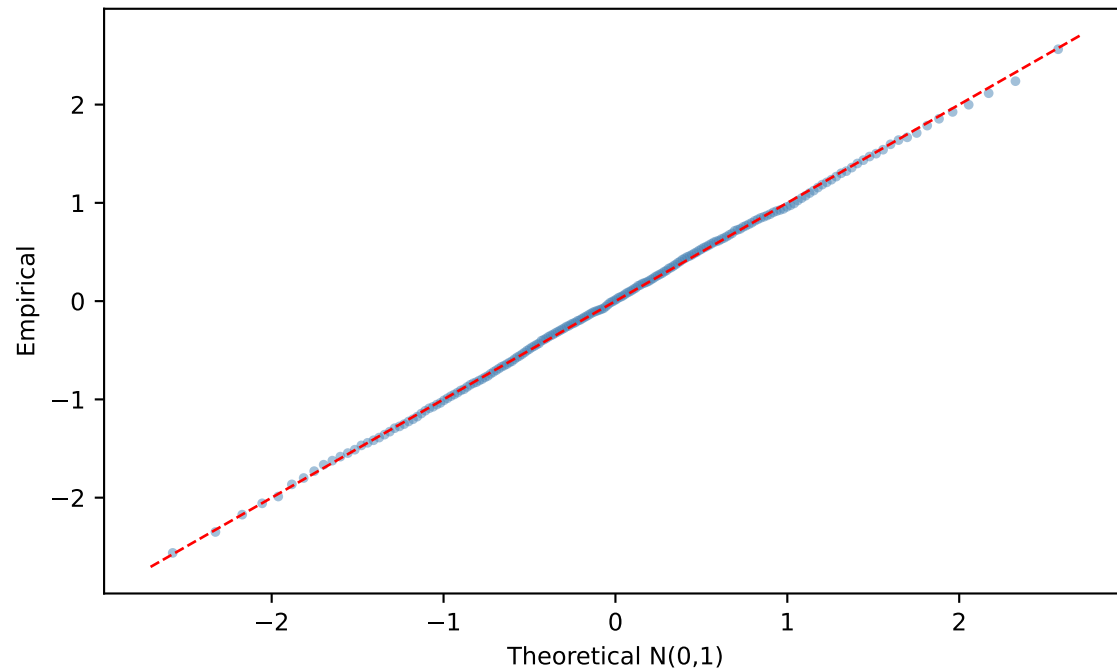


phi_temperature_short_lags — IS diagnostics

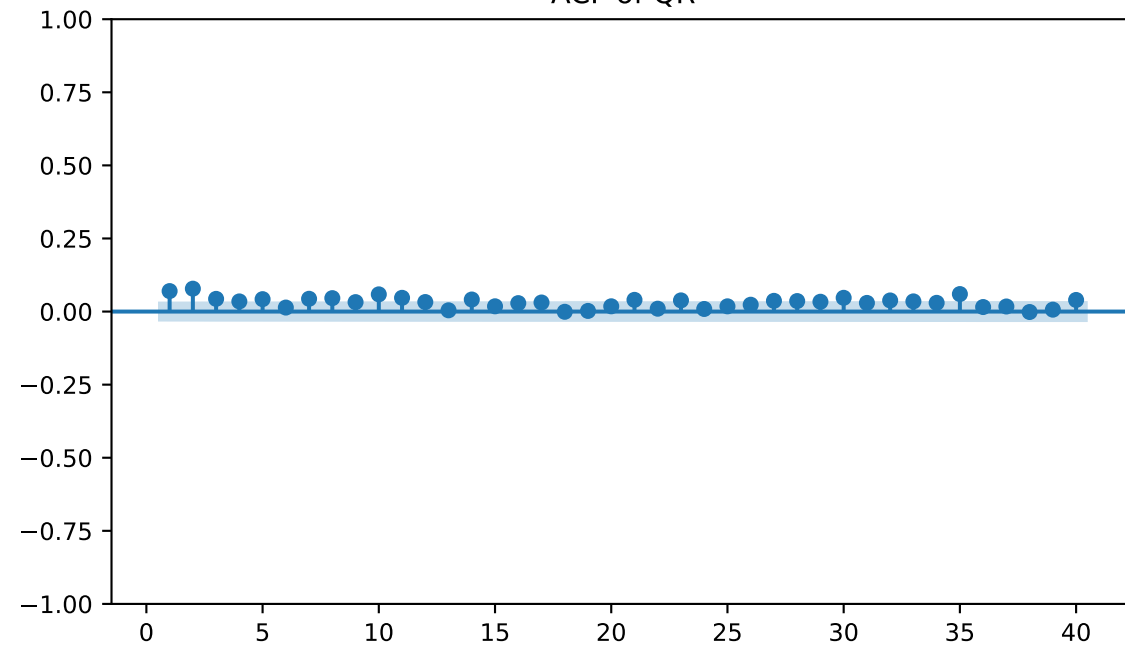
PIT histogram



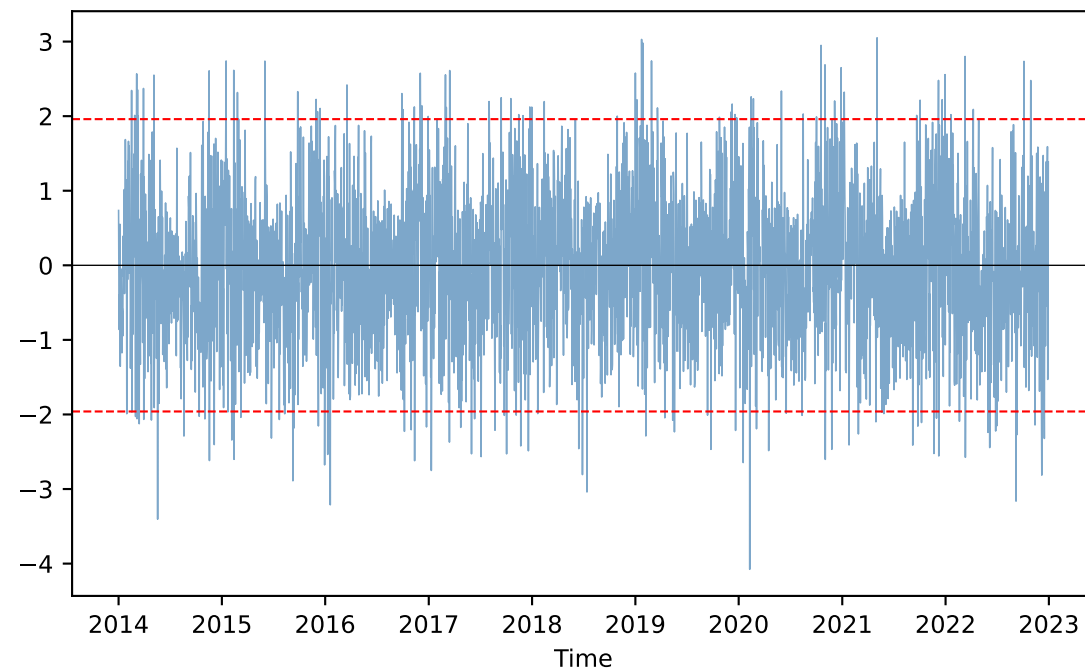
Normal QQ plot



ACF of QR

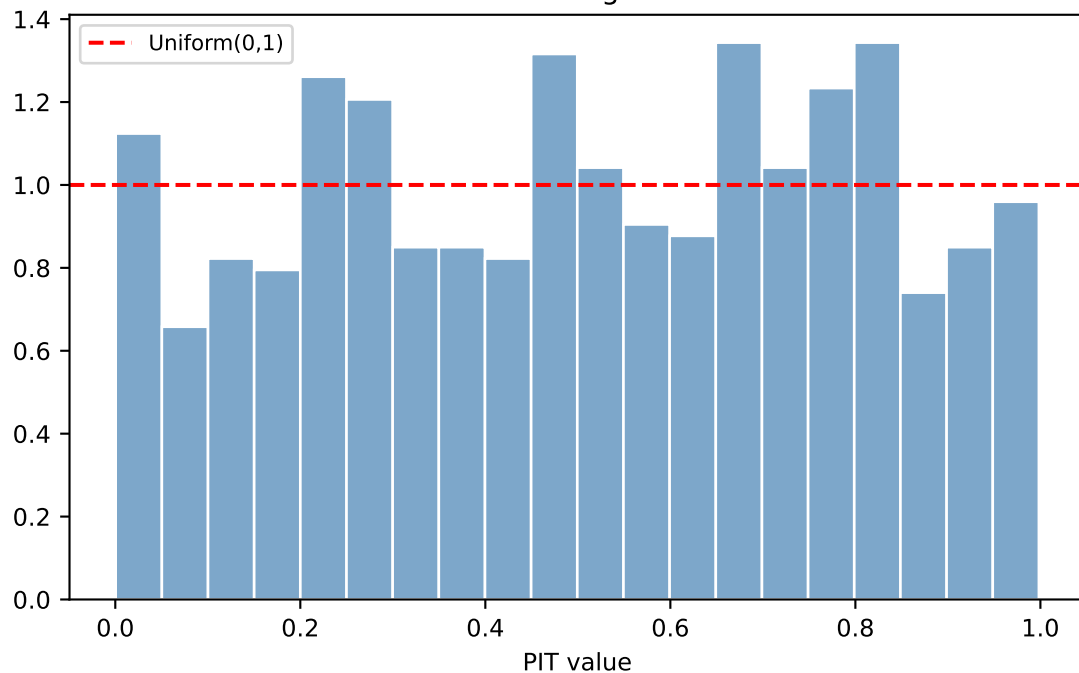


QR time series

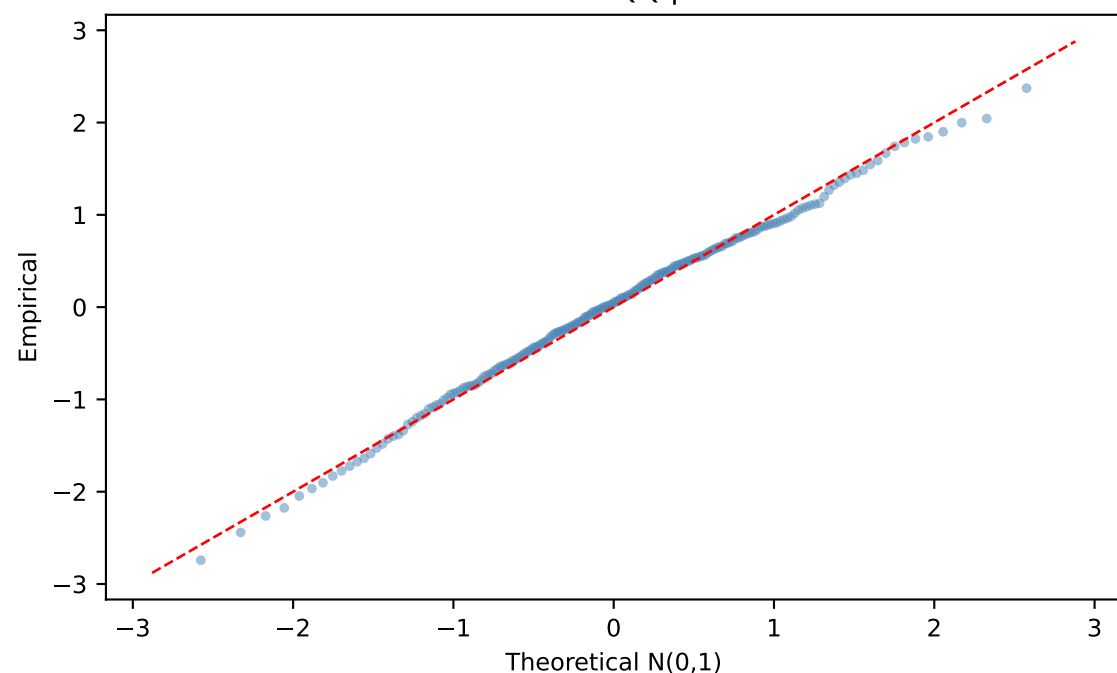


phi_temperature_short_lags — OOS diagnostics

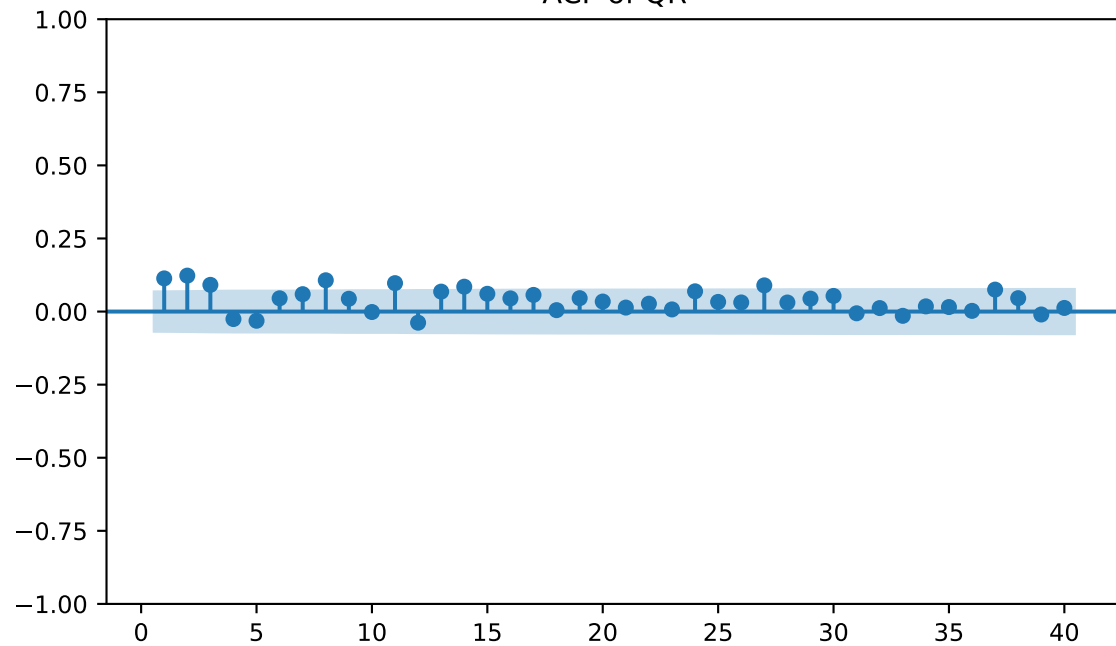
PIT histogram



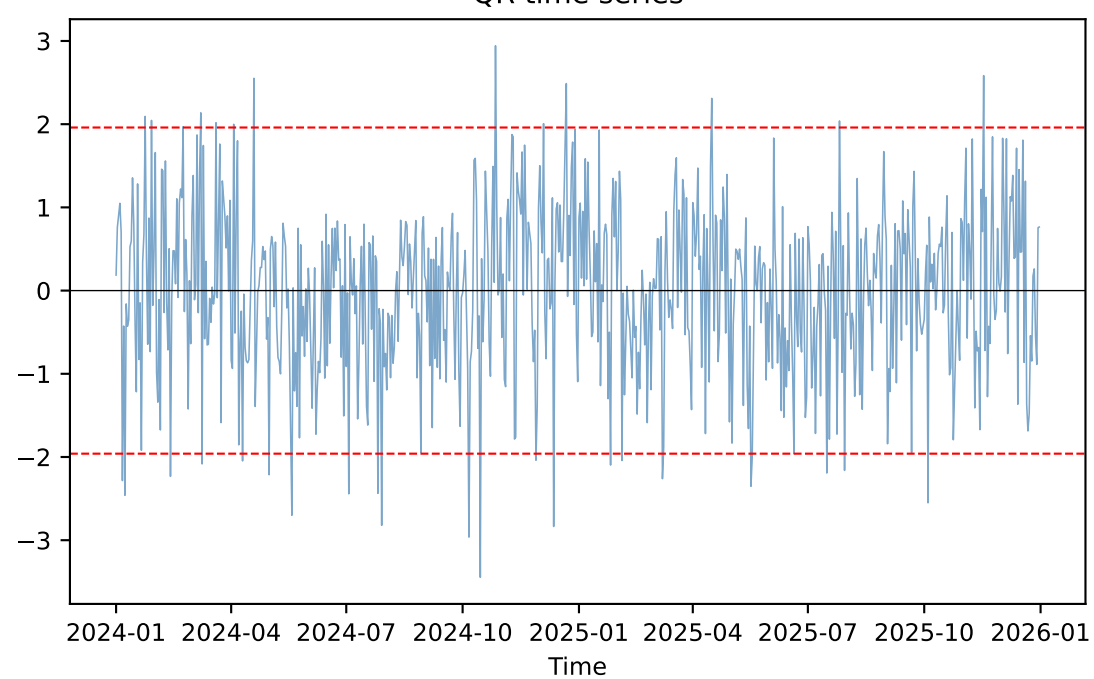
Normal QQ plot



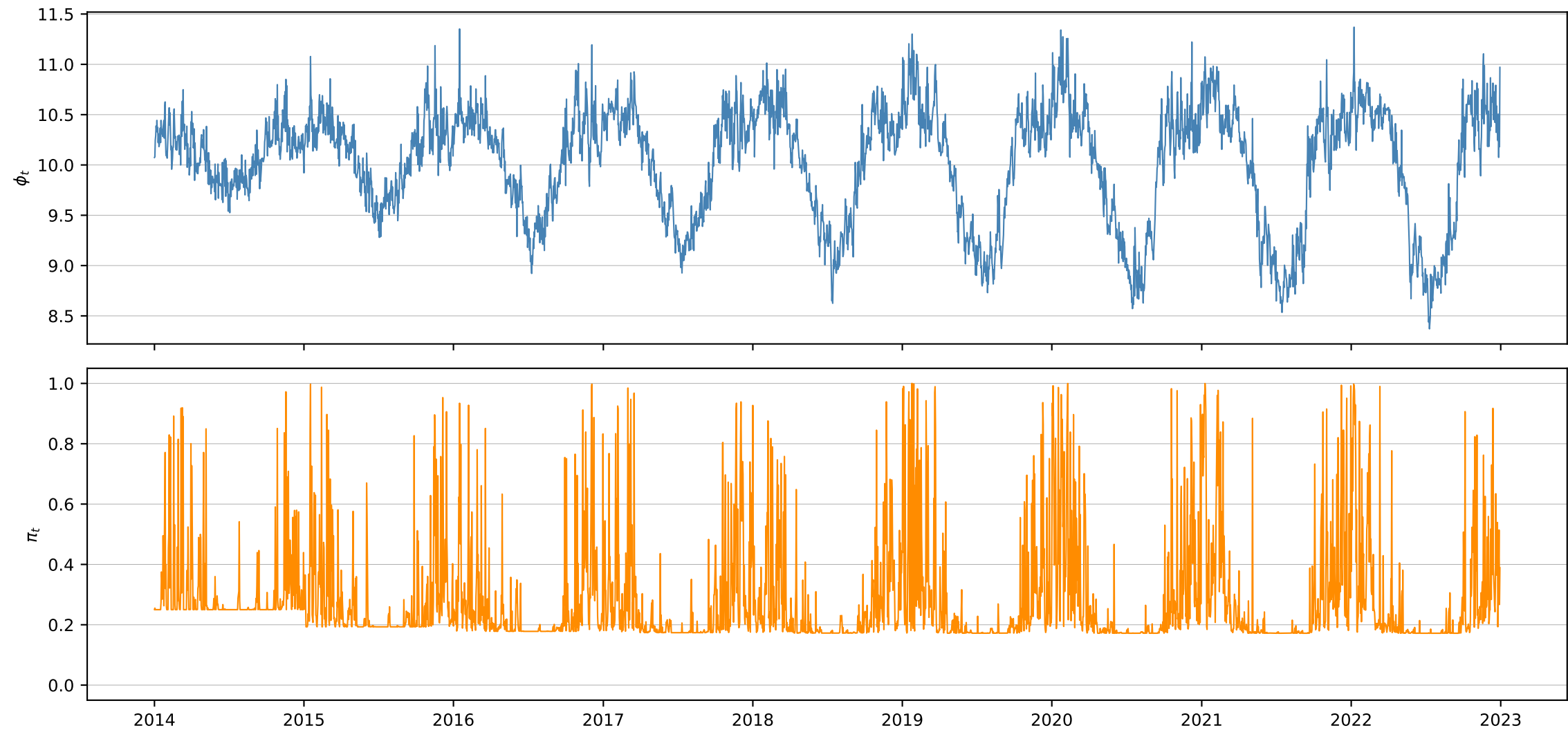
ACF of QR



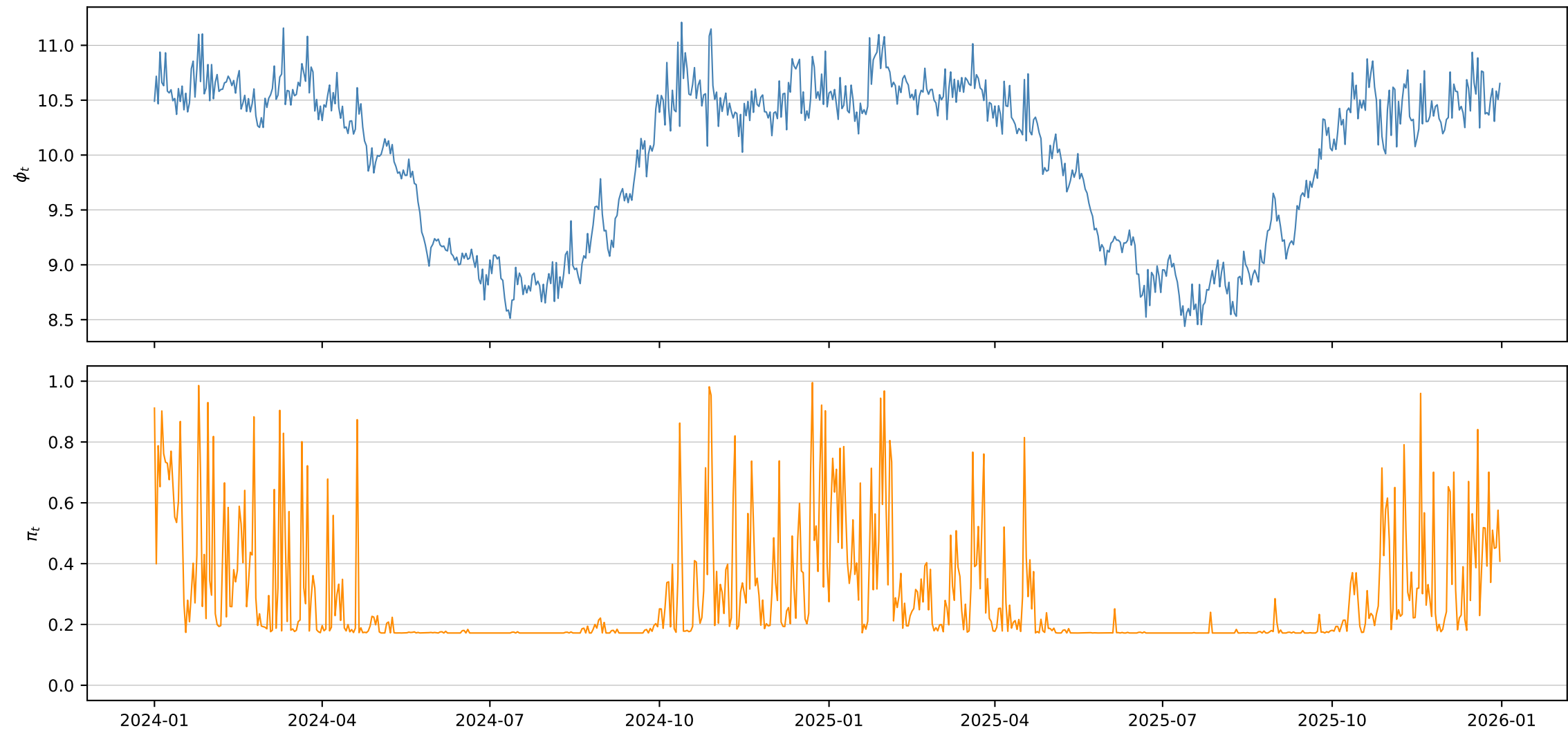
QR time series



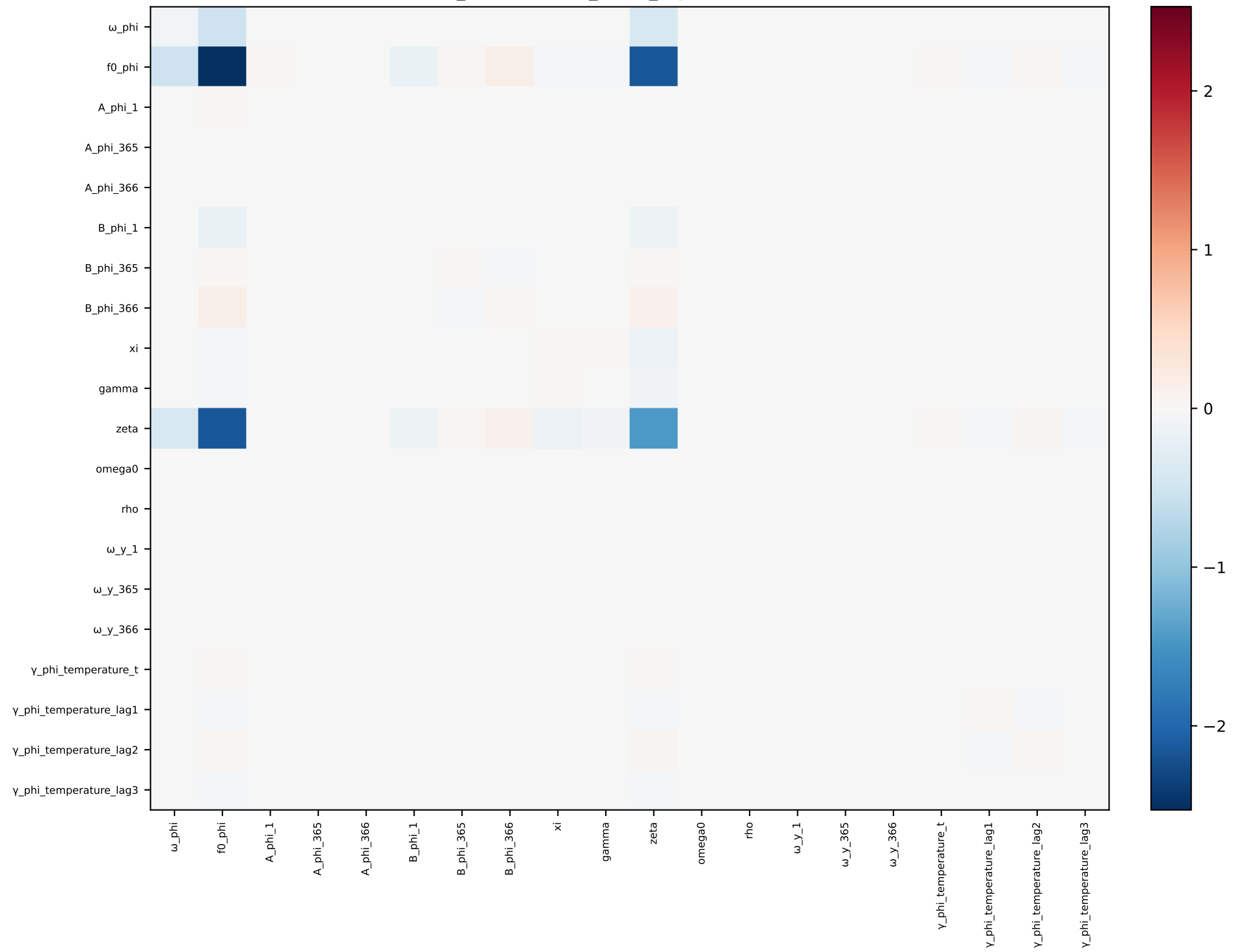
phi_temperature_short_lags — filtered states (IS)



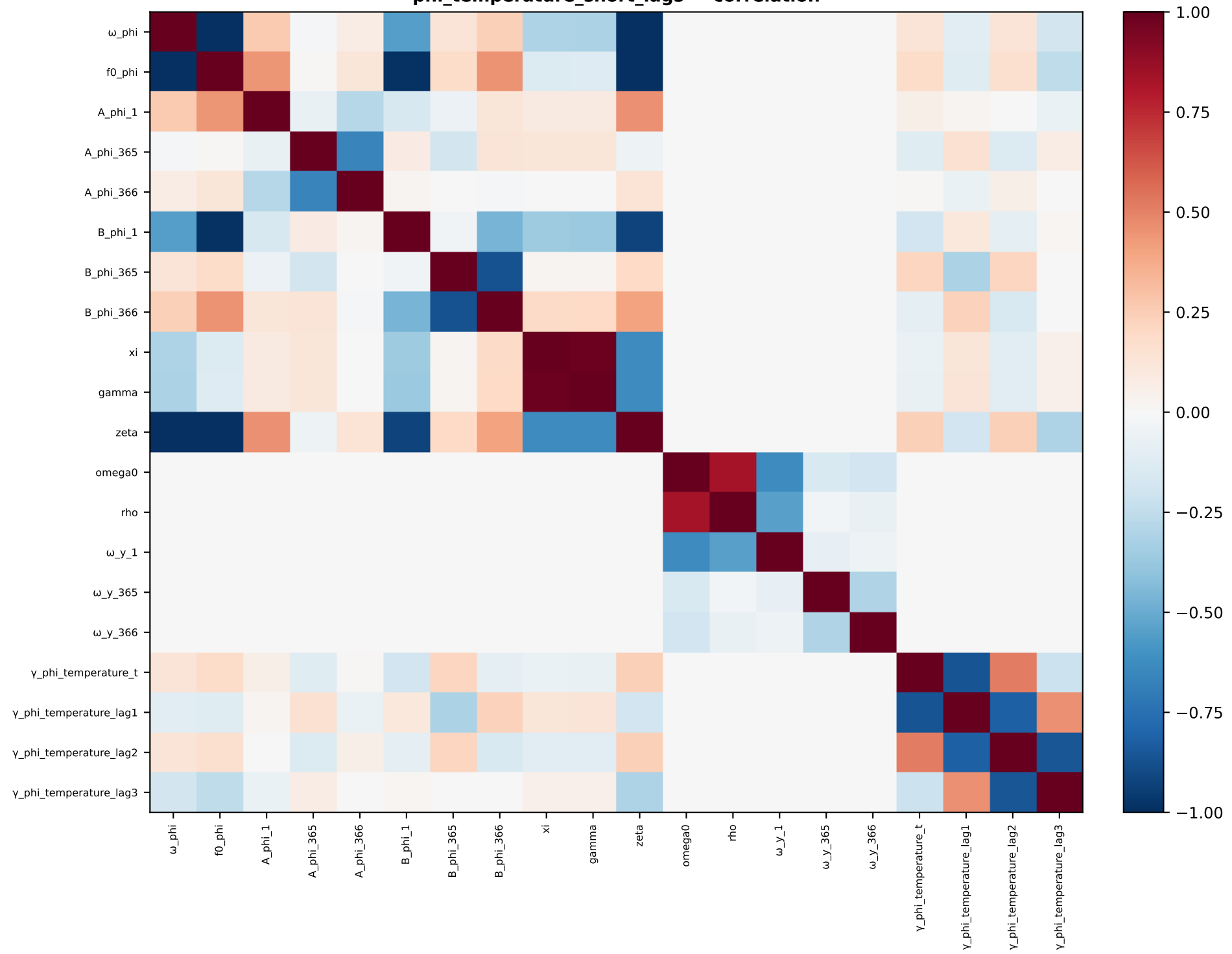
phi_temperature_short_lags — filtered states (OOS)



phi_temperature_short_lags — covariance

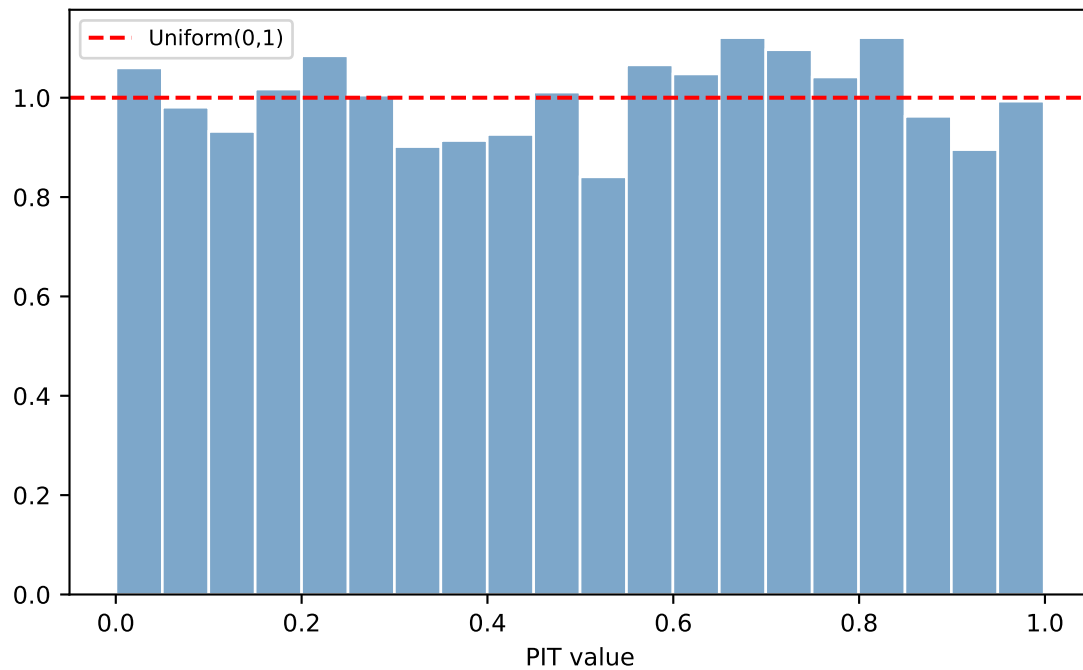


phi_temperature_short_lags — correlation

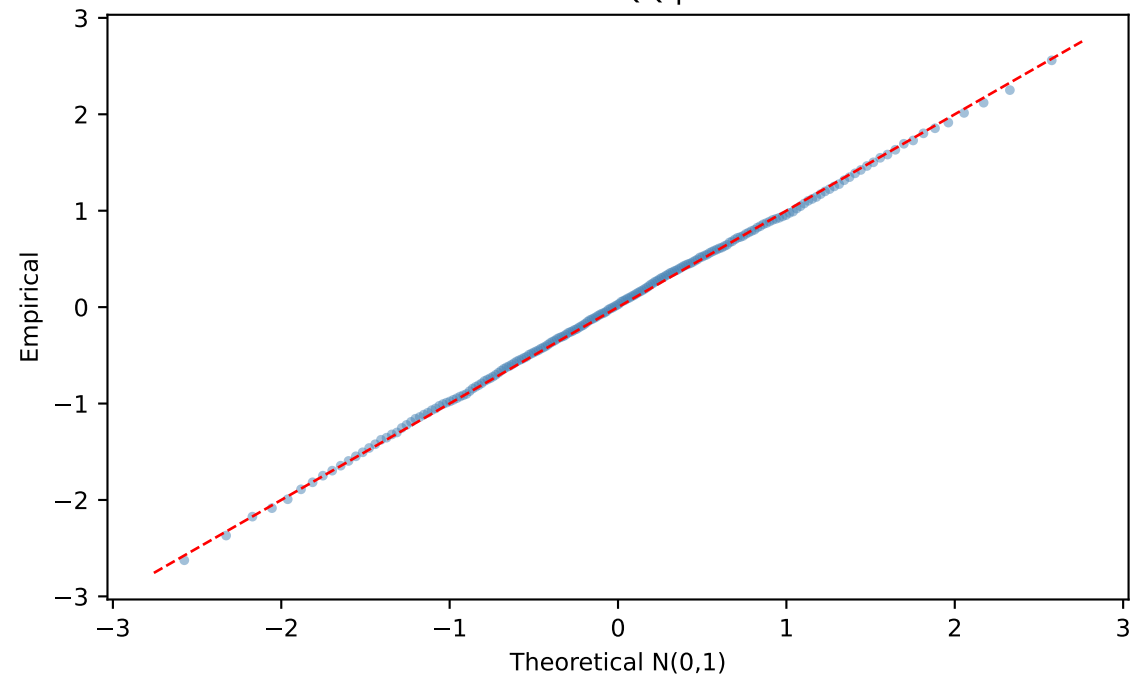


phi_temperature_seasonal_lags — IS diagnostics

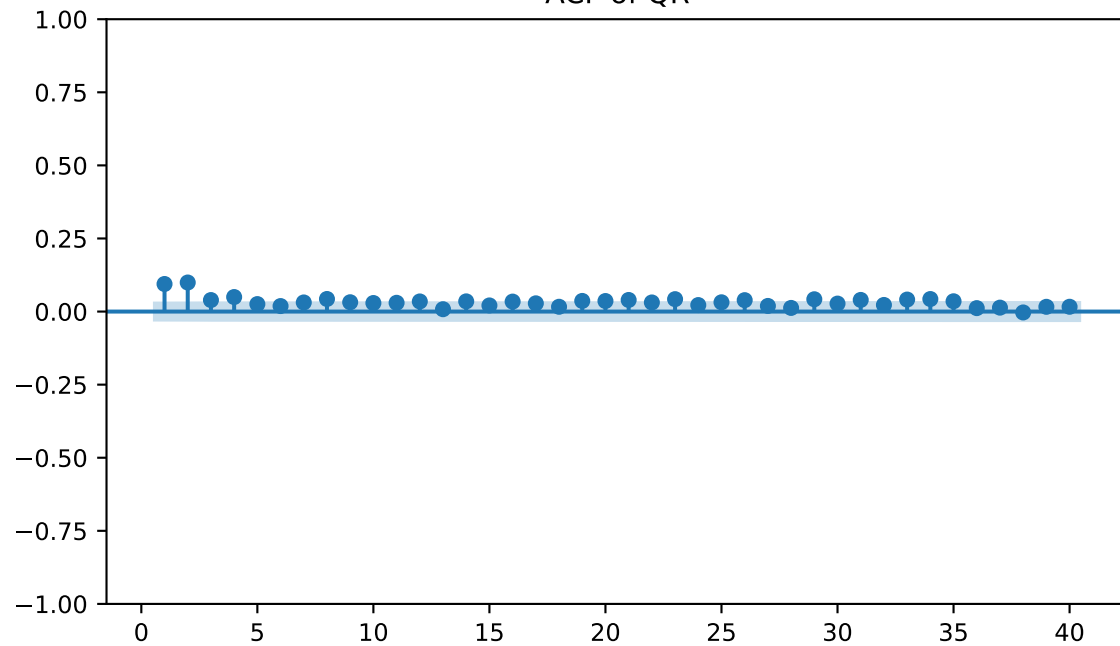
PIT histogram



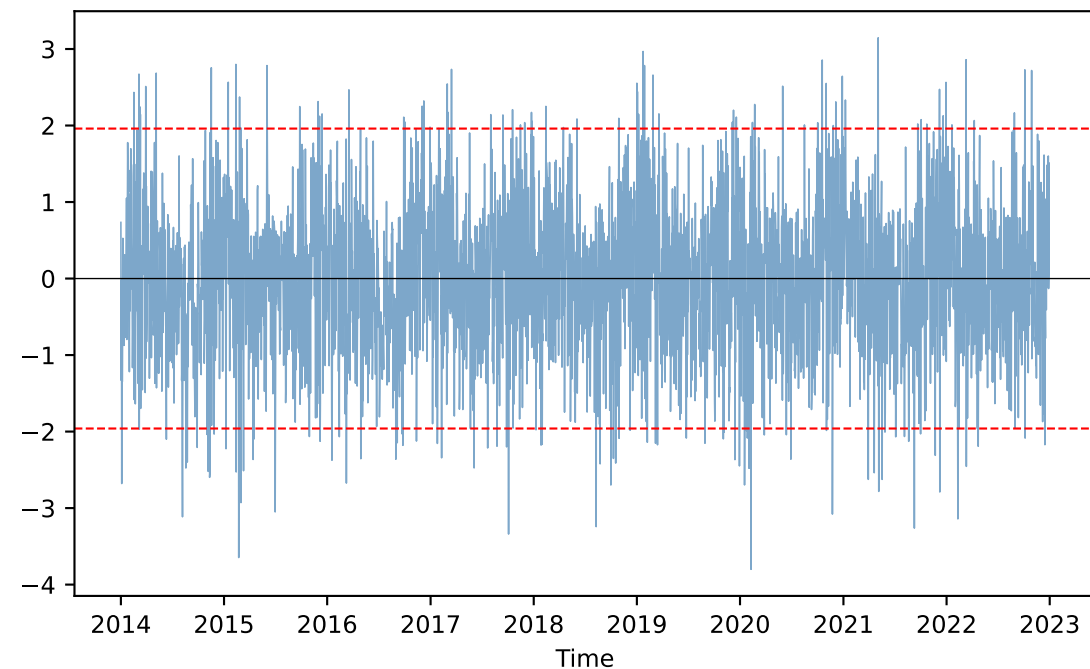
Normal QQ plot



ACF of QR

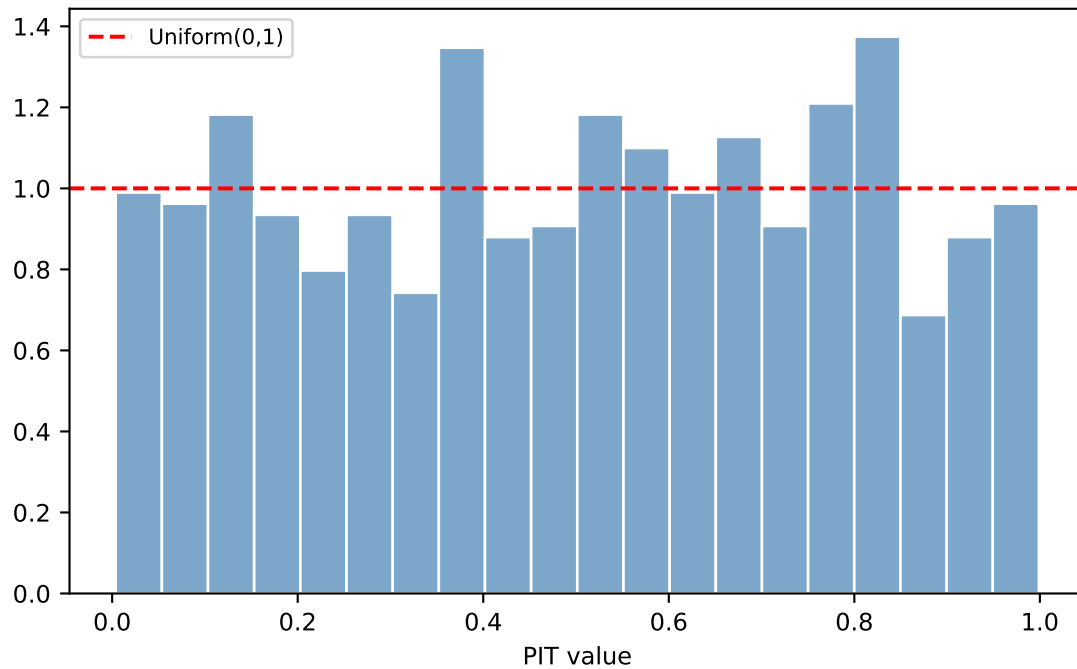


QR time series

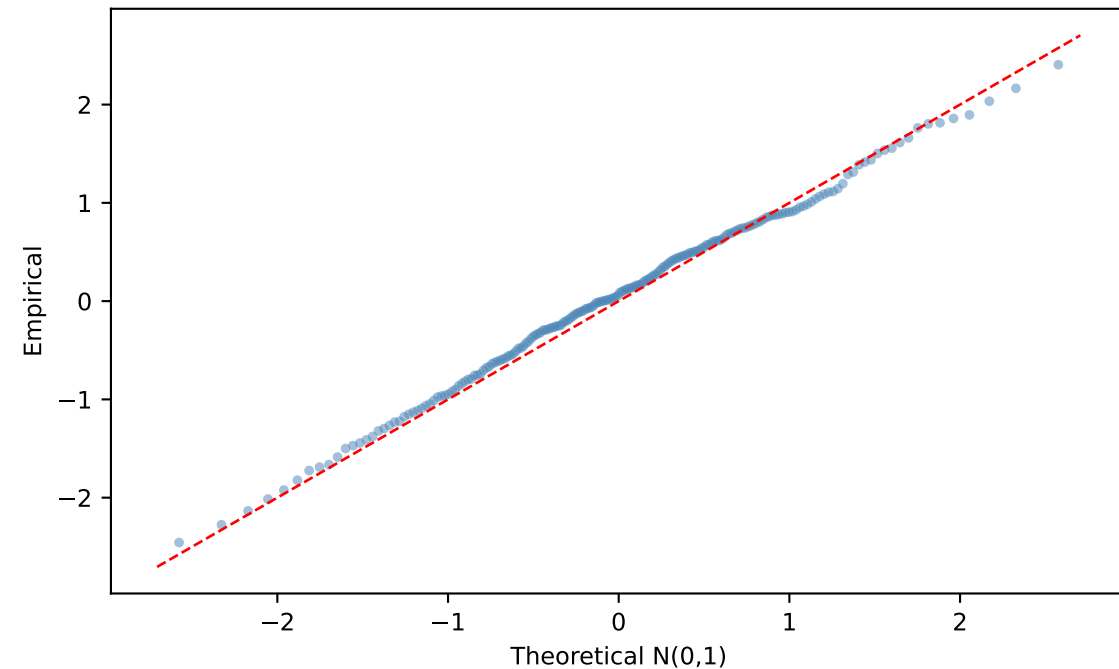


phi_temperature_seasonal_lags — OOS diagnostics

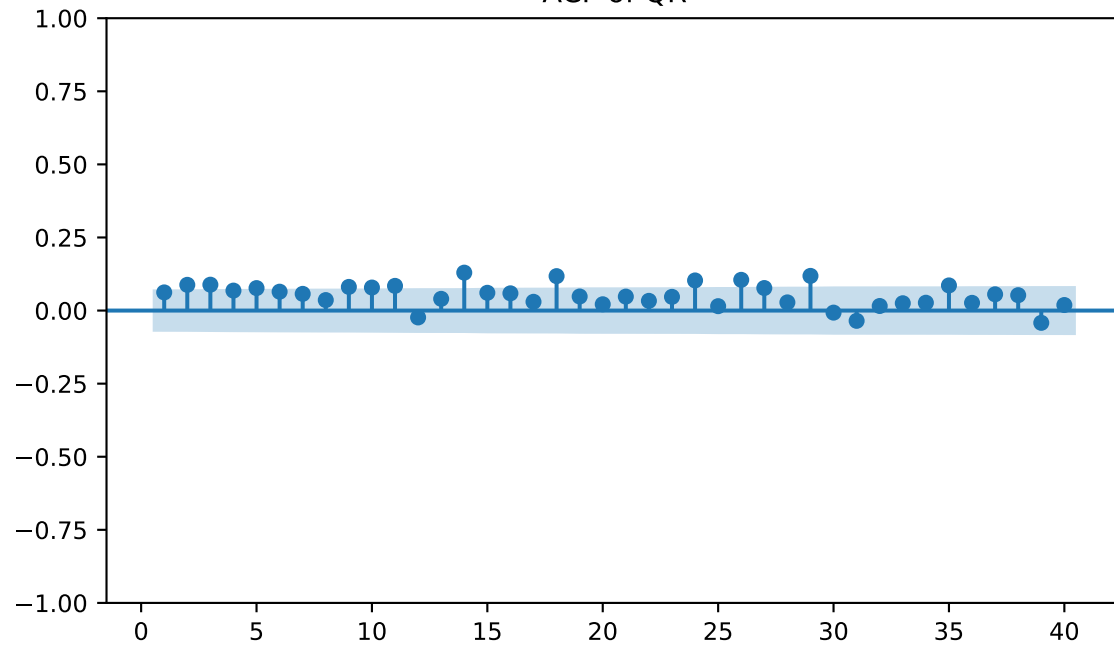
PIT histogram



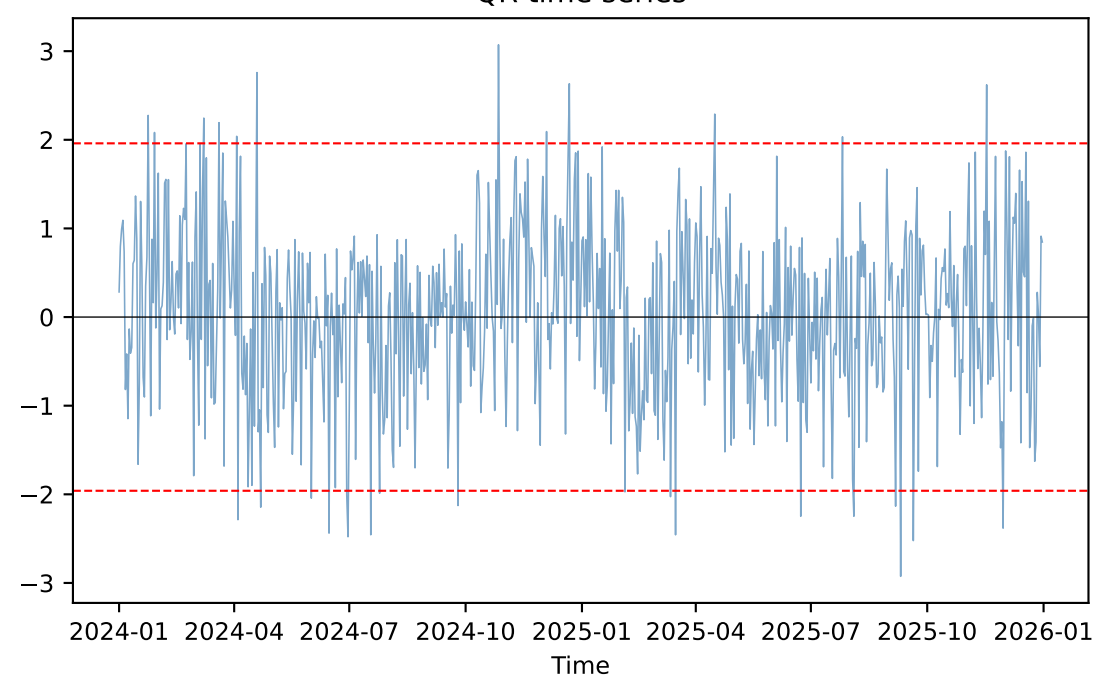
Normal QQ plot



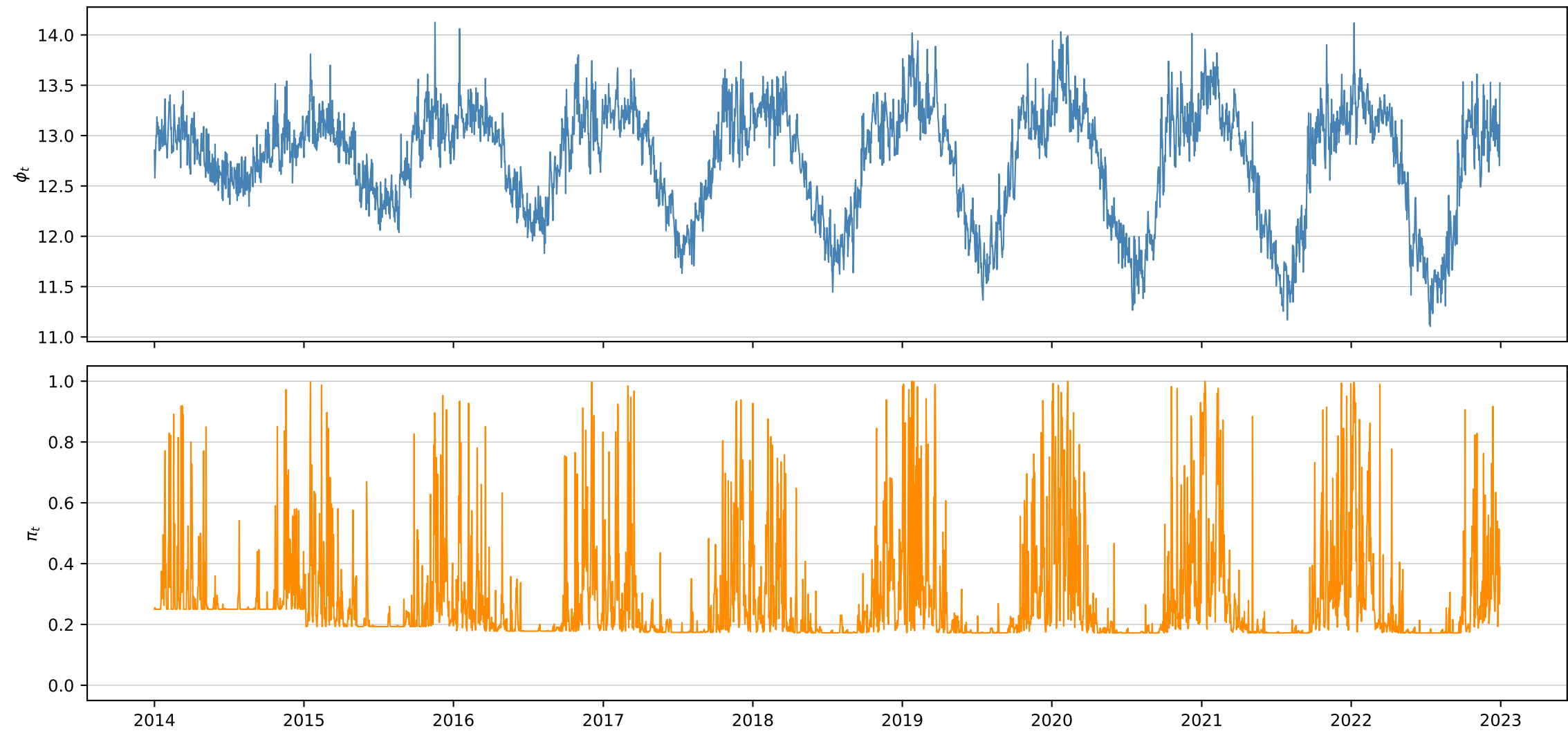
ACF of QR



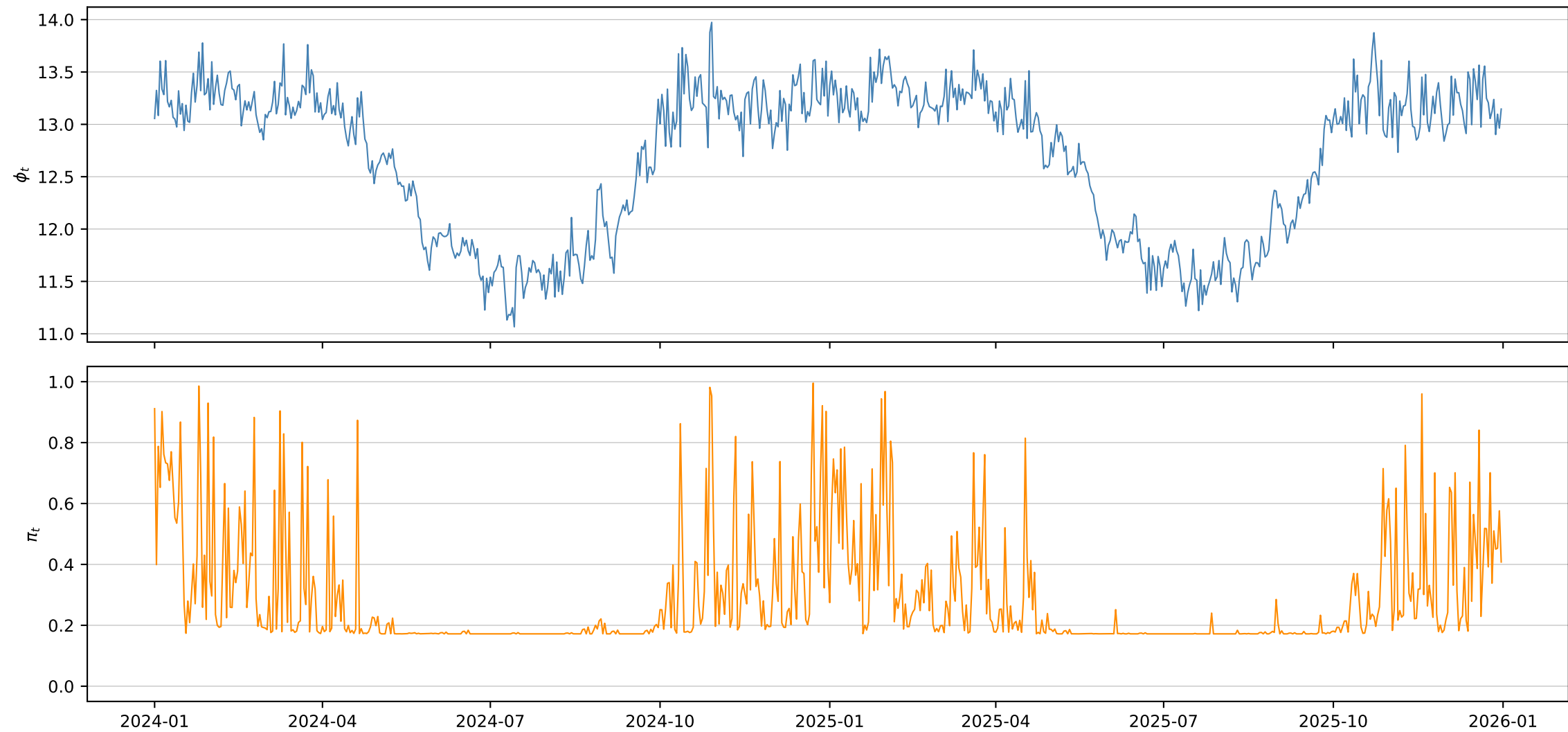
QR time series



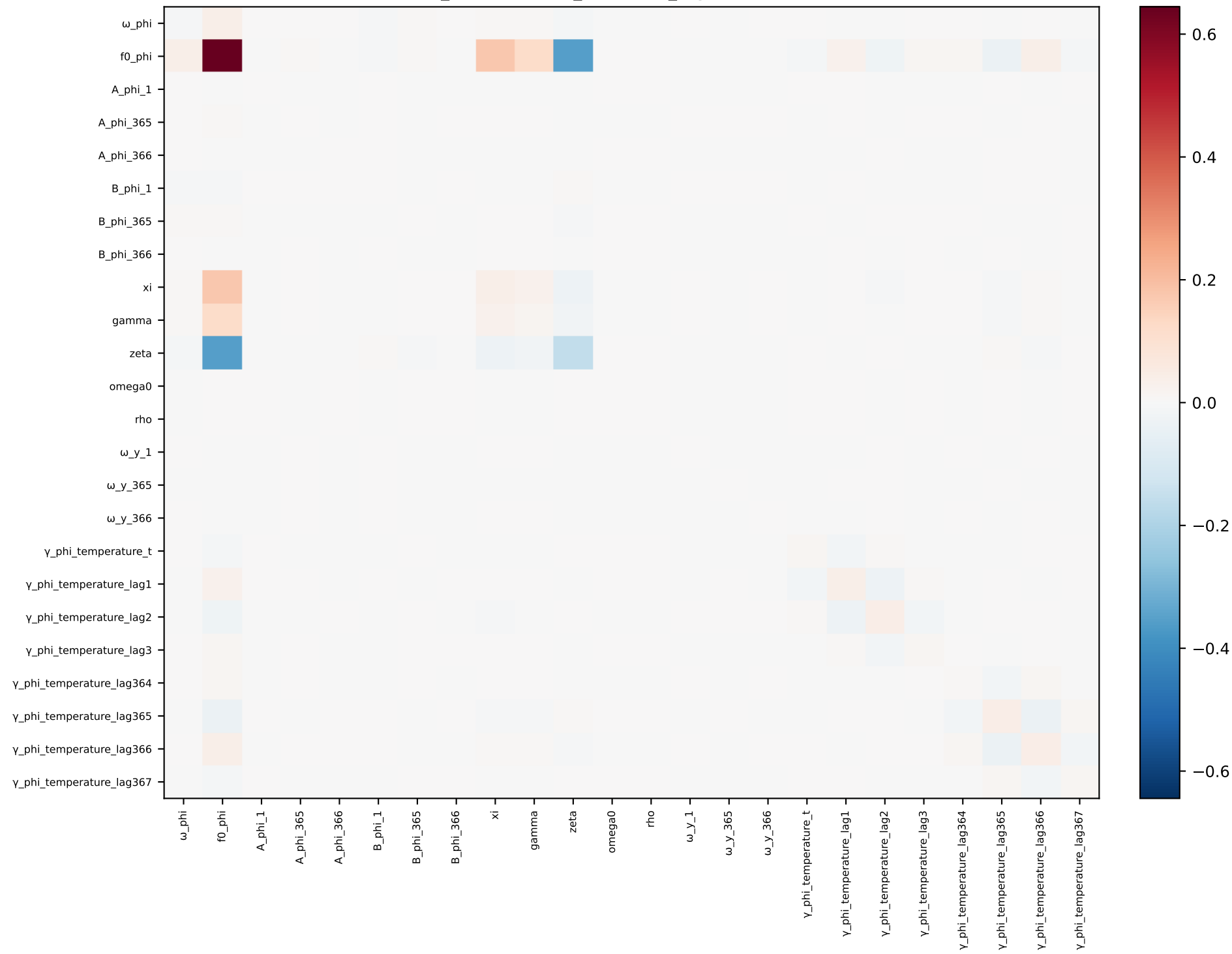
phi_temperature_seasonal_lags — filtered states (IS)



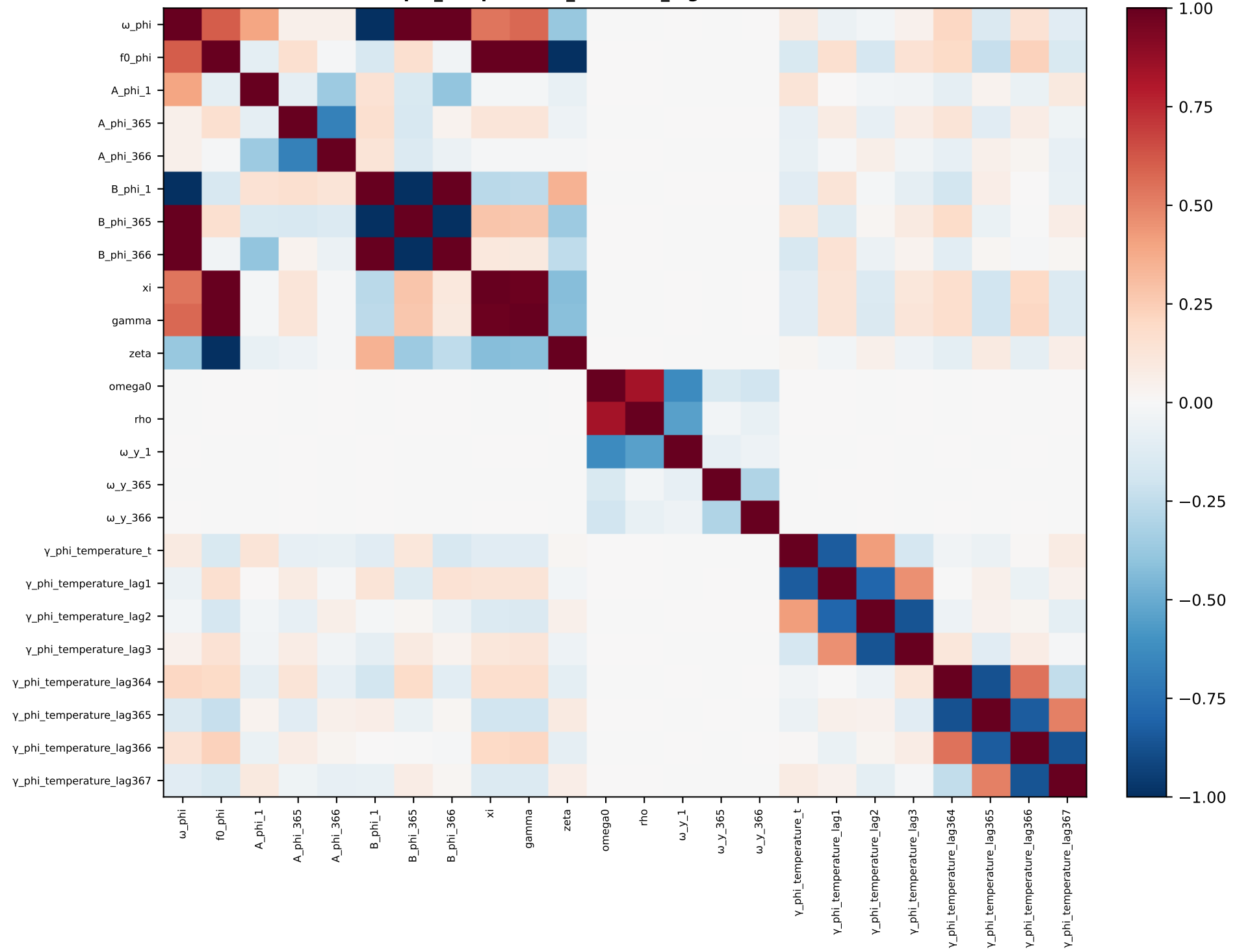
phi_temperature_seasonal_lags — filtered states (OOS)



phi_temperature_seasonal_lags — covariance

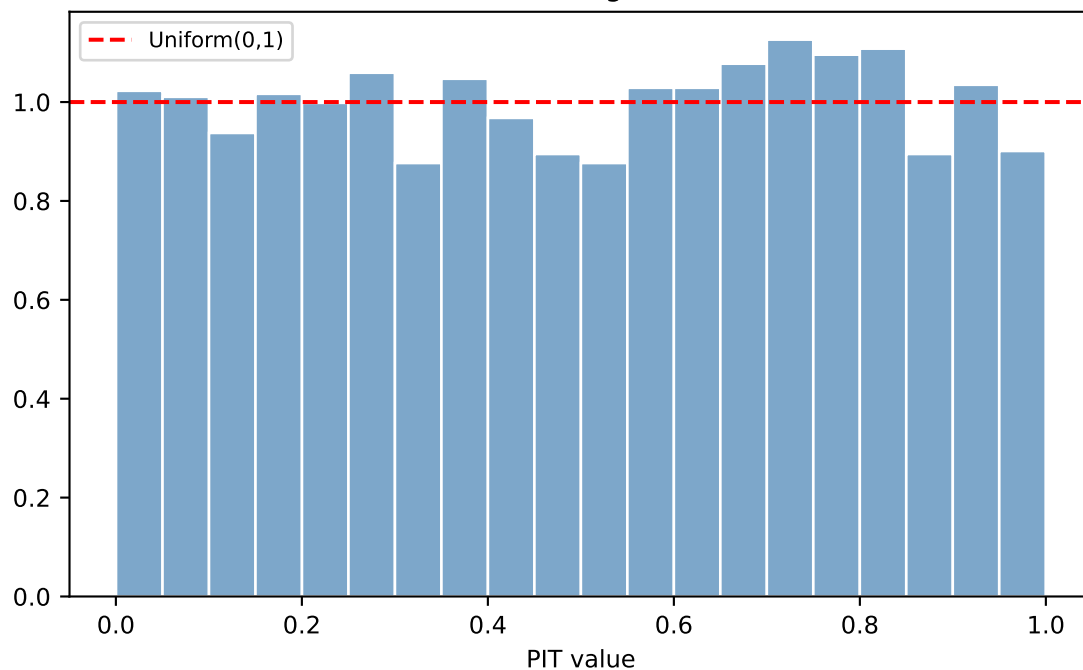


phi_temperature_seasonal_lags — correlation

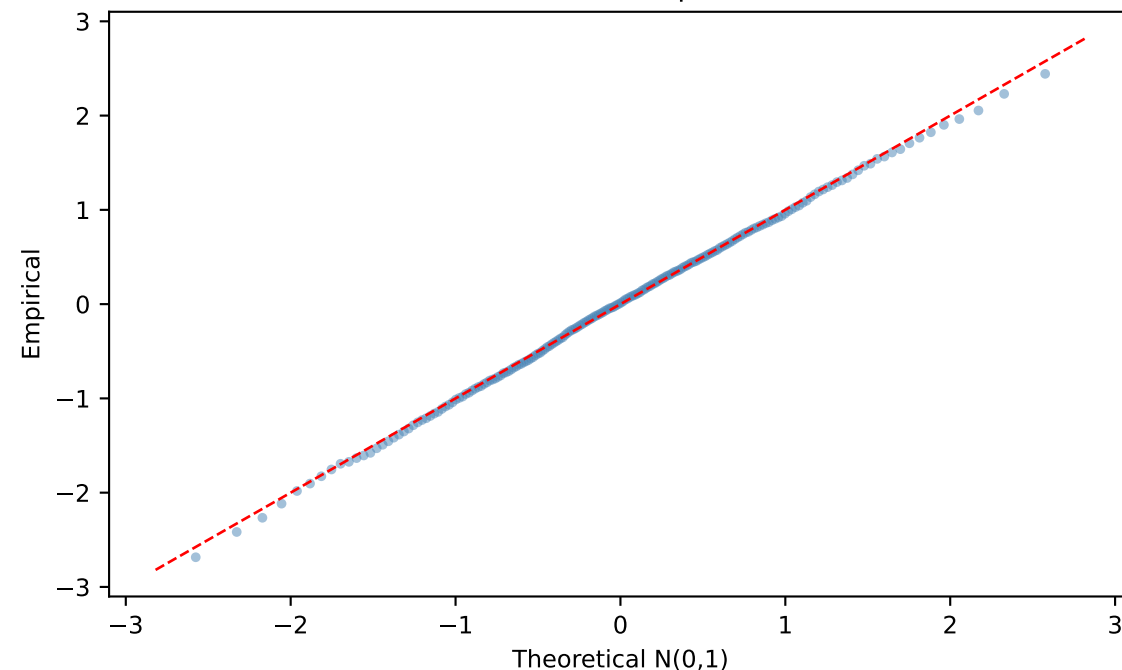


phi_dewpoint_temperature_t — IS diagnostics

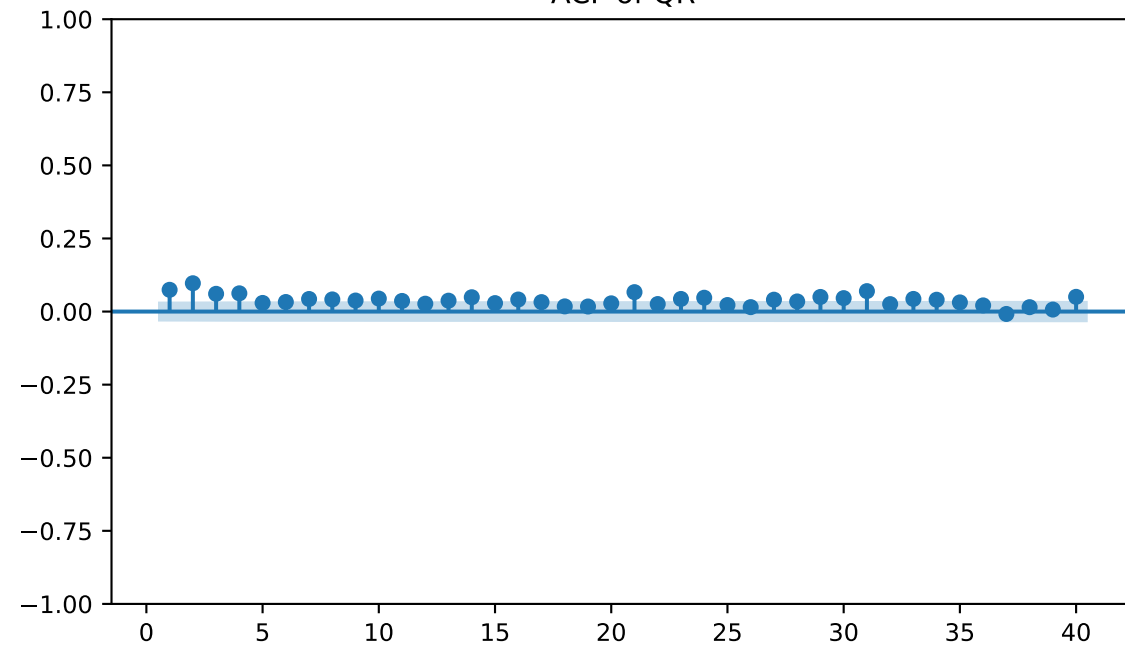
PIT histogram



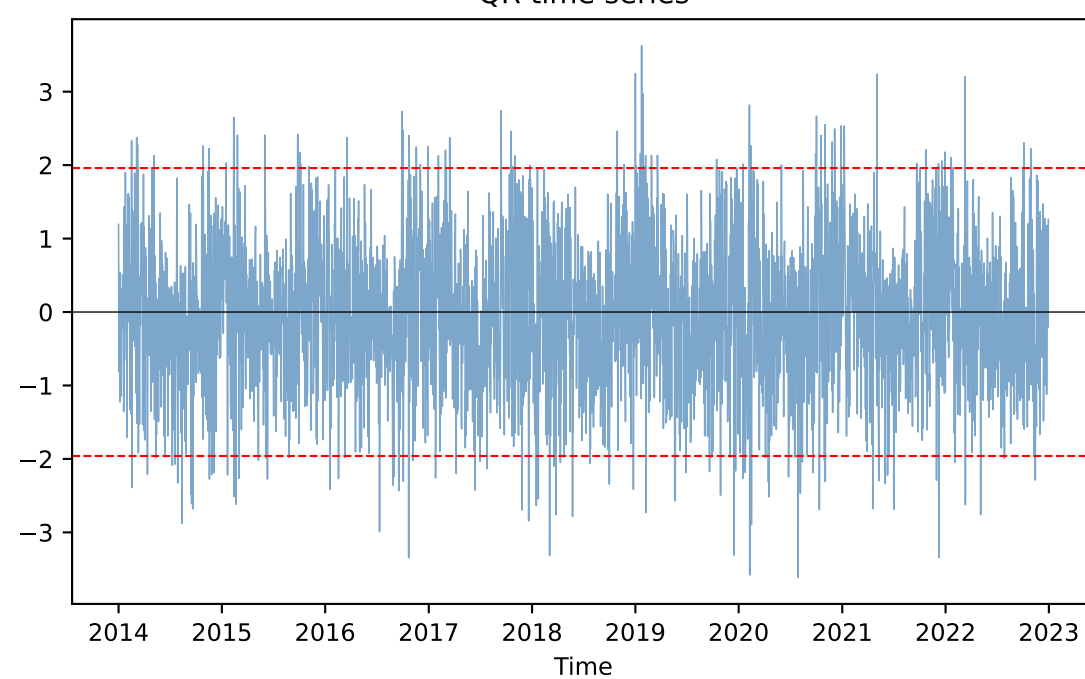
Normal QQ plot



ACF of QR

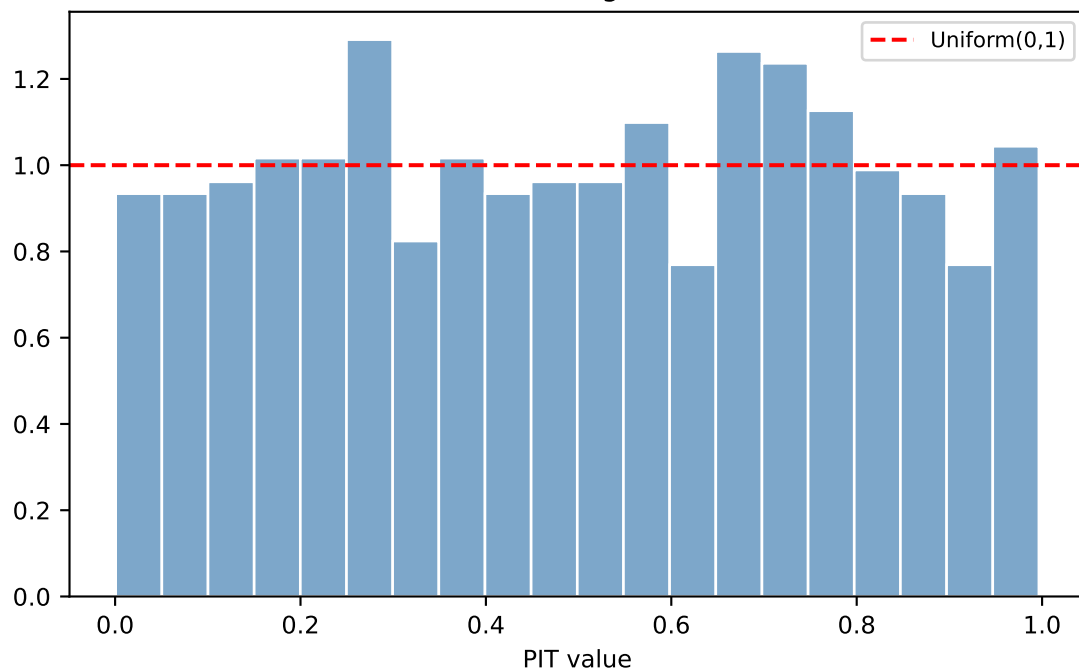


QR time series

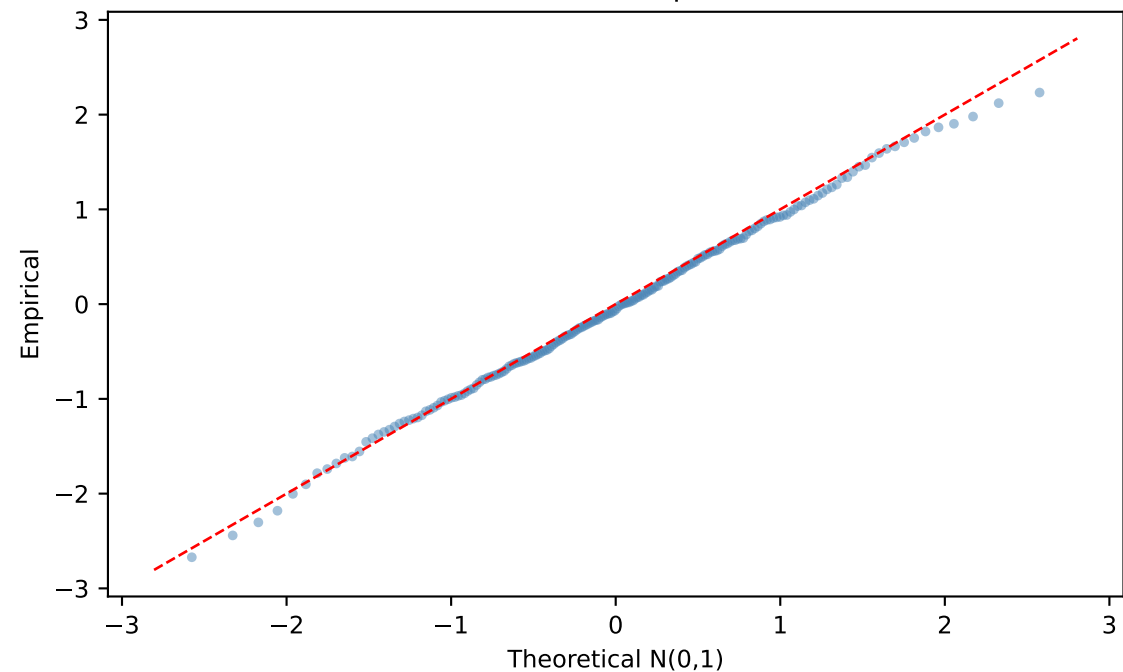


phi_dewpoint_temperature_t — OOS diagnostics

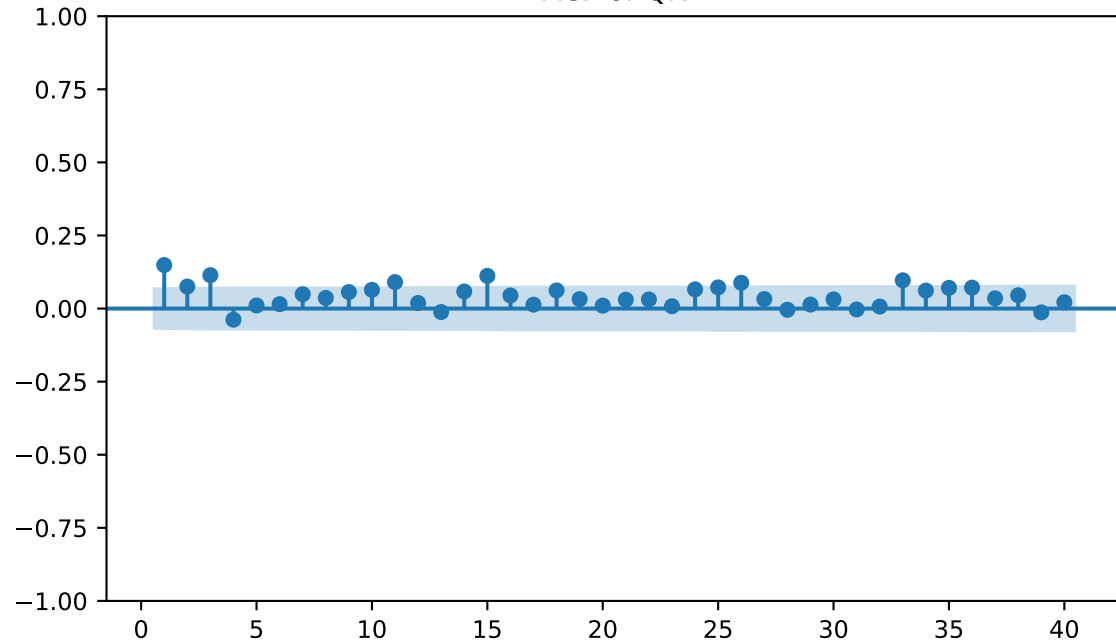
PIT histogram



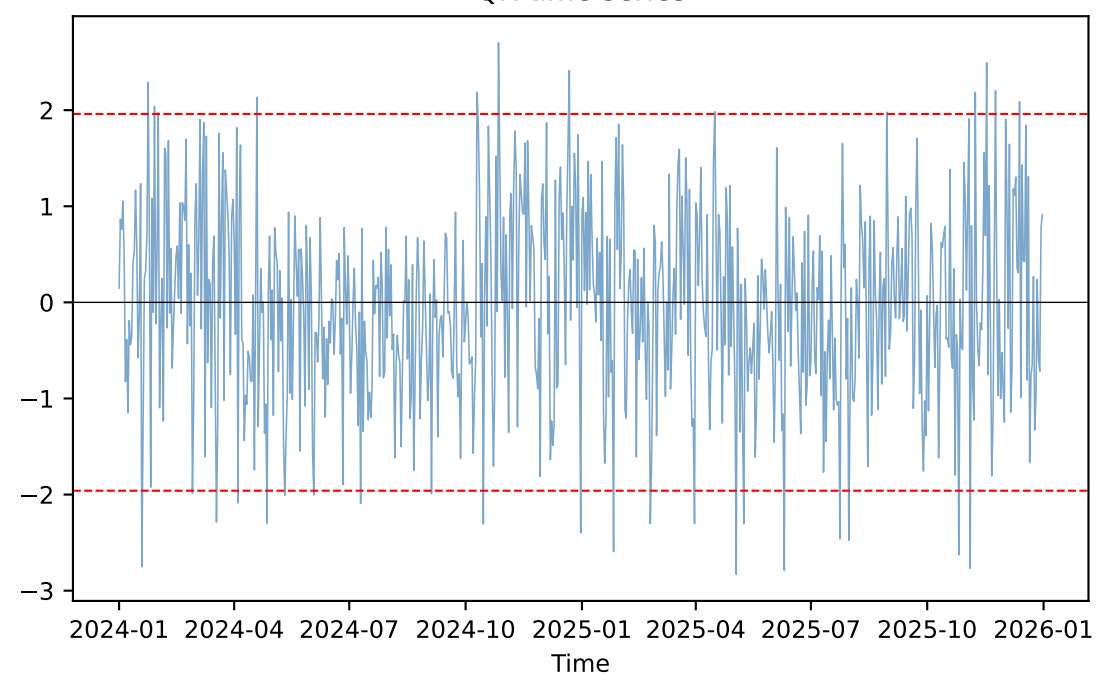
Normal QQ plot



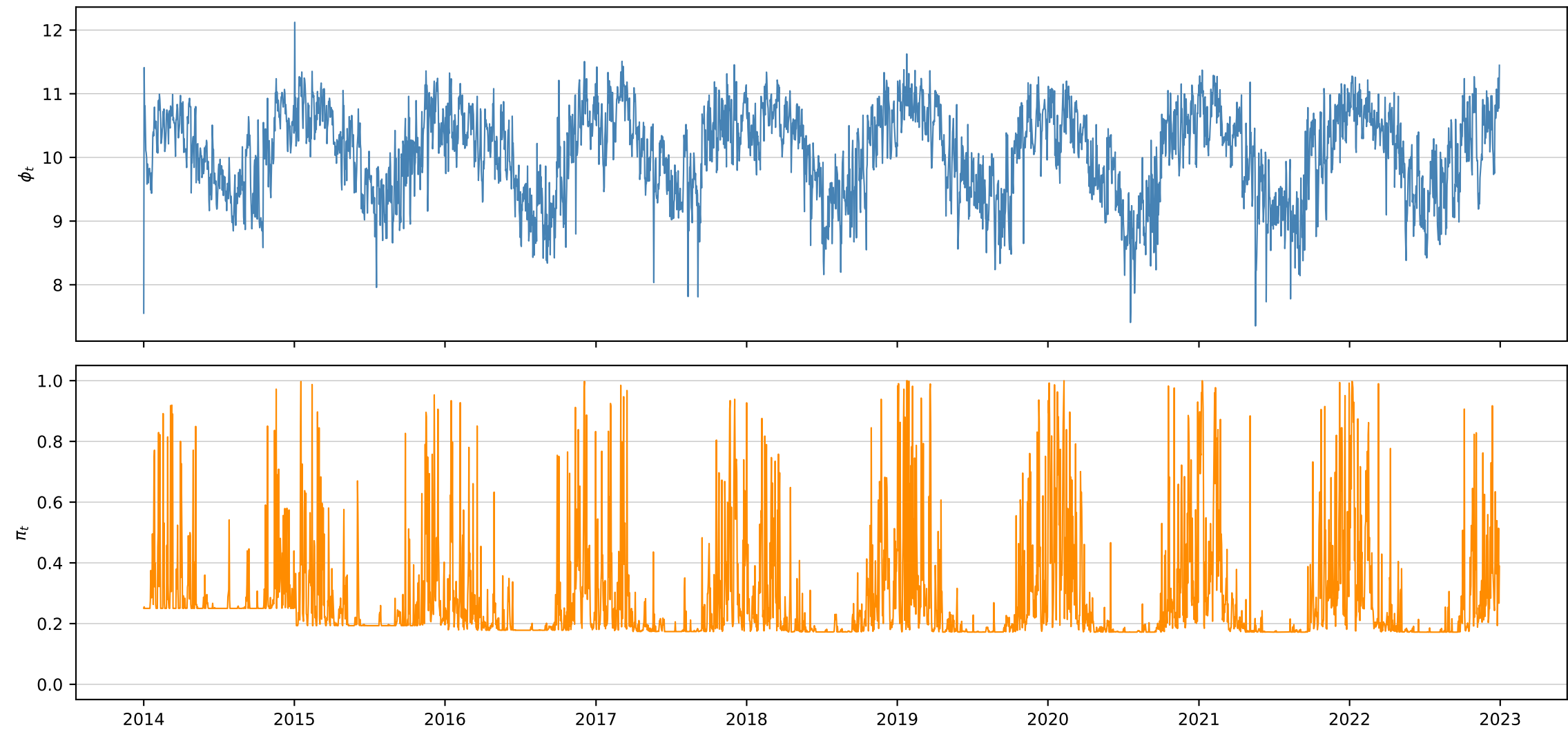
ACF of QR



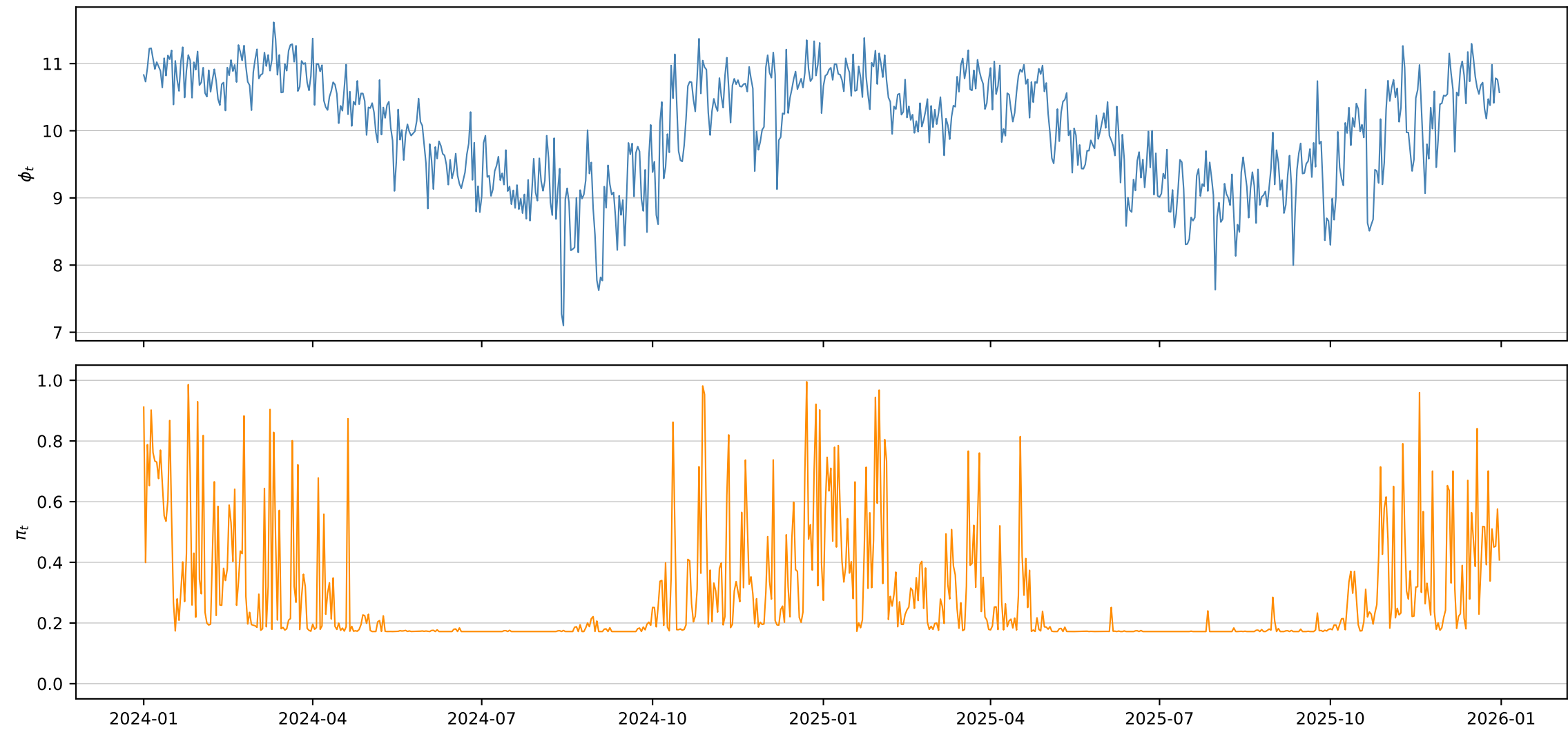
QR time series



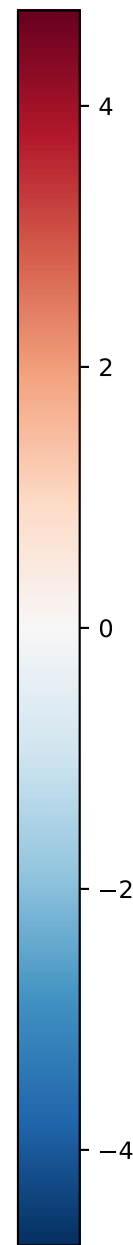
phi_dewpoint_temperature_t — filtered states (IS)



phi_dewpoint_temperature_t — filtered states (OOS)

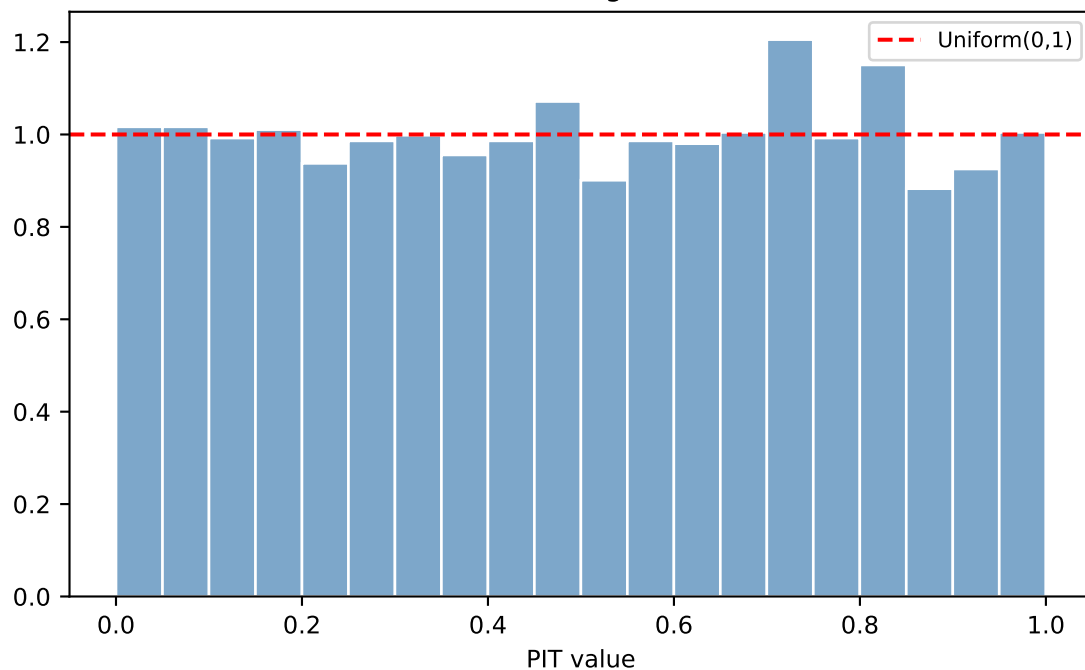


The heatmap displays the relationships between variables, with colors indicating the strength and direction of the correlation. Dark red indicates a strong positive correlation, while dark blue indicates a strong negative correlation. The plot shows several clusters of positive and negative relationships, with a prominent positive cluster in the top right and a negative cluster in the bottom left.

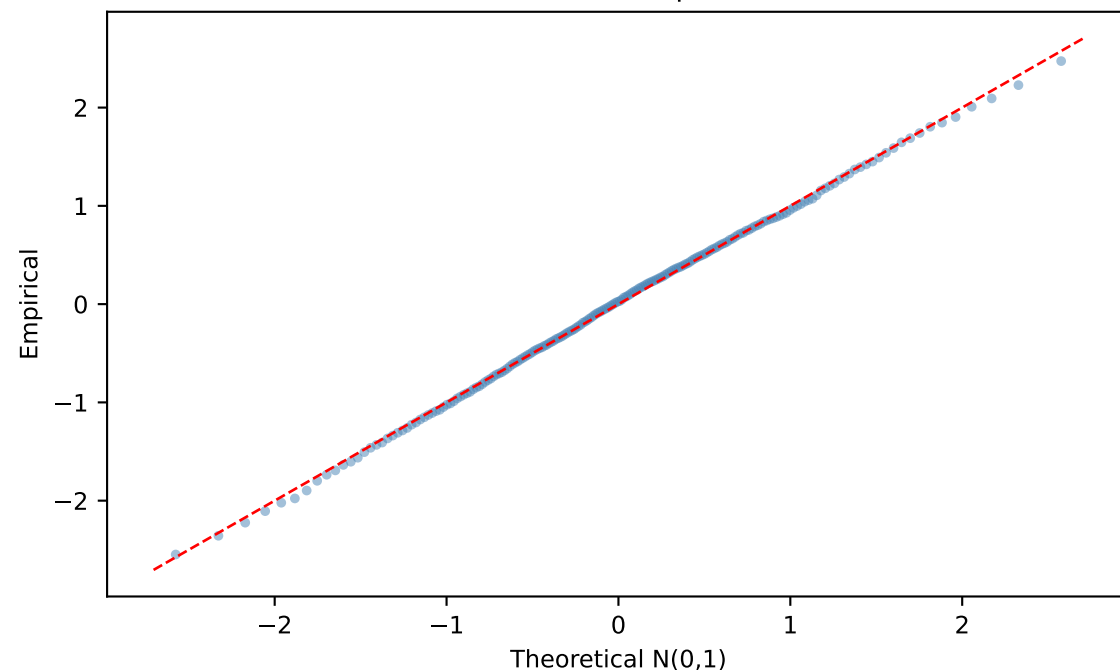


phi_dewpoint_temperature_lag1 — IS diagnostics

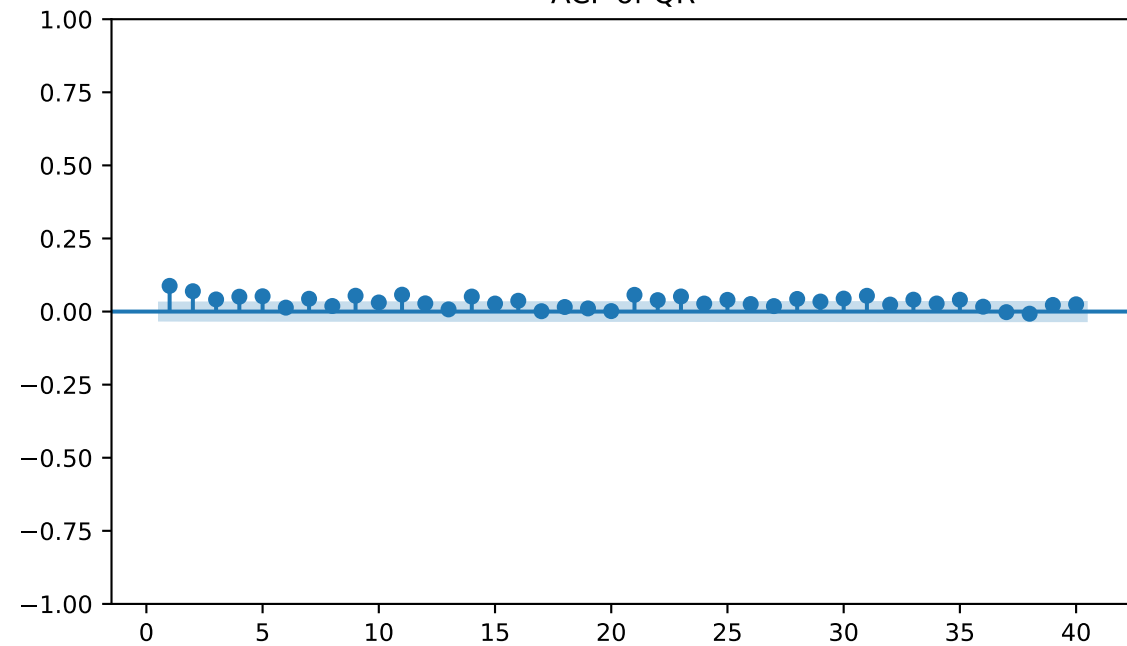
PIT histogram



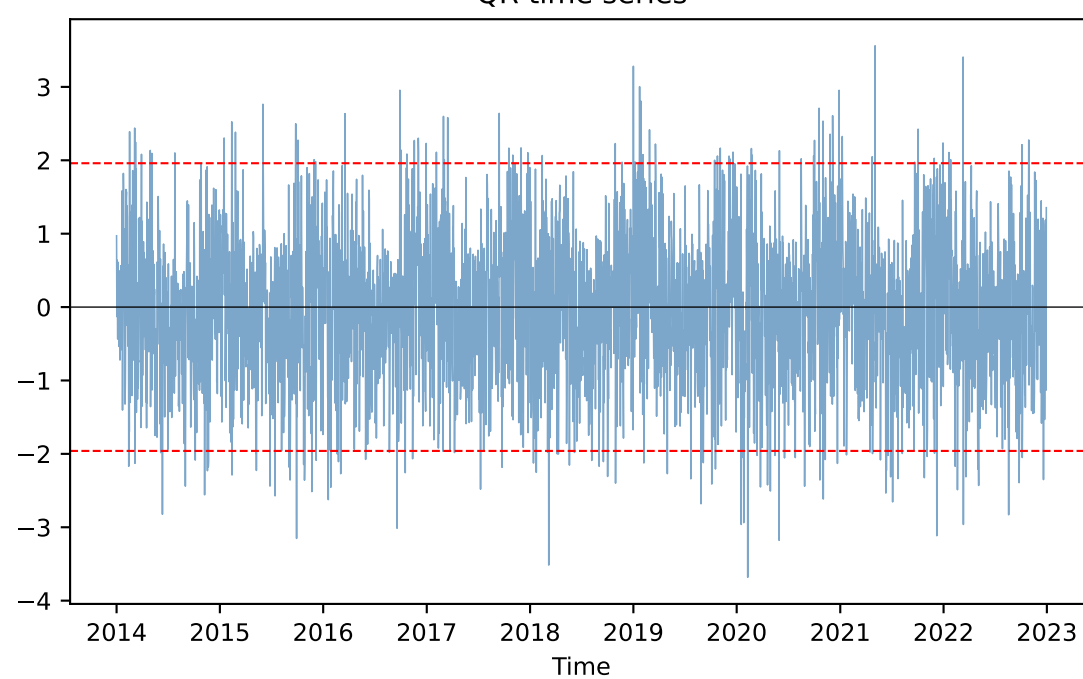
Normal QQ plot



ACF of QR

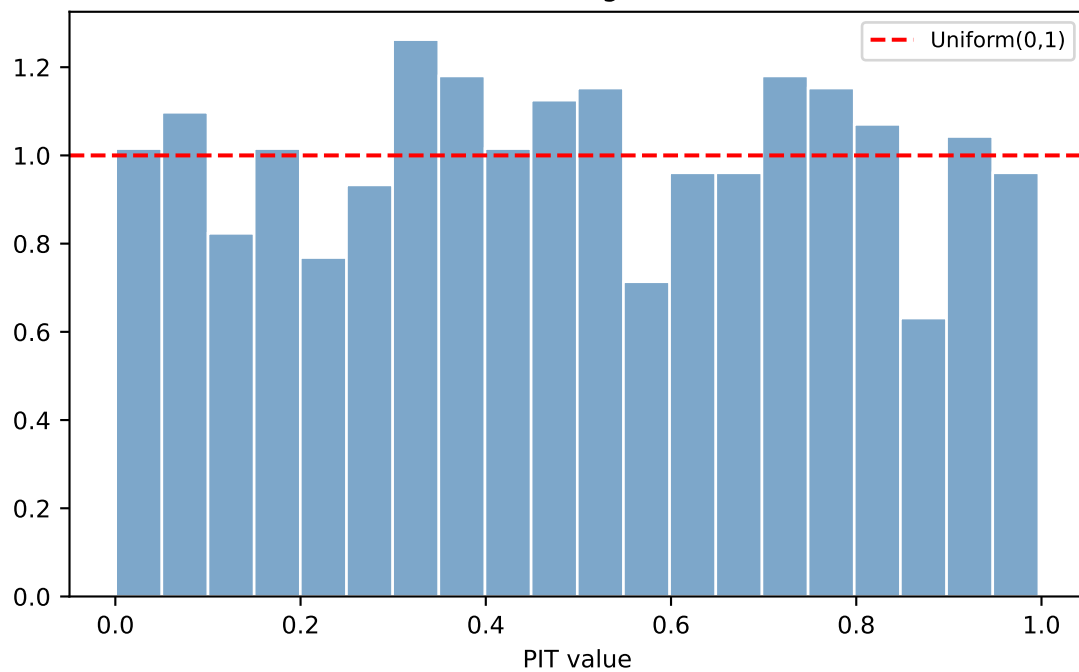


QR time series

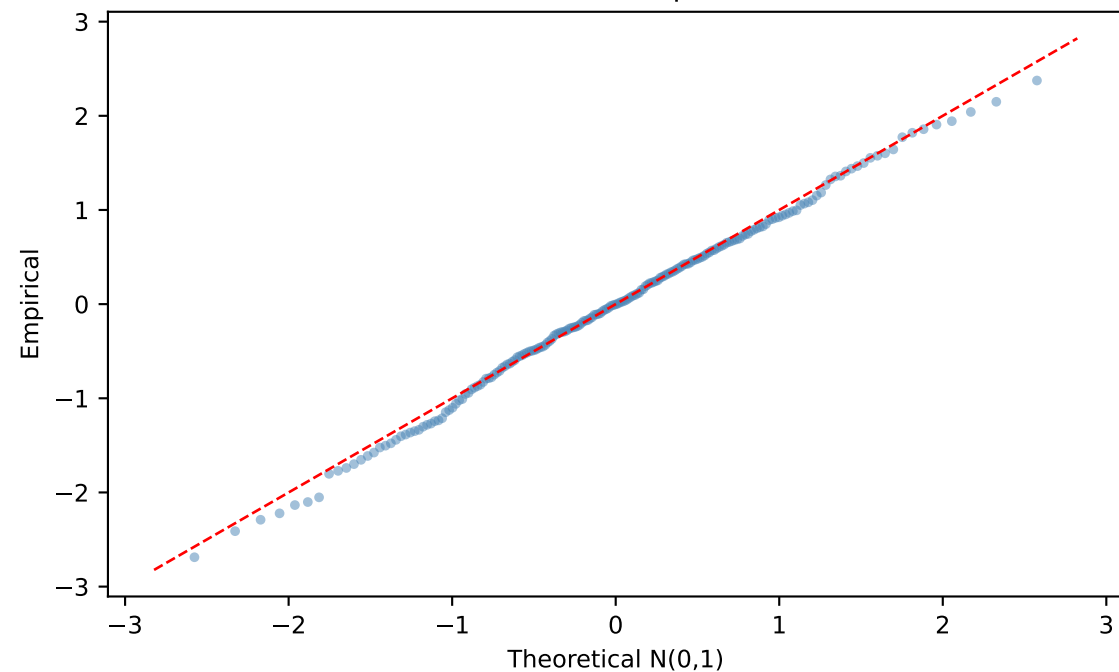


phi_dewpoint_temperature_lag1 — OOS diagnostics

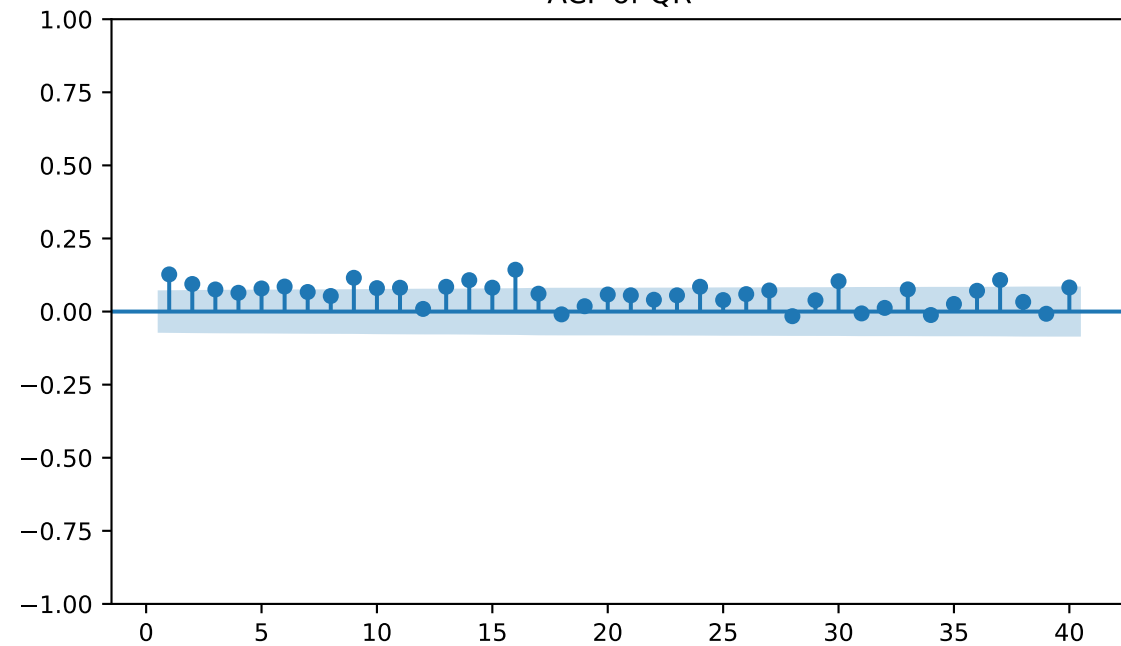
PIT histogram



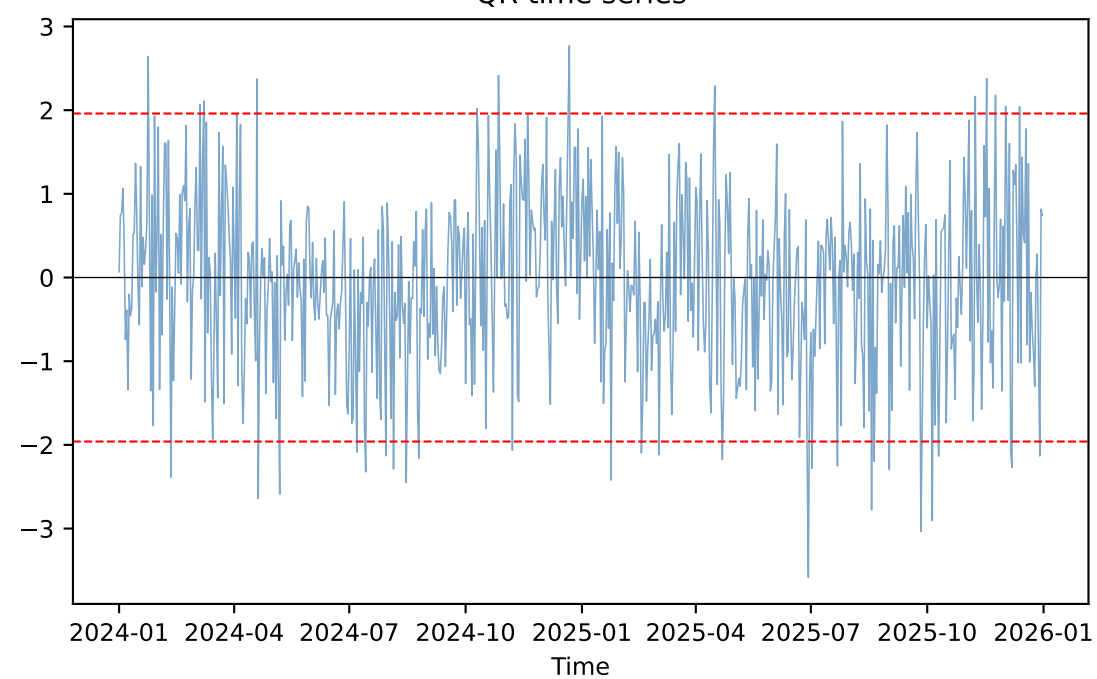
Normal QQ plot



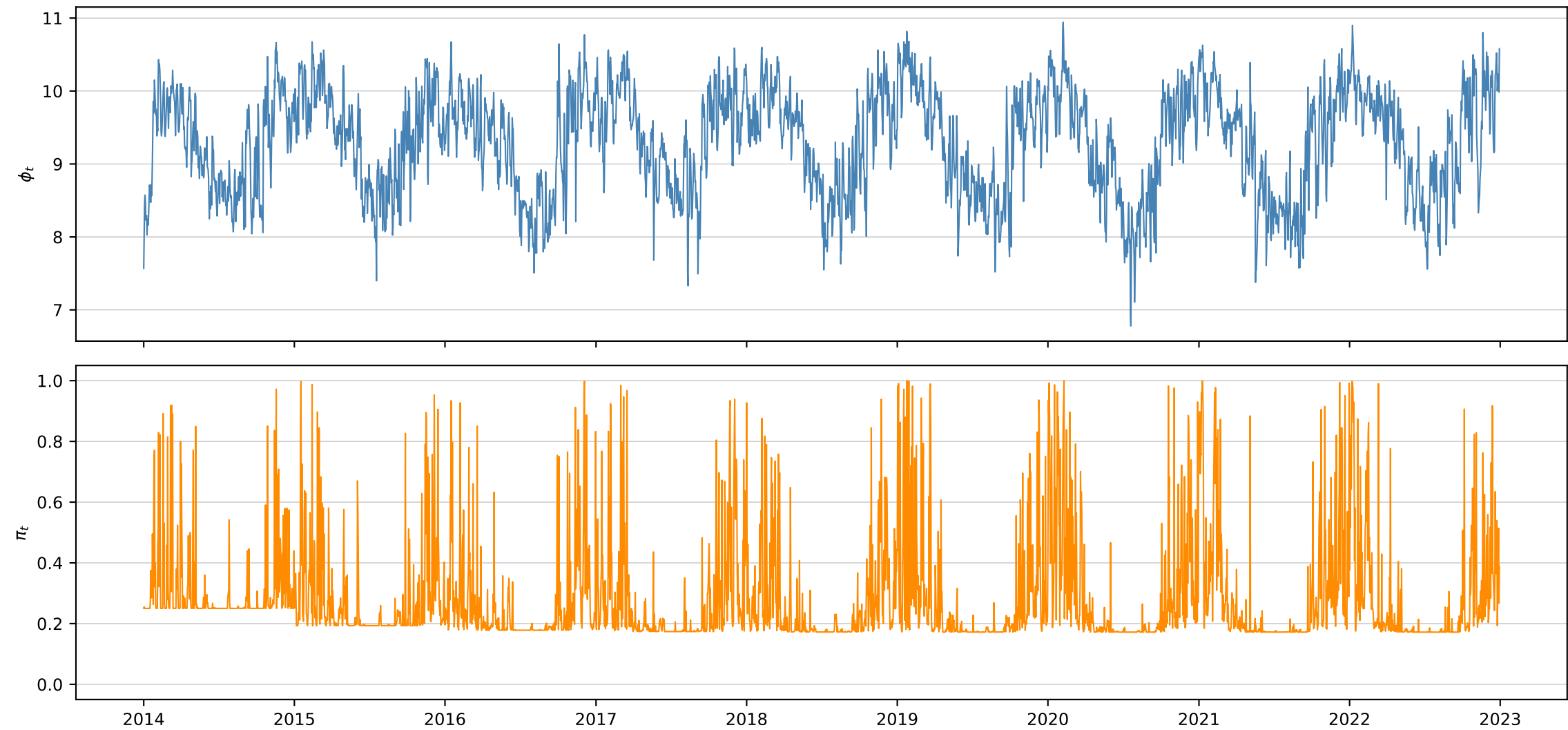
ACF of QR



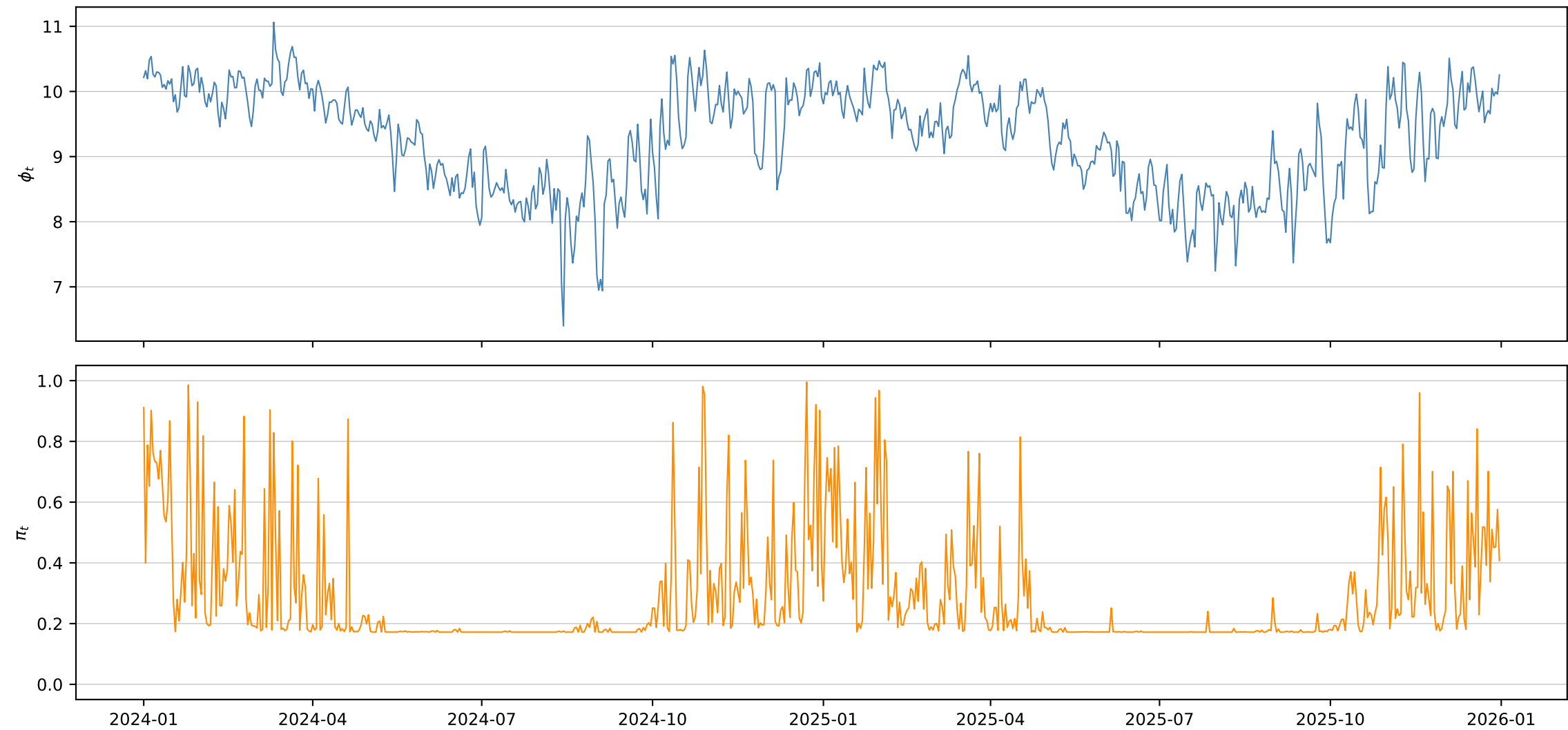
QR time series



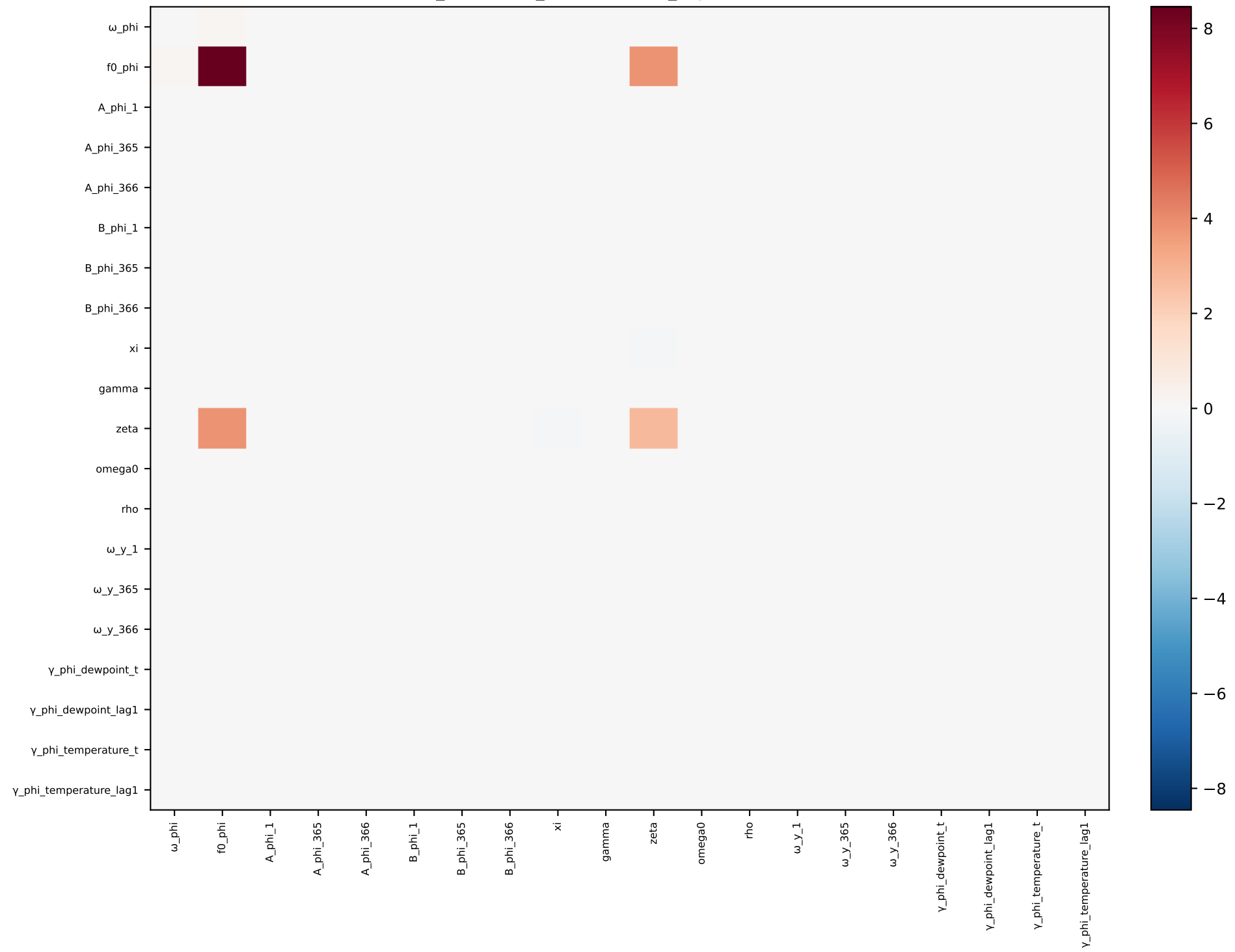
phi_dewpoint_temperature_lag1 — filtered states (IS)



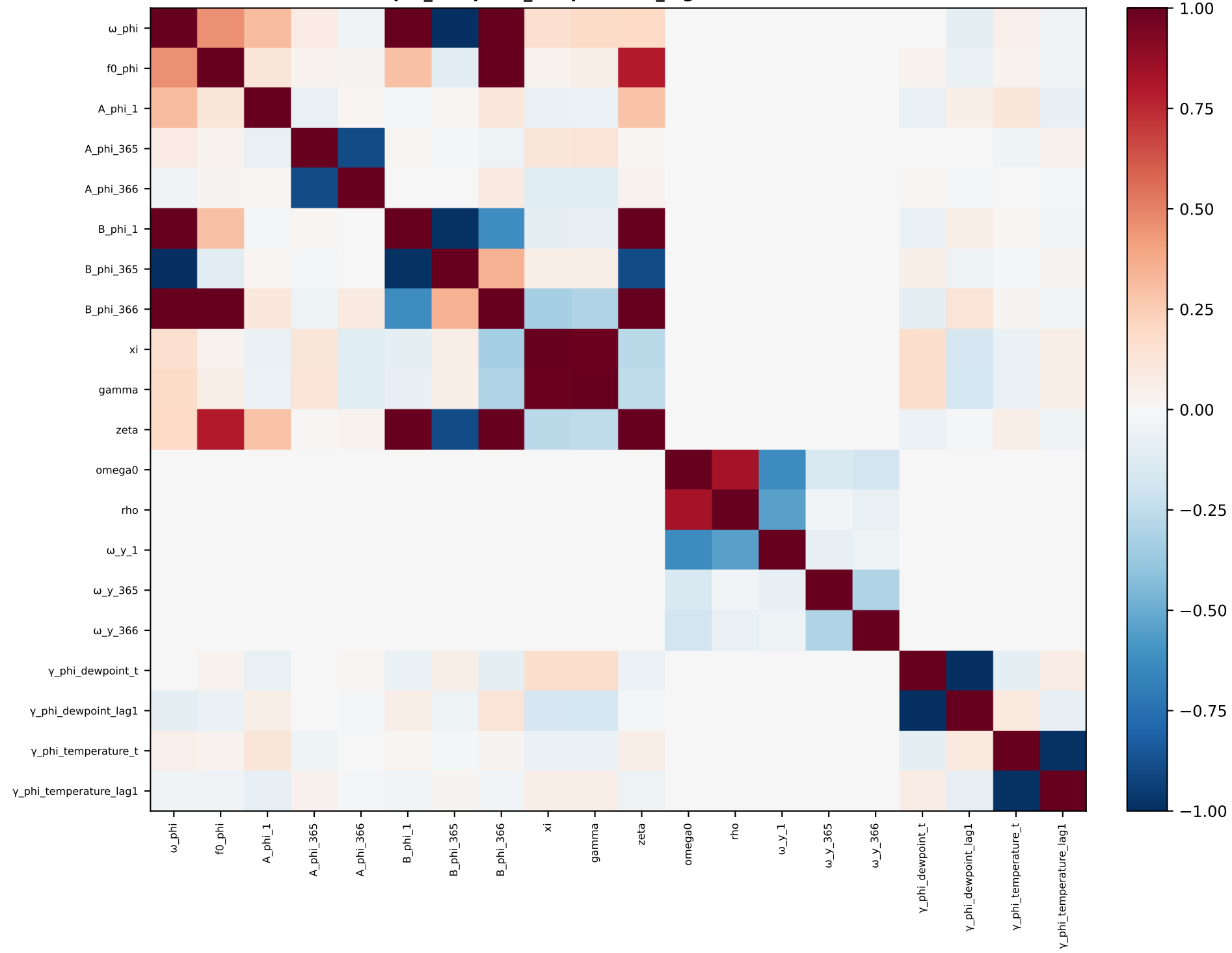
phi_dewpoint_temperature_lag1 — filtered states (OOS)



phi_dewpoint_temperature_lag1 — covariance

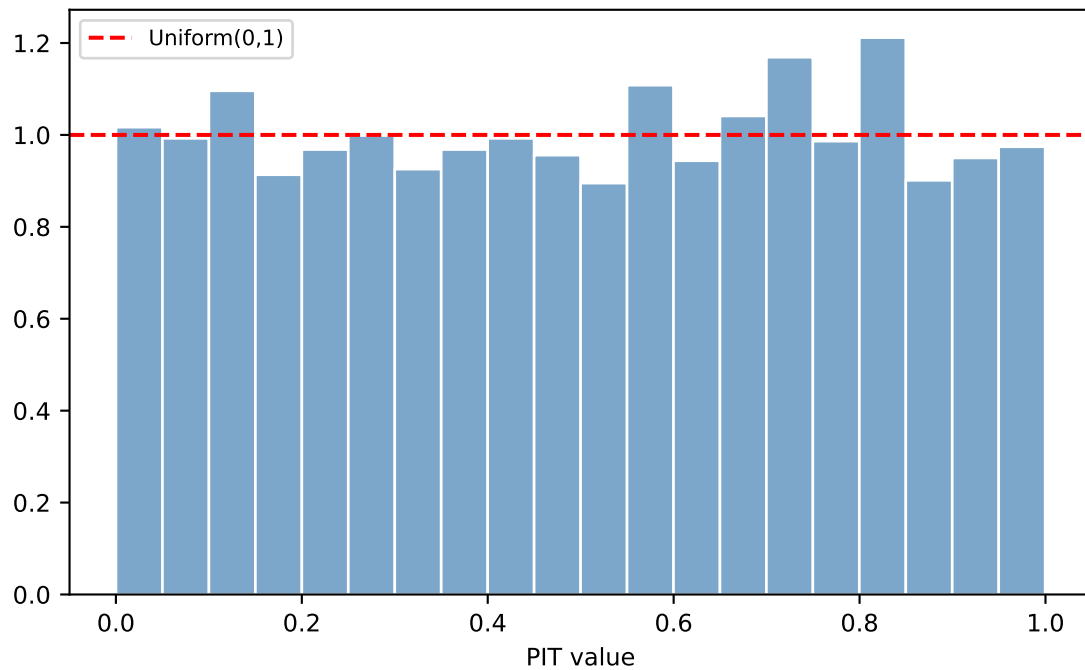


phi_dewpoint_temperature_lag1 — correlation

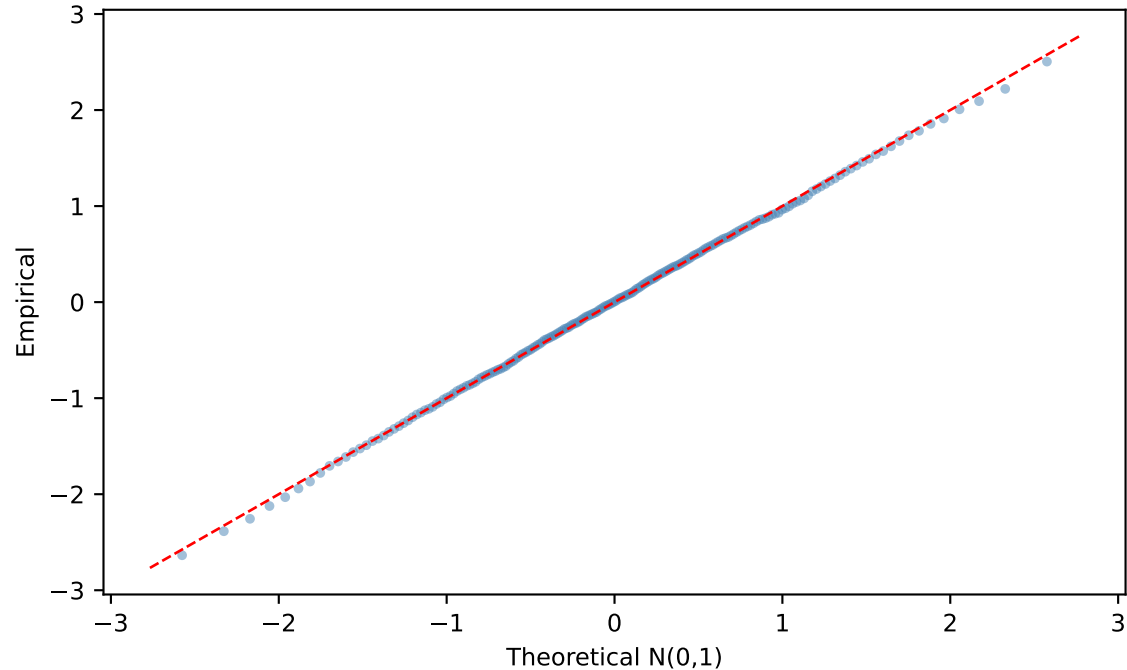


phi_dewpoint_temperature_short_lags — IS diagnostics

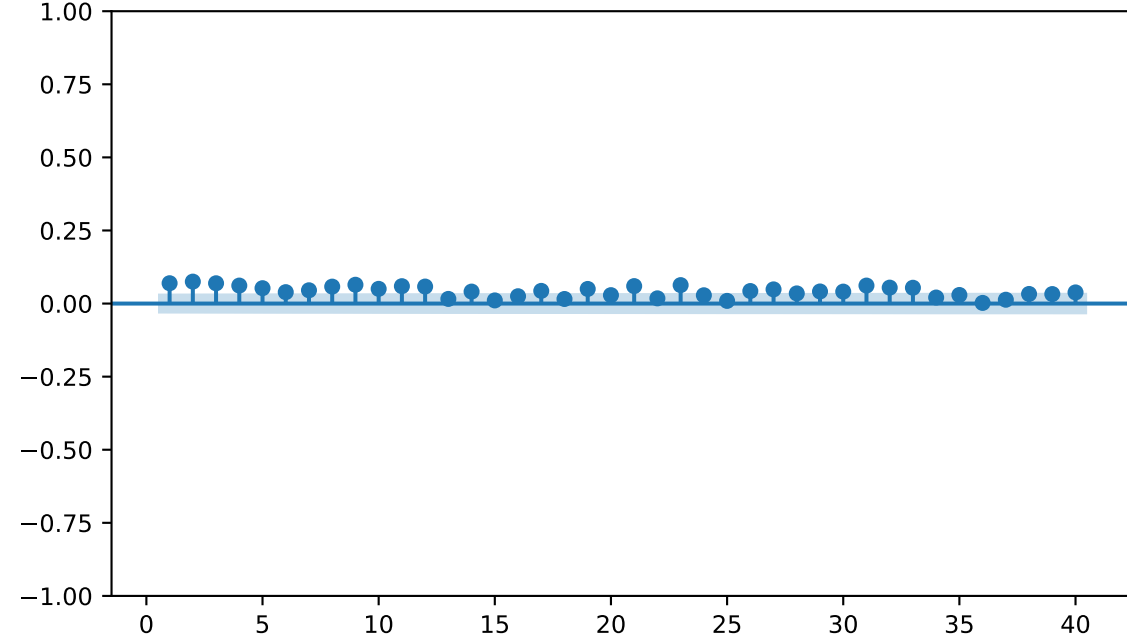
PIT histogram



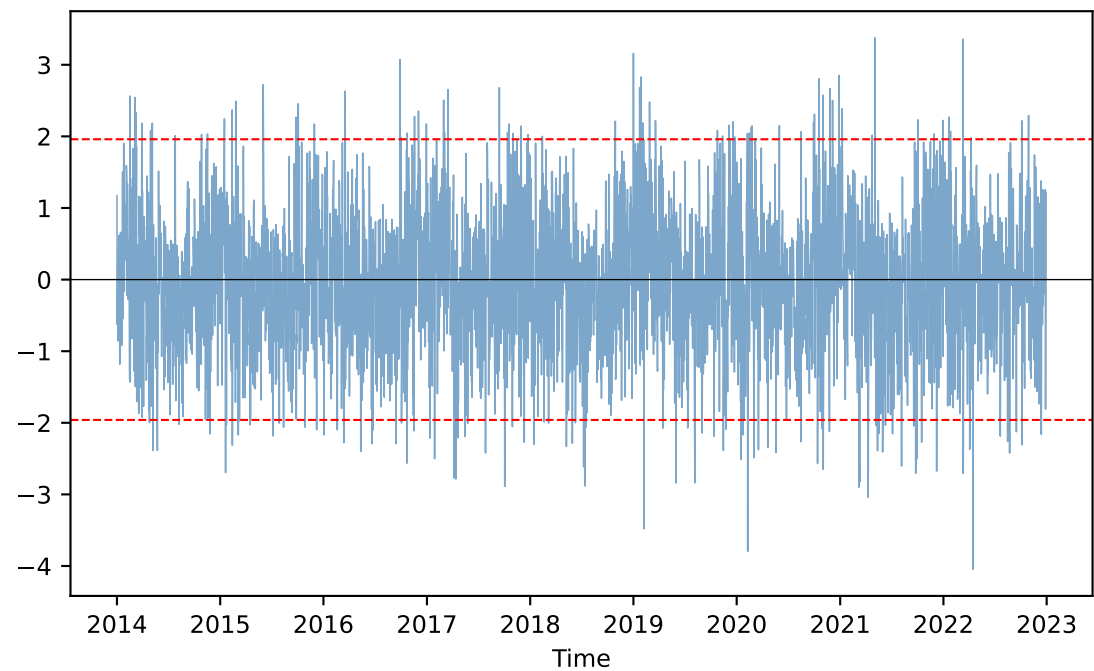
Normal QQ plot



ACF of QR

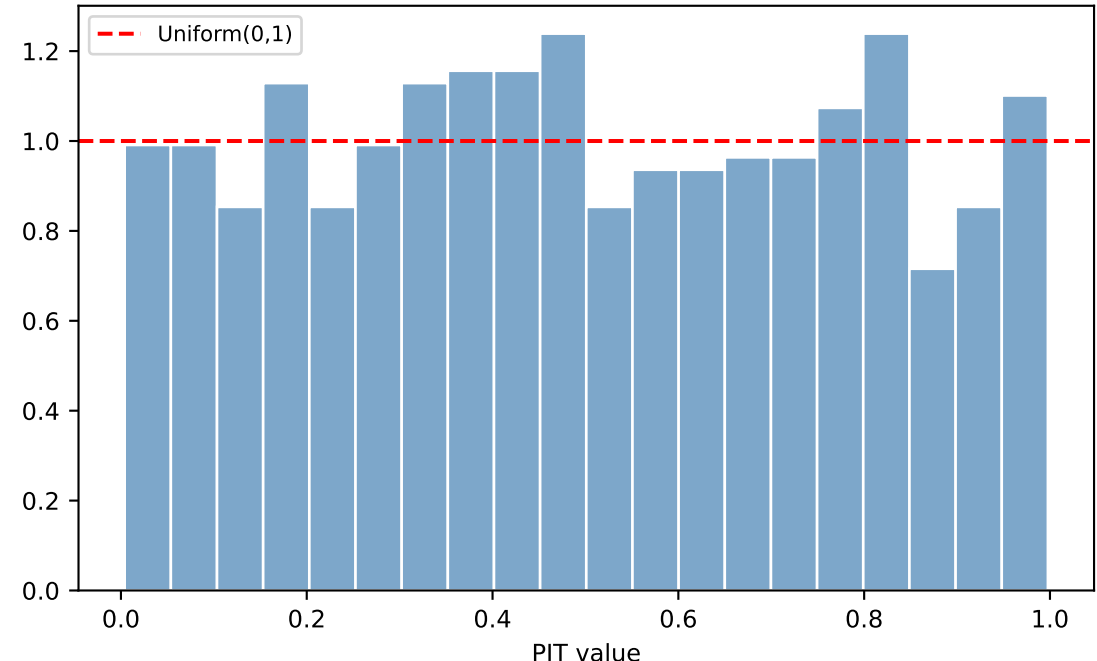


QR time series

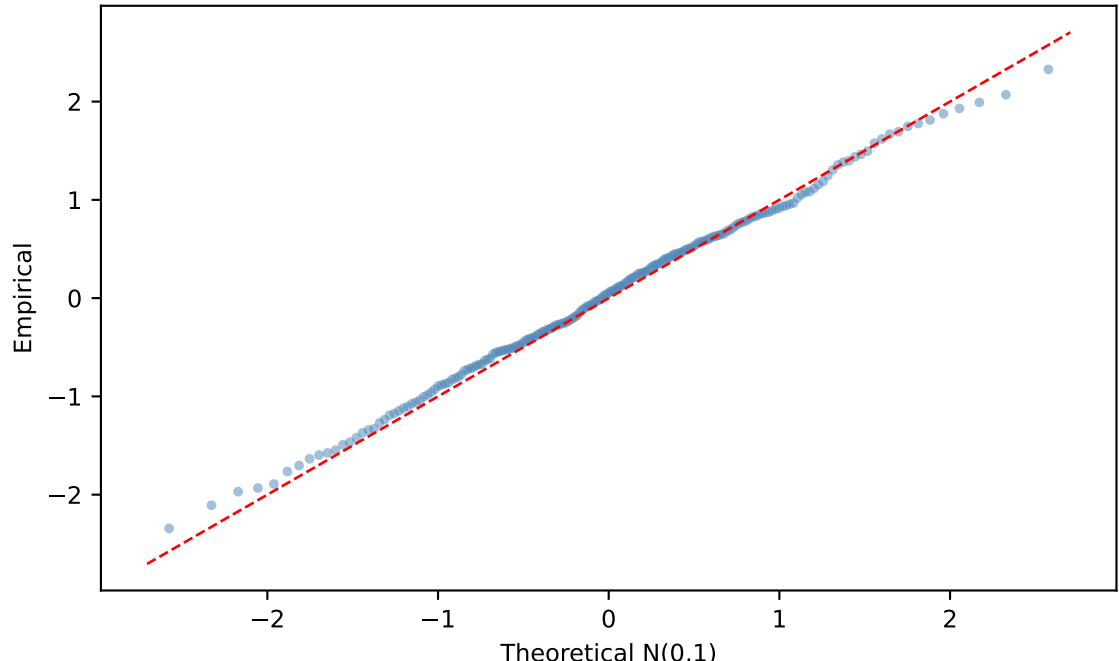


phi_dewpoint_temperature_short_lags — OOS diagnostics

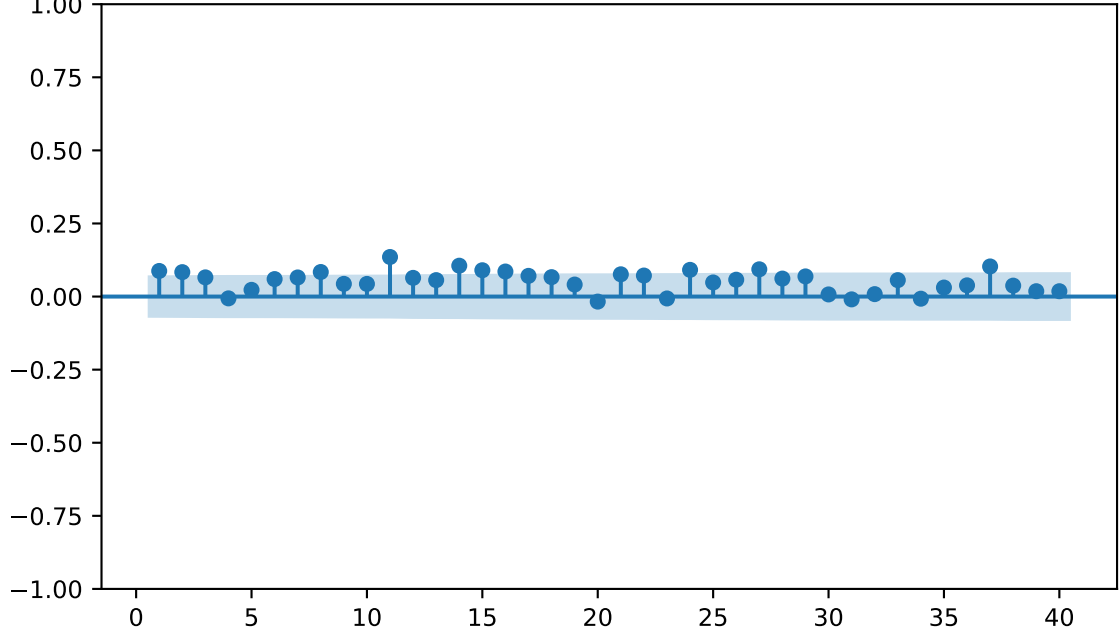
PIT histogram



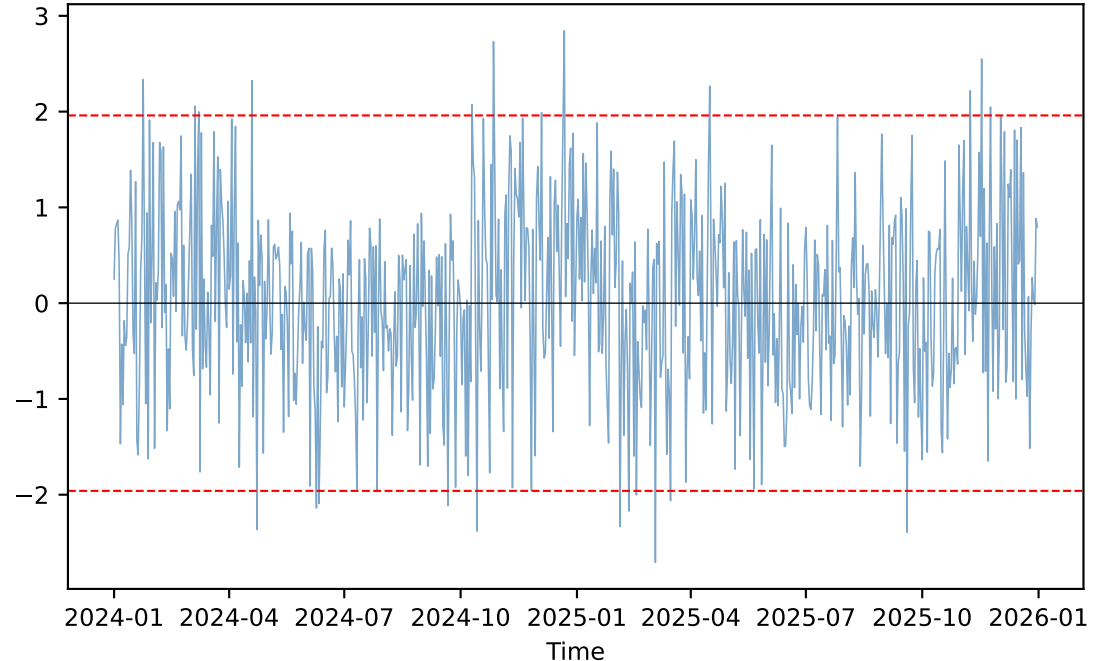
Normal QQ plot



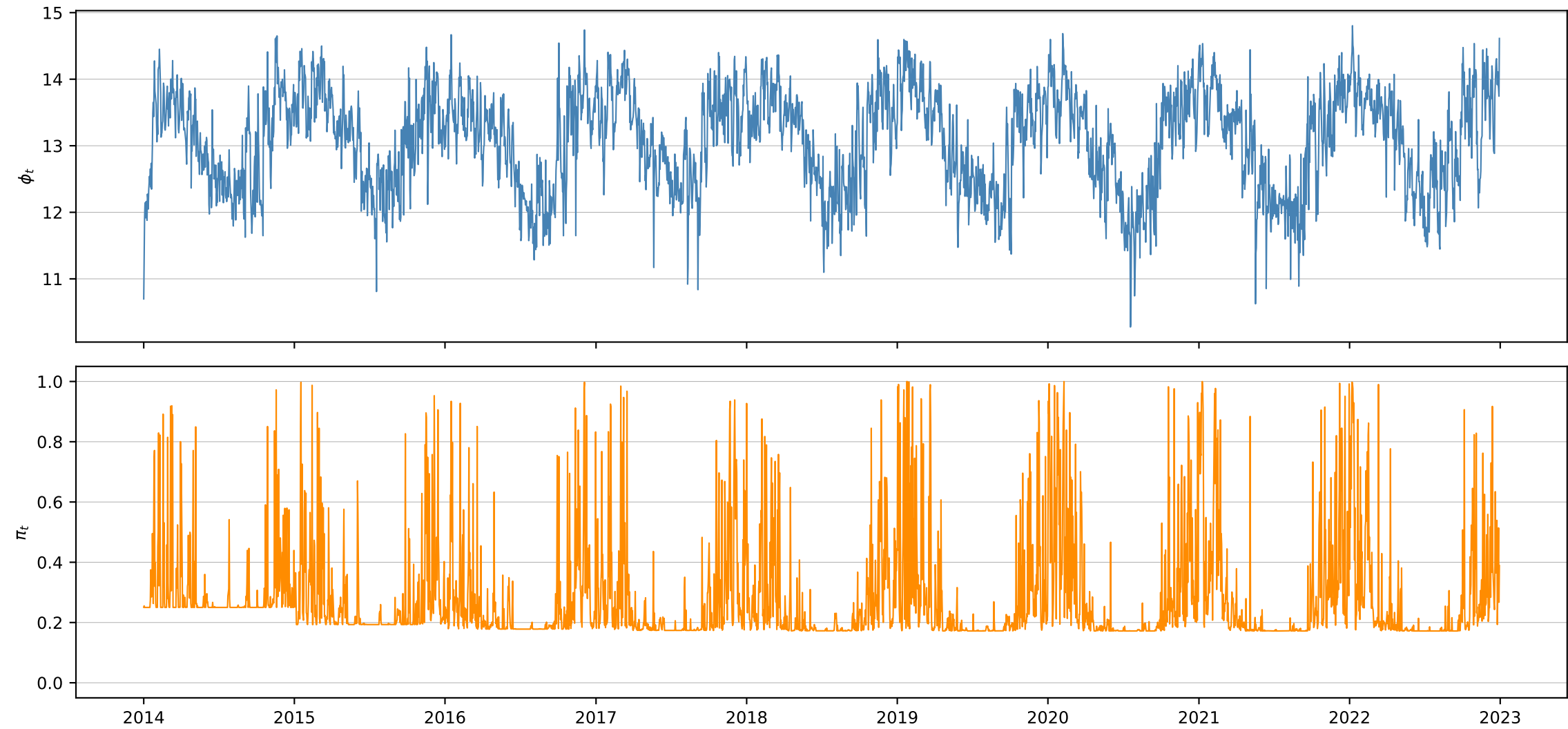
ACF of QR



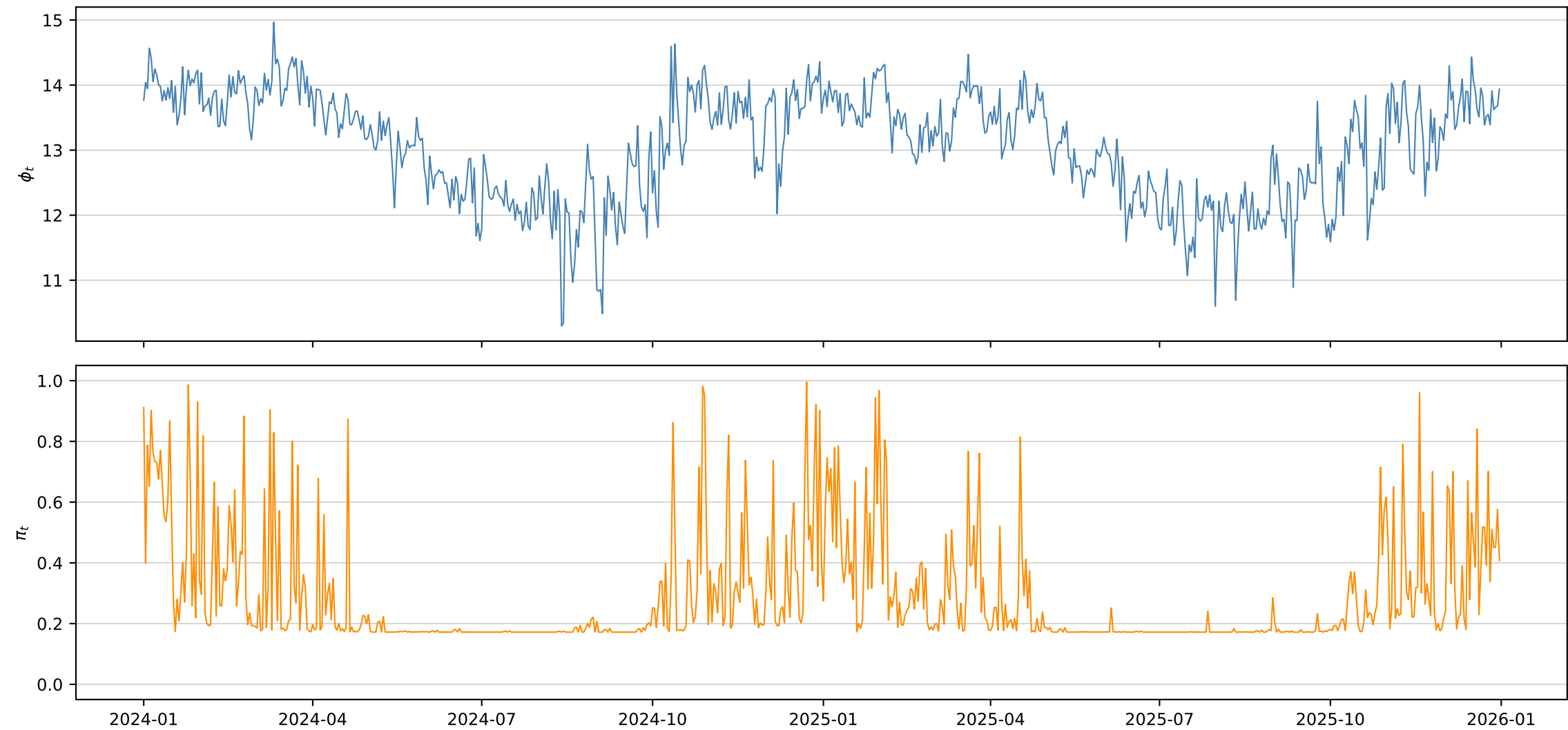
QR time series



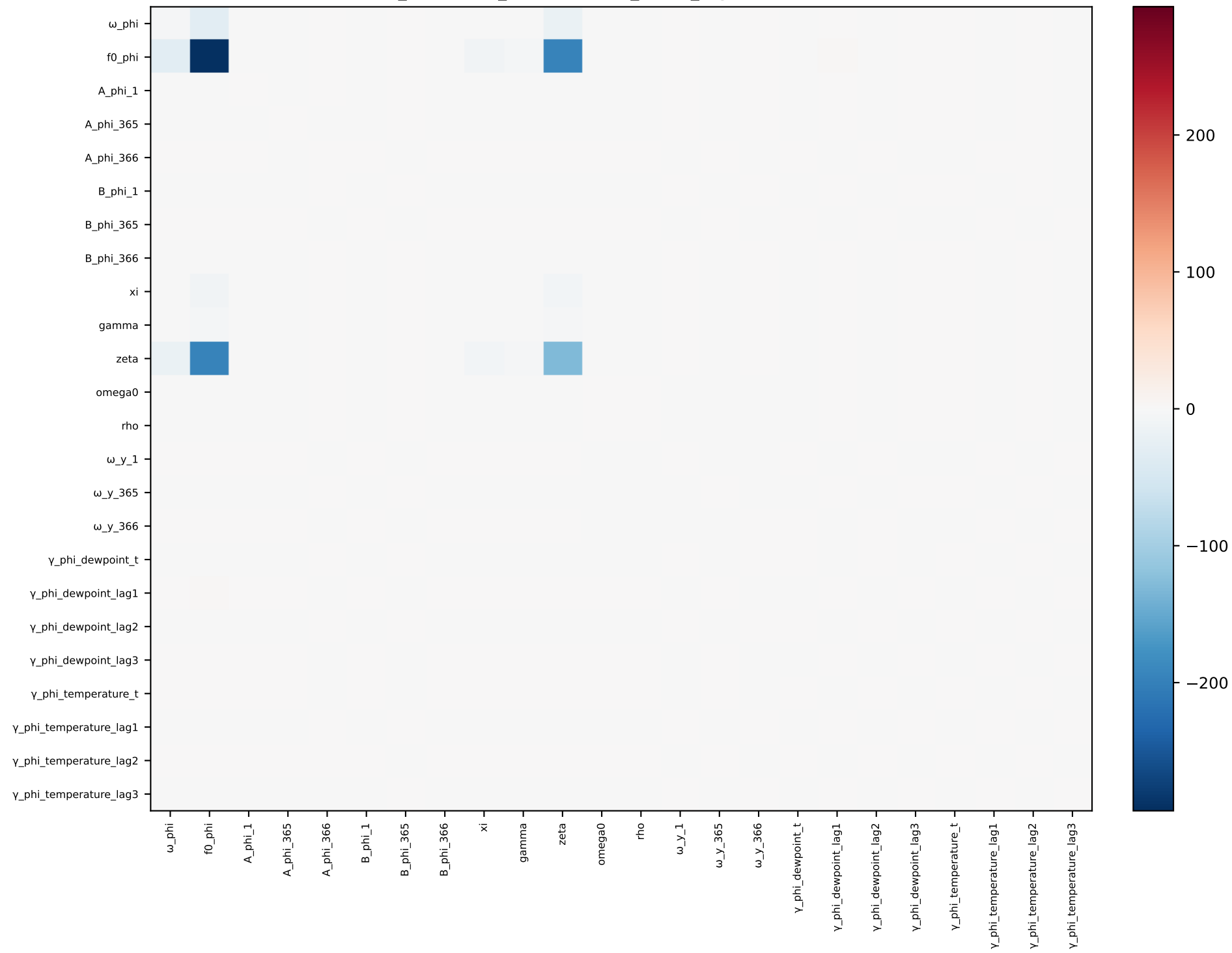
phi_dewpoint_temperature_short_lags — filtered states (IS)



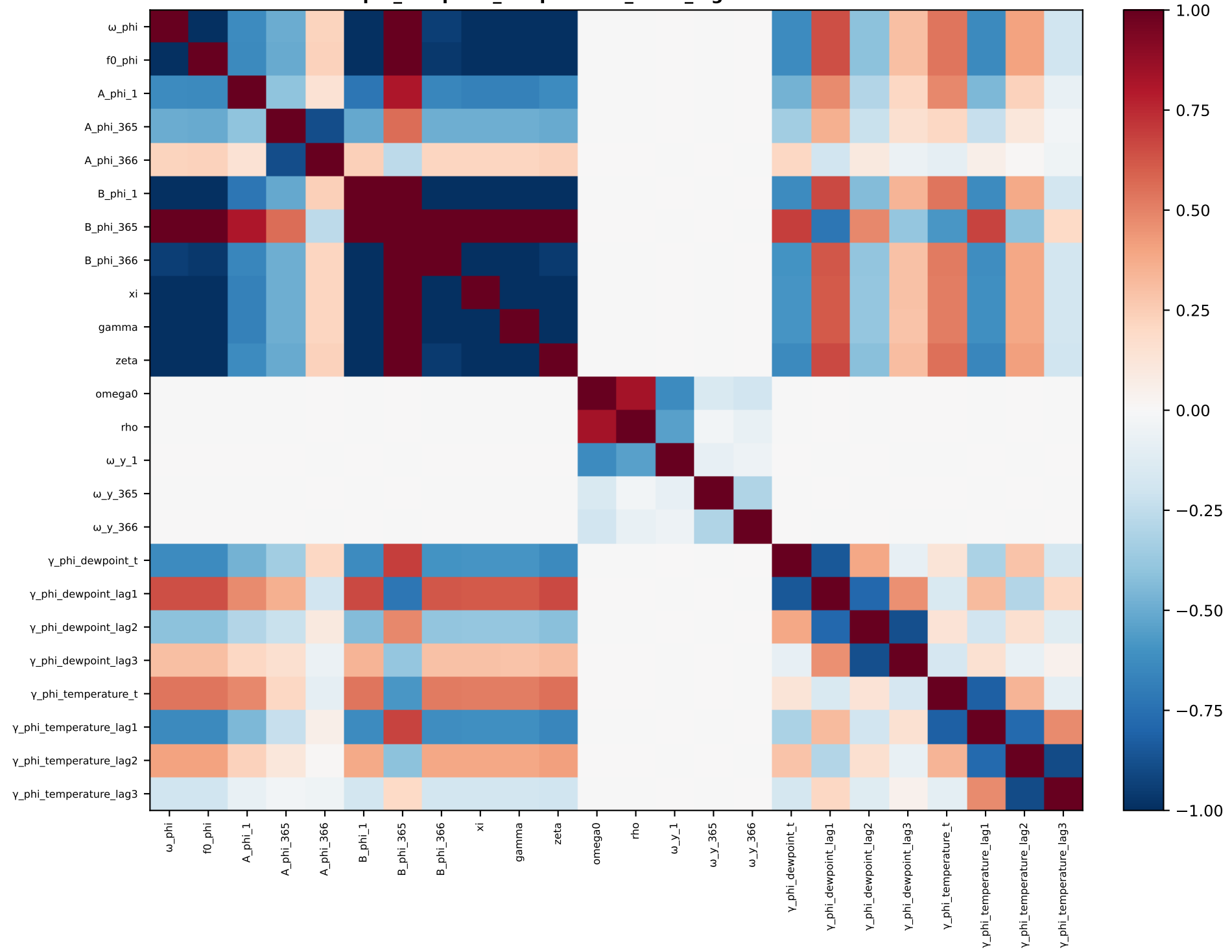
phi_dewpoint_temperature_short_lags — filtered states (OOS)



phi_dewpoint_temperature_short_lags – covariance

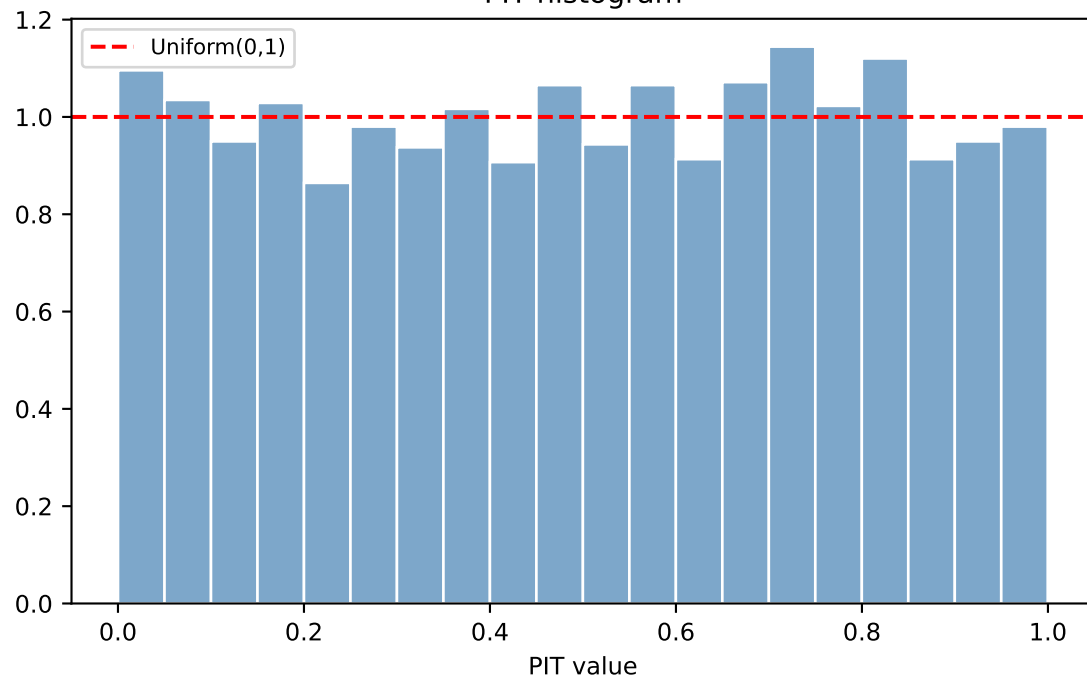


phi_dewpoint_temperature_short_lags — correlation

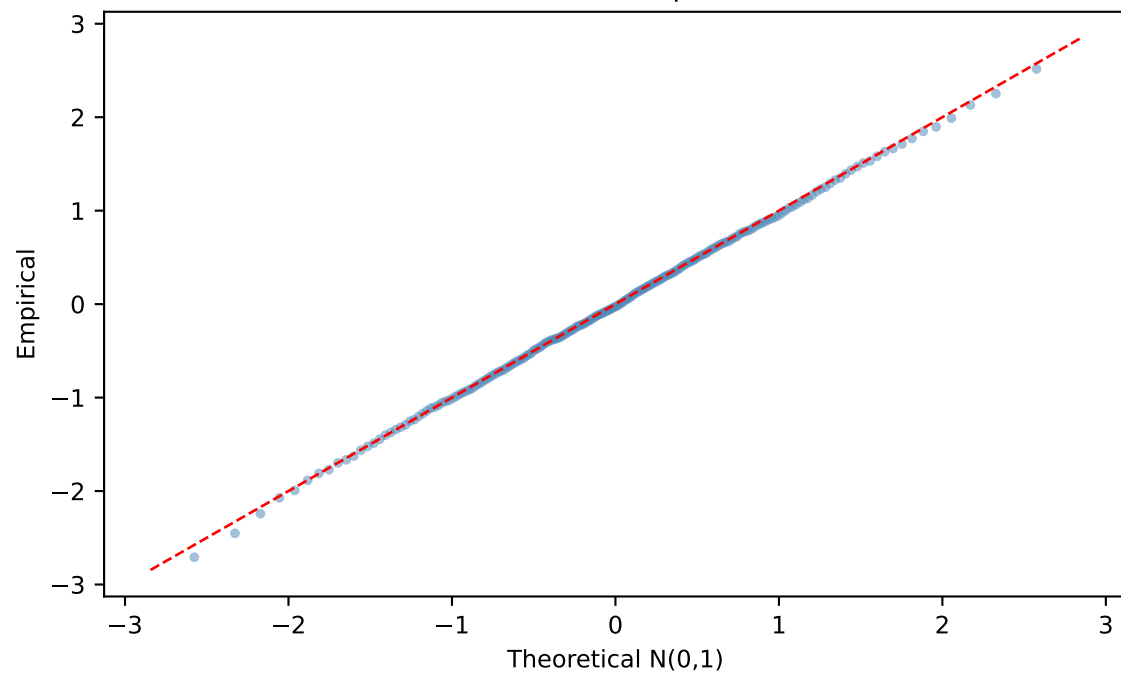


phi_dewpoint_temperature_seasonal_lags — IS diagnostics

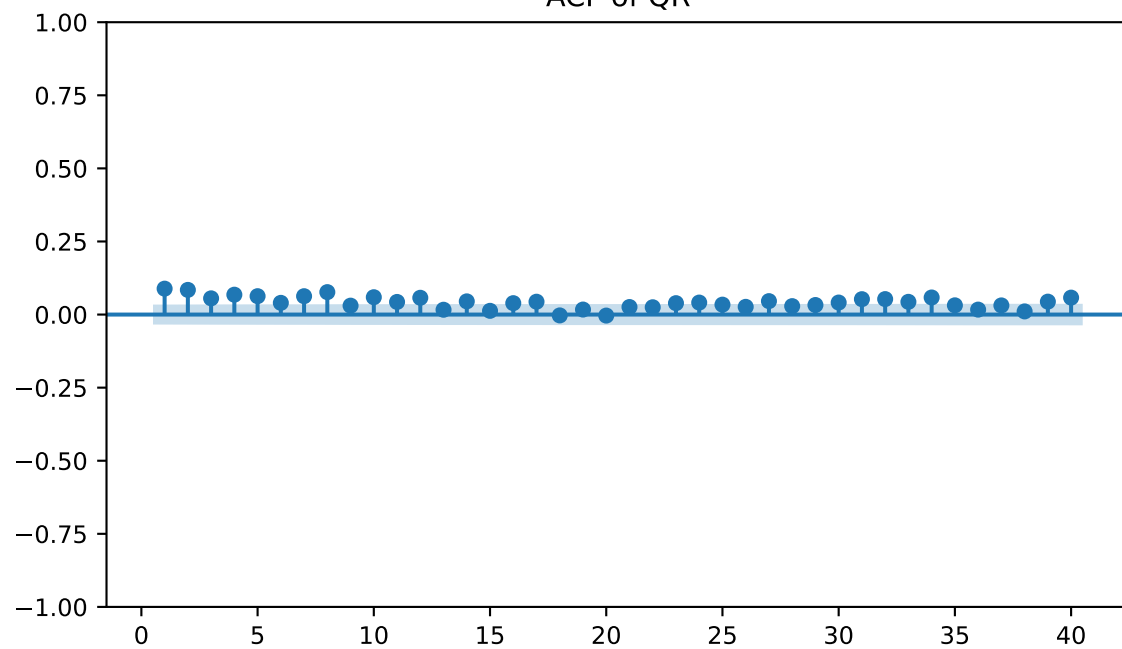
PIT histogram



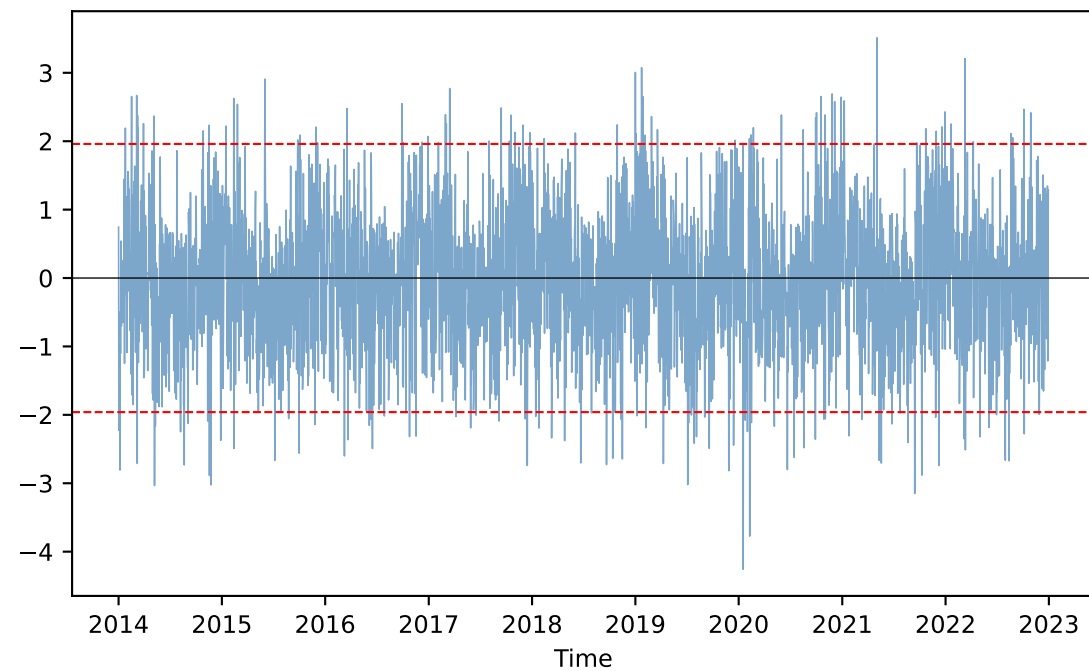
Normal QQ plot



ACF of QR

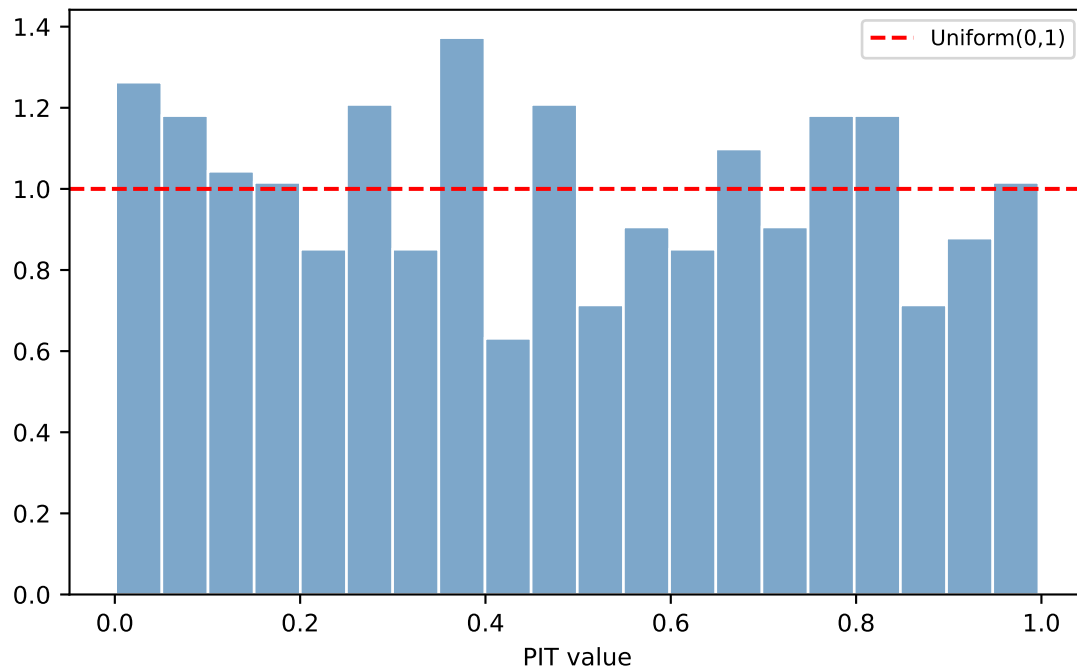


QR time series

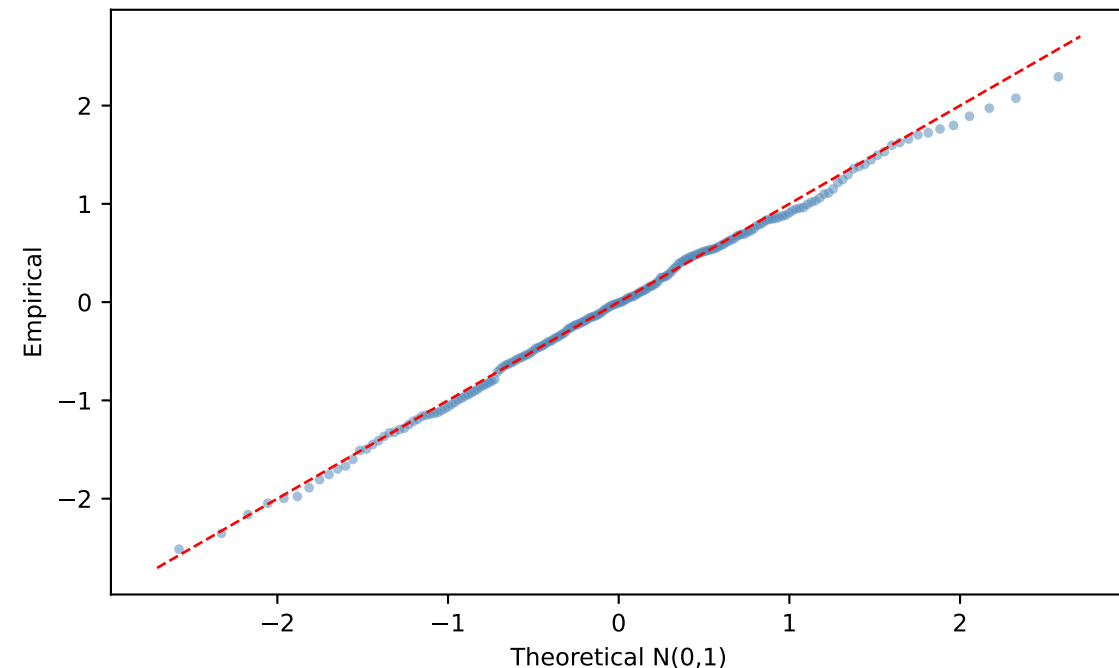


phi_dewpoint_temperature_seasonal_lags — OOS diagnostics

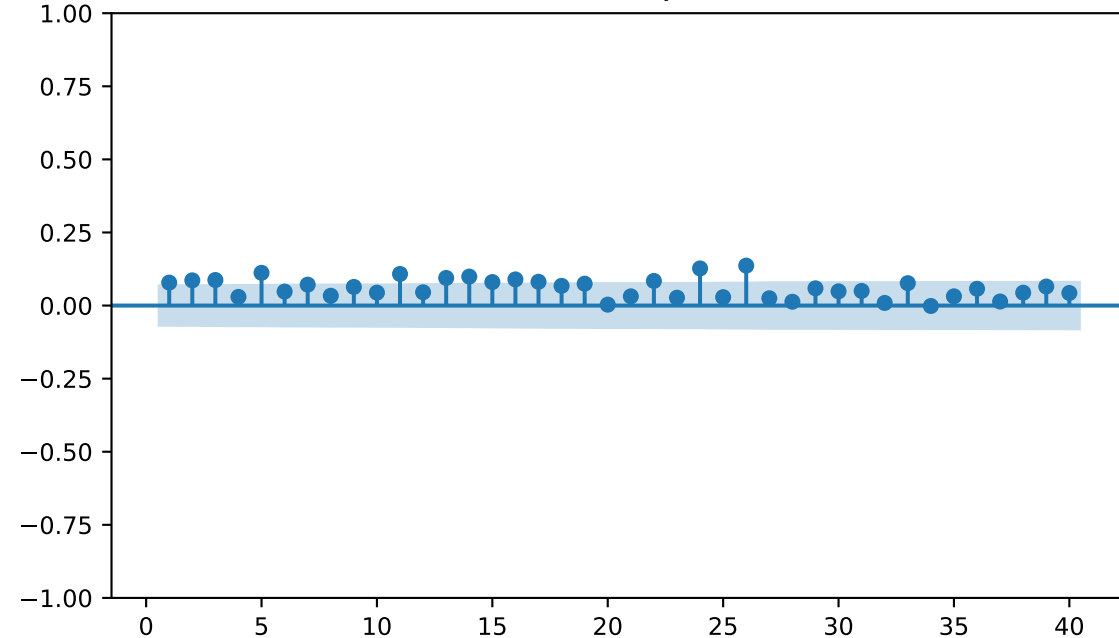
PIT histogram



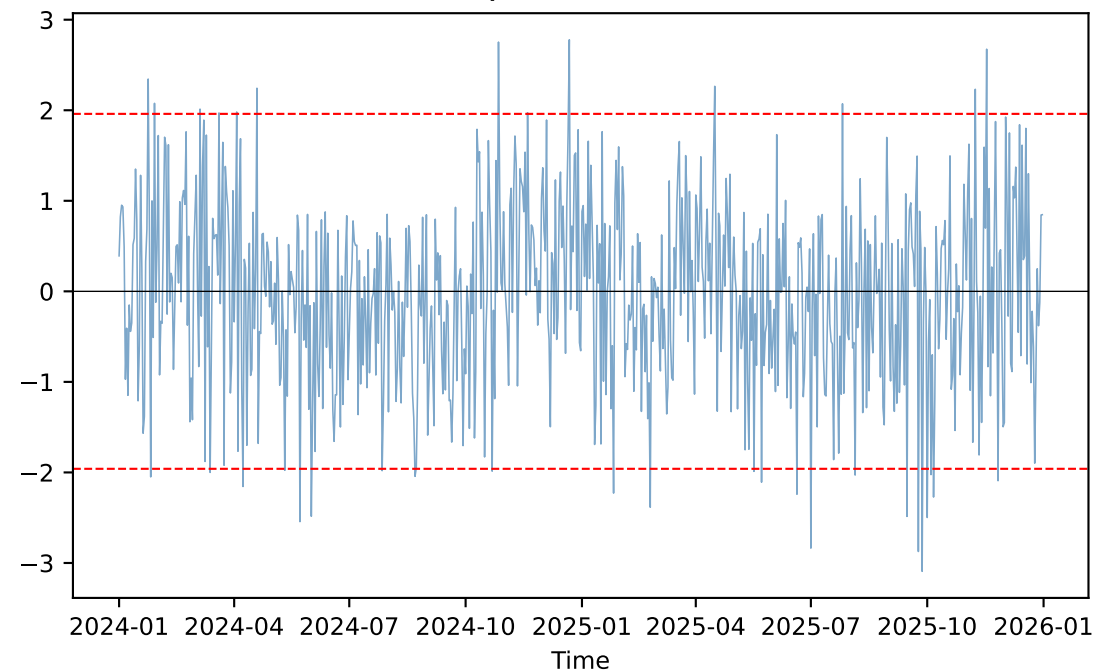
Normal QQ plot



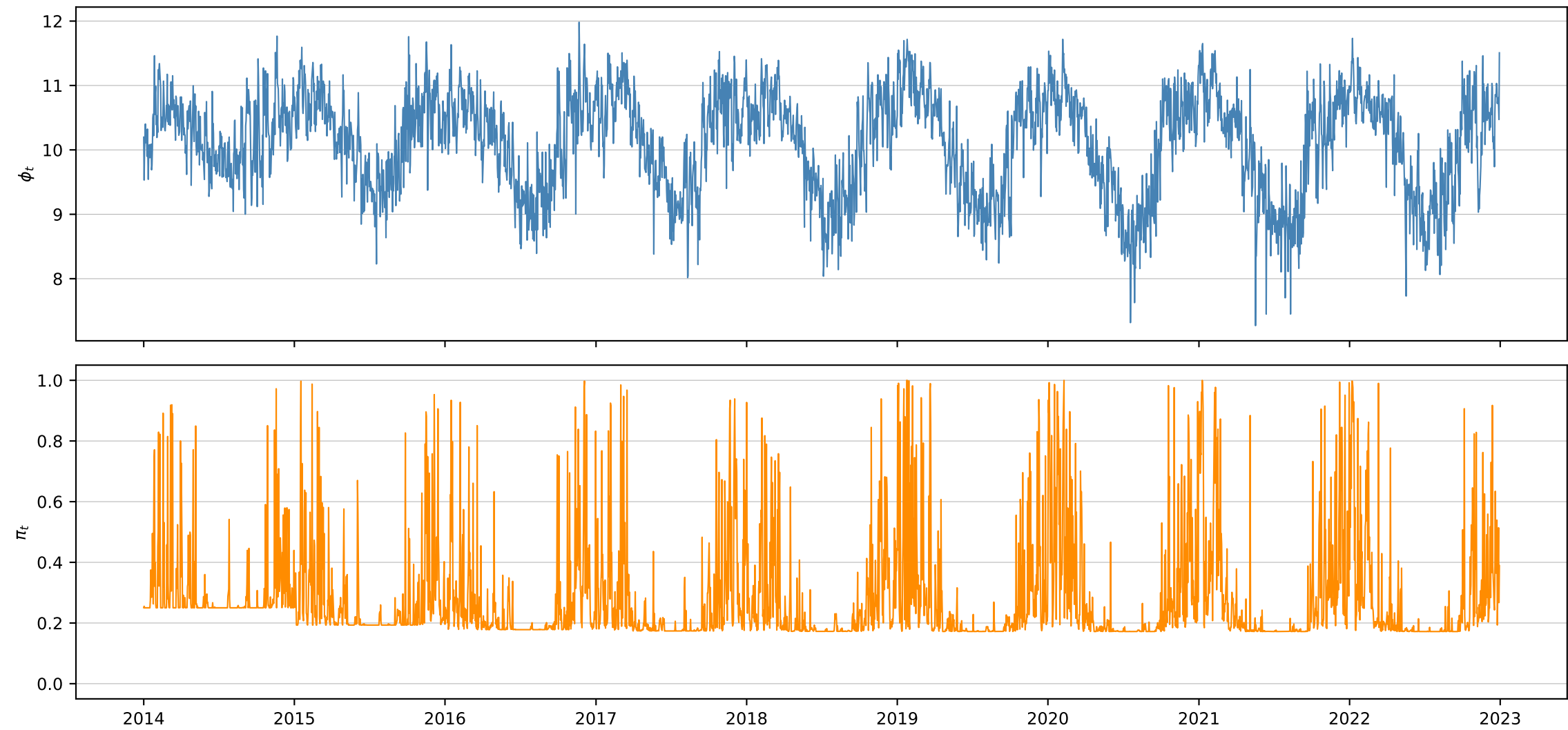
ACF of QR



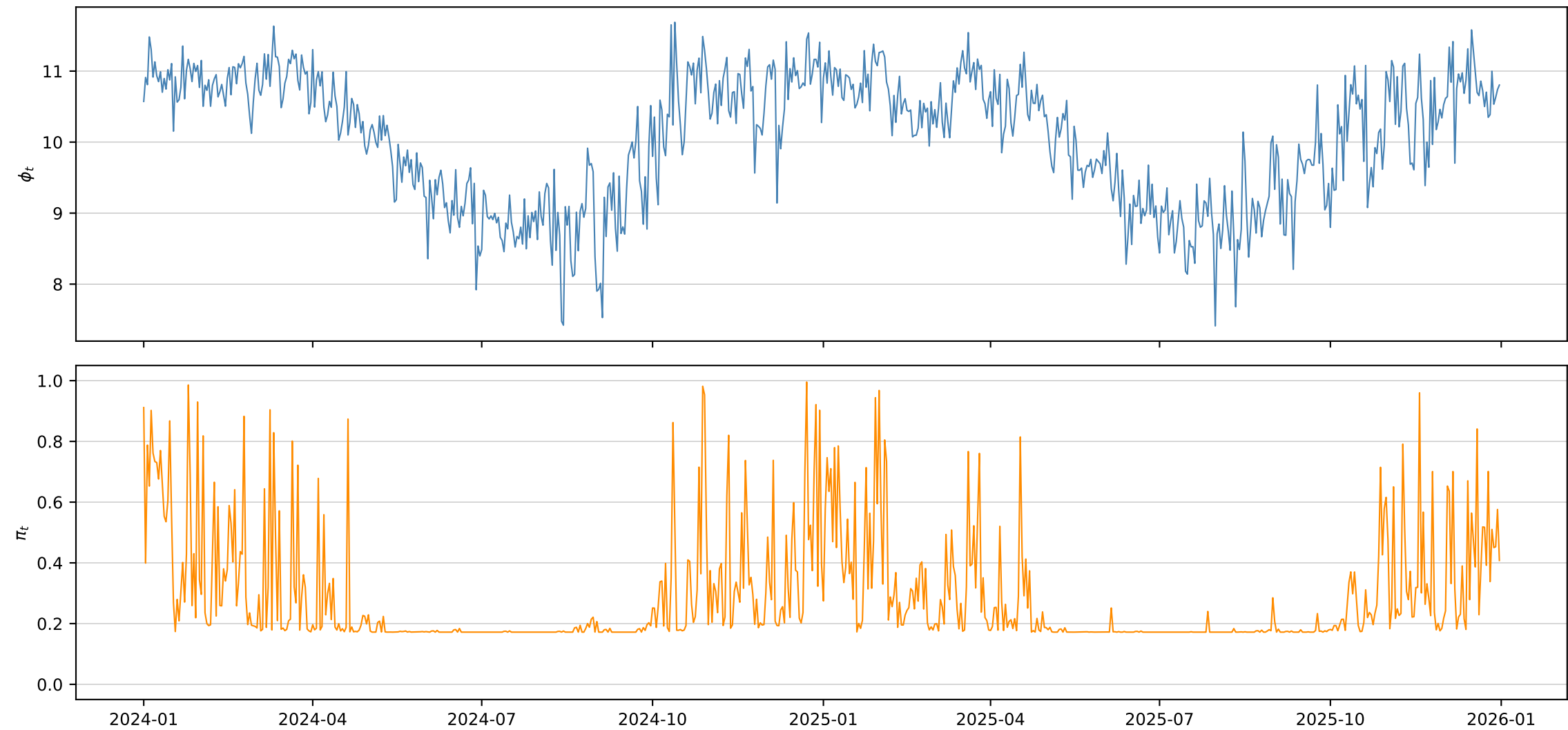
QR time series



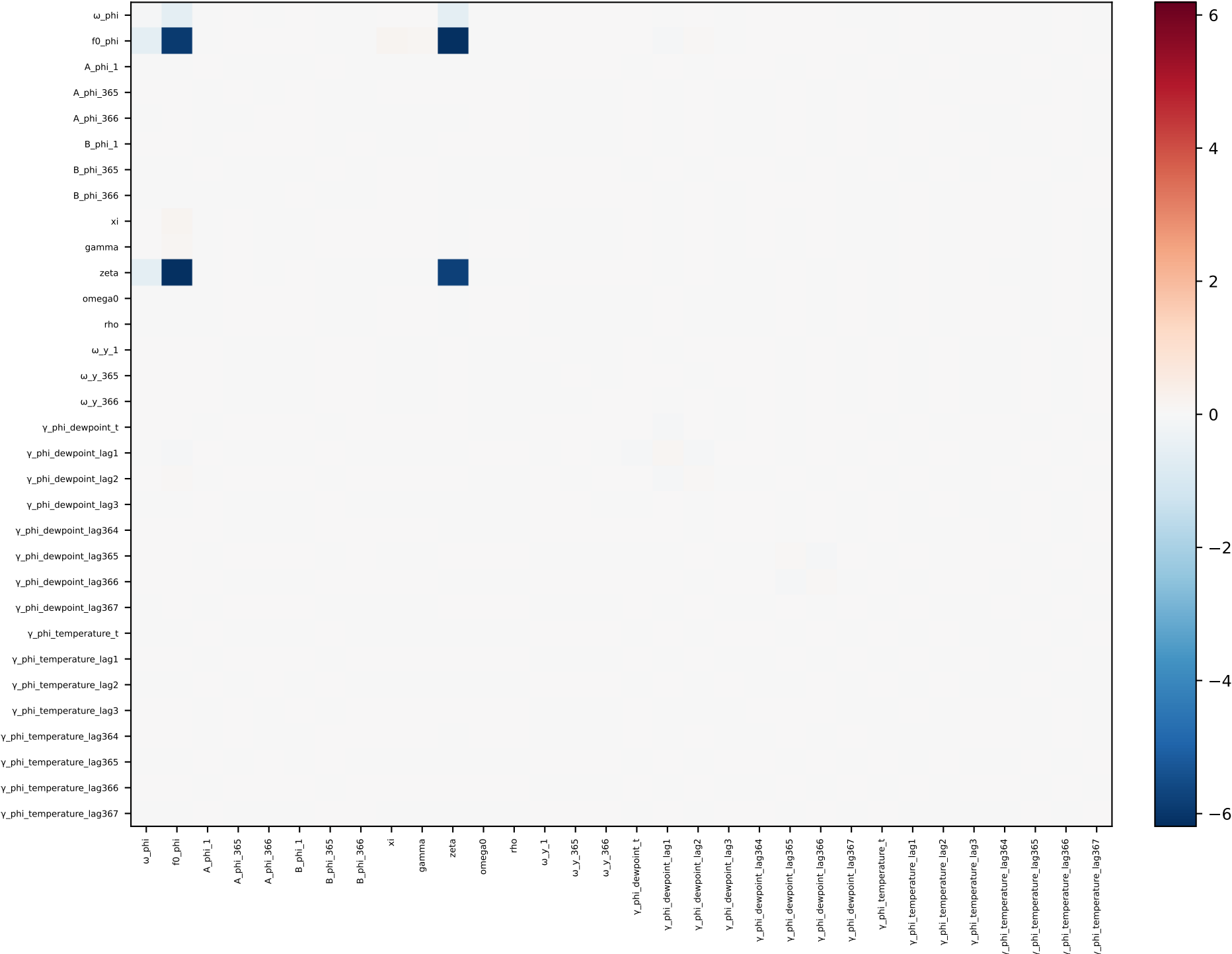
phi_dewpoint_temperature_seasonal_lags — filtered states (IS)



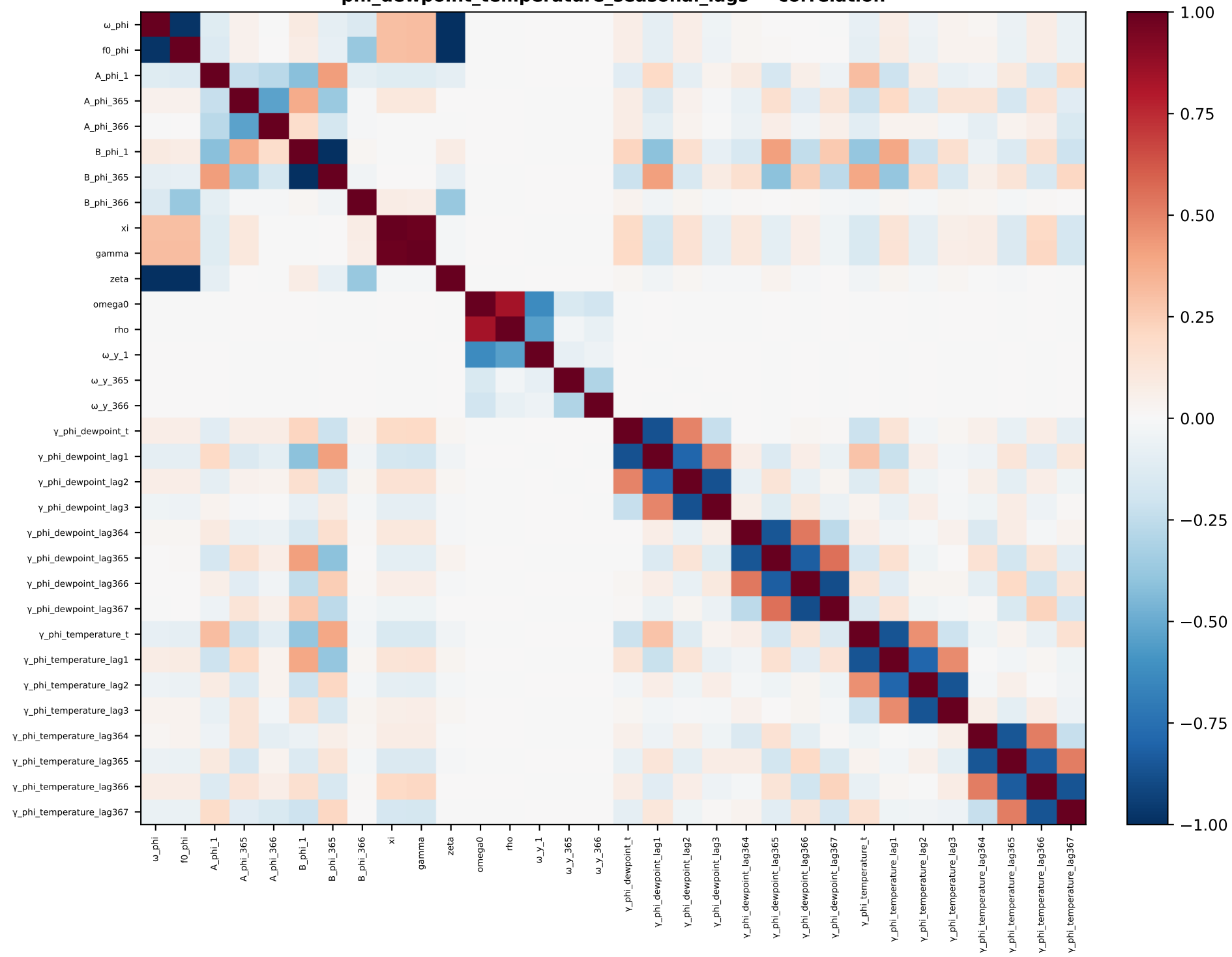
phi_dewpoint_temperature_seasonal_lags — filtered states (OOS)



phi_dewpoint_temperature_seasonal_lags – covariance

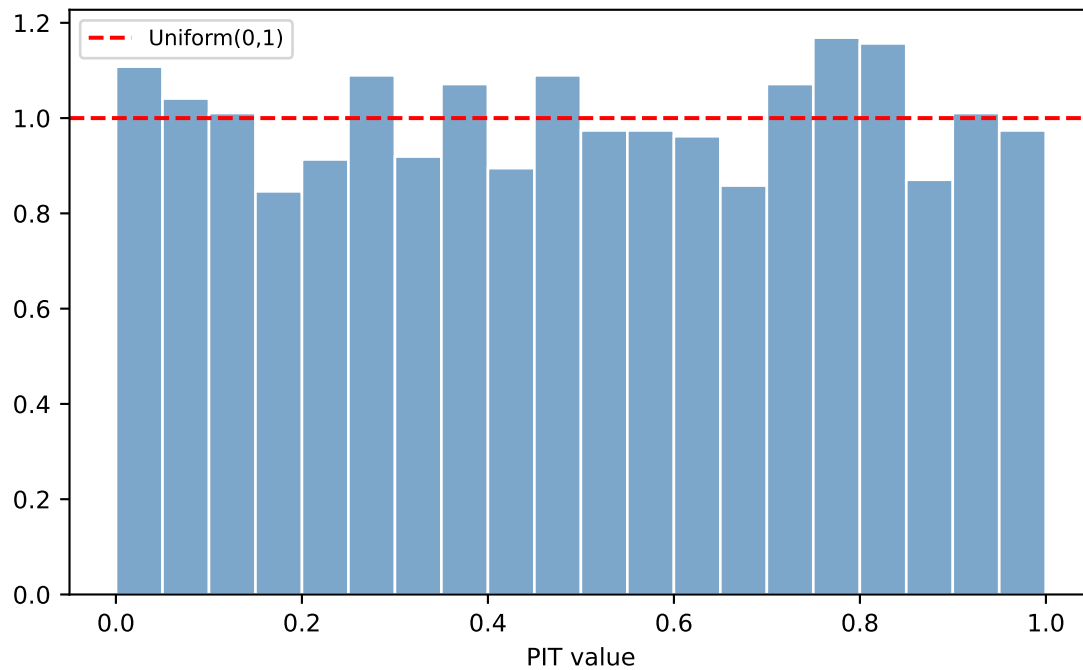


phi_dewpoint_temperature_seasonal_lags — correlation

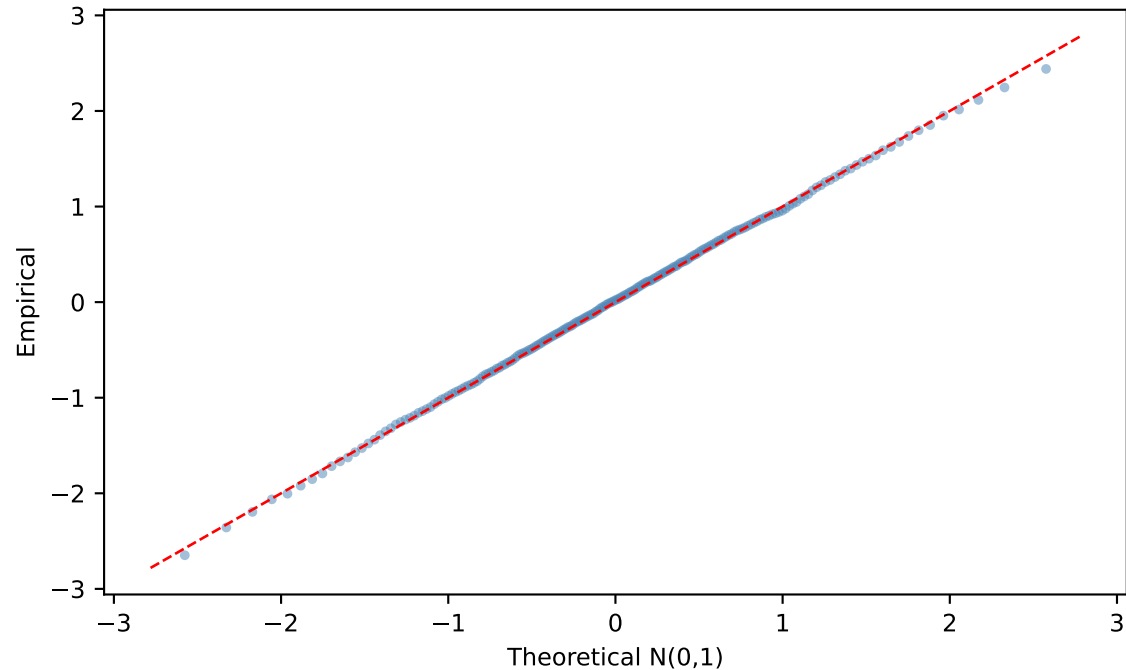


phi_mjo_t — IS diagnostics

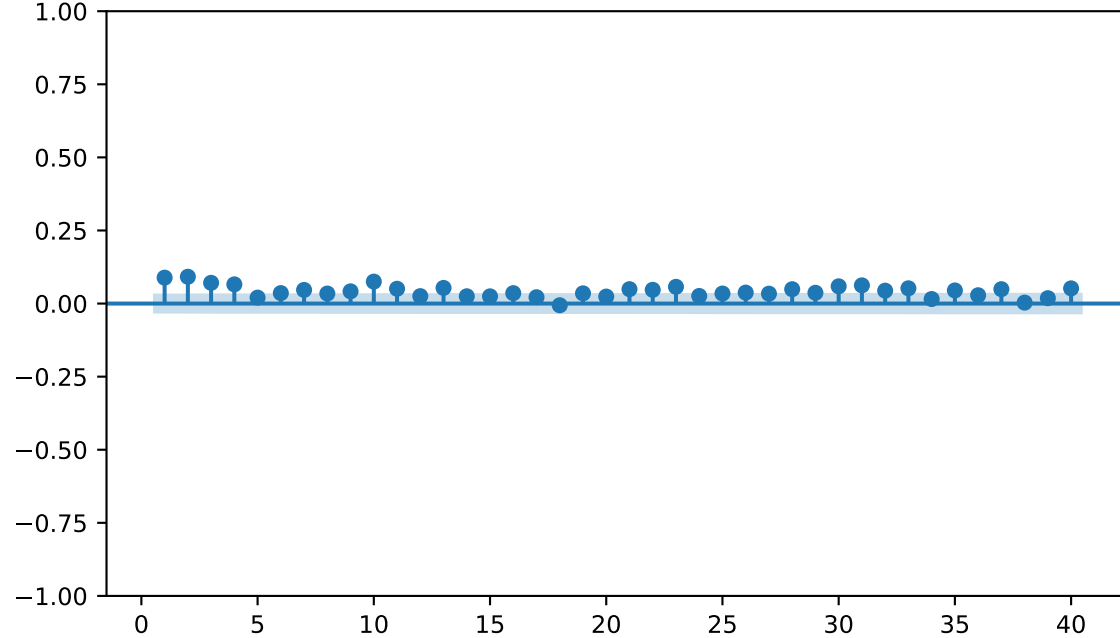
PIT histogram



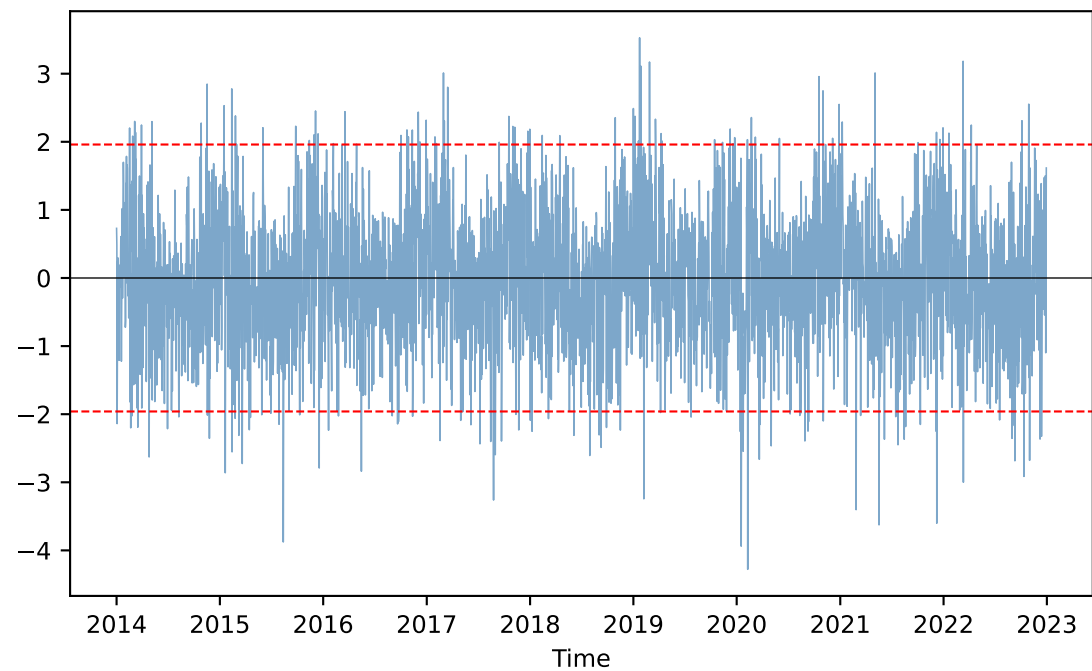
Normal QQ plot



ACF of QR

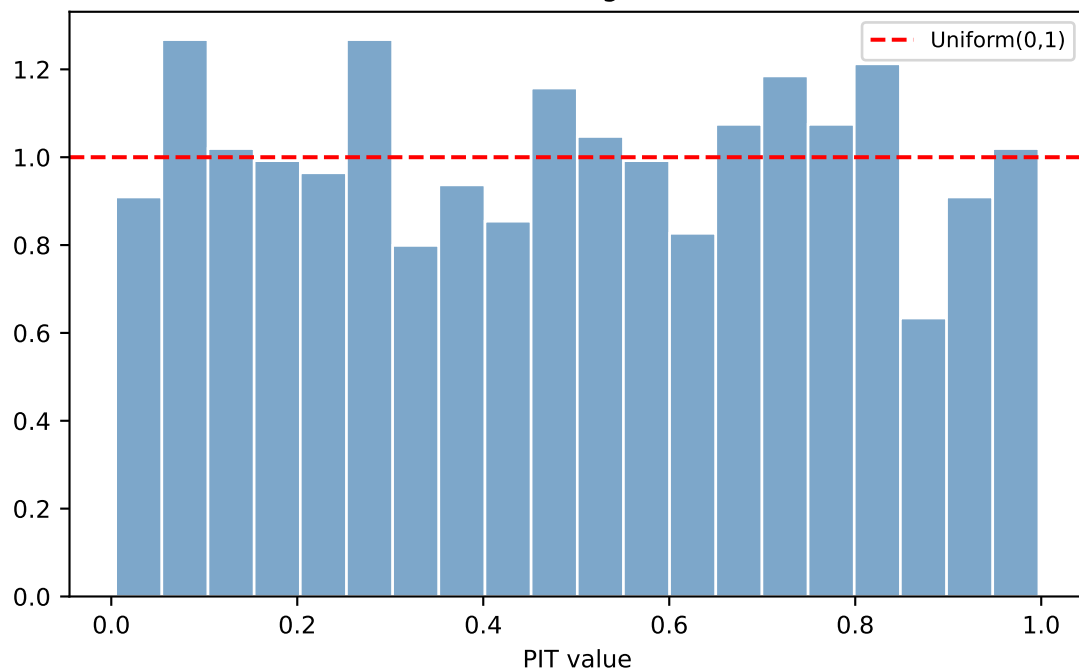


QR time series

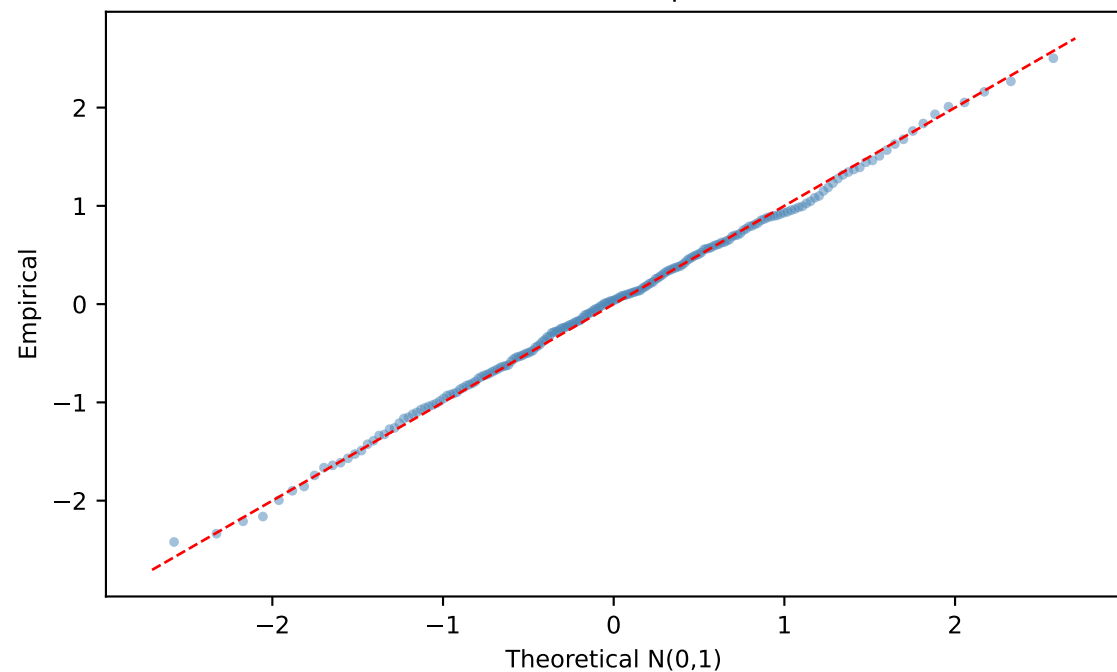


phi_mjo_t — OOS diagnostics

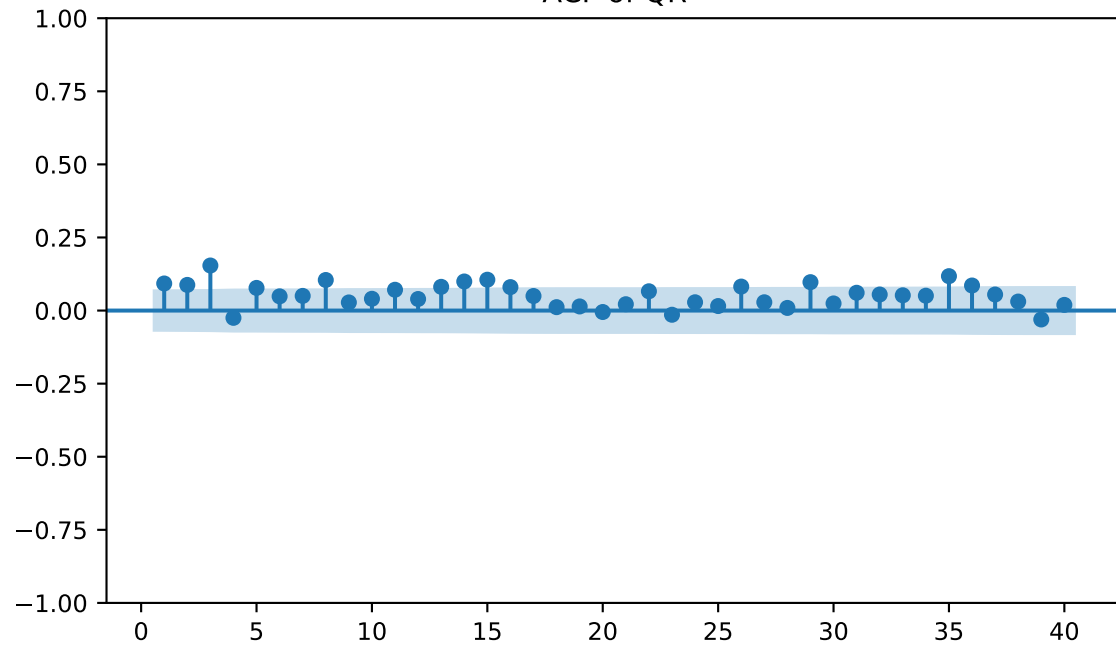
PIT histogram



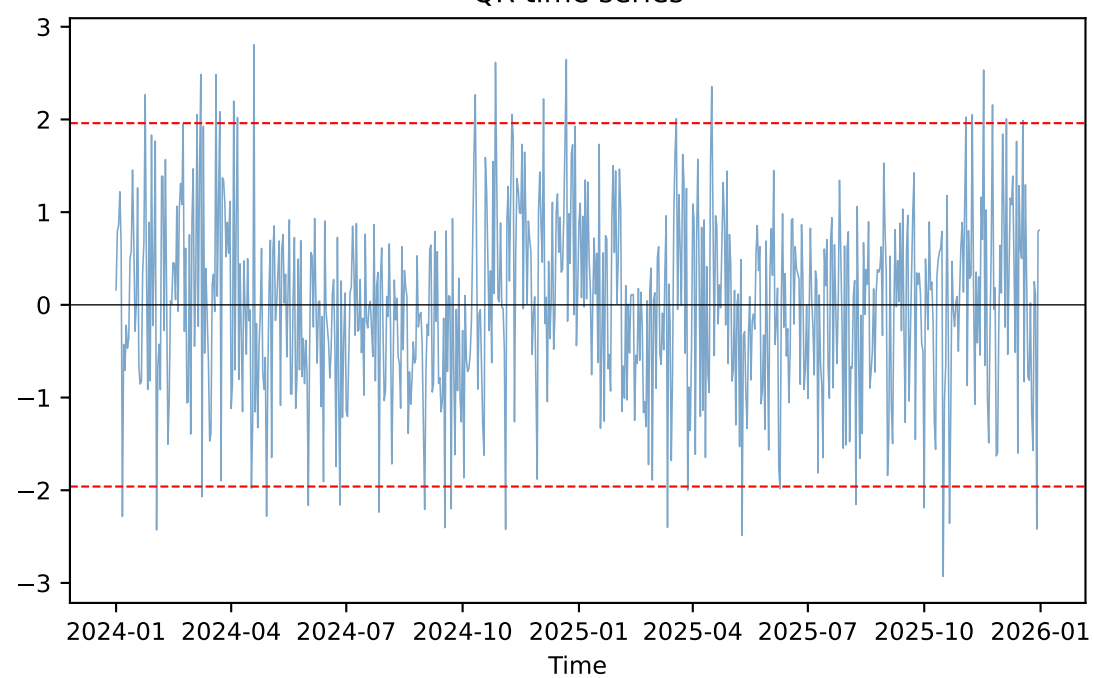
Normal QQ plot



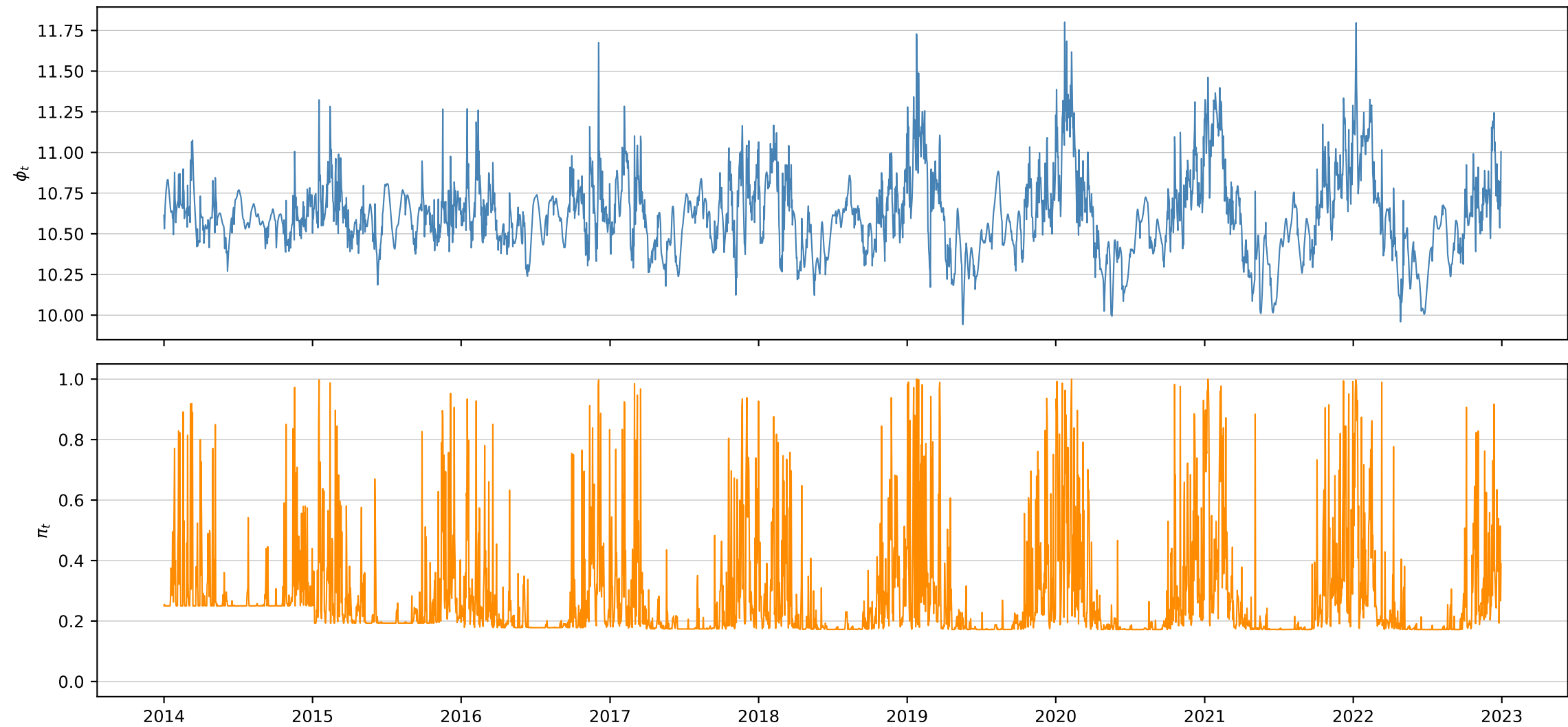
ACF of QR



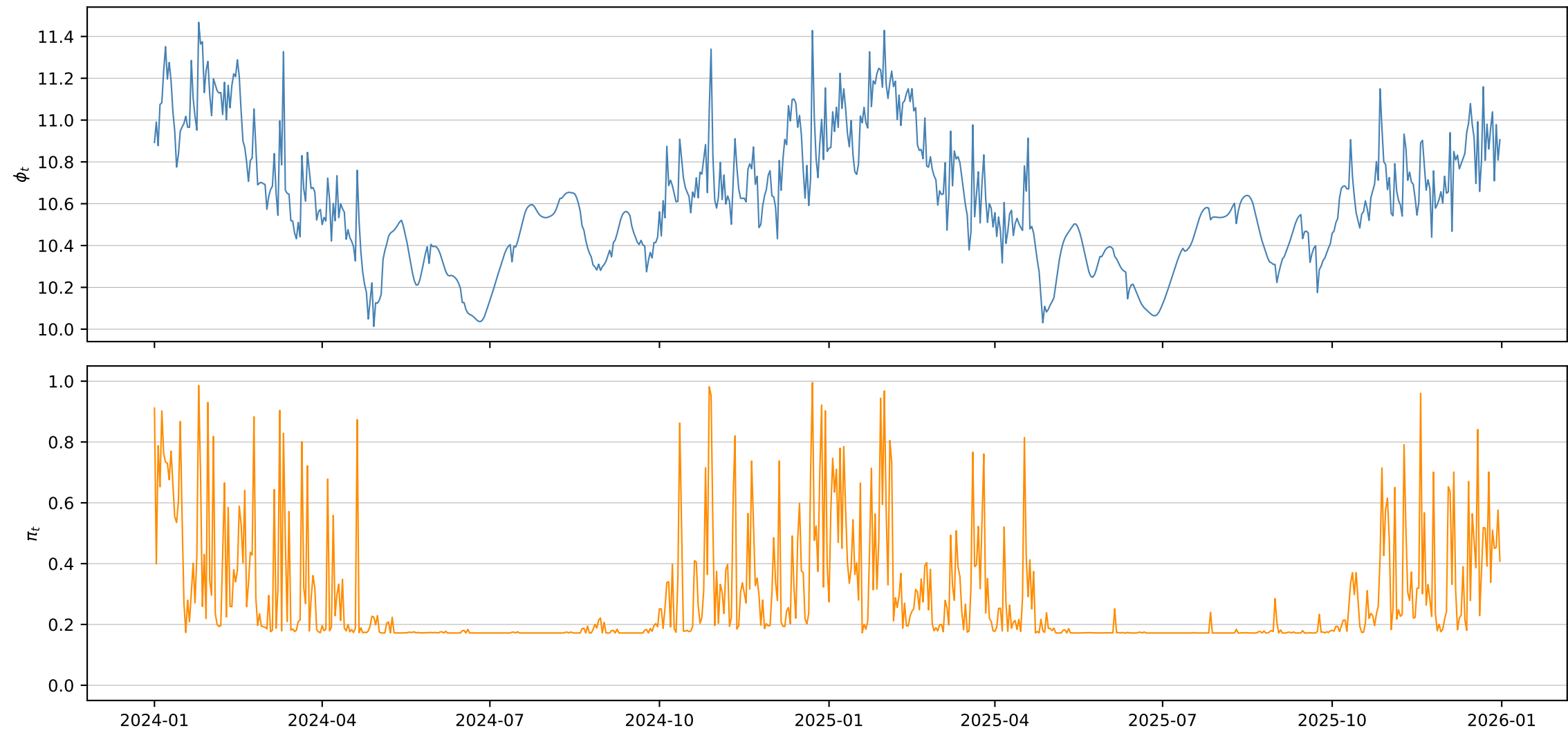
QR time series



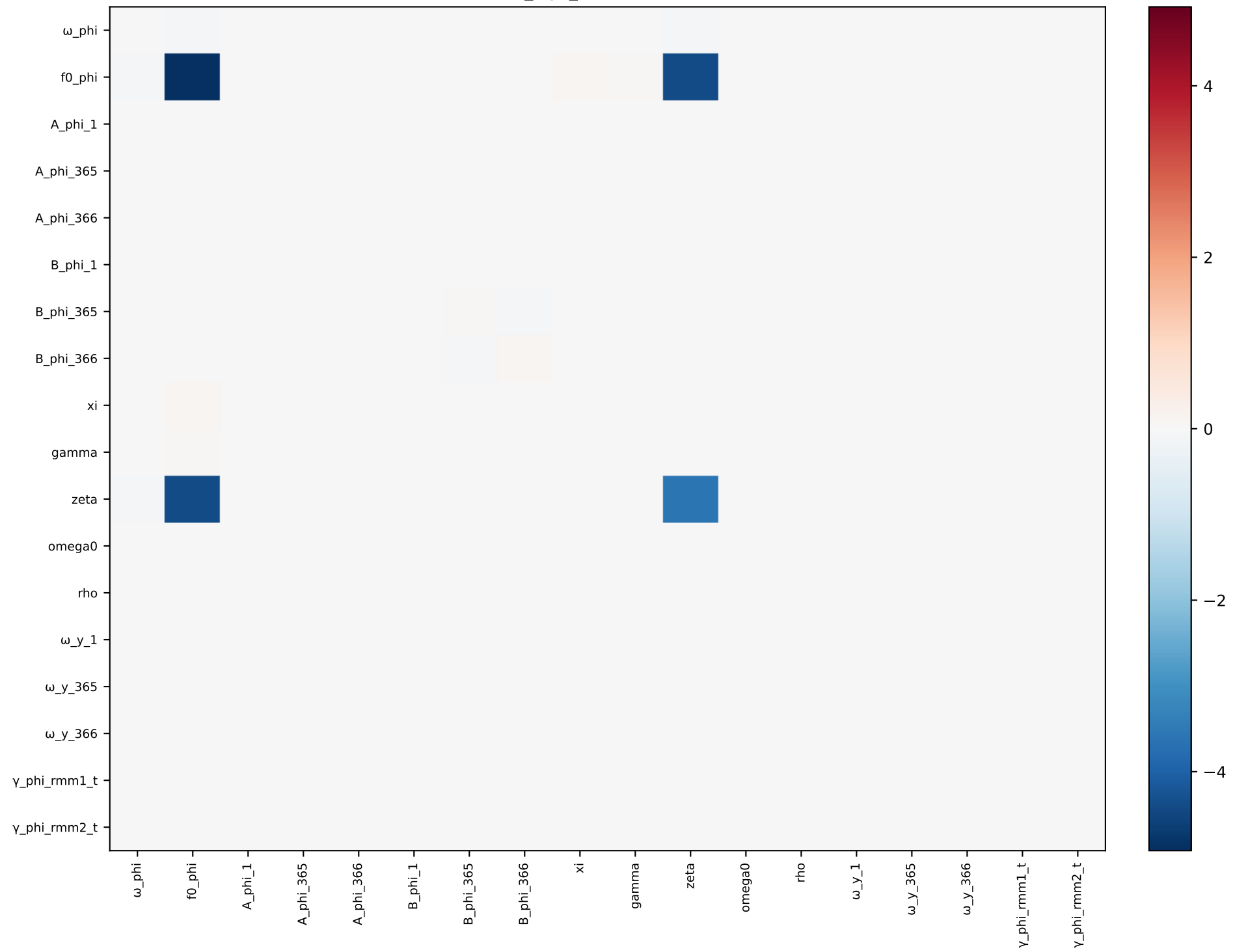
phi_mjo_t — filtered states (IS)



phi_mjo_t — filtered states (OOS)

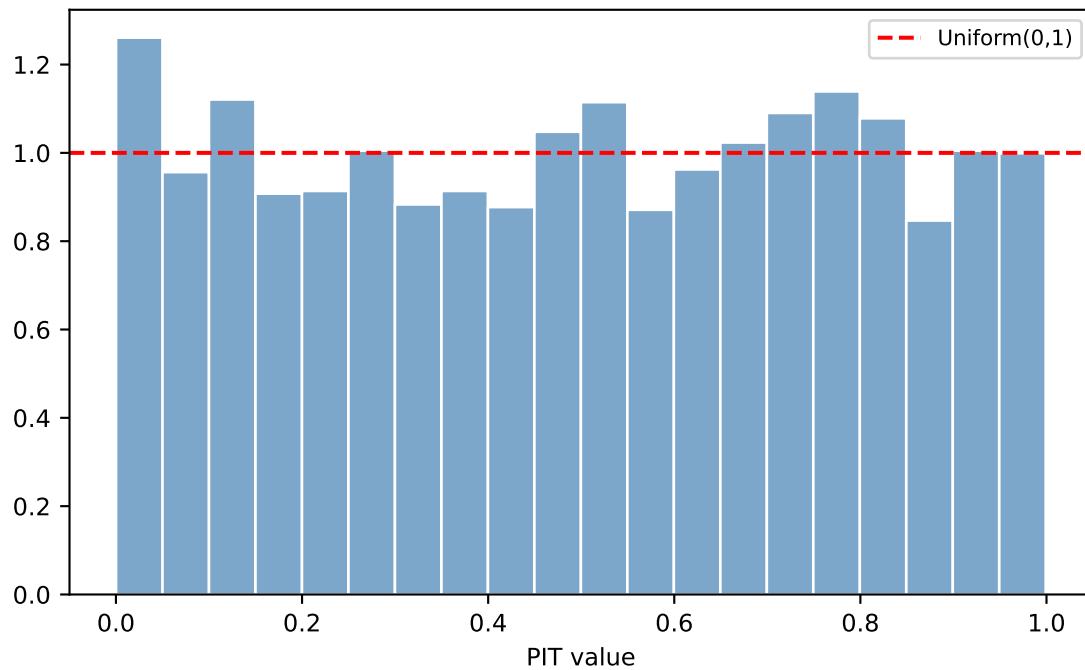


phi_mjo_t — covariance

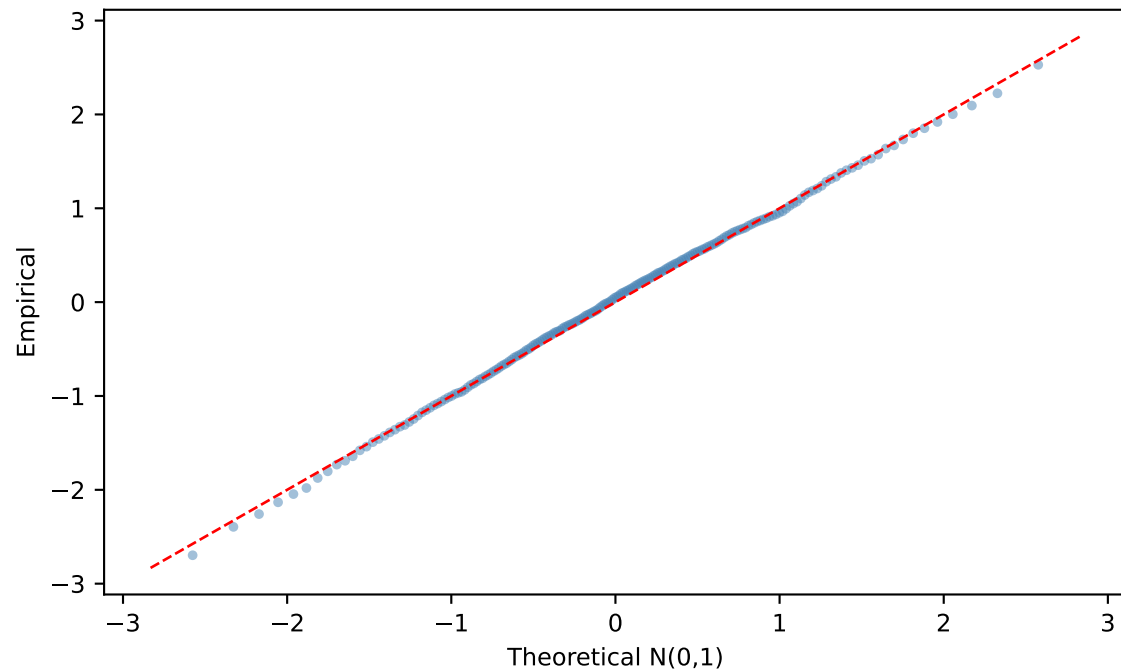


phi_mjo_lag1 — IS diagnostics

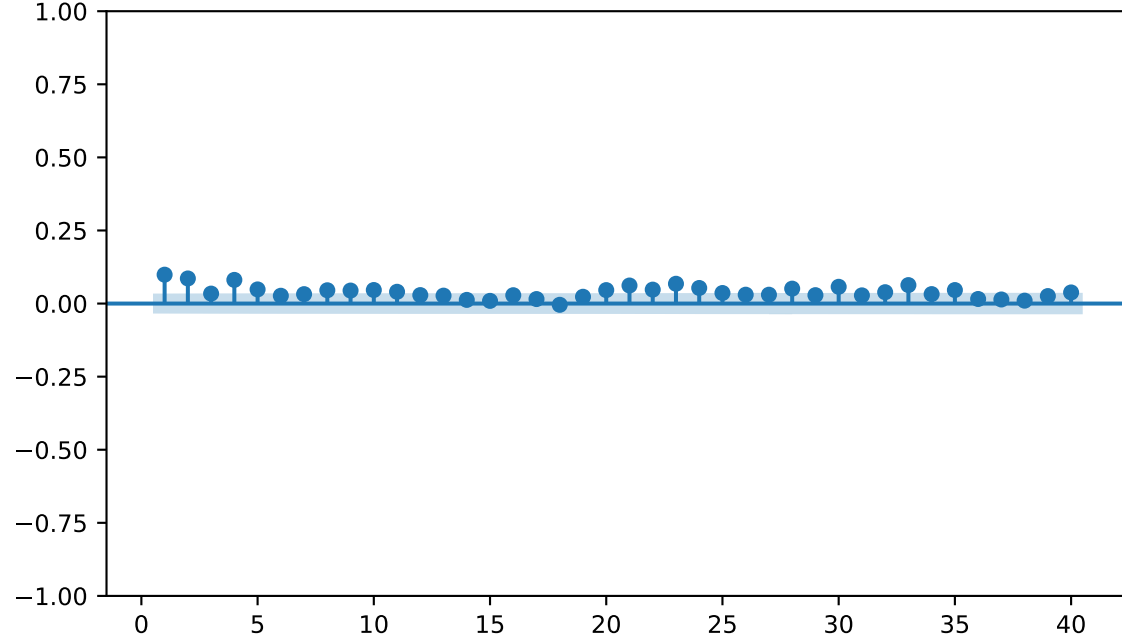
PIT histogram



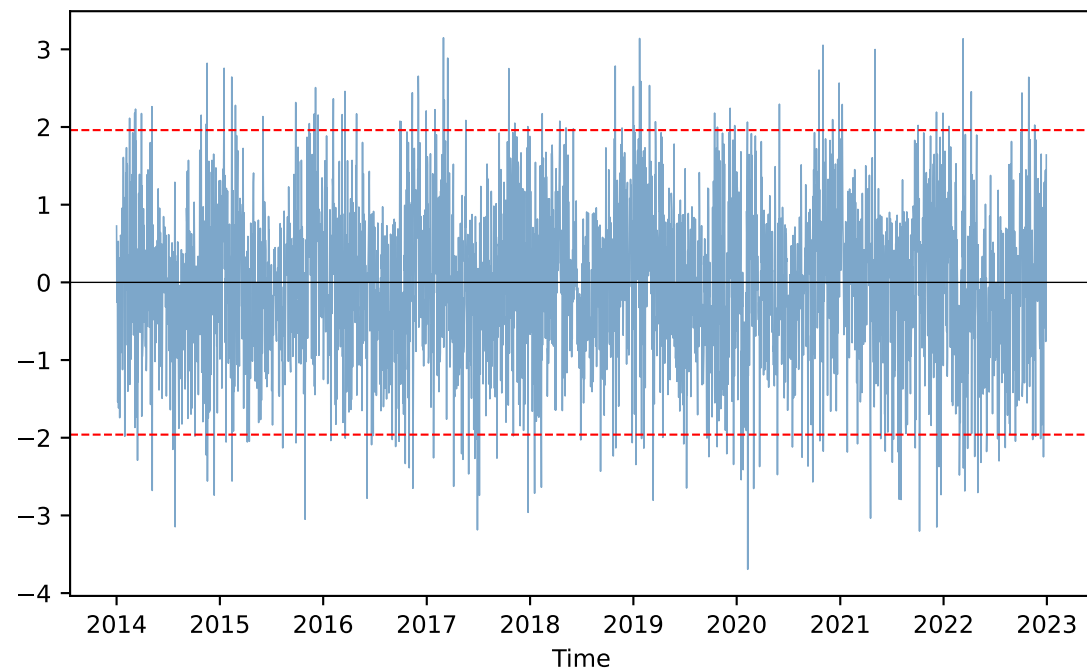
Normal QQ plot



ACF of QR

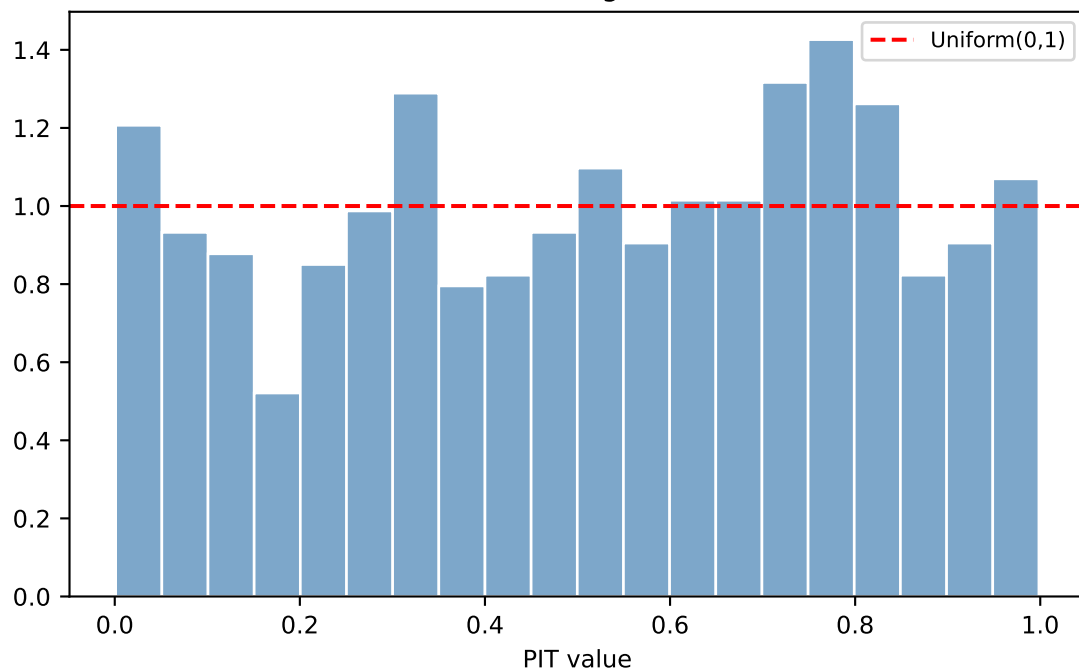


QR time series

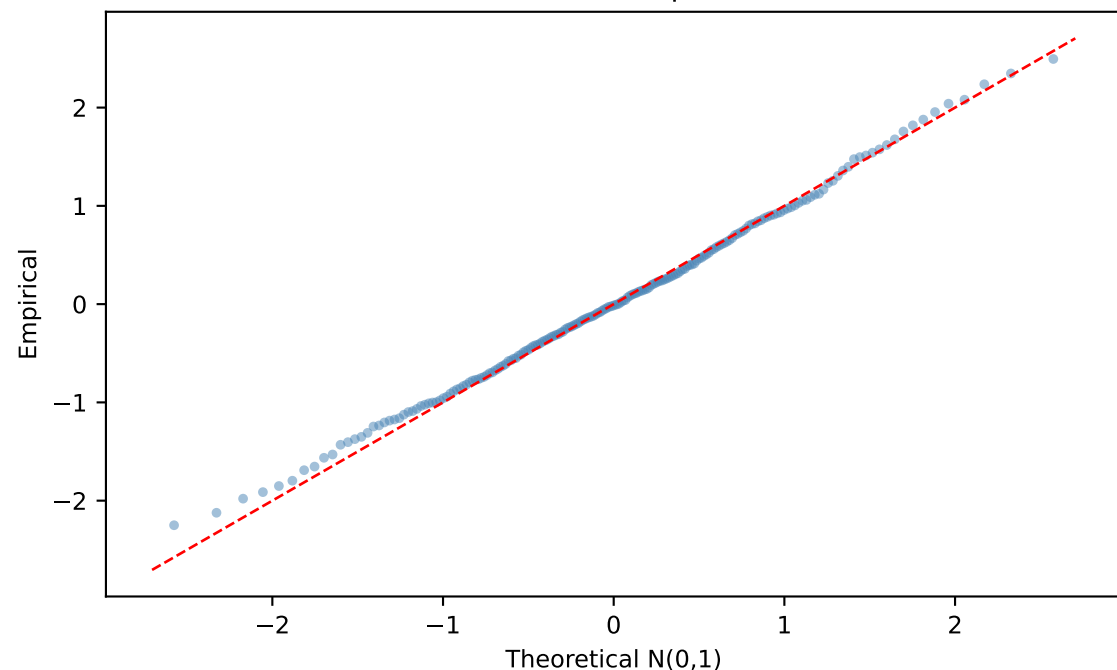


phi_mjo_lag1 — OOS diagnostics

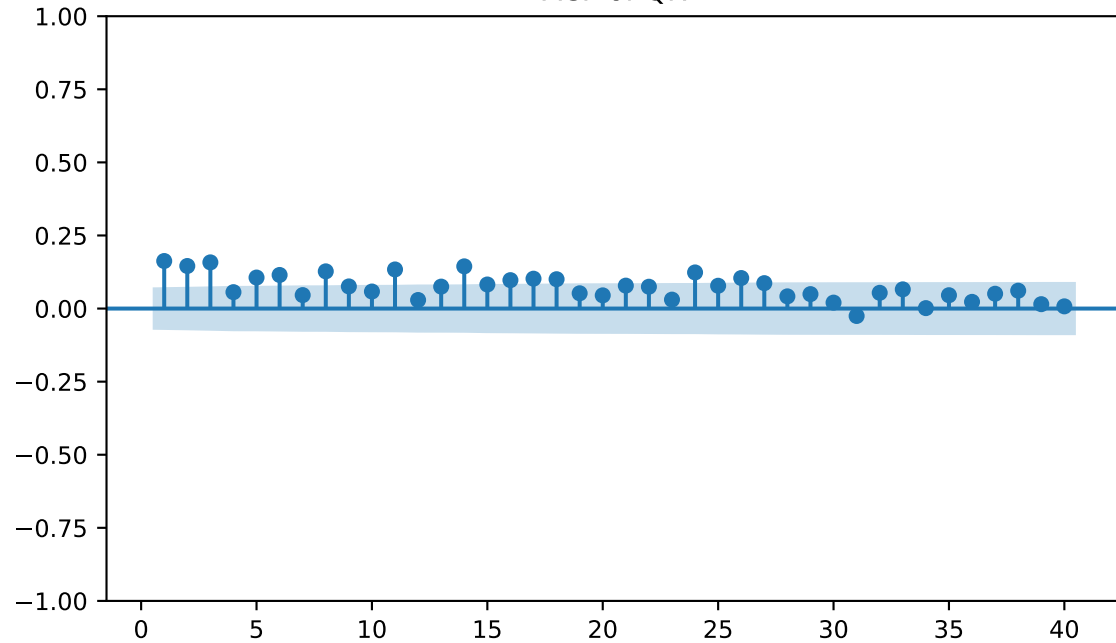
PIT histogram



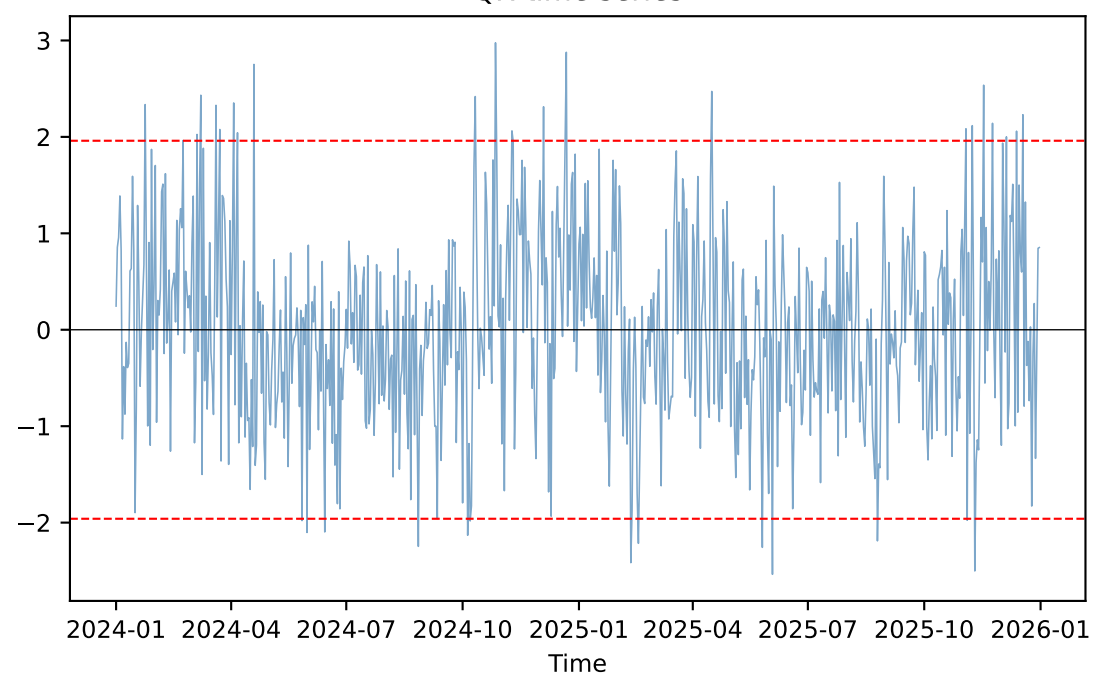
Normal QQ plot



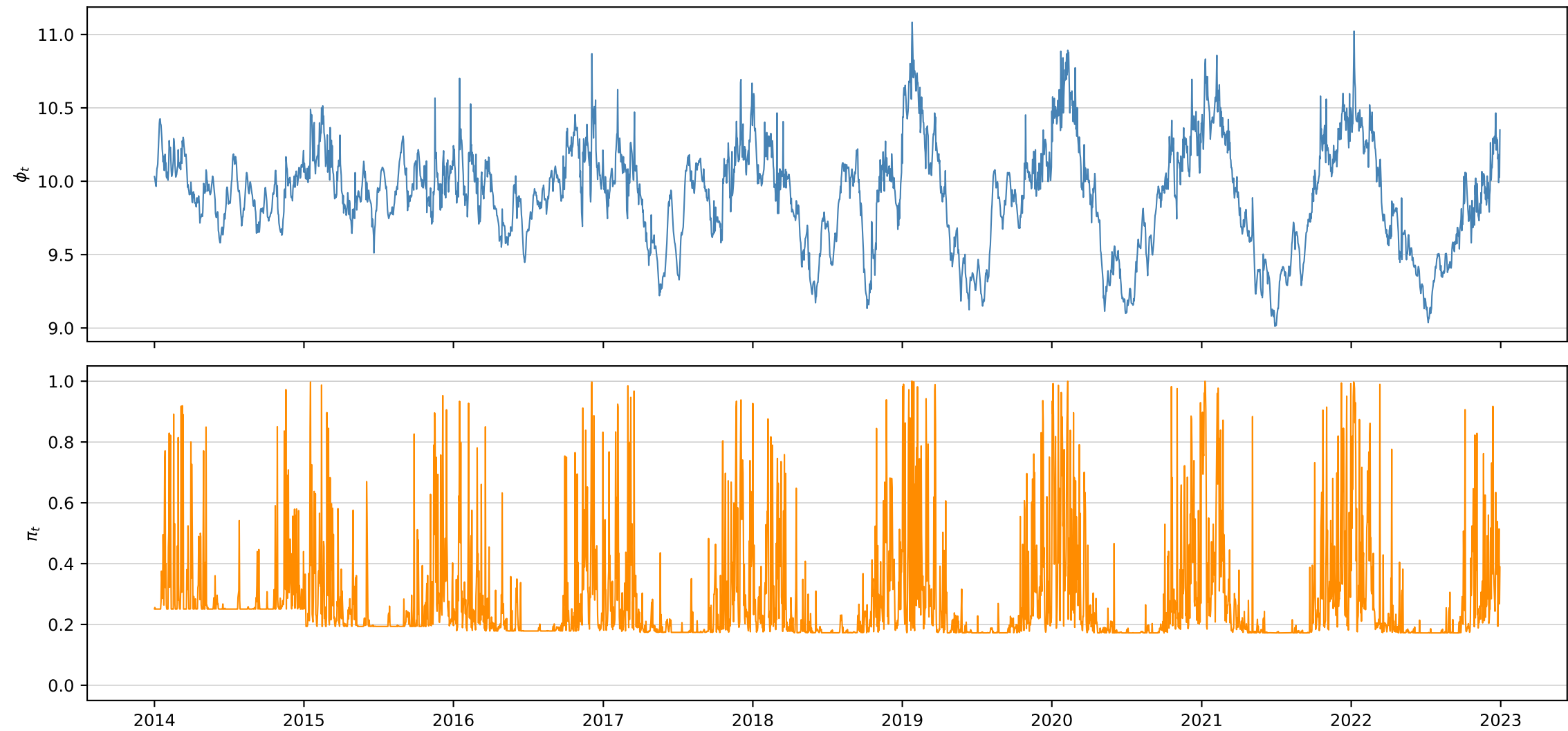
ACF of QR



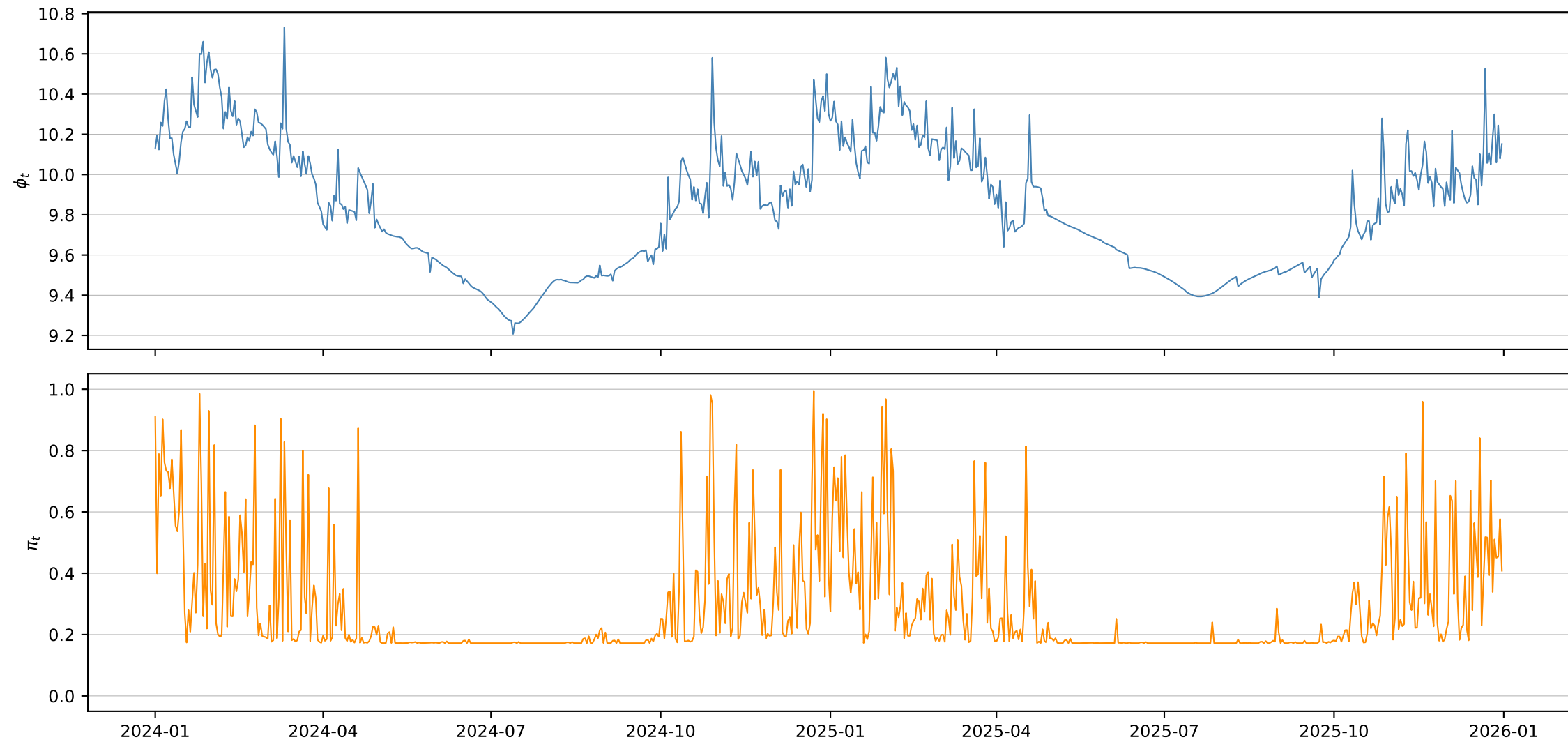
QR time series



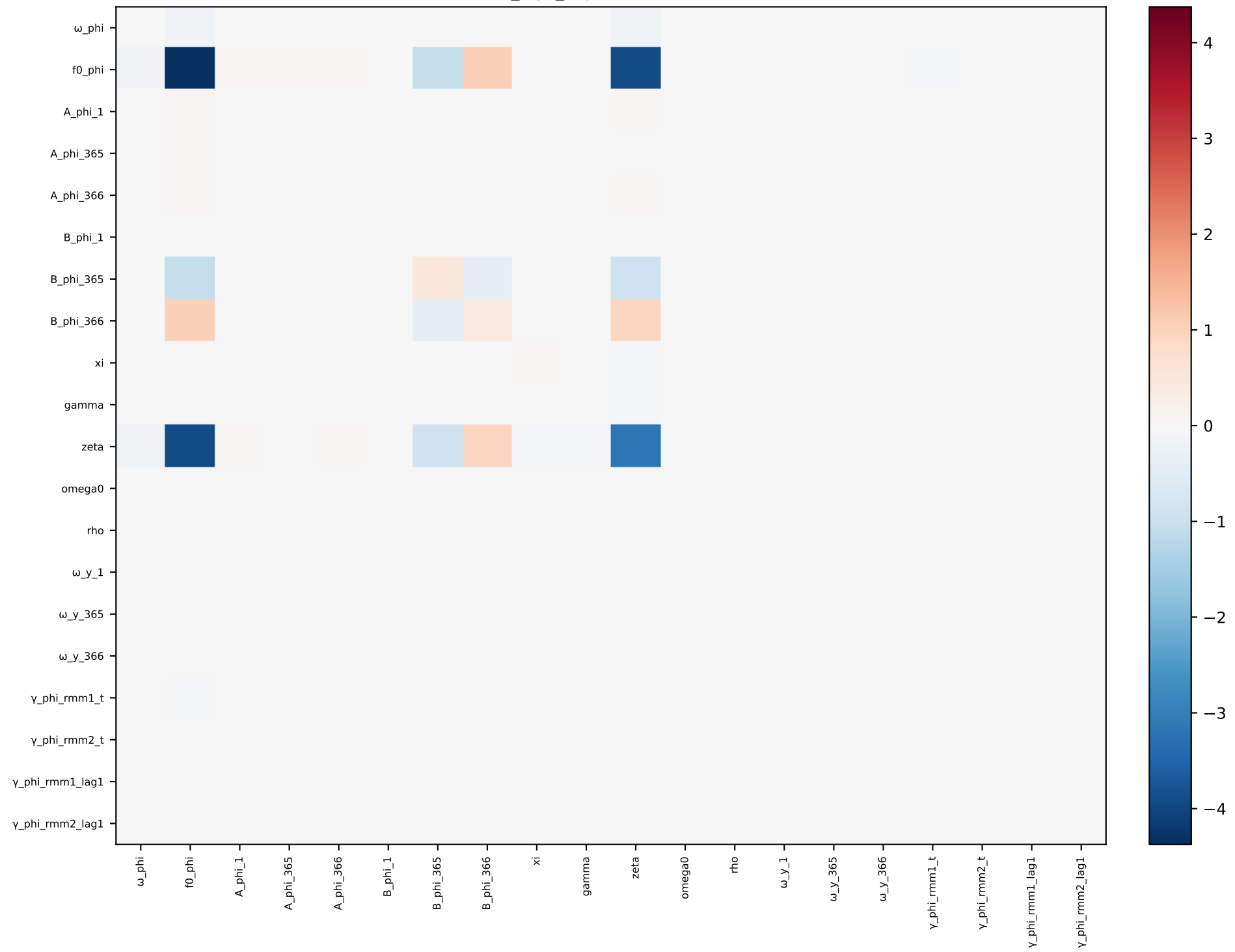
phi_mjo_lag1 — filtered states (IS)



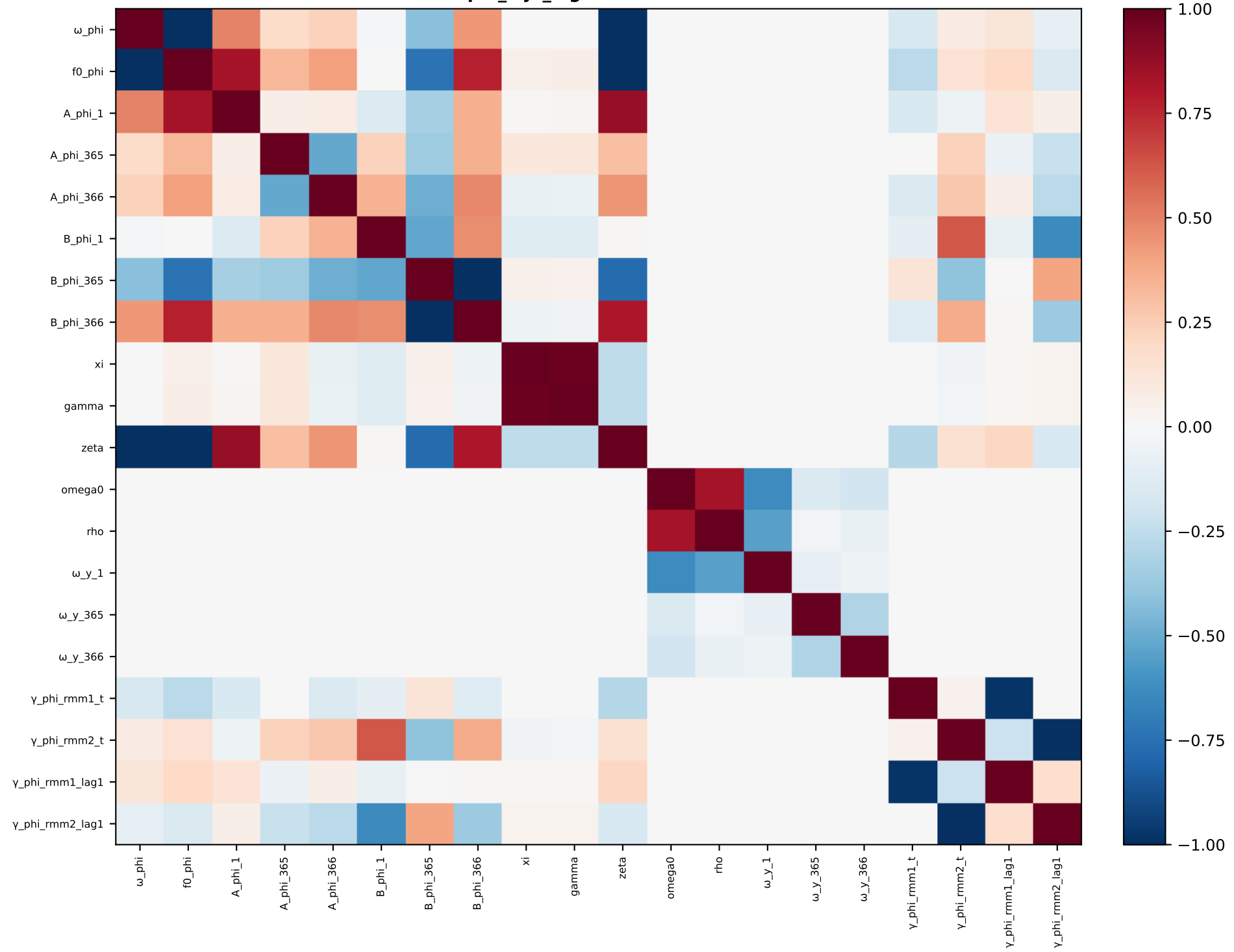
phi_mjo_lag1 — filtered states (OOS)



phi_mjo_lag1 — covariance

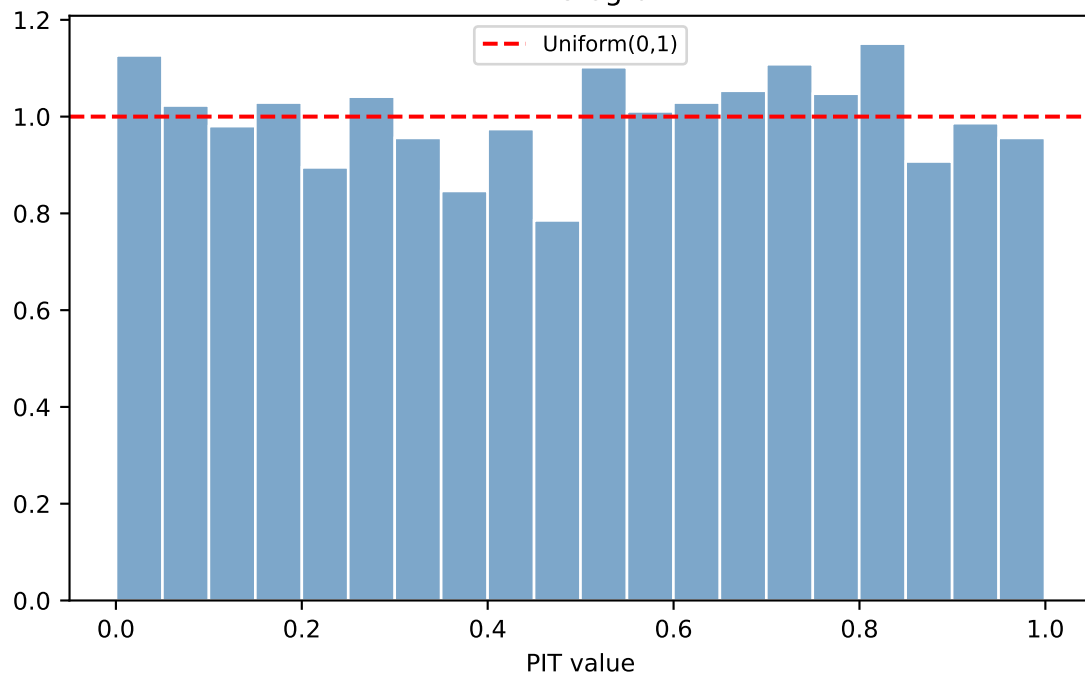


phi_mjo_lag1 — correlation

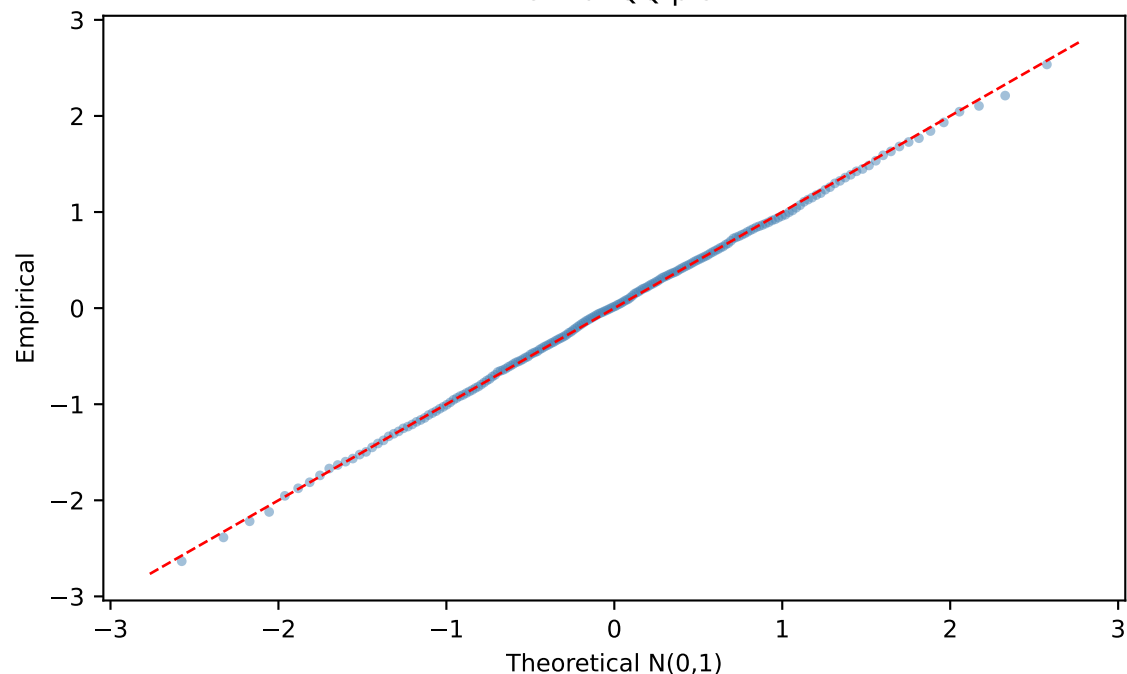


phi_mjo_short_lags — IS diagnostics

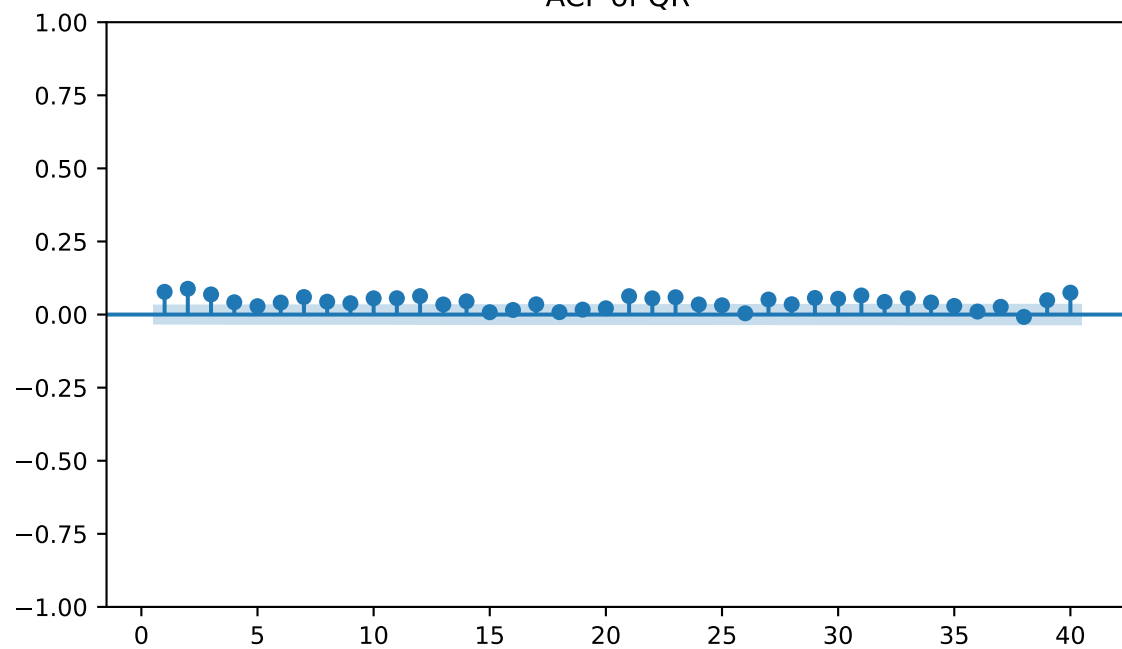
PIT histogram



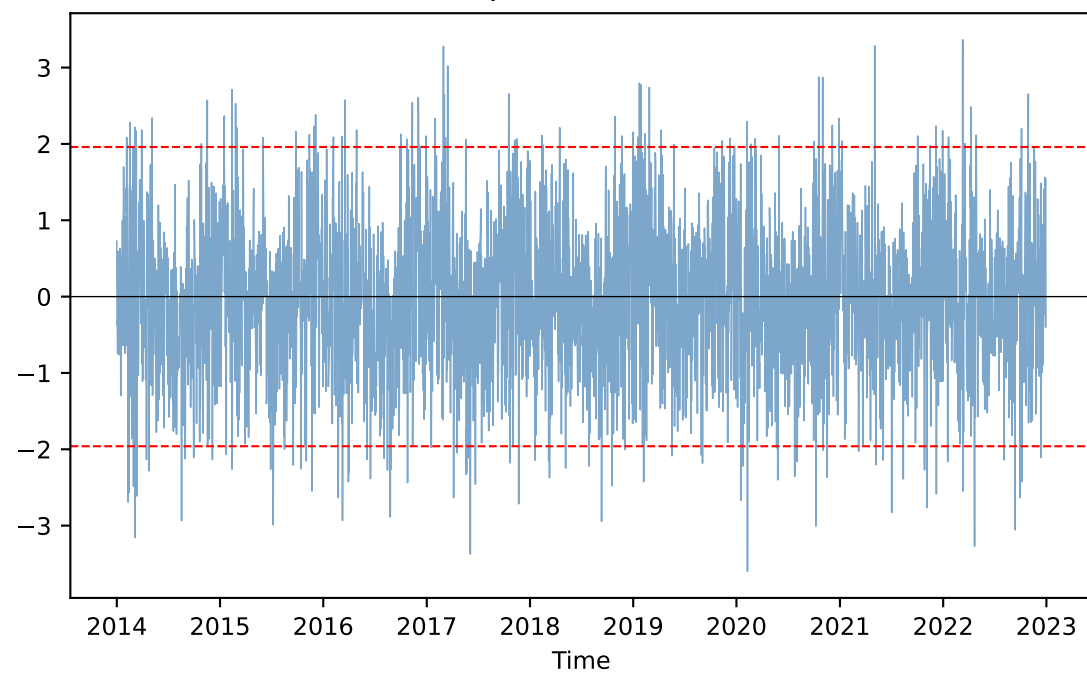
Normal QQ plot



ACF of QR

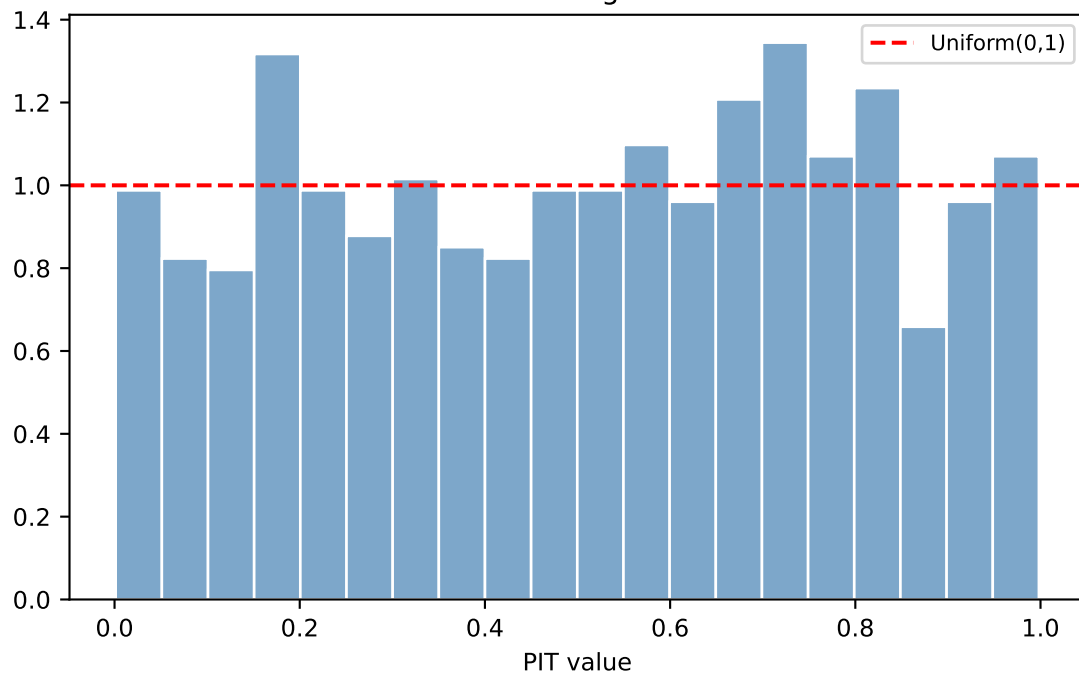


QR time series

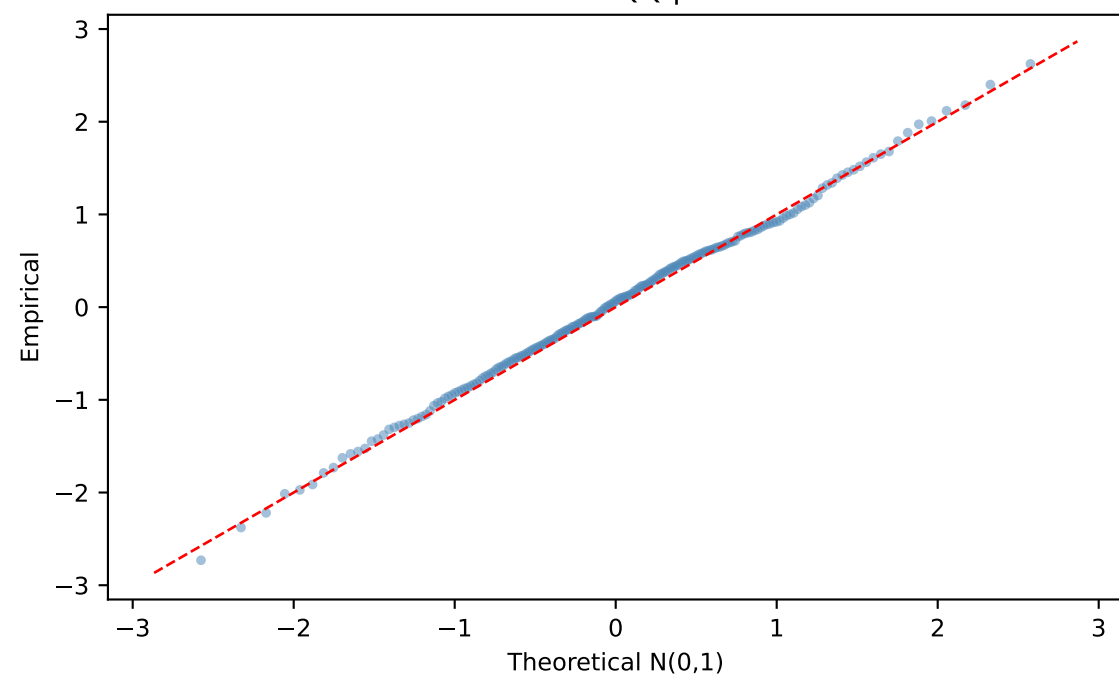


phi_mjo_short_lags — OOS diagnostics

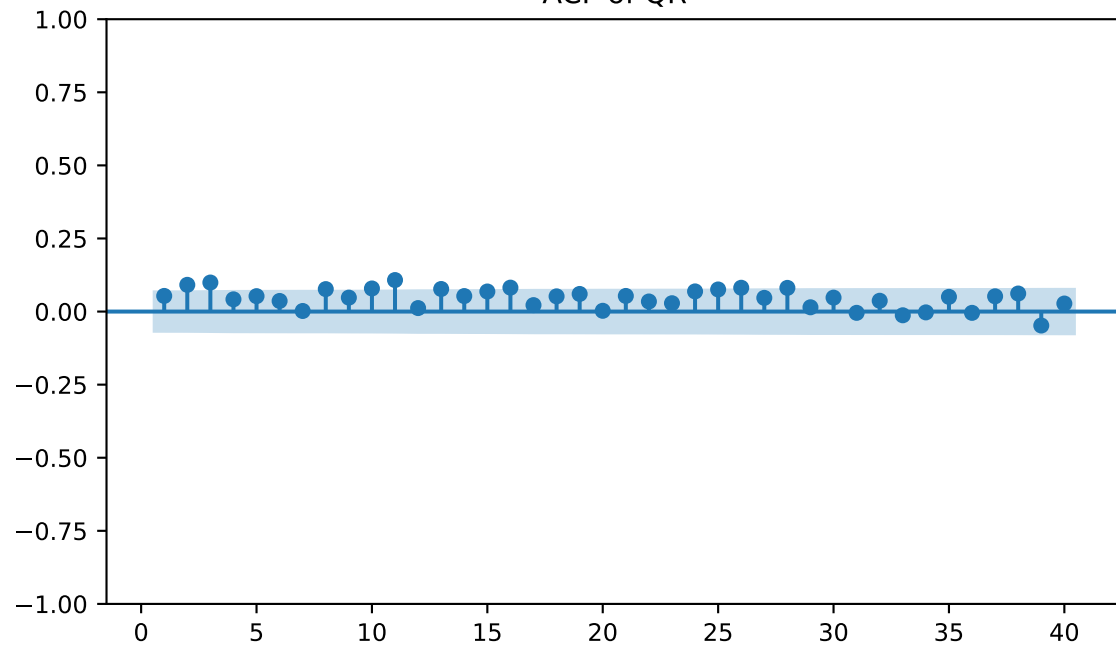
PIT histogram



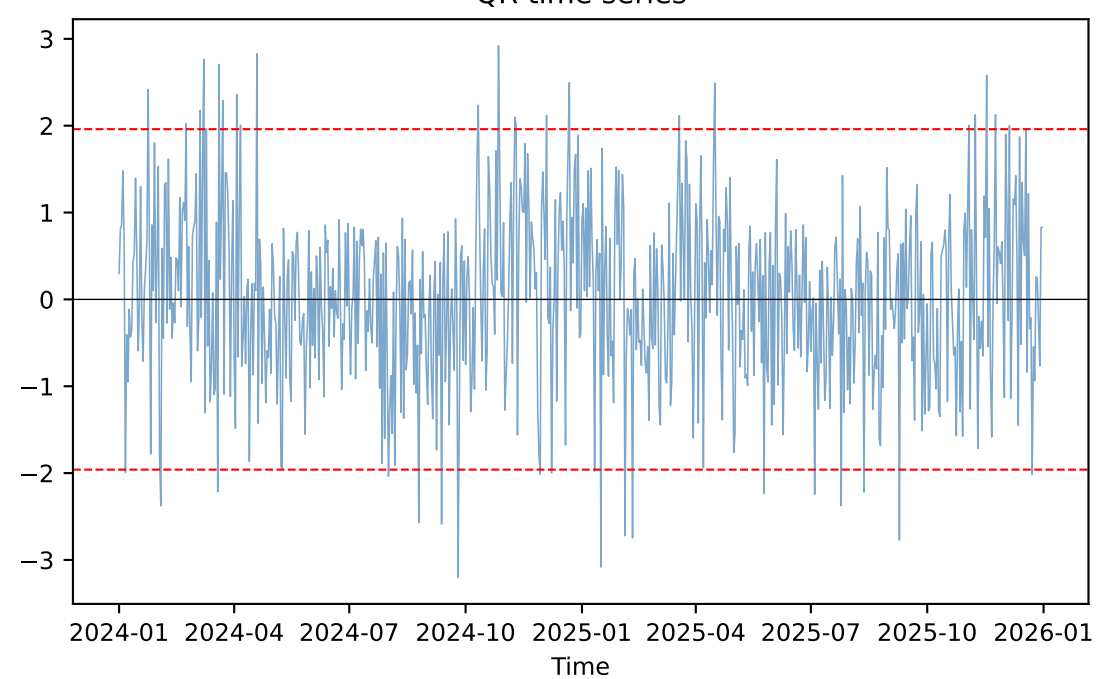
Normal QQ plot



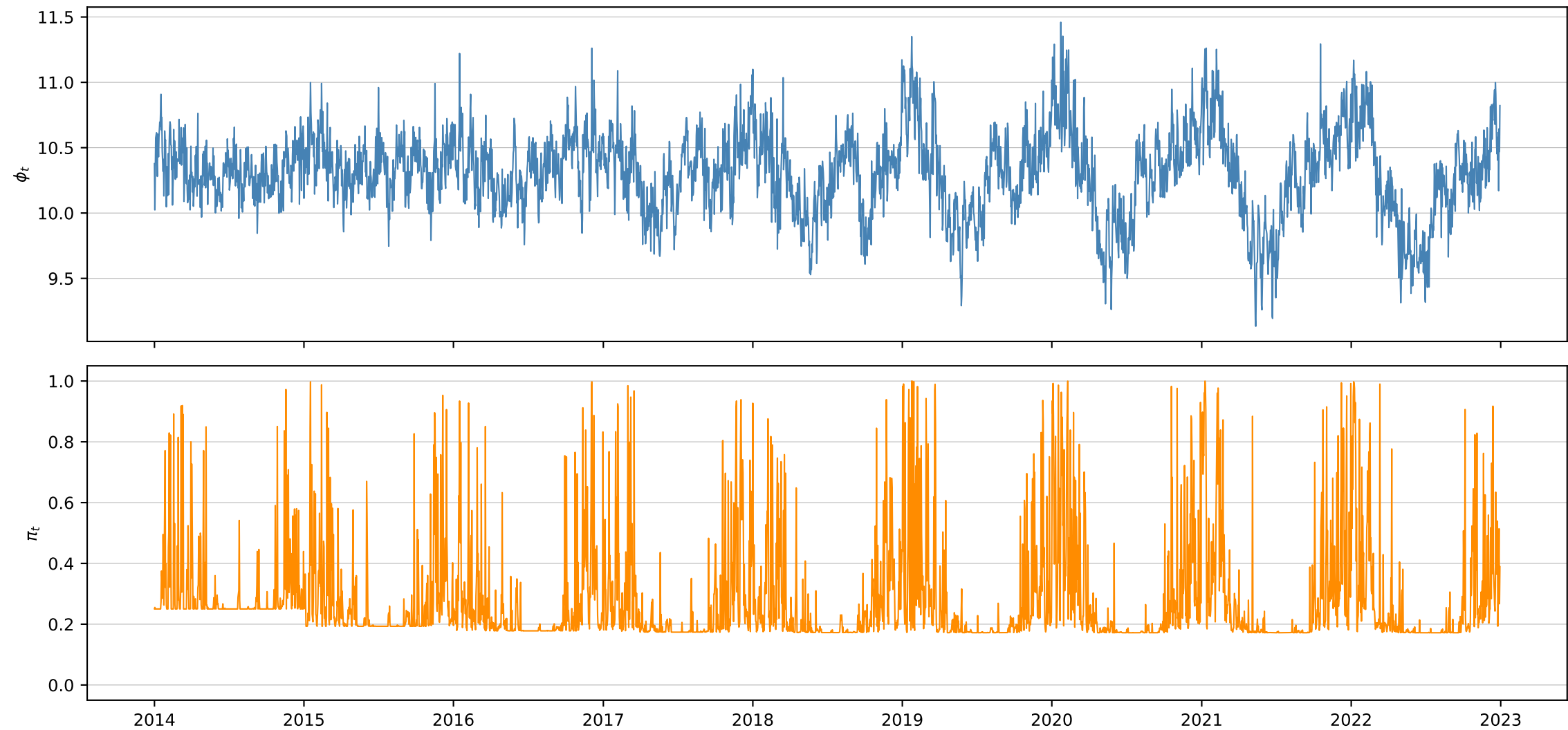
ACF of QR



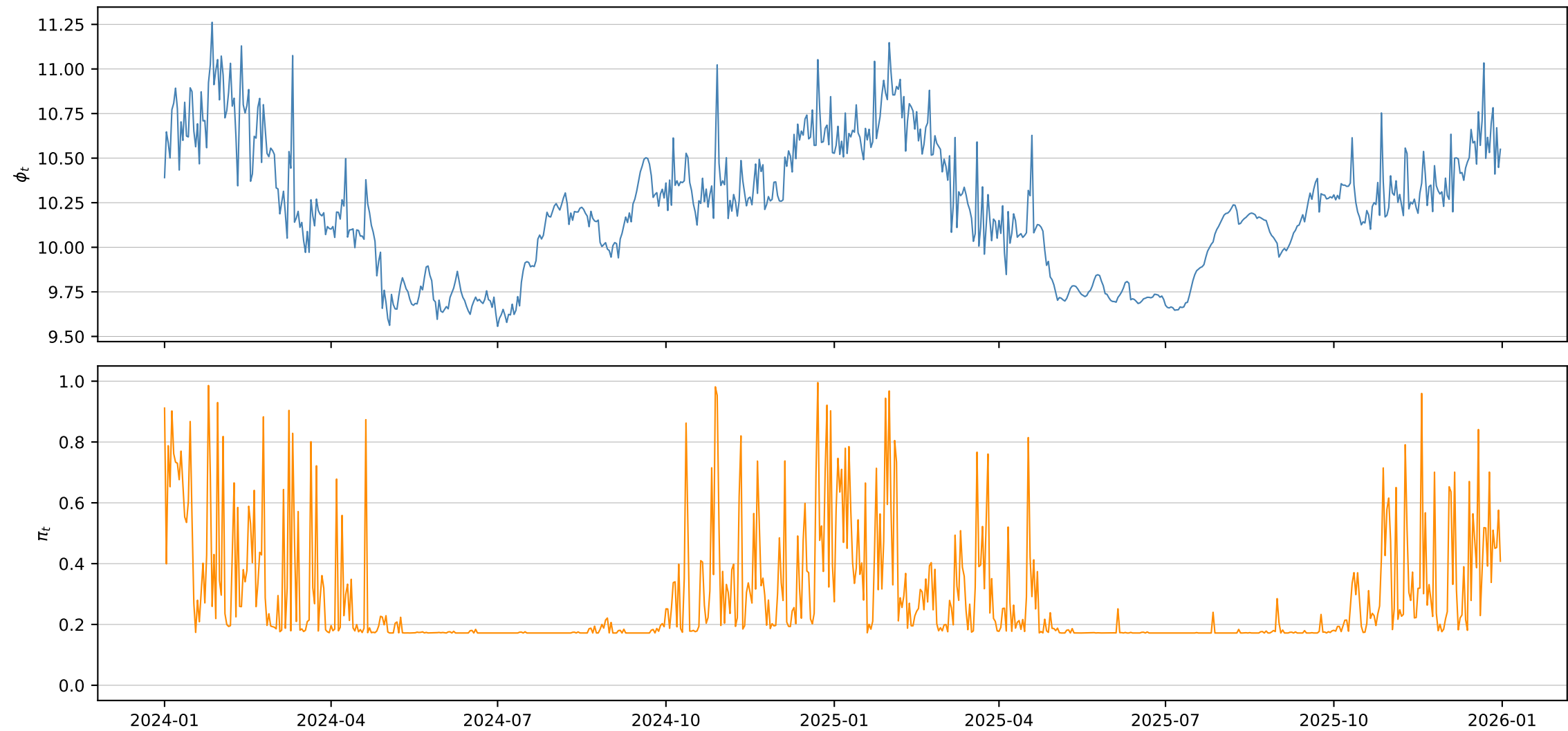
QR time series



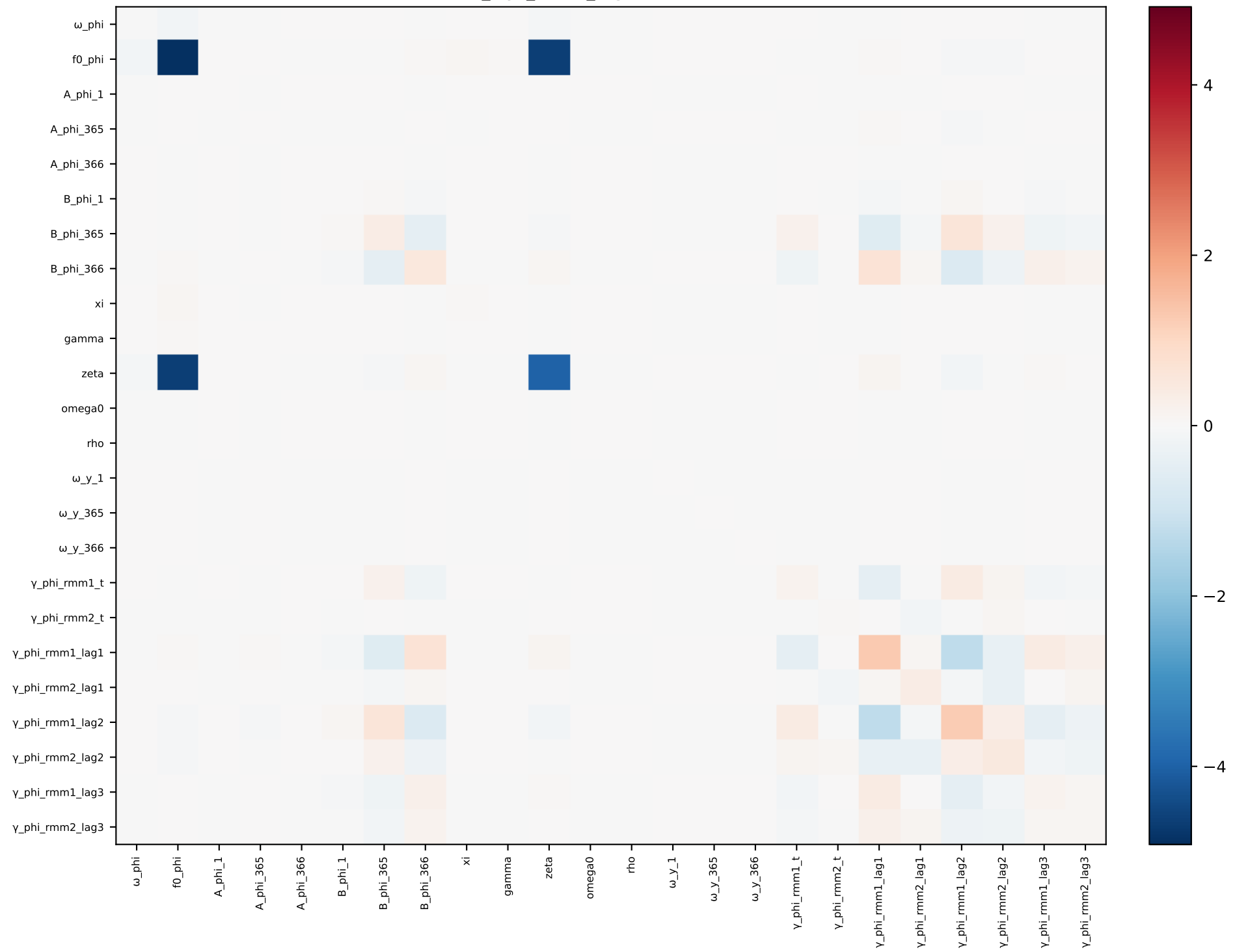
phi_mjo_short_lags — filtered states (IS)



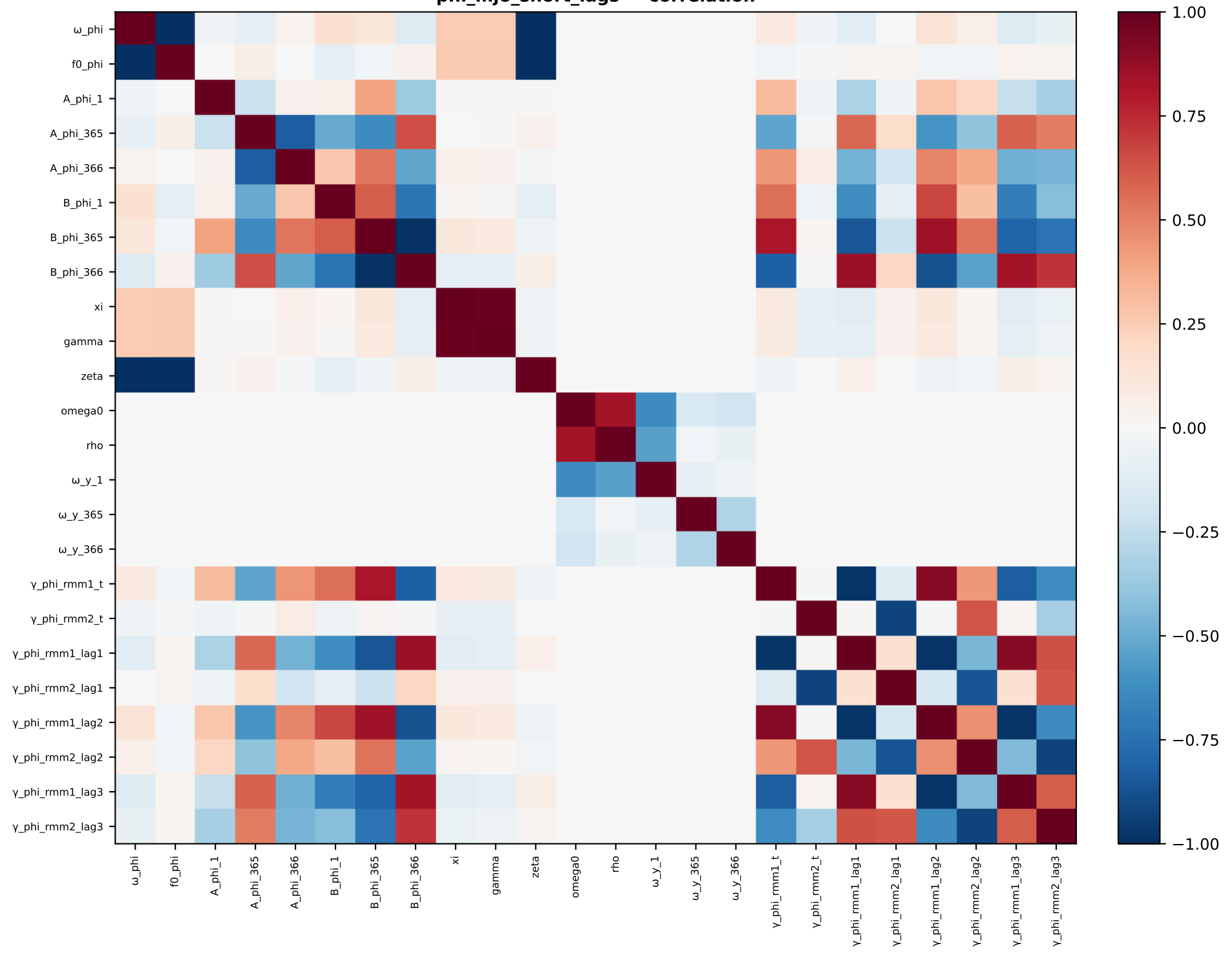
phi_mjo_short_lags — filtered states (OOS)



phi mjo short lags — covariance

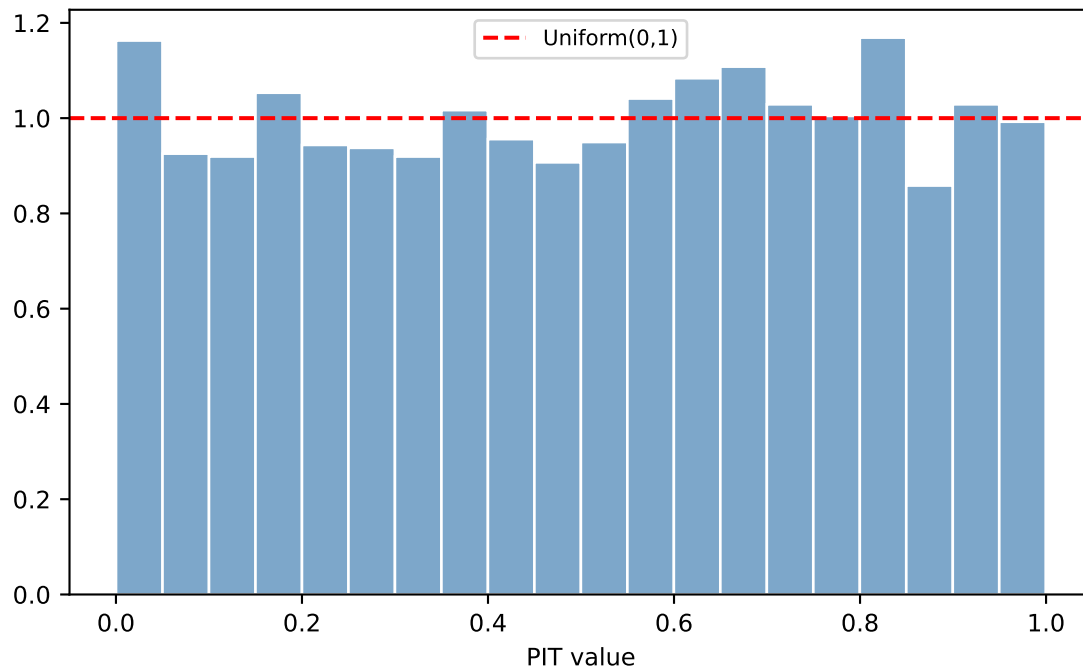


phi_mjo_short_lags — correlation

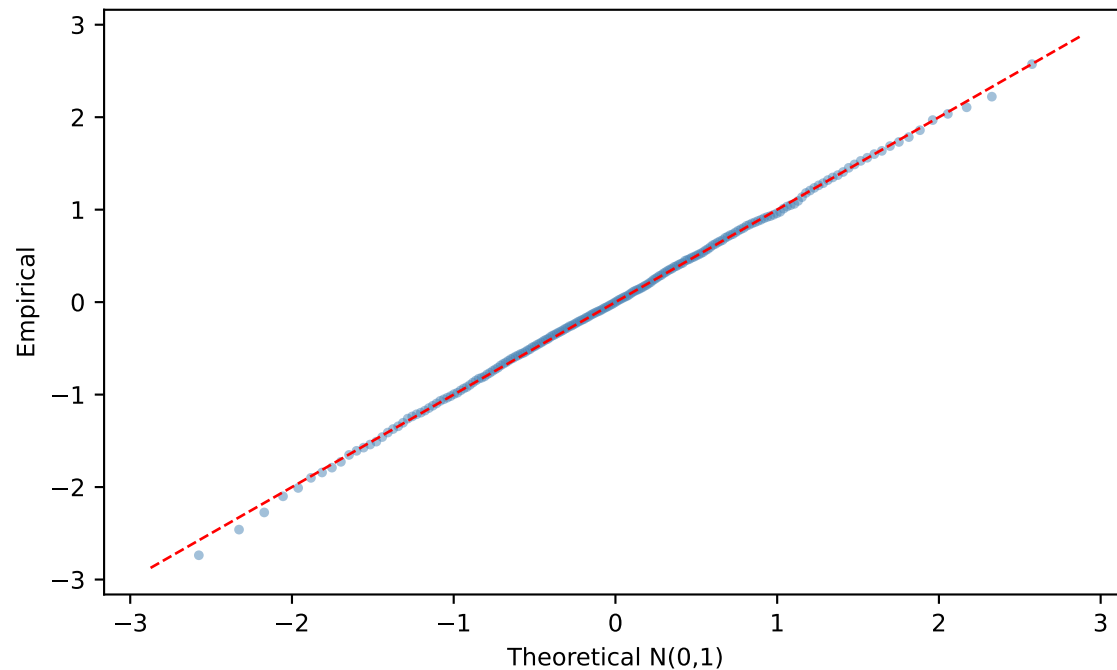


phi_nino34_t — IS diagnostics

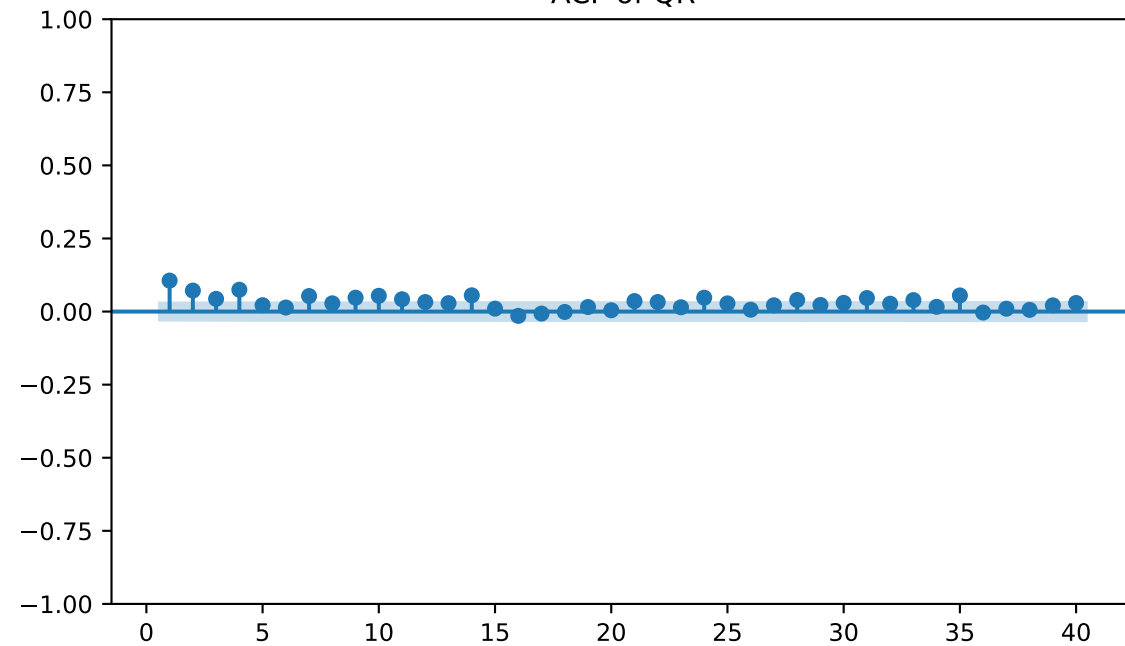
PIT histogram



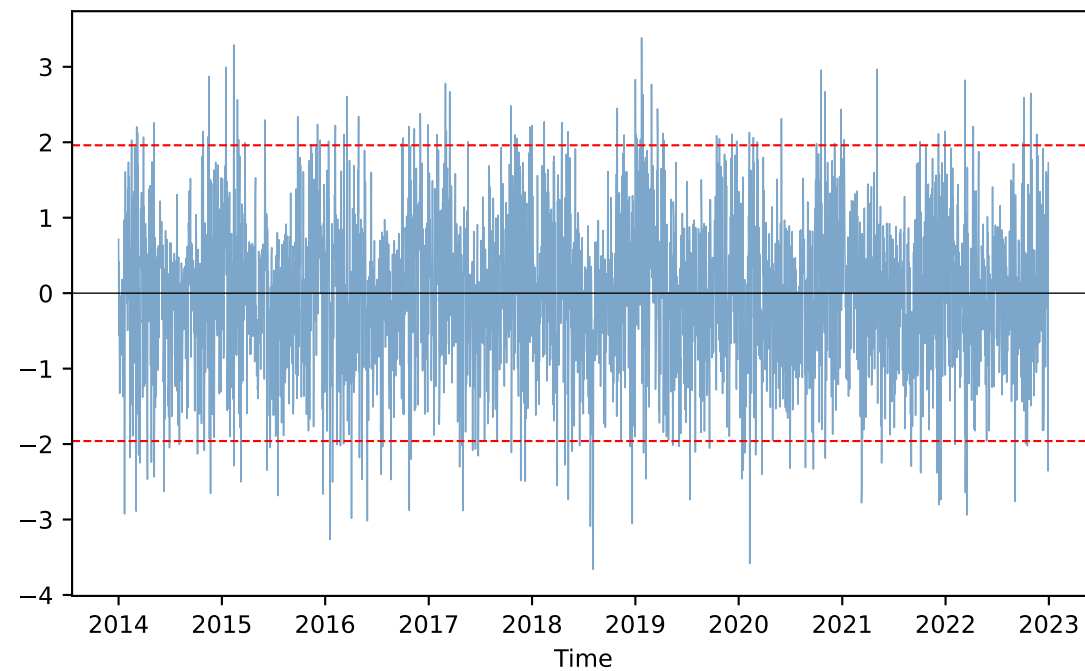
Normal QQ plot



ACF of QR

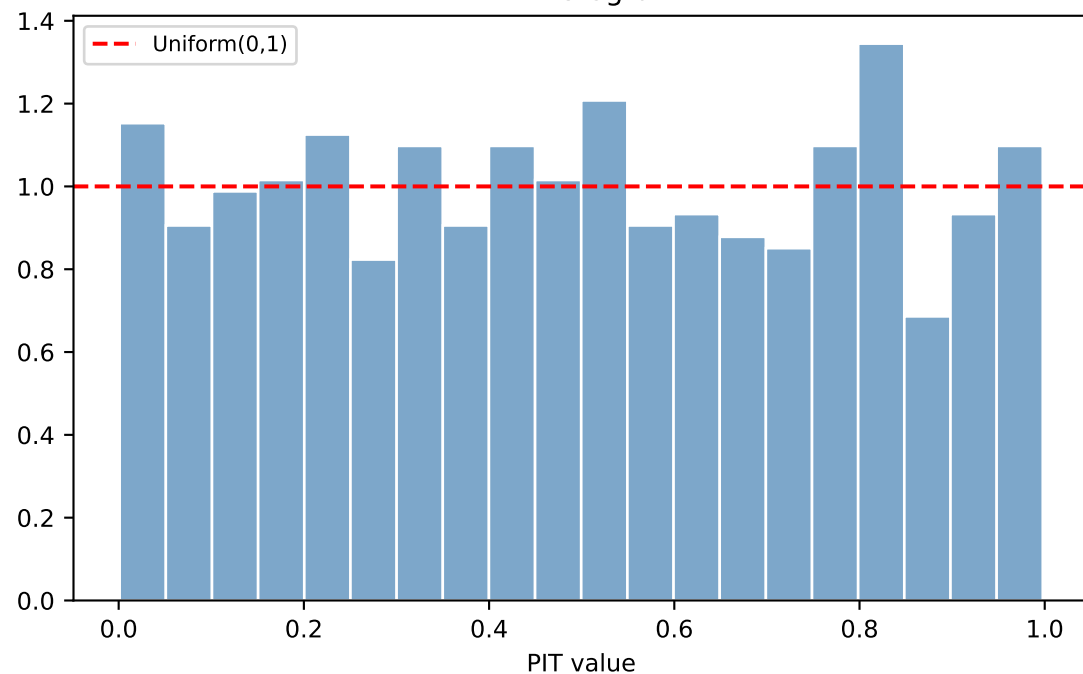


QR time series

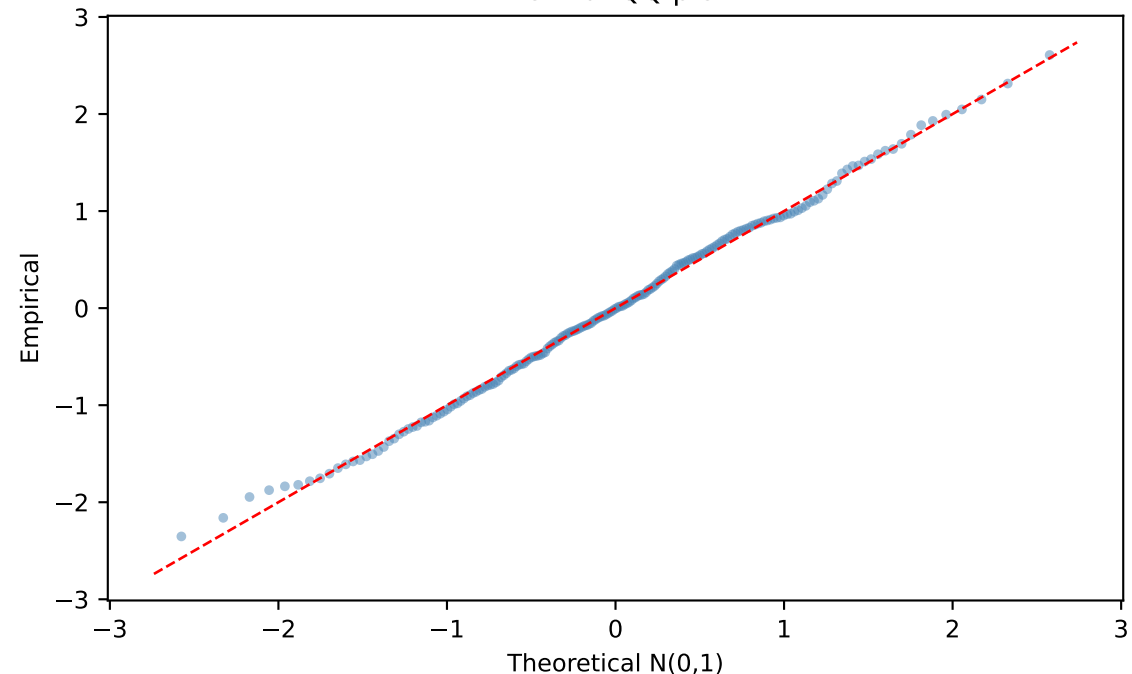


phi_nino34_t — OOS diagnostics

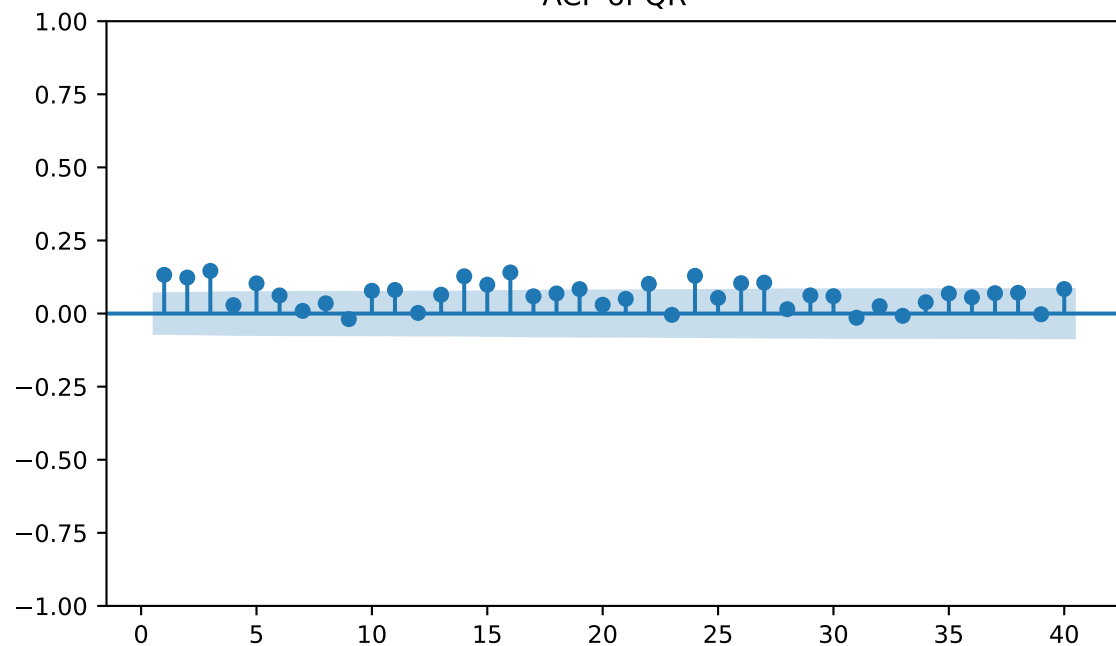
PIT histogram



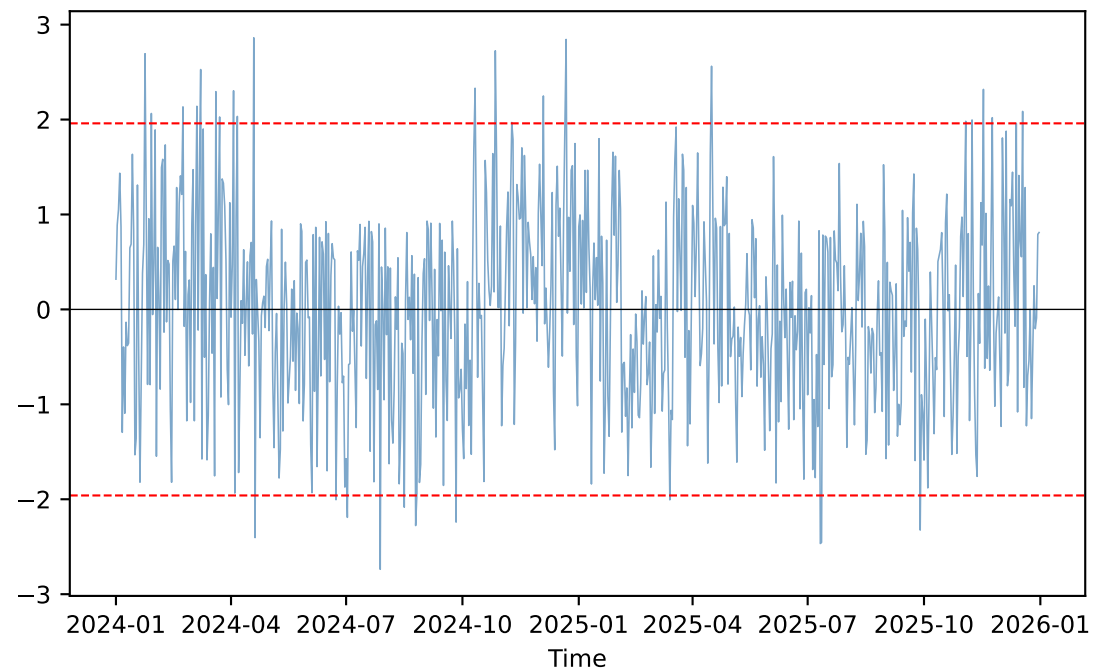
Normal QQ plot



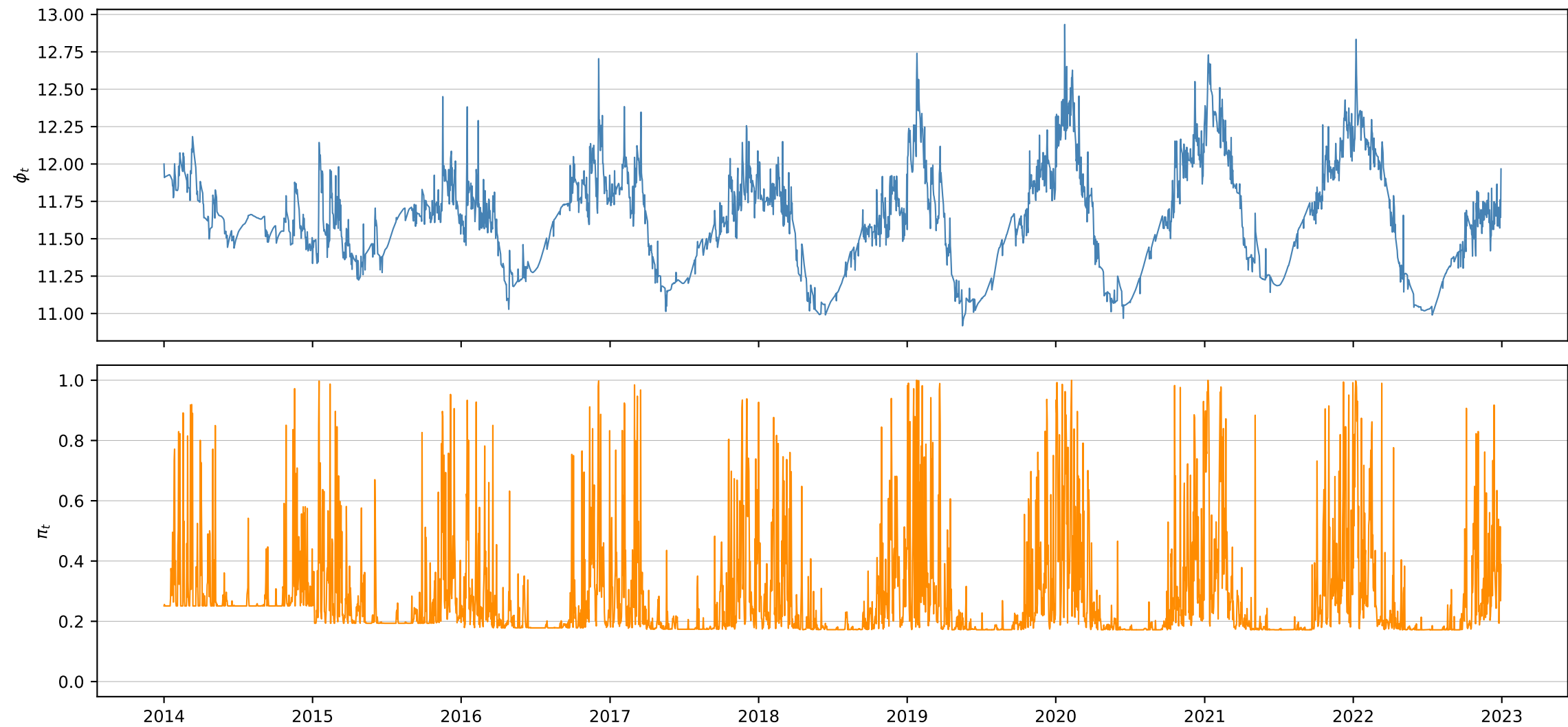
ACF of QR



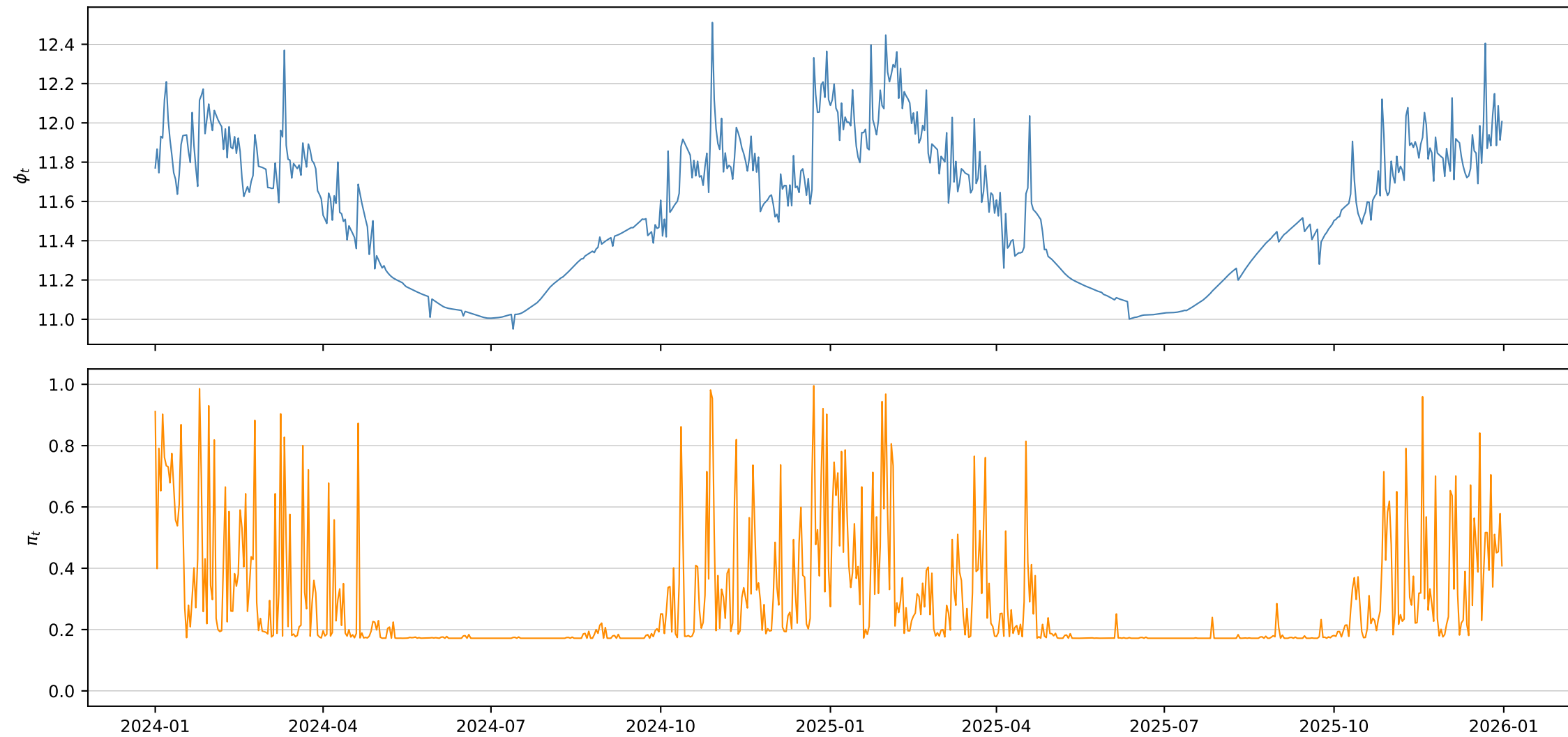
QR time series



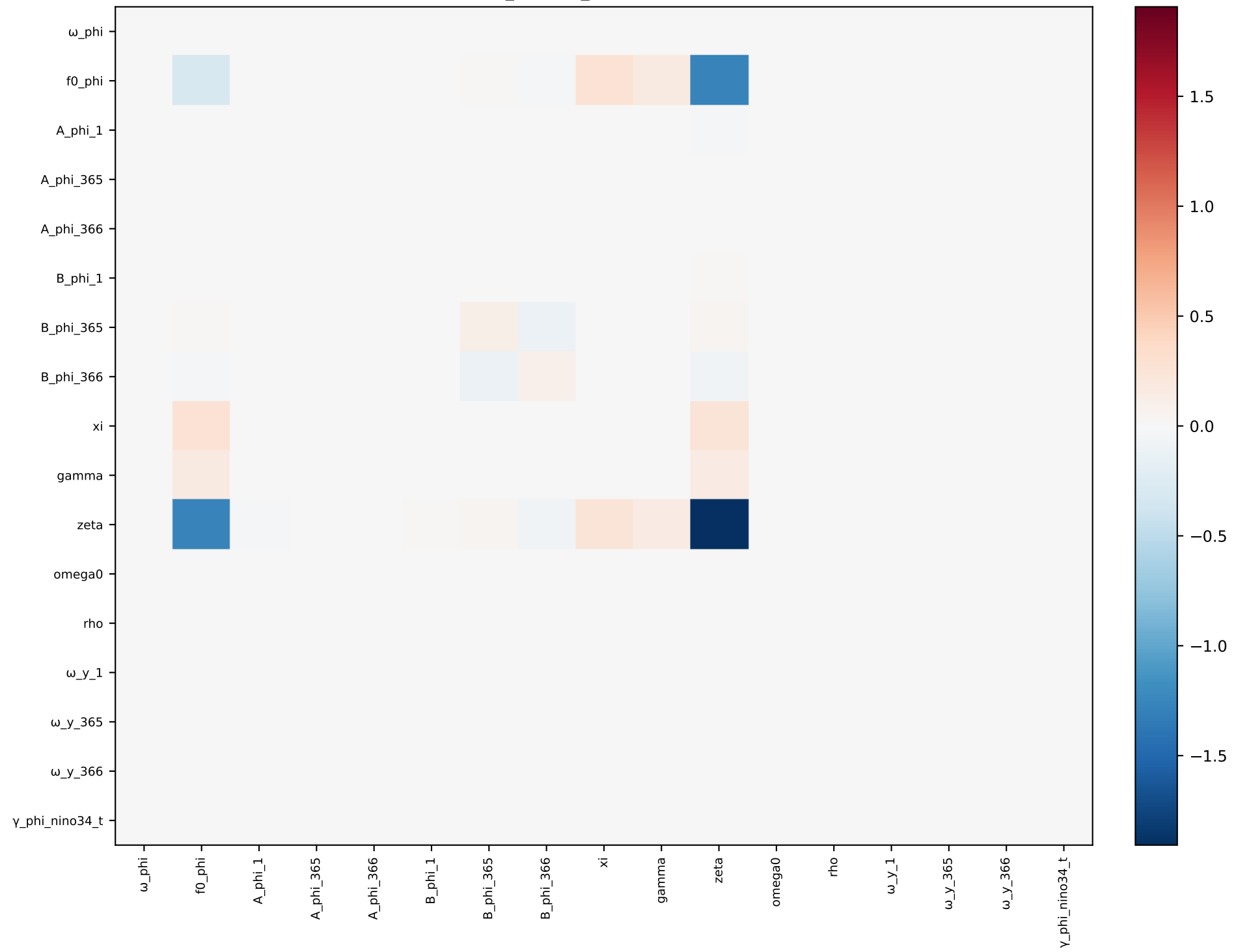
phi_nino34_t — filtered states (IS)



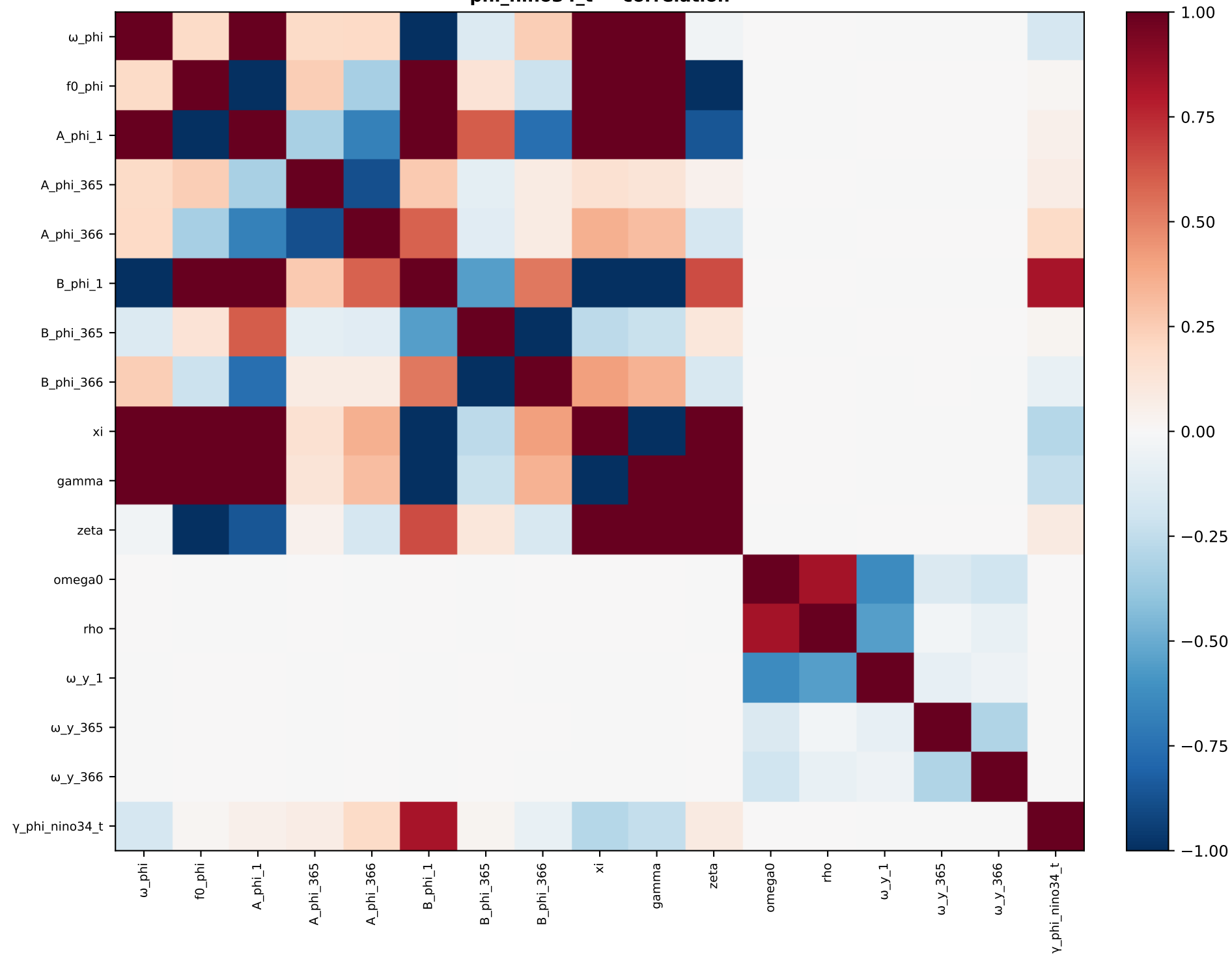
phi_nino34_t — filtered states (OOS)



phi_nino34_t – covariance

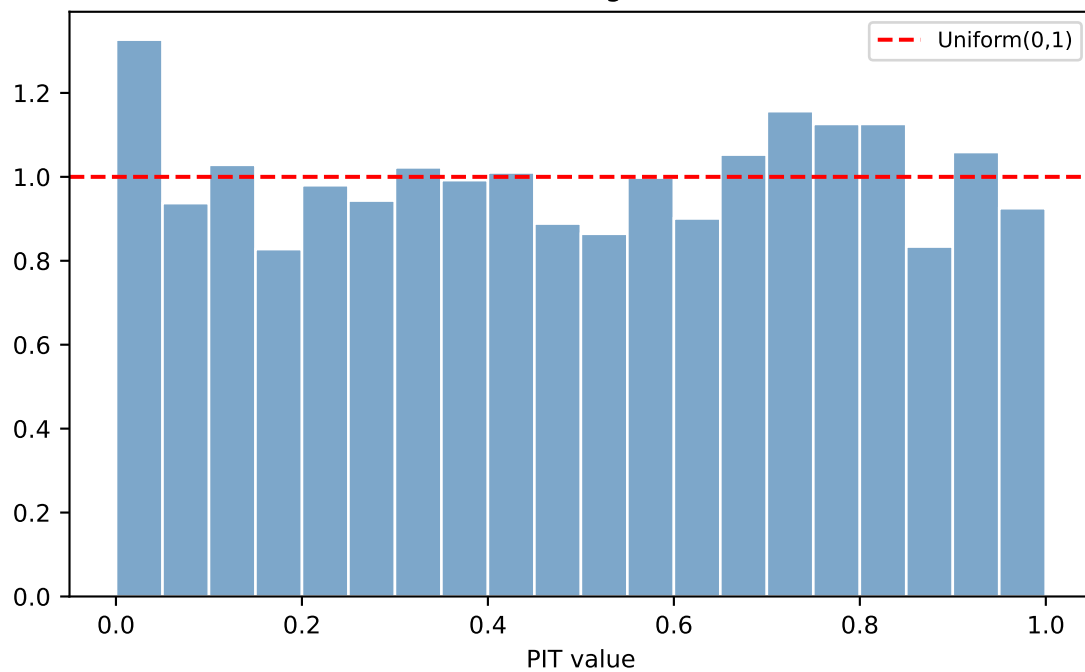


phi_nino34_t — correlation

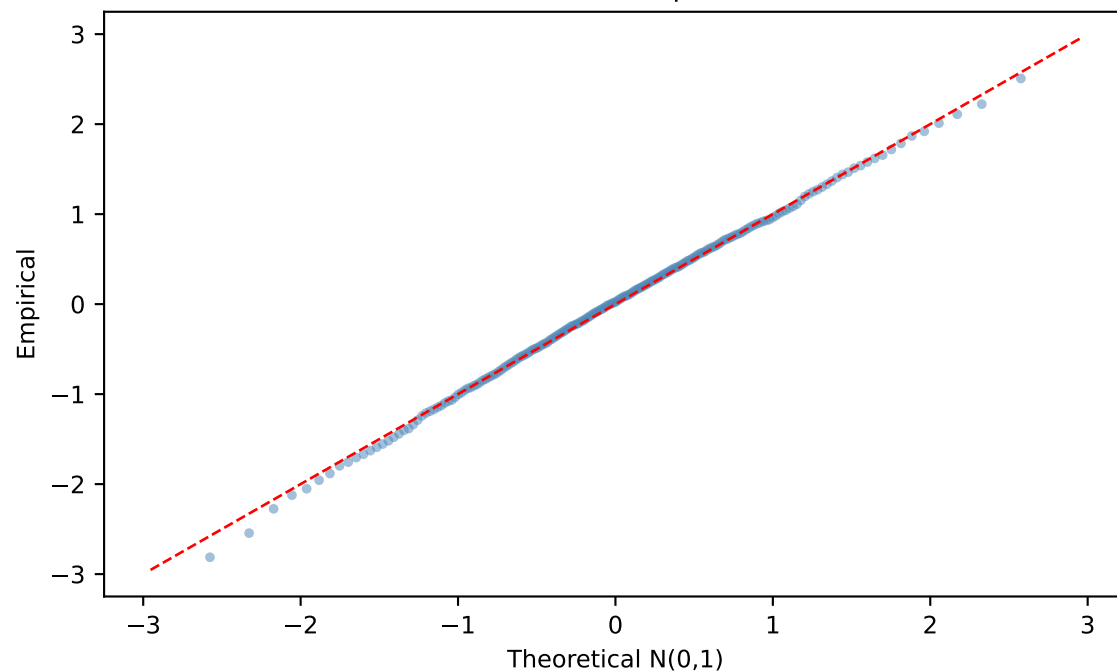


phi_nino34_lags — IS diagnostics

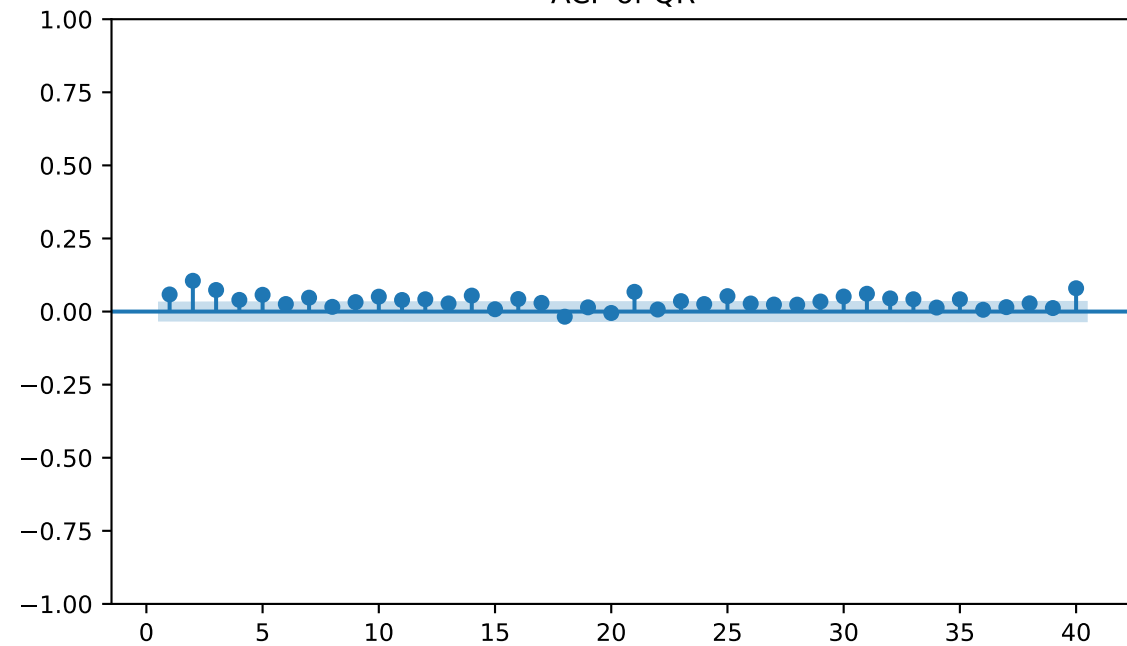
PIT histogram



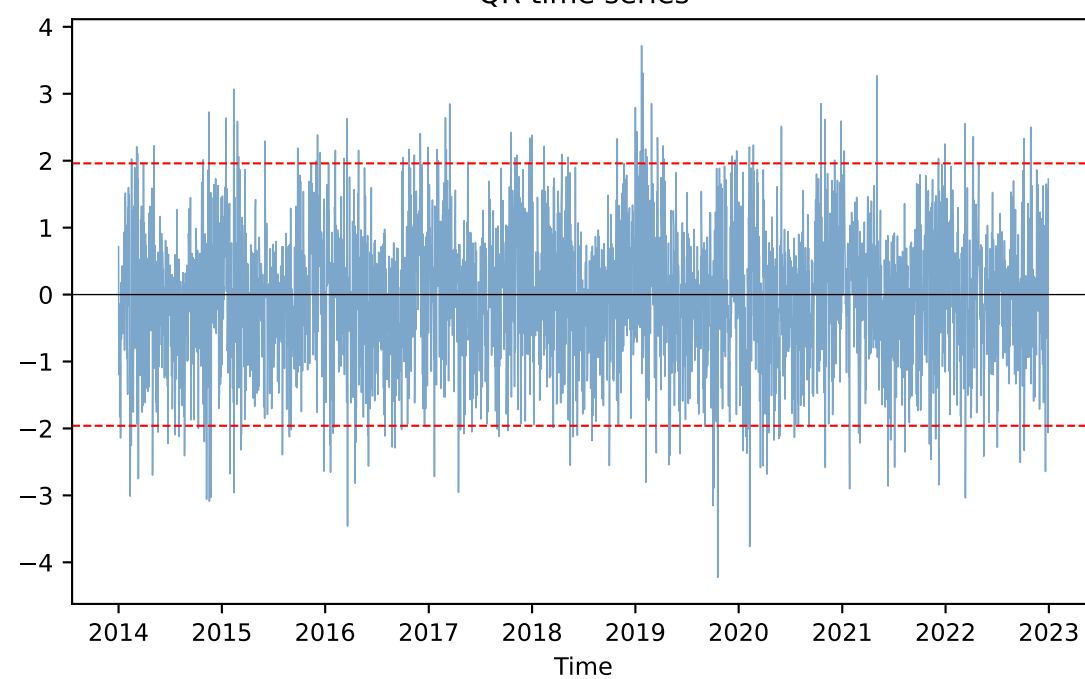
Normal QQ plot



ACF of QR

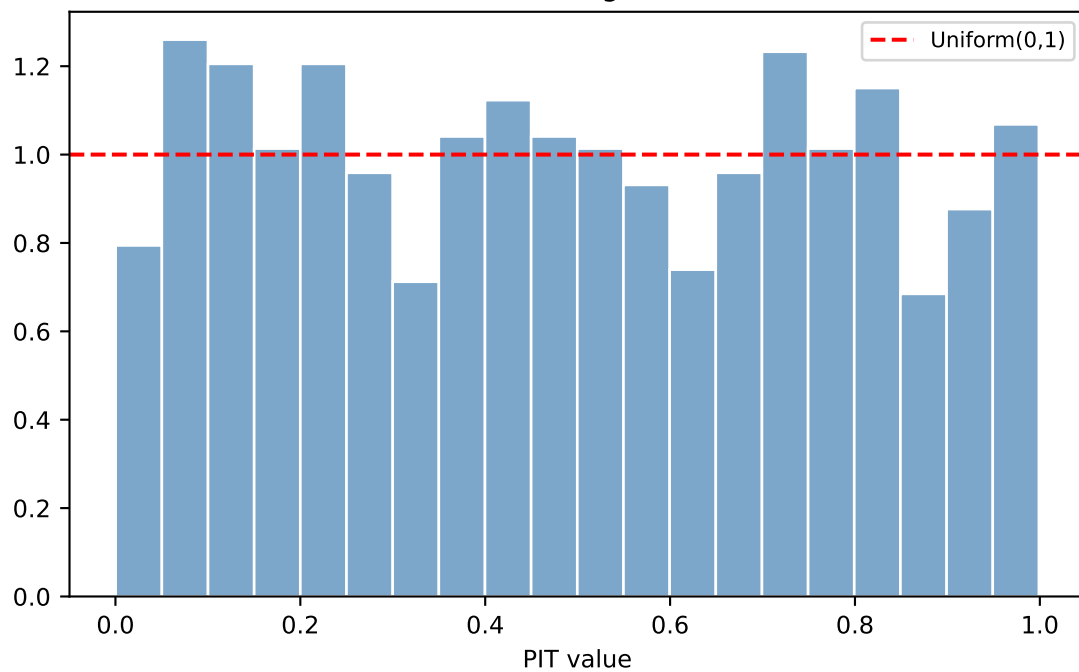


QR time series

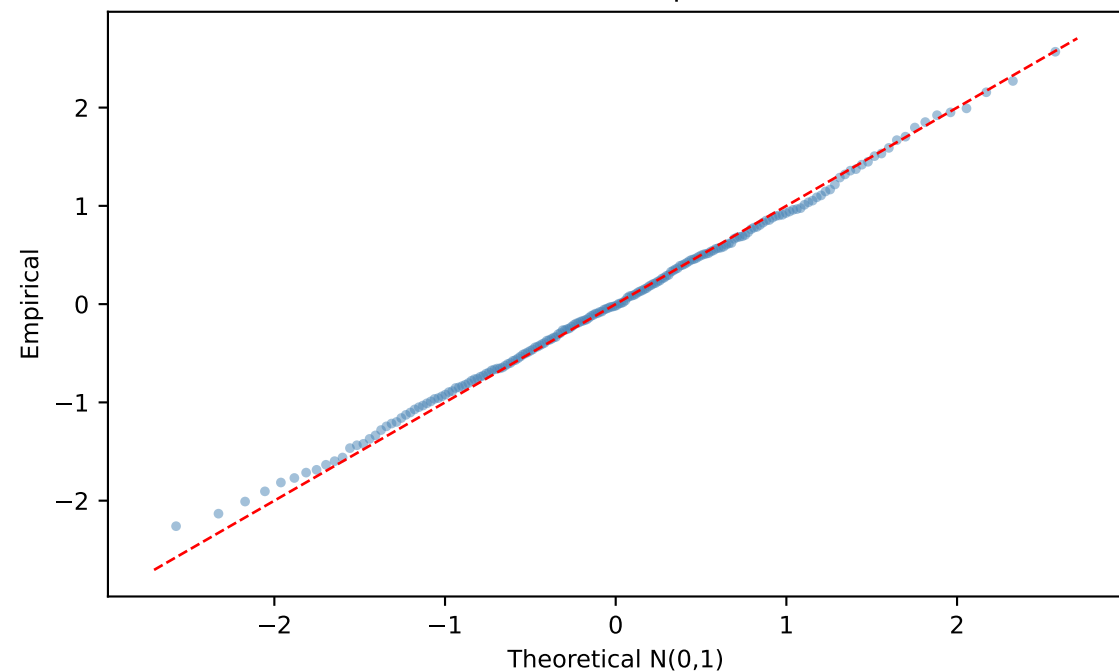


phi_nino34_lags — OOS diagnostics

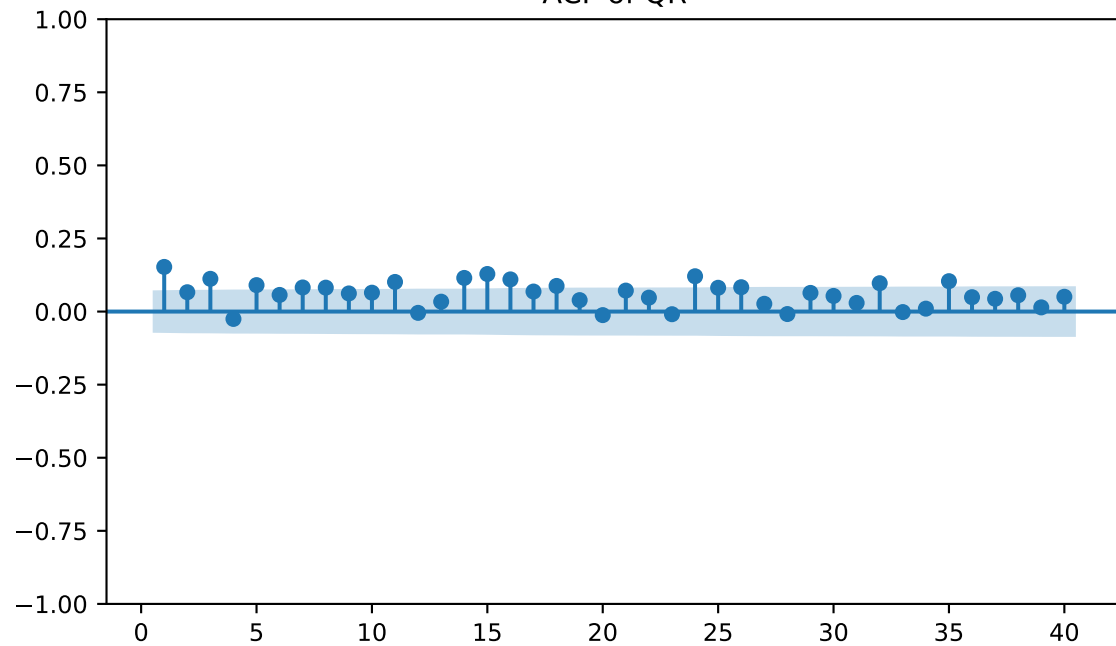
PIT histogram



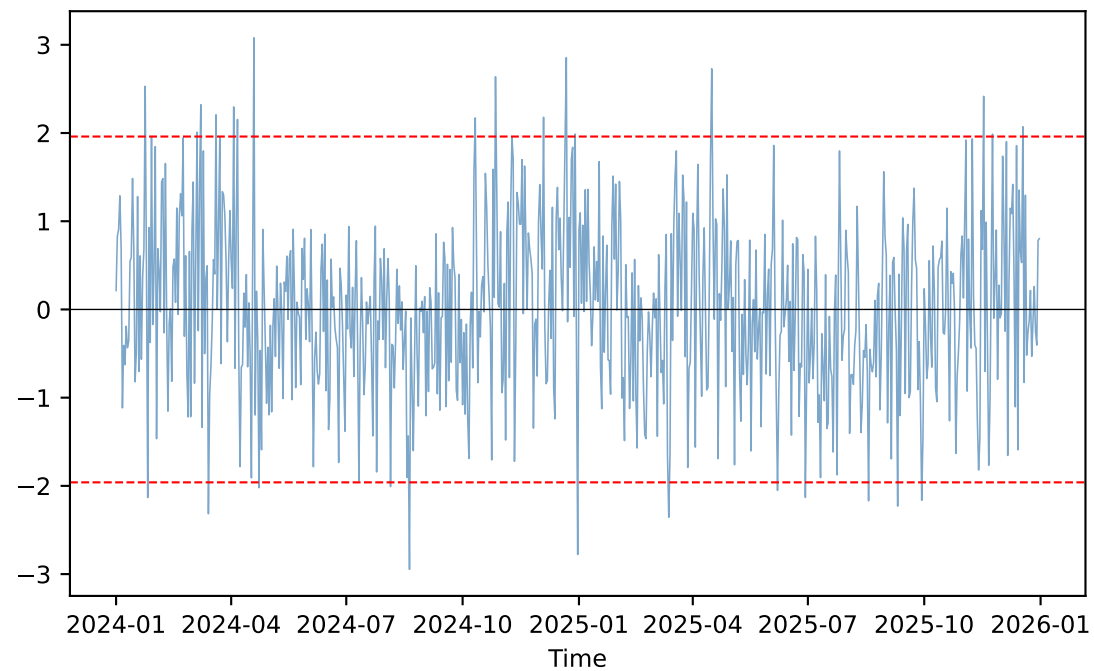
Normal QQ plot



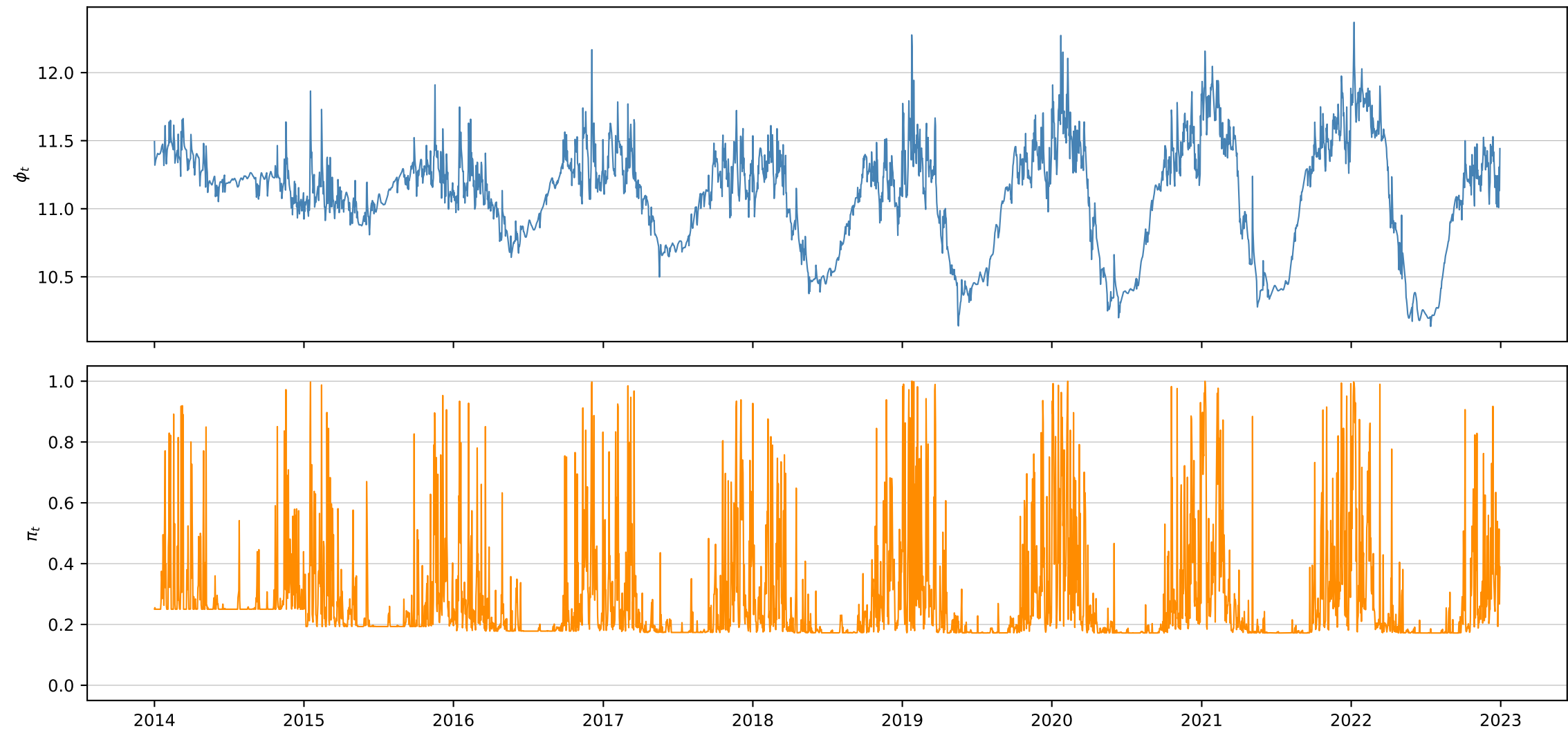
ACF of QR



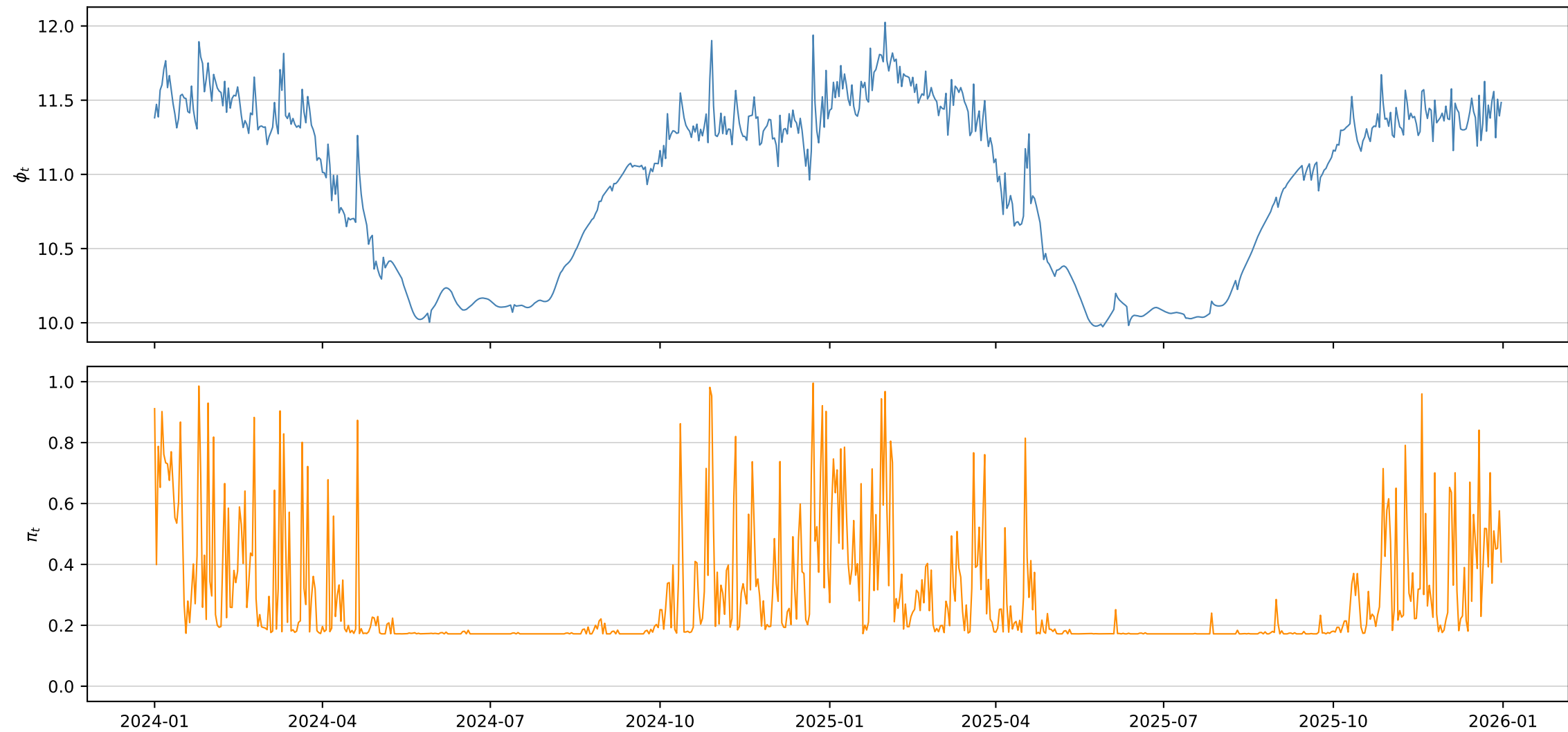
QR time series



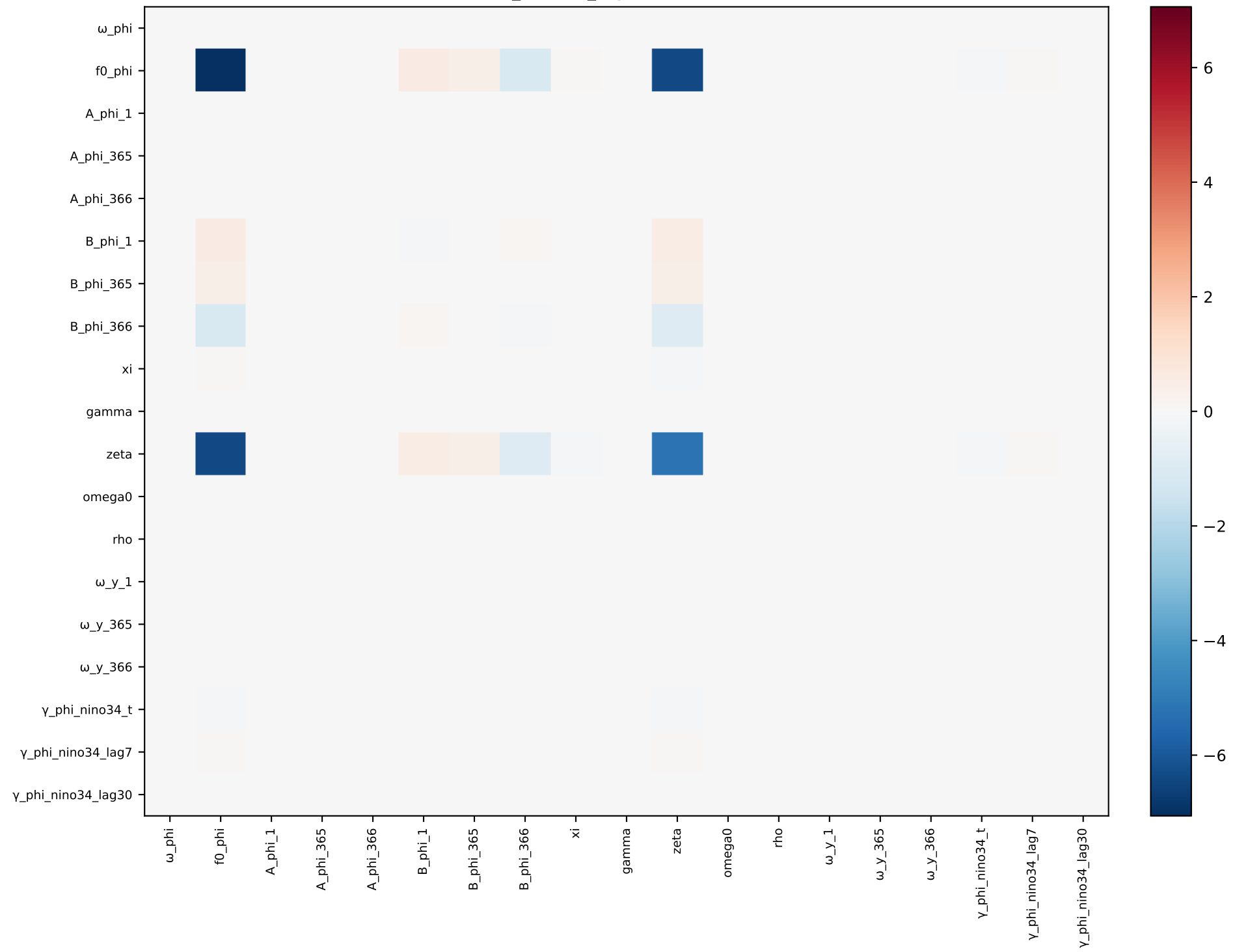
phi_nino34_lags — filtered states (IS)



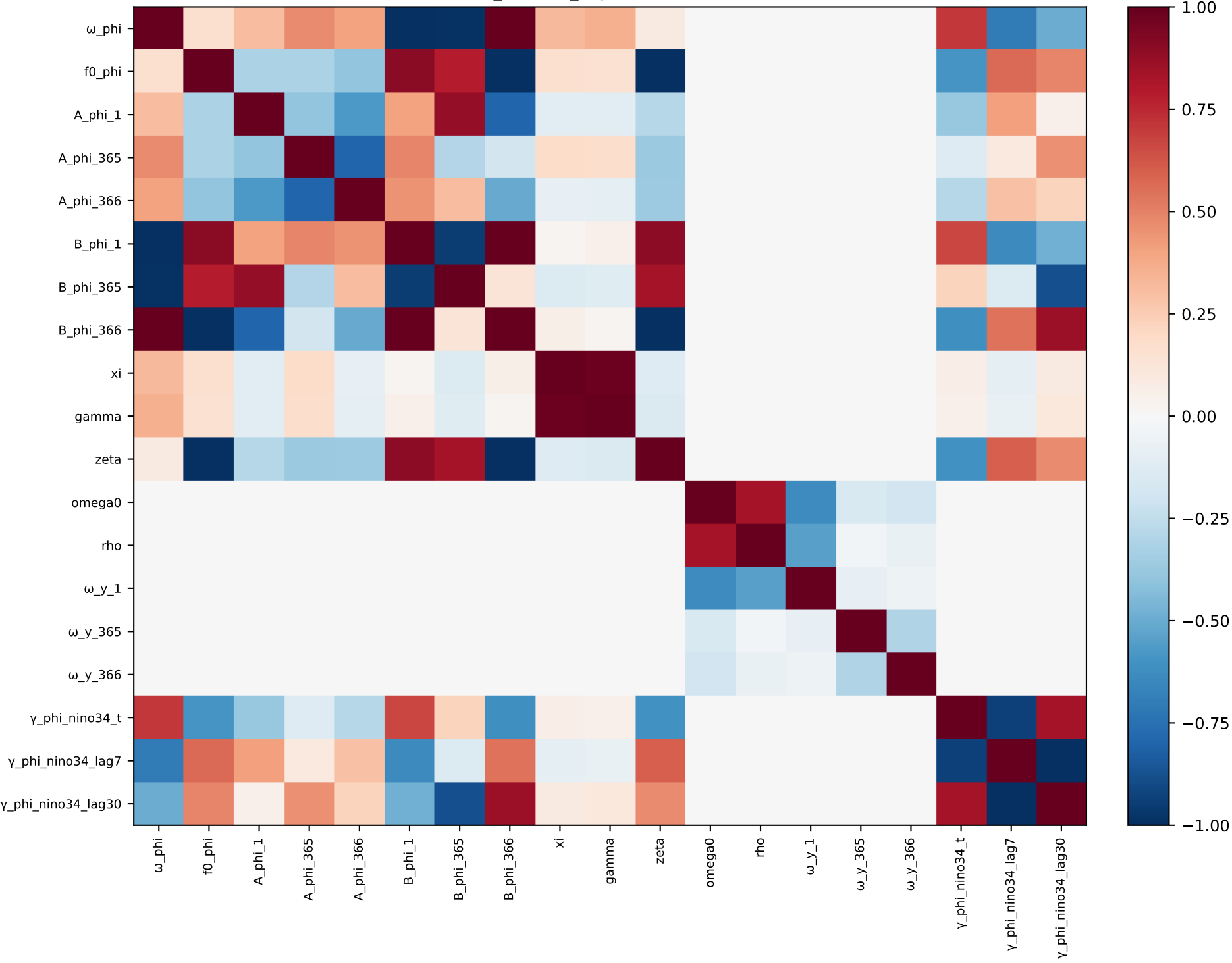
phi_nino34_lags — filtered states (OOS)



phi_nino34_lags — covariance

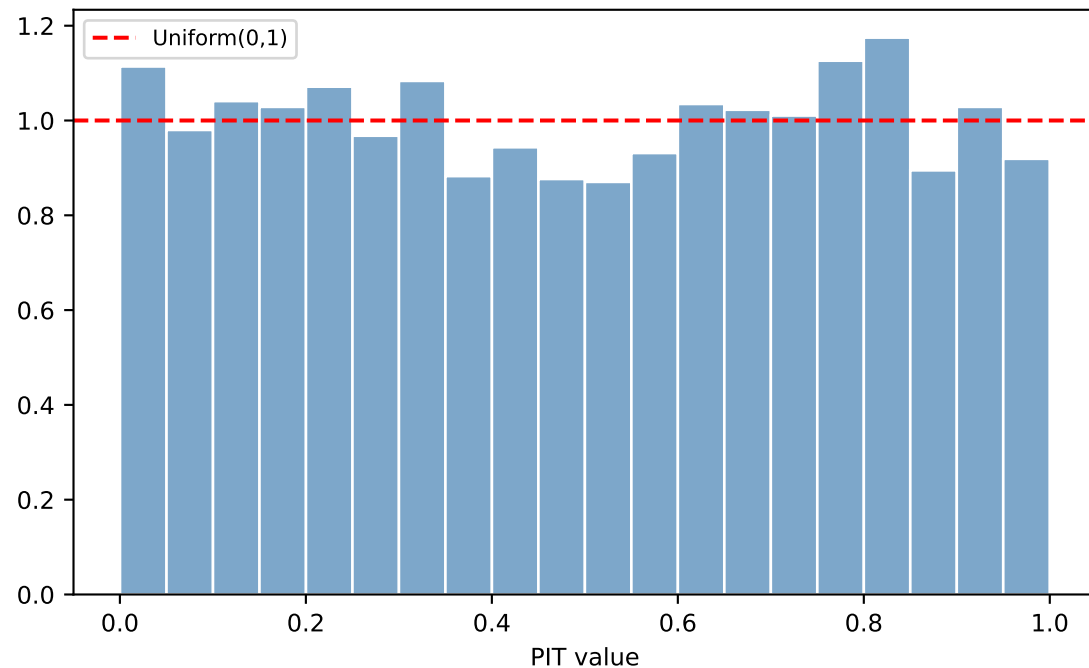


phi_nino34_lags — correlation

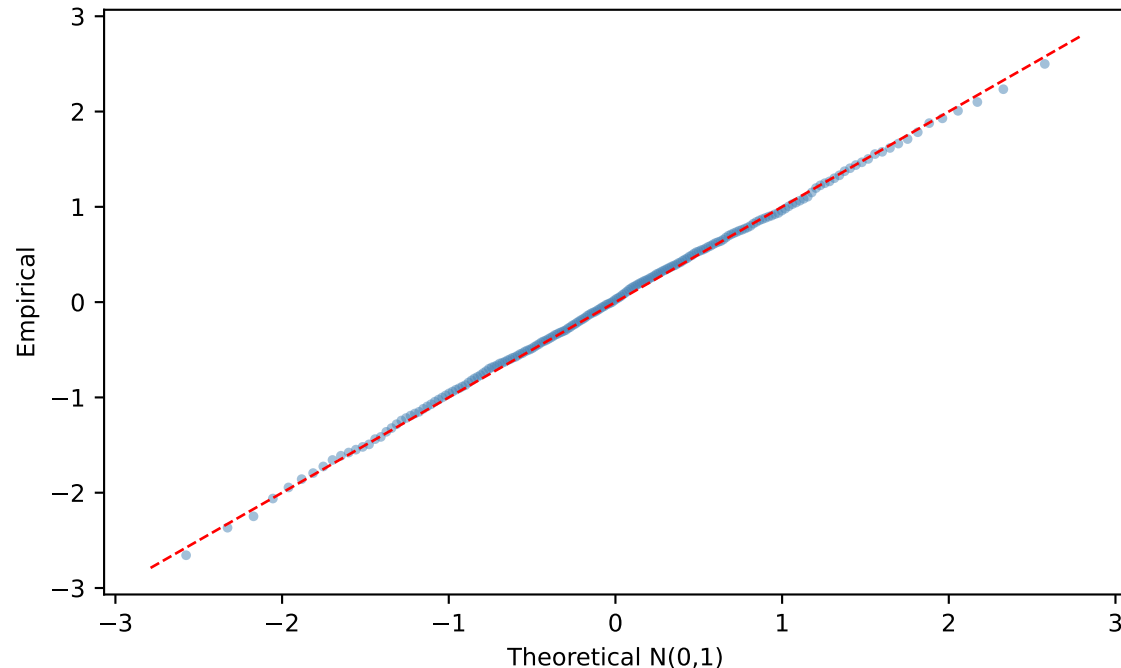


phi_nino34_rolling — IS diagnostics

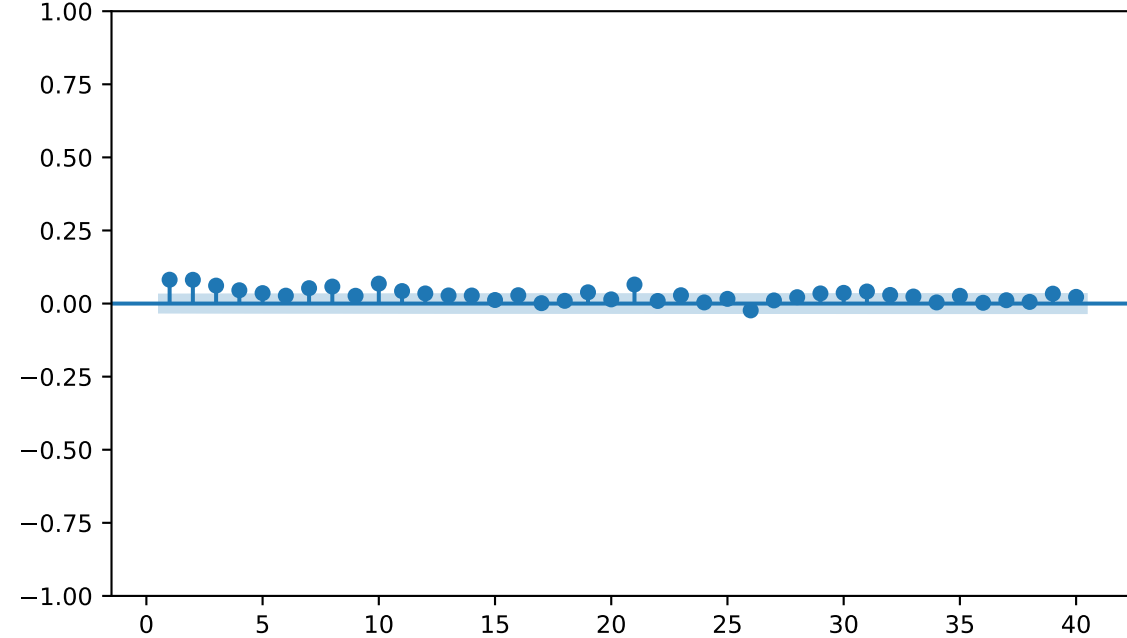
PIT histogram



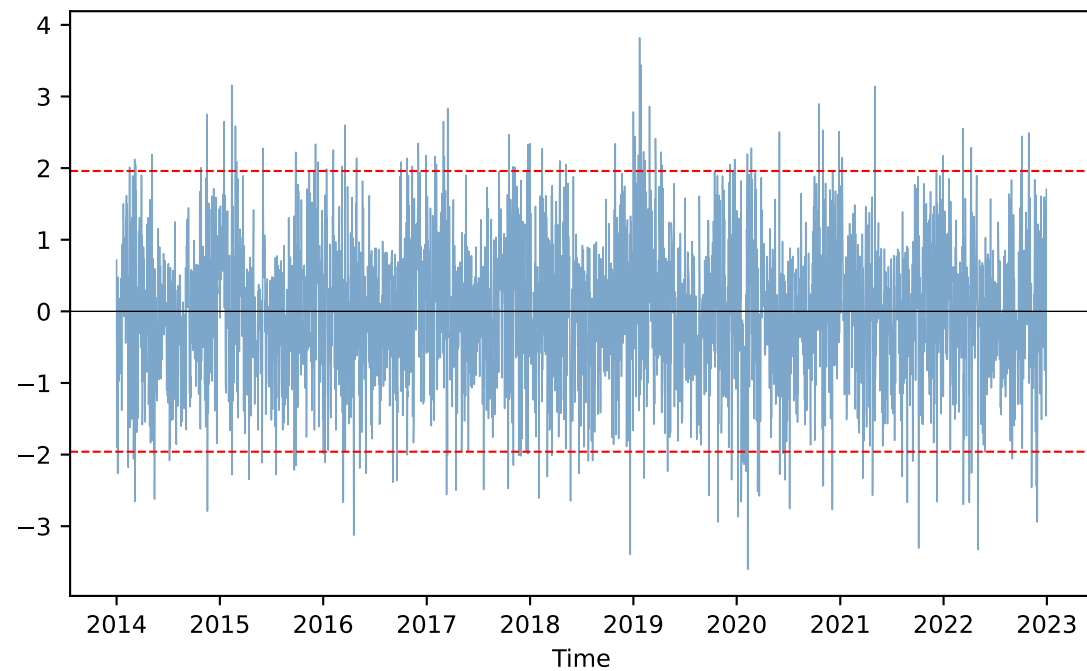
Normal QQ plot



ACF of QR

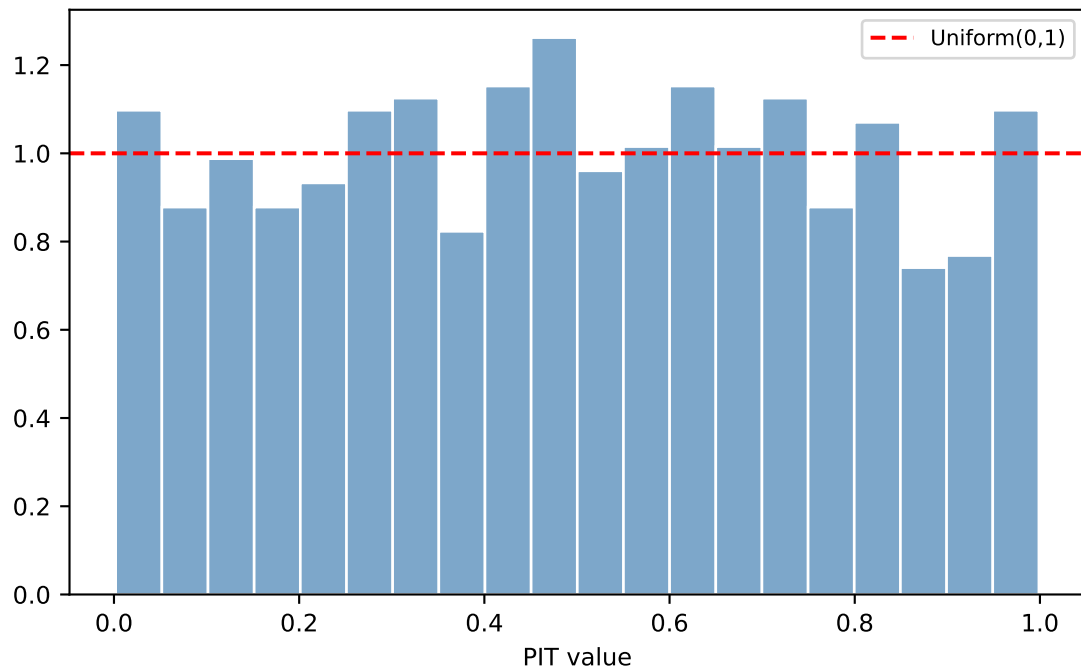


QR time series

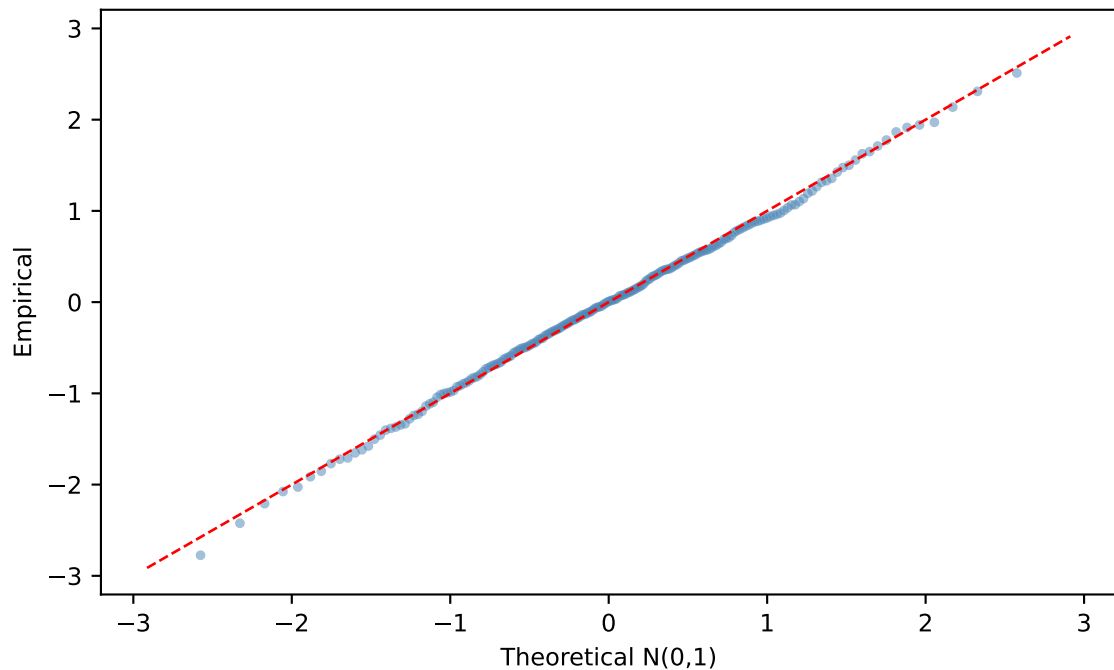


phi_nino34_rolling — OOS diagnostics

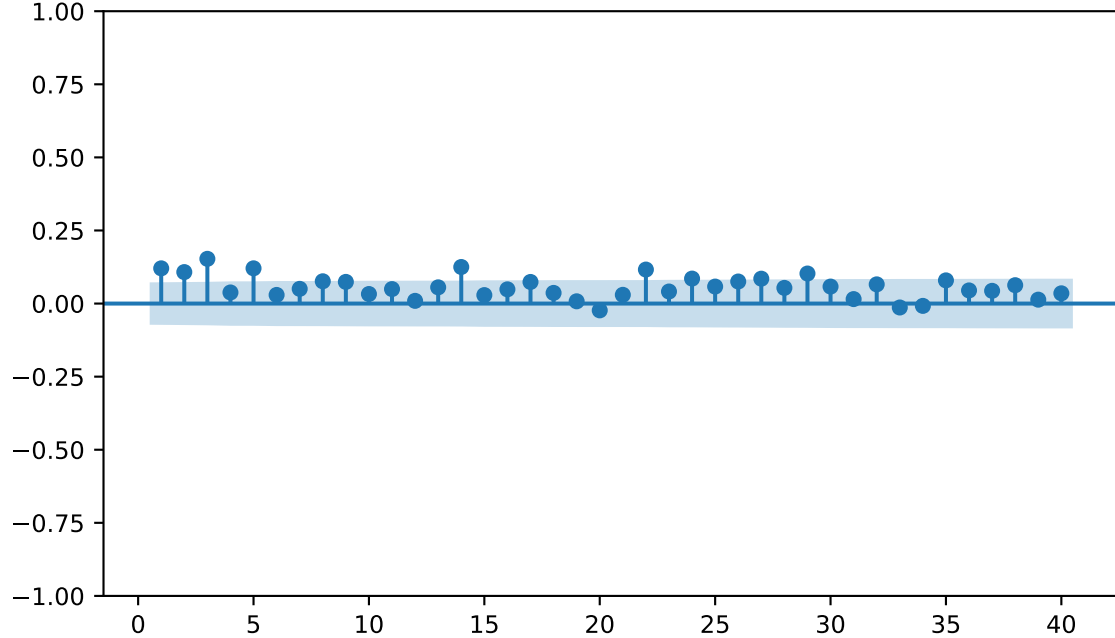
PIT histogram



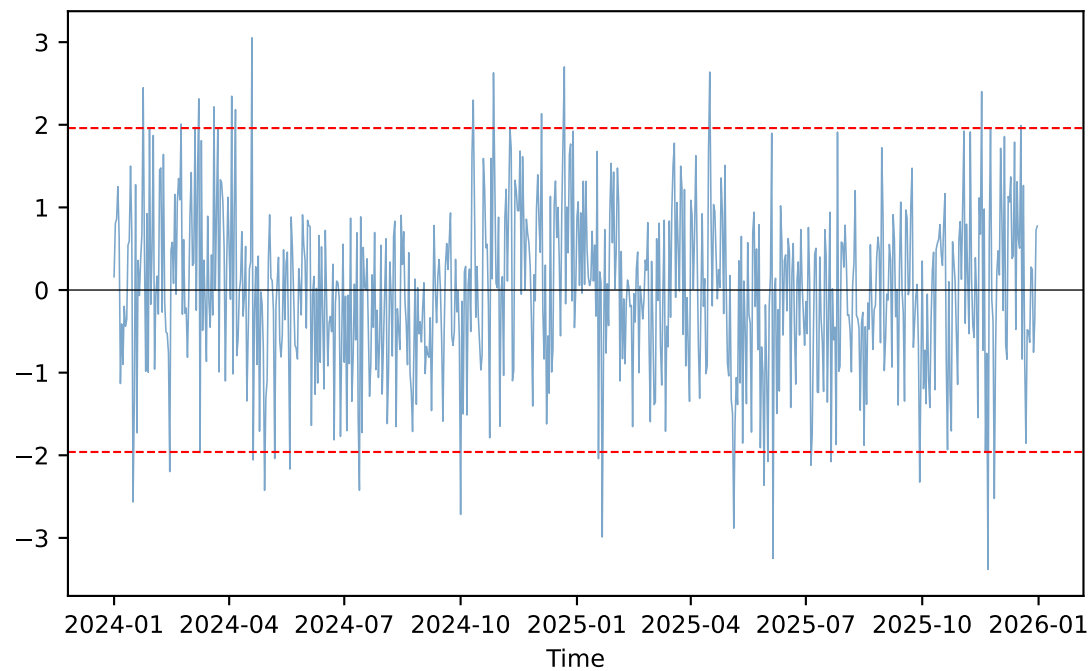
Normal QQ plot



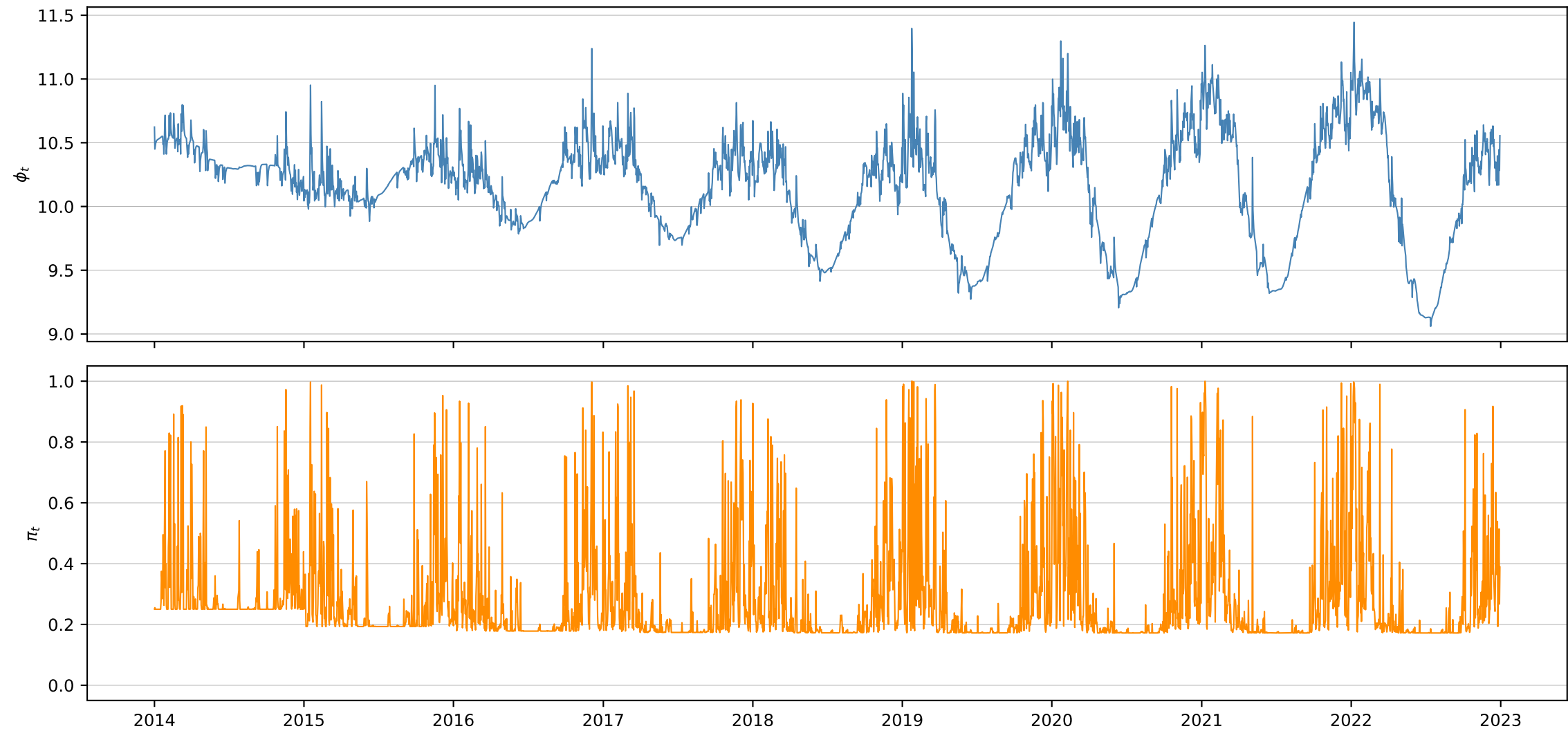
ACF of QR



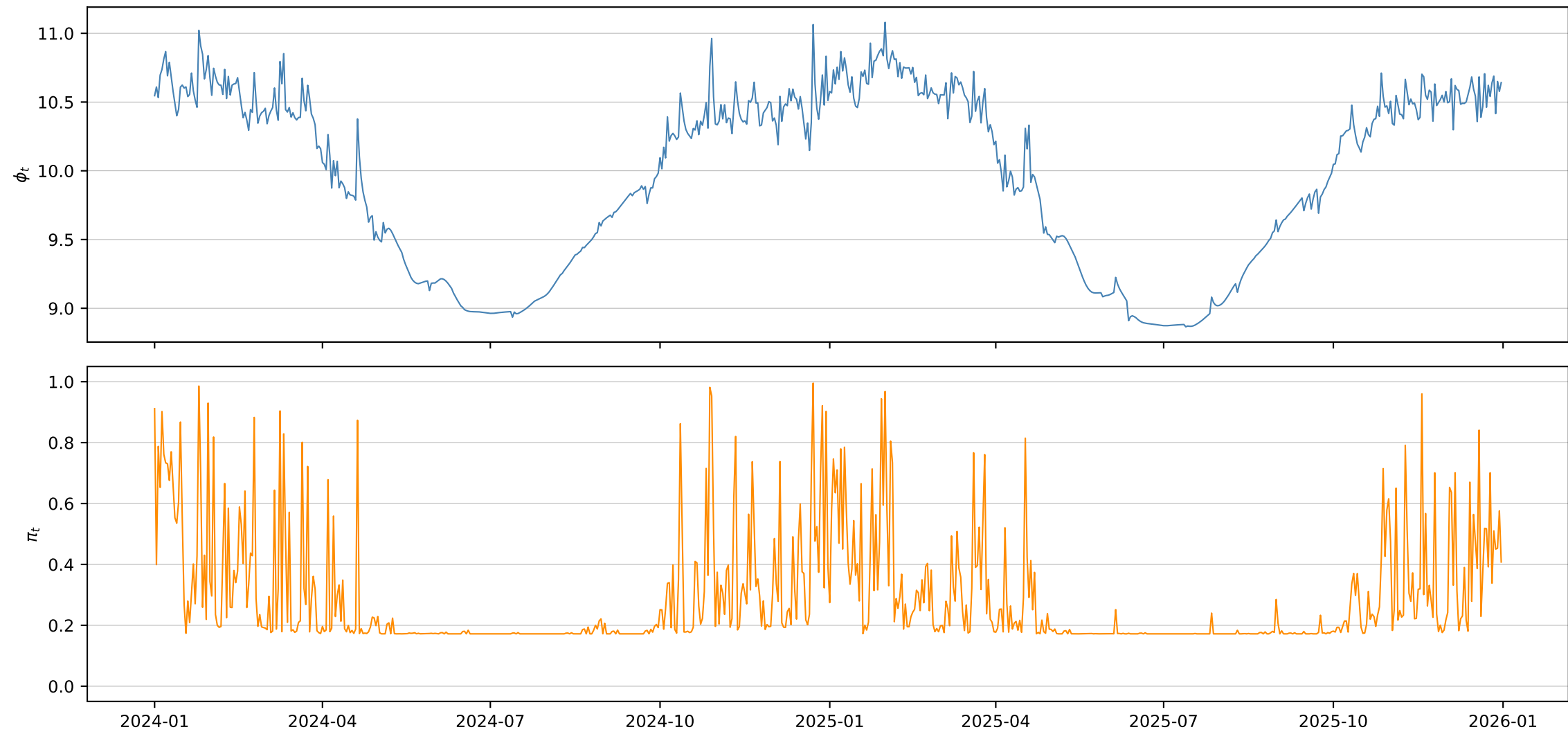
QR time series



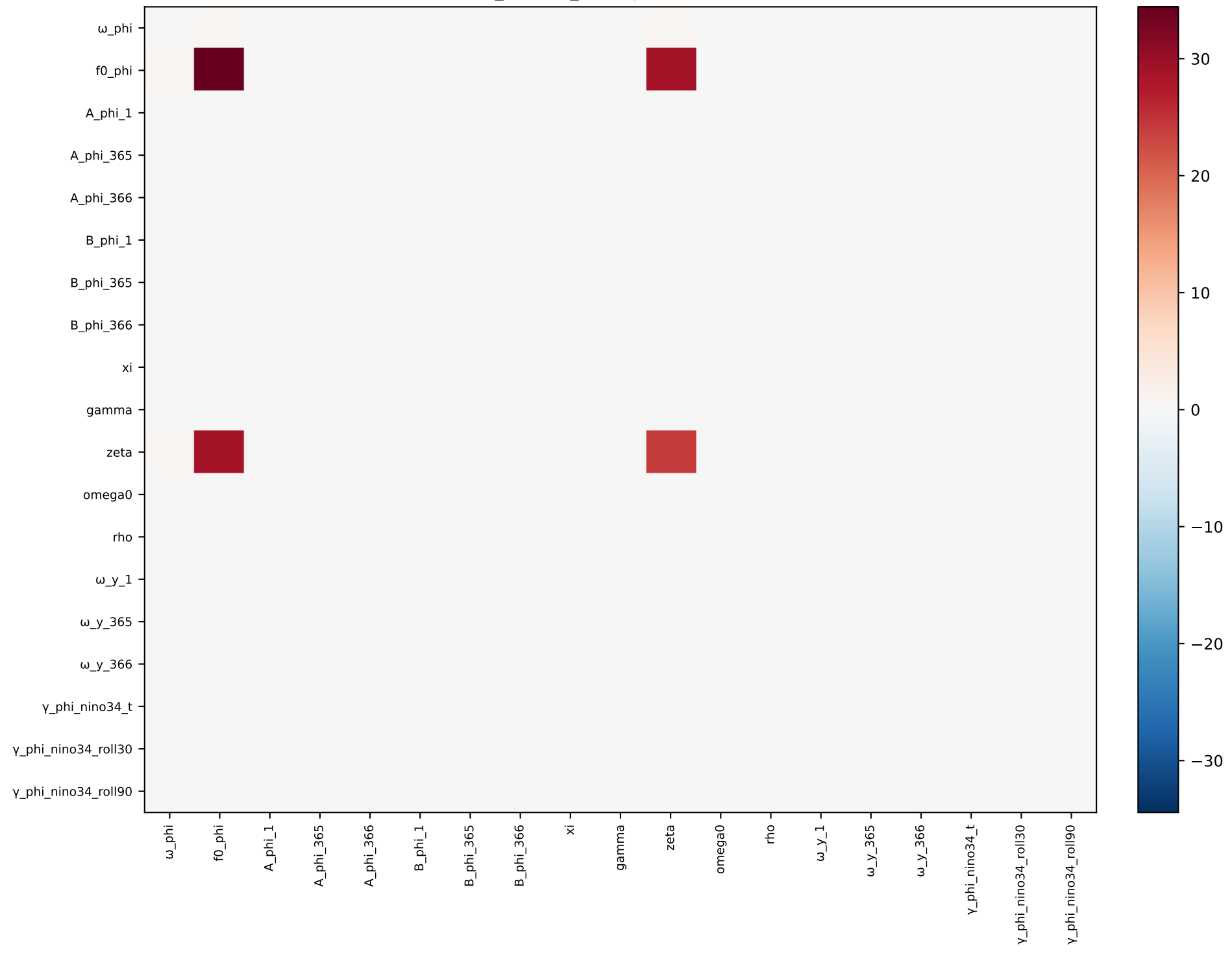
phi_nino34_rolling — filtered states (IS)



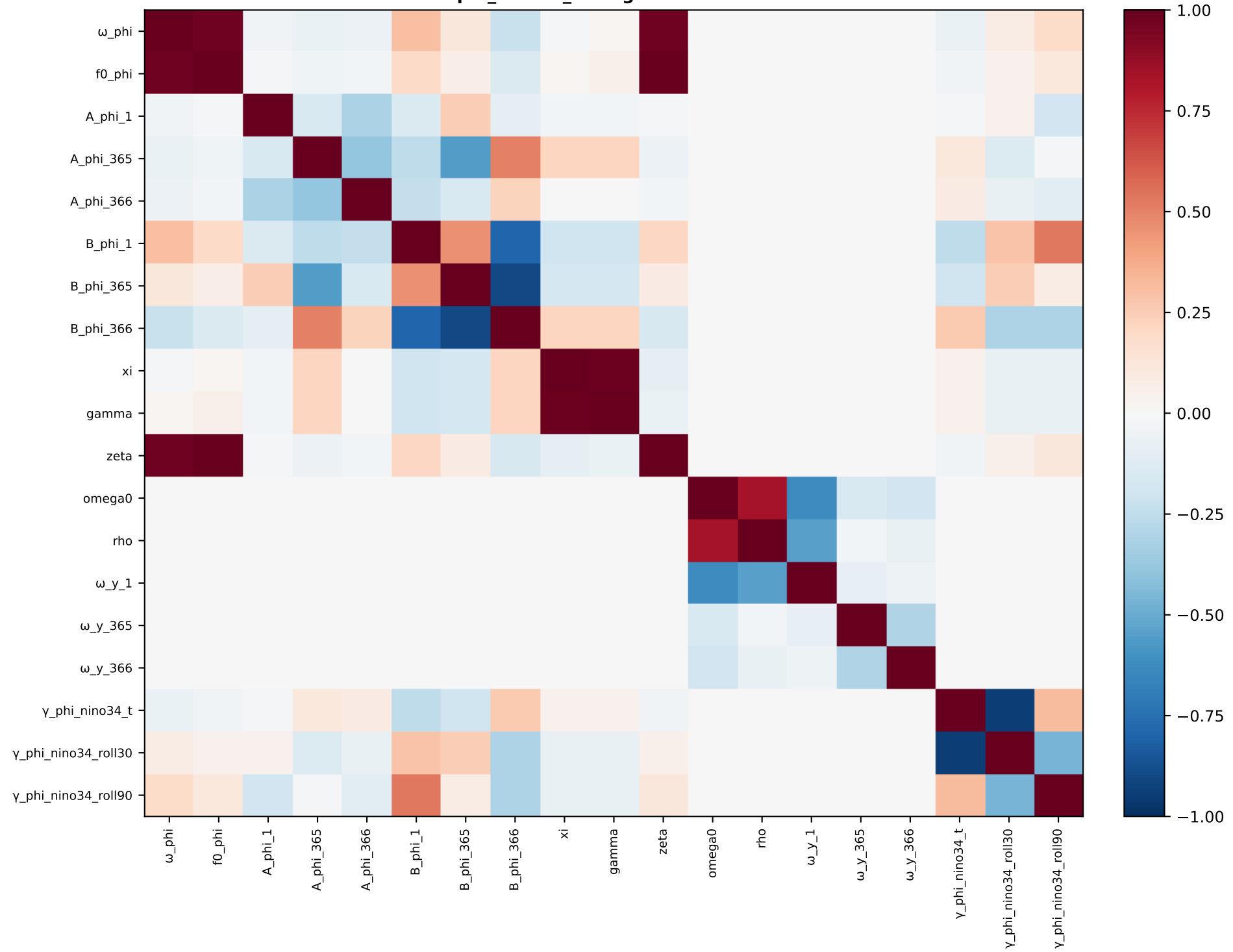
phi_nino34_rolling — filtered states (OOS)



phi_nino34_rolling — covariance

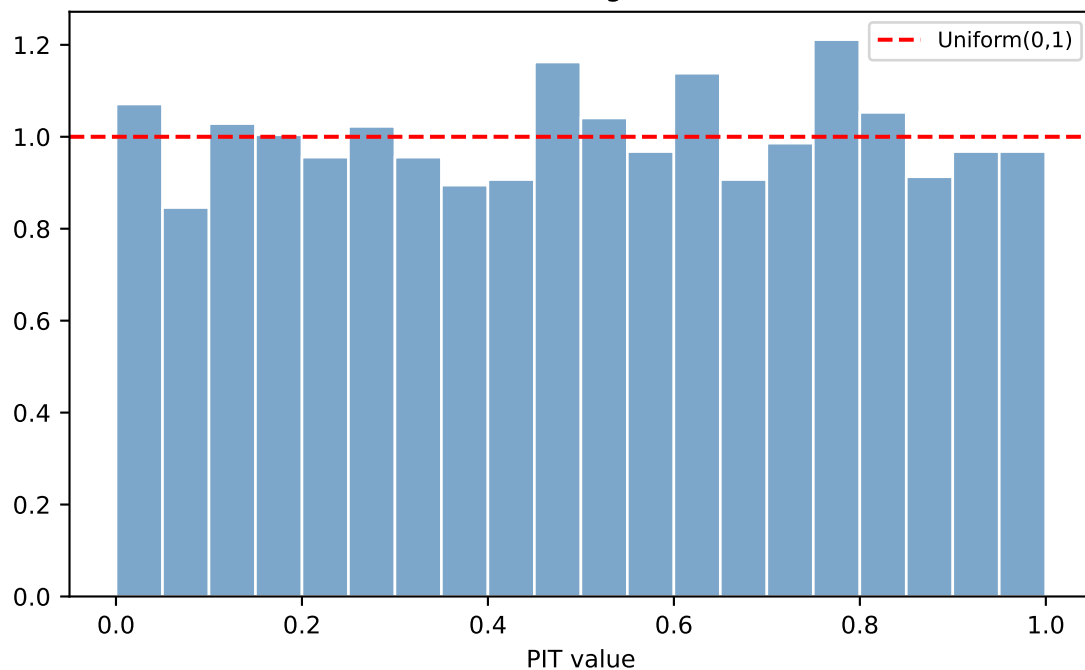


phi_nino34_rolling – correlation

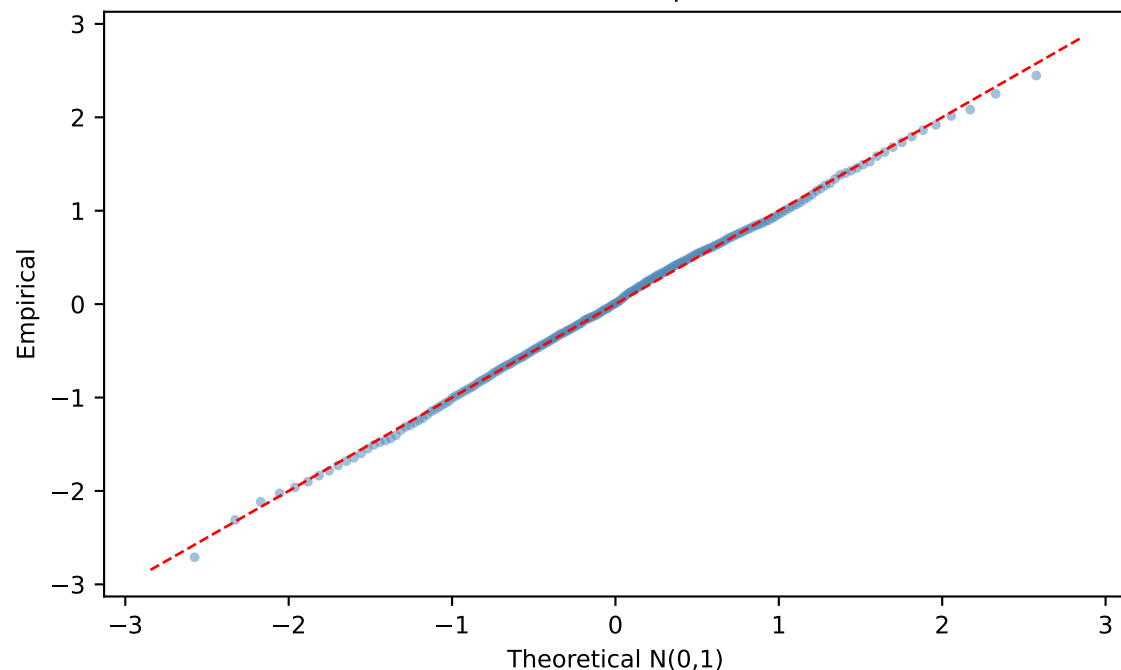


phi_interaction_dewpoint_temperature_t — IS diagnostics

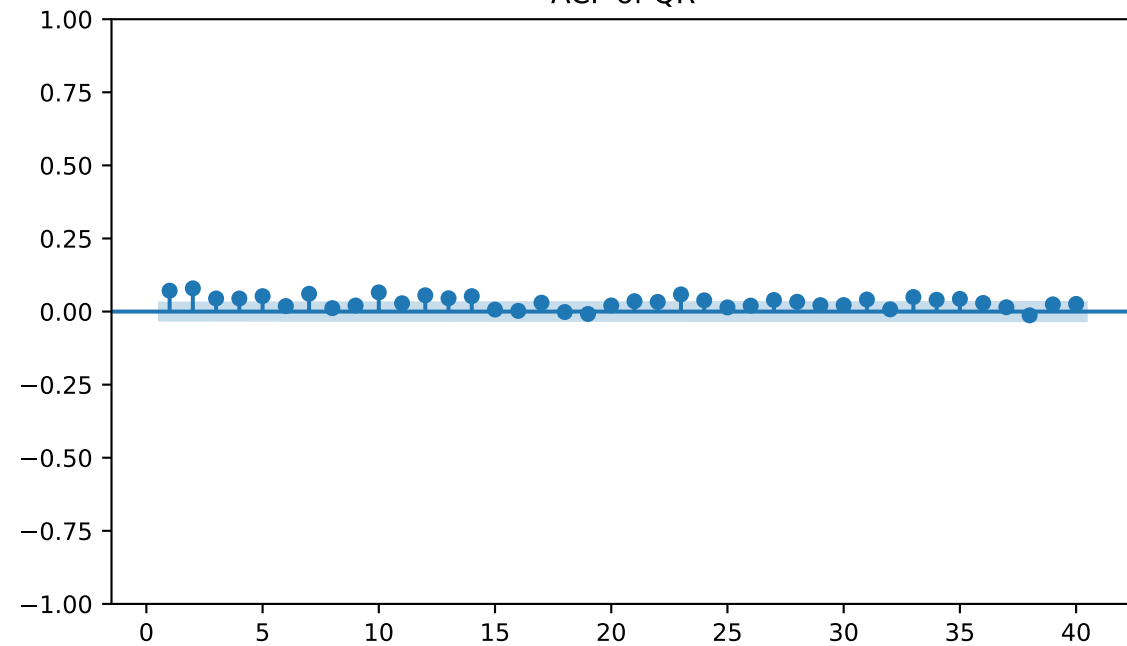
PIT histogram



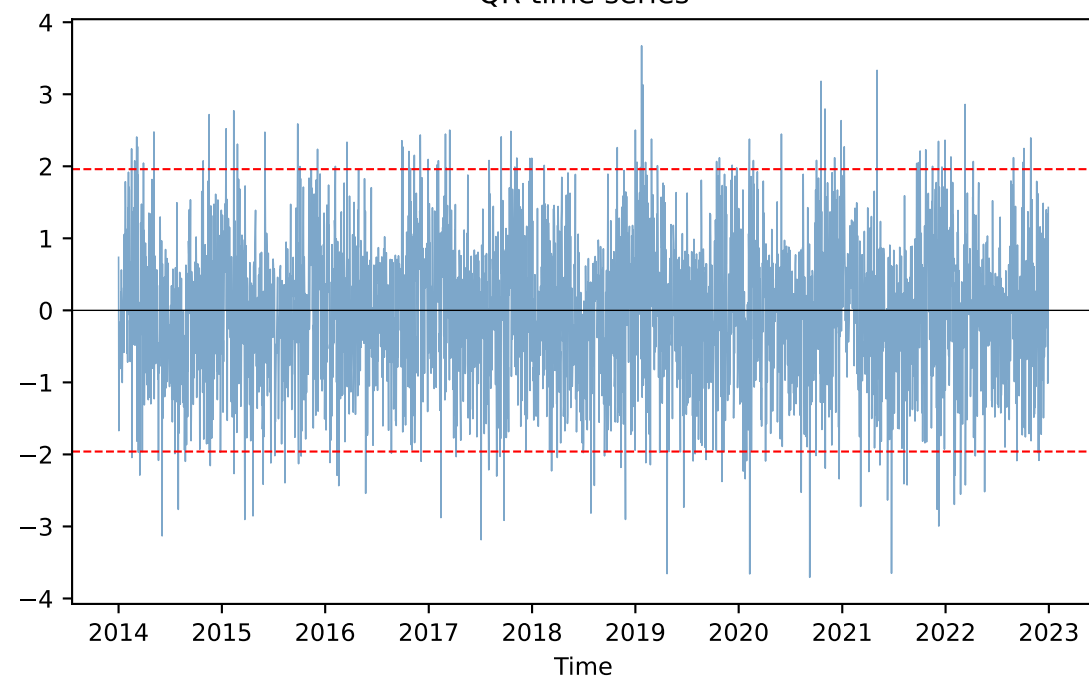
Normal QQ plot



ACF of QR

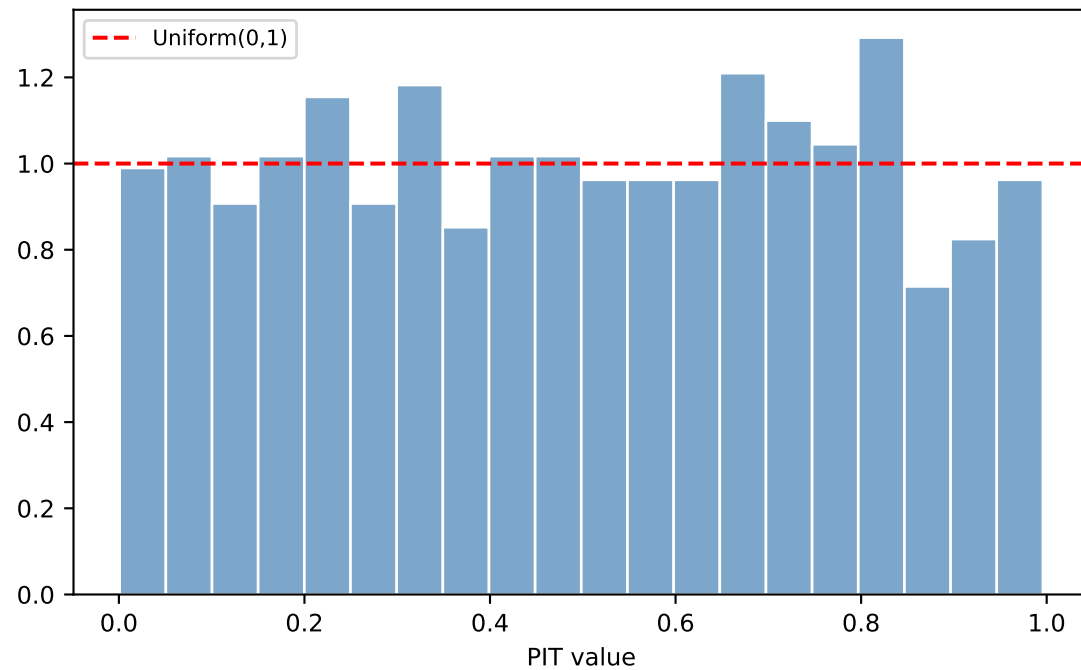


QR time series

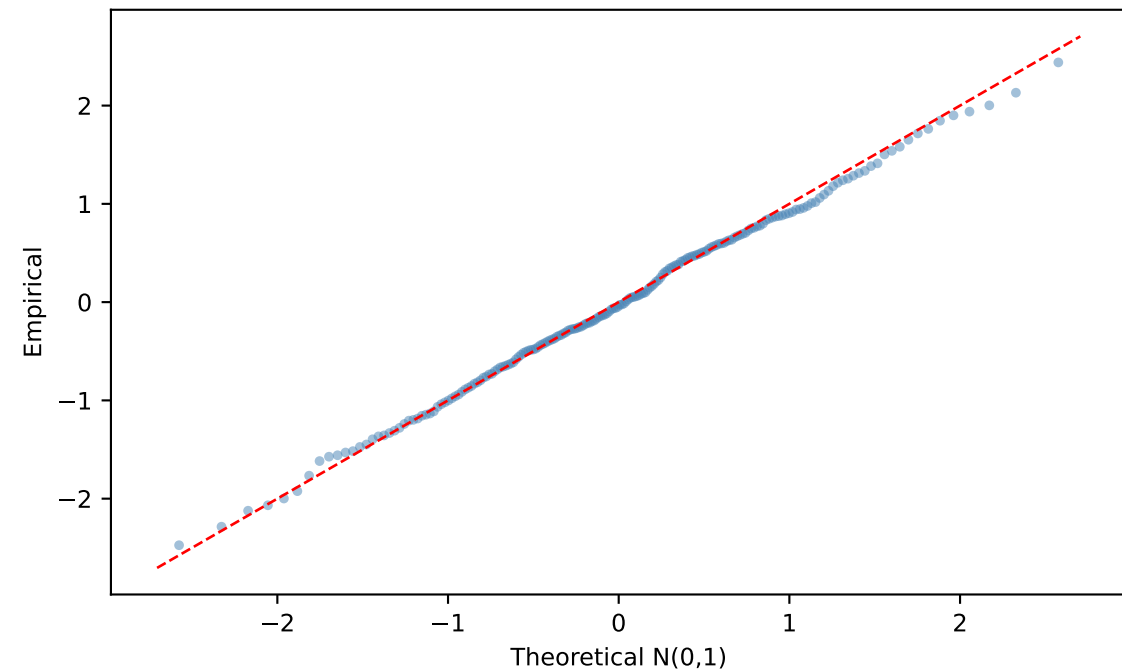


phi_interaction_dewpoint_temperature_t — OOS diagnostics

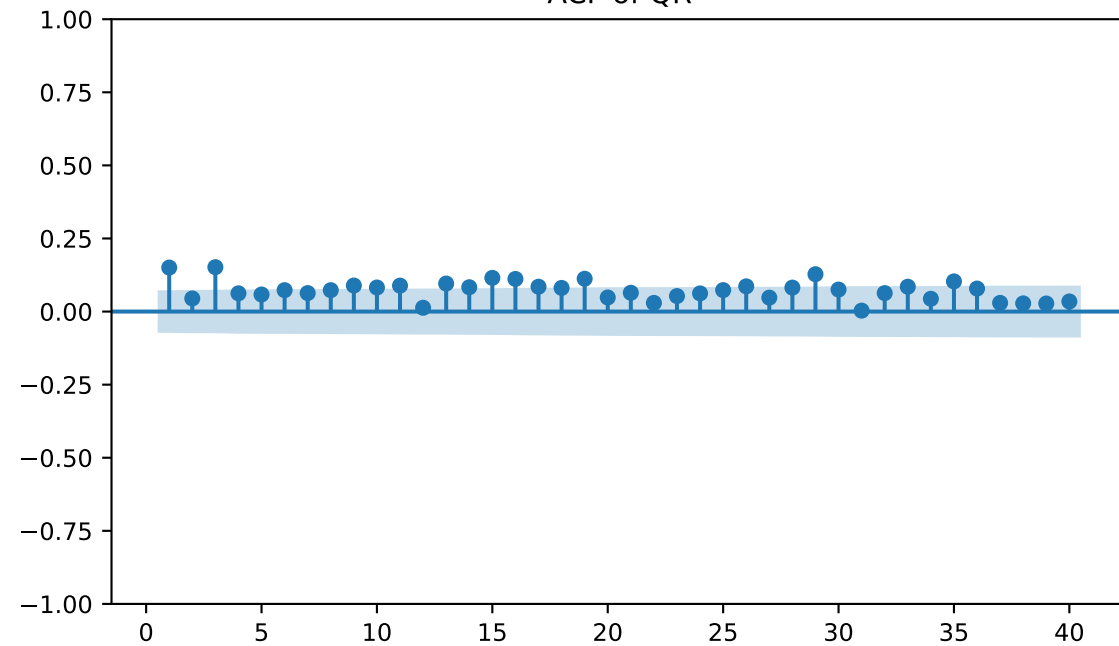
PIT histogram



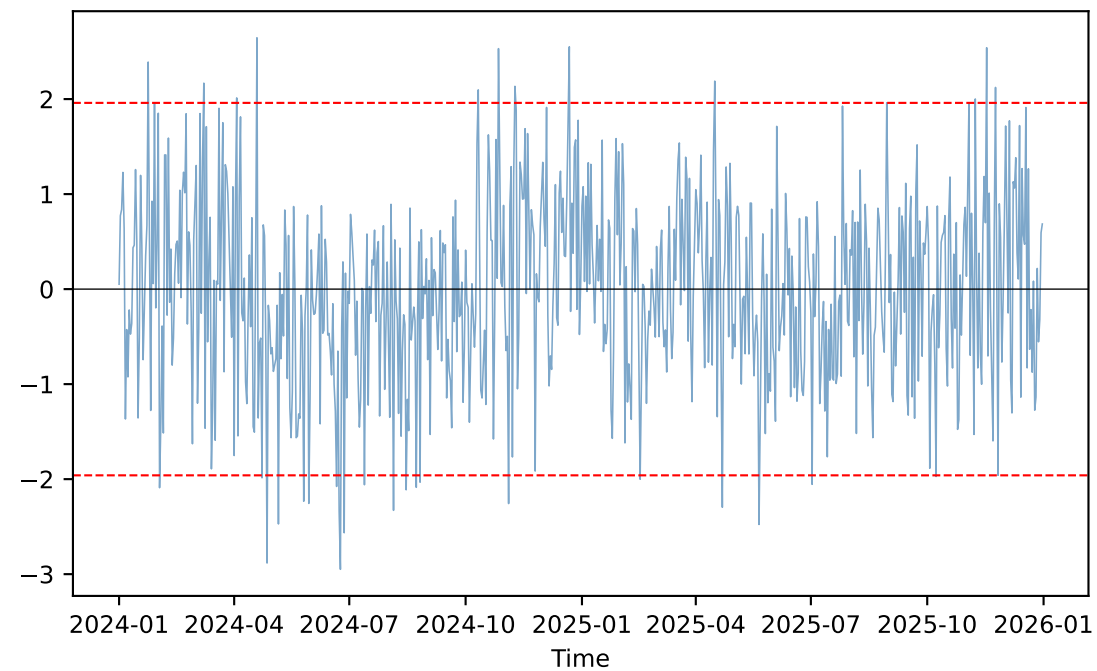
Normal QQ plot



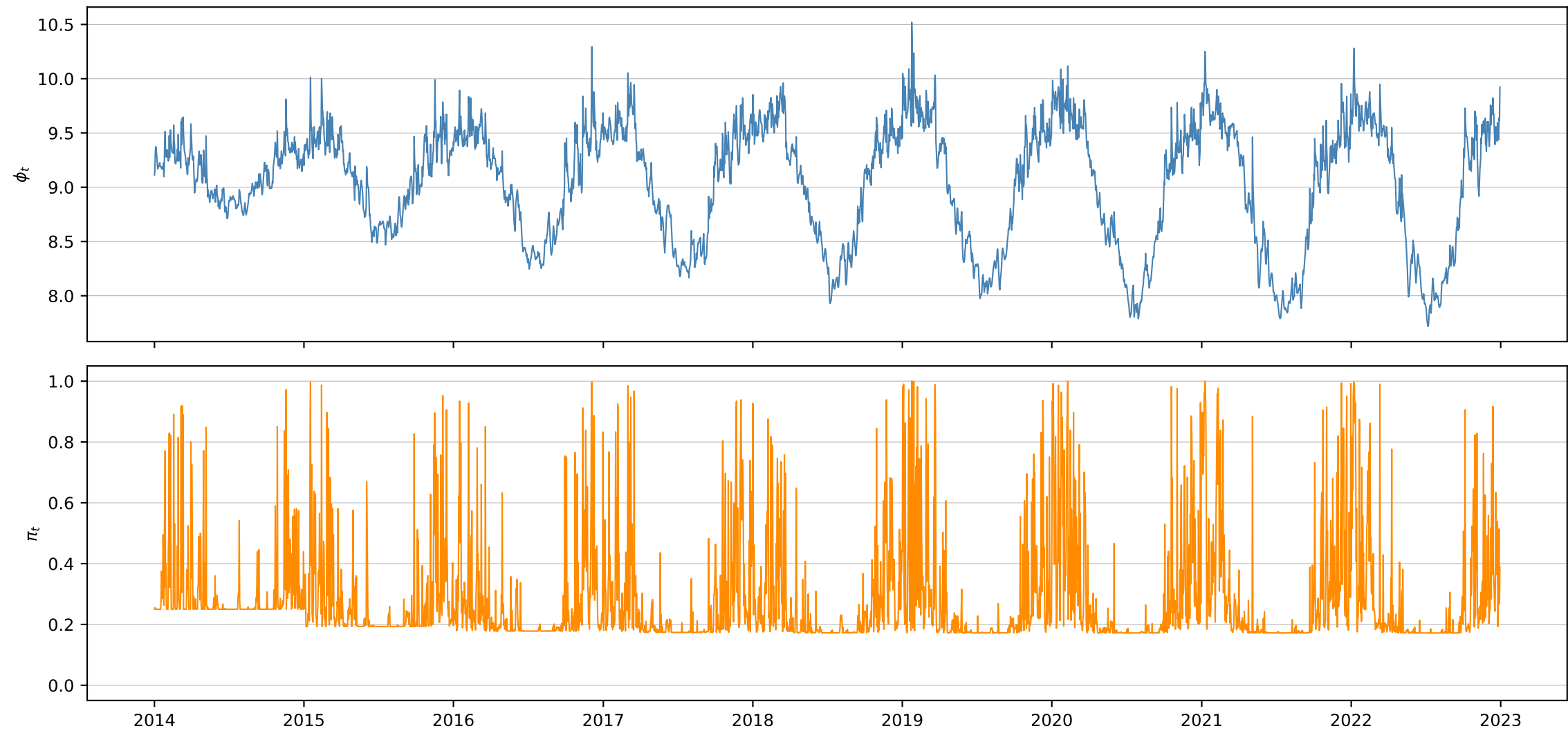
ACF of QR



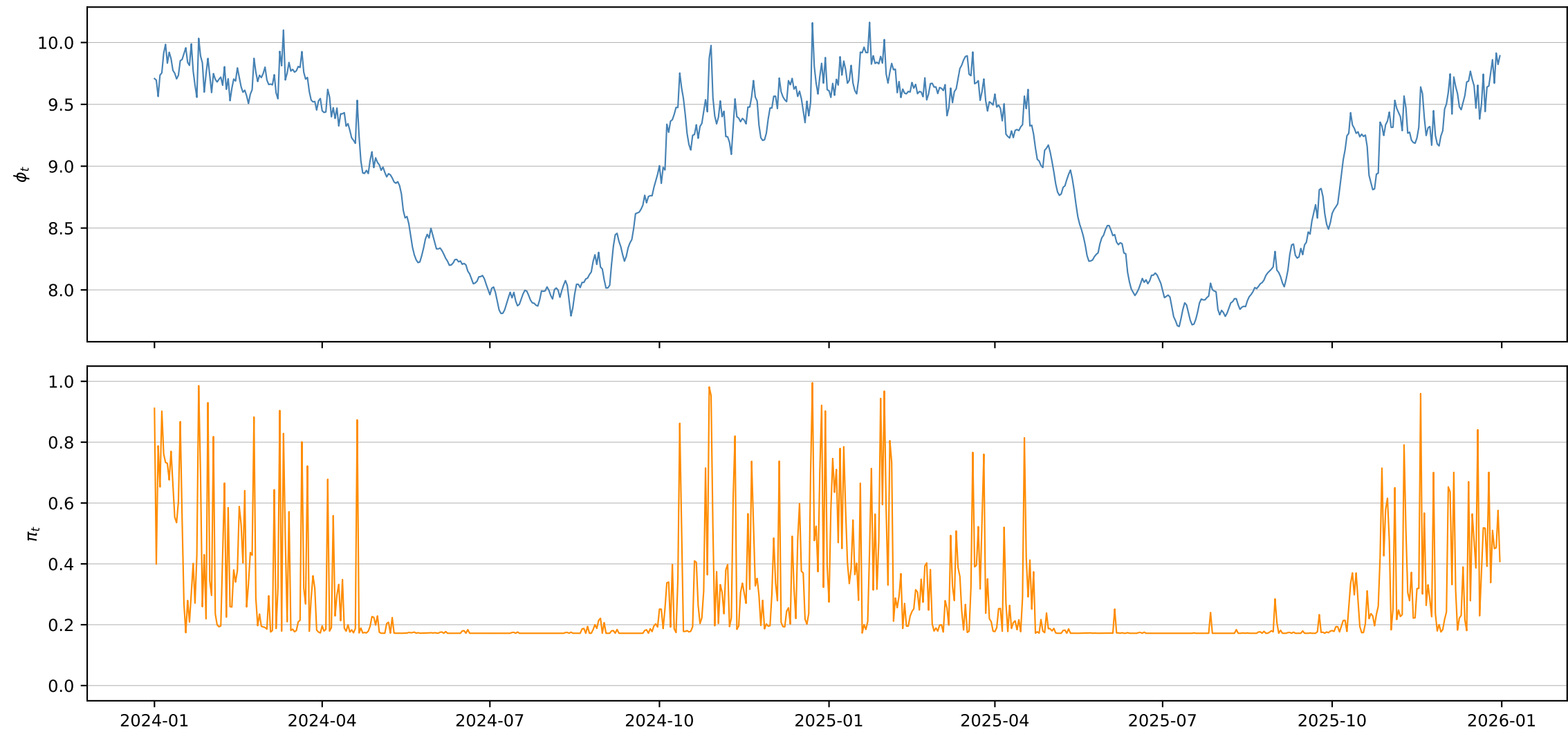
QR time series



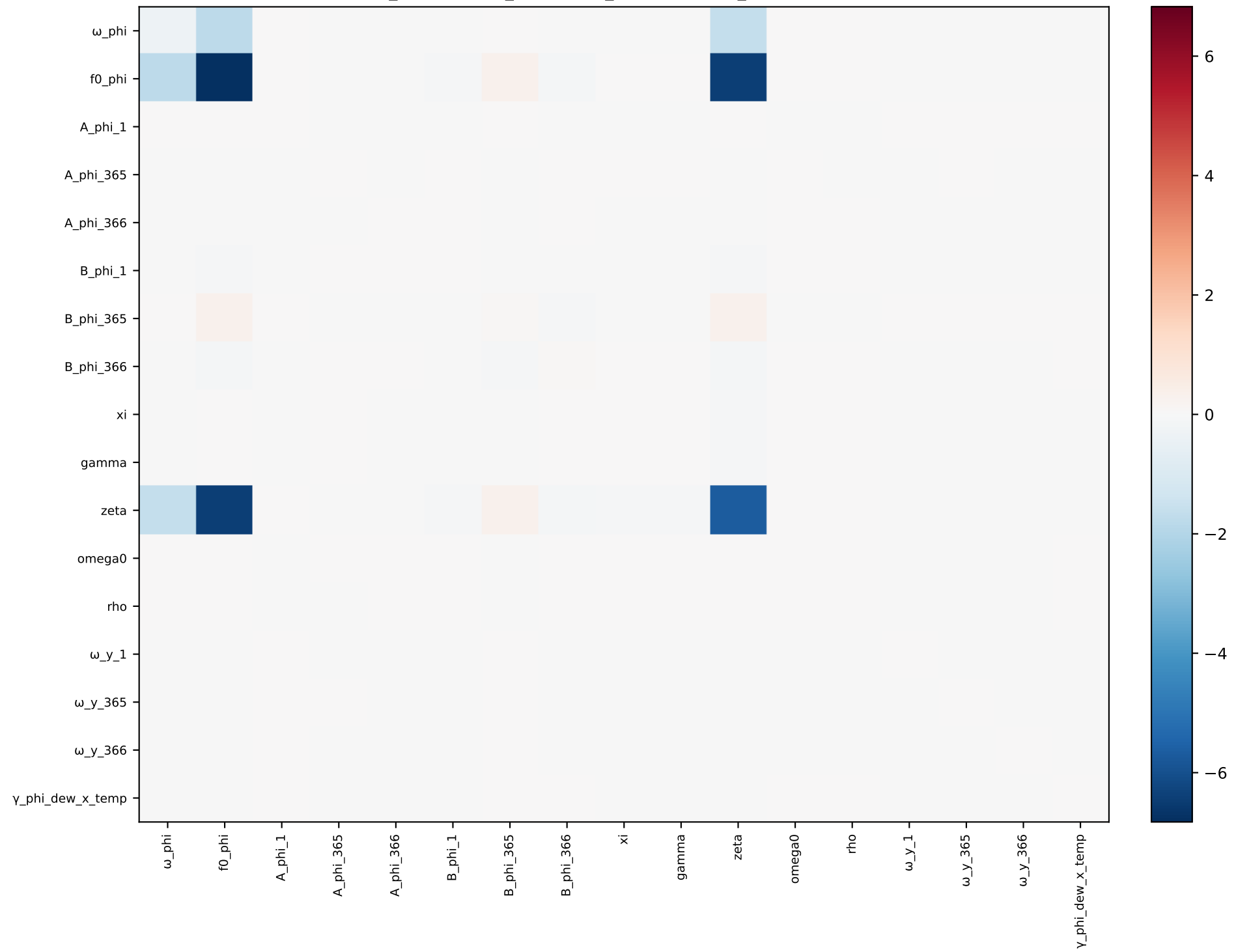
phi_interaction_dewpoint_temperature_t — filtered states (IS)

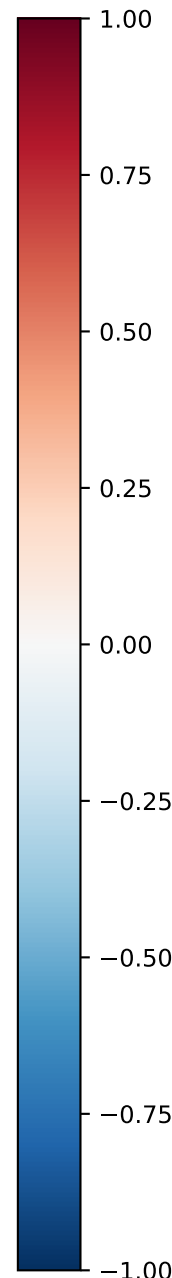


phi_interaction_dewpoint_temperature_t — filtered states (OOS)



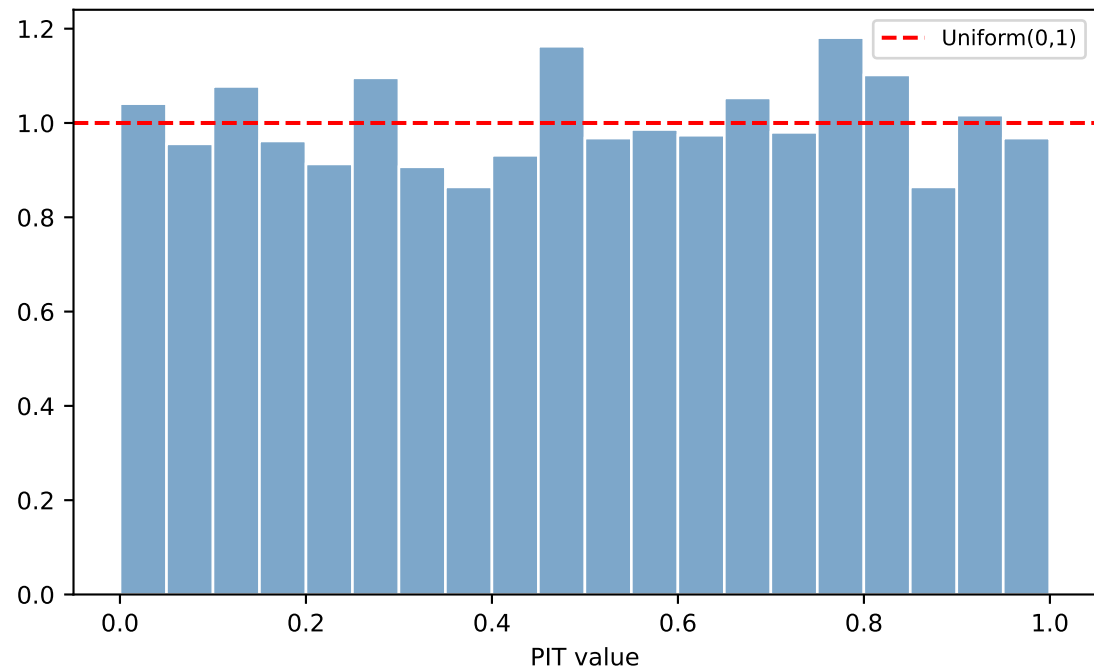
phi_interaction_dewpoint_temperature_t — covariance



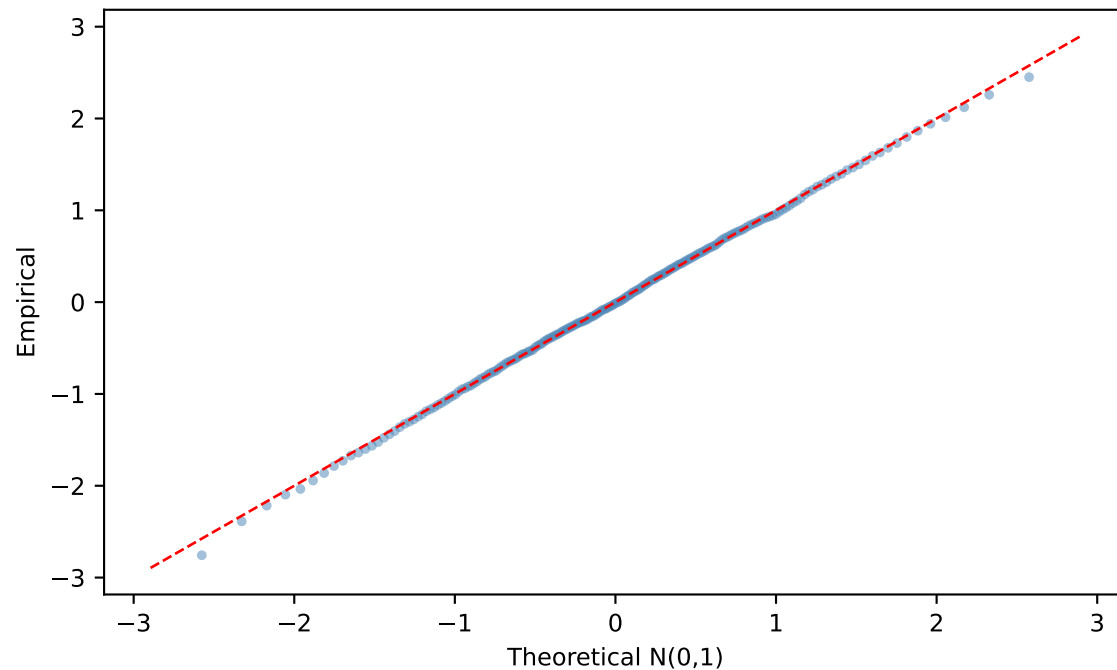


phi_interaction_dewpoint_mjo — IS diagnostics

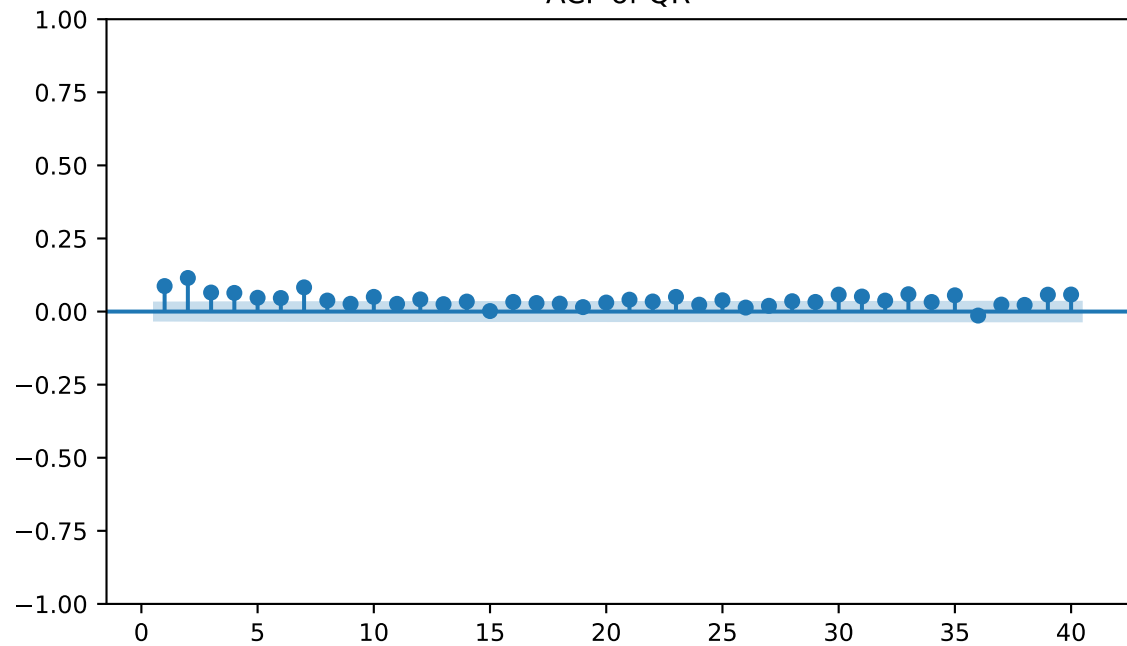
PIT histogram



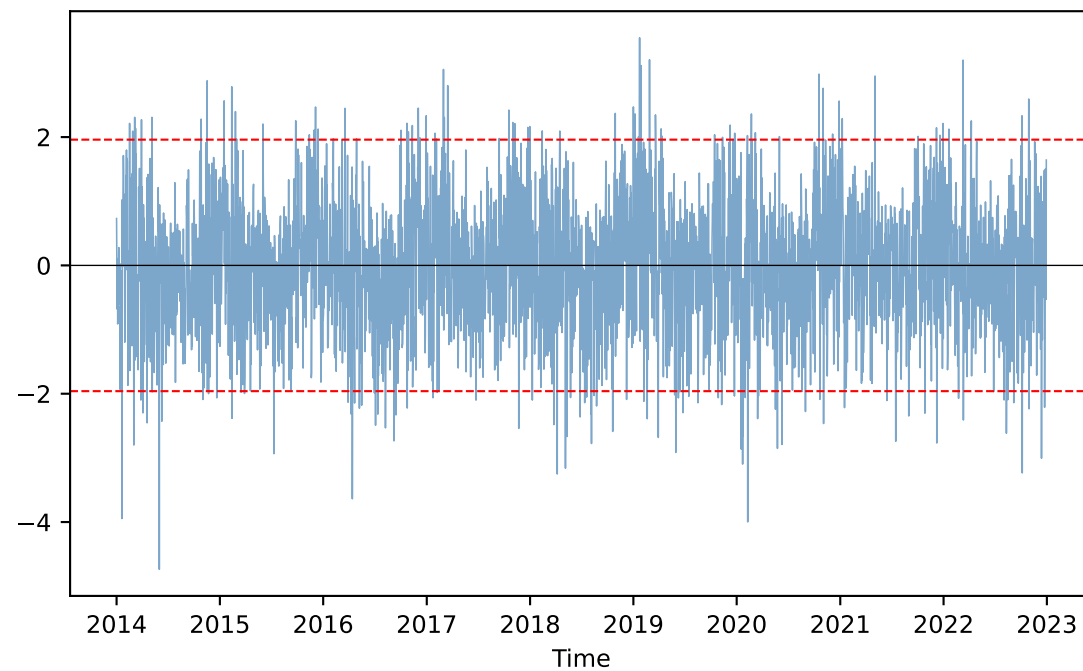
Normal QQ plot



ACF of QR

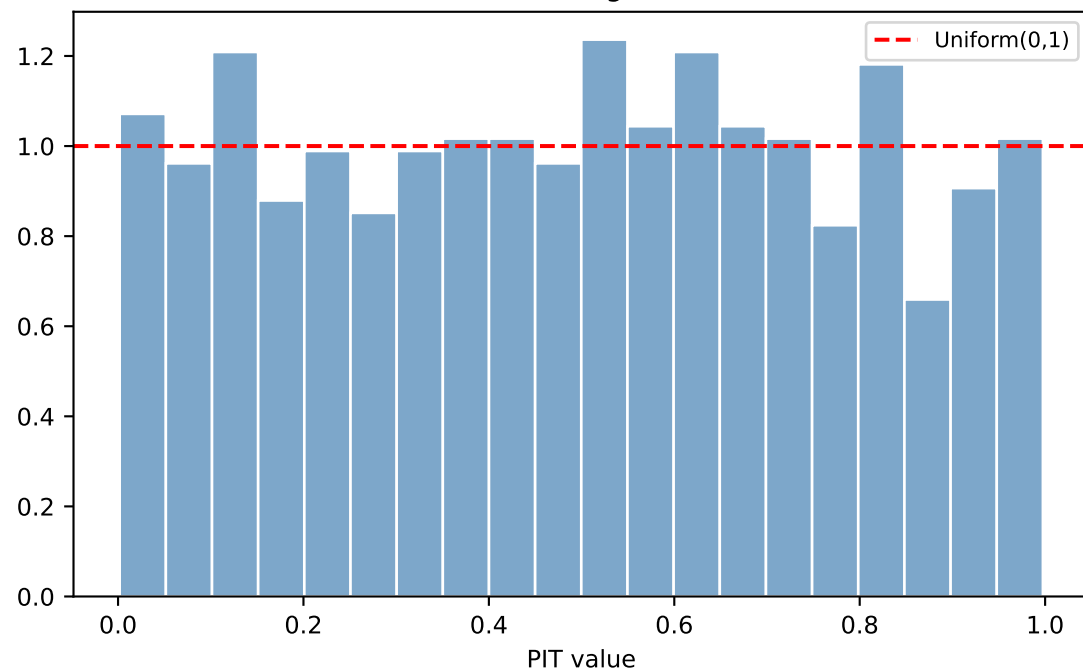


QR time series

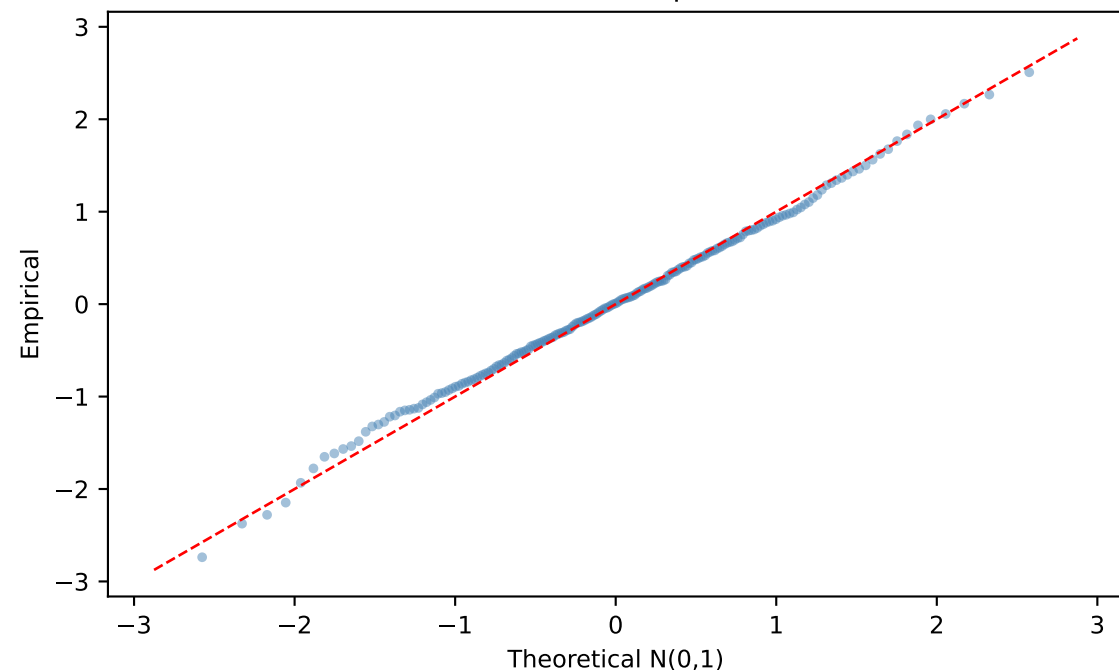


phi_interaction_dewpoint_mjo — OOS diagnostics

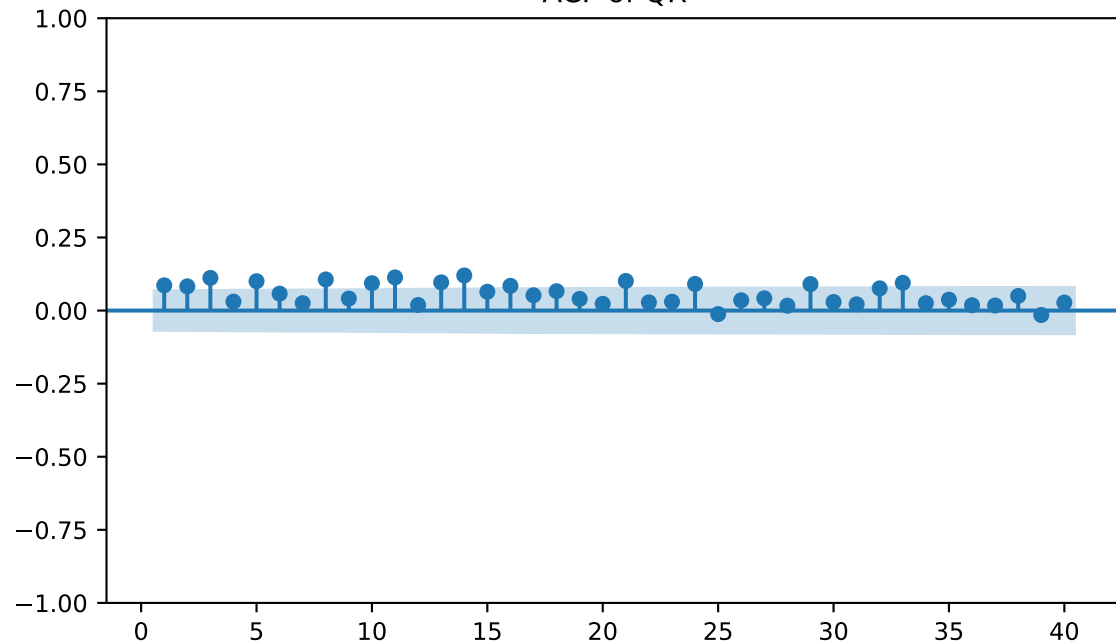
PIT histogram



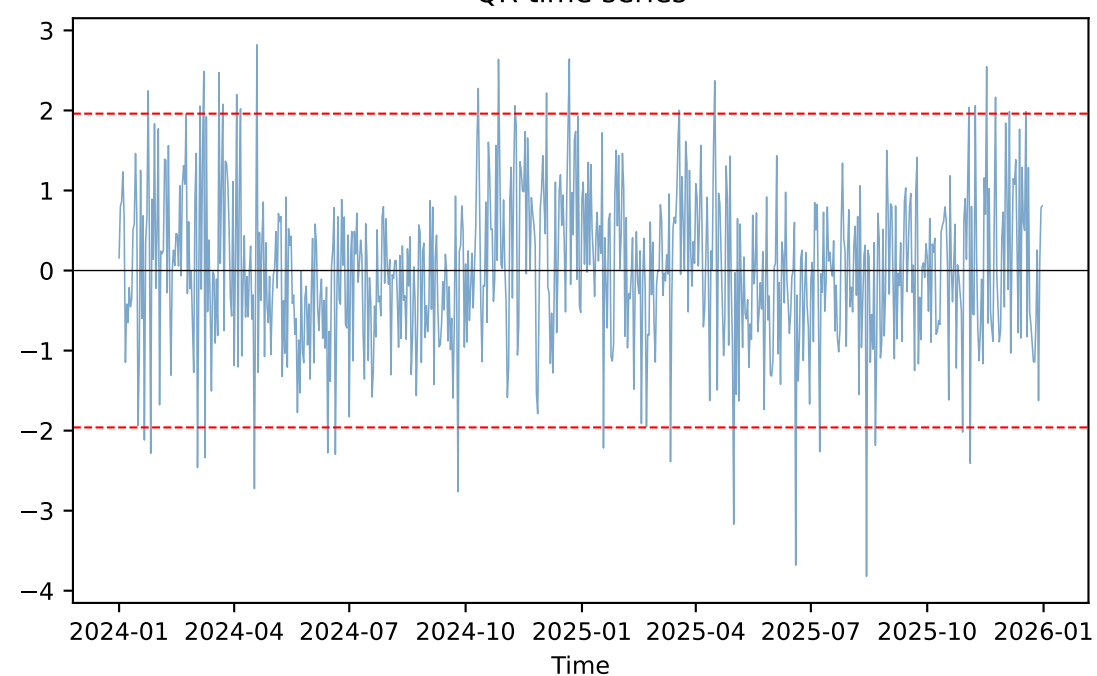
Normal QQ plot



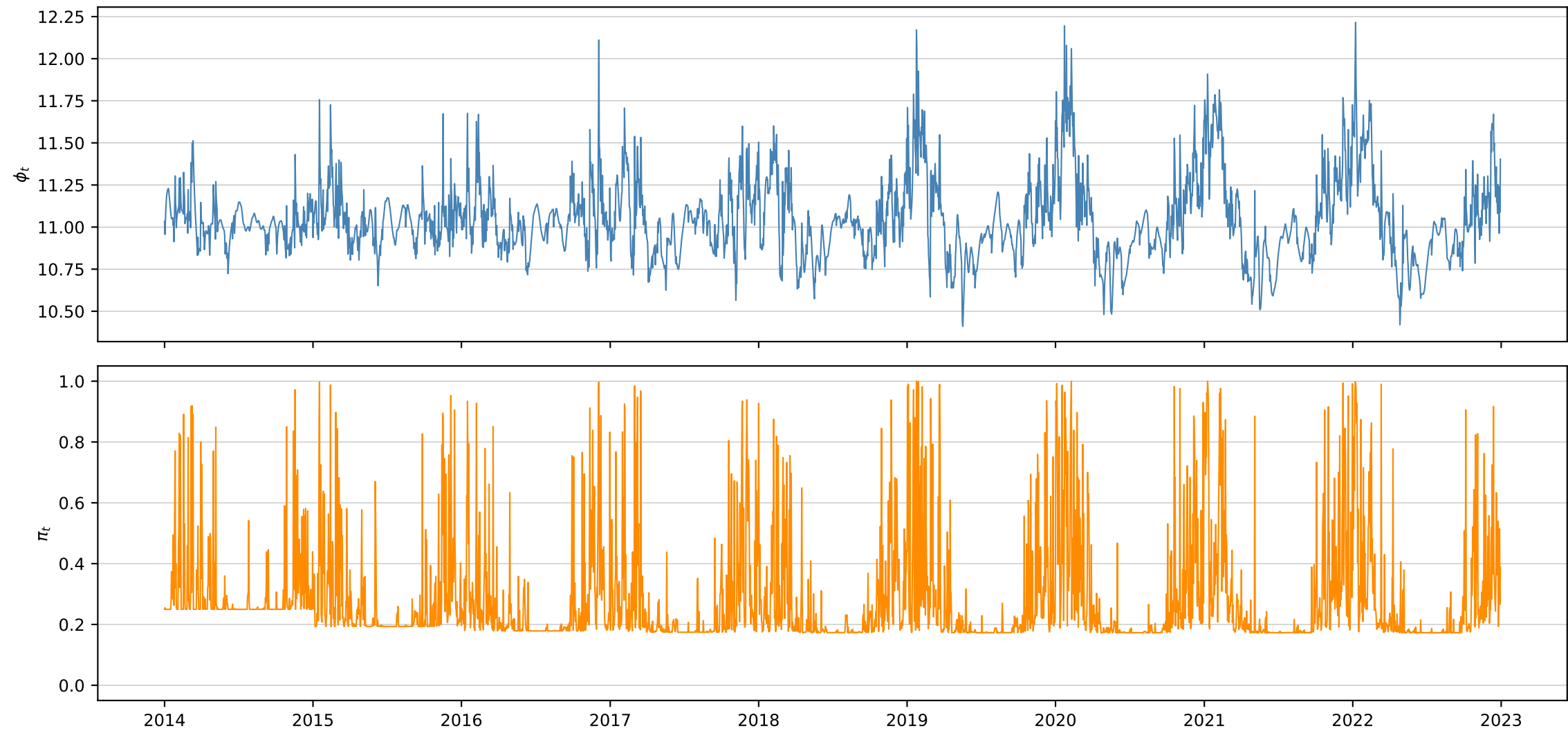
ACF of QR



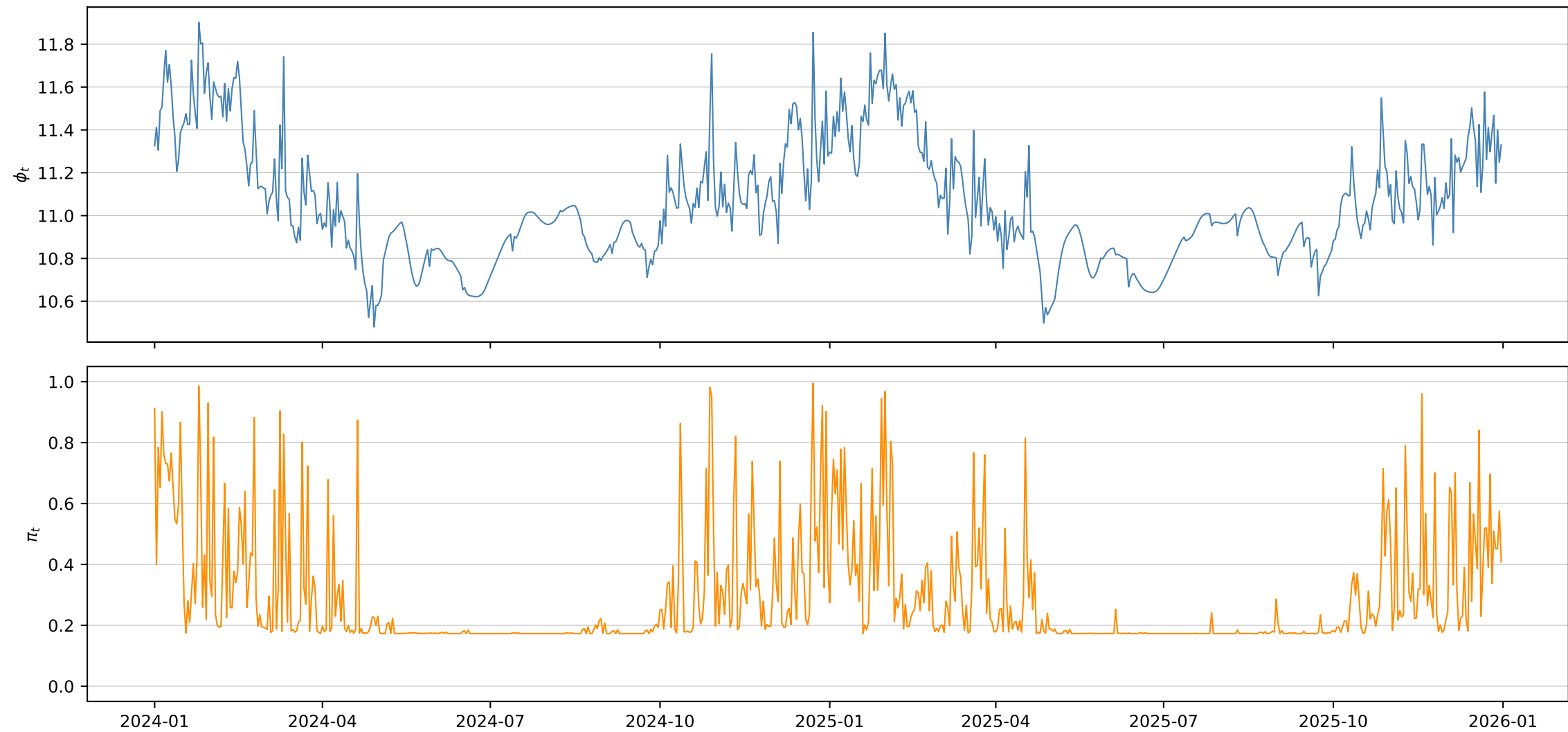
QR time series



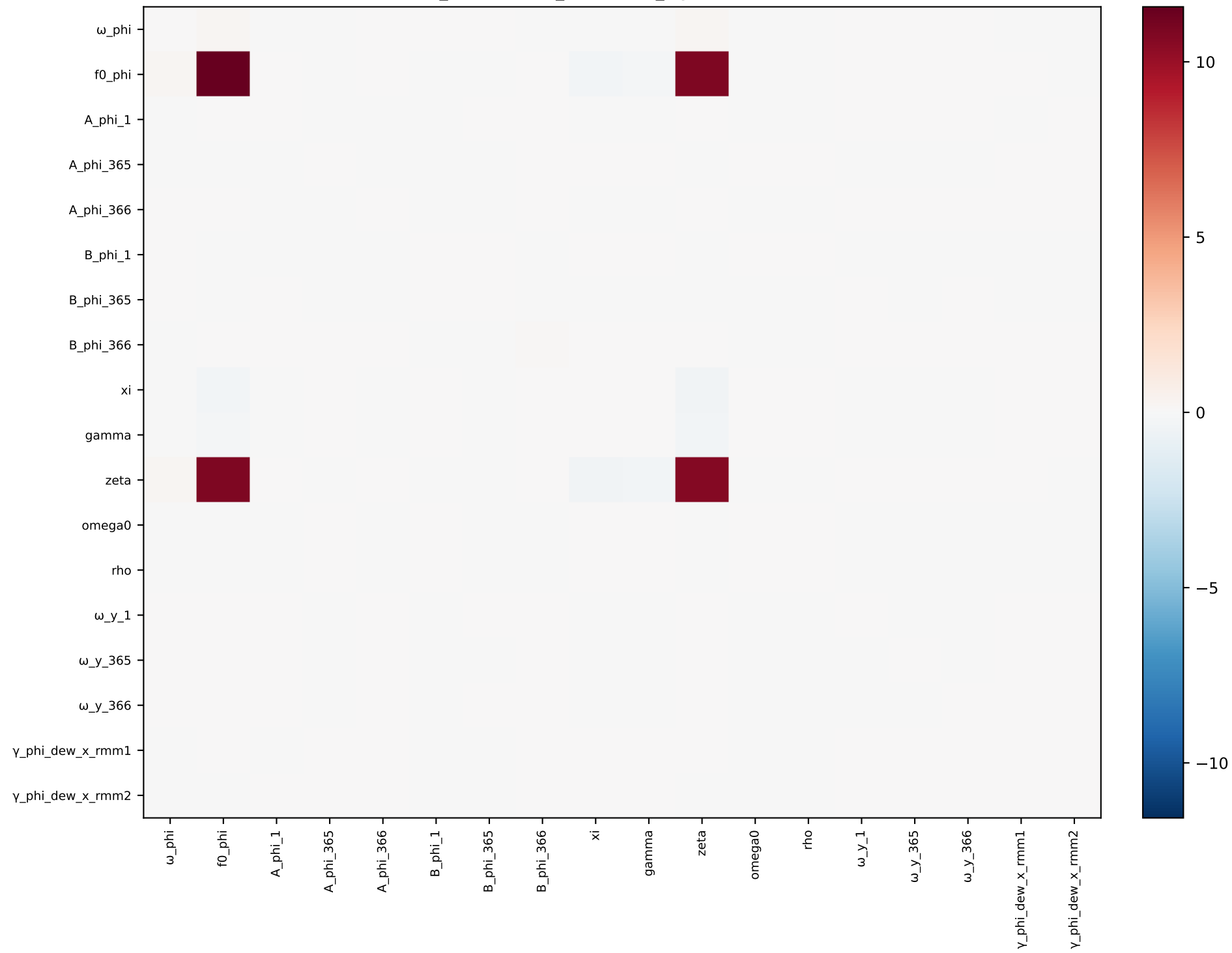
phi_interaction_dewpoint_mjo — filtered states (IS)



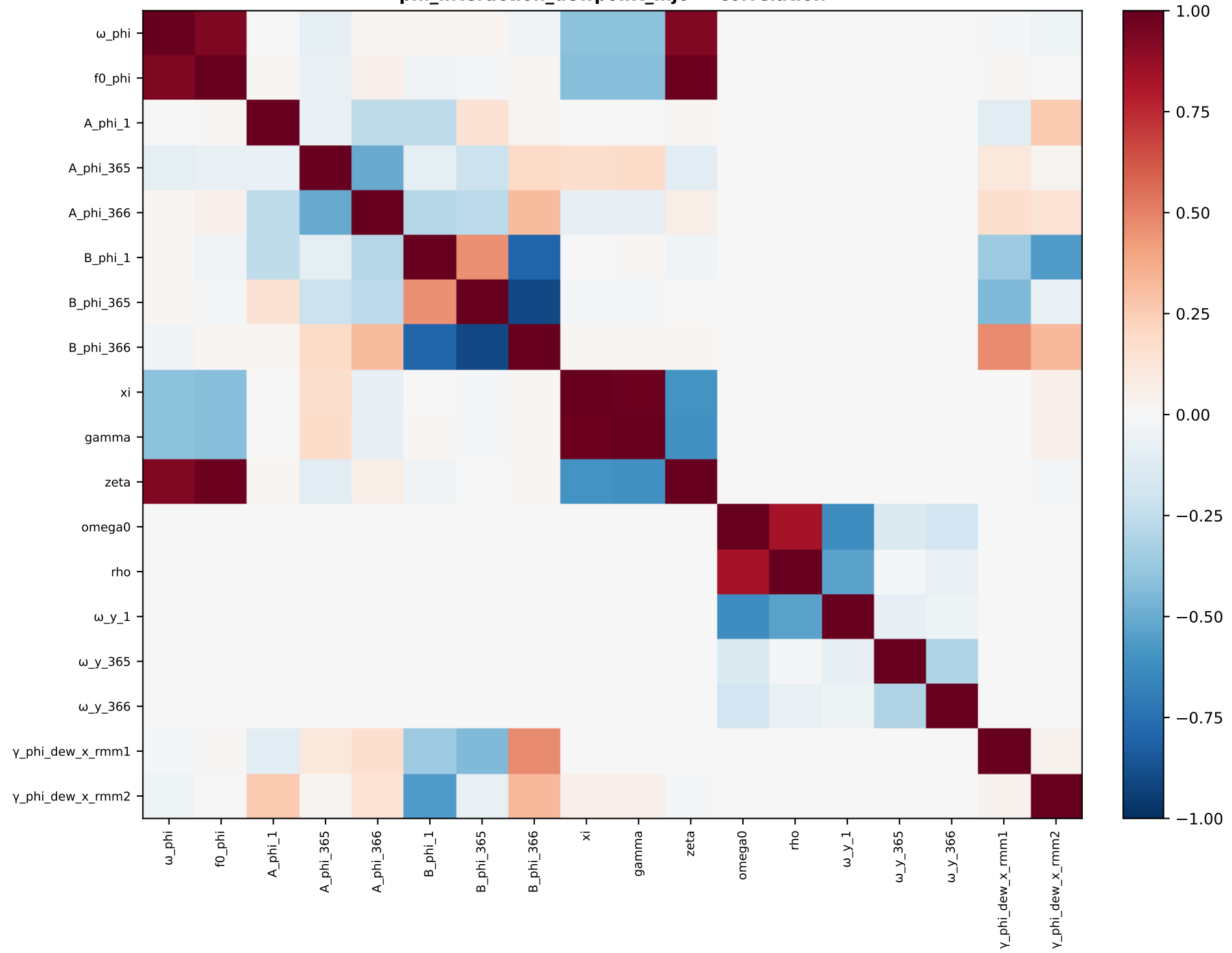
phi_interaction_dewpoint_mjo — filtered states (OOS)



phi_interaction_dewpoint_mjo — covariance

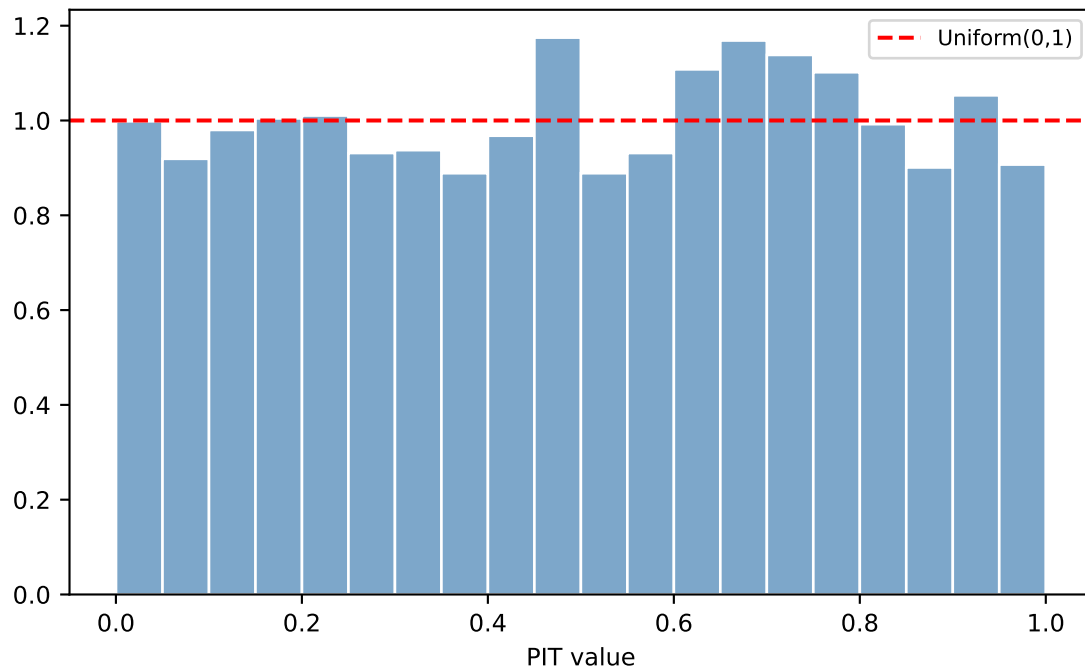


phi_interaction_dewpoint_mjo — correlation

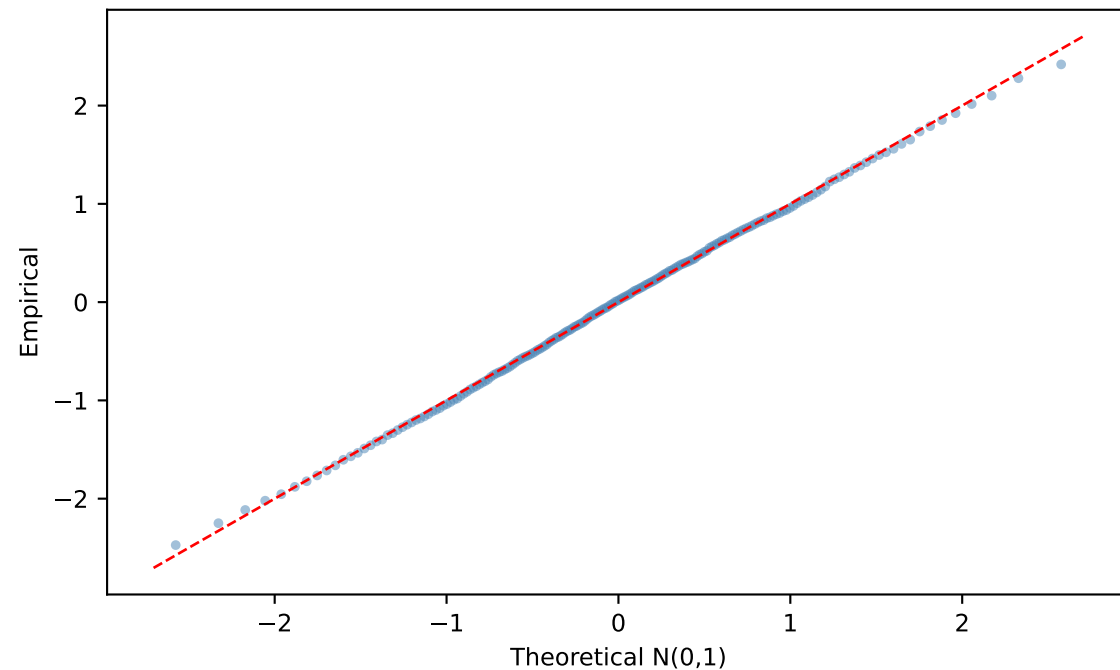


phi_interaction_dewpoint_nino34 — IS diagnostics

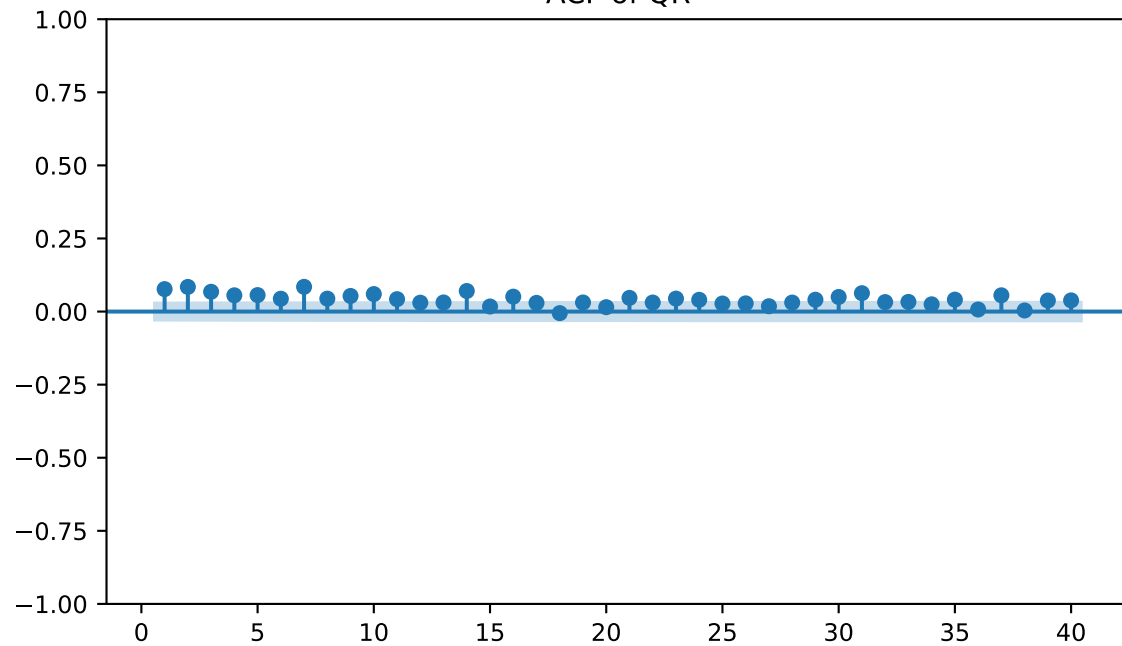
PIT histogram



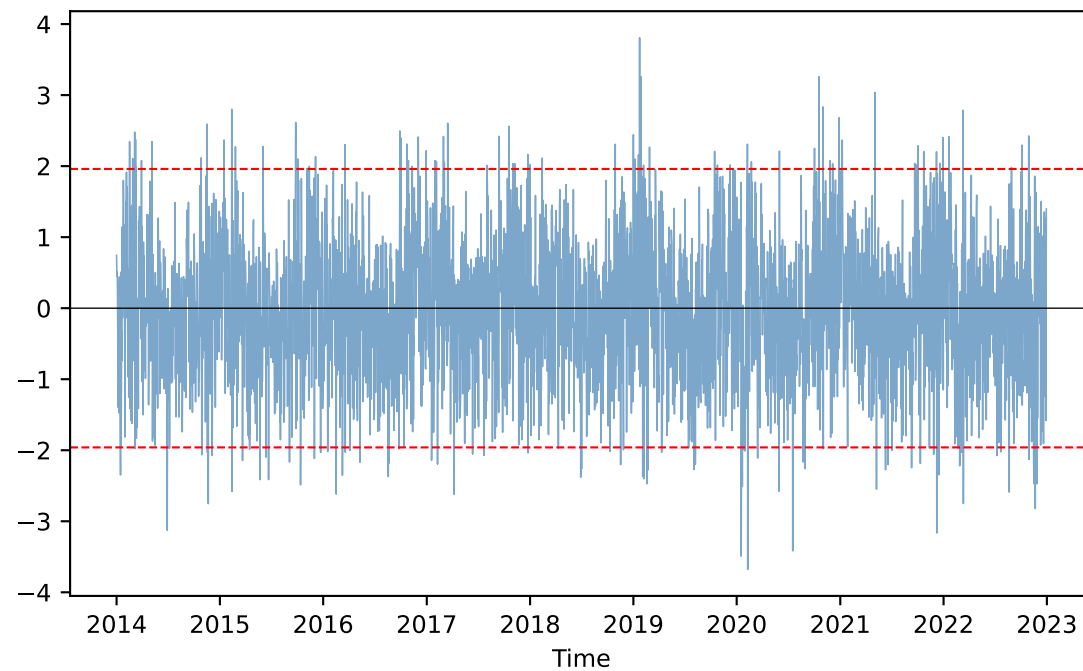
Normal QQ plot



ACF of QR

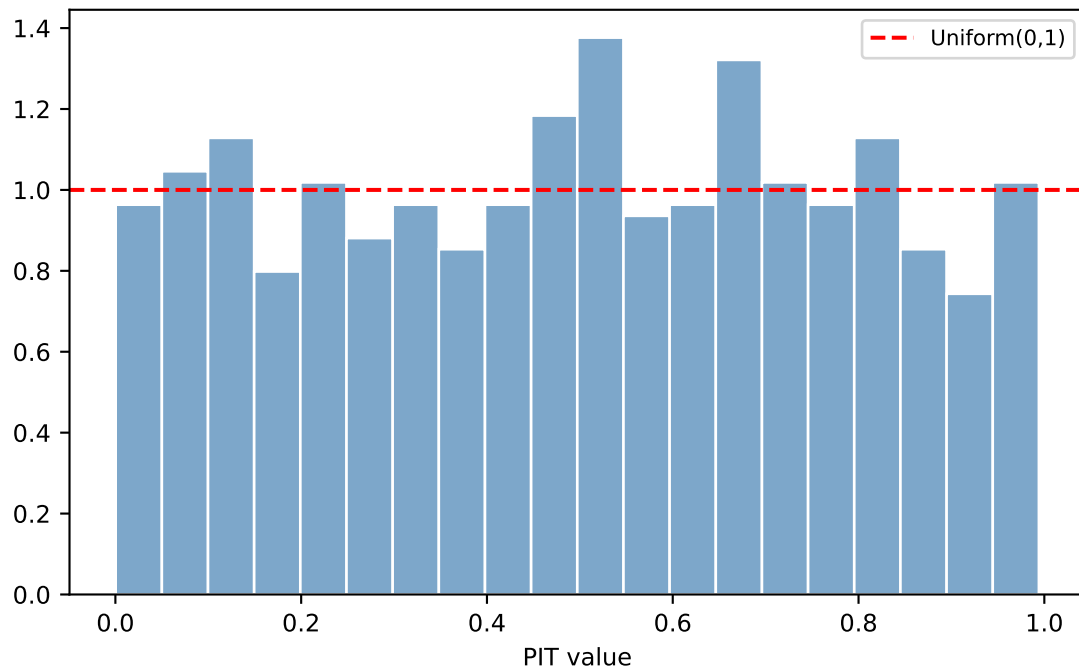


QR time series

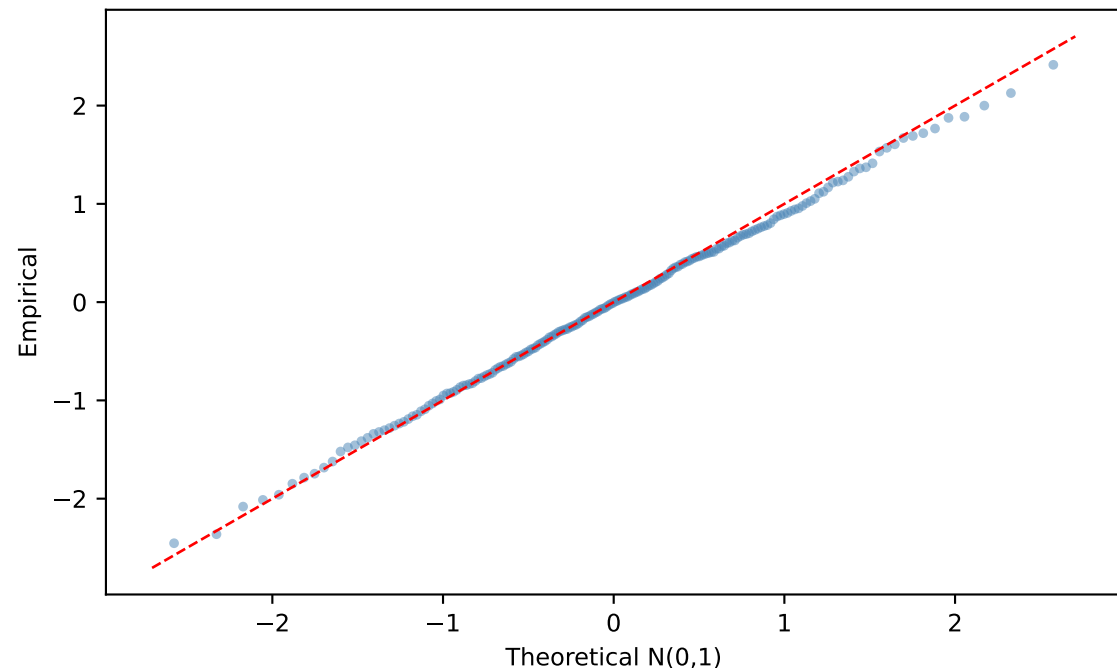


phi_interaction_dewpoint_nino34 — OOS diagnostics

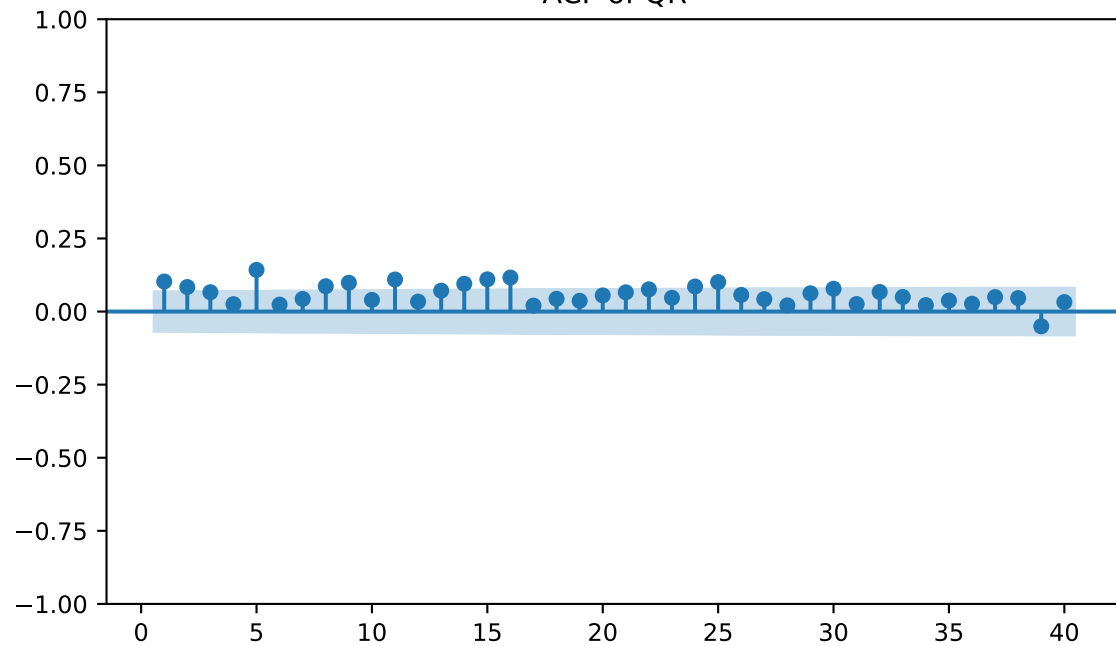
PIT histogram



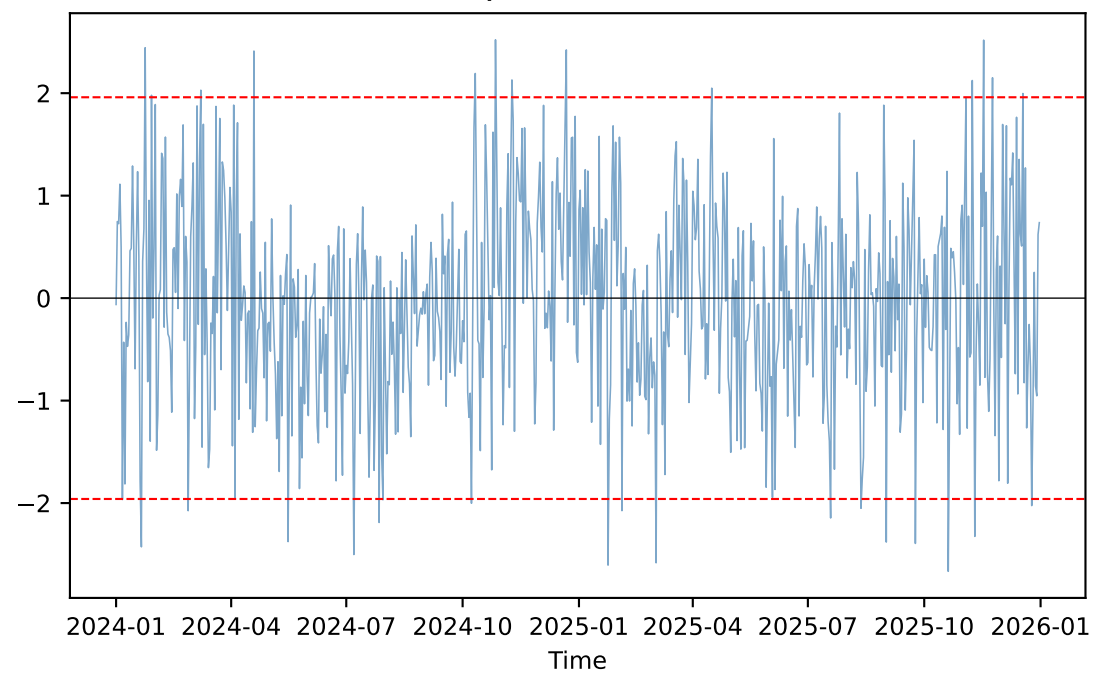
Normal QQ plot



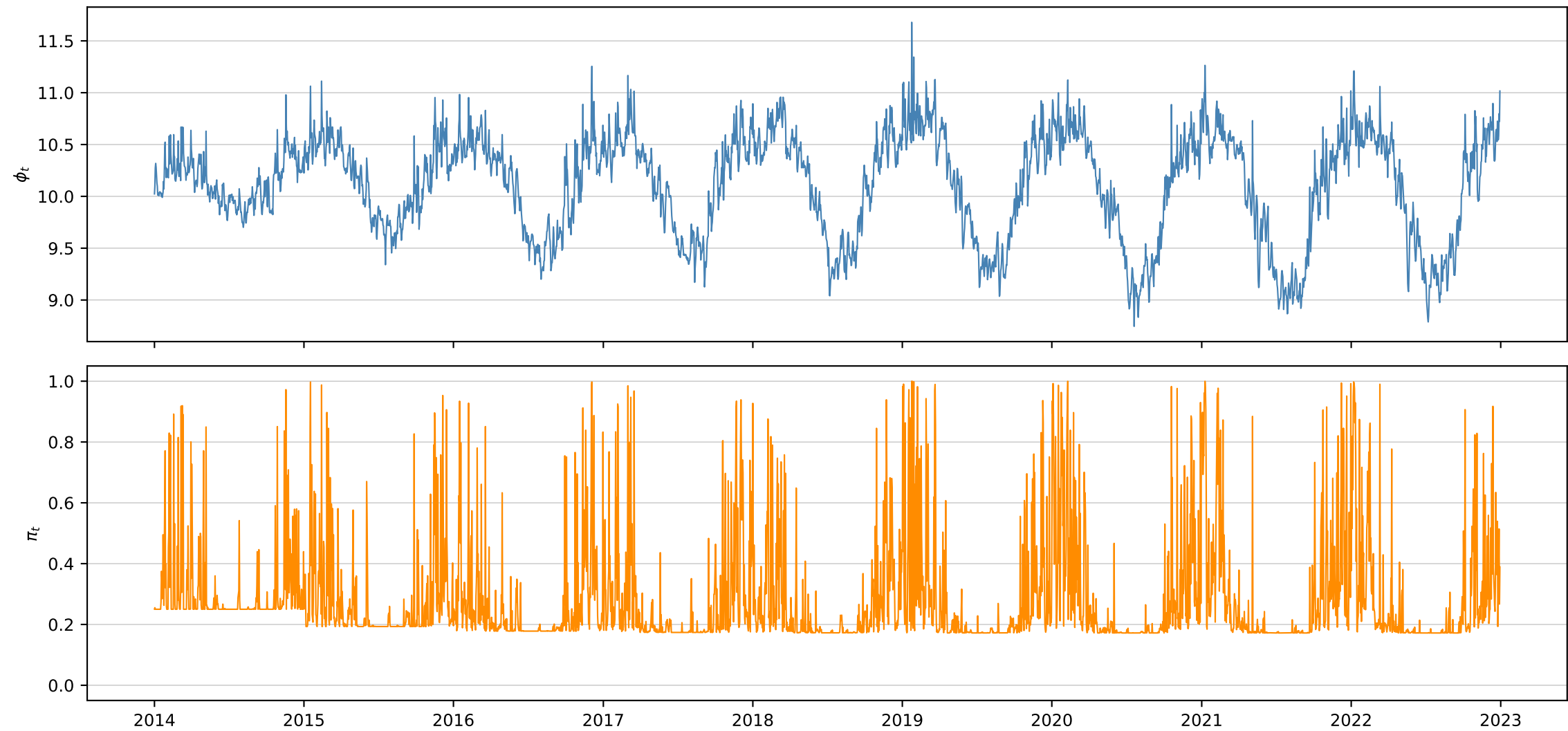
ACF of QR



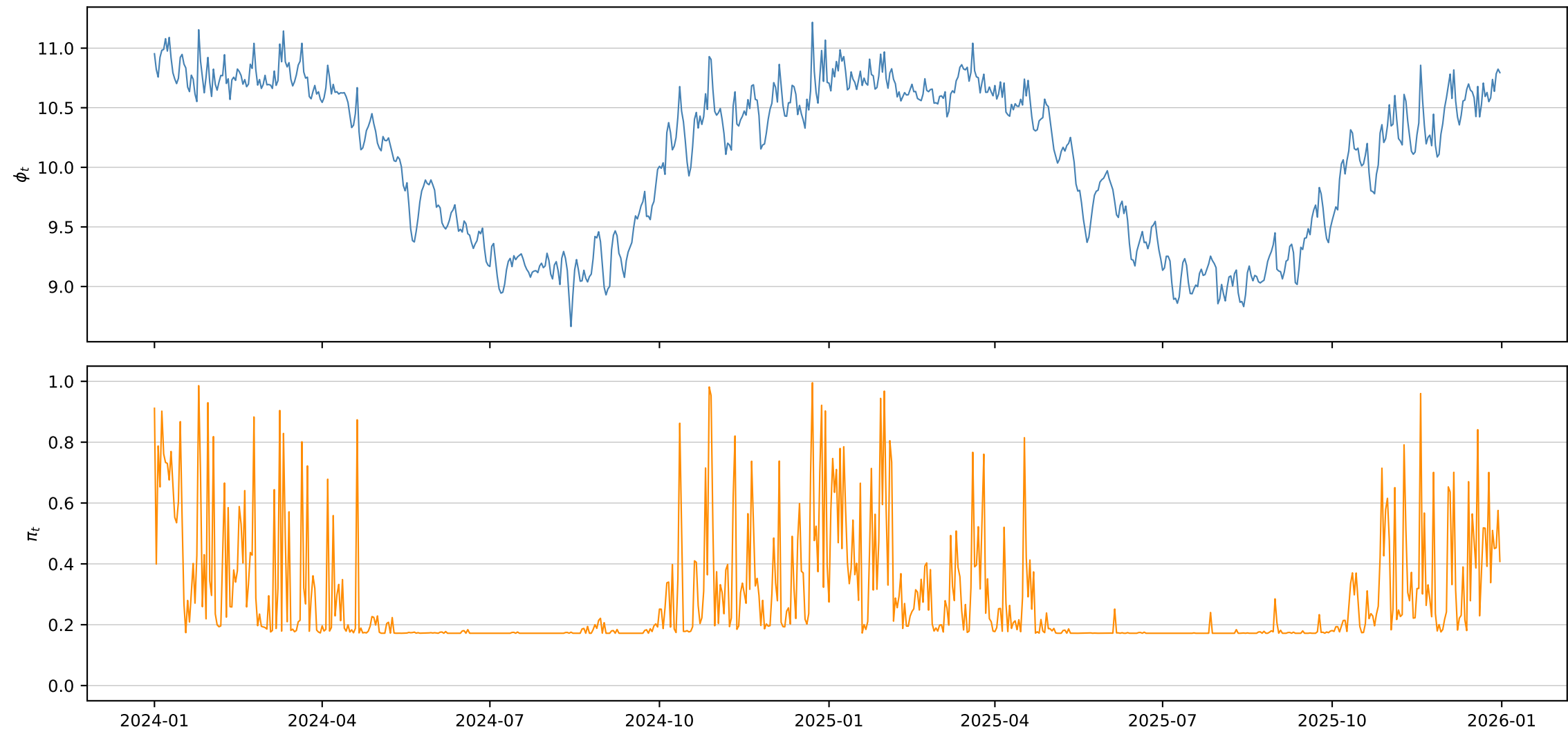
QR time series



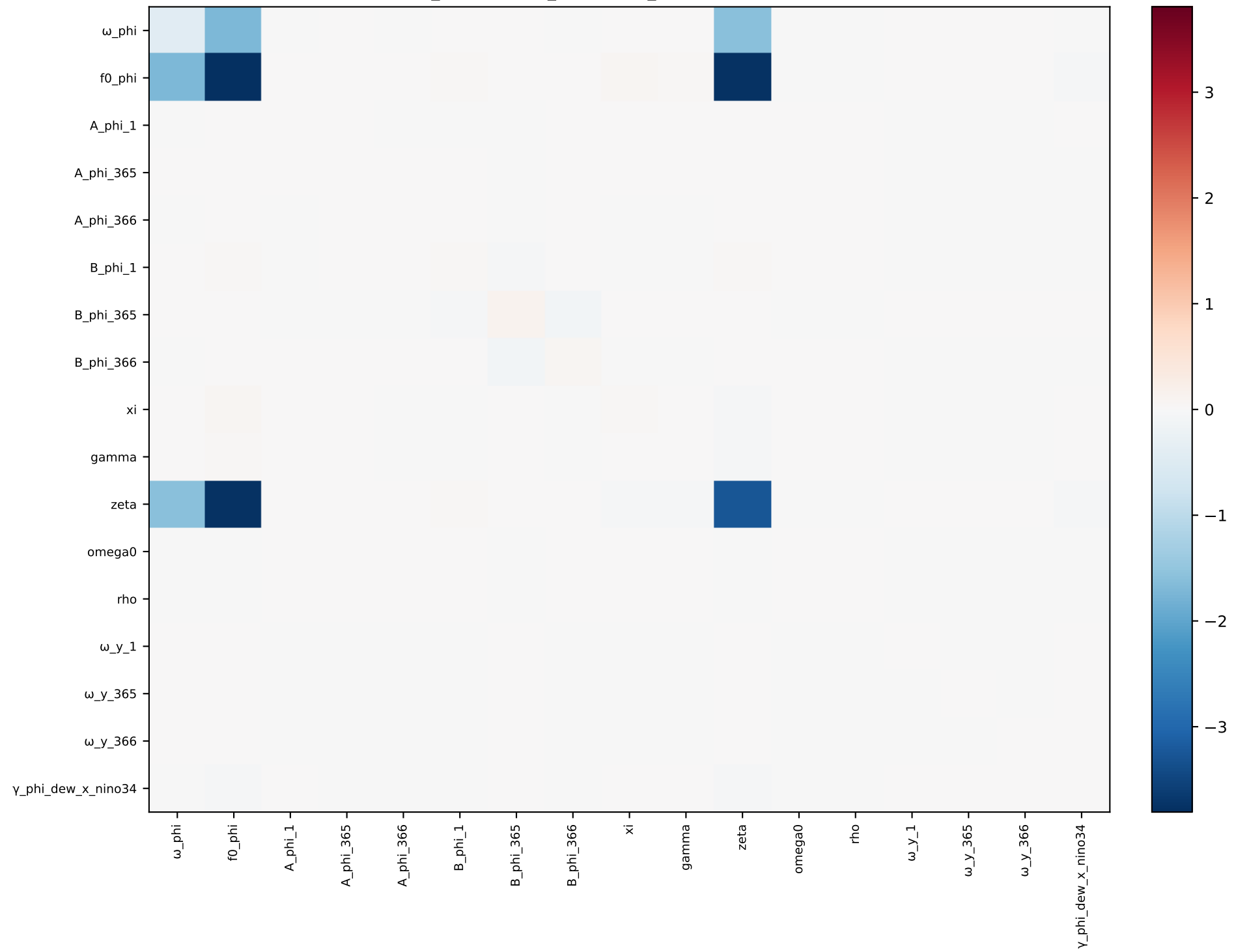
phi_interaction_dewpoint_nino34 — filtered states (IS)

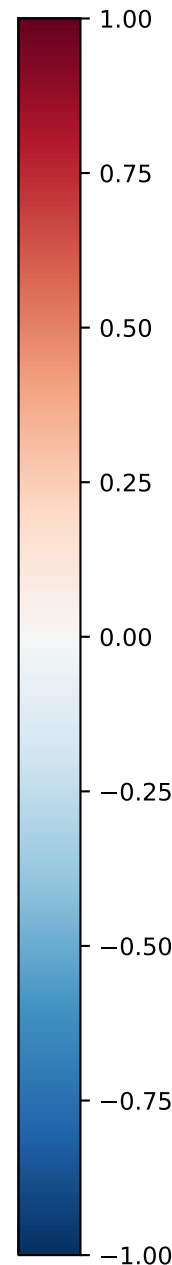


phi_interaction_dewpoint_nino34 — filtered states (OOS)



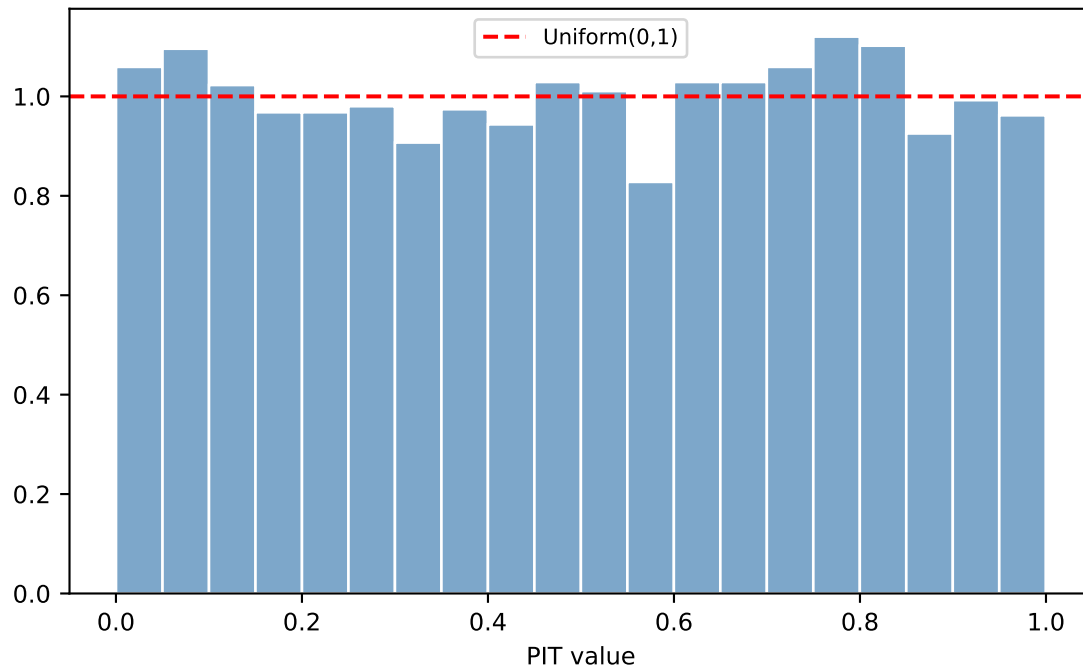
phi_interaction_dewpoint_nino34 — covariance



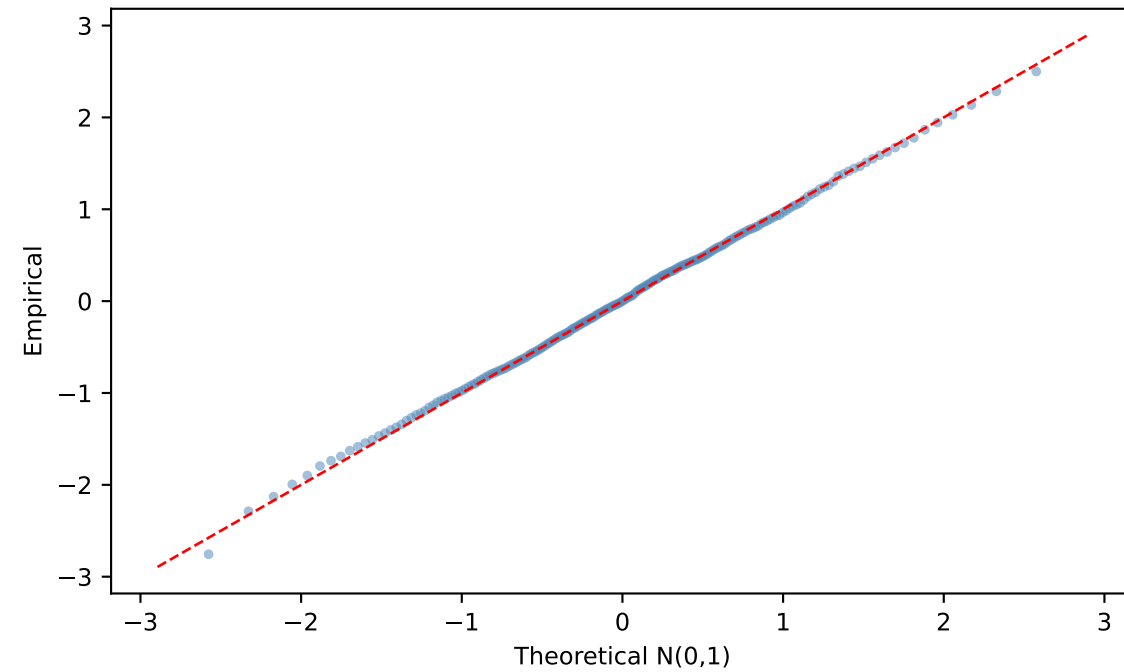


phi_interaction_temperature_nino34 — IS diagnostics

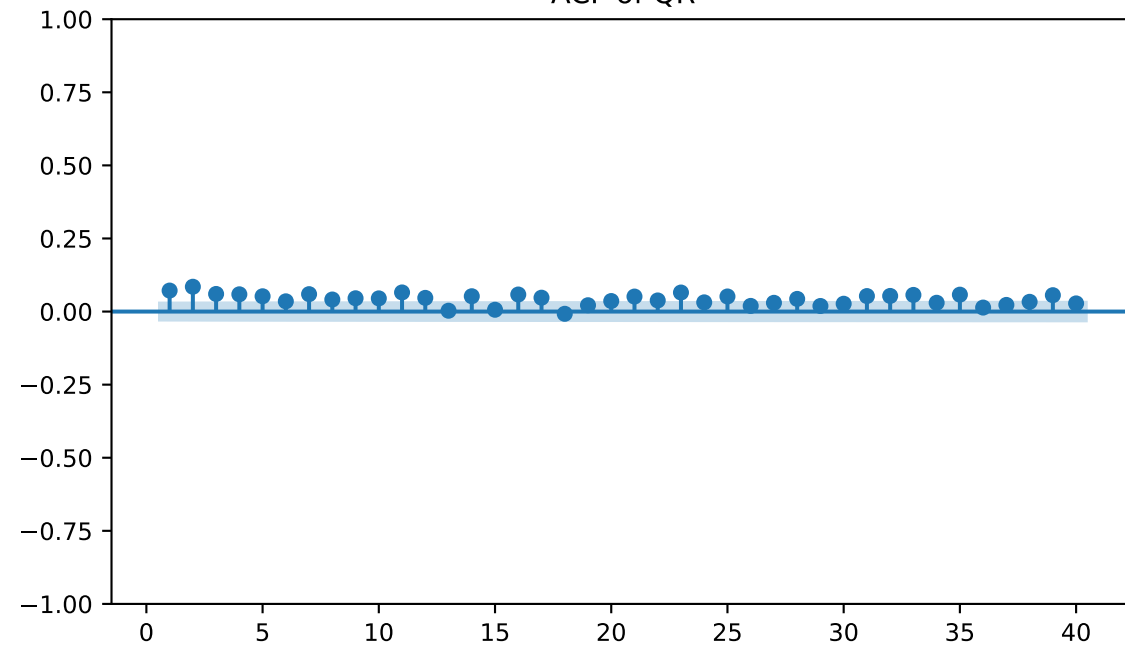
PIT histogram



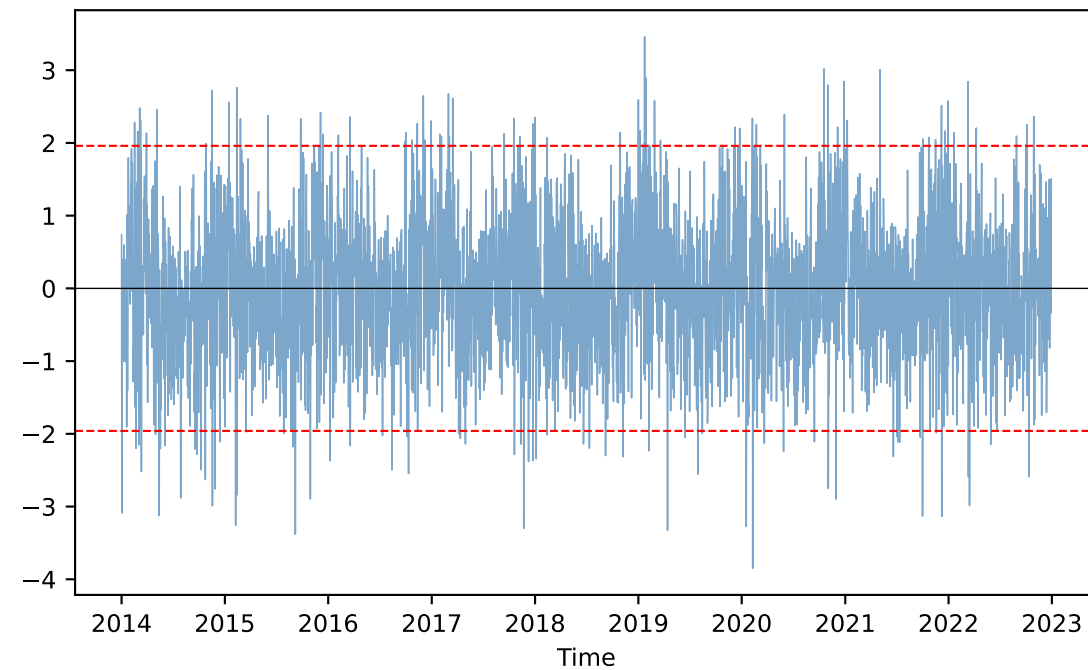
Normal QQ plot



ACF of QR

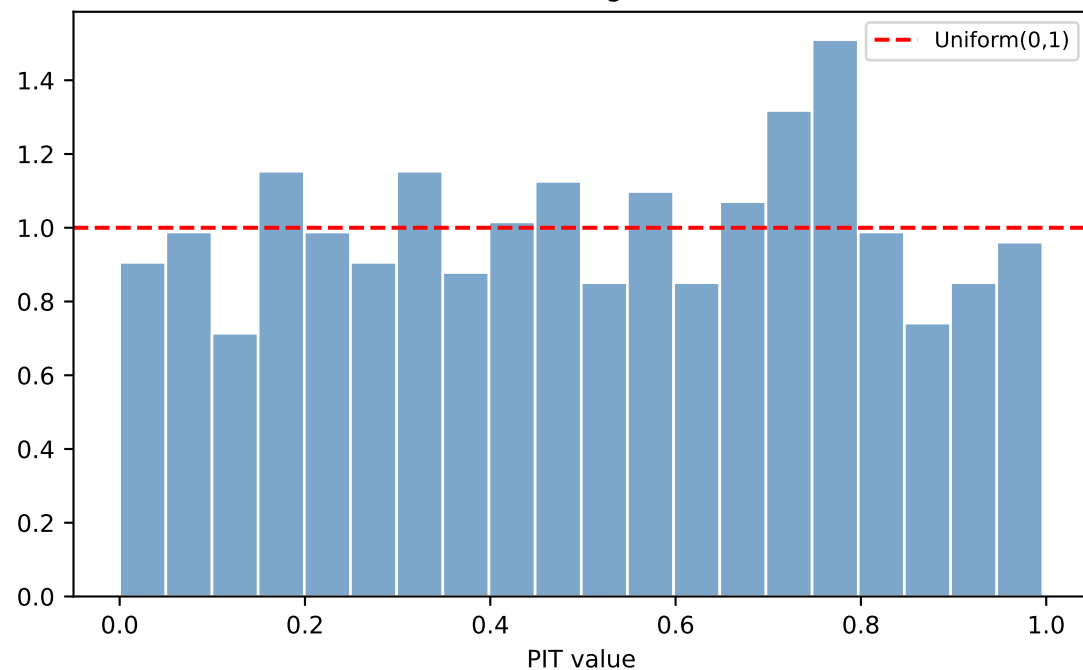


QR time series

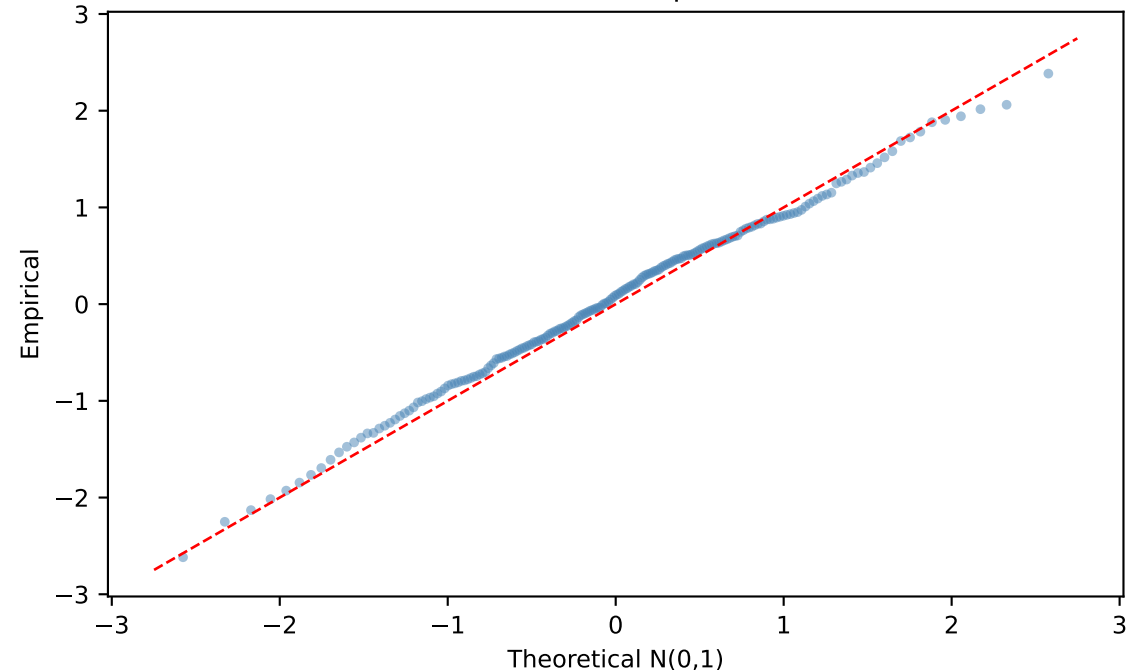


phi_interaction_temperature_nino34 — OOS diagnostics

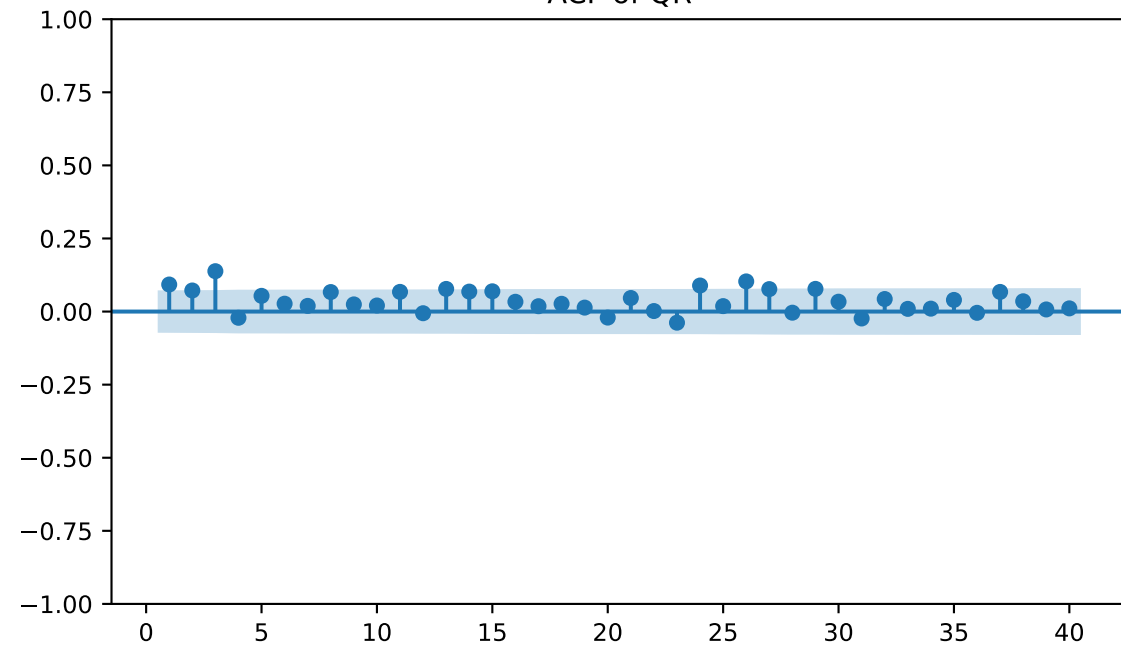
PIT histogram



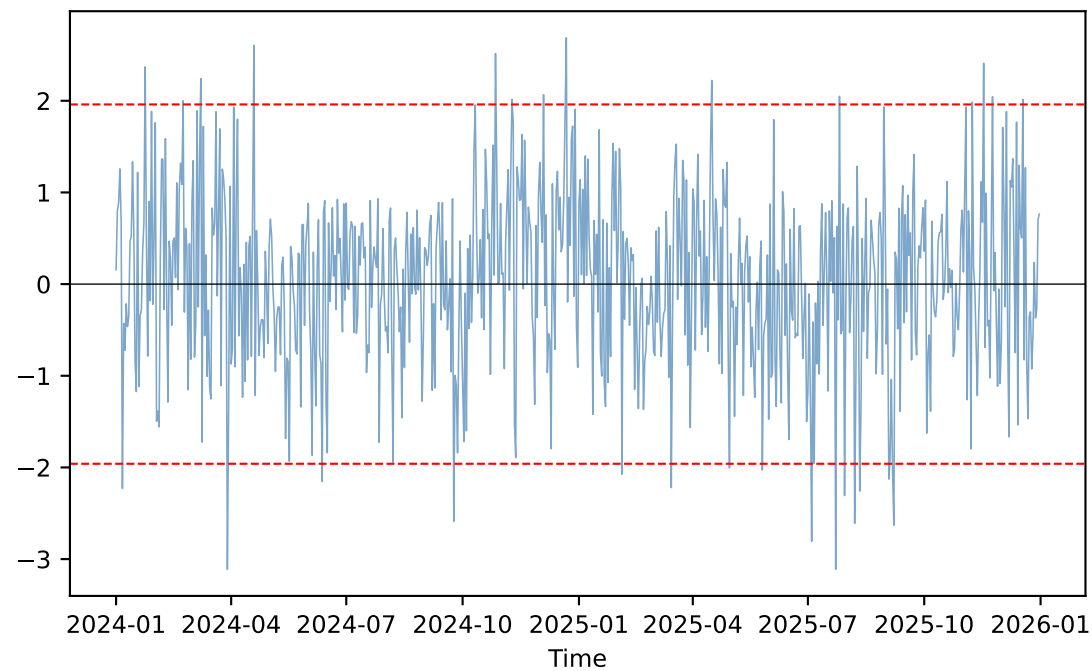
Normal QQ plot



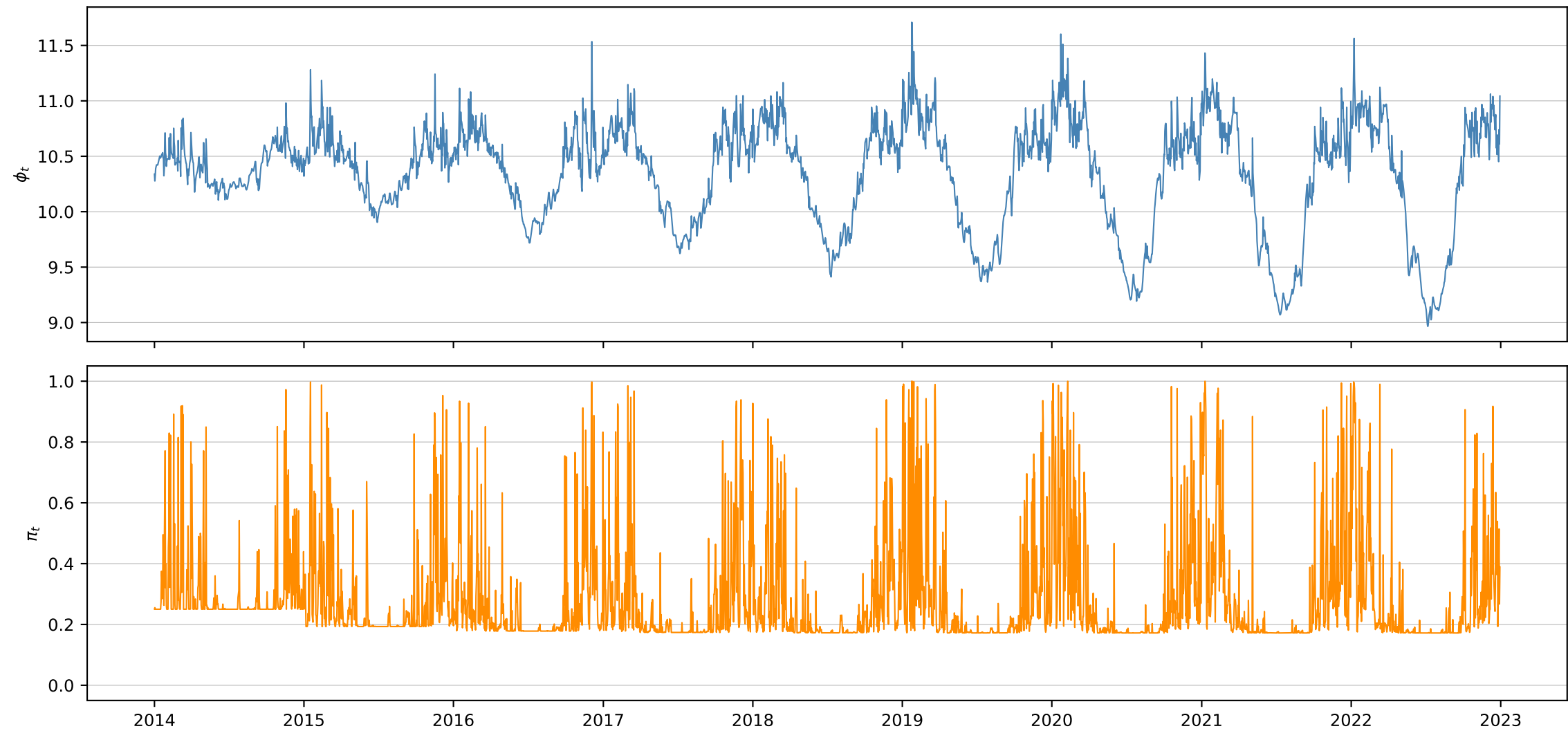
ACF of QR



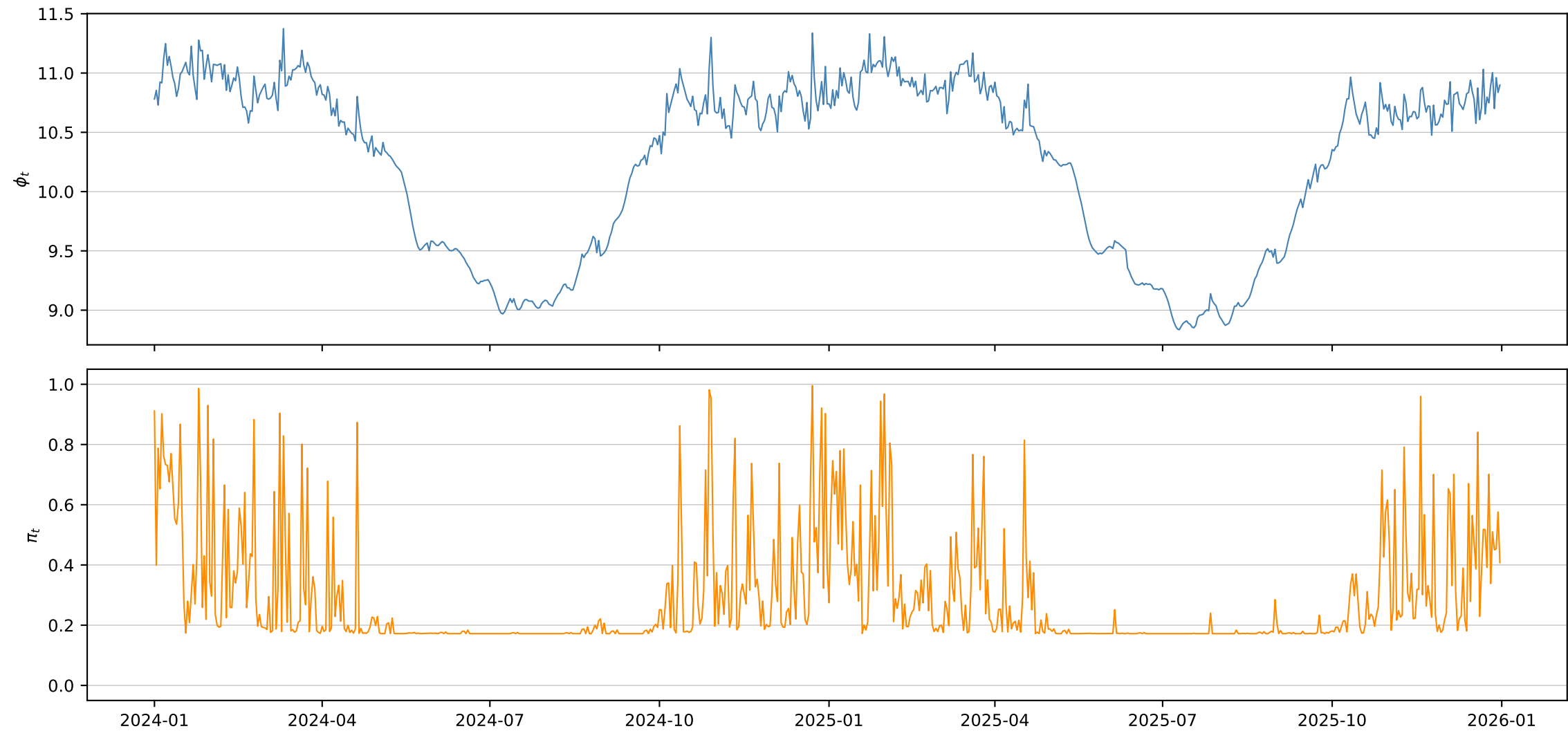
QR time series



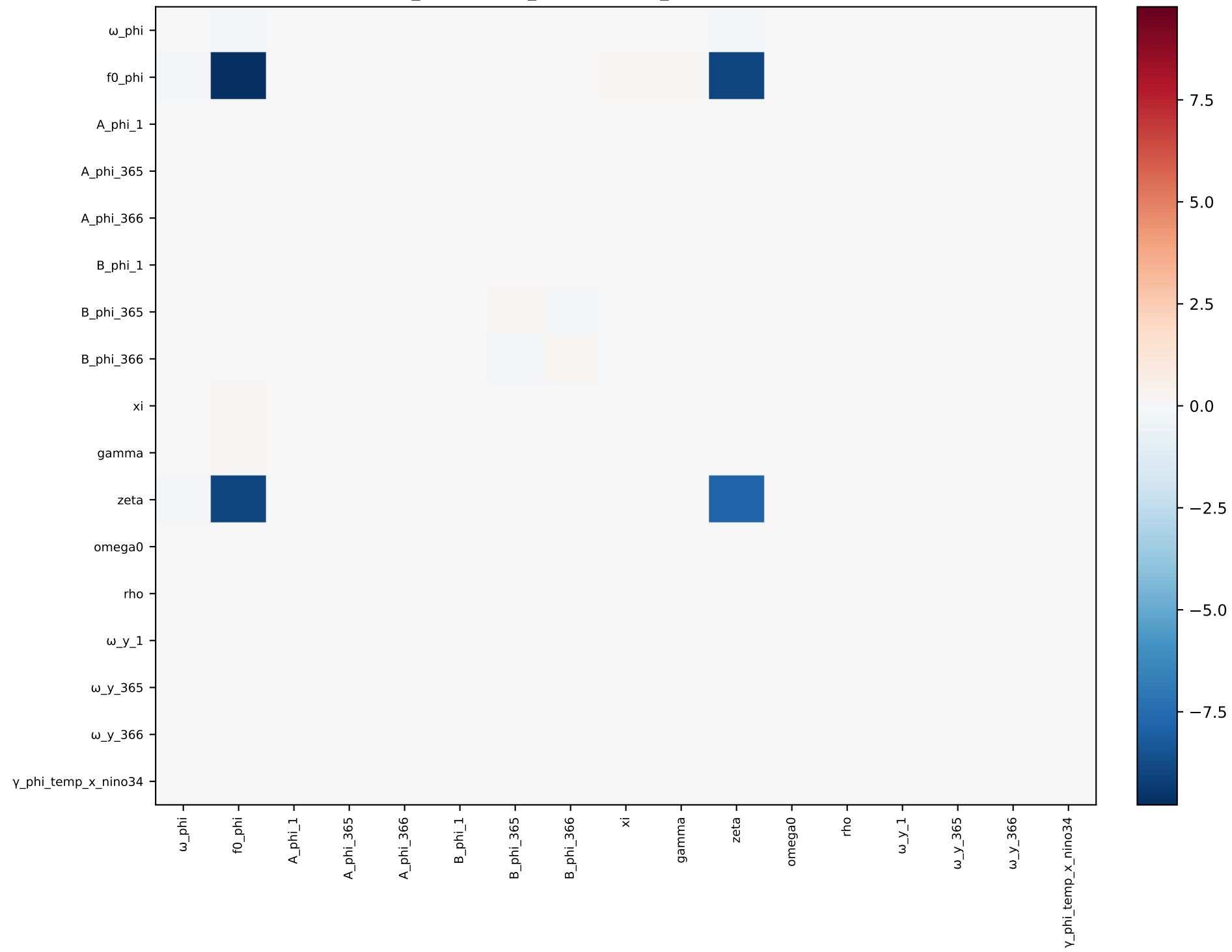
phi_interaction_temperature_nino34 — filtered states (IS)



phi_interaction_temperature_nino34 — filtered states (OOS)

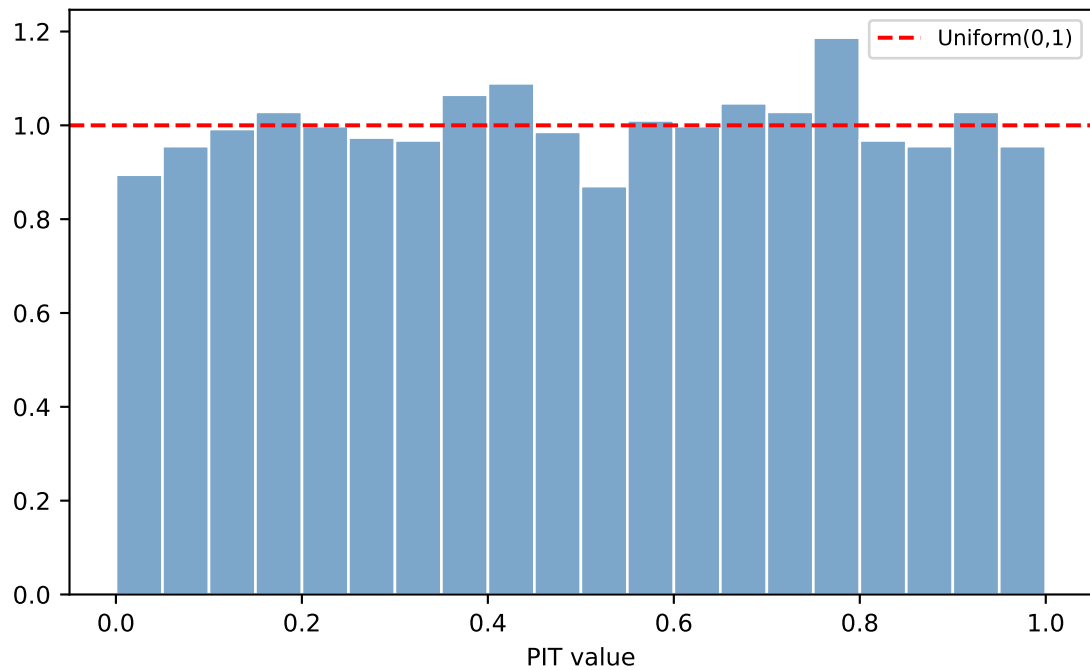


phi_interaction_temperature_nino34 — covariance

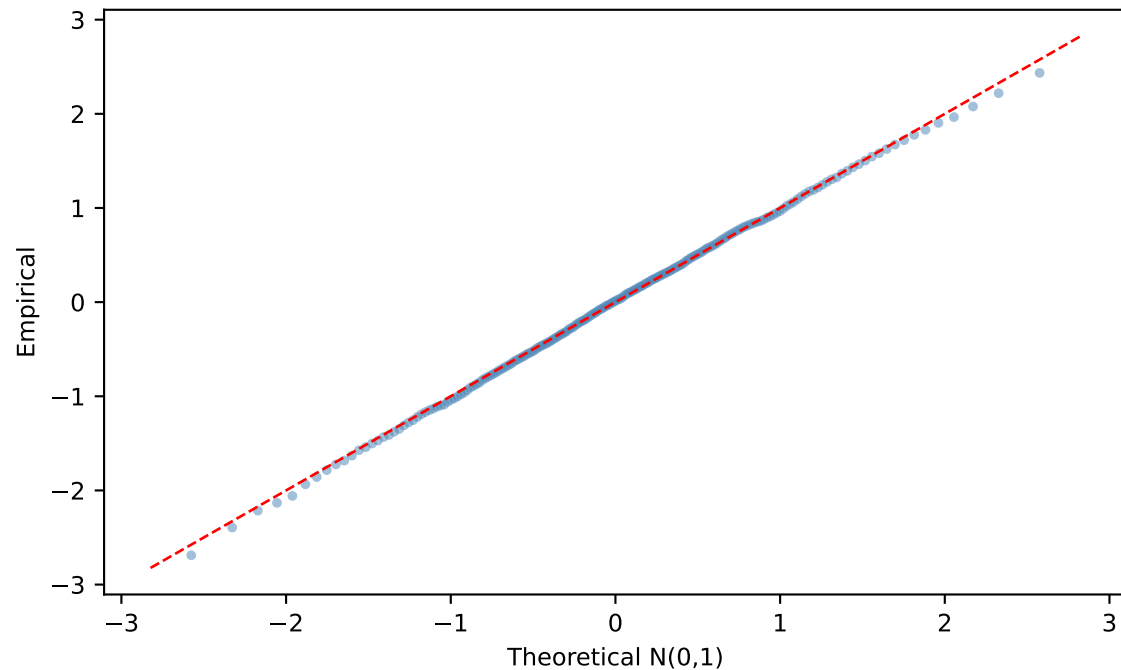


phi_xi_dewpoint_t — IS diagnostics

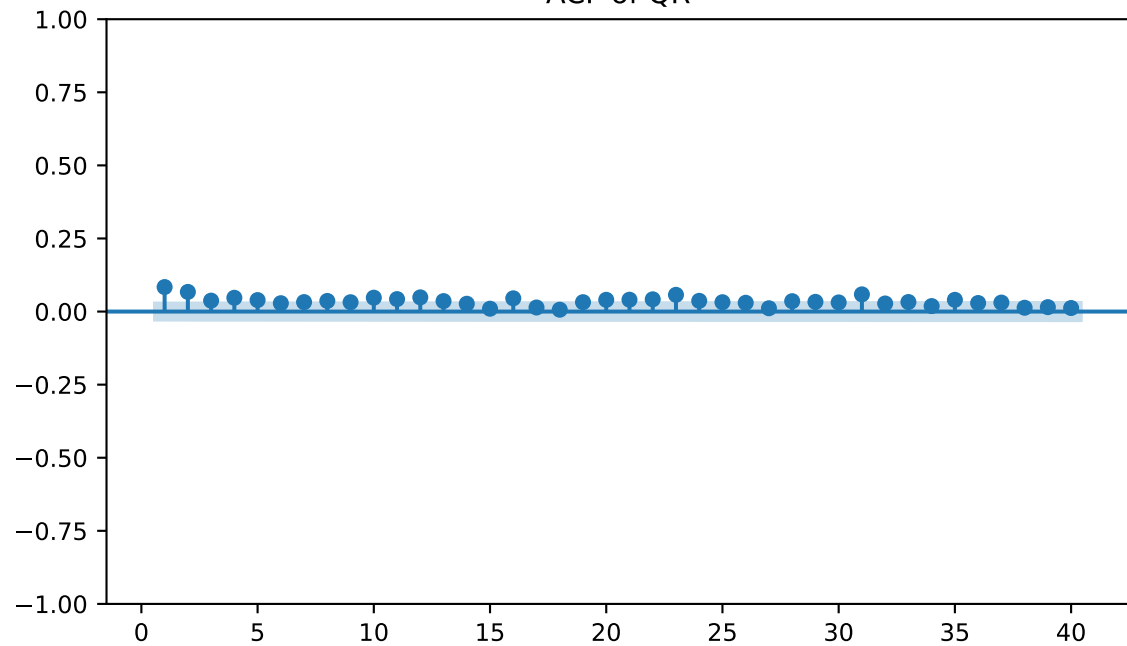
PIT histogram



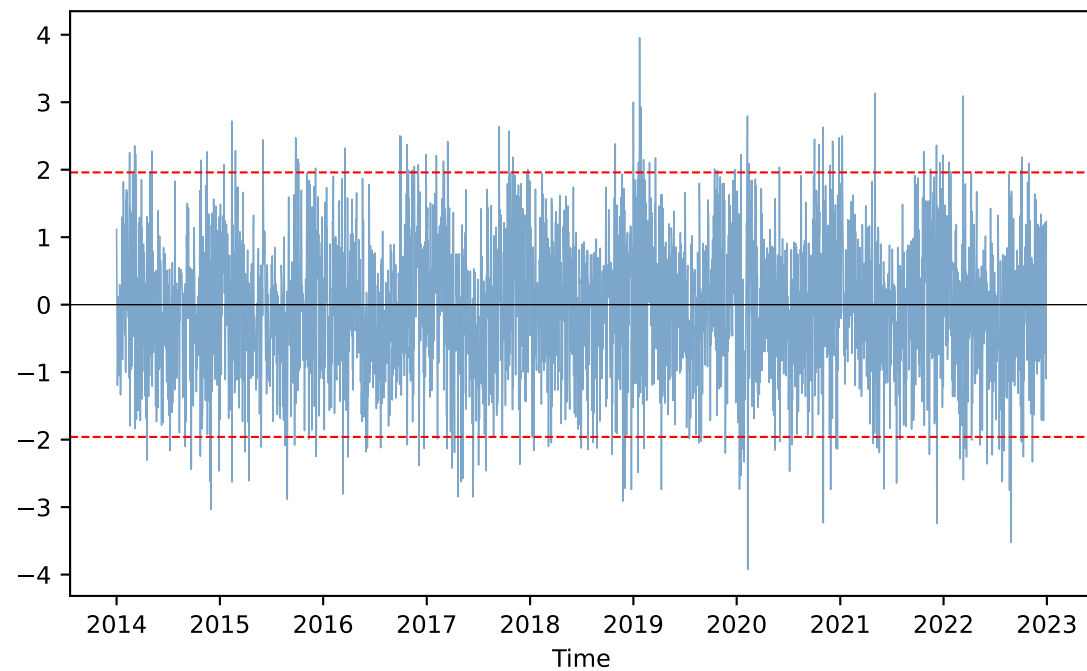
Normal QQ plot



ACF of QR

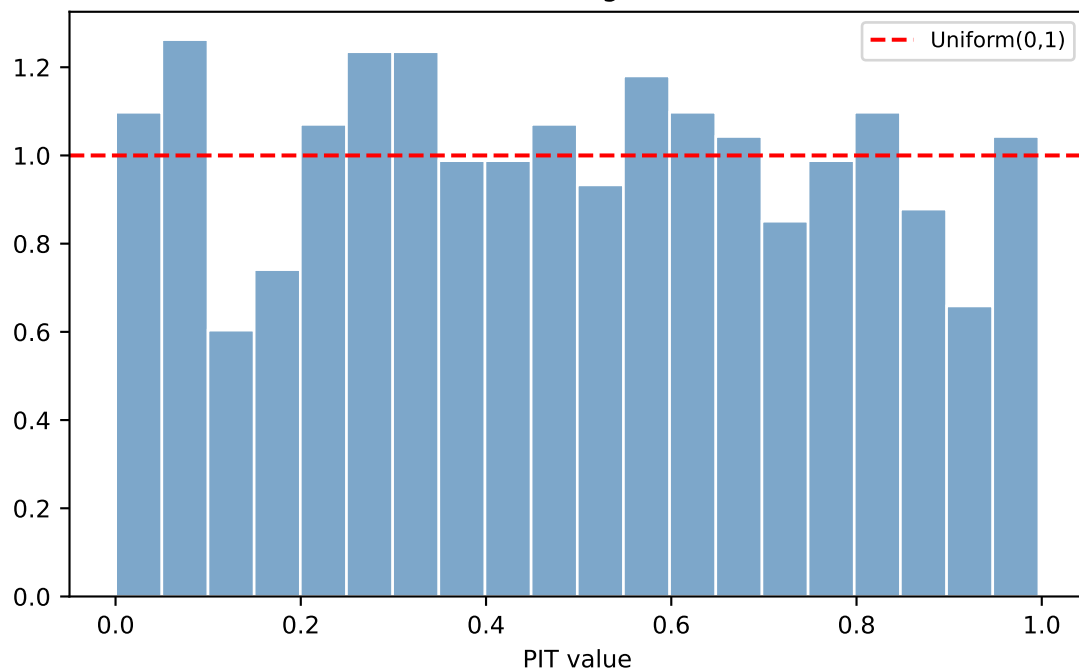


QR time series

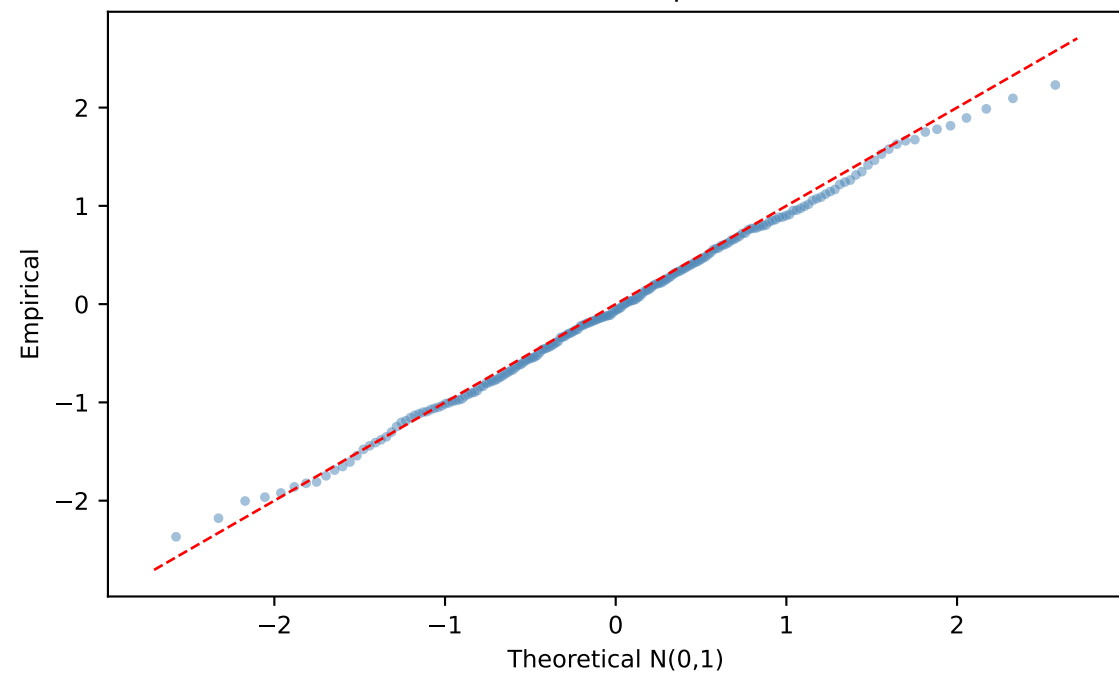


phi_xi_dewpoint_t — OOS diagnostics

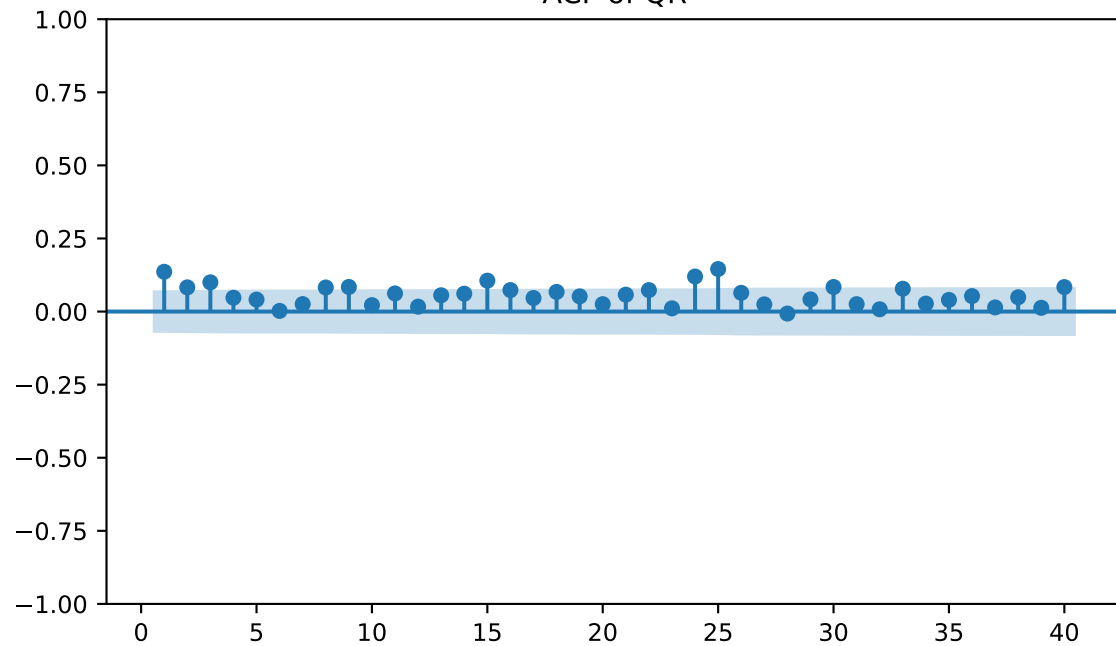
PIT histogram



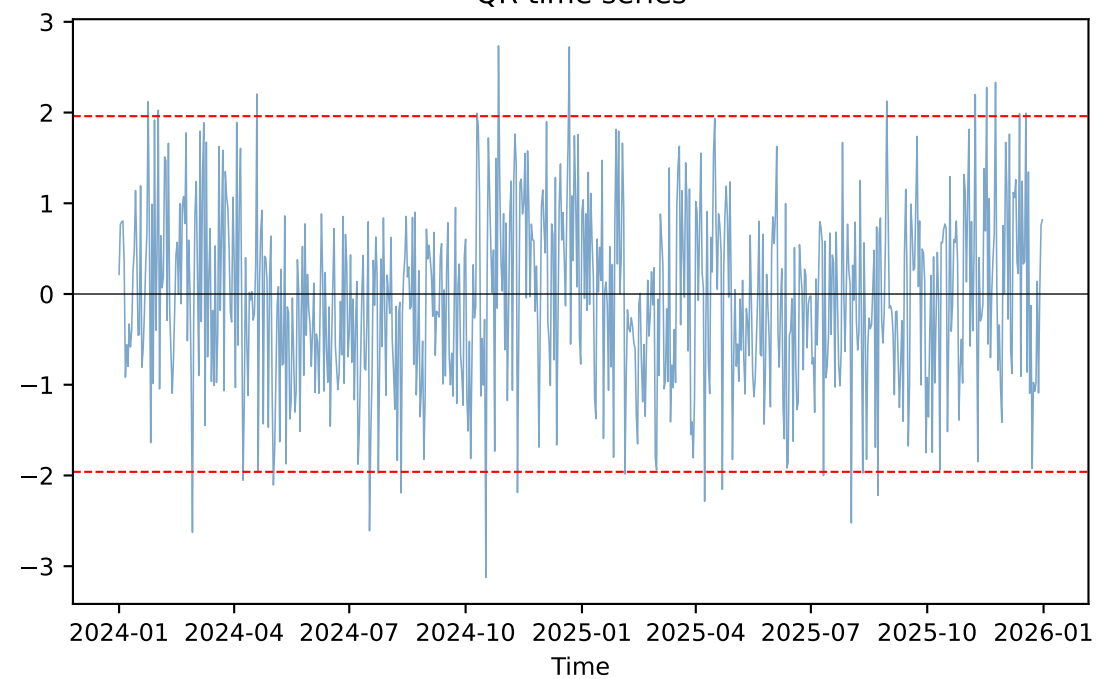
Normal QQ plot



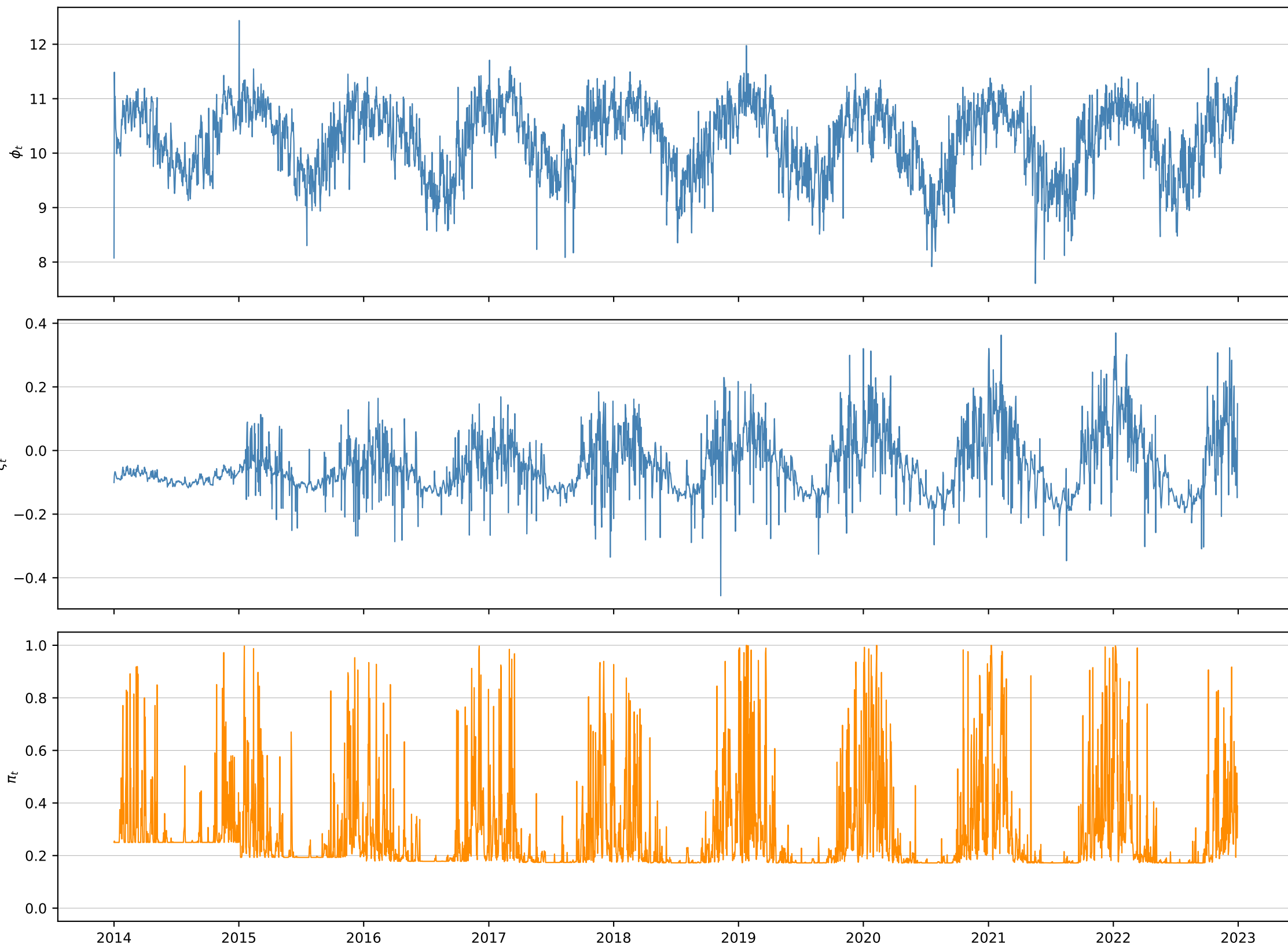
ACF of QR



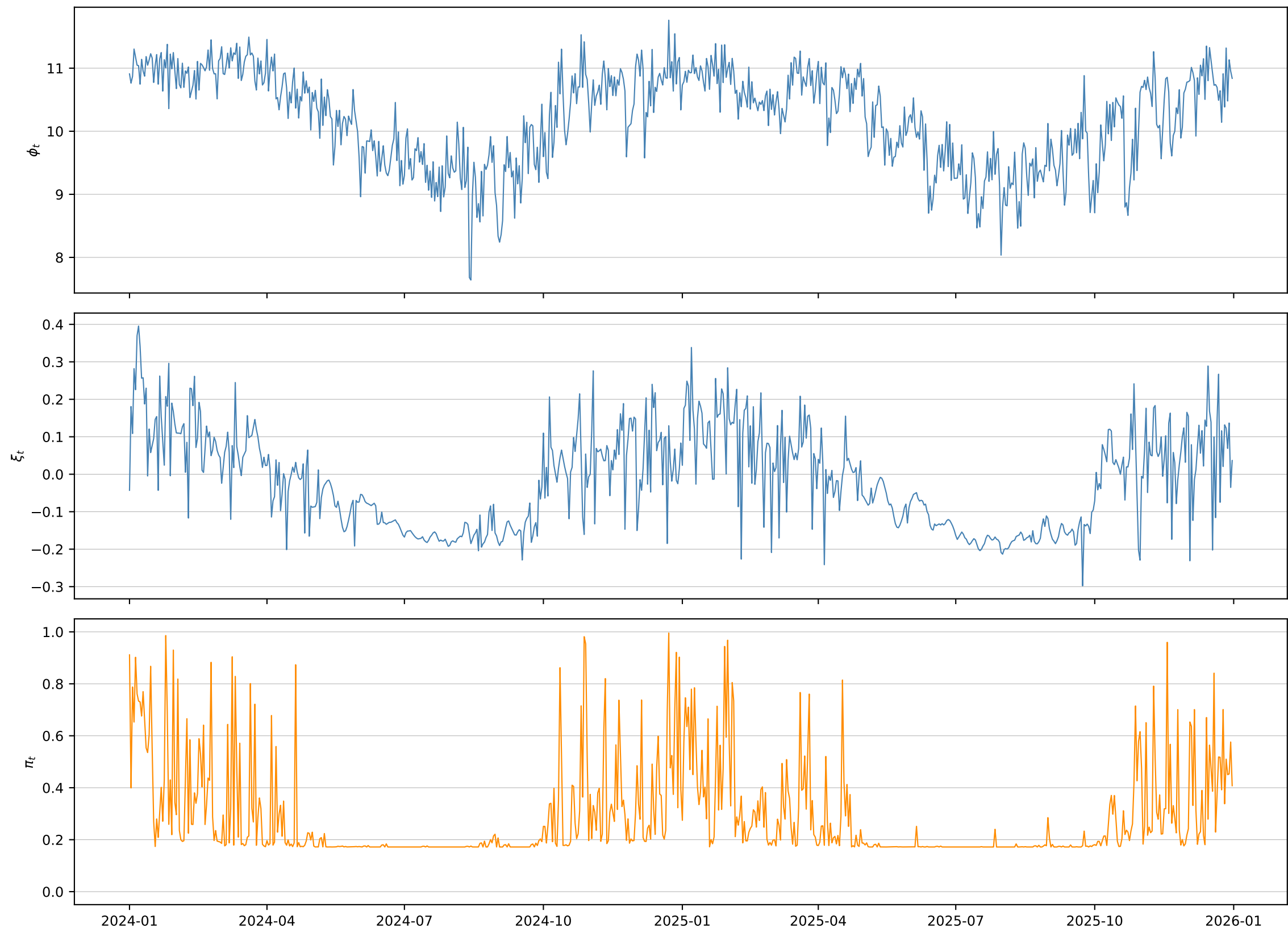
QR time series



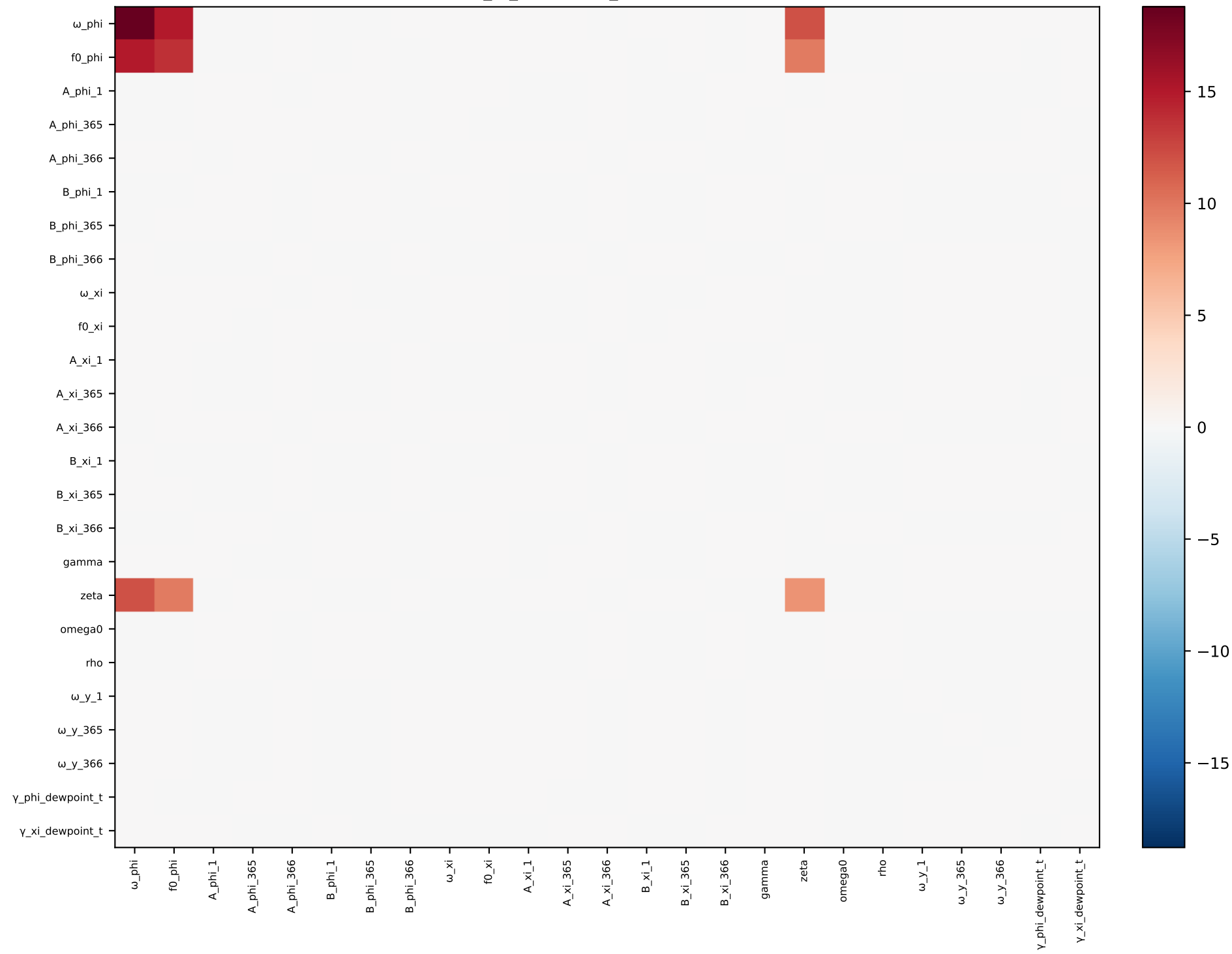
phi_xi_dewpoint_t — filtered states (IS)



phi_xi_dewpoint_t — filtered states (OOS)

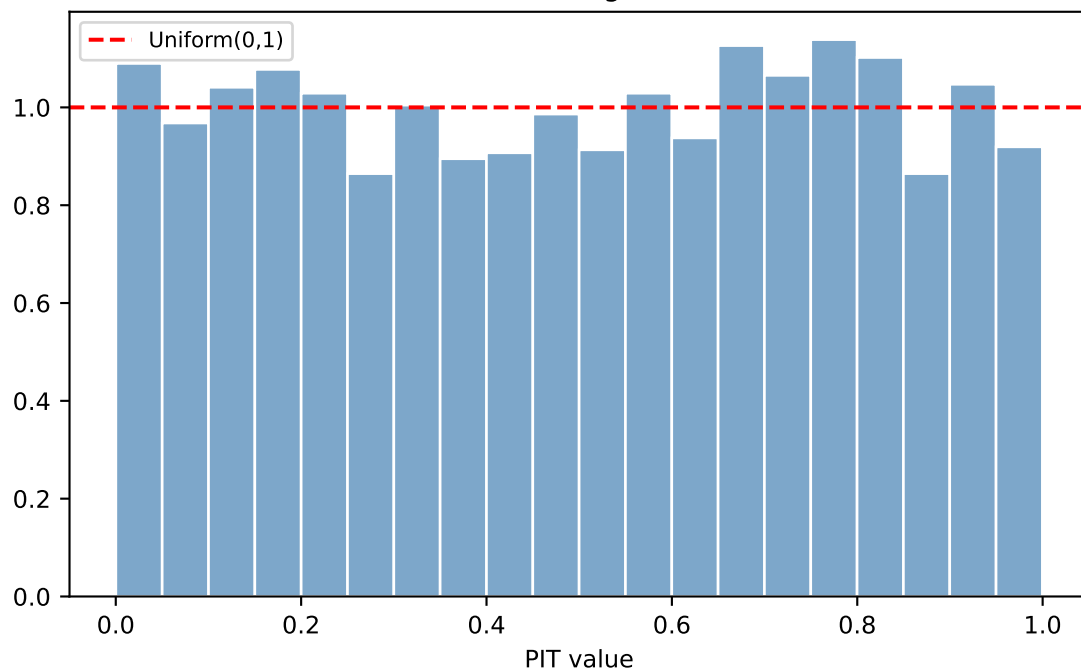


phi_xi_dewpoint_t — covariance

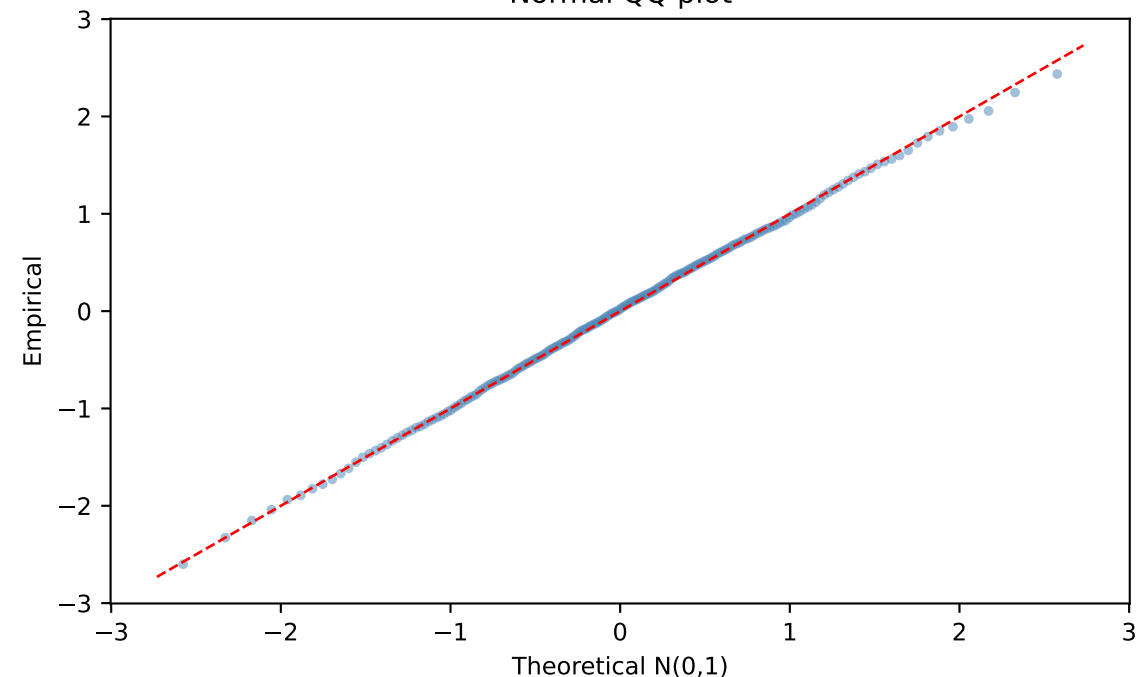


phi_xi_dewpoint_lag1 — IS diagnostics

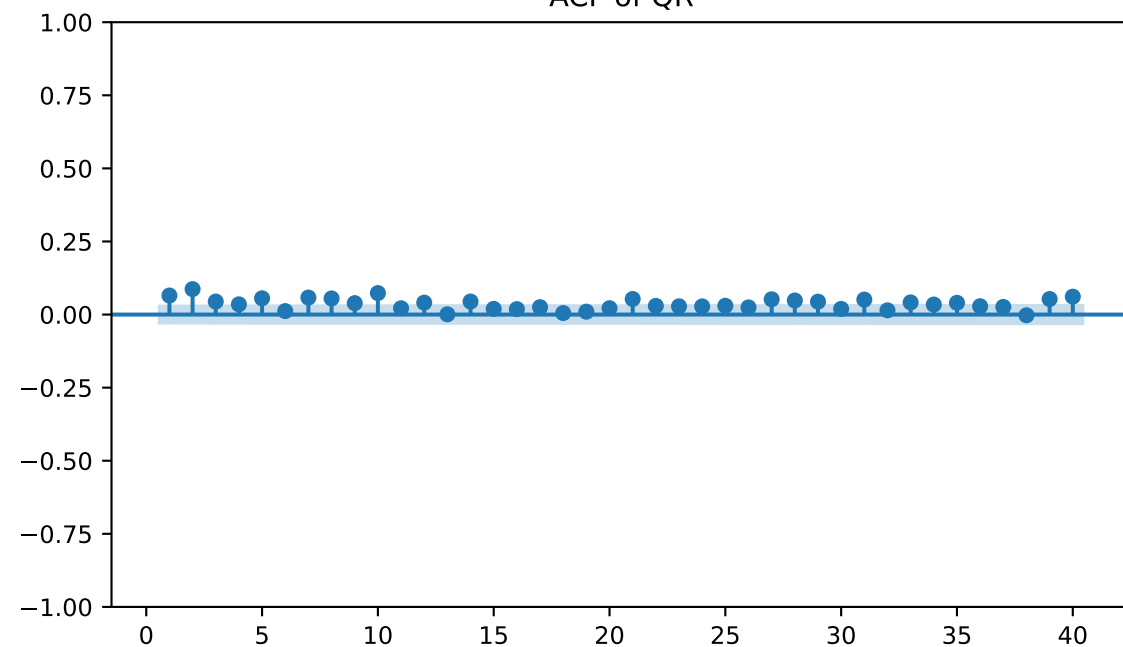
PIT histogram



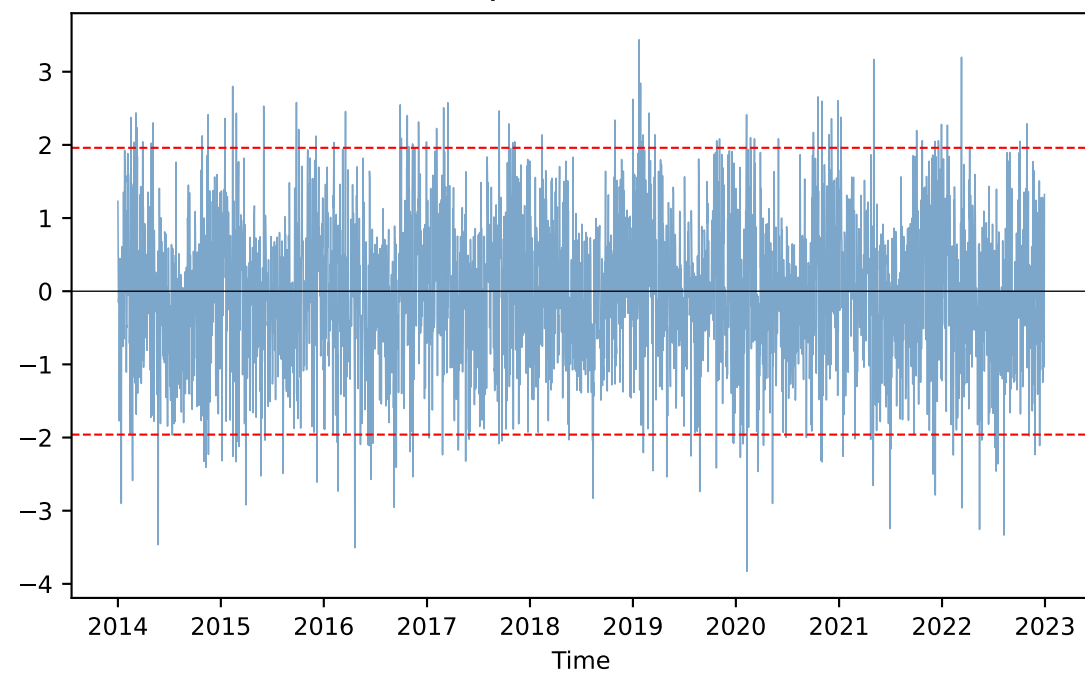
Normal QQ plot



ACF of QR

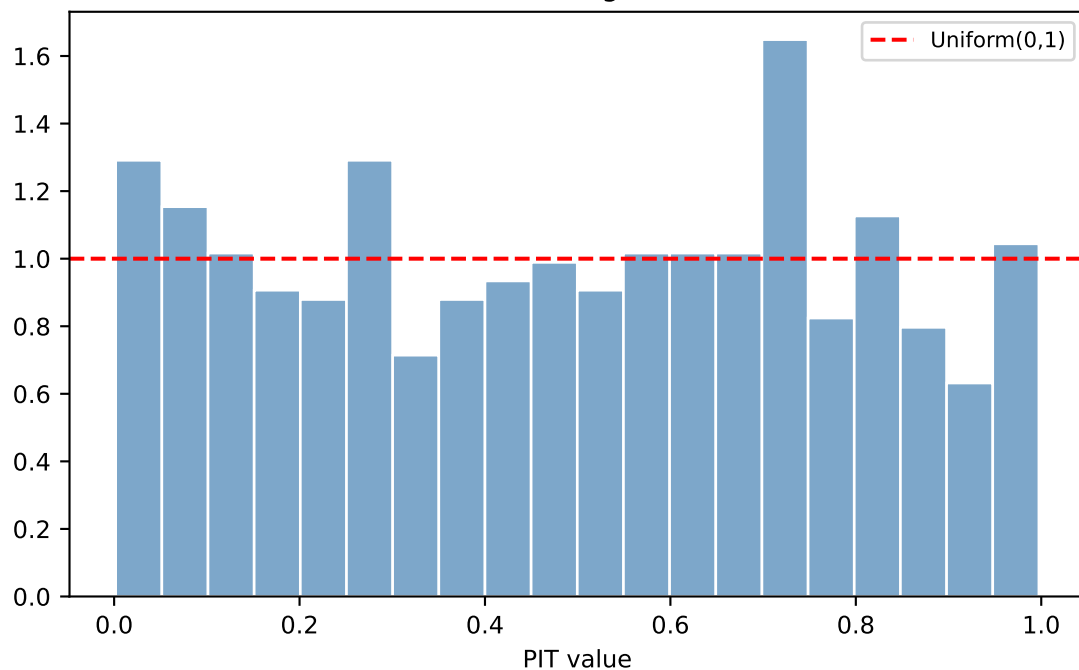


QR time series

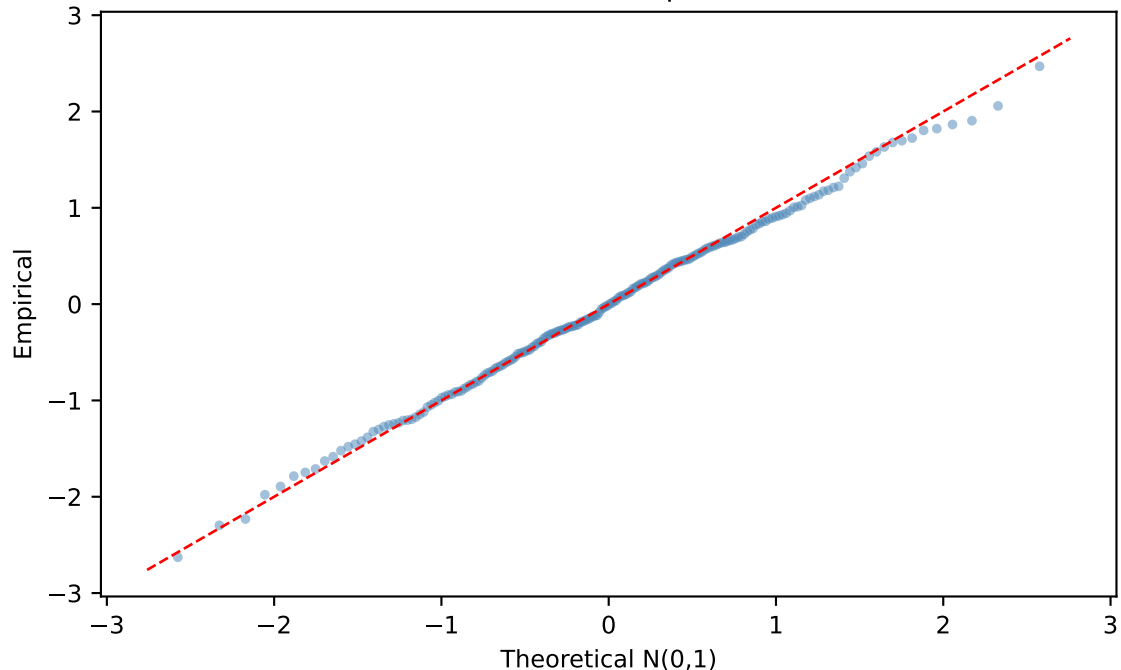


phi_xi_dewpoint_lag1 — OOS diagnostics

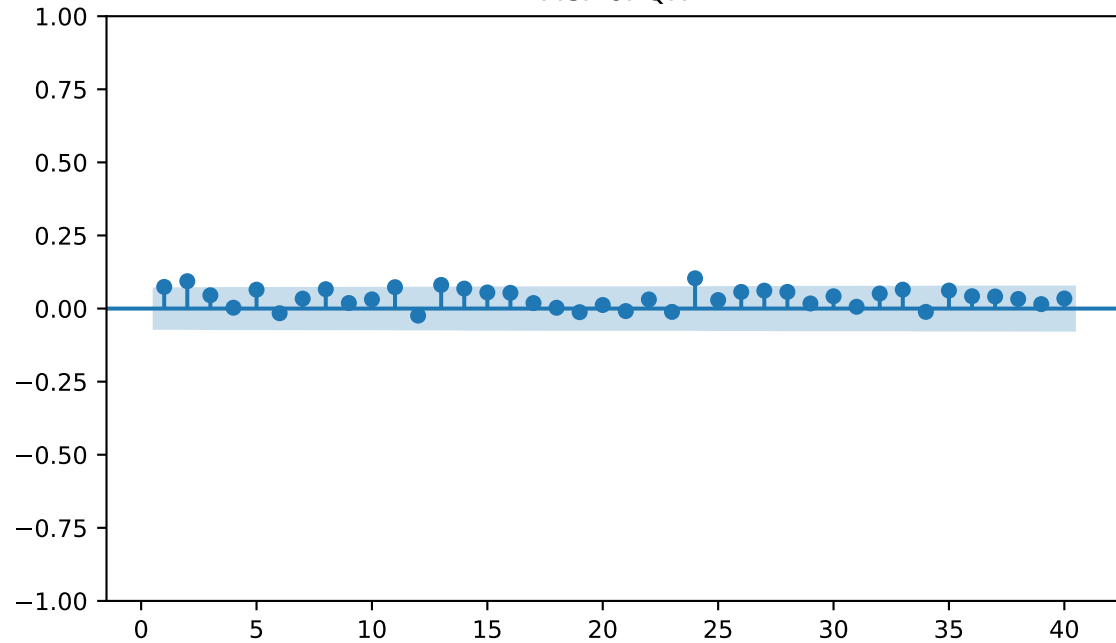
PIT histogram



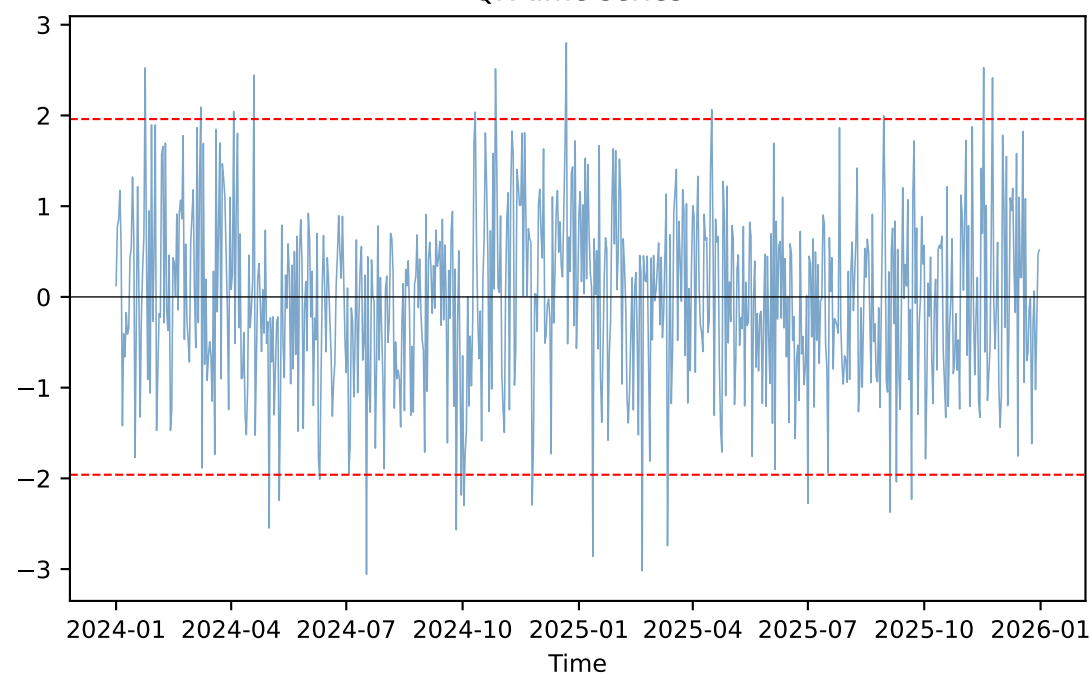
Normal QQ plot



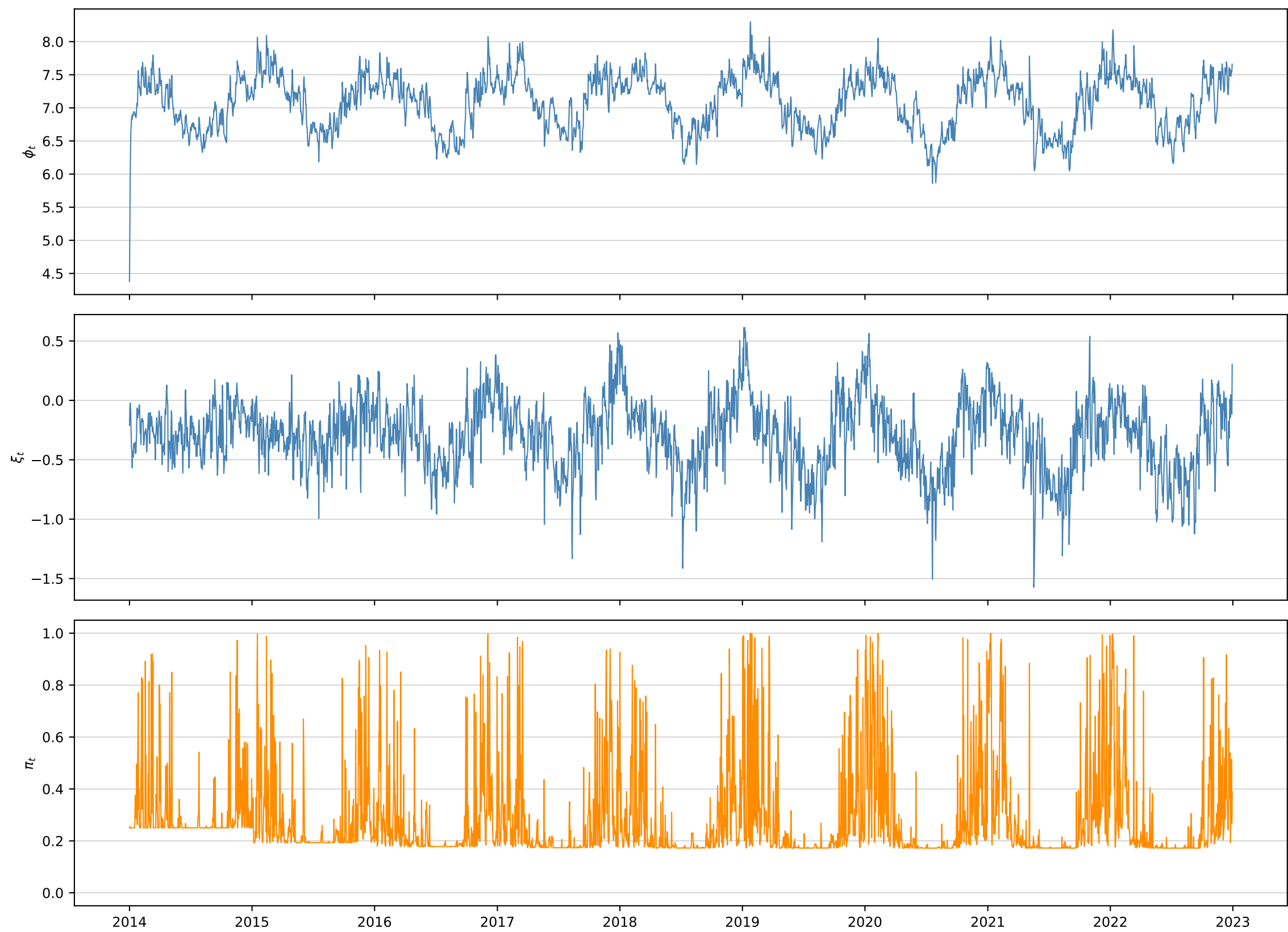
ACF of QR



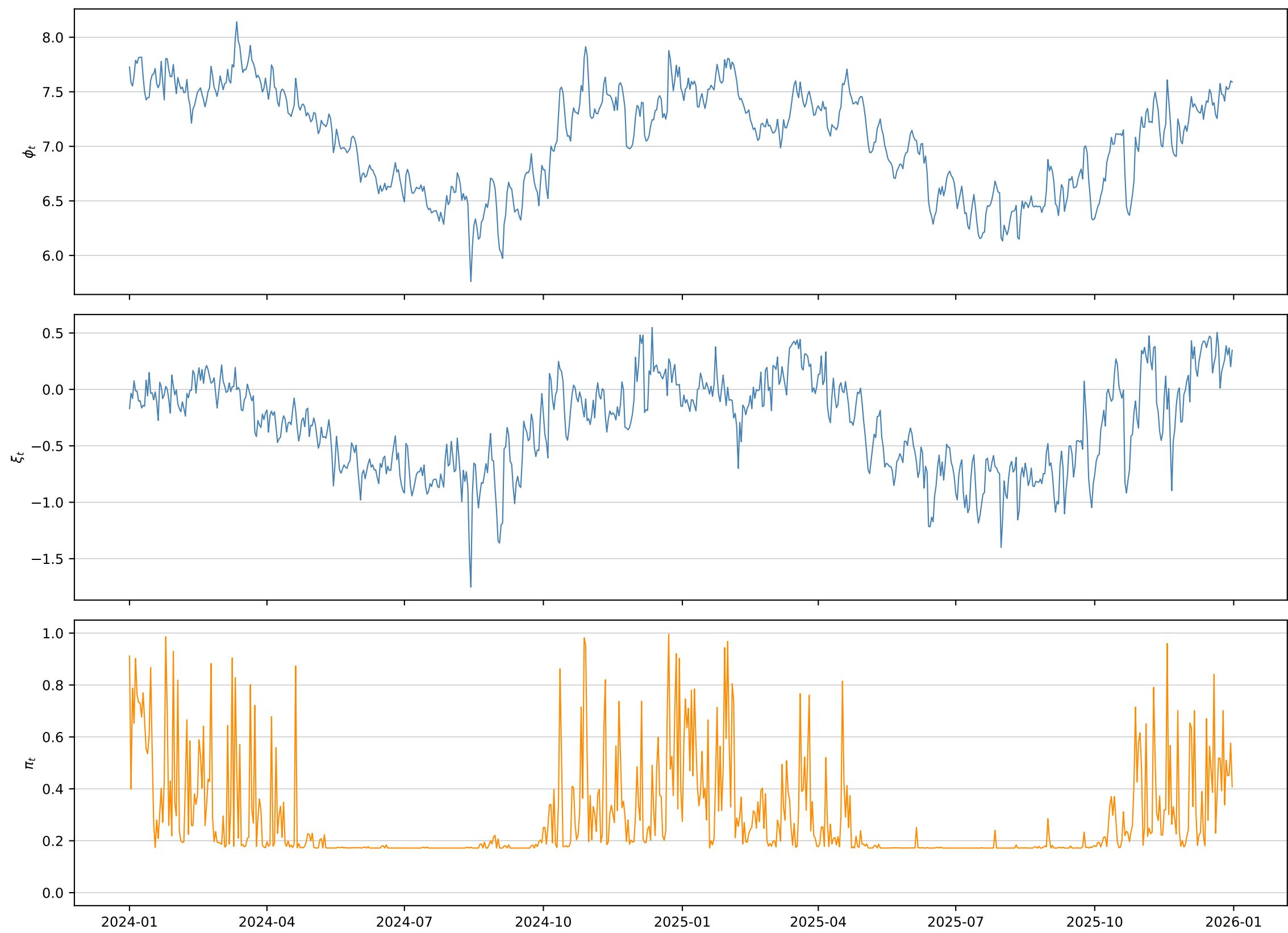
QR time series



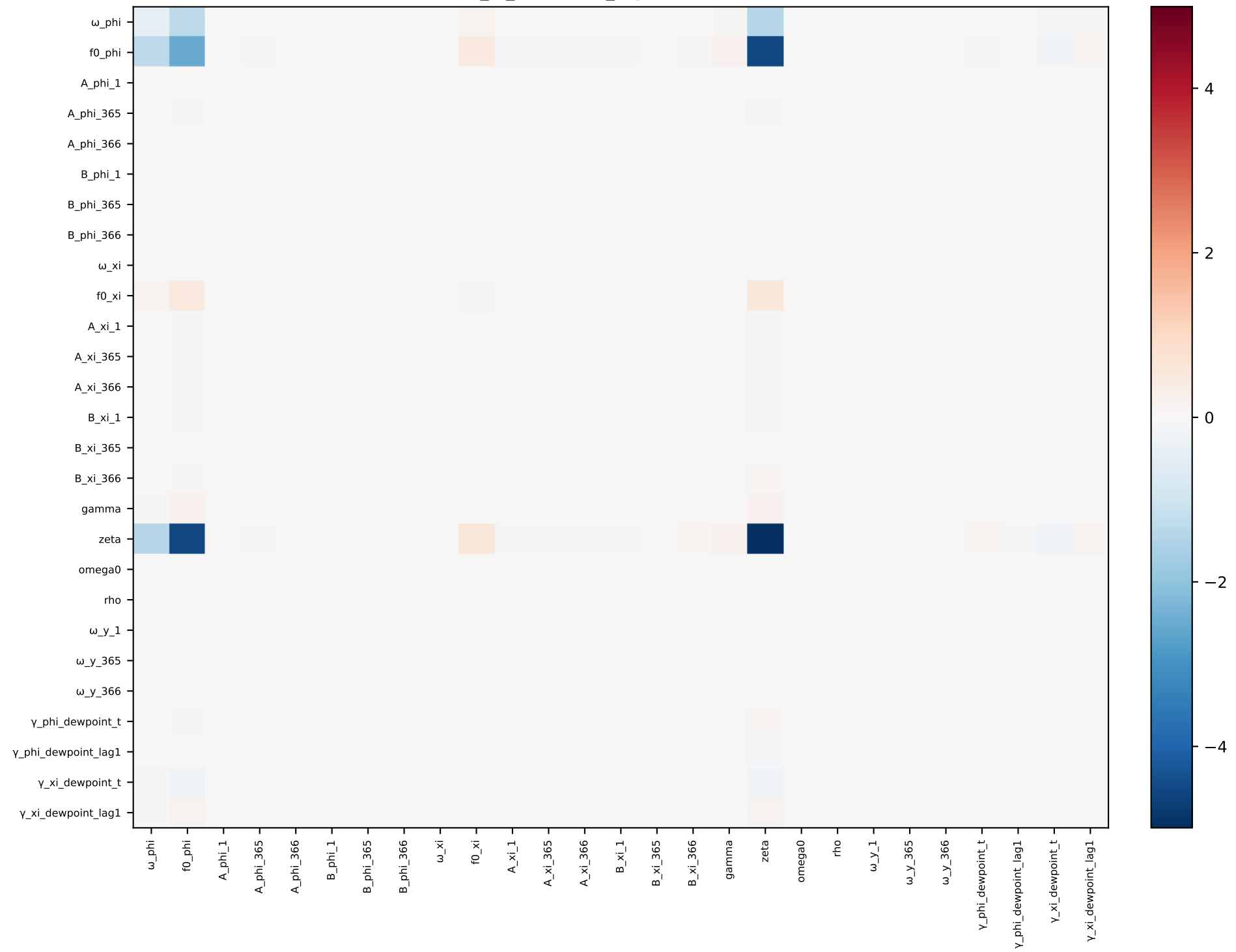
phi_xi_dewpoint_lag1 — filtered states (IS)



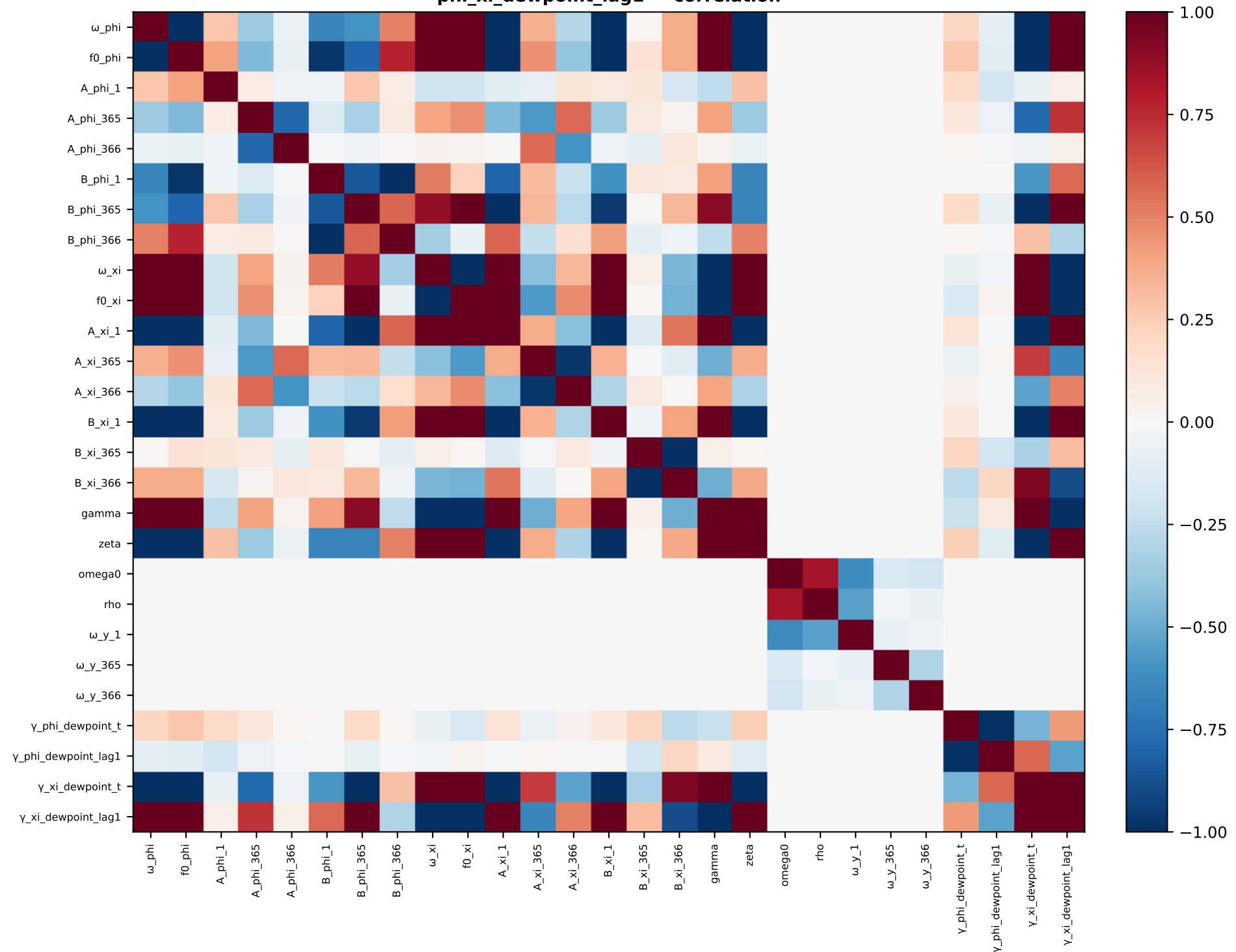
phi_xi_dewpoint_lag1 — filtered states (OOS)



phi_xi_dewpoint_lag1 — covariance

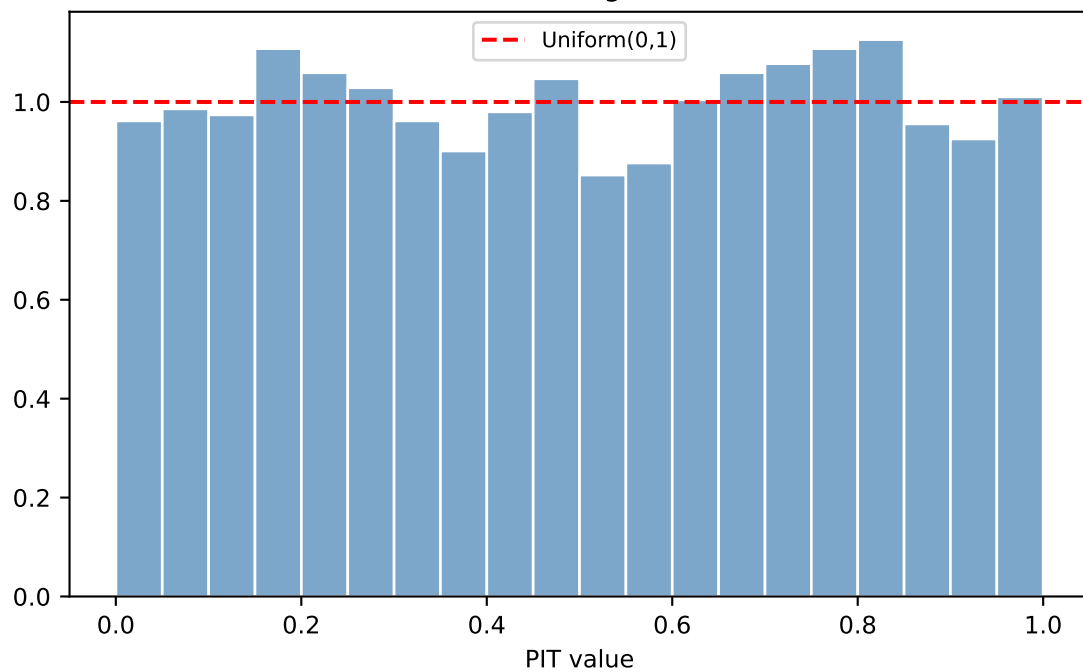


phi_xi_dewpoint_lag1 — correlation

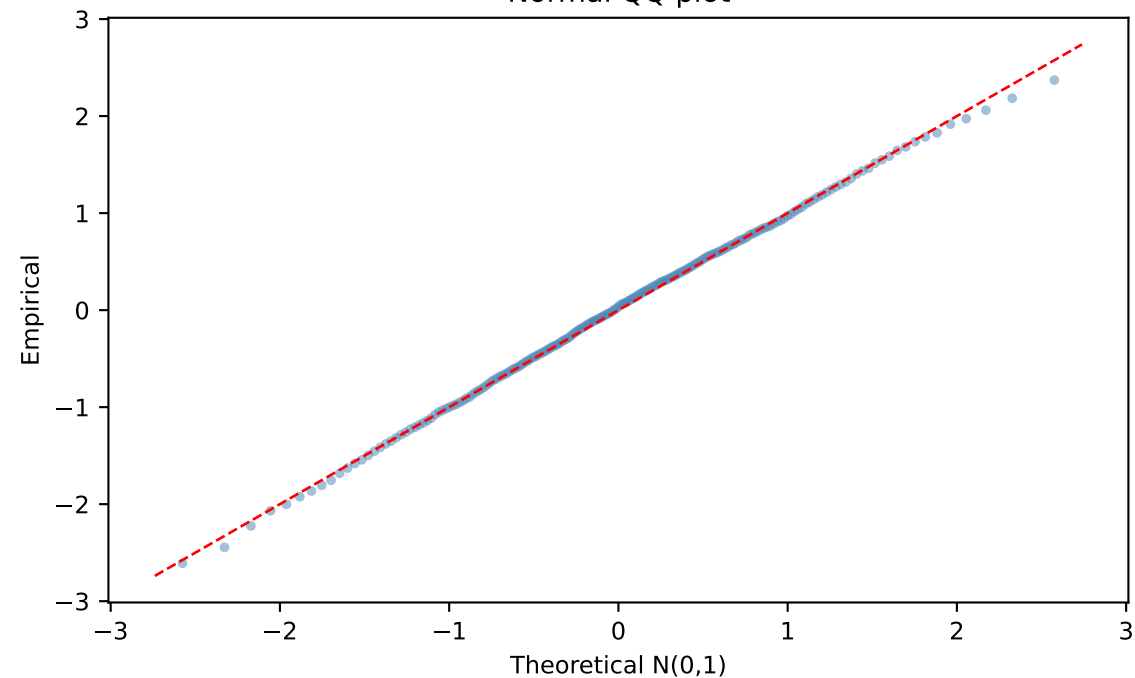


phi_xi_dewpoint_short_lags — IS diagnostics

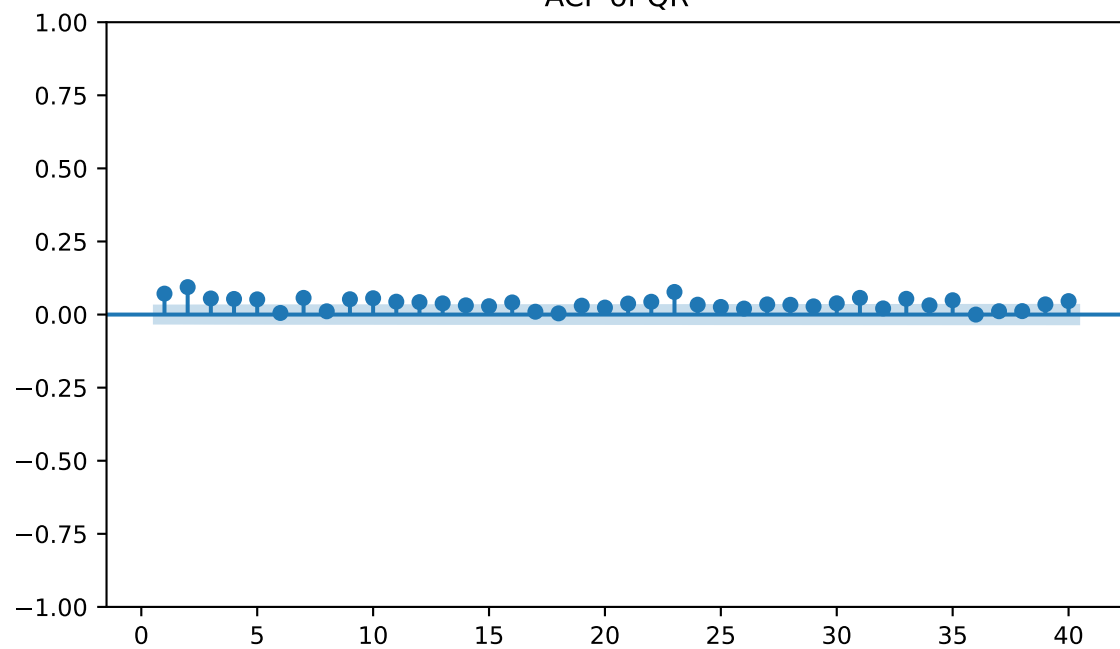
PIT histogram



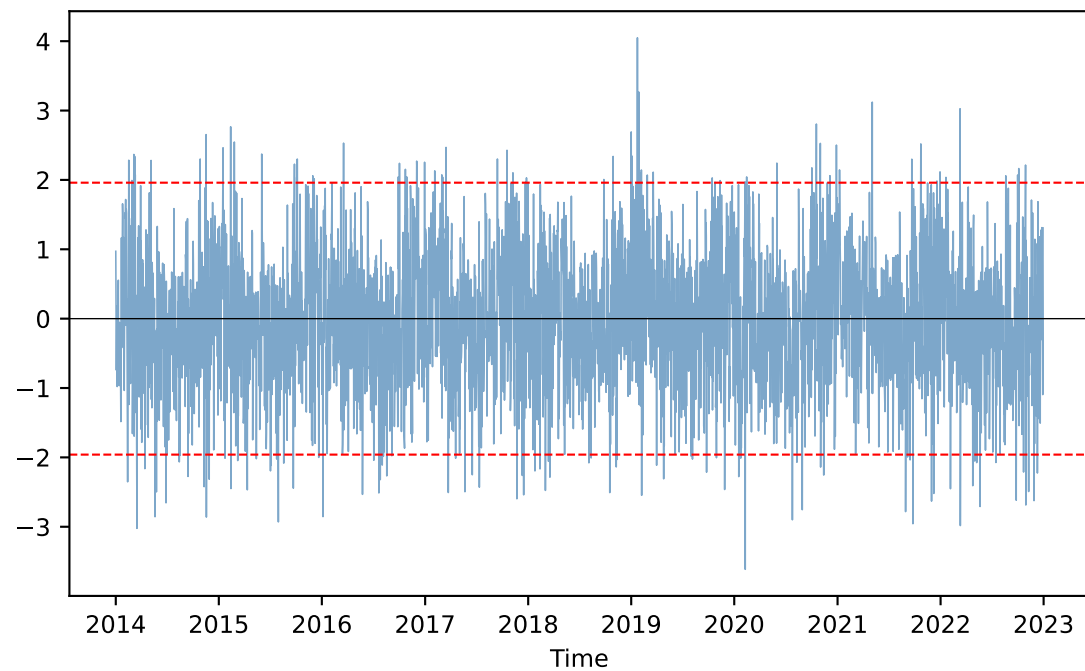
Normal QQ plot



ACF of QR

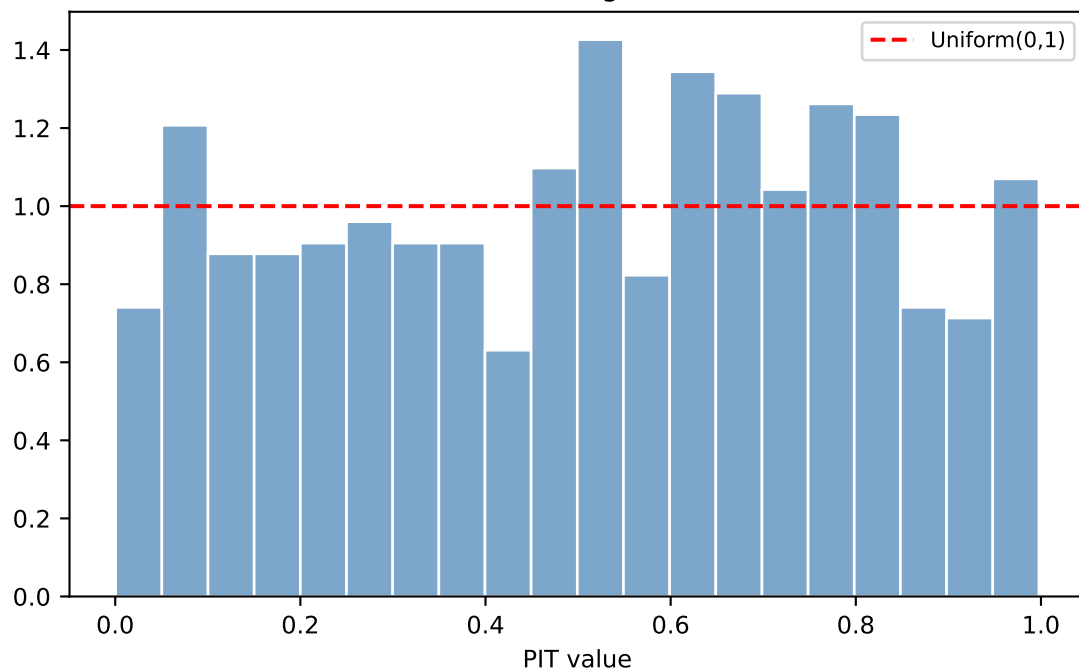


QR time series

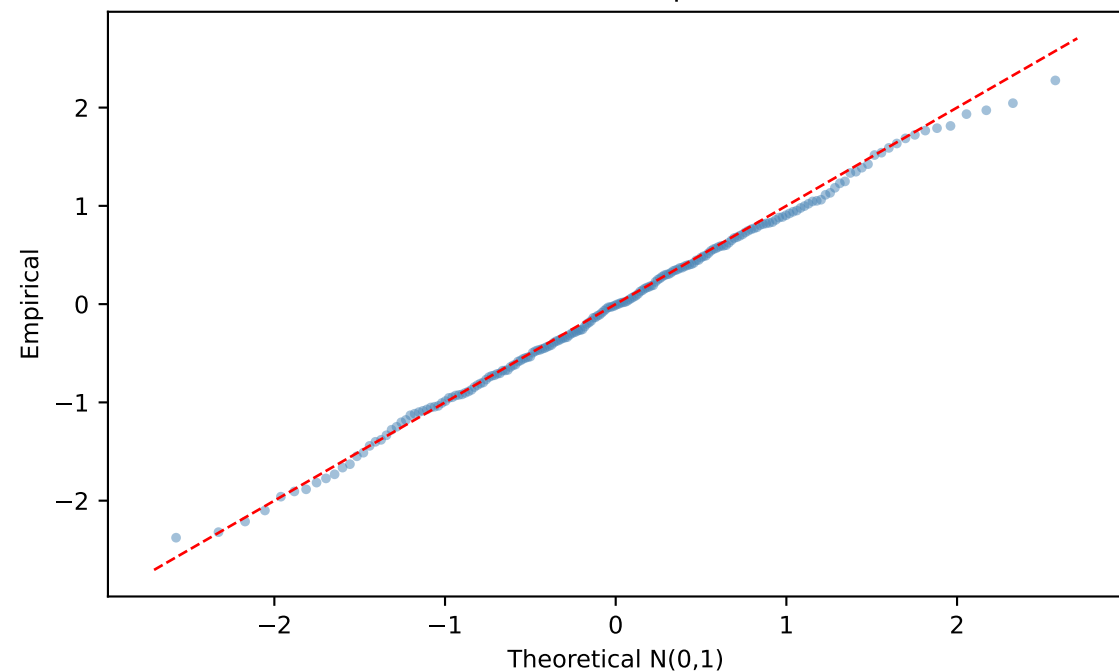


phi_xi_dewpoint_short_lags — OOS diagnostics

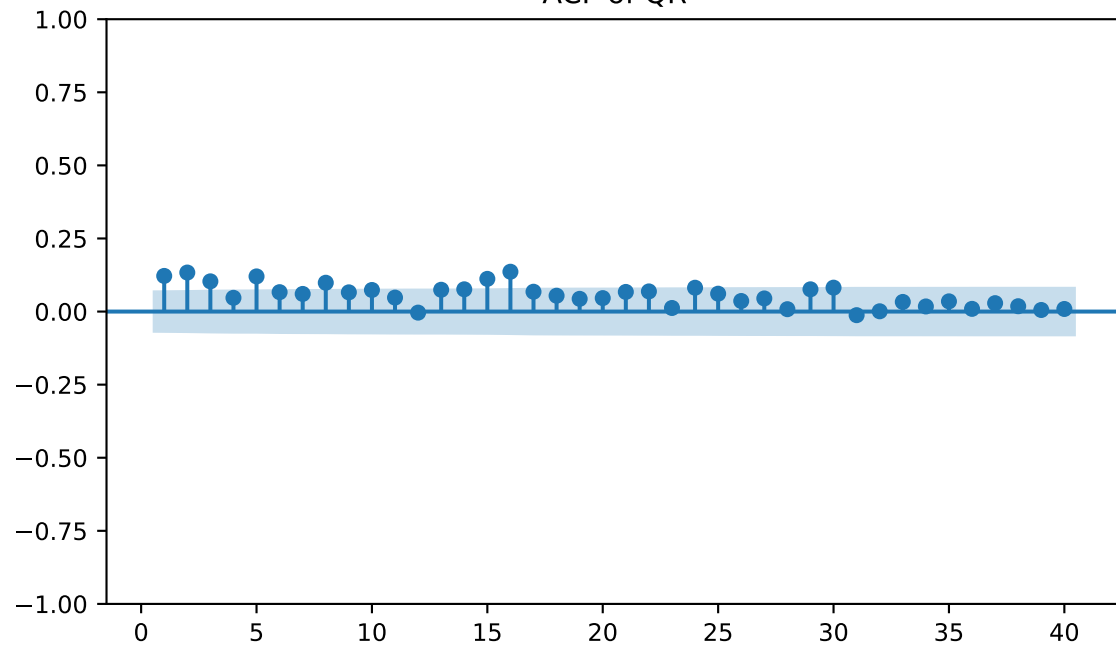
PIT histogram



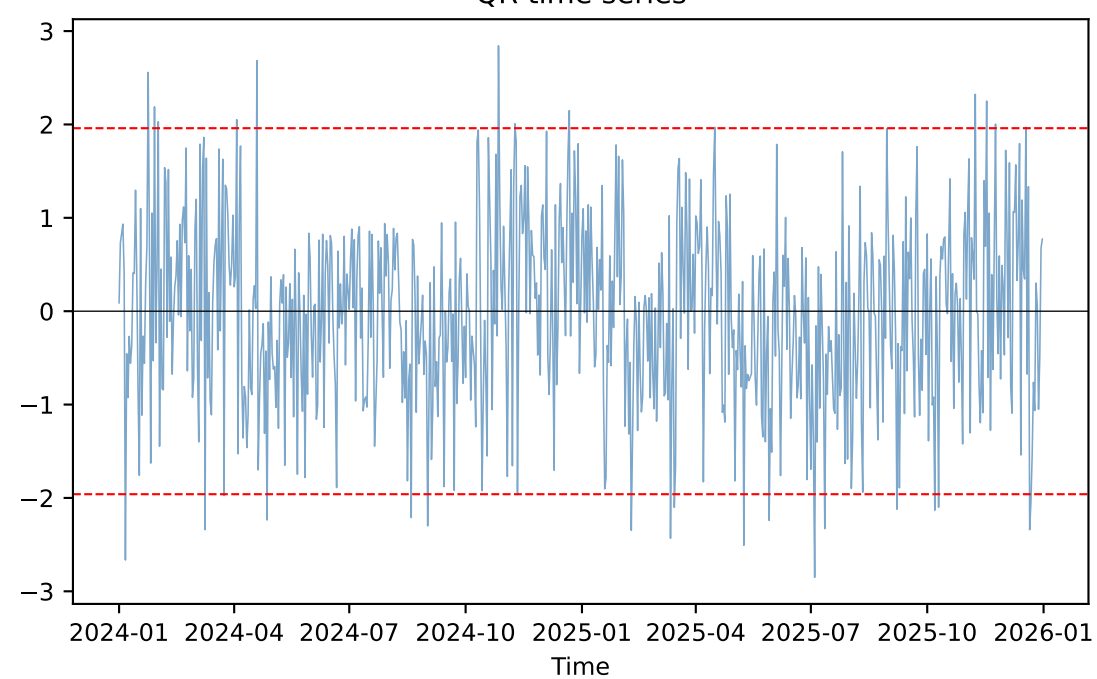
Normal QQ plot



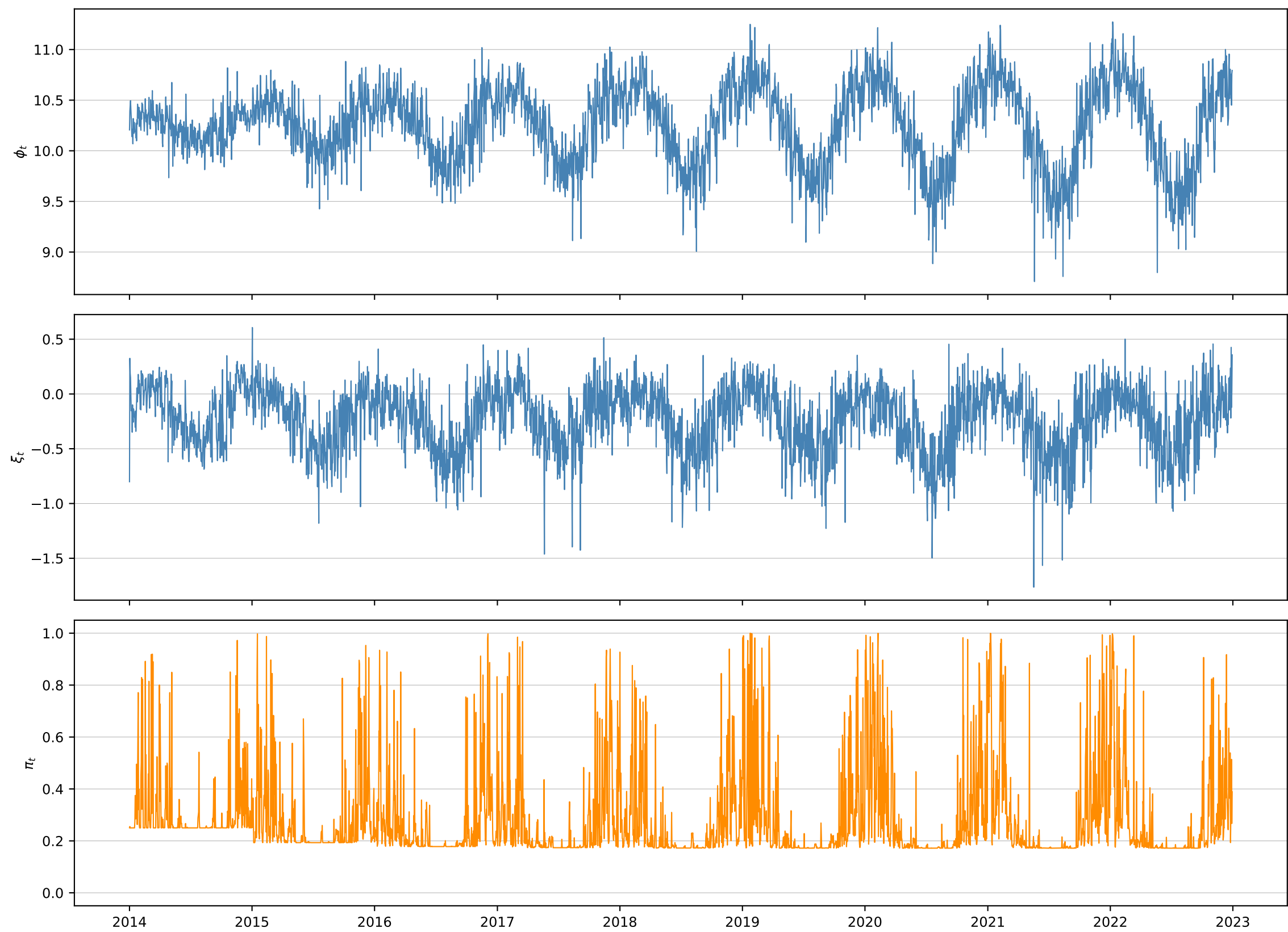
ACF of QR



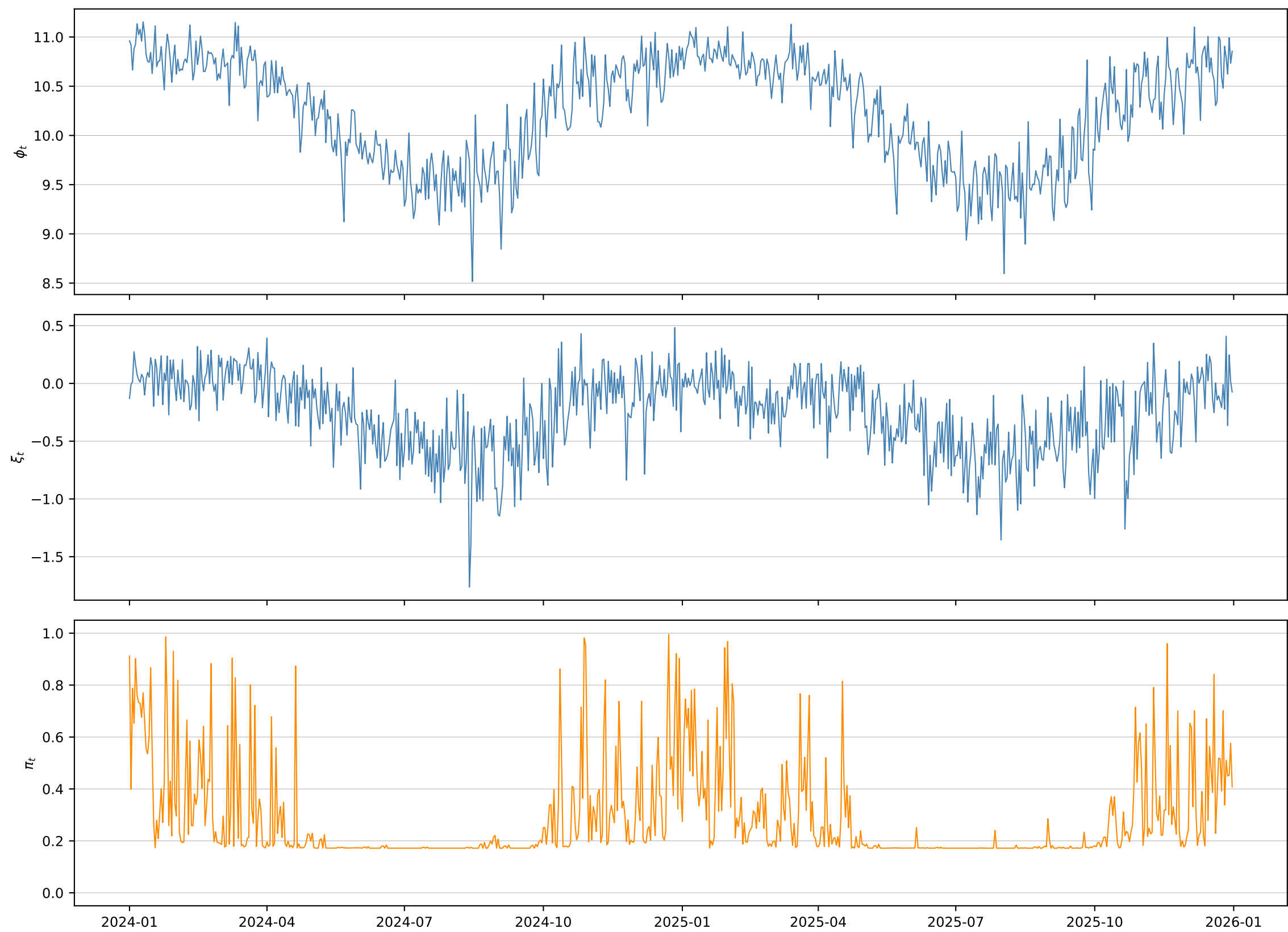
QR time series



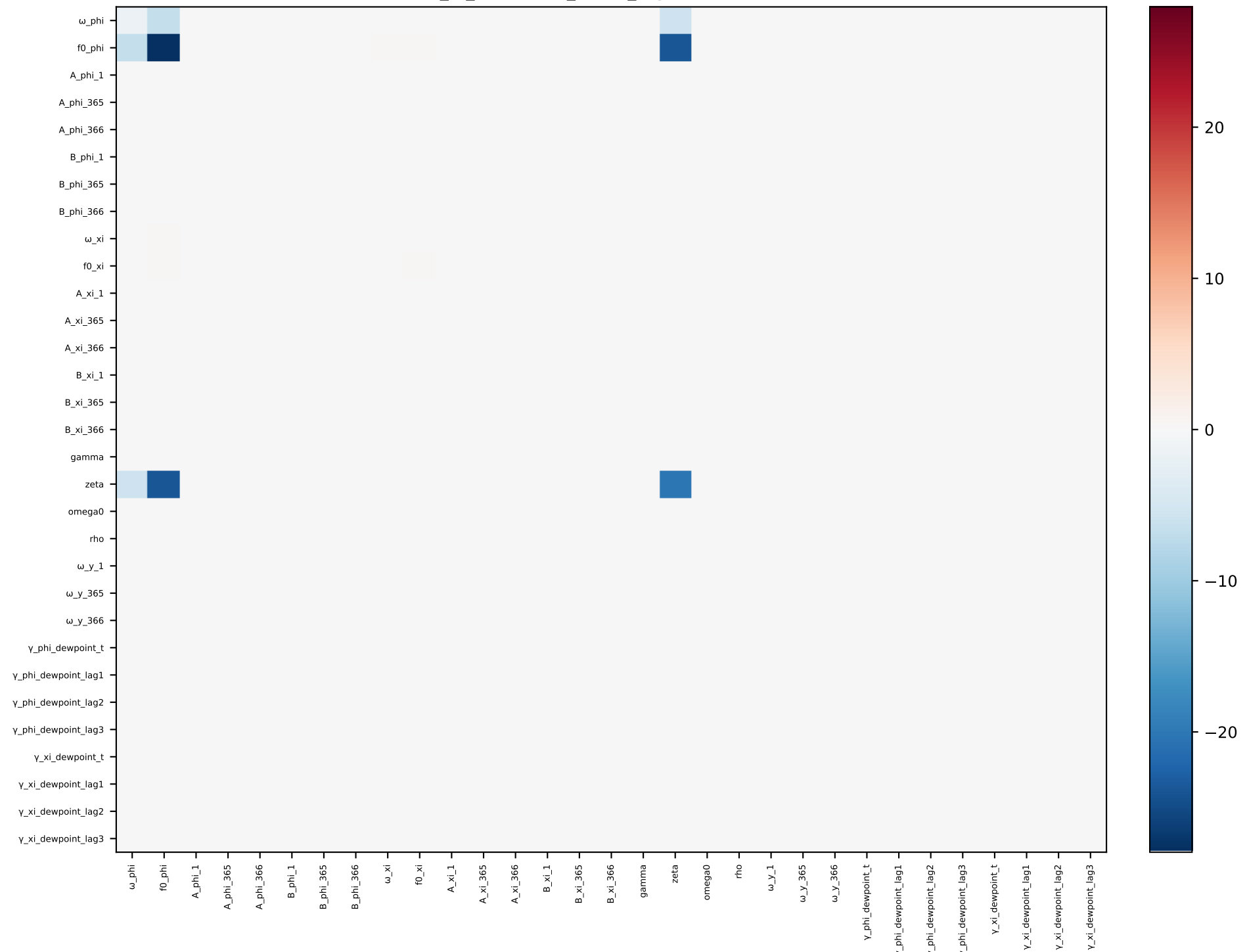
phi_xi_dewpoint_short_lags — filtered states (IS)



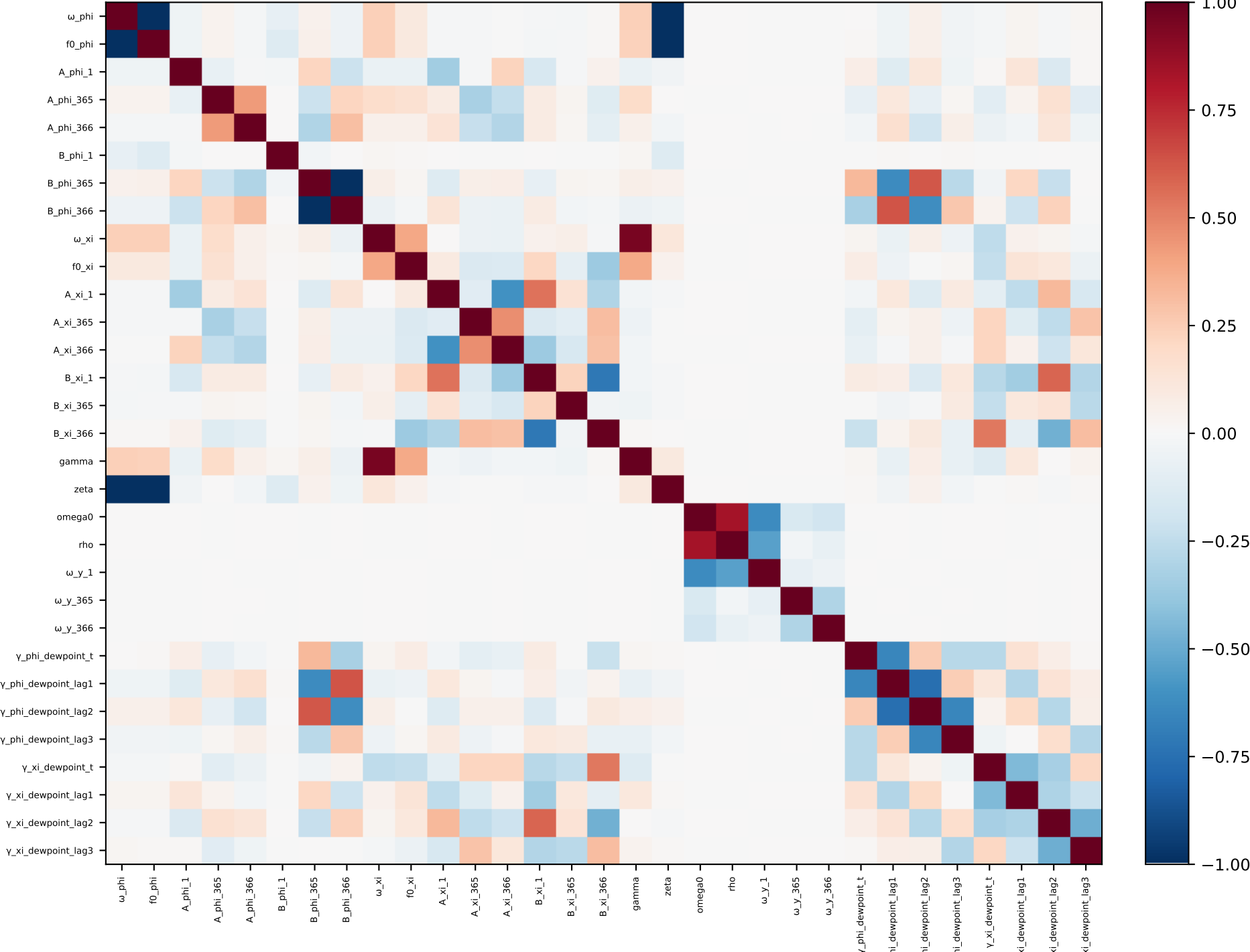
phi_xi_dewpoint_short_lags — filtered states (OOS)



phi_xi_dewpoint_short_lags – covariance

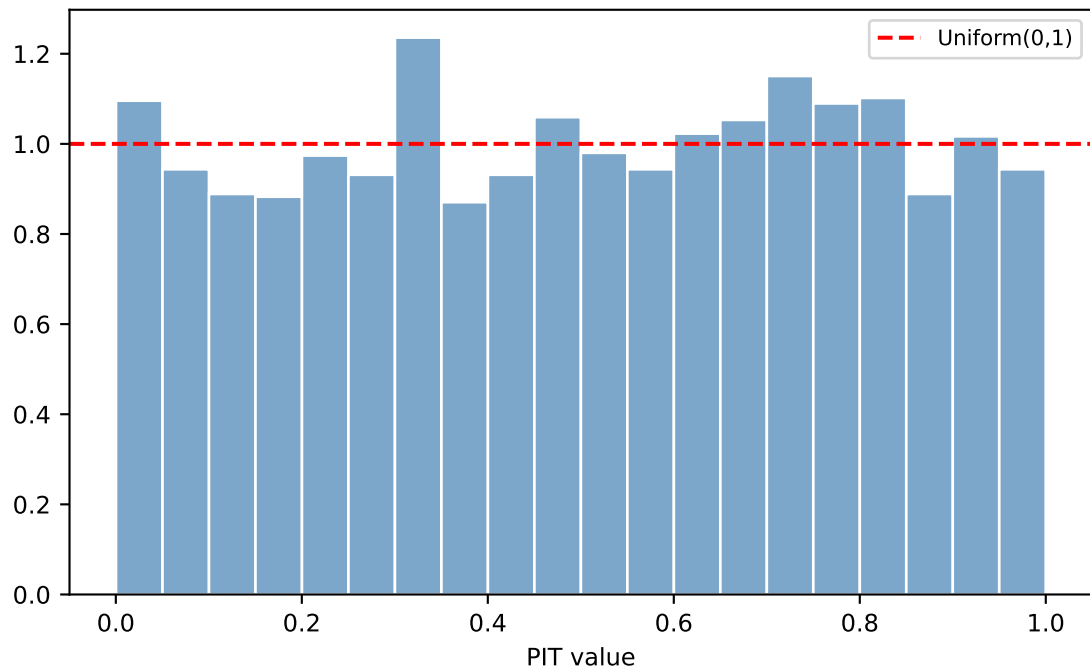


phi_xi_dewpoint_short_lags – correlation

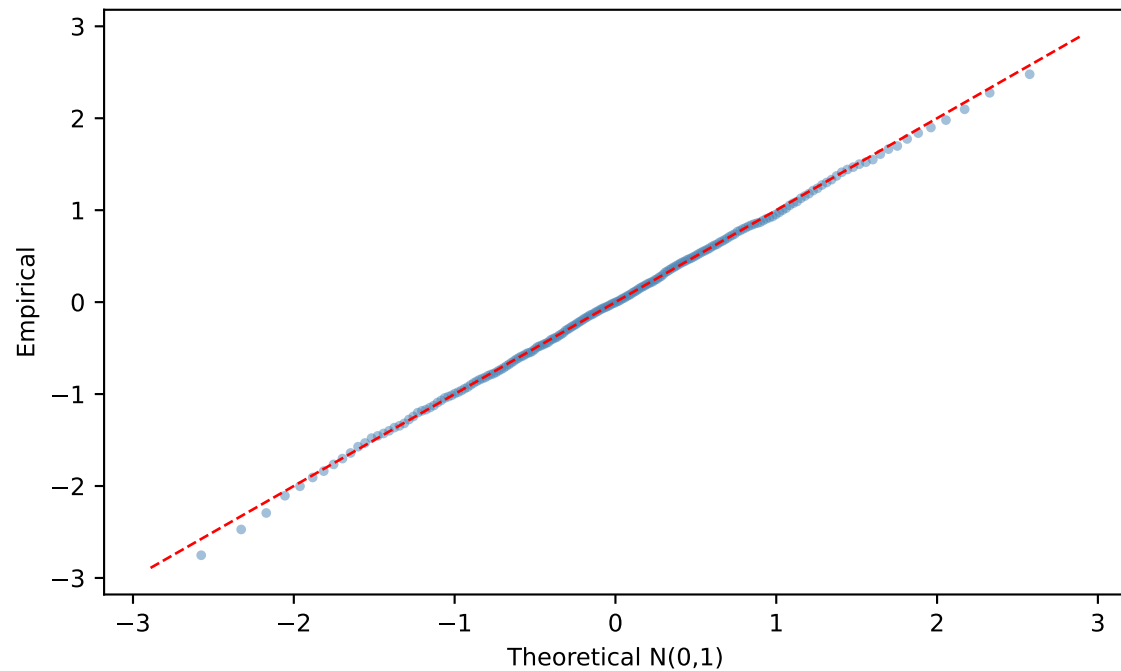


phi_xi_dewpoint_seasonal_lags — IS diagnostics

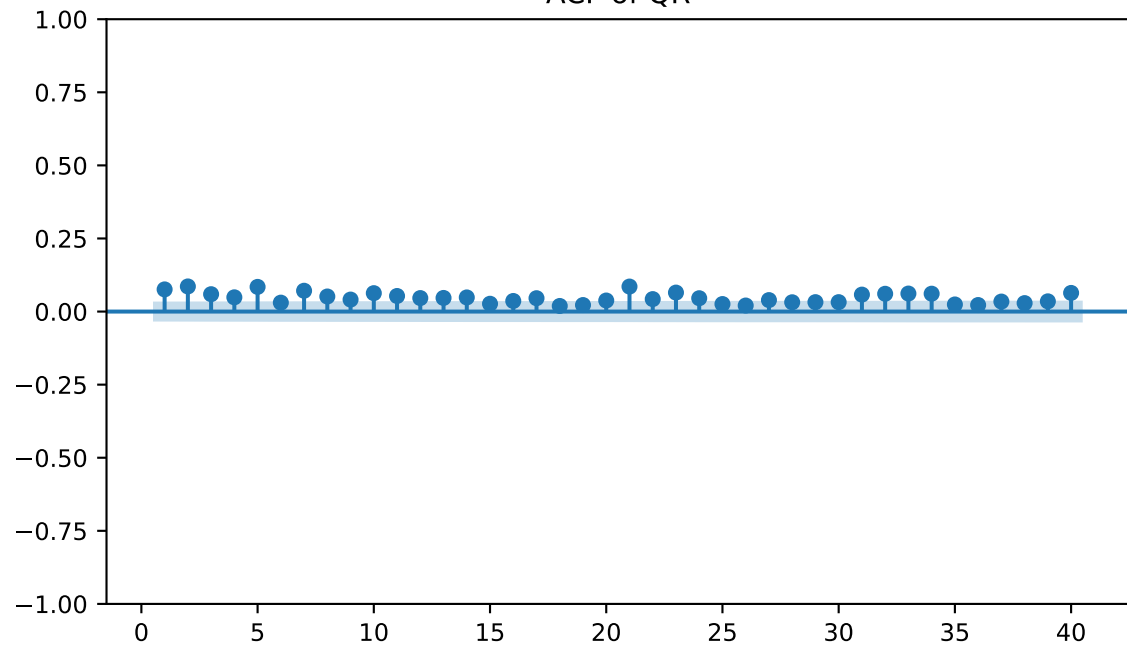
PIT histogram



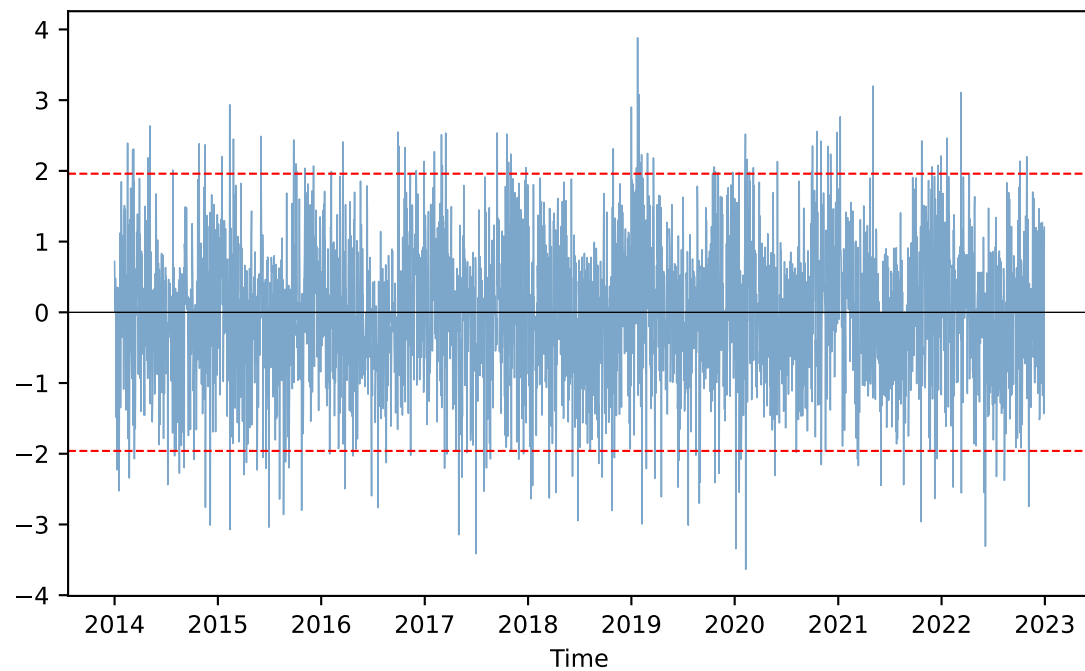
Normal QQ plot



ACF of QR

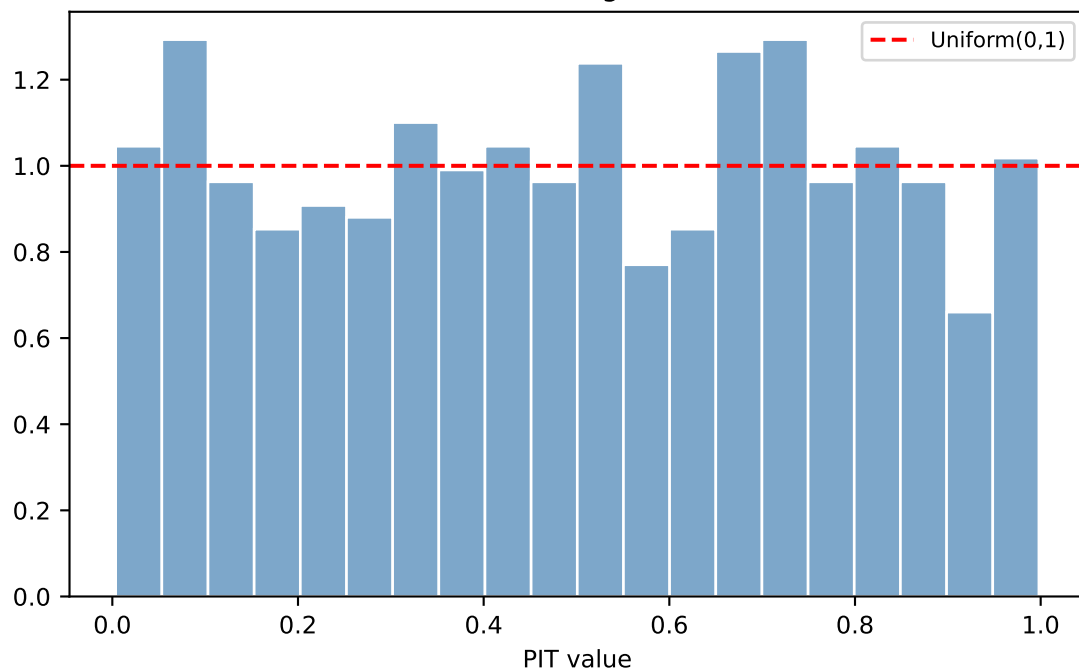


QR time series

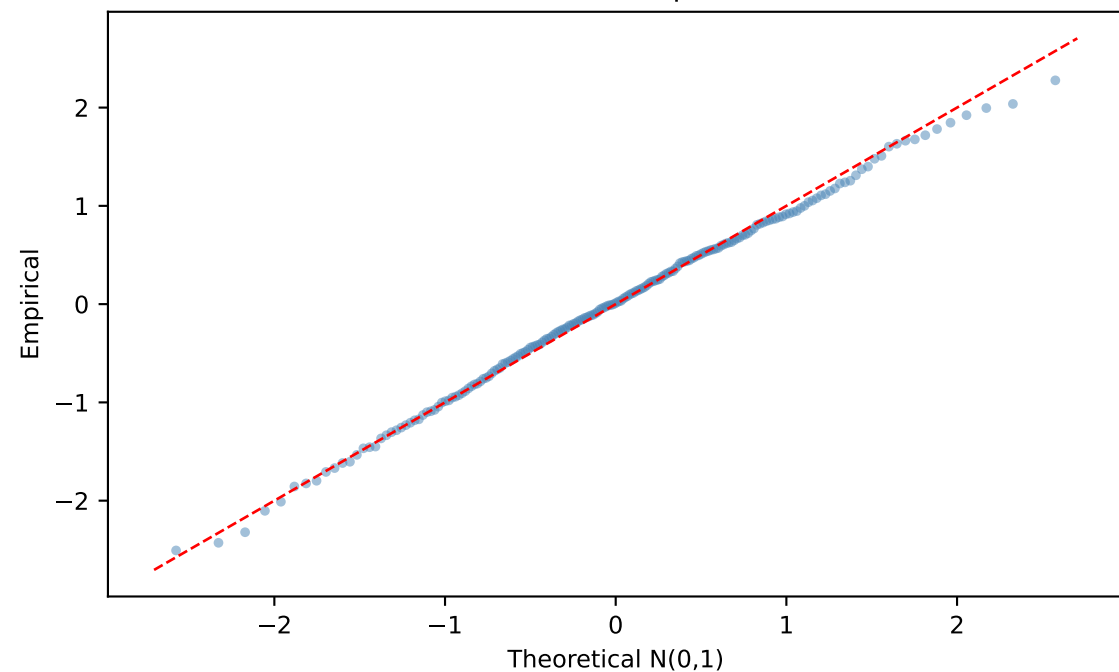


phi_xi_dewpoint_seasonal_lags — OOS diagnostics

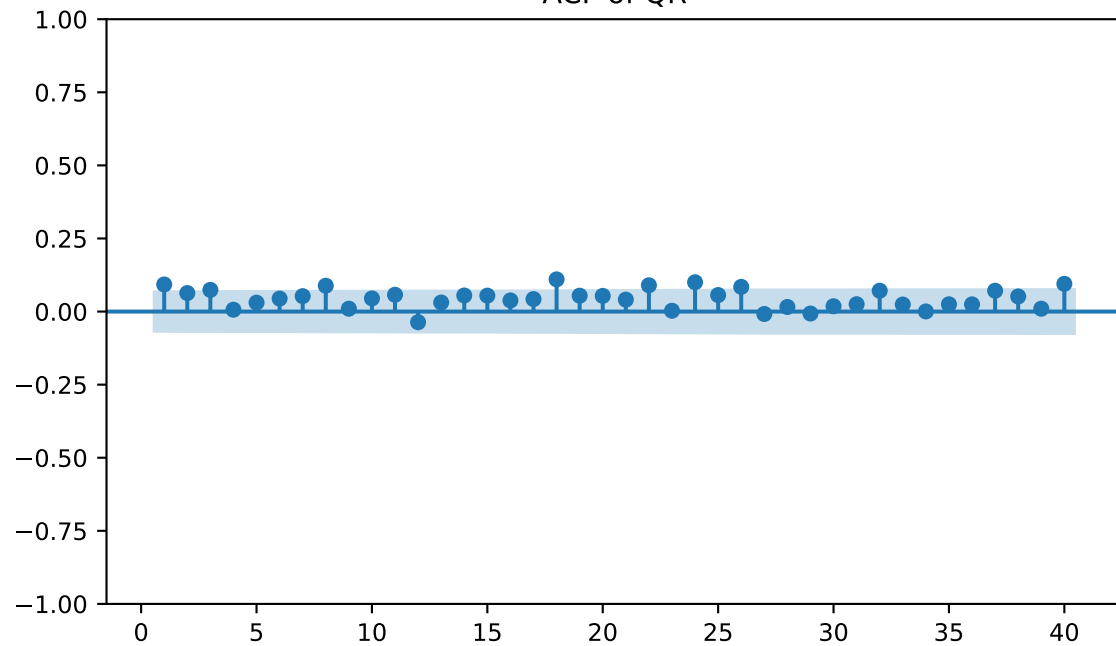
PIT histogram



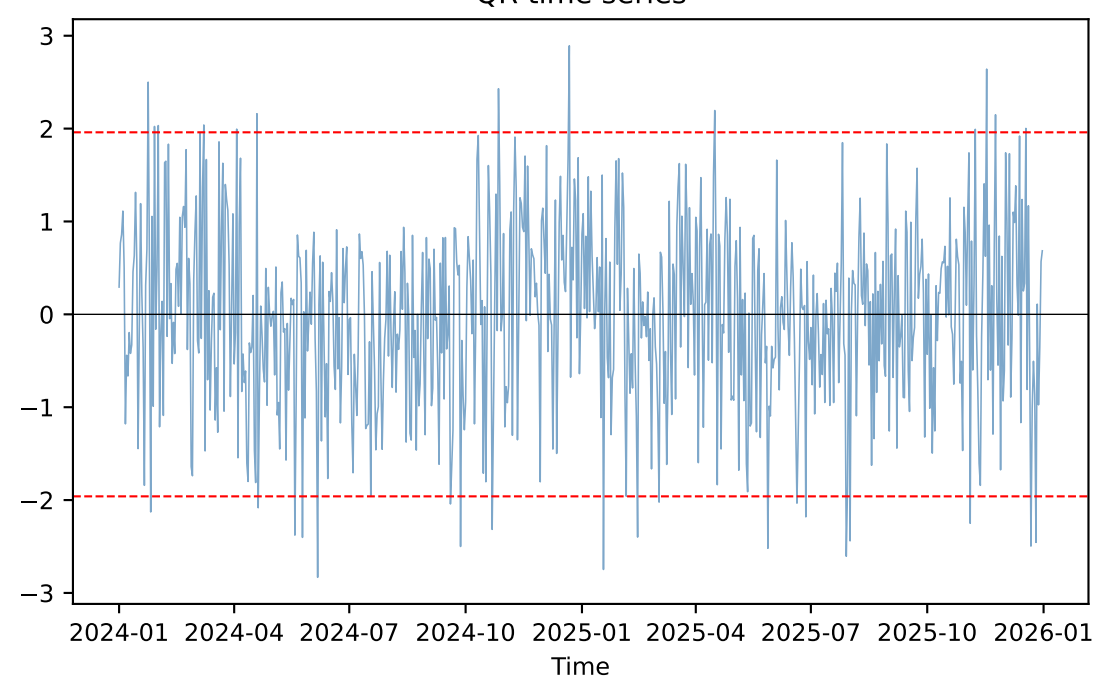
Normal QQ plot



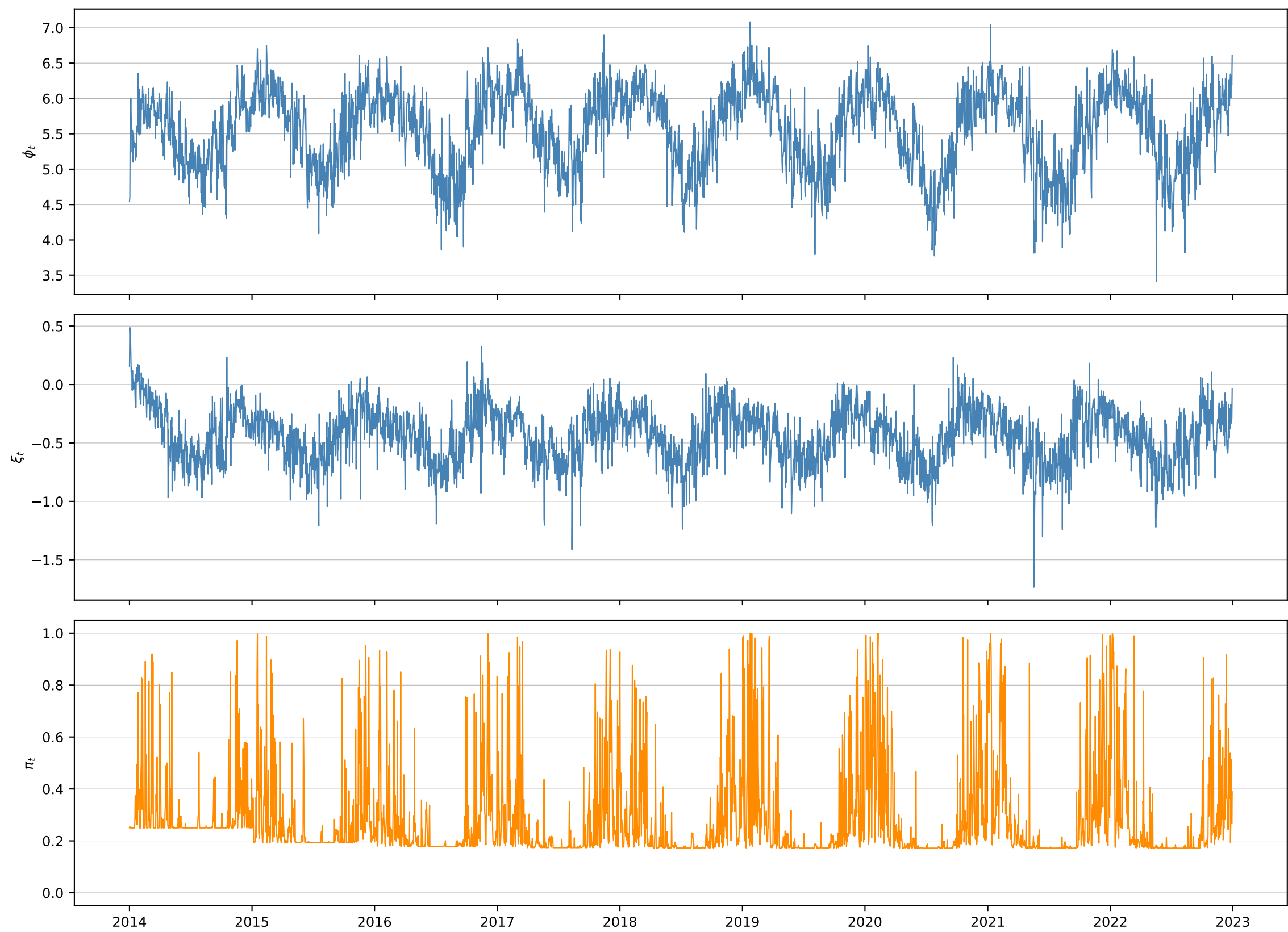
ACF of QR



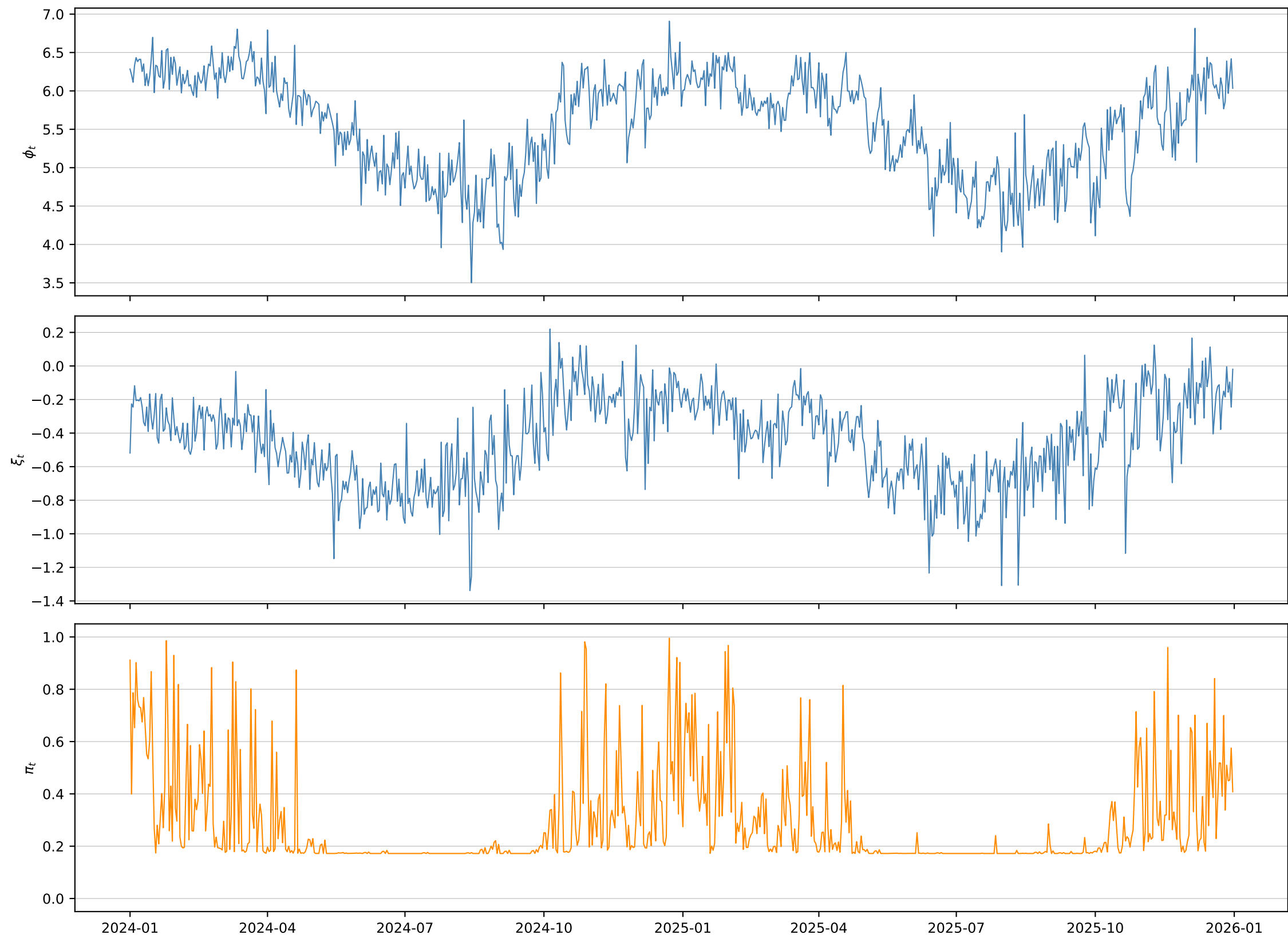
QR time series



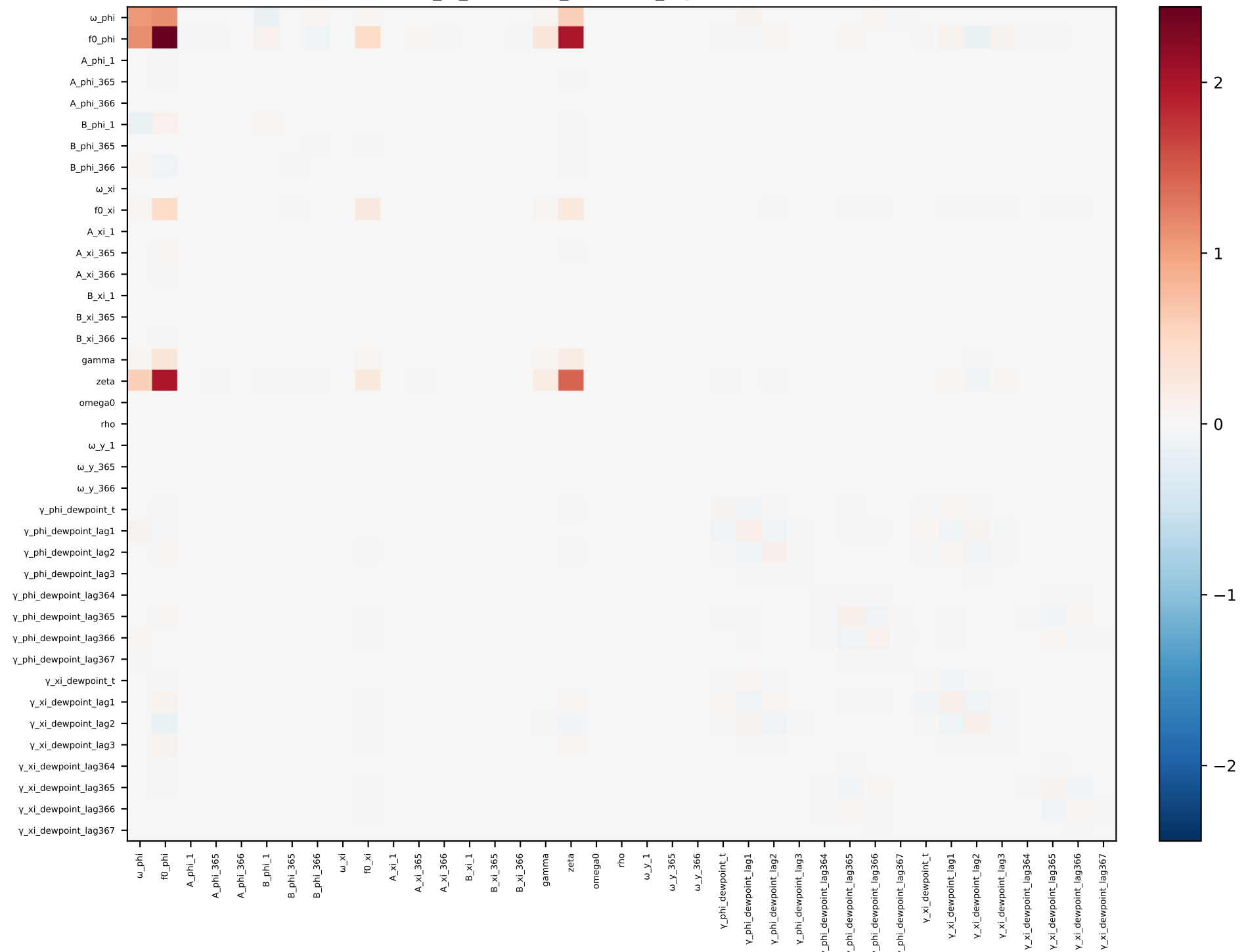
phi_xi_dewpoint_seasonal_lags — filtered states (IS)



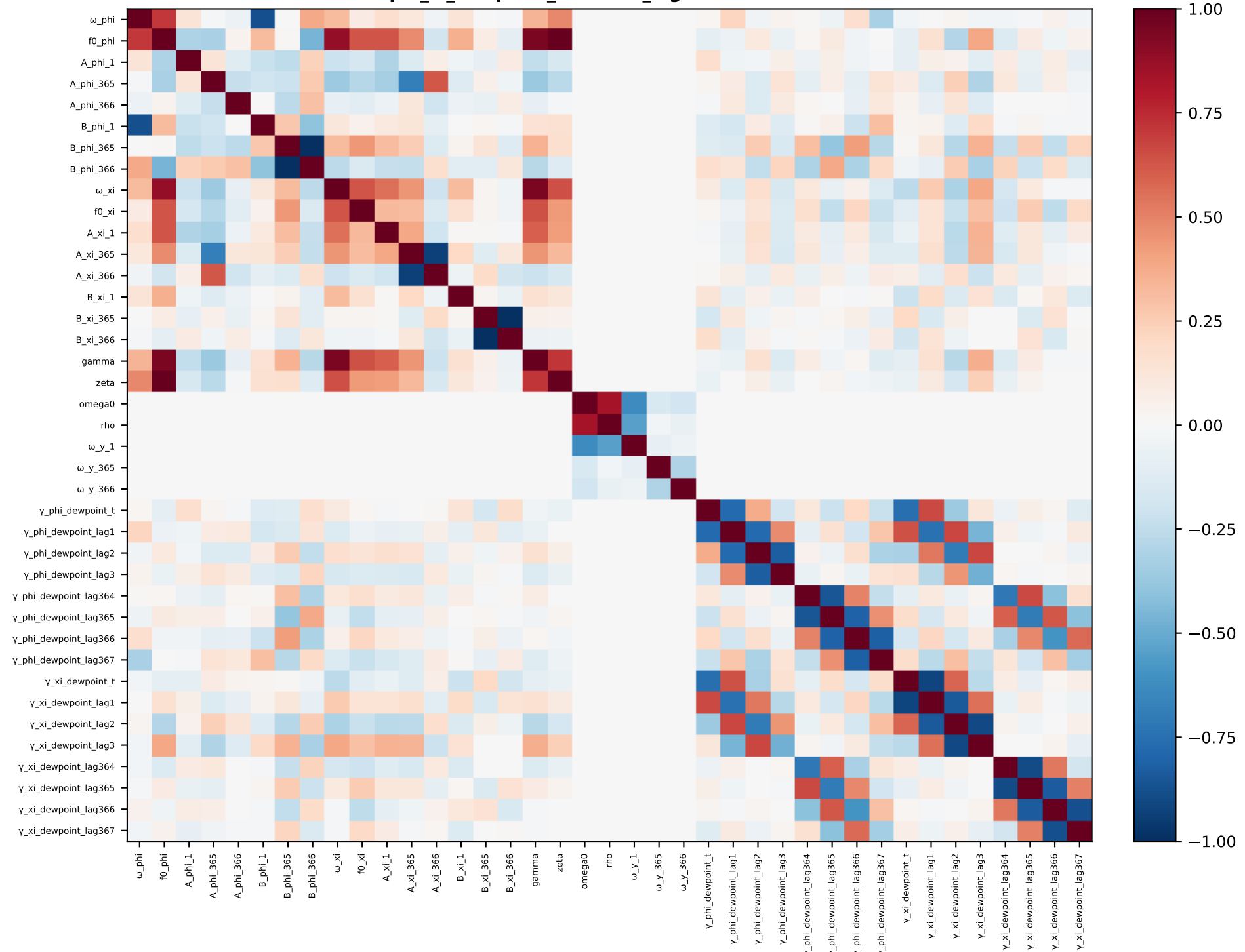
phi_xi_dewpoint_seasonal_lags — filtered states (OOS)



phi_xi_dewpoint_seasonal_lags — covariance

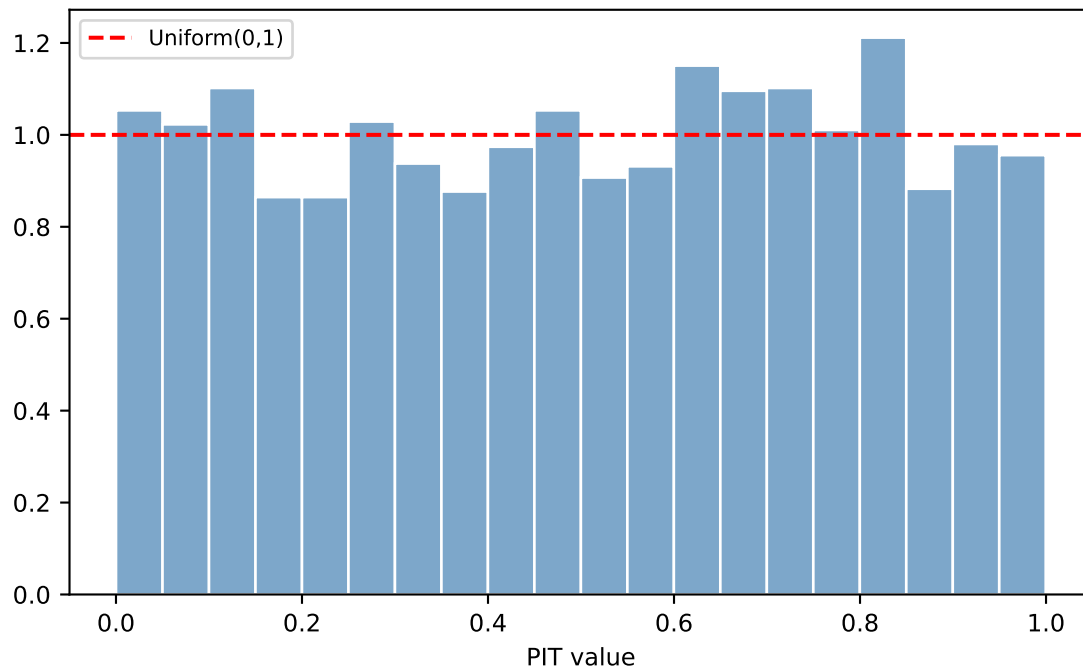


phi_xi_dewpoint_seasonal_lags — correlation

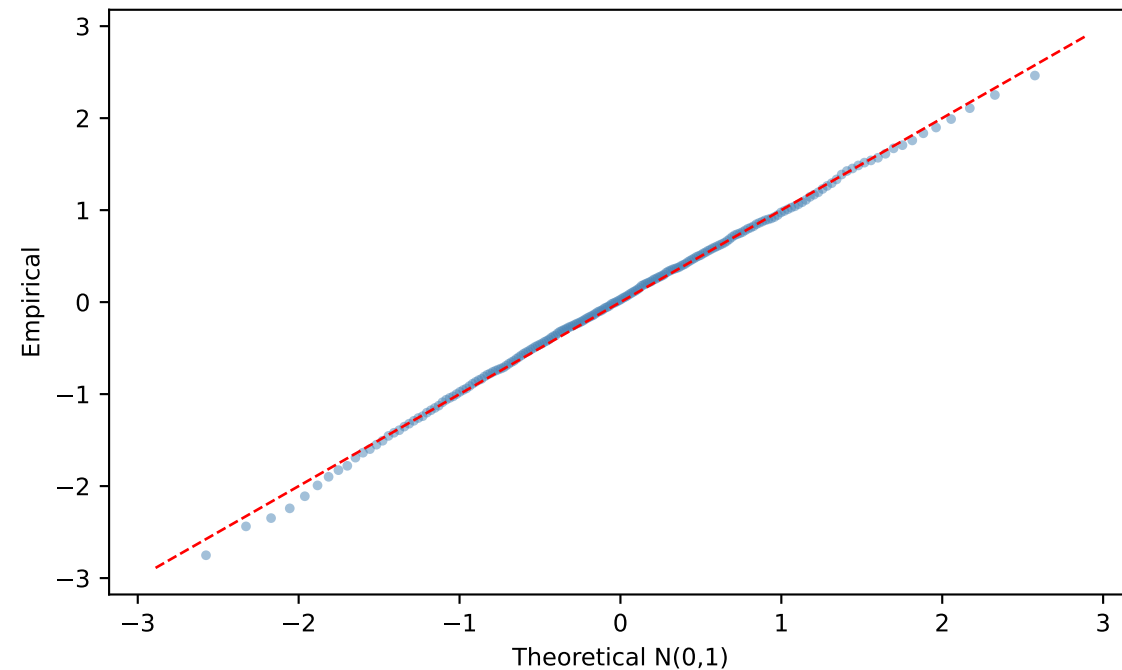


phi_xi_temperature_t — IS diagnostics

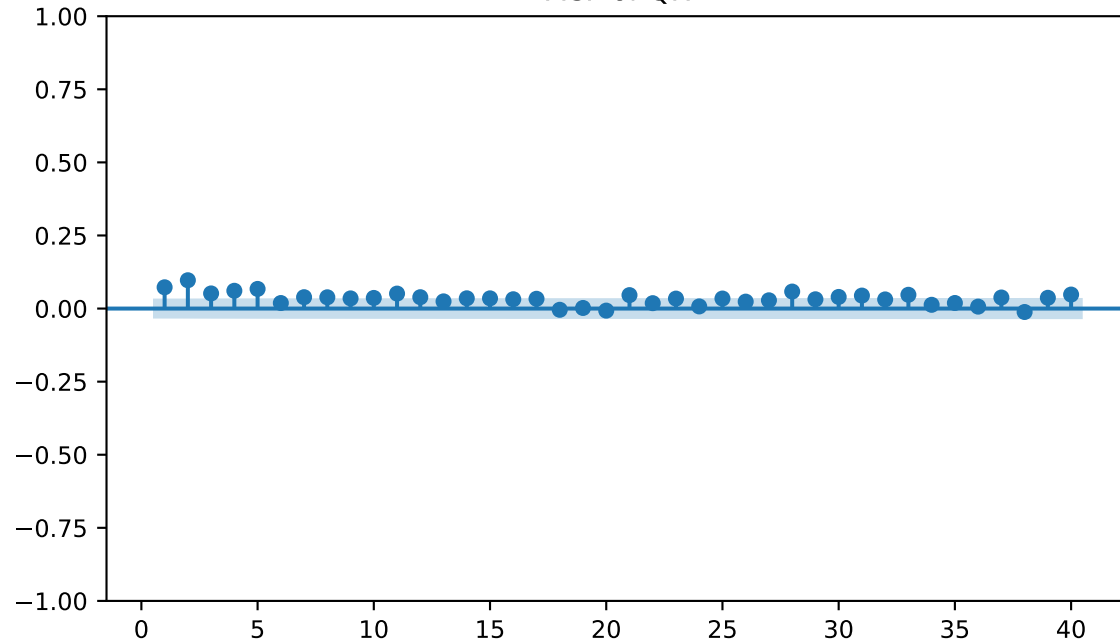
PIT histogram



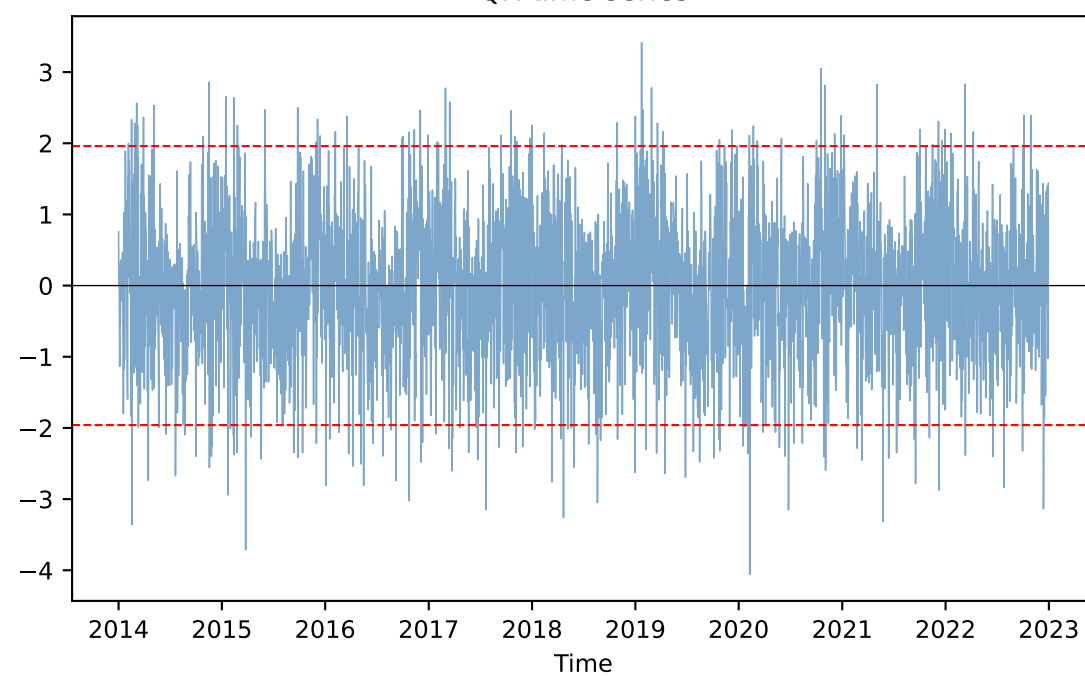
Normal QQ plot



ACF of QR

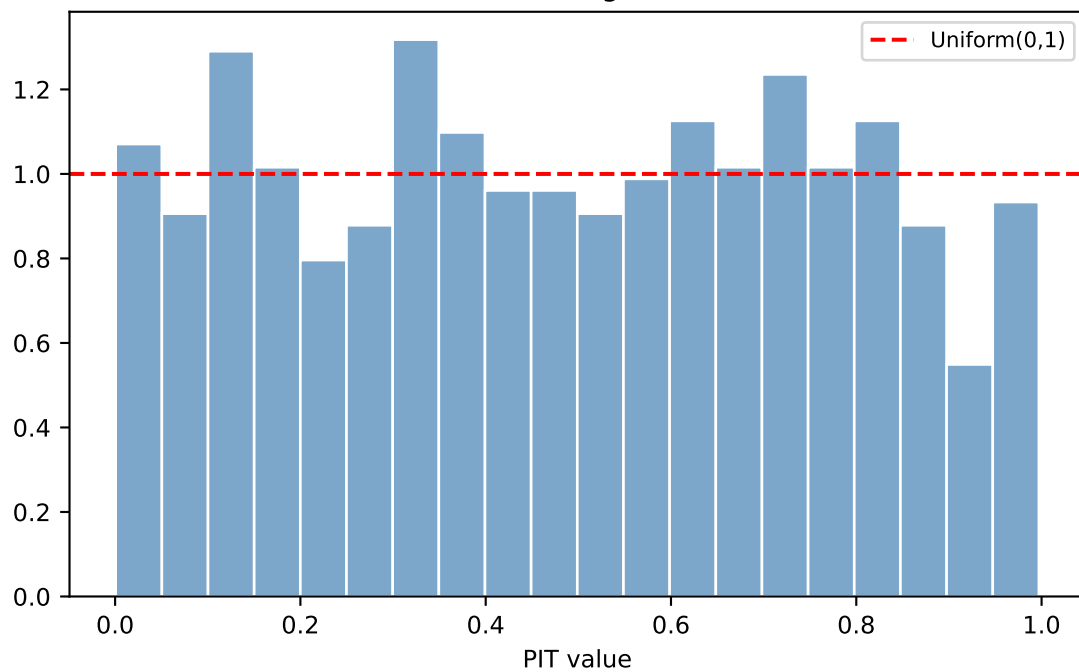


QR time series

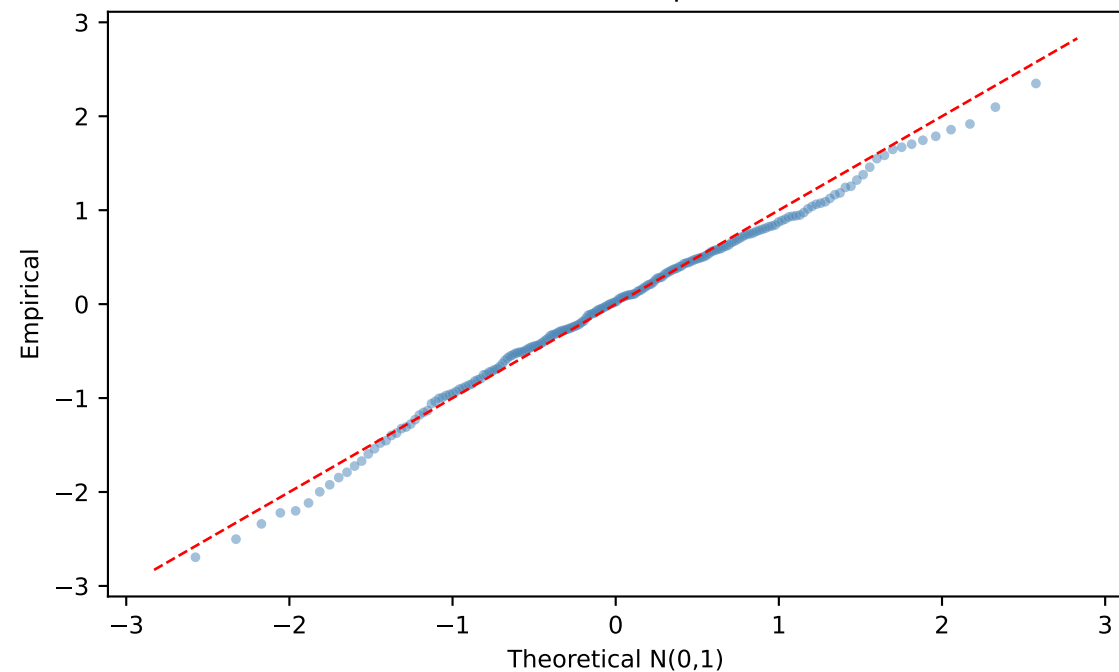


phi_xi_temperature_t — OOS diagnostics

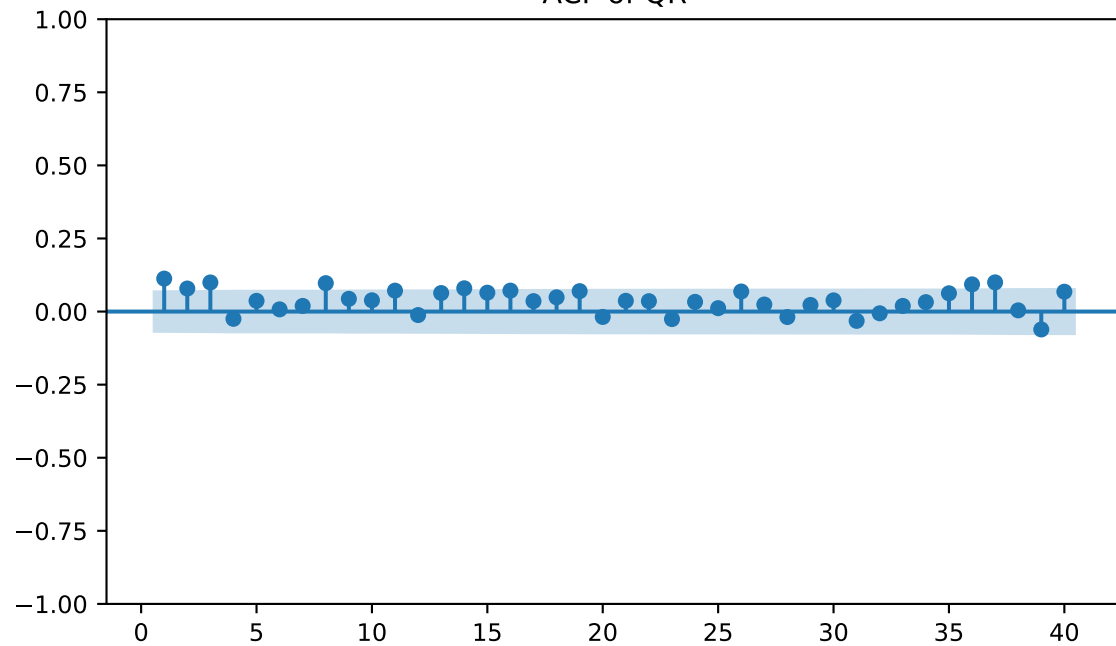
PIT histogram



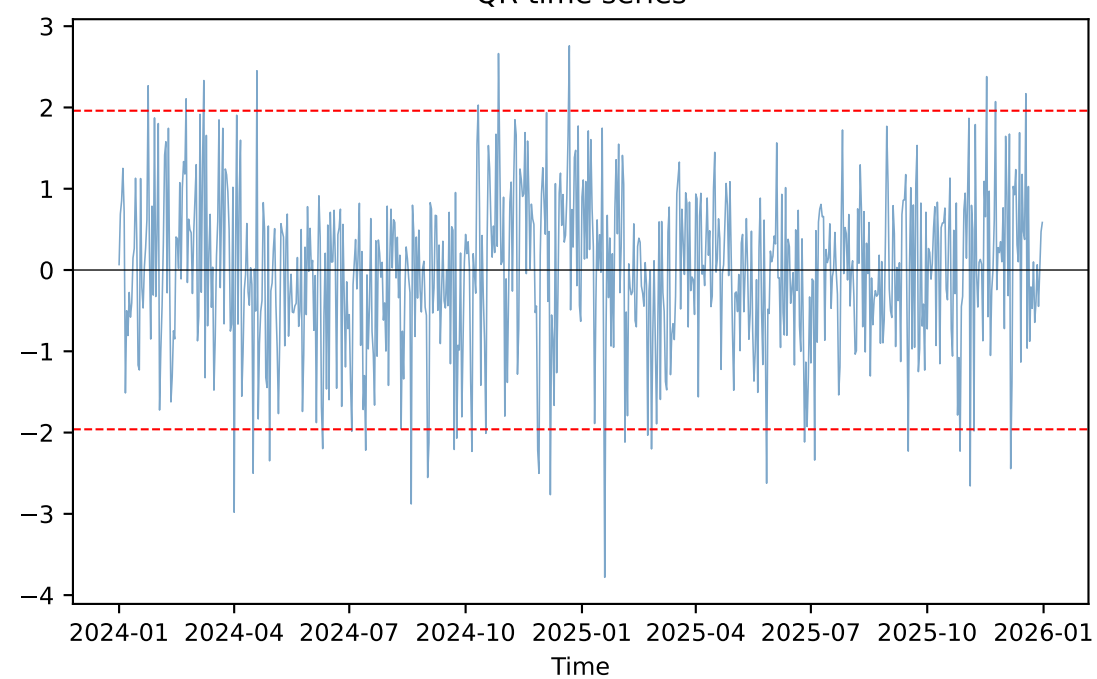
Normal QQ plot



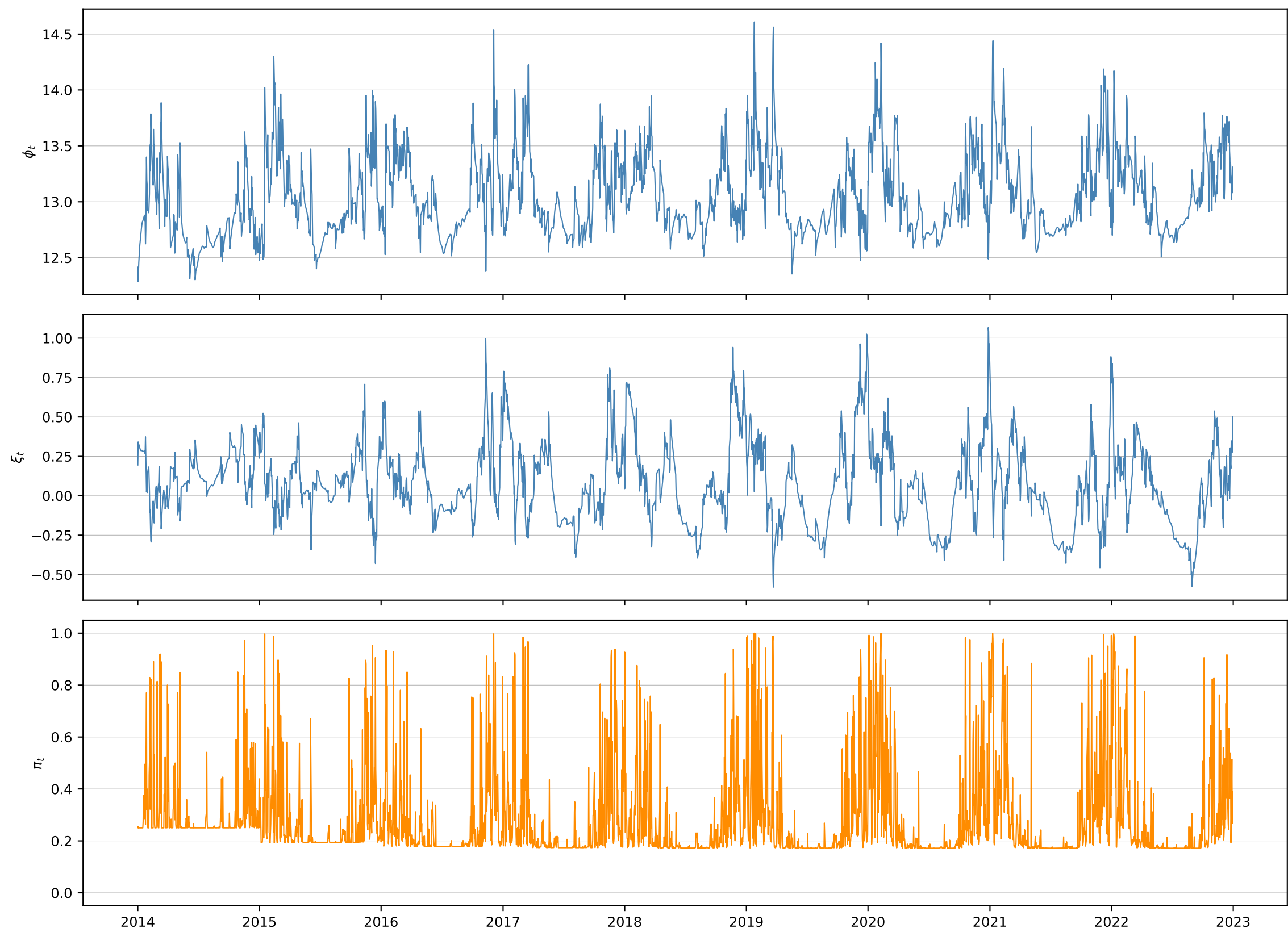
ACF of QR



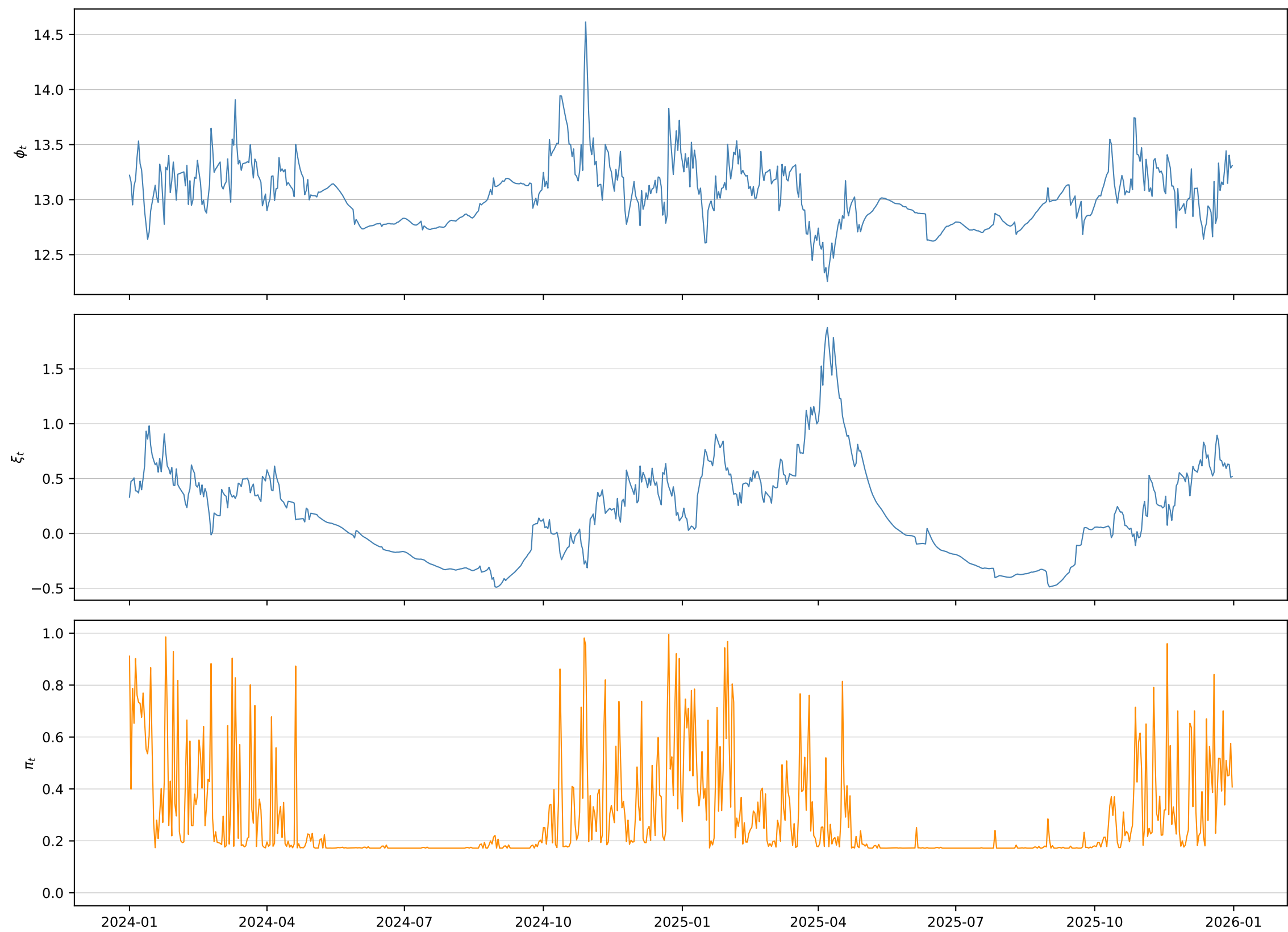
QR time series



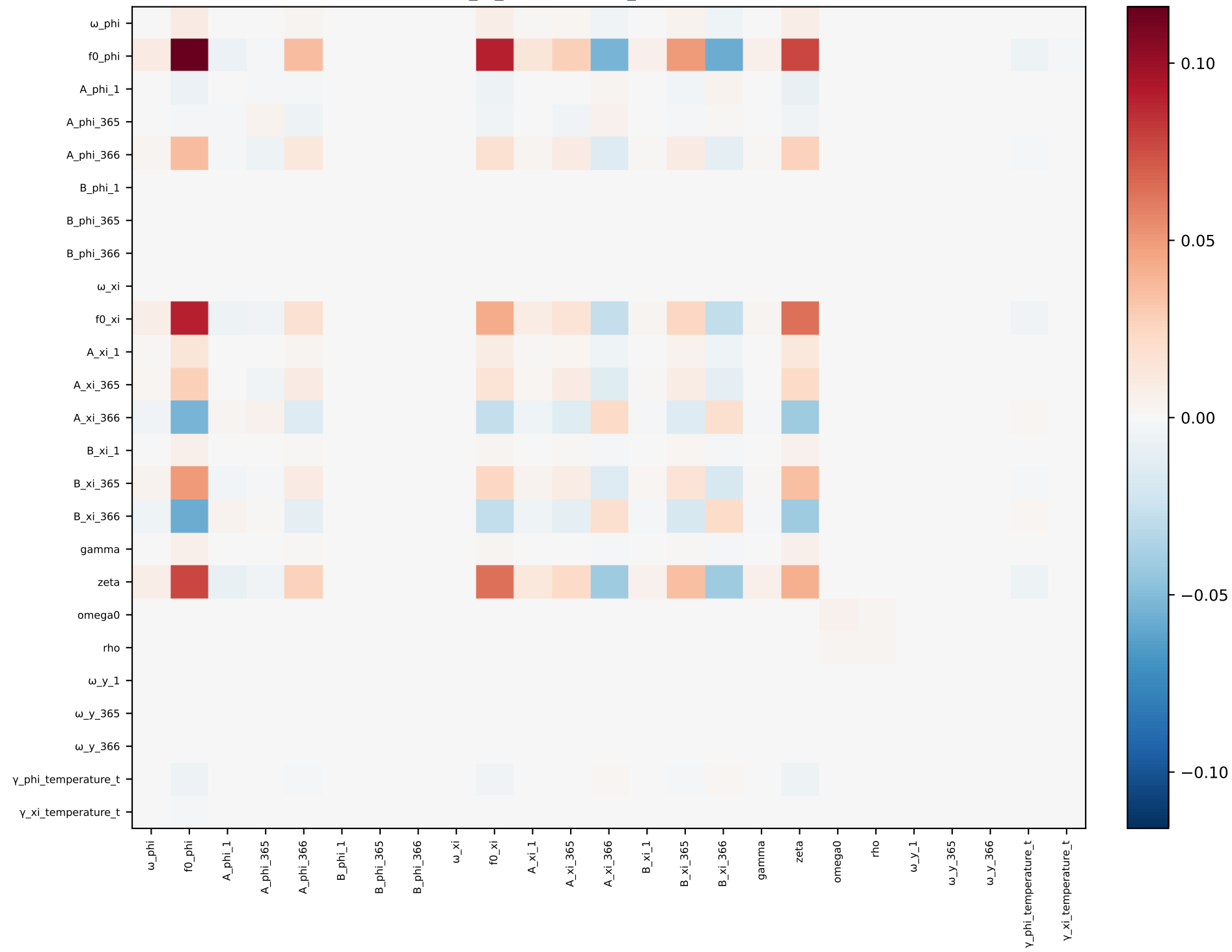
phi_xi_temperature_t — filtered states (IS)



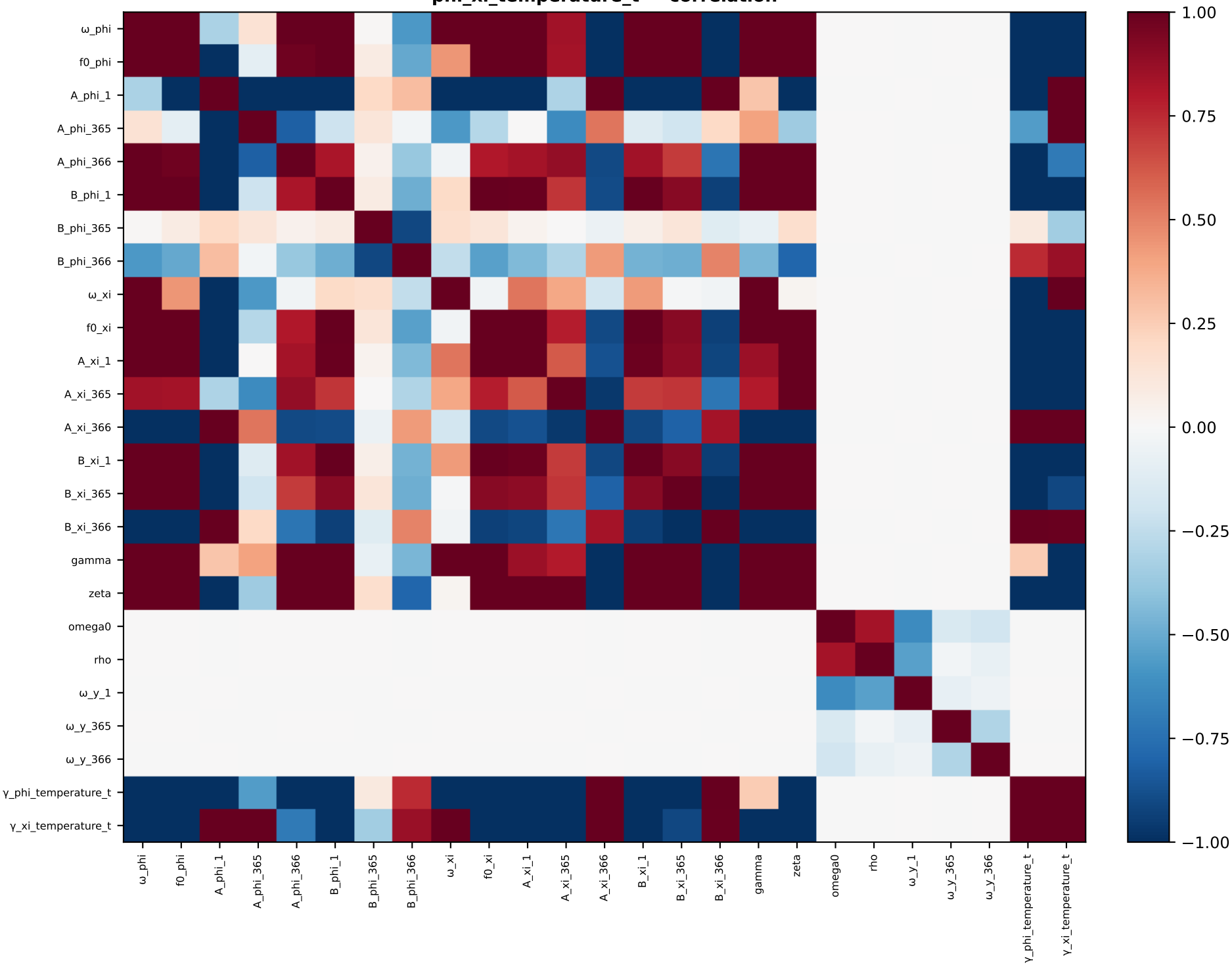
phi_xi_temperature_t — filtered states (OOS)



phi xi temperature t – covariance

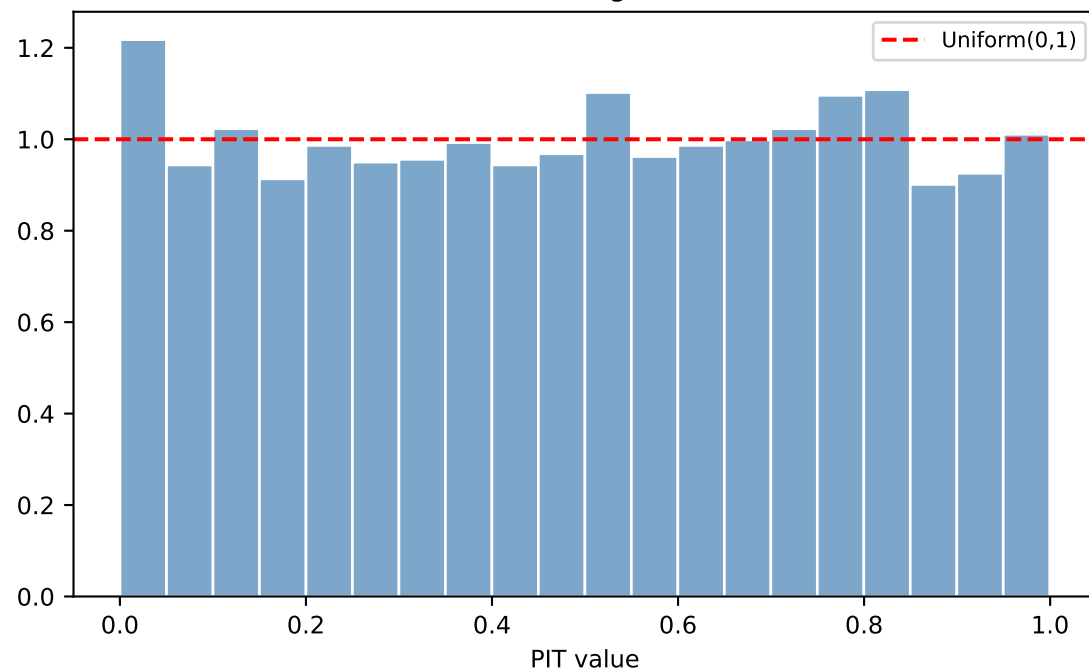


phi_xi_temperature_t — correlation

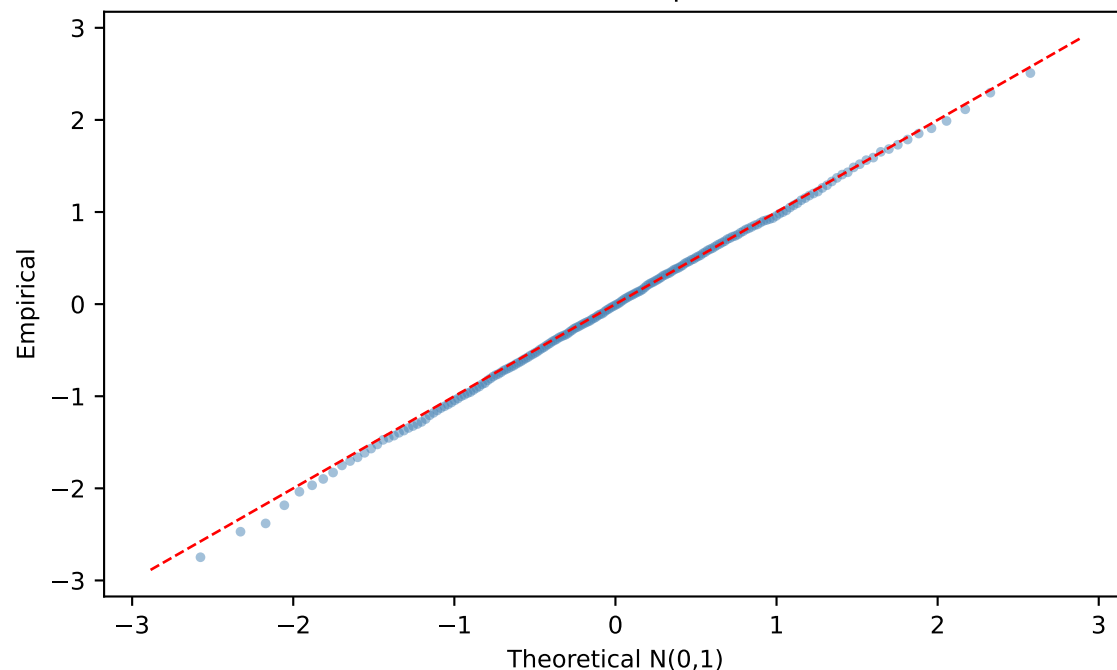


phi_xi_temperature_lag1 — IS diagnostics

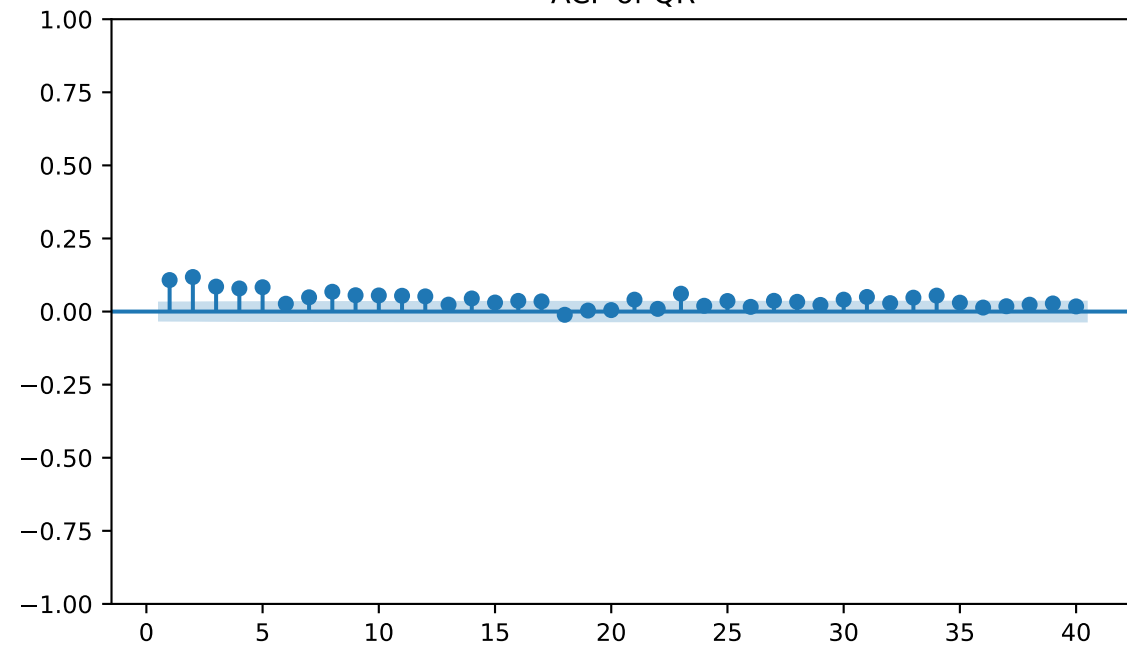
PIT histogram



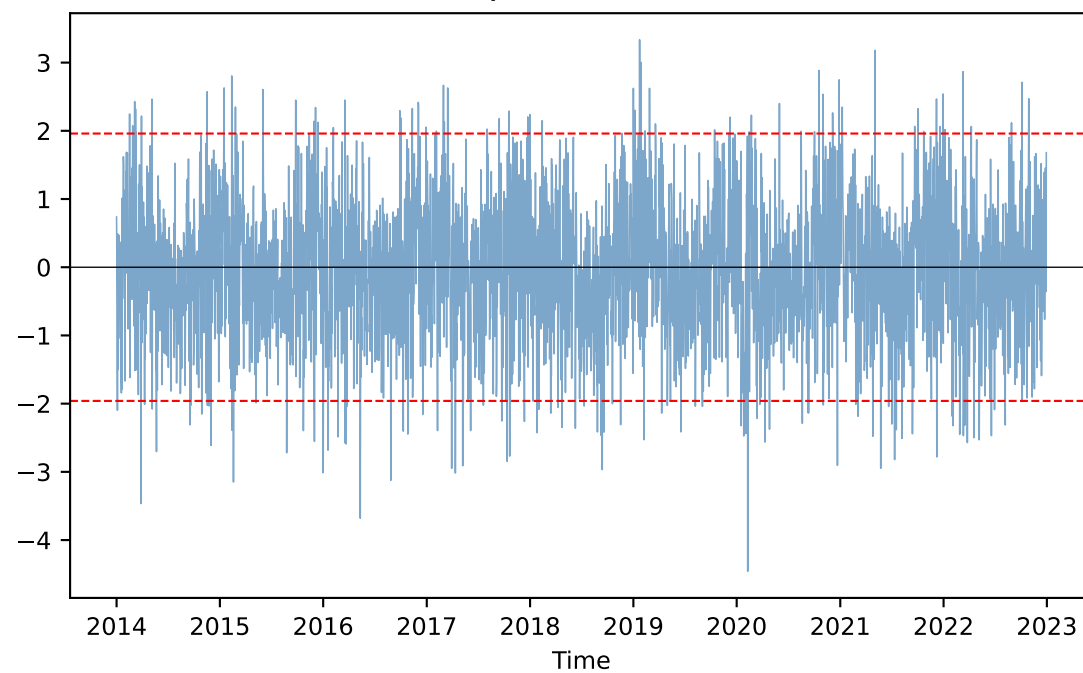
Normal QQ plot



ACF of QR

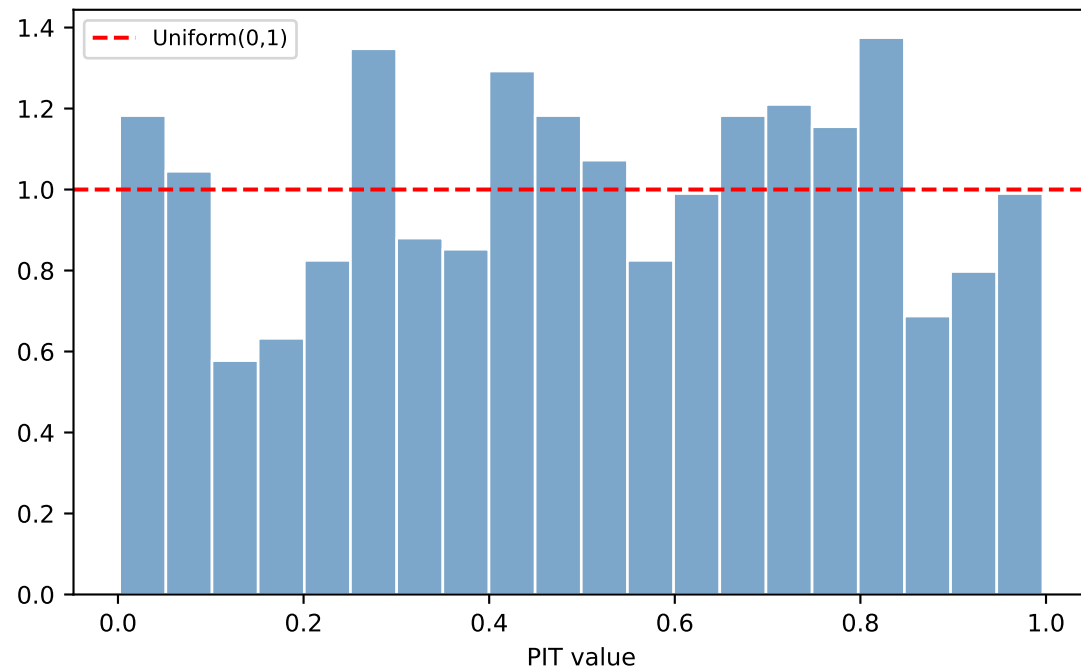


QR time series

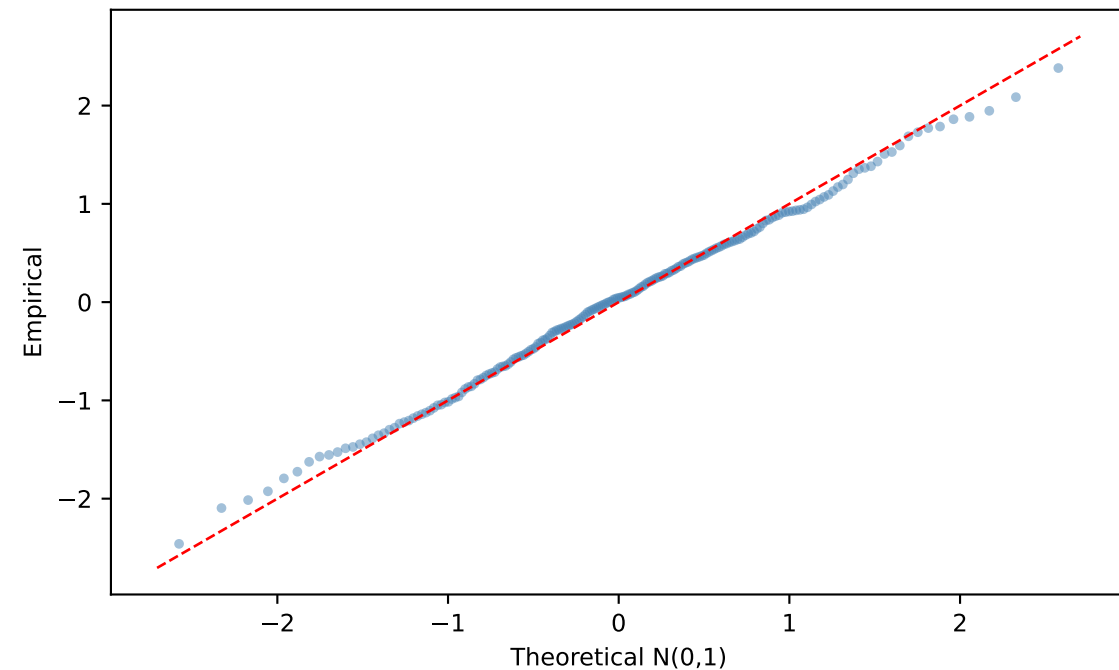


phi_xi_temperature_lag1 — OOS diagnostics

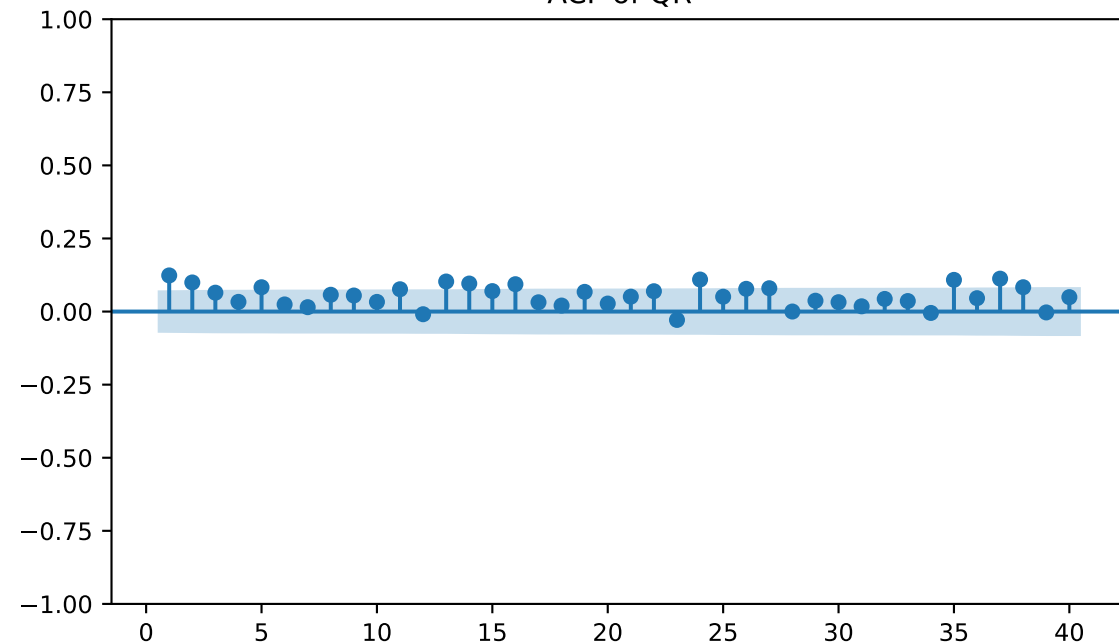
PIT histogram



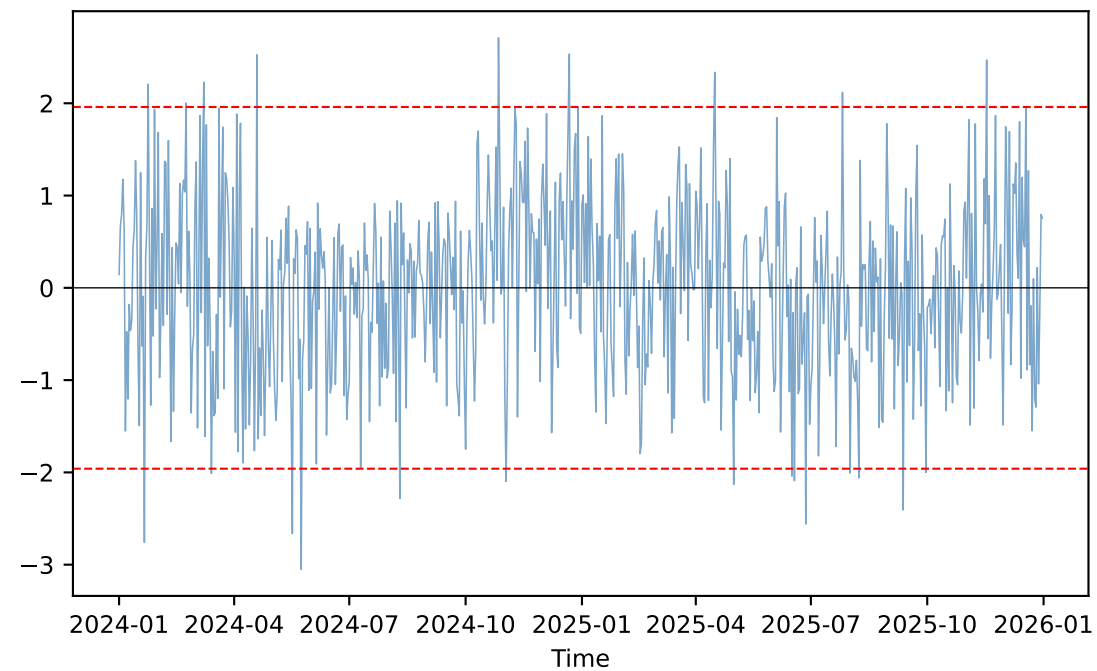
Normal QQ plot



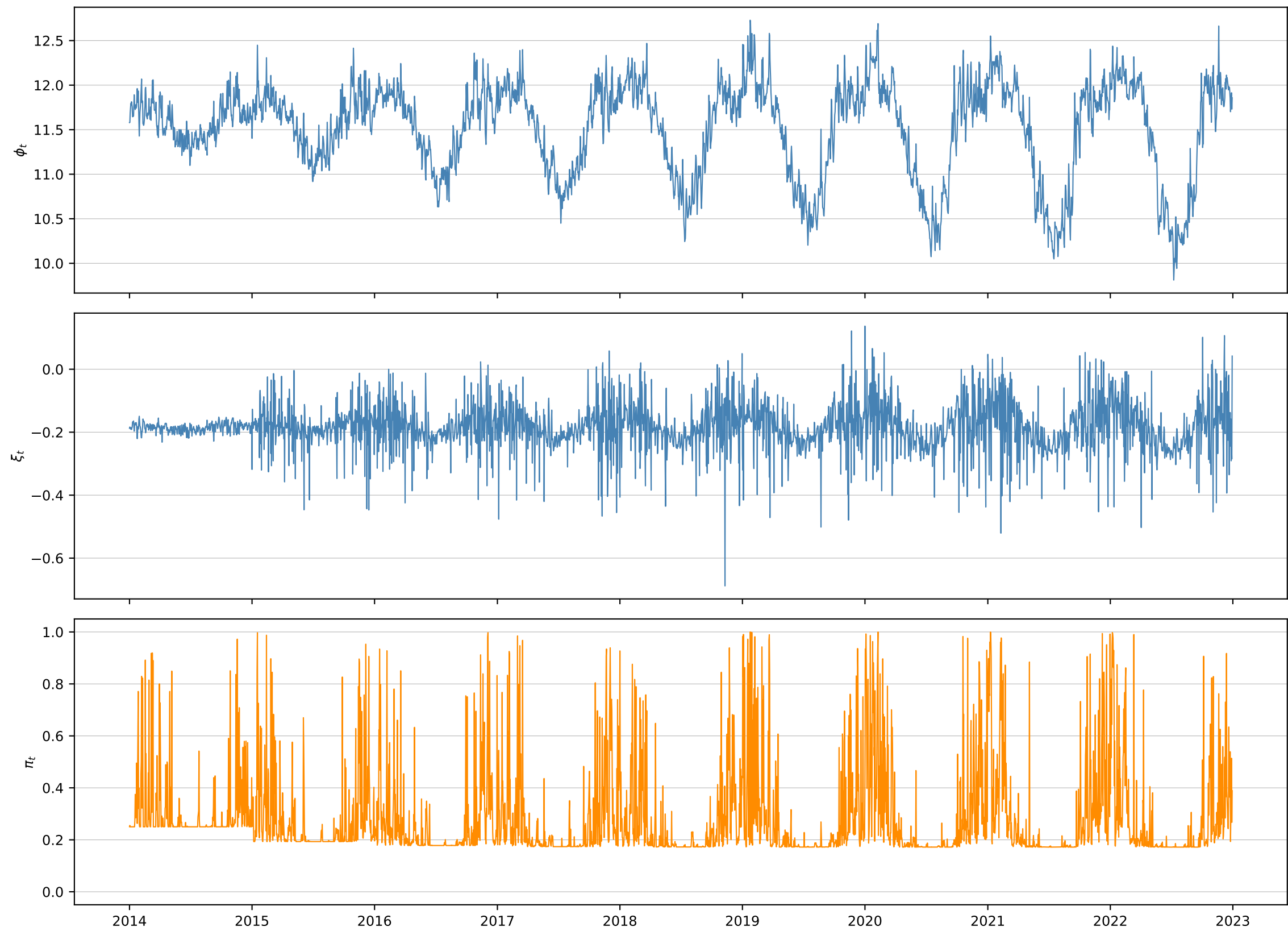
ACF of QR



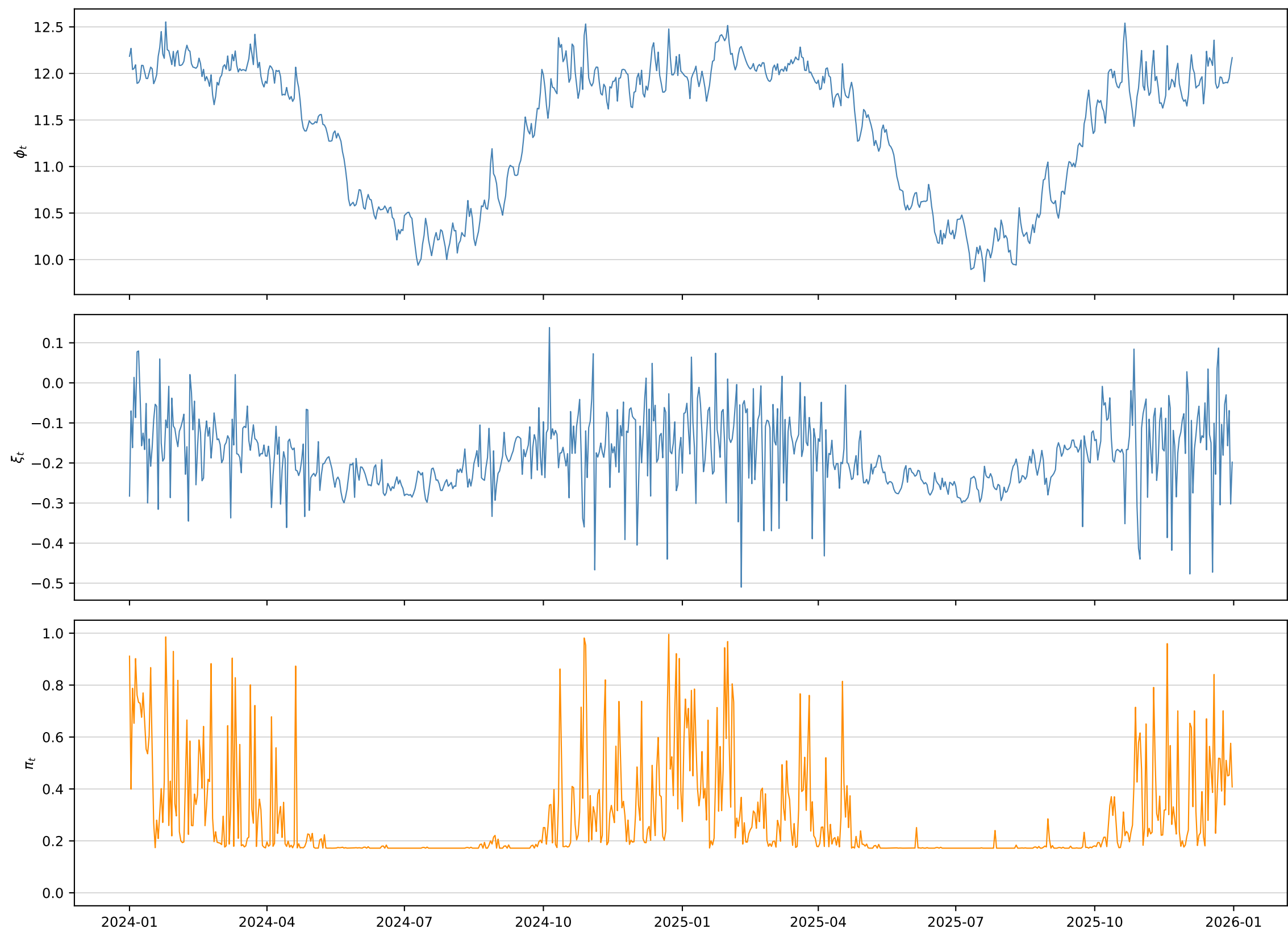
QR time series



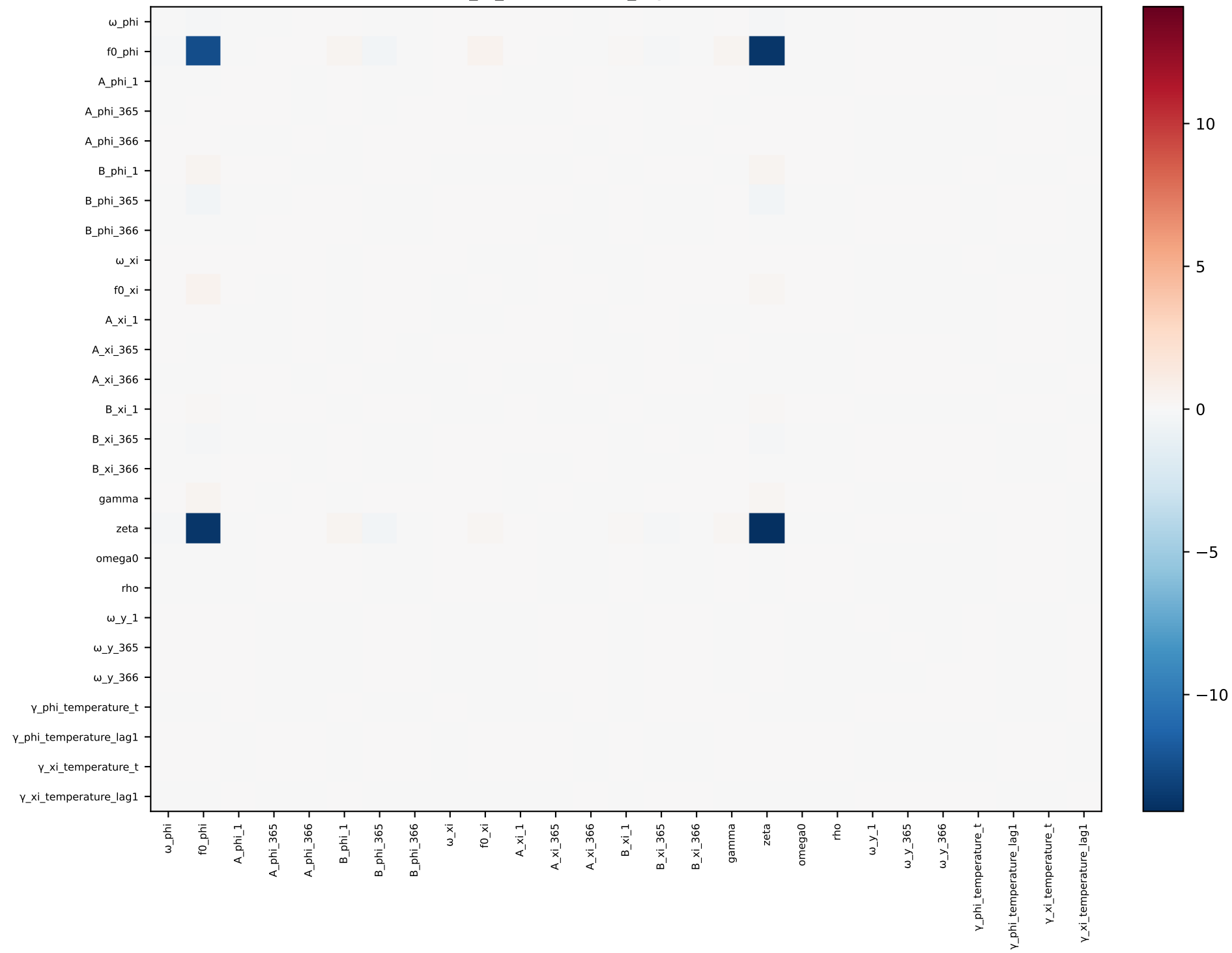
phi_xi_temperature_lag1 — filtered states (IS)



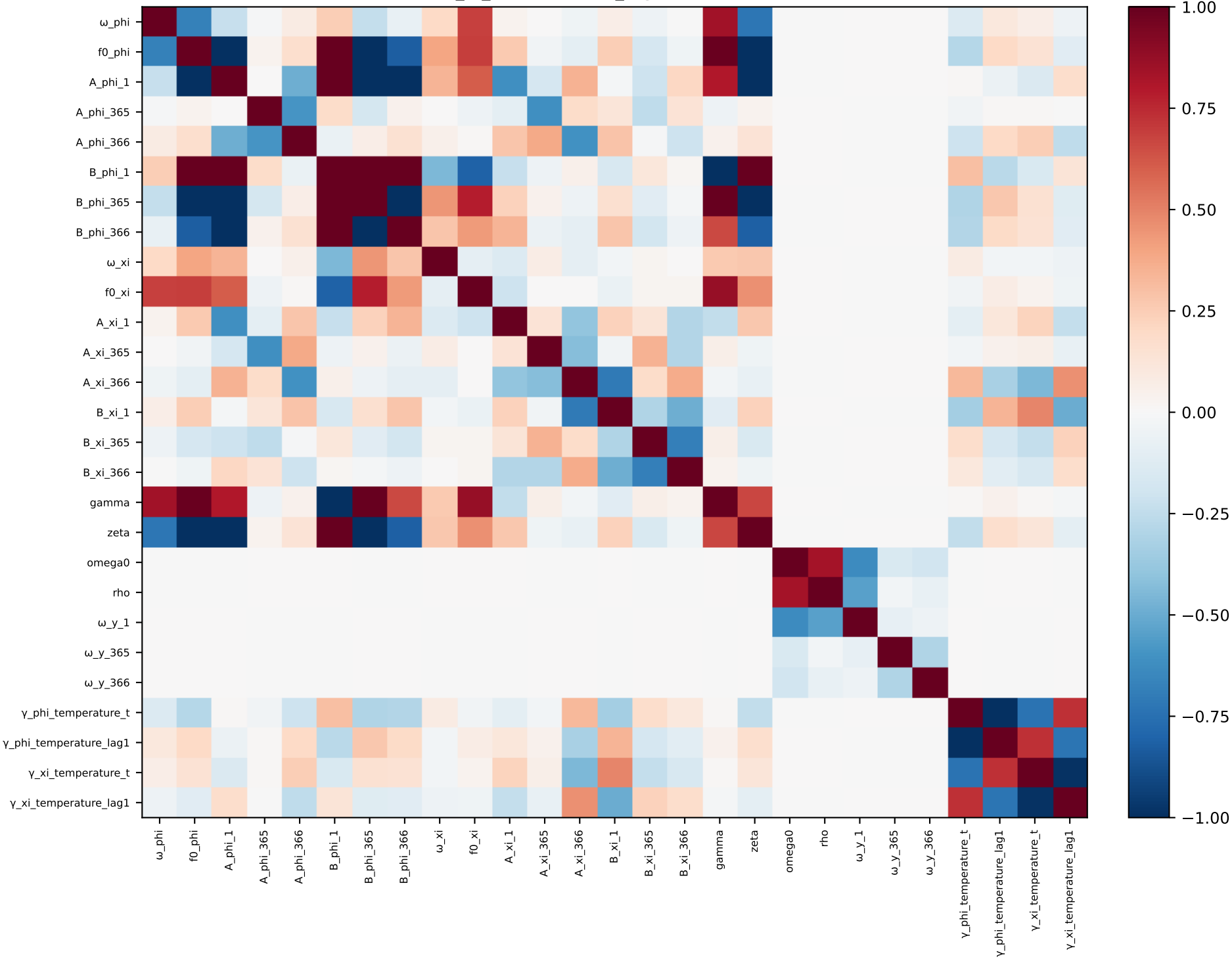
phi_xi_temperature_lag1 — filtered states (OOS)



phi_xi_temperature_lag1 — covariance

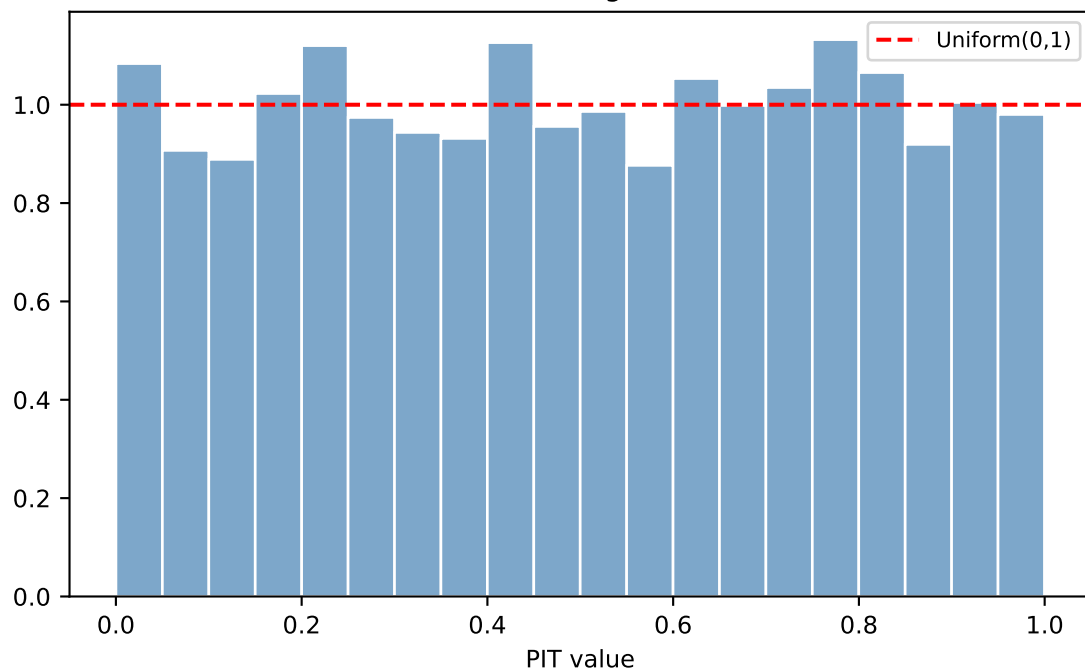


phi_xi_temperature_lag1 — correlation

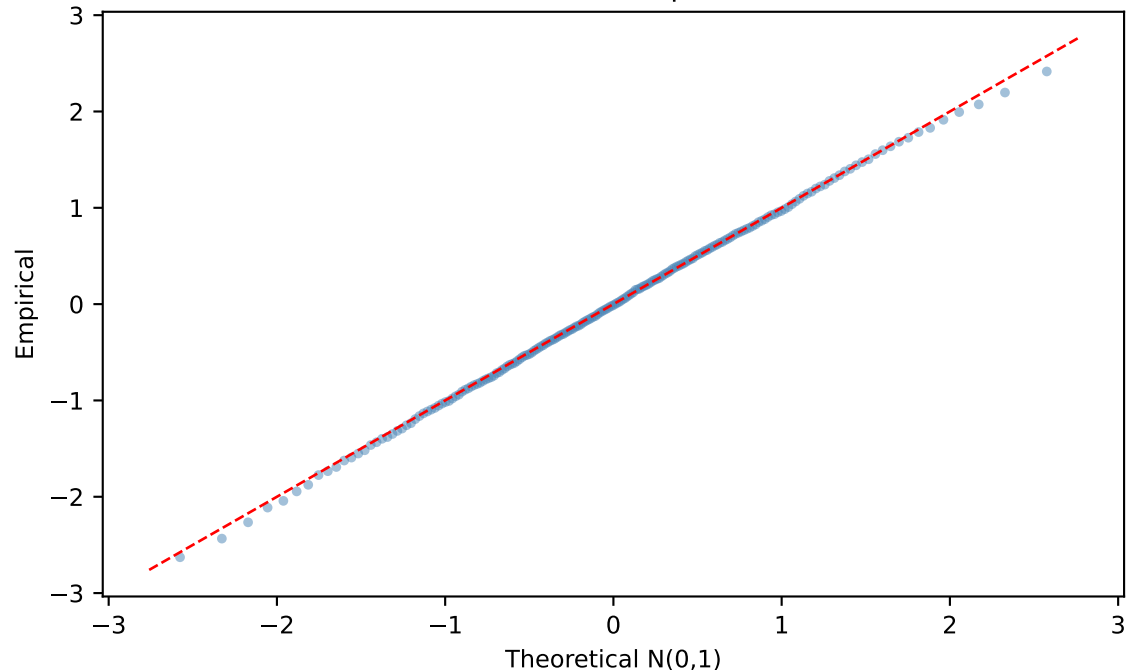


phi_xi_temperature_short_lags — IS diagnostics

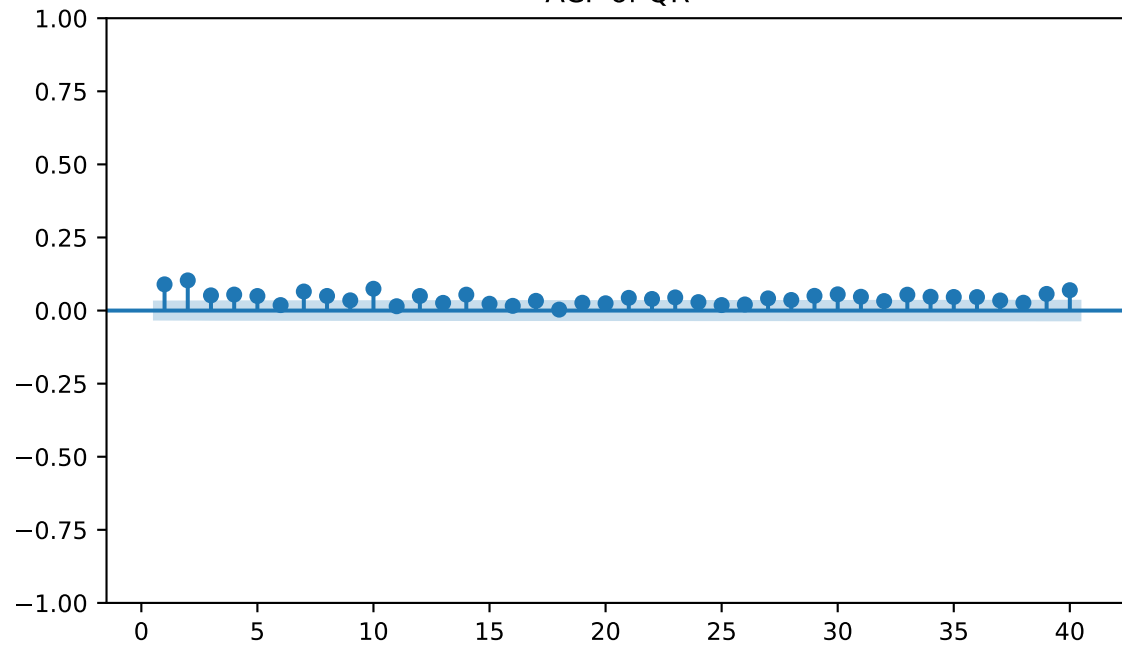
PIT histogram



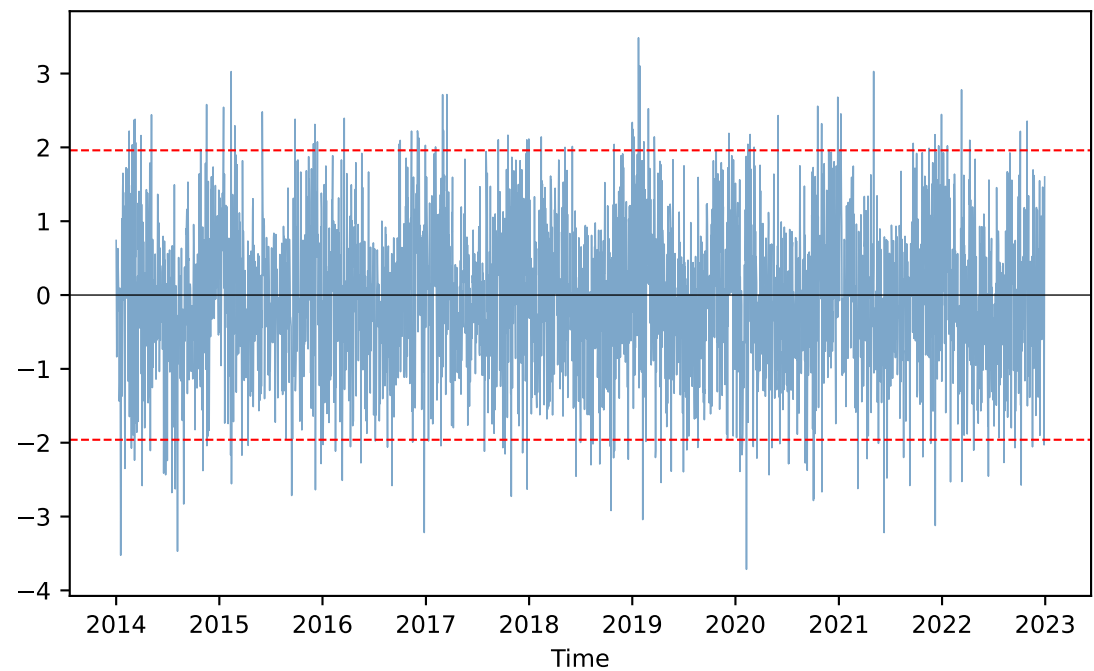
Normal QQ plot



ACF of QR

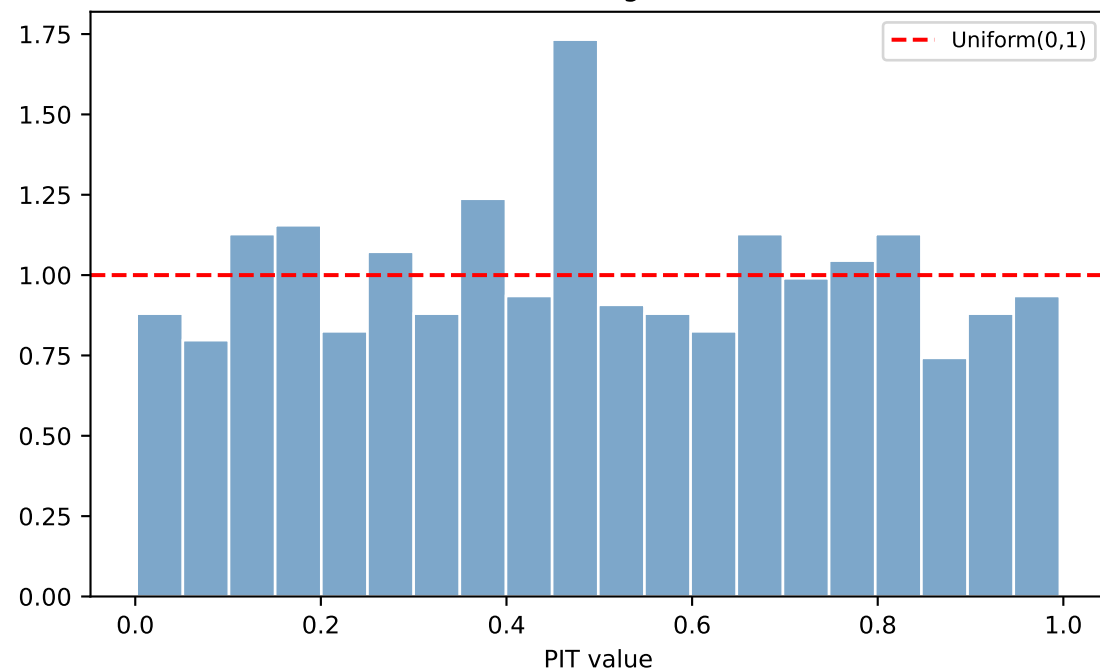


QR time series

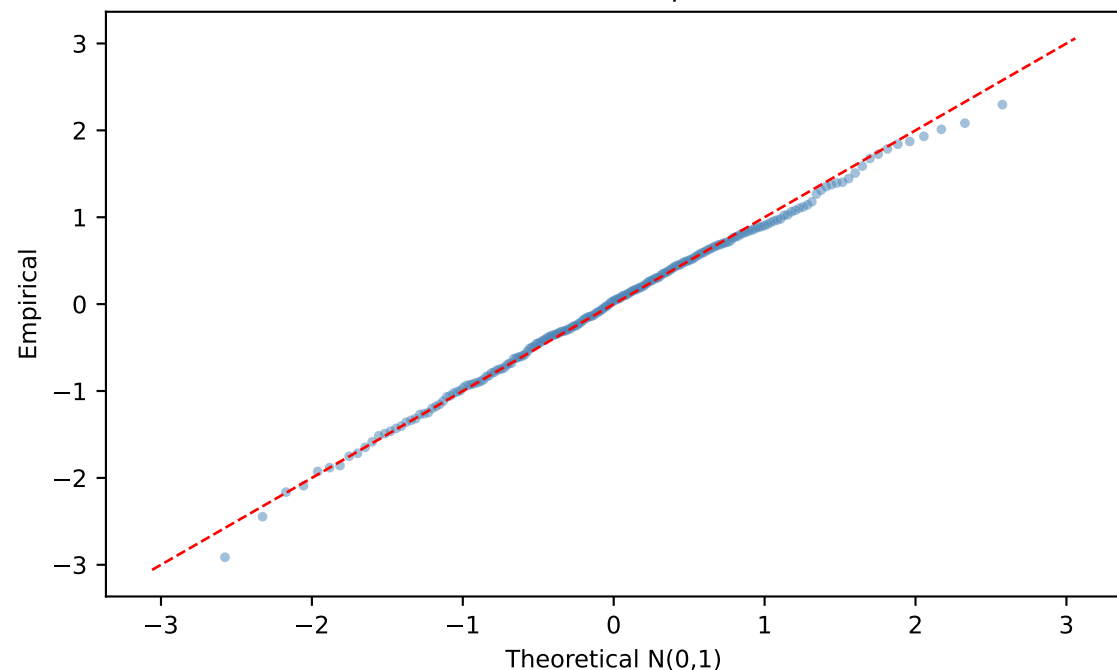


phi_xi_temperature_short_lags — OOS diagnostics

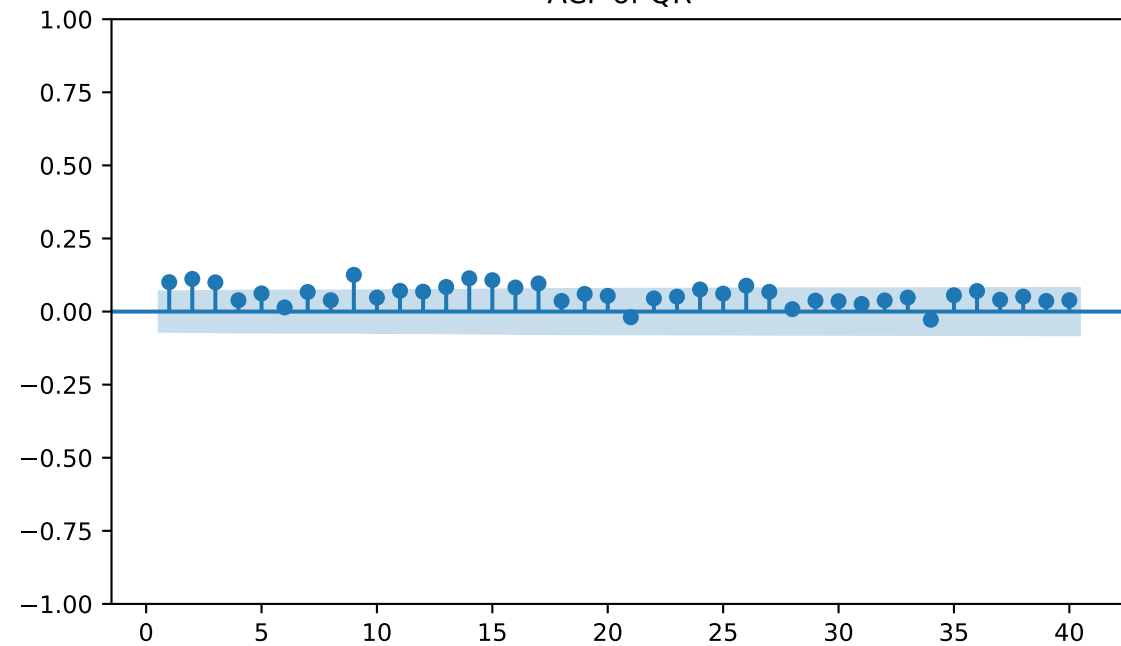
PIT histogram



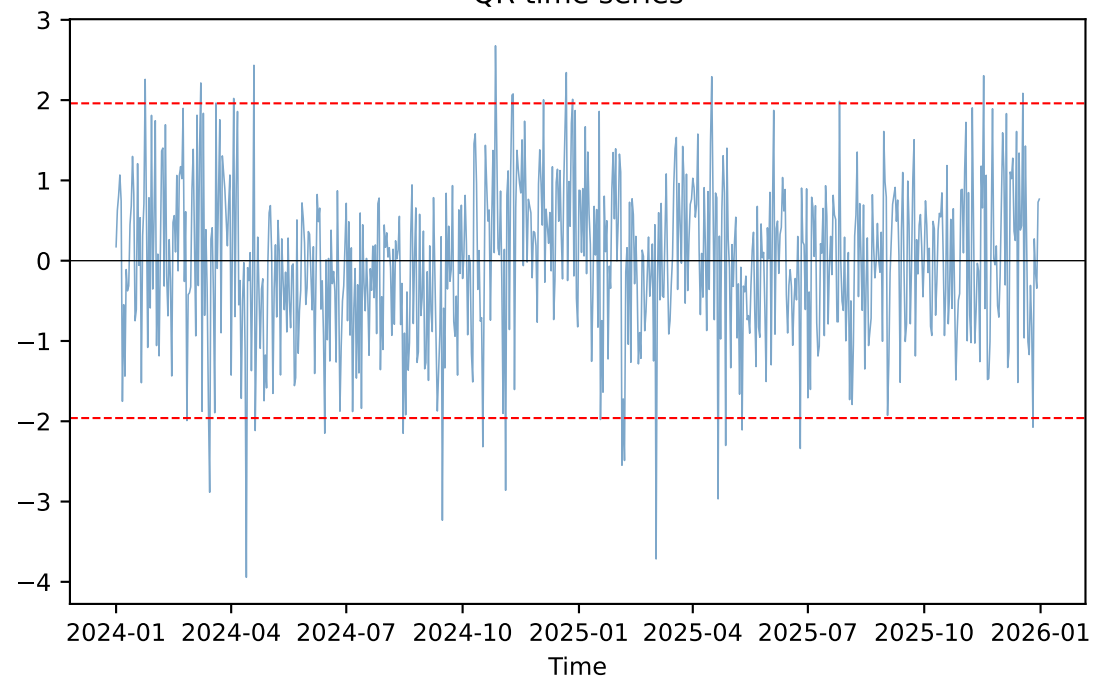
Normal QQ plot



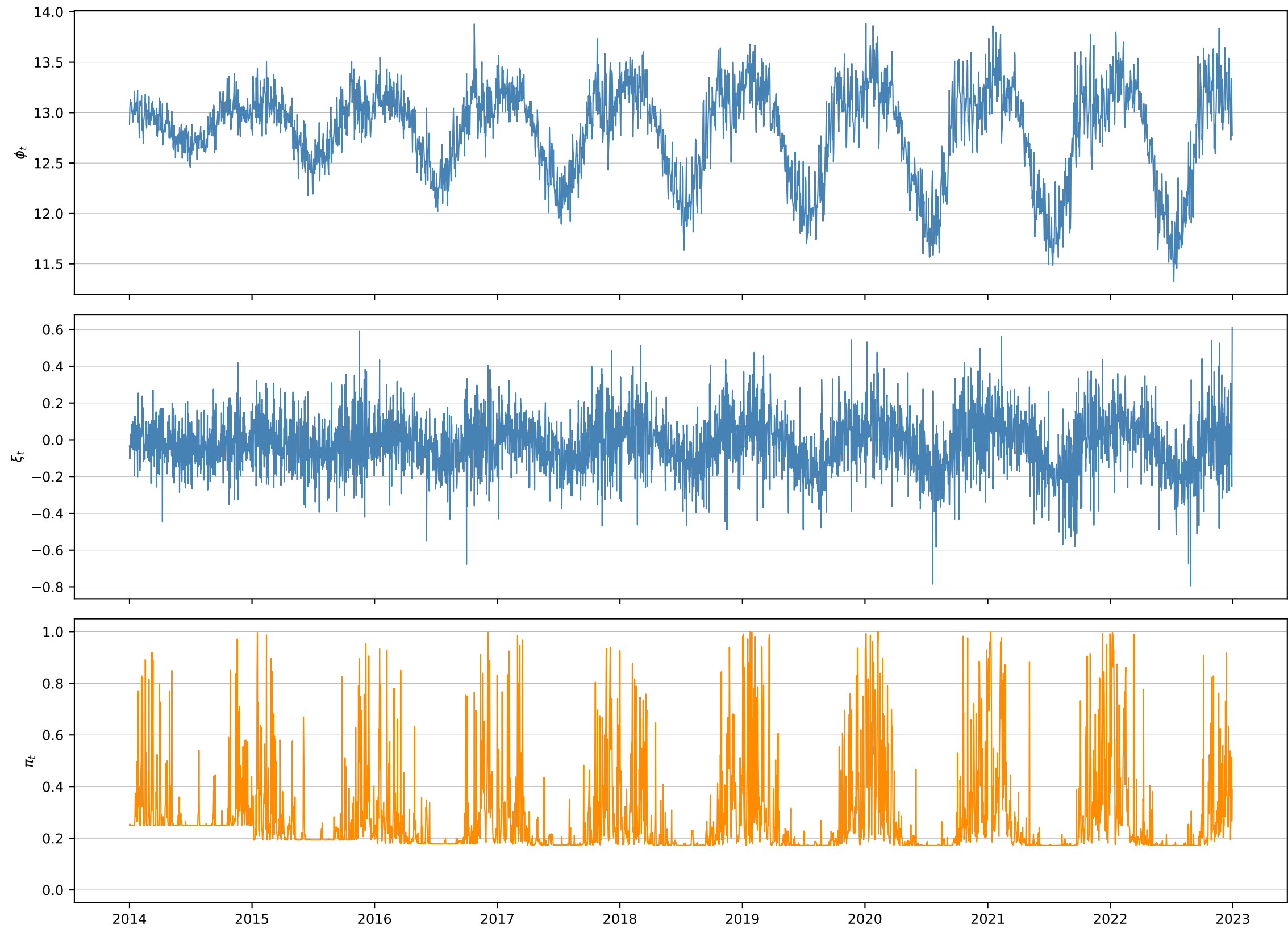
ACF of QR



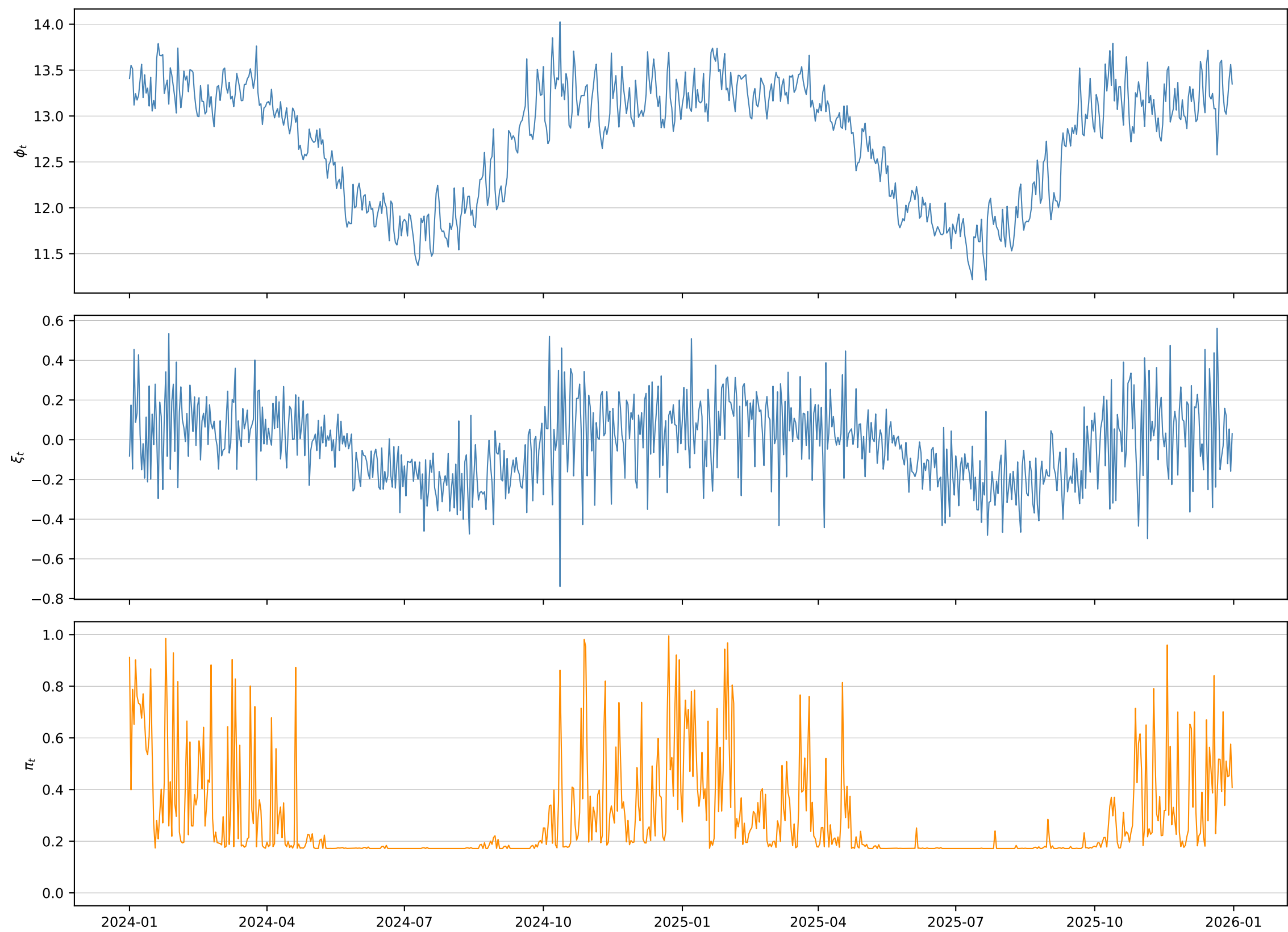
QR time series



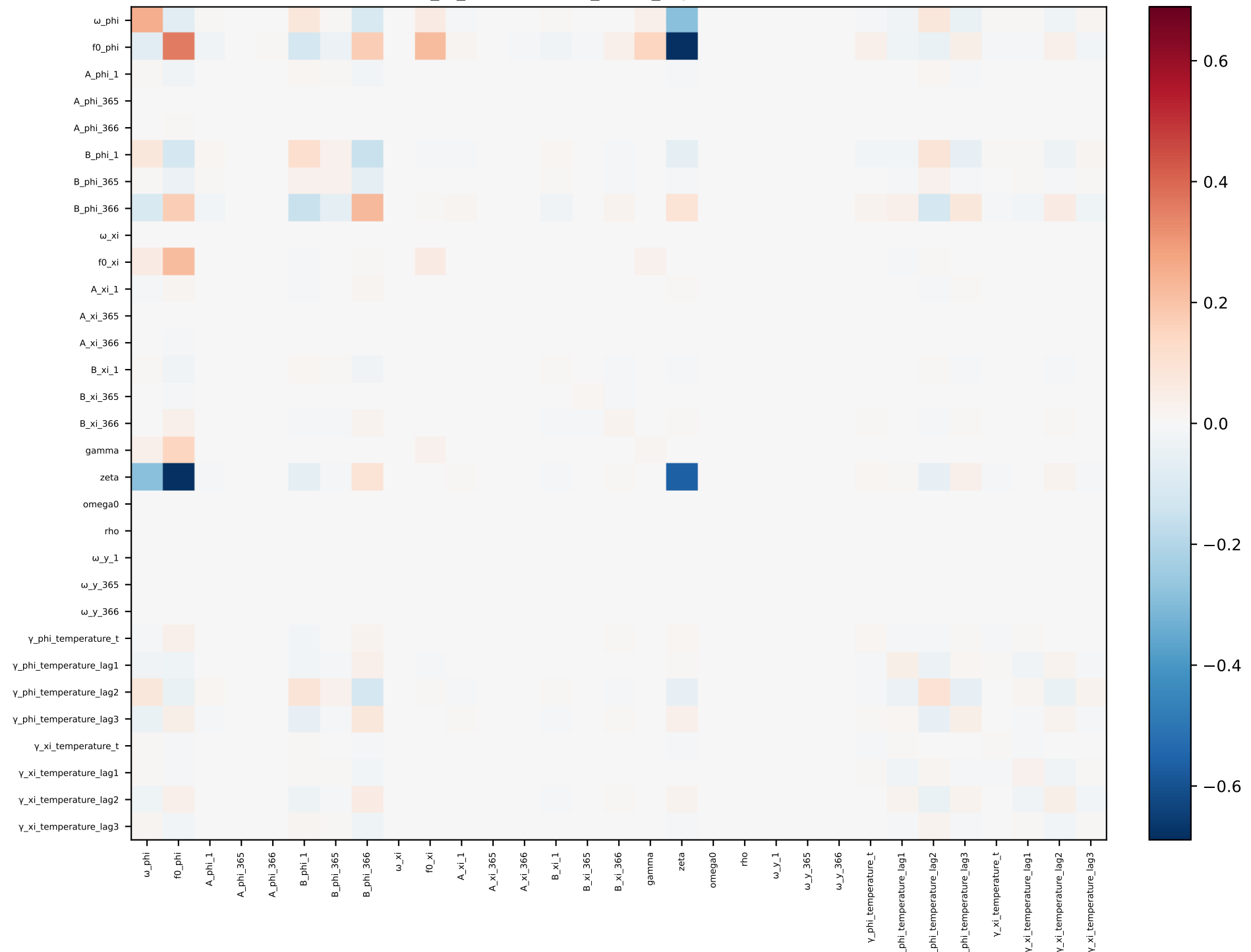
phi_xi_temperature_short_lags — filtered states (IS)



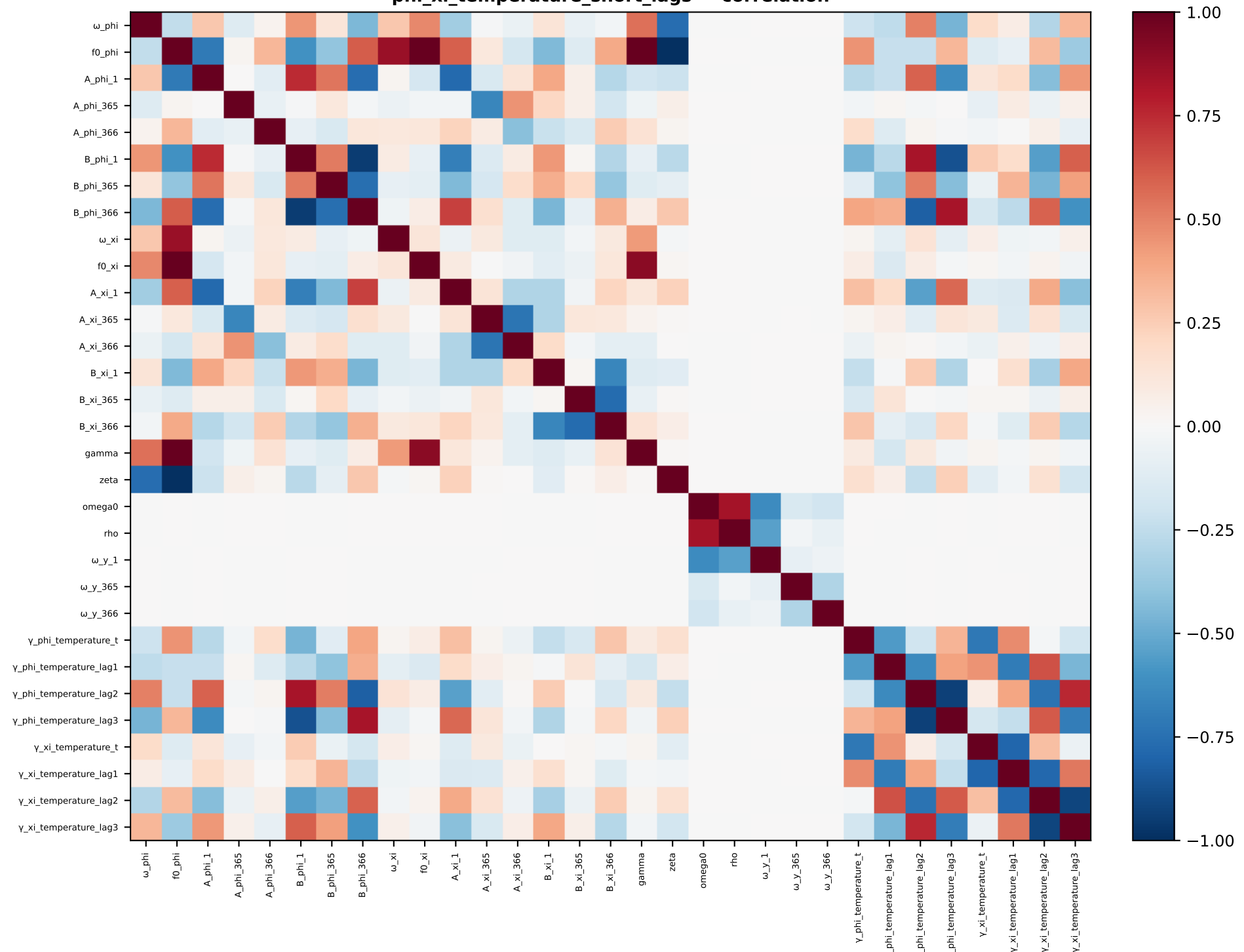
phi_xi_temperature_short_lags — filtered states (OOS)



phi_xi_temperature_short_lags — covariance

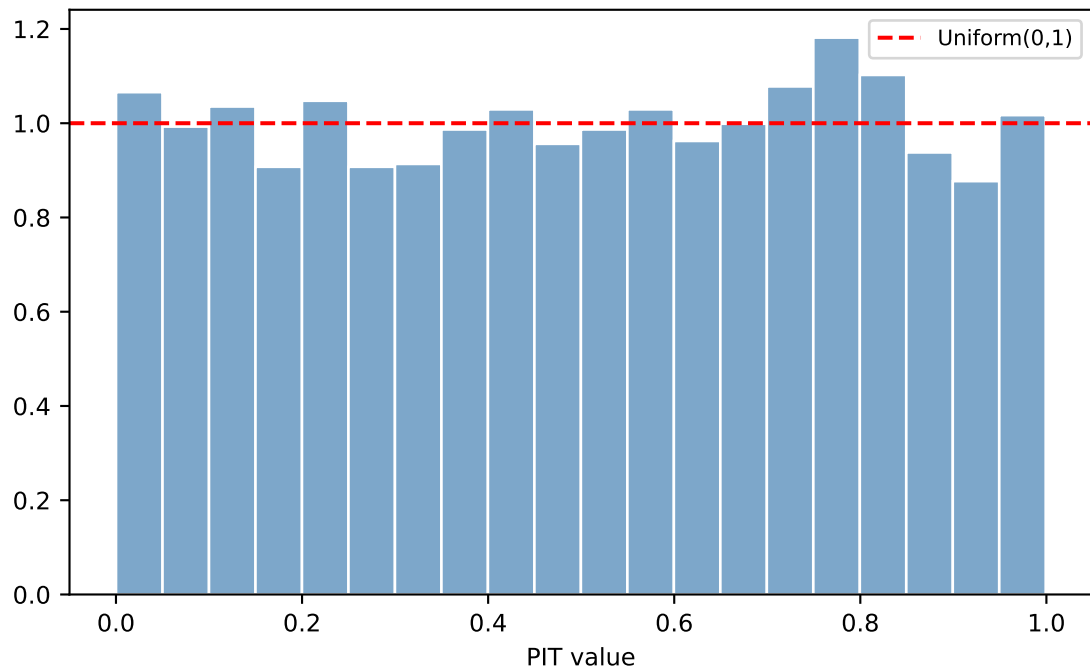


phi_xi_temperature_short_lags — correlation

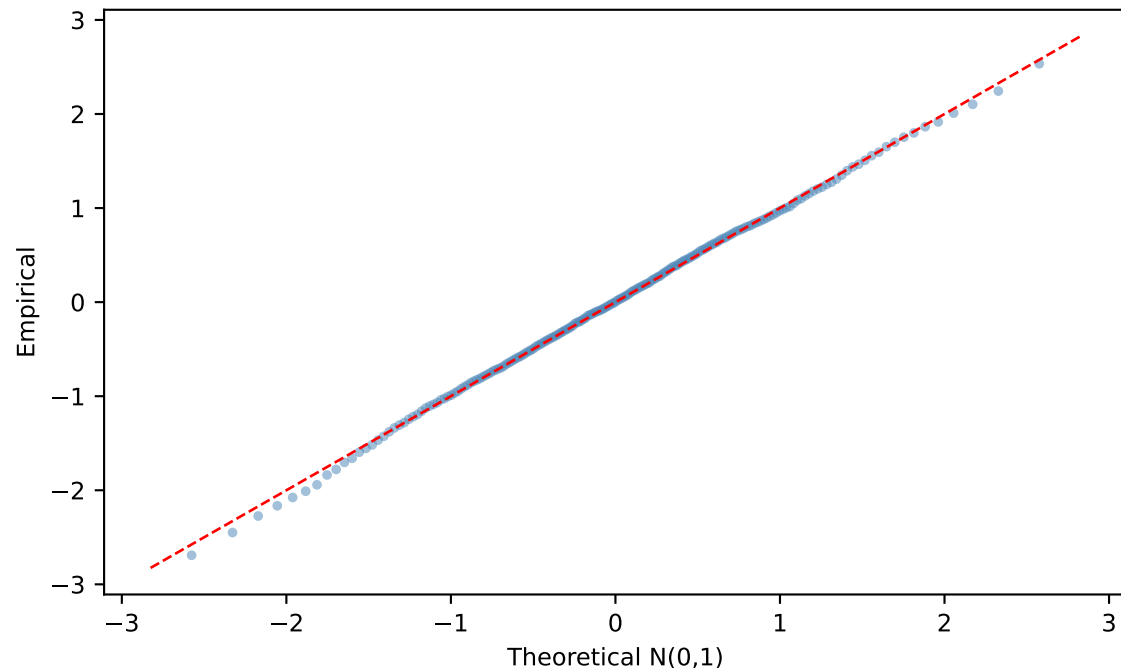


phi_xi_temperature_seasonal_lags — IS diagnostics

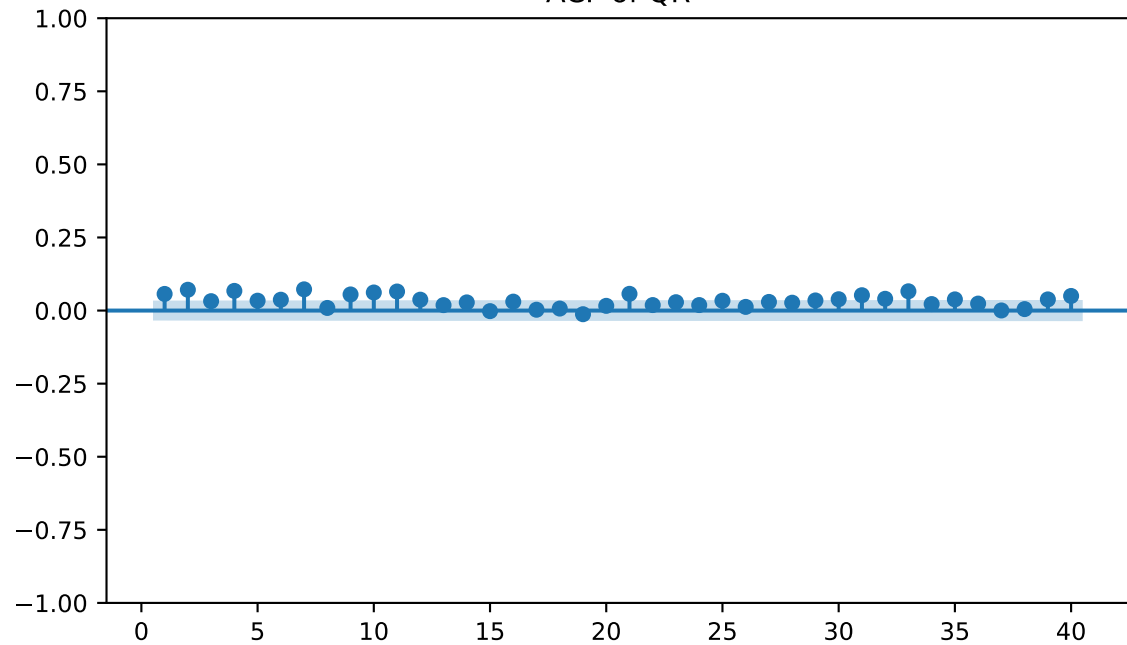
PIT histogram



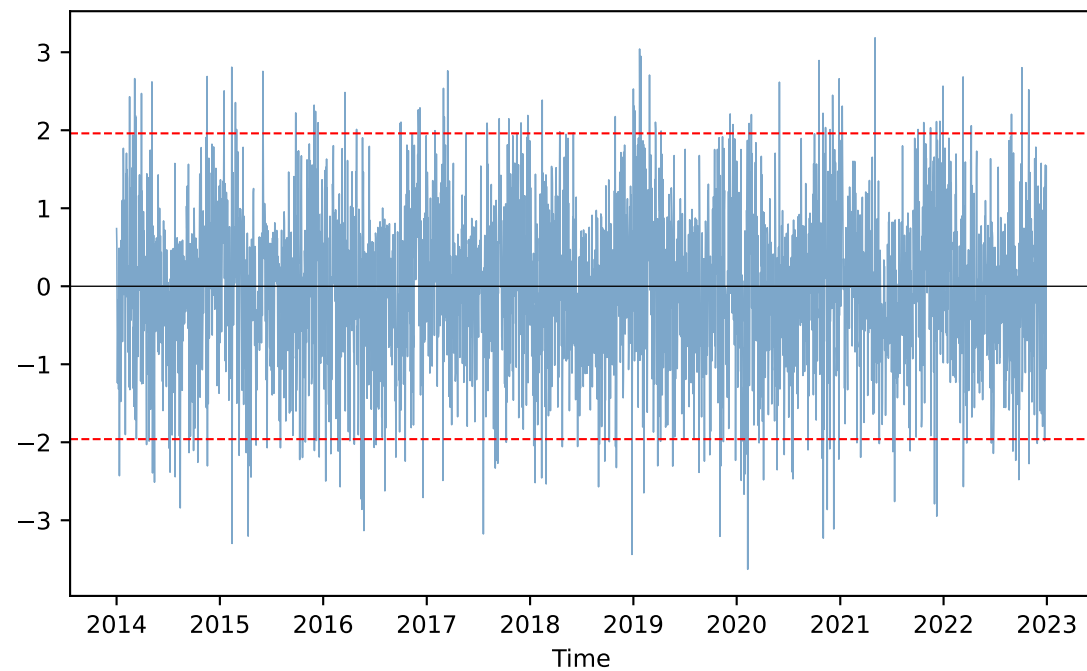
Normal QQ plot



ACF of QR

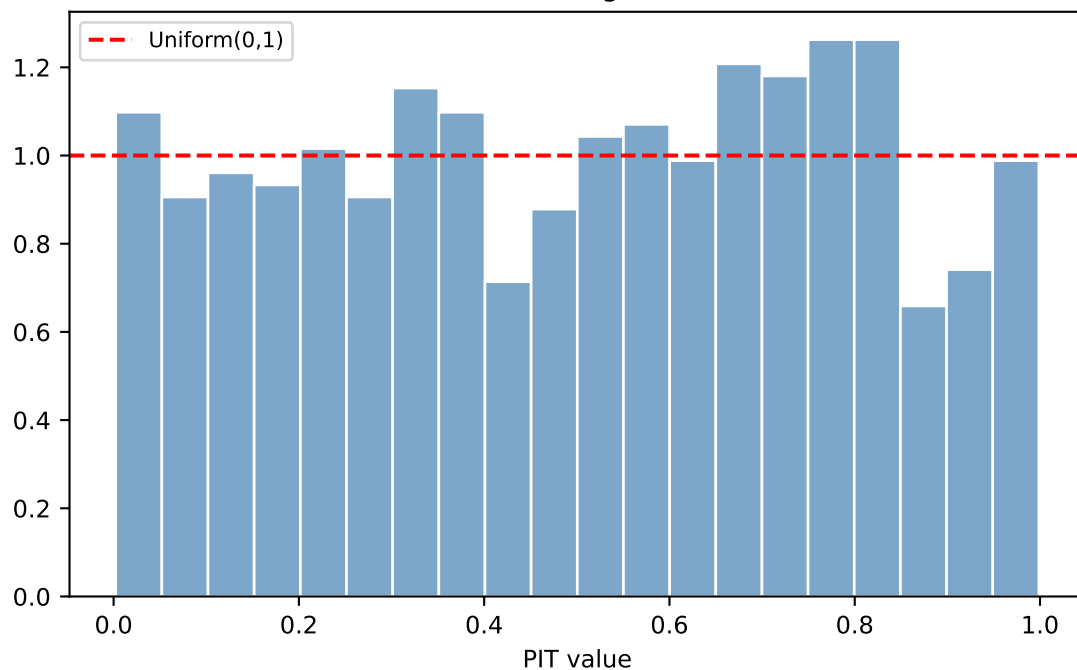


QR time series

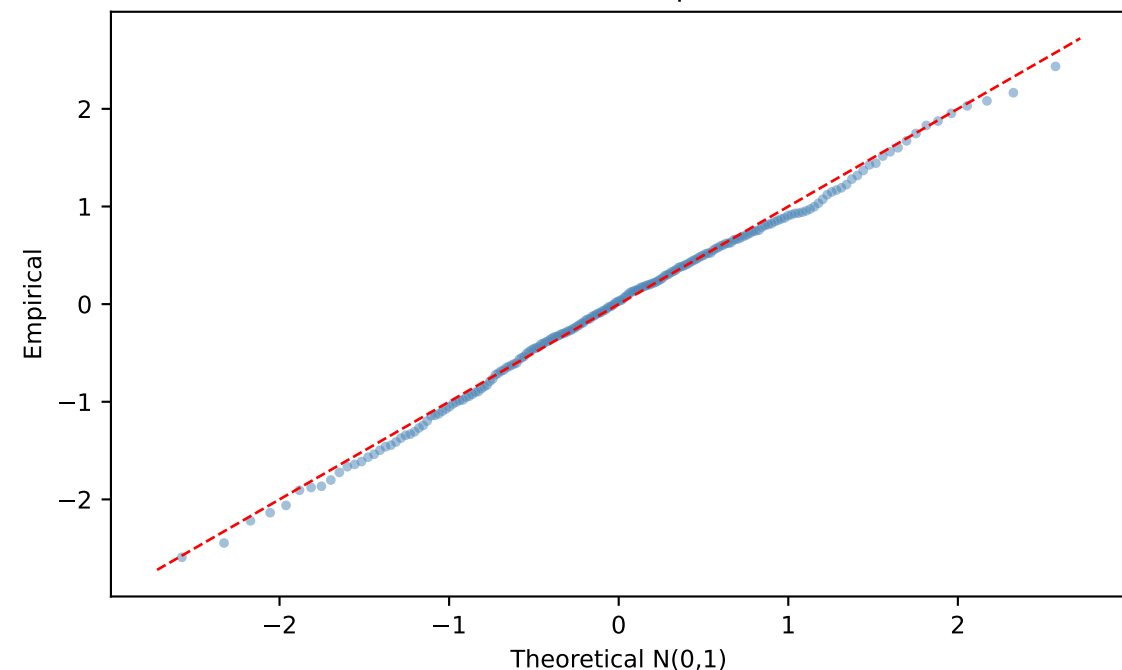


phi_xi_temperature_seasonal_lags — OOS diagnostics

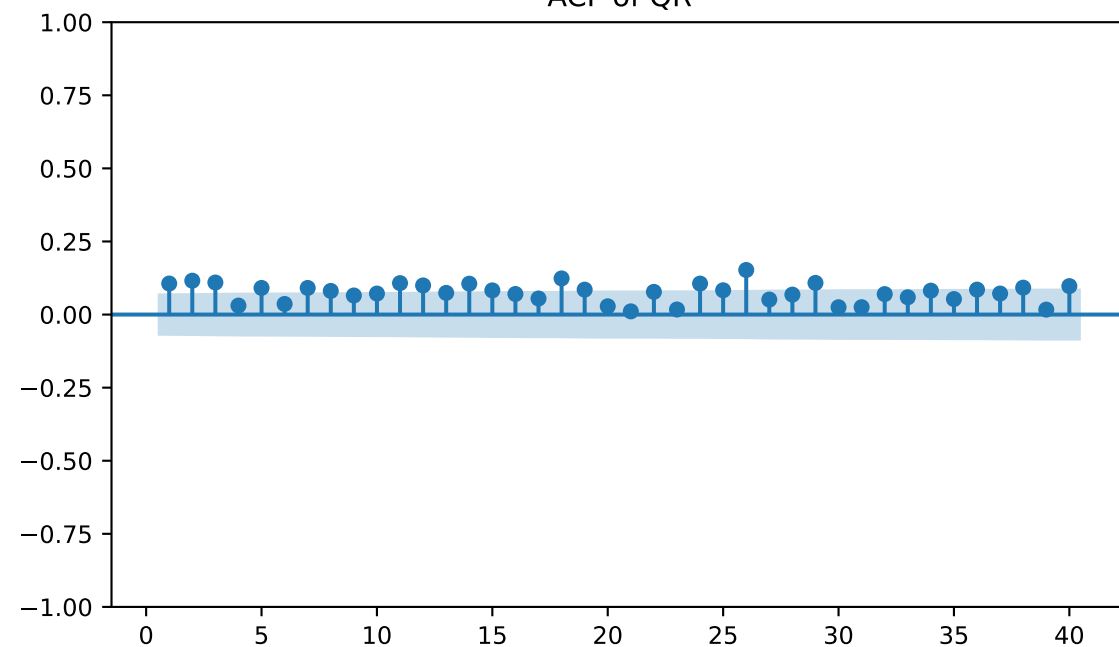
PIT histogram



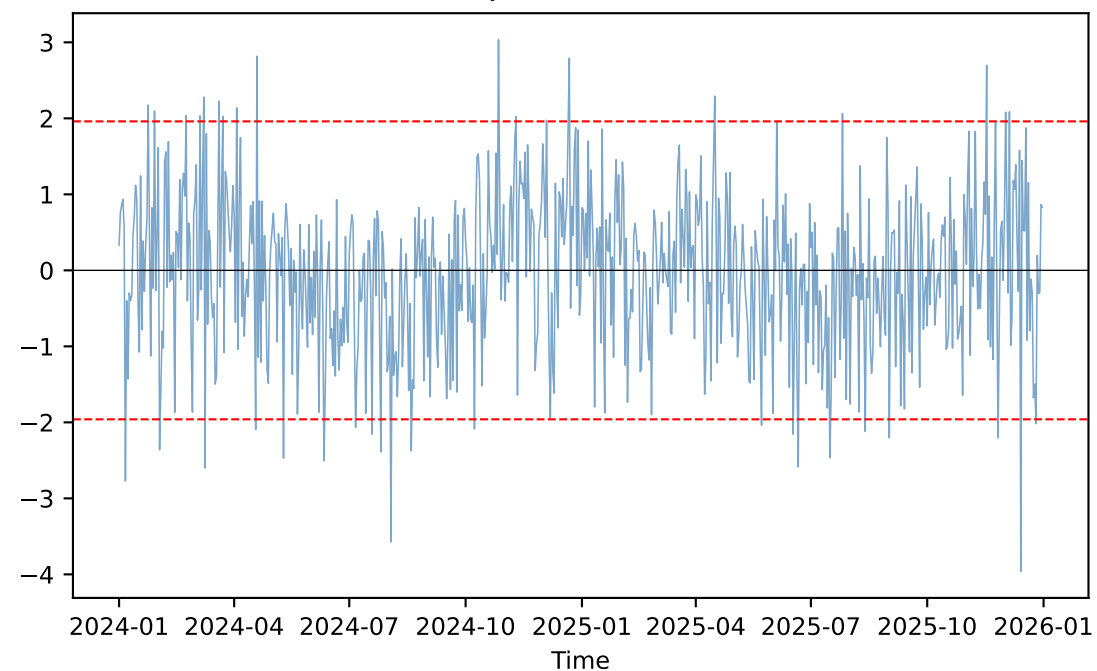
Normal QQ plot



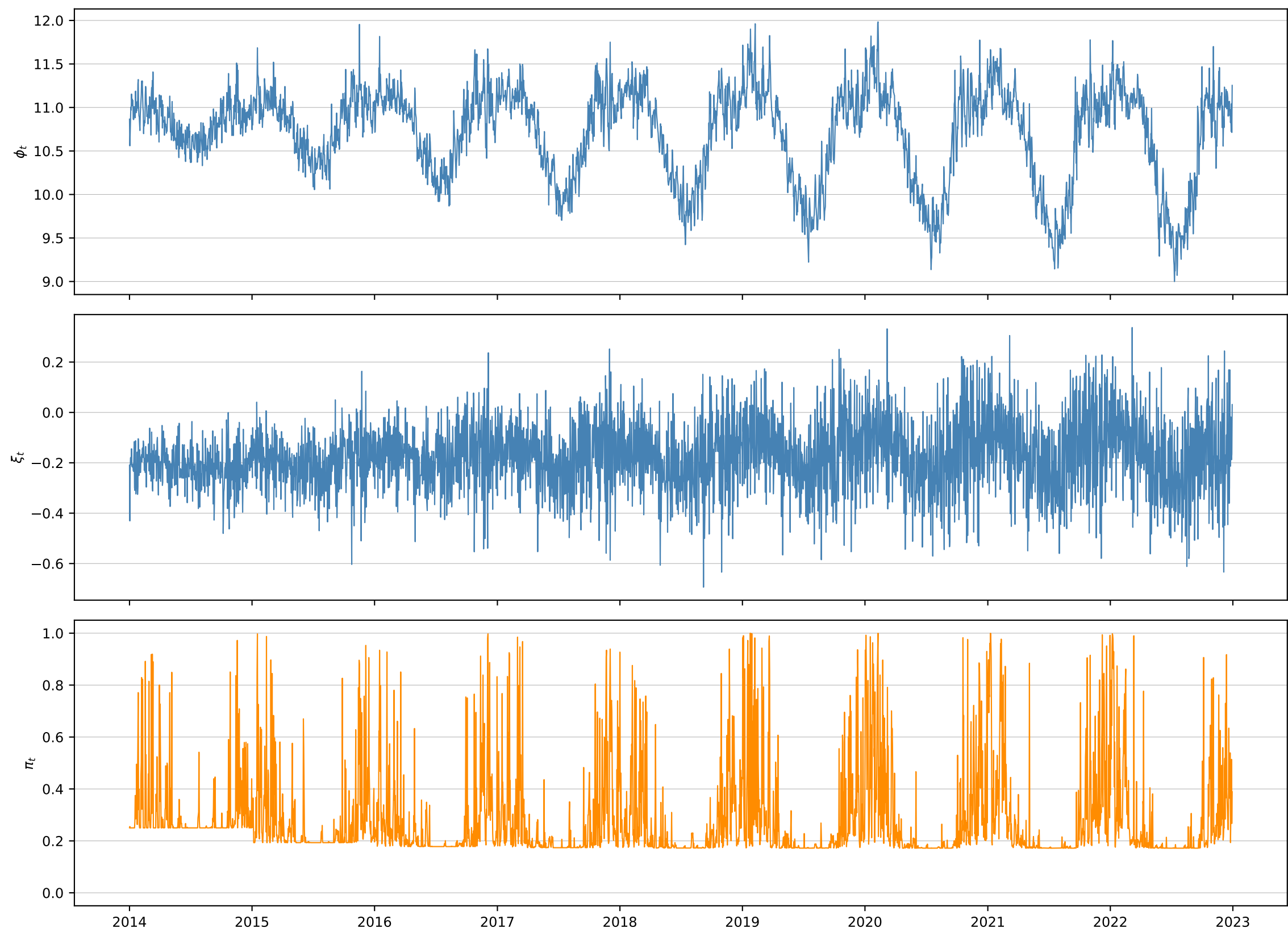
ACF of QR



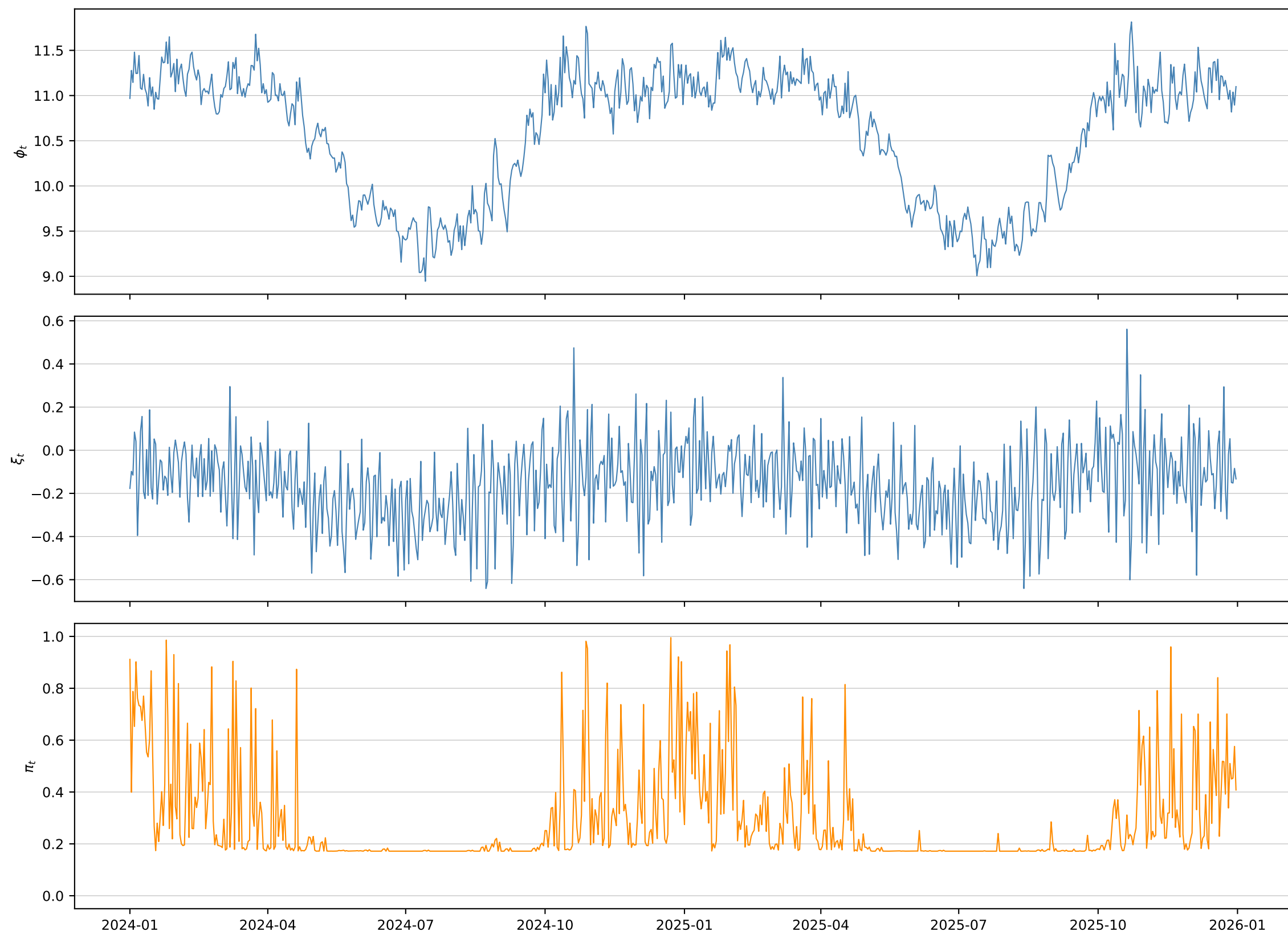
QR time series



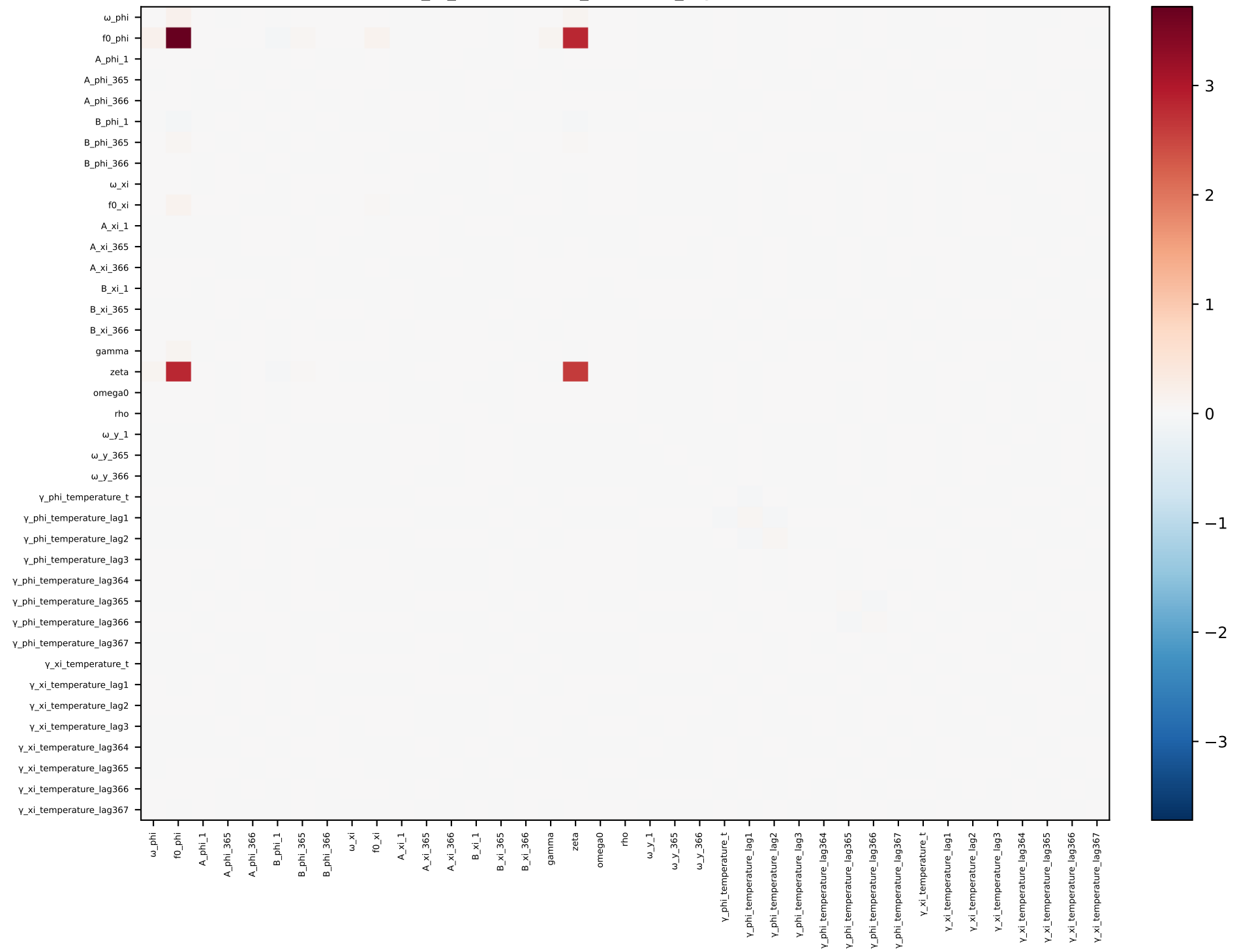
phi_xi_temperature_seasonal_lags — filtered states (IS)



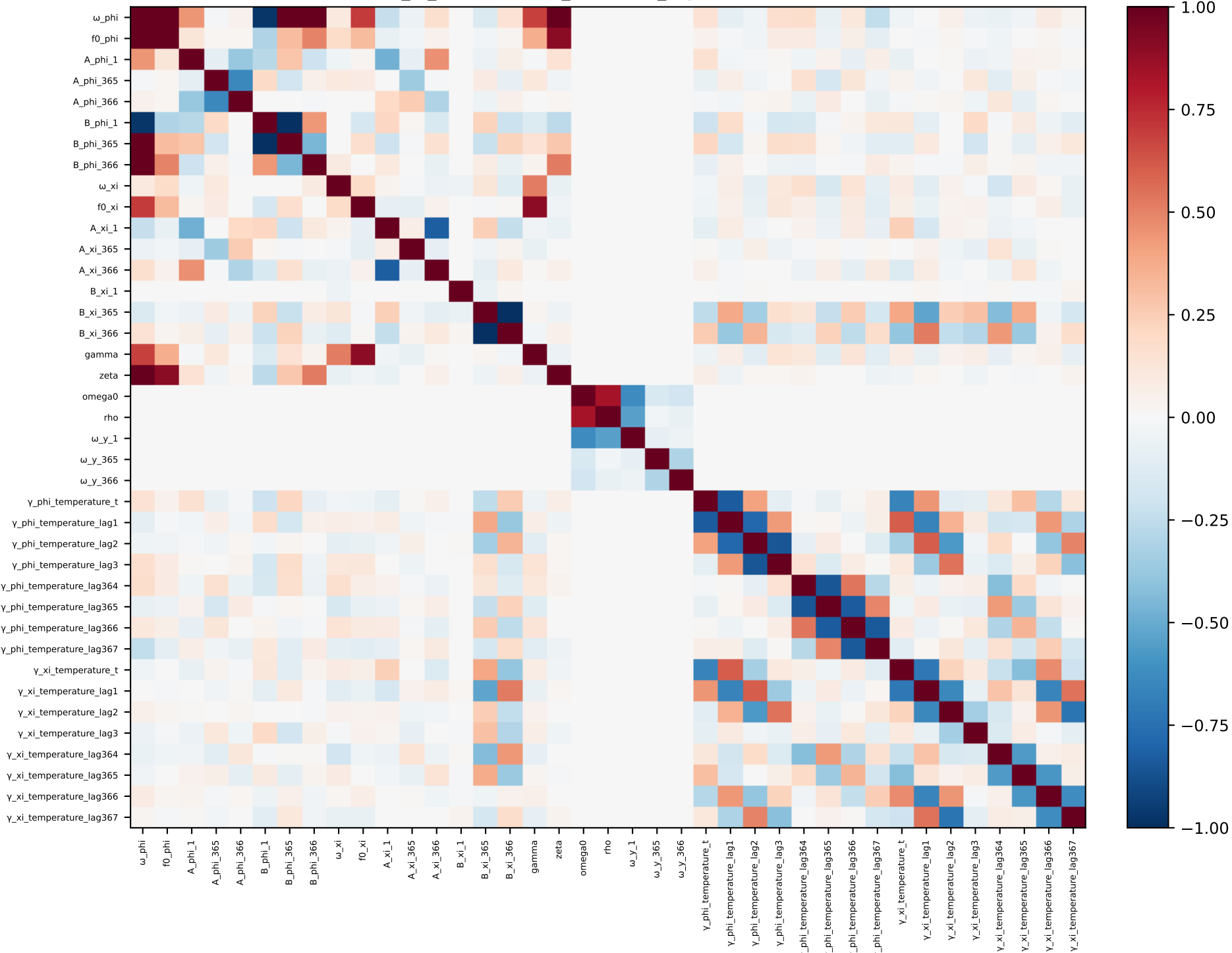
phi_xi_temperature_seasonal_lags — filtered states (OOS)



phi_xi_temperature_seasonal_lags — covariance

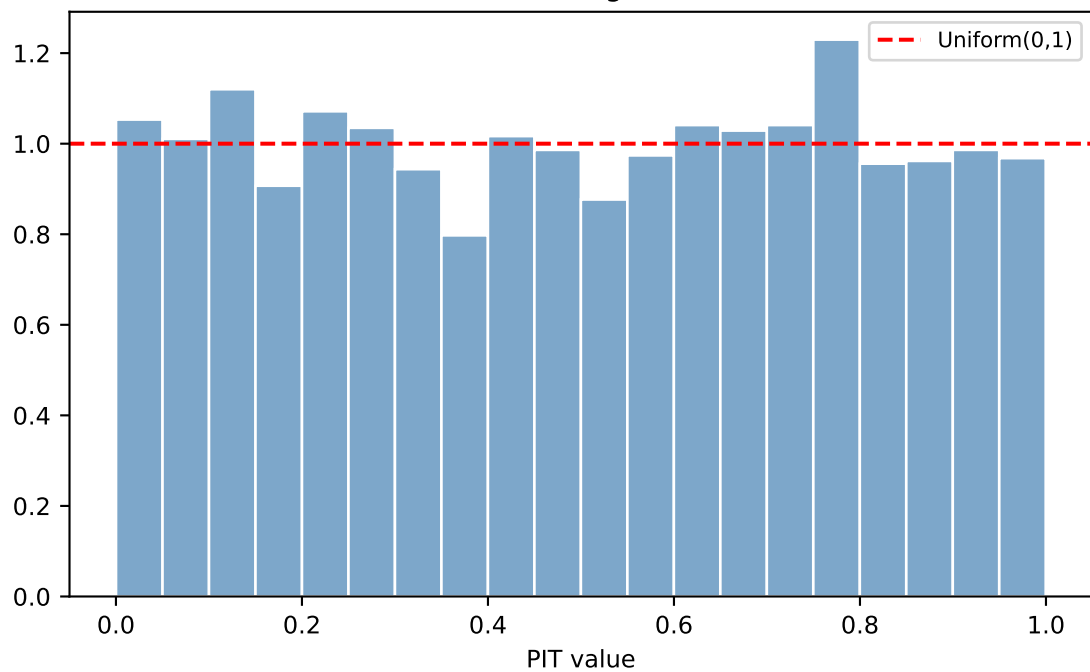


phi_xi_temperature_seasonal_lags — correlation

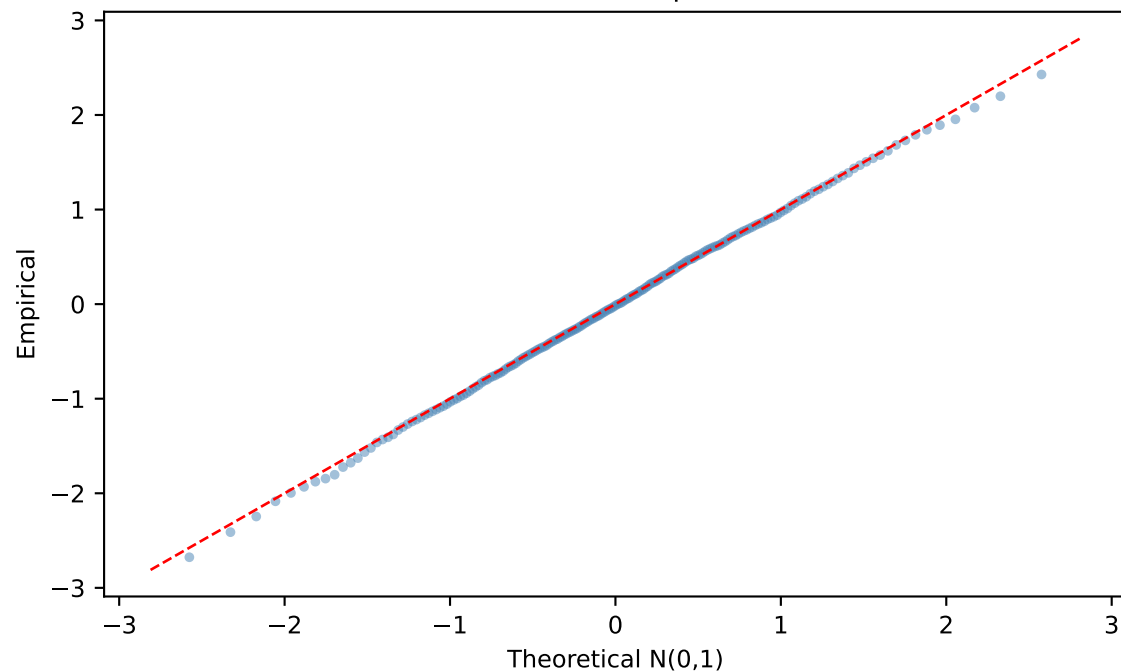


phi_xi_dewpoint_temperature_t — IS diagnostics

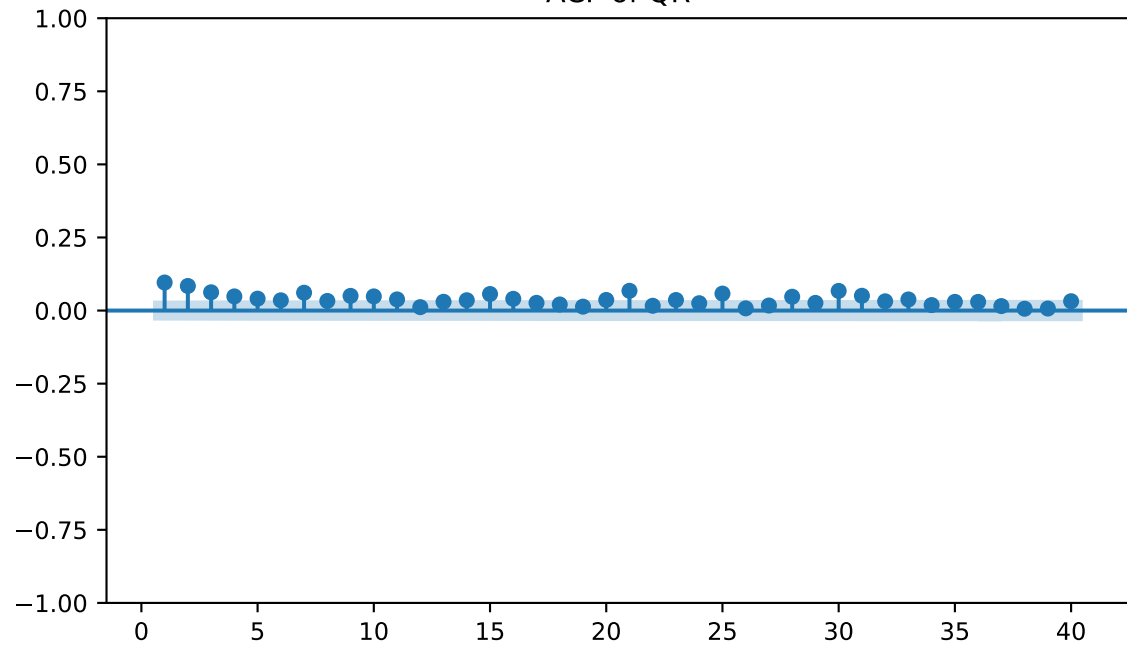
PIT histogram



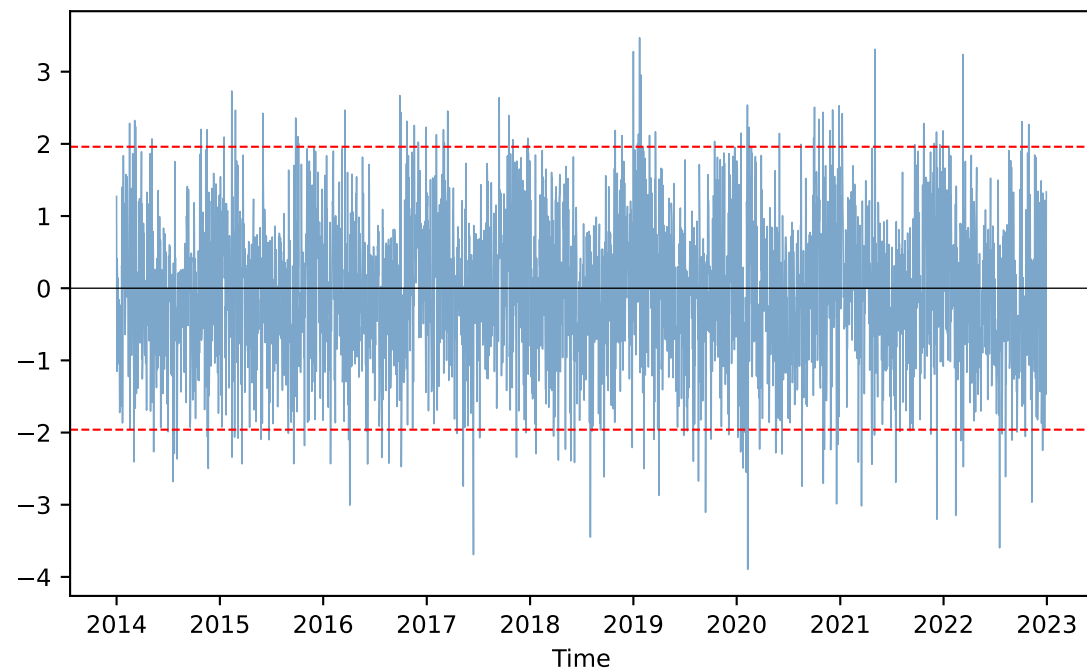
Normal QQ plot



ACF of QR

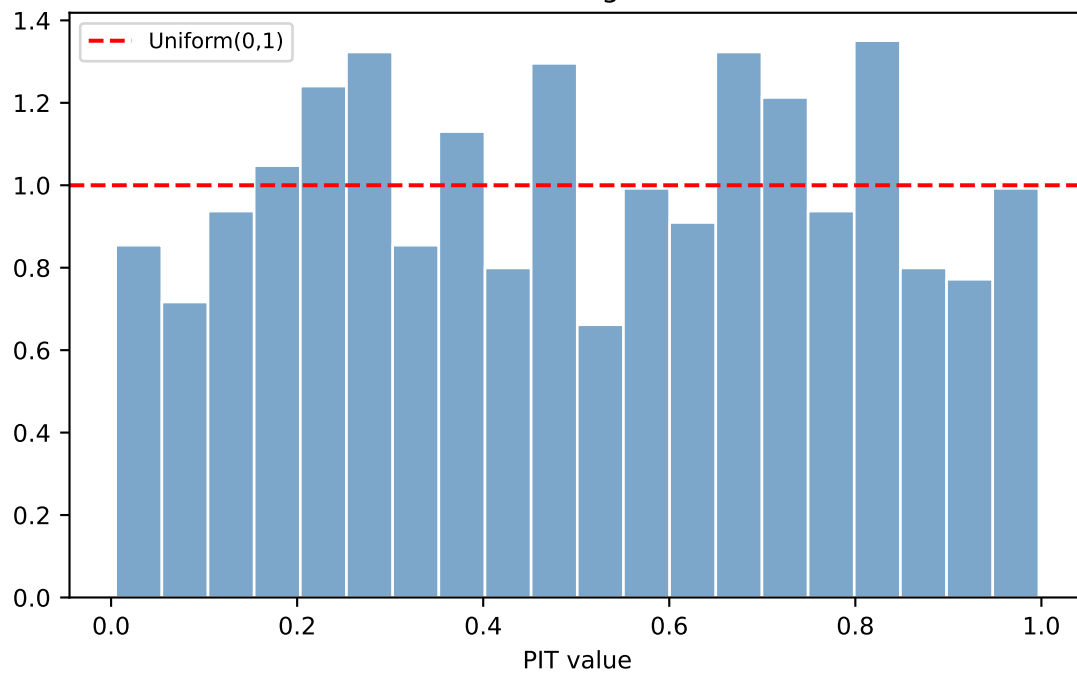


QR time series

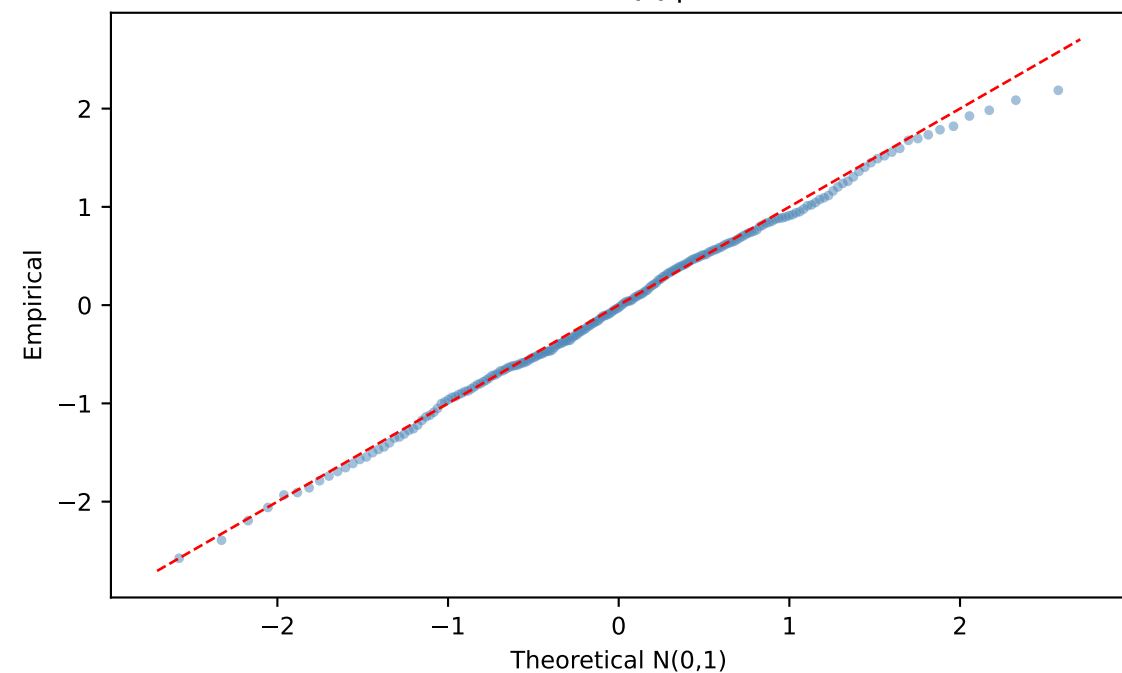


phi_xi_dewpoint_temperature_t — OOS diagnostics

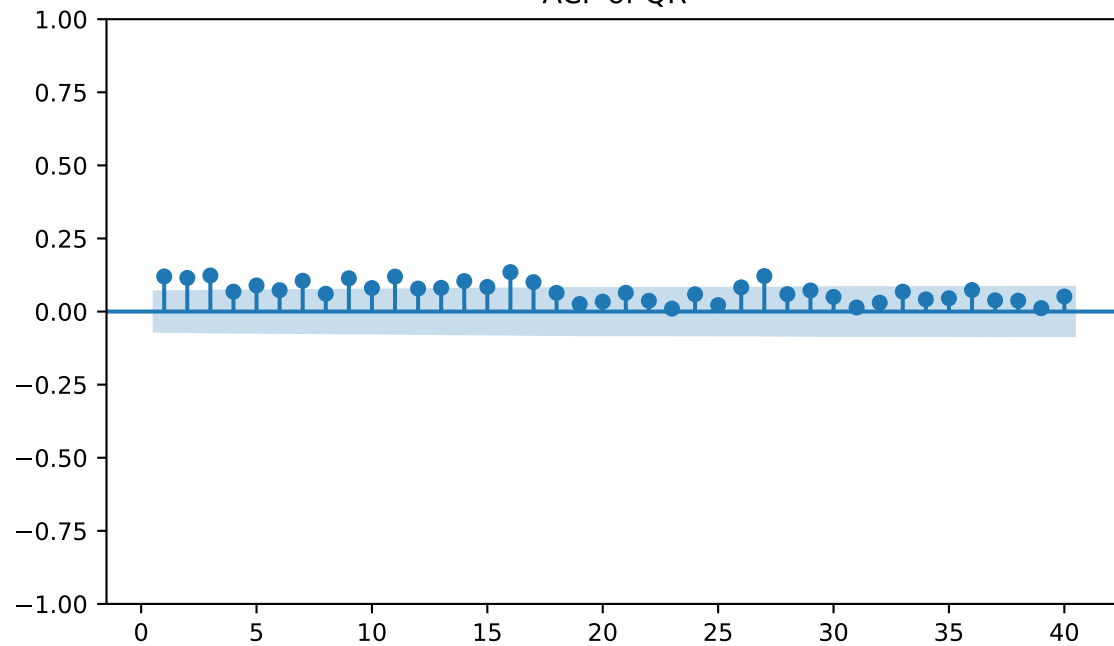
PIT histogram



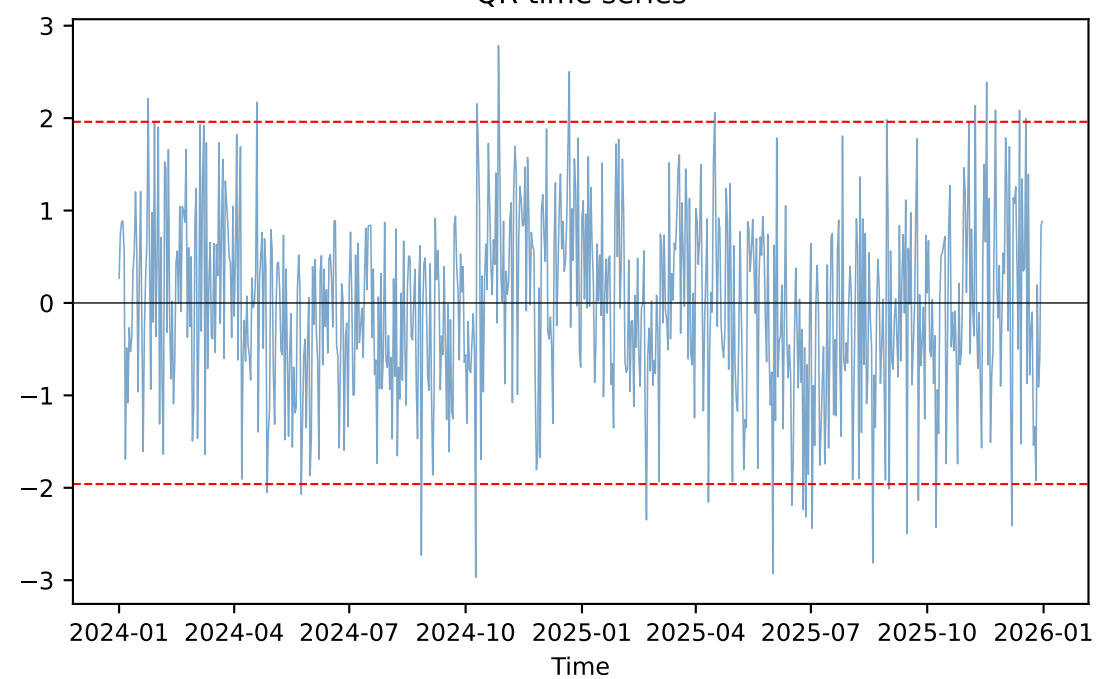
Normal QQ plot



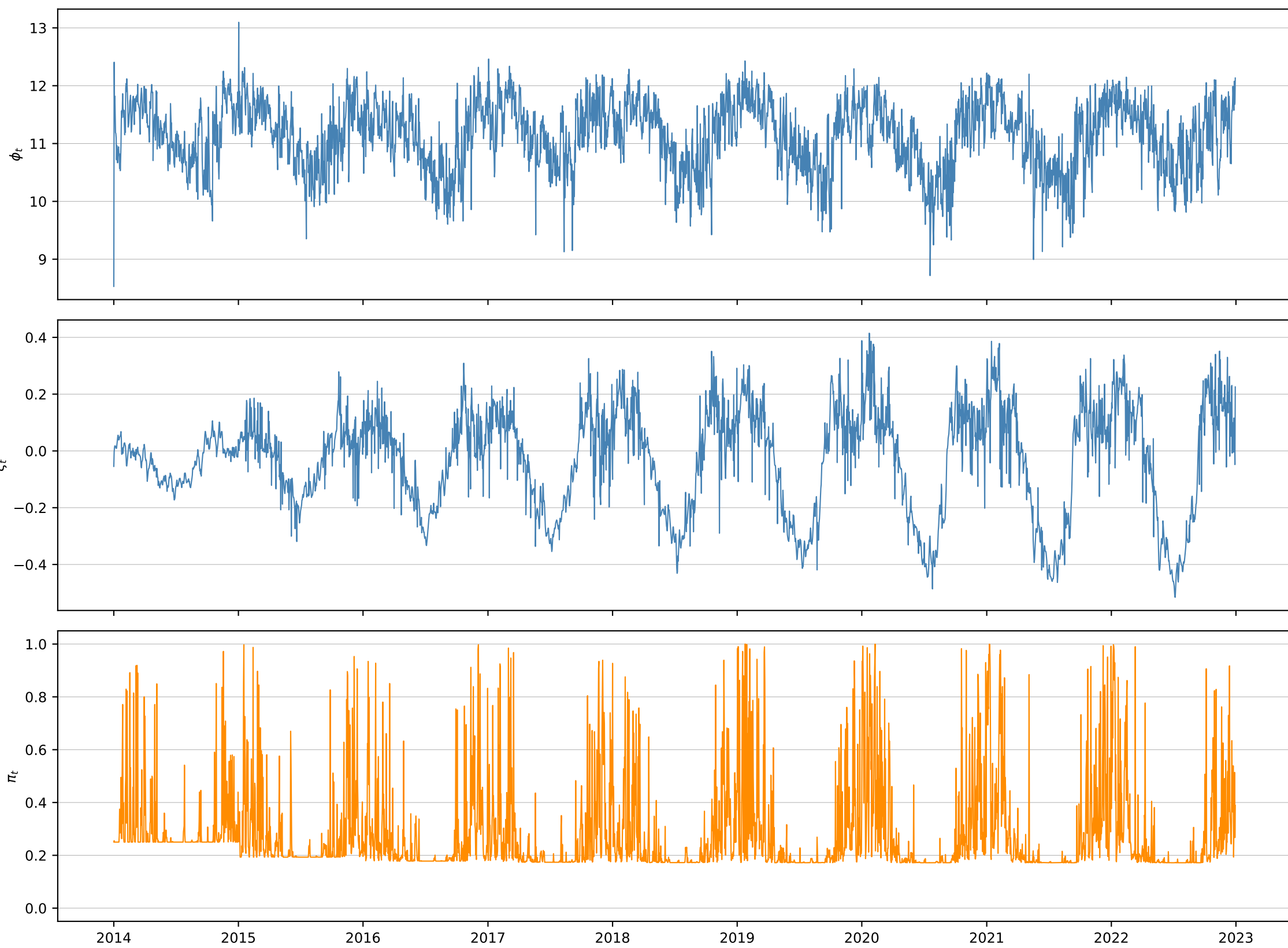
ACF of QR



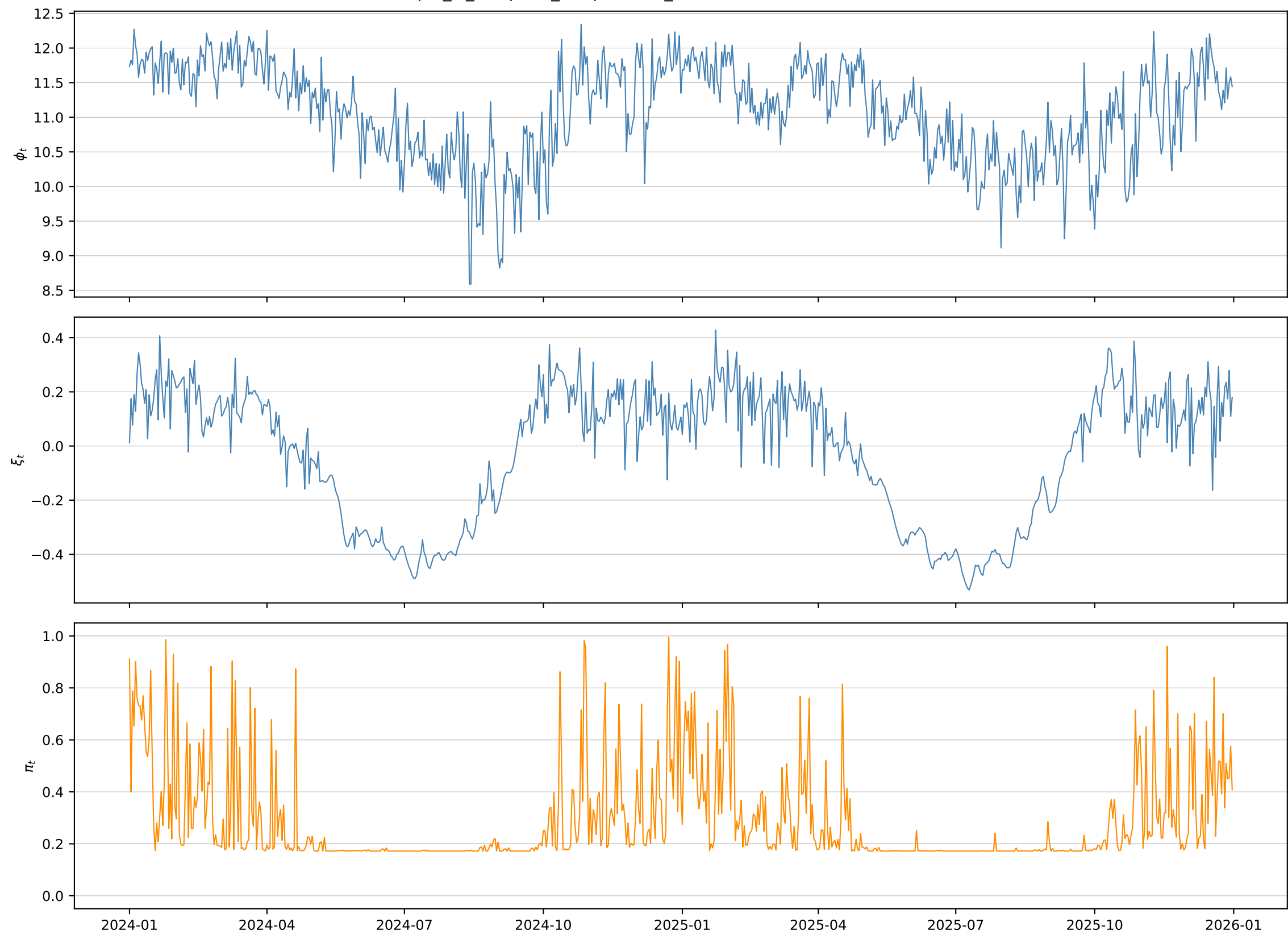
QR time series



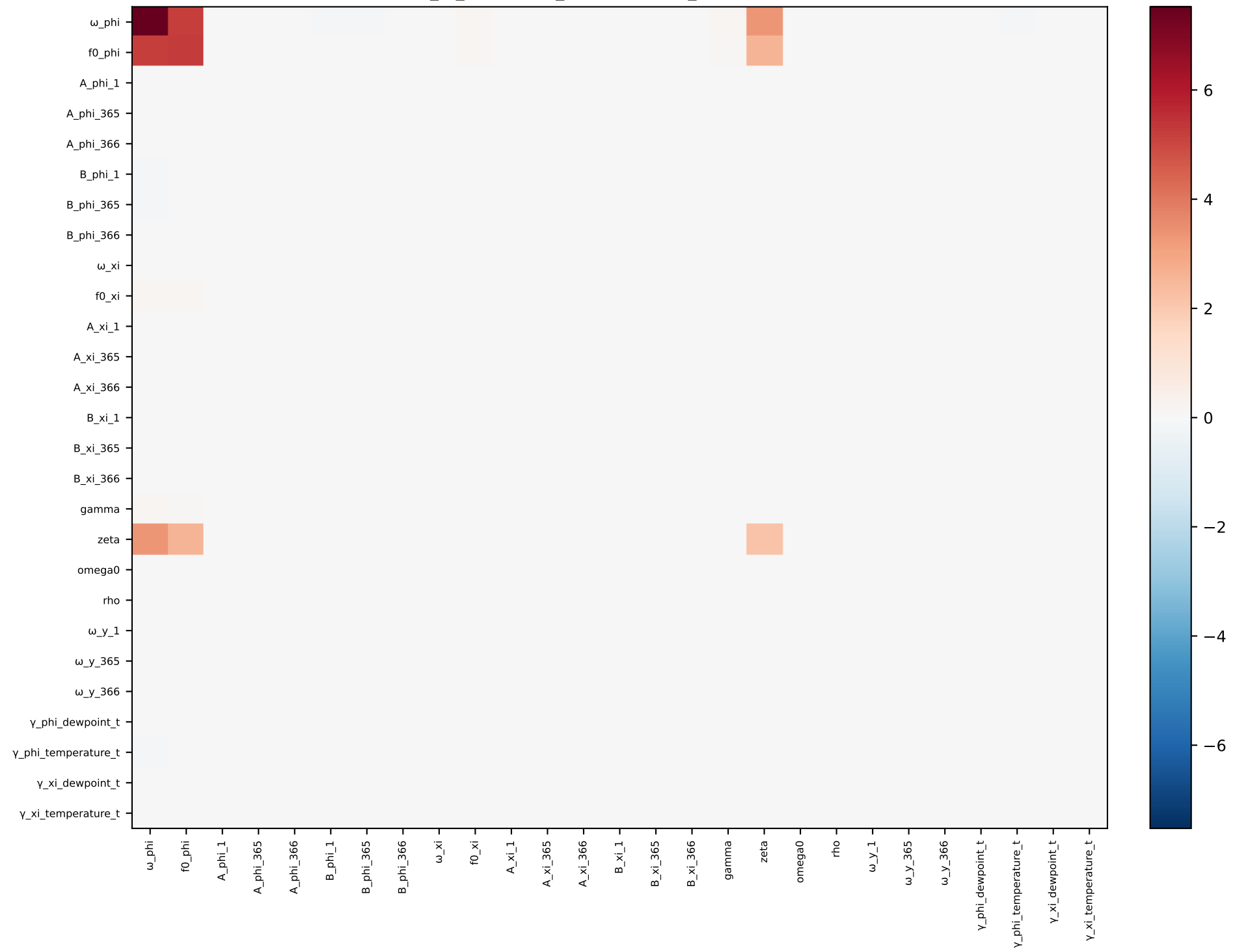
phi_xi_dewpoint_temperature_t — filtered states (IS)



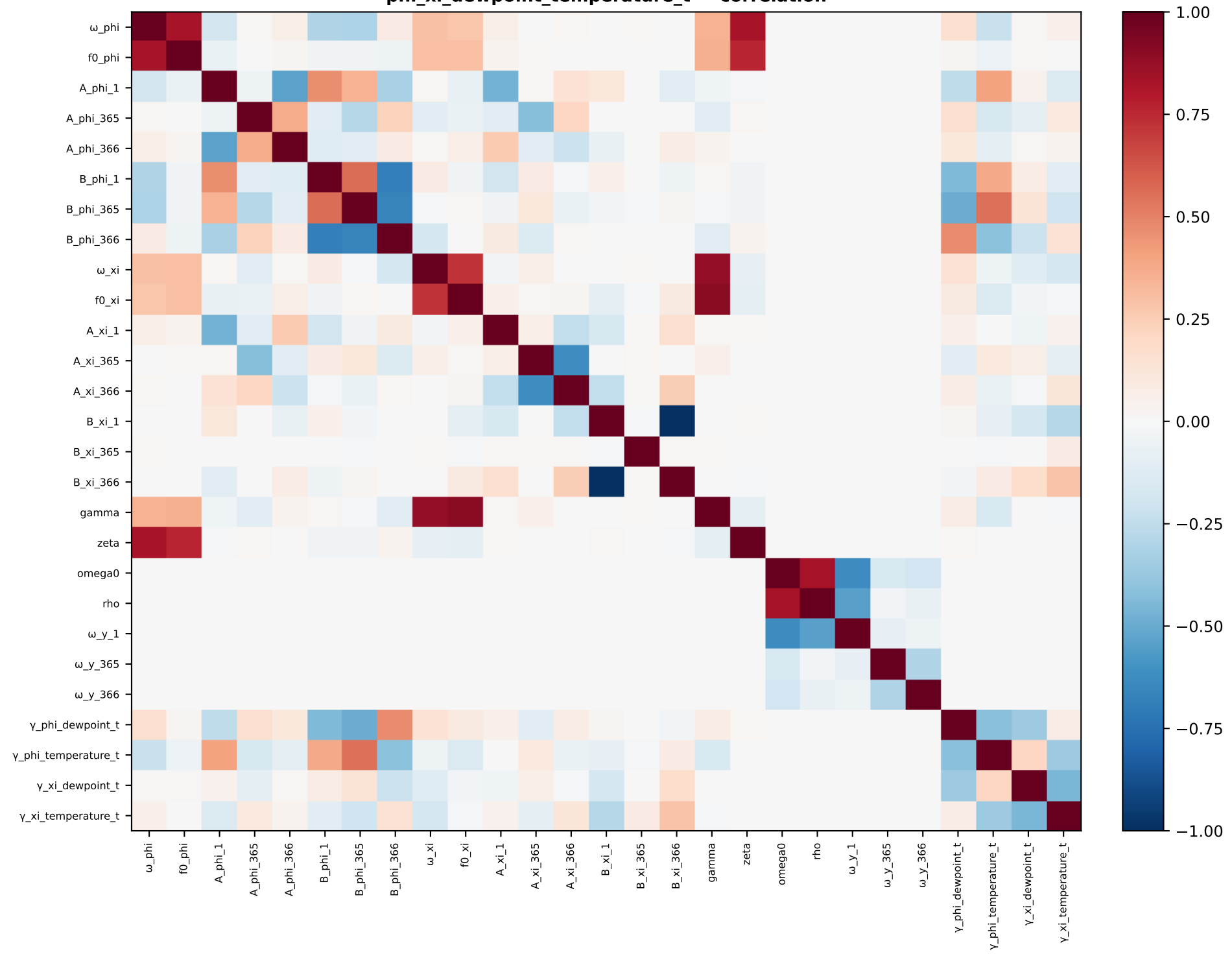
phi_xi_dewpoint_temperature_t — filtered states (OOS)



phi_xi_dewpoint_temperature_t — covariance

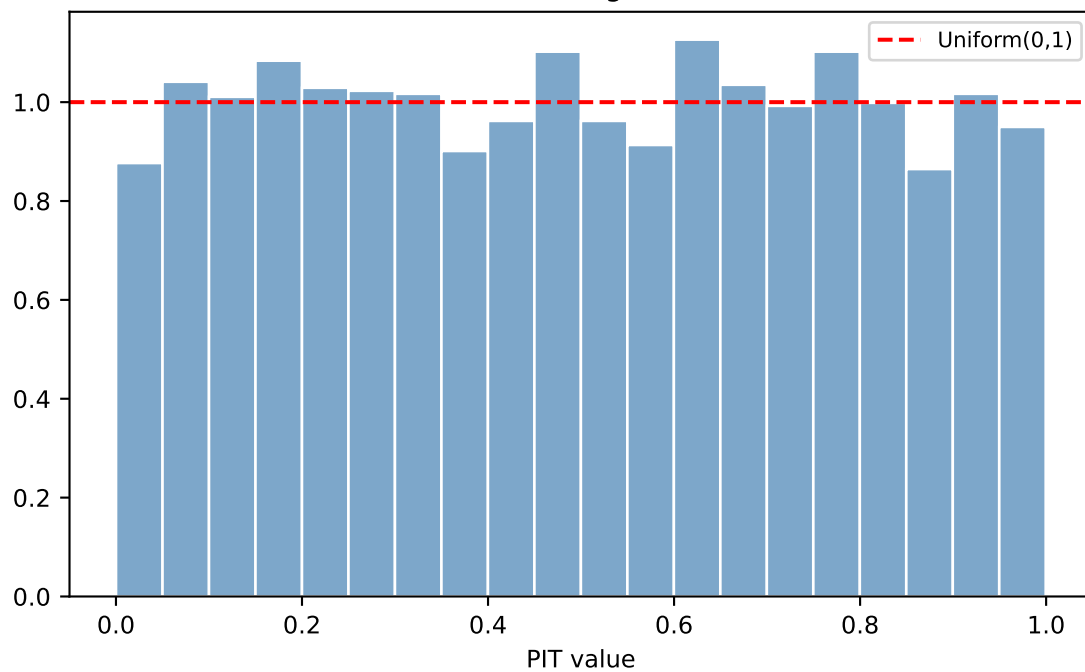


phi_xi_dewpoint_temperature_t — correlation

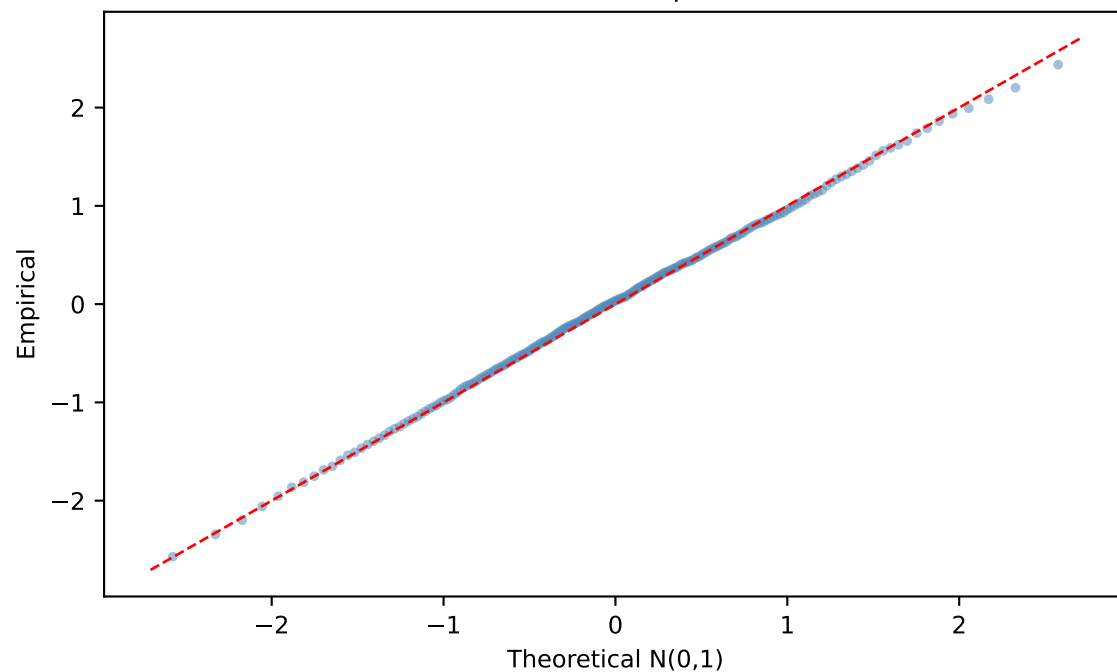


phi_xi_dewpoint_temperature_lag1 — IS diagnostics

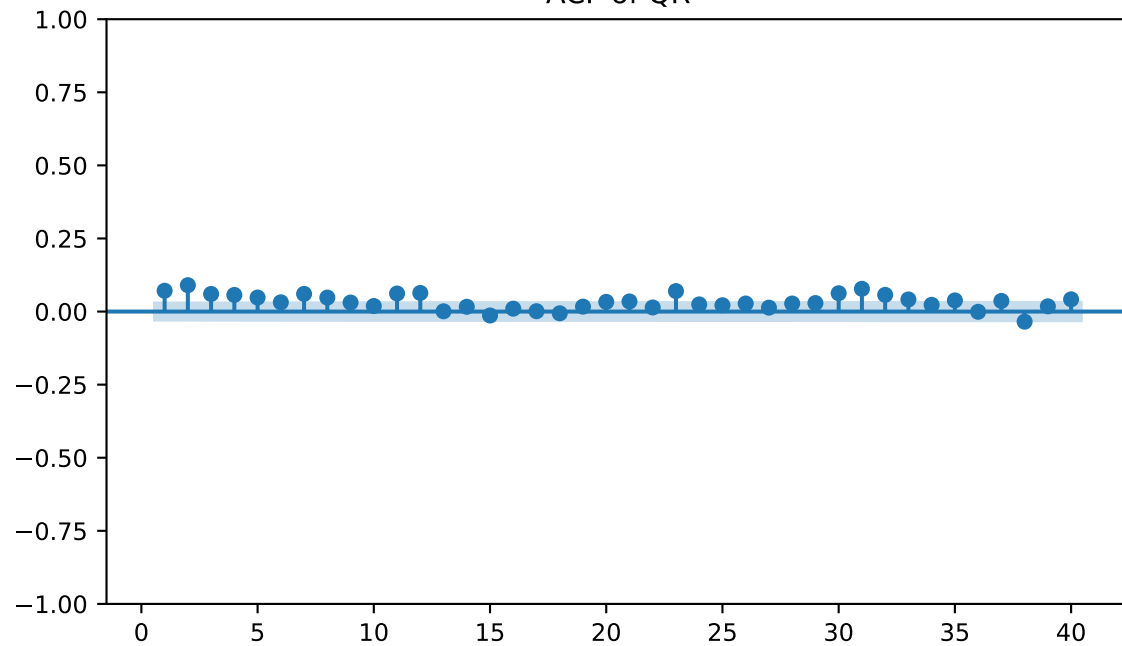
PIT histogram



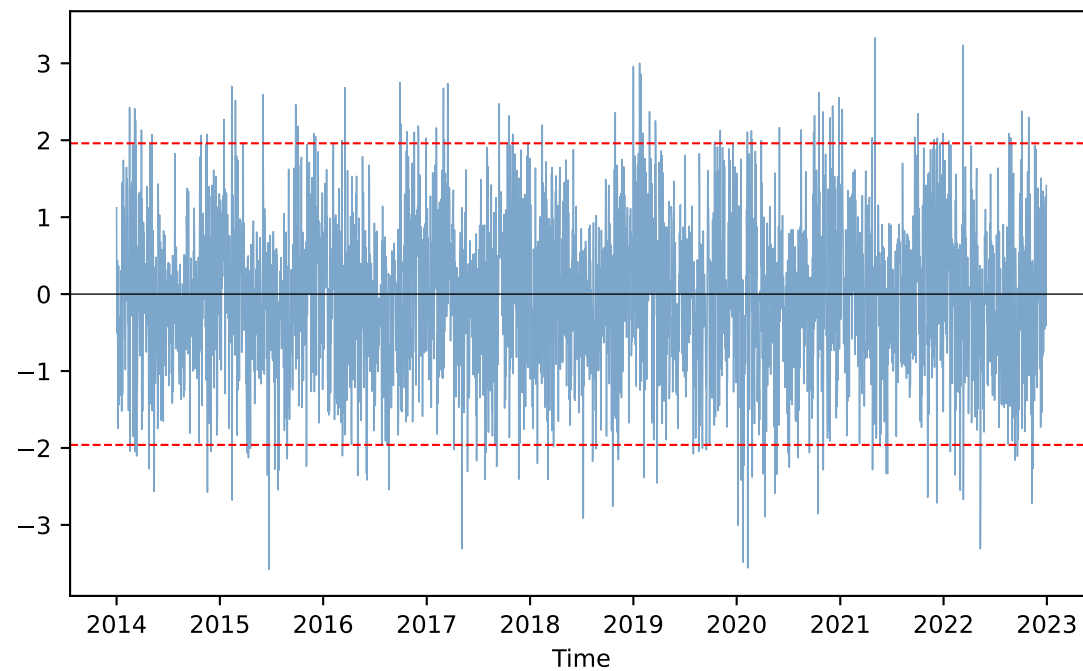
Normal QQ plot



ACF of QR

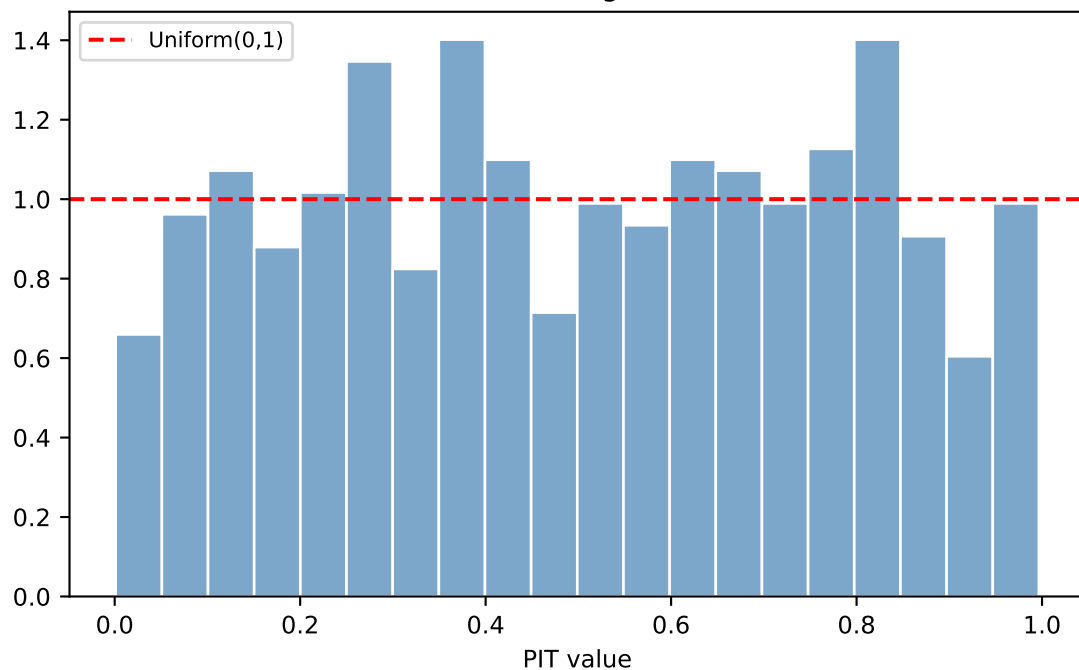


QR time series

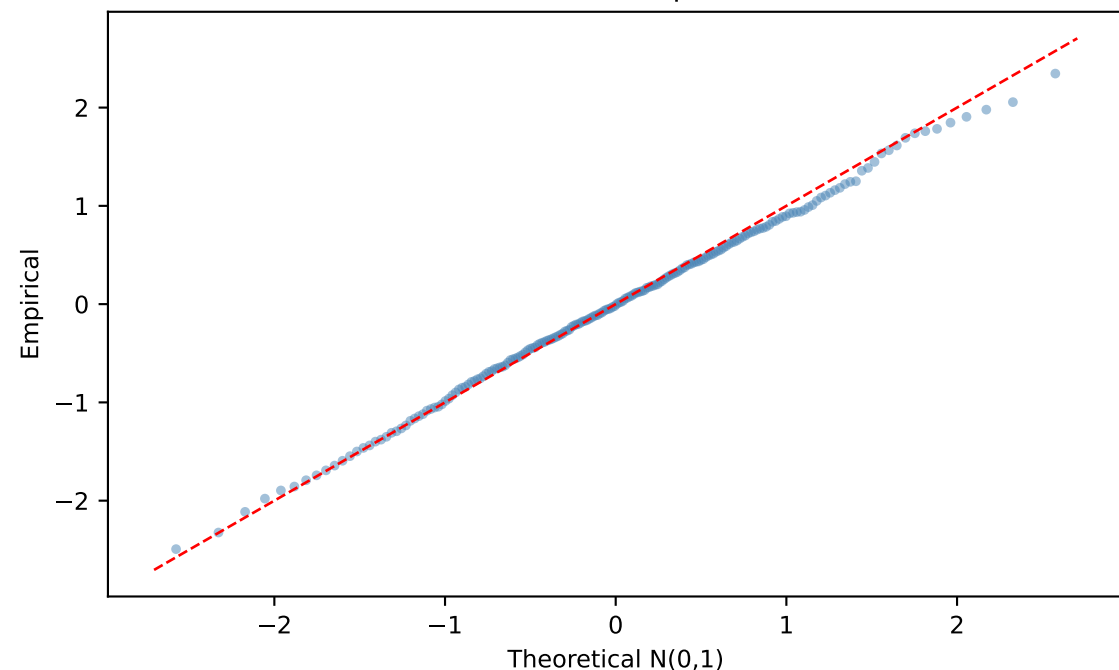


phi_xi_dewpoint_temperature_lag1 — OOS diagnostics

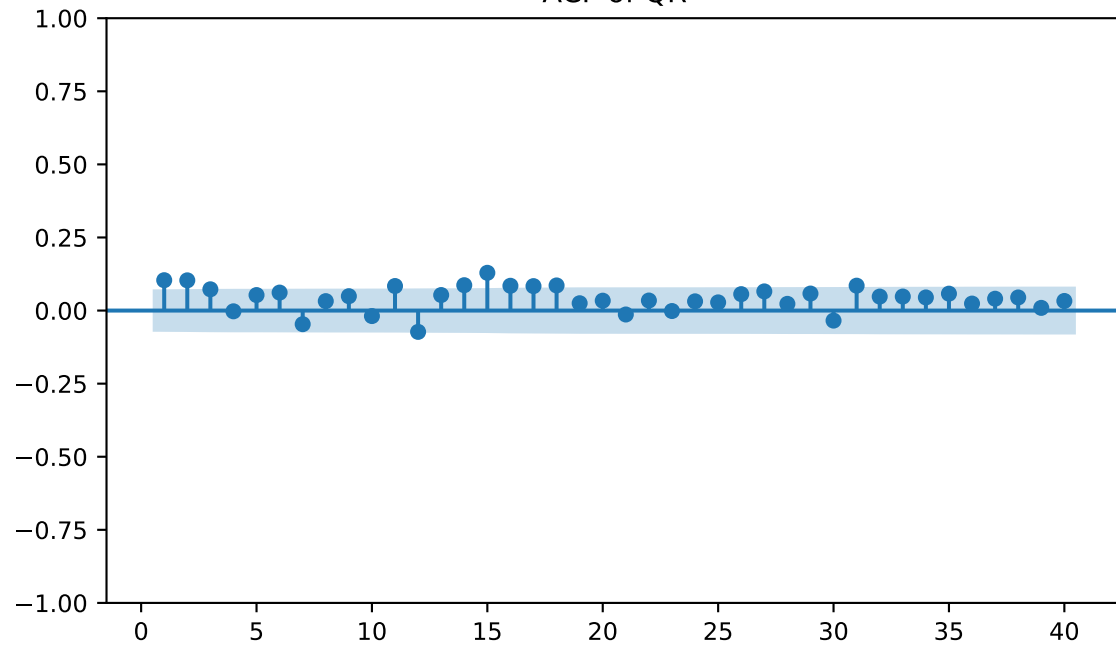
PIT histogram



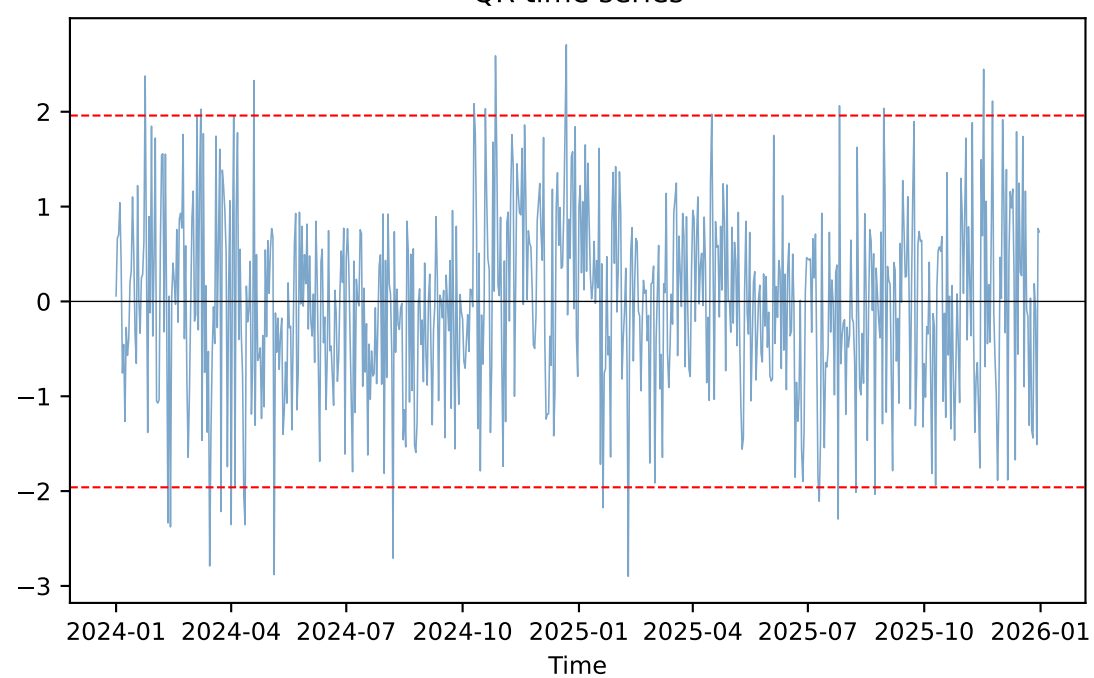
Normal QQ plot



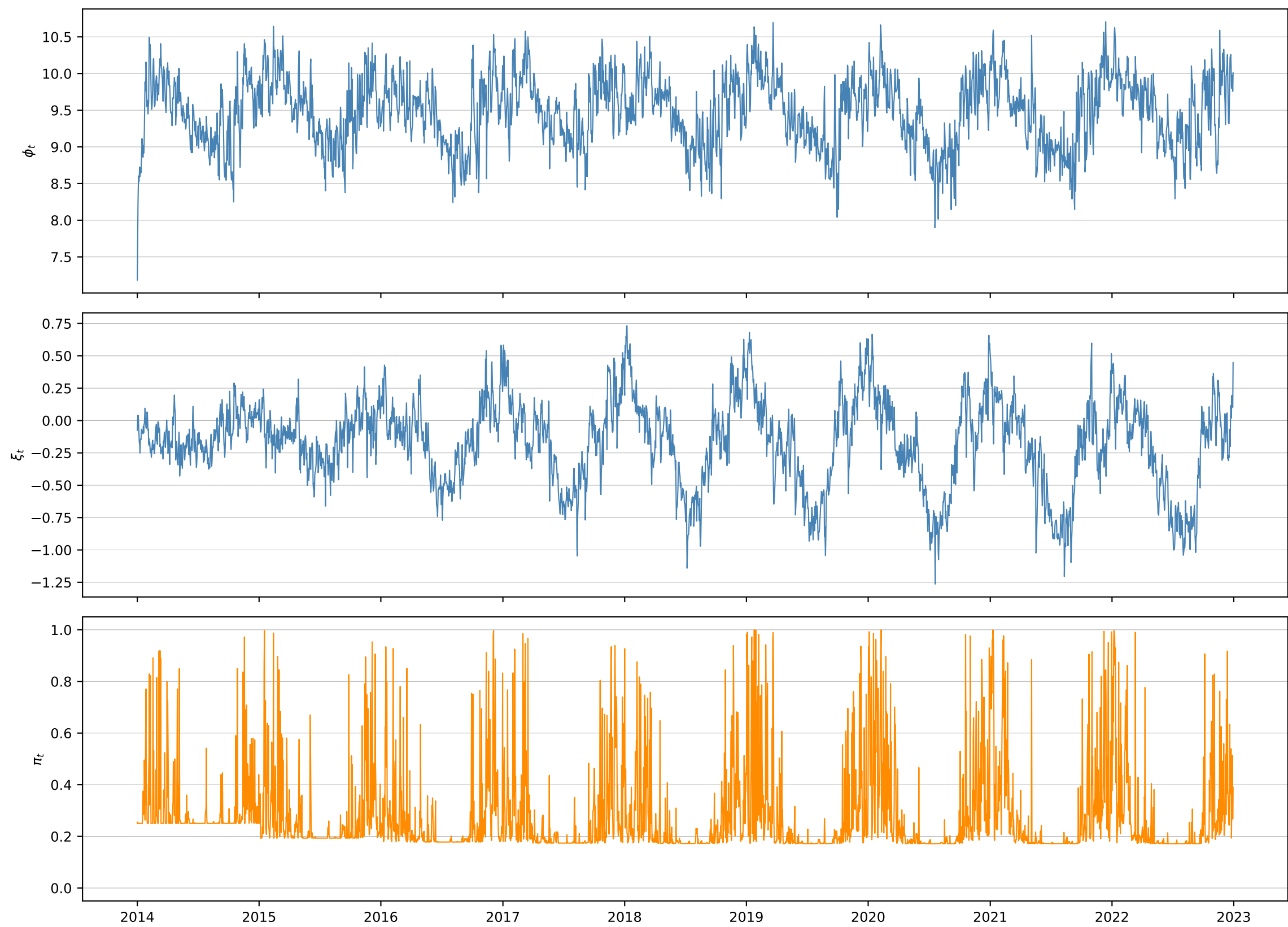
ACF of QR



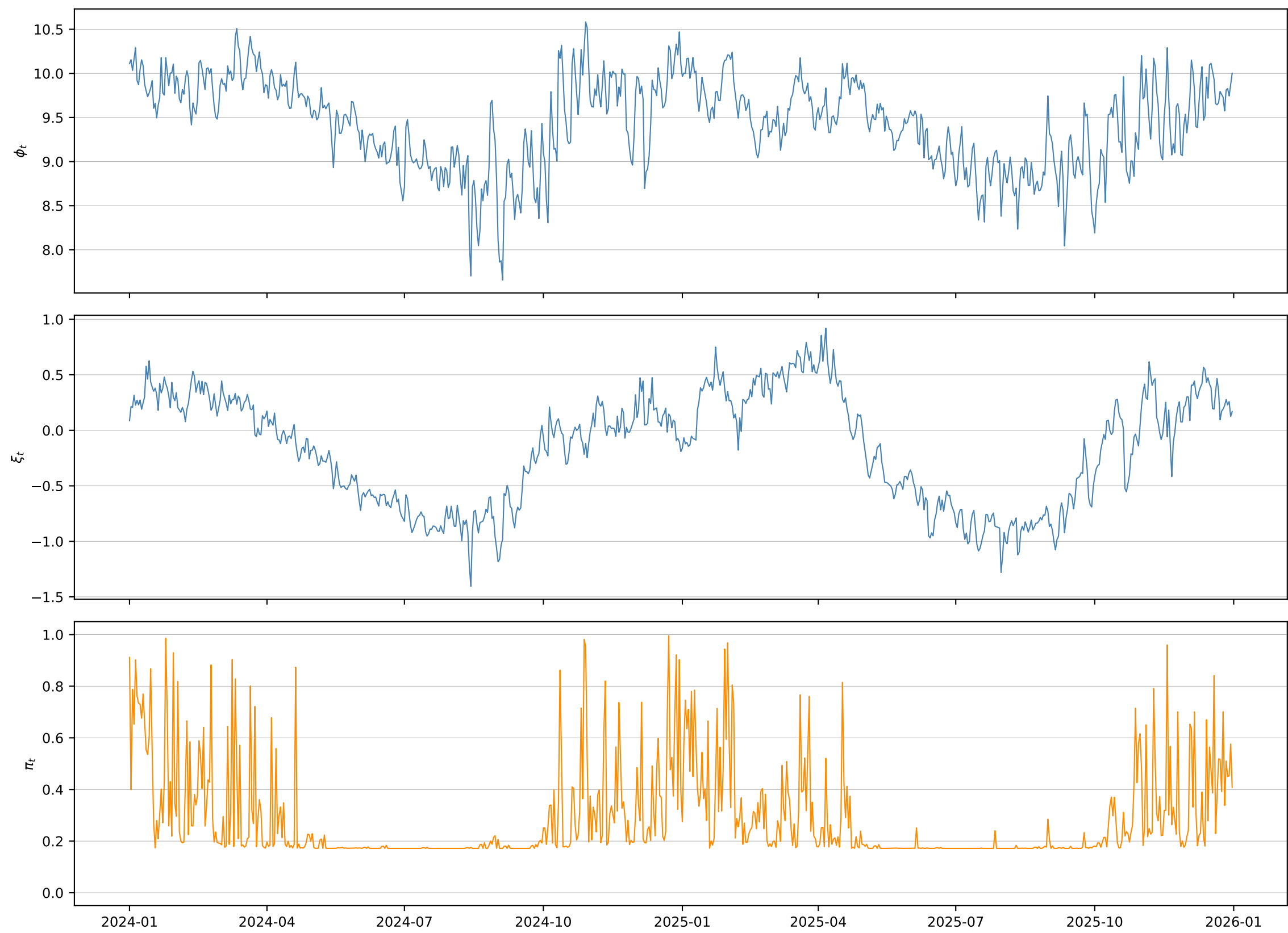
QR time series



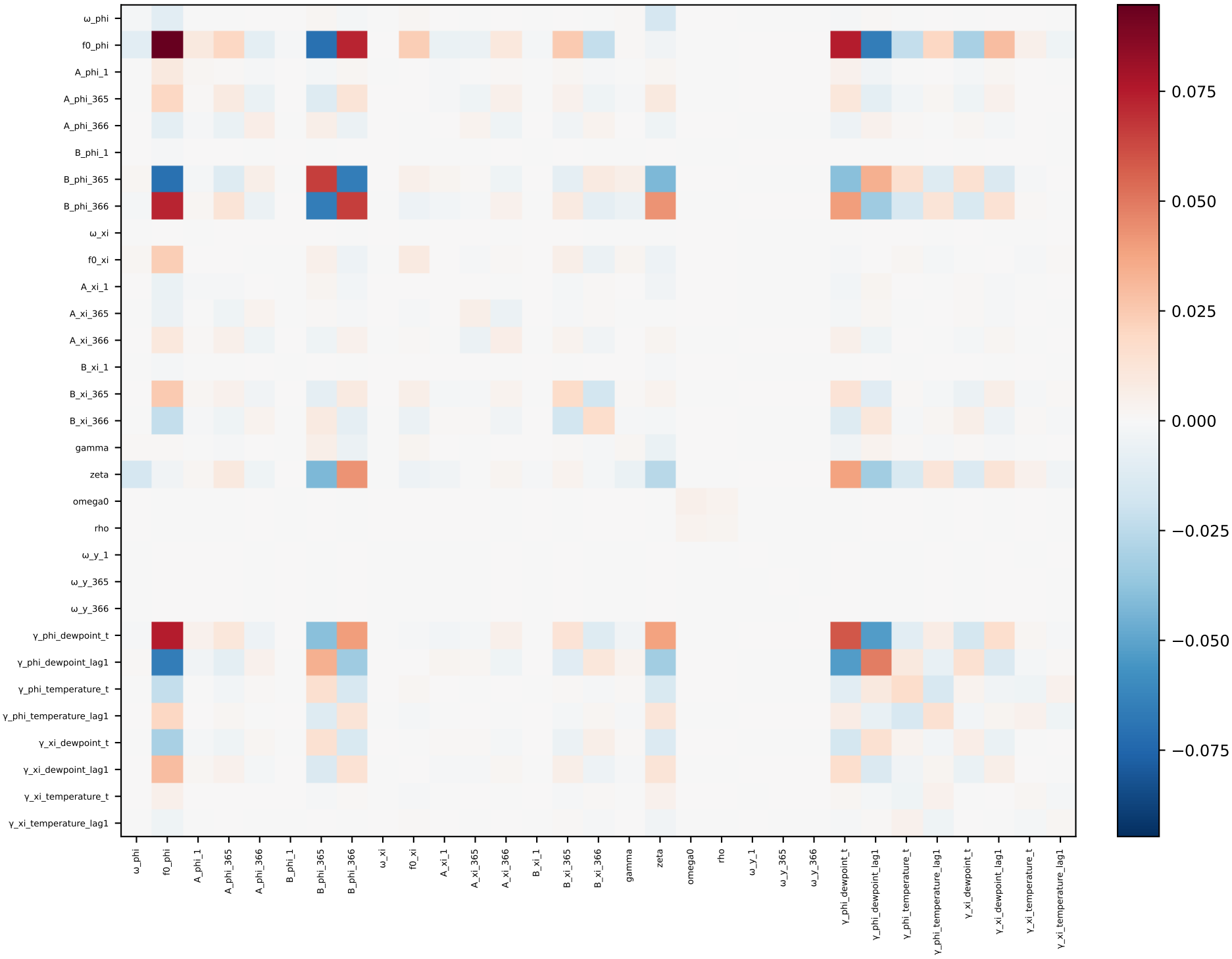
phi_xi_dewpoint_temperature_lag1 — filtered states (IS)



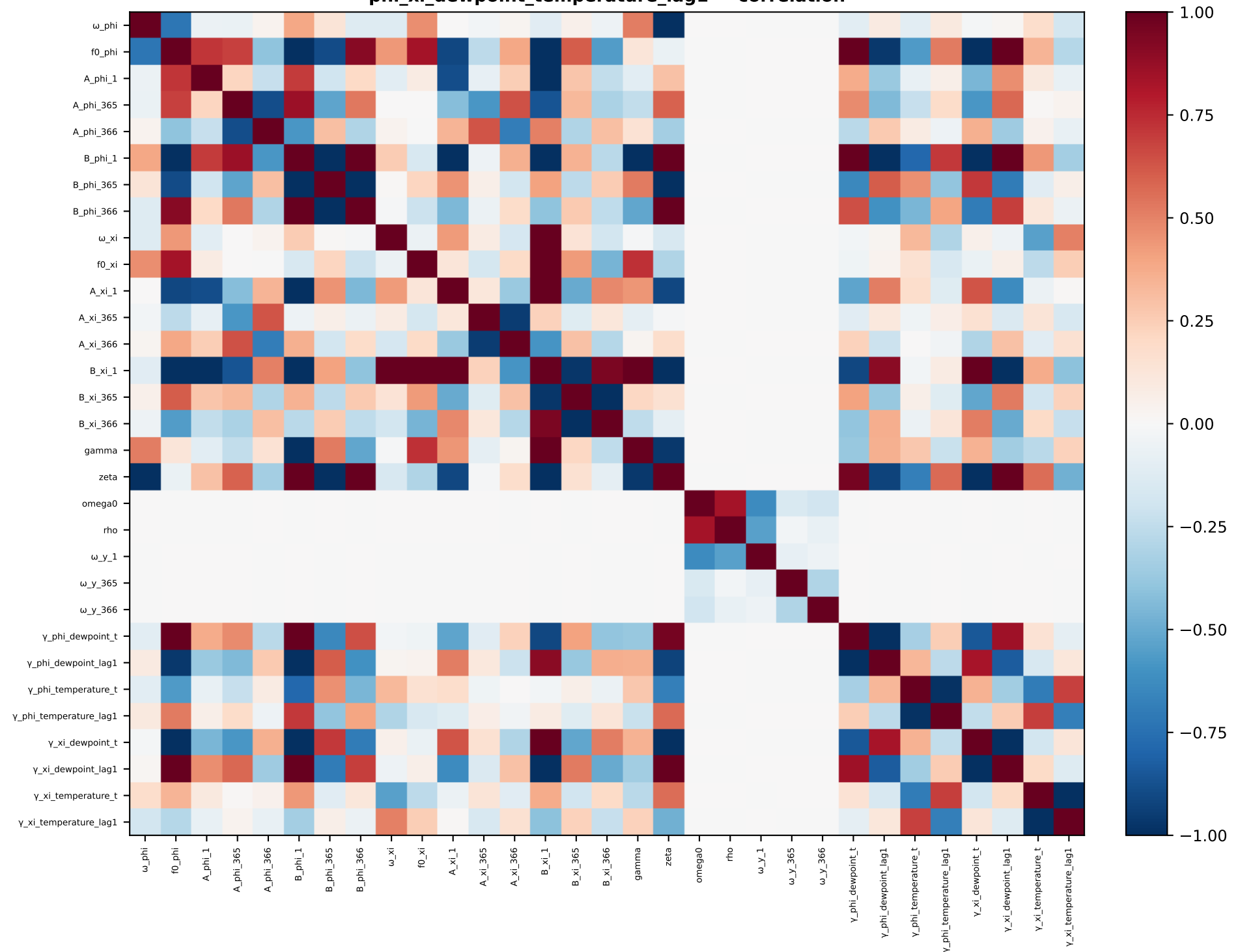
phi_xi_dewpoint_temperature_lag1 — filtered states (OOS)



phi_xi_dewpoint_temperature_lag1 — covariance

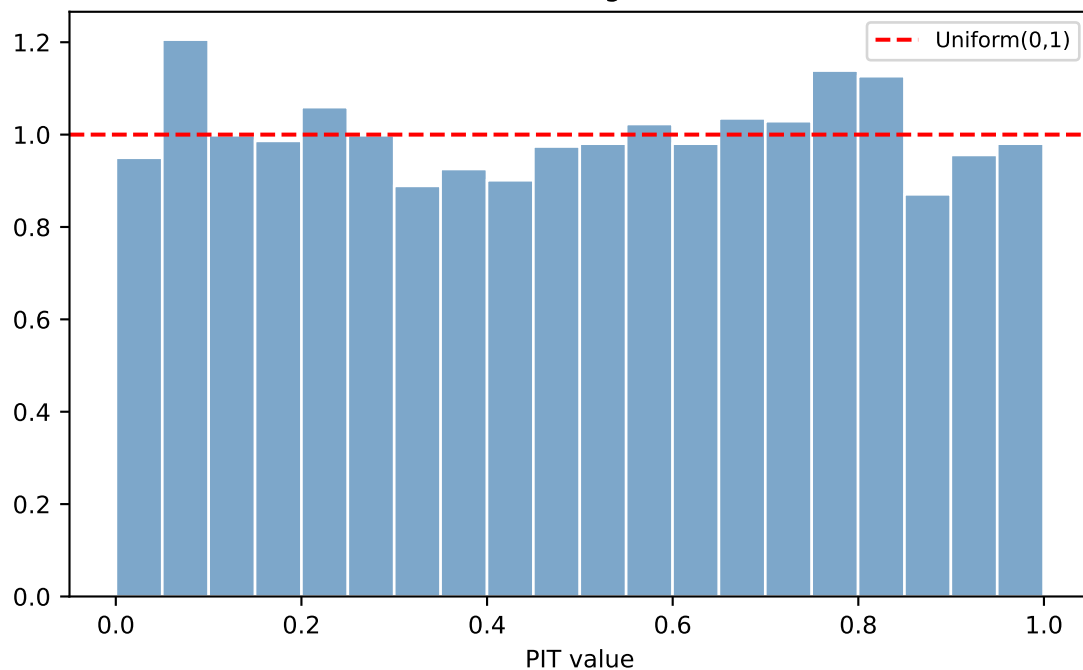


phi_xi_dewpoint_temperature_lag1 — correlation

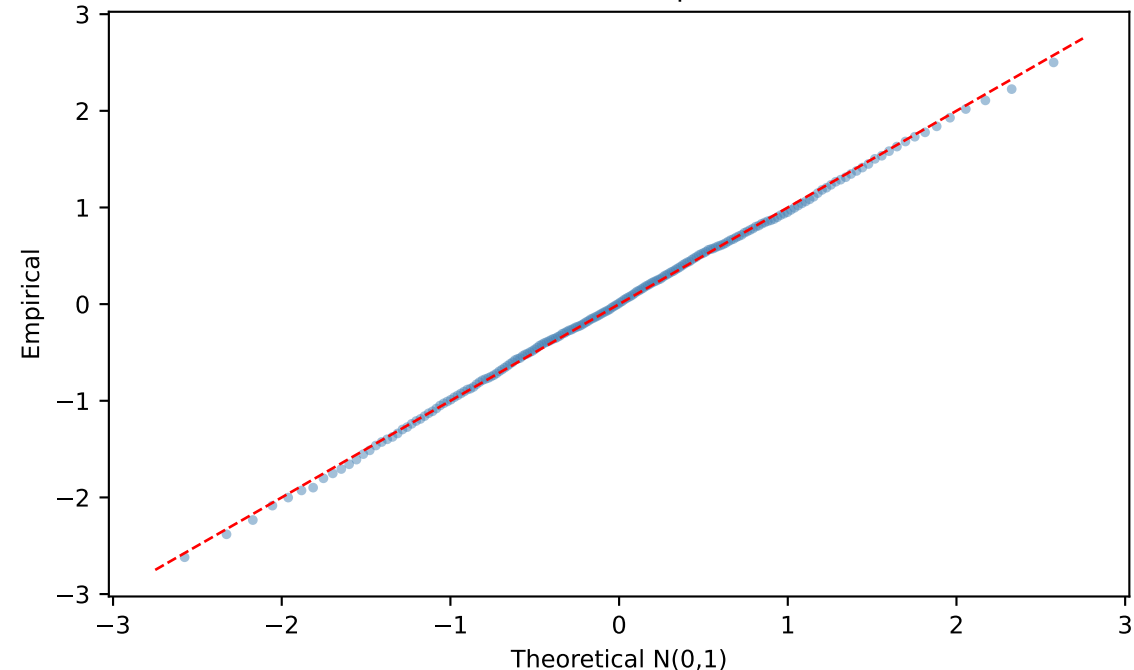


phi_xi_dewpoint_temperature_short_lags — IS diagnostics

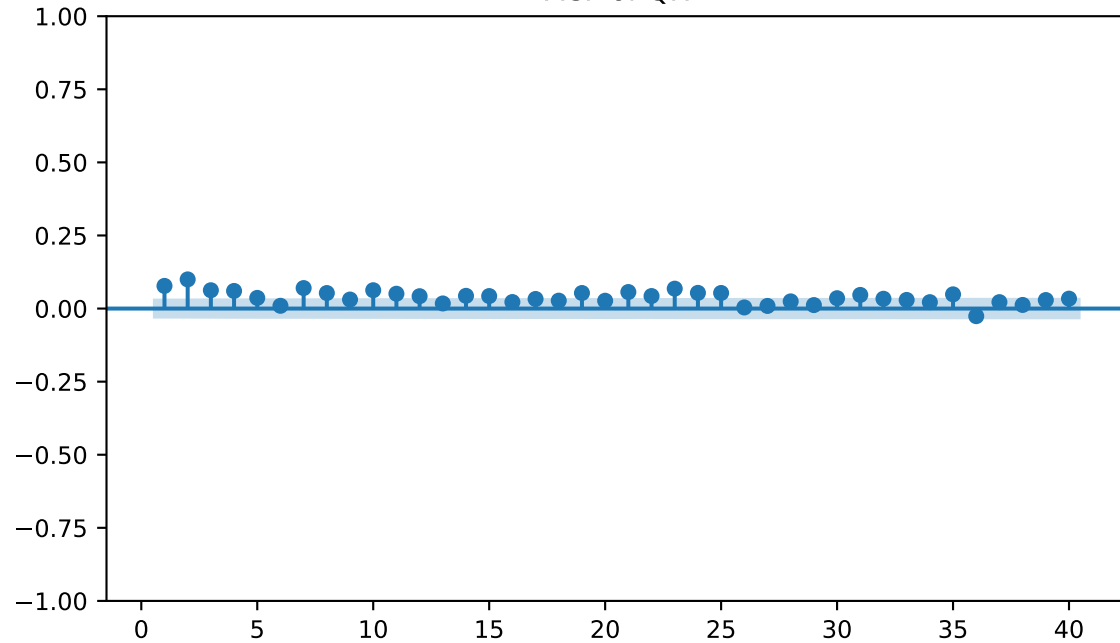
PIT histogram



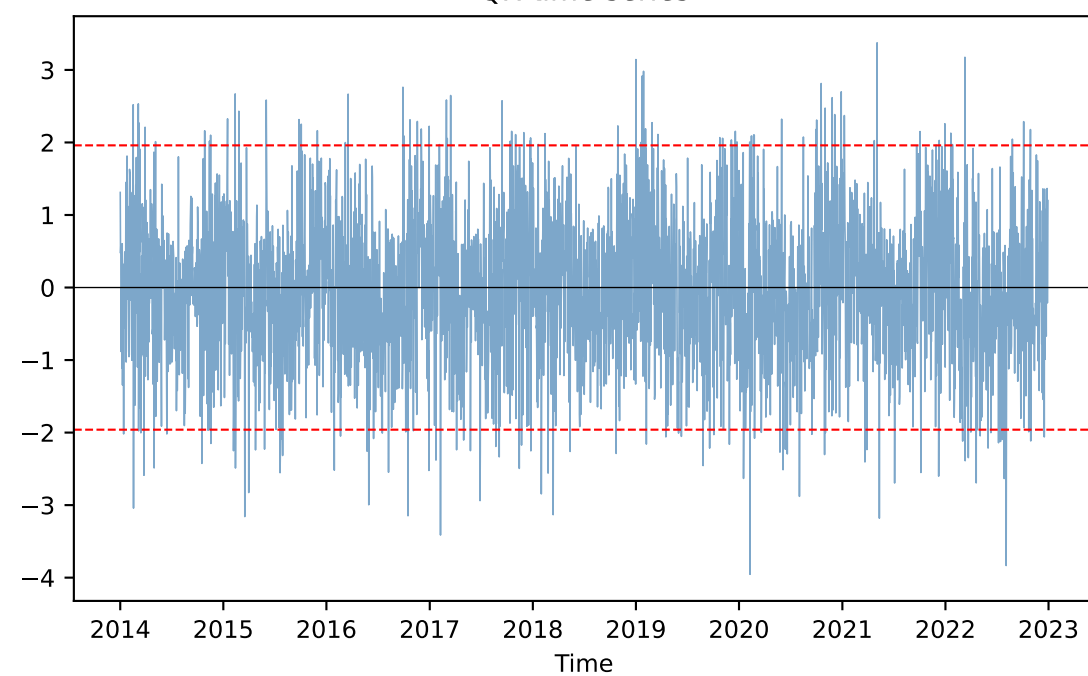
Normal QQ plot



ACF of QR

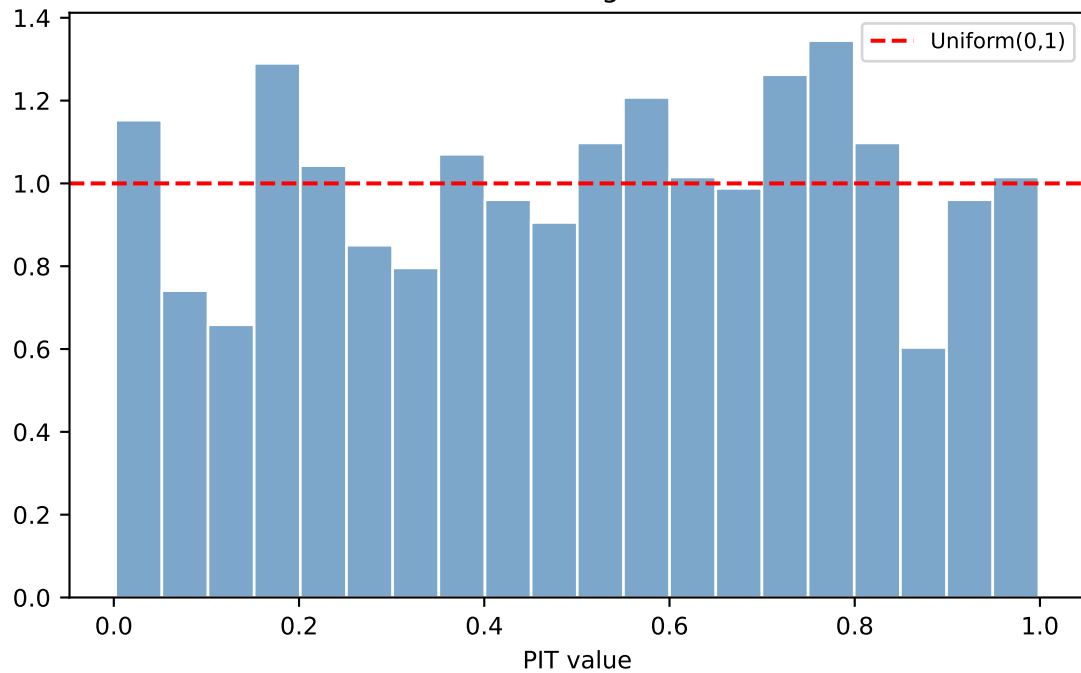


QR time series

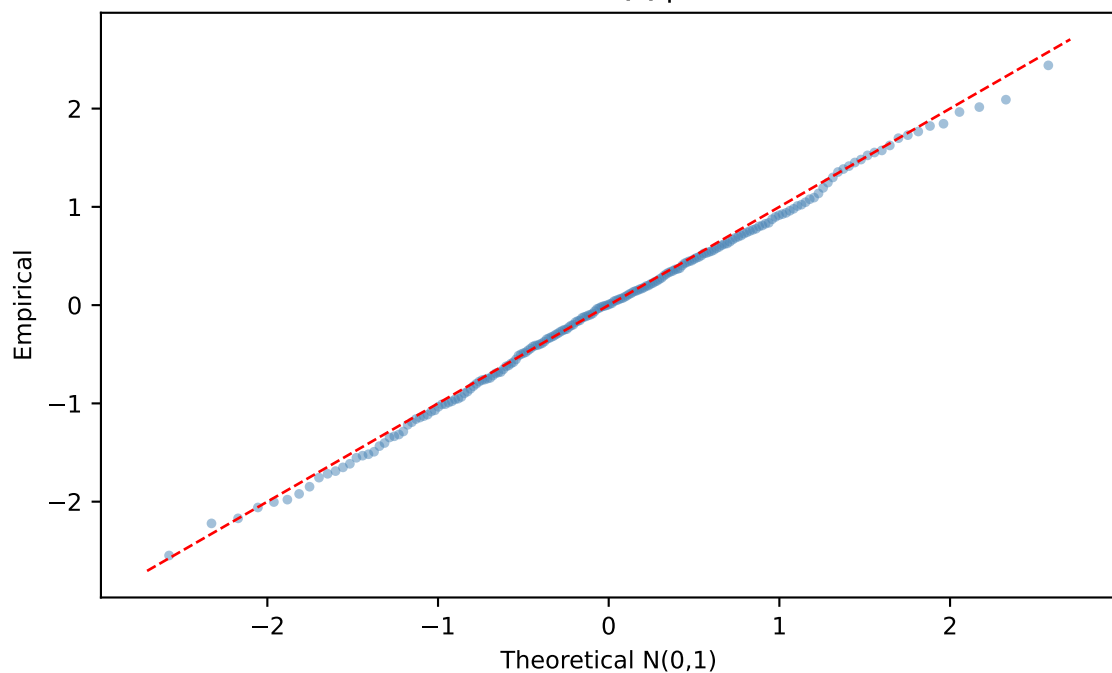


phi_xi_dewpoint_temperature_short_lags — OOS diagnostics

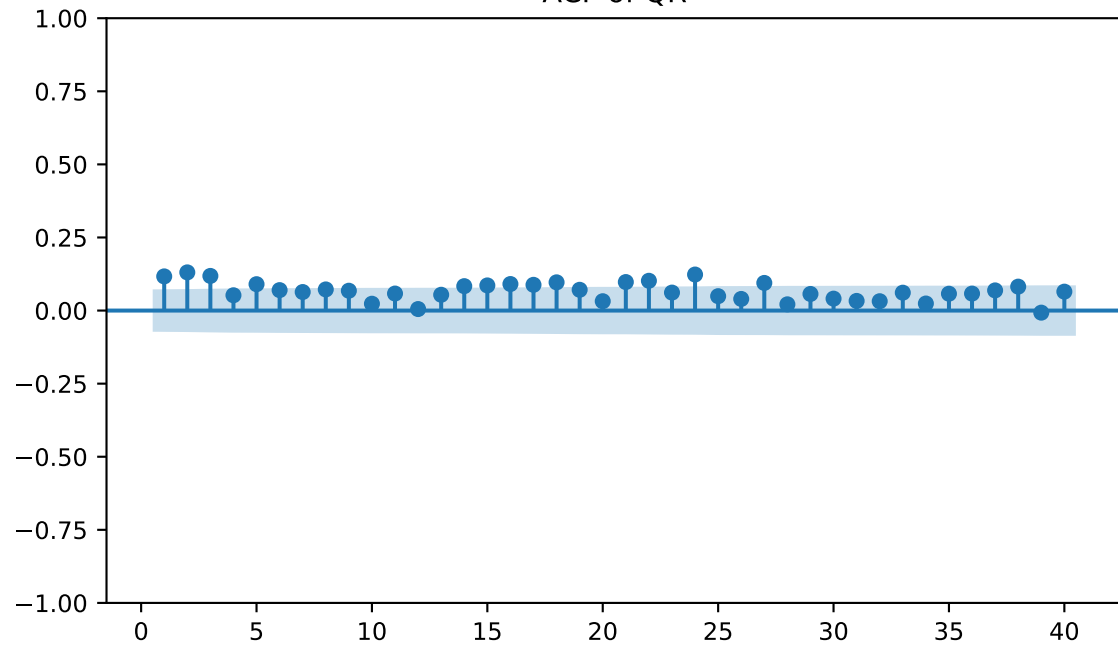
PIT histogram



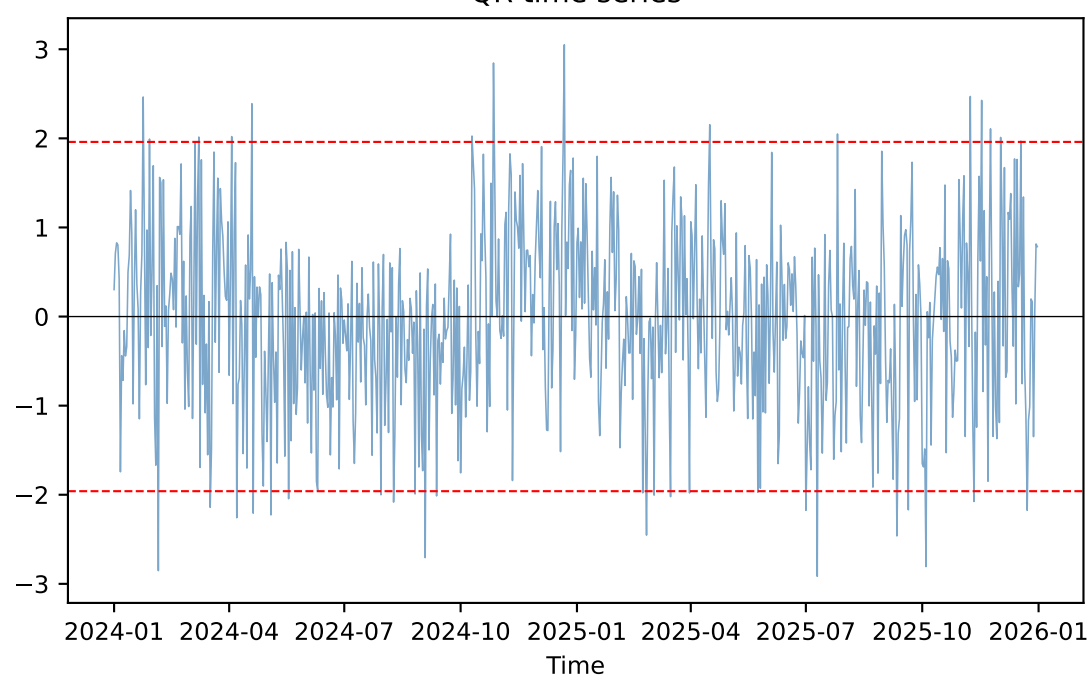
Normal QQ plot



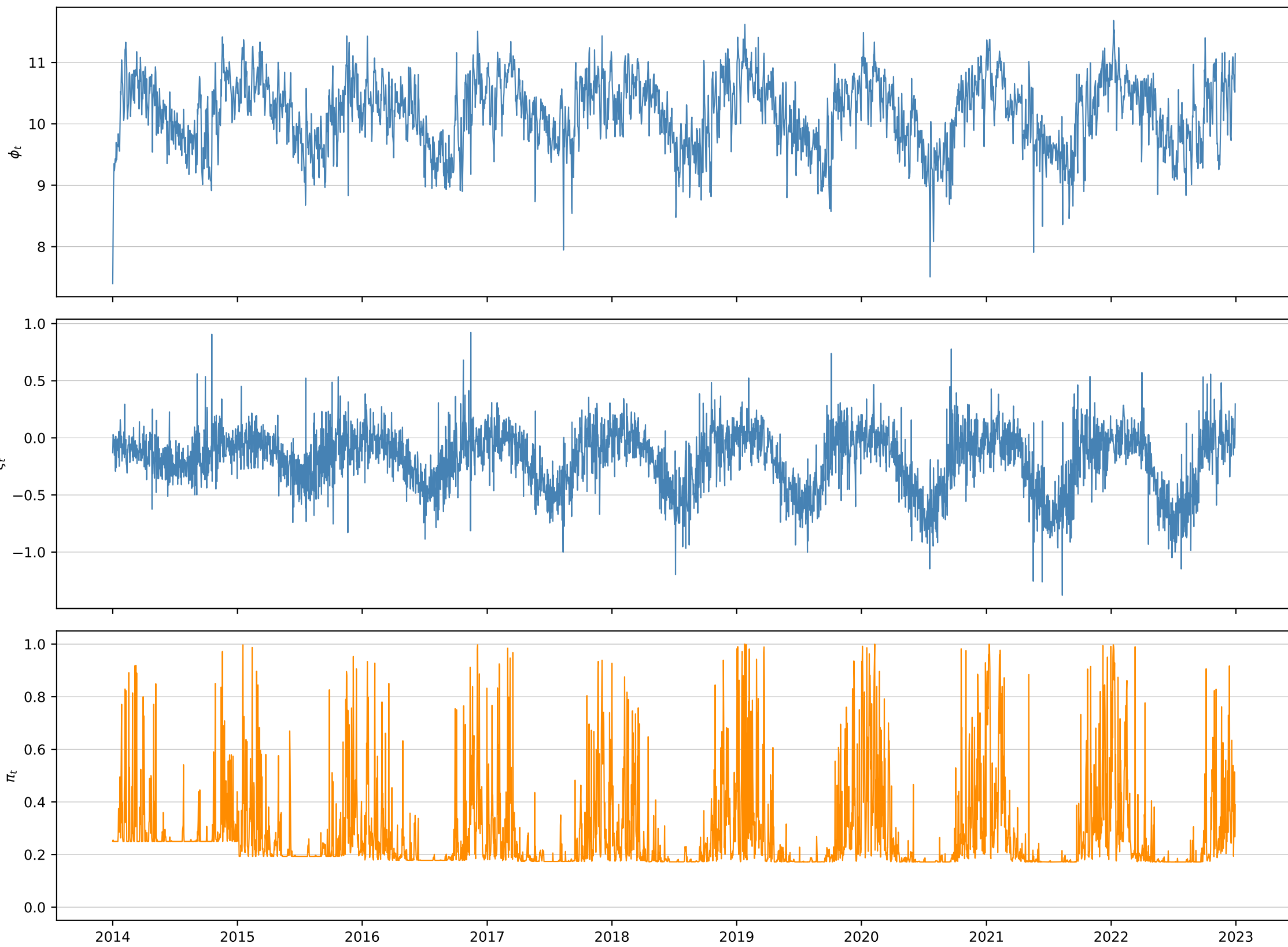
ACF of QR



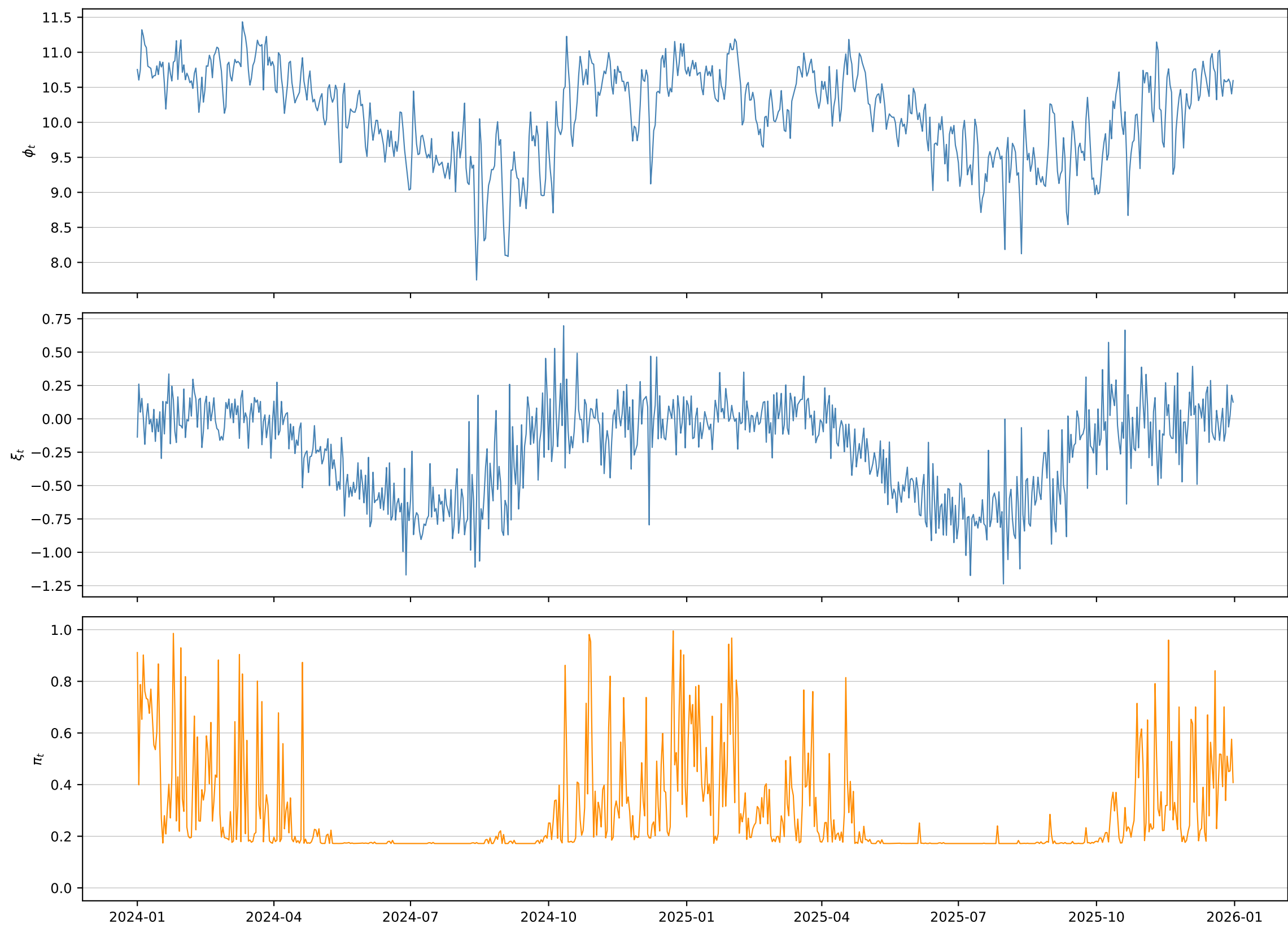
QR time series



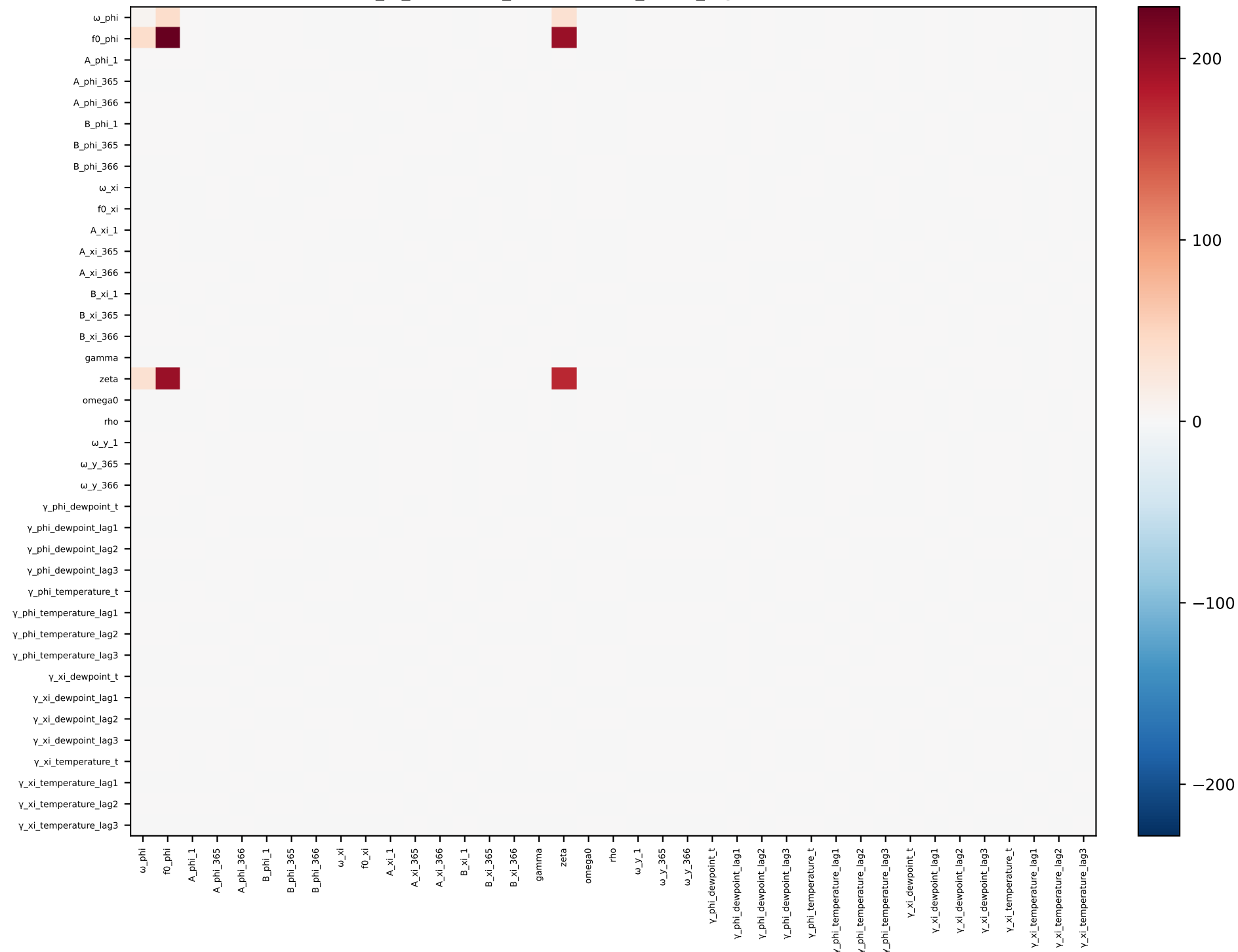
phi_xi_dewpoint_temperature_short_lags — filtered states (IS)



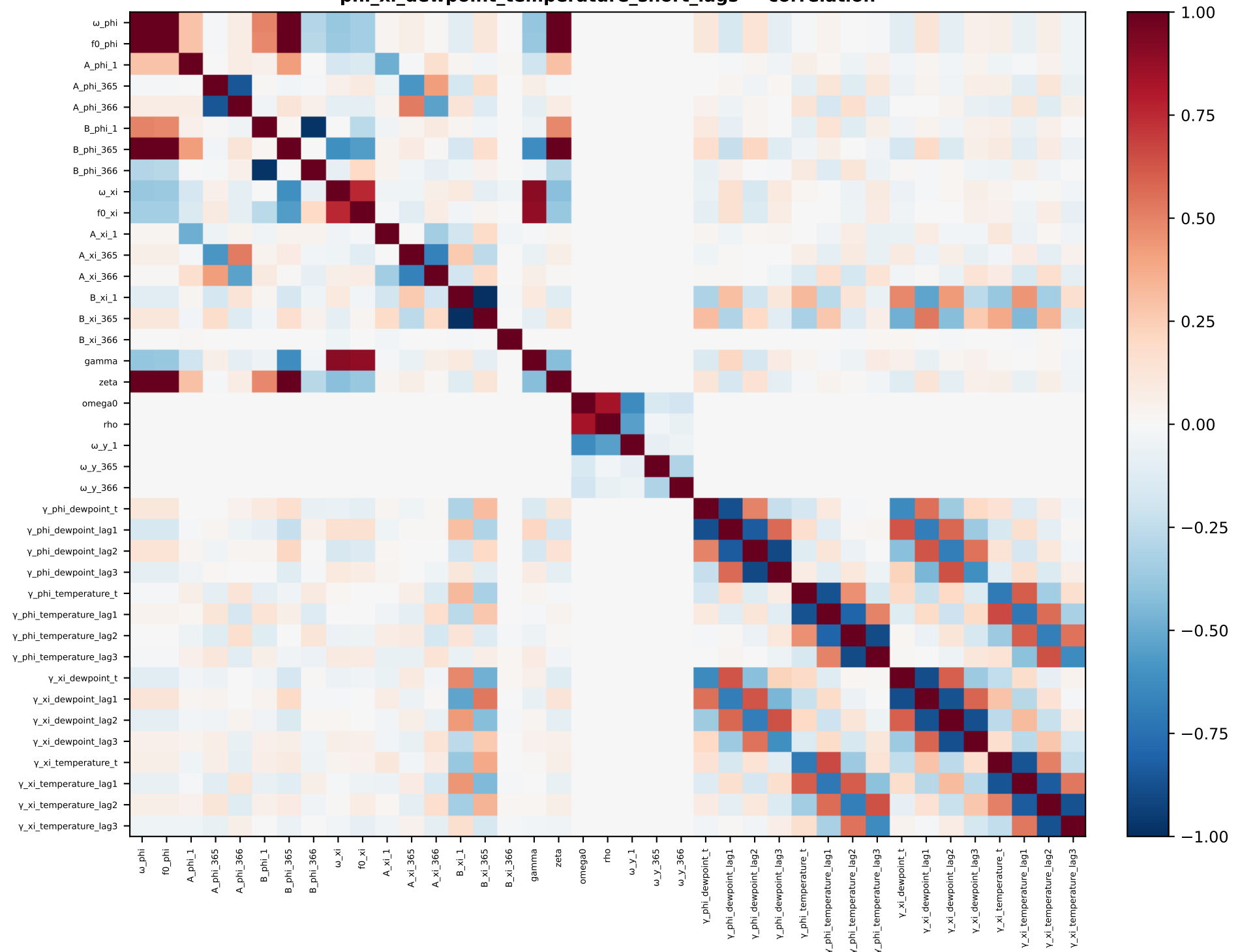
phi_xi_dewpoint_temperature_short_lags — filtered states (OOS)



phi_xi_dewpoint_temperature_short_lags — covariance

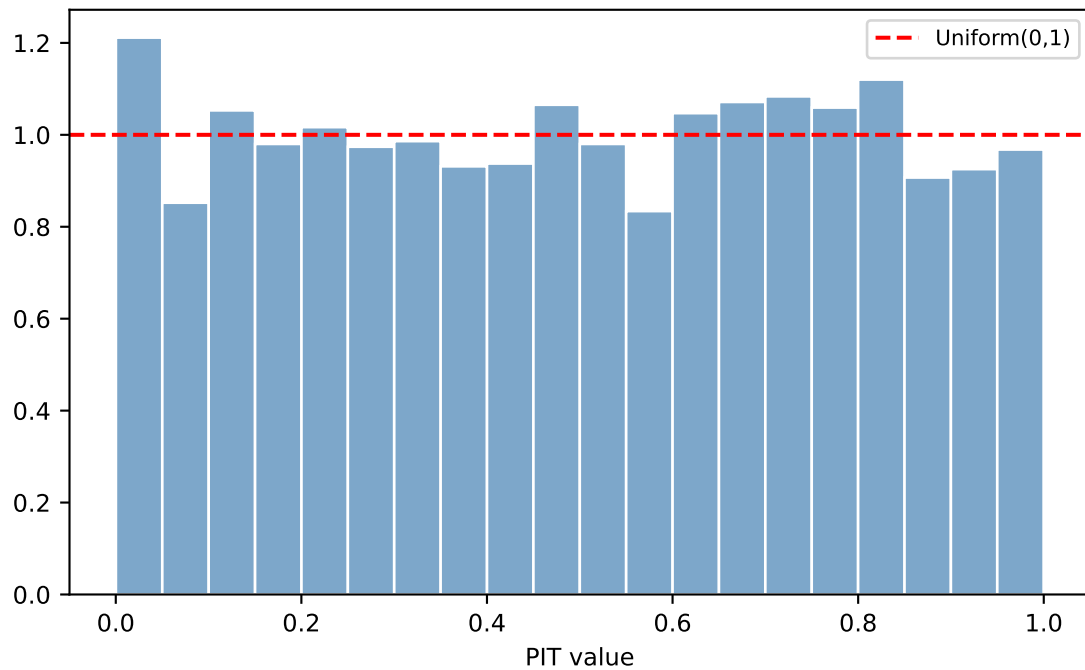


phi_xi_dewpoint_temperature_short_lags — correlation

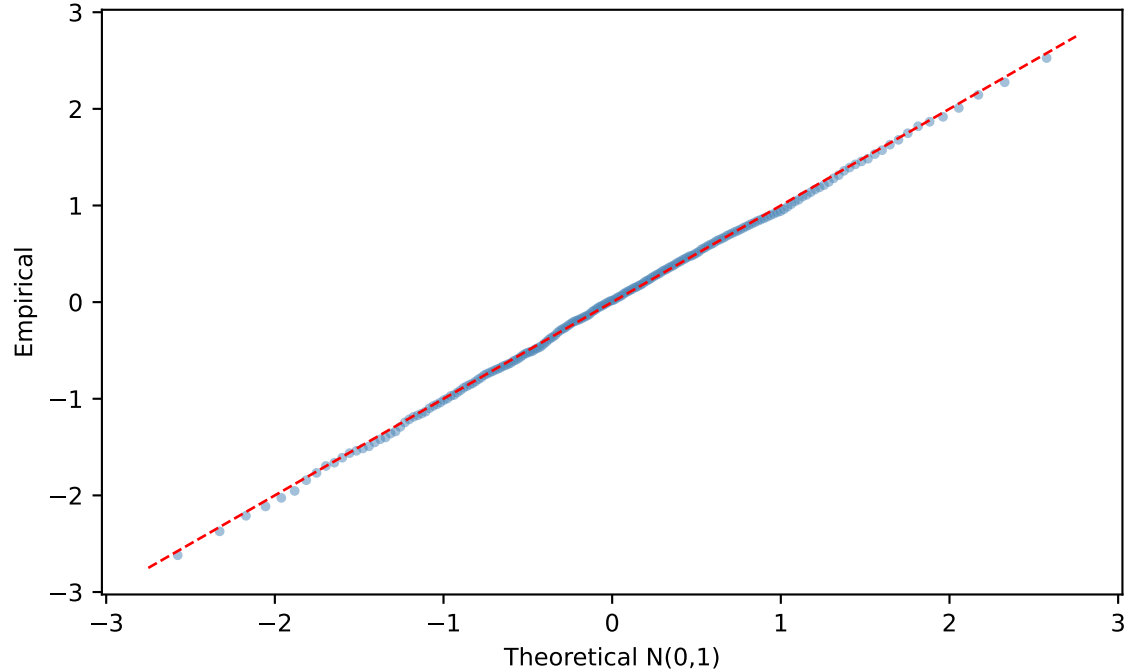


phi_xi_dewpoint_temperature_seasonal_lags — IS diagnostics

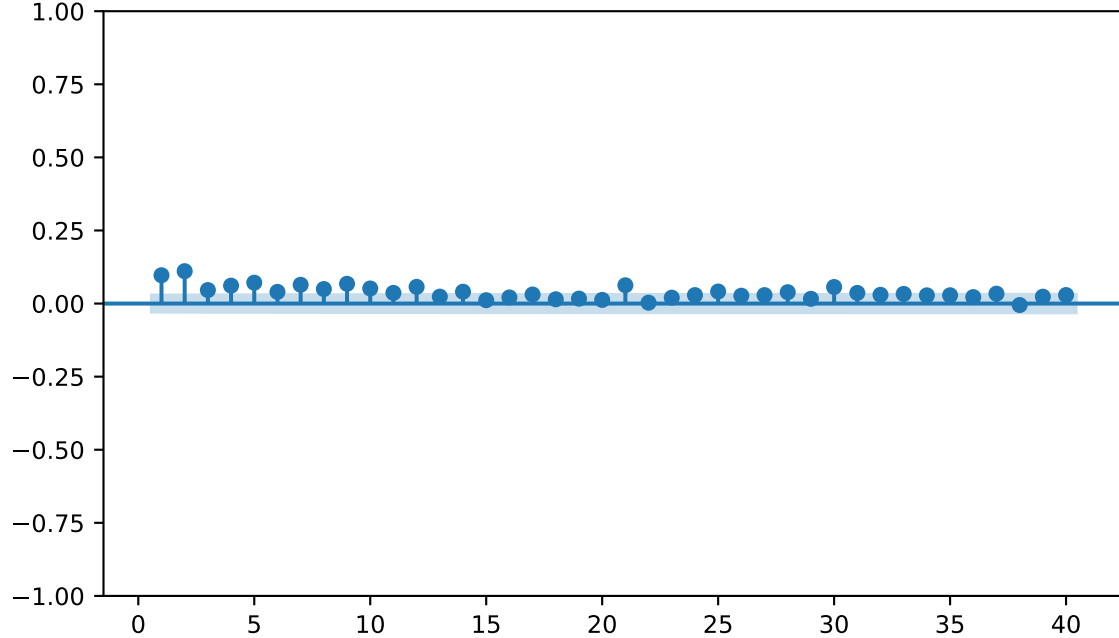
PIT histogram



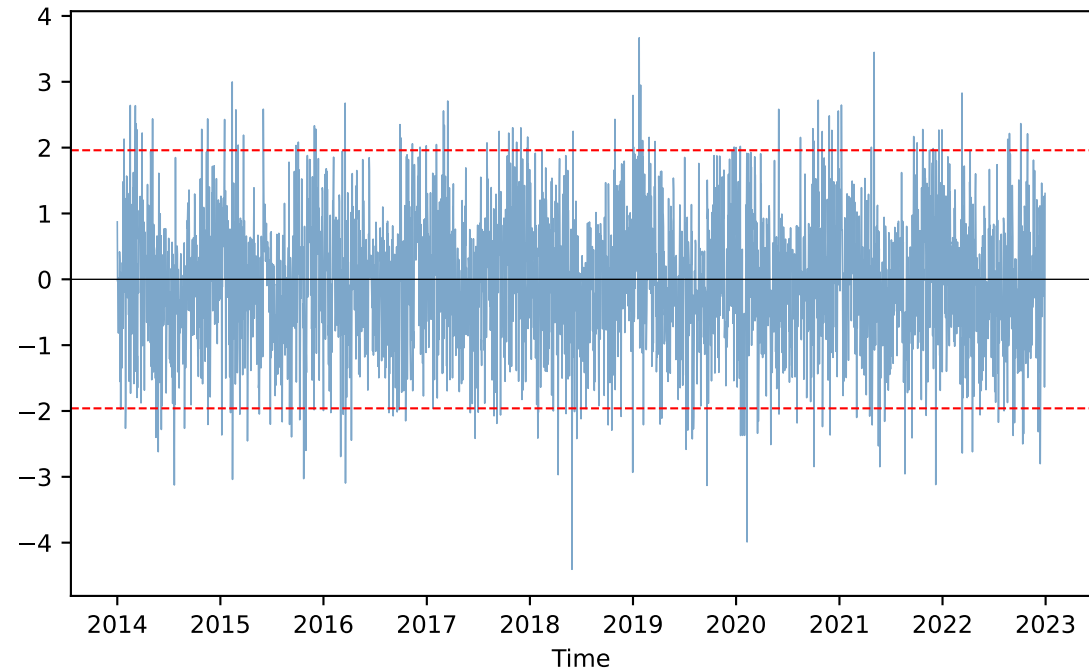
Normal QQ plot



ACF of QR

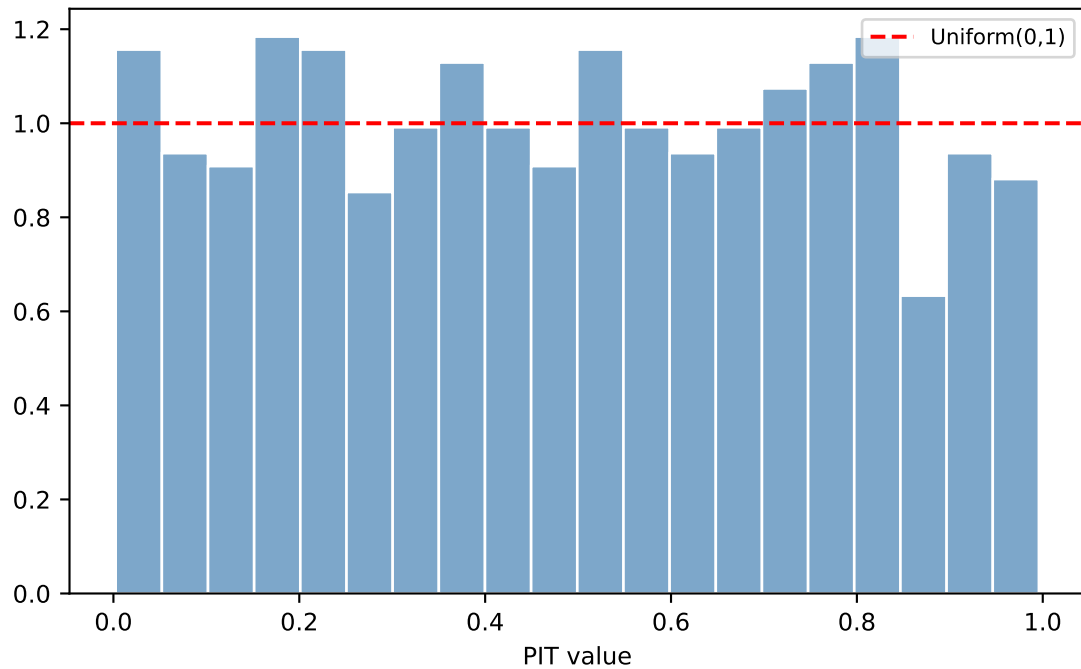


QR time series

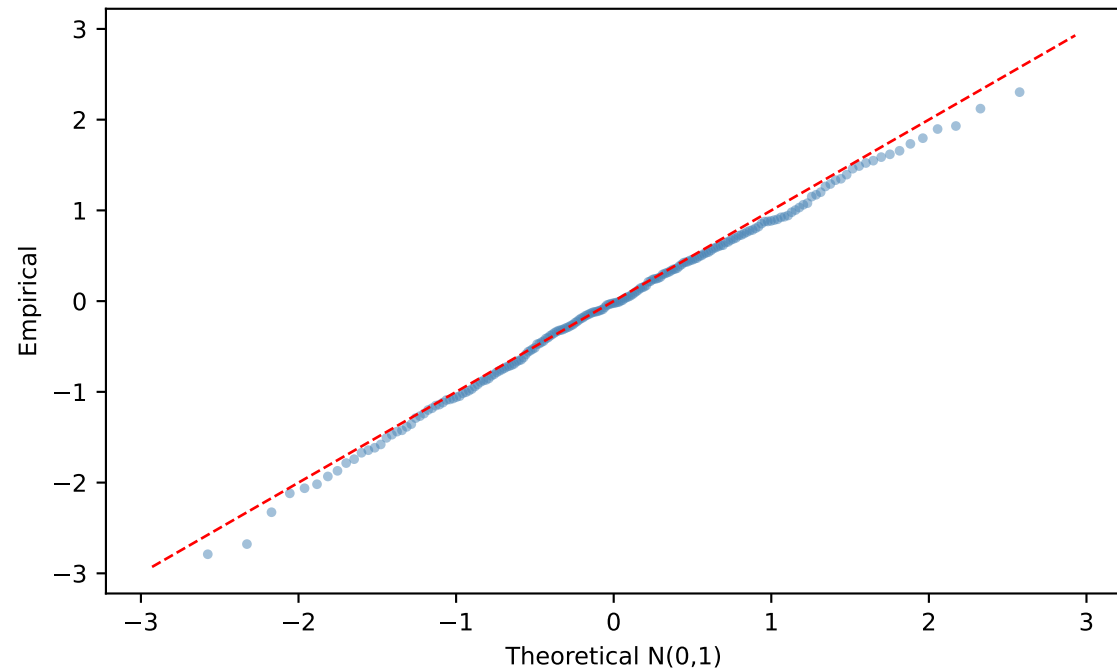


phi_xi_dewpoint_temperature_seasonal_lags — OOS diagnostics

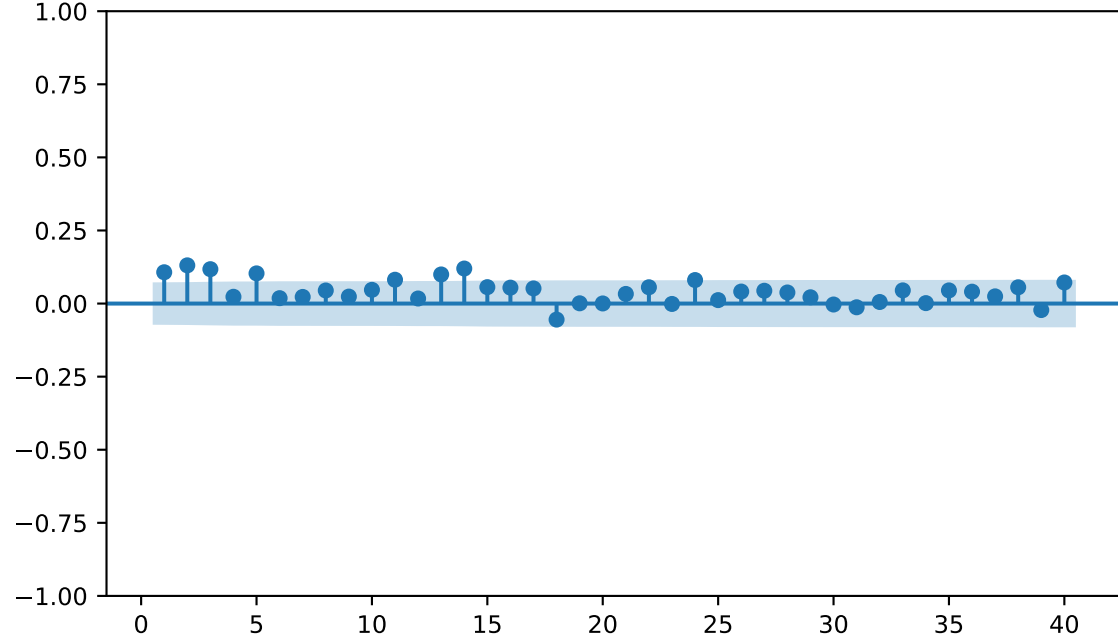
PIT histogram



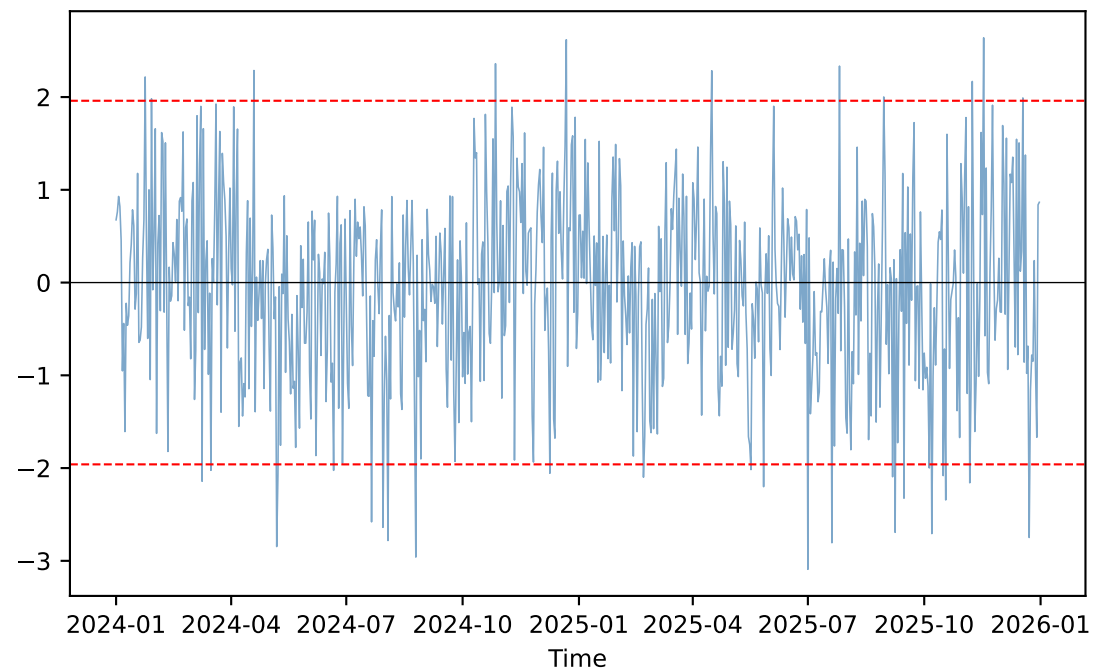
Normal QQ plot



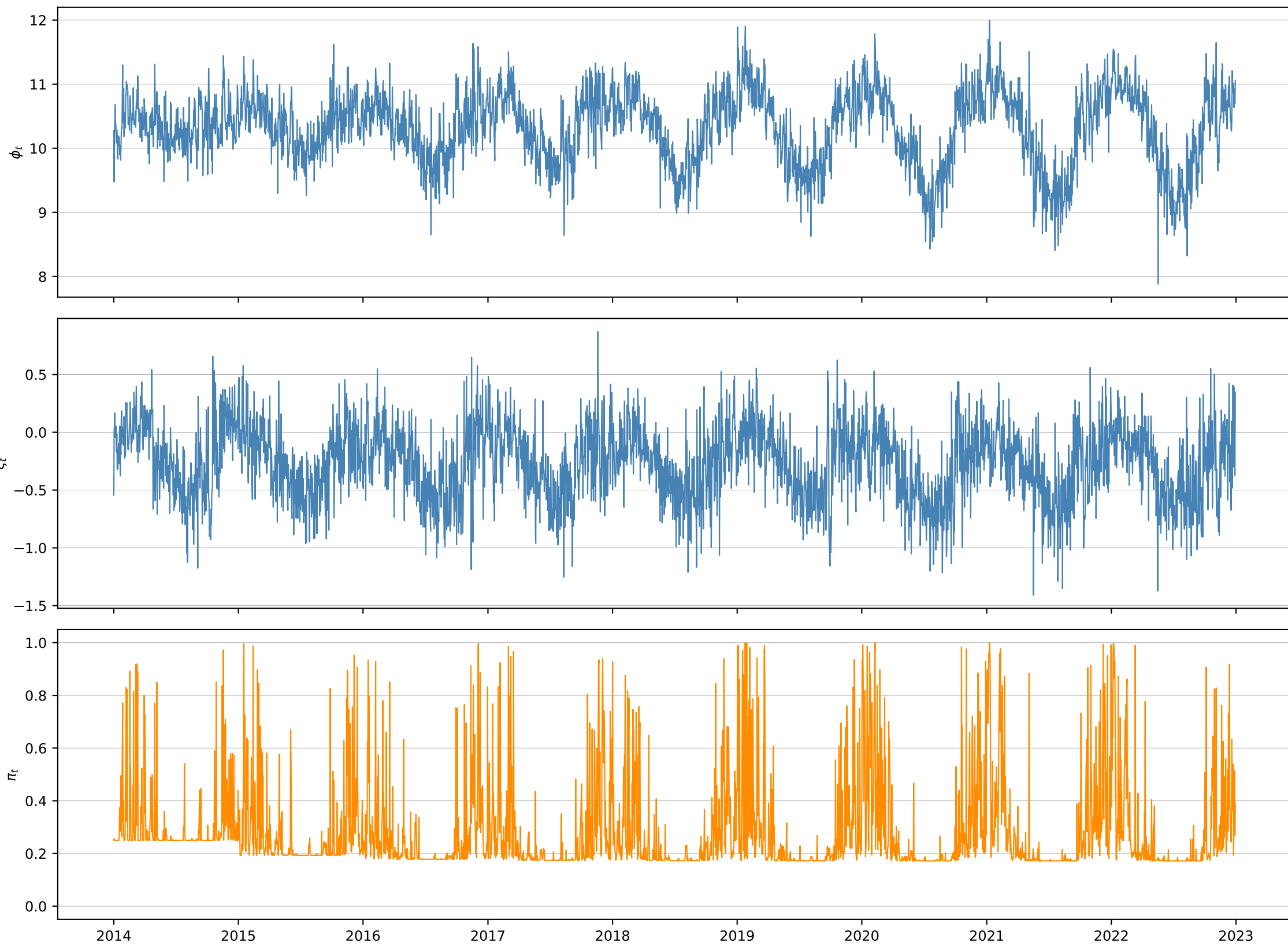
ACF of QR



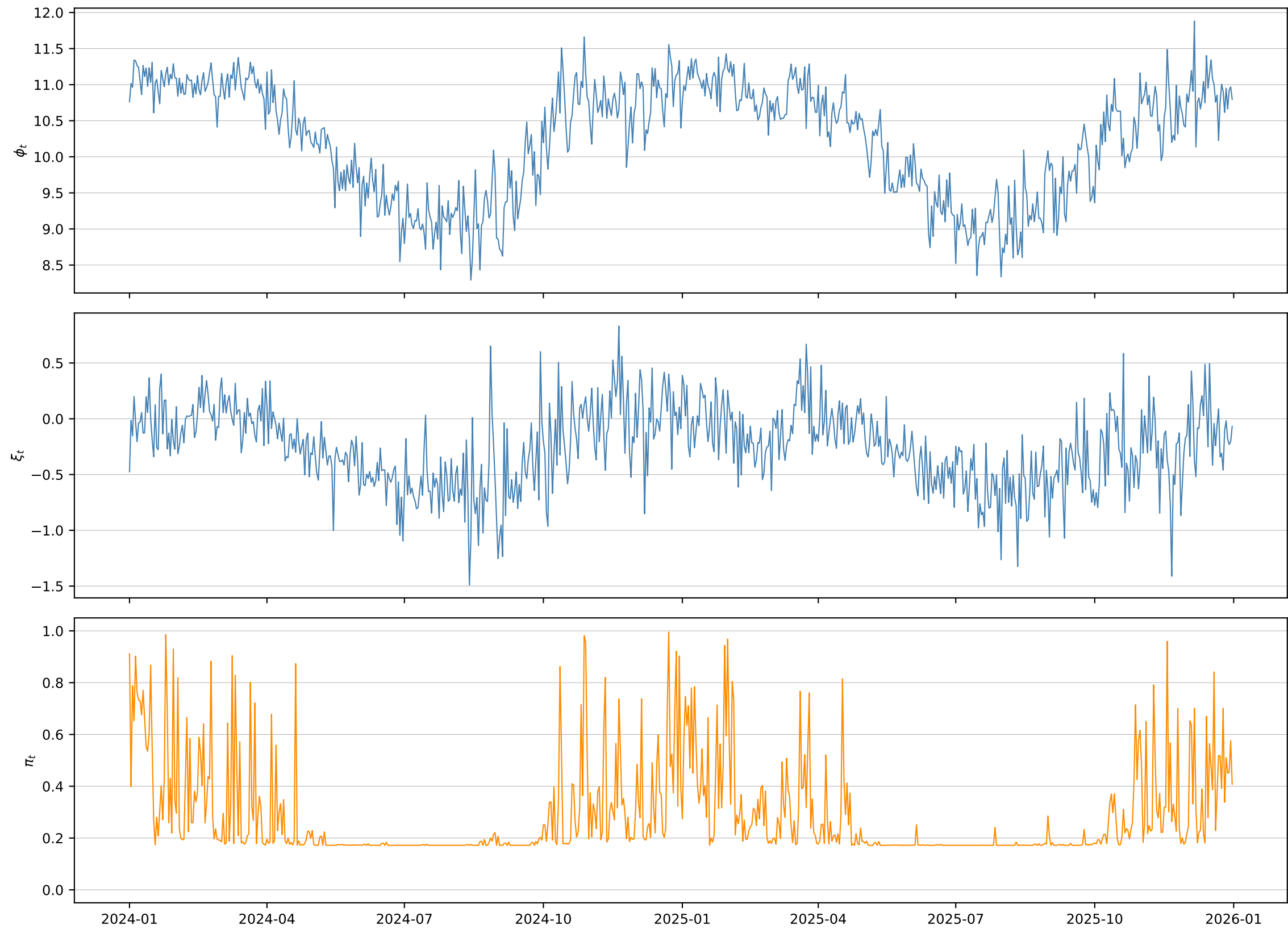
QR time series



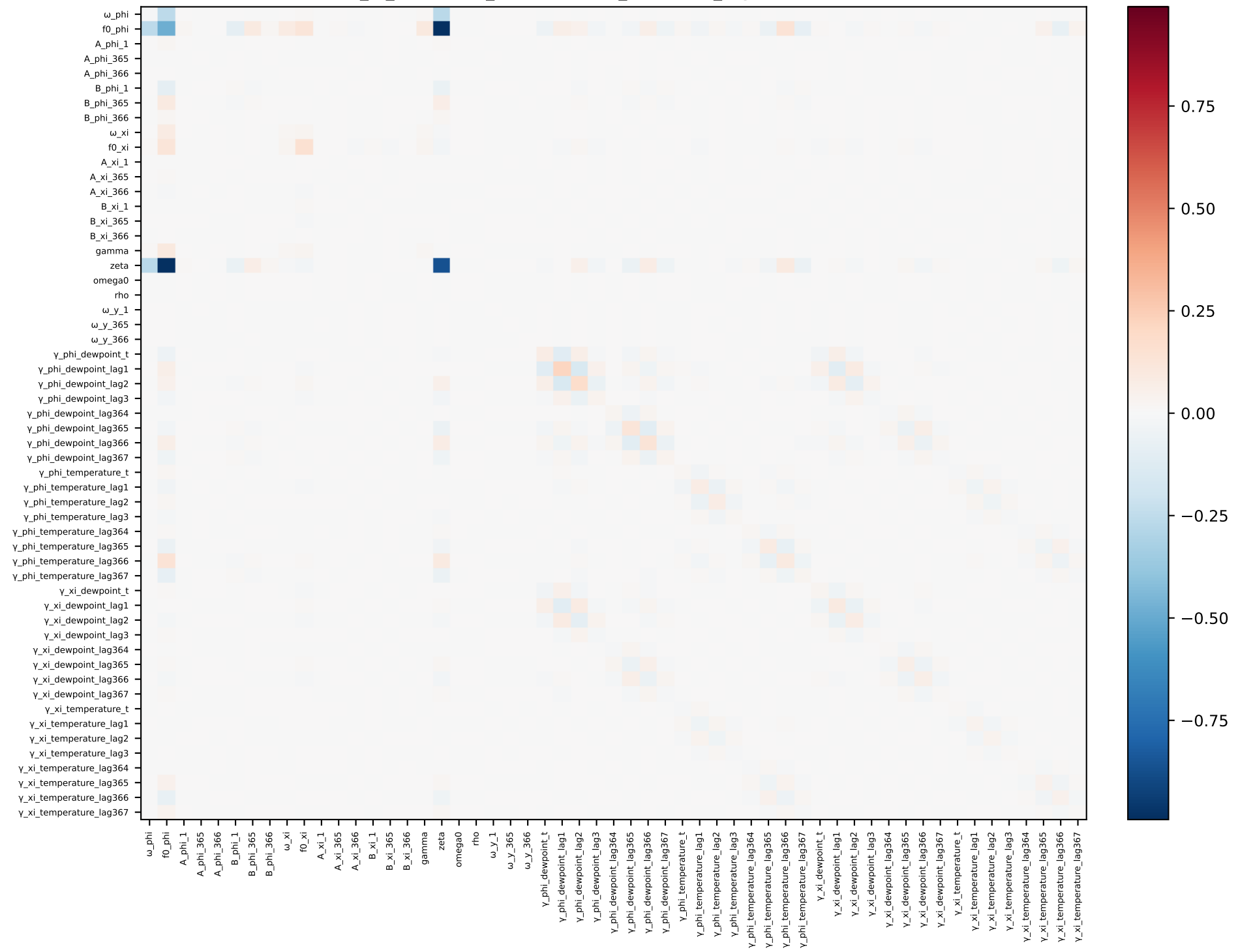
phi_xi_dewpoint_temperature_seasonal_lags — filtered states (IS)



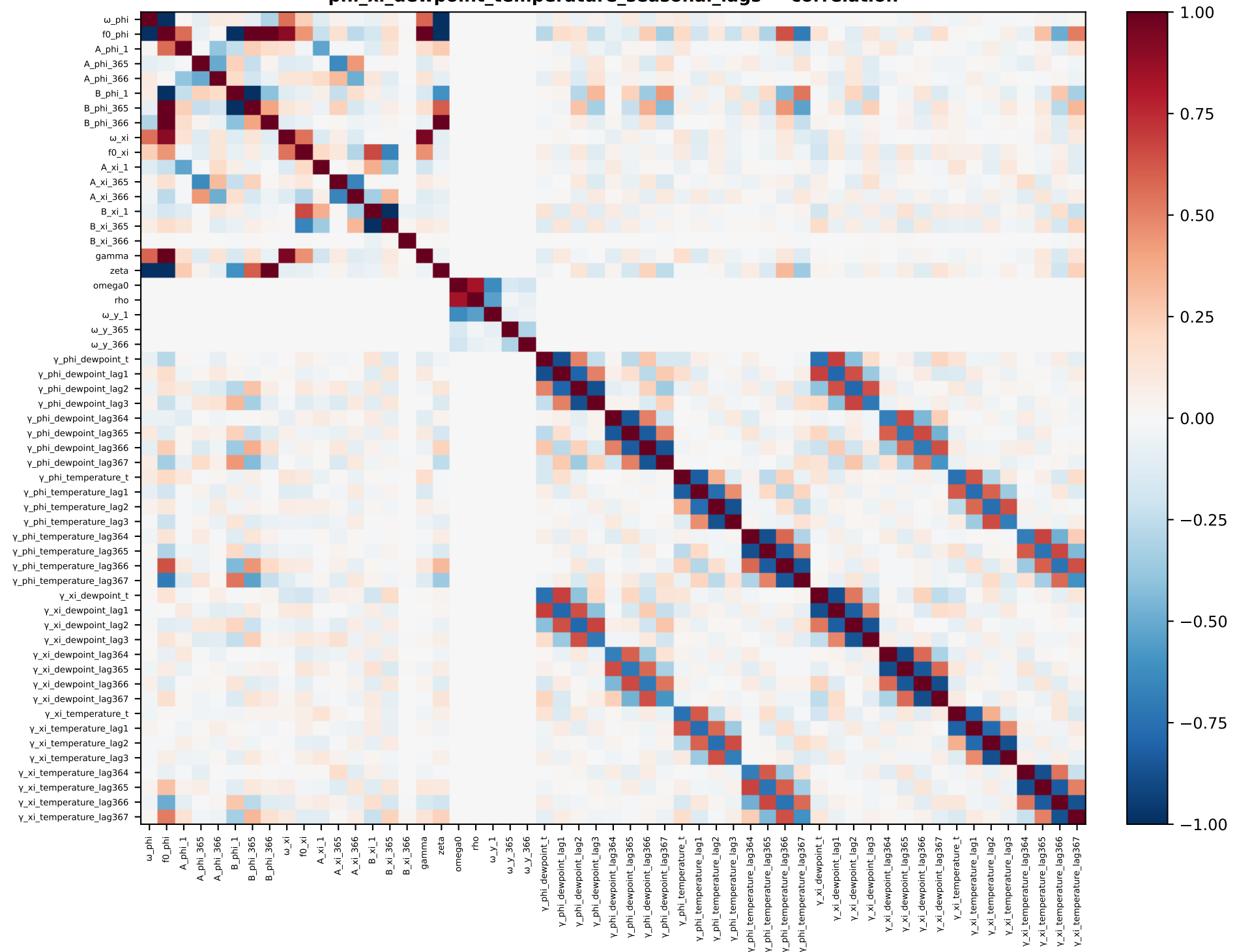
phi_xi_dewpoint_temperature_seasonal_lags — filtered states (OOS)



phi_xi_dewpoint_temperature_seasonal_lags – covariance

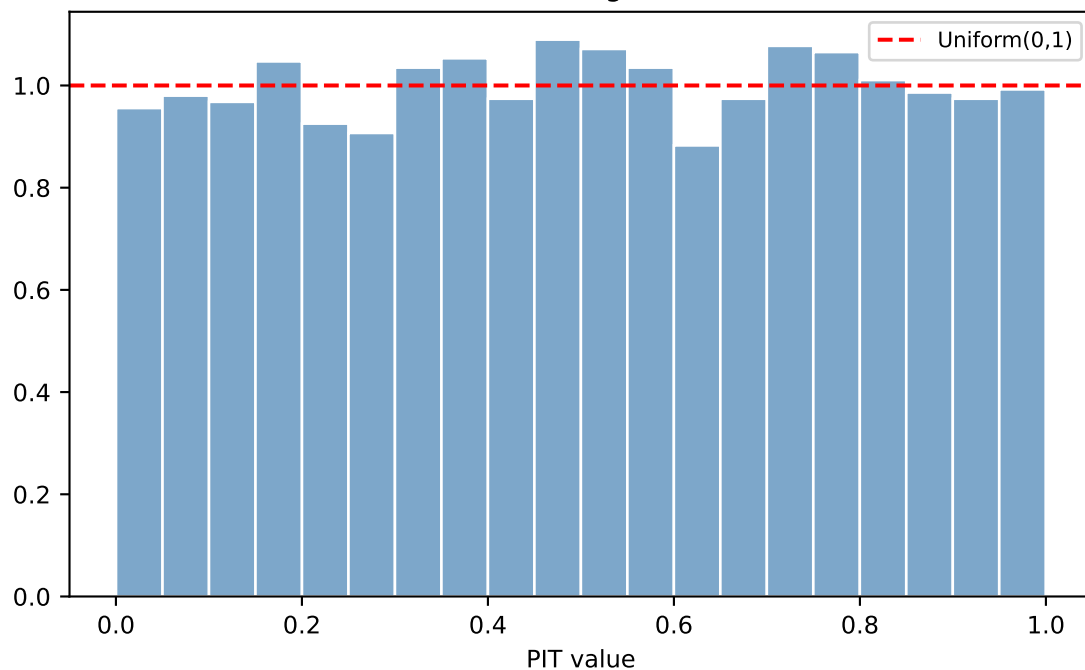


phi_xi_dewpoint_temperature_seasonal_lags — correlation

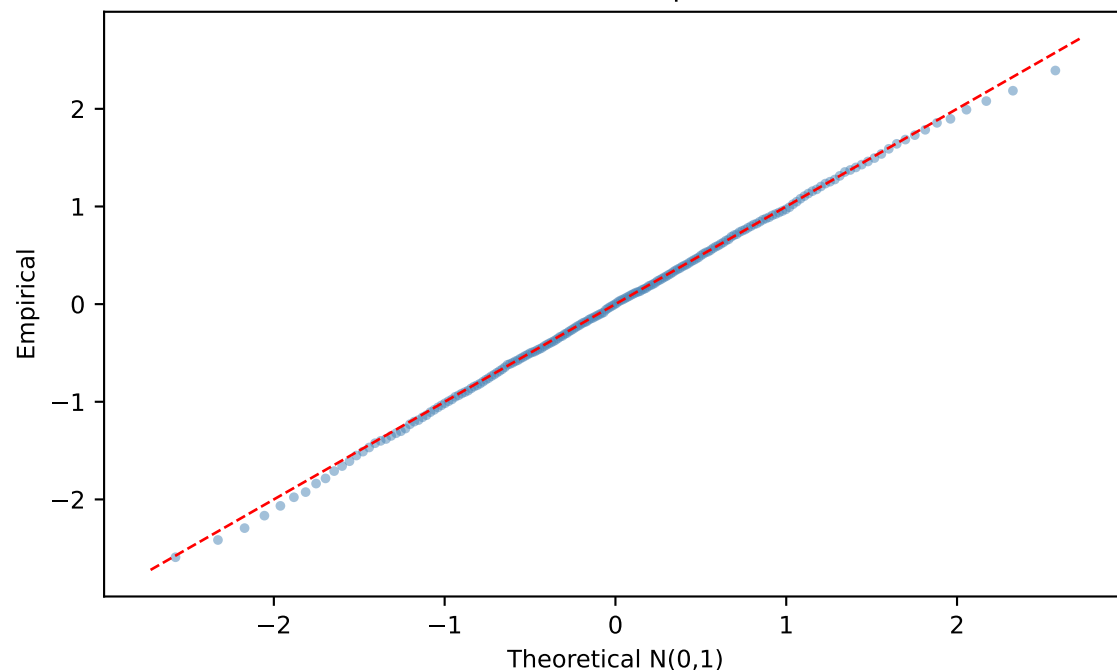


phi_xi_mjo_t — IS diagnostics

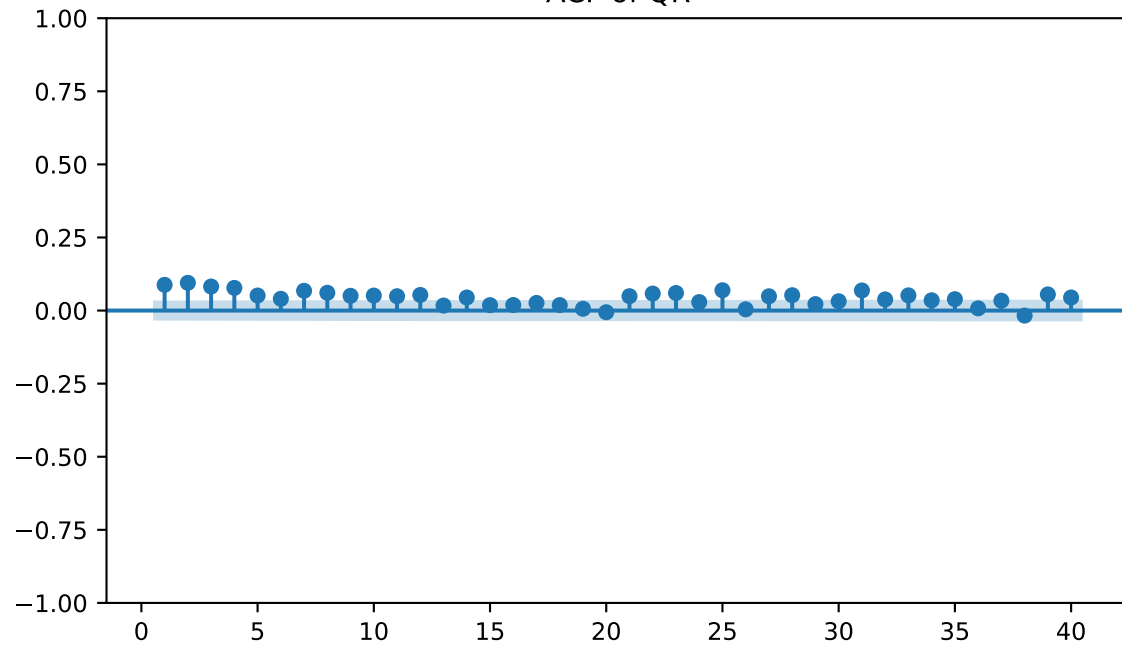
PIT histogram



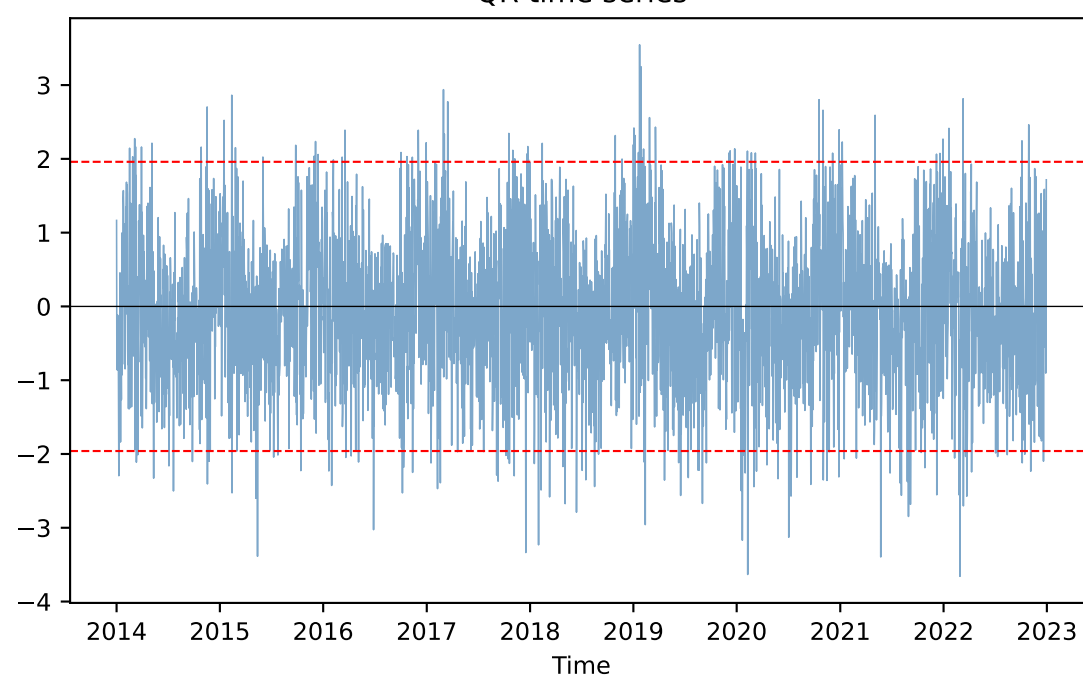
Normal QQ plot



ACF of QR

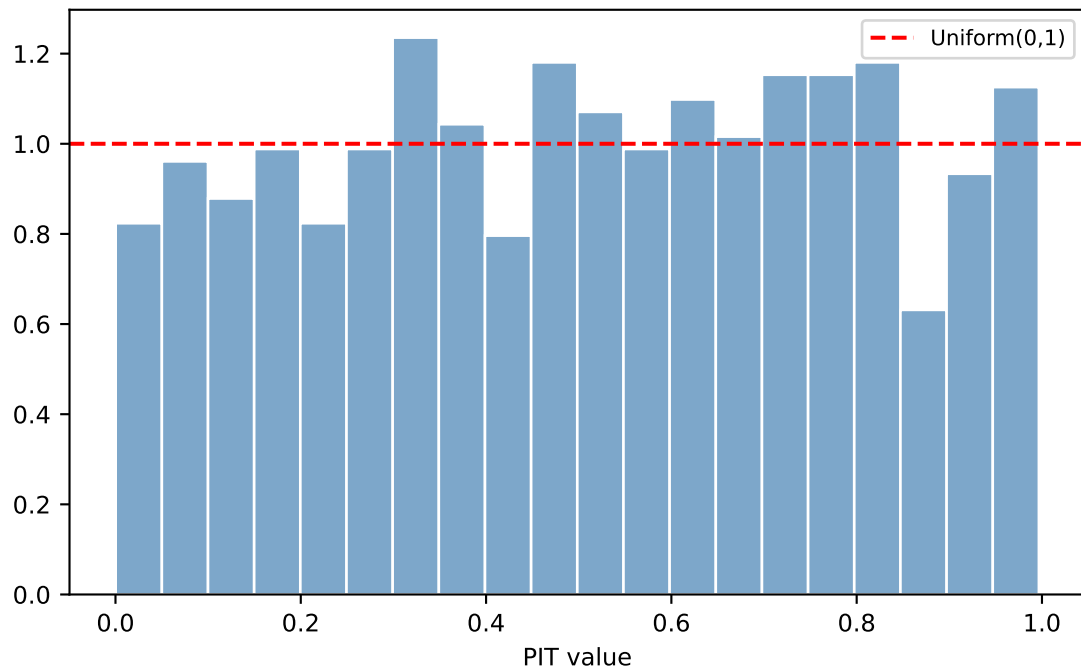


QR time series

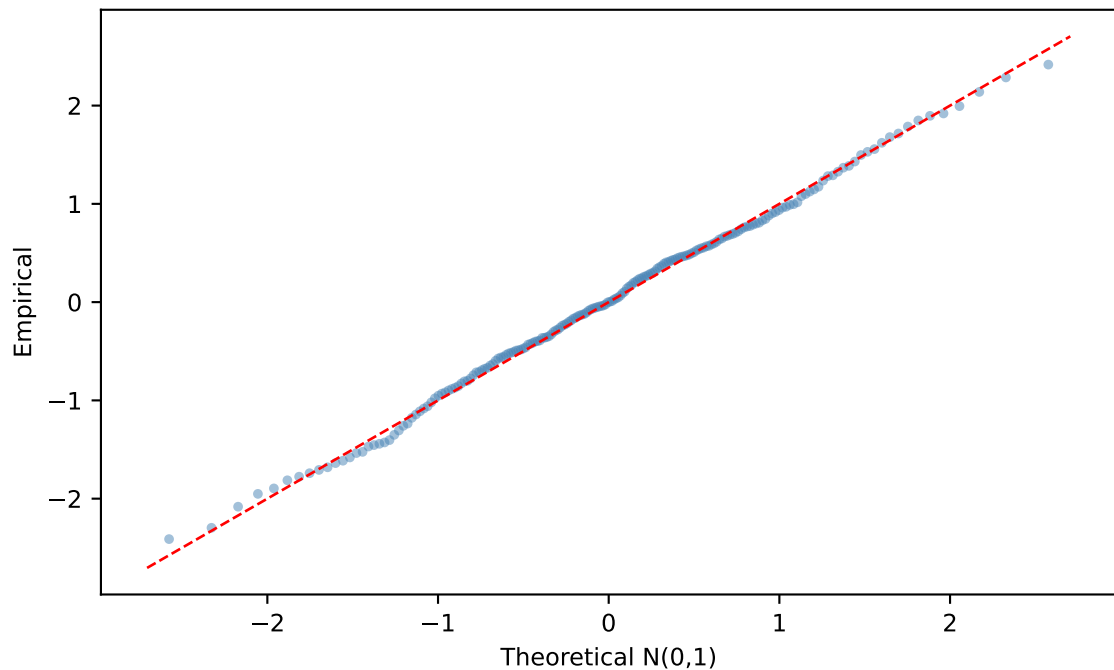


phi_xi_mjo_t — OOS diagnostics

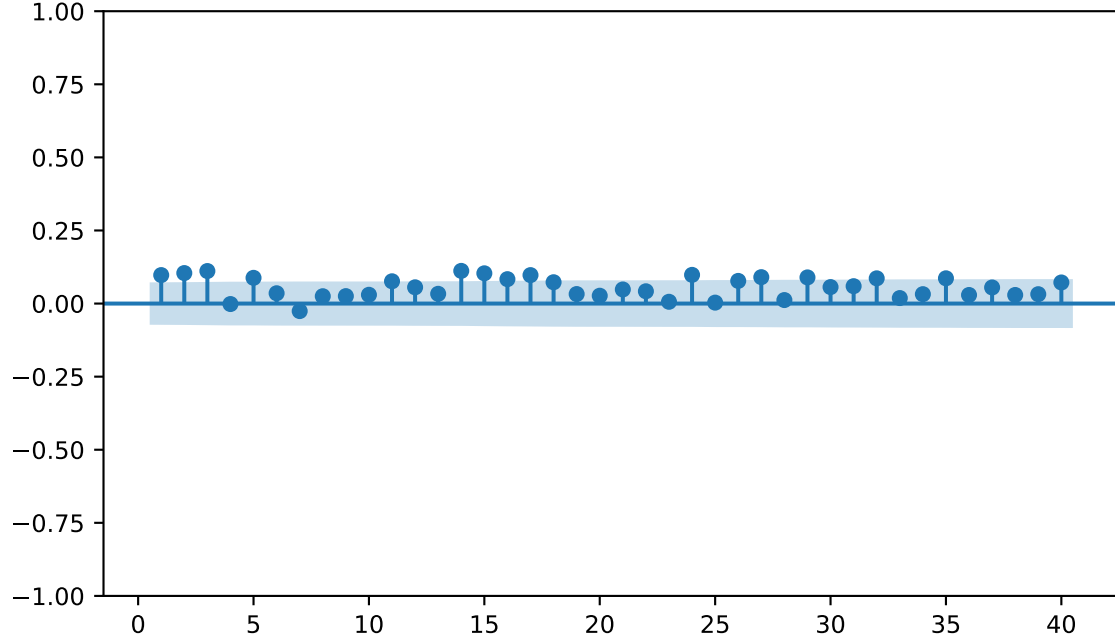
PIT histogram



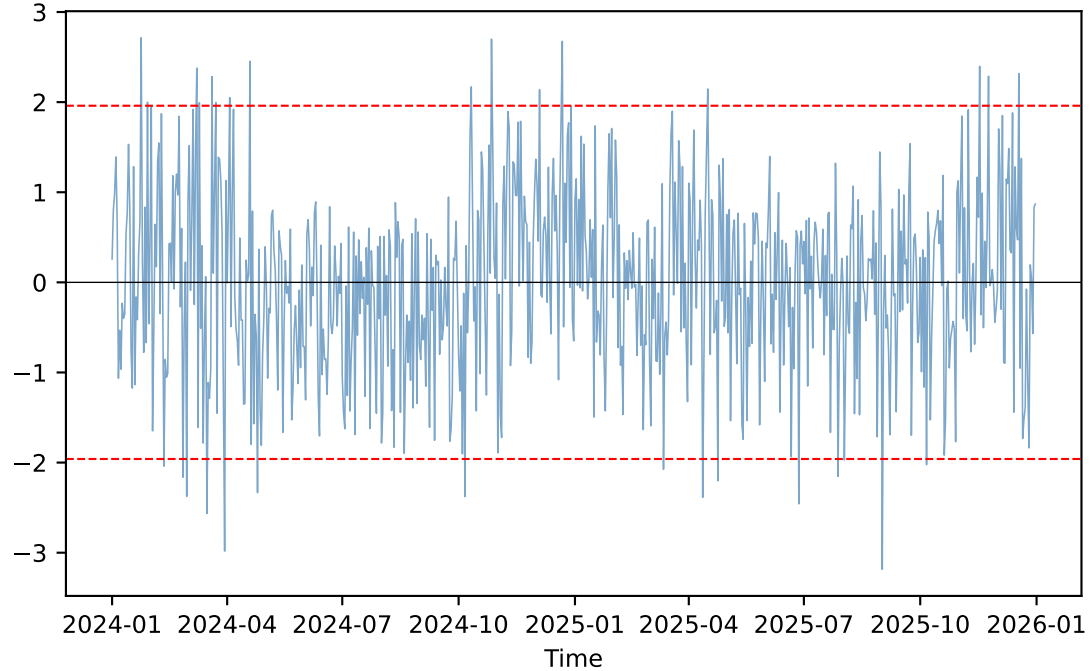
Normal QQ plot



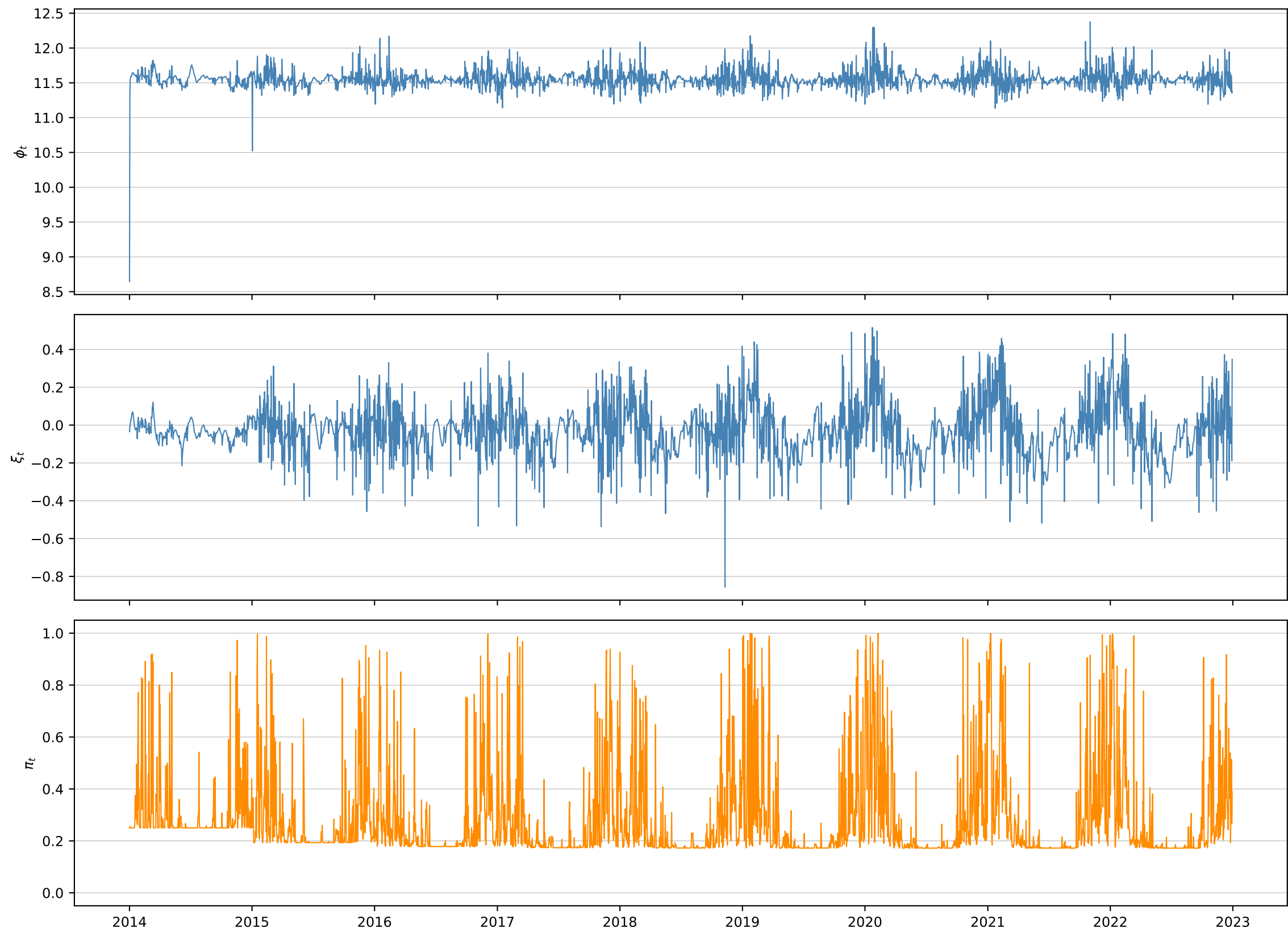
ACF of QR



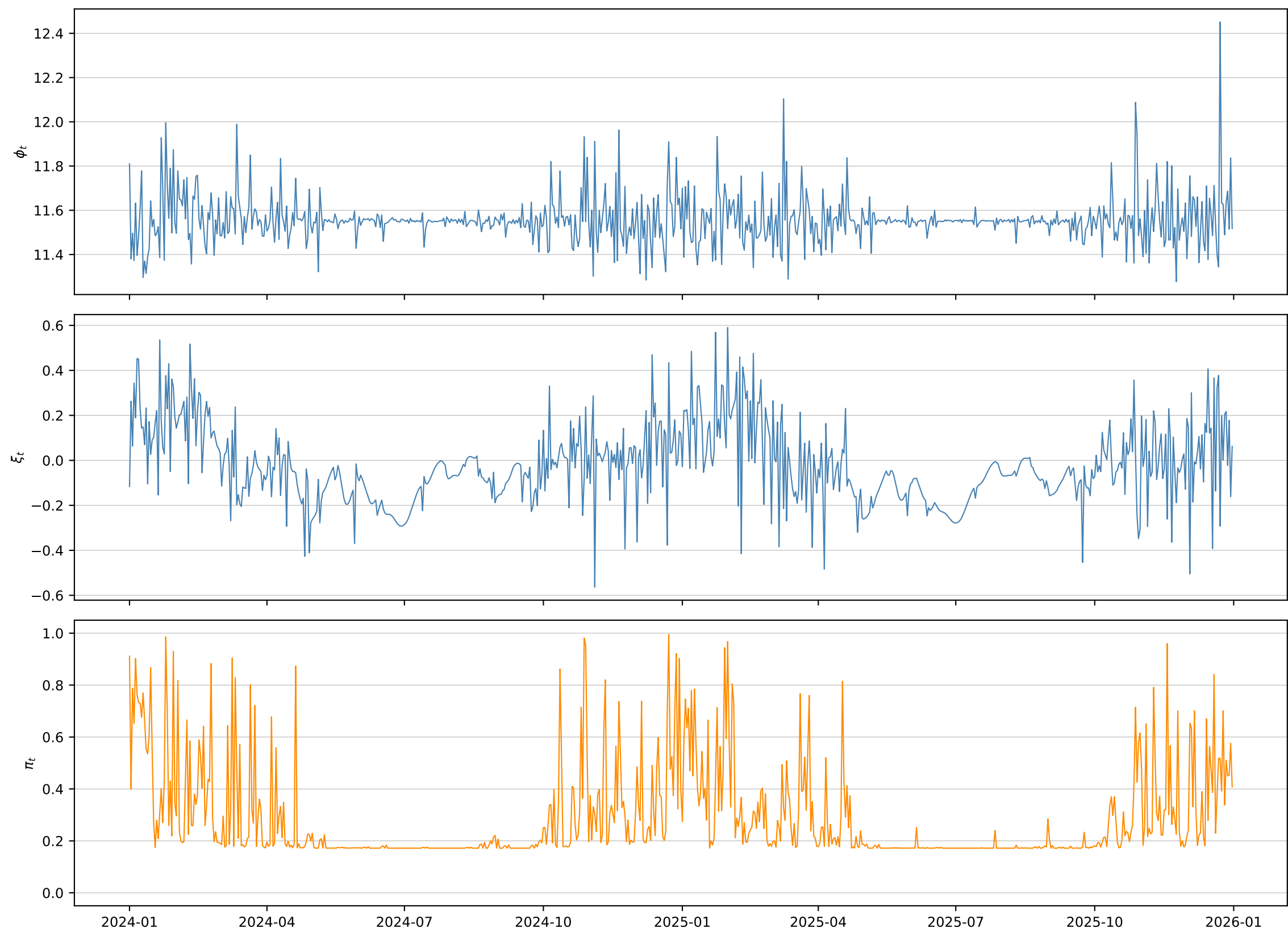
QR time series



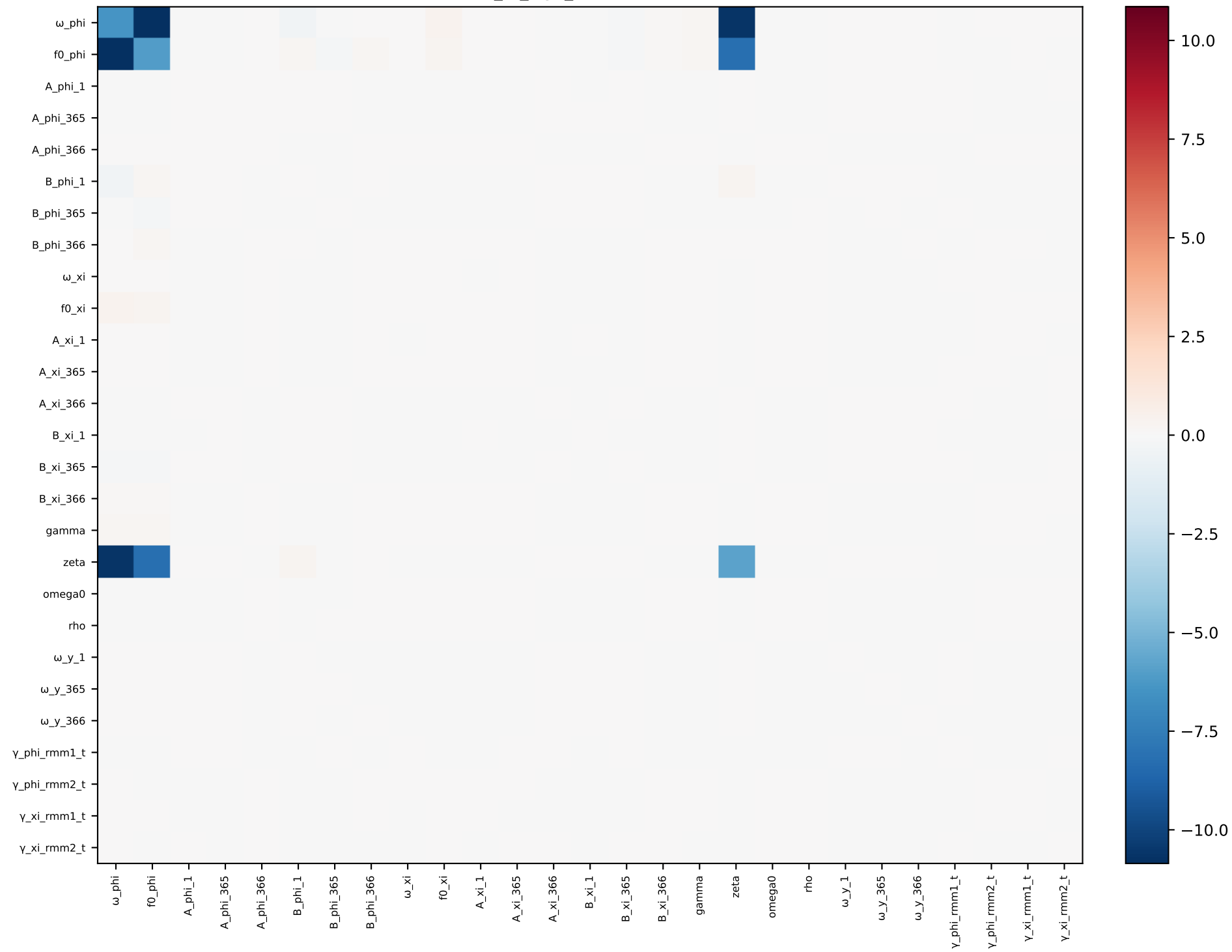
phi_xi_mjo_t — filtered states (IS)



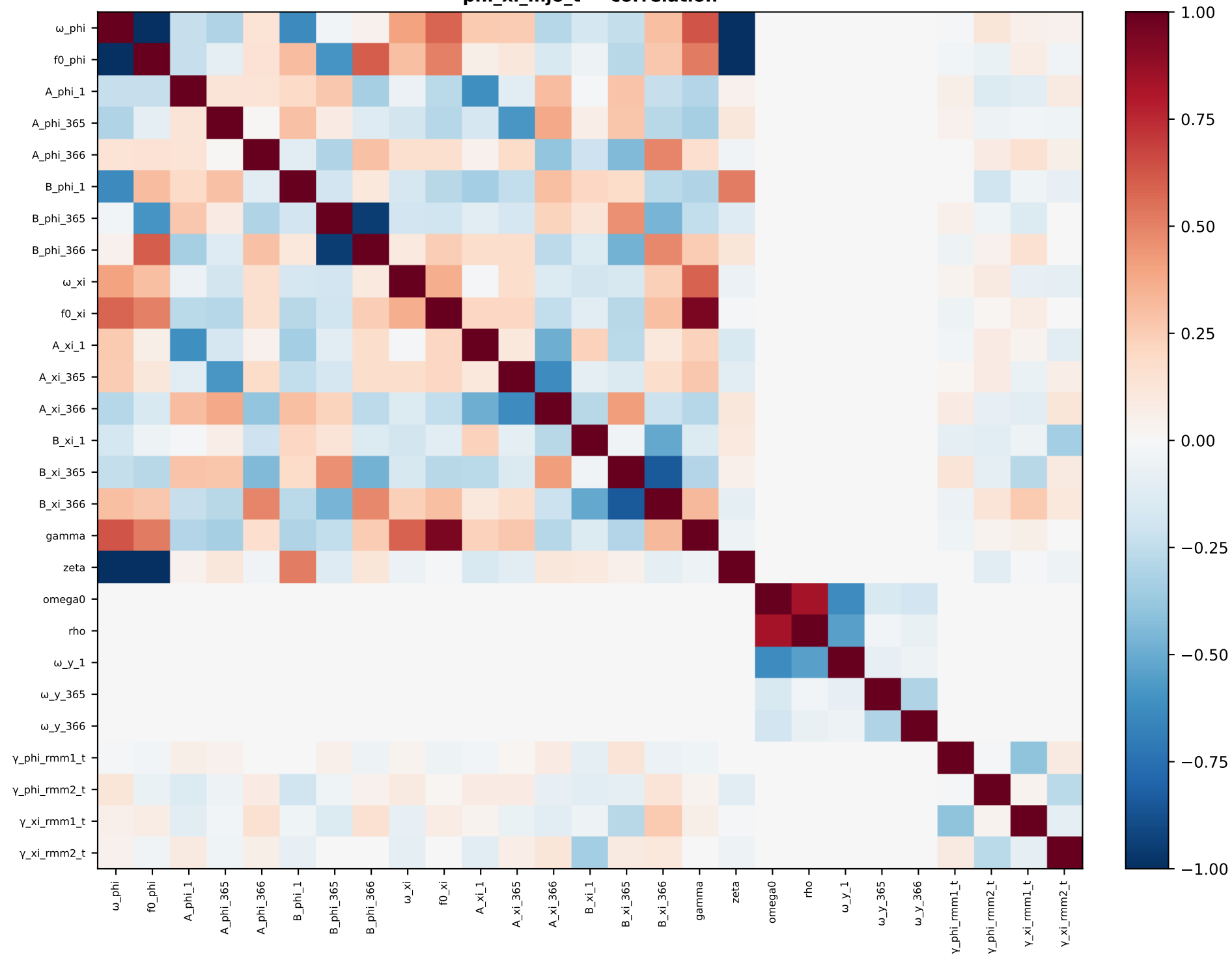
phi_xi_mjo_t — filtered states (OOS)



phi xi mjo t — covariance

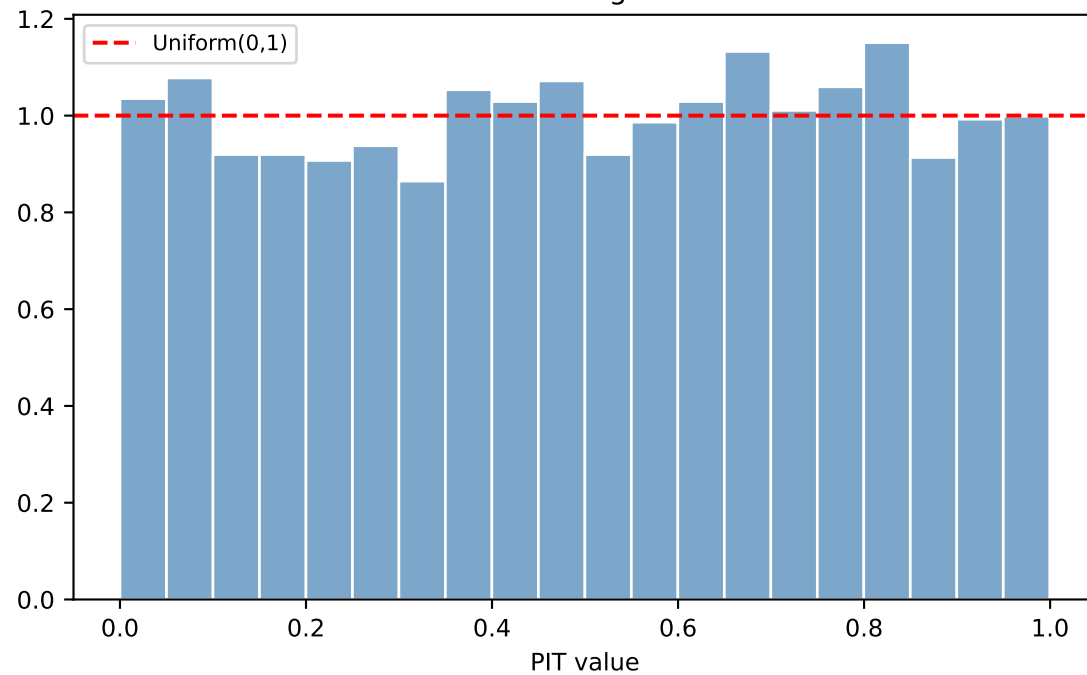


phi_xi_mjo_t — correlation

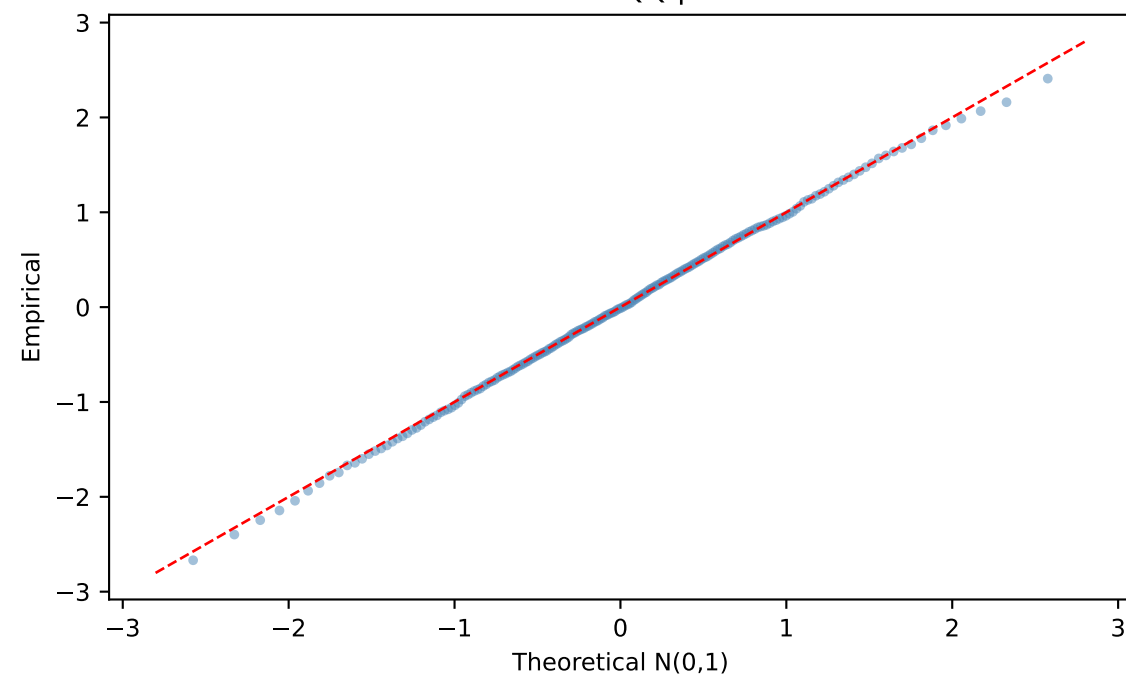


phi_xi_mjo_lag1 — IS diagnostics

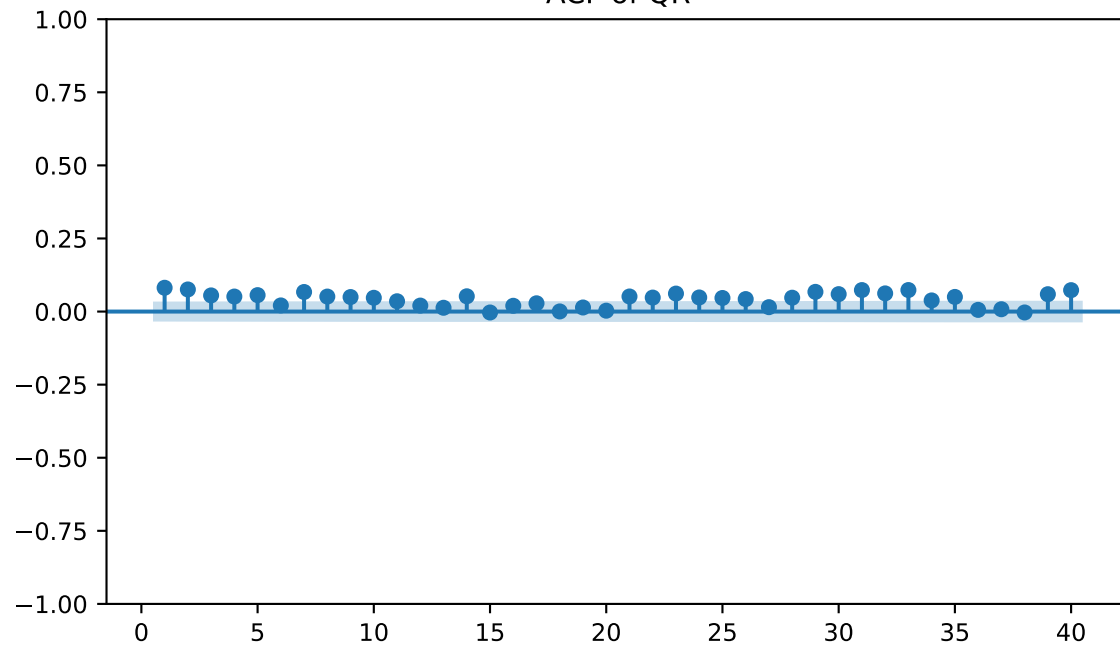
PIT histogram



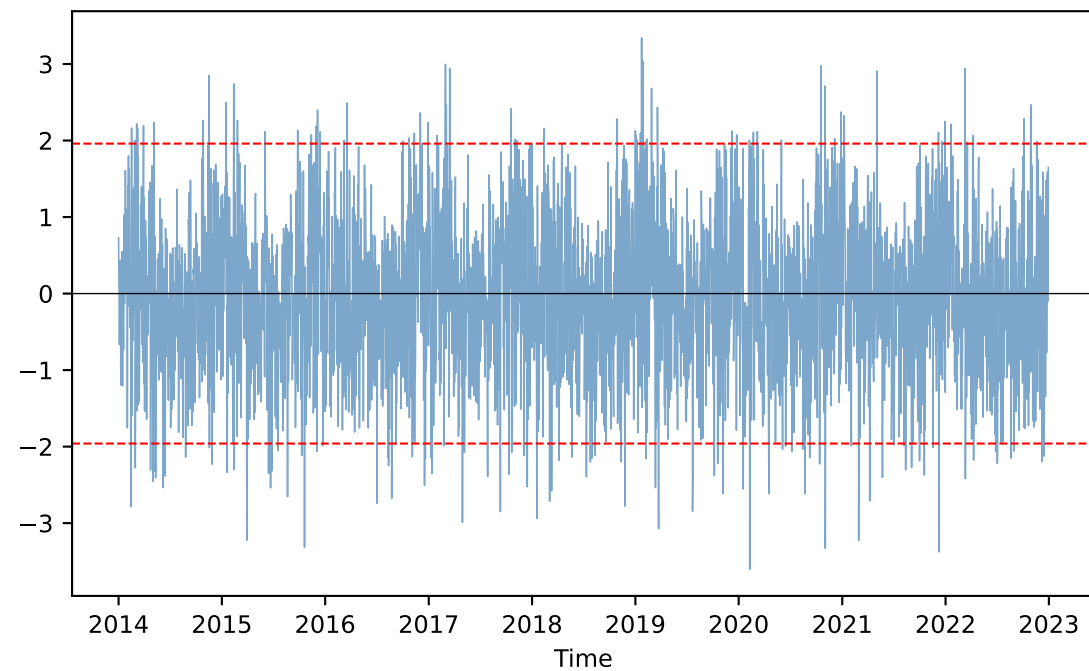
Normal QQ plot



ACF of QR

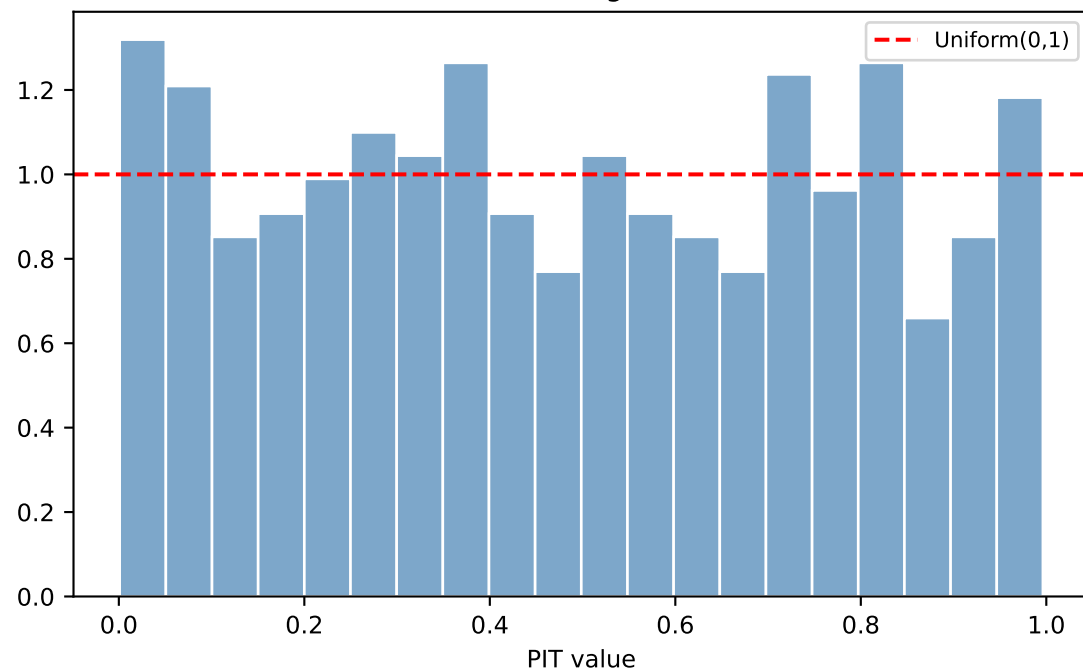


QR time series

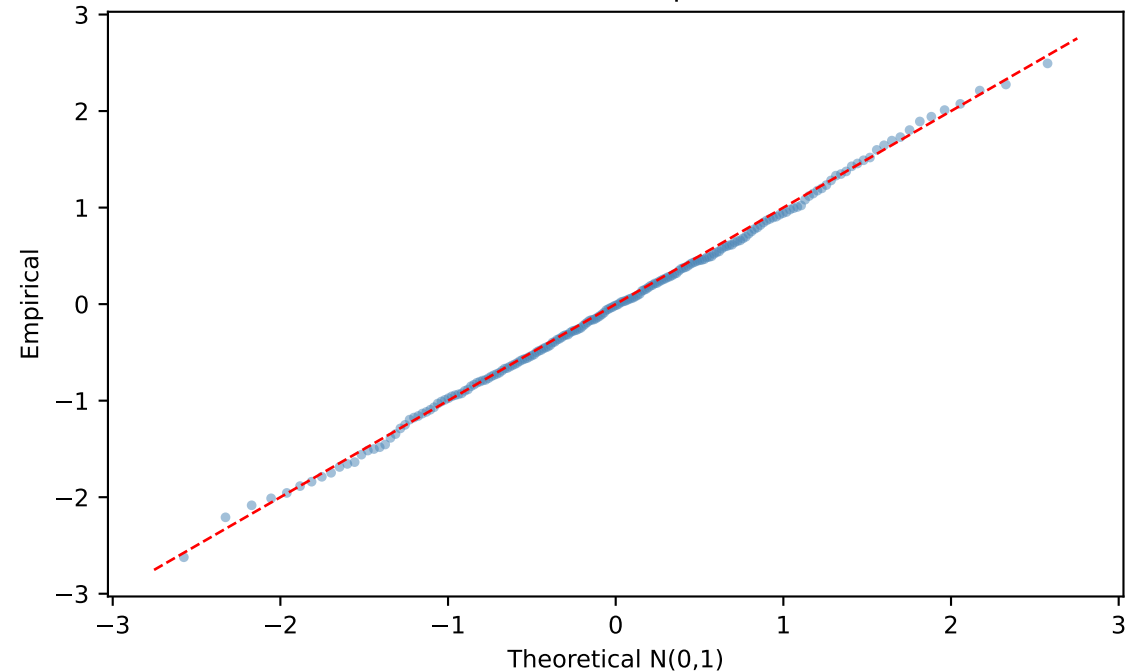


phi_xi_mjo_lag1 — OOS diagnostics

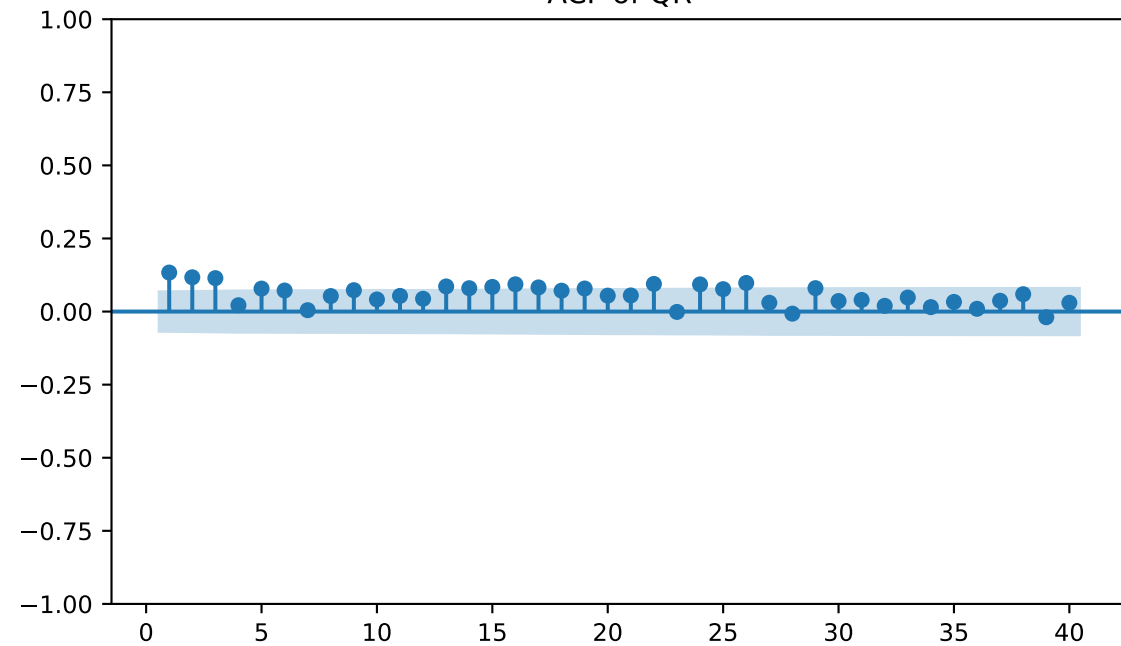
PIT histogram



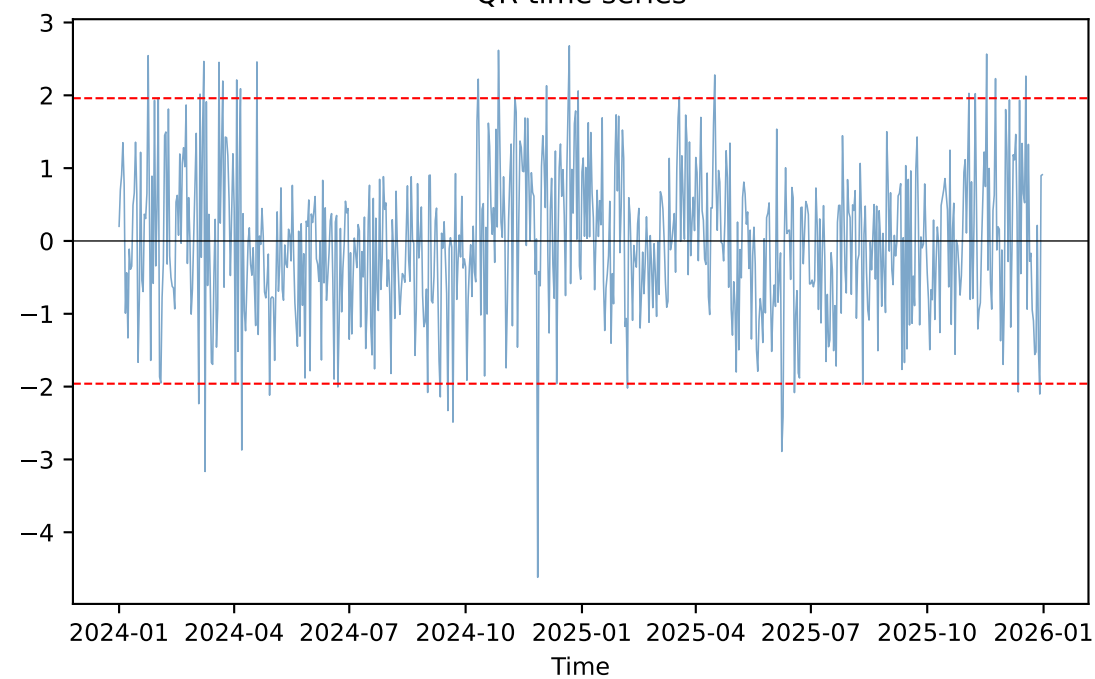
Normal QQ plot



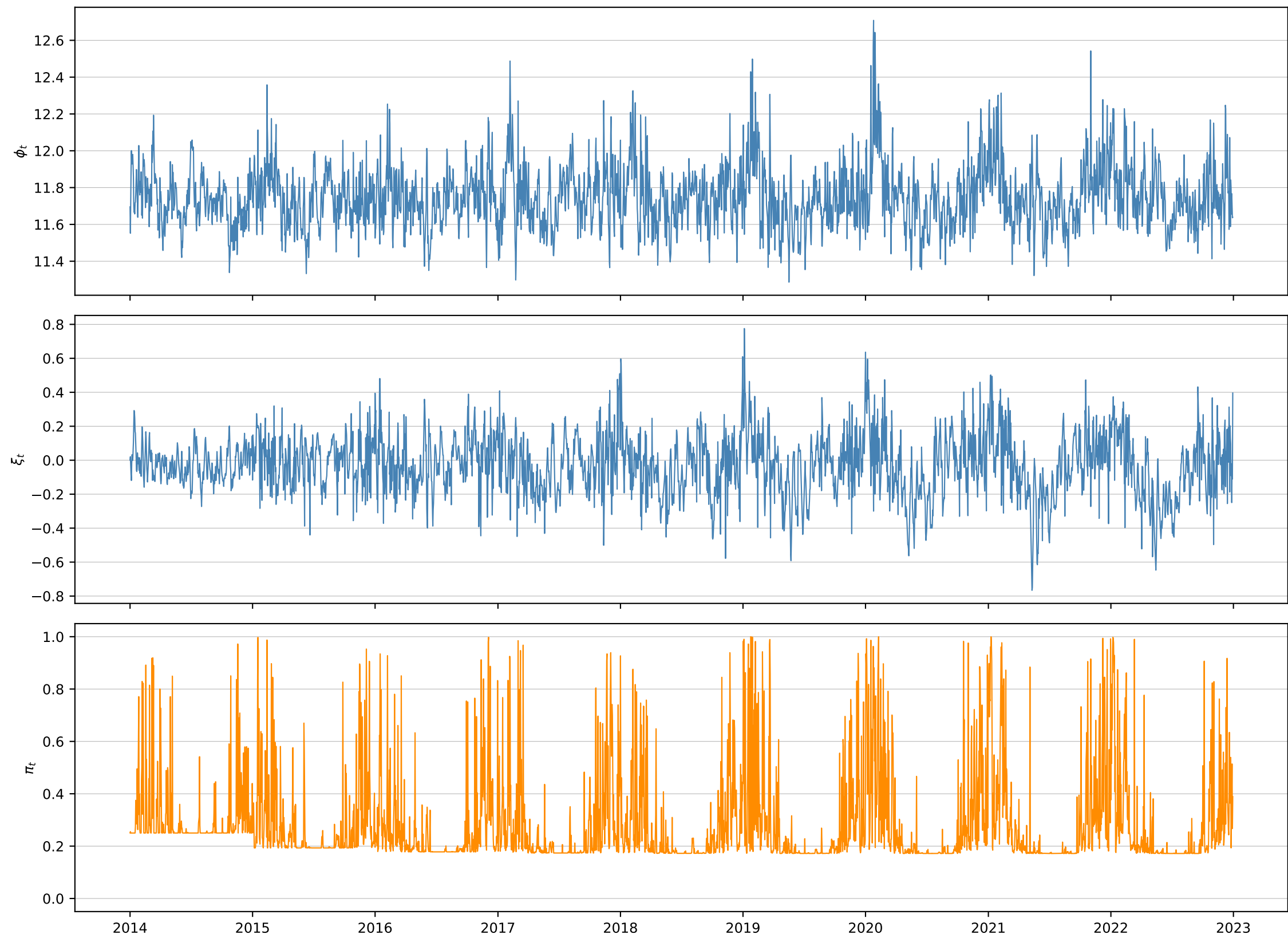
ACF of QR



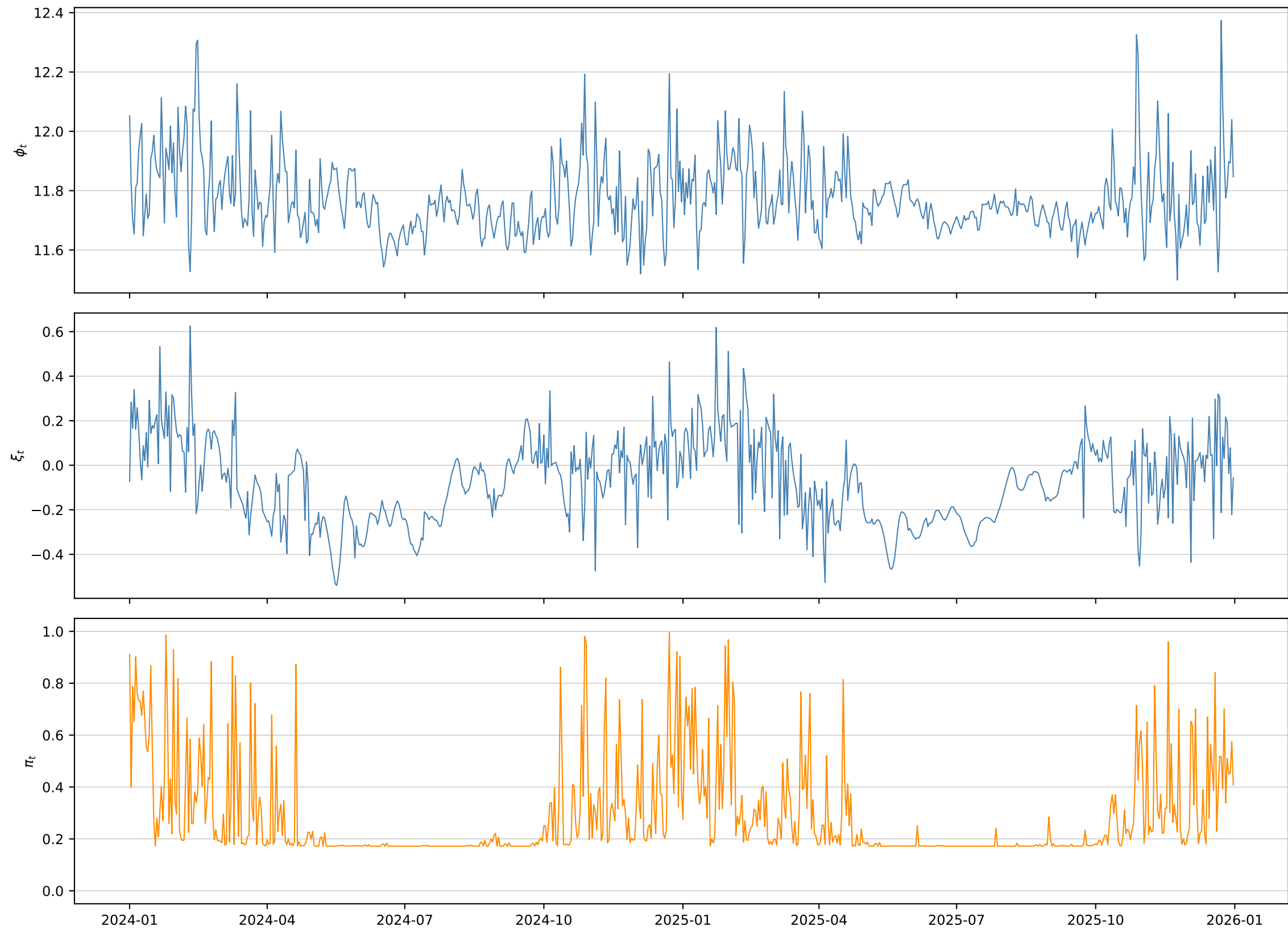
QR time series



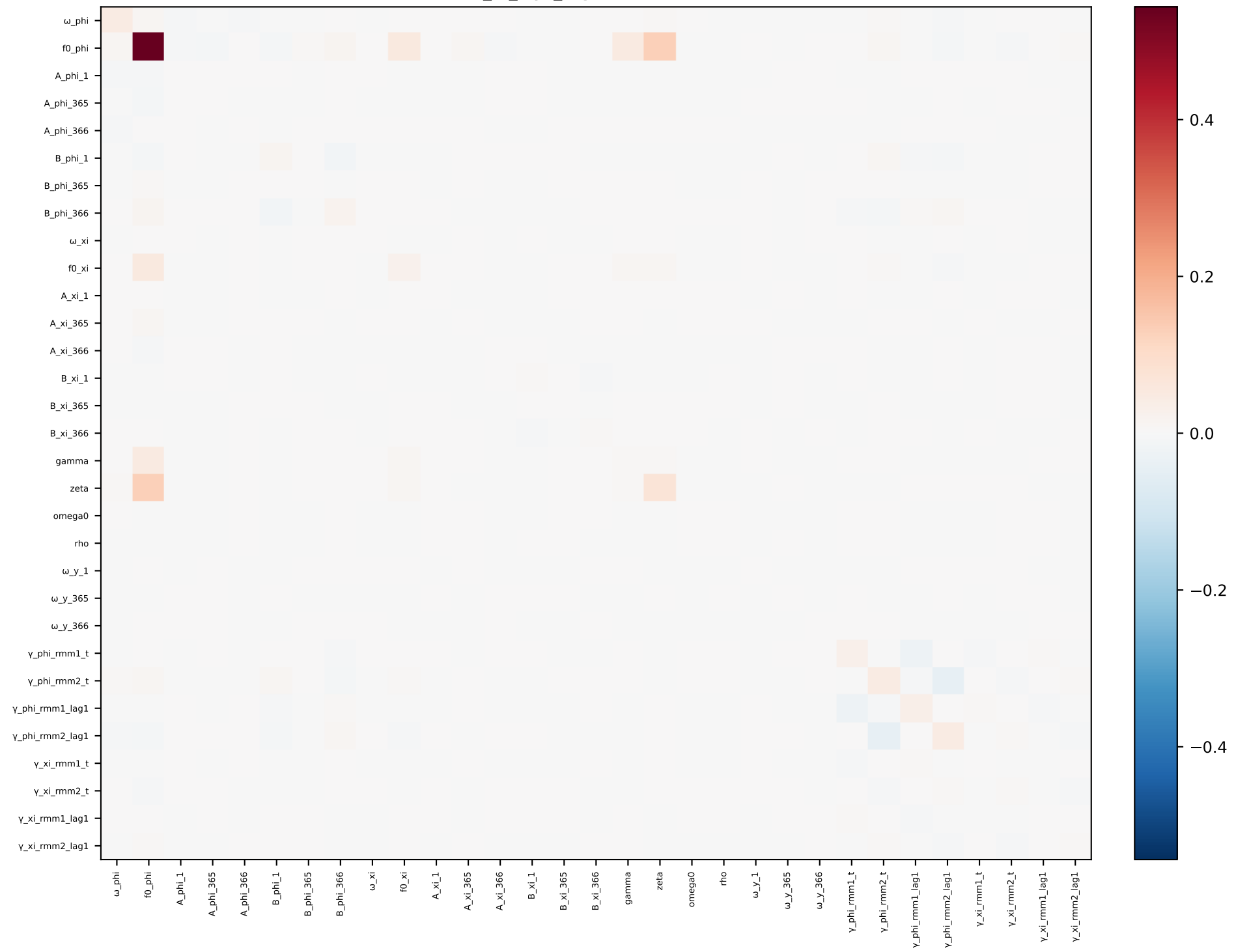
phi_xi_mjo_lag1 — filtered states (IS)



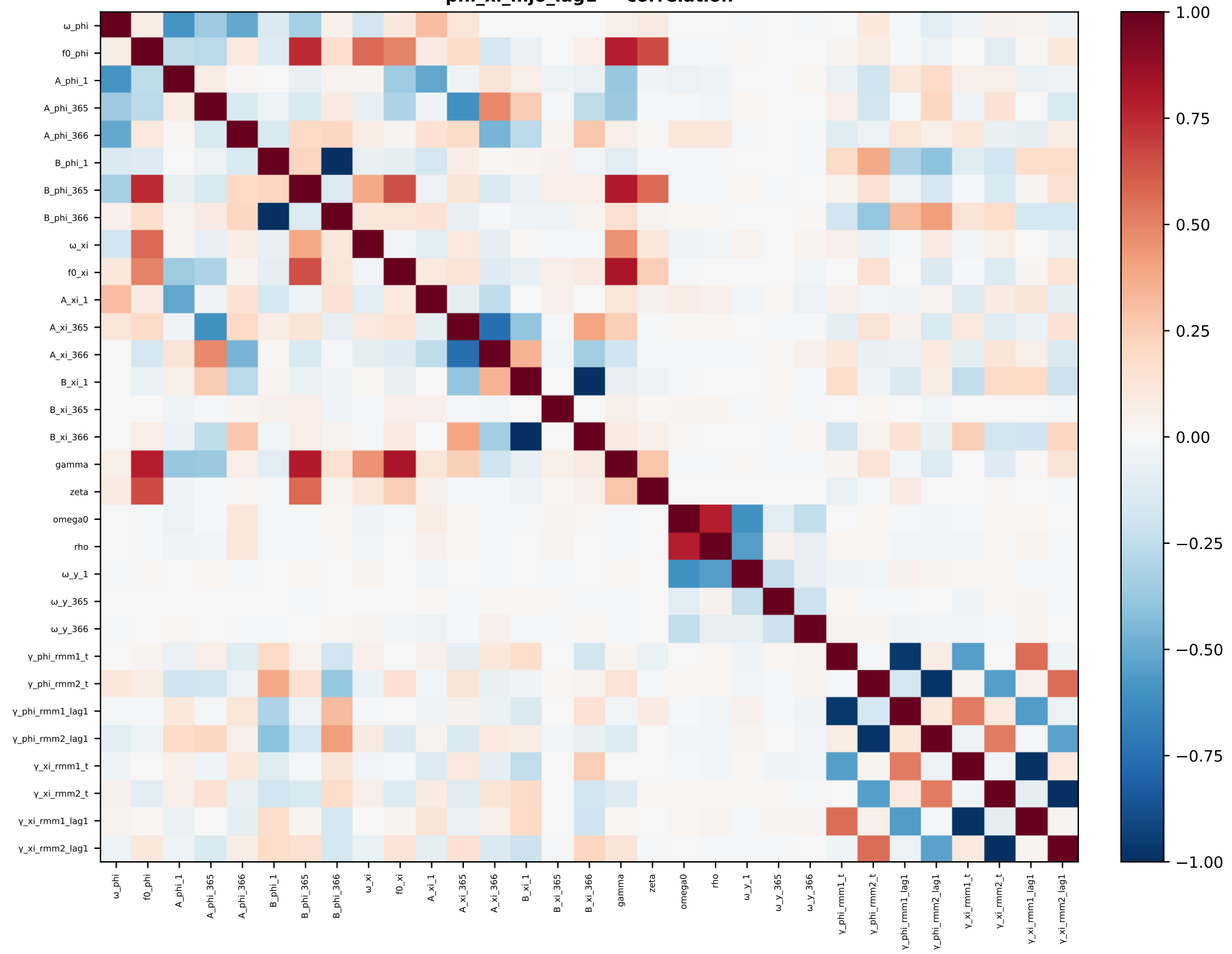
phi_xi_mjo_lag1 — filtered states (OOS)



phi_xi_mjo_lag1 — covariance

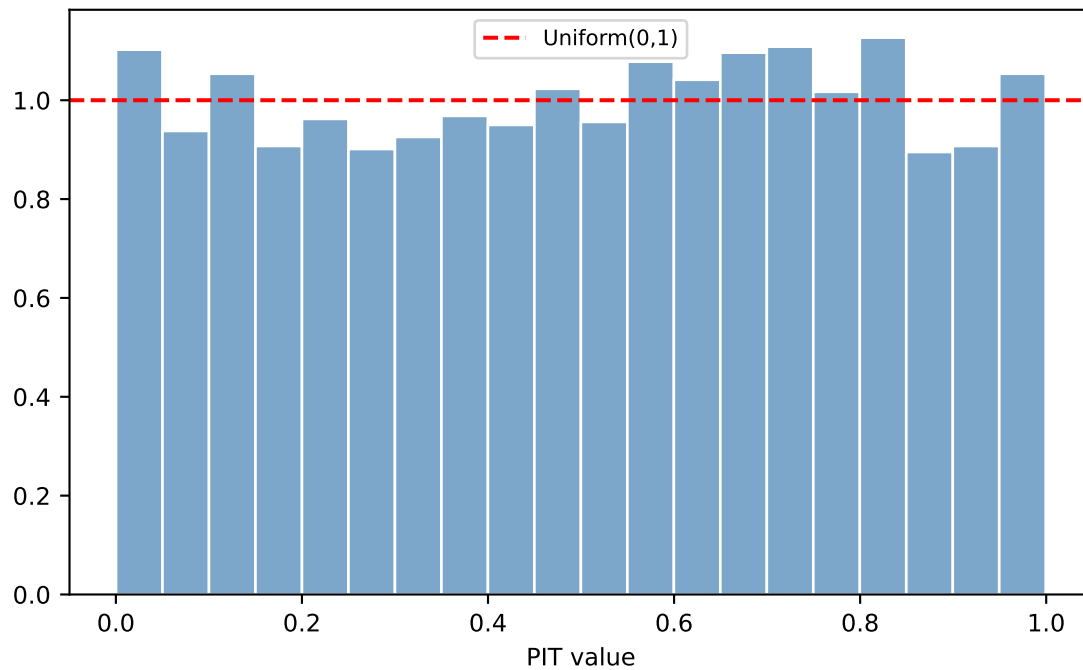


phi_xi_mjo_lag1 — correlation

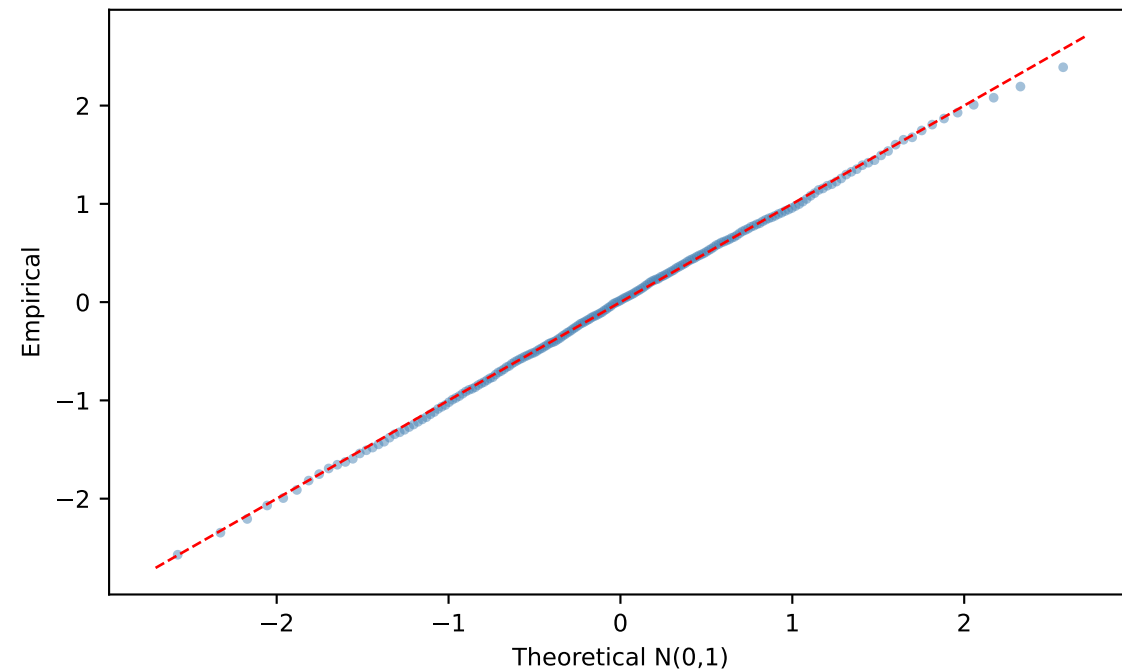


phi_xi_mjo_short_lags — IS diagnostics

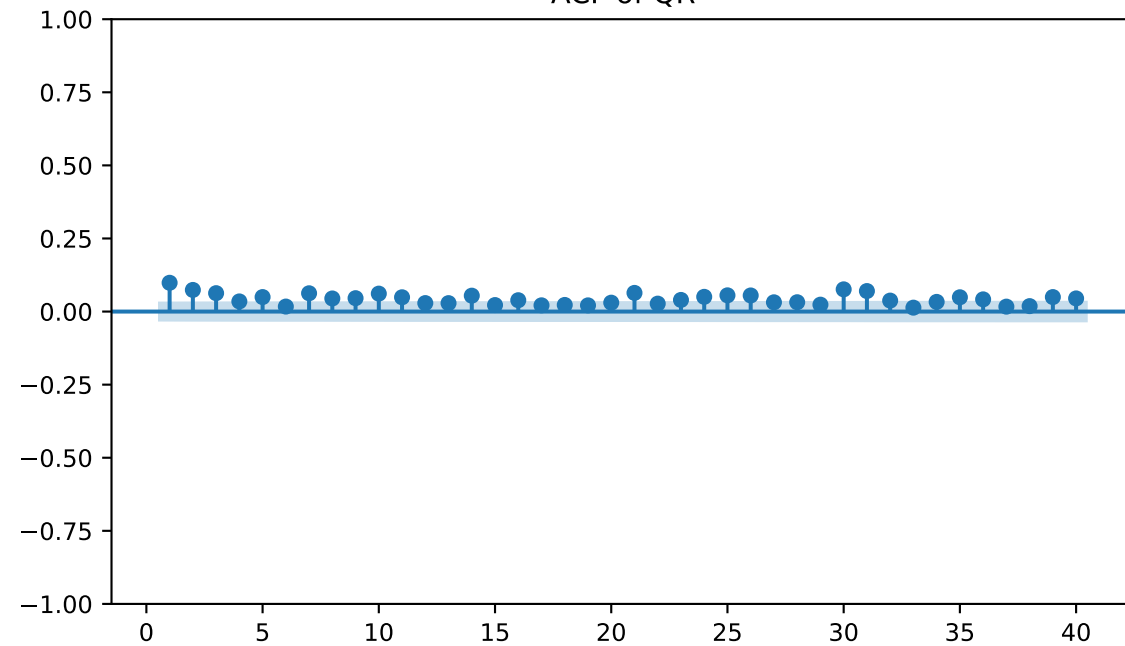
PIT histogram



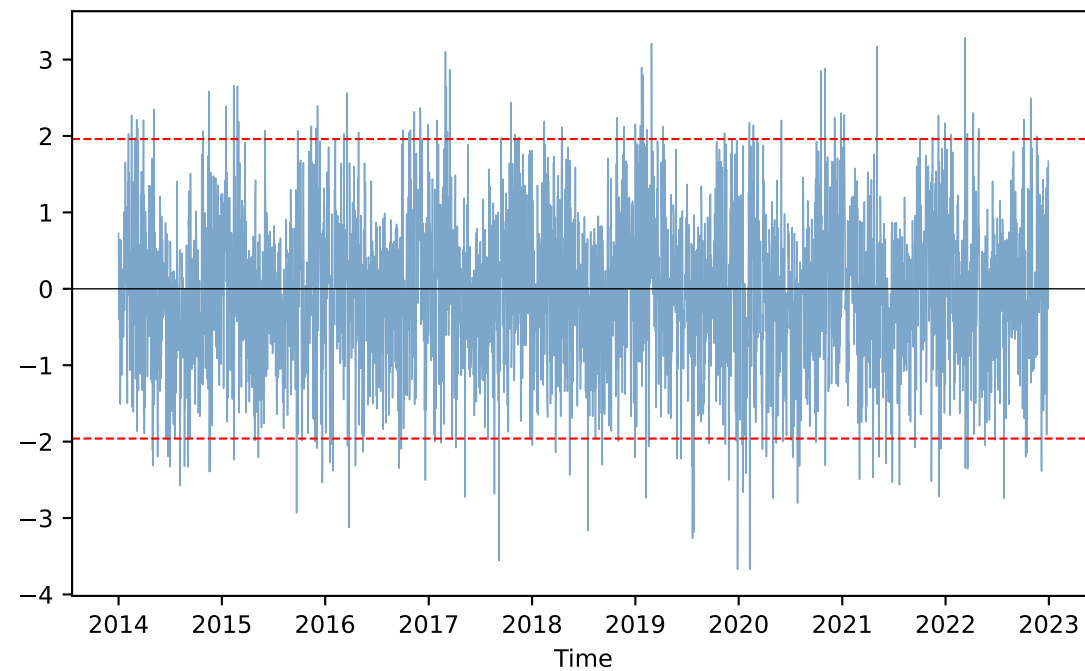
Normal QQ plot



ACF of QR

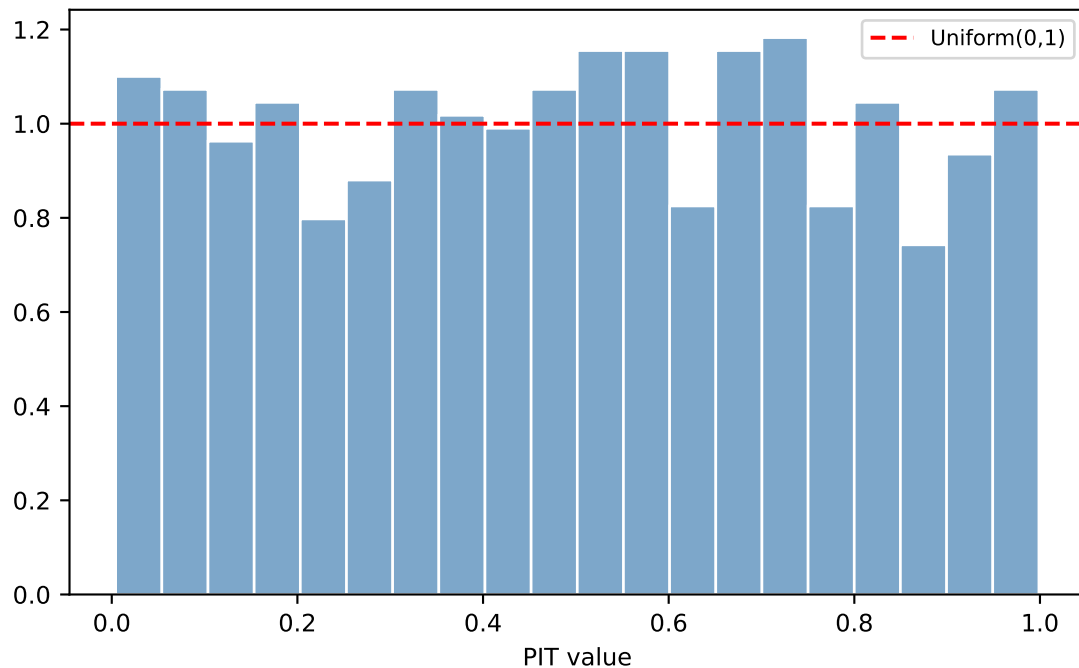


QR time series

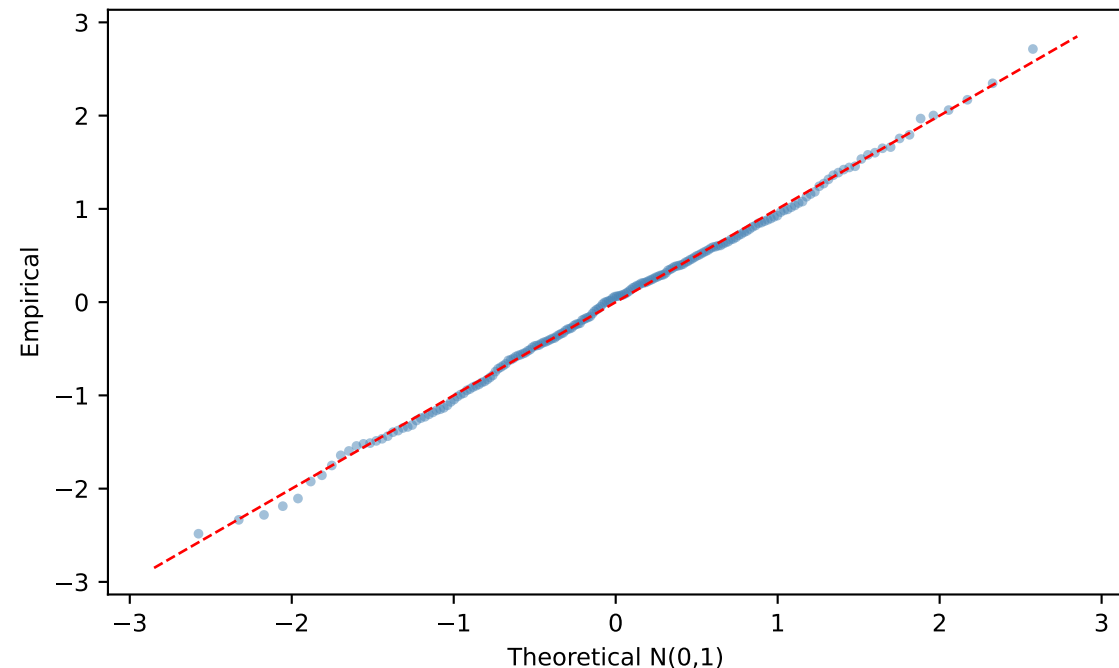


phi_xi_mjo_short_lags — OOS diagnostics

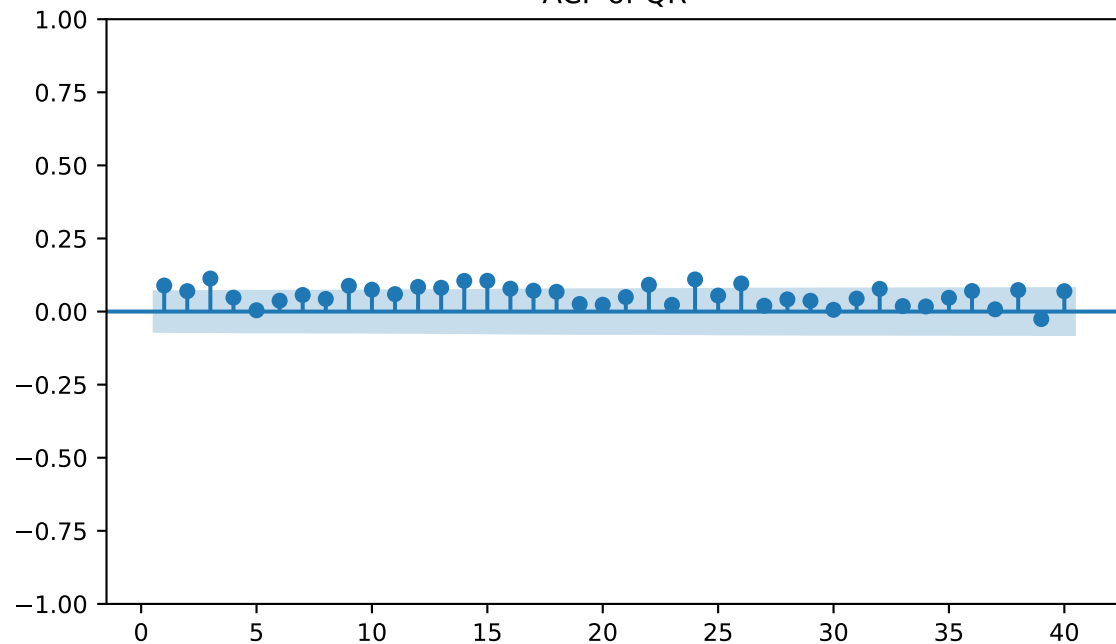
PIT histogram



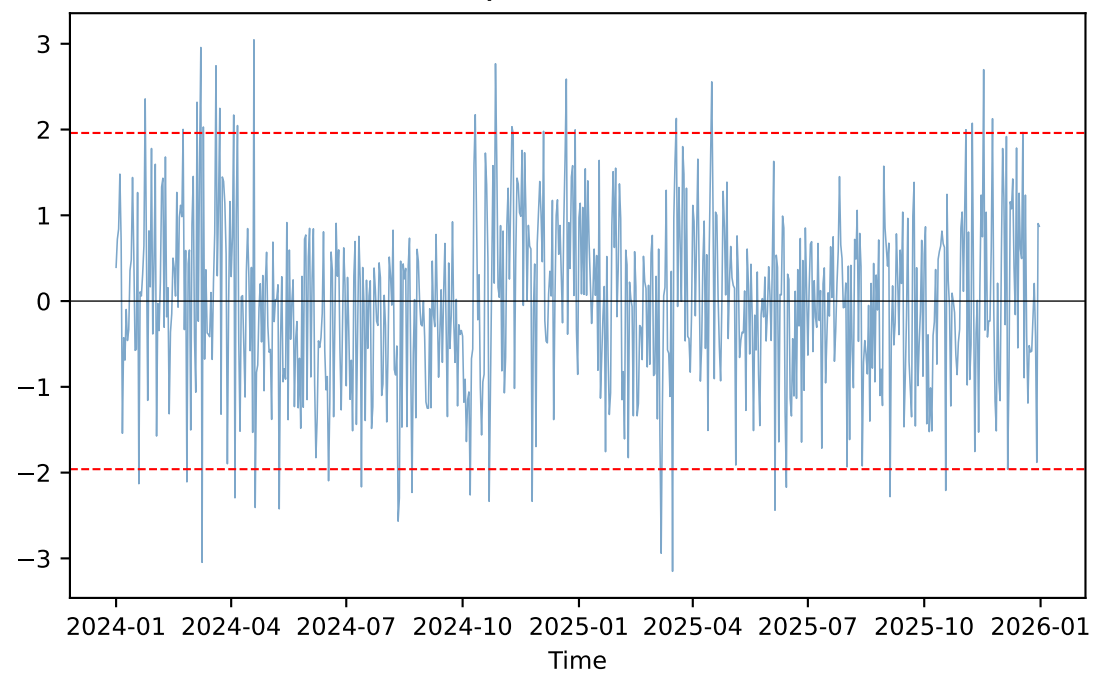
Normal QQ plot



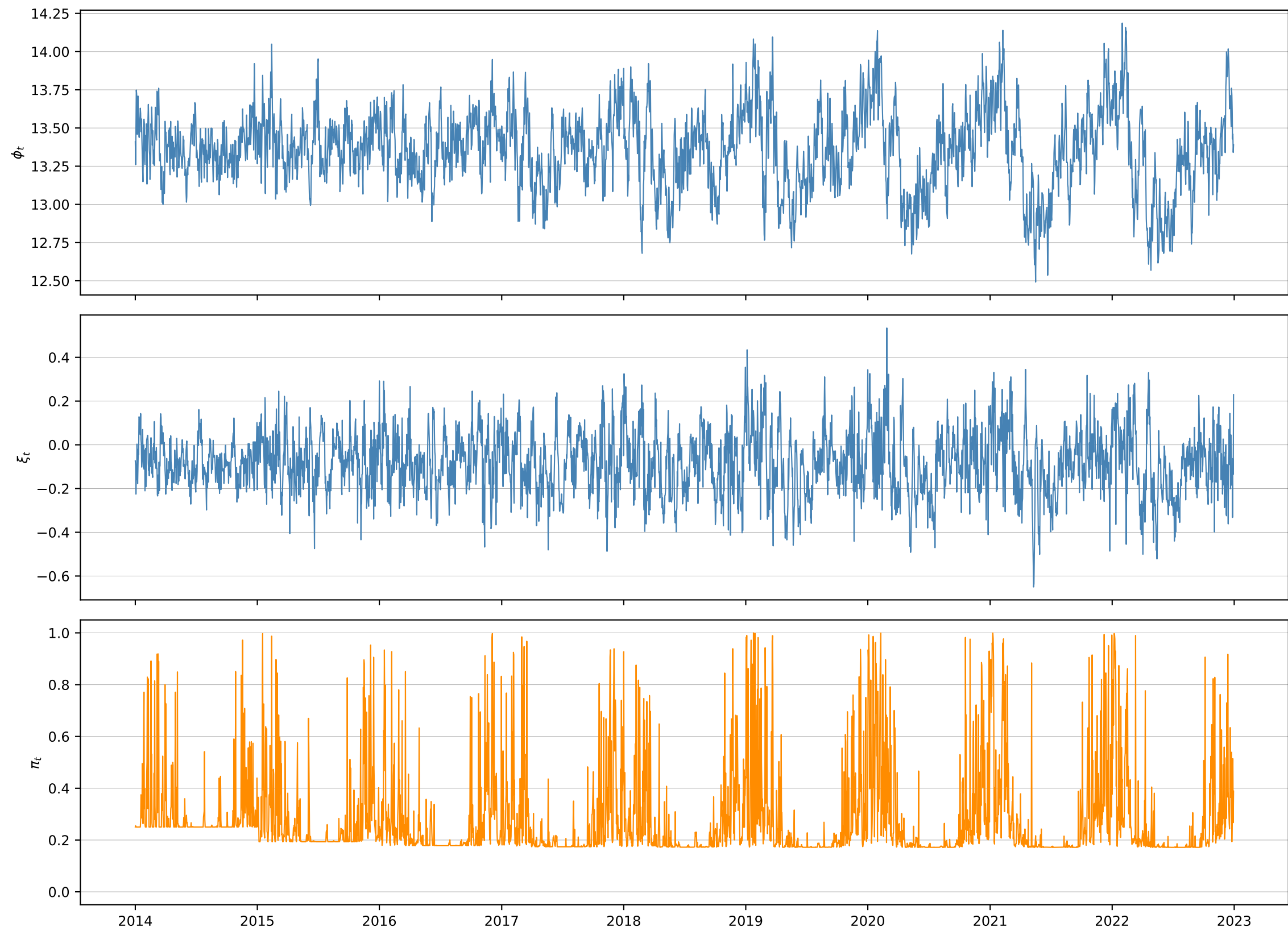
ACF of QR



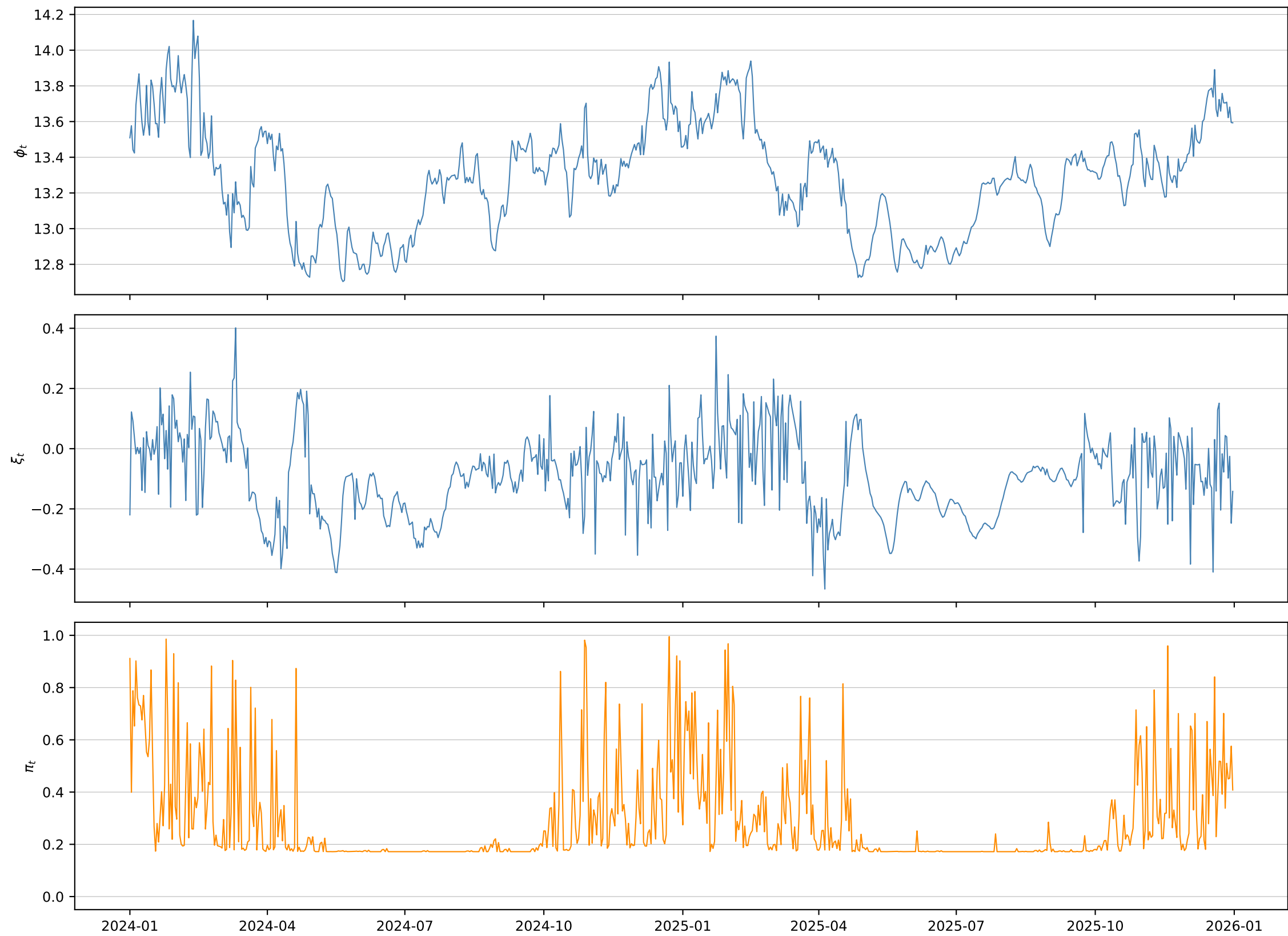
QR time series



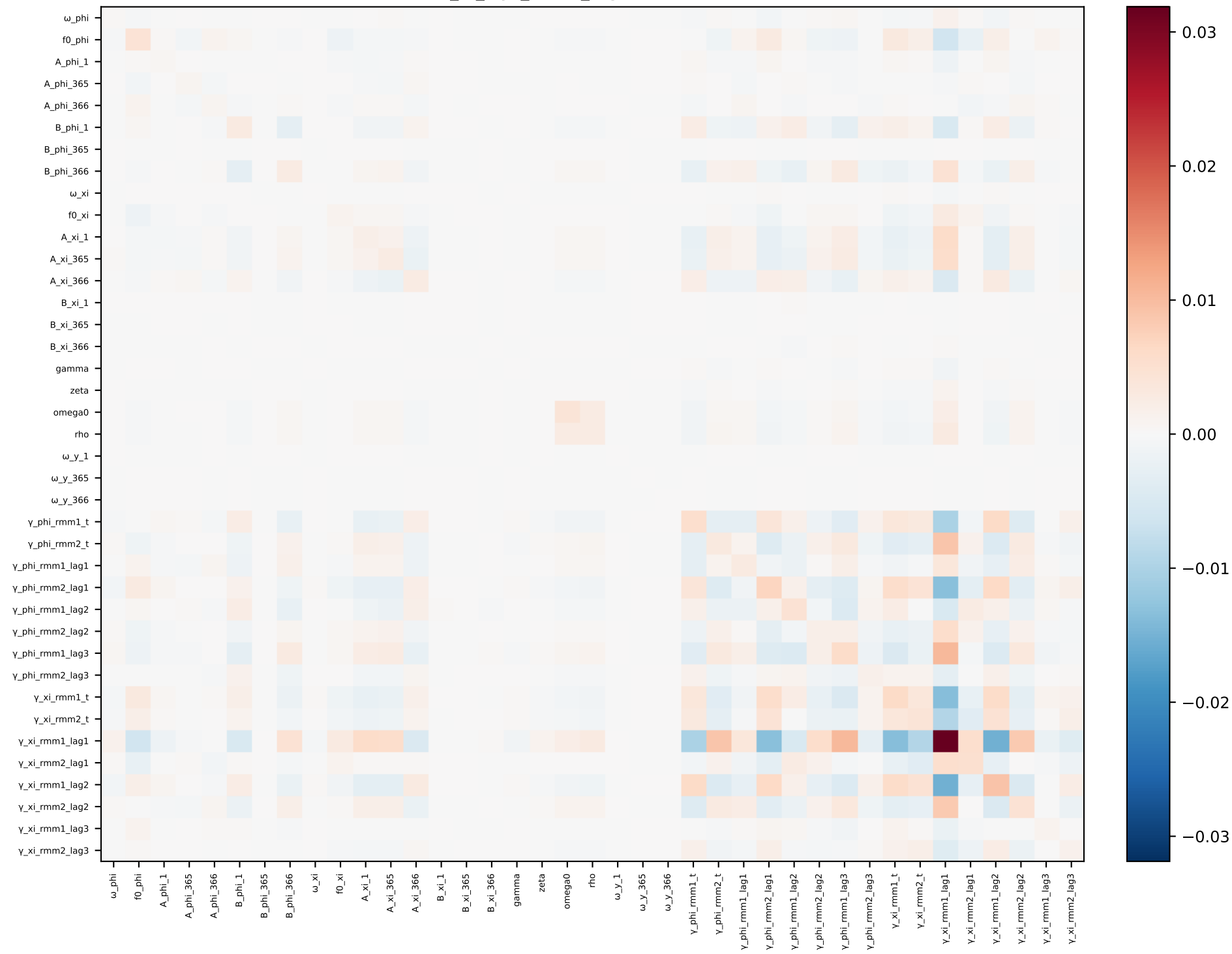
phi_xi_mjo_short_lags — filtered states (IS)



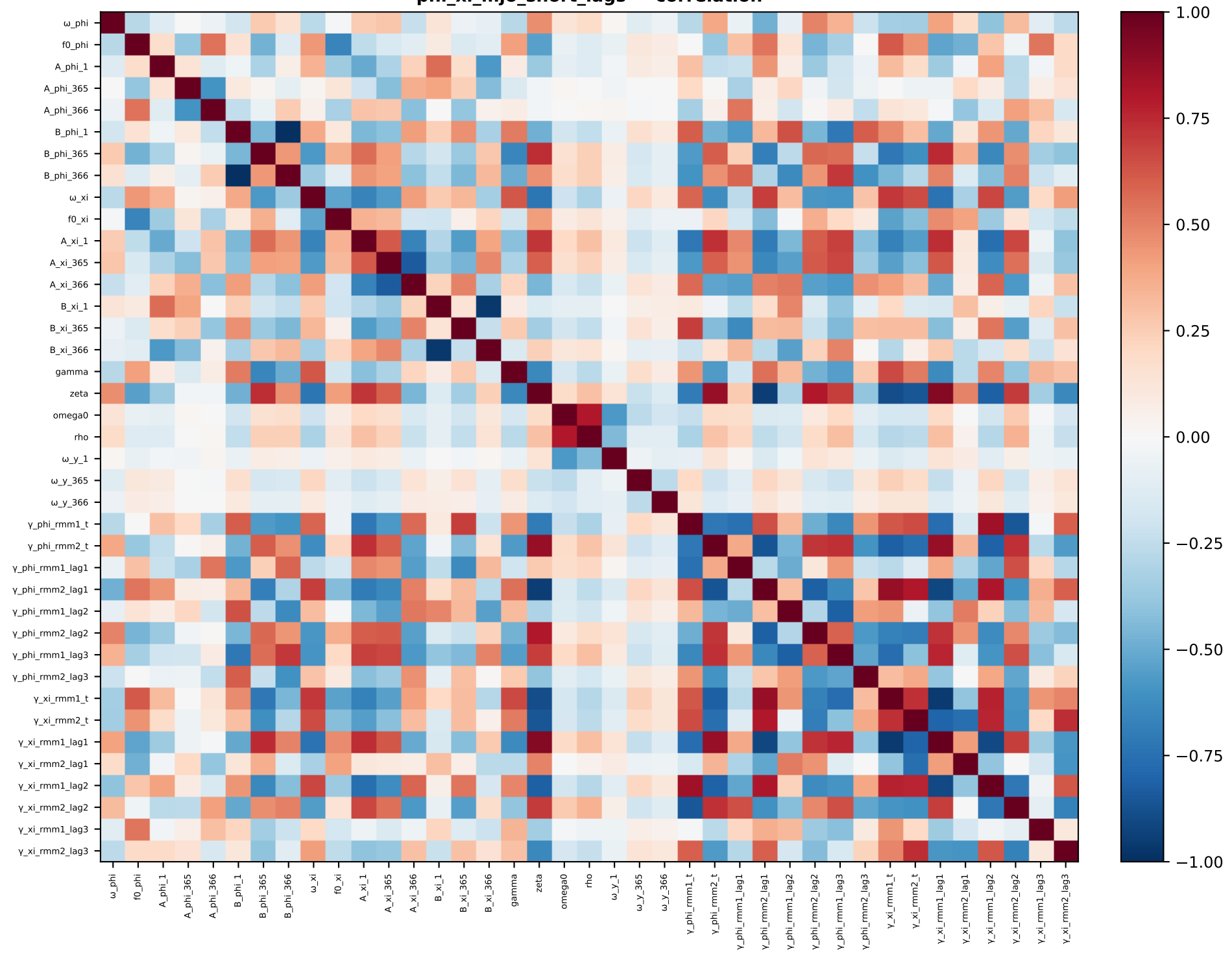
phi_xi_mjo_short_lags — filtered states (OOS)



phi_xi_mjo_short_lags — covariance

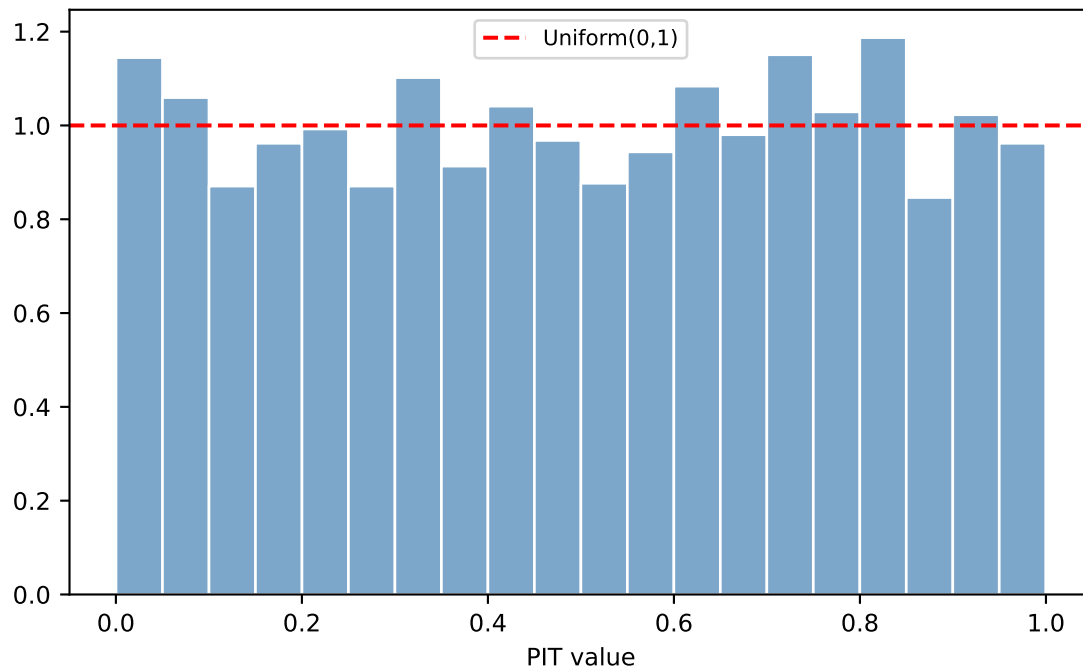


phi_xi_mjo_short_lags — correlation

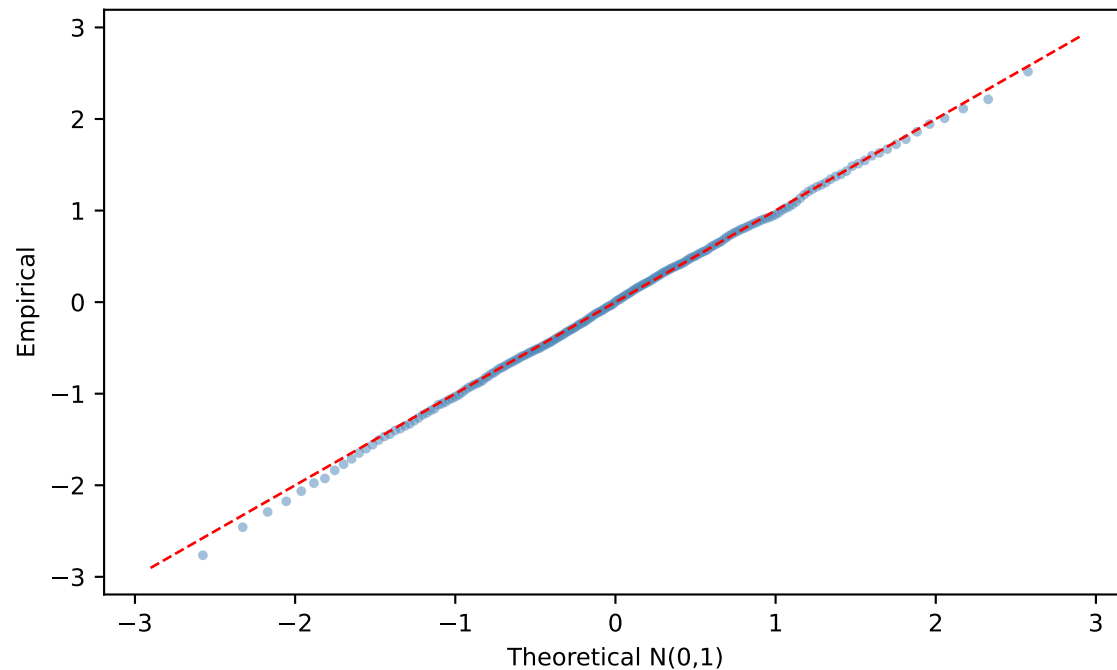


phi_xi_nino34_t — IS diagnostics

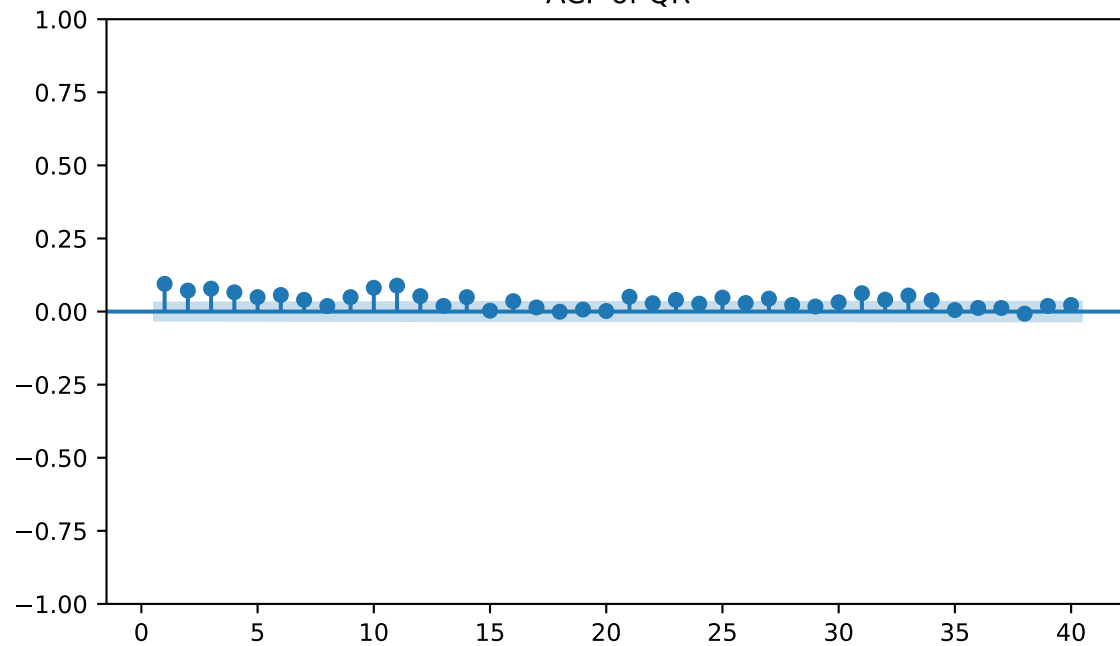
PIT histogram



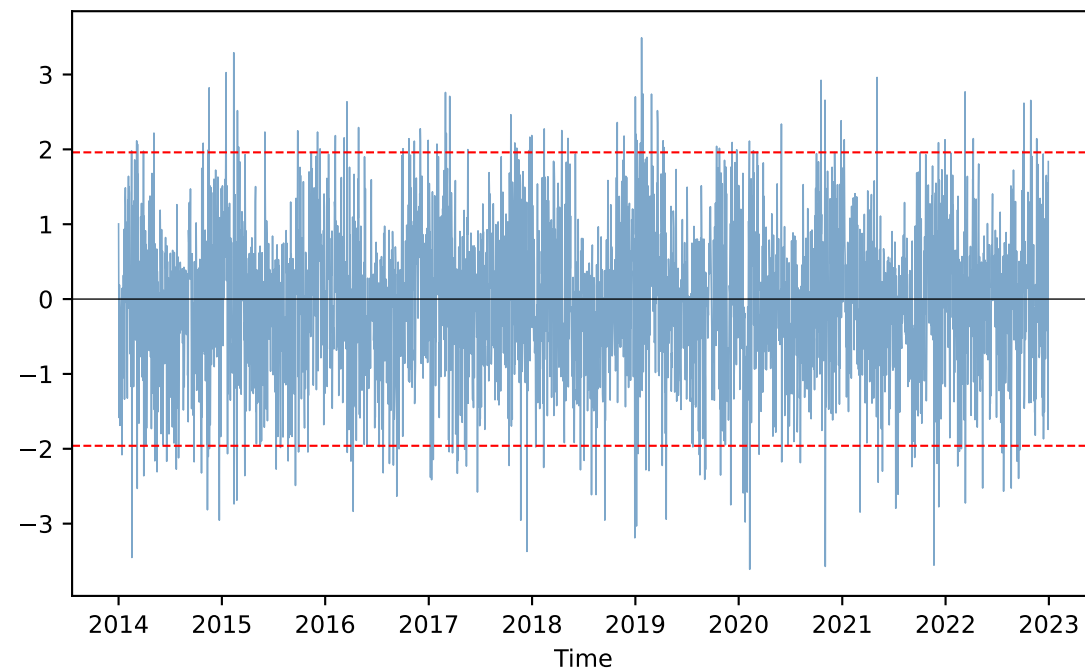
Normal QQ plot



ACF of QR

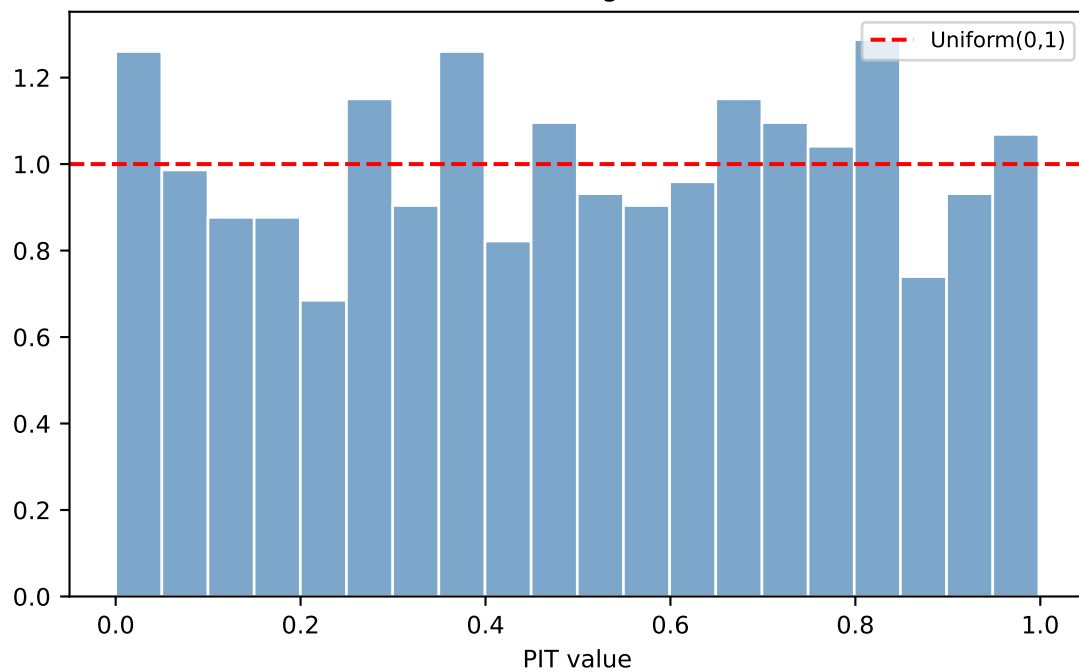


QR time series

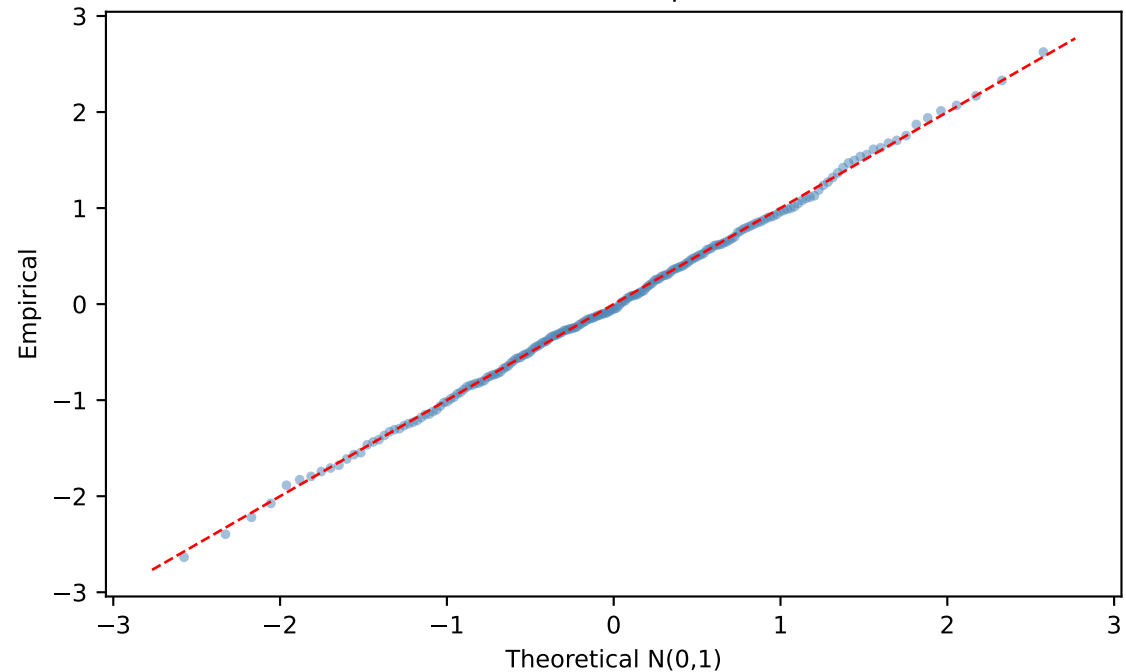


phi_xi_nino34_t — OOS diagnostics

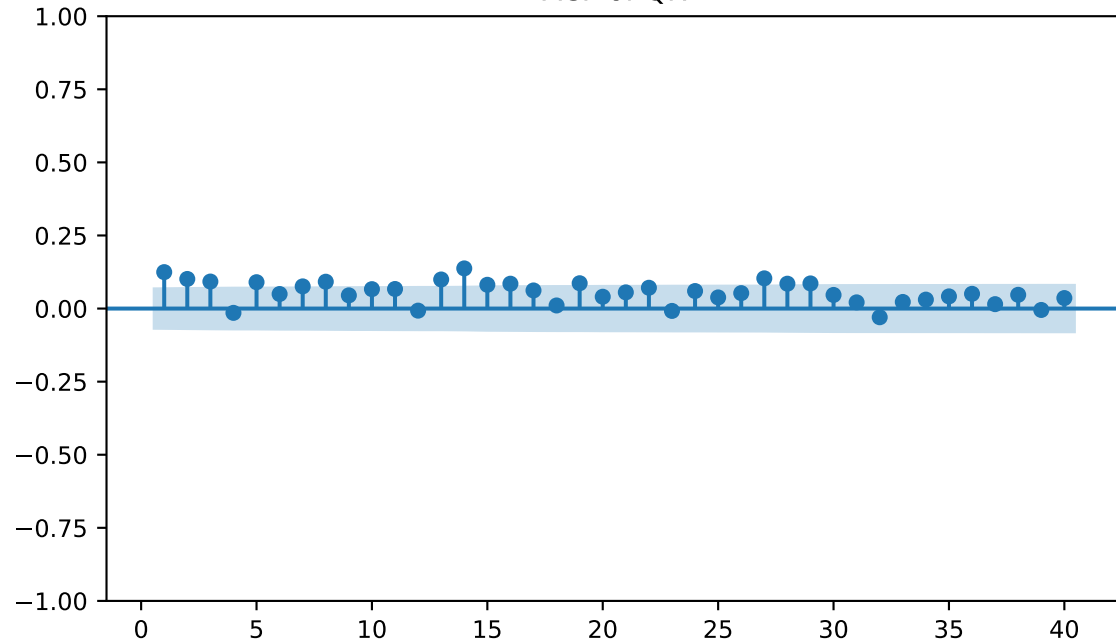
PIT histogram



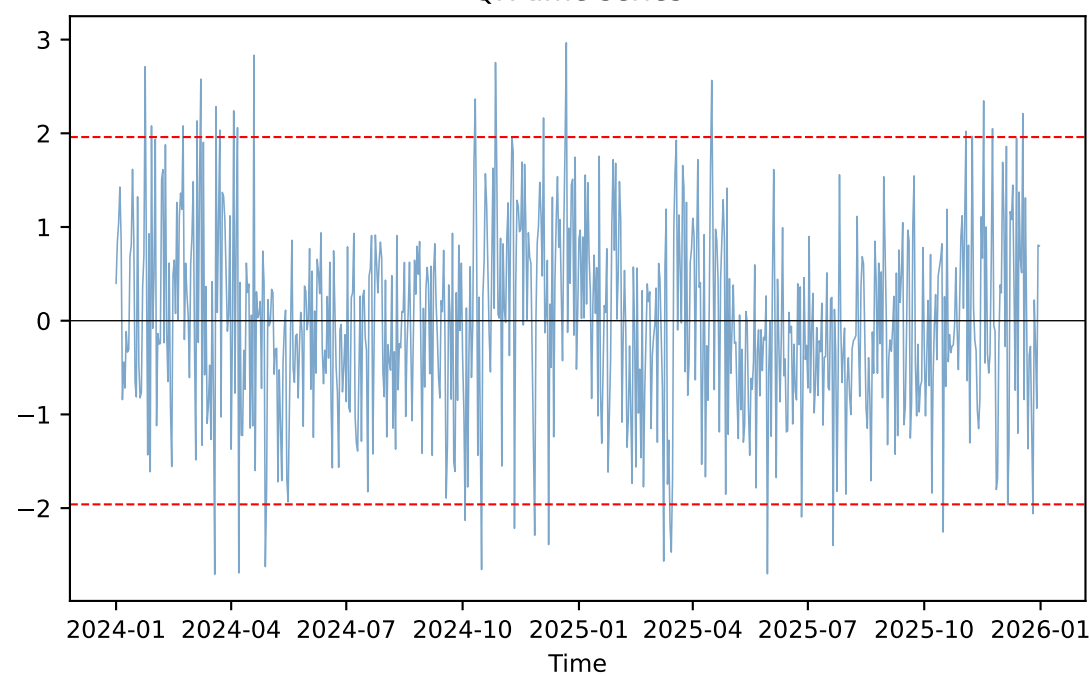
Normal QQ plot



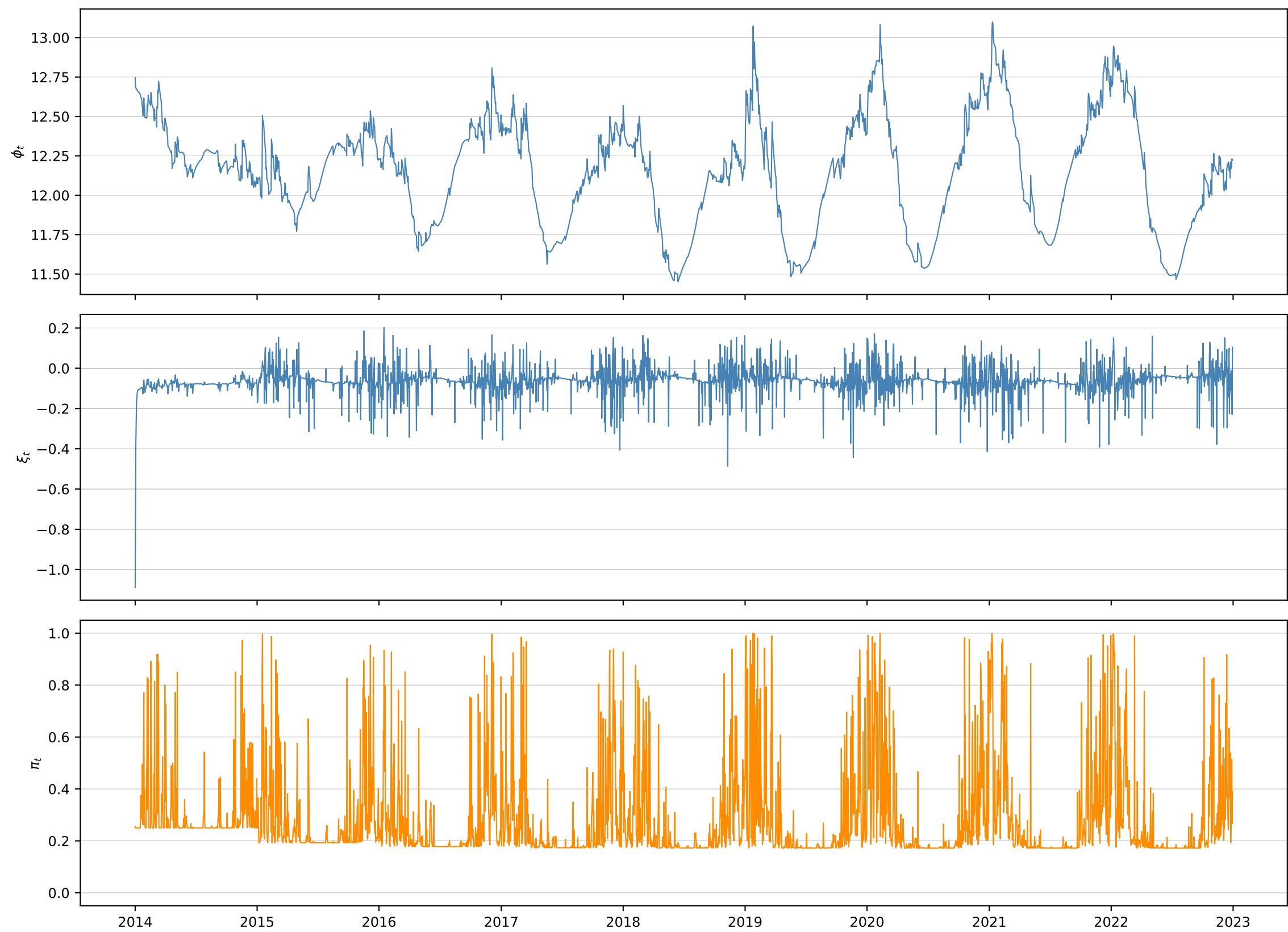
ACF of QR



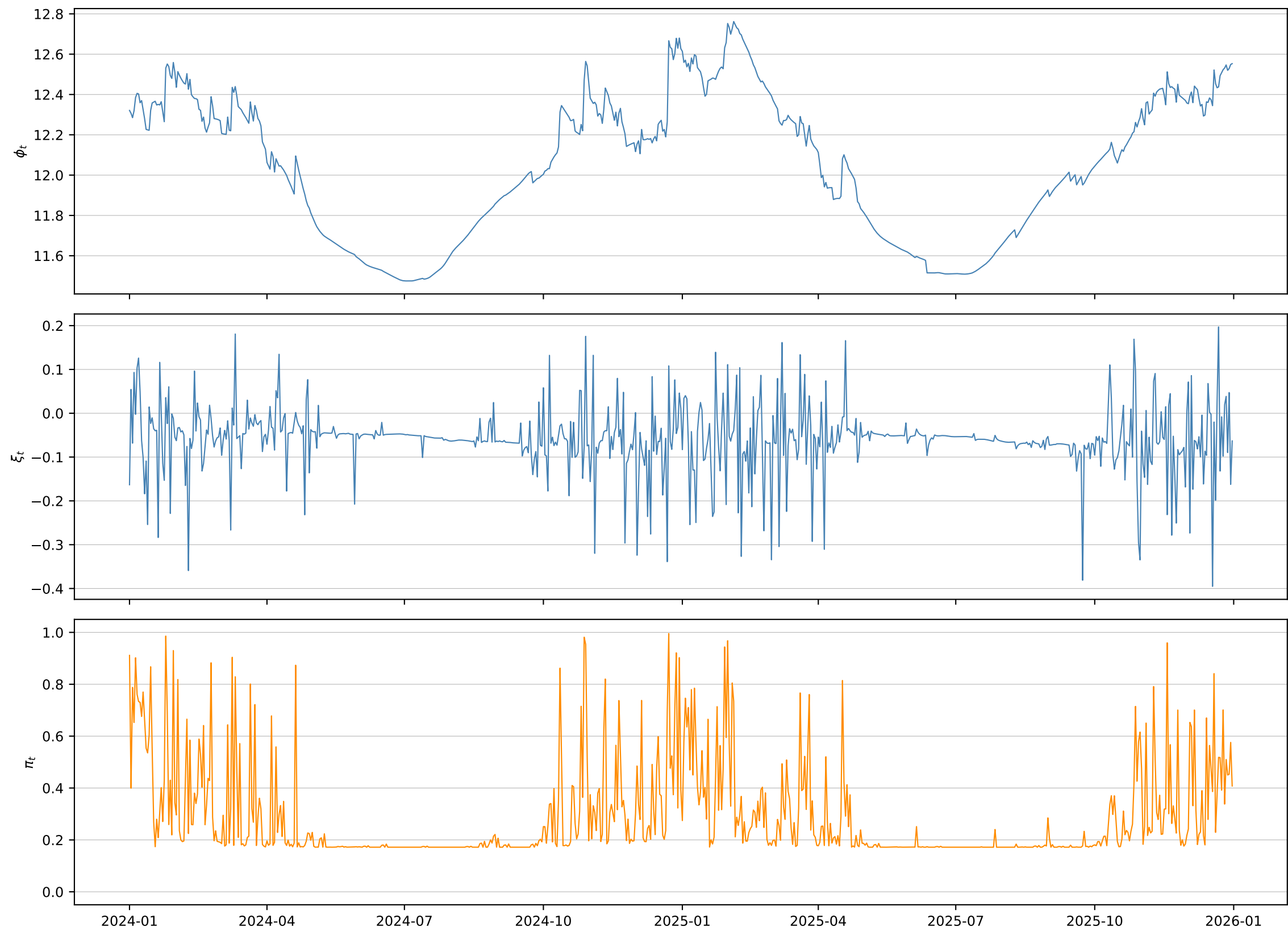
QR time series



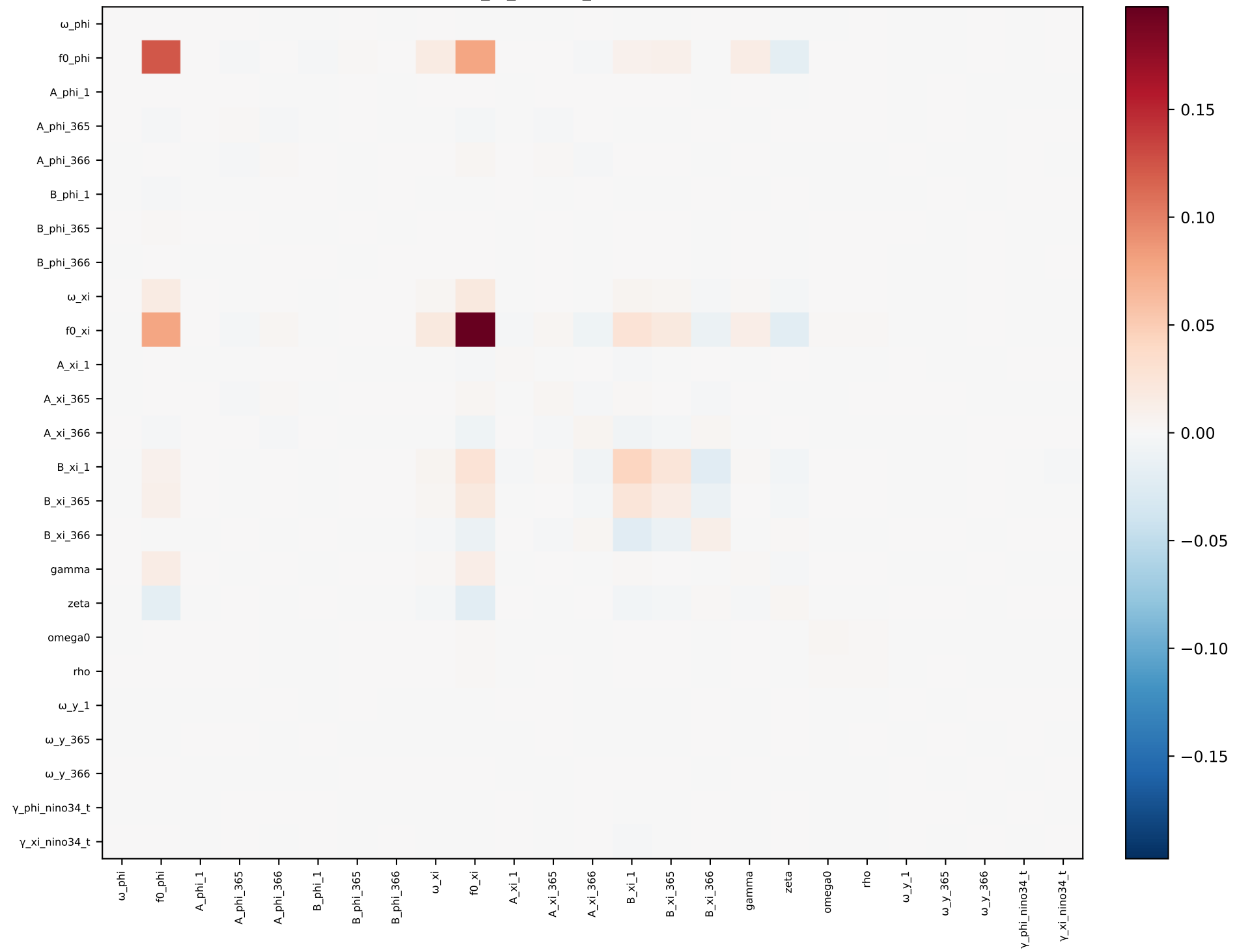
phi_xi_nino34_t — filtered states (IS)



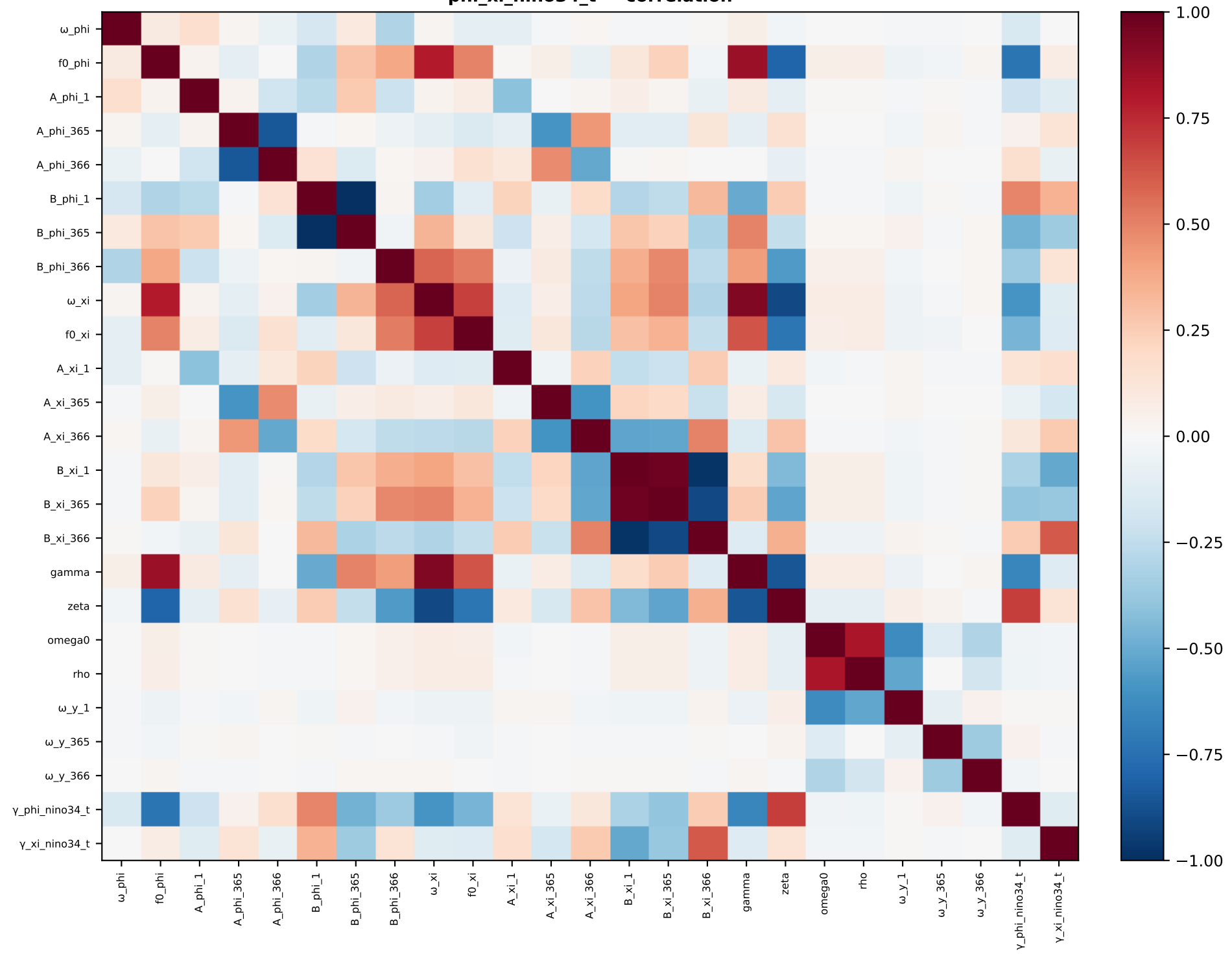
phi_xi_nino34_t — filtered states (OOS)



phi_xi_nino34_t — covariance

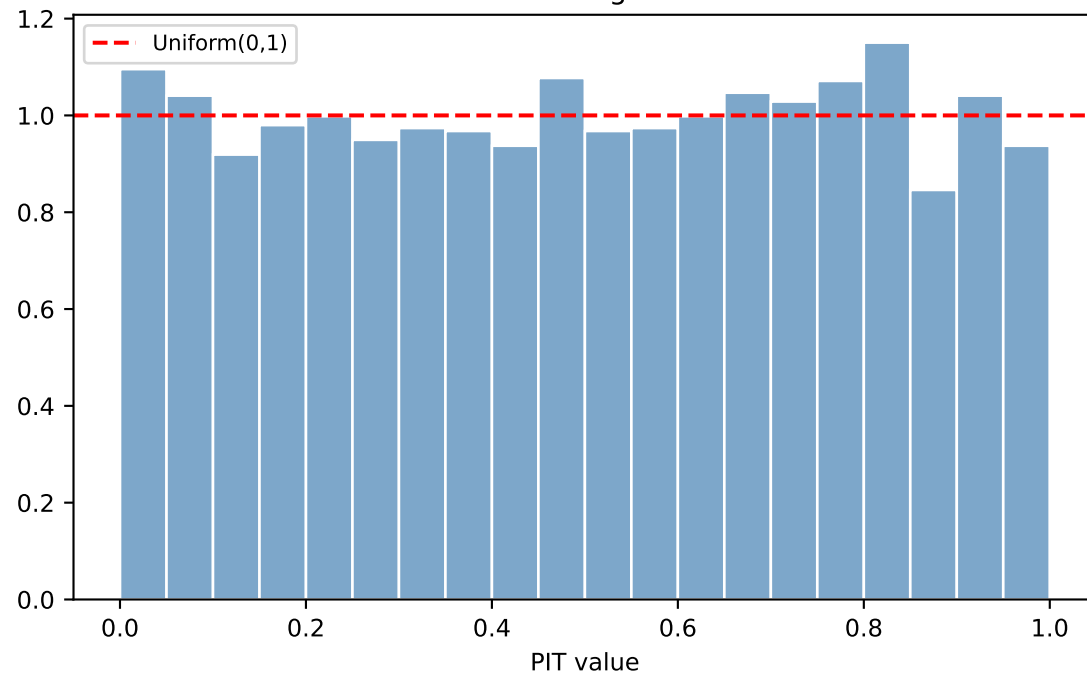


phi_xi_nino34_t — correlation

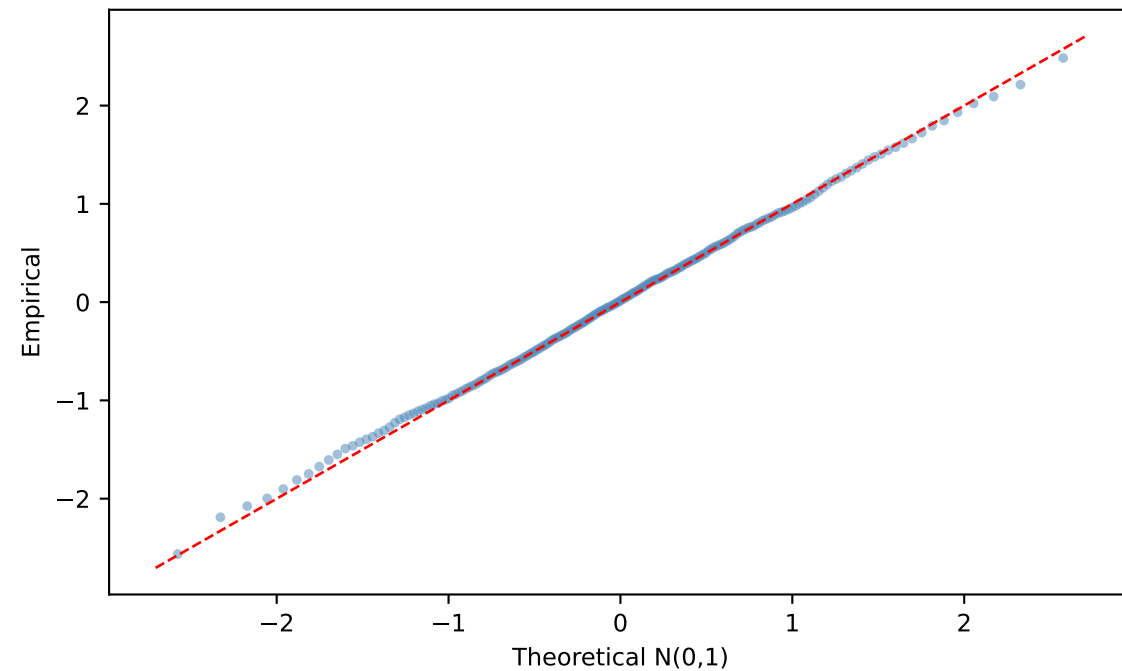


phi_xi_nino34_lags — IS diagnostics

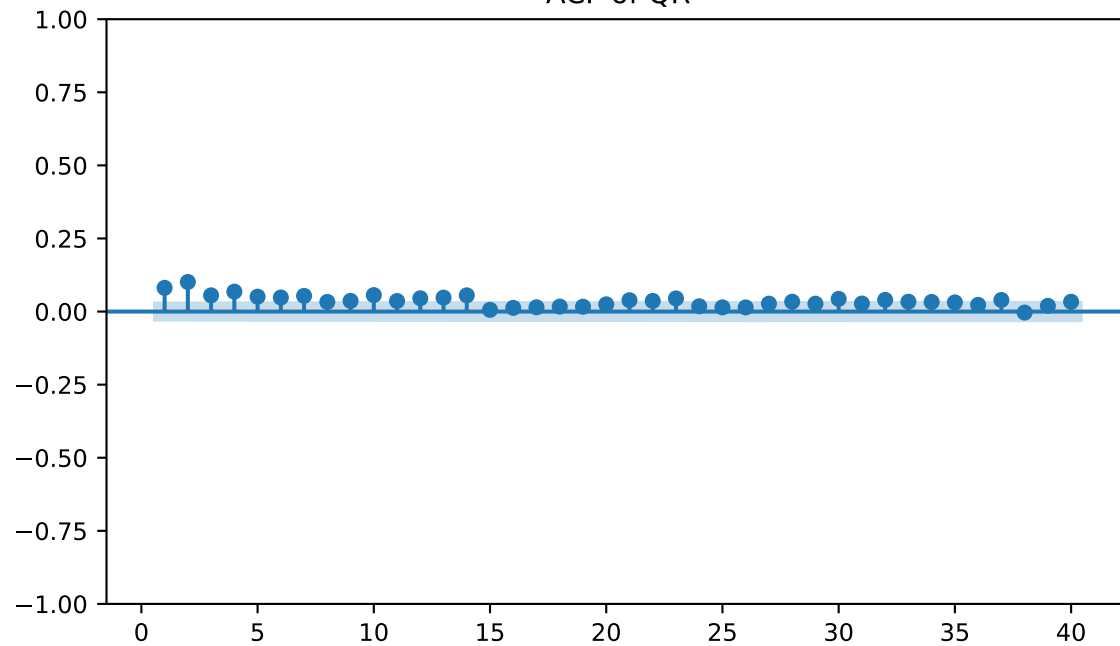
PIT histogram



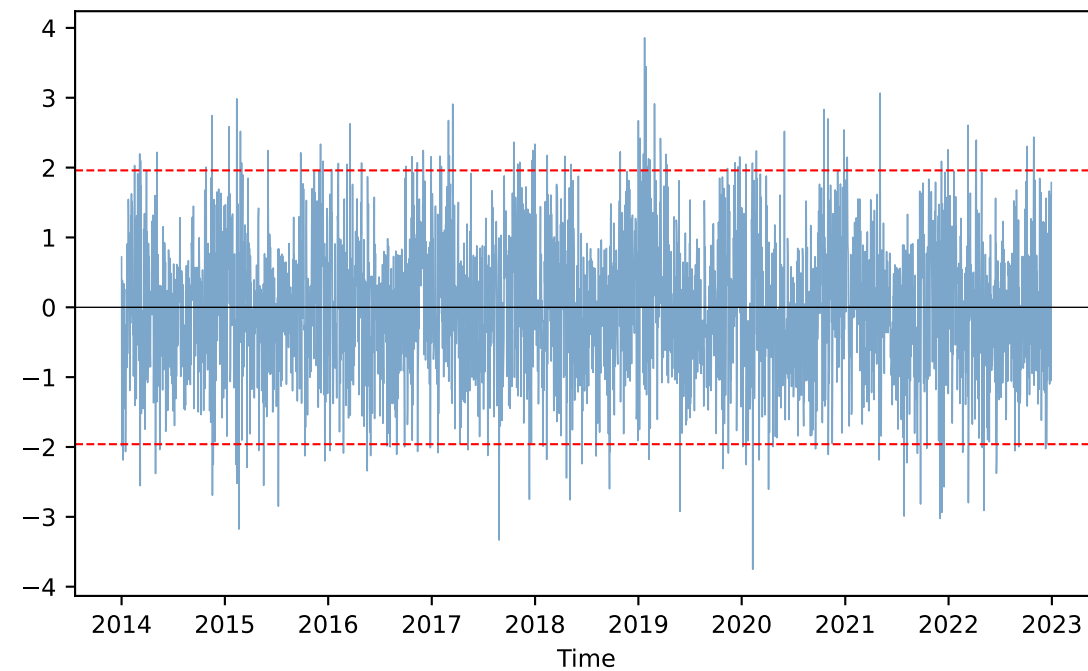
Normal QQ plot



ACF of QR

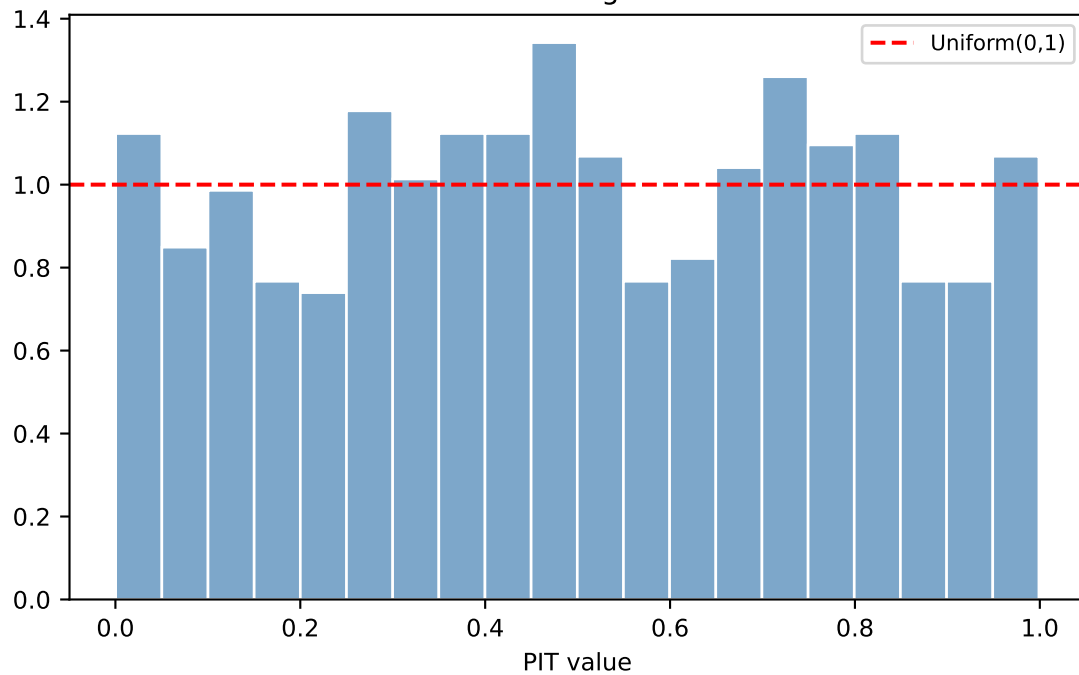


QR time series

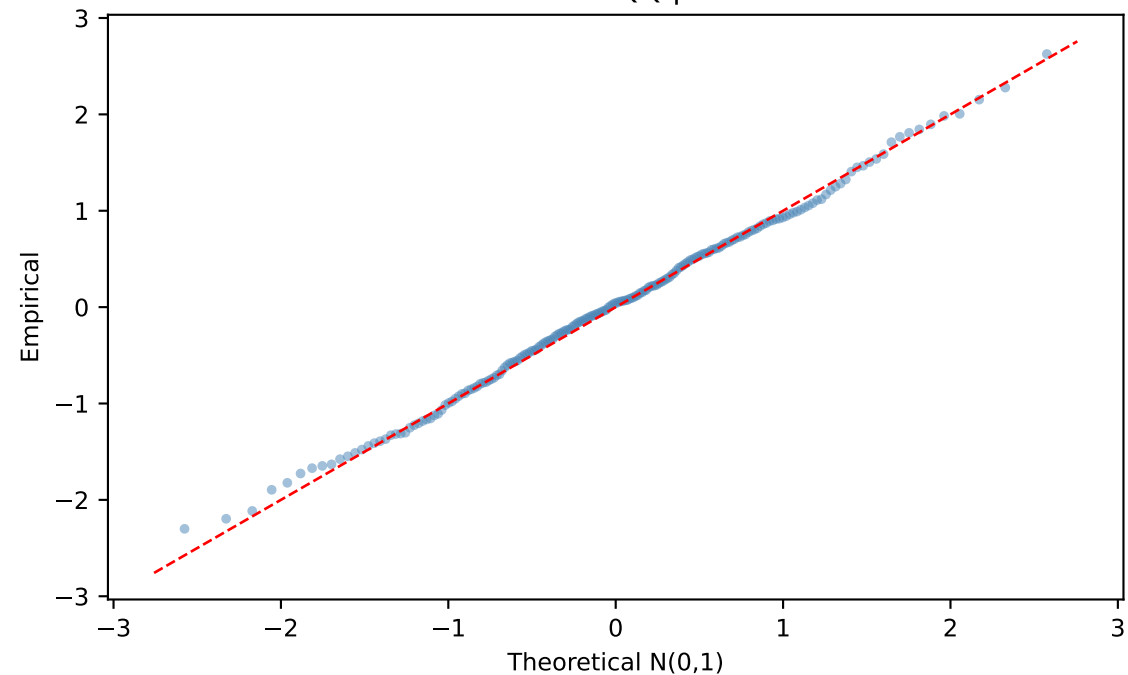


phi_xi_nino34_lags — OOS diagnostics

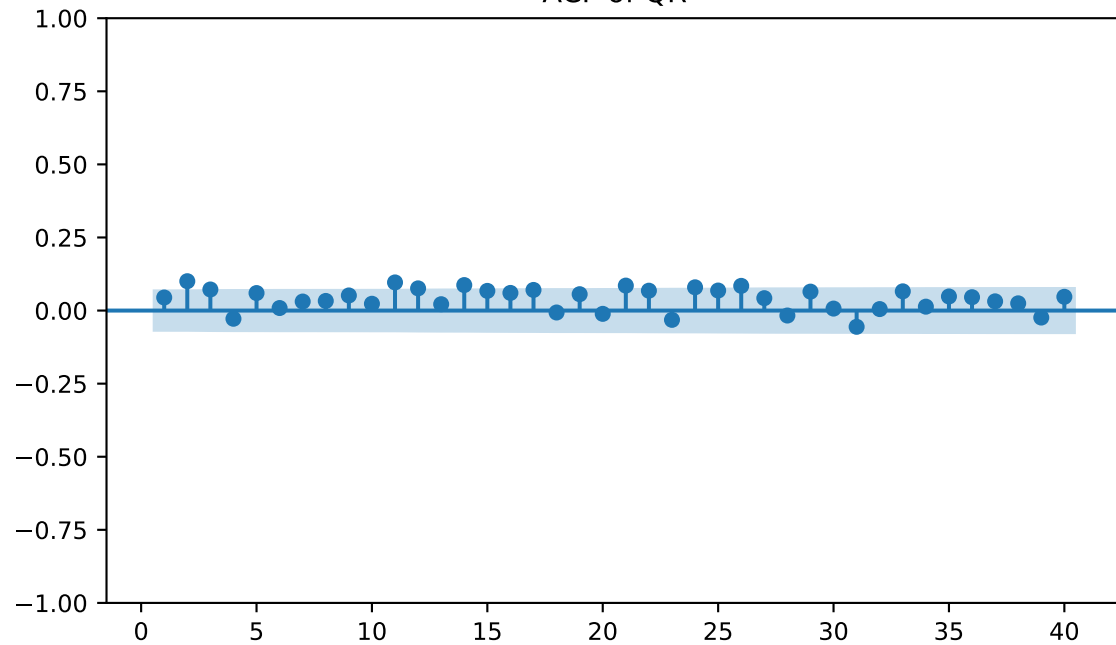
PIT histogram



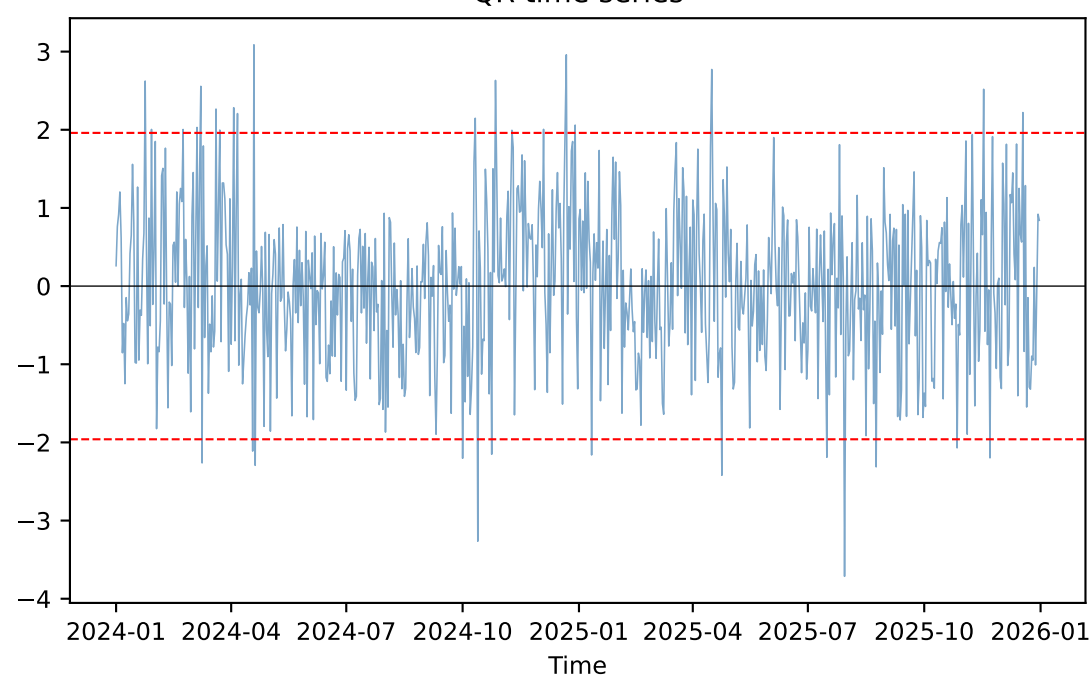
Normal QQ plot



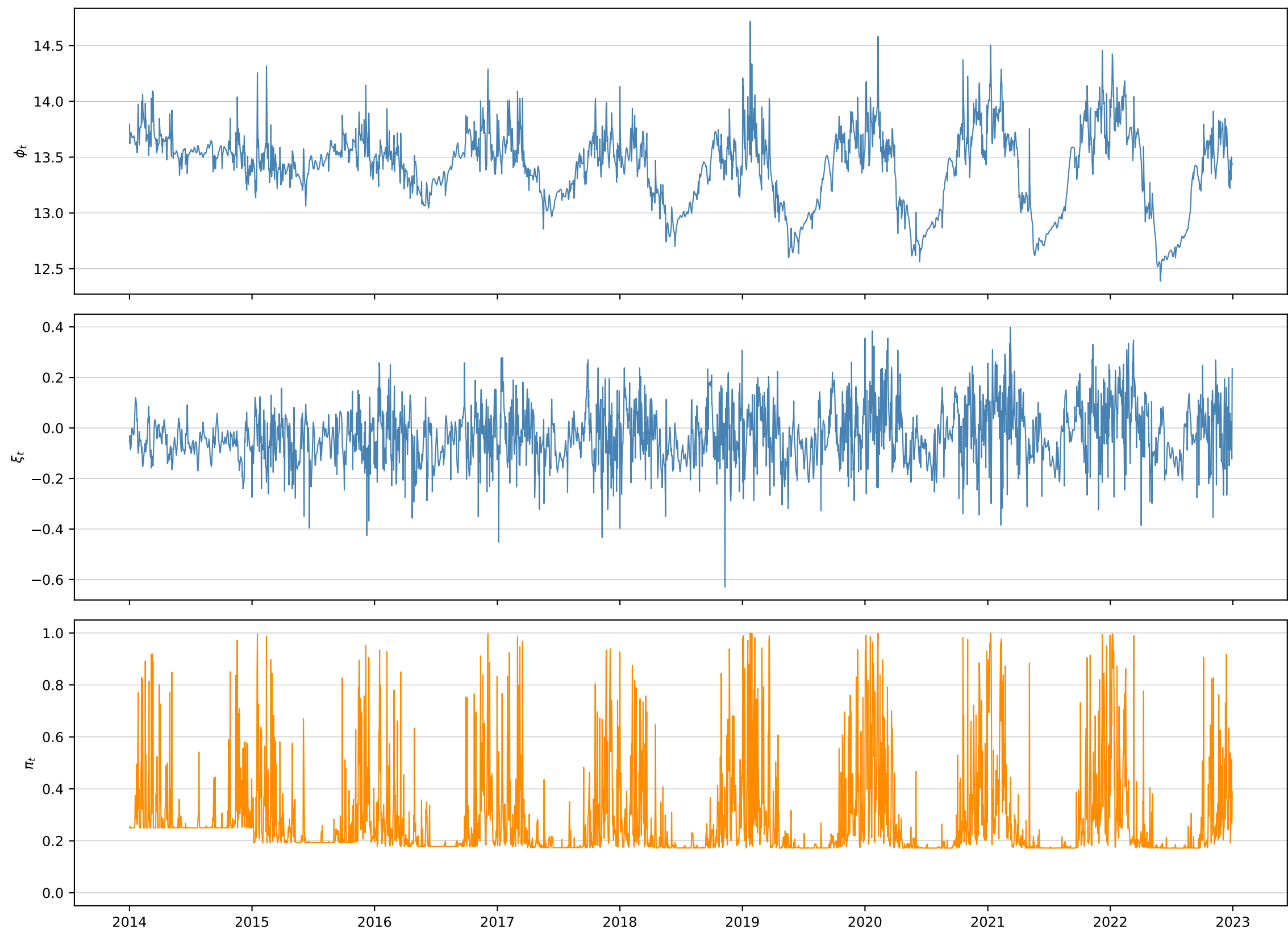
ACF of QR



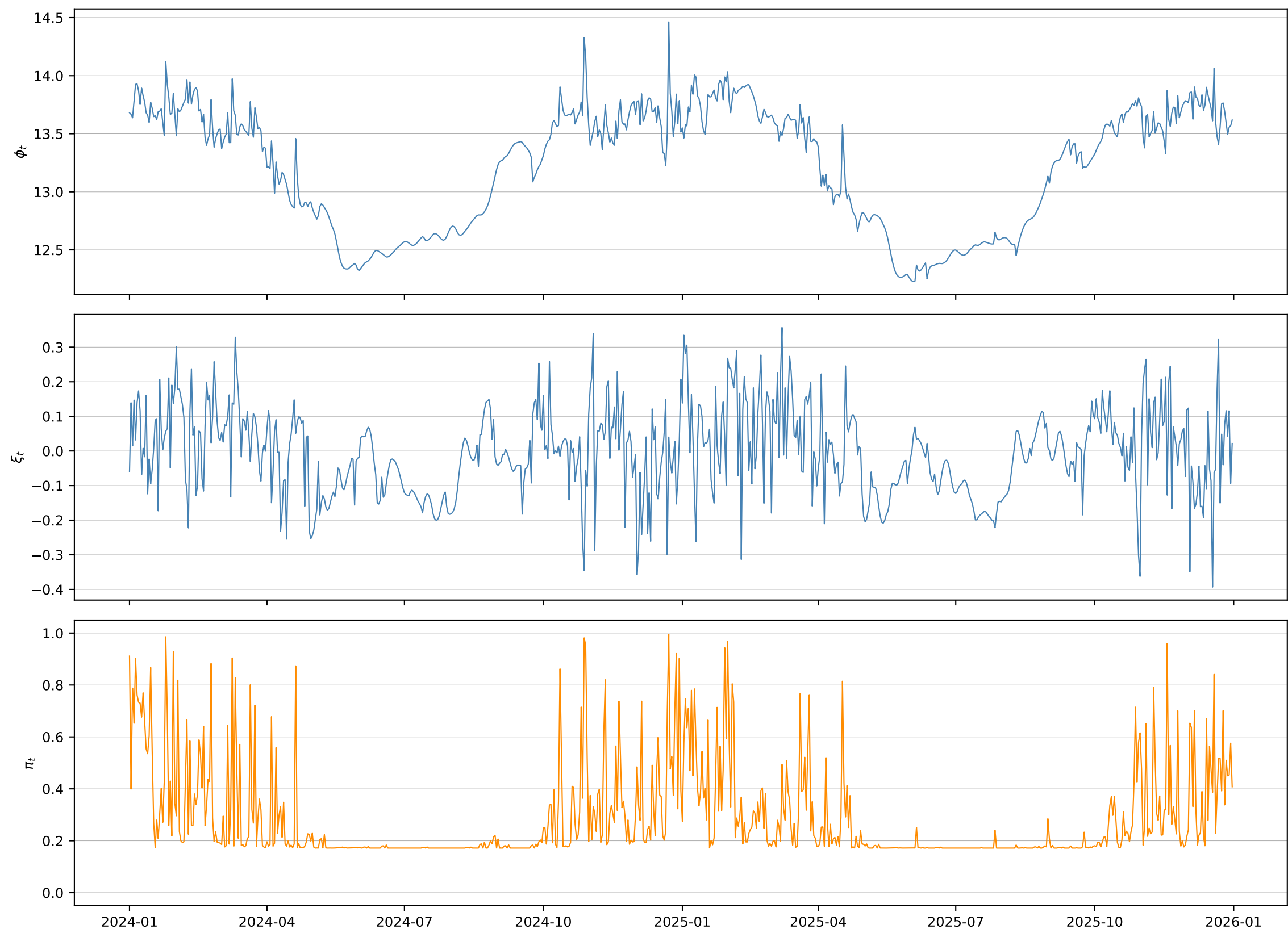
QR time series



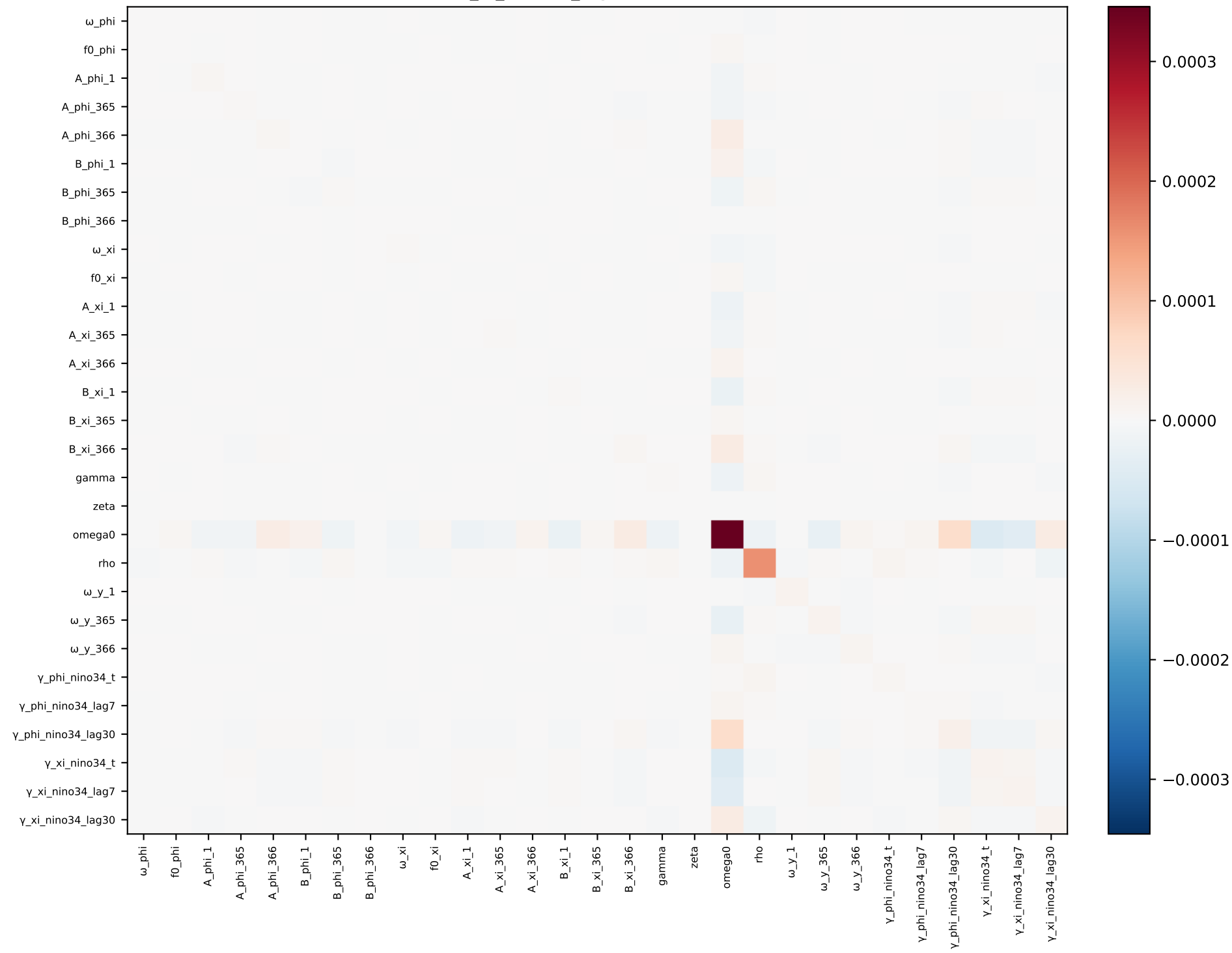
phi_xi_nino34_lags — filtered states (IS)



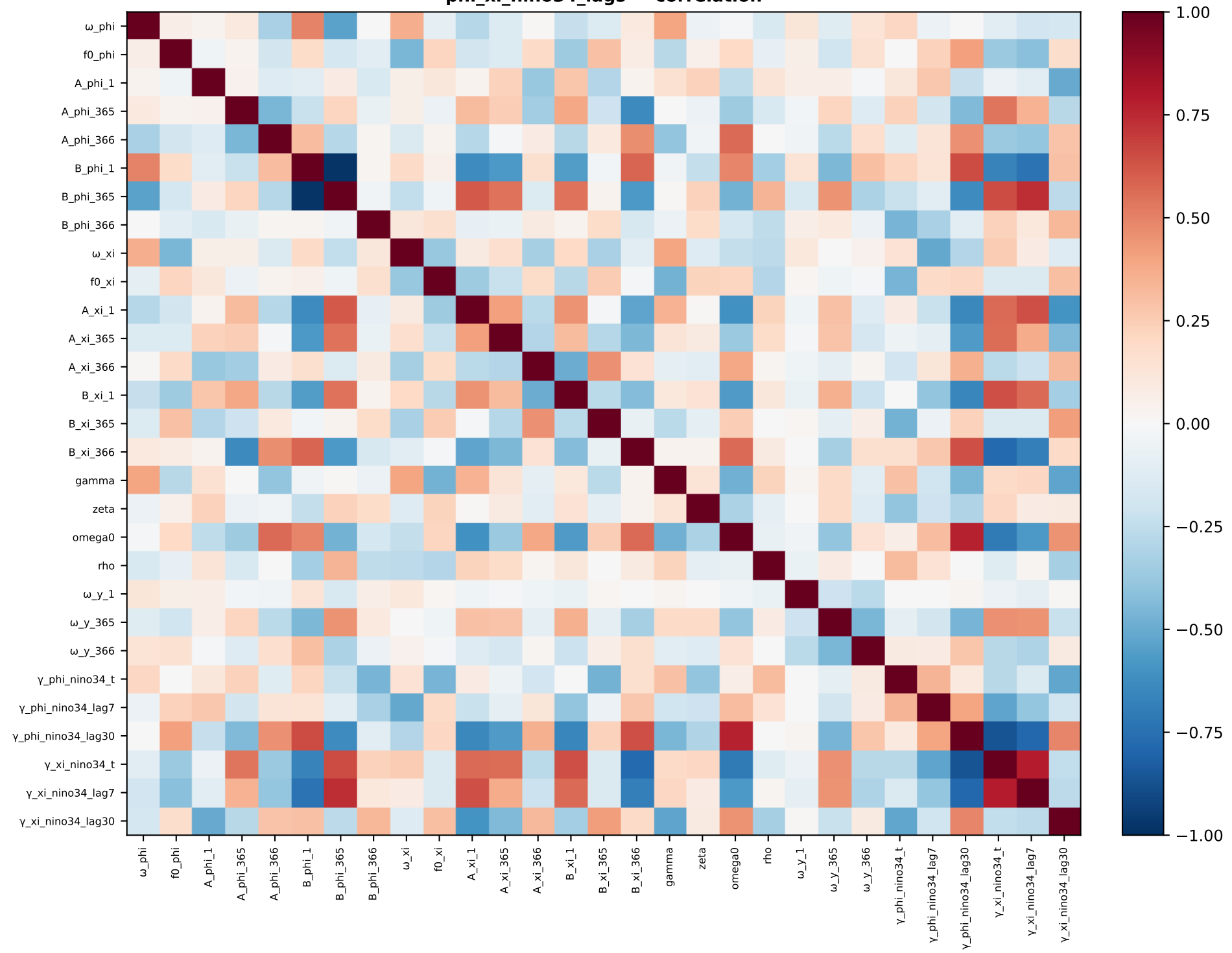
phi_xi_nino34_lags — filtered states (OOS)



phi_xi_nino34_lags — covariance

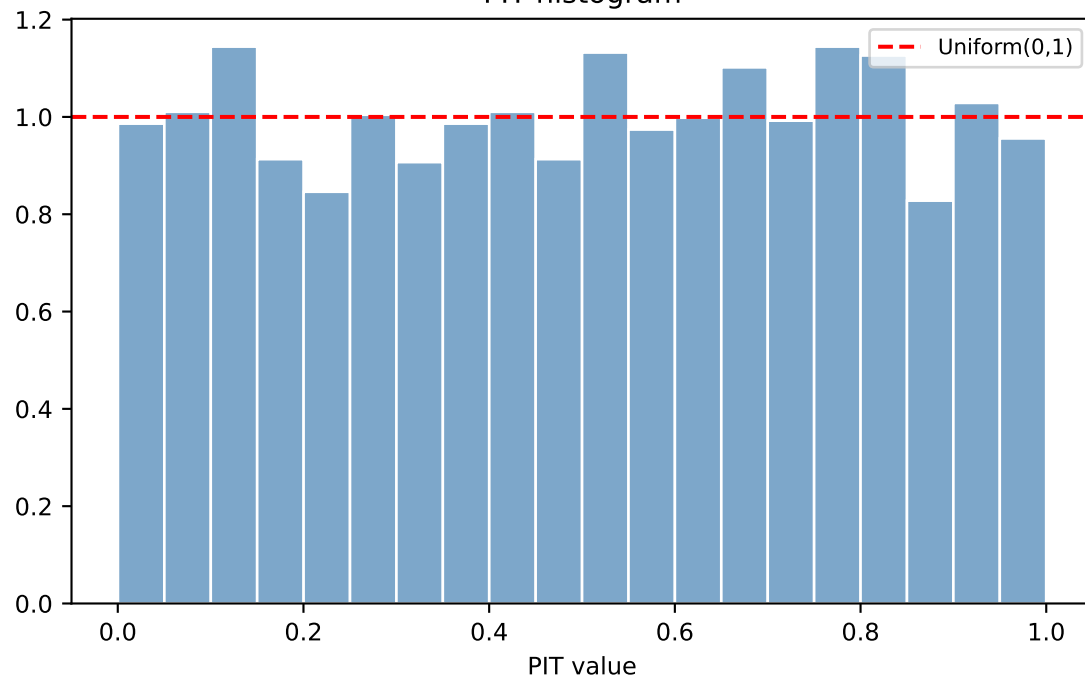


phi_xi_nino34_lags — correlation

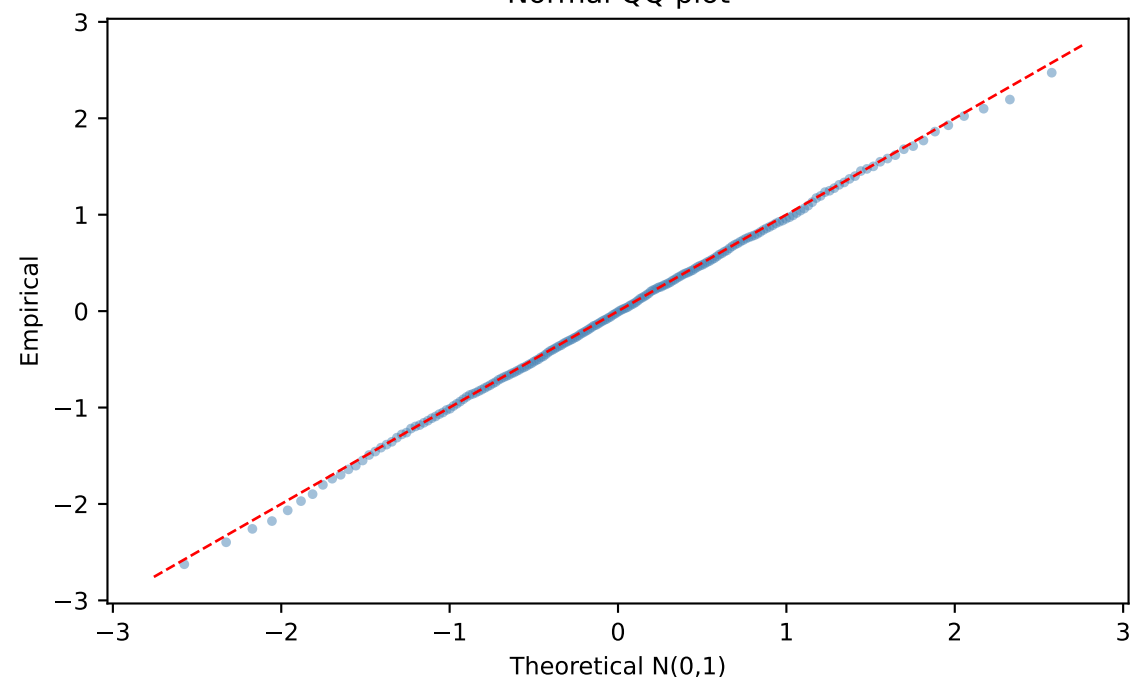


phi_xi_nino34_rolling — IS diagnostics

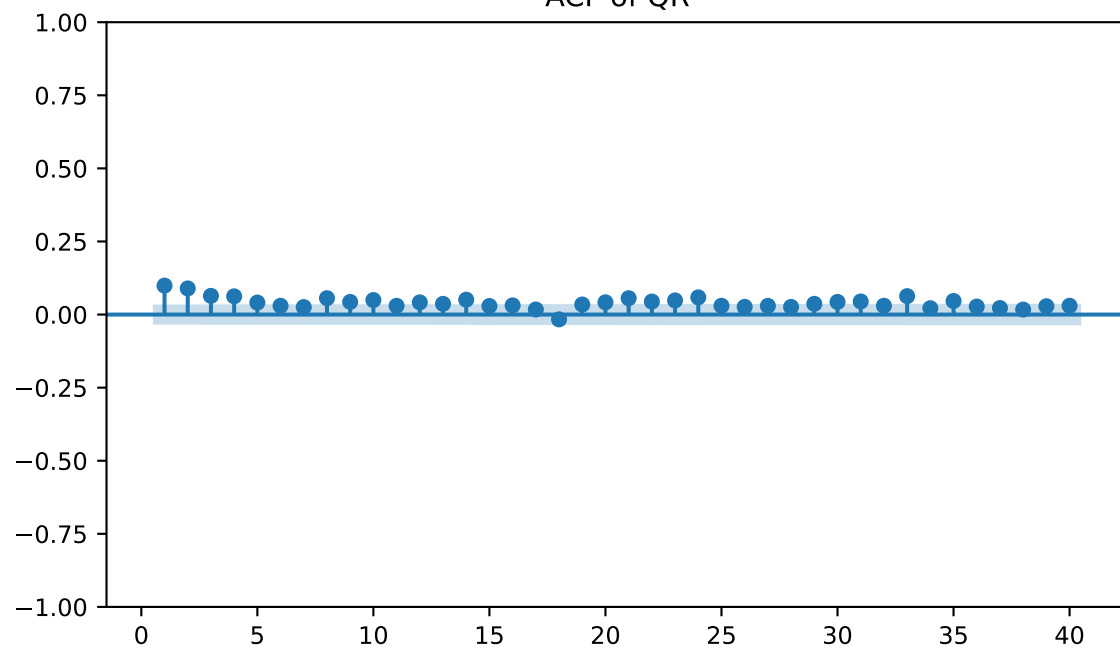
PIT histogram



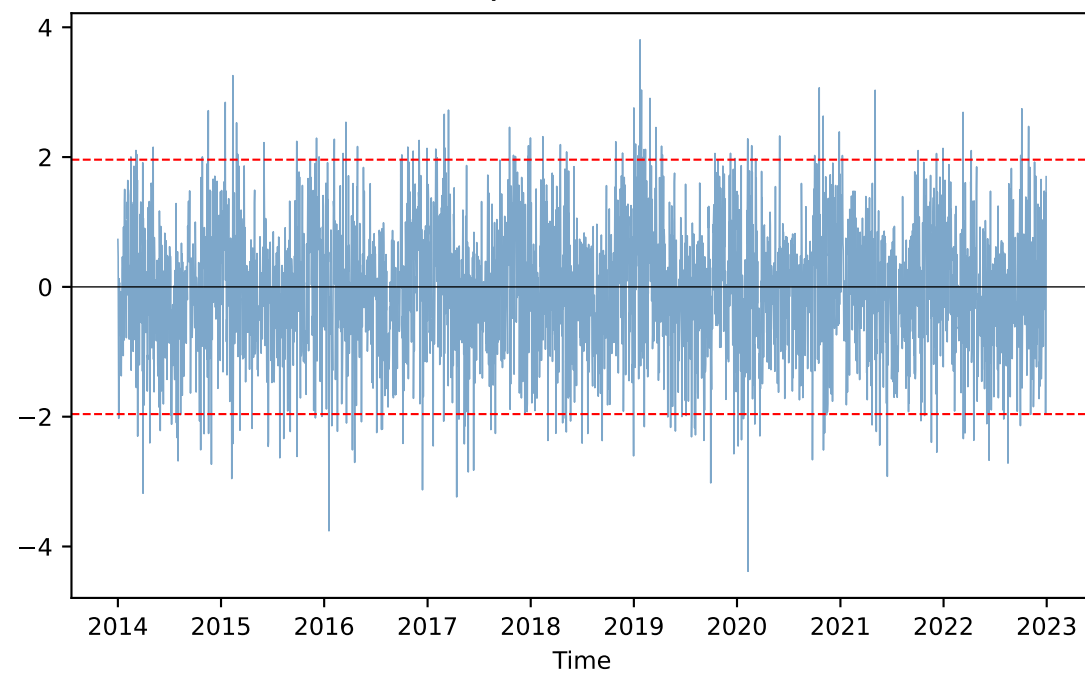
Normal QQ plot



ACF of QR

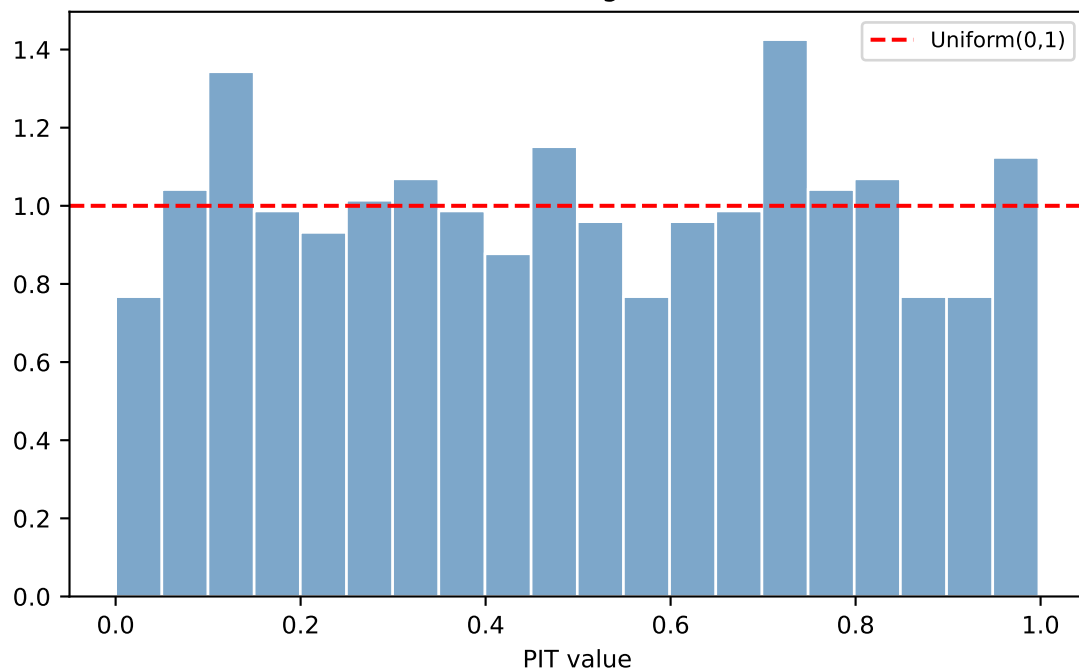


QR time series

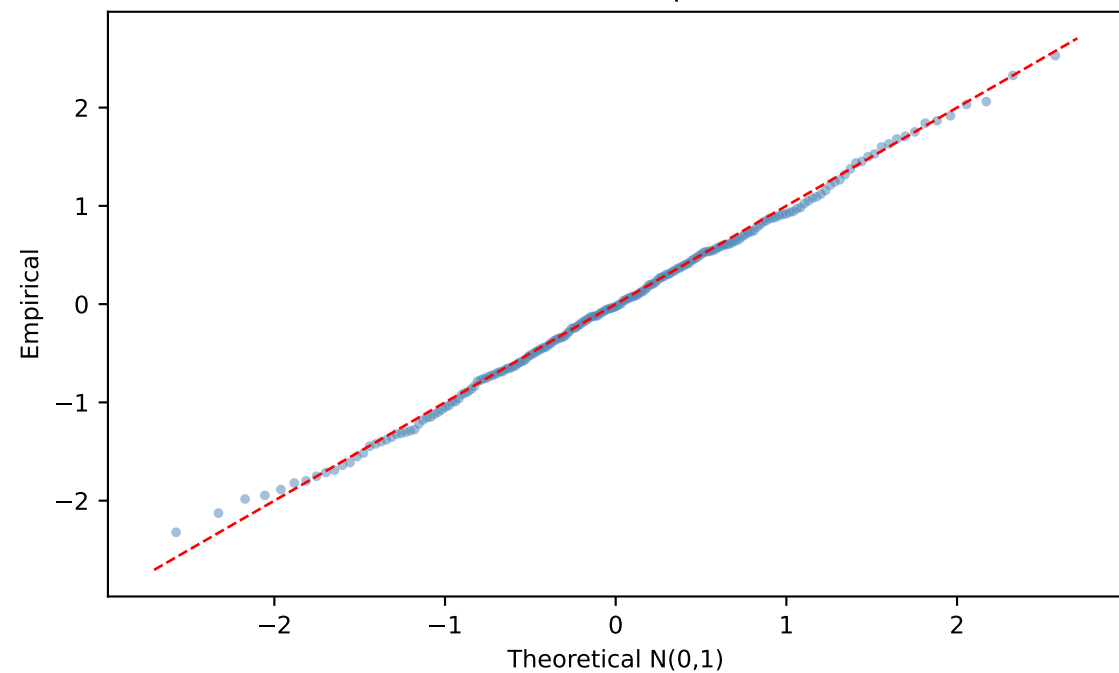


phi_xi_nino34_rolling — OOS diagnostics

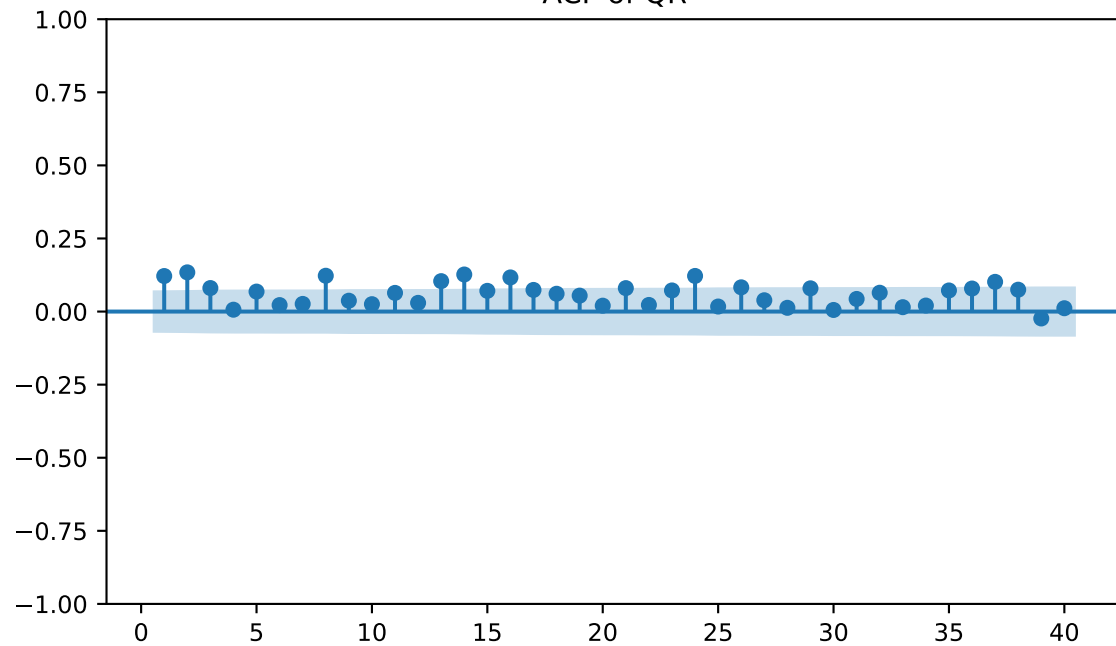
PIT histogram



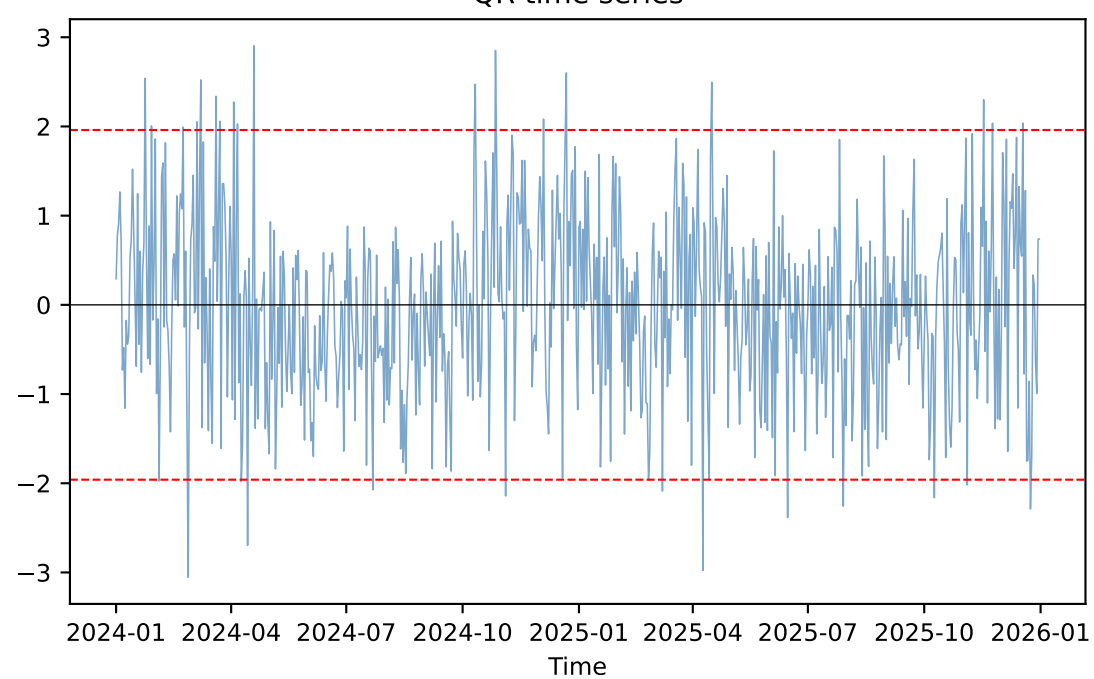
Normal QQ plot



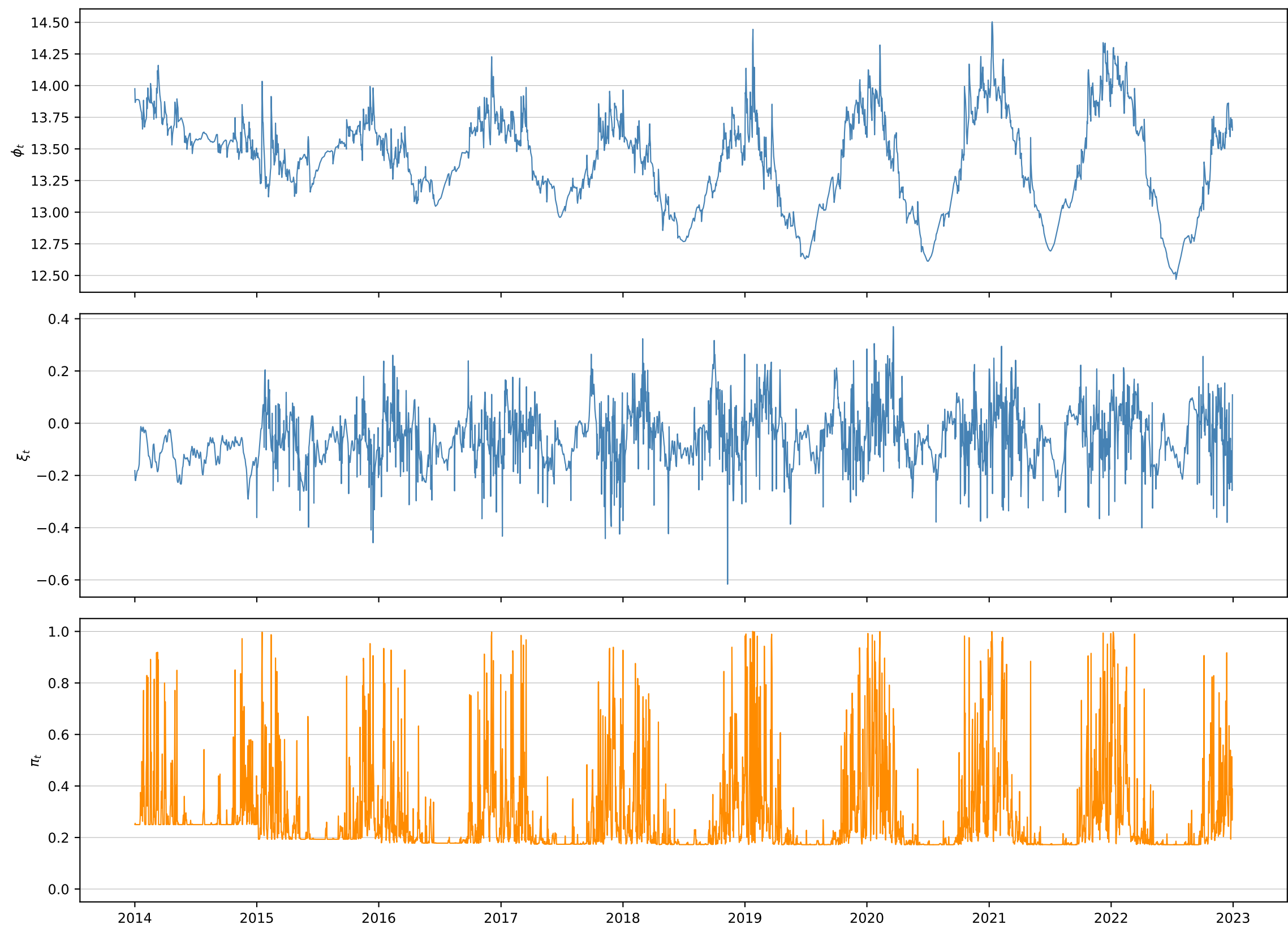
ACF of QR



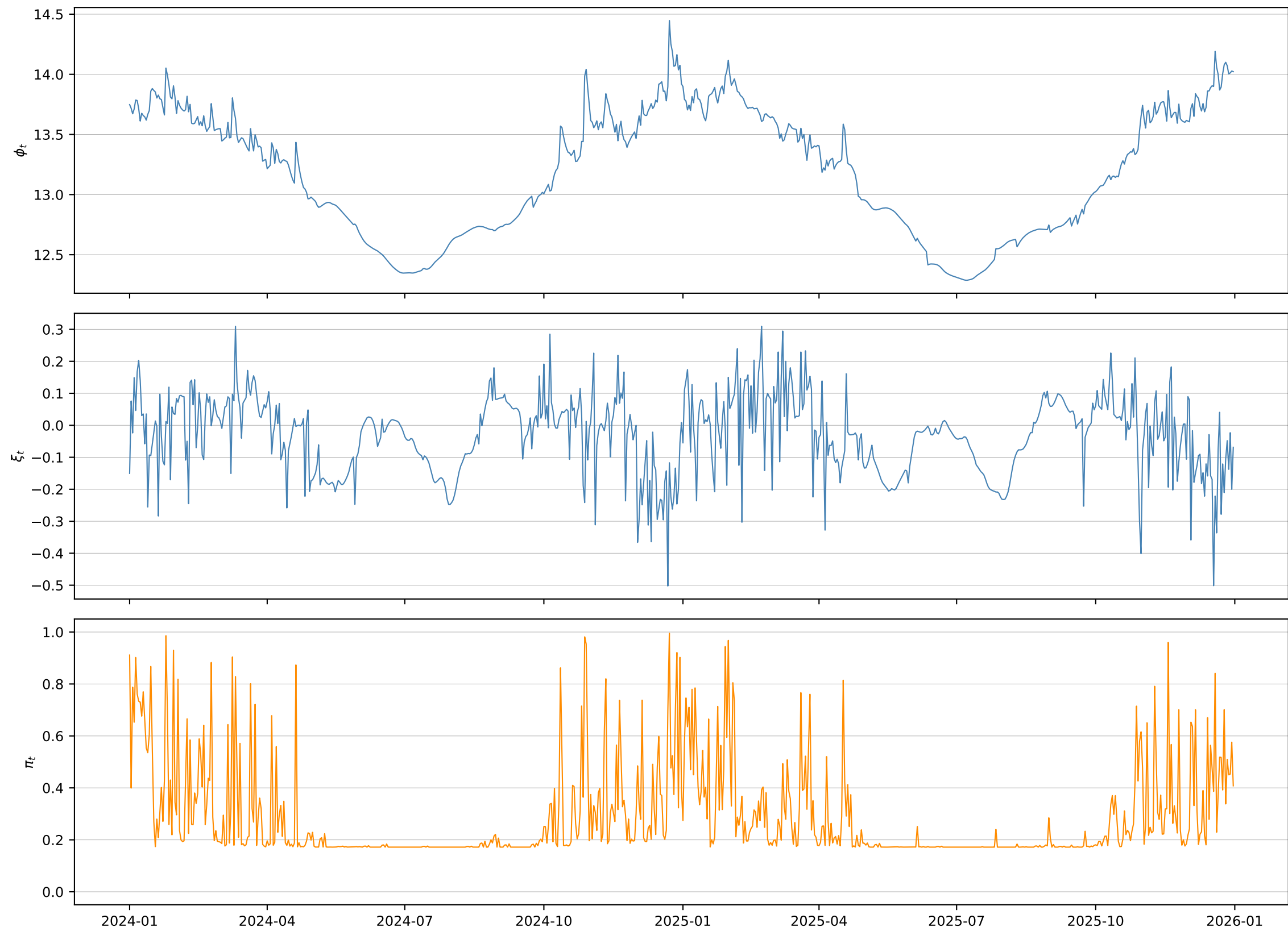
QR time series



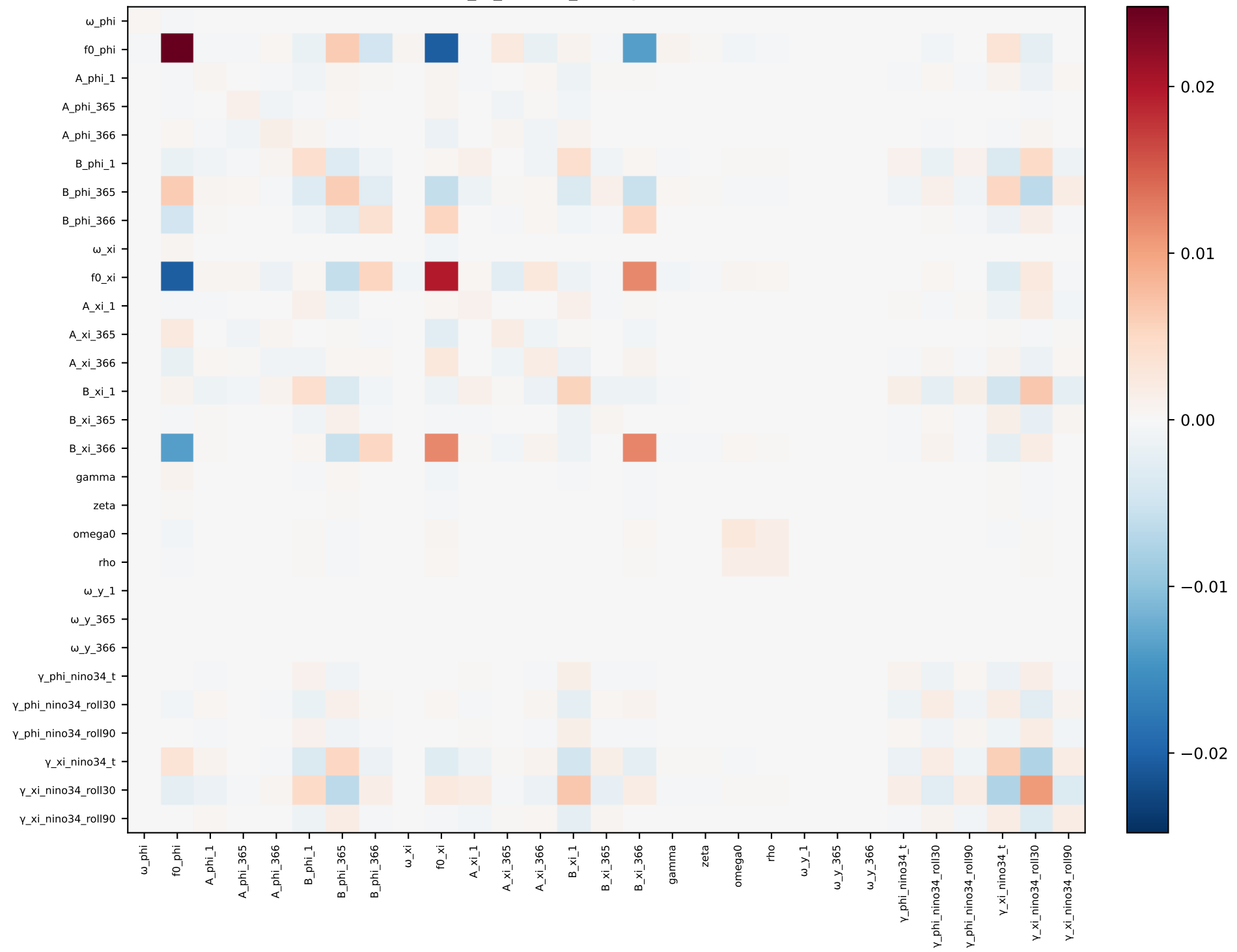
phi_xi_nino34_rolling — filtered states (IS)



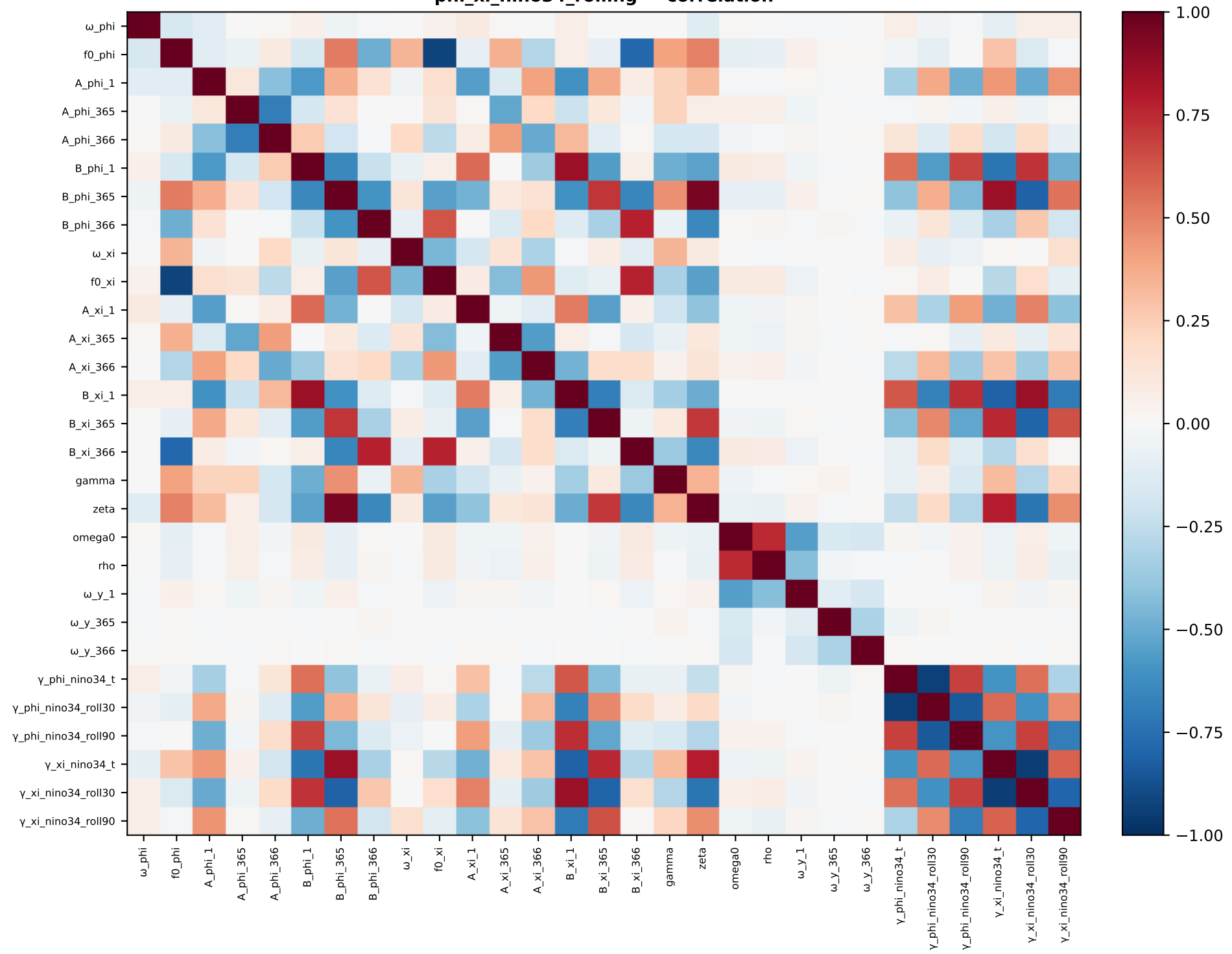
phi_xi_nino34_rolling — filtered states (OOS)



phi_xi_nino34_rolling — covariance

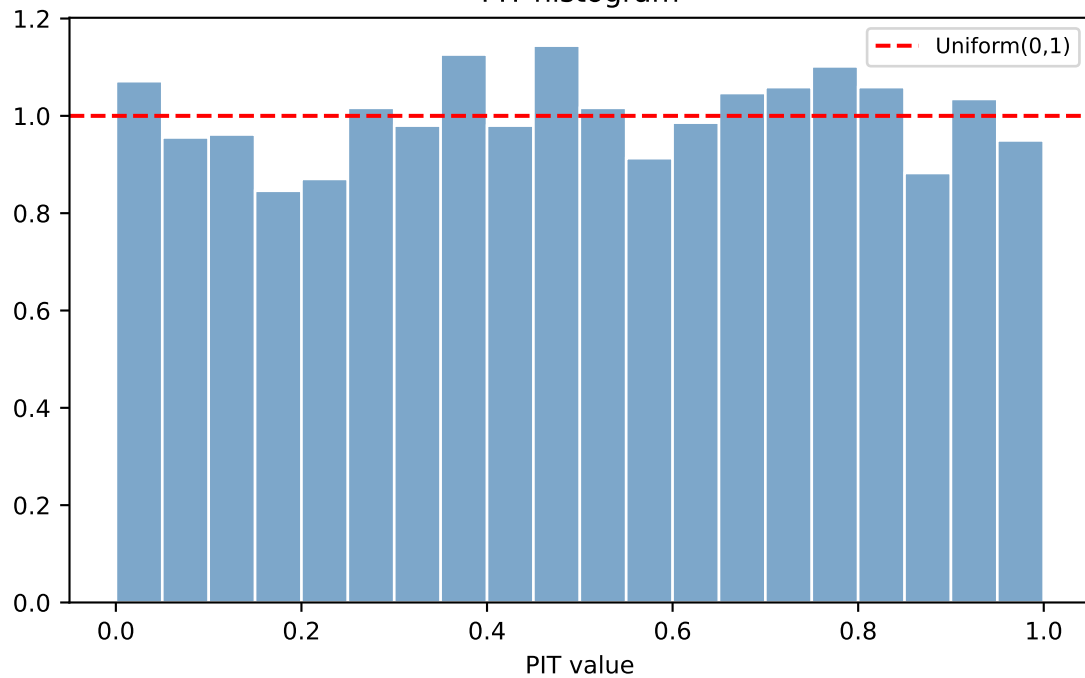


phi_xi_nino34_rolling — correlation

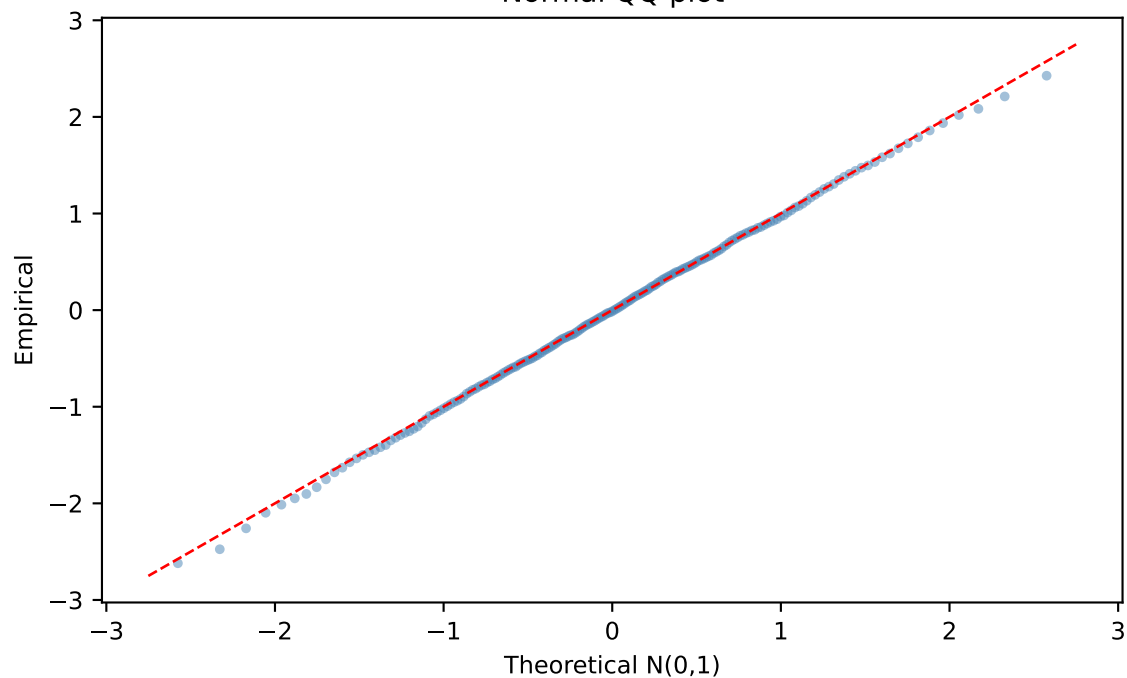


phi_xi_interaction_dewpoint_temperature_t — IS diagnostics

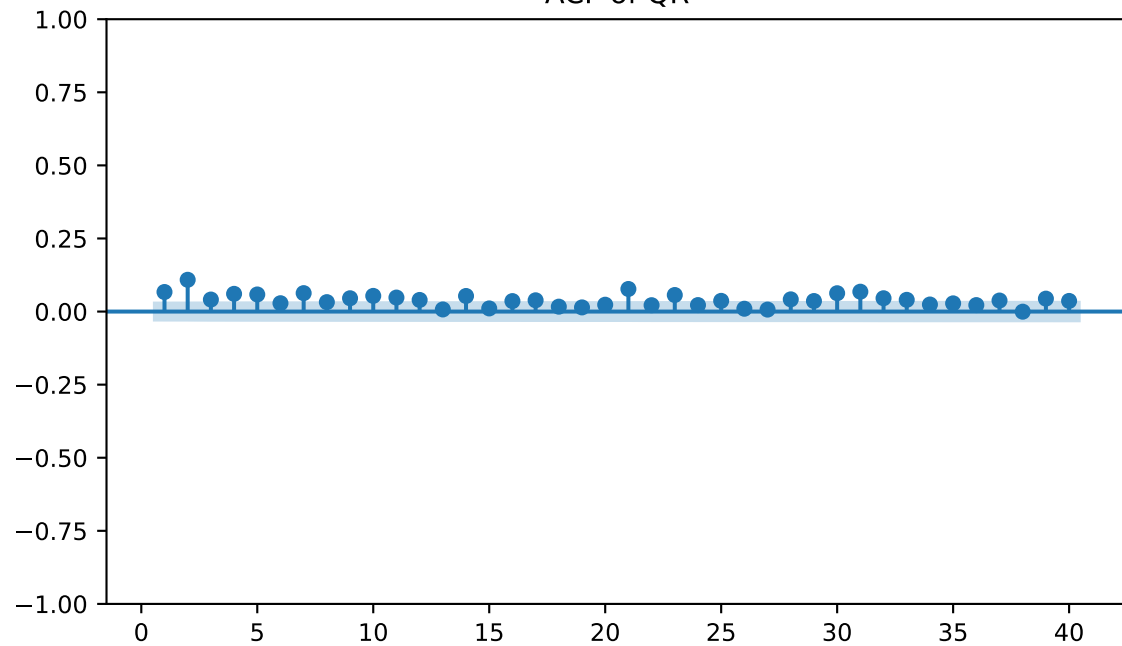
PIT histogram



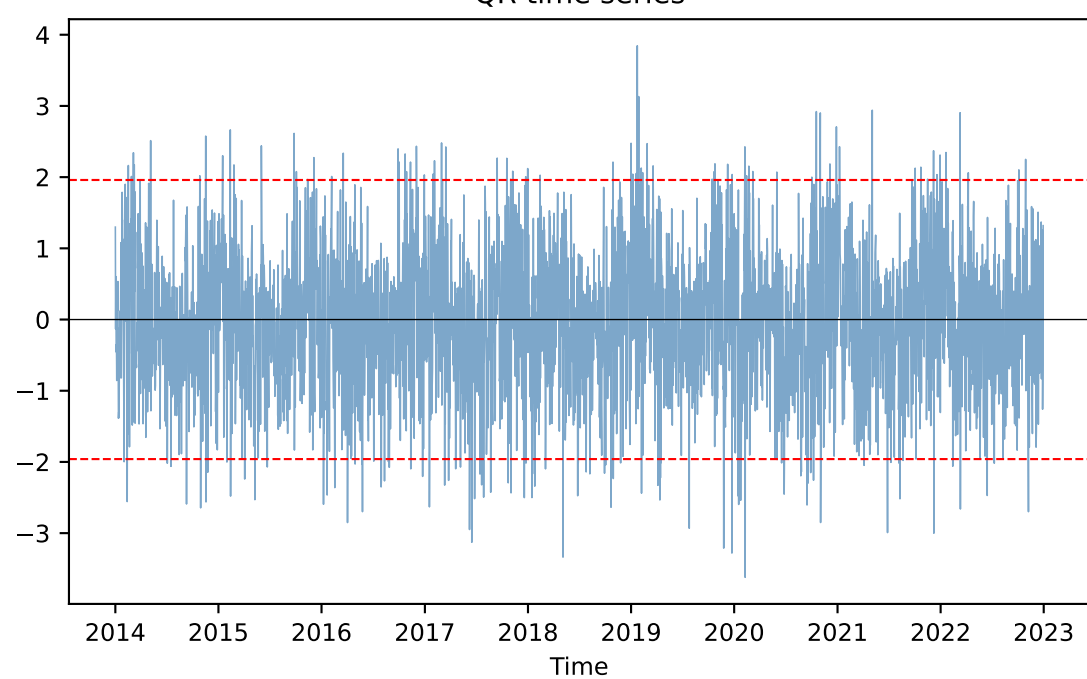
Normal QQ plot



ACF of QR

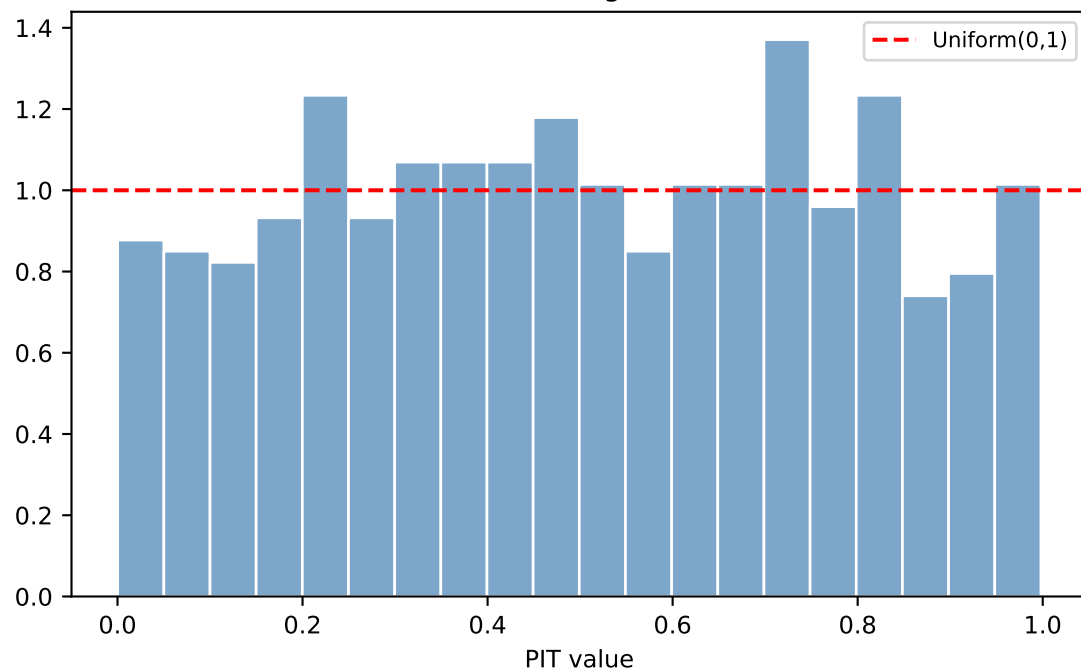


QR time series

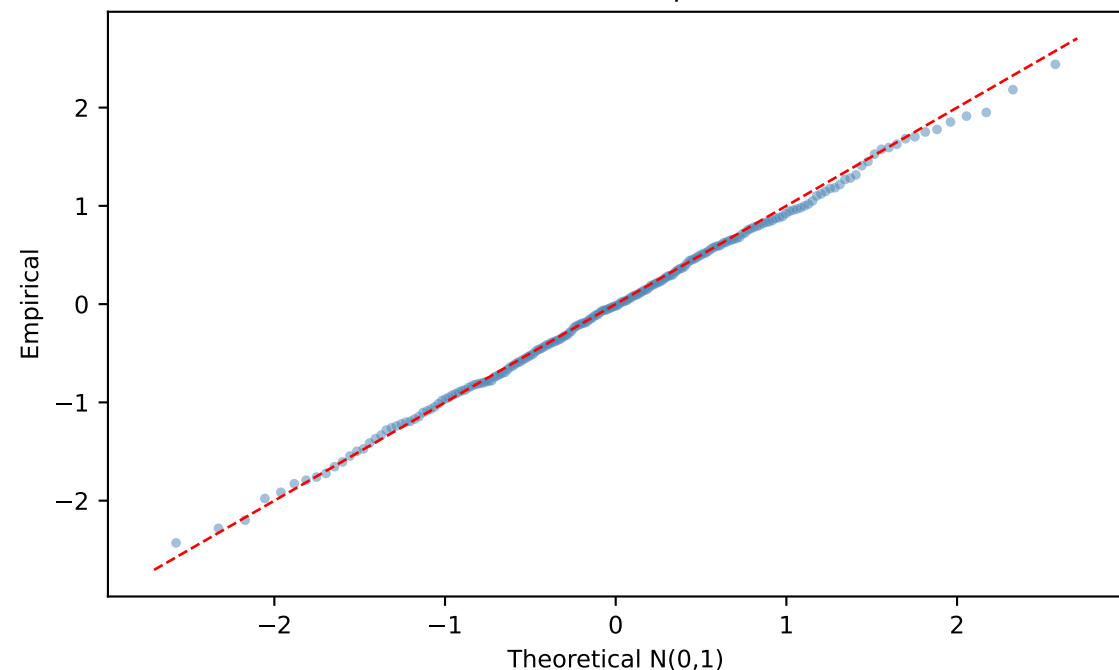


phi_xi_interaction_dewpoint_temperature_t — OOS diagnostics

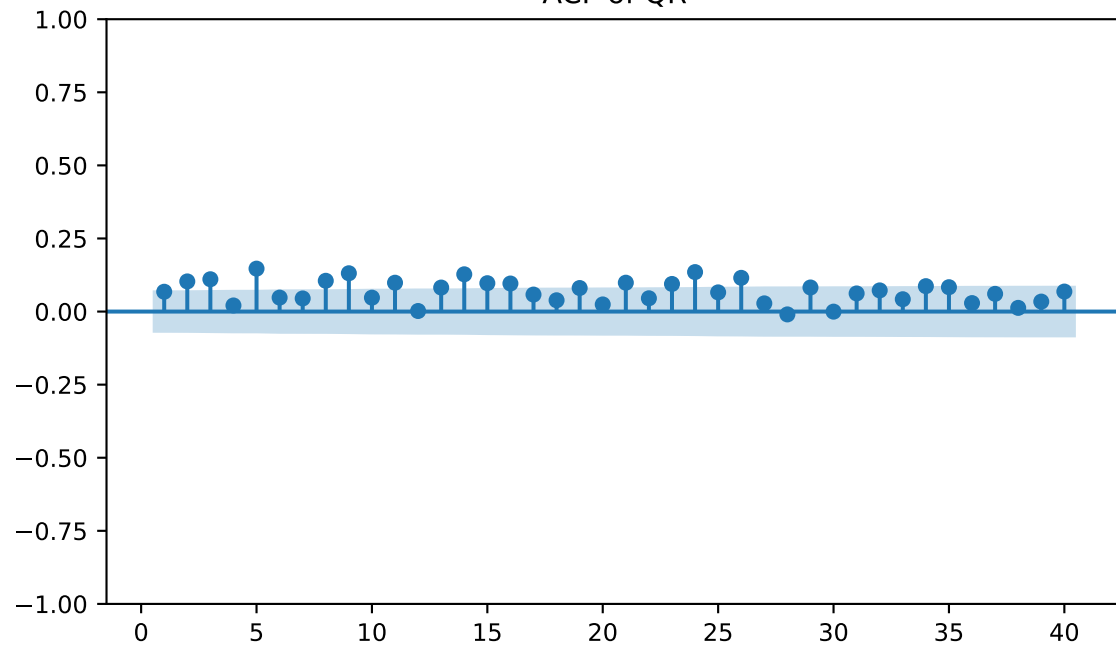
PIT histogram



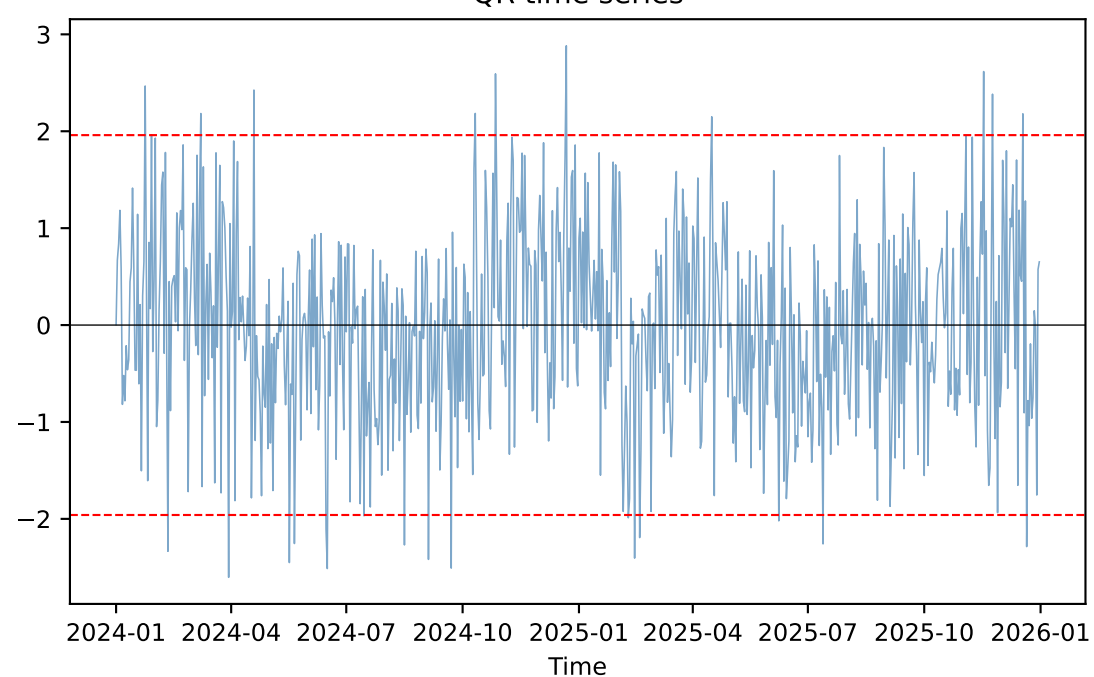
Normal QQ plot



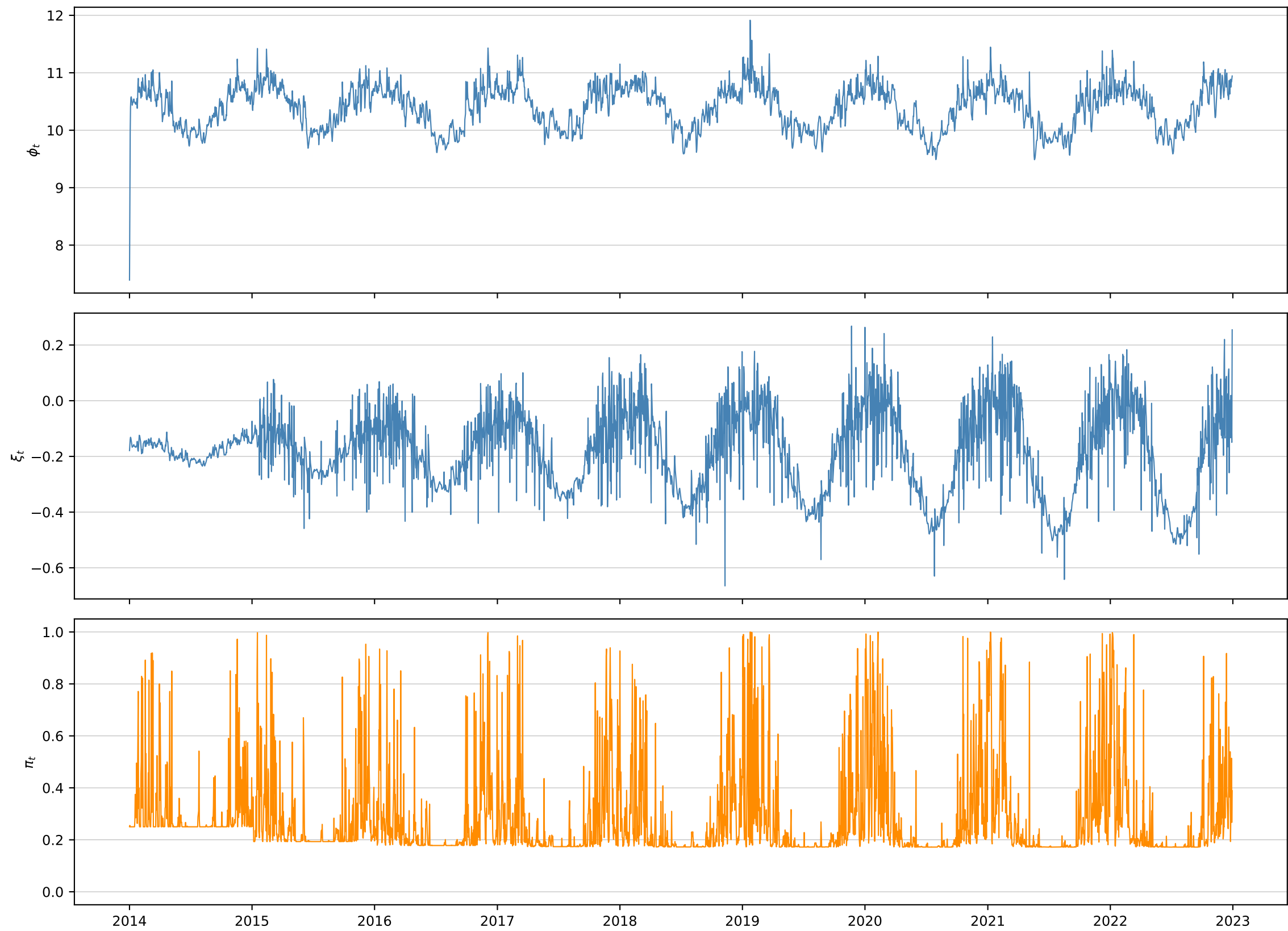
ACF of QR



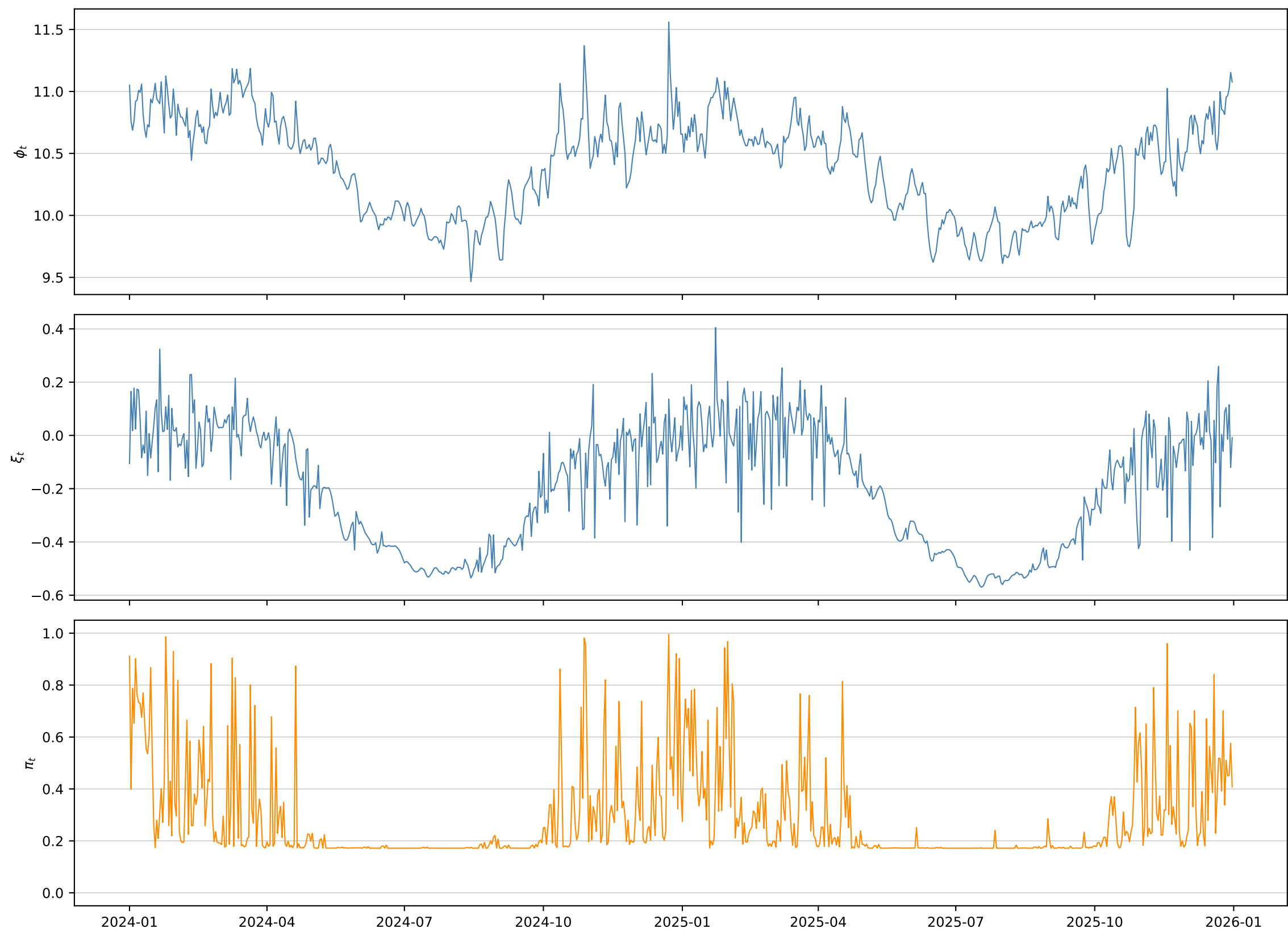
QR time series



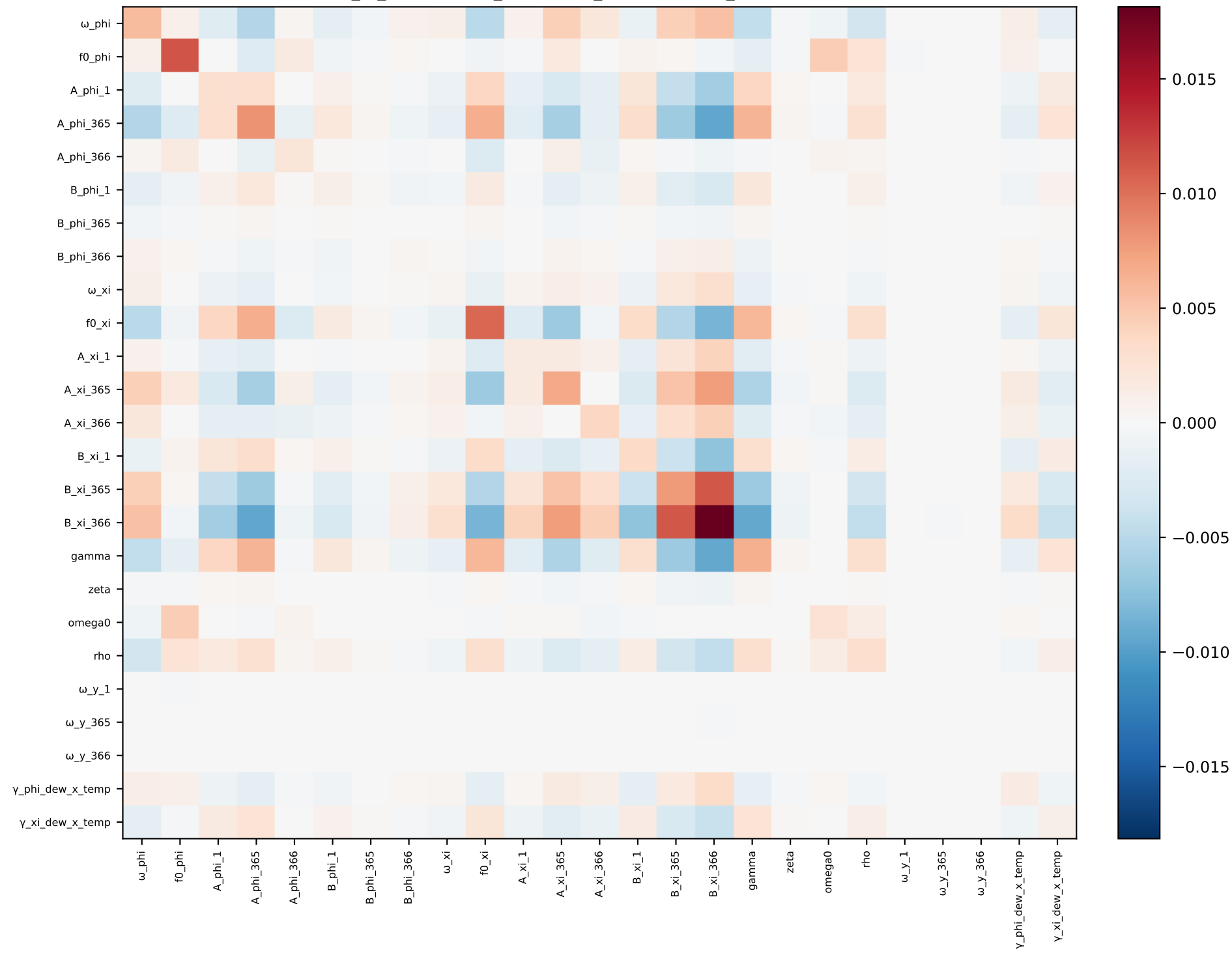
phi_xi_interaction_dewpoint_temperature_t — filtered states (IS)



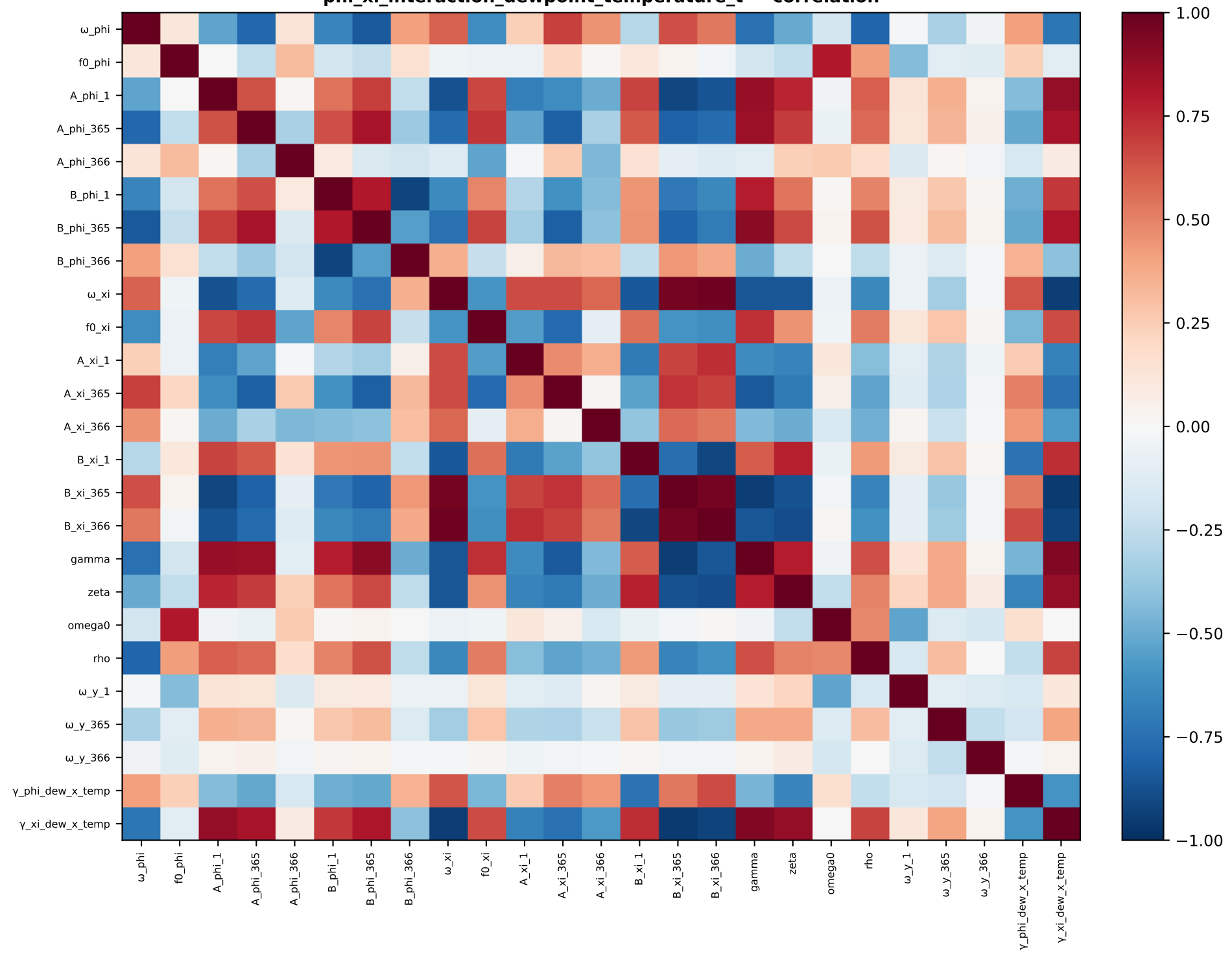
phi_xi_interaction_dewpoint_temperature_t — filtered states (OOS)



phi_xi_interaction_dewpoint_temperature_t — covariance

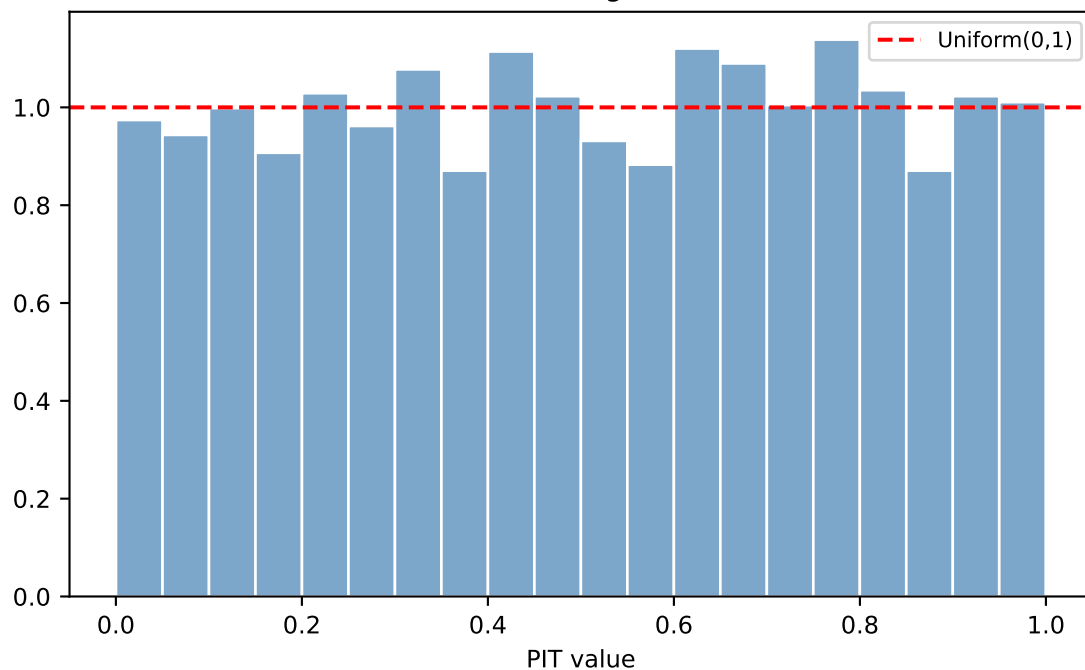


phi_xi_interaction_dewpoint_temperature_t — correlation

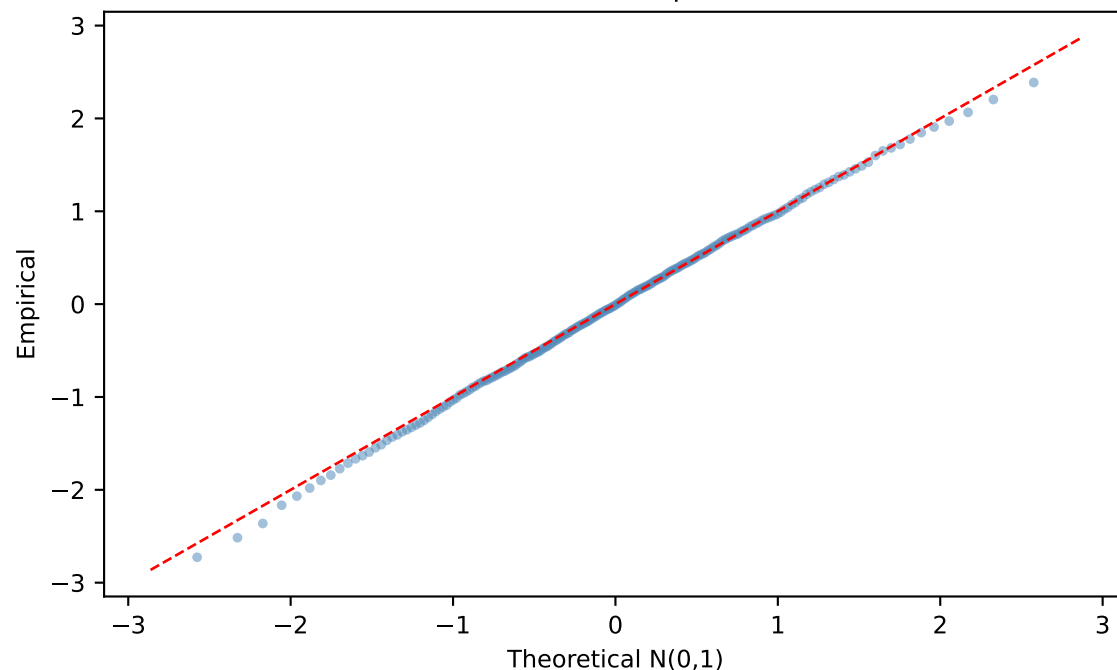


phi_xi_interaction_dewpoint_mjo — IS diagnostics

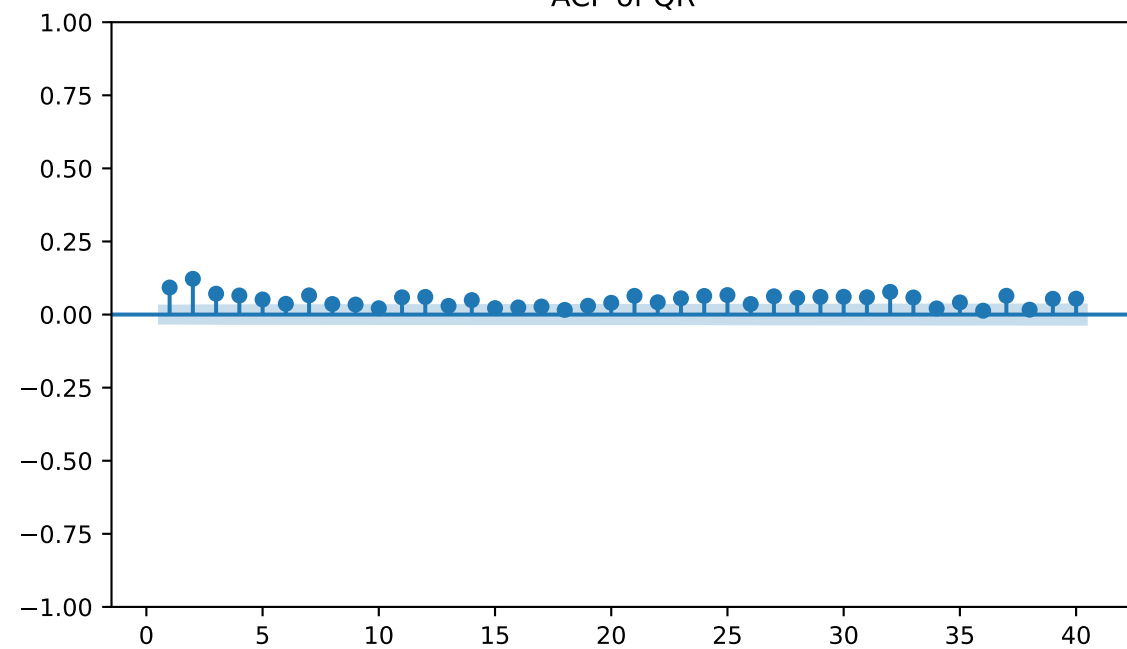
PIT histogram



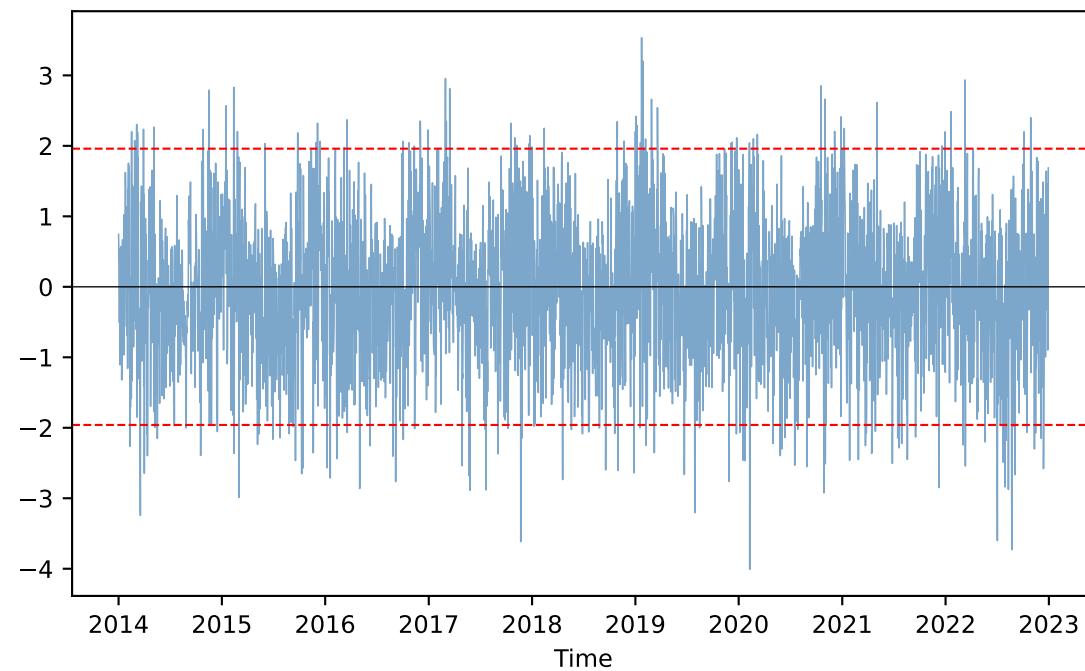
Normal QQ plot



ACF of QR

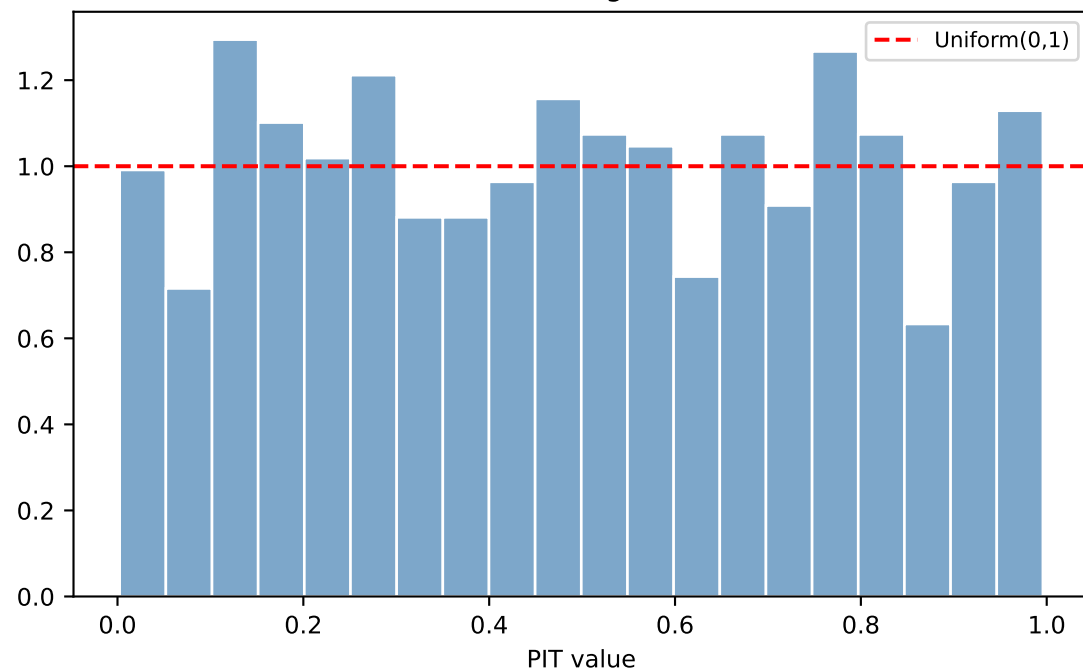


QR time series

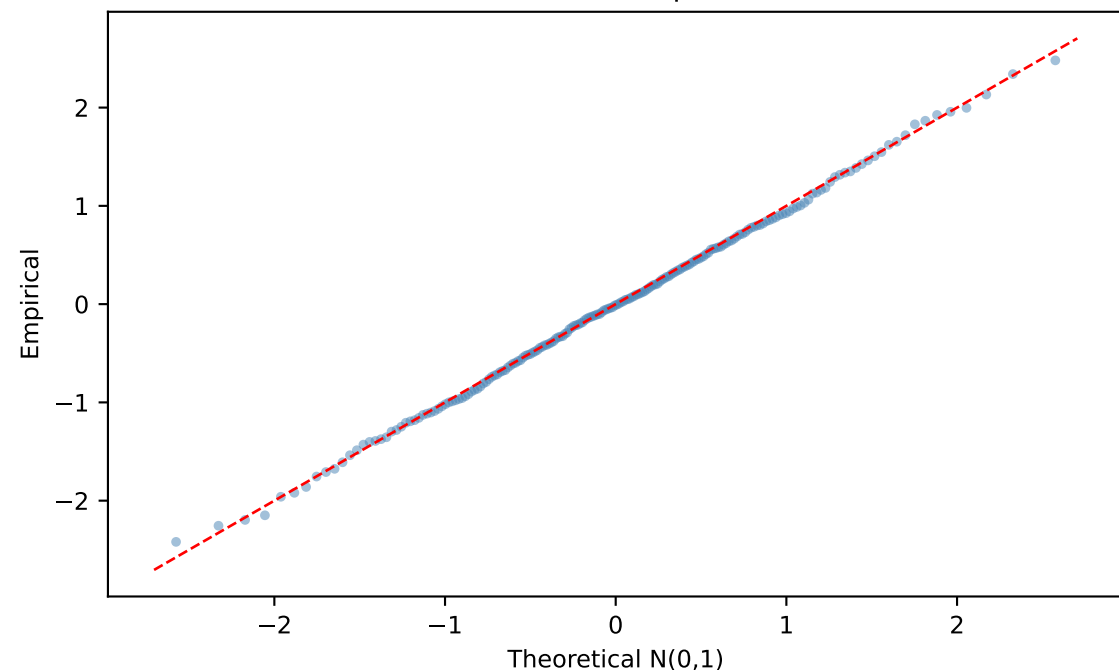


phi_xi_interaction_dewpoint_mjo — OOS diagnostics

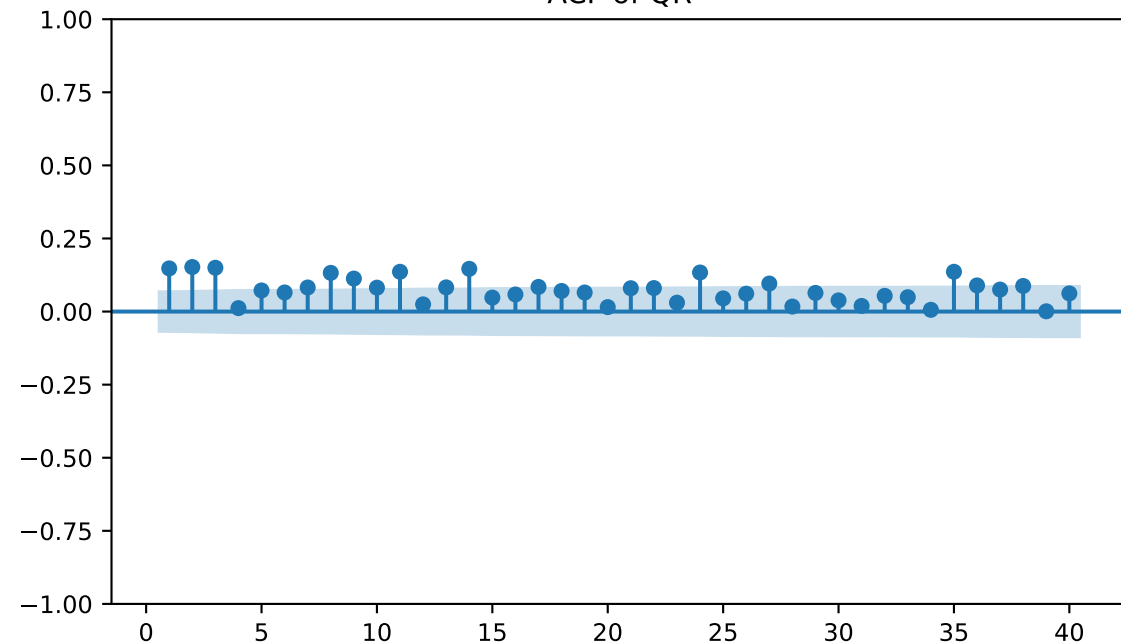
PIT histogram



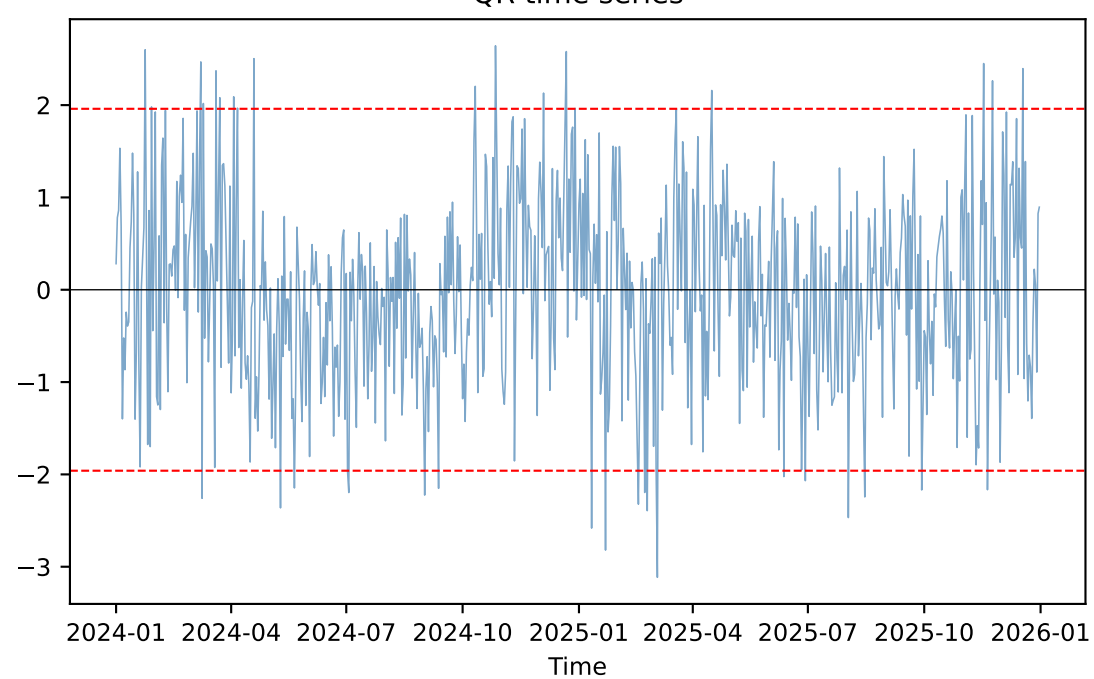
Normal QQ plot



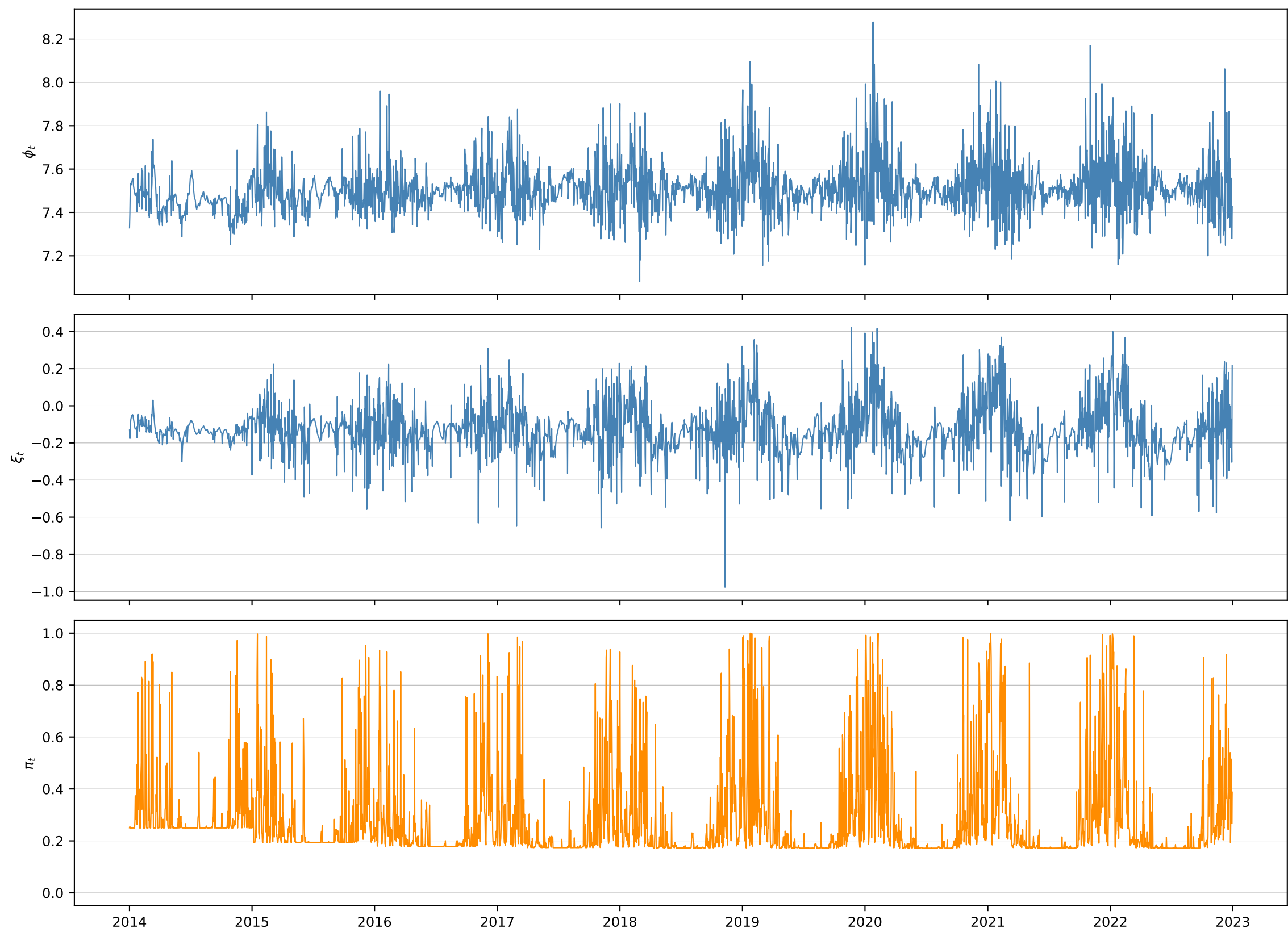
ACF of QR



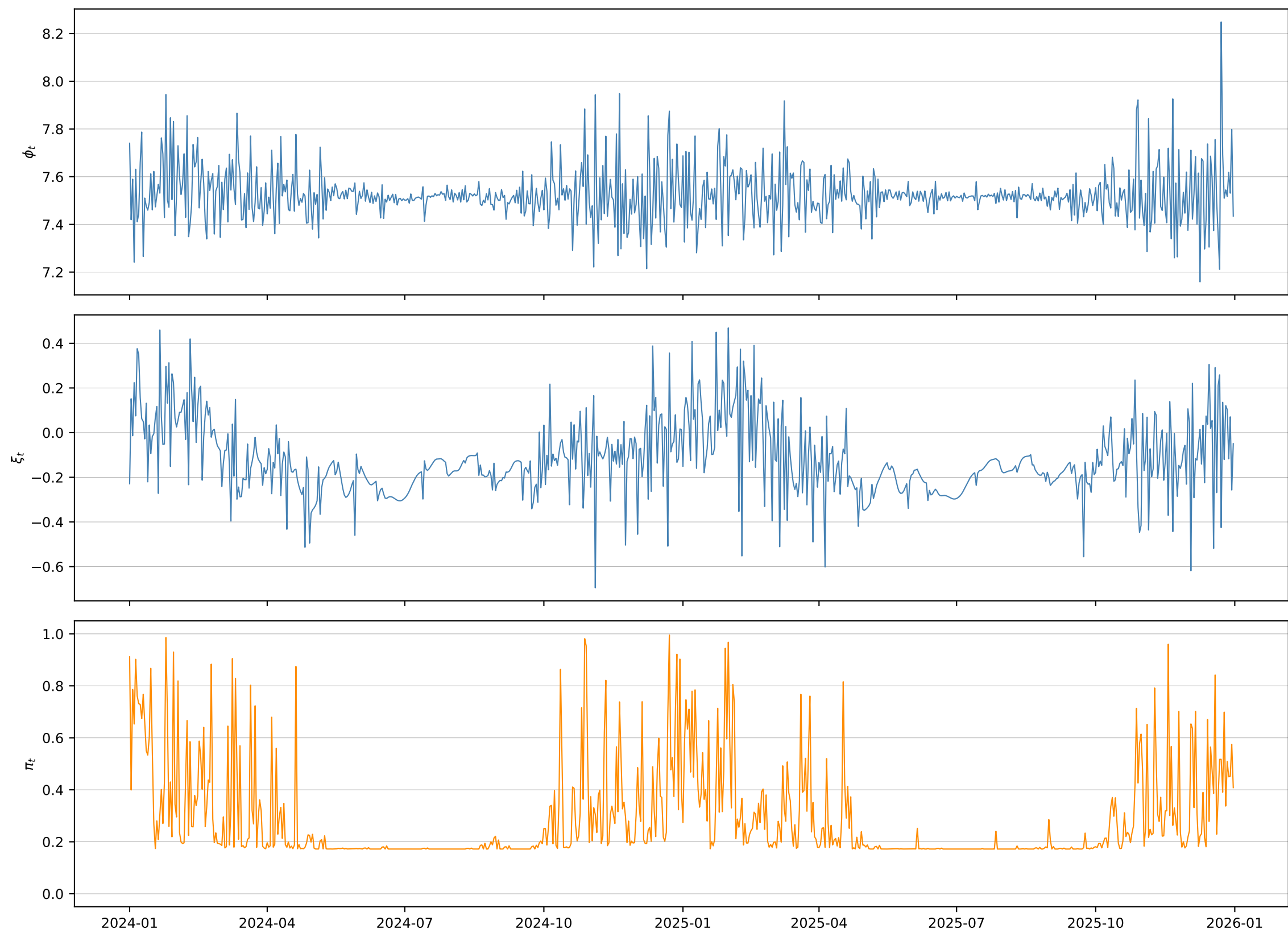
QR time series



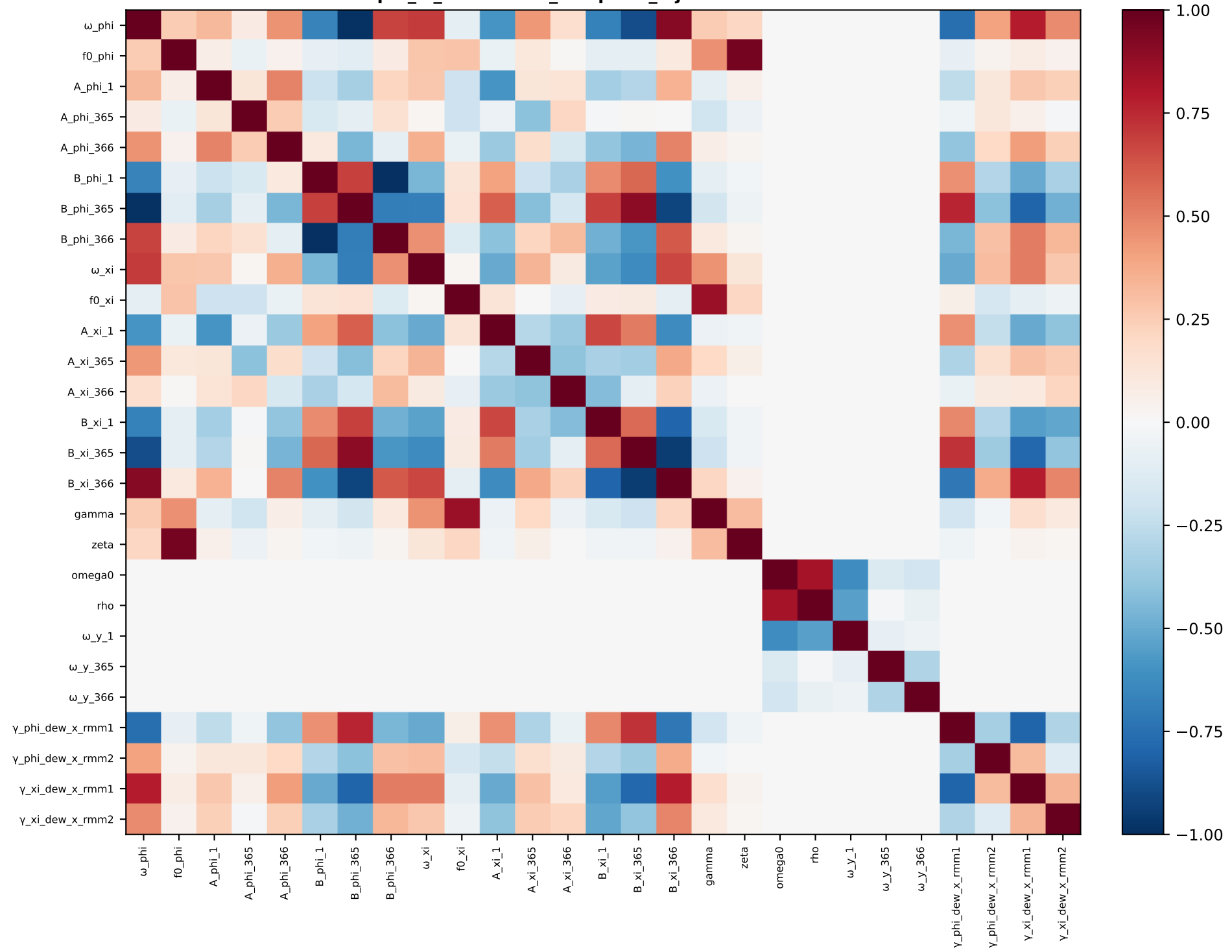
phi_xi_interaction_dewpoint_mjo — filtered states (IS)



phi_xi_interaction_dewpoint_mjo — filtered states (OOS)

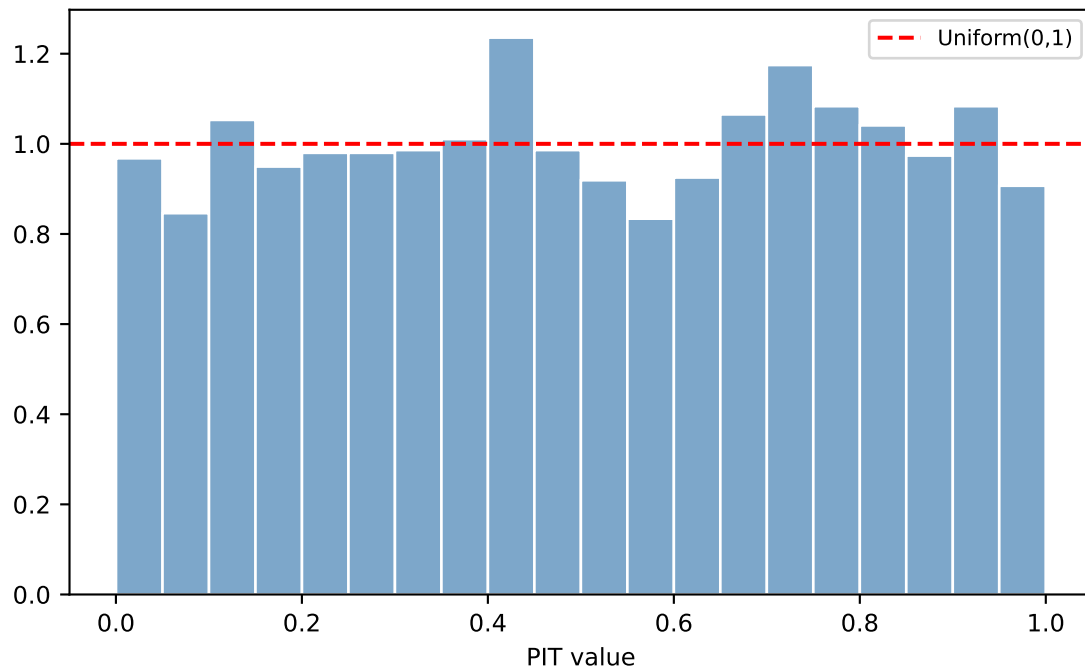


phi_xi_interaction_dewpoint_mjo — correlation

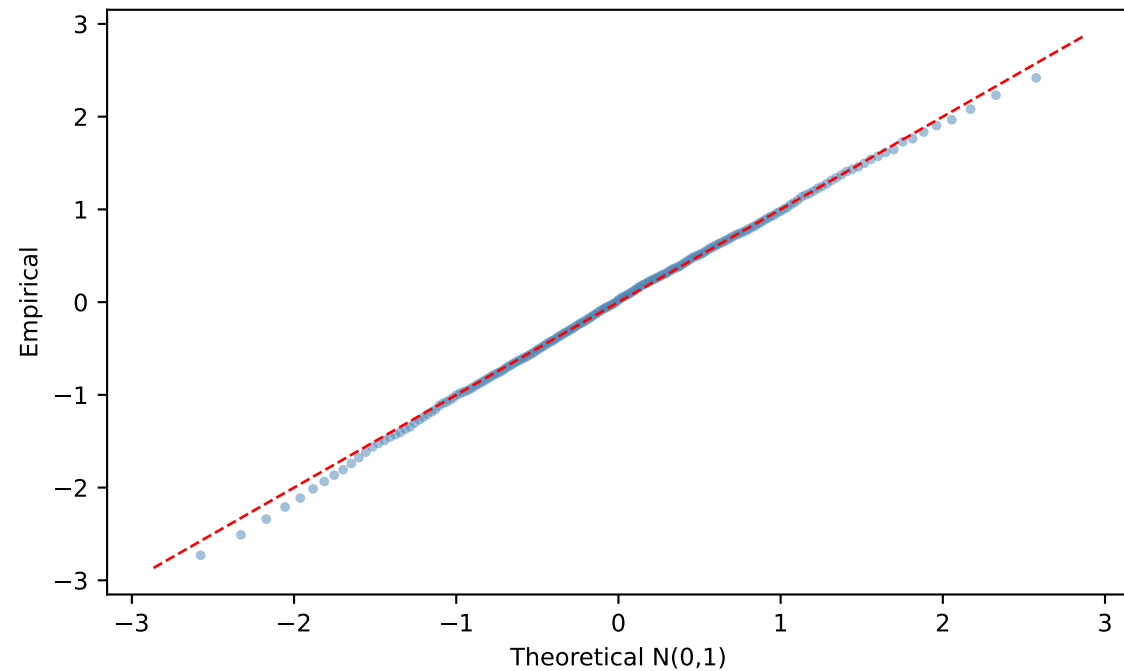


phi_xi_interaction_dewpoint_nino34 — IS diagnostics

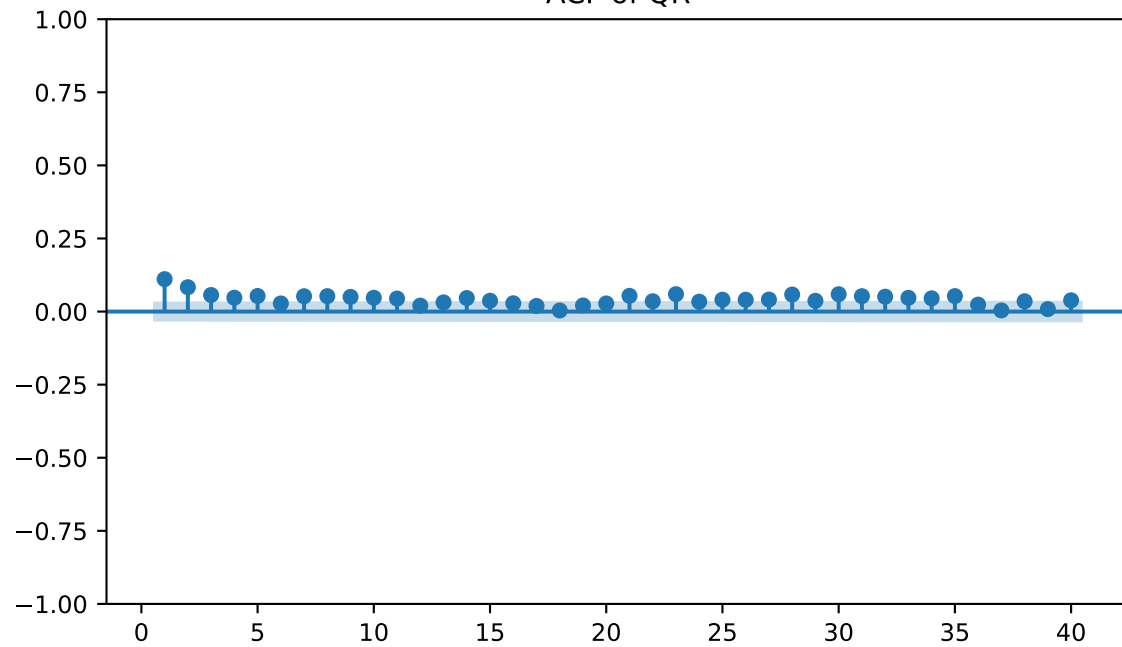
PIT histogram



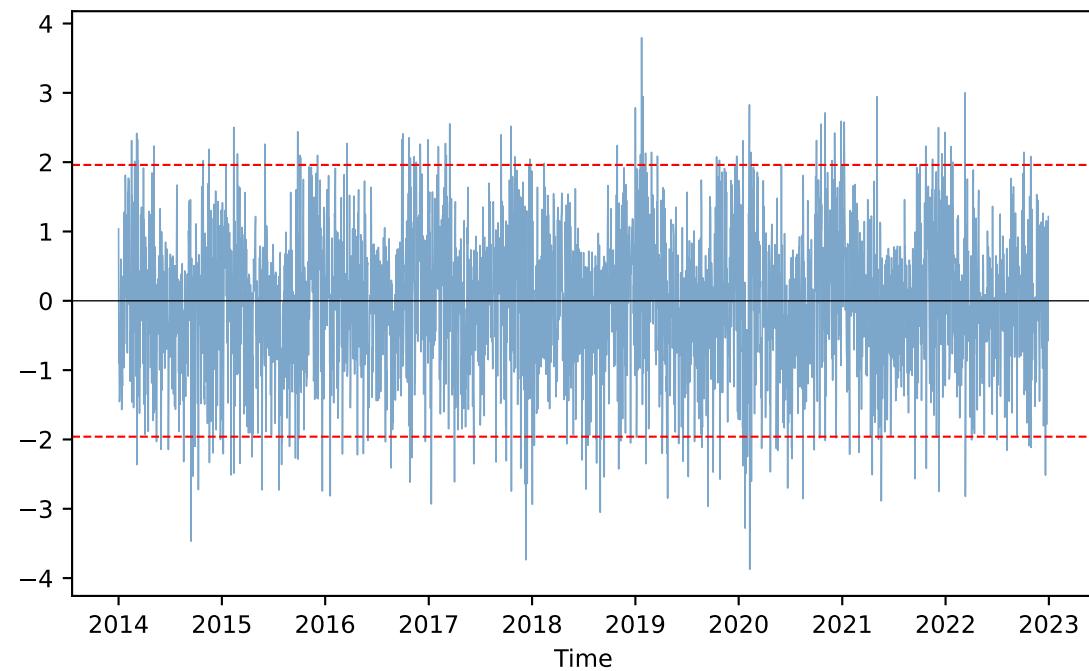
Normal QQ plot



ACF of QR

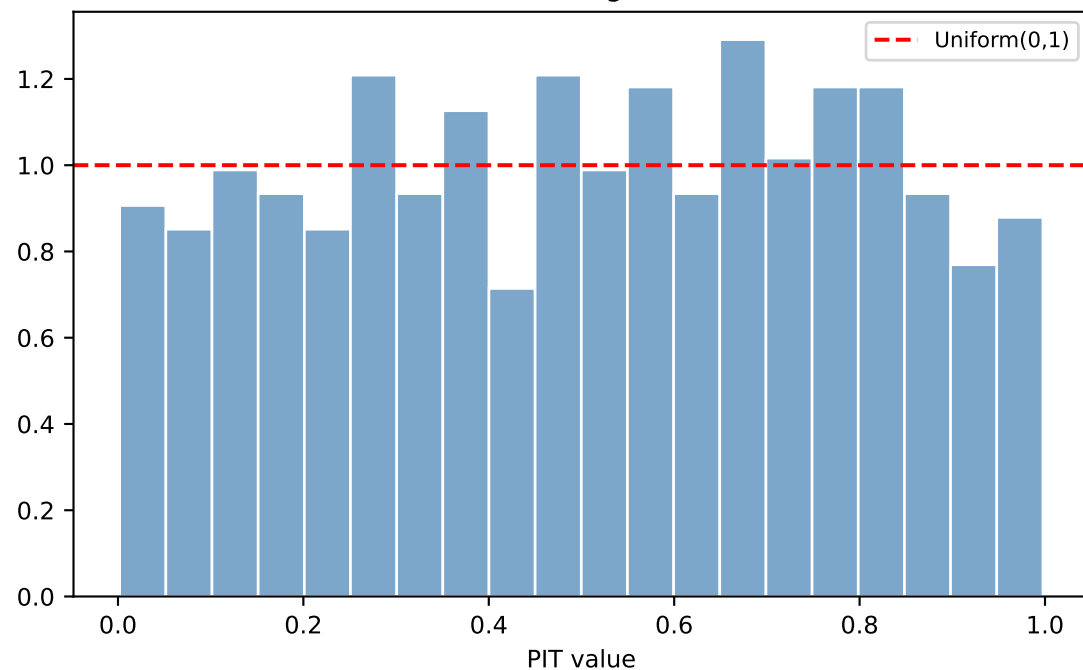


QR time series

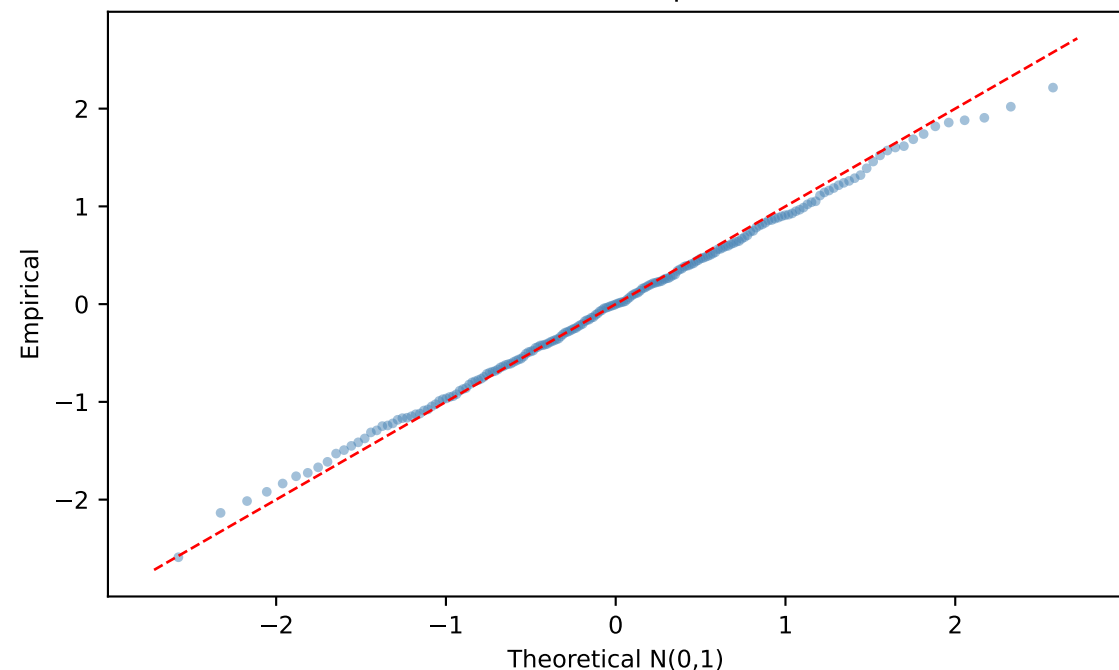


phi_xi_interaction_dewpoint_nino34 — OOS diagnostics

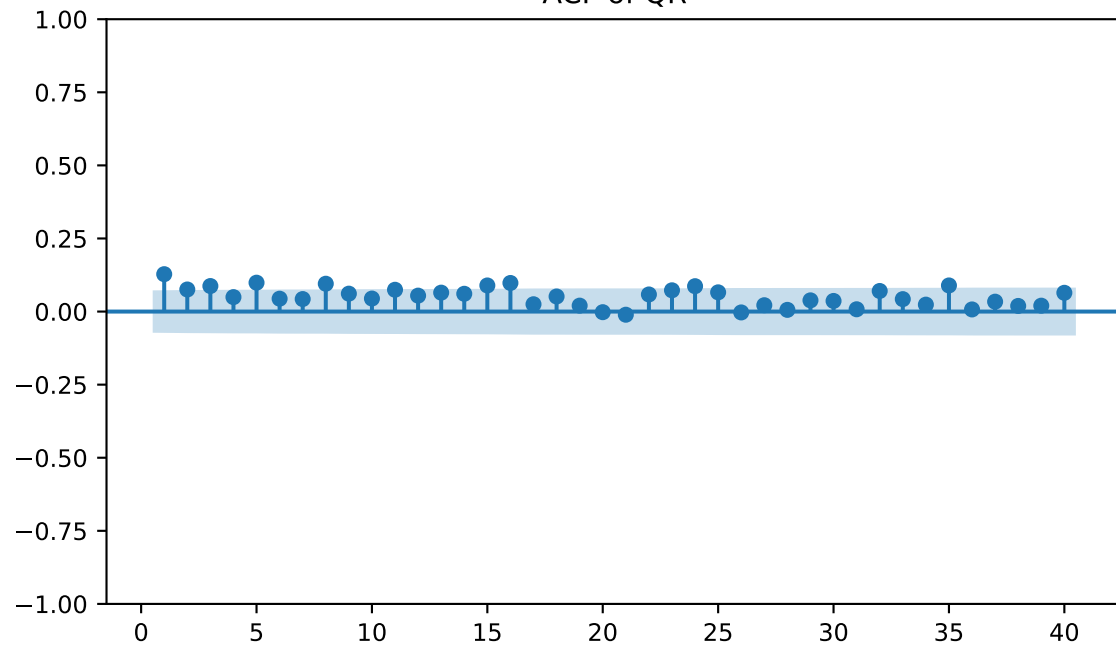
PIT histogram



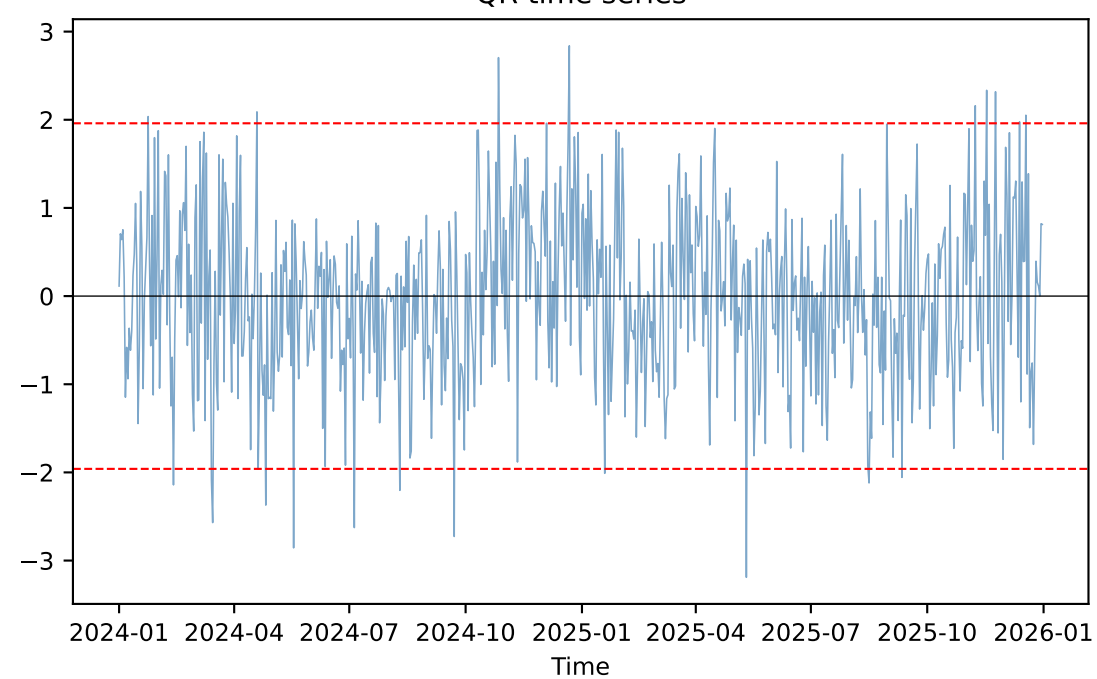
Normal QQ plot



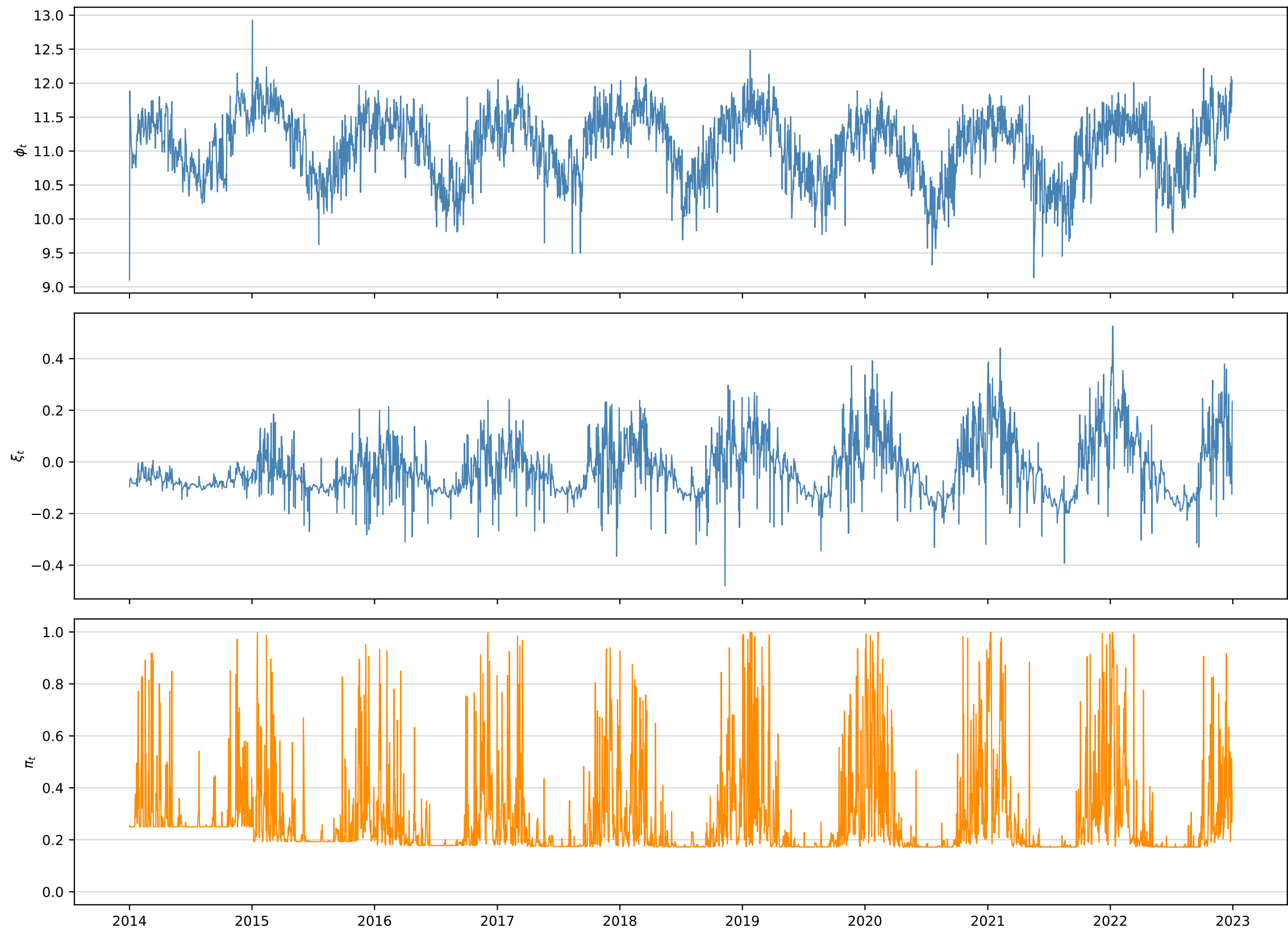
ACF of QR



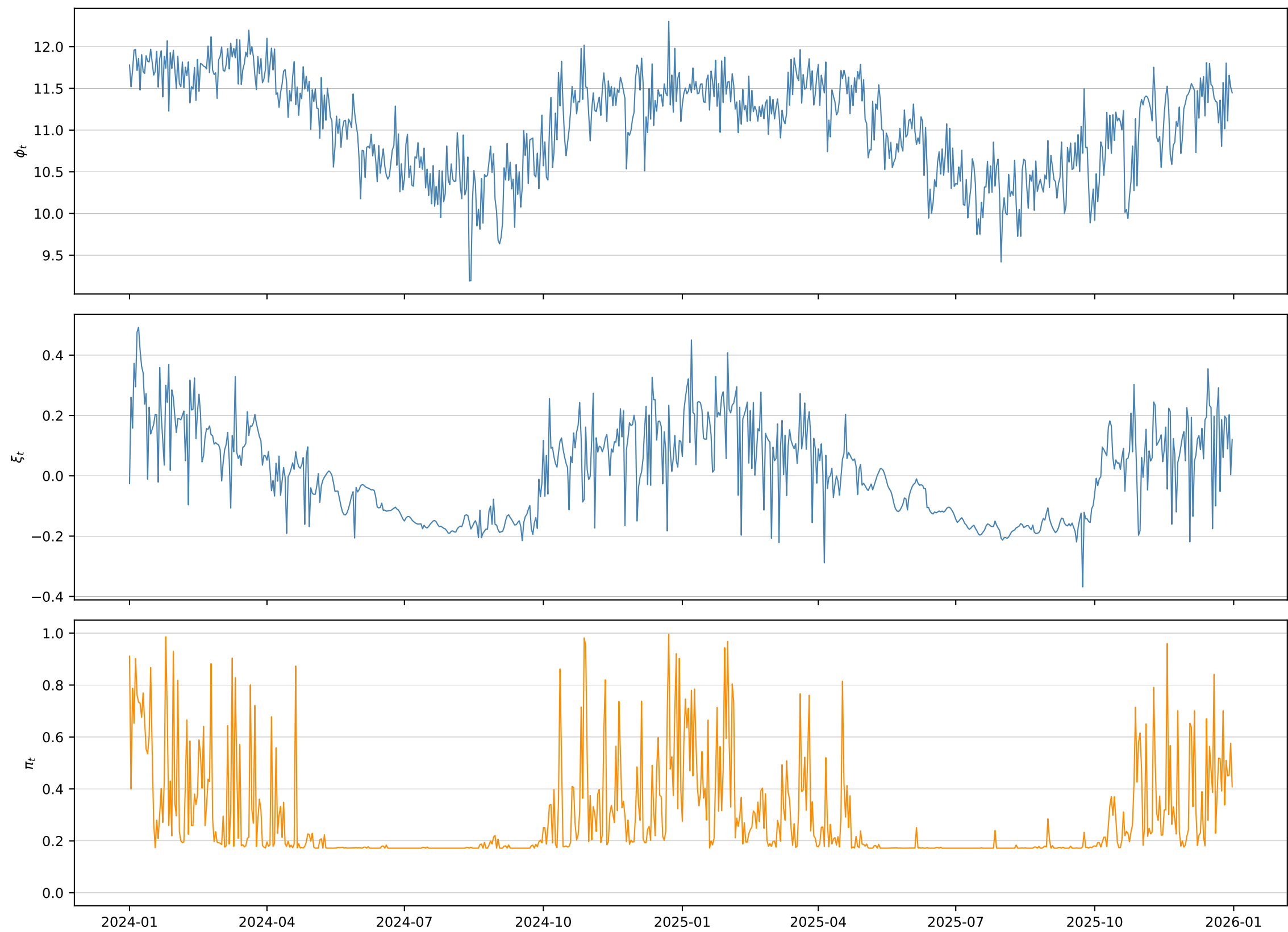
QR time series



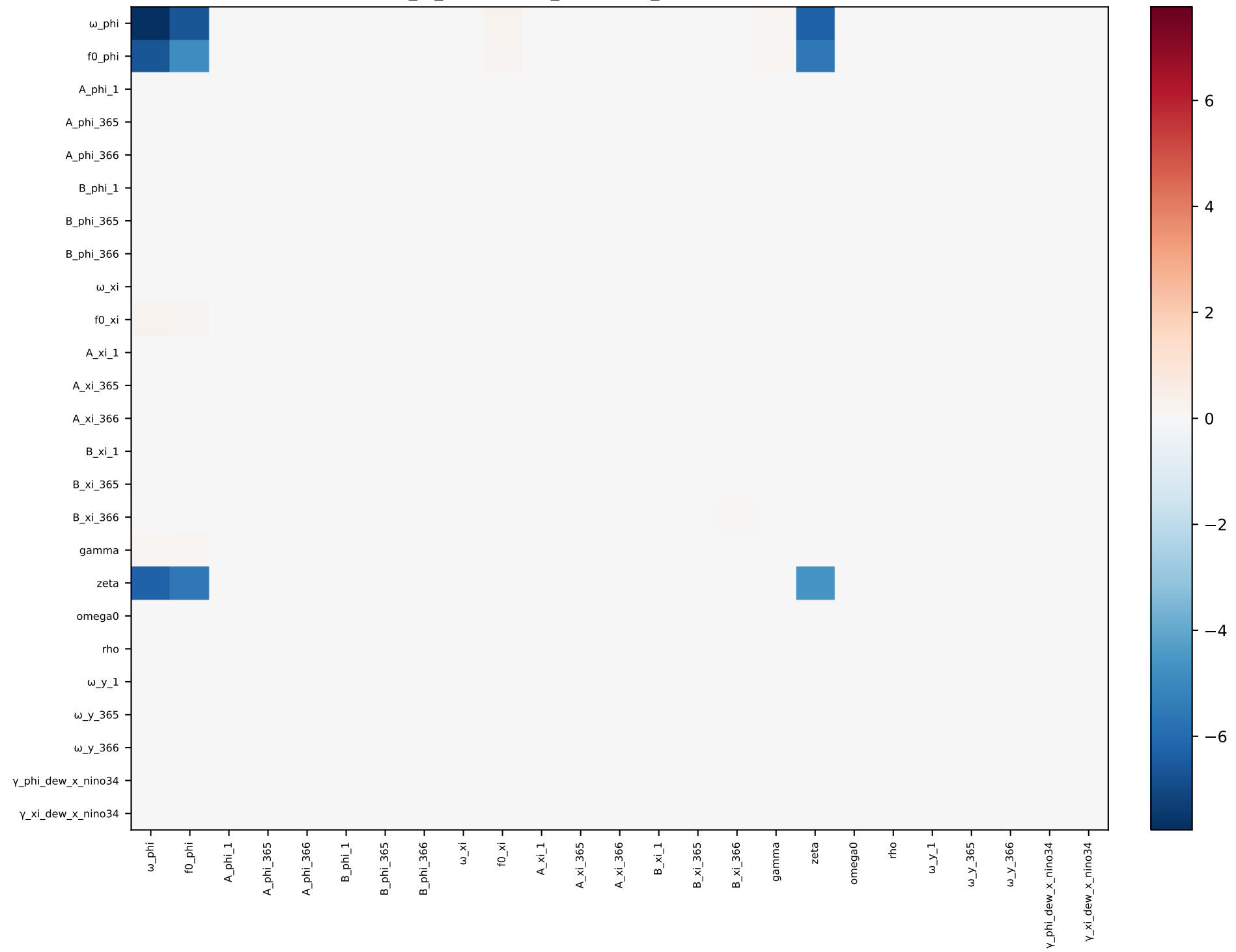
phi_xi_interaction_dewpoint_nino34 — filtered states (IS)



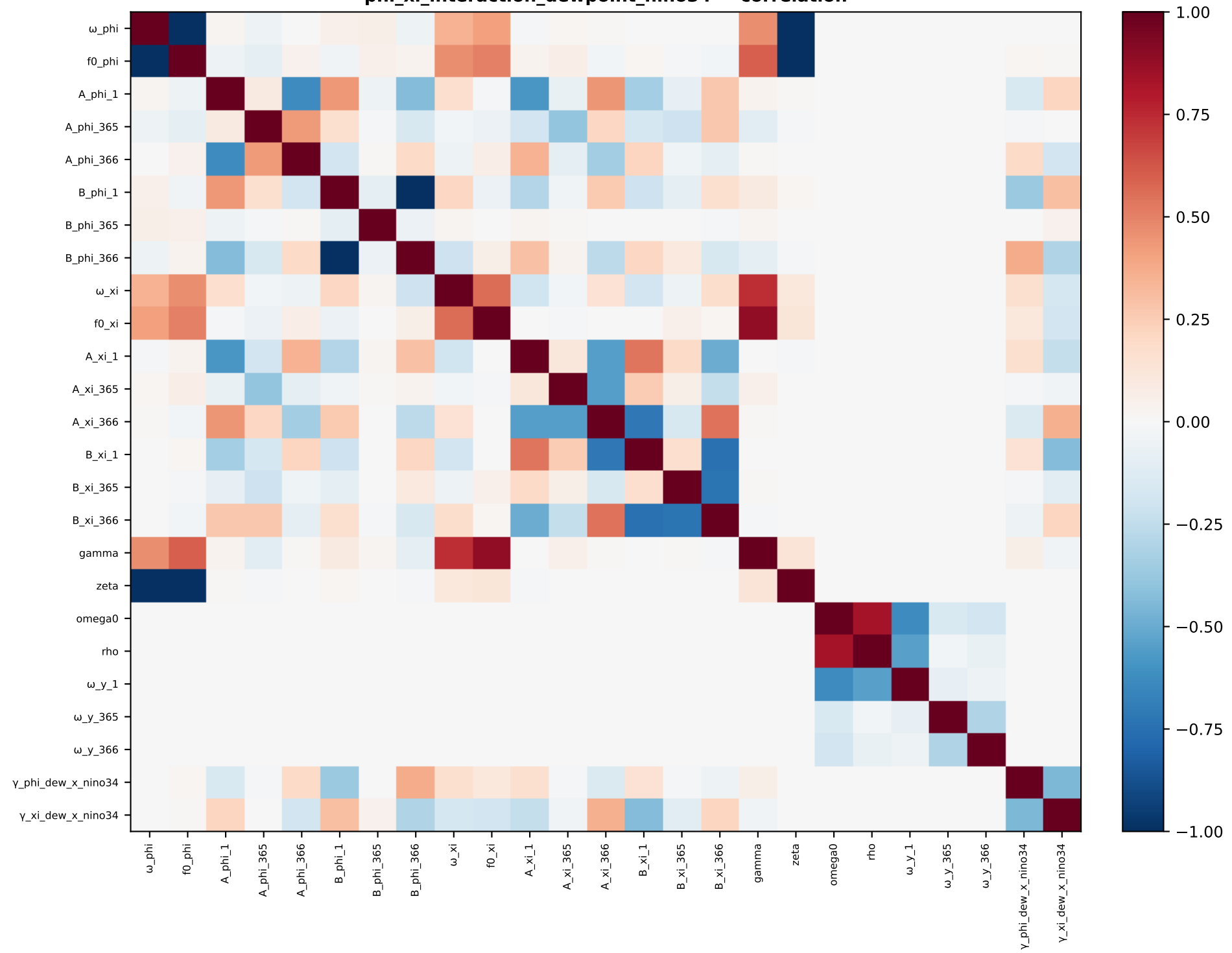
phi_xi_interaction_dewpoint_nino34 — filtered states (OOS)



phi_xi_interaction_dewpoint_nino34 — covariance

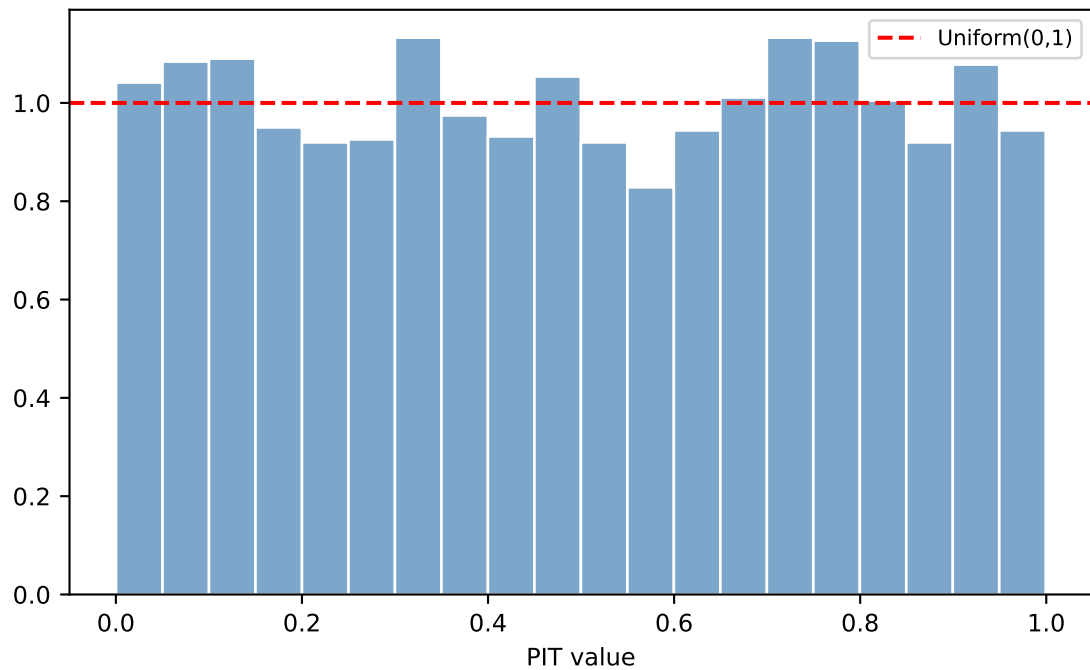


phi_xi_interaction_dewpoint_nino34 — correlation

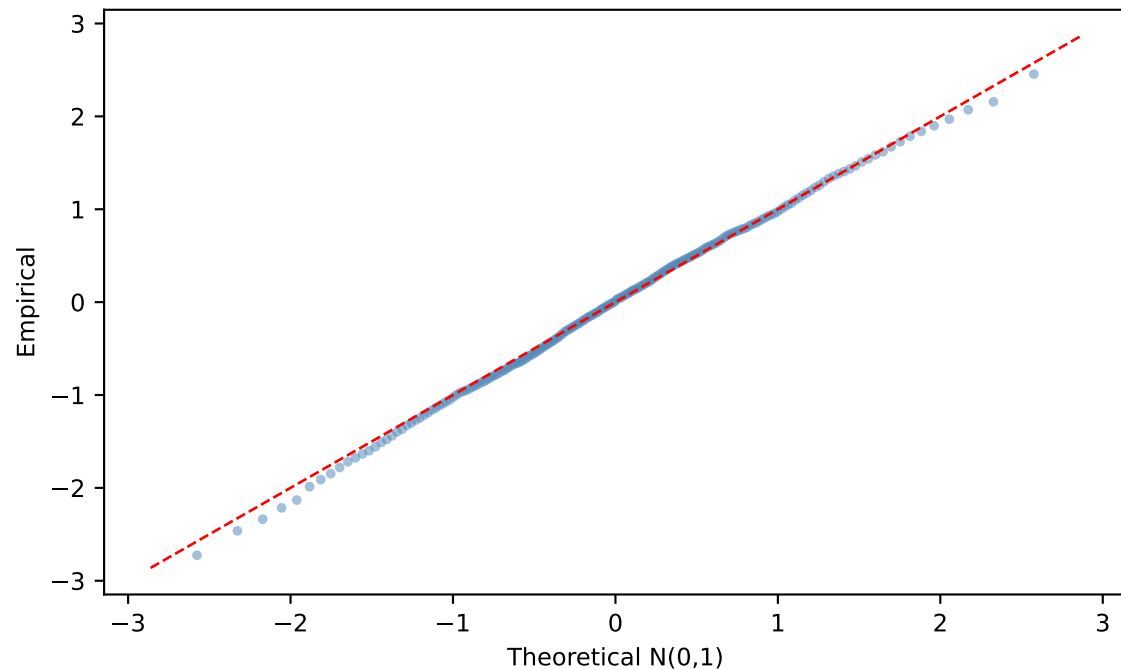


phi_xi_interaction_temperature_nino34 — IS diagnostics

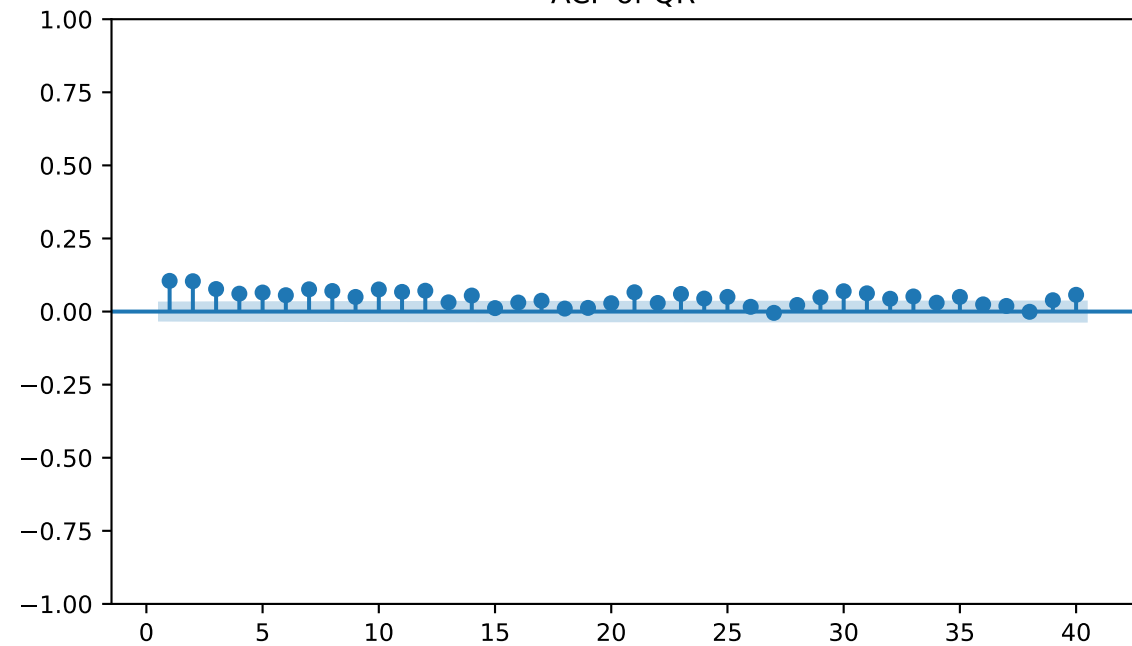
PIT histogram



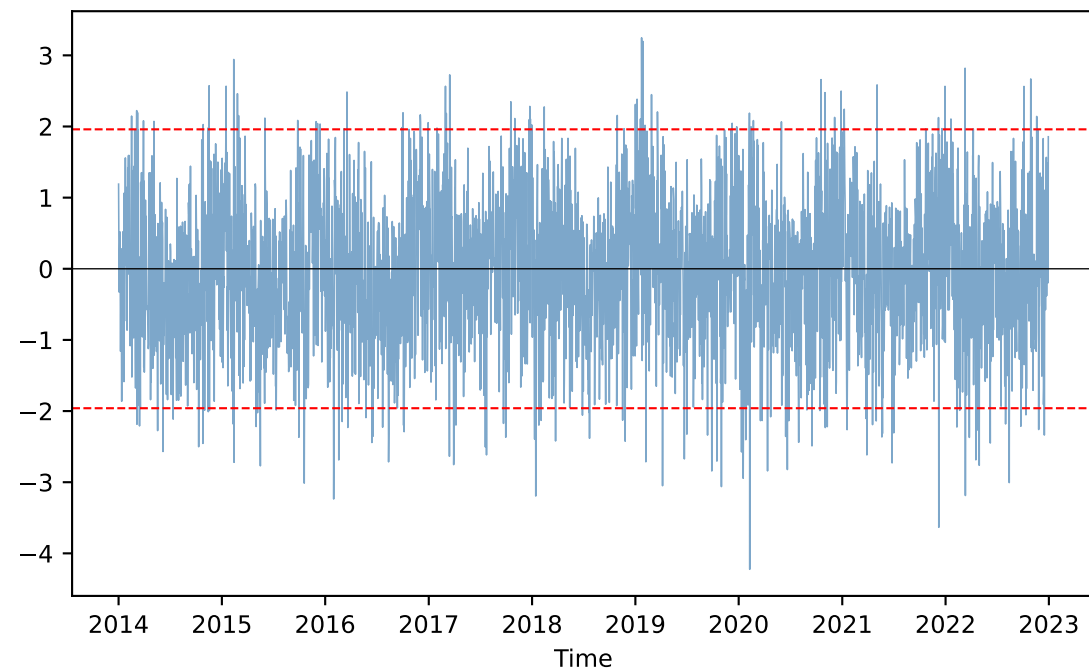
Normal QQ plot



ACF of QR

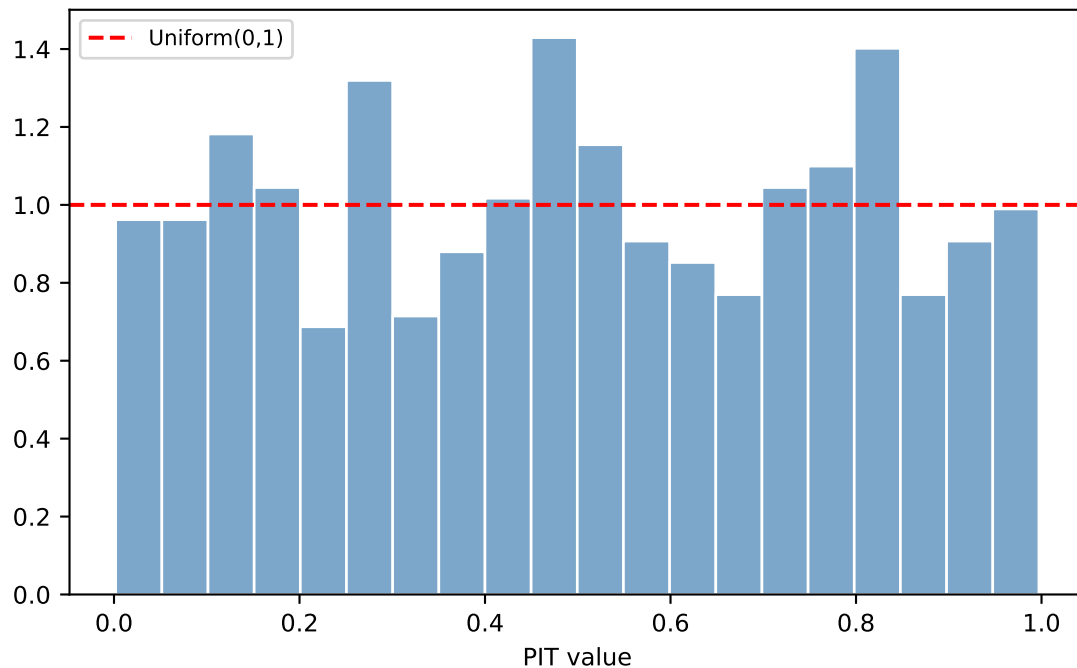


QR time series

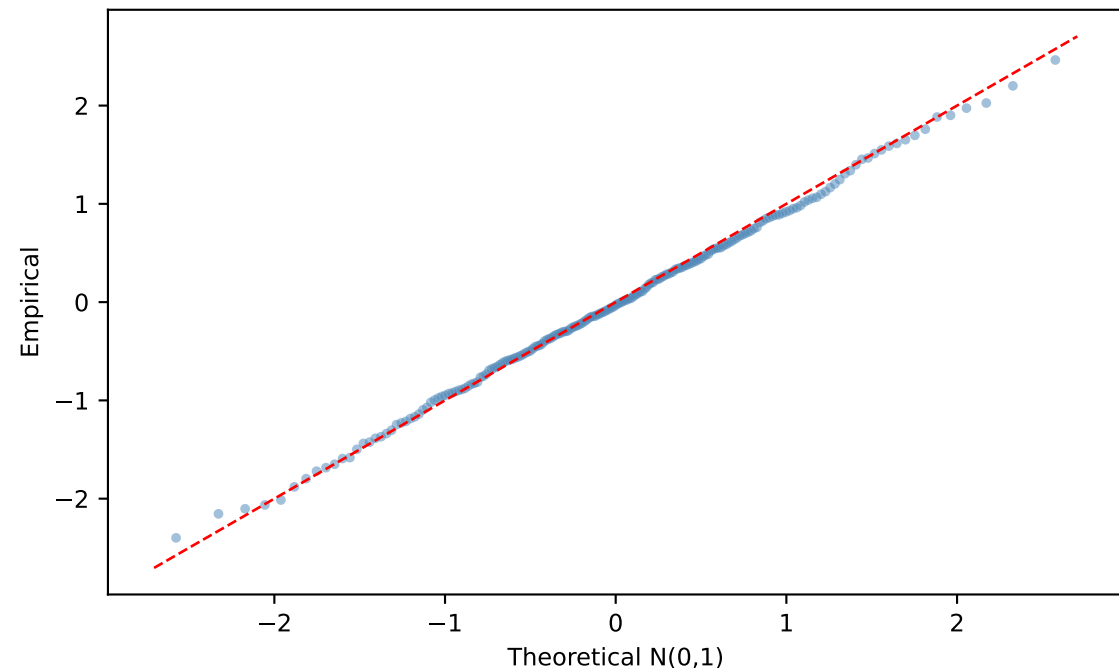


phi_xi_interaction_temperature_nino34 — OOS diagnostics

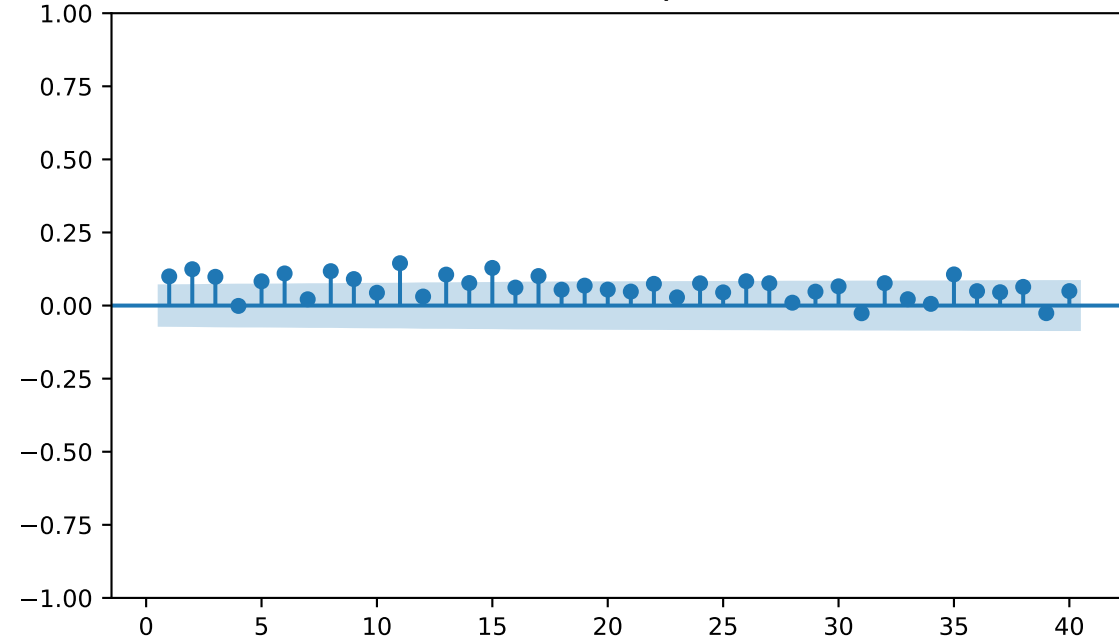
PIT histogram



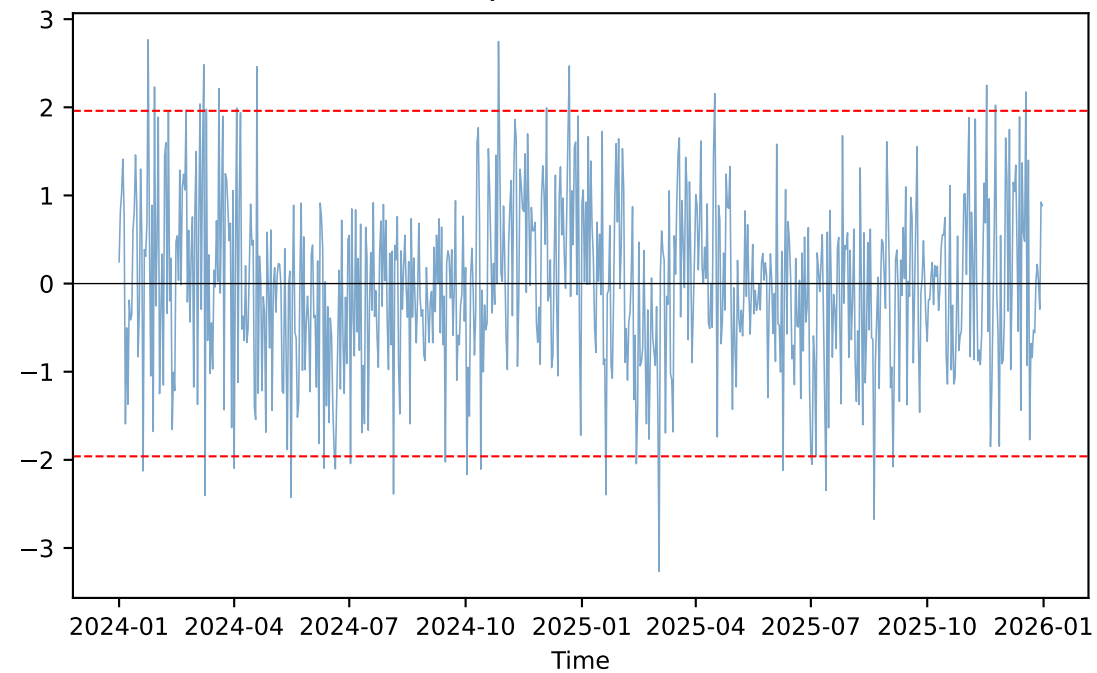
Normal QQ plot



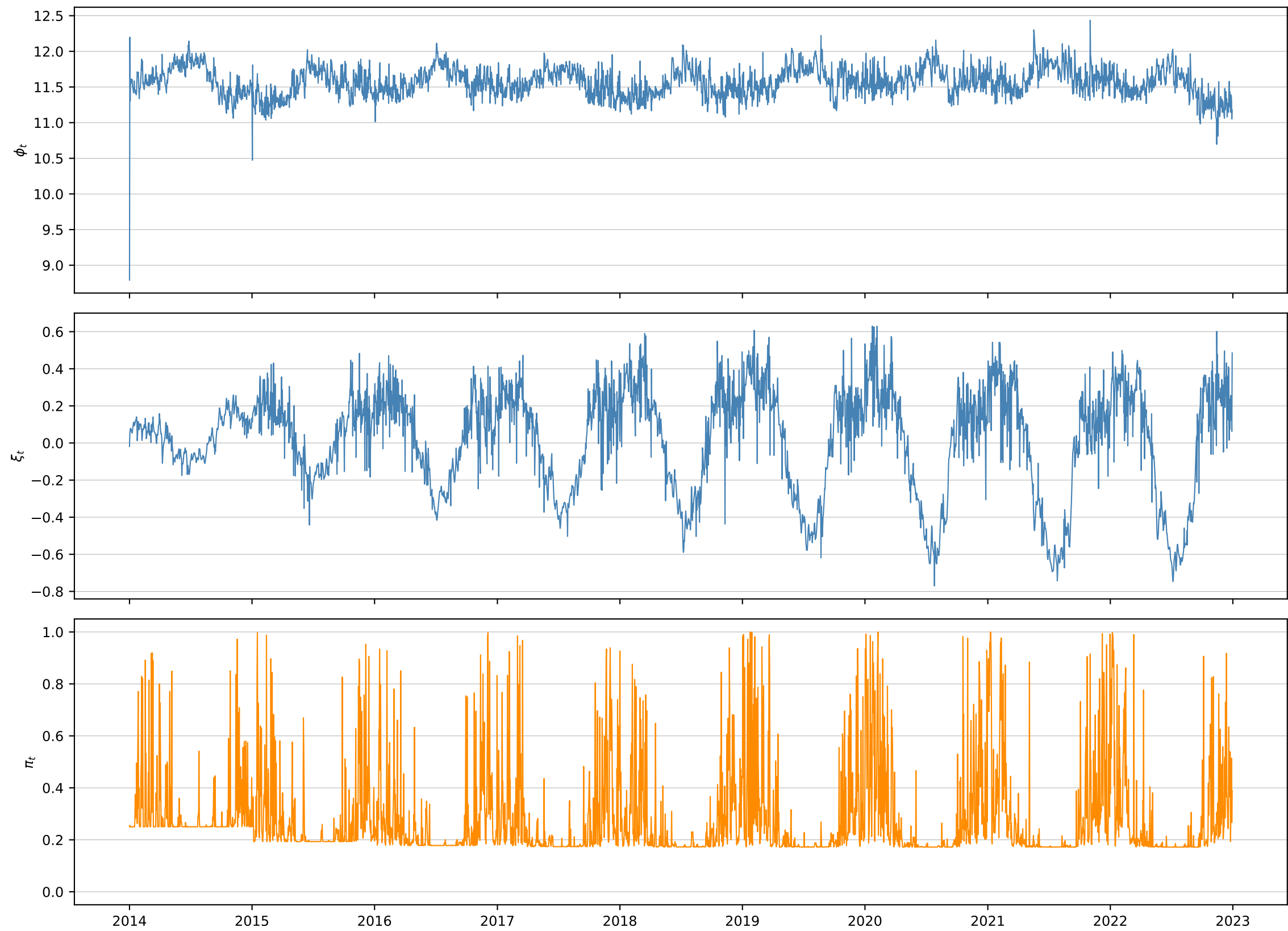
ACF of QR



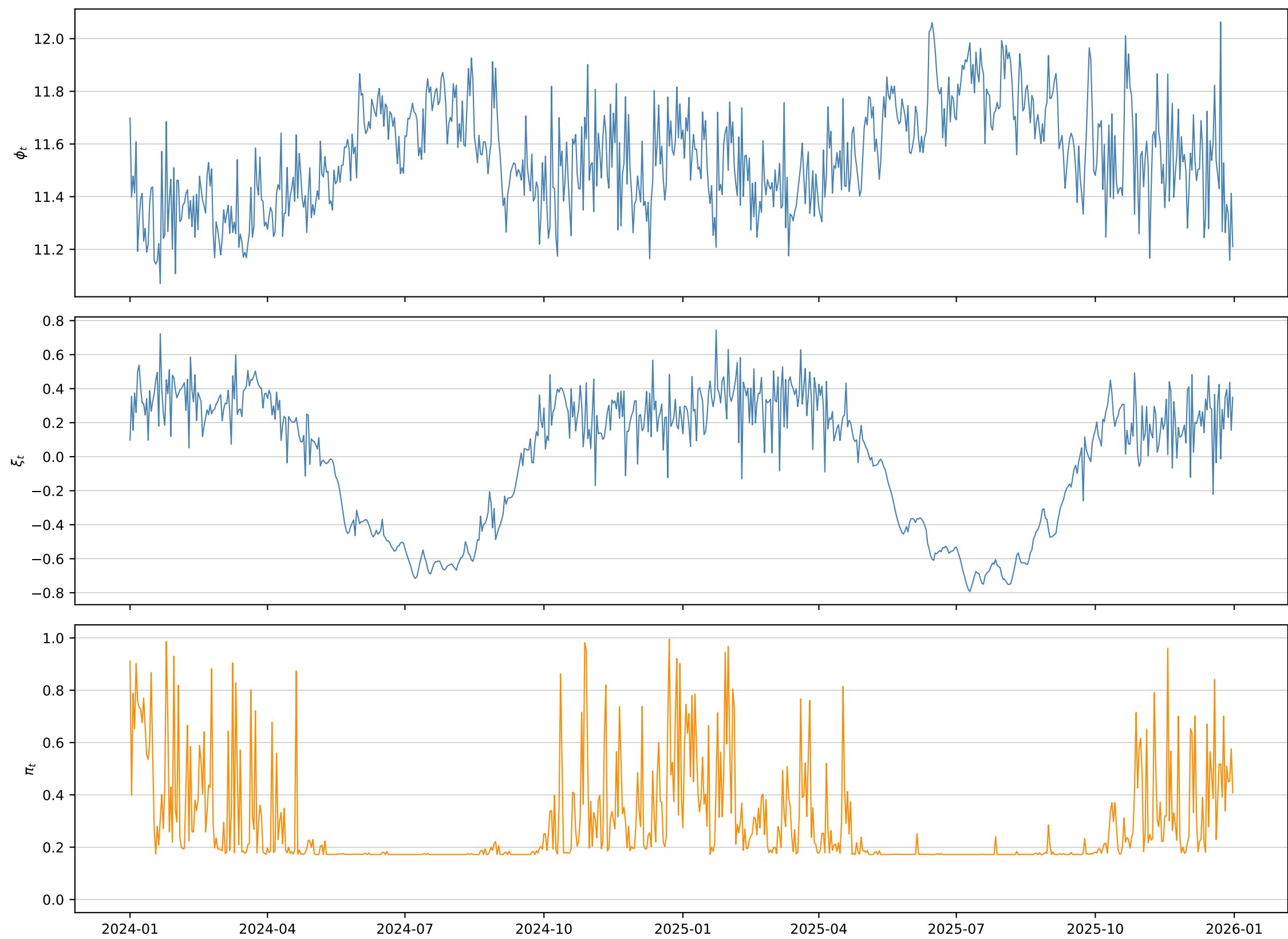
QR time series

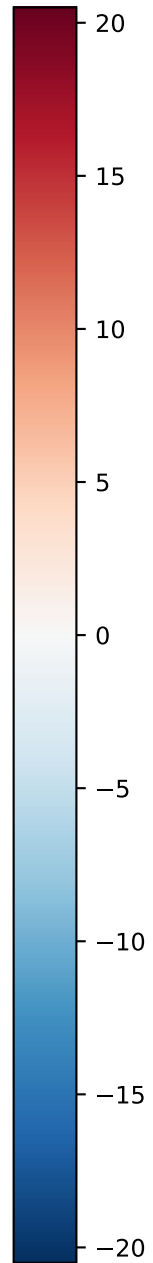


phi_xi_interaction_temperature_nino34 — filtered states (IS)

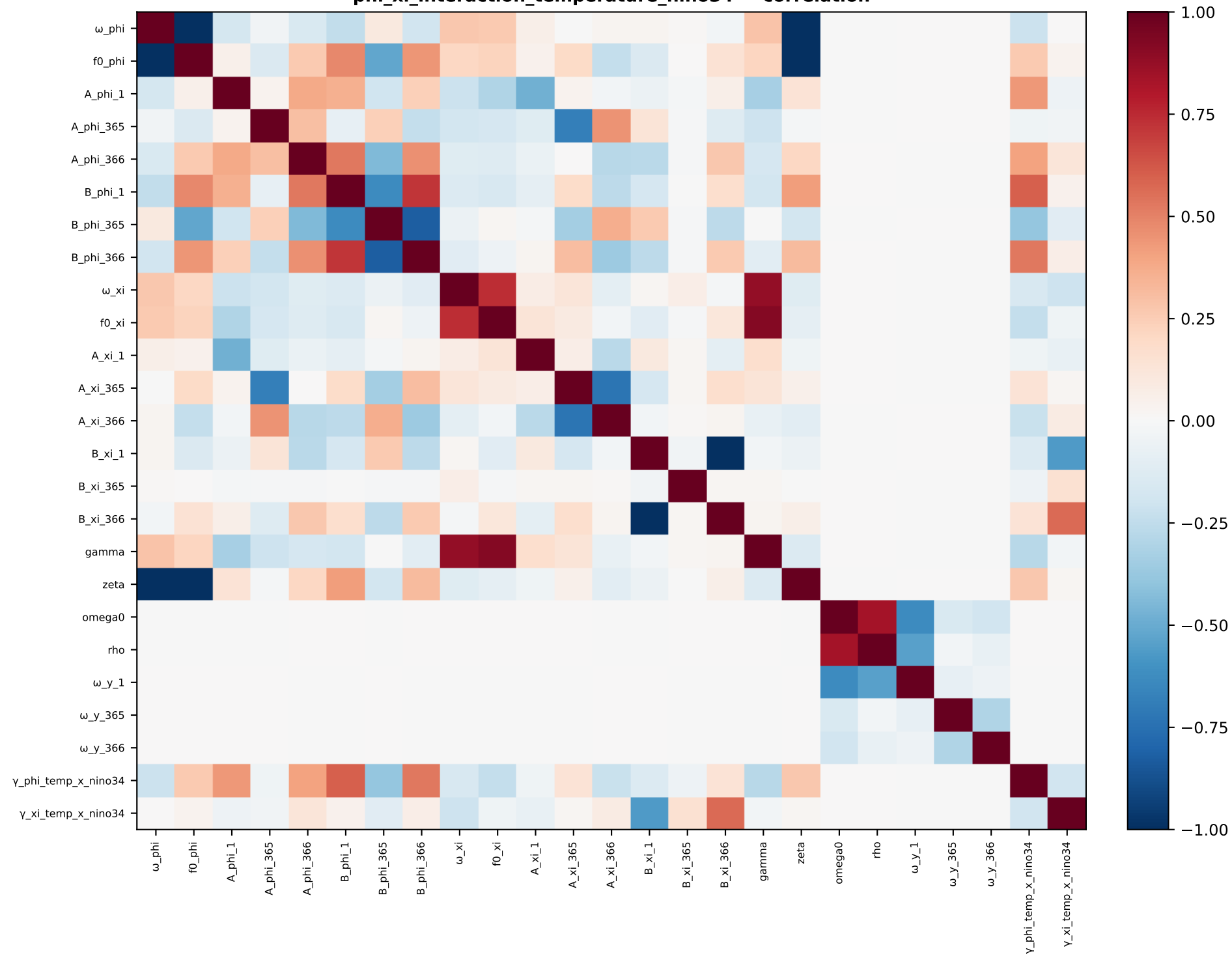


phi_xi_interaction_temperature_nino34 — filtered states (OOS)



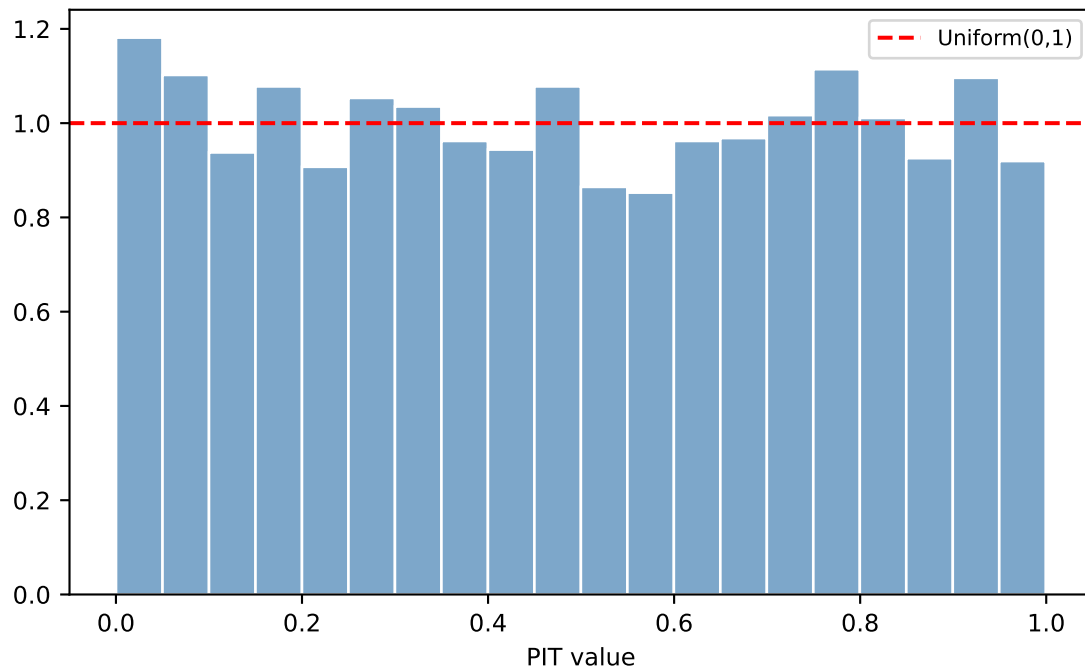


phi_xi_interaction_temperature_nino34 — correlation

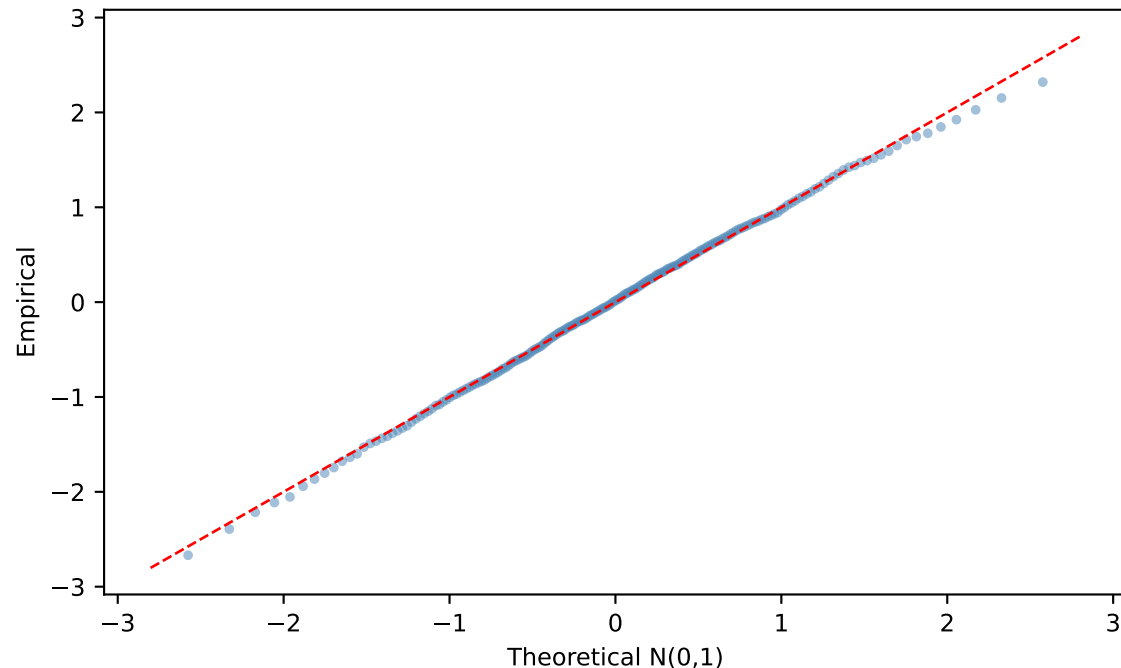


long_short_xi_basic — IS diagnostics

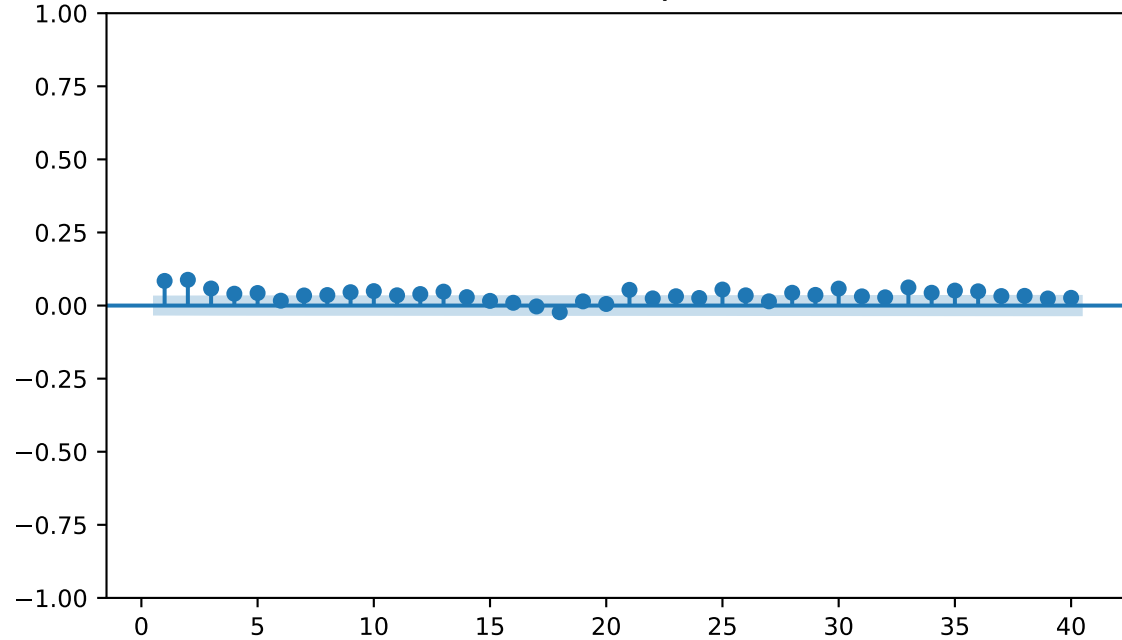
PIT histogram



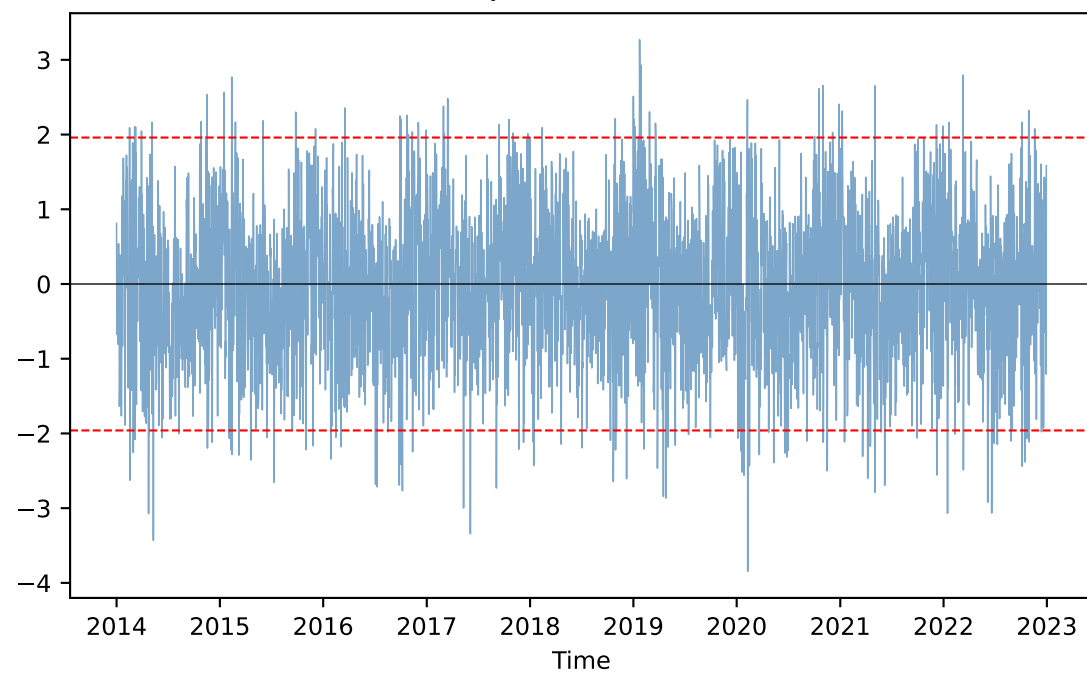
Normal QQ plot



ACF of QR

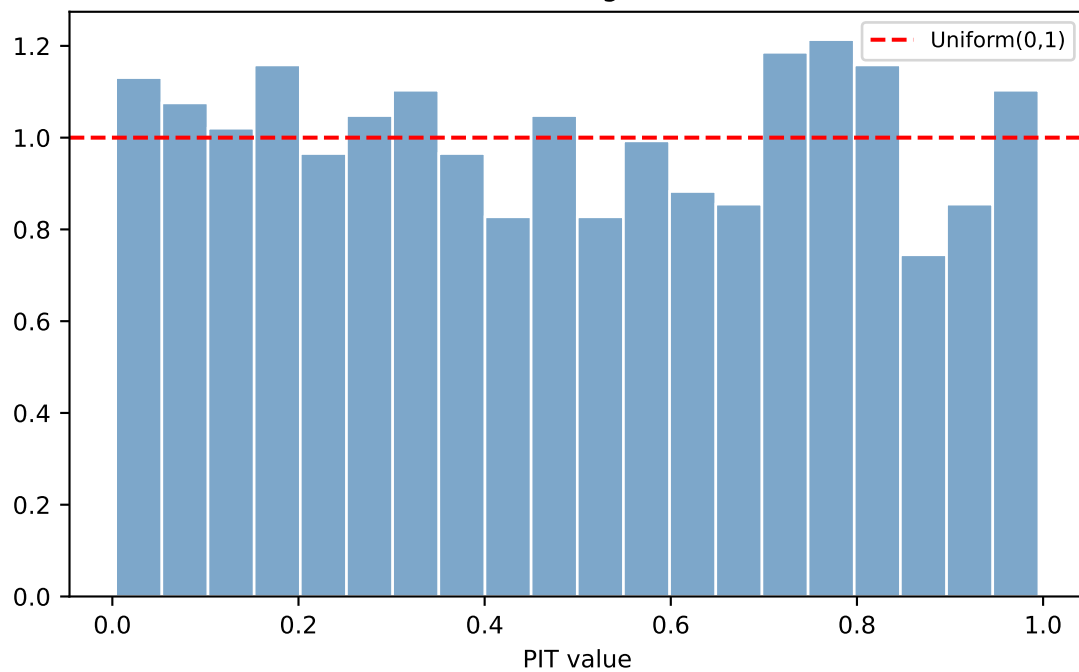


QR time series

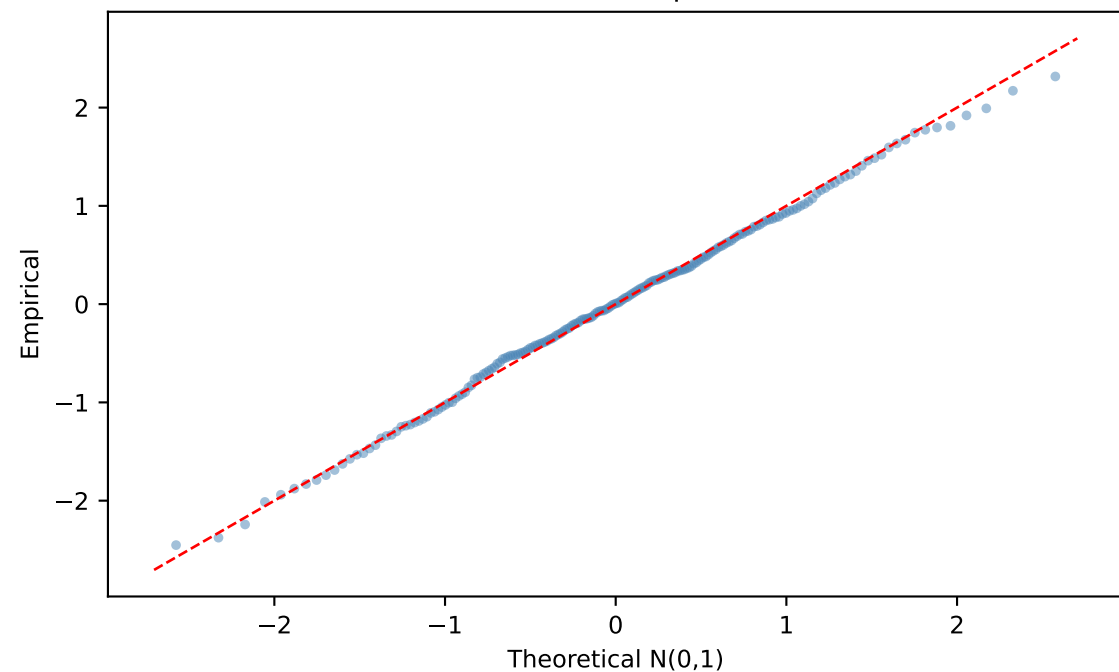


long_short_xi_basic — OOS diagnostics

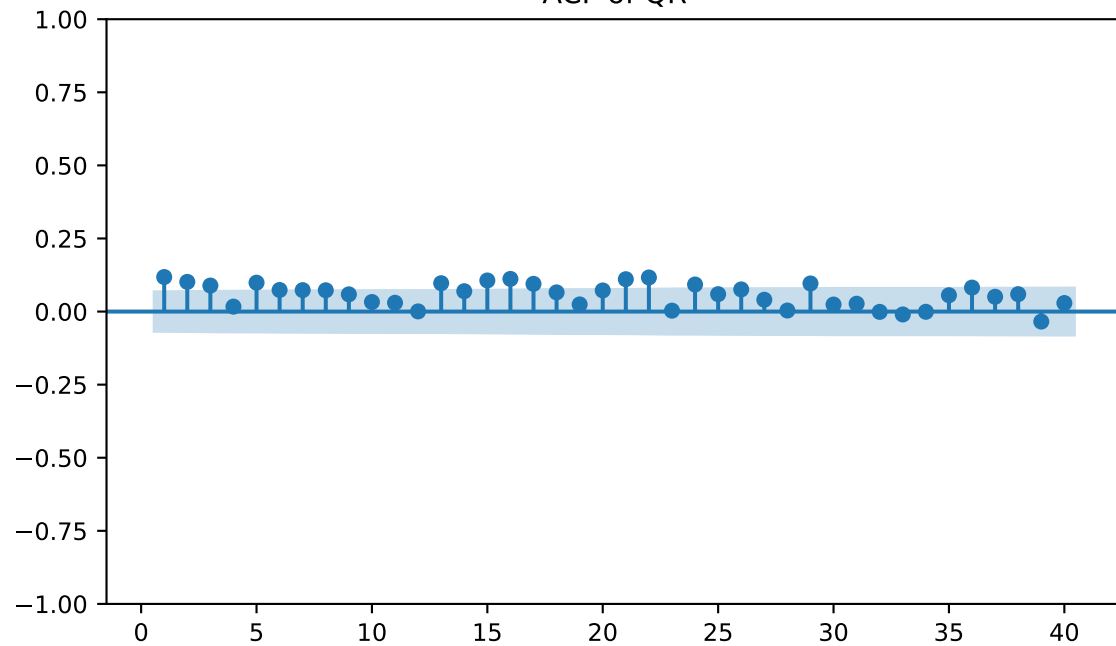
PIT histogram



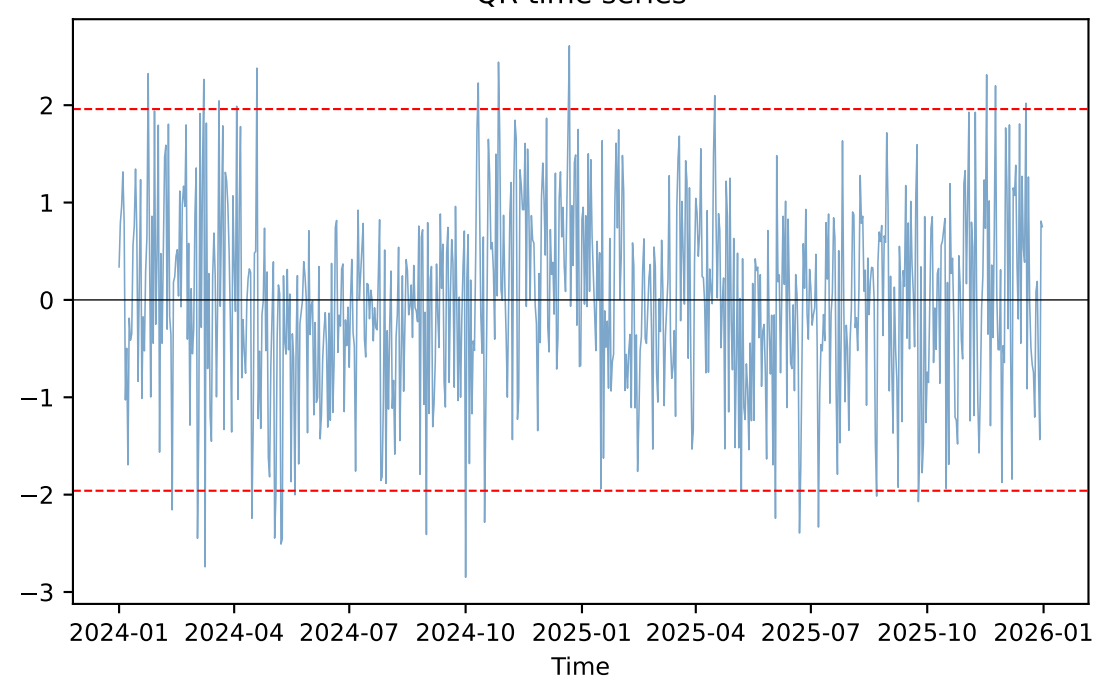
Normal QQ plot



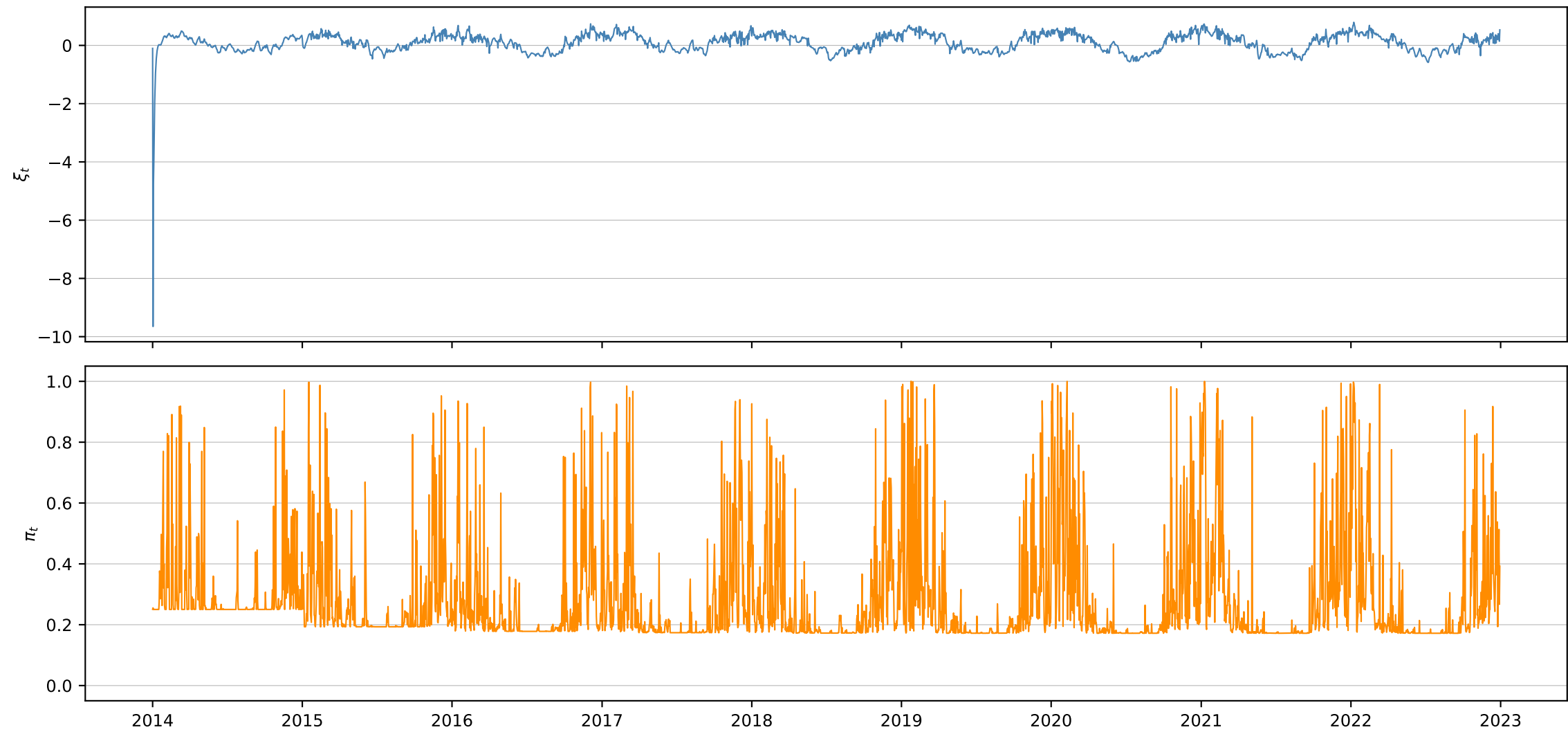
ACF of QR



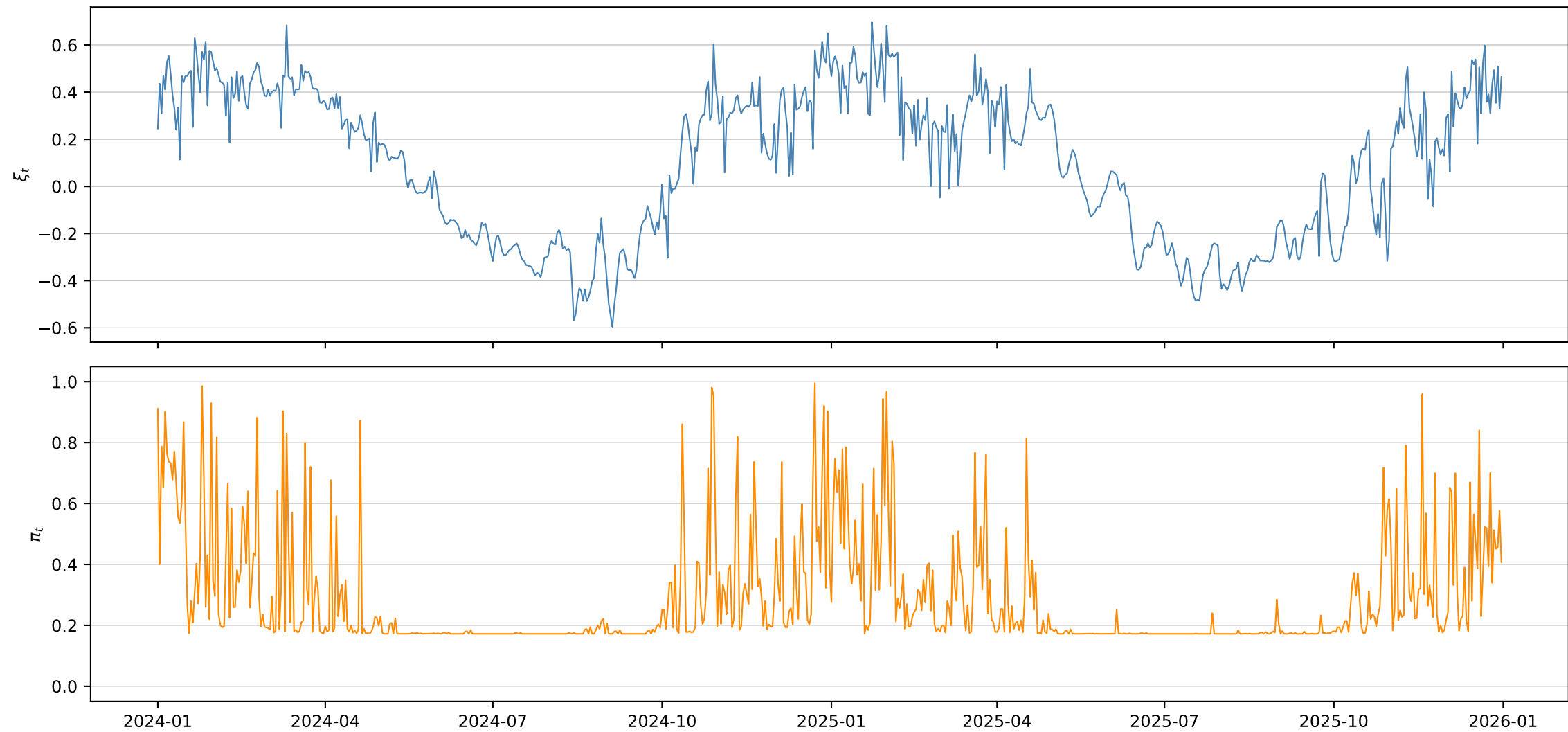
QR time series



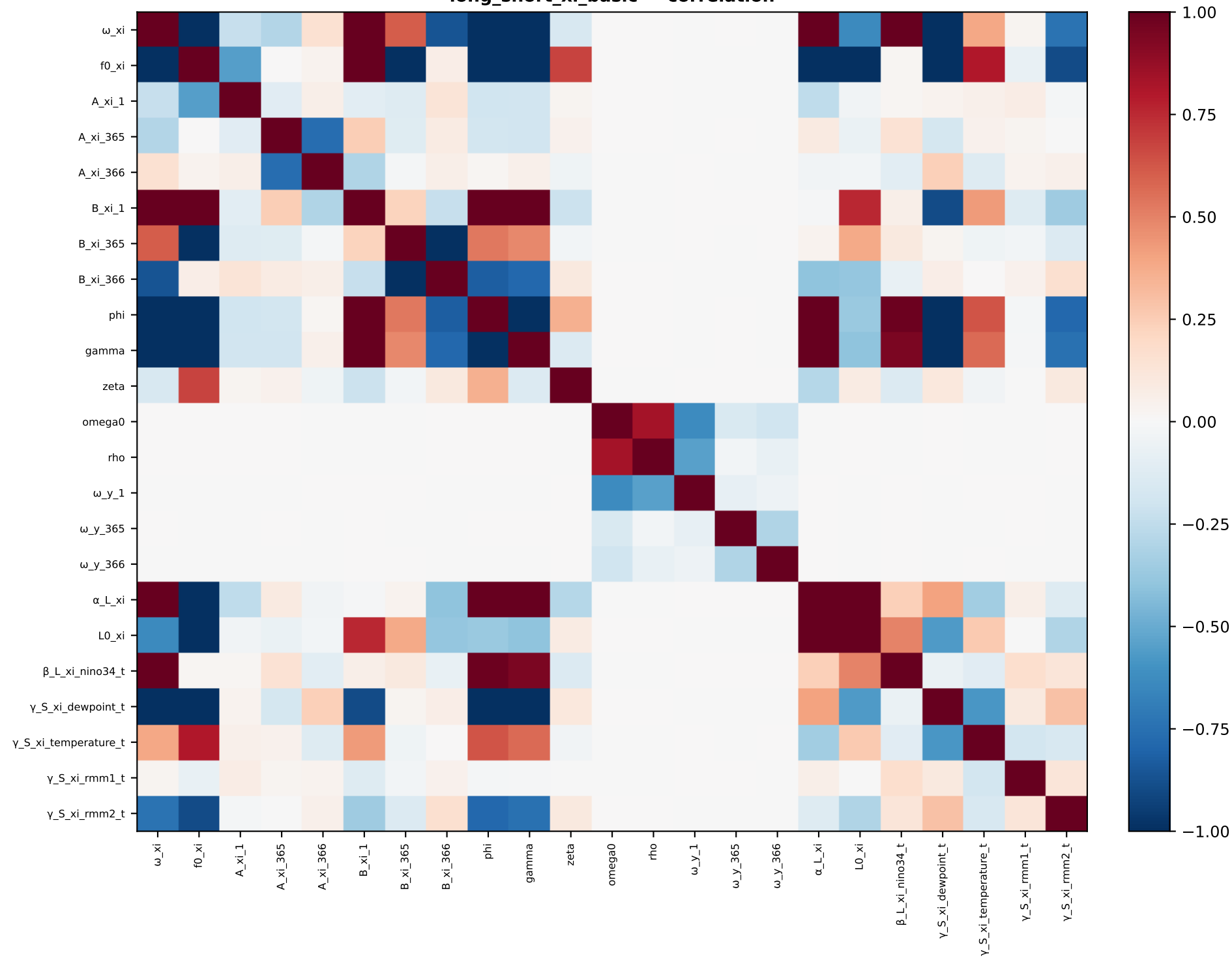
long_short_xi_basic — filtered states (IS)



long_short_xi_basic — filtered states (OOS)

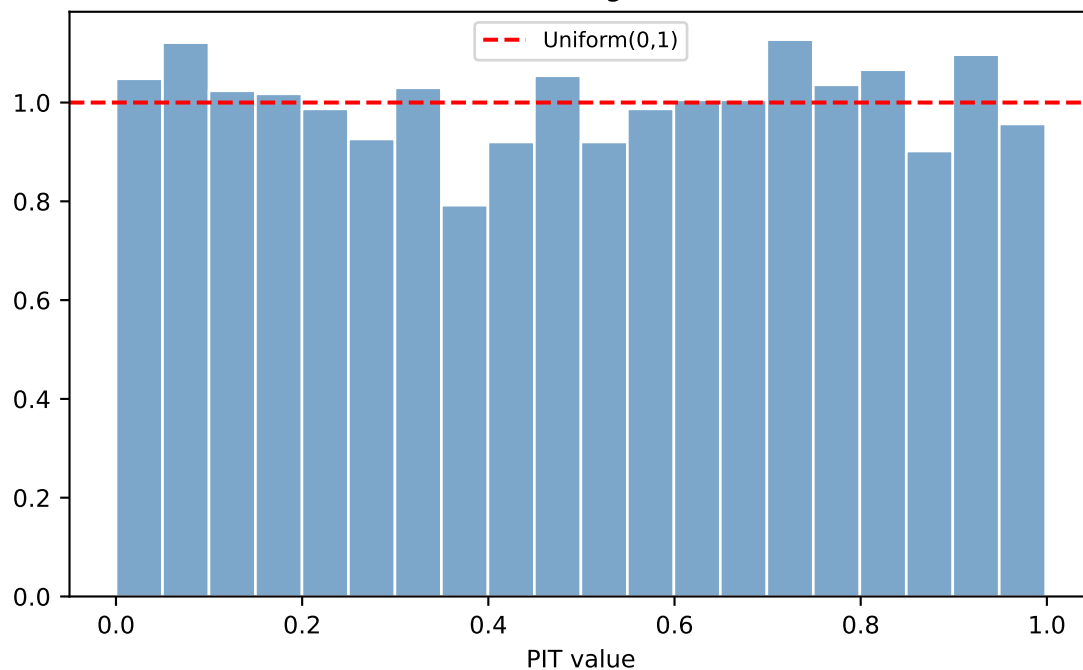


long_short_xi_basic — correlation

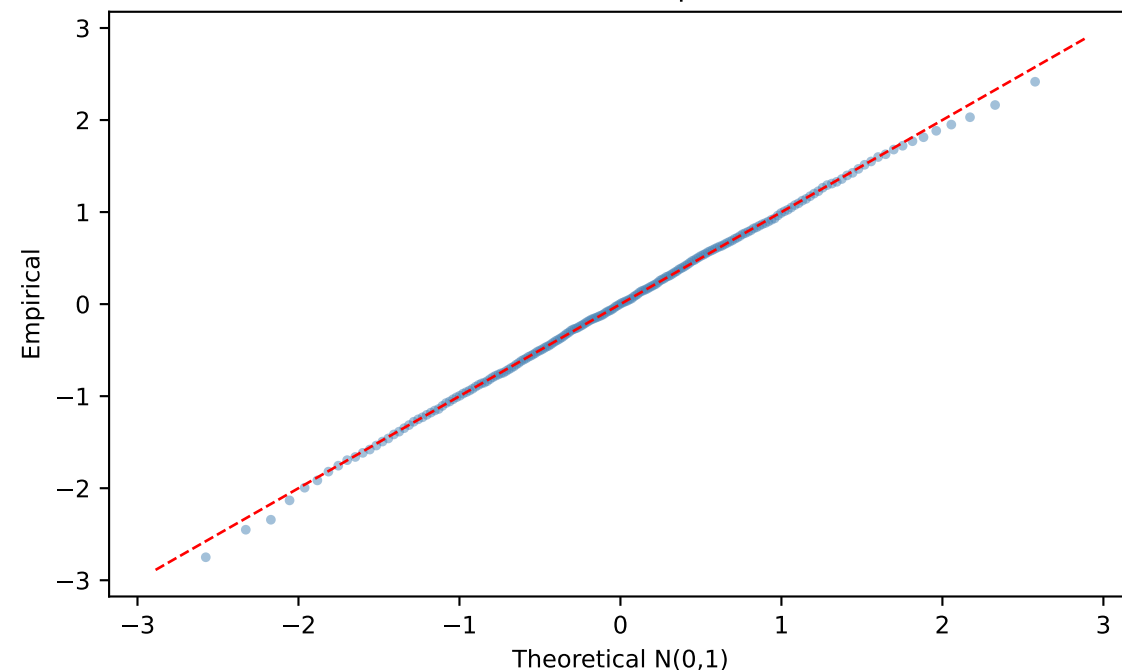


long_short_xi_short_lags — IS diagnostics

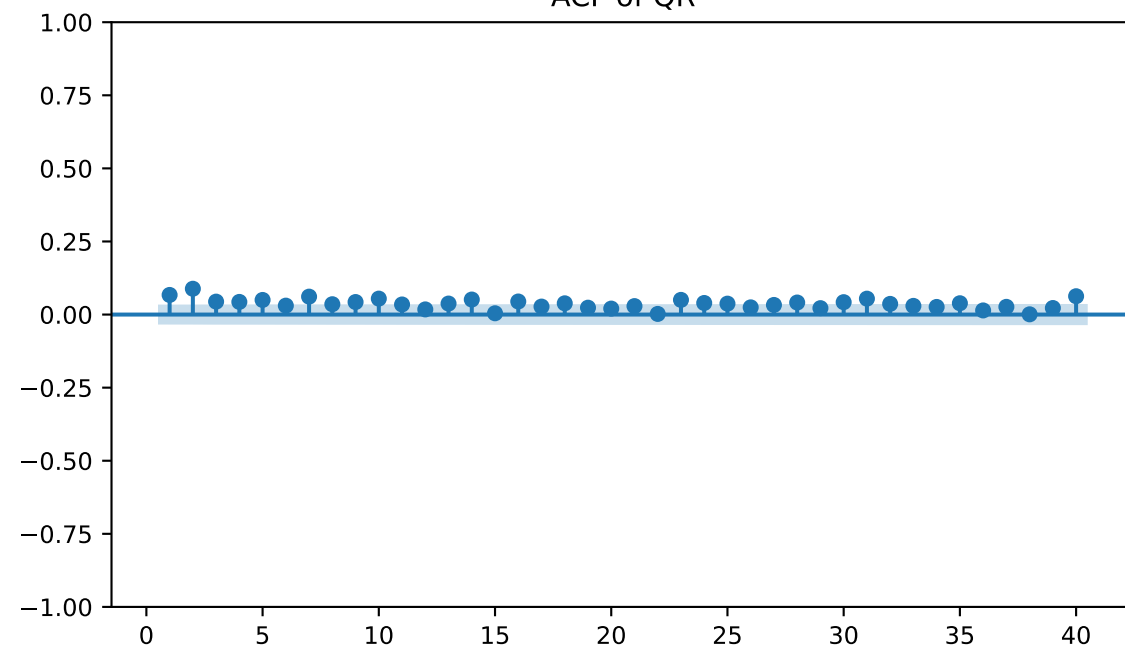
PIT histogram



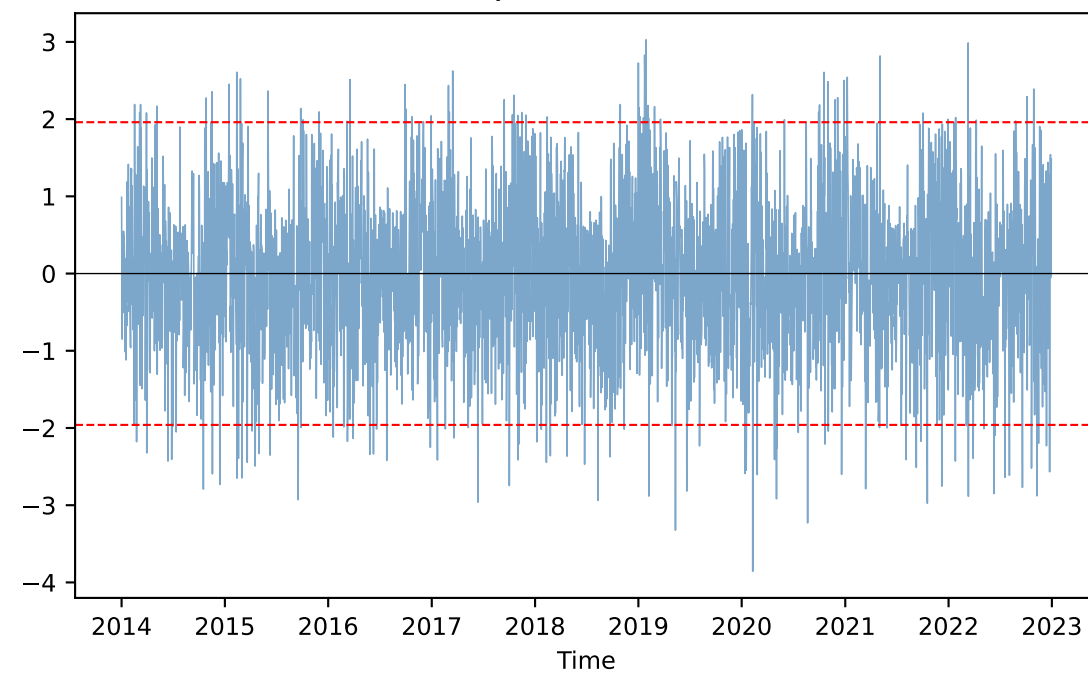
Normal QQ plot



ACF of QR

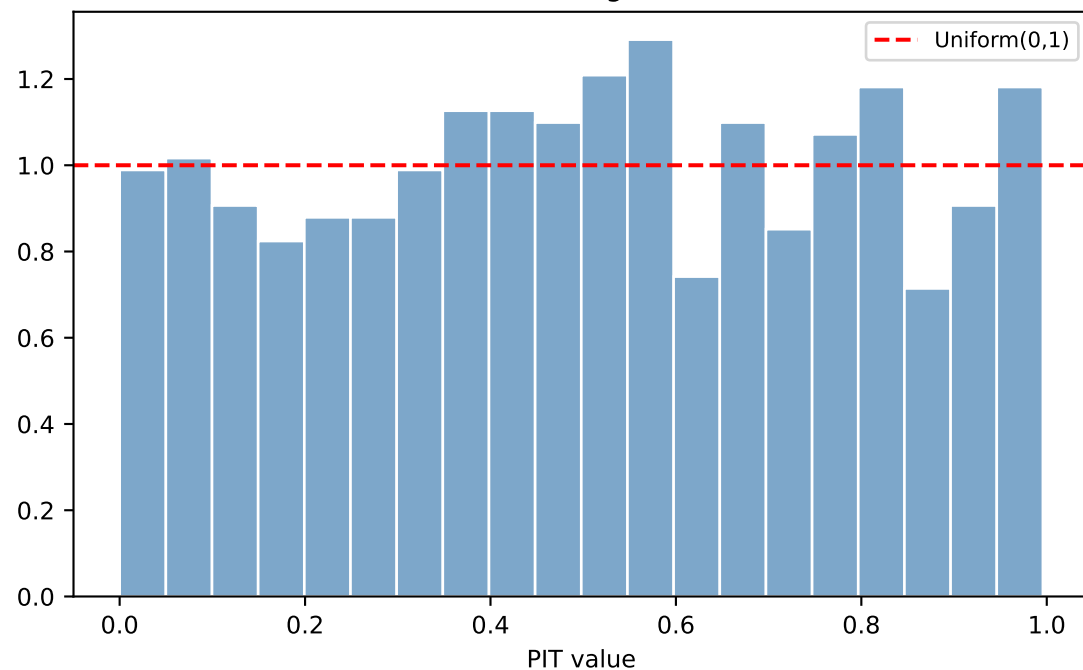


QR time series

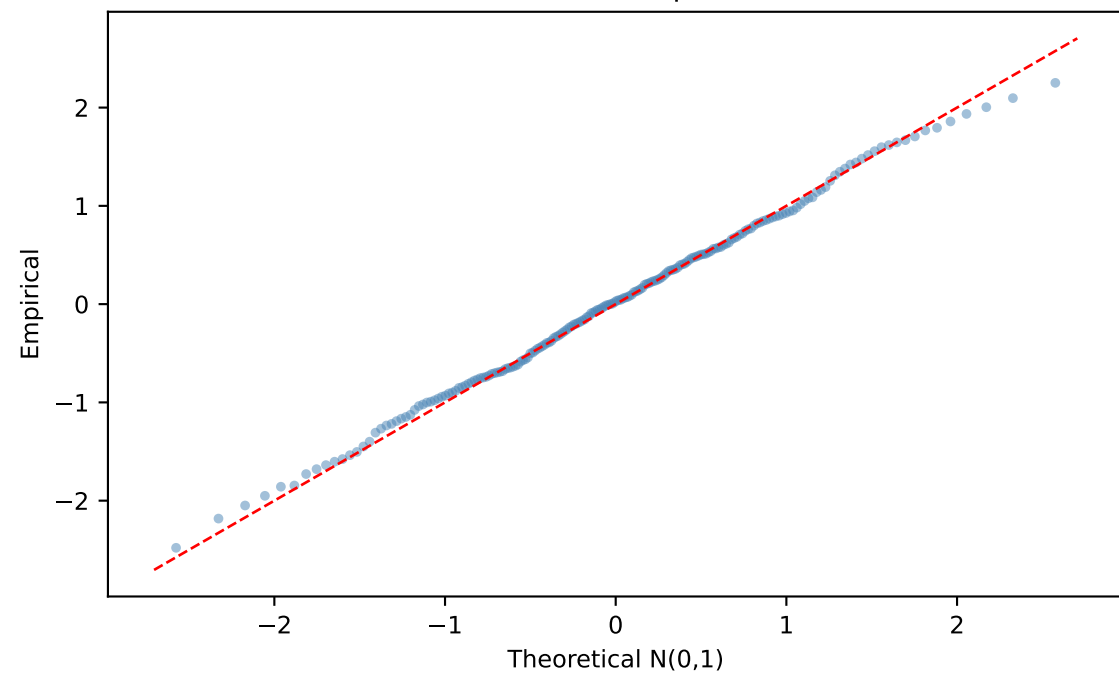


long_short_xi_short_lags — OOS diagnostics

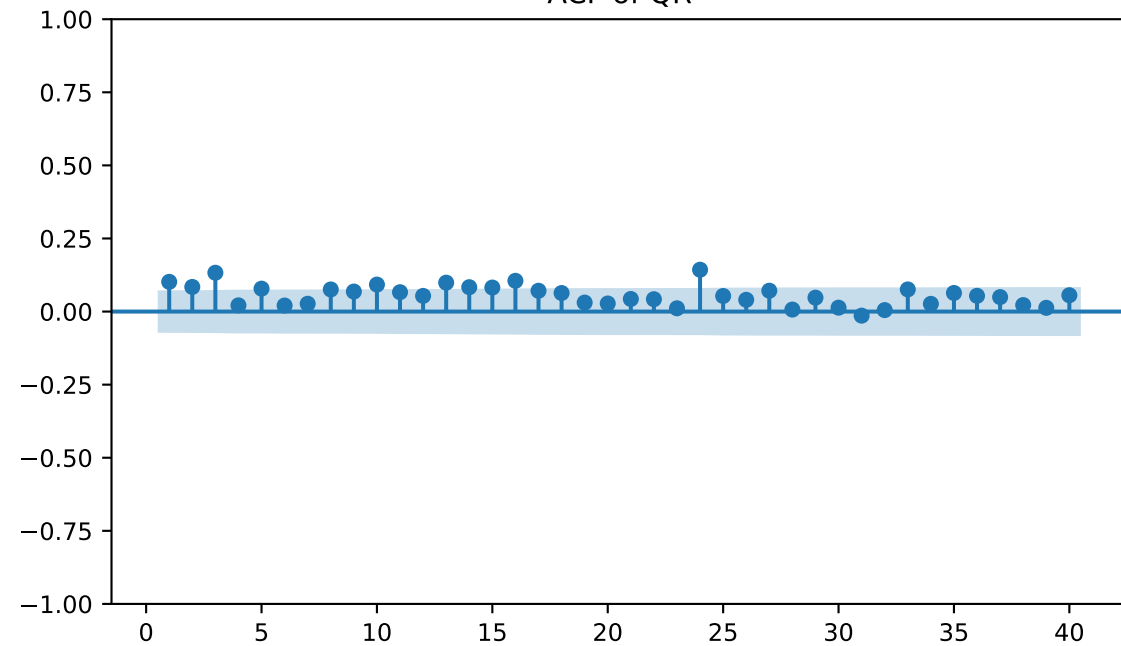
PIT histogram



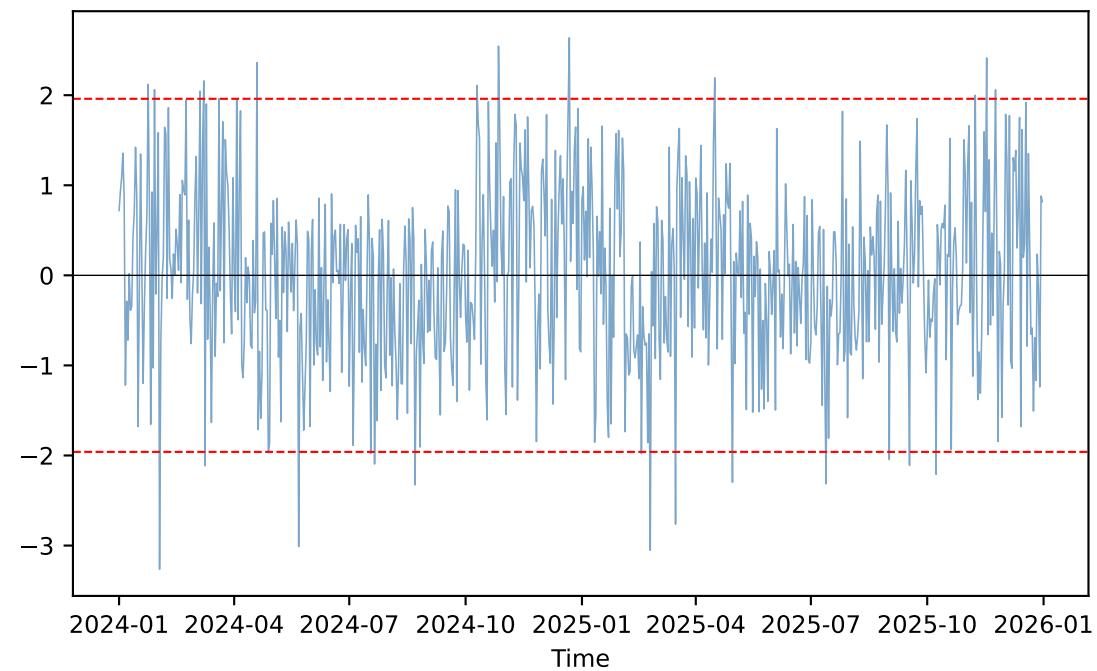
Normal QQ plot



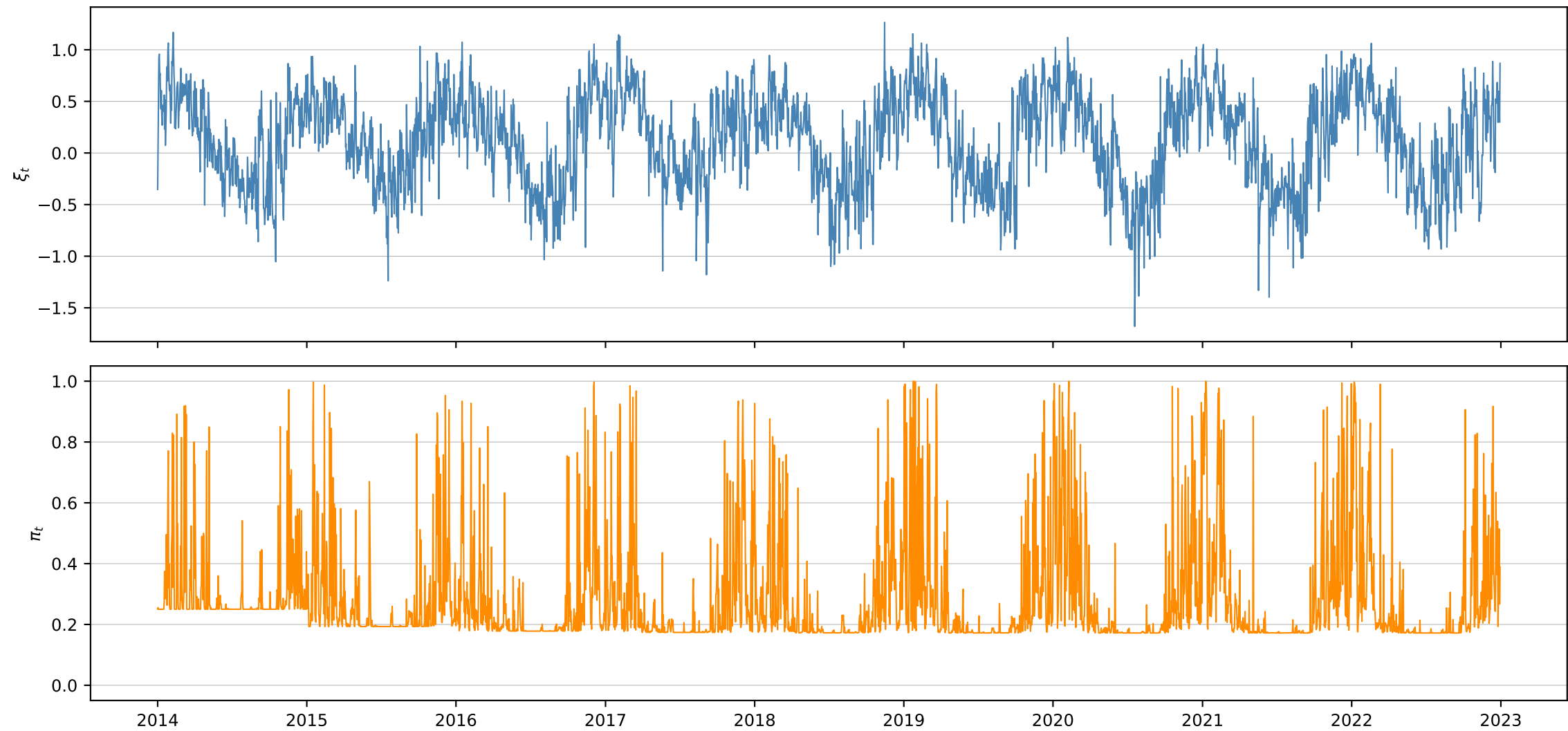
ACF of QR



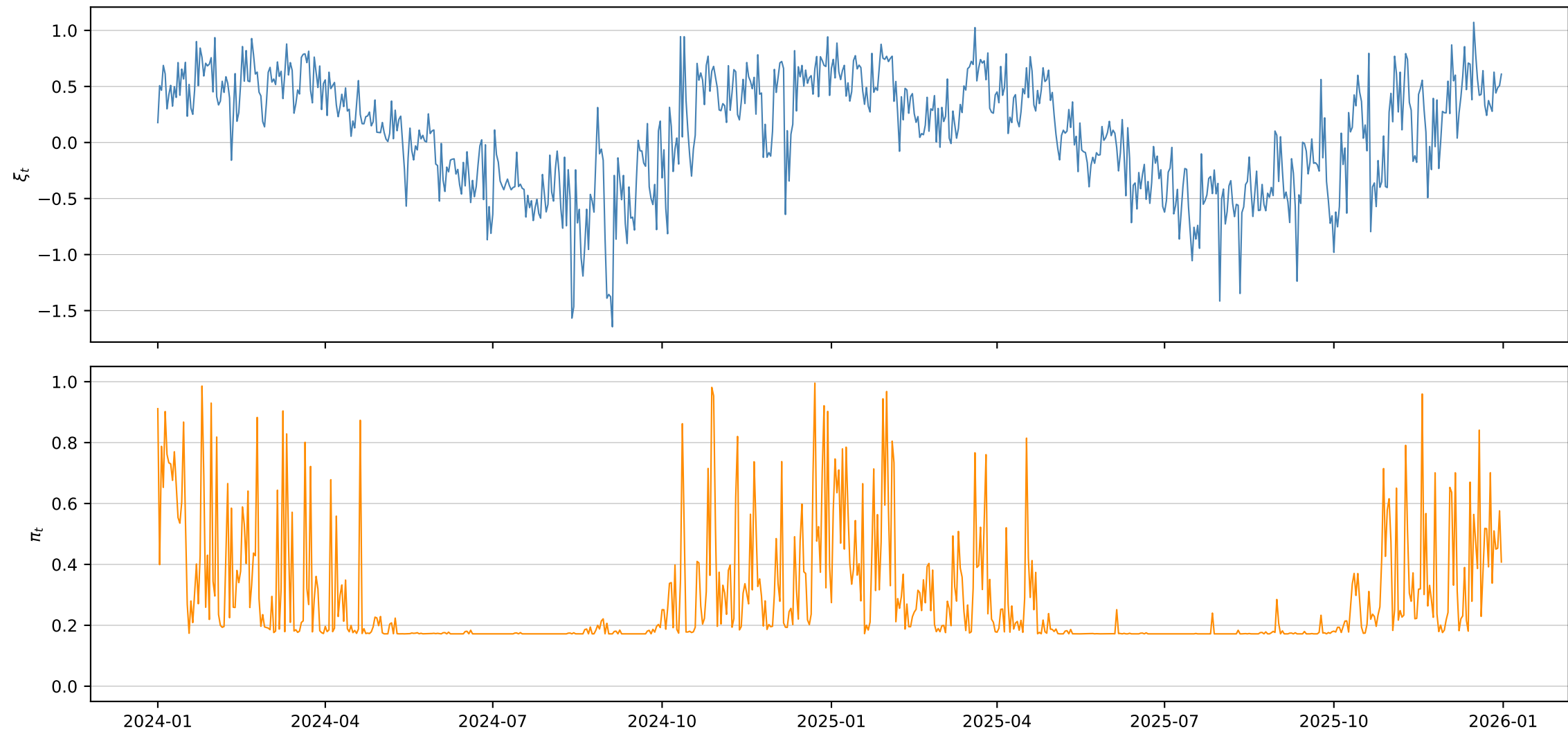
QR time series



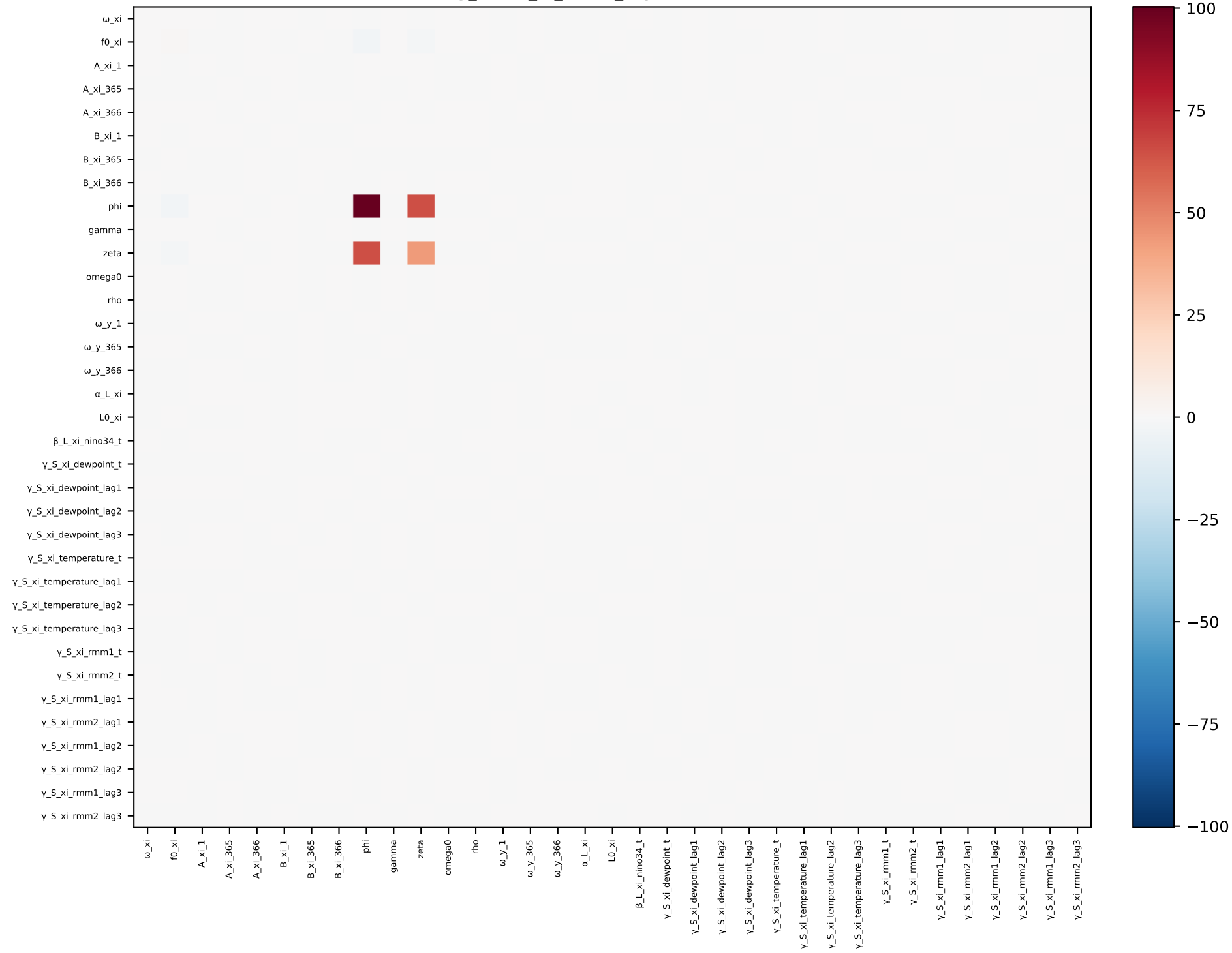
long_short_xi_short_lags — filtered states (IS)



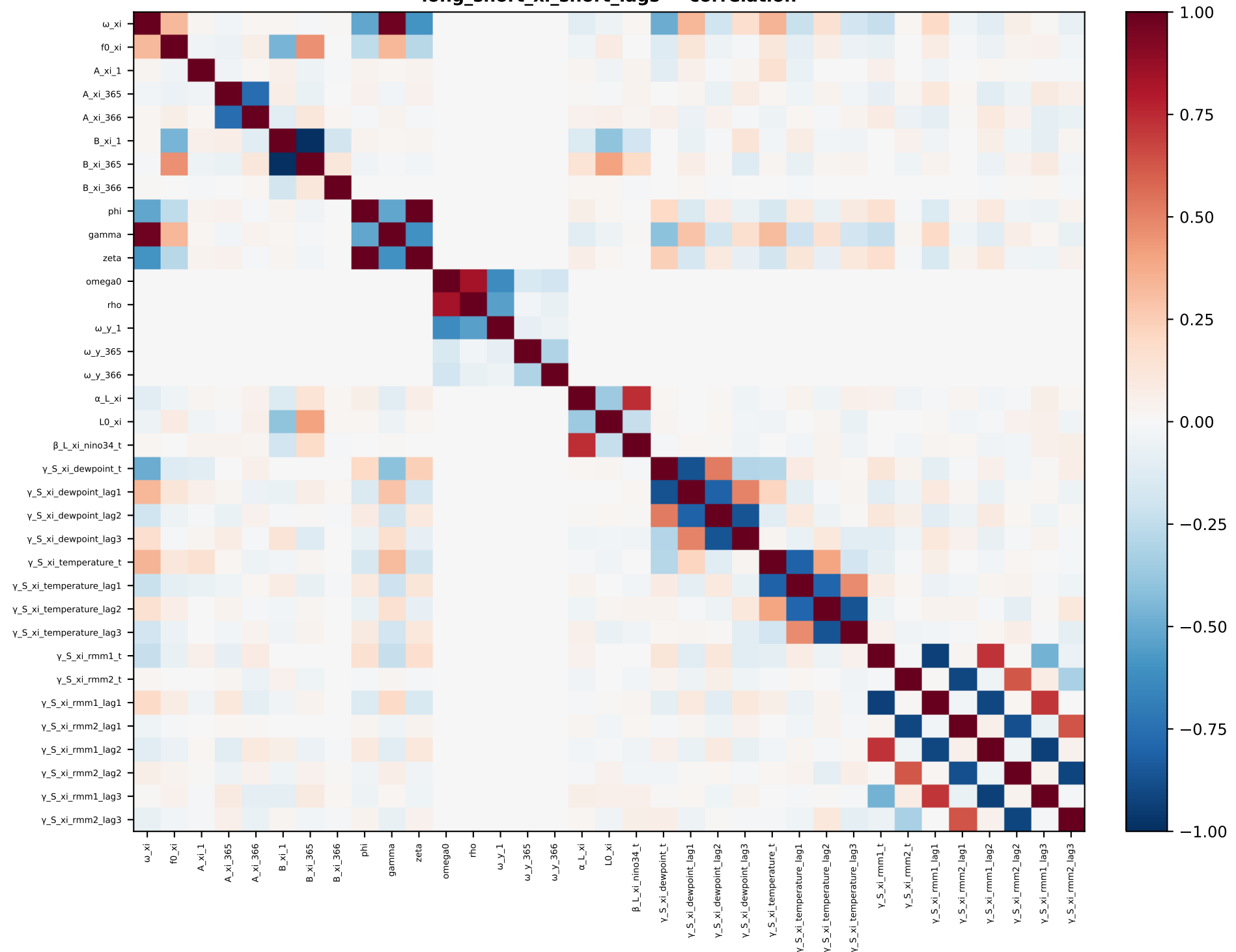
long_short_xi_short_lags — filtered states (OOS)



long_short_xi_short_lags — covariance

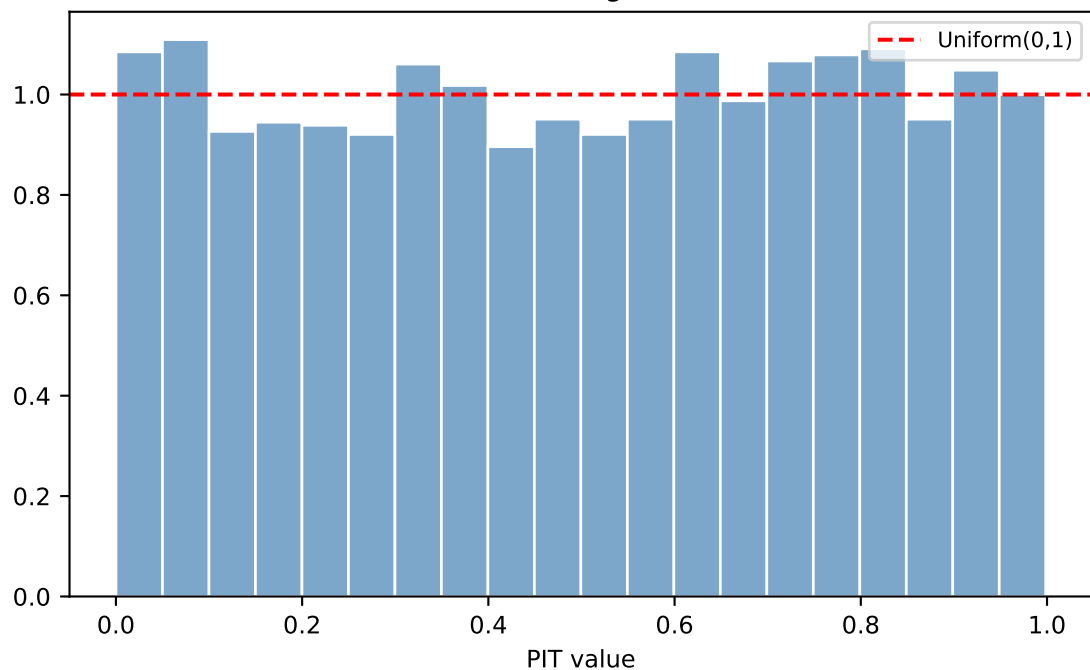


long_short_xi_short_lags — correlation

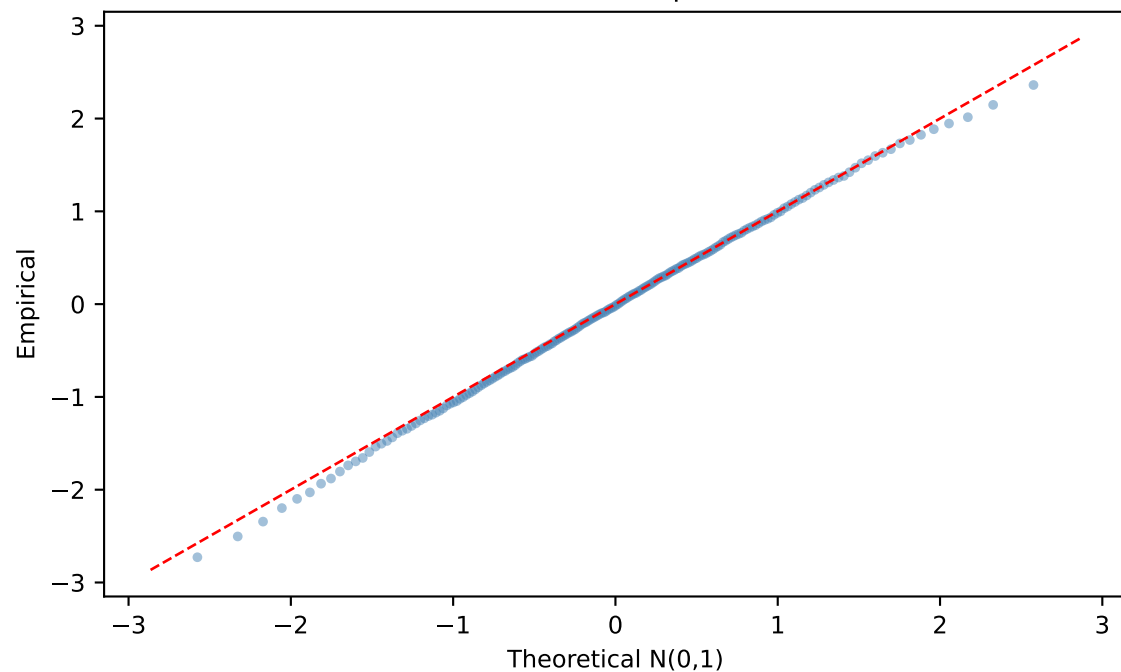


long_short_xi_weather_climate — IS diagnostics

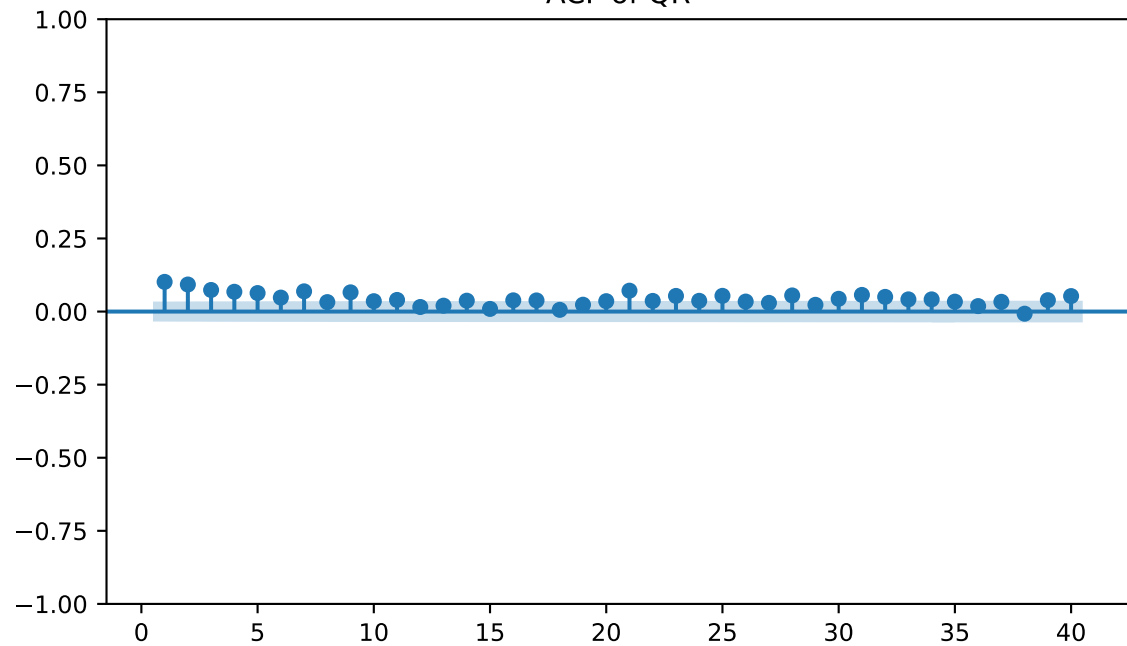
PIT histogram



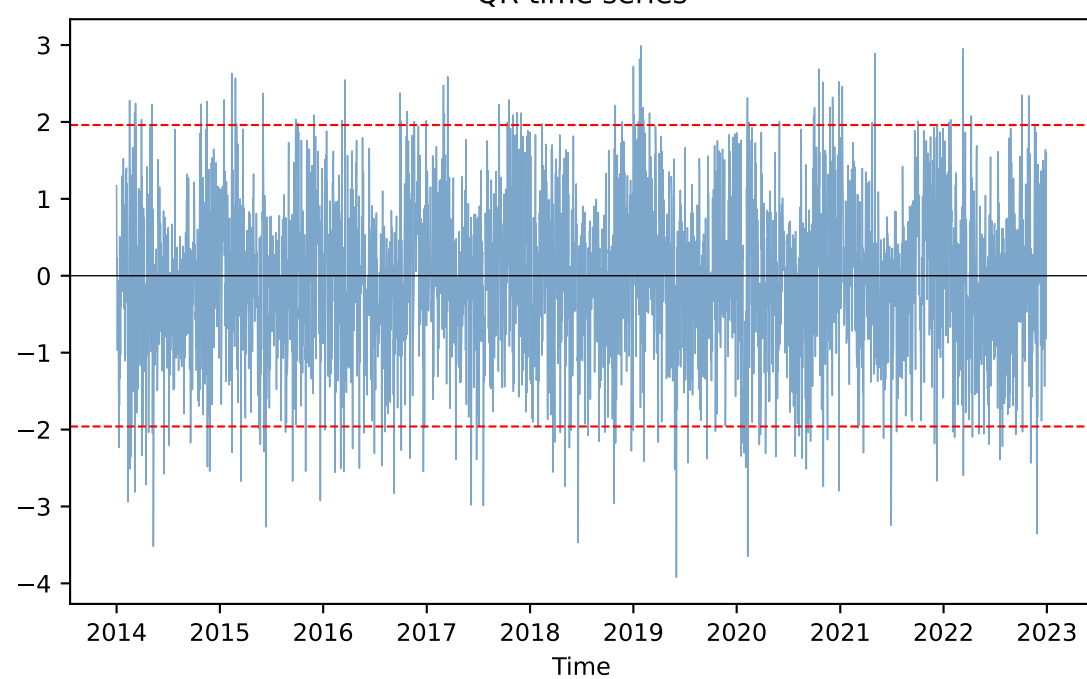
Normal QQ plot



ACF of QR

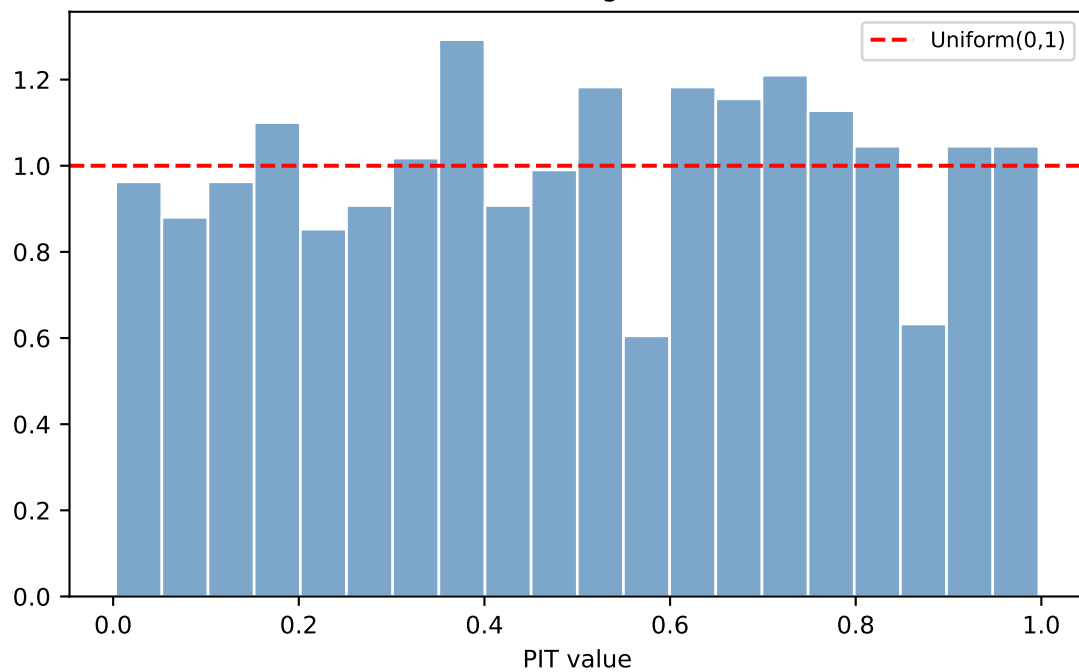


QR time series

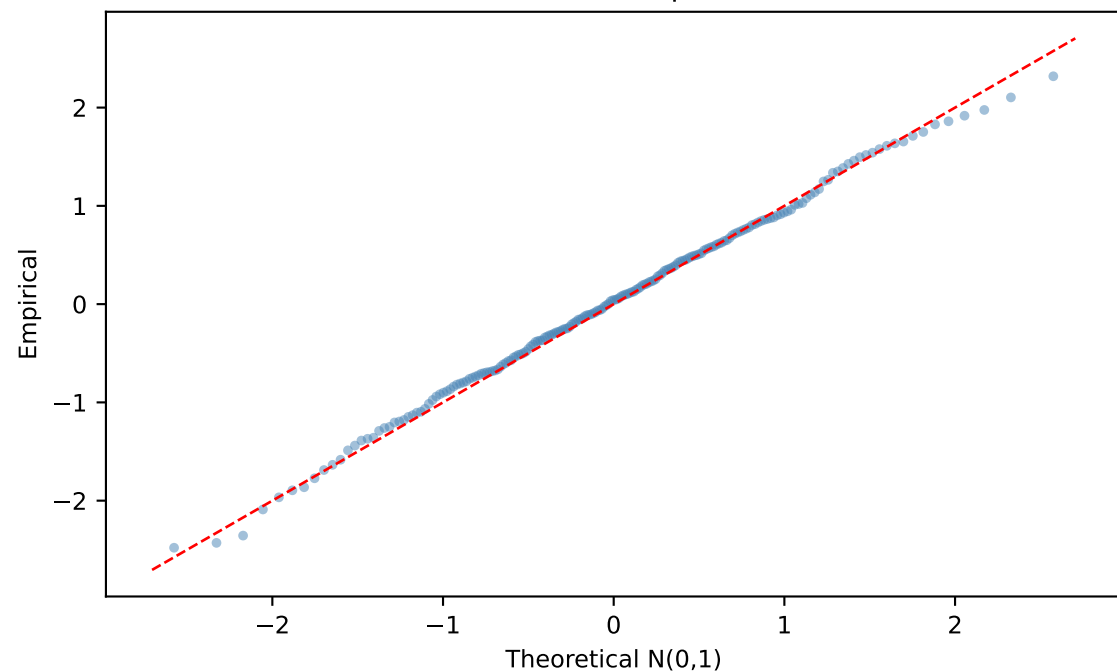


long_short_xi_weather_climate — OOS diagnostics

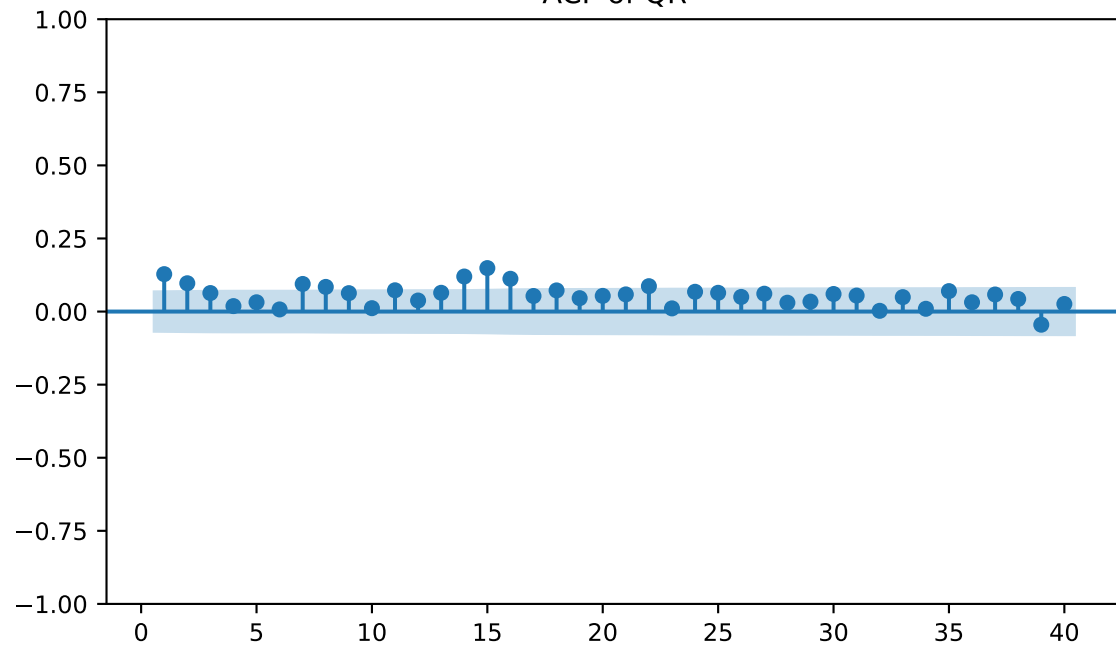
PIT histogram



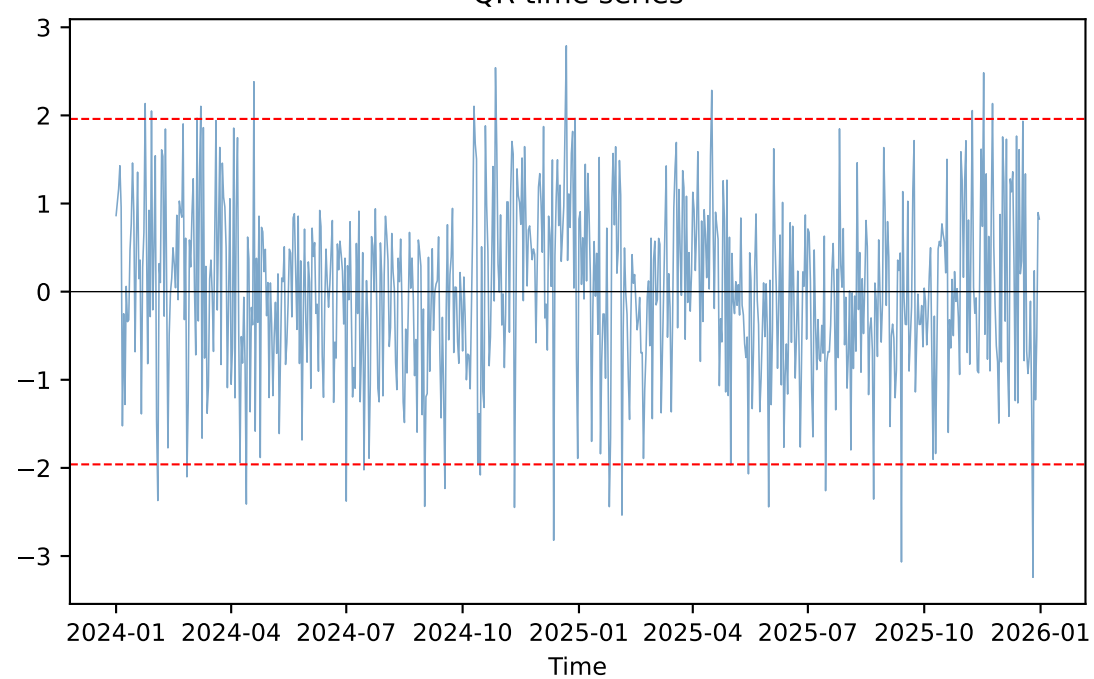
Normal QQ plot



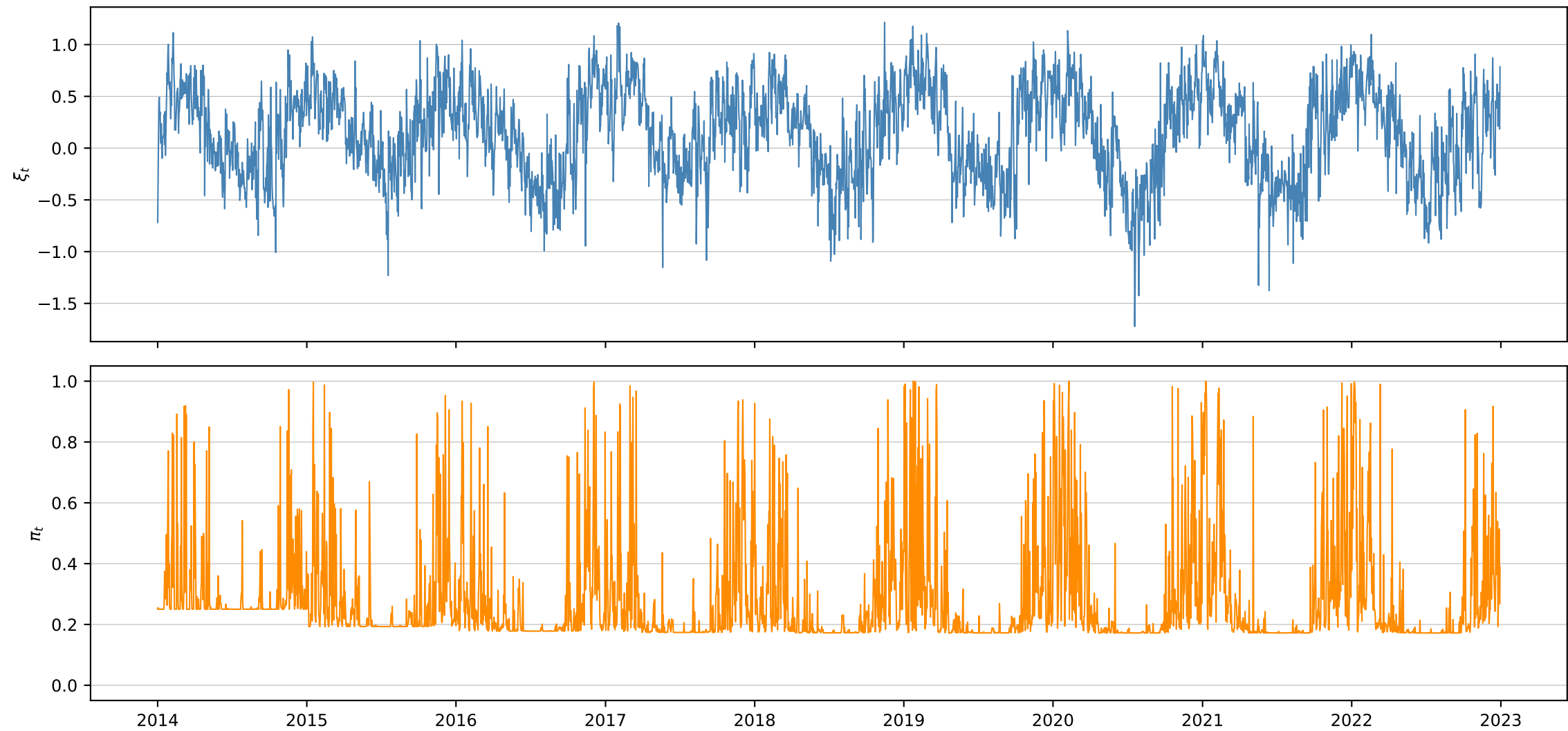
ACF of QR



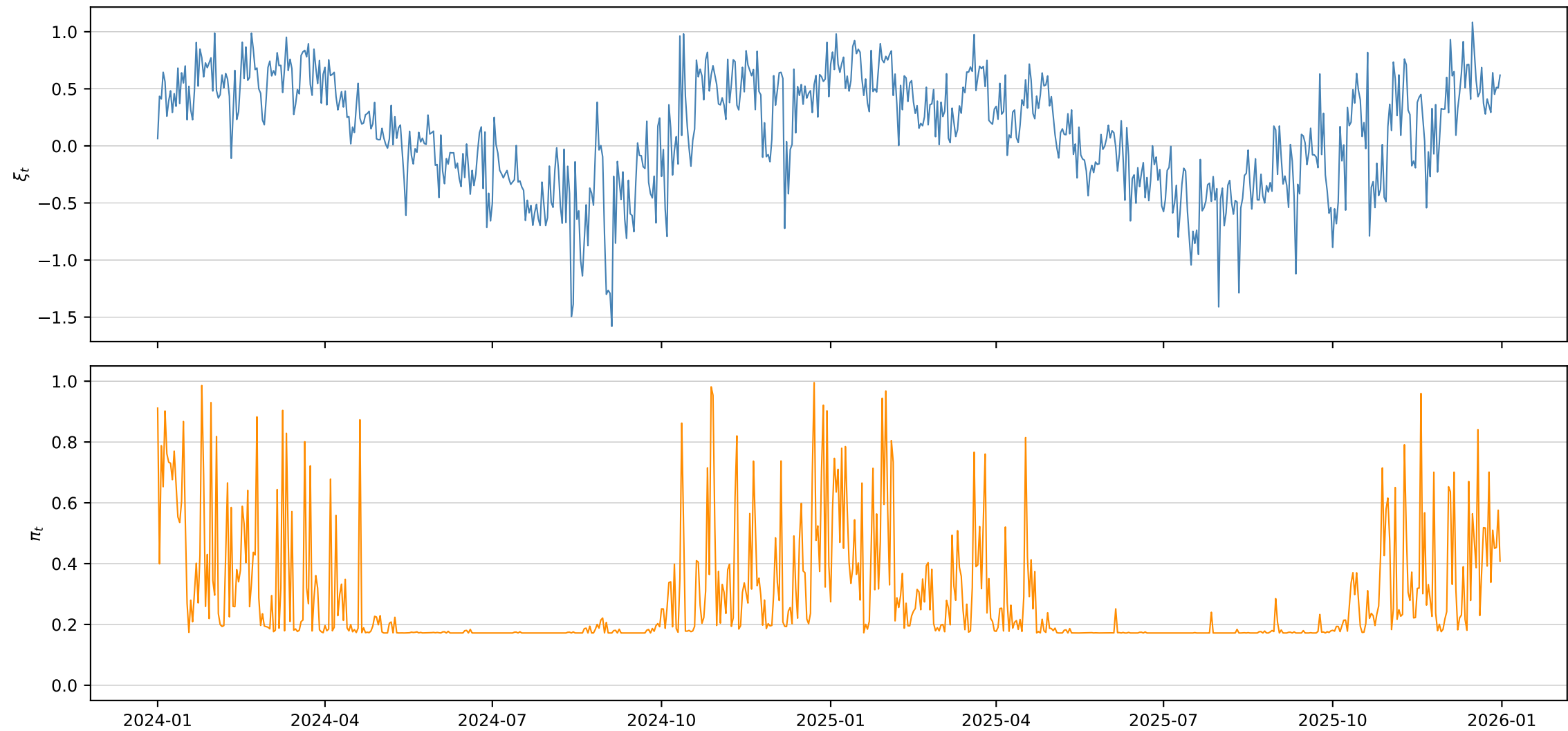
QR time series



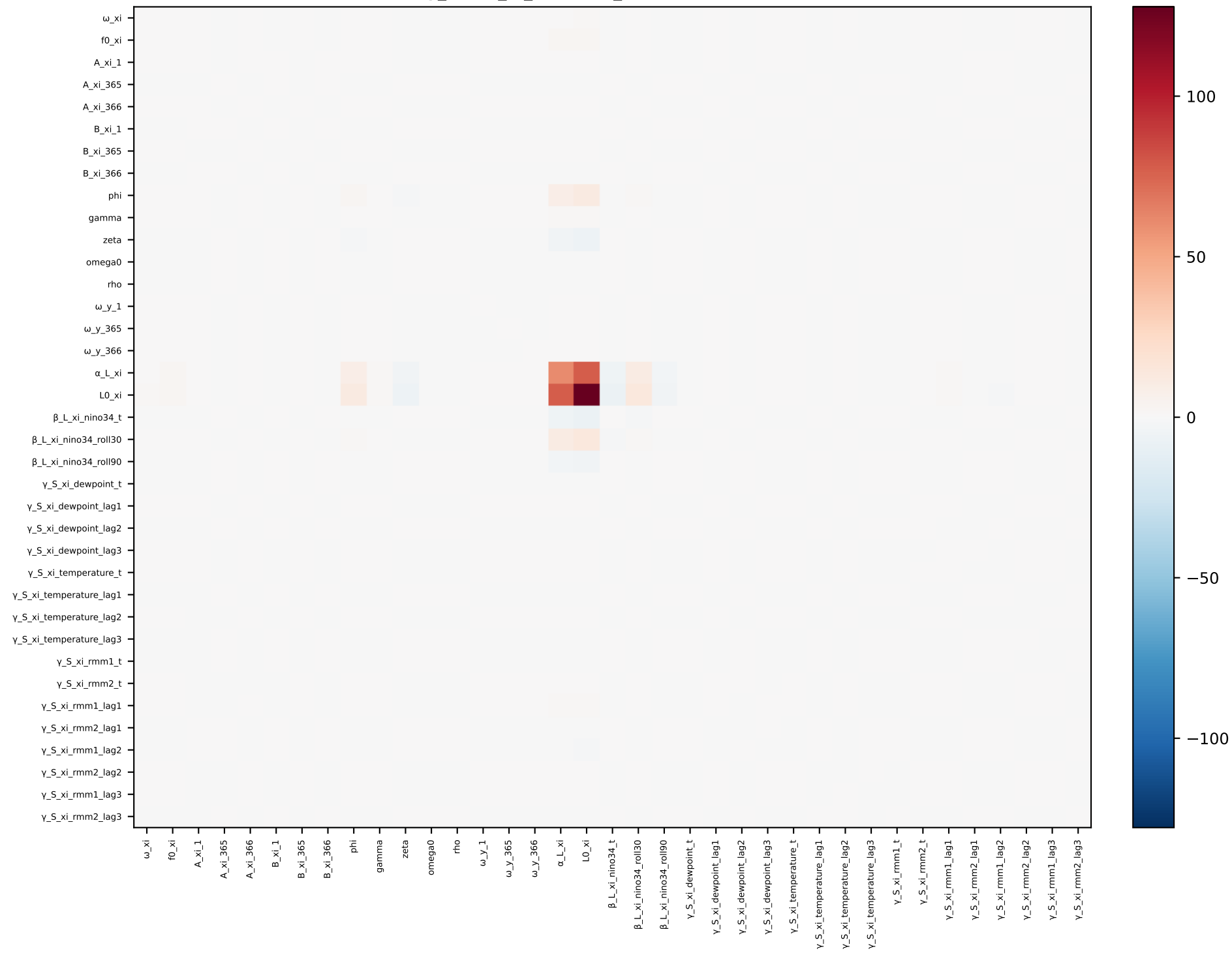
long_short_xi_weather_climate — filtered states (IS)



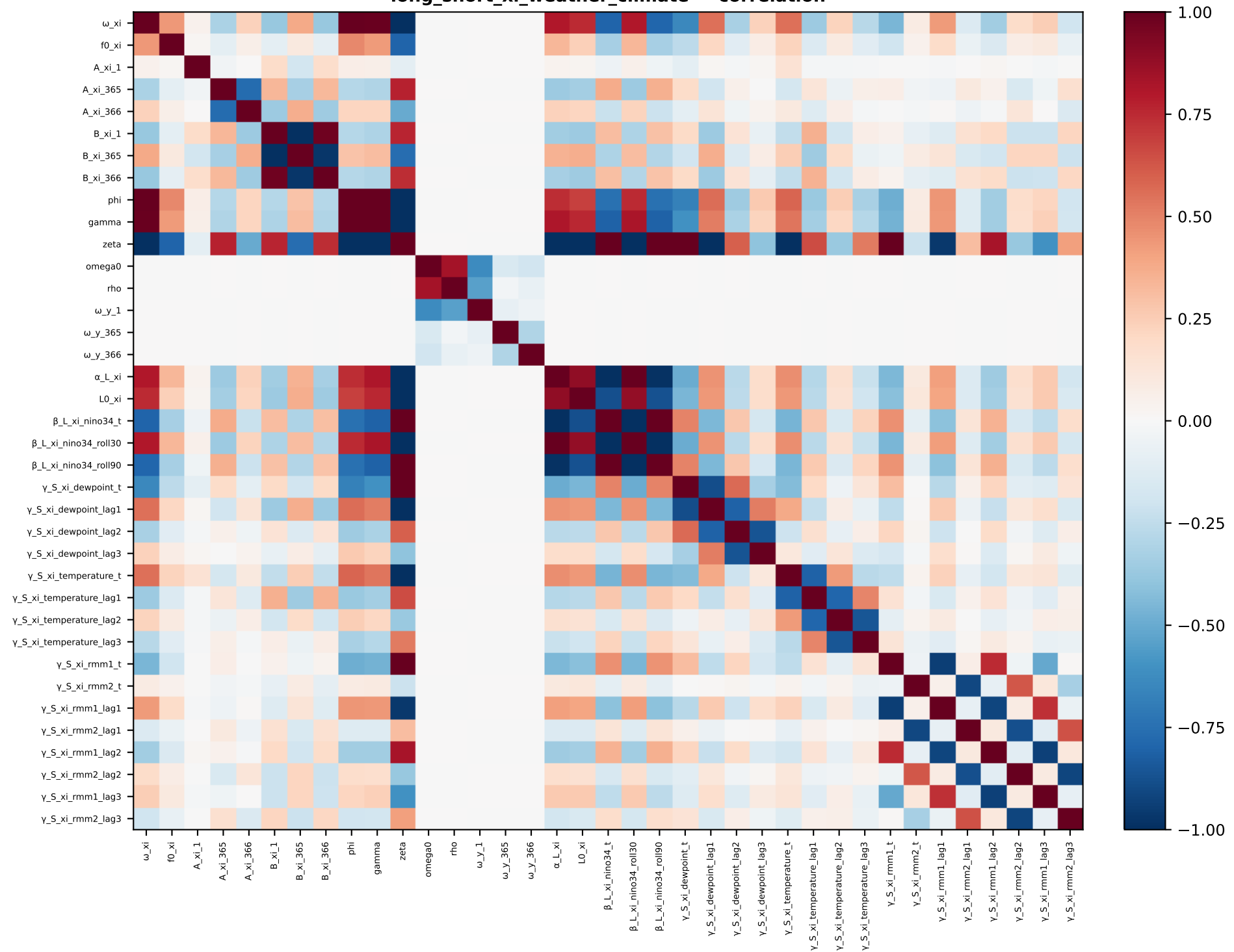
long_short_xi_weather_climate — filtered states (OOS)



long_short_xi_weather_climate — covariance

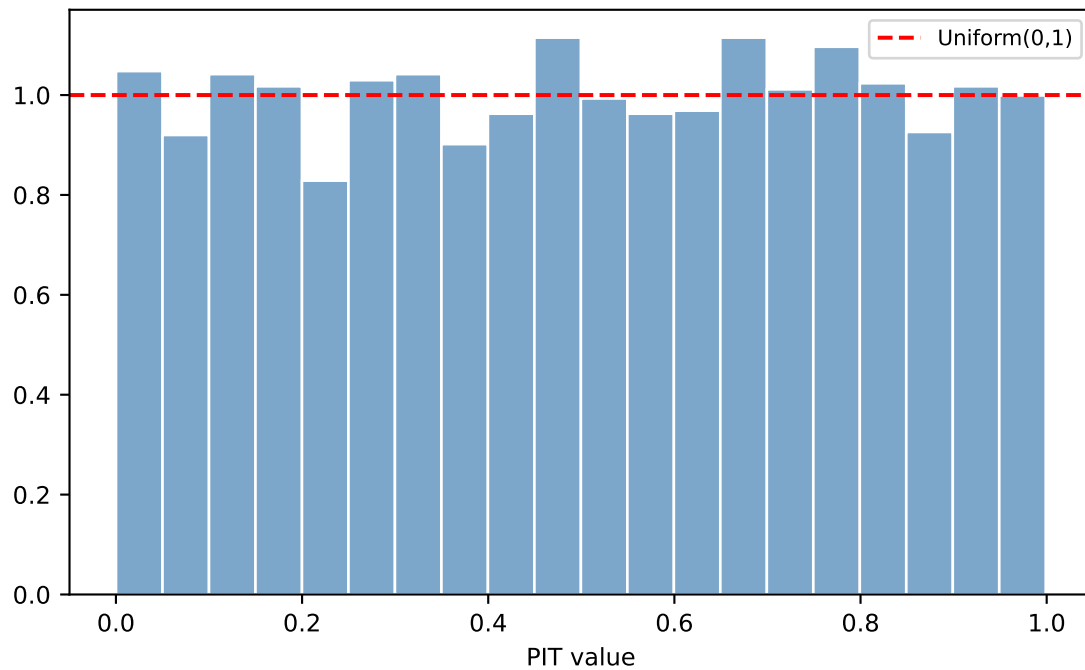


long_short_xi_weather_climate — correlation

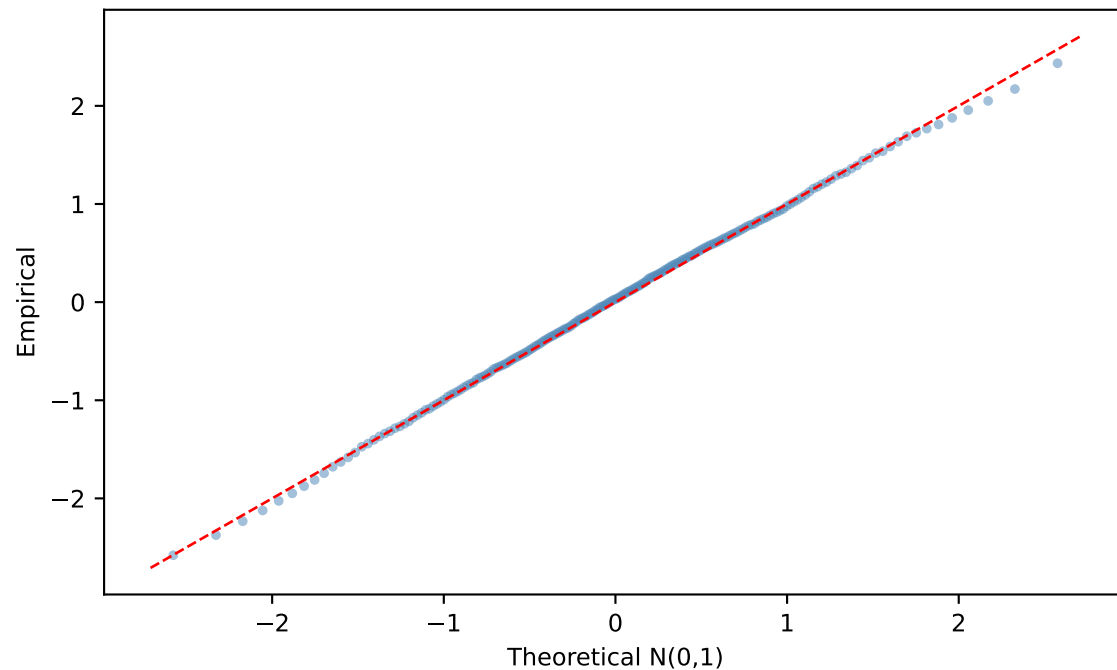


long_short_xi_interactions — IS diagnostics

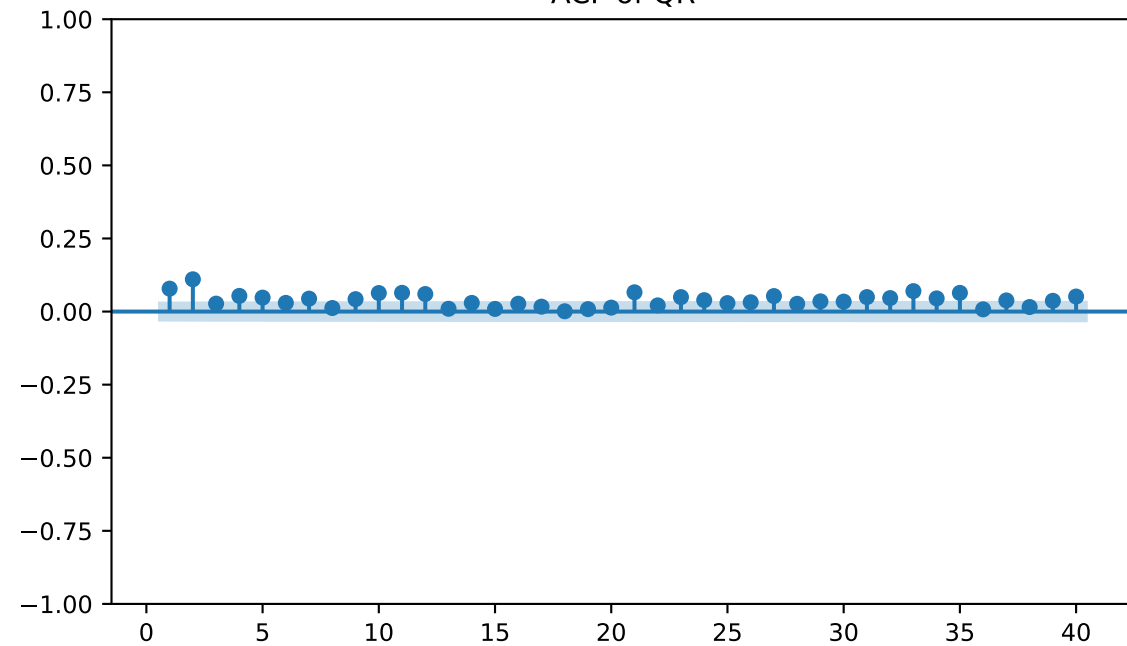
PIT histogram



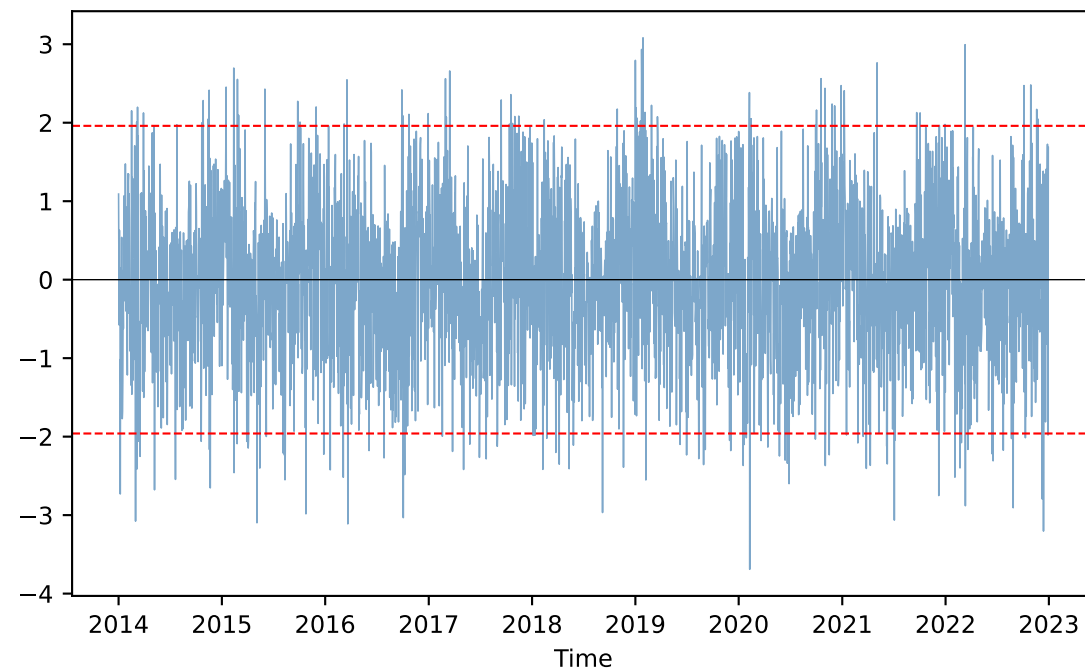
Normal QQ plot



ACF of QR

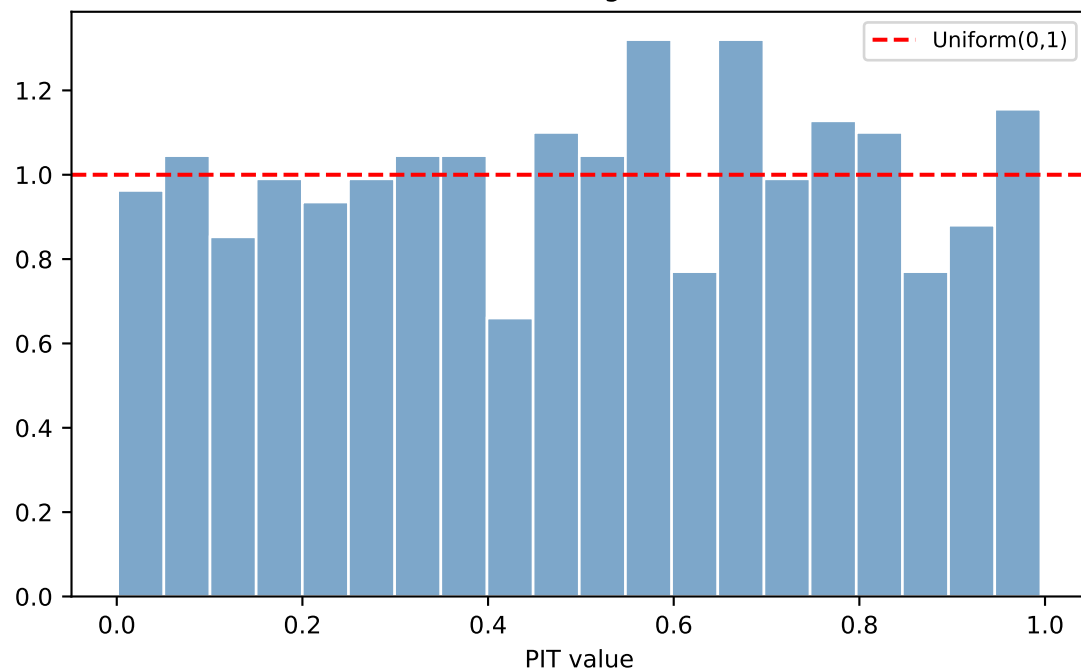


QR time series

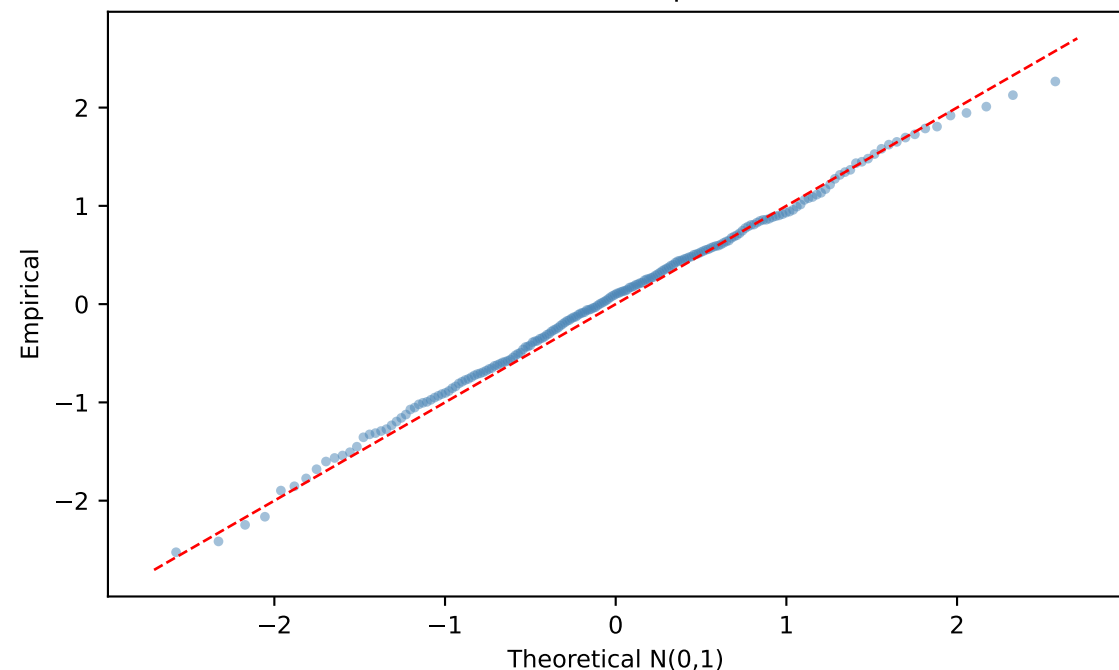


long_short_xi_interactions — OOS diagnostics

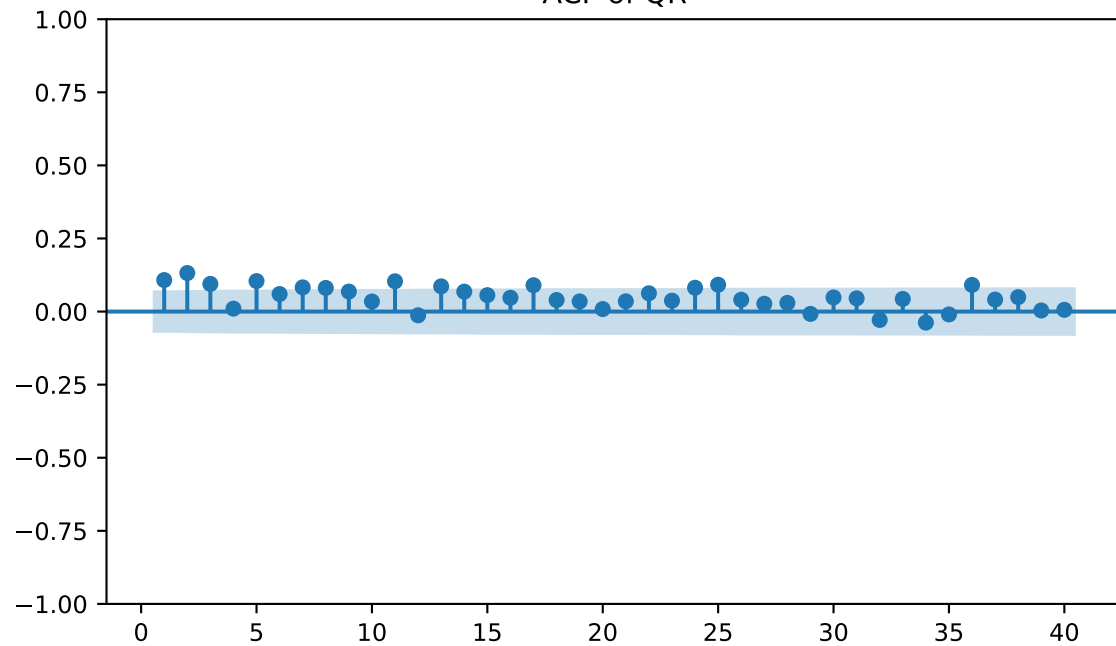
PIT histogram



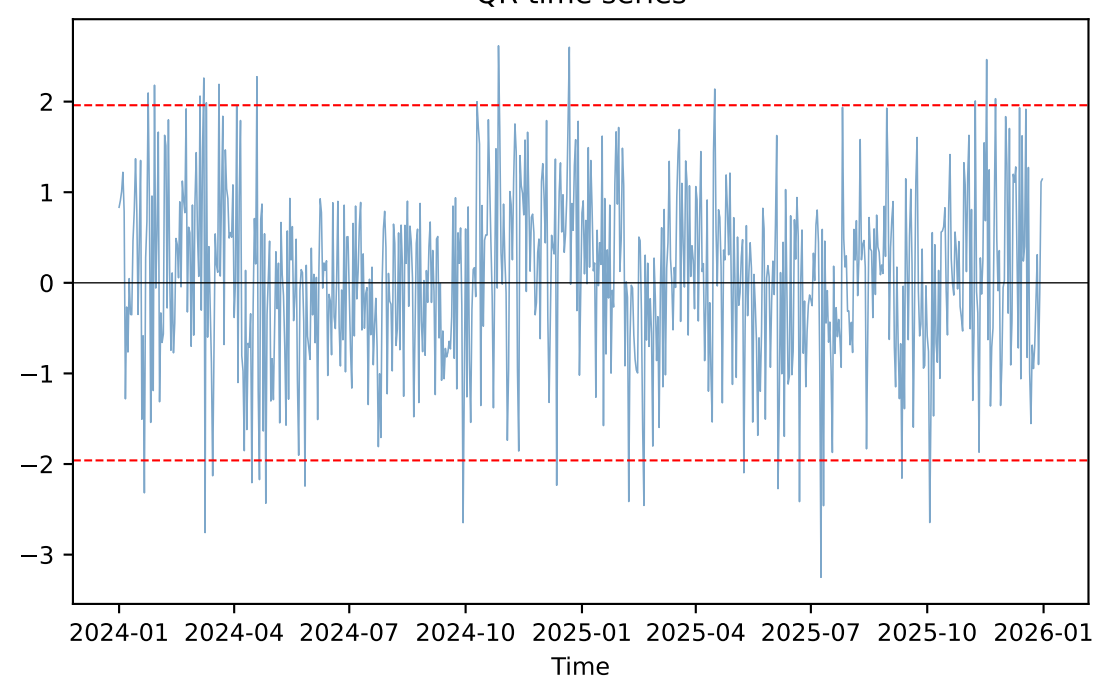
Normal QQ plot



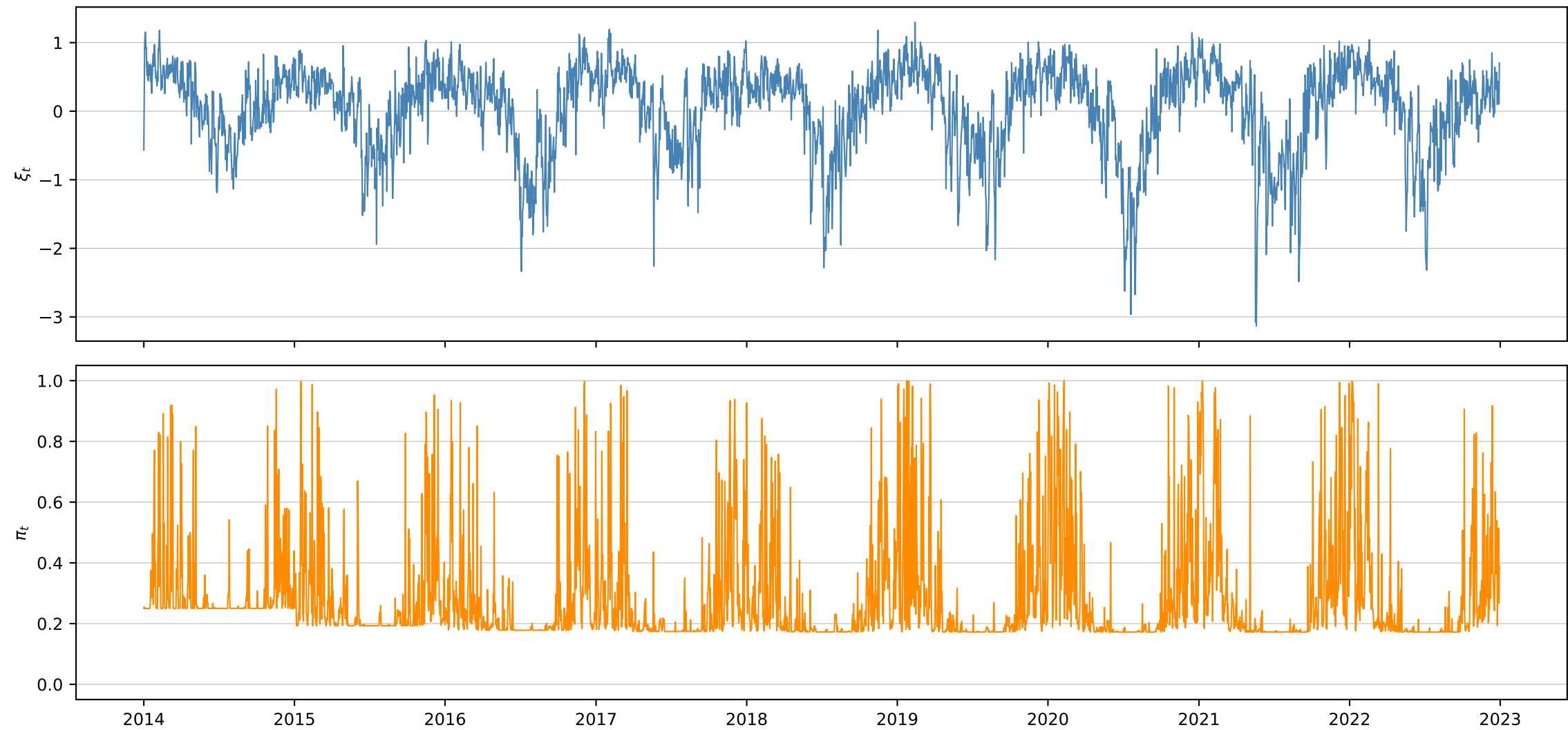
ACF of QR



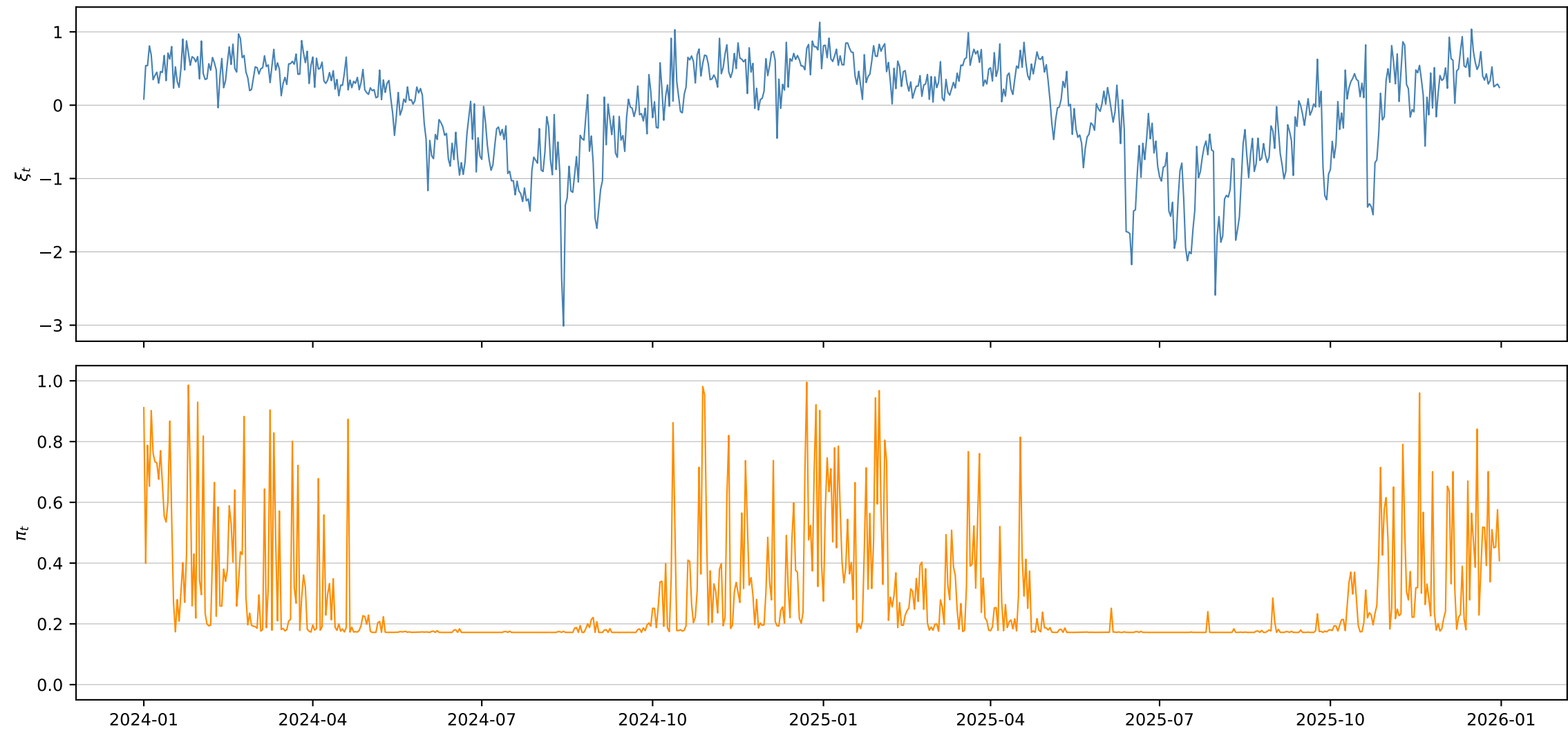
QR time series



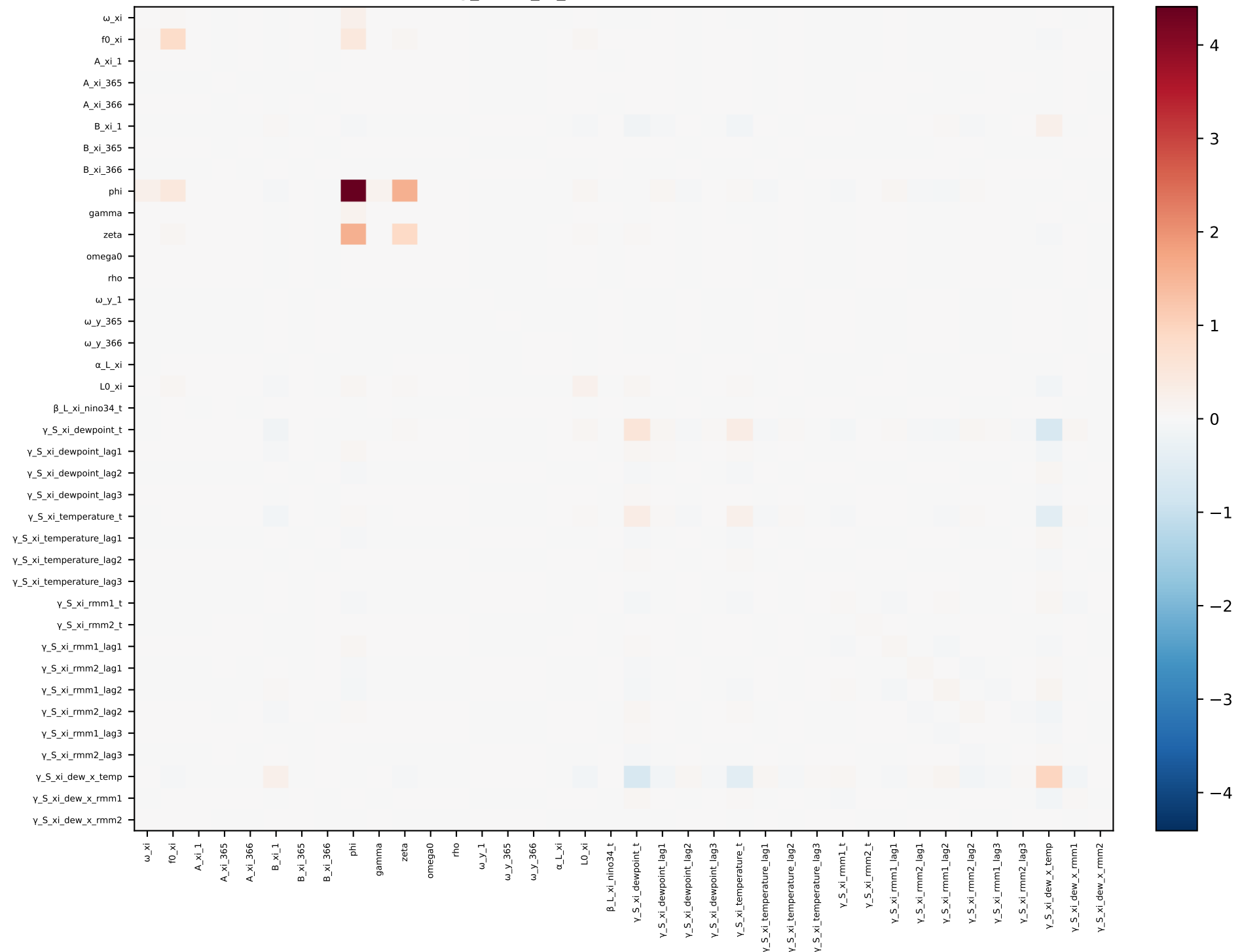
long_short_xi_interactions — filtered states (IS)



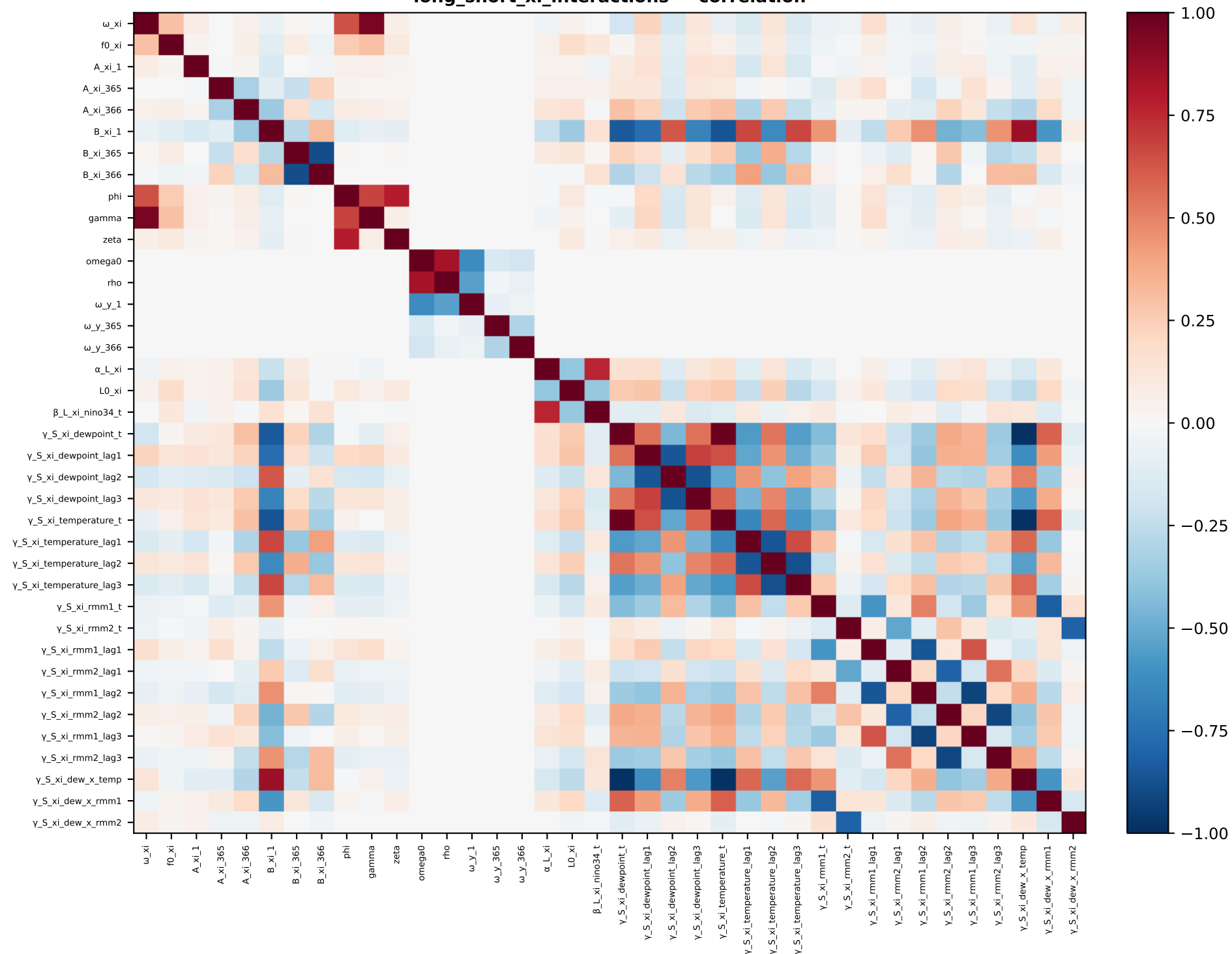
long_short_xi_interactions — filtered states (OOS)



long_short_xi_interactions — covariance

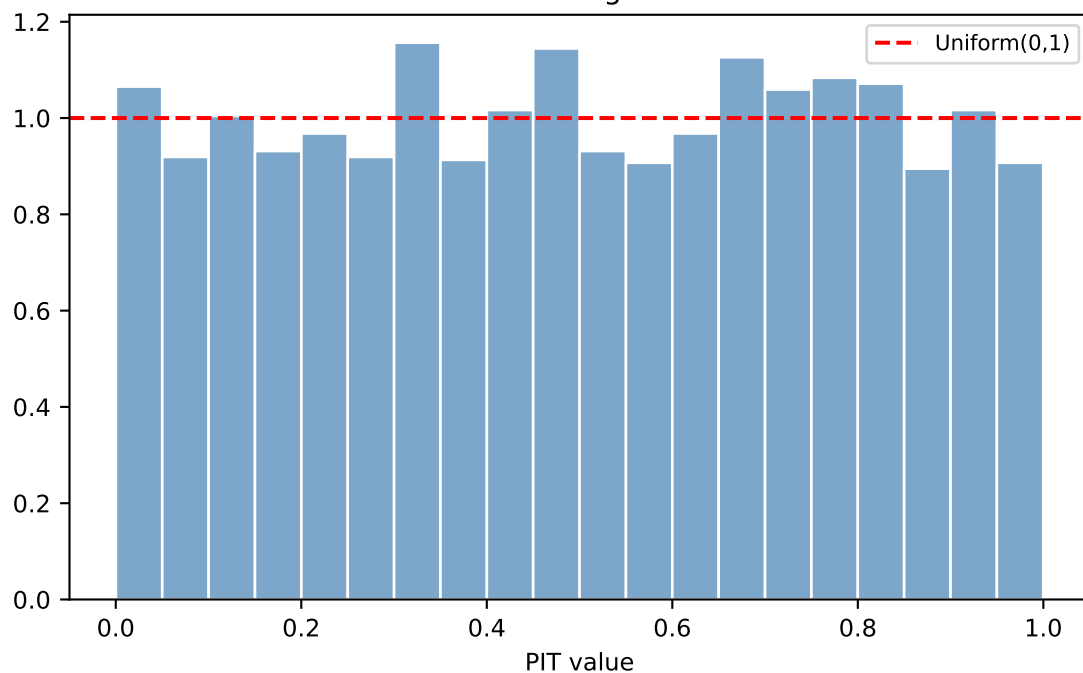


long_short_xi_interactions — correlation

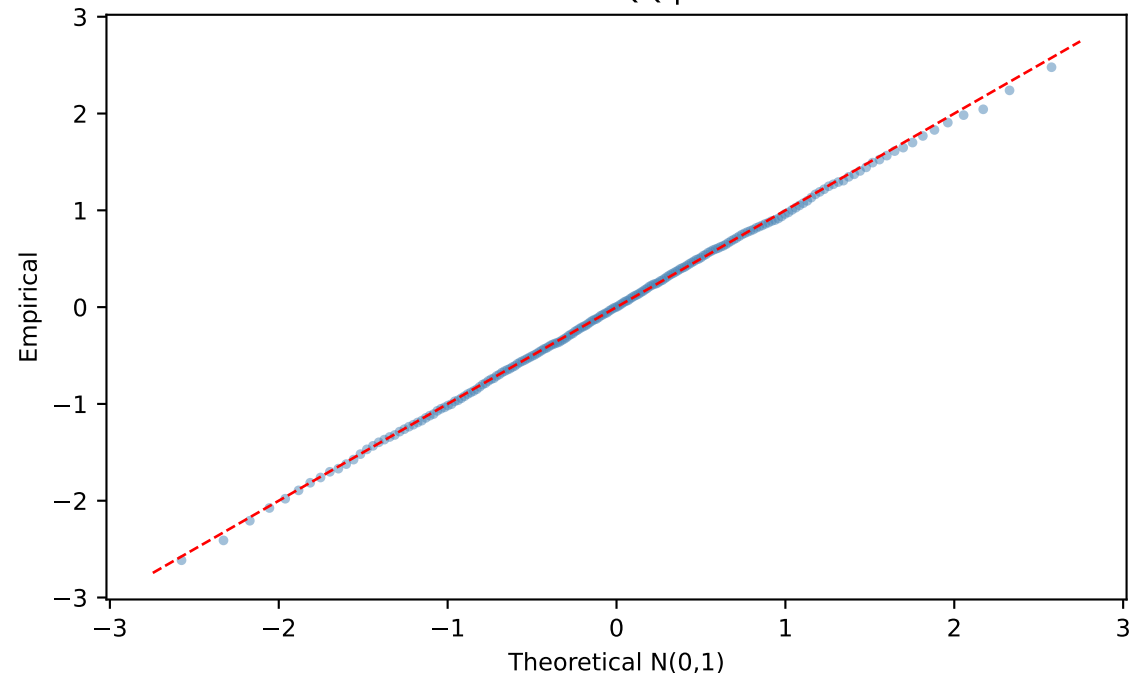


long_short_phi_basic — IS diagnostics

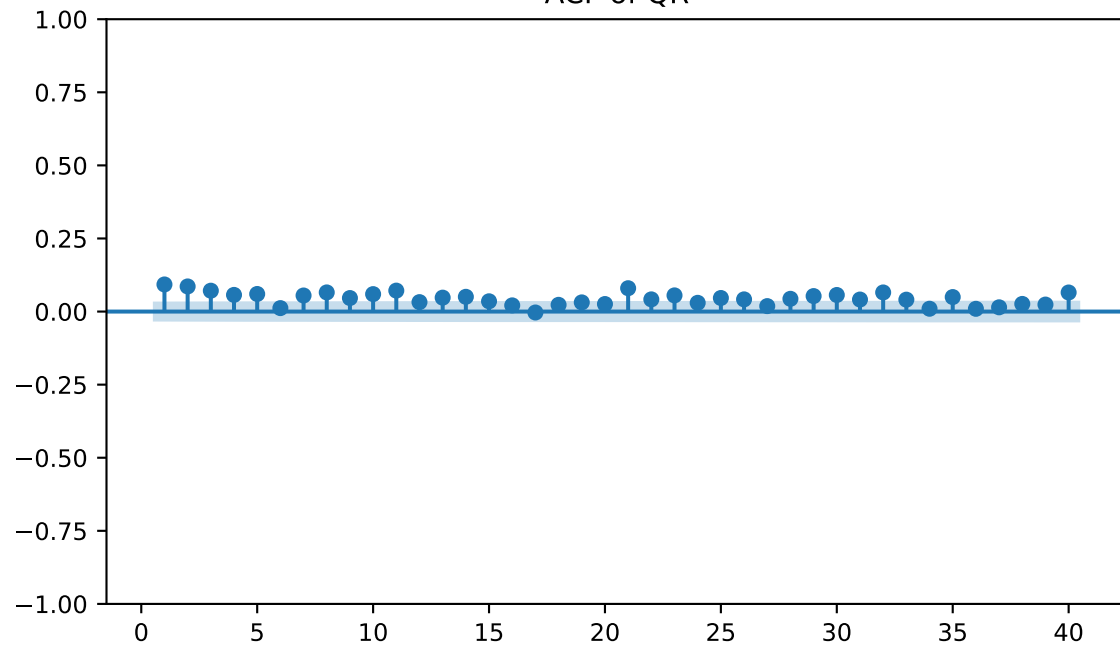
PIT histogram



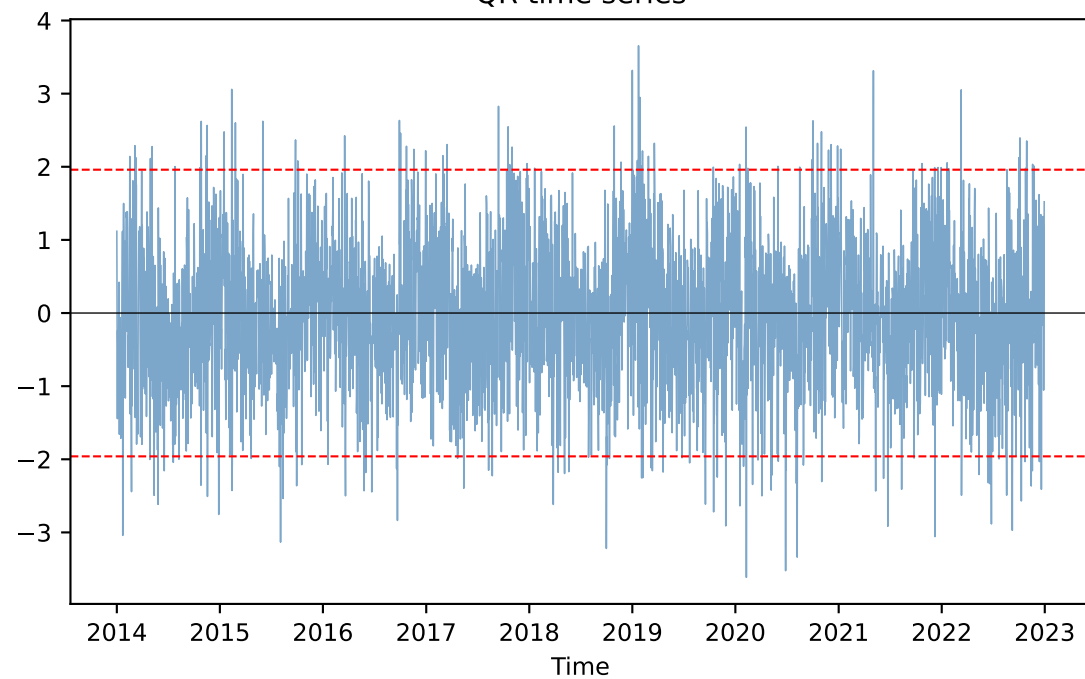
Normal QQ plot



ACF of QR

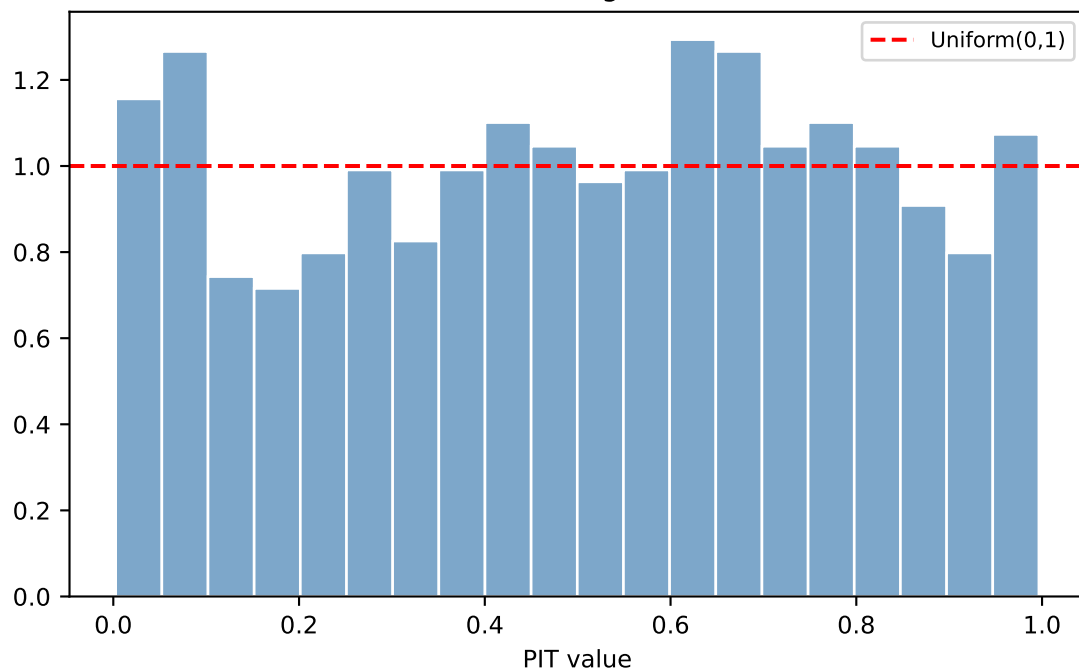


QR time series

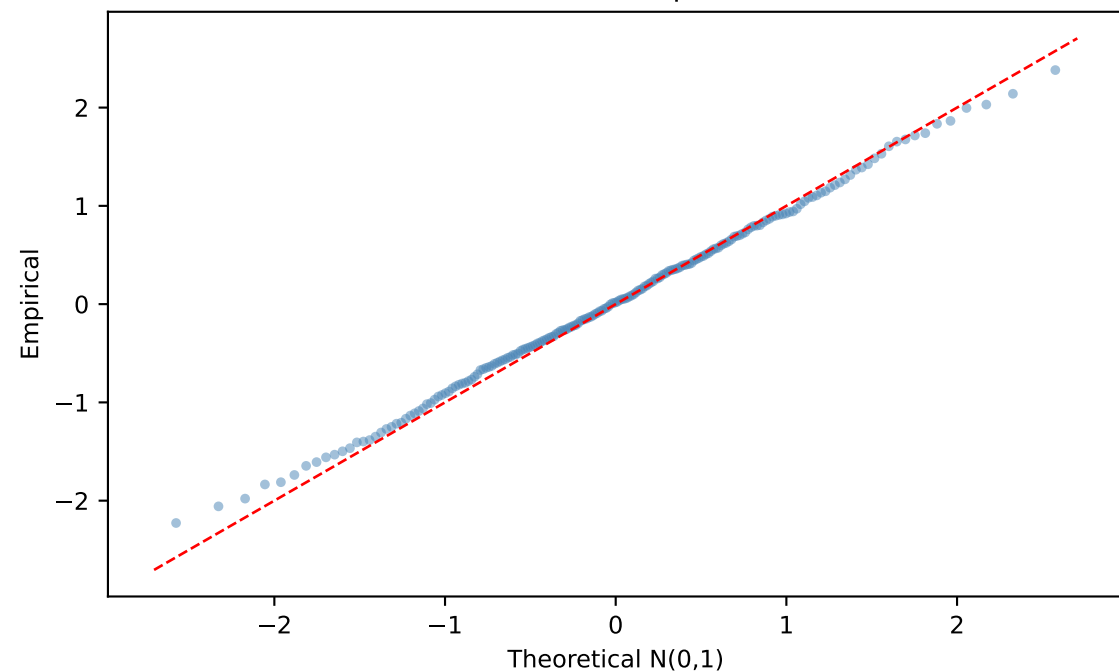


long_short_phi_basic — OOS diagnostics

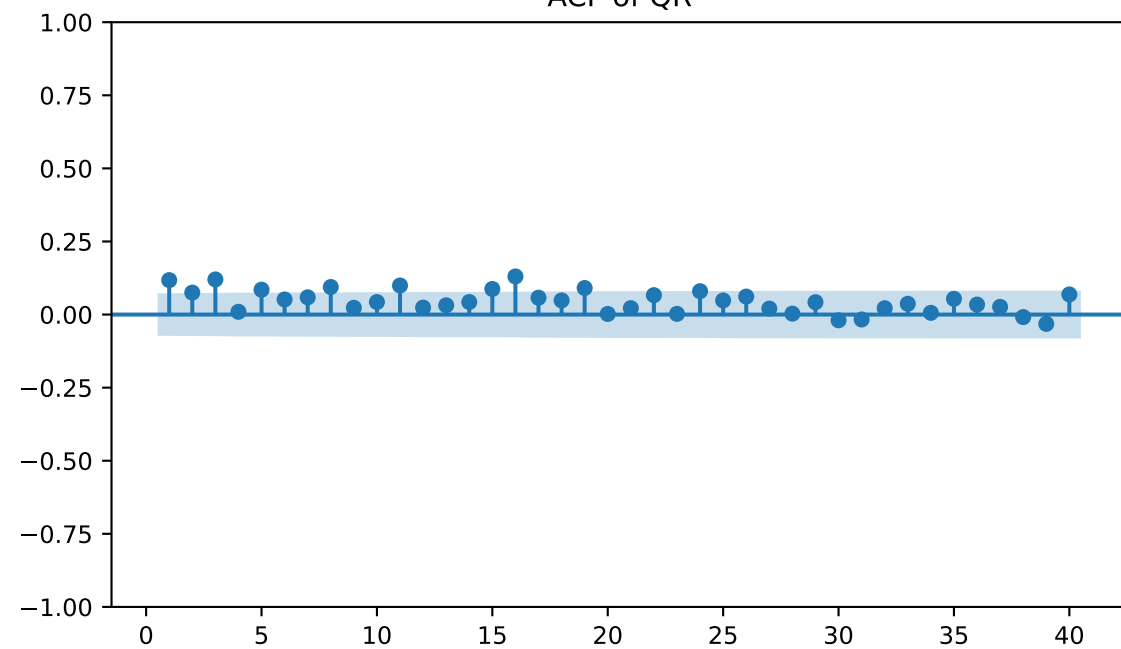
PIT histogram



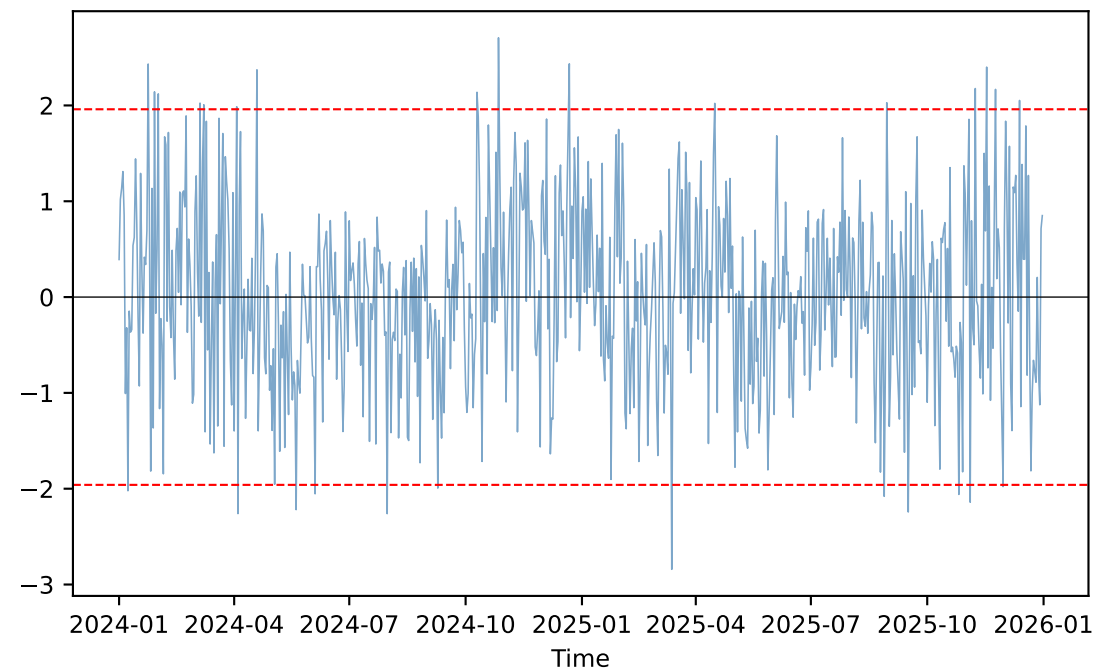
Normal QQ plot



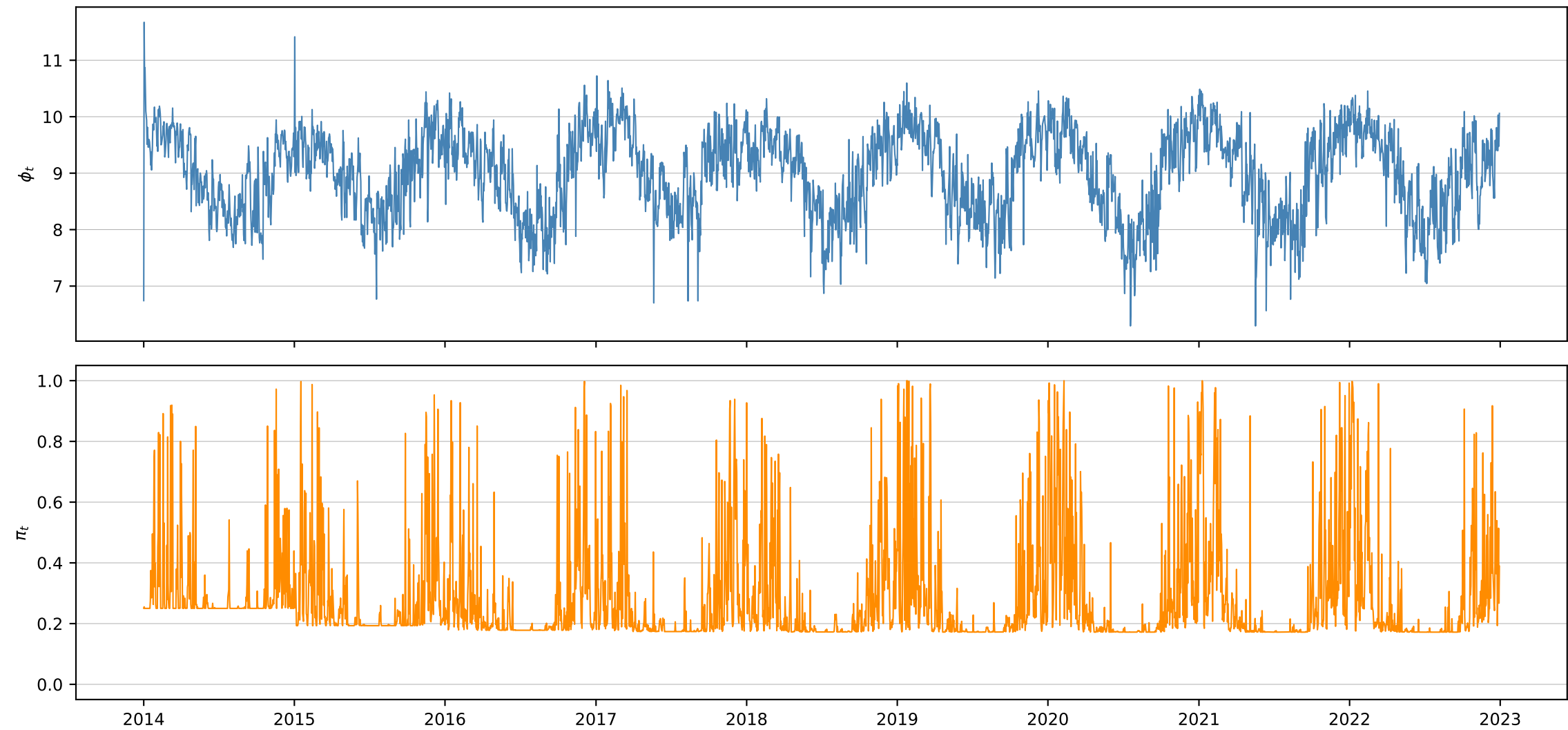
ACF of QR



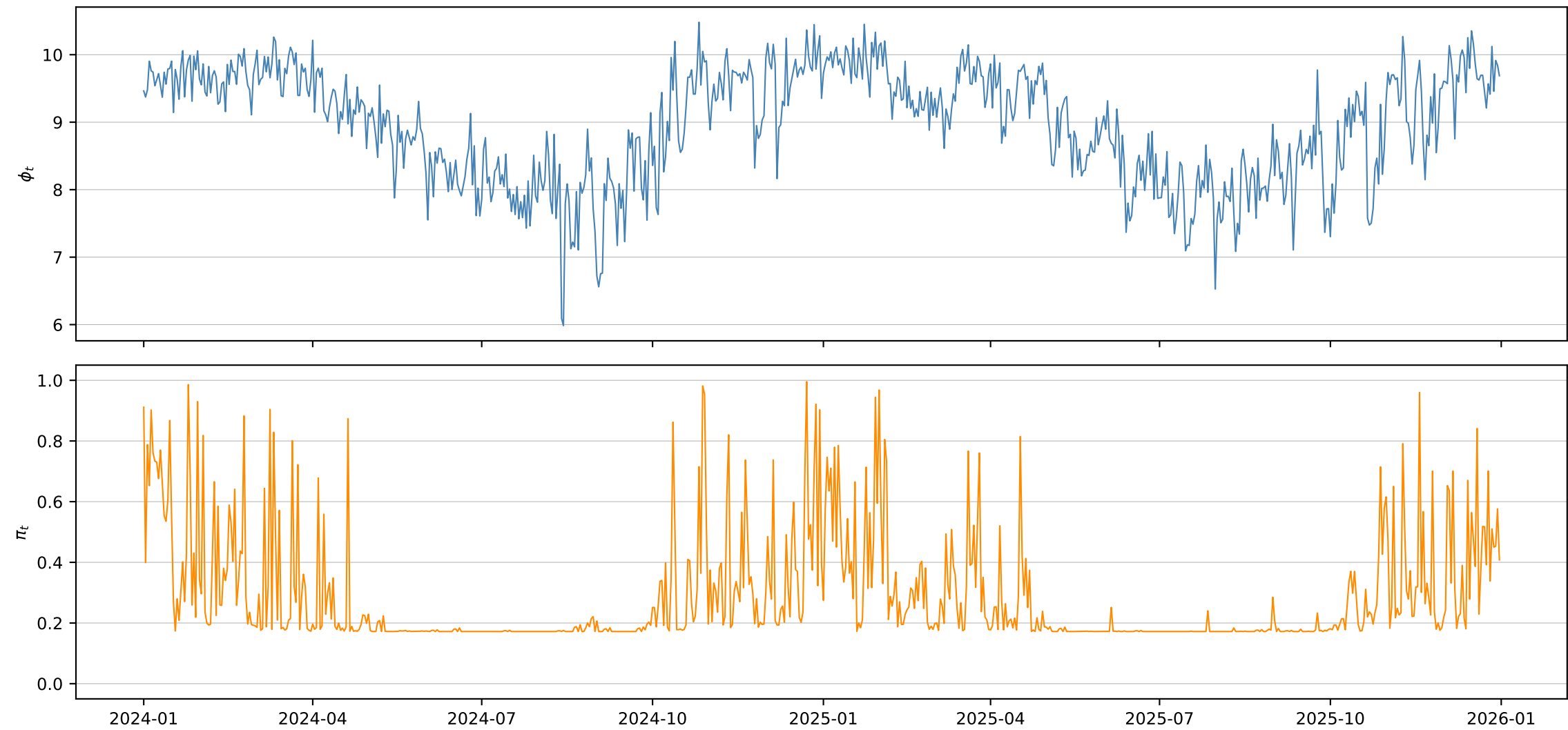
QR time series



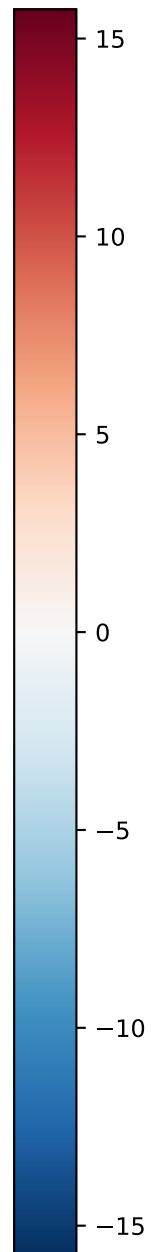
long_short_phi_basic — filtered states (IS)



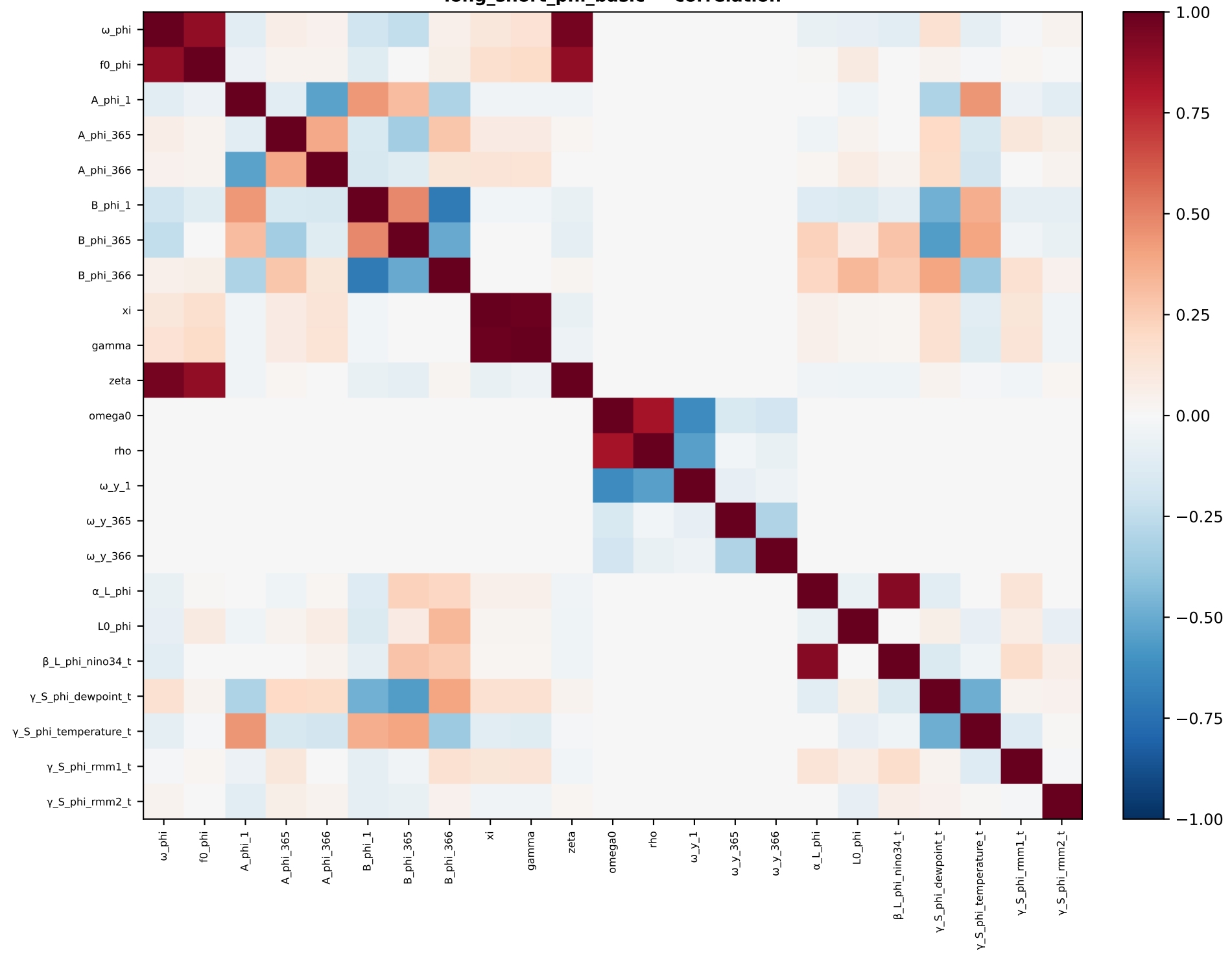
long_short_phi_basic — filtered states (OOS)



Heatmap visualization showing the correlation matrix for 20 variables. The variables are listed on both the x and y axes. The color scale ranges from -0.5 (dark blue) to 0.5 (dark red), with 0 being white. The diagonal elements are all 1.0 (white). The matrix is symmetric. Notable correlations include a strong positive correlation between ω_{ϕ} and $f0_{\phi}$ (dark red), and a strong negative correlation between ω_{ϕ} and $\alpha_L\phi$ (dark blue).

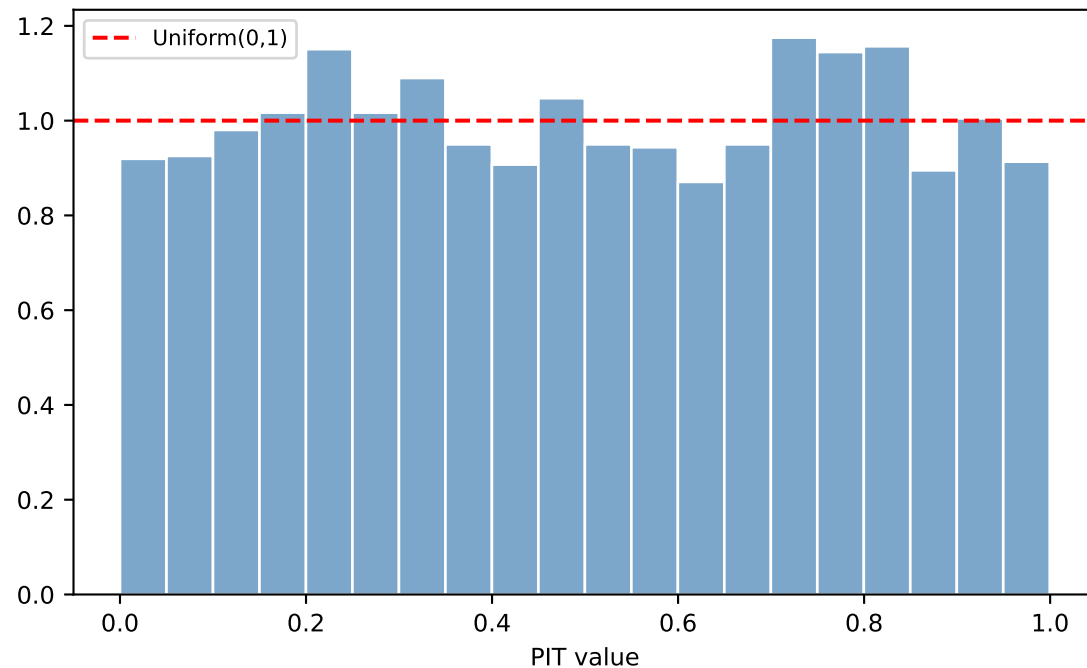


long_short_phi_basic — correlation

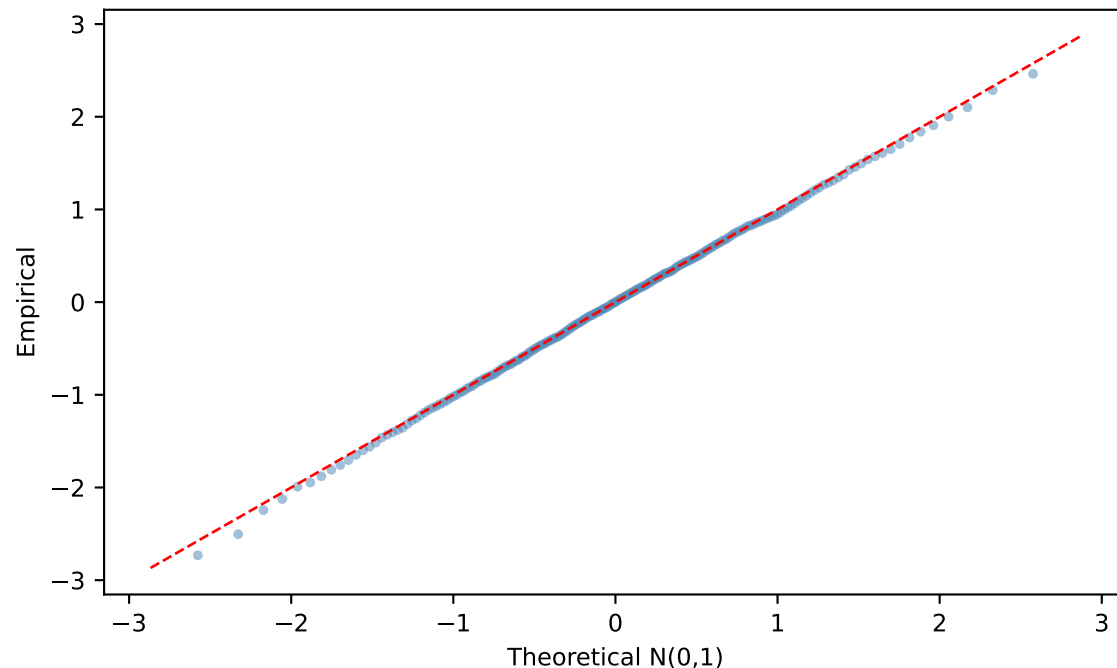


long_short_phi_short_lags — IS diagnostics

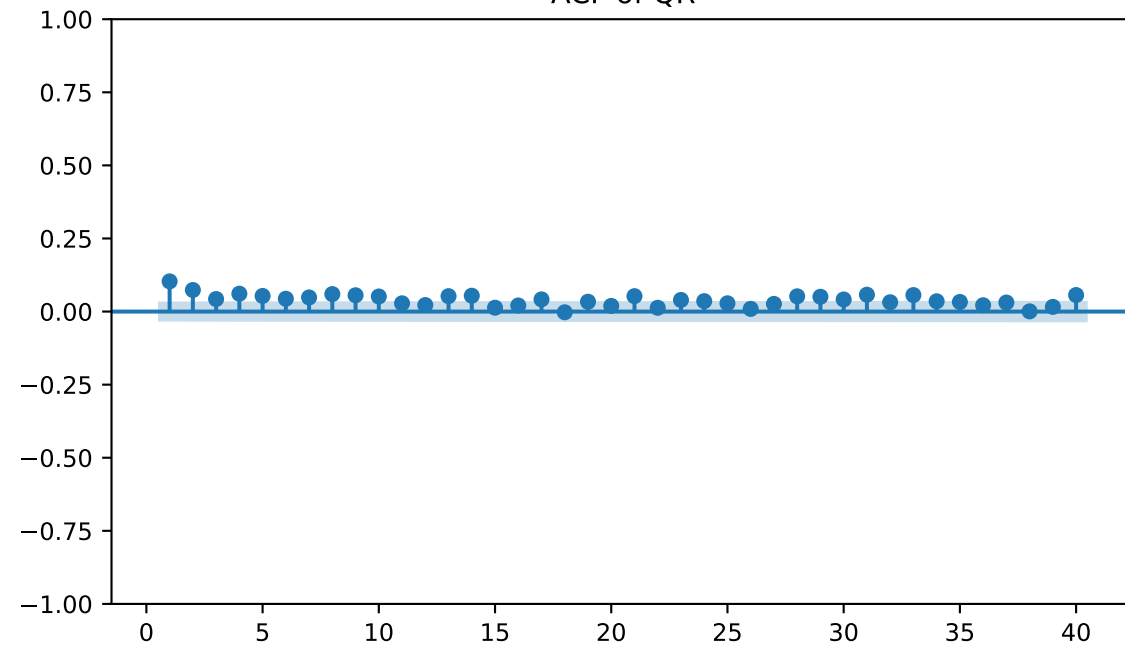
PIT histogram



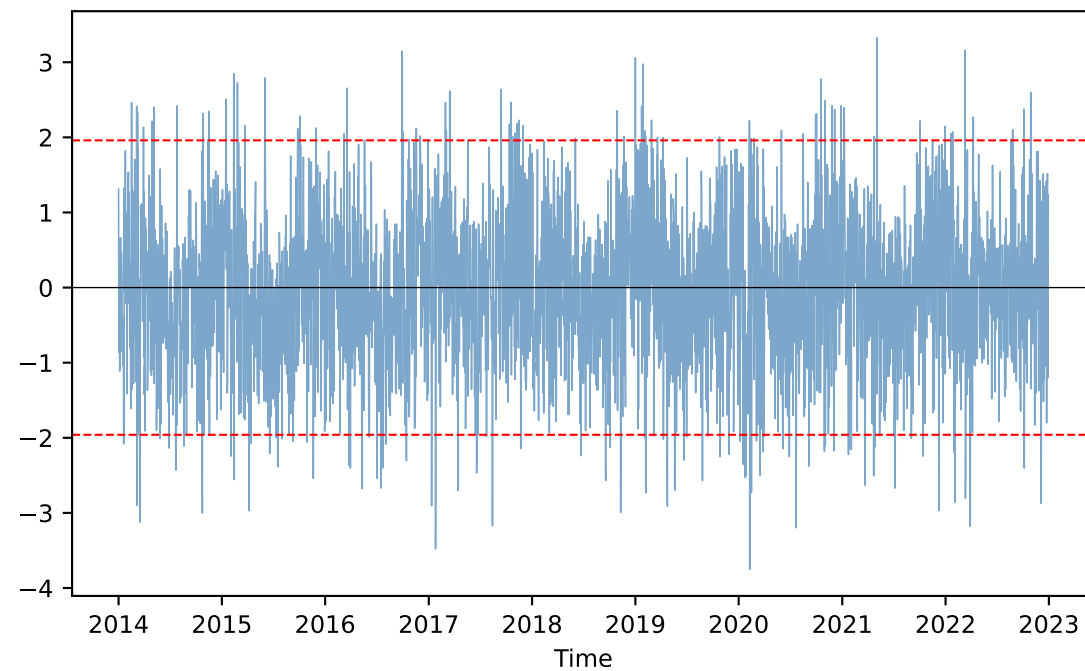
Normal QQ plot



ACF of QR

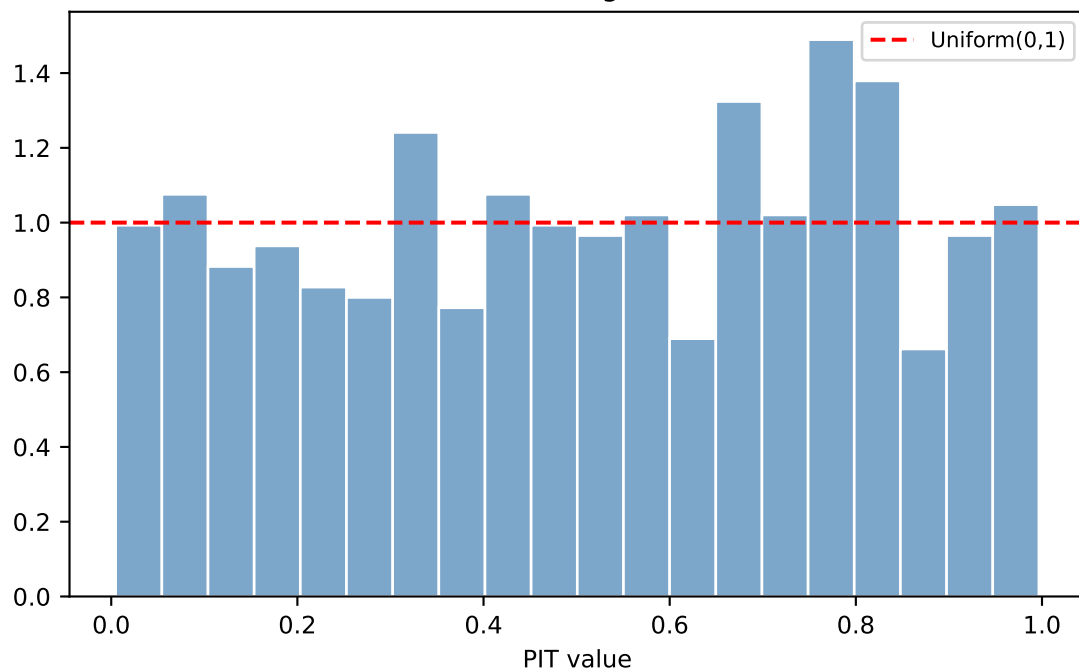


QR time series

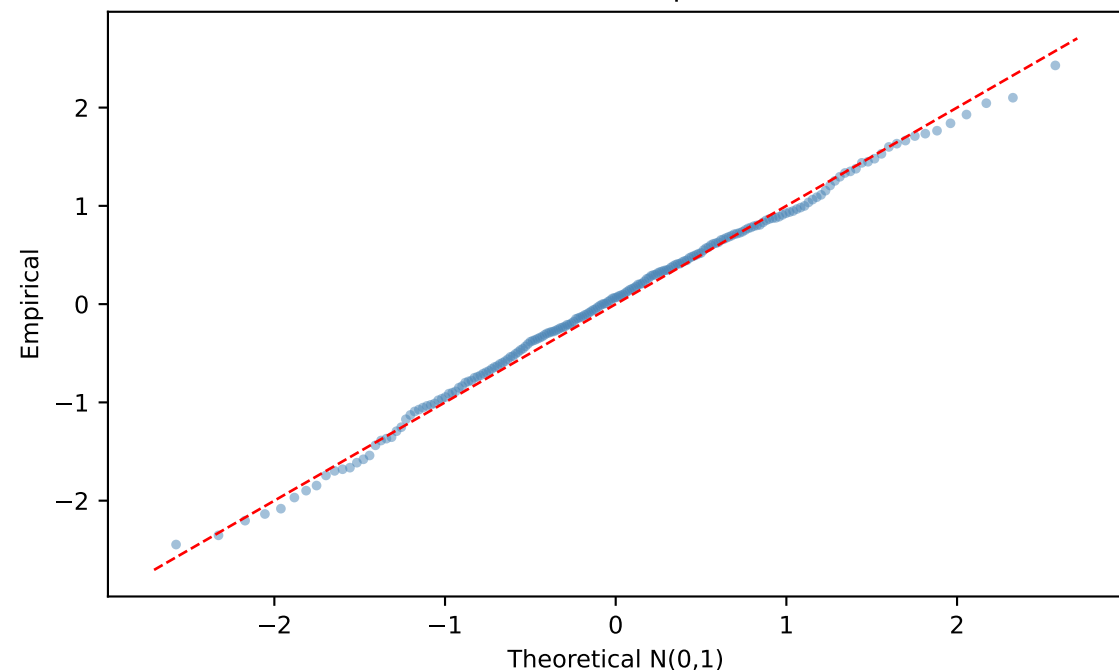


long_short_phi_short_lags — OOS diagnostics

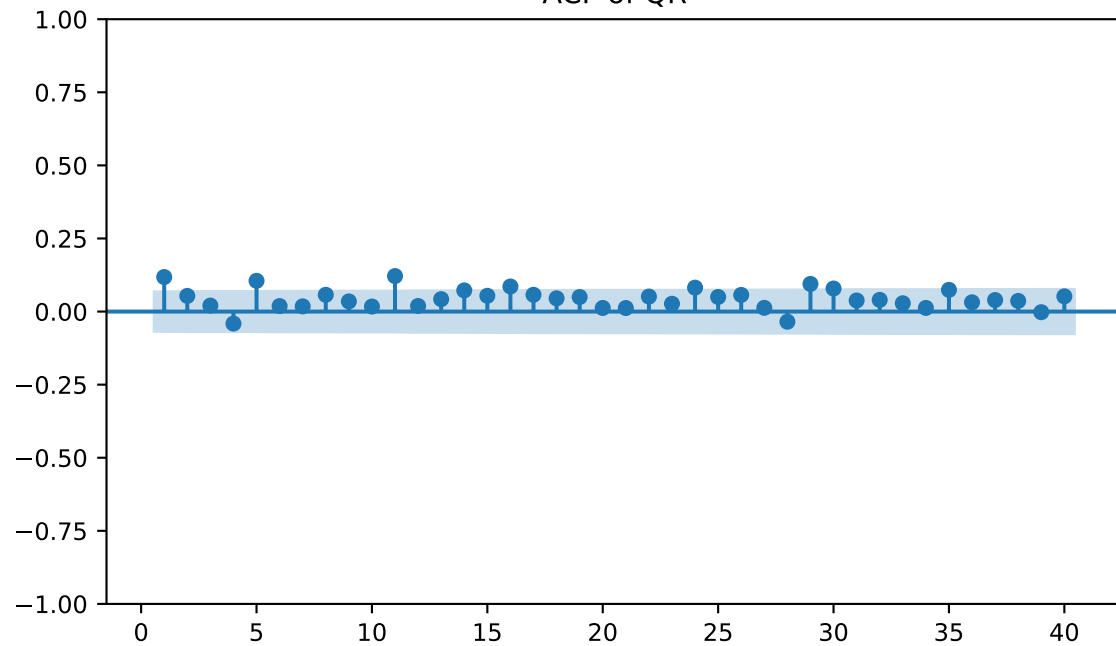
PIT histogram



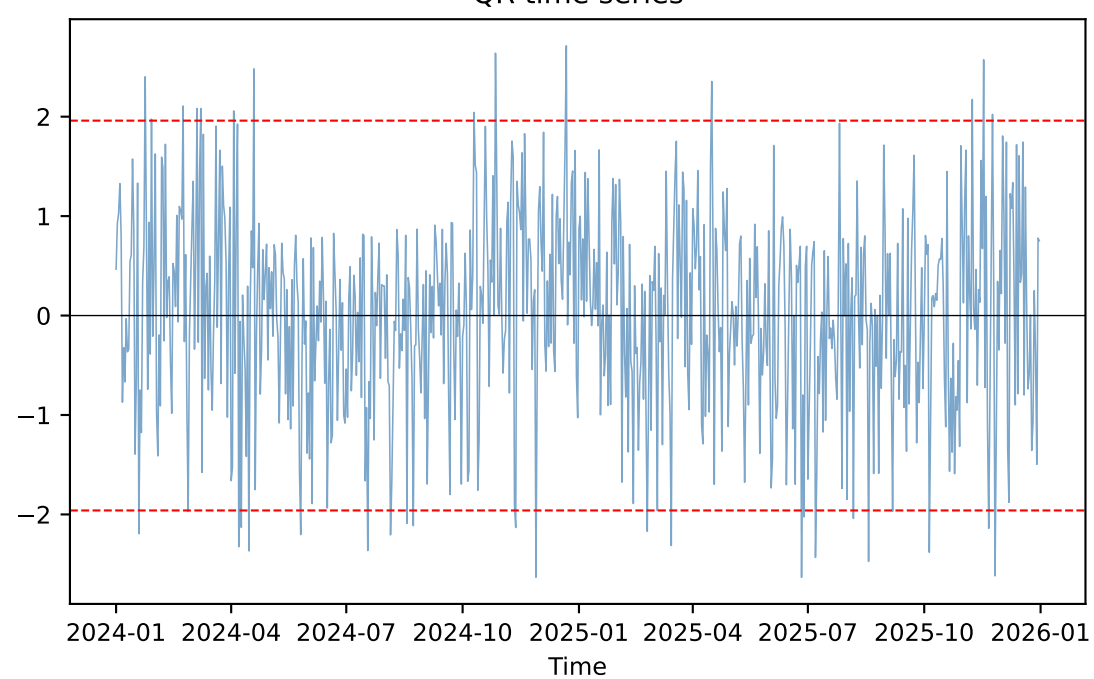
Normal QQ plot



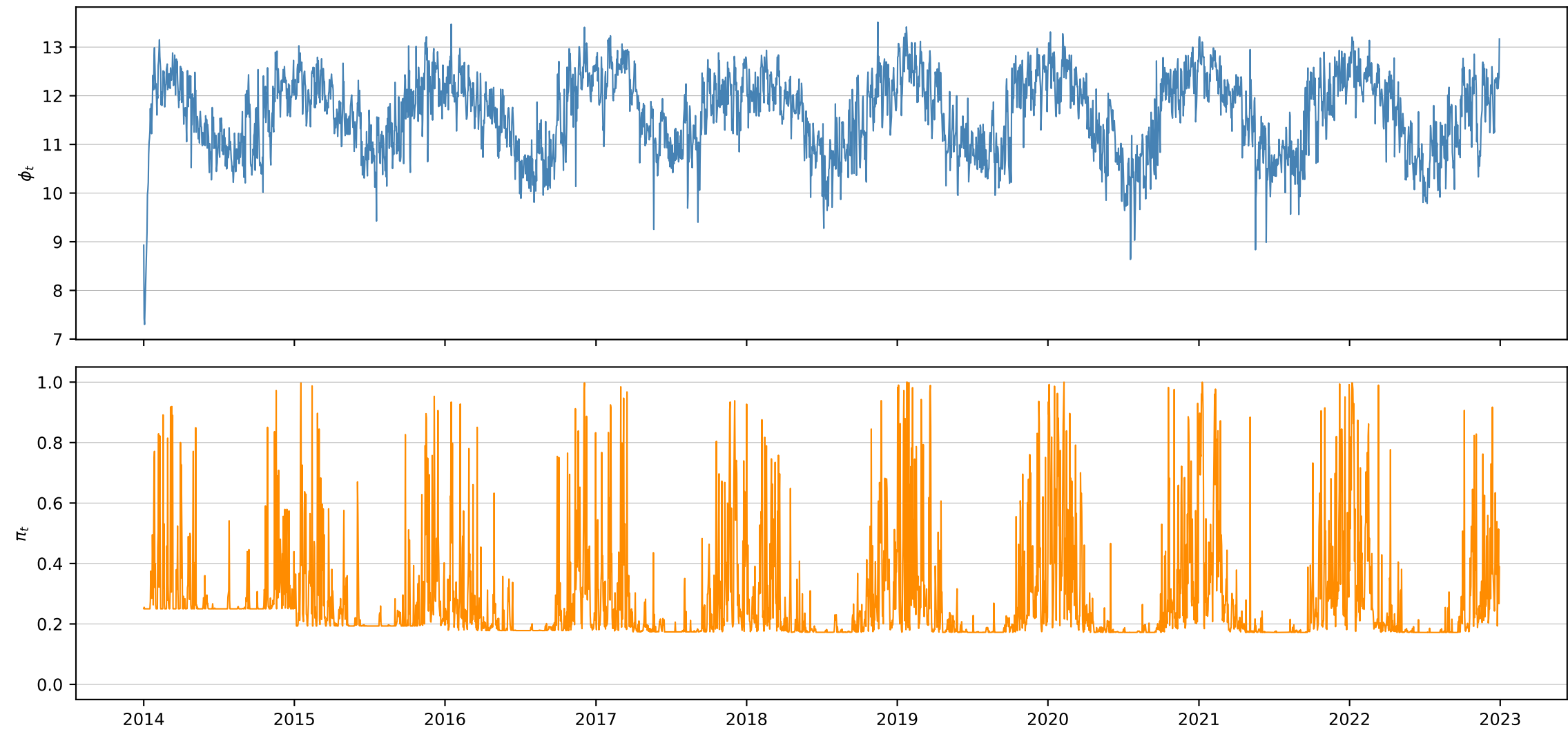
ACF of QR



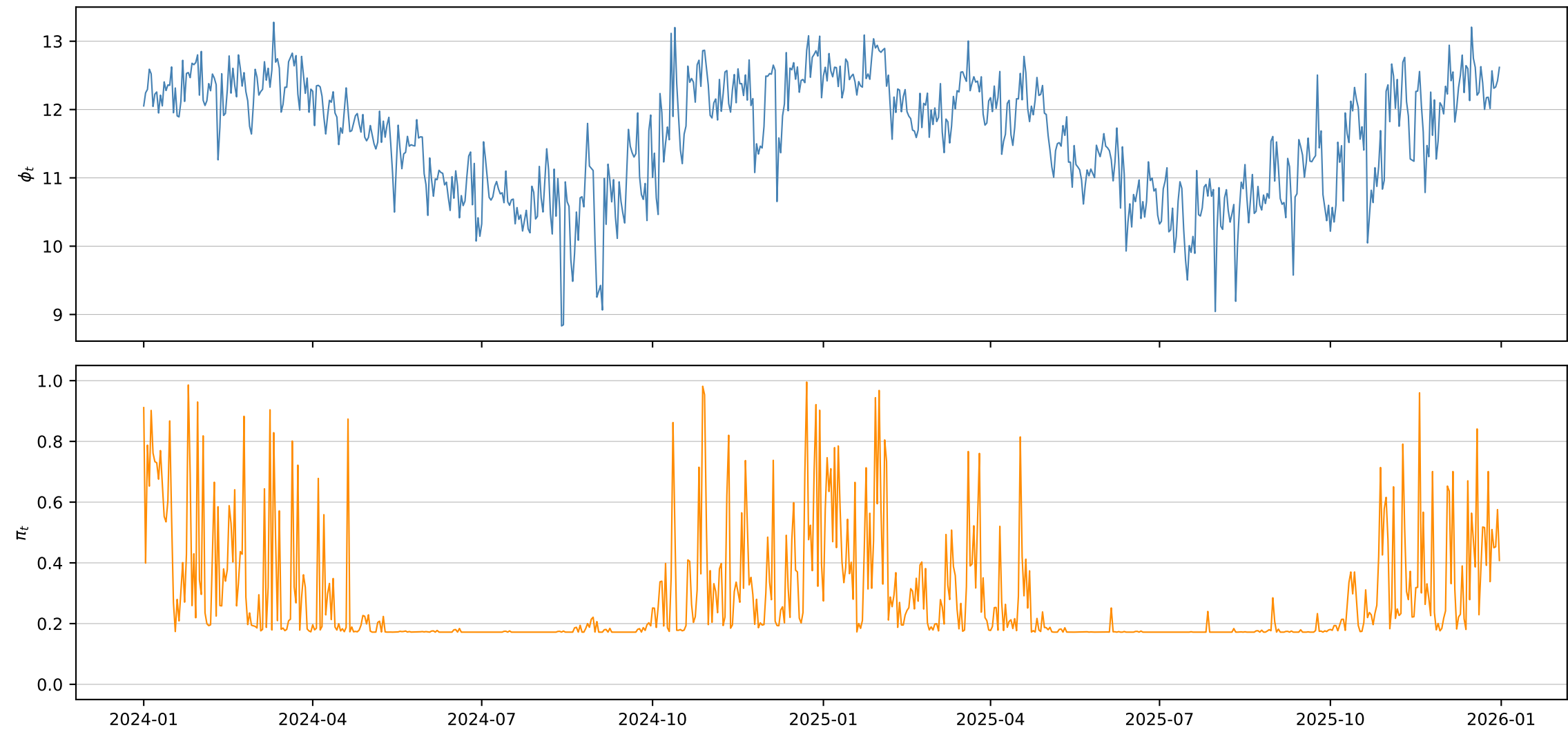
QR time series



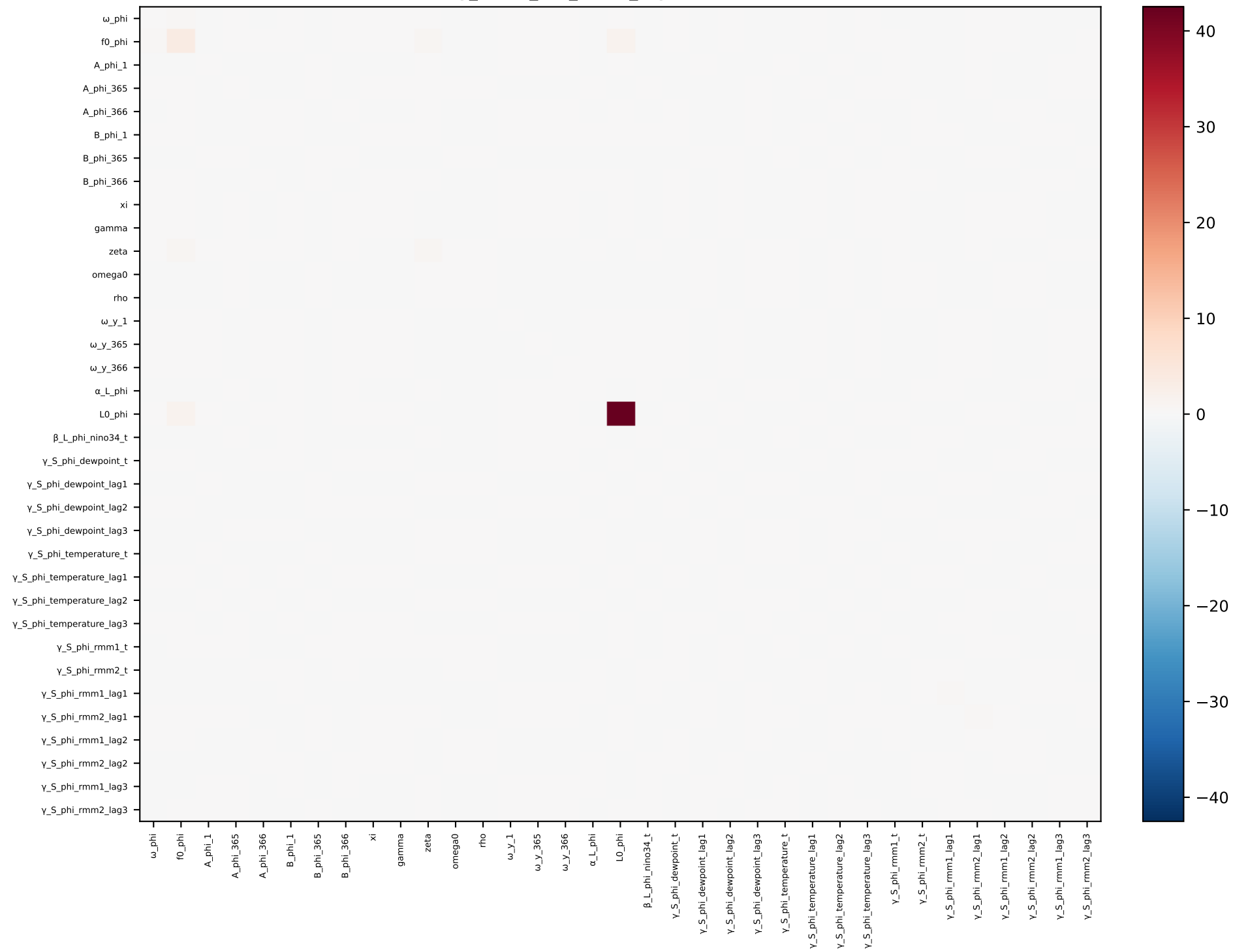
long_short_phi_short_lags — filtered states (IS)



long_short_phi_short_lags — filtered states (OOS)

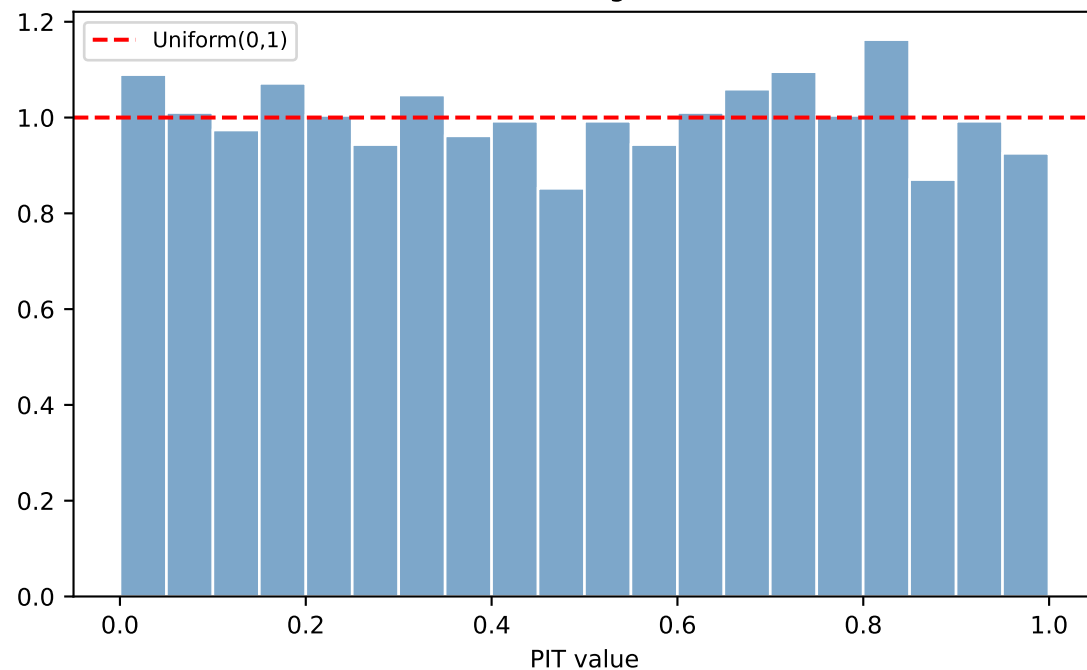


long_short_phi_short_lags — covariance

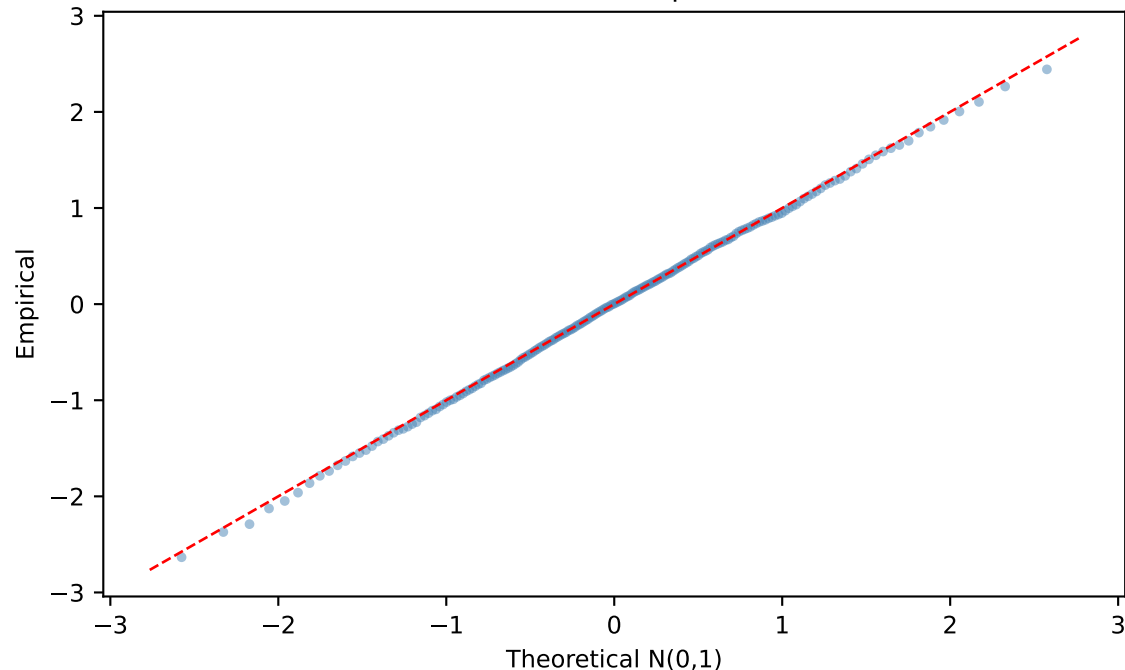


long_short_phi_weather_climate — IS diagnostics

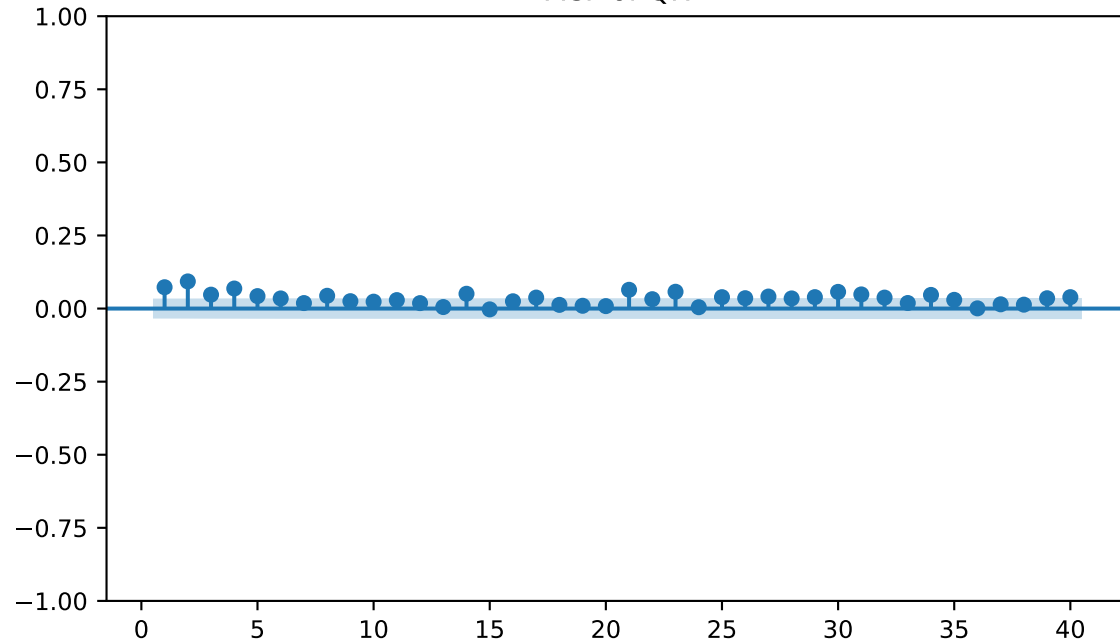
PIT histogram



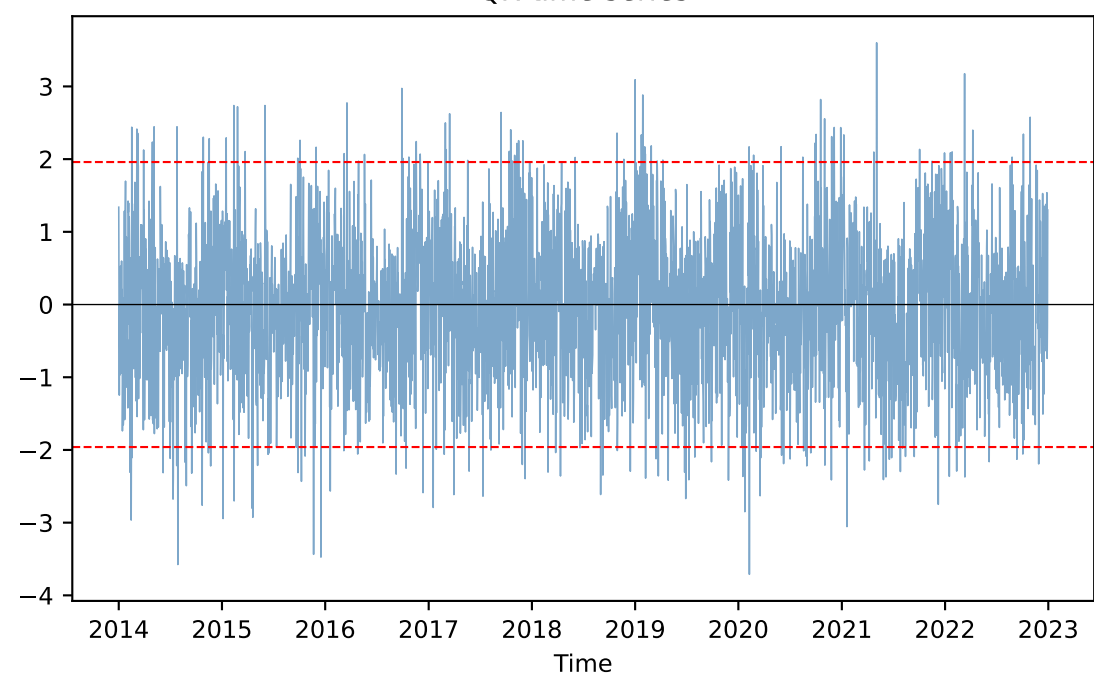
Normal QQ plot



ACF of QR

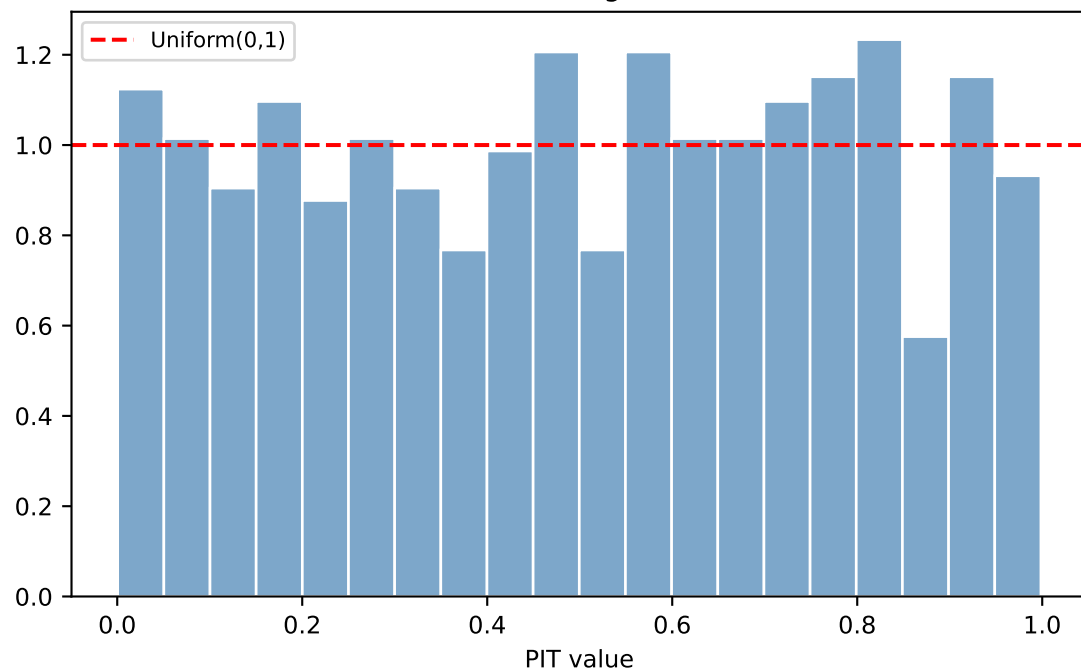


QR time series

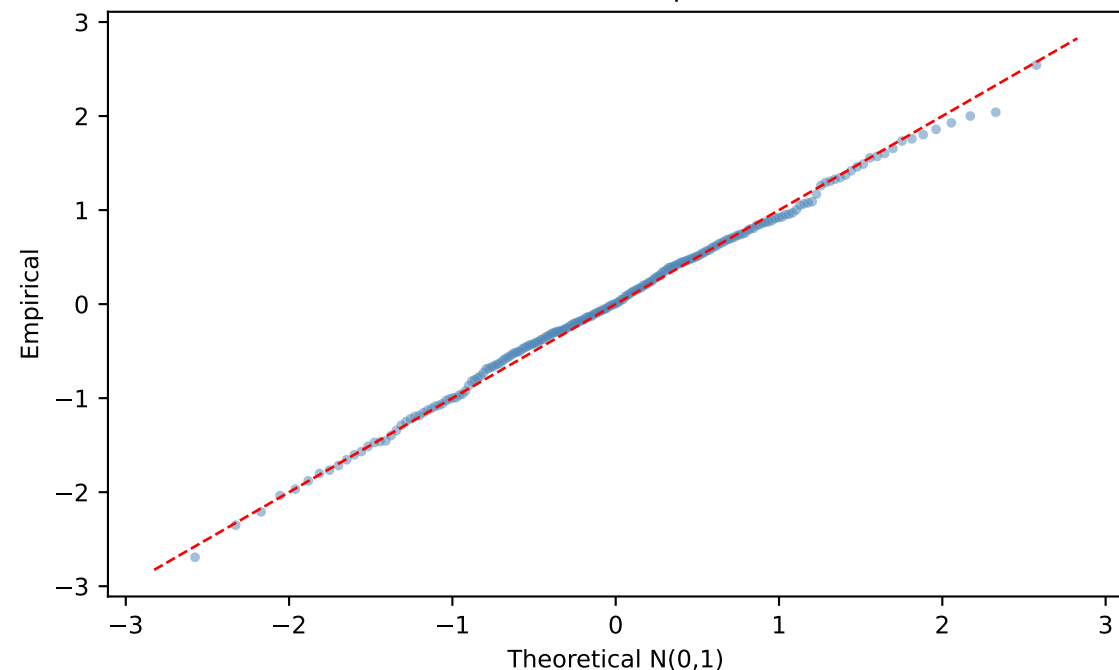


long_short_phi_weather_climate — OOS diagnostics

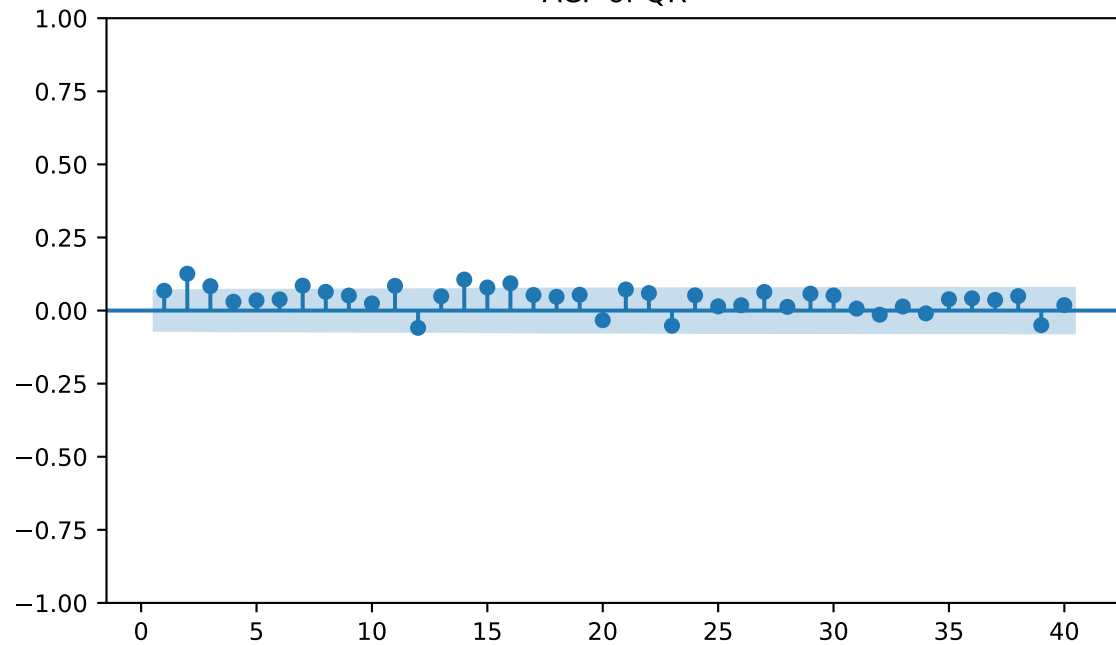
PIT histogram



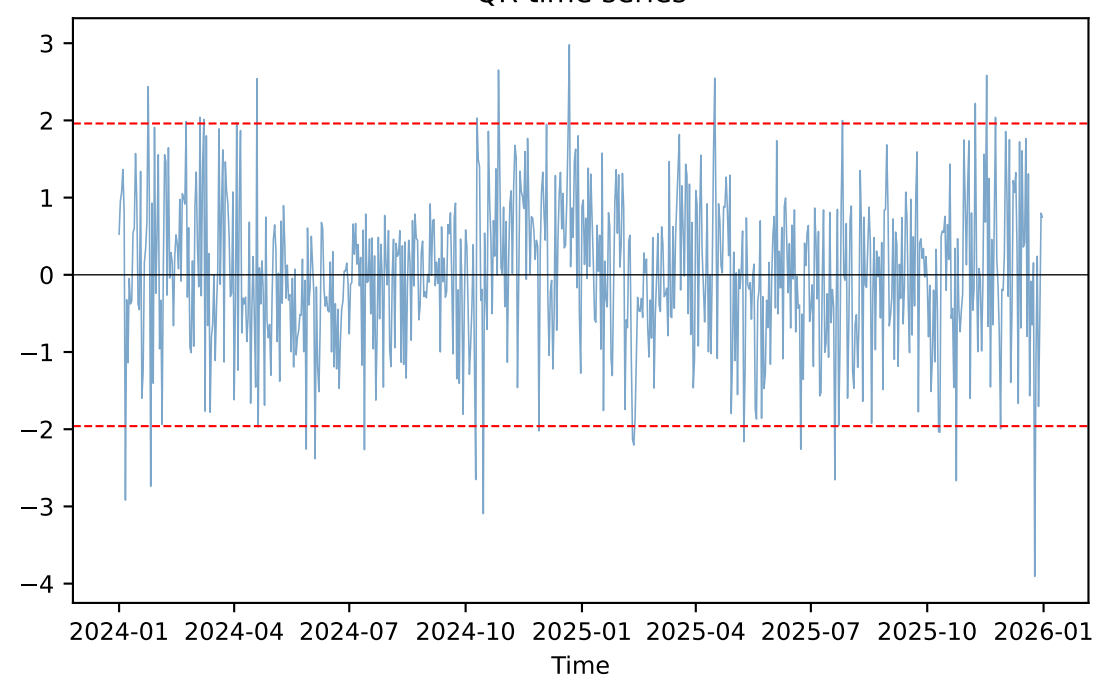
Normal QQ plot



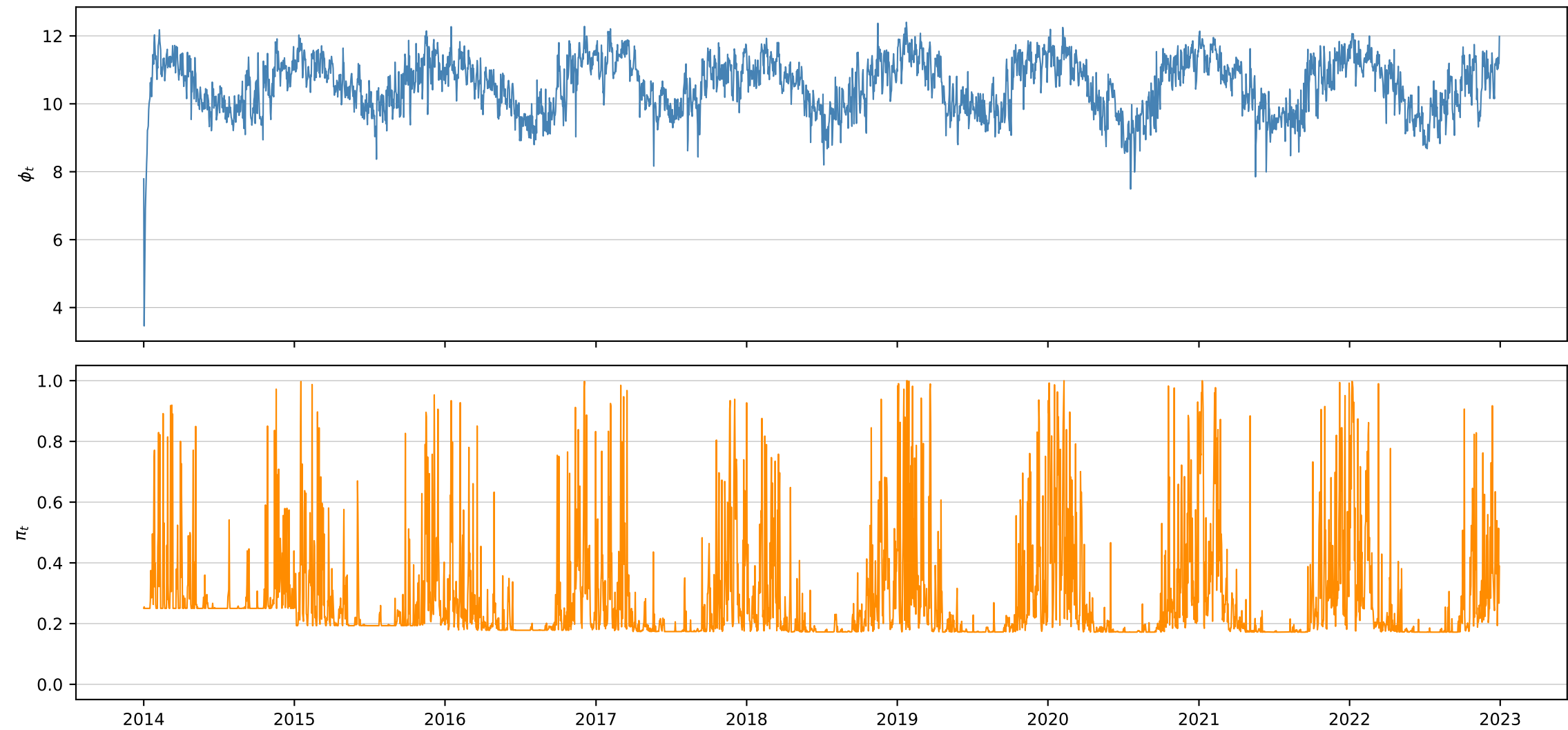
ACF of QR



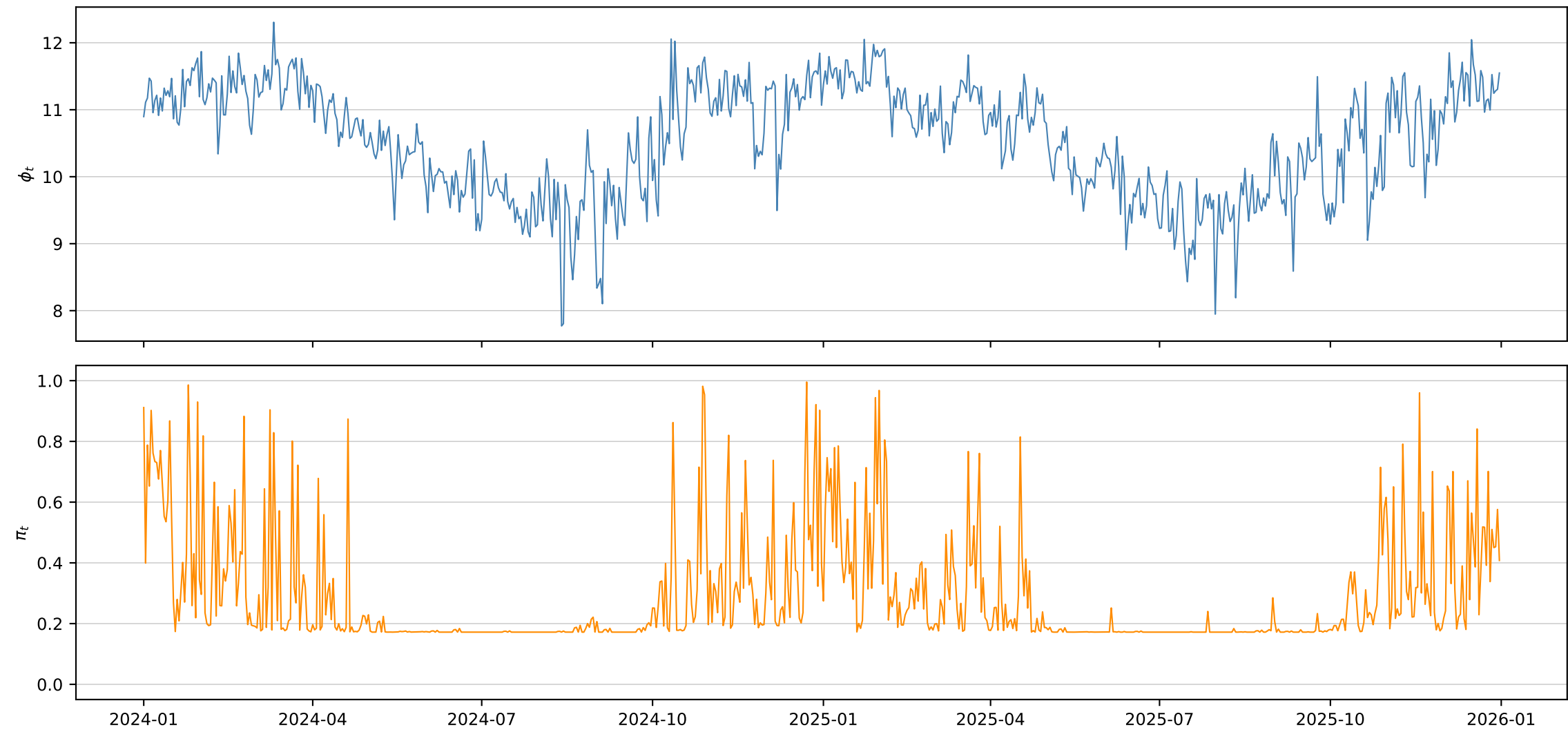
QR time series



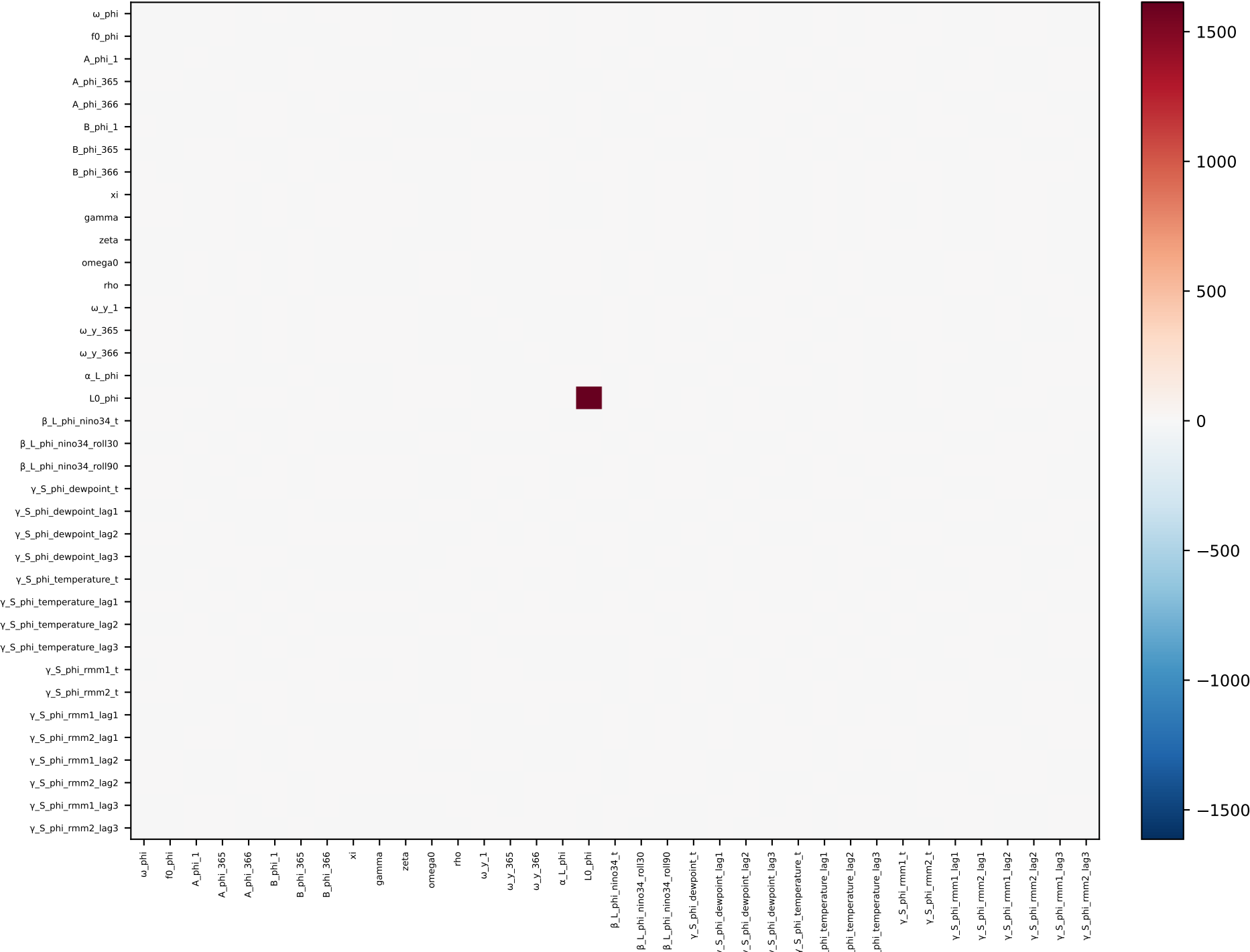
long_short_phi_weather_climate — filtered states (IS)



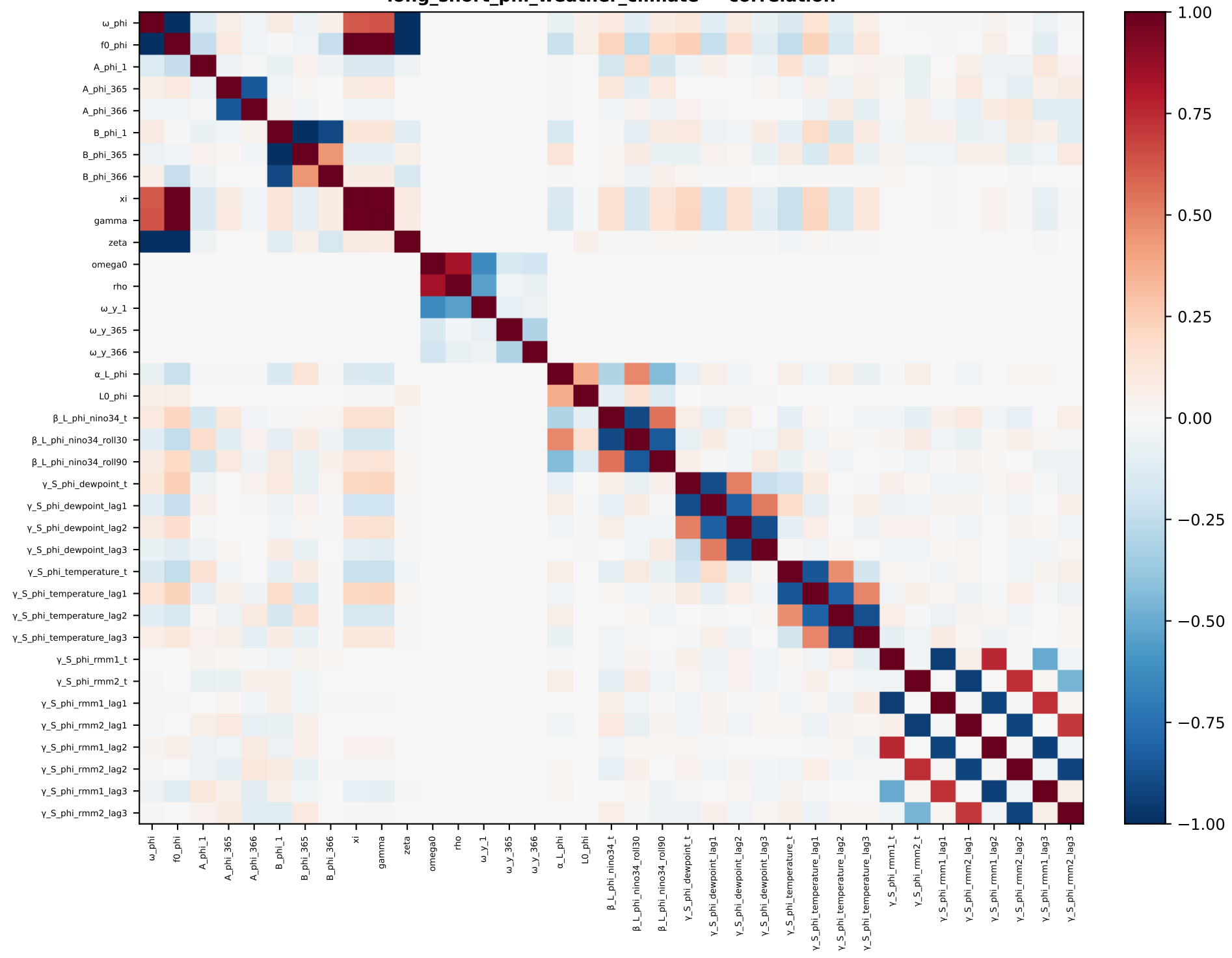
long_short_phi_weather_climate — filtered states (OOS)



long_short_phi_weather_climate – covariance

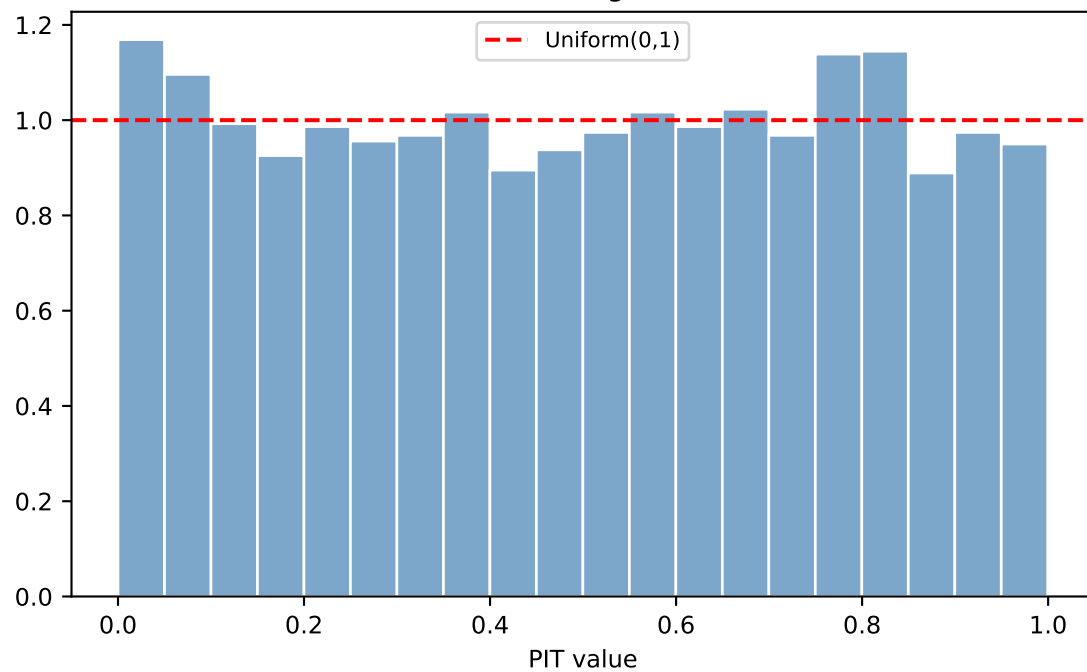


long_short_phi_weather_climate — correlation

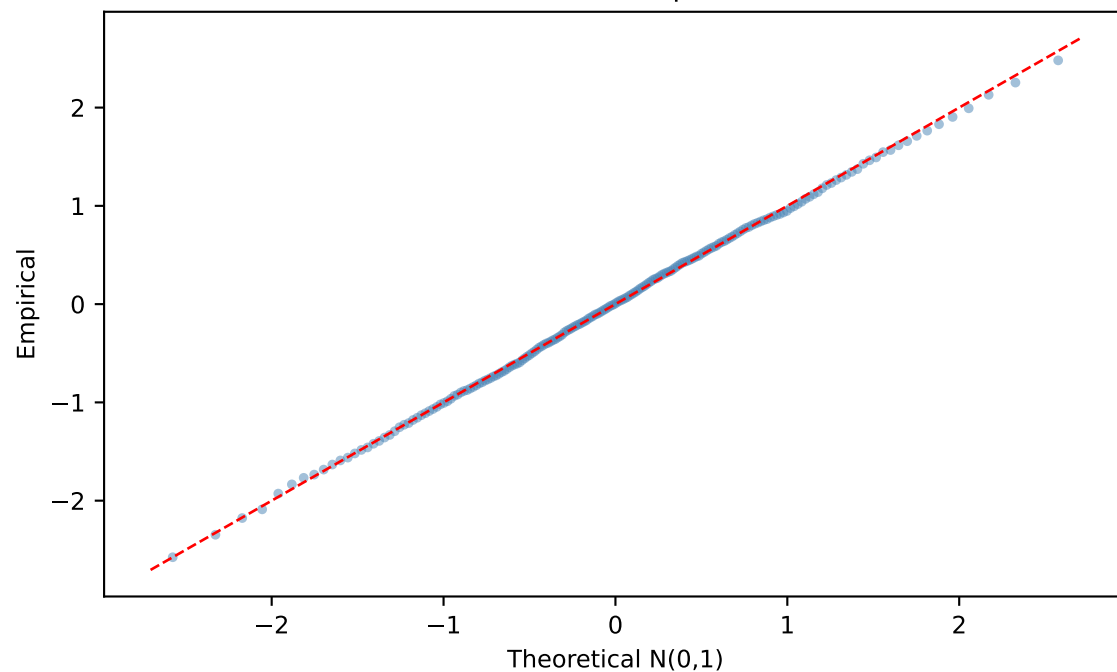


long_short_phi_interactions — IS diagnostics

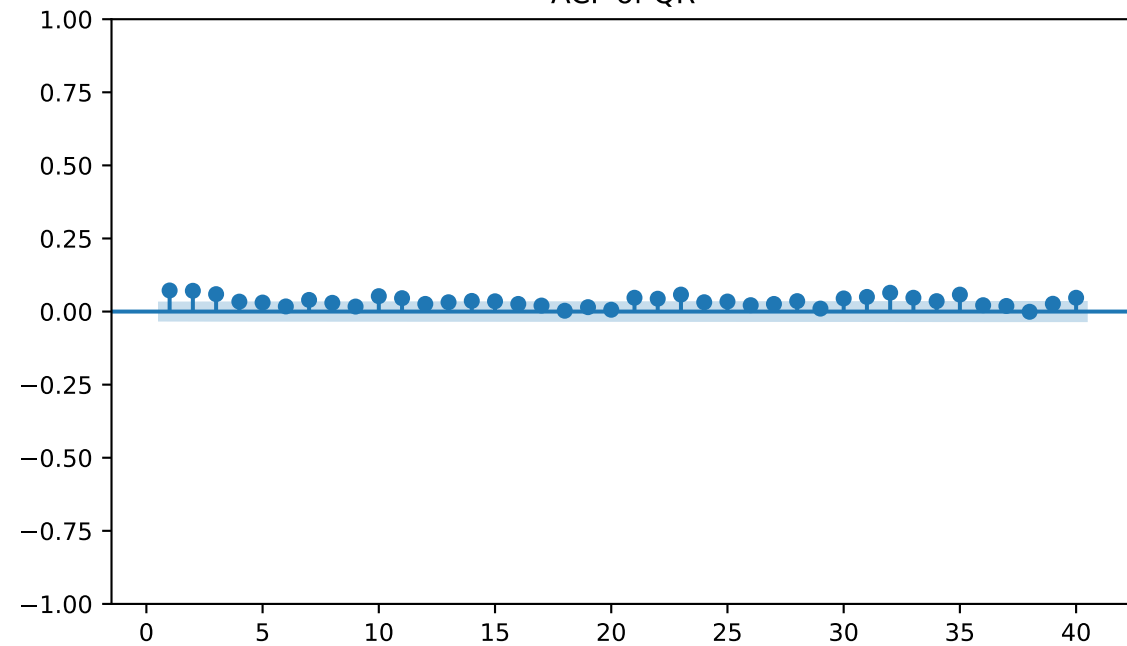
PIT histogram



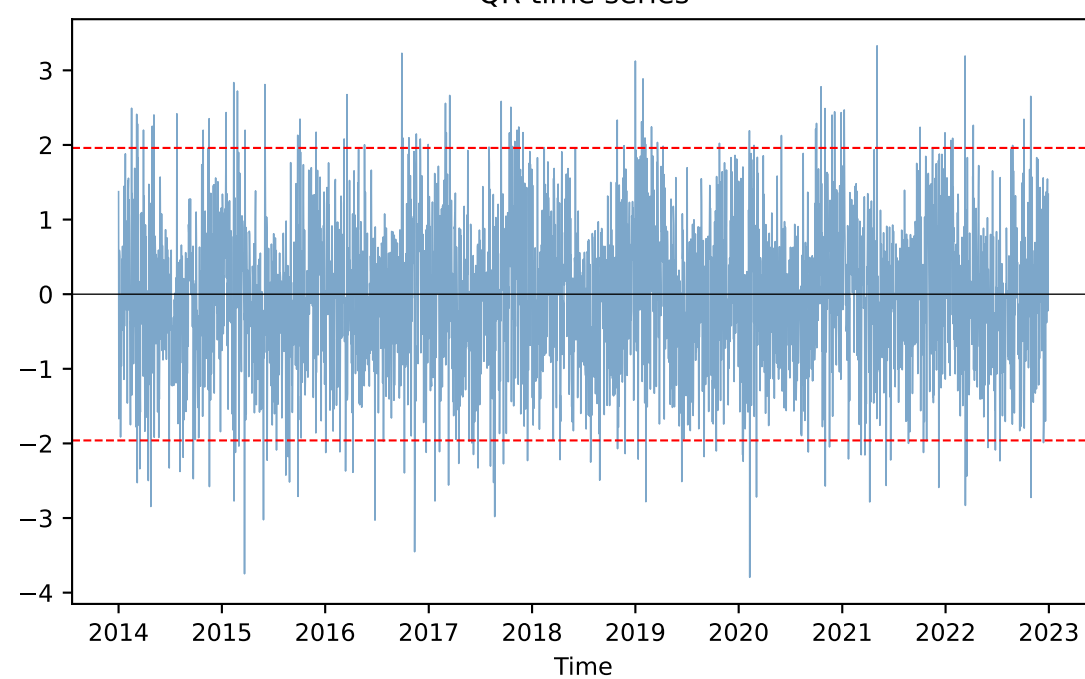
Normal QQ plot



ACF of QR

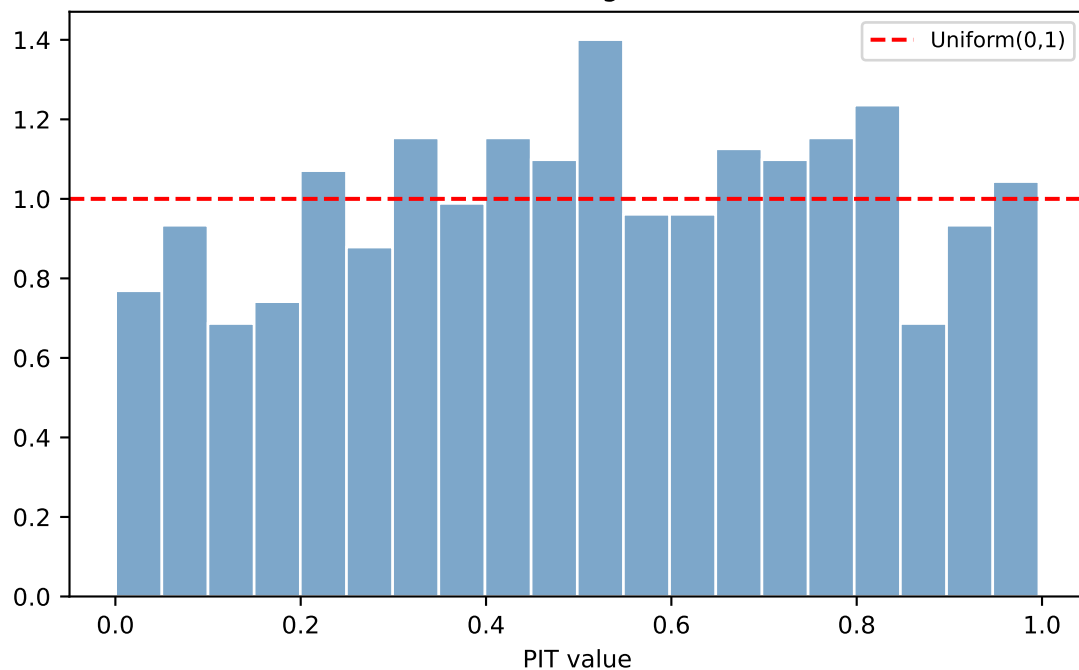


QR time series

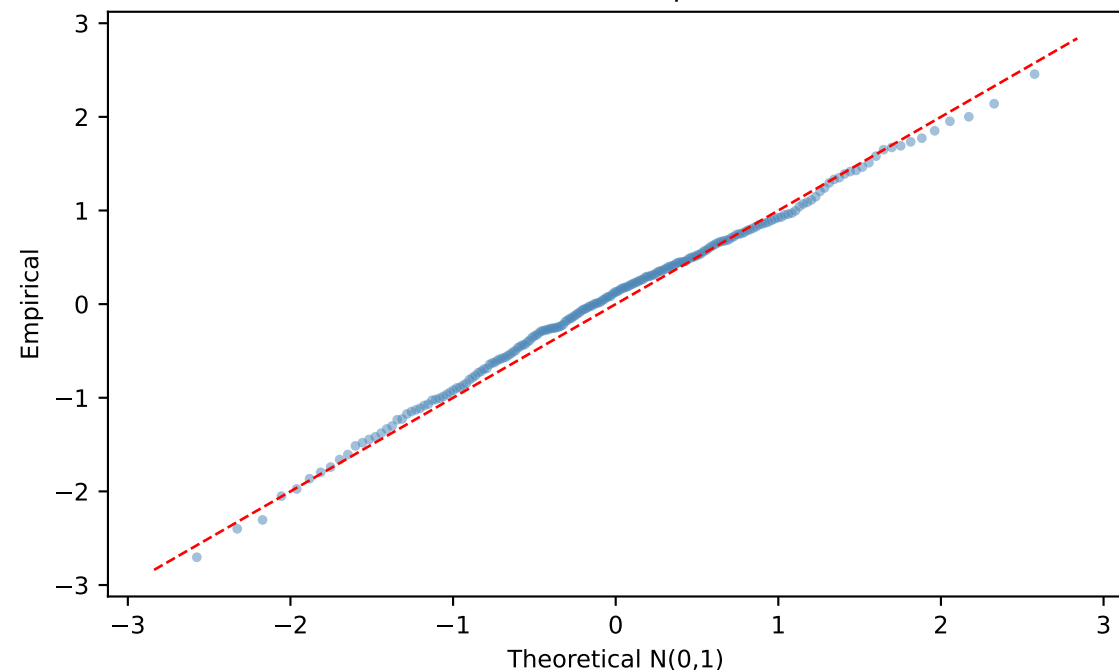


long_short_phi_interactions — OOS diagnostics

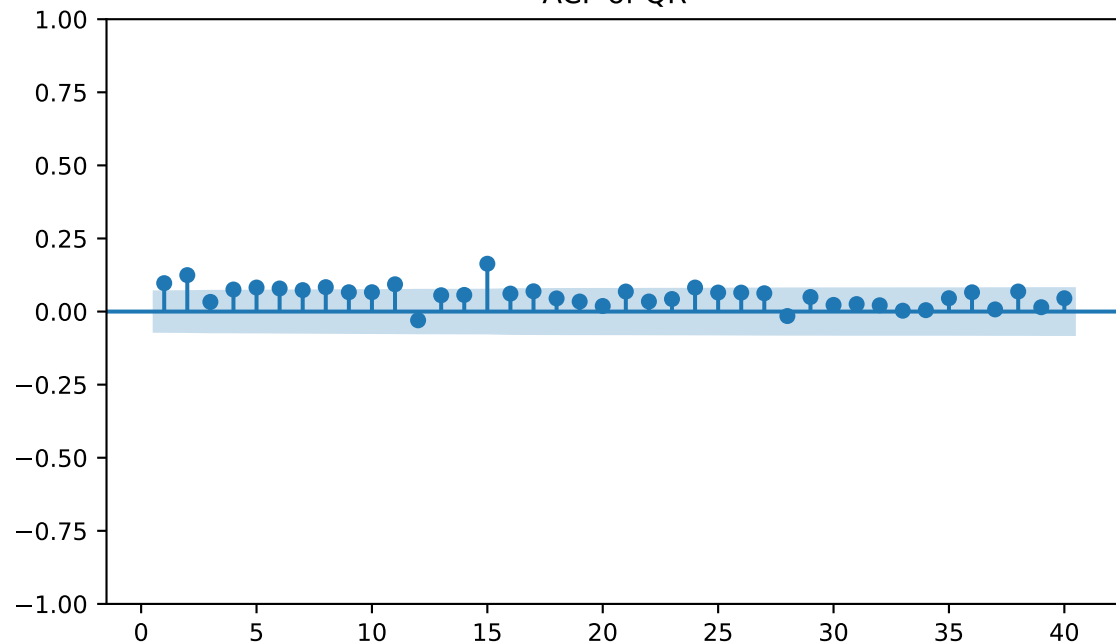
PIT histogram



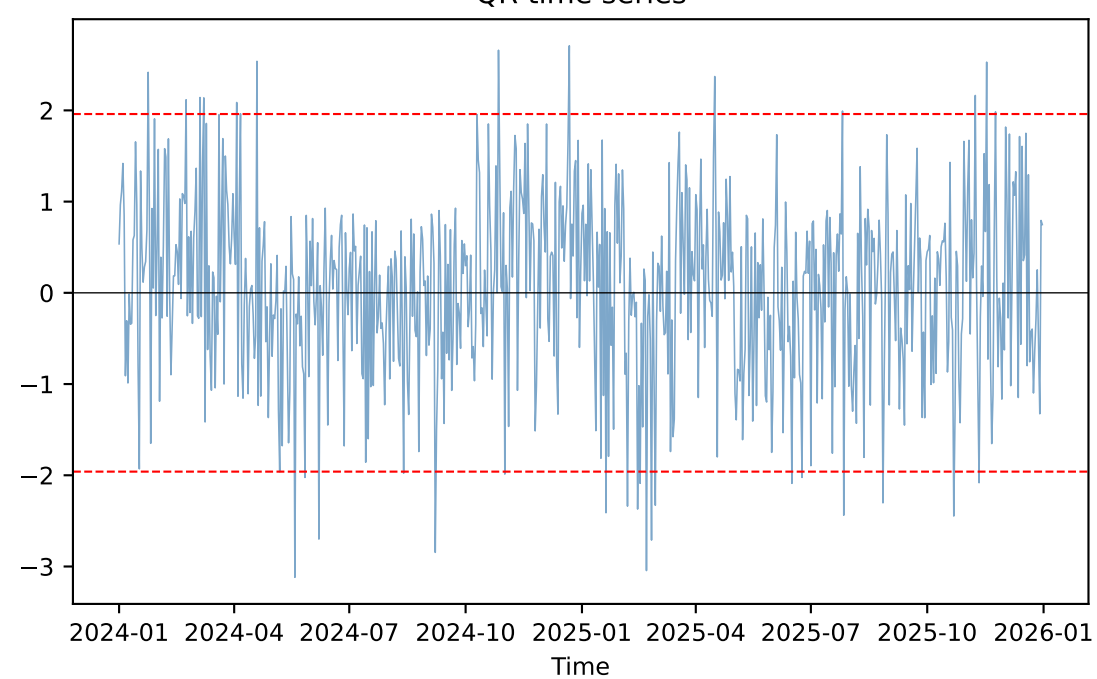
Normal QQ plot



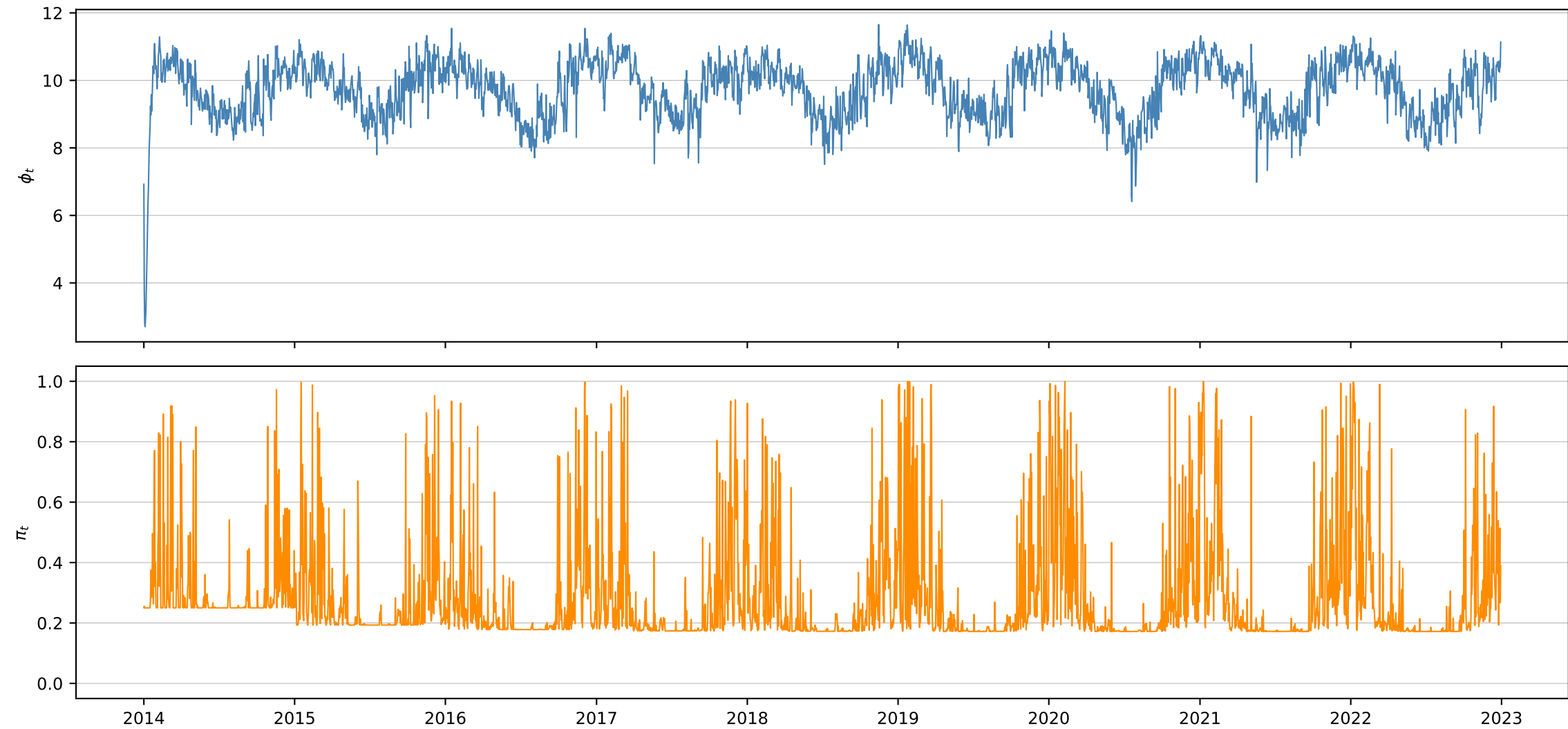
ACF of QR



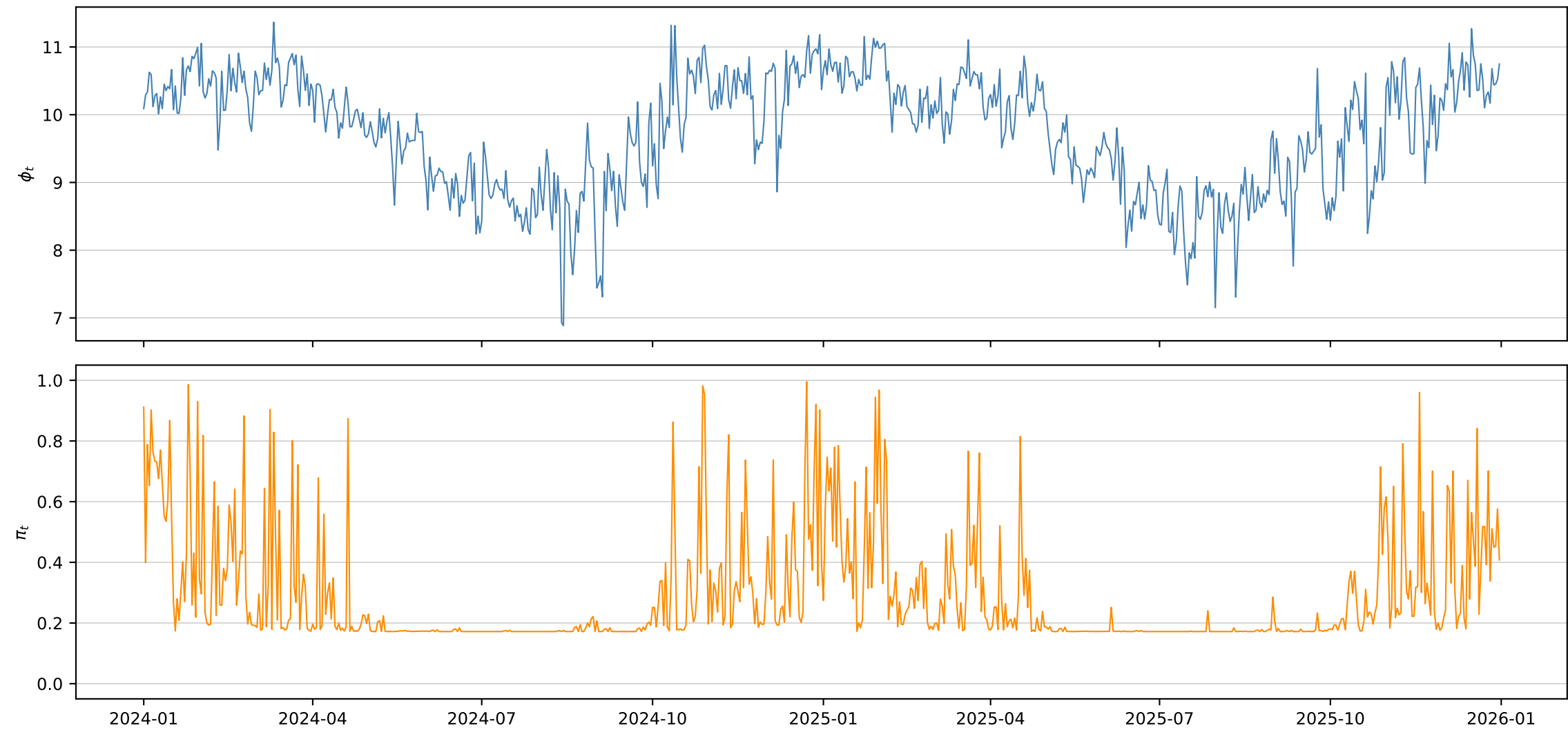
QR time series



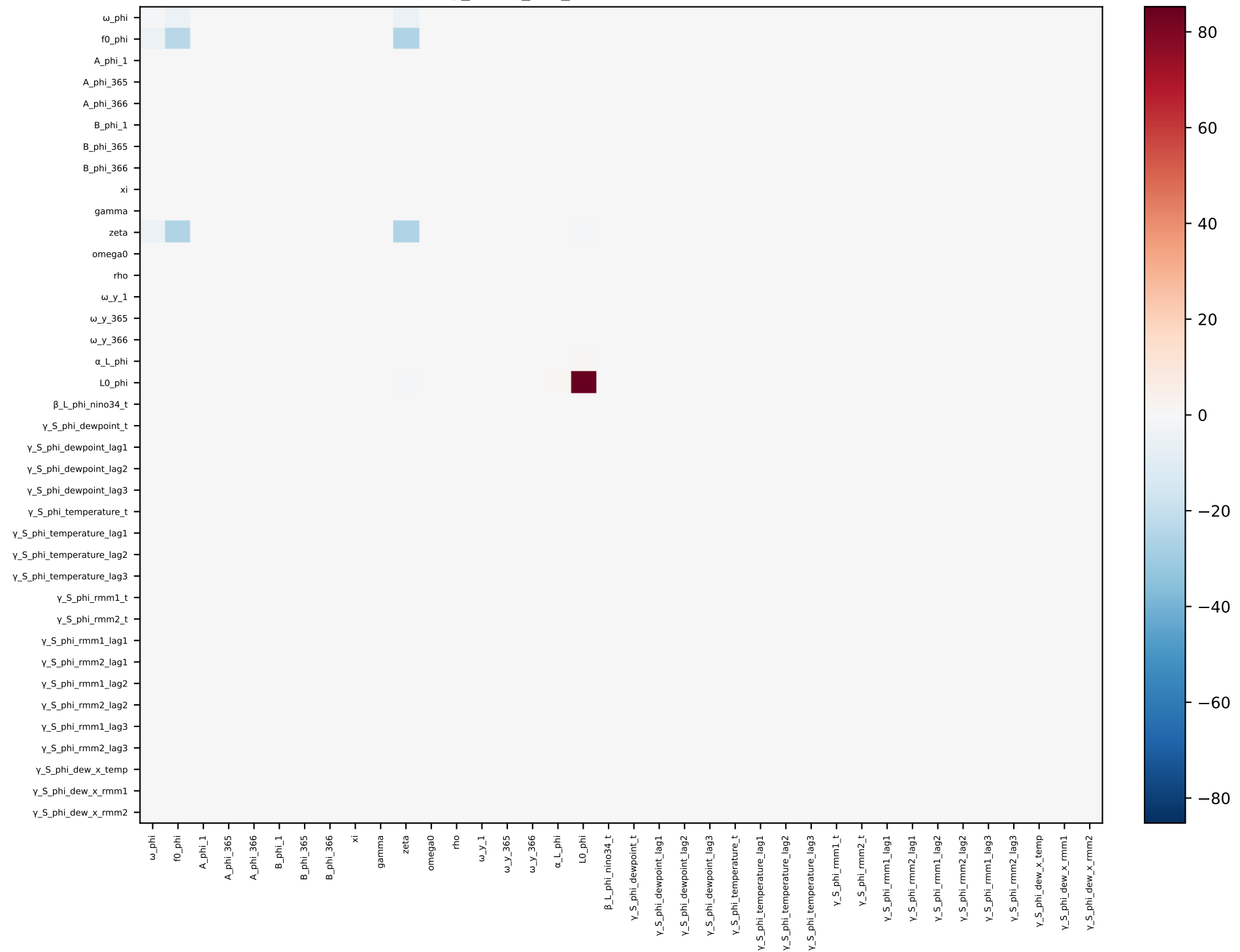
long_short_phi_interactions — filtered states (IS)



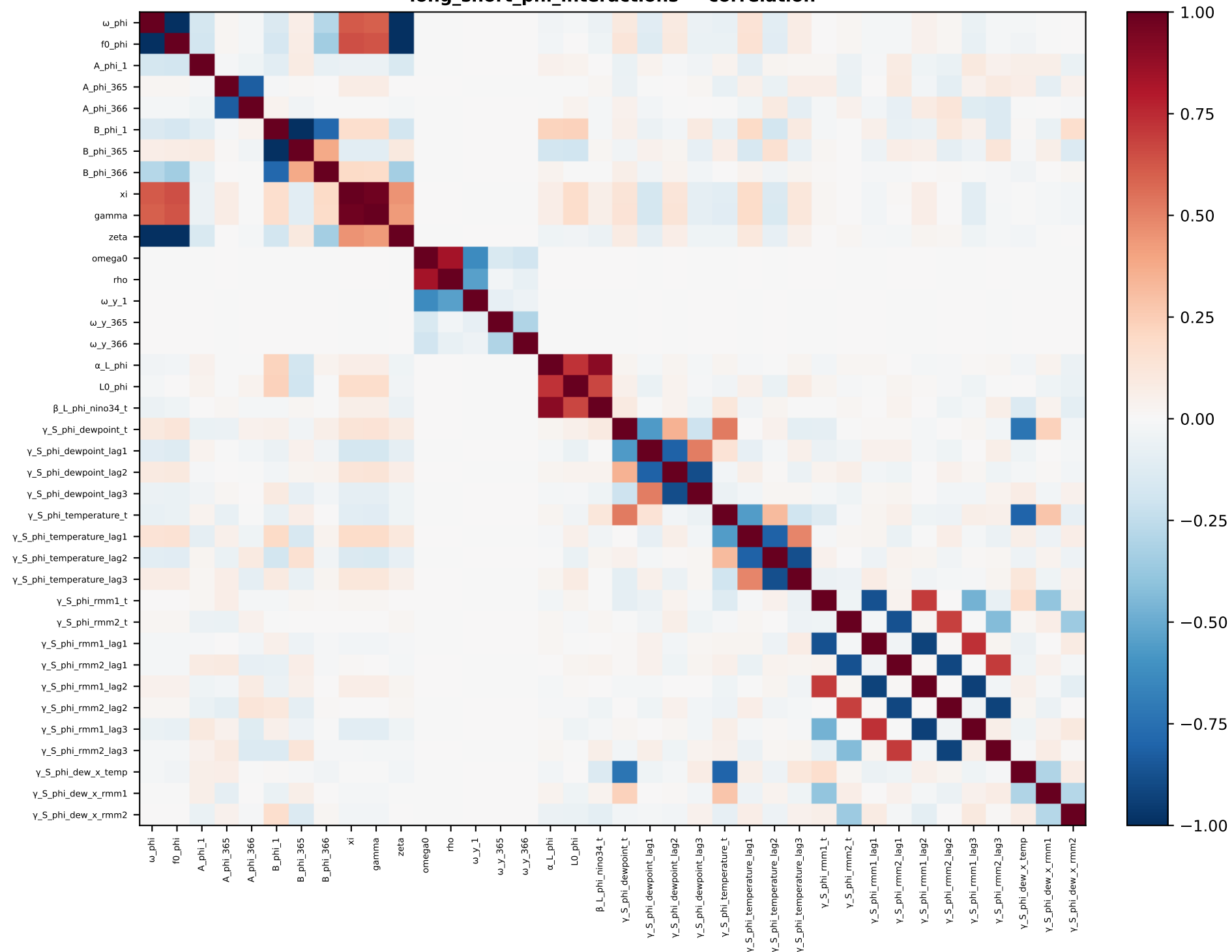
long_short_phi_interactions — filtered states (OOS)



long_short_phi_interactions — covariance

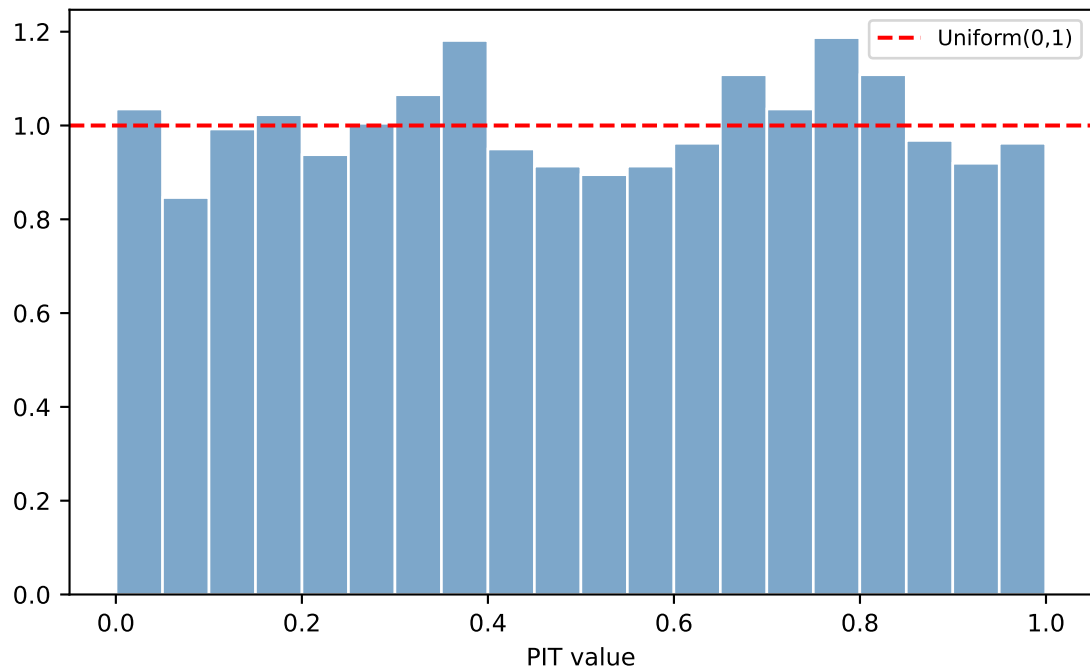


long_short_phi_interactions — correlation

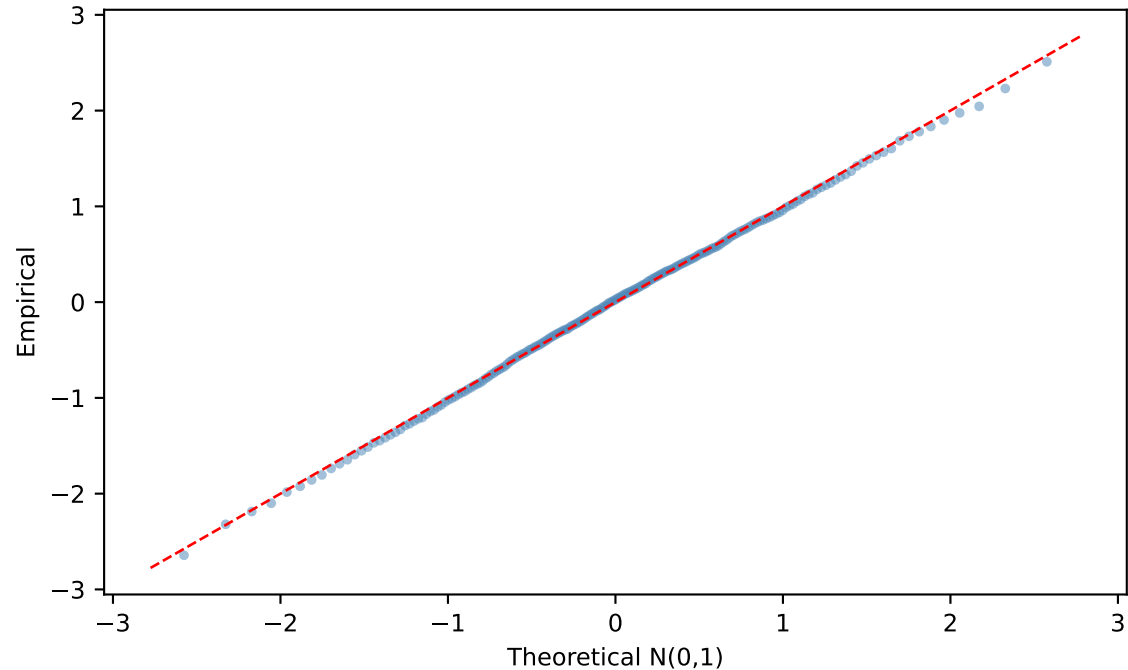


long_short_phi_xi_basic — IS diagnostics

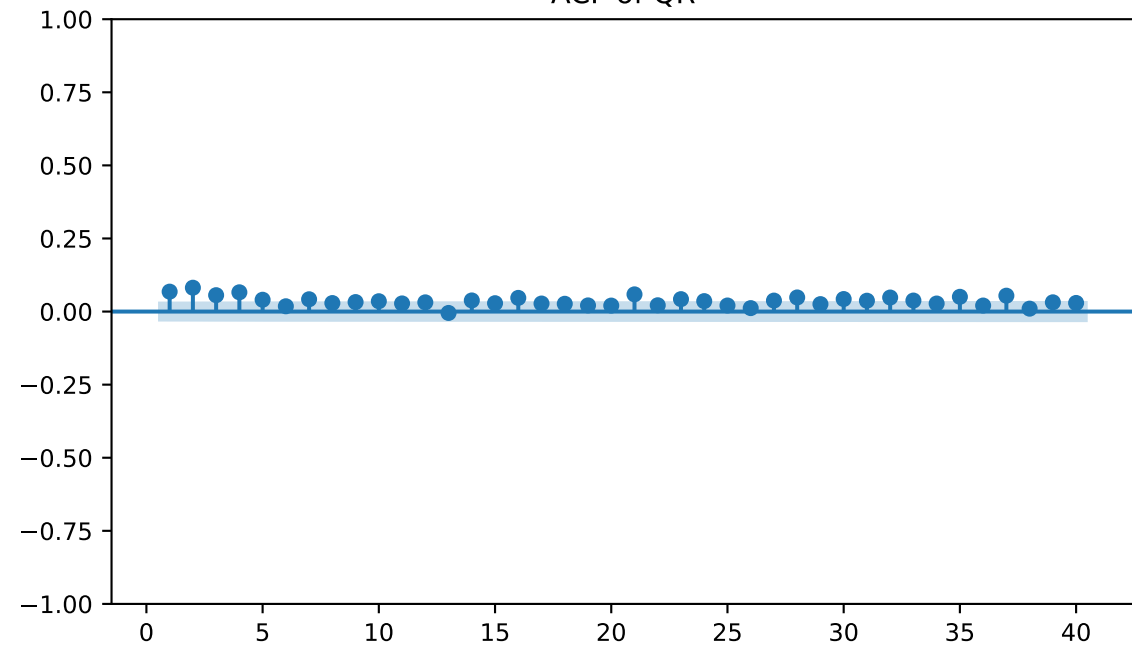
PIT histogram



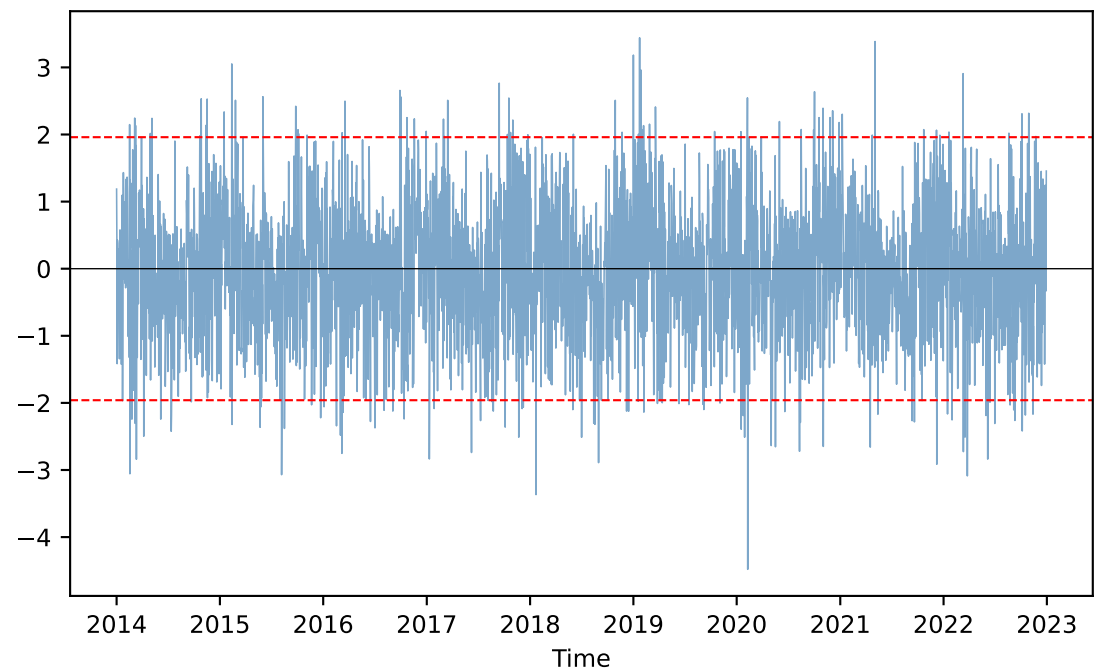
Normal QQ plot



ACF of QR

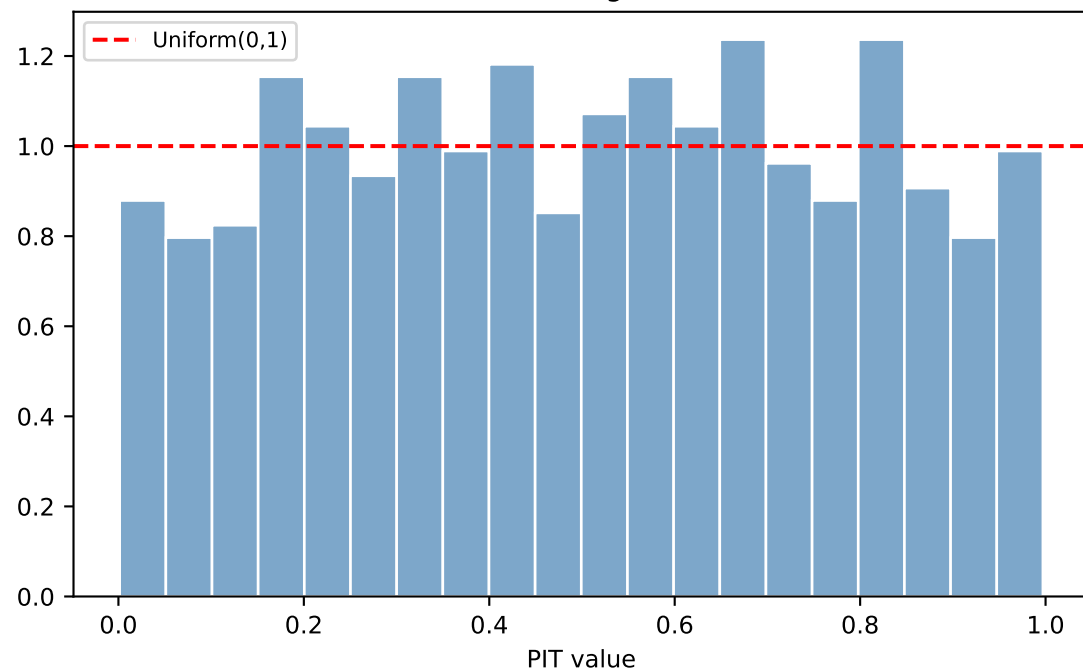


QR time series

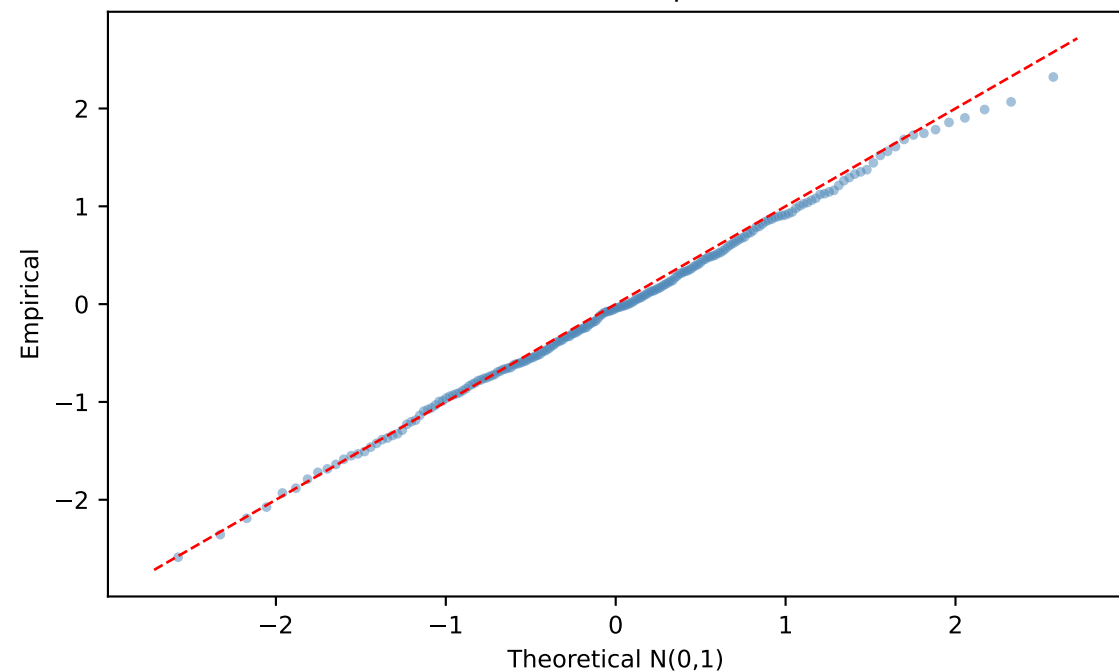


long_short_phi_xi_basic — OOS diagnostics

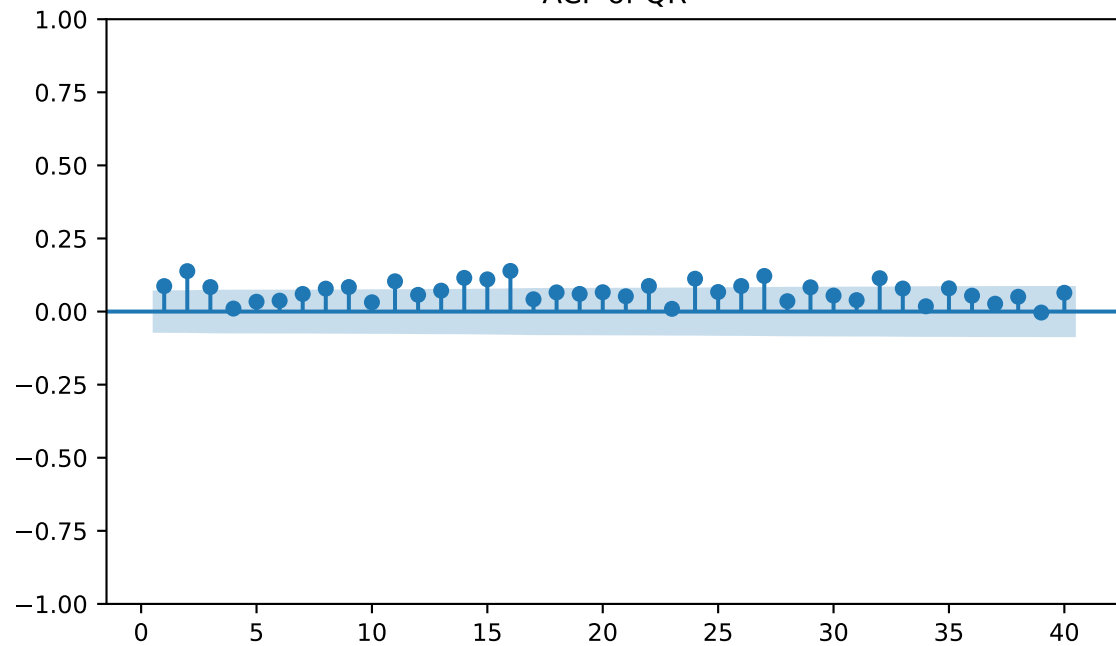
PIT histogram



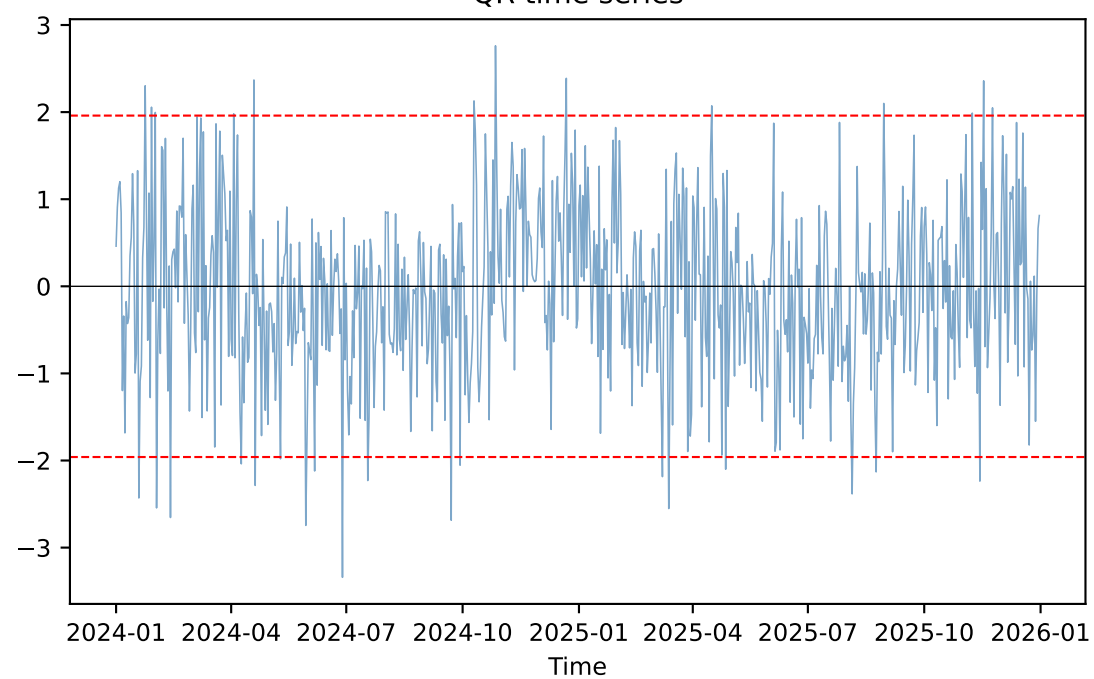
Normal QQ plot



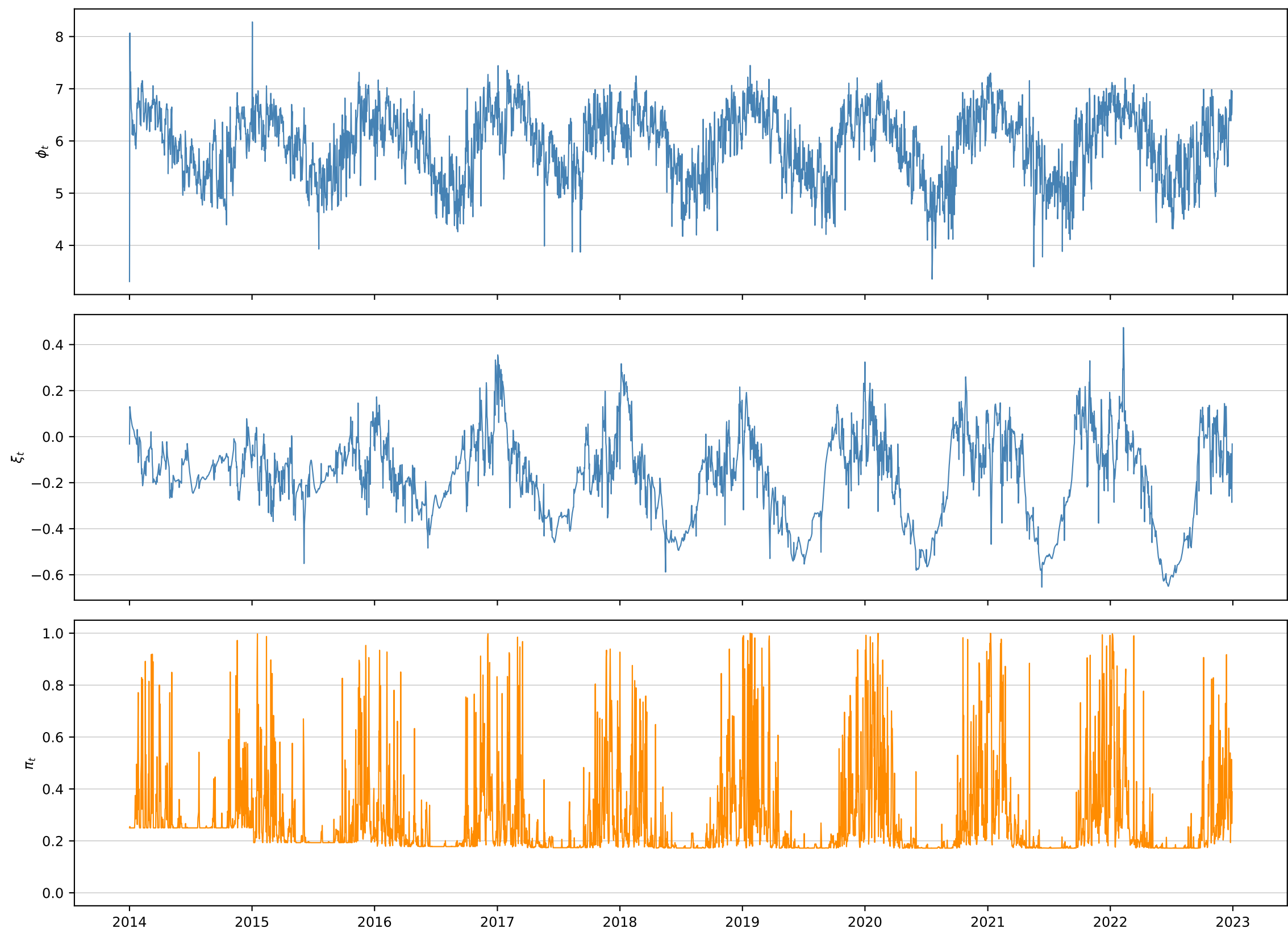
ACF of QR



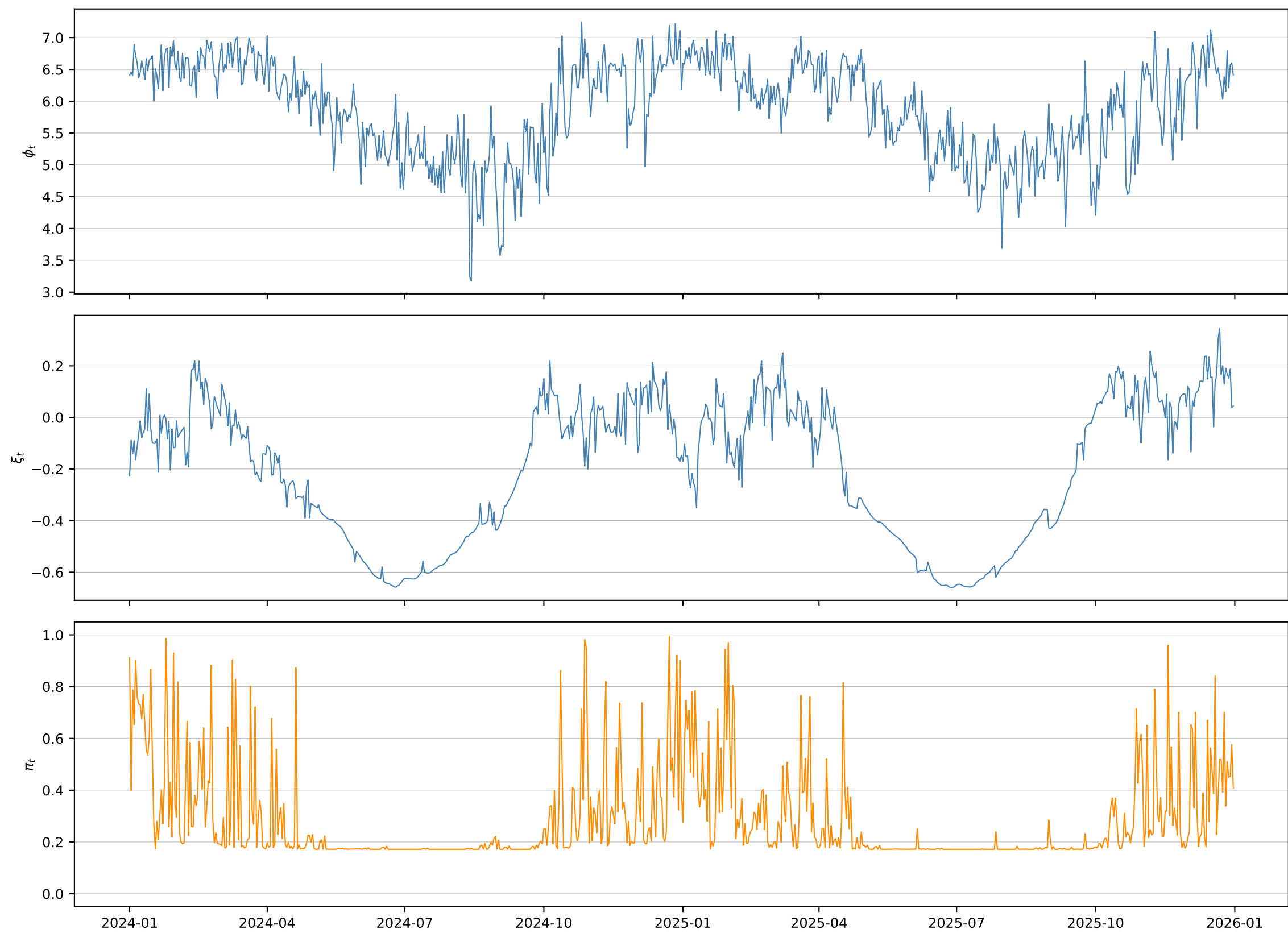
QR time series



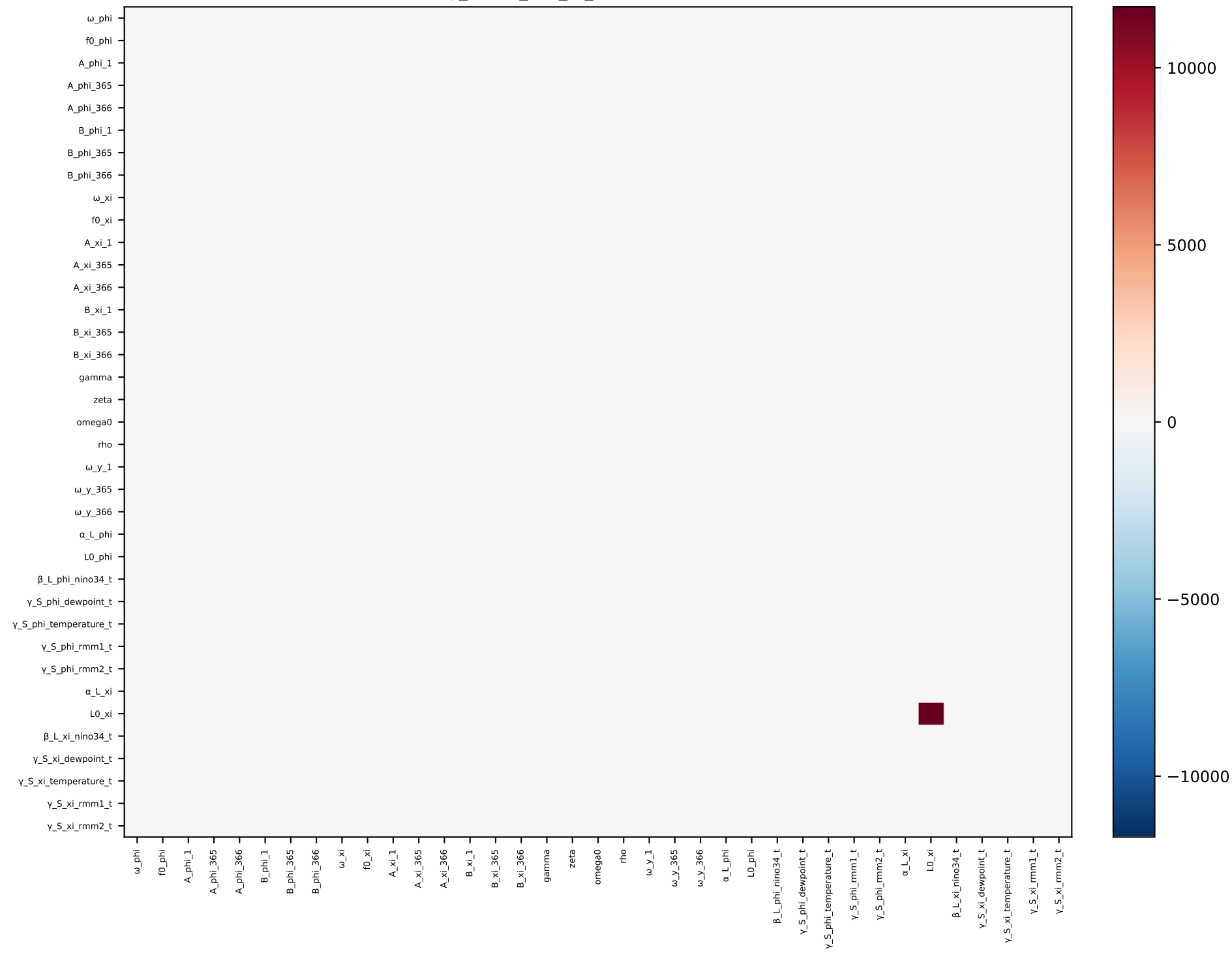
long_short_phi_xi_basic — filtered states (IS)



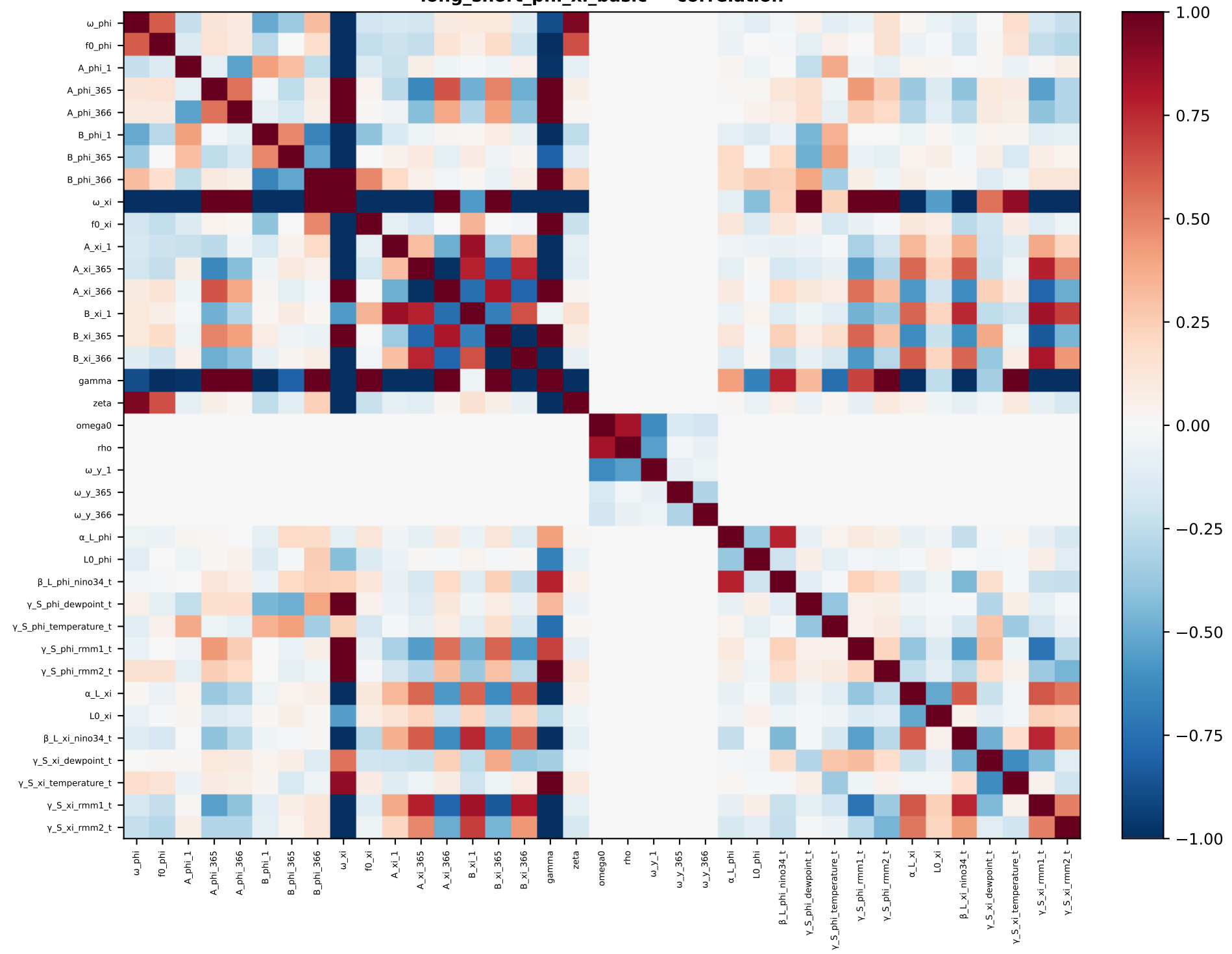
long_short_phi_xi_basic — filtered states (OOS)



long_short_phi_xi_basic — covariance

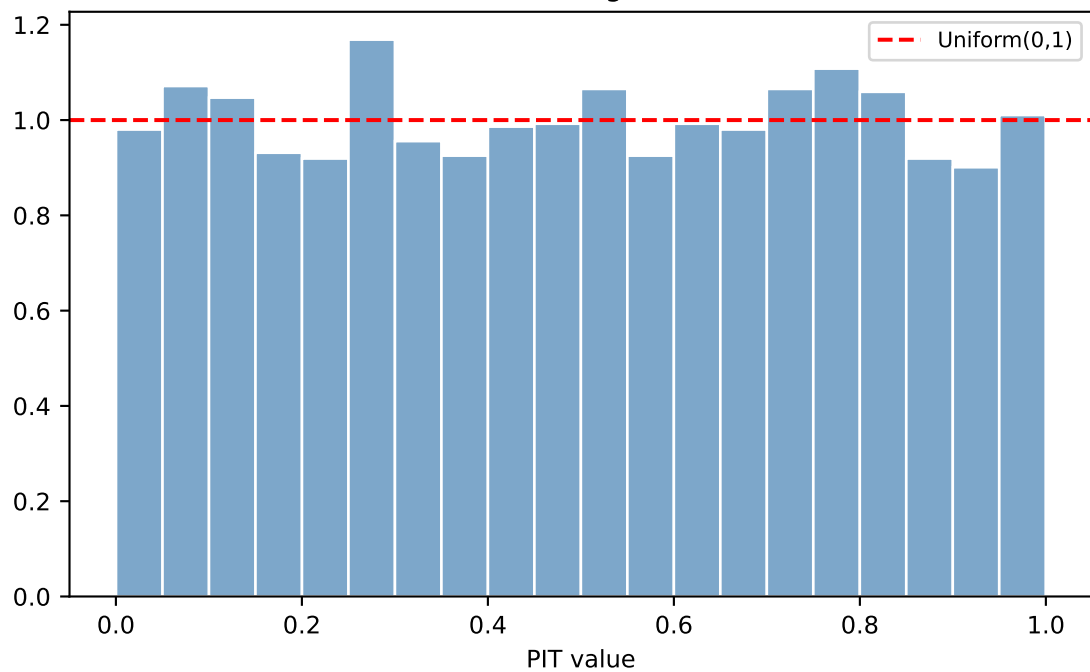


long_short_phi_xi_basic — correlation

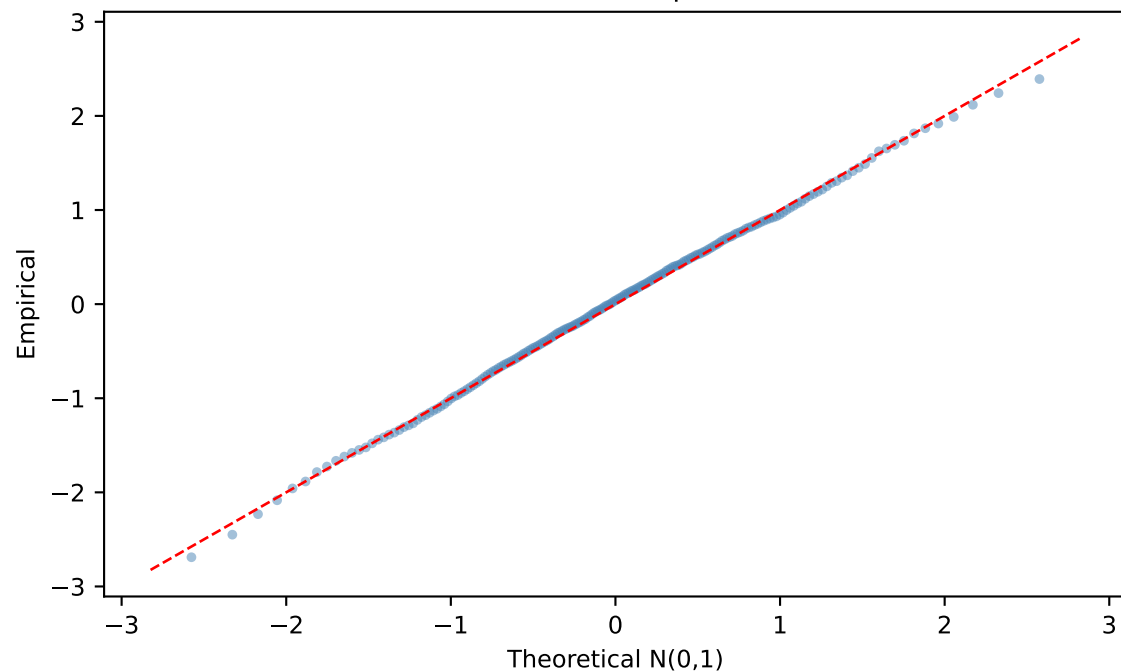


long_short_phi_xi_short_lags — IS diagnostics

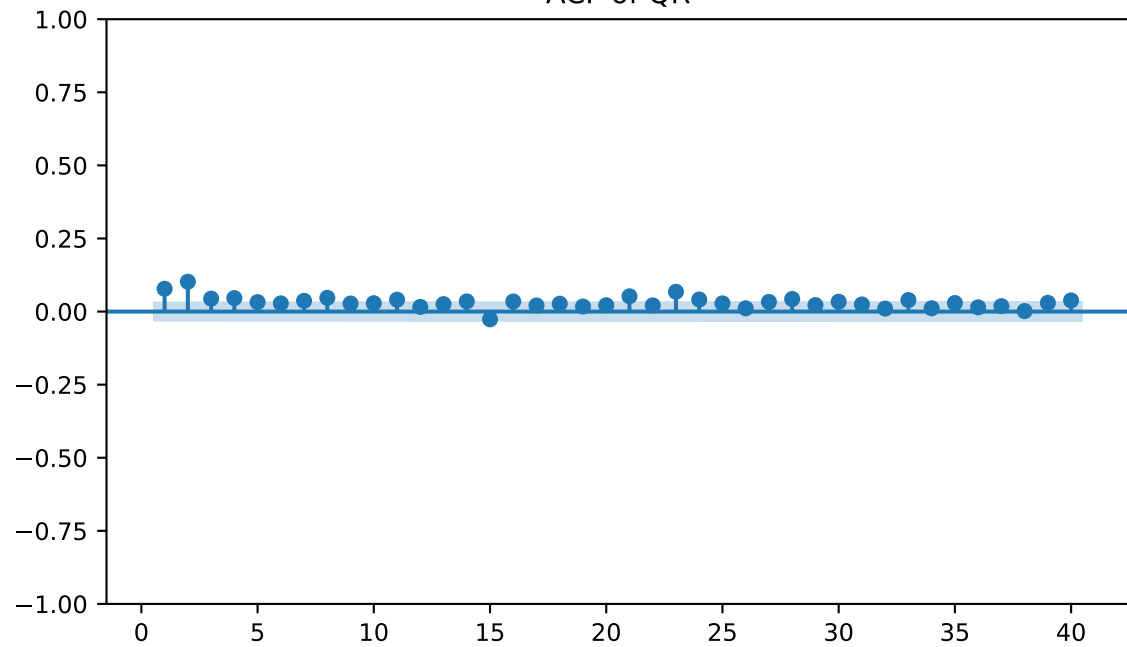
PIT histogram



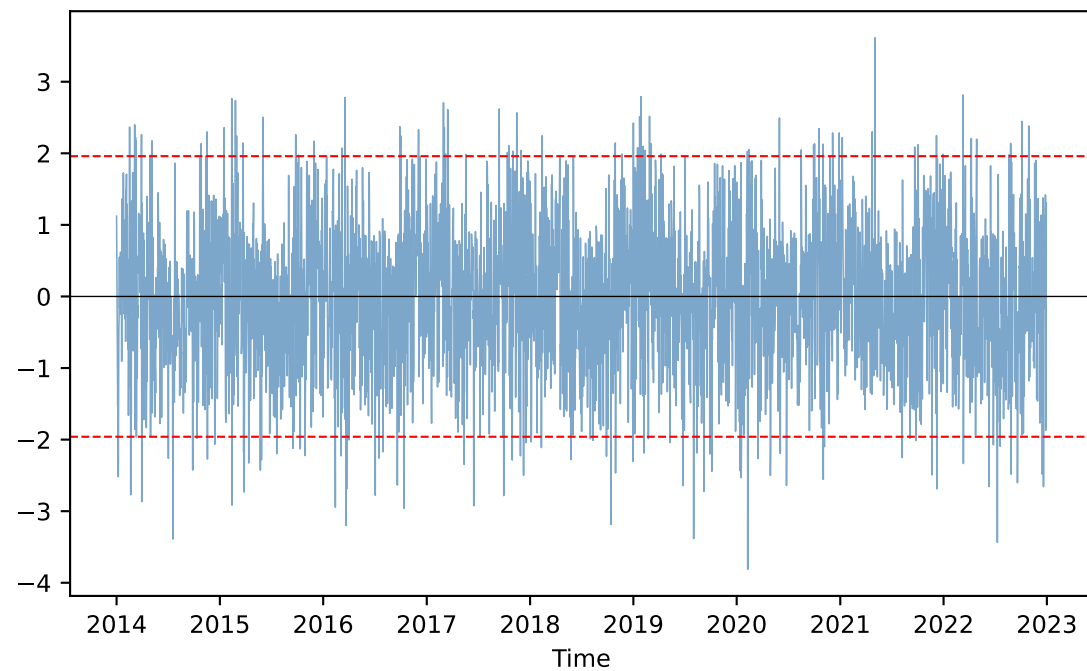
Normal QQ plot



ACF of QR

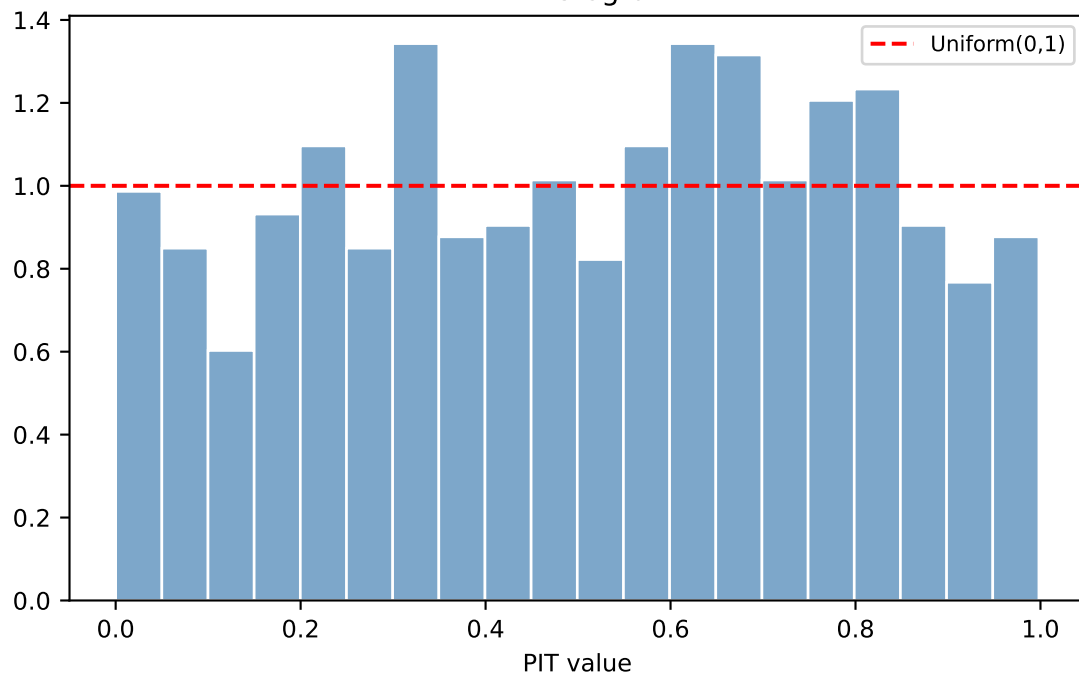


QR time series

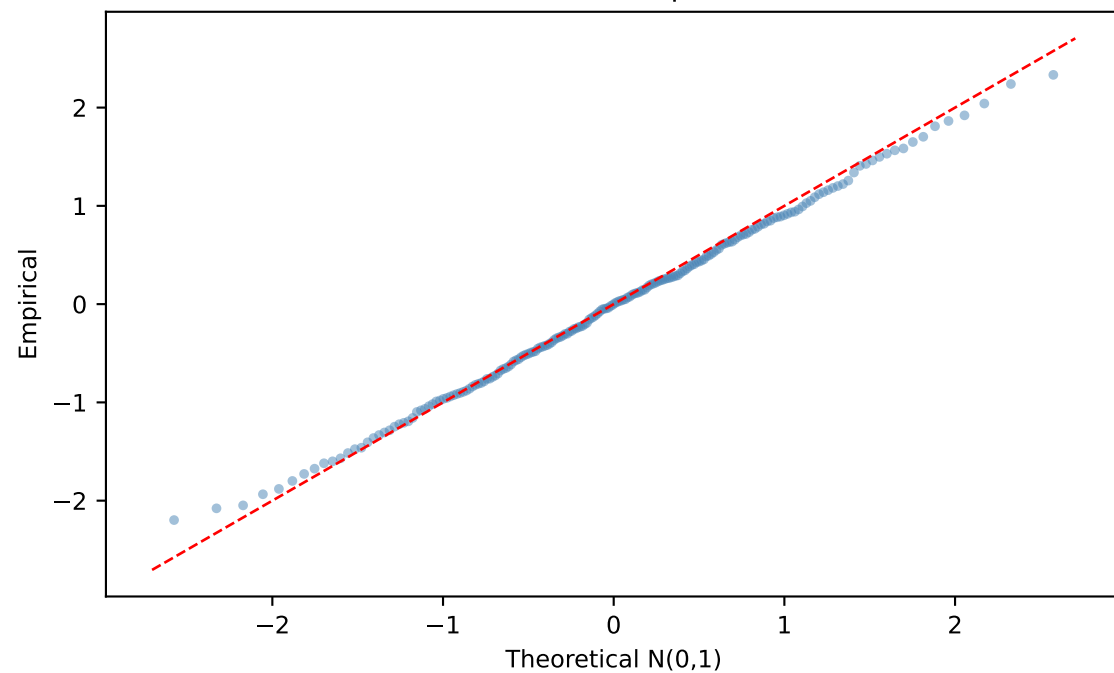


long_short_phi_xi_short_lags — OOS diagnostics

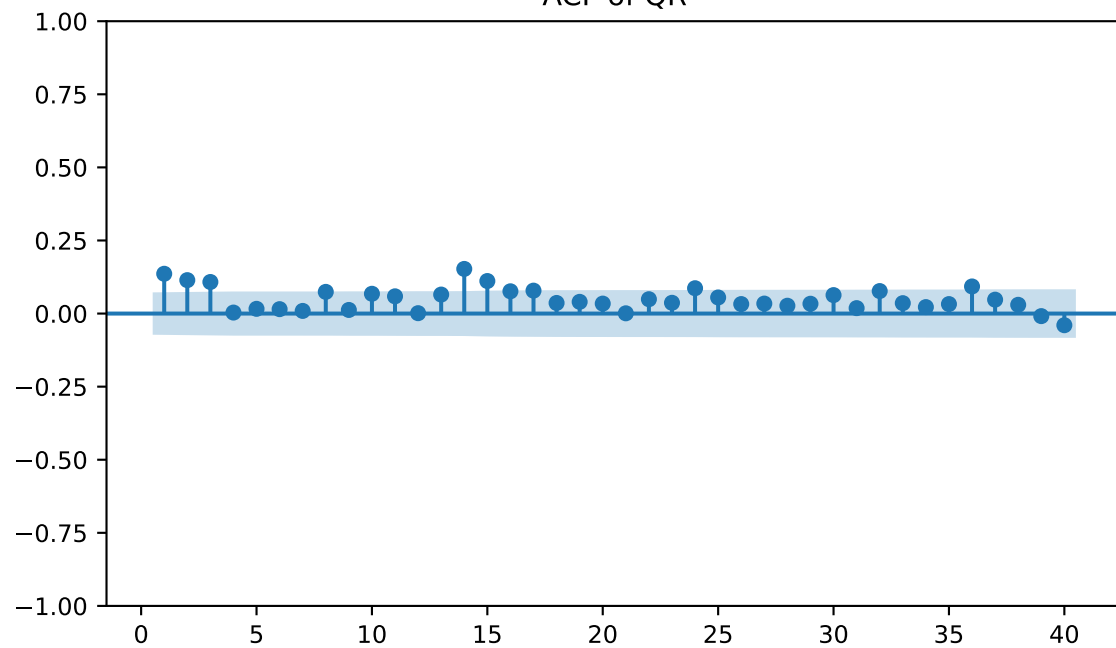
PIT histogram



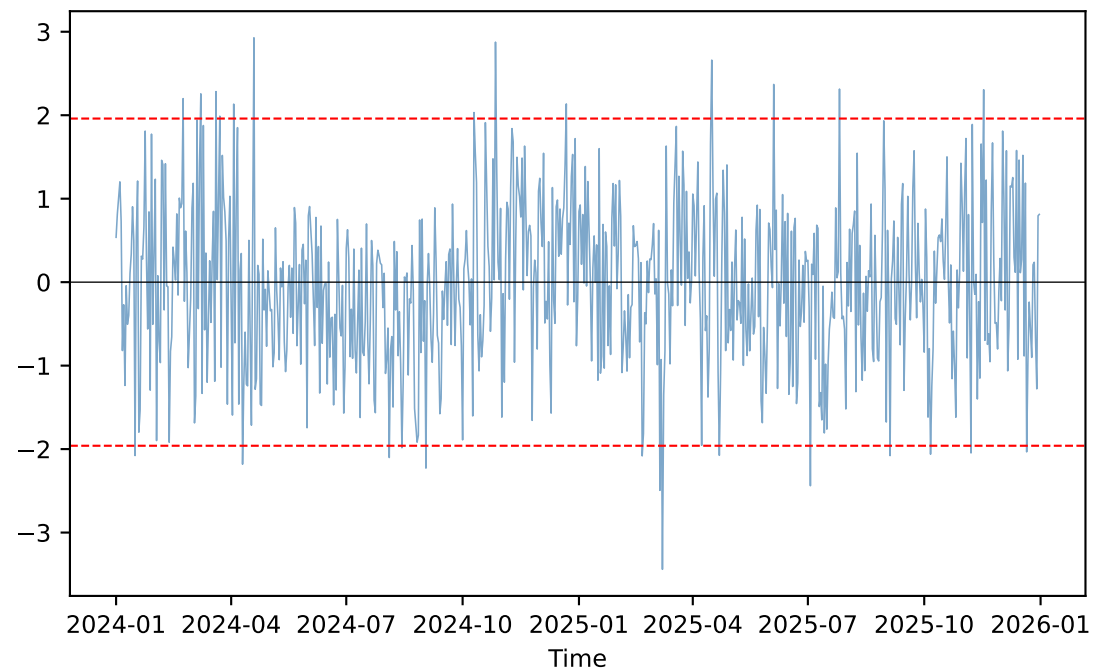
Normal QQ plot



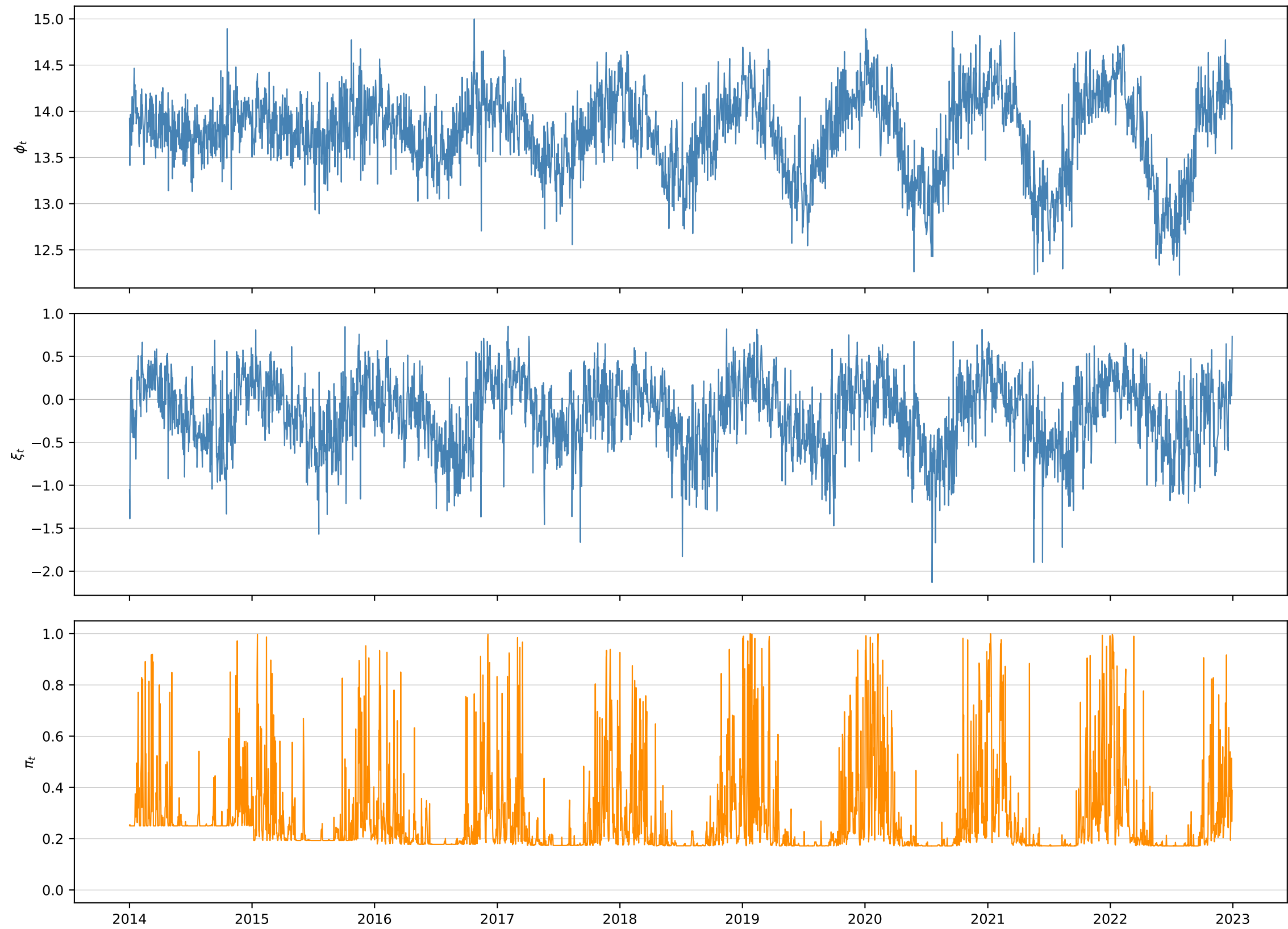
ACF of QR



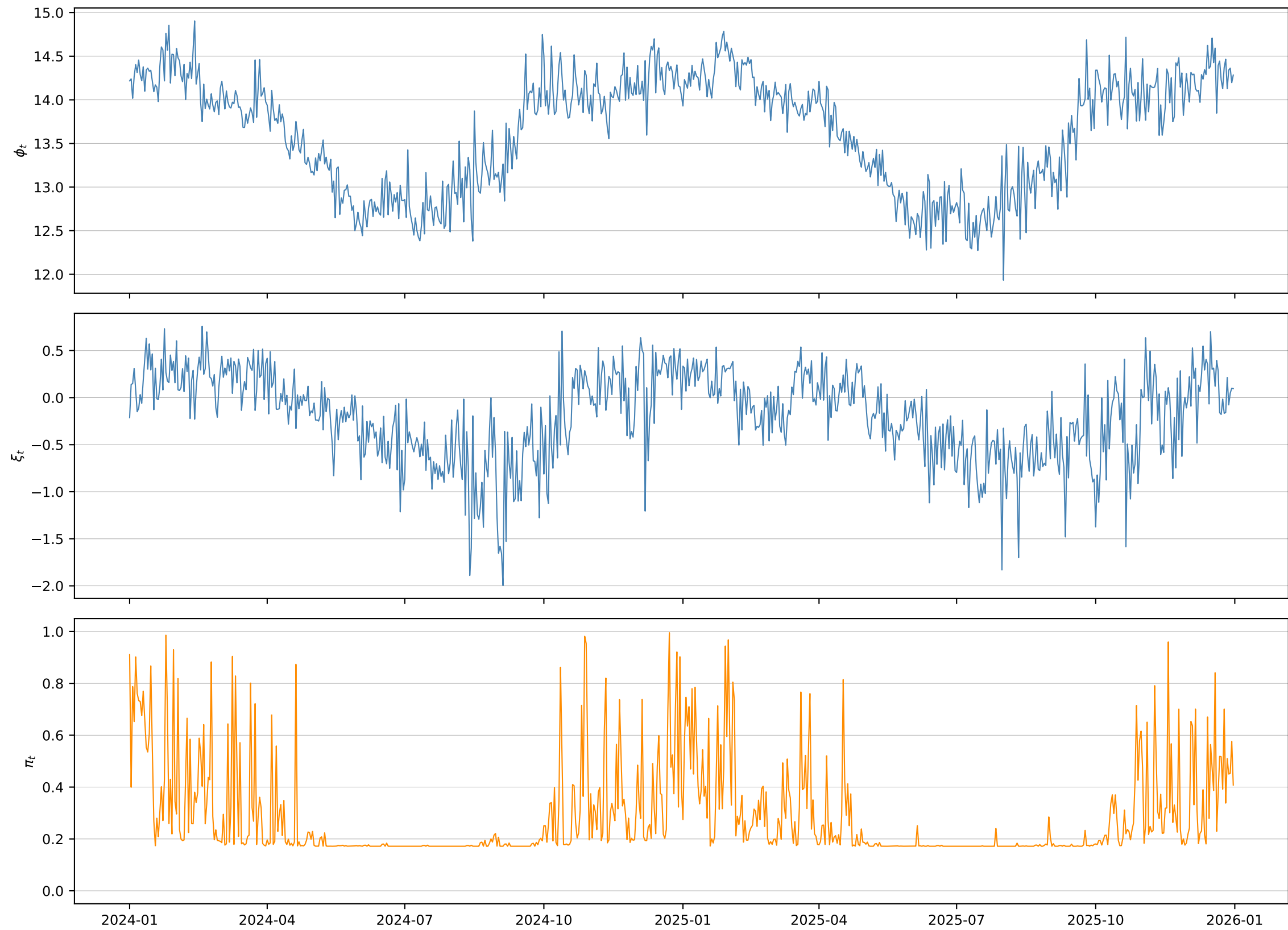
QR time series



long_short_phi_xi_short_lags — filtered states (IS)

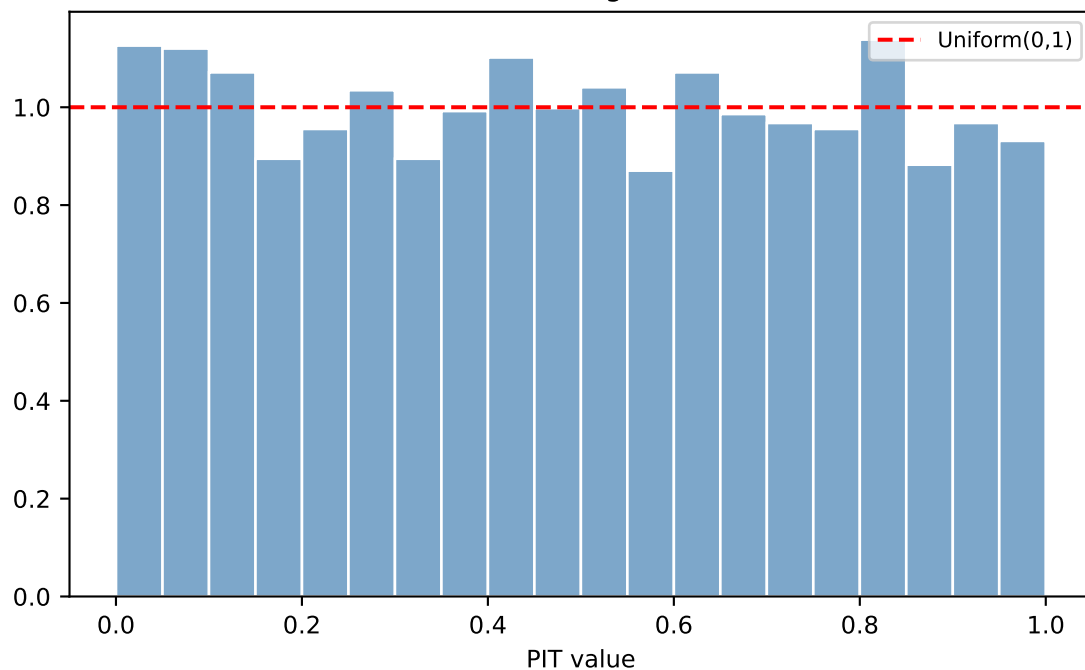


long_short_phi_xi_short_lags — filtered states (OOS)

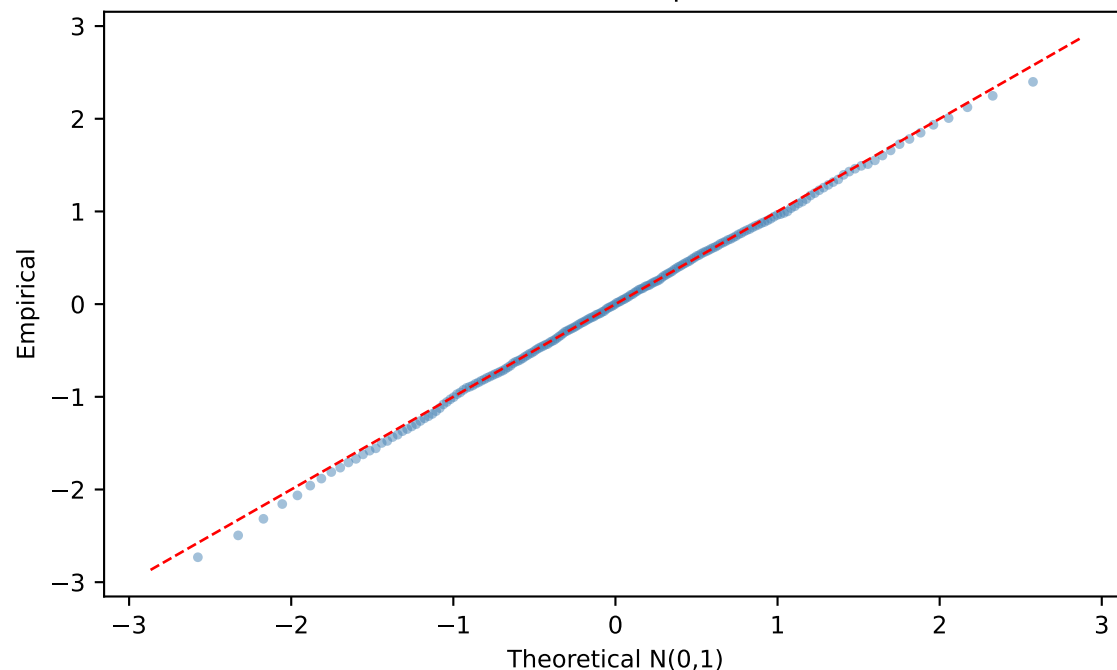


long_short_phi_xi_weather_climate — IS diagnostics

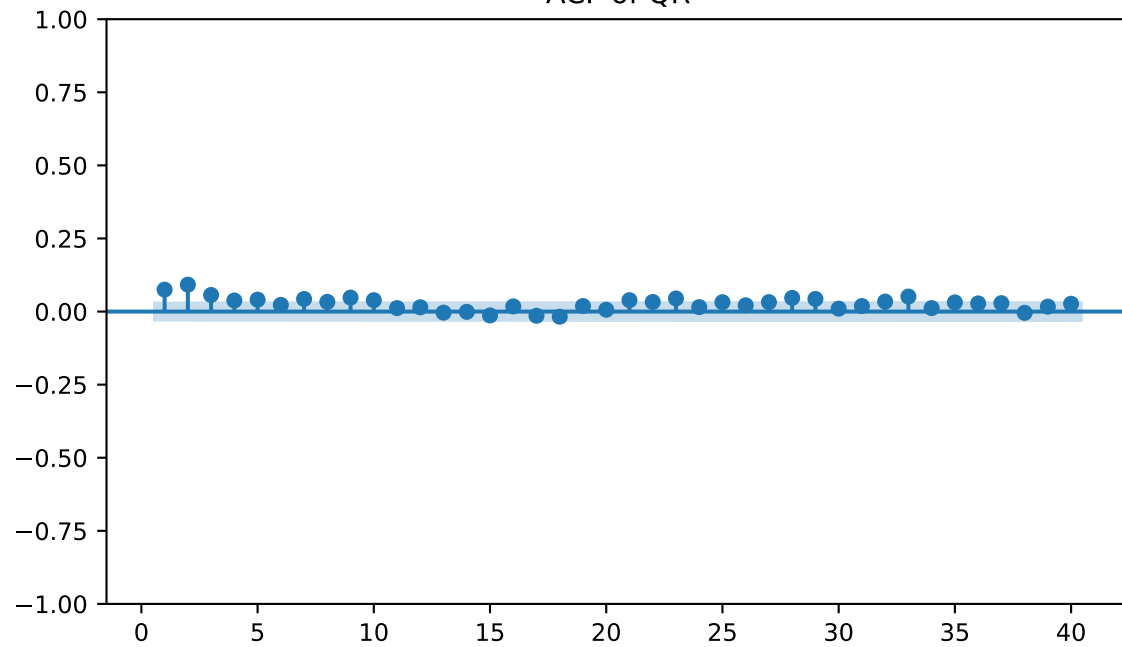
PIT histogram



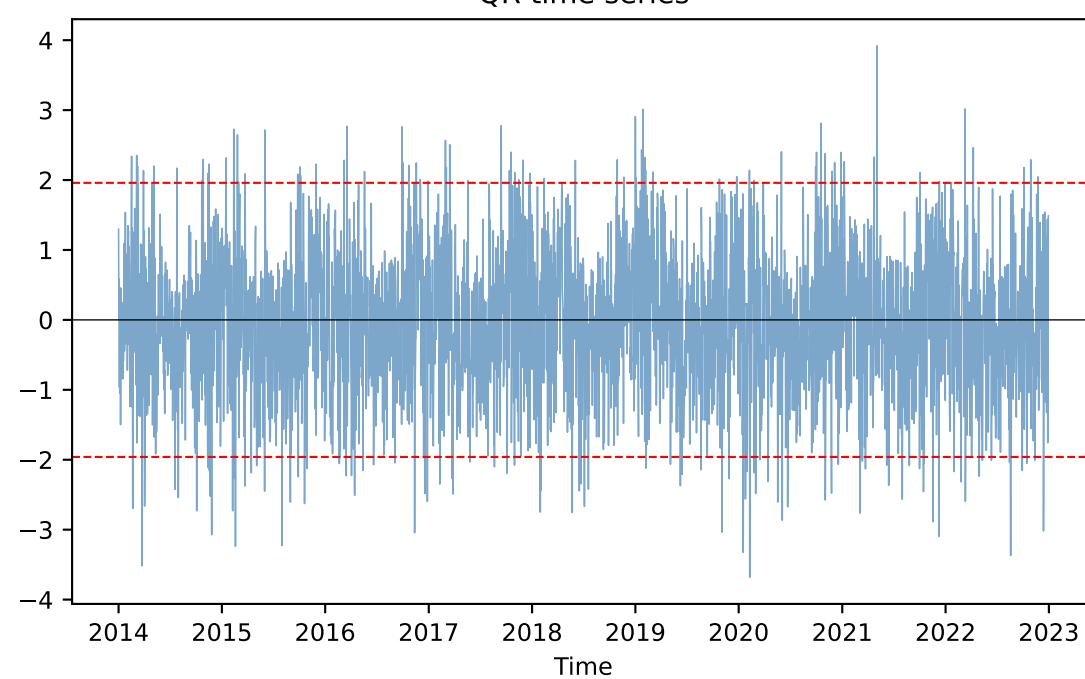
Normal QQ plot



ACF of QR

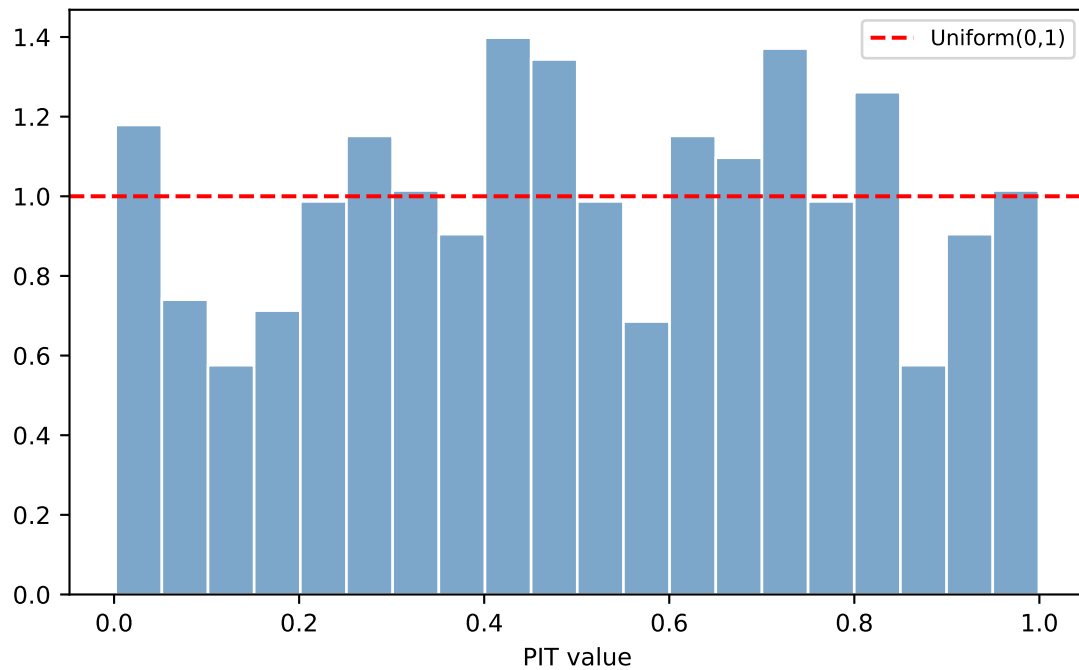


QR time series

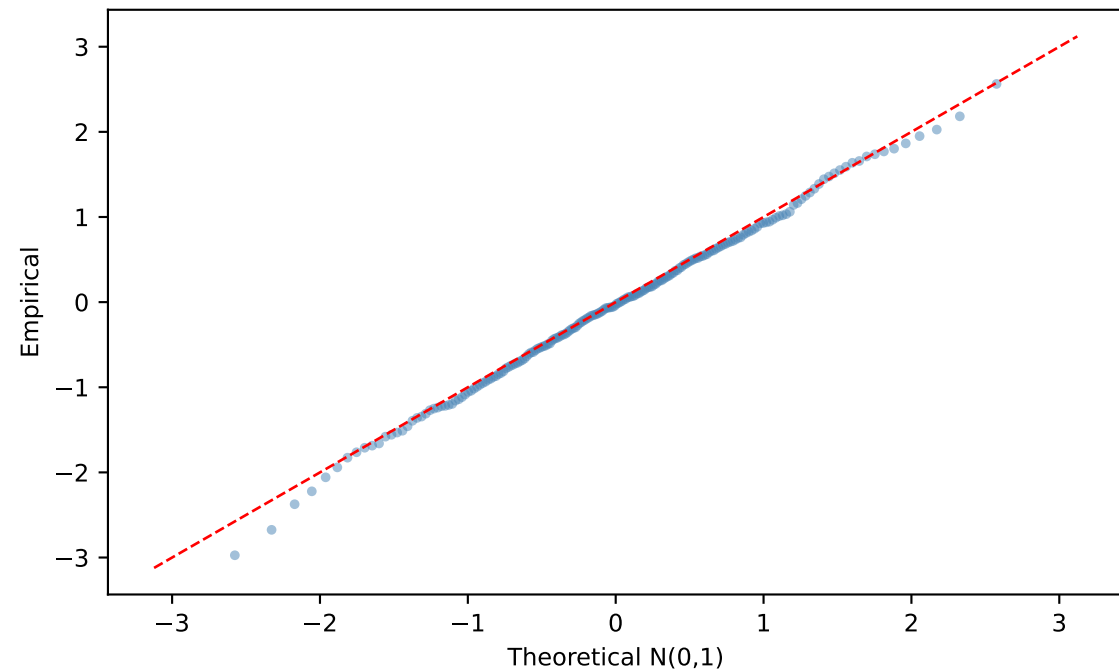


long_short_phi_xi_weather_climate — OOS diagnostics

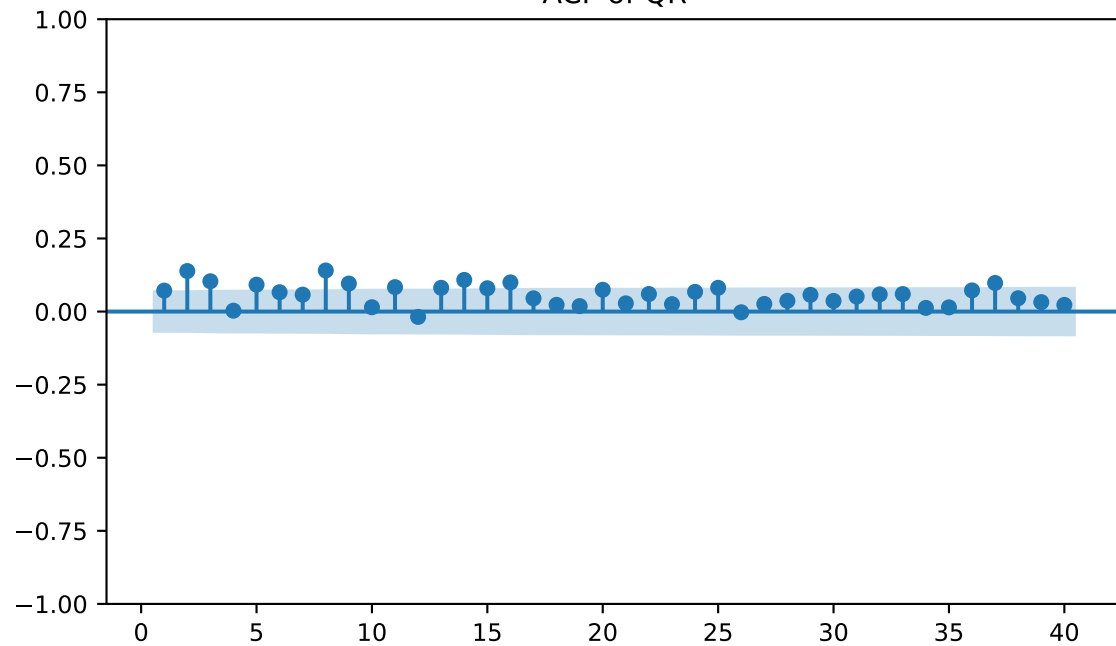
PIT histogram



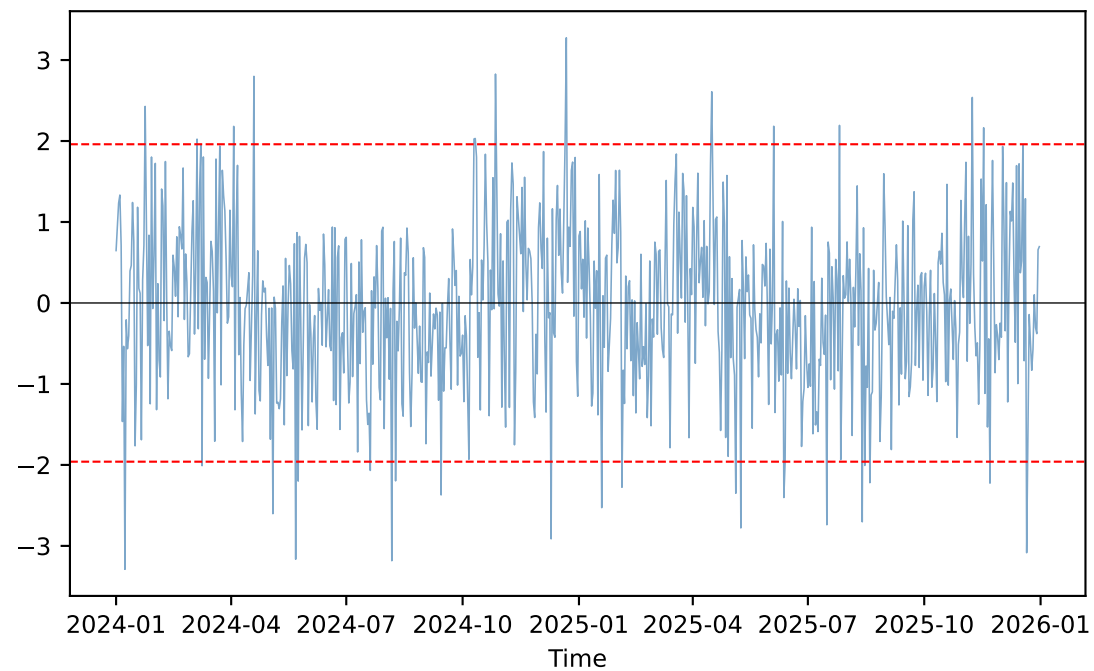
Normal QQ plot



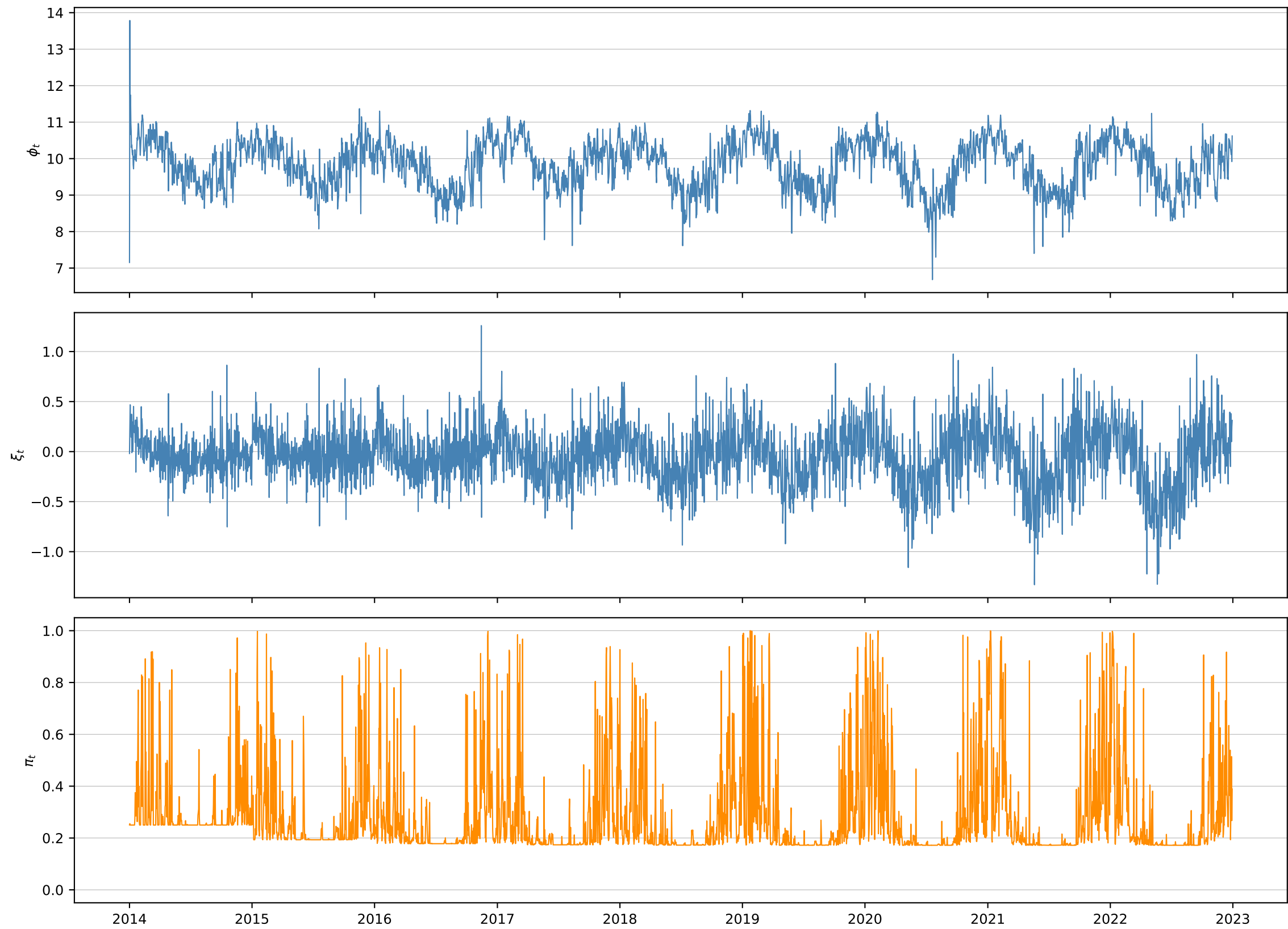
ACF of QR



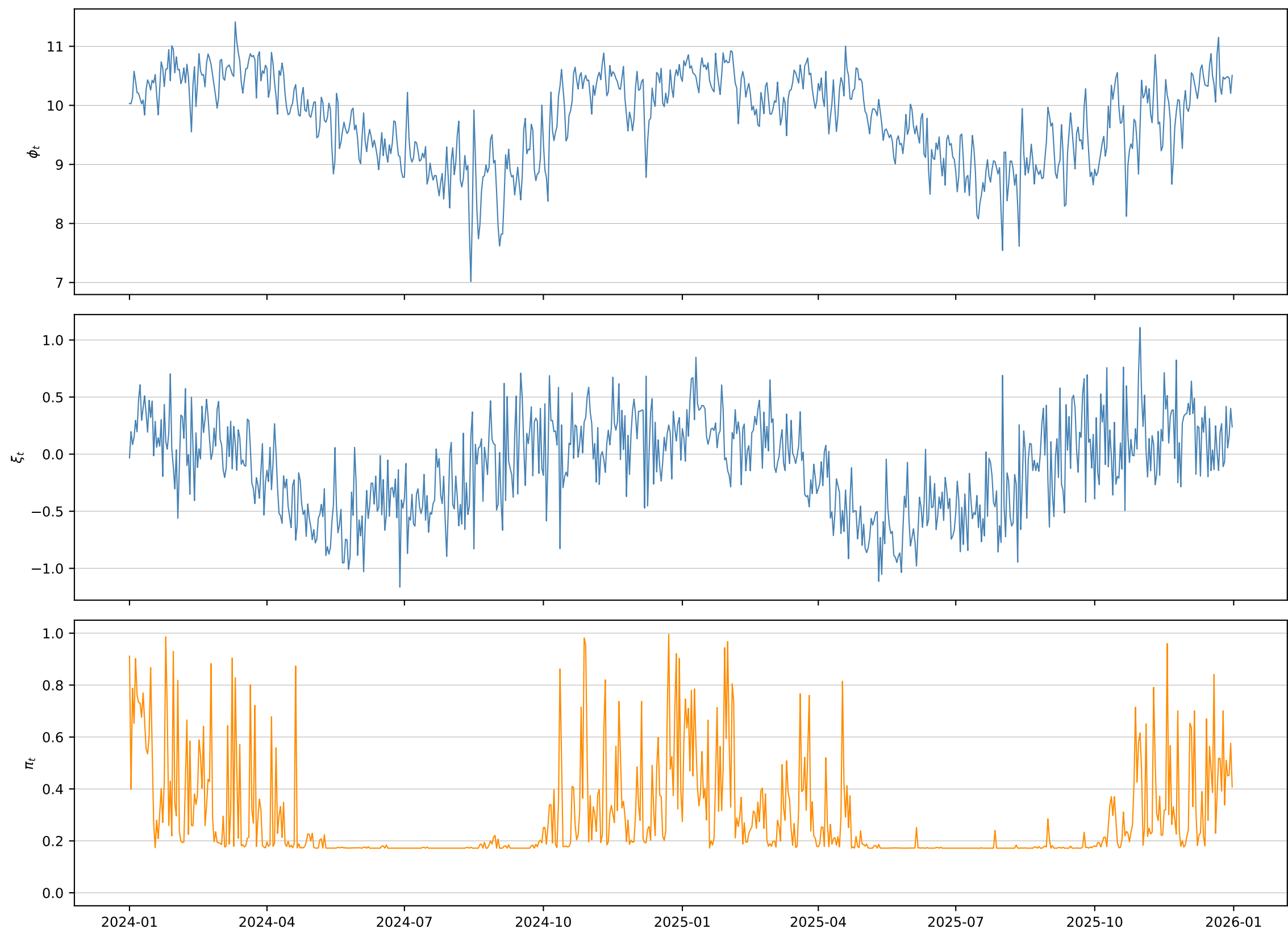
QR time series



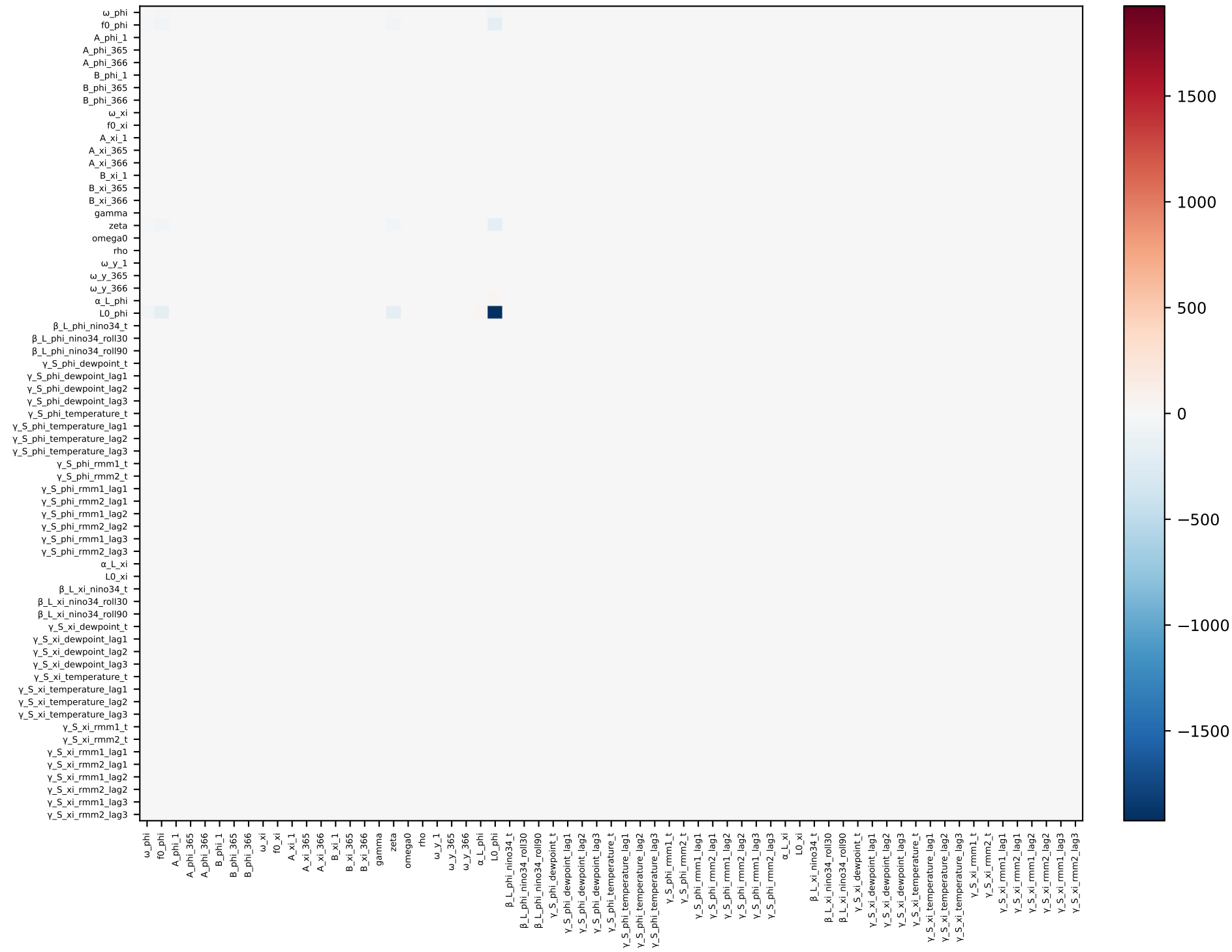
long_short_phi_xi_weather_climate — filtered states (IS)



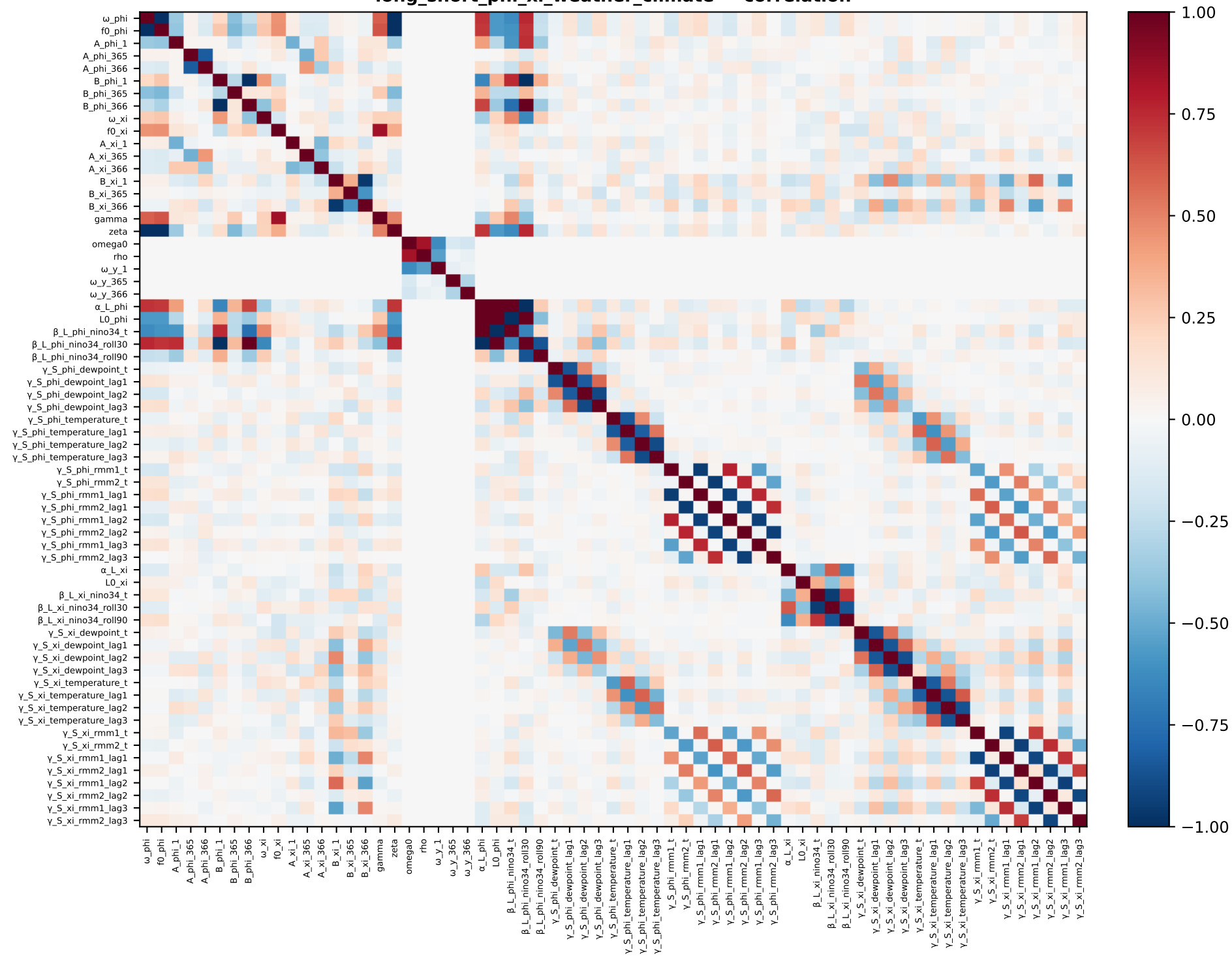
long_short_phi_xi_weather_climate — filtered states (OOS)



long_short_phi_xi_weather_climate — covariance

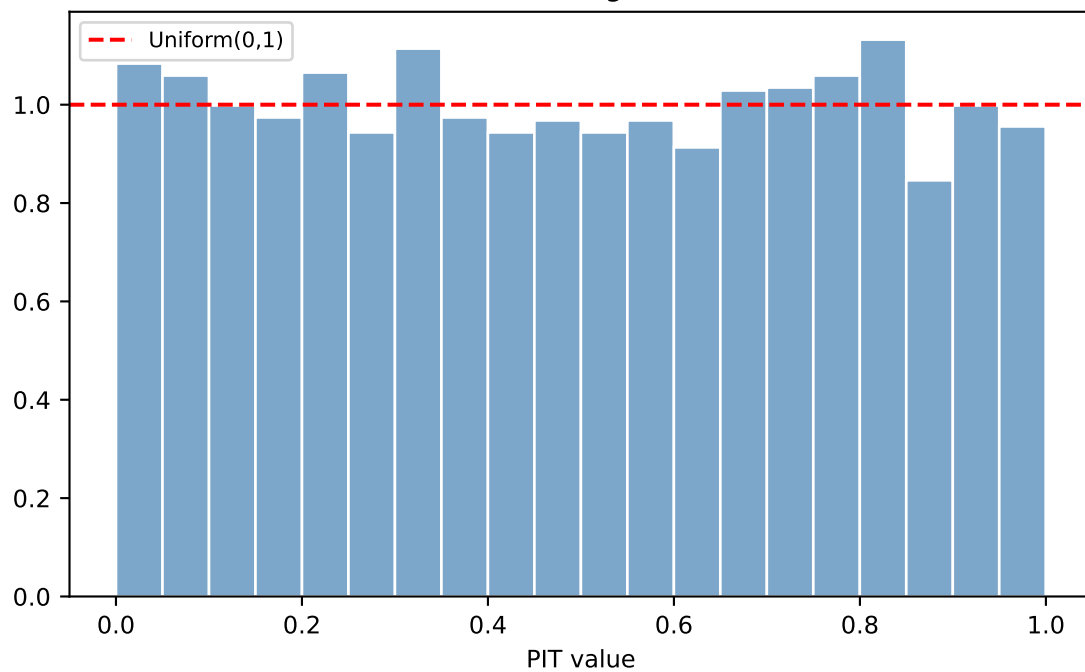


long_short_phi_xi_weather_climate — correlation

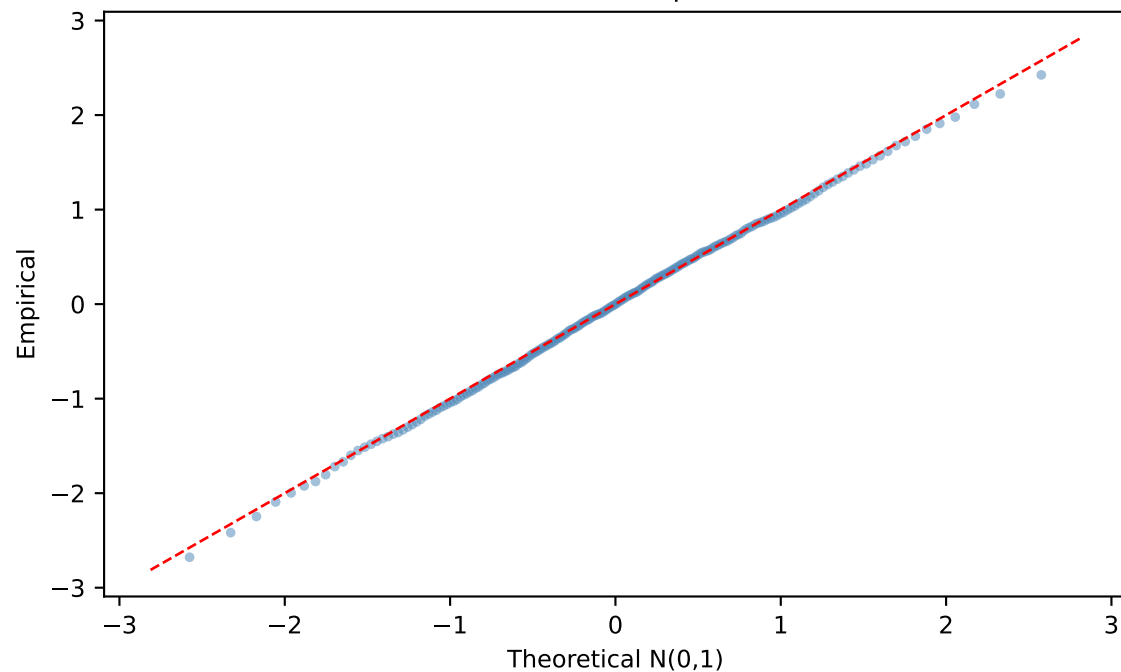


long_short_phi_xi_interactions — IS diagnostics

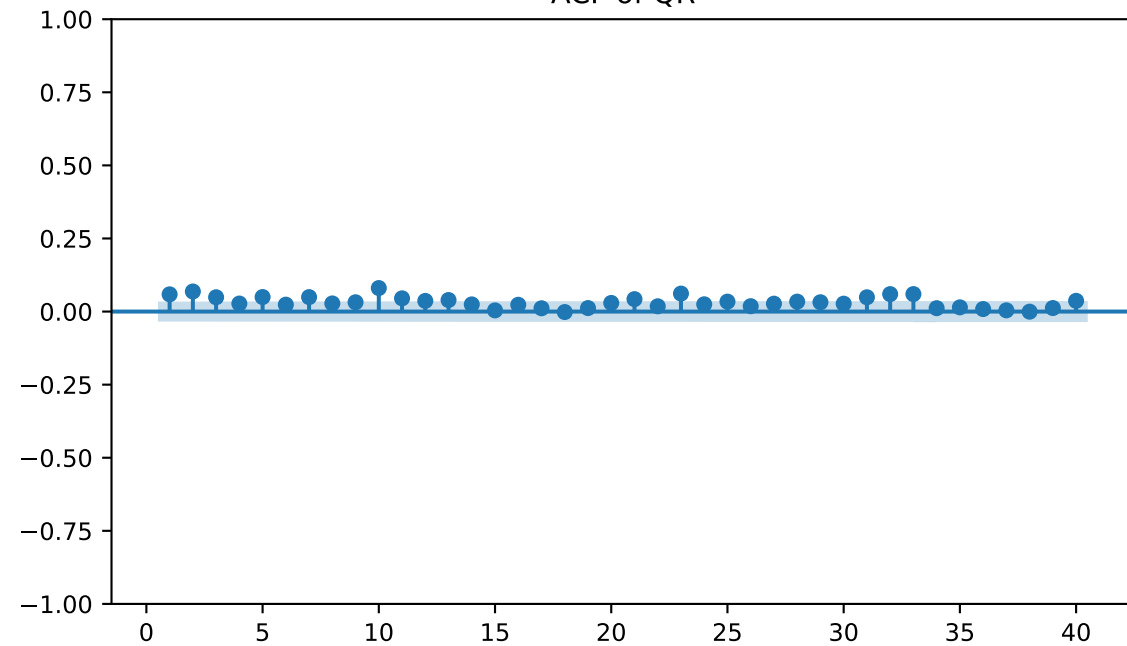
PIT histogram



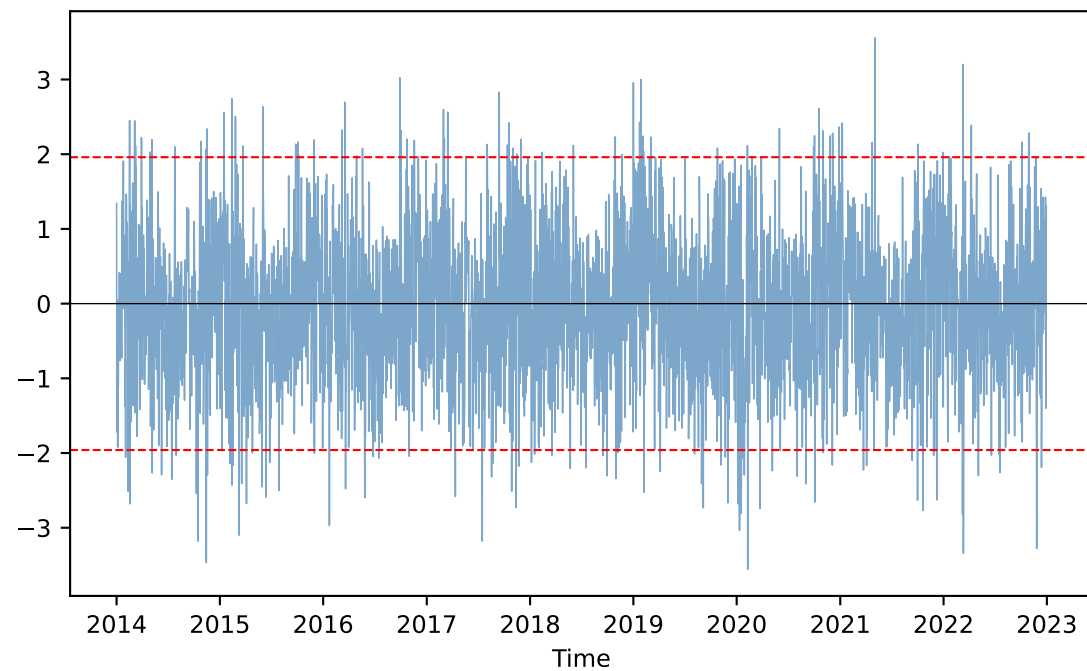
Normal QQ plot



ACF of QR

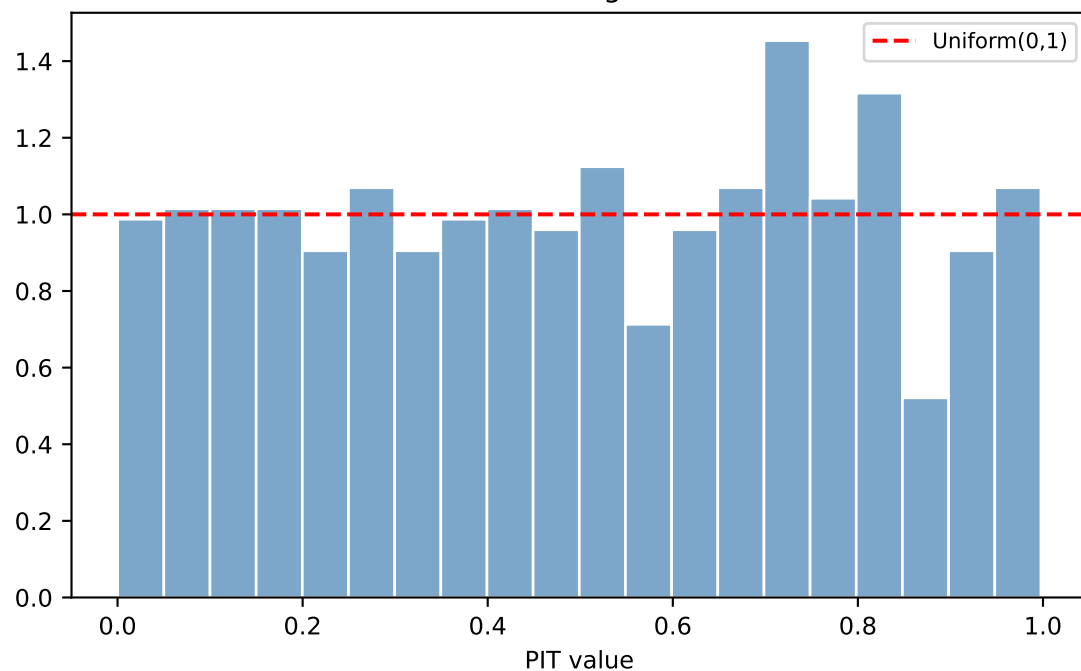


QR time series

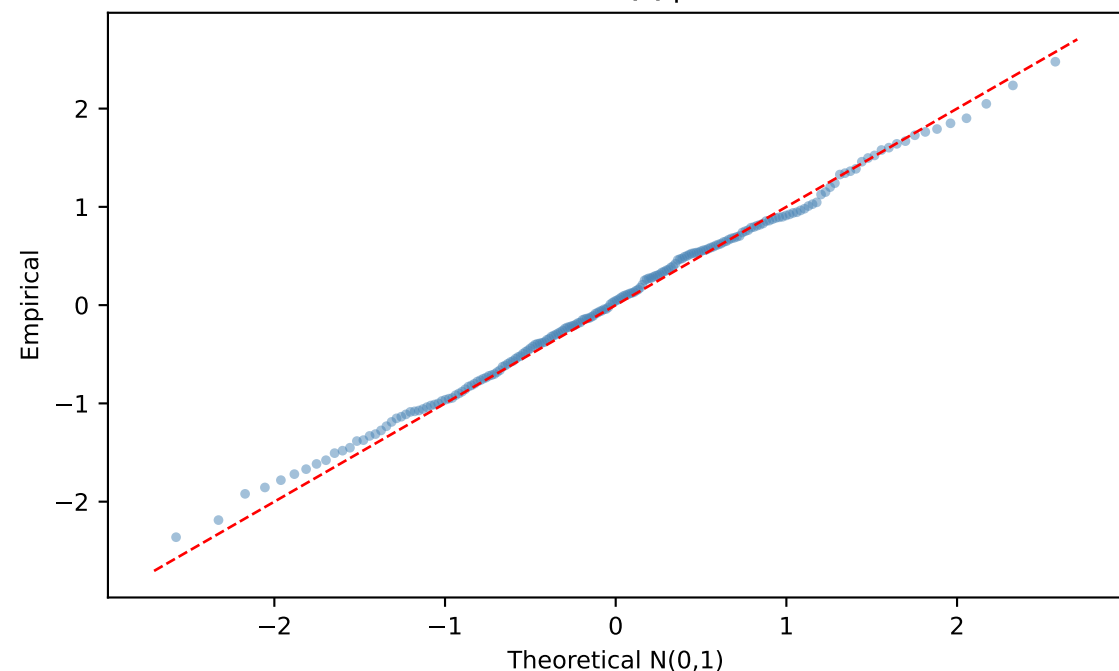


long_short_phi_xi_interactions — OOS diagnostics

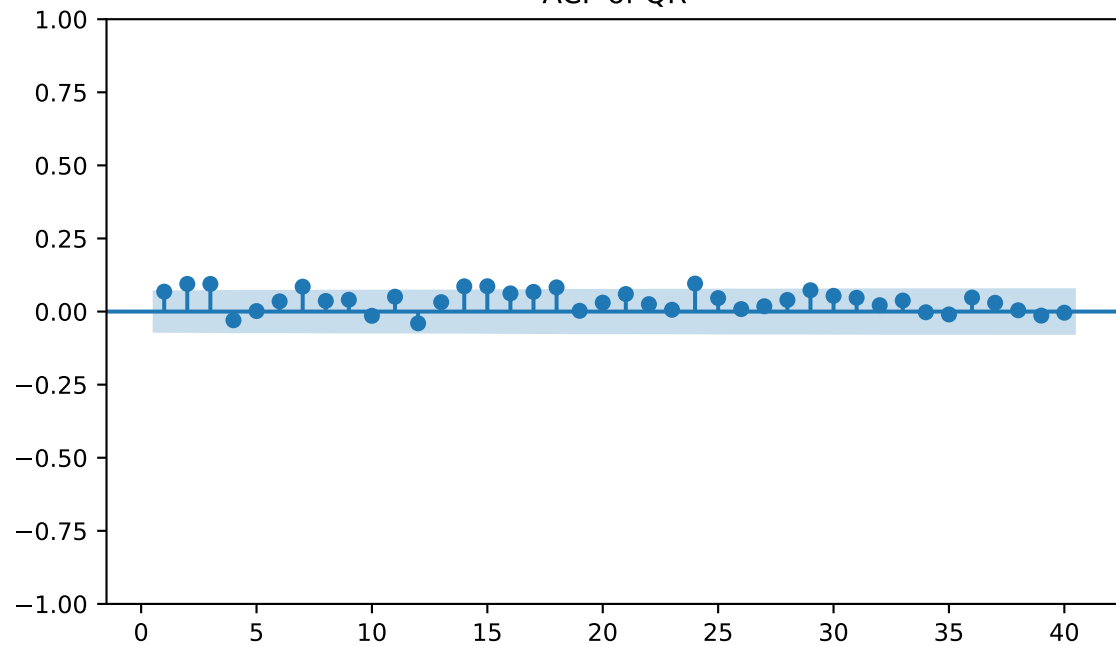
PIT histogram



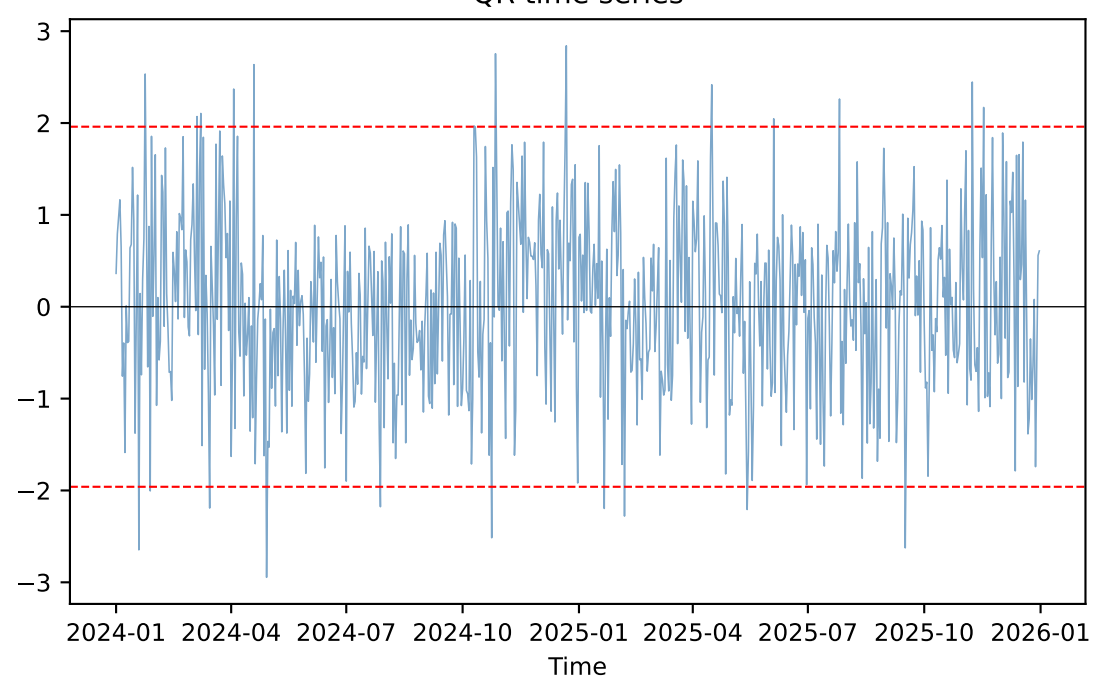
Normal QQ plot



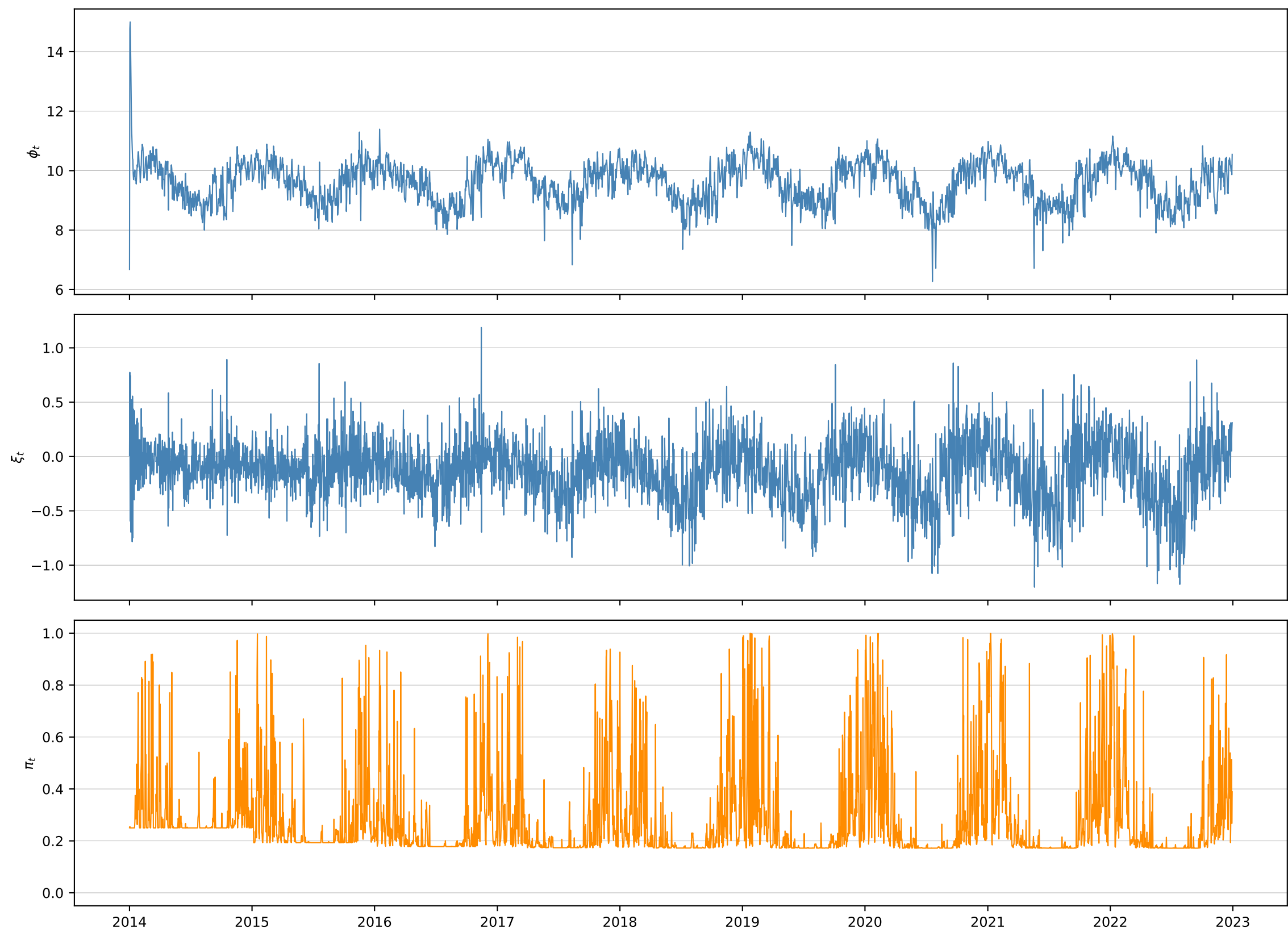
ACF of QR



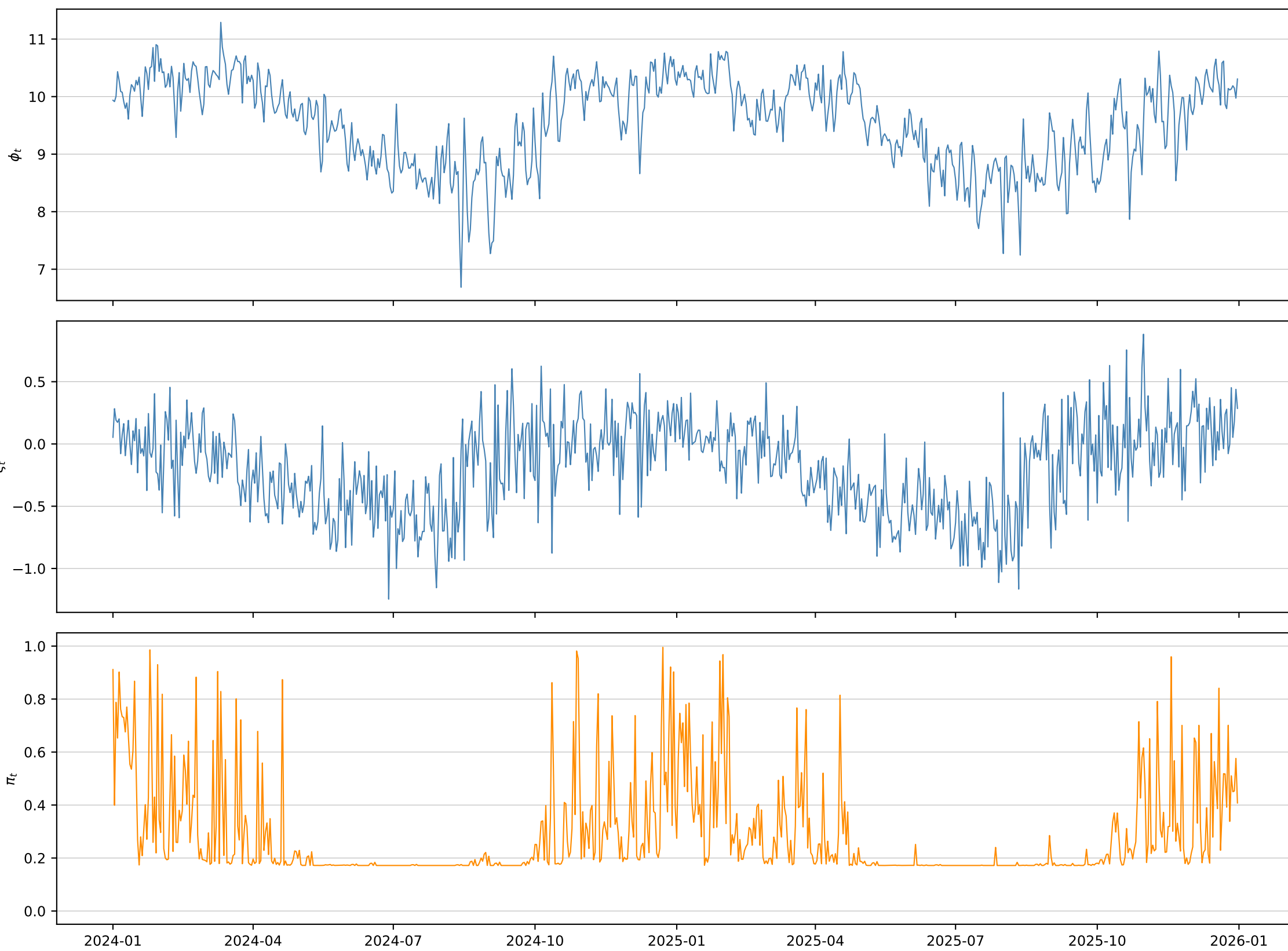
QR time series



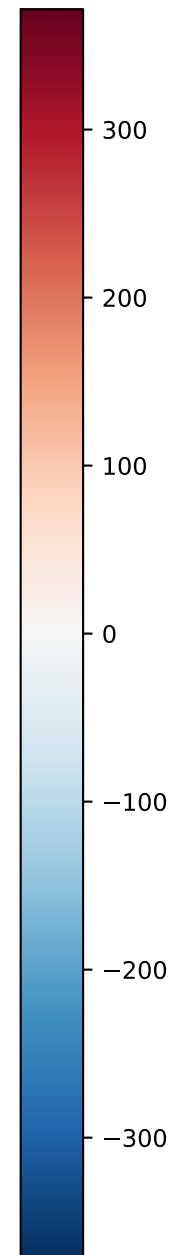
long_short_phi_xi_interactions — filtered states (IS)



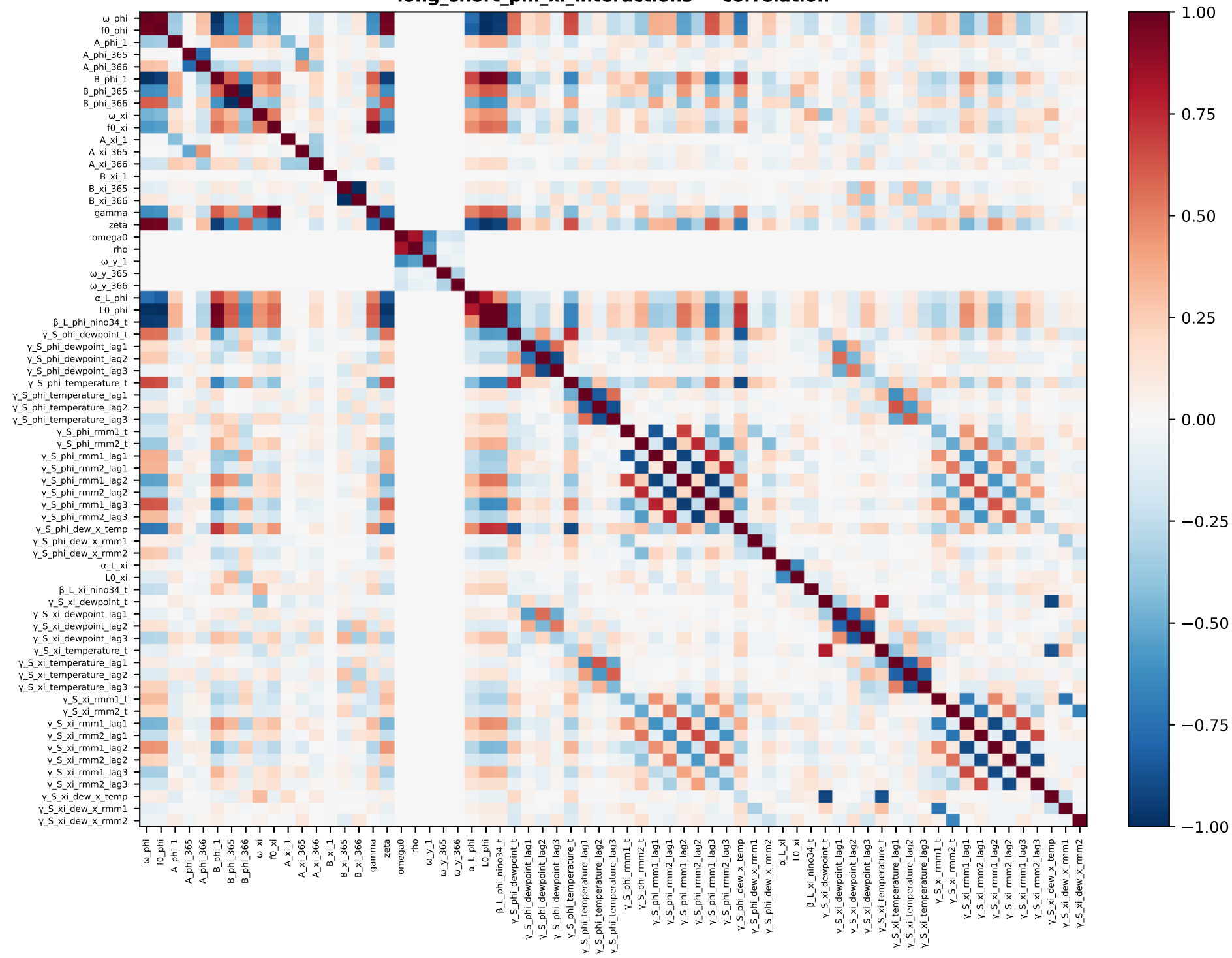
long_short_phi_xi_interactions — filtered states (OOS)



Heatmap visualization showing the correlation matrix for 50 variables. The variables are listed on both the x and y axes. The color scale ranges from -0.5 (dark blue) to 0.5 (dark red), with white representing 0.0. The diagonal is dark blue, indicating a correlation of 1.0. The upper triangle is mostly white, indicating low or no correlation. The lower triangle shows some positive correlations (light blue) and a few negative correlations (light orange/yellow). Notable negative correlations are seen between 'omega_phi' and 'alpha_L_phi', and 'omega_phi' and 'alpha_L_xi'.



long_short_phi_xi_interactions — correlation



Discussion

By OOS CRPS, the best-performing model is 'phi_xi_dewpoint_temperature_seasonal_lags' (CRPS = 2.4476) and the worst is 'xi_mjo_t' (CRPS = 2.7796). The lowest IS AIC belongs to 'phi_dewpoint_temperature_t' (AIC = 2278.8205). No model fails the Kupiec 95\% coverage test. Best tail-weighted CRPS (90% threshold): 'long_short_phi_interactions' (0.3562). Best tail-weighted CRPS (95% threshold): 'long_short_phi_interactions' (0.2619). Models with at least one covariate coefficient significant at 5\%:
'xi_dewpoint_lag1': gamma, gamma_xi_dewpoint_t, gamma_xi_dewpoint_lag1; 'xi_dewpoint_short_lags': gamma, gamma_xi_dewpoint_t, gamma_xi_dewpoint_lag1, gamma_xi_dewpoint_lag2; 'xi_dewpoint_seasonal_lags': gamma_xi_dewpoint_t, gamma_xi_dewpoint_lag1; 'xi_temperature_t': gamma, gamma_xi_temperature_t; 'xi_temperature_lag1': gamma, gamma_xi_temperature_lag1; 'xi_temperature_short_lags': gamma, gamma_xi_temperature_t, gamma_xi_temperature_lag1, gamma_xi_temperature_lag2.

Conclusion

Based on OOS CRPS, the recommended model is 'phi_xi_dewpoint_temperature_seasonal_lags'. For upper-tail performance specifically, 'long_short_phi_interactions' performs best by threshold-weighted CRPS at the 90th percentile. In general, long-short models are expected to improve upper-tail performance when Niño 3.4 carries predictive signal. Standard exogenous models are preferable when the covariate effect is linear and contemporaneous.